



Intelligent Security API (General)

Developer Guide

Legal Information

TO THE MAXIMUM EXTENT PERMITTED BY APPLICABLE LAW, THE DOCUMENT IS PROVIDED "AS IS" AND "WITH ALL FAULTS AND ERRORS". OUR COMPANY MAKES NO REPRESENTATIONS OR WARRANTIES, EXPRESS OR IMPLIED, INCLUDING BUT NOT LIMITED TO, WARRANTIES OF MERCHANTABILITY, FITNESS FOR A PARTICULAR PURPOSE OR NON-INFRINGEMENT. IN NO EVENT WILL OUR COMPANY BE LIABLE FOR ANY SPECIAL, CONSEQUENTIAL, INCIDENTAL, OR INDIRECT DAMAGES, INCLUDING, AMONG OTHERS, DAMAGES FOR LOSS OF BUSINESS PROFITS, BUSINESS INTERRUPTION OR LOSS OF DATA, CORRUPTION OF SYSTEMS, OR LOSS OF DOCUMENTATION, WHETHER BASED ON BREACH OF CONTRACT, TORT (INCLUDING NEGLIGENCE), OR OTHERWISE, IN CONNECTION WITH THE USE OF THE DOCUMENT, EVEN IF OUR COMPANY HAS BEEN ADVISED OF THE POSSIBILITY OF SUCH DAMAGES OR LOSS.

Contents

Chapter 1 Overview	1
1.1 Introduction	1
1.2 Update History	2
1.2.1 Version 20.07-Version 20.09	2
1.2.2 Version 20.04-Version 20.06	5
1.2.3 Version 20.01-Version 20.03	9
Chapter 2 ISAPI Description	13
2.1 Operation Method	13
2.2 URL Format	16
2.3 Message Format	17
2.4 Others	19
Chapter 3 Security	20
3.1 Authentication	20
3.2 User Permission	21
3.3 Encryption	21
3.3.1 Encrypt Stream	23
3.3.2 Encrypt Sensitive Information	25
3.4 Security Service	35
Chapter 4 Login	38
Chapter 5 Device Management	39
5.1 Device Activation	39
5.1.1 Directly Activate Device	39
5.1.2 Activate Network Camera via NVR	40
5.2 Reset Password by Setting Recovery Email	41
5.3 Online Upgrade Device	42
5.4 IoT Devices Access	44

5.4.1 Add IoT Devices to NVR/DVR	45
5.4.2 Configure Alarm of IoT Device	47
Chapter 6 Device Configuration and Maintenance	50
6.1 Basic	50
6.1.1 Securely Export and Import Configuration Files	51
6.2 Log	53
6.3 Status	53
6.4 Calibration	55
6.5 Auto Power On or Off	56
6.6 Advanced	56
Chapter 7 User Management	59
7.1 Configure Double Verification	60
Chapter 8 Video and Audio	65
8.1 Search for Recording Start and End Time by Channel	65
8.2 Live View and Playback	66
8.3 Start Two-Way Audio	73
8.4 Stream Encoding	74
Chapter 9 Capture and Recording	76
Chapter 10 PTZ Control	78
10.1 Control PTZ via Digital Channel	82
10.2 Configure Master-Slave Tracking	83
10.3 Configure Panorama Tracking	85
Chapter 11 Storage Management	86
11.1 Configure Log Storage Mode	86
11.2 HDD Management	87
11.3 Working Mode Configuration	89
11.4 SSD Management	90
11.5 Others	91

Chapter 12 File Management	92
12.1 Video Files	92
12.2 Audio Files	94
12.3 Log Files	95
Chapter 13 Alarm/Event	96
13.1 Configure Audio Detection Alarm	96
13.2 Configure Exception Alarm	99
13.3 Configure Certificate Expiry Alarm	99
13.4 Configure Network Traffic Pre-alarm	102
13.5 Configure Motion Detection Alarm	103
13.6 Configure Vibration Detection Alarm	105
13.7 Receive Alarm/Event in Arming Mode	108
13.8 Receive Alarm/Event in Listening Mode	109
13.9 Subscribe Alarm/Event in Arming Mode	111
Chapter 14 Parameter Configuration	114
14.1 I/O Settings	114
14.2 Video/Image Settings	116
14.3 Audio Settings	121
14.4 Channel Settings	122
14.5 Peripherals Settings	124
14.6 Picture Storage Server Settings	125
Chapter 15 System Configuration	126
15.1 Network Settings	126
15.1.1 Email	127
15.1.2 Network Access	127
15.1.3 Network Interface	130
15.1.4 Wireless Network	131
15.1.5 Network Service	134

15.2 Time Settings	136
15.3 System Diagnose	137
15.4 Data Replenishment	137
Appendix A. Request URIs	140
A.1 /ISAPI/AUXInfo	140
A.1.1 /ISAPI/AUXInfo/attributes/Channels	140
A.1.2 /ISAPI/AUXInfo/attributes/Channels/<ID>	140
A.2 /ISAPI/ContentMgmt	141
A.2.1 /ISAPI/ContentMgmt/capabilities	141
A.2.2 /ISAPI/ContentMgmt/download	141
A.2.3 /ISAPI/ContentMgmt/download/capabilities	141
A.2.4 /ISAPI/ContentMgmt/download/toUSB/<taskId>/progress?format=json	142
A.2.5 /ISAPI/ContentMgmt/download/toUSB/capabilities?format=json	142
A.2.6 /ISAPI/ContentMgmt/download/toUSB?format=json	143
A.2.7 /ISAPI/ContentMgmt/InputProxy/channels	143
A.2.8 /ISAPI/ContentMgmt/InputProxy/channels/<ID>	144
A.2.9 /ISAPI/ContentMgmt/InputProxy/channels/<ID>/capabilities	145
A.2.10 /ISAPI/ContentMgmt/InputProxy/channels/<ID>/chanCtrl	145
A.2.11 /ISAPI/ContentMgmt/InputProxy/channels/<ID>/deviceInfo	146
A.2.12 /ISAPI/ContentMgmt/InputProxy/channels/<ID>/portMapParam	147
A.2.13 /ISAPI/ContentMgmt/InputProxy/channels/<ID>/reboot	147
A.2.14 /ISAPI/ContentMgmt/InputProxy/channels/<ID>/status	148
A.2.15 /ISAPI/ContentMgmt/InputProxy/channels/<ID>/video/overlays	148
A.2.16 /ISAPI/ContentMgmt/InputProxy/channels/<ID>/video/overlays/capabilities	149
A.2.17 /ISAPI/ContentMgmt/InputProxy/channels/activate	149
A.2.18 /ISAPI/ContentMgmt/InputProxy/channels/activate/capabilities	150
A.2.19 /ISAPI/ContentMgmt/InputProxy/channels/capabilities	150
A.2.20 /ISAPI/ContentMgmt/InputProxy/channels/status	151

A.2.21 /ISAPI/ContentMgmt/InputProxy/deviceInfo	151
A.2.22 /ISAPI/ContentMgmt/InputProxy/search	151
A.2.23 /ISAPI/ContentMgmt/InputProxy/sourceCapability	152
A.2.24 /ISAPI/ContentMgmt/IOProxy/inputs	152
A.2.25 /ISAPI/ContentMgmt/IOProxy/inputs/<ID>	153
A.2.26 /ISAPI/ContentMgmt/IOProxy/inputs/<ID>/status	154
A.2.27 /ISAPI/ContentMgmt/IOProxy/outputs	154
A.2.28 /ISAPI/ContentMgmt/IOProxy/outputs/<ID>	155
A.2.29 /ISAPI/ContentMgmt/IOProxy/outputs/<ID>/status	156
A.2.30 /ISAPI/ContentMgmt/IOProxy/outputs/<ID>/trigger	156
A.2.31 /ISAPI/ContentMgmt/logConfig	157
A.2.32 /ISAPI/ContentMgmt/logConfig/capabilities	157
A.2.33 /ISAPI/ContentMgmt/logSearch	158
A.2.34 /ISAPI/ContentMgmt/logSearch?format=json	158
A.2.35 /ISAPI/ContentMgmt/logSearch/dataPackage	158
A.2.36 /ISAPI/ContentMgmt/PTZCtrlProxy/capabilities	159
A.2.37 /ISAPI/ContentMgmt/PTZCtrlProxy/channels	159
A.2.38 /ISAPI/ContentMgmt/PTZCtrlProxy/channels/<ID>	160
A.2.39 /ISAPI/ContentMgmt/PTZCtrlProxy/channels/<ID>/autoPan	161
A.2.40 /ISAPI/ContentMgmt/PTZCtrlProxy/channels/<ID>/capabilities	161
A.2.41 /ISAPI/ContentMgmt/PTZCtrlProxy/channels/<ID>/continuous	161
A.2.42 /ISAPI/ContentMgmt/PTZCtrlProxy/channels/<ID>/momentary	162
A.2.43 /ISAPI/ContentMgmt/PTZCtrlProxy/channels/<ID>/onpushfoucs/start	162
A.2.44 /ISAPI/ContentMgmt/PTZCtrlProxy/channels/<ID>/patrols	163
A.2.45 /ISAPI/ContentMgmt/PTZCtrlProxy/channels/<ID>/patrols/<ID>	164
A.2.46 /ISAPI/ContentMgmt/PTZCtrlProxy/channels/<ID>/patrols/<ID>/pause	165
A.2.47 /ISAPI/ContentMgmt/PTZCtrlProxy/channels/<ID>/patrols/<ID>/start	165
A.2.48 /ISAPI/ContentMgmt/PTZCtrlProxy/channels/<ID>/patrols/<ID>/status	166

A.2.49 /ISAPI/ContentMgmt/PTZCtrlProxy/channels/<ID>/patrols/<ID>/stop	166
A.2.50 /ISAPI/ContentMgmt/PTZCtrlProxy/channels/<ID>/position3D	167
A.2.51 /ISAPI/ContentMgmt/PTZCtrlProxy/channels/<ID>/presets	167
A.2.52 /ISAPI/ContentMgmt/PTZCtrlProxy/channels/<ID>/presets/<ID>	168
A.2.53 /ISAPI/ContentMgmt/PTZCtrlProxy/channels/<ID>/presets/<ID>/goto	169
A.2.54 /ISAPI/ContentMgmt/record/control/cleanUp/capabilities?format=json	169
A.2.55 /ISAPI/ContentMgmt/record/control/cleanUp?format=json	170
A.2.56 /ISAPI/ContentMgmt/record/control/locks/capabilities	170
A.2.57 /ISAPI/ContentMgmt/record/control/locks/name	171
A.2.58 /ISAPI/ContentMgmt/record/control/manual/start/tracks/<ID>	171
A.2.59 /ISAPI/ContentMgmt/record/control/manual/stop/tracks/<ID>	172
A.2.60 /ISAPI/ContentMgmt/record/control/manualRefresh/channels/<ID>	172
A.2.61 /ISAPI/ContentMgmt/record/control/remark/capabilities?format=json	173
A.2.62 /ISAPI/ContentMgmt/record/control/remark?format=json	173
A.2.63 /ISAPI/ContentMgmt/record/profile	173
A.2.64 /ISAPI/ContentMgmt/record/storageMounts	174
A.2.65 /ISAPI/ContentMgmt/record/tracks	174
A.2.66 /ISAPI/ContentMgmt/record/tracks/<ID>	175
A.2.67 /ISAPI/ContentMgmt/record/tracks/<ID>/capabilities	176
A.2.68 /ISAPI/ContentMgmt/record/tracks/<ID>/dailyDistribution	176
A.2.69 /ISAPI/ContentMgmt/search	177
A.2.70 /ISAPI/ContentMgmt/search/profile	179
A.2.71 /ISAPI/ContentMgmt/security/logSearch	179
A.2.72 /ISAPI/ContentMgmt/SmartSearch	179
A.2.73 /ISAPI/ContentMgmt/SmartSearch/capabilities	180
A.2.74 /ISAPI/ContentMgmt/storage	180
A.2.75 /ISAPI/ContentMgmt/Storage/capabilities	181
A.2.76 /ISAPI/ContentMgmt/Storage/dataReconstruction/progress?format=json	181

A.2.77 /ISAPI/ContentMgmt/Storage/dataReconstruction?format=json	182
A.2.78 /ISAPI/ContentMgmt/Storage/extension	182
A.2.79 /ISAPI/ContentMgmt/Storage/extension/capabilities	183
A.2.80 /ISAPI/ContentMgmt/Storage/ExtraInfo	183
A.2.81 /ISAPI/ContentMgmt/Storage/ExtraInfo/capabilities	184
A.2.82 /ISAPI/ContentMgmt/Storage/hdd	184
A.2.83 /ISAPI/ContentMgmt/Storage/hdd/<ID>	184
A.2.84 /ISAPI/ContentMgmt/Storage/hdd/<ID>/BadSectorsTest/pause	185
A.2.85 /ISAPI/ContentMgmt/Storage/hdd/<ID>/BadSectorsTest/resume	185
A.2.86 /ISAPI/ContentMgmt/Storage/hdd/<ID>/BadSectorsTest/start	186
A.2.87 /ISAPI/ContentMgmt/Storage/hdd/<ID>/BadSectorsTest/status	186
A.2.88 /ISAPI/ContentMgmt/Storage/hdd/<ID>/BadSectorsTest/stop	187
A.2.89 /ISAPI/ContentMgmt/Storage/hdd/<ID>/encryptFormat?format=json	187
A.2.90 /ISAPI/ContentMgmt/Storage/hdd/<ID>/encryptVerify?format=json	188
A.2.91 /ISAPI/ContentMgmt/Storage/hdd/<ID>/formatStatus	188
A.2.92 /ISAPI/ContentMgmt/Storage/hdd/<ID>/format?formatType=	189
A.2.93 /ISAPI/ContentMgmt/Storage/hdd/<ID>/logProTest?format=json	189
A.2.94 /ISAPI/ContentMgmt/Storage/hdd/<ID>/SMARTTest/start	190
A.2.95 /ISAPI/ContentMgmt/Storage/hdd/<ID>/SMARTTest/status	190
A.2.96 /ISAPI/ContentMgmt/Storage/hdd/capabilities	191
A.2.97 /ISAPI/ContentMgmt/storage/hdd/clearingSpace/capabilities?format=json	191
A.2.98 /ISAPI/ContentMgmt/storage/hdd/clearingSpace?format=json	192
A.2.99 /ISAPI/ContentMgmt/Storage/hdd/format	192
A.2.100 /ISAPI/ContentMgmt/Storage/hdd/SMARTTest/config	193
A.2.101 /ISAPI/ContentMgmt/Storage/hdd/<ID>/syncStatus?format=json	193
A.2.102 /ISAPI/ContentMgmt/Storage/hdd/specifyHddFormat?format=json	194
A.2.103 /ISAPI/ContentMgmt/Storage/quota	194
A.2.104 /ISAPI/ContentMgmt/Storage/quota/<ID>	194

A.2.105 /ISAPI/ContentMgmt/Storage/ssd	195
A.2.106 /ISAPI/ContentMgmt/Storage/ssd/<ID>	195
A.2.107 /ISAPI/ContentMgmt/Storage/ssd/<ID>/capabilities?format=json	196
A.2.108 /ISAPI/ContentMgmt/Storage/ssd/<ID>/SMARTTest/status	196
A.2.109 /ISAPI/ContentMgmt/Storage/ssd/<ID>/SMARTTest/start	197
A.2.110 /ISAPI/ContentMgmt/Storage/ssd/<ID>/upgrade/status?format=json	197
A.2.111 /ISAPI/ContentMgmt/Storage/ssd/<ID>/upgrade?format=json	198
A.2.112 /ISAPI/ContentMgmt/StreamingProxy/channels/<ID>/Picture	198
A.2.113 /ISAPI/ContentMgmt/StreamingProxy/channels/<ID>/capabilities	199
A.2.114 /ISAPI/ContentMgmt/StreamingProxy/channels/<ID>/PictureByUrl/capabilities? format=json	200
A.2.115 /ISAPI/ContentMgmt/StreamingProxy/channels/<ID>/PictureByUrl?format=json	200
A.2.116 /ISAPI/ContentMgmt/time/search/capabilities?format=json	201
A.2.117 /ISAPI/ContentMgmt/time/search?format=json	201
A.3 /ISAPI/Event	202
A.3.1 /ISAPI/Event/capabilities	202
A.3.2 /ISAPI/Event/channels/<ID>/capabilities	202
A.3.3 /ISAPI/Event/channels/capabilities	202
A.3.4 /ISAPI/Event/IOT/channels/<ID>/capabilities?format=json	203
A.3.5 /ISAPI/Event/notification/alertStream	203
A.3.6 /ISAPI/Event/notification/httpHosts	204
A.3.7 /ISAPI/Event/notification/httpHosts/<ID>	205
A.3.8 /ISAPI/Event/notification/httpHosts/<ID>/test	207
A.3.9 /ISAPI/Event/notification/httpHosts/capabilities	207
A.3.10 /ISAPI/Event/notification/subscribeEvent	208
A.3.11 /ISAPI/Event/notification/subscribeEvent/<ID>	208
A.3.12 /ISAPI/Event/notification/subscribeEventCap	209
A.3.13 /ISAPI/Event/notification/unSubscribeEvent	209

A.3.14 /ISAPI/Event/schedules/motionDetections	210
A.3.15 /ISAPI/Event/schedules/motionDetections/<ID>	210
A.3.16 /ISAPI/Event/schedules/vibrationDetection/<ID>	211
A.3.17 /ISAPI/Event/schedules/<EventType>/<ID>	212
A.3.18 /ISAPI/Event/triggers/hdBadBlock	212
A.3.19 /ISAPI/Event/triggers/hdImpact	213
A.3.20 /ISAPI/Event/triggers/highHDTemperature	213
A.3.21 /ISAPI/Event/triggers/lowHDTemperature	214
A.3.22 /ISAPI/Event/triggers/severeHDFailure	215
A.3.23 /ISAPI/Event/triggers/<eventType>-<ID>	215
A.3.24 /ISAPI/Event/triggers/<ID>/notifications	216
A.3.25 /ISAPI/Event/triggersCap/IOT	217
A.3.26 /ISAPI/Event/triggers/notifications/AudioAlarm/<audioID>/test?format=json	217
A.3.27 /ISAPI/Event/triggers/notifications/AudioAlarm/capabilities?format=json	218
A.3.28 /ISAPI/Event/triggers/notifications/AudioAlarm/customAudio/<customAudioID>? format=json	218
A.3.29 /ISAPI/Event/triggers/notifications/AudioAlarm/customAudio?format=json	219
A.3.30 /ISAPI/Event/triggers/notifications/AudioAlarm/customAudioInfo/ <customAudioID>?format=json	219
A.3.31 /ISAPI/Event/triggers/notifications/AudioAlarm/customAudioInfo?format=json ..	220
A.3.32 /ISAPI/Event/triggers/notifications/AudioAlarm/customAudioTypeInfo/channels? format=json	221
A.3.33 /ISAPI/Event/triggers/notifications/AudioAlarm?format=json	221
A.3.34 /ISAPI/Event/triggers/notifications/whiteLightAlarm?format=json	222
A.3.35 /ISAPI/Event/triggers/notifications/whiteLightAlarm/capabilities?format=json	222
A.3.36 /ISAPI/Event/triggersCap	223
A.3.37 http://<ipAddress>:<portNo>/<url>	223
A.4 /ISAPI/Image	224
A.4.1 /ISAPI/Image/channels	224

A.4.2 /ISAPI/Image/channels/<ID>	224
A.4.3 /ISAPI/Image/channels/<ID>/BLC	225
A.4.4 /ISAPI/Image/channels/<ID>/capabilities	226
A.4.5 /ISAPI/Image/channels/<ID>/capturemode	226
A.4.6 /ISAPI/Image/channels/<ID>/color	227
A.4.7 /ISAPI/Image/channels/<ID>/color/capabilities	228
A.4.8 /ISAPI/Image/channels/<ID>/corridor	228
A.4.9 /ISAPI/Image/channels/<ID>/dehaze	229
A.4.10 /ISAPI/Image/channels/<ID>/EPTZ	229
A.4.11 /ISAPI/Image/channels/<ID>/EPTZ/mode/capabilities?format=json	230
A.4.12 /ISAPI/Image/channels/<ID>/EPTZ/mode?format=json	231
A.4.13 /ISAPI/Image/channels/<ID>/exposure	231
A.4.14 /ISAPI/Image/channels/<ID>/focusConfiguration	232
A.4.15 /ISAPI/Image/channels/<ID>/focusConfiguration/capabilities	232
A.4.16 /ISAPI/Image/channels/<ID>/gain	233
A.4.17 /ISAPI/Image/channels/<ID>/imageCap	234
A.4.18 /ISAPI/Image/channels/<ID>/ImageFlip	234
A.4.19 /ISAPI/Image/channels/<ID>/imageMode/<ID>	235
A.4.20 /ISAPI/Image/channels/<ID>/imageModes	235
A.4.21 /ISAPI/Image/channels/<ID>/IrcutFilter	236
A.4.22 /ISAPI/Image/channels/<ID>/ISPMode	236
A.4.23 /ISAPI/Image/channels/<ID>/lensDistortionCorrection	237
A.4.24 /ISAPI/Image/channels/<ID>/lensDistortionCorrection/capabilities	238
A.4.25 /ISAPI/Image/channels/<ID>/mountingScenario	238
A.4.26 /ISAPI/Image/channels/<ID>/mountingScenario/capabilities	239
A.4.27 /ISAPI/Image/channels/<ID>/noiseReduce	239
A.4.28 /ISAPI/Image/channels/<ID>/Palettes	240
A.4.29 /ISAPI/Image/channels/<ID>/powerLineFrequency	241

A.4.30 /ISAPI/Image/channels/<ID>/regionalFocus	241
A.4.31 /ISAPI/Image/channels/<ID>/reset	242
A.4.32 /ISAPI/Image/channels/<ID>/restore	242
A.4.33 /ISAPI/Image/channels/<ID>/sharpness	243
A.4.34 /ISAPI/Image/channels/<ID>/shutter	244
A.4.35 /ISAPI/Image/channels/<ID>/supplementLight	244
A.4.36 /ISAPI/Image/channels/<ID>/supplementLight/capabilities	245
A.4.37 /ISAPI/Image/channels/<ID>/targetEnhancement	245
A.4.38 /ISAPI/Image/channels/<ID>/targetEnhancement/capabilities	246
A.4.39 /ISAPI/Image/channels/<ID>/tempRange	247
A.4.40 /ISAPI/Image/channels/<ID>/tempRange/capabilities	247
A.4.41 /ISAPI/Image/channels/<ID>/WDR	248
A.4.42 /ISAPI/Image/channels/<ID>/whiteBalance	249
A.4.43 /ISAPI/Image/channels/capabilities	249
A.4.44 /ISAPI/Image/channels/imageModes	250
A.4.45 /ISAPI/Image/channels/imageModes/<ID>	250
A.5 /ISAPI/MasterSlaveTracking	251
A.5.1 /ISAPI/MasterSlaveTracking/capabilities	251
A.5.2 /ISAPI/MasterSlaveTracking/channels/<ID>/slaveCamera/<ID>	251
A.5.3 /ISAPI/MasterSlaveTracking/channels/<ID>/slaveCamera/<ID>/tracking	252
A.5.4 /ISAPI/MasterSlaveTracking/channels/<ID>/slaveCamera/<ID>/trackingRatio	252
A.5.5 /ISAPI/MasterSlaveTracking/channels/<ID>/slaveCamera/capabilities	253
A.5.6 /ISAPI/MasterSlaveTracking/channels/<ID>/slaveCameraStatus	253
A.5.7 /ISAPI/MasterSlaveTracking/channels/<ID>/tracking/capabilities	254
A.5.8 /ISAPI/MasterSlaveTracking/linkedTracking/capabilities?format=json	254
A.5.9 /ISAPI/MasterSlaveTracking/linkedTracking?format=json	254
A.6 /ISAPI/SDT	255
A.6.1 /ISAPI/SDT/Management/capabilities?format=json	255

A.6.2 /ISAPI/SDT/Management/EventSearch?format=json	255
A.6.3 /ISAPI/SDT/Management/EventSearch/capabilities?format=json	256
A.6.4 /ISAPI/SDT/Management/IntelligentSearch/capabilities?format=json	256
A.6.5 /ISAPI/SDT/Management/IntelligentSearch?format=json	257
A.7 /ISAPI/PTZCtrl	257
A.7.1 /ISAPI/PTZCtrl/channels/<ID>	257
A.7.2 /ISAPI/PTZCtrl/channels/<ID>/auxcontrols	258
A.7.3 /ISAPI/PTZCtrl/channels/<ID>/auxcontrols/<ID>	258
A.7.4 /ISAPI/PTZCtrl/channels/<ID>/capabilities	259
A.7.5 /ISAPI/PTZCtrl/channels/<ID>/EagleFocusing/auto/capabilities?format=json	260
A.7.6 /ISAPI/PTZCtrl/channels/<ID>/EagleFocusing/auto?format=json	260
A.7.7 /ISAPI/PTZCtrl/channels/<ID>/EagleFocusing/capabilities	260
A.7.8 /ISAPI/PTZCtrl/channels/<ID>/GPS?format=json	261
A.7.9 /ISAPI/PTZCtrl/channels/<ID>/gyroVerify/capabilities?format=json	261
A.7.10 /ISAPI/PTZCtrl/channels/<ID>/gyroVerify?format=json	262
A.7.11 /ISAPI/PTZCtrl/channels/<ID>/lensCorrection/capabilities?format=json	262
A.7.12 /ISAPI/PTZCtrl/channels/<ID>/lensCorrection?format=json	263
A.7.13 /ISAPI/PTZCtrl/channels/<ID>/lockPTZ	263
A.7.14 /ISAPI/PTZCtrl/channels/<ID>/maxelevation	264
A.7.15 /ISAPI/PTZCtrl/channels/<ID>/maxelevation/capabilities	265
A.7.16 /ISAPI/PTZCtrl/channels/<ID>/moveAutoTracking?format=json	265
A.7.17 /ISAPI/PTZCtrl/channels/<ID>/onpushfoucs/reset	266
A.7.18 /ISAPI/PTZCtrl/channels/<ID>/parkaction	266
A.7.19 /ISAPI/PTZCtrl/channels/<ID>/parkaction/capabilities	267
A.7.20 /ISAPI/PTZCtrl/channels/<ID>/presets	267
A.7.21 /ISAPI/PTZCtrl/channels/<ID>/presets/<ID>	268
A.7.22 /ISAPI/PTZCtrl/channels/<ID>/PTZOSDDisplay	269
A.7.23 /ISAPI/PTZCtrl/channels/<ID>/save	270

A.7.24 /ISAPI/PTZCtrl/channels/<ID>/save?format=json	270
A.7.25 /ISAPI/PTZCtrl/channels/<ID>/status	271
A.7.26 /ISAPI/PTZCtrl/channels/<ID>/trackingRatio/capabilities?format=json	271
A.7.27 /ISAPI/PTZCtrl/channels/<ID>/trackingRatio?format=json	271
A.7.28 /ISAPI/PTZCtrl/channels/<ID>/zoomFocus	272
A.8 /ISAPI/Security	273
A.8.1 /ISAPI/Security/adminAccesses	273
A.8.2 /ISAPI/Security/adminAccesses/<ID>	273
A.8.3 /ISAPI/Security/adminAccesses/capabilities	274
A.8.4 /ISAPI/Security/advanced?format=json	275
A.8.5 /ISAPI/Security/capabilities	275
A.8.6 /ISAPI/Security/certificate/select/<functinName>?format=json	276
A.8.7 /ISAPI/Security/certificate/select/capabilities?format=json	276
A.8.8 /ISAPI/Security/challenge	277
A.8.9 /ISAPI/Security/CommuMode/capabilities?format=json	277
A.8.10 /ISAPI/Security/CommuMode?format=json	277
A.8.11 /ISAPI/Security/deviceCertificate	278
A.8.12 /ISAPI/Security/deviceCertificate/capabilities?format=json	279
A.8.13 /ISAPI/Security/deviceCertificate/certificateRevocation/capabilities?format=json	279
A.8.14 /ISAPI/Security/deviceCertificate/certificateRevocation?format=json	280
A.8.15 /ISAPI/Security/deviceCertificate/certificates/<customID>?format=json	280
A.8.16 /ISAPI/Security/deviceCertificate/certificates/<ID>	281
A.8.17 /ISAPI/Security/deviceCertificate/certificates/<ID>/recreate?format=json	281
A.8.18 /ISAPI/Security/deviceCertificate/certificates/<ID>/status?format=json	282
A.8.19 /ISAPI/Security/deviceCertificate/certificates/capabilities?format=json	282
A.8.20 /ISAPI/Security/deviceCertificate/certificates/recreate?format=json	283
A.8.21 /ISAPI/Security/deviceCertificate/certificates/status?format=json	283

A.8.22 /ISAPI/Security/deviceCertificate/certificates?format=json	283
A.8.23 /ISAPI/Security/deviceCertificate/selfSignCert	284
A.8.24 /ISAPI/Security/deviceCertificate?customID=	284
A.8.25 /ISAPI/Security/doubleVerification/users/<ID>?format=json	285
A.8.26 /ISAPI/Security/doubleVerification/users/capabilities?format=json	286
A.8.27 /ISAPI/Security/doubleVerification/users?format=json	287
A.8.28 /ISAPI/Security/doubleVerification/UsersPermission/<ID>?format=json	288
A.8.29 /ISAPI/Security/doubleVerification/UsersPermission/capabilities?format=json	288
A.8.30 /ISAPI/Security/doubleVerification?format=json	289
A.8.31 /ISAPI/Security/email/parameter/capabilities?format=json	289
A.8.32 /ISAPI/Security/email/parameter?format=json	290
A.8.33 /ISAPI/Security/email/qrCode?format=json	291
A.8.34 /ISAPI/Security/emailCertification?format=json	292
A.8.35 /ISAPI/Security/extern/capabilities	292
A.8.36 /ISAPI/Security/illegalLoginLock	292
A.8.37 /ISAPI/Security/loginLinkNum?format=json	293
A.8.38 /ISAPI/Security/onlineUser	294
A.8.39 /ISAPI/Security/questionConfiguration	294
A.8.40 /ISAPI/Security/RTSPCertificate	295
A.8.41 /ISAPI/Security/RTSPCertificate/capabilities	296
A.8.42 /ISAPI/Security/serverCertificate/capabilities?format=json	296
A.8.43 /ISAPI/Security/serverCertificate/certificate	296
A.8.44 /ISAPI/Security/serverCertificate/certificate?customID=	297
A.8.45 /ISAPI/Security/serverCertificate/certificates?format=json	298
A.8.46 /ISAPI/Security/serverCertificate/certificates/<customID>?format=json	299
A.8.47 /ISAPI/Security/serverCertificate/certSignReq	299
A.8.48 /ISAPI/Security/serverCertificate/certSignReq?customID=	300
A.8.49 /ISAPI/Security/serverCertificate/downloadCertSignReq	301

A.8.50 /ISAPI/Security/serverCertificate/downloadCertSignReq?customID=	301
A.8.51 /ISAPI/Security/serverCertificate/selfSignCert?customID=	302
A.8.52 /ISAPI/Security/userCheck	302
A.8.53 /ISAPI/Security/UserPermission	303
A.8.54 /ISAPI/Security/UserPermission/<ID>	303
A.8.55 /ISAPI/Security/UserPermission/<ID>/localPermission	304
A.8.56 /ISAPI/Security/UserPermission/<ID>/remotePermission	305
A.8.57 /ISAPI/Security/UserPermission/adminCap	305
A.8.58 /ISAPI/Security/UserPermission/anonymouslogin	306
A.8.59 /ISAPI/Security/UserPermission/capabilities	306
A.8.60 /ISAPI/Security/UserPermission/operatorCap	307
A.8.61 /ISAPI/Security/UserPermission/viewerCap	307
A.8.62 /ISAPI/Security/users	307
A.8.63 /ISAPI/Security/users/<ID>	311
A.8.64 /ISAPI/Security/users/<ID>/capabilities	313
A.8.65 /ISAPI/Security/webCertificate	313
A.8.66 /ISAPI/SecurityCP/ReportCenterCfg/capabilities?format=json	314
A.8.67 /ISAPI/SecurityCP/ReportCenterCfg/<ID>?format=json	314
A.9 /ISAPI/Smart	315
A.9.1 /ISAPI/Smart/capabilities	315
A.9.2 /ISAPI/Smart/AudioDetection/channels/<ID>	316
A.9.3 /ISAPI/Smart/AudioDetection/channels/<ID>/capabilities	316
A.9.4 /ISAPI/Smart/AudioDetection/channels/<ID>/status	317
A.10 /ISAPI/Snapshot	317
A.10.1 /ISAPI/Snapshot/channels	317
A.10.2 /ISAPI/Snapshot/channels/<ID>	318
A.10.3 /ISAPI/Snapshot/channels/<ID>/capabilities	318
A.11 /ISAPI/Streaming	319

A.11.1 /ISAPI/Streaming/channels	319
A.11.2 /ISAPI/Streaming/channels/<ID>	320
A.11.3 /ISAPI/Streaming/channels/<ID>/capabilities	322
A.11.4 /ISAPI/Streaming/channels/<ID>/dynamicCap	324
A.11.5 /ISAPI/Streaming/channels/<ID>/picture	324
A.11.6 /ISAPI/Streaming/channels/<ID>/regionClip	325
A.11.7 /ISAPI/Streaming/channels/<ID>/regionClip/capabilities	326
A.11.8 /ISAPI/Streaming/channels/<ID>/smartOverlap/capabilities?format=json	326
A.11.9 /ISAPI/Streaming/channels/<ID>/smartOverlap?format=json	327
A.11.10 /ISAPI/Streaming/channels/<ID>/status	327
A.11.11 /ISAPI/Streaming/encryption/capabilities?format=json	328
A.11.12 /ISAPI/Streaming/encryption/secretKey?format=json	328
A.11.13 /ISAPI/Streaming/encryption?format=json	329
A.11.14 /ISAPI/Streaming/status	330
A.11.15 rtsp://<host>[:port]/ISAPI/Streaming/channels/<ID>	330
A.11.16 rtsp://<host>[:port]/ISAPI/Streaming/channels/<ID>?npq=	332
A.11.17 rtsp://<host>[:port]/ISAPI/Streaming/tracks/<ID>?starttime=&endtime=	333
A.12 /ISAPI/System	335
A.12.1 /ISAPI/System/accessDevice/associatedChannel?format=json	335
A.12.2 /ISAPI/System/AcsUpdate/capabilities	335
A.12.3 /ISAPI/System/activate	336
A.12.4 /ISAPI/System/algorithmsInfo/capabilities?format=json	336
A.12.5 /ISAPI/System/algorithmsInfo?format=json	336
A.12.6 /ISAPI/System/algorithmsVersion	337
A.12.7 /ISAPI/System/Audio/AudioIn	337
A.12.8 /ISAPI/System/Audio/AudioIn/capabilities	338
A.12.9 /ISAPI/System/Audio/AudioIn/channels/<ID>	338
A.12.10 /ISAPI/System/Audio/AudioIn/channels/<ID>/capabilities	339

A.12.11 /ISAPI/System/Audio/AudioOut/channels/<ID>	339
A.12.12 /ISAPI/System/Audio/AudioOut/channels/<ID>/capabilities	340
A.12.13 /ISAPI/System/Audio/capabilities	340
A.12.14 /ISAPI/System/Audio/channels	341
A.12.15 /ISAPI/System/Audio/channels/<ID>	341
A.12.16 /ISAPI/System/Audio/channels/<ID>/dynamicCap	341
A.12.17 /ISAPI/System/autoMaintenance/capabilities?format=json	342
A.12.18 /ISAPI/System/autoMaintenance?format=json	342
A.12.19 /ISAPI/System/autoSwitch/capabilities?format=json	343
A.12.20 /ISAPI/System/autoSwitch?format=json	343
A.12.21 /ISAPI/System/Bluetooth/capabilities	344
A.12.22 /ISAPI/System/Bluetooth/ibeaconParam	344
A.12.23 /ISAPI/System/Bluetooth/ibeaconParam/capabilities	345
A.12.24 /ISAPI/System/capabilities	345
A.12.25 /ISAPI/System/channels/<ID>/vibrationDetection/capabilities?format=json	346
A.12.26 /ISAPI/System/channels/<ID>/vibrationDetection?format=json	346
A.12.27 /ISAPI/System/configurationData?secretkey=	347
A.12.28 /ISAPI/System/configurationData?type=	348
A.12.29 /ISAPI/System/deviceInfo	349
A.12.30 /ISAPI/System/deviceInfo/capabilities	350
A.12.31 /ISAPI/System/DeviceLanguage	350
A.12.32 /ISAPI/System/DeviceLanguage/capabilities	351
A.12.33 /ISAPI/System/diagnosedData	351
A.12.34 /ISAPI/System/diagnosedData/exportStatus	352
A.12.35 /ISAPI/System/diagnosedData/parameter	352
A.12.36 /ISAPI/System/diagnosedData/parameter/capabilities	353
A.12.37 /ISAPI/System/diagnosis/capabilities?format=json	353
A.12.38 /ISAPI/System/diagnosis?format=json	354

A.12.39 /ISAPI/System/discoveryMode	354
A.12.40 /ISAPI/System/discoveryMode/capabilities	355
A.12.41 /ISAPI/System/doubleLensParking/capabilities	355
A.12.42 /ISAPI/System/dumpData	355
A.12.43 /ISAPI/System/exportLogToUSB/capabilities?format=json	356
A.12.44 /ISAPI/System/exportLogToUSB/mode?format=json	356
A.12.45 /ISAPI/System/exportLogToUSB/status?format=json	357
A.12.46 /ISAPI/System/exportUSBLog/capabilities?format=json	357
A.12.47 /ISAPI/System/exportUSBLog/status?format=json	357
A.12.48 /ISAPI/System/exportUSBLog?format=json	358
A.12.49 /ISAPI/System/externalDevice/capabilities	358
A.12.50 /ISAPI/System/factoryReset?mode=	359
A.12.51 /ISAPI/System/fileExport/capabilities?format=json	359
A.12.52 /ISAPI/System/fileExport?format=json	360
A.12.53 /ISAPI/System/firmwareCodeV2	360
A.12.54 /ISAPI/System/GPSCalibration/channels/<ID>/capabilities?format=json	360
A.12.55 /ISAPI/System/GPSCalibration/channels/<ID>/points/<PID>?format=json	361
A.12.56 /ISAPI/System/GPSCalibration/channels/<ID>/points/status?format=json	362
A.12.57 /ISAPI/System/GPSCalibration/channels/<ID>/points?format=json	362
A.12.58 /ISAPI/System/GPSPTZCalibration/channels/<ID>/calibration?format=json ..	363
A.12.59 /ISAPI/System/GPSVerification/channels/<ID>/display?format=json	364
A.12.60 /ISAPI/System/GPSVerification/channels/<ID>/points/<ID>?format=json	364
A.12.61 /ISAPI/System/GPSVerification/channels/<ID>/points/capabilities?format=json	365
A.12.62 /ISAPI/System/GPSVerification/channels/<ID>/points/resultInfo?format=json ..	366
A.12.63 /ISAPI/System/GPSVerification/channels/<ID>/points?format=json	366
A.12.64 /ISAPI/System/guardAgainstTheft	367
A.12.65 /ISAPI/System/guardAgainstTheft/capabilities	368

A.12.66 /ISAPI/System/guideConfig/<guideEvent>/capabilities?format=json	368
A.12.67 /ISAPI/System/guideConfig/<guideEvent>?format=json	369
A.12.68 /ISAPI/System/Hardware	369
A.12.69 /ISAPI/System/Hardware/defog	370
A.12.70 /ISAPI/System/installationAngleCalibration/channels/<ID>/capabilities?format=json	371
A.12.71 /ISAPI/System/installationAngleCalibration/channels/<ID>?format=json	371
A.12.72 /ISAPI/System/IO	372
A.12.73 /ISAPI/System/IO/capabilities	372
A.12.74 /ISAPI/System/IO/inputs	372
A.12.75 /ISAPI/System/IO/inputs/<ID>	373
A.12.76 /ISAPI/System/IO/inputs/<ID>/status	373
A.12.77 /ISAPI/System/IO/inputs/capabilities	374
A.12.78 /ISAPI/System/IO/outputs	374
A.12.79 /ISAPI/System/IO/outputs/capabilities	375
A.12.80 /ISAPI/System/IO/outputs/<ID>	375
A.12.81 /ISAPI/System/IO/outputs/<ID>/status	376
A.12.82 /ISAPI/System/IO/outputs/<ID>/trigger	376
A.12.83 /ISAPI/System/IO/status	376
A.12.84 /ISAPI/System/IOT/channelConfig?format=json	377
A.12.85 /ISAPI/System/IOT/channels/<ID>/<EventType>/capabilities?format=json	378
A.12.86 /ISAPI/System/IOT/channels/<ID>/<EventType>?format=json	378
A.12.87 /ISAPI/System/IOT/channels/<ID>/alarmStatistics?format=json	379
A.12.88 /ISAPI/System/IOT/channels/<ID>/all?format=json	379
A.12.89 /ISAPI/System/IOT/channels/<ID>/basicParam?format=json	380
A.12.90 /ISAPI/System/IOT/channels/<ID>/OSD/capabilities?format=json	380
A.12.91 /ISAPI/System/IOT/channels/<ID>/OSD?format=json	380
A.12.92 /ISAPI/System/IOT/channels/<ID>/status?format=json	381

A.12.93 /ISAPI/System/IOT/channels/<ID>?format=json	382
A.12.94 /ISAPI/System/IOT/channels/status?format=json	383
A.12.95 /ISAPI/System/IOT/channels/status?format=json&deviceInductiveType=	383
A.12.96 /ISAPI/System/IOT/channels?format=json	384
A.12.97 /ISAPI/System/IOT/channels?format=json&deviceInductiveType=	385
A.12.98 /ISAPI/System/IOT/linkageChannels?format=json	386
A.12.99 /ISAPI/System/IOT/search?format=json	386
A.12.100 /ISAPI/System/IOT/sourceCapabilities?format=json	387
A.12.101 /ISAPI/System/IOT/sourceSupport/capabilities?format=json	387
A.12.102 /ISAPI/System/laserOpticalAxis/capabilities?format=json	387
A.12.103 /ISAPI/System/laserOpticalAxis/goto?format=json	388
A.12.104 /ISAPI/System/laserOpticalAxis?format=json	388
A.12.105 /ISAPI/System/logServer	389
A.12.106 /ISAPI/System/logServer/capabilities	389
A.12.107 /ISAPI/System/logServer/test	390
A.12.108 /ISAPI/System/Network/adaption/capabilities?format=json	390
A.12.109 /ISAPI/System/Network/adaption?format=json&streamType=	391
A.12.110 /ISAPI/System/Network/Bond	391
A.12.111 /ISAPI/System/Network/Bond/<ID>	392
A.12.112 /ISAPI/System/Network/capabilities	392
A.12.113 /ISAPI/System/Network/channels/<ID>/buletooth/status	393
A.12.114 /ISAPI/System/Network/DDNS	394
A.12.115 /ISAPI/System/Network/DDNS/capabilities	394
A.12.116 /ISAPI/System/Network/DDNS/<ID>	395
A.12.117 /ISAPI/System/Network/DDNS/<ID>/capabilities	395
A.12.118 /ISAPI/System/Network/Ehome	396
A.12.119 /ISAPI/System/Network/Ehome/capabilities	396
A.12.120 /ISAPI/System/Network/Ehome?centerID=	397

A.12.121 /ISAPI/System/Network/EZVIZ	398
A.12.122 /ISAPI/System/Network/EZVIZ/QRCode	398
A.12.123 /ISAPI/System/Network/EZVIZ/QRCode?format=json	399
A.12.124 /ISAPI/System/Network/EZVIZ/capabilities	400
A.12.125 /ISAPI/System/Network/EZVIZ/secretKey?format=json	400
A.12.126 /ISAPI/System/Network/EZVIZ/unbind?format=json	400
A.12.127 /ISAPI/System/Network/ftp	401
A.12.128 /ISAPI/System/Network/ftp/<ID>	402
A.12.129 /ISAPI/System/Network/ftp/capabilities	402
A.12.130 /ISAPI/System/Network/ftp/test	403
A.12.131 /ISAPI/System/Network/interfaces	403
A.12.132 /ISAPI/System/Network/interfaces/<ID>	404
A.12.133 /ISAPI/System/Network/interfaces/<ID>/capabilities	405
A.12.134 /ISAPI/System/Network/interfaces/<ID>/discovery	405
A.12.135 /ISAPI/System/Network/interfaces/<ID>/discovery/capabilities	406
A.12.136 /ISAPI/System/Network/interfaces/<ID>/dynamicHostName?format=json	406
A.12.137 /ISAPI/System/Network/interfaces/<ID>/dynamicHostName/capabilities? format=json	407
A.12.138 /ISAPI/System/Network/interfaces/<ID>/ieee802.1x	407
A.12.139 /ISAPI/System/Network/interfaces/<ID>/ieee802.1x/capabilities	408
A.12.140 /ISAPI/System/Network/interfaces/<ID>/ipAddress	408
A.12.141 /ISAPI/System/Network/interfaces/<ID>/ipAddress/capabilities	410
A.12.142 /ISAPI/System/Network/interfaces/<ID>/link	410
A.12.143 /ISAPI/System/Network/interfaces/<ID>/link/capabilities	411
A.12.144 /ISAPI/System/Network/interfaces/<ID>/NetworkMode	411
A.12.145 /ISAPI/System/Network/interfaces/<ID>/wireless	412
A.12.146 /ISAPI/System/Network/interfaces/<ID>/wireless/accessDeviceList/<ID>	413
A.12.147 /ISAPI/System/Network/interfaces/<ID>/wireless/accessDeviceList/capabilities	413

A.12.148 /ISAPI/System/Network/interfaces/<ID>/wireless/accessPointList	414
A.12.149 /ISAPI/System/Network/interfaces/<ID>/wireless/accessPointList/<ID>	414
A.12.150 /ISAPI/System/Network/interfaces/<ID>/wireless/capabilities	414
A.12.151 /ISAPI/System/Network/interfaces/<ID>/wirelessServer	415
A.12.152 /ISAPI/System/Network/interfaces/<ID>/wirelessServer/accessDeviceList	416
A.12.153 /ISAPI/System/Network/interfaces/<ID>/wirelessServer/capabilities	416
A.12.154 /ISAPI/System/Network/interfaces/<ID>/wirelessServerStatus	416
A.12.155 /ISAPI/System/Network/interfaces/<ID>/wirelessServerStatus/capabilities	417
A.12.156 /ISAPI/System/Network/interfaces/<ID>/wirelessStatus	418
A.12.157 /ISAPI/System/Network/interfaces/<ID>/wirelessStatus/capabilities	418
A.12.158 /ISAPI/System/Network/interfaces/<ID>/WPS	419
A.12.159 /ISAPI/System/Network/interfaces/<ID>/WPS/ApPinCode	420
A.12.160 /ISAPI/System/Network/interfaces/<ID>/WPS/AutoConnect	420
A.12.161 /ISAPI/System/Network/interfaces/<ID>/WPS/devicePinCode	421
A.12.162 /ISAPI/System/Network/interfaces/<ID>/WPS/devicePinCodeUpdate	421
A.12.163 /ISAPI/System/Network/interfaces/capabilities	422
A.12.164 /ISAPI/System/Network/ipFilter	422
A.12.165 /ISAPI/System/Network/ipFilter/capabilities	423
A.12.166 /ISAPI/System/Network/MACFilter	423
A.12.167 /ISAPI/System/Network/MACFilter/capabilities	424
A.12.168 /ISAPI/System/Network/mailing	424
A.12.169 /ISAPI/System/Network/mailing/<ID>	424
A.12.170 /ISAPI/System/Network/mailing/capabilities	425
A.12.171 /ISAPI/System/Network/mailing/test	425
A.12.172 /ISAPI/System/Network/pingtest	426
A.12.173 /ISAPI/System/Network/POE/capabilities?format=json	426
A.12.174 /ISAPI/System/Network/POE?format=json	427
A.12.175 /ISAPI/System/Network/resourceStatistics?format=json	427

A.12.176 /ISAPI/System/Network/SIP	428
A.12.177 /ISAPI/System/Network/SIP/<ID>	428
A.12.178 /ISAPI/System/Network/SIP/<ID>/capabilities	429
A.12.179 /ISAPI/System/Network/SIP/<ID>/SIPInfo	429
A.12.180 /ISAPI/System/Network/SIP/<ID>/SIPInfo/capabilities	430
A.12.181 /ISAPI/System/Network/SIP/<ID>/SIPInfo/multiInfo	430
A.12.182 /ISAPI/System/Network/SIP/capabilities	431
A.12.183 /ISAPI/System/Network/SIP/voiceTalkCtrl/<ID>?format=json	431
A.12.184 /ISAPI/System/Network/SIP/voiceTalkCtrl/capabilities?format=json	432
A.12.185 /ISAPI/System/Network/SNMP	433
A.12.186 /ISAPI/System/Network/SNMP/advanced	433
A.12.187 /ISAPI/System/Network/SNMP/advanced/trapReceivers	434
A.12.188 /ISAPI/System/Network/SNMP/advanced/trapReceivers/<ID>	435
A.12.189 /ISAPI/System/Network/socketIP	436
A.12.190 /ISAPI/System/Network/ssh	436
A.12.191 /ISAPI/System/Network/ssh/capabilities	437
A.12.192 /ISAPI/System/Network/StaticRoute	437
A.12.193 /ISAPI/System/Network/telnetd	438
A.12.194 /ISAPI/System/Network/telnetd/capabilities	439
A.12.195 /ISAPI/System/Network/UPnP	439
A.12.196 /ISAPI/System/Network/UPnP/capabilities	440
A.12.197 /ISAPI/System/Network/UPnP/ports	440
A.12.198 /ISAPI/System/Network/UPnP/ports/status	441
A.12.199 /ISAPI/System/Network/UPnP/ports/<ID>	441
A.12.200 /ISAPI/System/Network/UPnP/ports/<ID>/status	442
A.12.201 /ISAPI/System/Network/verificationCodeCheck	442
A.12.202 /ISAPI/System/Network/WirelessDial/Interfaces	442
A.12.203 /ISAPI/System/Network/WirelessDial/Interfaces/<ID>	443

A.12.204 /ISAPI/System/Network/WirelessDial/Interfaces/<ID>/dial	444
A.12.205 /ISAPI/System/Network/WirelessDial/Interfaces/<ID>/dial/capabilities	445
A.12.206 /ISAPI/System/Network/WirelessDial/Interfaces/<ID>/dialstatus	445
A.12.207 /ISAPI/System/Network/WirelessDial/Interfaces/<ID>/keepAlive/capabilities? format=json	446
A.12.208 /ISAPI/System/Network/WirelessDial/Interfaces/<ID>/keepAlive/test?format=json	446
A.12.209 /ISAPI/System/Network/WirelessDial/Interfaces/<ID>/keepAlive?format=json	447
A.12.210 /ISAPI/System/Network/WirelessDial/Interfaces/<ID>/messageConfig	447
A.12.211 /ISAPI/System/Network/WirelessDial/Interfaces/<ID>/messageConfig/ messageConfigCap	448
A.12.212 /ISAPI/System/Network/WirelessDial/Interfaces/<ID>/messageConfig/WhiteList	448
A.12.213 /ISAPI/System/Network/WirelessDial/Interfaces/<ID>/messageConfig/WhiteList/ <ID>	449
A.12.214 /ISAPI/System/Network/WirelessDial/Interfaces/<ID>/schedule	450
A.12.215 /ISAPI/System/Network/WirelessDial/Interfaces/<ID>/trafficMonitor/capabilities? format=json	451
A.12.216 /ISAPI/System/Network/WirelessDial/Interfaces/<ID>/trafficMonitor/status? format=json	451
A.12.217 /ISAPI/System/Network/WirelessDial/Interfaces/<ID>/trafficMonitor?format=json	452
A.12.218 /ISAPI/System/Network/WPS/capabilities	452
A.12.219 /ISAPI/System/onlineUpgrade/capabilities	453
A.12.220 /ISAPI/System/onlineUpgrade/deviceParameter?format=json	453
A.12.221 /ISAPI/System/onlineUpgrade/downloadPackage/pause?format=json	454
A.12.222 /ISAPI/System/onlineUpgrade/downloadPackage/resume?format=json	454
A.12.223 /ISAPI/System/onlineUpgrade/downloadPackage/status?format=json	454
A.12.224 /ISAPI/System/onlineUpgrade/downloadPackage?format=json	455
A.12.225 /ISAPI/System/onlineUpgrade/ignoreCurrentVersion?format=json	455

A.12.226 /ISAPI/System/onlineUpgrade/server	456
A.12.227 /ISAPI/System/onlineUpgrade/status	456
A.12.228 /ISAPI/System/onlineUpgrade/upgradeWithoutDownload?format=json	457
A.12.229 /ISAPI/System/onlineUpgrade/version	457
A.12.230 /ISAPI/System/PictureServer/capabilities?format=json	457
A.12.231 /ISAPI/System/PictureServer?format=json	458
A.12.232 /ISAPI/System/reboot	459
A.12.233 /ISAPI/System/Serial/authentication/capabilities?format=json	459
A.12.234 /ISAPI/System/Serial/authentication?format=json	459
A.12.235 /ISAPI/System/Serial/capabilities	460
A.12.236 /ISAPI/System/Serial/ports	460
A.12.237 /ISAPI/System/Serial/ports/<ID>	461
A.12.238 /ISAPI/System/Serial/ports/<ID>/capabilities	461
A.12.239 /ISAPI/System/Serial/ports/capabilities	462
A.12.240 /ISAPI/System/Serial/ports/command	462
A.12.241 /ISAPI/System/Serial/ports/command/capabilities	463
A.12.242 /ISAPI/System/serialLogCfg/capabilities?format=json	464
A.12.243 /ISAPI/System/serialLogCfg?format=json	464
A.12.244 /ISAPI/System/shutdown?format=json	465
A.12.245 /ISAPI/System/Software/channels/<ID>	465
A.12.246 /ISAPI/System/Software/channels/<ID>/capabilities	466
A.12.247 /ISAPI/System/SSDFileSystem/capacity?format=json	466
A.12.248 /ISAPI/System/SSDFileSystem/format?format=json	467
A.12.249 /ISAPI/System/SSDFileSystem/format/status?format=json	467
A.12.250 /ISAPI/System/SSDFileSystem/upgrade?format=json	468
A.12.251 /ISAPI/System/SSDFileSystem/upgrade/status?format=json	468
A.12.252 /ISAPI/System/status	469
A.12.253 /ISAPI/System/syncSignalOutput/<ID>	469

A.12.254 /ISAPI/System/syncSignalOutput/<ID>/capabilities	470
A.12.255 /ISAPI/System/time	470
A.12.256 /ISAPI/System/time/capabilities	471
A.12.257 /ISAPI/System/time/localTime	471
A.12.258 /ISAPI/System/time/ntpServers	472
A.12.259 /ISAPI/System/time/ntpServers/<ID>	473
A.12.260 /ISAPI/System/time/ntpServers/<ID>/capabilities	473
A.12.261 /ISAPI/System/time/ntpServers/capabilities	474
A.12.262 /ISAPI/System/time/ntpServers/test	474
A.12.263 /ISAPI/System/time/timeType?format=json	475
A.12.264 /ISAPI/System/time/timeZone	475
A.12.265 /ISAPI/System/TwoWayAudio/channels	476
A.12.266 /ISAPI/System/TwoWayAudio/channels/<ID>	477
A.12.267 /ISAPI/System/TwoWayAudio/channels/<ID>/audioData	477
A.12.268 /ISAPI/System/TwoWayAudio/channels/<ID>/capabilities	479
A.12.269 /ISAPI/System/TwoWayAudio/channels/<ID>/close	479
A.12.270 /ISAPI/System/TwoWayAudio/channels/<ID>/open	480
A.12.271 /ISAPI/System/TwoWayAudio/channels/capabilities	481
A.12.272 /ISAPI/System/unitConfig/capabilities?format=json	482
A.12.273 /ISAPI/System/unitConfig?format=json	482
A.12.274 /ISAPI/System/updateFirmware	483
A.12.275 /ISAPI/System/updateFirmware?type=&id=	484
A.12.276 /ISAPI/System/updateFirmware?type=&moduleAddress=	484
A.12.277 /ISAPI/System/upgradeStatus	485
A.12.278 /ISAPI/System/usb/capabilities?format=json	485
A.12.279 /ISAPI/System/usb?format=json	486
A.12.280 /ISAPI/System/userLock/config/capabilities?format=json	486
A.12.281 /ISAPI/System/userLock/config?format=json	487

A.12.282 /ISAPI/System/userLock/lockedUsers?format=json	487
A.12.283 /ISAPI/System/userLock/unlockUser?format=json	488
A.12.284 /ISAPI/System/Video/capabilities	488
A.12.285 /ISAPI/System/Video/inputs	489
A.12.286 /ISAPI/System/Video/inputs/channels	489
A.12.287 /ISAPI/System/Video/inputs/channels/<ID>	489
A.12.288 /ISAPI/System/Video/inputs/channels?format=json	490
A.12.289 /ISAPI/System/Video/inputs/channels/<ID>/focus	491
A.12.290 /ISAPI/System/Video/inputs/channels/<ID>?format=json	491
A.12.291 /ISAPI/System/Video/inputs/channels/<ID>/capabilities	491
A.12.292 /ISAPI/System/Video/inputs/channels/<ID>/iris	492
A.12.293 /ISAPI/System/Video/inputs/channels/<ID>/motionDetection	492
A.12.294 /ISAPI/System/Video/inputs/channels/<ID>/motionDetection/capabilities	493
A.12.295 /ISAPI/System/Video/inputs/channels/<ID>/motionDetection/layout	494
A.12.296 /ISAPI/System/Video/inputs/channels/<ID>/motionDetection/layout/capabilities	494
A.12.297 /ISAPI/System/Video/inputs/channels/<ID>/motionDetection/layout/gridLayout	495
A.12.298 /ISAPI/System/Video/inputs/channels/<ID>/overlays	496
A.12.299 /ISAPI/System/Video/inputs/channels/<ID>/overlays/BatteryPowerOverlay	498
A.12.300 /ISAPI/System/Video/inputs/channels/<ID>/overlays/BatteryPowerOverlay/ capabilities	498
A.12.301 /ISAPI/System/Video/inputs/channels/<ID>/overlays/capabilities	499
A.12.302 /ISAPI/System/Video/inputs/channels/<ID>/overlays/channelNameOverlay ...	499
A.12.303 /ISAPI/System/Video/inputs/channels/<ID>/overlays/dateTimeOverlay	500
A.12.304 /ISAPI/System/Video/inputs/channels/<ID>/overlays/dateTimeOverlay/ capabilities	501
A.12.305 /ISAPI/System/Video/inputs/channels/<ID>/overlays/text	501
A.12.306 /ISAPI/System/Video/inputs/channels/<ID>/overlays/text/<ID>	502

A.12.307 /ISAPI/System/Video/inputs/channels/<ID>/privacyMask	503
A.12.308 /ISAPI/System/Video/inputs/channels/<ID>/privacyMask/privacyMaskCap	504
A.12.309 /ISAPI/System/Video/inputs/channels/<ID>/privacyMask/regions	504
A.12.310 /ISAPI/System/Video/inputs/channels/<ID>/privacyMask/regions/<ID>	505
A.12.311 /ISAPI/System/Video/inputs/channels/<ID>/VCAResource	506
A.12.312 /ISAPI/System/Video/inputs/channels/<ID>/VCAResource/capabilities	507
A.12.313 /ISAPI/System/Video/inputs/channels/capabilities	507
A.12.314 /ISAPI/System/Video/inputs/channels/counting/collection/capabilities? format=json	508
A.12.315 /ISAPI/System/Video/inputs/channels/counting/collection?format=json	508
A.12.316 /ISAPI/System/Video/inputs/channels/heatMap/collection/capabilities? format=json	509
A.12.317 /ISAPI/System/Video/inputs/channels/heatMap/collection?format=json	509
A.12.318 /ISAPI/System/Video/inputs/OSDLanguage	509
A.12.319 /ISAPI/System/Video/outputs	510
A.12.320 /ISAPI/System/Video/outputs/channels	510
A.12.321 /ISAPI/System/Video/outputs/channels/<ID>	511
A.12.322 /ISAPI/System/Video/outputs/channels/<ID>/capabilities	512
A.12.323 /ISAPI/System/Video/outputs/PreviewSwitch/capabilities	512
A.12.324 /ISAPI/System/Video/outputs/PreviewSwitch/capabilities?videoOutType=	512
A.12.325 /ISAPI/System/Video/outputs/PreviewSwitch?groupID=&videoOutType=	513
A.12.326 /ISAPI/System/Video/outputs/PreviewSwitch? groupID=&videoOutType=&previewFrameNo=	514
A.12.327 /ISAPI/System/wiresharkToUSB/capabilities?format=json	515
A.12.328 /ISAPI/System/wiresharkToUSB/mode?format=json	515
A.12.329 /ISAPI/System/wiresharkToUSB/status?format=json	516
A.12.330 /ISAPI/System/workingstatus/hdStatus?format=json	516
A.12.331 /ISAPI/System/workingstatus/chanStatus?format=json	517
A.12.332 /ISAPI/System/workingstatus/capabilities?format=json	517

A.12.333 /ISAPI/System/workingstatus?format=json	518
A.12.334 /ISAPI/System/workingstatus/IOStatus?format=json	518
A.12.335 /ISAPI/System/zeroBiasCalibration/channels/<ID>?format=json	519
A.13 /ISAPI/Thermal	519
A.13.1 /ISAPI/Thermal/capabilities	519
A.13.2 /ISAPI/Thermal/temperature/collection/capabilities?format=json	520
A.13.3 /ISAPI/Thermal/temperature/collection?format=json	520
Appendix B. Appendixes	521
B.1 Request and Response Messages	521
B.1.1 JSON Messages	521
B.1.2 XML Messages	664
B.2 Log Types for ISAPI	948
B.3 Supported Alarm/Event Types	968
B.4 Response Codes of Text Protocol	972

Chapter 1 Overview

1.1 Introduction

The OPEN Intelligent Security Application Programming Interface (ISAPI) is a text protocol in RESTful style based on HTTP for communicating between security devices/servers (e.g., cameras, DVR, NVR, etc.) and client software/system. It defines the communication standard between device/server and client software/system via the Internet Protocol (IP), see the network topology and communication protocol layer framework in the figures below.

 Note

REST (REpresentational State Transfer) is a protocol design method which abstracts all information as the resources. The abstracted resources are marked by the uniform identifies, i.e., URI (Uniform Resource Identifiers) for simple and extendable management.

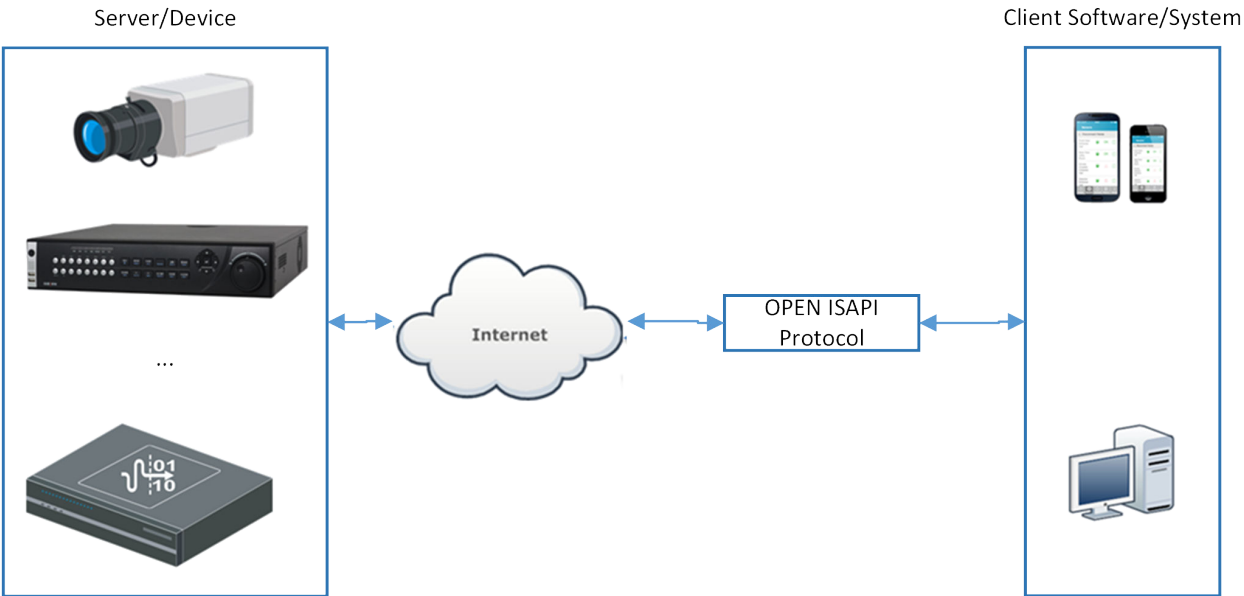


Figure 1-1 Network Topology

Search Online Device	Interaction Commands	Streaming
SADP	HTTP	RTSP/RTP/RTCP
MCAST	TCP	TCP/UDP
IP		

Figure 1-2 Communication Protocol Layer Framework

1.2 Update History

1.2.1 Version 20.07-Version 20.09

Summary of Changes in Version 2.6_Aug., 2020

Related Product: iDS-9632NXI-I8/X-AI and iDS-6708NXI-I/X-AI series DeepinMind NVR with Software Version 4.1.80

1. Extended audible warning configuration capability **JSON_AudioAlarmCap** (related URI: **/ISAPI/Event/triggers/notifications/AudioAlarm/capabilities?format=json**):
added one audio type 16 (AI custom audio file) to node **audioTypeListCap**.
2. Extended information of custom alarm audio file **JSON_GetCustomAudio** (related URI: **/ISAPI/Event/triggers/notifications/AudioAlarm/customAudioInfo?format=json**):
added one node **AudioType** (audio file type).
3. Extended the information of custom alarm audio file to be imported **JSON_ImportCustomAudio** (related URI: **/ISAPI/Event/triggers/notifications/AudioAlarm/customAudio?format=json**):
added two nodes **AudioType** (audio file type) and **isCover** (whether to cover the audio file with same name).
4. Extended the device capability **XML_DeviceCap** (related URI: **/ISAPI/System/capabilities**):
added two nodes **<isSupportAIOpenPlatform>** (whether the device supports AI open platform capabilities) and **<isSupportPictureDownloadError>** (whether the device supports reporting picture download failure).
5. Extended **Response Codes of Text Protocol** :
added five sub status codes to invalid operation: 0x400019C5 (AITargetBPCaptureFail), 0x400019C6 (AITargetBPToDSPFail), 0x400019C7 (AITargetBPDuplicateName), 0x400019D0 (audioFileNameWrong), and 0x400019D1 (audioFileImportFail) ;
added two sub status codes to invalid content: 0x60001135 (AudioFileNameDuplicate) and 0x60001136 (CurrentAudioFileAI RuleInUseAlreadyDelete).
6. Added seven minor operation log types (see **Log Types for ISAPI** for details), i.e., AITargetBPAdd, AITargetBPDelete, AITargetBPSearch, AITargetBPUpdate, AIRuleConfigTrigger, AudioFileImport, and AudioFileDownload.

Summary of Changes in Version 2.6_Aug., 2020

Related Product: DS-K1T680DX Series Face Recognition Terminal with Software Version 3.1.2

1. Extended the network capability message **XML_NetworkCap** (related URI: **/ISAPI/System/Network/capabilities**):
added two nodes **<isSupportEZVIZUnbind>** (whether it supports unbinding devices from the Hik-Connect account) and **<isSupportEZVIZQRcode>** (whether it supports getting the Hik-Connect QR code).
2. Extended Hik-Connect access parameter message **XML_EZVIZ** (related URI: **/ISAPI/System/Network/EZVIZ**):

added a node **<bindStatus>** (Hik-Connect account binding status of the current device).

3. Added a URI of getting the Hik-Connect QR code of the device: GET ***/ISAPI/System/Network/EZVIZ/QRCode*** .

Summary of Changes in Version 2.6_July, 2020

Related Product: iDS-96064NXI series, iDS-96128NXI series, and iDS-96256NXI series DeepinMind NVR with software version 4.22.200

1. Extended the capability of digital video channel proxy ***XML_Cap_StreamingChannel*** (related URI: ***/ISAPI/ContentMgmt/StreamingProxy/channels/<ID>/capabilities***):
added one node **<isSupportPictureByUrl>** (whether the device supports capture picture in URL format).
2. Added URIs for capturing the picture in URL format:
Get capability: GET ***/ISAPI/ContentMgmt/StreamingProxy/channels/<ID>/PictureByUrl/capabilities?format=json*** ;
Capture the picture in URL format: POST ***/ISAPI/ContentMgmt/StreamingProxy/channels/<ID>/PictureByUrl?format=json*** .
3. Extended the Hik-Connect access configuration capability ***XML_Cap_EZVIZ*** (related URI: ***/ISAPI/System/Network/EZVIZ/capabilities***):
added one node **<isSupportEZVIZQRCode>** (whether the device supports getting the Hik-Connect QR code).
4. Added the URI for getting Hik-Connect QR code: ***/ISAPI/System/Network/EZVIZ/QRCode?format=json*** .
5. Extended the HDD parameters ***XML_hdd*** and HDD list parameters ***XML_hddList*** (related URI: ***/ISAPI/ContentMgmt/Storage/hdd*** and ***/ISAPI/ContentMgmt/Storage/hdd/<ID>***):
added three values to node **<status>** (HDD status): "synching" (synchronizing), "syncError" (synchronization exception), and "unloaded".
6. Added the URI for getting the HDD data sync status: GET ***/ISAPI/ContentMgmt/Storage/hdd/<ID>/syncStatus?format=json*** .
7. Extended the device storage capability ***XML_RacmCap*** (related URI: ***/ISAPI/ContentMgmt/capabilities***):
added one node **<isSupportSSDSMARTTest>** (whether the device supports SSD S.M.A.R.T detection).
8. Extended the device capability ***XML_DeviceCap*** (related URI: ***/ISAPI/System/capabilities***):
added three nodes: **<isSupportSSDFileSystemUpgrade>** (whether the device supports SSD file system upgrade), **<isSupportSSDFileSystemFormat>** (whether the device supports SSD file system formatting), and **<isSupportSSDFileSystemCapacity>** (whether the device supports getting space distribution information of SSD file system).
9. Added URIs for SSD management, see ***SSD Management*** for details.
10. Added one sub status code 0x400020F3-"NotSupportWithVideoTask" (the function is not supported) to status code 4 (Invalid Operation) in ***Response Codes of Text Protocol*** .
11. Extended the log types ***Log Types for ISAPI*** :
added one minor exception log type "ezvizUpgradeException" (Hik-Connect upgrade exception);

added four minor operation log types: "localSSDOperateStart" (Local SSD operation (firmware operations) started), "localSSDOperateStop" (Local SSD operation (firmware operations) ended), "remoteSSDOperateStart" (Remote SSD operation (firmware operations) started), "remoteSSDOperateStop" (Remote SSD operation (firmware operations) ended).

Summary of Changes in Version 2.6_July, 2020

Related Product: CTRL-NEX-001A Control Panel with Software Version 1.0.0

1. Extended wireless dial-up capability message **XML_Cap_Dial** and wireless dial-up parameter message **XML_Dial** (related URIs: **/ISAPI/System/Network/WirelessDial/Interfaces/<ID>/dial/capabilities** and **/ISAPI/System/Network/WirelessDial/Interfaces/<ID>/dial**):
added four nodes: **<ISPName>** (ISP name), **<IMEINo>** (IMEI code), **<ICCID>** (ICCID code), and **<netType>** (network type).
2. Extended condition message of searching for specific resources **XML_VideoPic_CMSearchDescription** (related URI: **/ISAPI/ContentMgmt/search**):
added a sub node **pircamID** (pircam search ID) to the node **contentTypeList**.
3. Extended event linkage notification message **XML_EventTriggerNotification** (related URI: **/ISAPI/Event/triggers/<ID>/notifications**):
added two linkage actions "siren" (siren linkage) and "output" (relay linkage) to the sub node **EventTriggerNotification** of the node **EventTriggerNotificationList**.
4. Extended alarm linkage capability message **XML_EventTriggersCap** (related URI: **/ISAPI/Event/triggersCap**):
added two sub nodes to the node **DiskfullTriggerCap**, i.e., **isSupportSiren** (whether it supports siren linkage) and **isSupportOutput** (whether it supports relay linkage).
5. Extended parameter message of an HDD quota **XML_diskQuota** (related URI: **/ISAPI/ContentMgmt/Storage/quota/<ID>**):
added a node **<reservedTimeCfg>** (video saving time).
6. Extended parameter message of a specific SIP server **XML_SIPServer** (related URI: **/ISAPI/System/Network/SIP/<ID>**):
added four sub nodes to the node **<Standard>**, i.e., **<serverAddressType>** (server address type), **<domainName>** (domain name), **<networkType>** (network type), and **<calledTargetUserName>** (called user name).
7. Extended configuration capability of all SIP servers **XML_SIPServerListCap** (related URI: **/ISAPI/System/Network/SIP/capabilities**):
added five sub nodes to the node **Standard** of **SIPServer**, i.e., **<isSptSIPVoiceTalk>** (whether it supports SIP intercom), **<serverAddressType>** (server address type), **<domainName>** (domain name), **<networkType>** (network type), and **<calledTargetUserName>** (called user name).
8. Added URIs of configuring SIP intercom parameters:
Get configuration capability: GET **/ISAPI/System/Network/SIP/voiceTalkCtrl/capabilities?format=json** ;
Get or set parameters: GET or PUT **/ISAPI/System/Network/SIP/voiceTalkCtrl/<ID>?format=json** .
9. Added 8 sub status codes to status code 4 (Invalid Operation) in **Response Codes of Text Protocol** :

0x40008038-"overAudioFileNumLimit" (The number of audio files exceeds the limit),
0x40008039-"audioFileNamesIsLong" (The audio file name is too long),
0x4000803a-"audioFormatIsWrong" (The audio file format is invalid),
0x4000803b-"audioFileIsLarge" (The size of the audio file exceeds the limit),
0x4000803c-"pircamCapTimeOut" (Capturing of pircam timed out),
0x4000803d-"pircamCapFail" (Capturing of pircam failed), 0x4000803e-"pircamIsCaping" (The pircam is capturing), and 0x4000803f-"audioFileHasExisted" (The audio file already exists).

10. Extended the log types **Log Types for ISAPI** :

added 10 alarm log types: "soundIntensityMutation" (Sudden Increase of Sound Intensity Detection), "soundIntensityMutationStop" (Sudden Increase of Sound Intensity Detection Ended), "soundIntensitySteepFall" (Sudden Decrease of Sound Intensity Detection), "soundIntensitySteepFallStop" (Sudden Decrease of Sound Intensity Detection Ended), "moveAlarm" (Motion Alarm), "moveAlarmRestored" (Motion Alarm Restored), "lowTemperatureAlarm" (Low Temperature Alarm), "lowTemperatureAlarmRestored" (Low Temperature Alarm Restored), "highTemperatureAlarm" (High Temperature Alarm), and "highTemperatureAlarmRestored" (High Temperature Alarm Restored);
added 20 operation log types: "SSHEnabled" (SSH Enabled), "SSHDisabled" (SSH Disabled), "installationModeEntered" (Installation Mode Enabled), "installationModeExited" (Installation Mode Disabled), "diagnosisModeConfigured" (Diagnosis Mode Configured), "fileExported" (File Exported), "audioFileUploaded" (Audio File Uploaded), "audioFileDeleted" (Audio File Deleted), "PIRCAMCapture" (PIRCAM Captured), "SIPIntercomStarted" (SIP Intercom Started), "SIPIntercomEnded" (SIP Intercom Ended), "enrollmentModeEntered" (Registration Mode Enabled), "enrollmentModeExited" (Registration Mode Disabled), "videoAudioSelfCheckStarted" (Self-Test of Audio and Video Started), "videoAudioSelfCheckStopped" (Self-Test of Audio and Video Stopped), "cardReaderunlocked" (Card Reader Unlocked), "cardReaderlocked" (Card Reader Locked), "videoAudioSelfCheckEnded" (Self-Test of Audio and Video Ended), "previewStart" (Live View Started), and "previewStop" (Live View Stopped)..

1.2.2 Version 20.04-Version 20.06

Summary of Changes in Version 2.6_July, 2020

Related Products: DS-2CV2, DS-2CD2, and DS-2CD3 Series Network Camera in Software Version 5.5.150

1. Extended the message about the Hik-Connect access configuration capability **XML_Cap_EZVIZ** (related URI: **/ISAPI/System/Network/EZVIZ/capabilities**):
added two nodes, i.e, <isSupportUpISAPI> (whether the device supports uplink transmitting ISAPI message) and <isSupportDownISAPI> (whether the device supports downlink transmitting ISAPI message).
2. Extended the message about the Hik-Connect access configuration parameters **XML_EZVIZ** (related URI: **/ISAPI/System/Network/EZVIZ**):
added a node <upISAPIStatus> (status of uplink transmitting ISAPI message).

Summary of Changes in Version 2.6_July, 2020

Related Product: Decoder with Model DS-6901UDI, DS-6904UDI, DS-6908UDI, DS-6910UDI, DS-6912UDI, DS-6916UDI in Software Version 2.6.0

1. Extended the message about the capability of exporting files from the device **JSON_FileExportCap** (related URI: **/ISAPI/System/fileExport/capabilities?format=json**): added a file type "errorInfo" (error information) to the node "fileType".
2. Extended the message about the configuration parameters of exporting files from the device **JSON_FileExport** (related URI: **/ISAPI/System/fileExport?format=json**): added a file type "errorInfo" (error information) to the node "fileType"; added two nodes, i.e., "devType" (device type) and "subSys" (partition No.).

Summary of Changes in Version 2.6_June, 2020

Related Product: Radar PTZ Camera with Model of iDS-2SR8141XS-AB(40X) in Software Version 5.5.21

Extended the tracking parameters **JSON_TrackingRatio** and its configuration capability **JSON_TrackingRatioCap** (related URIs: **/ISAPI/PTZCtrl/channels/<ID>/trackingRatio?format=json** and **/ISAPI/PTZCtrl/channels/<ID>/trackingRatio/capabilities?format=json**): added a node "trackingTakeoverEnabled" (whether to enable tracking takeover).

Summary of Changes in Version 2.6_June, 2020

Related Product: iDS-8104AHQXLI-F4 Series ATM DVR

1. Extended the device storage capability **XML_RacmCap** (related URI: GET **/ISAPI/ContentMgmt/capabilities**): added four search conditions to node <pictureSearchType> and <recordSearchType>: "ATMPanel" (ATM panel mode), "ATMSurround" (ATM environment mode), ATMFace (ATM human face mode), and "ATMSafetyCabin" (ATM safety cabin mode).
2. Extended the search conditions of video and picture **XML_VideoPic_CMSearchDescription** (related URI: **/ISAPI/ContentMgmt/search**): added four search conditions to the sub node <metadataDescriptor> of <metadataList>: "ATMPanel" (ATM panel mode), "ATMSurround" (ATM environment mode), ATMFace (ATM human face mode), and "ATMSafetyCabin" (ATM safety cabin mode).
3. Extended the log types, see details in **Log Types for ISAPI**: added eight minor types to alarm logs: "ATMPanelStart" (ATM panel alarm started), "ATMPanelStop" (ATM panel alarm ended), "ATMSurroundStart" (ATM surround alarm started), "ATMSurroundStop" (ATM surround alarm ended), "ATMFaceStart" (ATM face alarm started), "ATMFaceStop" (ATM face alarm ended), "ATMSafetyCabinStart" (ATM safety cabin alarm started), and "ATMSafetyCabinStop" (ATM safety cabin alarm ended).

Summary of Changes in Version 2.6_June, 2020

Related Products: NVR with Models of iDS-9632NXI-I8/BA and iDS-6704NXI-I/BA in Software Version 4.1.73

1. Extended the device storage capability ***XML_RacmCap*** (related URI: ***/ISAPI/ContentMgmt/capabilities***):
added a value "loitering" (loitering detection) to the nodes ***<perimeterEventPicSearch>*** (perimeter protection events that support picture search) and ***<pictureSearchType>*** (picture search conditions).
2. Added two error codes, i.e., 0x400012C8-"ruleEventTypeDuplicate" and 0x400015C2-"polygonNotAllowedCrossing", to ***Response Codes of Text Protocol***.

Summary of Changes in Version 2.6_June, 2020

Related Product: DS-2TD1217B series and DS-2TD2636B series Thermal & Optical Bi-spectrum Network Camera

1. Extended the channel attributes ***XML_ChannelInfo*** (related URI: GET ***/ISAPI/AUXInfo/attributes/Channels/<ID>***):
added one sub node ***<FaceSnapThermometry>*** (upload the captured face picture with temperature information) to node ***<Thermal>***.
2. Extended the audible warning configuration capability ***JSON_AudioAlarmCap*** (related URI: GET ***/ISAPI/Event/triggers/notifications/AudioAlarm/capabilities?format=json***) and audible warning configurations ***JSON_AudioAlarm*** (related URI: ***/ISAPI/Event/triggers/notifications/AudioAlarm?format=json***):
added one alarm type "noMaskDetection" (no wearing mask detection) to node ***alarmType***.
3. Extended the device event capability ***XML_EventCap*** (related URI: GET ***/ISAPI/Event/capabilities***):
added two nodes ***<isSupportNoMaskDetection>*** (whether device supports no wearing mask detection) and ***<isSupportTMPA>*** (whether device supports temperature measurement pre-alarm).

Summary of Changes in Version 2.6_May, 2020

Related Product: iDS-2SK8144IXS-D/J(40X/2812) Smart Linkage PTZ Camera with software version 5.5.40

1. Extended device capability ***XML_DeviceCap*** (related URI: GET ***/ISAPI/System/capabilities***):
added five nodes: ***<isSupportPersonArmingTrack>*** (whether device supports auto person arming), ***<isSupportManualPersonArmingTrack>*** (whether device supports manual person arming), ***<isSupportGPSCalibrationMode>*** (whether device supports GPS calibration), ***<isSupportGPSVerification>*** (whether device supports GPS verification), and ***<isSupportHBDLib>*** (whether device supports human body picture library).
2. Extended message of event capability supported by a single channel ***XML_ChannelEventCap*** (related URI: GET ***/ISAPI/Event/channels/<ID>/capabilities***):
added two event types "personArmingTrack" (auto person arming) and "manualPersonArmingTrack" (manual person arming) to the node ***<eventType>***.
3. Extended the linkage capabilities of different alarm categories ***XML_EventTriggersCap*** (related URI: GET ***/ISAPI/Event/triggersCap***):
added one node ***<HBDLibTriggerCap>*** (alarm linkage configuration capability of human body picture library).

4. Added two event types "personArmingTrack" (auto person arming) and "manualPersonArmingTrack" (manual person arming) to **Supported Alarm/Event Types**.
5. Extended the master-slave tracking capability **XML_MasterSlaveTrackingCap** (related URI: GET **/ISAPI/MasterSlaveTracking/capabilities**):
added one node <isSupportLinkedTracking> (whether device supports advanced parameters configuration of linked tracking capture).
6. Added a function of configuring advanced parameters of linked tracking capture.
Get capability: GET **/ISAPI/MasterSlaveTracking/linkedTracking/capabilities?format=json** ;
Get or set advanced parameters of linked tracking capture: GET or PUT **/ISAPI/MasterSlaveTracking/linkedTracking?format=json**.
7. Extended **Response Codes of Text Protocol** :
added three sub status codes to status code 4 (Invalid Operation):
0x40002092-"HBDLibNumOverLimit" (the number of human body picture libraries reaches the upper limit), 0x4000209C-pointsOnStraightLine (calibrating failed. The calibration points cannot be one the same line), and 0x4000209D-TValueLessThanOrEqualZero (calibration failed. The T value of the calibration points should be larger than 0);
added nine sub status codes to status code 6 (Invalid Message Content):
0x6000008b-"customHBPIDError" (incorrect ID of custom human body picture library),
0x6000008c-"customHBPIDRepeat" (duplicated ID of custom human body picture library),
0x6000008d-"dataVersionsInHBDLibMismatches" (database versions mismatches of human body picture library), 0x6000008e-"invalidHBPID" (invalid human body picture PID),
0x6000008f-"invalidHBDID" (invalid ID of human body picture library),
0x60000090-"humanLibraryError" (error of human body picture library),
0x60000091-"humanLibraryNumError" (no more human body picture library can be added),
0x60000092-"humanImagesNumError" (no more human body picture can be added), and
0x60000093-"noHumanInThePicture" (modeling failed, no human body in the picture).
8. Extended the capability of exporting device diagnose information
XML_Cap_DiagnosedDataParameter (related URI: **/ISAPI/System/diagnosedData/parameter/capabilities**) and device diagnose information exporting parameters
XML_DiagnosedDataParameter (related URI: **/ISAPI/System/diagnosedData/parameter**):
added three nodes <HardwareInfo> (whether to enable exporting hardware information), <SoftwareInfo> (whether to enable exporting software information), and <logInfo> (whether to enable exporting startup log).
9. Added the URIs for verifying GPS calibration, refer to **GPS Calibration** for details.
10. Extended the device protocol capability **XML_Cap_AdminAccessProtocolList** (related URI: GET **/ISAPI/Security/adminAccesses/capabilities**) and device supported protocol parameters
XML_AdminAccessProtocol (related URI: **/ISAPI/Security/adminAccesses**):
added one protocol "Bonjour" to node <protocol>.
11. Extended the device security capability **XML_SecurityCap** (related URI: GET **/ISAPI/Security/capabilities**) and **XML_IllegalLoginLock** (related URI: **/ISAPI/Security/illegalLoginLock**):
added one node <maxIllegalLoginLockTime> (lock duration when maximum illegal login attempts reached the upper limit).

12. Extended the PTZ channel capability ***XML_PTZChanelCap*** (related URI: GET ***/ISAPI/PTZCtrl/channels/<ID>/capabilities***):
added one nodes ***<isSupportGyroVerify>*** (whether device supports gyroscope attitude calibration).
13. Added URIs for calibrating gyroscope attitude:
Get capability: GET ***/ISAPI/PTZCtrl/channels/<ID>/gyroVerify/capabilities?format=json***
Set gyroscope attitude calibration parameters: PUT ***/ISAPI/PTZCtrl/channels/<ID>/gyroVerify?format=json***.
14. Extended the intelligent resource switch capability ***XML_Cap_VCResource*** (related URI: GET ***/ISAPI/System/Video/inputs/channels/<ID>/VCResource/capabilities***) and intelligent resource switch parameters ***XML_VCResource*** (related URI: ***/ISAPI/System/Video/inputs/channels/<ID>/VCResource***):
added one intelligent resource types "personArming" (person arming) to node ***<type>***.

Summary of Changes in Version 2.6_May, 2020

Related Product: TB-4117-3-S Temperature Screening Thermal Camera Module with Software Version 5.5.2

Added URIs for configuring device USB parameters:

Get USB parameters configuration capability: GET ***/ISAPI/System/usb/capabilities?format=json*** ;

Get or set USB parameters: GET or PUT ***/ISAPI/System/usb?format=json***.

Summary of Changes in Version 2.6_April, 2020

Related Product: DS-2DE Series IR Network Speed Dome with software version 5.5.23

Extended the supplement light configuration capability ***XML_Cap_SupplementLight*** and supplement light configuration parameters ***XML_SupplementLight*** (related URIs: ***/ISAPI/Image/channels/<ID>/supplementLight/capabilities*** and ***/ISAPI/Image/channels/<ID>/supplementLight***):

added two nodes: ***<irLightbrightnessLimit>*** (IR light brightness limit) and ***<whiteLightbrightnessLimit>*** (white light brightness limit).

1.2.3 Version 20.01-Version 20.03

Summary of Changes in Version 2.6_Mar., 2020

Related Product: iDS-SR8141MXS-A/T2(40×) Radar PTZ Camera with Software Version 5.5.20

1. Extended device capability message ***XML_DeviceCap*** (related URI: ***/ISAPI/System/capabilities***):
added two nodes: ***<radarIPDForm>*** (radar form) and ***<mixedTargetDetectionWebNoDisplay>*** (whether to enable not displaying multi-target-type recognition).
2. Extended PTZ control capability message ***XML_PTZChanelCap*** (related URI: ***/ISAPI/PTZCtrl/channels/<ID>/capabilities***):
added a node ***<isSupportTrackingRatioCfg>*** (whether it supports configuring tracking parameters).
3. Added two URIs of configuring tracking parameters:

Get configuration capability: GET */ISAPI/PTZCtrl/channels/<ID>/trackingRatio/capabilities?format=json* ;

Get or set parameters: GET or PUT */ISAPI/PTZCtrl/channels/<ID>/trackingRatio?format=json* .

4. Extended sub status codes in **Response Codes of Text Protocol** :

added a sub status code to status code 3 (Device Error): 0x30002051-"calibrationTimeout" (Calibration timed out);

added four sub status codes to status code 4 (Invalid Operation):

0x4000206A-"completeTheLevelCalibrationFirst" (The level calibration is not set),

0x4000206B-"completeTheRadarCameraCalibrationFirst" (The radar-camera calibration is not

set), 0x40002093-"theShieldRegionError" (Saving failed. The shielded area should be the ground area where the shielded object is located), and 0x40002094-"theDetectionAreaError" (Saving

failed. The detection area should only cover the ground area);

added two sub status codes to status code 6 (Invalid Content):

0x60008016-"uploadImageSizeError" (The size of the uploaded picture is larger than 5 MB) and

0x6000202C-"width/heightRatioOfPictureError" (The width/height ratio of the uploaded picture should be in the range from 1:2 to 2:1).

Summary of Changes in Version 2.6_Mar., 2020

Related Product: iDS-2CD70 series box camera, iDS-2CD70 series shield, iDS-2CD71 and iDS-2CD75 series dome camera, iDS-2CD7A series bullet camera with software version 5.5.120

1. Added URIs of configuring vibration detection alarm, see **Configure Vibration Detection Alarm** for details.
2. Extended device capability message **XML_DeviceCap** (related URI: */ISAPI/System/capabilities*):
added one node **<isSupportVibrationDetection>** (whether it supports vibration detection).
3. Extended message about event capability of all channels **XML_ChannelEventCapList** and message about event capability of a single channel **XML_ChannelEventCap** (related URIs: */ISAPI/Event/channels/capabilities* and */ISAPI/Event/channels/<ID>/capabilities*):
added one event type "vibrationDetection" (vibration detection) to the node **<eventType>**.
4. Extended message about linkage capabilities of different alarm categories **XML_EventTriggersCap** (related URI: */ISAPI/Event/triggersCap*):
added one node **<VibrationDetectionTriggerCap>** (alarm linkage capability of vibration detection).
5. Extended linkage parameter message **XML_EventTrigger** (related URI: */ISAPI/Event/triggers/<eventType>-<ID>*):
added one event type "vibrationDetection" (vibration detection) to the node **<eventType>**.
6. Extended device storage capability message **XML_RacmCap** (related URI: */ISAPI/ContentMgmt/capabilities*):
added a type "vibrationDetection" (vibration detection) to the node **<pictureSearchType>** (picture search condition) and **<recordSearchType>** (video search condition).
7. Extended capability message **XML_Cap_VCResource** and configuration message **XML_VCResource** about intelligent resources switch (related URIs: */ISAPI/System/Video/inputs/channels/<ID>/VCResource/capabilities* and */ISAPI/System/Video/inputs/channels/<ID>/VCResource*):

- added one intelligent resource type "faceCounting" (face counting) to the node **<type>**.
- Extended the log types, see details in **Log Types for ISAPI** :
added two minor types "vibrationDetectionStart" (Vibration Detection Alarm Started) and "vibrationDetectionStop" (Vibration Detection Alarm Stopped) to Alarm Logs.
 - Extended error codes, see details in **Response Codes of Text Protocol** :
added one error code 0x40002096-"invalidLaneLine" (Saving failed. Invalid lane line.) to status code 4 (Invalid Operation).

Summary of Changes in Version 2.6_Feb., 2020

Related Product: Thermal & Optical Bi-spectrum Network Speed Dome; Thermal & Optical Bi-spectrum Positioning System; Thermal & Optical Bi-spectrum Network Stable PTZ Camera in Version 5.5.25

Added URIs of configuring laser optical axis:

Get capability: GET **/ISAPI/System/laserOpticalAxis/capabilities?format=json** ;

Get or set laser optical axis parameters: GET or PUT **/ISAPI/System/laserOpticalAxis?format=json** ;

Adjust laser optical axis: PUT **/ISAPI/System/laserOpticalAxis/goto?format=json** .

Summary of Changes in Version 2.6_Feb., 2020

Related Product: DS-K1T804A Series and DS-K1T8003 Series Fingerprint Access Control Terminal with Software Version 1.3.0; DS-K1A802A Series and DS-K1A8503 Series Fingerprint Time Attendance Terminal with Software Version 1.3.0

- Extended configuration capability message **XML_Cap_DeviceLanguage** and parameter message **XML_DeviceLanguage** of device language (related URIs: **/ISAPI/System/DeviceLanguage/capabilities** and **/ISAPI/System/DeviceLanguage**):
added a language supported by the device "Ukrainian" to the node **<language>**.
- Extended Wi-Fi configuration capability message **XML_Cap_Wireless** and Wi-Fi parameter message **XML_Wireless** of a specific network interface (related URIs: **/ISAPI/System/Network/interfaces/<ID>/wireless/capabilities** and **/ISAPI/System/Network/interfaces/<ID>/wireless**):
added a security mode "WPA/WPA2" to the sub node **<securityMode>** of the node **<WirelessSecurity>**.

Summary of Changes in Version 2.6_Jan., 2020

Related Product: iDS-9632NXI-I8/X(B) DeepinMind NVR and iDS-6708NXI-I/8F(B) DeepinMind NVS with Software Version V4.1.72

- Extended the log types in **Log Types for ISAPI** :
added two alarm minor log types: "sleepOnDutyStart" (Sleeping on Duty Alarm Started) and "sleepOnDutyStop" (Sleeping on Duty Alarm Stopped).
- Extended message about device storage capability **XML_RacmCap** (related URI: **/ISAPI/ContentMgmt/capabilities**):
added two nodes **<behaviorEventPicSearch>** (behavior analysis events supported by picture search) and **<perimeterEventPicSearch>** (perimeter events supported by picture search).
added one search condition "playCellphone" (playing mobile phone) to the node **<pictureSearchType>**.

- added 6 video types "running", "failDown", "violentMotion", "retention", "allPerimeterEvent", "allBehaviorEvent" to the node **<recordSearchType>** (video file type to be searched for).
3. Extended message about version information of algorithm library ***XML_AlgorithmsVersion*** (related URI: ***/ISAPI/System/algorithmsVersion***):
added two algorithm library names "streetBehavior" (street behavior) and "workBehavior" (work behavior)
to the node **<name>** of **<AlgorithmsVersionInfo>**.
4. Extended error codes in ***Response Codes of Text Protocol*** :
added sub error codes to status code 4 (Invalid Operation):
0x400012a6-"vehicleEnginesNoResource" (Insufficient vehicle engine resources),
0x400050df-"noDetectionArea" (No detection area),
0x4000128A-"noMorePeopleNumChangeRulesAdded" (Maximum number of people number changing rules reached), 0x4000128D-"noMoreViolentMotionRulesAdded" (Maximum number of violent motion rules reached), 0x4000128E-"noMoreLeavePositionRulesAdded" (Maximum number of leaving position rules reached), 0x400012a9-"noMoreRunningRulesAdded" (Maximum number of running rules reached), 0x400012aa-"noMoreGroupRulesAdded" (Maximum number of people gathering rules reached),
0x400012ab-"noMoreFailDownRulesAdded" (Maximum number of people falling down rules reached), 0x400012ac-"noMorePlayCellphoneRulesAdded" (Maximum number of playing cellphone rules reached), 0x400015ad-"noMoreRetentionRulesAdded" (Maximum number of people retention rules reached), 0x400015ae-"noMoreSleepOnDutyRulesAdded" (Maximum number of sleeping on duty rules reached).
added sub errors to status code 6 (Invalid Message Content):
0x60002023-"noSafetyHelmetRegion" (The hard hat detection area is not configured),
0x60001053-"behaviorAnalysisRuleInfoError" (Behavior analysis rule parameters error),
0x60001054-"safetyHelmetParamError" (Hard hat parameters error),
0x60001077-"OneChannelOnlyCanBindOneEngine" (No more engines can be bound).

Summary of Changes in Version 2.6_Jan., 2020

Related Product: DS-K1T341AM, DS-K1T341AMF, DS-K1T642M, DS-K1T642MF, DS-K1T642E, DS-K1T642EF, DS-K1T642MW, DS-K1T642MFW, DS-K1T642EW, and DS-K1T642EFW Face Recognition Terminal with Software Version 1.0

- Added two URIs of getting the algorithm library information:
Get capability: GET ***/ISAPI/System/algorithmsInfo/capabilities?format=json*** ;
Get algorithm library information: GET ***/ISAPI/System/algorithmsInfo?format=json*** .
- Extended device capability message ***XML_DeviceCap*** (related URI: ***/ISAPI/System/capabilities***):
added two nodes: **<isSupportCertificateCaptureEvent>** (whether it supports certificate capture and comparison events) and **<isSupportAlgorithmsInfo>** (whether it supports getting the algorithm library version information).

Chapter 2 ISAPI Description

The design of Intelligent Security API (hereafter referred as to ISAPI) adopts RESTful style, so this part introduces the predefined resource operation methods, API (URL) format, interaction message format, time format, namespace, and error processing method.

2.1 Operation Method

The methods to operate resources via ISAPI are same as those of HTTP (Hyper Text Transport Protocol) and RTSP (Real Time Streaming Protocol).

Note

The RTSP operation methods are mainly used to get the real-time stream for live view, two-way audio, and playback in this manual. For details about HTTP and RTSP, please refer to <https://tools.ietf.org/html/rfc2612> and <https://tools.ietf.org/html/rfc2326>.

Table 2-1 HTTP Operation Method

Method	Description
POST	Create resources. This method is only available for adding resource that does not exist before.
GET	Retrieve resources. This method cannot change the system status, only return data as the response to the requester.
PUT	Update resources. This method is usually for update the resource that already exists, but it can also be used to create the resource if the specific resource does not exist.
DELETE	Delete resources.

Table 2-2 RTSP Operation Method

Method	Description
OPTIONS	Get the supported RTSP operation methods. See the request and response message format below when interacting between client software and server. <pre>OPTIONS %s RTSP/1.0\r\n //Request URL CSeq:%u\r\n //Command No. User-Agent:%s\r\n //Client software name \r\n /*Succeeded*/ RTSP/1.0 200 OK\r\n CSeq: %u\r\n //Command No.</pre>

Method	Description
	Public: %s\r\n //Supported operation methods Date: %s\r\n //Date and time \r\n /*Failed*/ RTSP/1.0 4XX/5XX %s\r\n CSeq: %u\r\n //Command No. Date: %s\r\n //Date and time \r\n
DESCRIBE	Transfer basic information by SDP (Session Description Protocol, see https://tools.ietf.org/html/rfc2327) files, such as URL with SETUP command and so on. See the request and response message format below when interacting between client software and server. DESCRIBE %s RTSP/1.0\r\n //URL CSeq: %u\r\n //Command No. Accept: application/sdp\r\n //The SDP description is accepted Authorization: %s\r\n //Authentication information User-Agent: %s\r\n //Client software name \r\n /*Succeeded*/ RTSP/1.0 200 OK\r\n CSeq: %u\r\n //Command No. Content-Type: application/sdp\r\n //The SDP description exists behind the command Content-Base: %s\r\n //URL Content-Length: %d\r\n //The length of contents behind the command \r\n [content] //SDP description /*Failed*/ RTSP/1.0 4XX/5XX %s\r\n CSeq: %u\r\n //Command No. \r\n
SETUP	Interact the session information, such as transmission mode, port number, and so on. See the request and response message format below when interacting between client software and server. SETUP %s RTSP/1.0\r\n //URL CSeq: %u\r\n //Command No. Authorization: %s\r\n //Authentication information Session: %s\r\n //Session ID is only returned at the even number of times Transport: %s\r\n //Transmission protocol User-Agent: %s\r\n //Client software name \r\n /*Succeeded*/ RTSP/1.0 200 OK \r\n

Method	Description
	<pre> CSeq: %u\r\n Session:%s\r\n //Session ID Transport: s% //Transmission method Date: s% //Date and time /*Failed*/ RTSP/1.0 4XX/5XX %s\r\n CSeq: %u\r\n //Command No. \r\n </pre>
PLAY	Start the stream transmission. See the request and response message format below when interacting between client software and server. <pre> PLAY %s RTSP/1.0\r\n //URL CSeq:%u\r\n //Command No. Authorization:%s\r\n //Authentication information Session:%s\r\n //Session ID Range: npt=%f-%f\r\n //Determine the play range User-Agent:%s\r\n //Client software name \r\n /*Succeeded*/ RTSP/1.0 200 OK \r\n CSeq: %u\r\n Session:%s\r\n RTP-Info:%s Date: %s /*Failed*/ RTSP/1.0 4XX/5XX %s\r\n CSeq: %u\r\n //Command No. Session:%s\r\n \r\n </pre>
PAUSE	Pause the stream transmission.
TEARDOWN	Stop the stream transmission. See the request and response message format below when interacting between client software and server. <pre> TEARDOWN %s RTSP/1.0\r\n //URL CSeq: %u\r\n //Command No. Authorization:%s\r\n //Authentication information Session:%s\r\n //Session ID User-Agent:%s\r\n //Client software name \r\n /*Succeeded*/ RTSP/1.0 200 OK \r\n CSeq: %u\r\n Session:%s\r\n </pre>

Method	Description
	Date:%s\r\n\r\n /*Failed*/ RTSP/1.0 4XX/5XX %s\r\n CSeq: %u\r\n //Command No. Session:%s\r\n\r\n

2.2 URL Format

URL (Uniform Resource Locator) is a further class of URIs, it can identify a resource and locate the resource by describing its primary access mechanism.

The format of URL is defined as the follows: **<protocol>://<host>[:port][abs_path [?query]]**.

protocol

Protocol types, i.e., HTTP (version 1.1) and RTSP (version 1.0).

host

Host name, IP address, or the FQDN (Fully Qualified Domain Name) of network devices.

port

Port number of host service for listening the connection status of TCP (Transmission Control Protocol, see <https://tools.ietf.org/html/rfc793>) or UDP (User Datagram Protocol, see <https://tools.ietf.org/html/rfc768>). If this field is not configured, the default port number will be adopted. For HTTP, the default port number is 80, and for RTSP, the default port number is 554.

abs_path

Resource URI: /ServiceName/ResourceType/resource. Here, the **ServiceName** is ISAPI; the **ResourceType** is predefined with upper camel case according to different functions, see details in the following table; the resource is defined with lower camel case and can be extended in actual applications. E.g., /ISAPI/System/Network/interfaces.

Predefined URI Model	Description
/ISAPI/System/...	System related resources
/ISAPI/Security/...	Security related resources
/ISAPI/Event...	Event/alarm related resources
/ISAPI/Image/...	Video encoding and image related resources
/ISAPI/ContentMgmt/...	Storage management related resources

query

Strings for describing resources information, including related parameters. The parameter names and values must be listed as the following format in this field: ?p1=v1&p2=v2&...&pn=vn.



Note

- To locate the connected device, when operating lower-level device via the URL, the **query** field should be filled as ?devIndex=uuid&p1=v1&p2=v2&...&pn=vn. The uuid (or guid) is a 32-byte (128 bits) random number, which is unique and generated by operating system when adding device, and its format is "xxxxxxxx-xxxx-xxxx-xxxx-xxxxxxxxxxxx".
- For message in JSON format, the **query** field should be filled as ?format=json&p1=v1&p2=v2&...&pn=vn. For details about message format, refer to the next section below. E.g., <http://10.17.132.22/ISAPI/System/time?format=json&devIndex=550e8400e29b41d4a716446655440000>.

2.3 Message Format

During the ISAPI integration, the request and response messages generated among the interaction between devices and platform are data in XML format or JSON format.



Note

The message format here is only available for URLs based on HTTP.

XML Format

- For the previous integration, XML is a common format which may only cause a little changes in the later integration.
- Generally, for configuration information, the **Content-Type** in the XML format message is "application/xml; charset='UTF-8'", see details below.

```
//Request Message
GET /ISAPI/System/status HTTP/1.1
...

//Response Message
HTTP/1.1 200 OK
...
Content-Type: application/xml; charset="UTF-8"
...
<?xml version="1.0" encoding="UTF-8"?>
<DeviceStatus version="2.0" xmlns="http://www.isapi.org/ver20/XMLSchema">
...
</DeviceStatus>
```

For data (e.g., firmware, configuration files), the **Content-Type** in the XML format message is "application/octet-stream", see details below.

```
//Request Message
PUT /ISAPI/System/configurationData HTTP/1.1
```



```
...
Content-Type: application/octet-stream
...
[proprietary configuration file data content]

//Response Message
HTTP/1.1 200 OK
...

Content-Type: application/xml; charset="UTF-8"
...
<?xml version="1.0" encoding="UTF-8"?>
<ResponseStatus version="2.0" xmlns="http://www.isapi.org/ver20/XMLSchema">
...
</ResponseStatus>
```

JSON Format Message

- The leaf node (without any sub node) in the message is named by lower camel case, while the non-leaf node in the message is named by upper camel case.
- To communicate by the messages in JSON format, the devices must support the public specifications in <http://www.ecma-international.org/publications/files/ECMA-ST/ECMA-404.pdf> and HTTP with version 1.1.



Note

JSON is a lightweight data format which is a subset of JavaScript language and is small, fast, and easy to be parsed.

- Generally, for configuration information, the **Content-Type** of message is "application/json", see the example below:

```
//Request message
GET /ISAPI/System/status HTTP/1.1
...

//Response message
HTTP/1.1 200 OK
...
Content-Type: application/json
...
"DeviceStatus":""
...
```

For data (e.g., firmware, configuration files), the **Content-Type** of message is "application/octet-stream", see the example below:

```
//Request message
PUT /ISAPI/System/configurationData HTTP/1.1
...
Content-Type: application/octet-stream
...
[proprietary configuration file data content]
```

```
//Response message
HTTP/1.1 200 OK
...
Content-Type: application/json
...
"ResponseStatus":""
...
```

2.4 Others

Time Format

The time format during ISAPI integration adopts ISO8601 standard (see details in <http://www.w3.org/TR/NOTE-datetime-970915>), that is, YYYY-MM-DDThh:mm:ss.sTZD (e.g., 2017-08-16T20:17:06+08:00).

Namespace

For message in XML format, namespace is required. The following namespaces are available:

- xmlns=http://www.isapi.org/ver20/XMLSchema
- xmlns:xs="http://www.w3.org/2001/XMLSchema"
- xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance"
- xmlns:xlink="http://www.w3.org/1999/xlink"

Error Processing

During the integration applications of ISAPI protocol, when the error of URL based on HTTP occurred, the ResponseStatus message (in XML or JSON format) which contains error code will be returned. If the error of URL based on RTSP occurs, the corresponding status code will directly be returned, for details, refer to <https://tools.ietf.org/html/rfc2326> .

Chapter 3 Security

This part mainly introduces the authentication, user permission, and encryption in the integration applications of ISAPI.

3.1 Authentication

When communicating via ISAPI protocol, the digest of the session must be authenticated.



- The authentication must be based on *HTTP Authentication: Basic and Digest Access Authentication*, see <https://tools.ietf.org/html/rfc2617> for details.
- The request session must contain authentication information, otherwise, the device will return 401 error code.

The message digest, which contains user name, password, specific nonce value, HTTP or RTSP operation methods, and request URL, is generated by the MD5 algorithm, see the calculation rules below.

qop=Undefined

Digest=MD5(MD5(A1):<nonce>:MD5(A2))

qop="auth:"

Digest=MD5(MD5(A1):<nonce>:<nc>:<cnonce>:<qop>:MD5(A2))

qop="auth-int:"

Digest=MD5(MD5(A1):<nonce>:<nc>:<cnonce>:<qop>:MD5(A2))



- The **qop** is a value for determining whether the authentication is required.
- A1 and A2 are two data blocks required for digest calculation.
 - A1: Data block about security, which contains user name, password, security domain, random number, and so on. If the digest calculation algorithm is MD5, A1=<user>:<realm>:<password>; if the algorithm is MD5-sess, A1=MD5(<user>:<realm>:<password>):<nonce>:<cnonce>.
 - A2: Data block about message, such as URL, repeated requests, message body, and so on, it helps to prevent repeated, and realize the resource/message tamper-proof. If the **qop** is not defined or it is "auth:", A2=<request-method>:<uri-directive-value>; if the **qop** is "auth-int:", A2=<request-method>:<uri-directive-value>:MD5(<request-entity-body>).
- The **nonce** is the random number generated by service, the following generation formula is suggested: nonce = BASE64(time-stamp MD5(time-stamp ":" ETag ":" private-key)). The **time-stamp** in the formula is the time stamp generated by service or the unique serial No.; the **ETag** is

the value of HTTP ETag header in the request message; the **private-key** is the data that only known by service.

If authentication failed, the device will return the ***XML_ResponseStatus_AuthenticationFailed*** message, and the remaining authentication attempts will also be returned. If the remaining attempts is 0, the user will be locked at the next authentication attempt.

3.2 User Permission

Three user types with different permissions are adopted in the security specifications to control the access.

Administrator

The account of this user type is called "admin", which has the permissions to access all supported resources. And it must be activated at any time.

Operator

The account of this user type has the permissions to access common resources and less advanced resources, see details in the description of each resource.

Viewer

The account of this user type has the permissions to access common resources only, see details in the description of each resource.

3.3 Encryption

Two encryption modes are available, including all information encryption (based on HTTPS) and sensitive information (e.g., user name, password, card No. and so on) encryption.



- The sensitive information varies with different applications, see details in the description of protocol messages.
 - Encryption will not affect the escape behavior of contents in XML format, which means that the contents should be escaped before encryption.
-

Encryption Capability

Before encrypting the information, the encryption capability of devices or client software should be obtained via the URI: GET ***/ISAPI/Security/capabilities*** to know the available encryption algorithm. The encryption algorithm information is returned by the **securityVersion** node in response message ***XML_SecurityCap***.

Note

Different values of **securityVersion** node correspond to different encrypted contents and encryption modes. If node value is 1, it indicates that the encryption mode is AES128 CBC, and only the password will be encrypted; if the node value is 2, the AES256 CBC mode is adopted, and both the user name and password will be encrypted.

Encryption Flow

Flow of encrypting source data: **Source Data** → **Encode by Base64 Method** → **Encrypt in AES128/AEC256 CBC Mode** .

Note

Encoding data by Base64 method is to be compatible with binary data and get the data size before encrypting by AES128/AEC256 CBC mode (the data size is not real after encrypting as 0 will be supplied).

Encryption Key

The key used for encryption is generated by iteratively encrypting the (Password + "AaBbCcDd1234!@#\$") according to SHA256 algorithm. E.g., if the device password is abcd1234, then the key is SHA(SHA(SHA("abcd1234AaBbCcDd1234!@#\$"))...).

Note

- The number of iterations is determined by device capability, see the node **keyIterateNum** in **XML_SecurityCap** message.
 - If the password is changed when generating the key, the previous password will be used.
 - The device gets the key when it is authenticated, and the client software/system calculates the key after logging in.
-

Encryption URI Format

The encryption modes are displayed by adding **query** parameters, i.e., **security** and **iv**, to the URI (e.g., /ISAPI/xxx?security=1&iv=xxx).

Note

- The parameter **security** is the version No. of encryption scheme. When **security** does not exist, it indicates that the data is not encrypted; when **security**=1, it indicates that the nodes of sensitive information in the message are encrypted in AES128 CBC mode; when **security**=2, it indicates that the nodes of sensitive information in the message are encrypted in AES256 CBC mode.
- The parameter **iv** is the initialization vector, and it is required when **security**=1 or 2. The 16-byte hexadecimal iv of AES128/AES256 CBC mode is generated by client software or system, and it will be transformed to 32-byte characters when transmitting it to devices via URI. When encrypting in AES128/AES256 CBC mode, loop encryption by data block size is not required,

and after encrypting for once, the **iv** will be changed, so if there are two or more nodes to be encrypted, you should copy the iv during encoding to make sure the iv is same for encrypting each node.

- When you want to determine the value of **security** in the URI, you must compare the encryption scheme version No. of client software and device. If the supported versions are different, you should adopt the lower version. E.g., if the client software supports **security=2**, the device supports **security=1**, then **security=1** will be adopted.
-

Suggestion About Extension

If there are other existing nodes in the message are regarded as sensitive information, you can add values to the parameter **security** in the URI, e.g., **security=3** (encrypt added sensitive information in AES128 CBC mode) and **security=4** (encrypt added sensitive information in AES256 CBC mode).

3.3.1 Encrypt Stream

To improve the security of real-time or history stream from devices and comply with the laws and regulations of some parts of oversea markets, we have provided the stream encryption function based on AES algorithm to encrypt the stream.

Steps

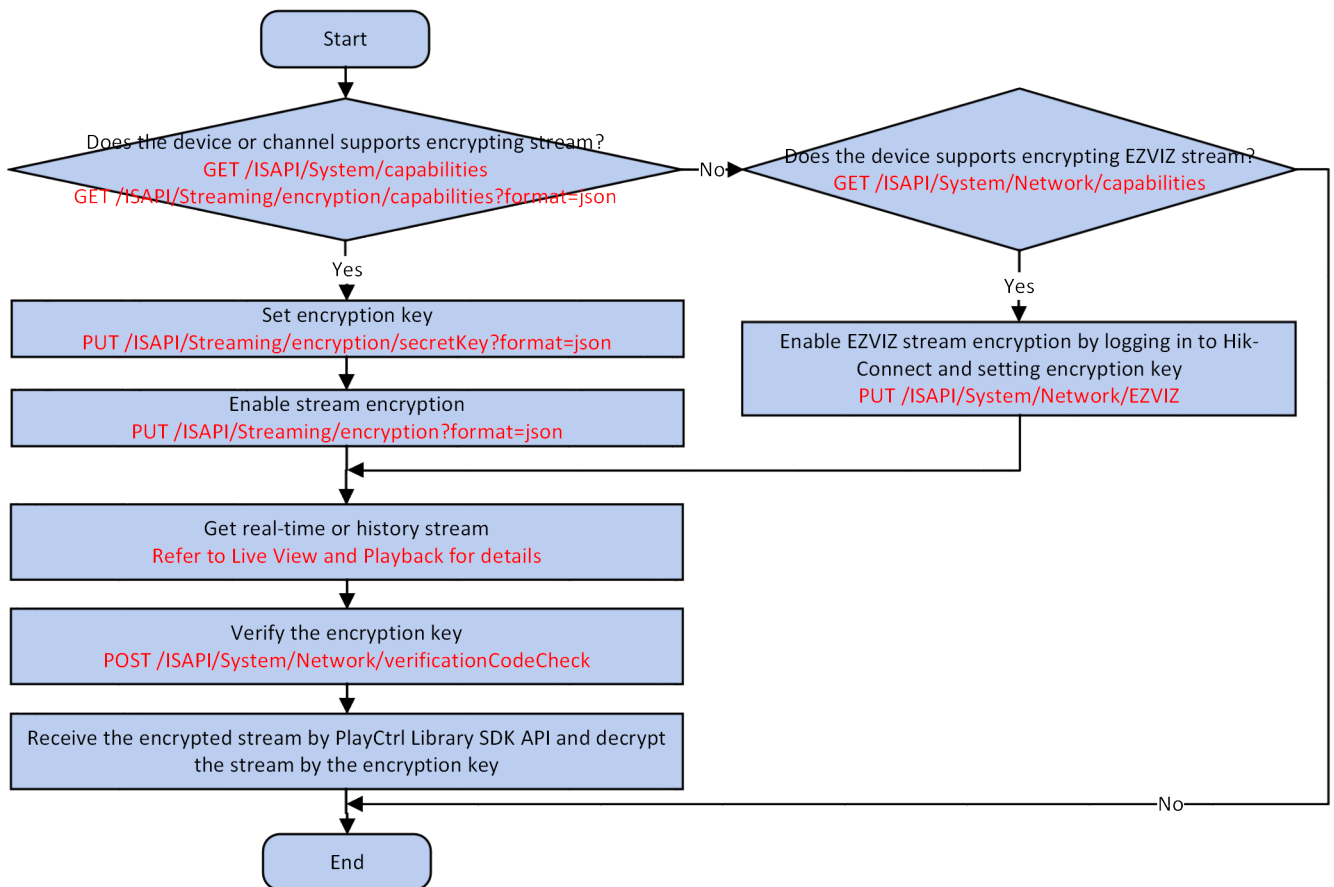


Figure 3-1 Programming Flow of Encrypting Stream

Note

- For devices with Hisilicon chips, up to 16 channels' stream can be encrypted. If the number of channels is larger than 16, the error "insufficient resources" will be returned when starting live view or playback.
- For Sky series devices, all channels' stream can be encrypted.
- The PlayCtrl library will not check if the stream is encrypted. If the stream is encrypted, but the decryption key is not configured or incorrect decryption key is configured, the display image will be black.
- If the encryption algorithm of the external stream is known, you can decrypt and play the stream without the PlayCtrl library.

1. Call **`/ISAPI/System/capabilities`** or **`/ISAPI/Streaming/encryption/capabilities?format=json`** by GET method to get device capability or encryption capability to check if the device or channel supports stream encryption.

- The node **<isSupportEncryption>** is not returned in the capability message ***XML_DeviceCap*** or the value of node **<enable>** in the capability message ***JSON_EncryptionCap*** is "false": perform the steps below.
 - a. Call ***/ISAPI/System/Network/capabilities*** by GET method to get network capability to check if the device supports encrypting EZVIZ stream.

Note

If the node **<isSupportEZVIZ>** is returned in the capability message ***XML_NetworkCap*** and its value is "true", it indicates that encrypting EZVIZ stream is supported. Otherwise, please end this task.

- b. Call ***/ISAPI/System/Network/EZVIZ*** by PUT method to log in to Hik-Connect and set encryption key to enable EZVIZ stream encryption.
 - The node **<isSupportEncryption>** is returned in the capability message ***XML_DeviceCap*** and its value is "true" or the value of node **<enable>** in the capability message ***JSON_EncryptionCap*** is "true": the device or channel supports stream encryption, and continue to perform the steps below.
 - a. Set encryption key: PUT ***/ISAPI/Streaming/encryption/secretKey?format=json*** .
 - b. Enable stream encryption: PUT ***/ISAPI/Streaming/encryption?format=json*** .
2. Get real-time or history stream, refer to ***Live View and Playback*** for details.
 3. Call ***/ISAPI/System/Network/verificationCodeCheck*** by POST method to verify the encryption key.

Note

To check if verifying verification code is supported, you can call ***/ISAPI/System/Network/capabilities*** by GET method to get the device network capability (***XML_NetworkCap***). If supports, the node **<isSupportVerificationCodeCheck>** is returned in the capability and its value is "true".

4. Receive the encrypted stream by calling encrypted stream callback function (PlayM4_SetEncryptTypeCallBack) of PlayCtrl Library SDK and decrypt the stream by the encryption key.

Note

Refer to *Cross-Platform PlayCtrl Library SDK User Manual* for details of the encrypted stream callback function and playing of the decrypted stream.

3.3.2 Encrypt Sensitive Information

Some sensitive information (such as user name, password, and so on) in the messages should be encrypted to improve security during interaction with devices via ISAPI.

Steps

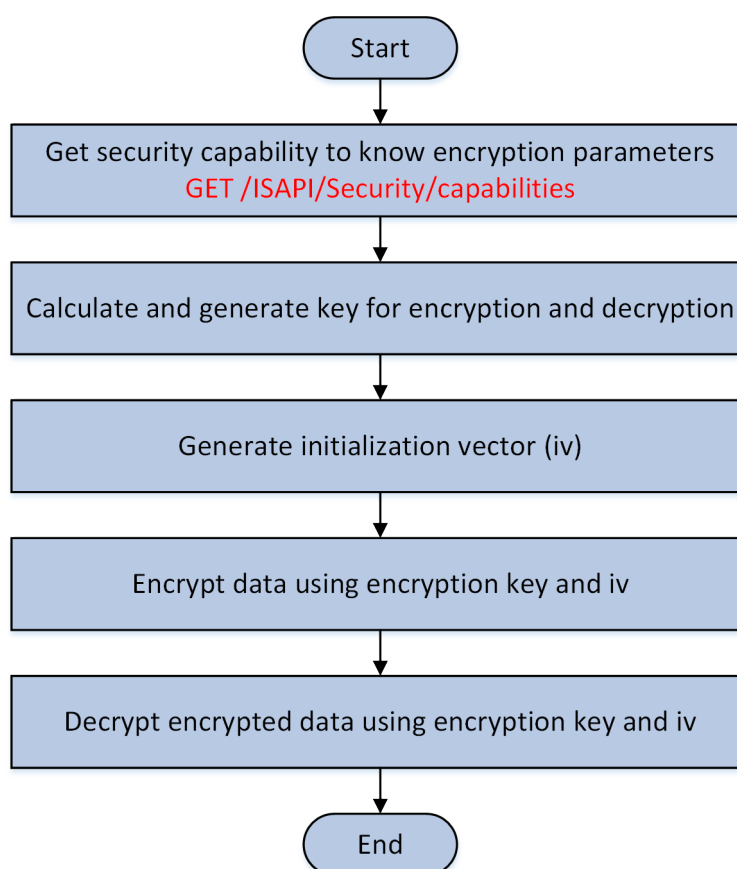


Figure 3-2 Programming Flow of Encrypting Sensitive Information

Note

Values of the following parameters in query parameters, XML messages, and JSON messages are sensitive information that needs to be encrypted.

Table 3-1 Sensitive Information to be Encrypted

Format	Parameters
Query	identityKey (identity key, which is the key of the private face picture library) and secretkey (secret key)
XML	<userName> (user name), <password> (password), <loginPassword> (login password), <snmpAuthenticationPassword> (SNMP authentication password), <snmpPrivacyPassword> (SNMP encryption password), <accountName> (account name), <encryptionKey> (encryption key), <sharedKey> (shared key), <Username> (user name), <passwd>(password), <name> (name), <authPasswd> (authentication

Format	Parameters
	password), <passWord> (password), <key> (key), <tokenKey> (token key), <oldPwd> (old password), <newPwd> (new password), and <terminalPwd> (terminal password)
JSON	"userName" (user name), "password" (password), "loginPassword" (login password), "snmpAuthenticationPassword" (SNMP authentication password), "snmpPrivacyPassword" (SNMP encryption password), "accountName" (account name), "encryptionKey" (encryption key), "sharedKey" (shared key), "Username" (user name), "passwd" (password), "authPasswd" (authentication password), "passWord" (password), "key" (key), "tokenKey" (token key), "randomCode" (random code), "AK" (access key), "SK" (encryption key), "Card" (card No.), "Password" (password), "FingerPrint" (fingerprint), "TemporaryPassword" (temporary password), "RemoteControl" (remote control), and "WirelessInfo" (wireless information)

1. Call **/ISAPI/Security/capabilities** by GET method to get the device security capability to know the encryption parameters.

The security capability is returned in the message **XML_SecurityCap** . Nodes of encryption parameters include <securityVersion> (encryption scheme version), <keyIterateNum> (iteration times), <isIrreversible> (whether the password is irreversible), and <salt> (salt value).



Note

The salt value will be returned only when the request URL contains the query parameter **username**.

2. Calculate and generate the key for encryption and decryption according to the encryption parameters.
 - If <isIrreversible> is "true" and <salt> is also configured, the key is calculated and generated by iteratively encrypting the value of (user name+salt+password) with SHA256 algorithm.



Note

- The value of <salt> should be different for different users.
 - The password encrypted by this method is irreversible.
- If <isIrreversible> is "false" or does not exist, and <salt> is null, the key is calculated and generated by iteratively encrypting the password with SHA256 algorithm.



Note

The password encrypted by this method is reversible.

3. Generate a byte array containing 16 elements randomly as the initialization vector (iv) and covert the byte array to character array.
4. Encrypt the data.
 - 1) Convert iv from character array to byte array.

- 2) Encode the data to be encrypted by using base64.
- 3) Set the AES key based on the encryption key and the encryption version.



Note

Currently, only AES-128 encryption is supported, which indicates that the **securityVersion** is 1.

- 4) Encrypt the data by using iv and AES key in AES-CBC mode.
 - 5) Convert the encrypted data to character array.
5. Decrypt the encrypted data.
- 1) Convert iv from character array to byte array.
 - 2) Convert the encrypted data from character array to byte array.
 - 3) Set the AES key based on the encryption key and the encryption version.



Note

Currently, only AES-128 encryption is supported, which indicates that the **securityVersion** is 1.

- 4) Decrypt the encrypted data by using iv and AES key in AES-CBC mode.
- 5) Decode the decrypted data by using base64.

Example

Sample Code of Encrypting Sensitive Information

```
using System;
using System.Collections.Generic;
using System.Linq;
using System.Security.Cryptography;
using System.Text;
using System.Threading.Tasks;

namespace SystemManagement.InformationEncryption
{
    class InfoEncryption
    {
        /// <summary>
        /// Generate initialization vector (iv)
        /// </summary>
        /// <param name="strInitVector"></param>
        /// <returns></returns>
        public static void GetInitVector(out string strInitVector)
        {
            byte[] szInitVector = new byte[16];
            Random ra = new Random();
            ra.NextBytes(szInitVector);
            byte[] byHexAes = AESEncryption.converByteArrayToCharArray(szInitVector, szInitVector.Length);
            strInitVector = Encoding.UTF8.GetString(byHexAes).ToLower();//This must be in lowercase.
        }

        /// <summary>
        /// Encrypt data with SHA256 algorithm. The data encrypted by this algorithm is reversible.
    }
}
```

```
/// </summary>
/// <param name="strData"></param>
/// <returns></returns>
public static string sha256(string strData)
{
    byte[] szData = Encoding.UTF8.GetBytes(strData);
    byte[] szHash = SHA256Managed.Create().ComputeHash(szData);
    byte[] szSha256 = AESEncryption.converByteArrayToCharArray(szHash, 32); //The standard algorithm is Hash. The
length 32 is set according to the sample document.
    return Encoding.UTF8.GetString(szSha256);
}

/// <summary>
/// Encrypt (user name+salt+password) with SHA256 algorithm. The password encrypted by this algorithm is
irreversible.
/// </summary>
/// <param name="strUser"></param>
/// <param name="szSalt"></param>
/// <param name="strPassword"></param>
/// <returns></returns>
public static string calcSha256(string strUser, byte[] szSalt, string strPassword)
{
    string strSrcData = strUser;
    byte[] szRealSalt = new byte[64]; //The size of the salt value is 64 bits.
    if (szSalt.Length > 64)
    {
        return null;
    }
    Array.Copy(szSalt, szRealSalt, szSalt.Length);
    strSrcData = strSrcData + Encoding.UTF8.GetString(szRealSalt) + strPassword;
    return sha256(strSrcData);
}

/// <summary>
/// Generate encryption key
/// </summary>
/// <param name="strUserName"></param>
/// <param name="strSalt"></param>
/// <param name="strPassword"></param>
/// <param name="szOut"></param>
/// <param name="iKeyIterateNum"></param>
/// <param name="blIrreversible"></param>
/// <returns></returns>
public static void getEncryptKey(string strUserName, string strSalt, string strPassword, out byte[] szOut, int
iKeyIterateNum, bool blIrreversible)
{
    byte[] szSalt = null;
    if (strSalt != null)
    {
        szSalt = Encoding.UTF8.GetBytes(strSalt);
    }
    string strSrcData = string.Empty;
```

```
if (blrreversible && szSalt.Length > 0)
{
    string strIrrPsw = calcSha256(strUserName, szSalt, strPassword);
    if (strIrrPsw.Length > 64)
    {
        strSrcData = strIrrPsw.Substring(0,64);
    }
    else
    {
        strSrcData = strIrrPsw;
    }
}
else
{
    if (strPassword.Length > 64)
    {
        strSrcData = strPassword.Substring(0,64);
    }
    else
    {
        strSrcData = strPassword;
    }
}
strSrcData += "AaBbCcDd1234!@#$$";
if (iKeyIterateNum <= 0)//Iterations
{
    iKeyIterateNum = 100;
}

//For the SHA256 iteration, iKeyIterateNum represents the number of iterations returned by the capability set.
for (int i = 0; i < iKeyIterateNum; i++)
{
    strSrcData = sha256(strSrcData);
}
byte[] szSHA256 = Encoding.UTF8.GetBytes(strSrcData);

//The result calculated with SHA256 algorithm is converted to a byte array for the last time.
byte[] szByteArray = AESEncryption.convertCharArrayToByteArray(szSHA256, szSHA256.Length);

if (szByteArray.Length > 32)
{
    szOut = new byte[32];
    Array.Copy(szByteArray, szOut, 32);
}
else
{
    szOut = new byte[szByteArray.Length];
    Array.Copy(szByteArray, szOut, szByteArray.Length);
}
}

/// <summary>
```

```
/// Encrypt content (sensitive information)
/// </summary>
/// <param name="strInitVexor"></param>
/// <param name="szAESKey"></param>
/// <param name="strSrcContent"></param>
/// <param name="strOut"></param>
/// <param name="iSecurityVersion"></param>
/// <returns></returns>
public static void getEncryptContent(string strInitVexor, byte[] szAESKey, string strSrcContent, out string strOut, int
iSecurityVersion)
{
    if (iSecurityVersion != 1)//1 refers to the AES-128 algorithm. Currently only 1 is supported.
    {
        strOut = strSrcContent;
        return;
    }

    //Convert to UTF-8
    byte[] szInitVexor = Encoding.UTF8.GetBytes(strInitVexor);
    byte[] szInitVexorByteArray = AESEncryption.convertCharArrayToByteArray(szInitVexor, szInitVexor.Length);
    byte[] szSrcBytes = Encoding.UTF8.GetBytes(strSrcContent);
    string strSrcBase64 = Convert.ToBase64String(szSrcBytes);

    //AES encryption
    byte[] szAesData = AESEncryption.AesEncrypt(strSrcBase64, szAESKey, szInitVexorByteArray);
    byte[] szOut = AESEncryption.converByteArrayToCharArray(szAesData, szAesData.Length);
    strOut = Encoding.UTF8.GetString(szOut);
}

/// <summary>
/// Decrypt content (sensitive information)
/// </summary>
/// <param name="strInitVexor"></param>
/// <param name="szAESKey"></param>
/// <param name="strSrcContent"></param>
/// <param name="strOut"></param>
/// <param name="iSecurityVersion"></param>
/// <returns></returns>
public static void getDecryptContent(string strInitVexor, byte[] szAESKey, string strSrcContent, out string strOut, int
iSecurityVersion)
{
    if (iSecurityVersion != 1)//1 refers to the AES-128 algorithm. Currently only 1 is supported.
    {
        strOut = strSrcContent;
        return;
    }

    //Convert to UTF-8
    byte[] szInitVexor = Encoding.UTF8.GetBytes(strInitVexor);
    byte[] szInitVexorByteArray = AESEncryption.convertCharArrayToByteArray(szInitVexor, szInitVexor.Length);
    byte[] szSrcBytes = Encoding.UTF8.GetBytes(strSrcContent);
    byte[] szSrcByteArray = AESEncryption.convertCharArrayToByteArray(szSrcBytes, szSrcBytes.Length);
```

```
string strAesData = AESEncryption.AesDecrypt(szSrcByteArray, szAESKey, szInitVextorByteArray);
byte[] szOut = Convert.FromBase64String(strAesData);
strOut = Encoding.UTF8.GetString(szOut);
}
}
}
```

Note

The sample code of the class AESEncryption is as shown below.

```
using System;
using System.Collections.Generic;
using System.IO;
using System.Linq;
using System.Security.Cryptography;
using System.Text;
using System.Threading.Tasks;

namespace SystemManagement.InformationEncryption
{
    class AESEncryption
    {
        /// <summary>
        /// Convert hexadecimal to binary
        /// </summary>
        /// <param name="chstr"></param>
        /// <returns></returns>
        public static byte hexToBinary(byte chstr)
        {
            char crtn = '\0';
            if (('0' <= chstr) && ('9' >= chstr))
            {
                crtn = (char)(chstr & 0x0F);
            }
            else if (('A' <= chstr) && ('F' >= chstr))
            {
                crtn = (char)(chstr - 'A' + 10);
            }
            else if (('a' <= chstr) && ('f' >= chstr))
            {
                crtn = (char)(chstr - 'a' + 10);
            }
            return (byte)crtn;
        }

        /// <summary>
        /// Convert a character array to a byte array
        /// </summary>
        /// <param name="pSrc"></param>
        /// <param name="nSrcLen"></param>
```

```
/// <returns></returns>
public static byte[] convertCharArrayToByteArray(byte[] pSrc, int nSrcLen)
{
    byte[] byChallengeDst2 = new byte[nSrcLen / 2];
    for (int i = 0; i < nSrcLen; i = i + 2)
    {
        byChallengeDst2[i / 2] = (byte)(hexToBinary(pSrc[i]) << 4);
        byChallengeDst2[i / 2] += (byte)hexToBinary(pSrc[i + 1]);
    }
    return byChallengeDst2;
}

/// <summary>
/// Convert a byte array to a character array
/// </summary>
/// <param name="pSrc"></param>
/// <param name="nSrcLen"></param>
/// <returns></returns>
public static byte[] converByteArrayToCharArray(byte[] pSrc, int nSrcLen)
{
    StringBuilder strB = new StringBuilder();
    for (int i = 0; i < nSrcLen; i++)
    {
        strB.Append(pSrc[i].ToString("x2")); //Here x must be in lowercase, which means converting to a lowercase
        hexadecimal character.
    }
    return Encoding.UTF8.GetBytes(strB.ToString());
}

/// <summary>
/// Encrypt in AES-CBC mode
/// </summary>
/// <param name="strSrcContent"></param>
/// <param name="szAESKey"></param>
/// <param name="iv"></param>
/// <returns></returns>
public static byte[] AesEncrypt(string strSrcContent, byte[] szAESKey, byte[] iv)
{
    RijndaelManaged rijndaelCipher = new RijndaelManaged();
    rijndaelCipher.Mode = CipherMode.CBC;
    rijndaelCipher.Padding = PaddingMode.Zeros; //The end is all-zero padding
    rijndaelCipher.KeySize = 128;
    rijndaelCipher.BlockSize = 128;
    byte[] keyBytes = new byte[16];
    int len = szAESKey.Length;
    if (len > keyBytes.Length)
    {
        len = keyBytes.Length;
    }
    System.Array.Copy(szAESKey, keyBytes, len);
}
```



```
rijndaelCipher.Key = keyBytes;
rijndaelCipher.IV = iv;
ICryptoTransform transform = rijndaelCipher.CreateEncryptor();
byte[] szSrcContent = Encoding.UTF8.GetBytes(strSrcContent);
byte[] szDstContent = transform.TransformFinalBlock(szSrcContent, 0, szSrcContent.Length);
return szDstContent;
}

/// <summary>
/// Decrypt in AES-CBC mode
/// </summary>
/// <param name="szSrcContent"></param>
/// <param name="szAESKey"></param>
/// <param name="iv"></param>
/// <returns></returns>
public static string AesDecrypt(byte[] szSrcContent, byte[] szAESKey, byte[] iv)
{
    RijndaelManaged rijndaelCipher = new RijndaelManaged();
    rijndaelCipher.Mode = CipherMode.CBC;
    rijndaelCipher.Padding = PaddingMode.Zeros; //The end is all-zero padding
    rijndaelCipher.KeySize = 128;
    rijndaelCipher.BlockSize = 128;
    byte[] keyBytes = new byte[16];
    int len = szAESKey.Length;
    if (len > keyBytes.Length)
    {
        len = keyBytes.Length;
    }
    System.Array.Copy(szAESKey, keyBytes, len);
    rijndaelCipher.Key = keyBytes;
    rijndaelCipher.IV = iv;
    ICryptoTransform transform = rijndaelCipher.CreateDecryptor();
    byte[] szDstContent = transform.TransformFinalBlock(szSrcContent, 0, szSrcContent.Length);

    //After decrypting the last 16 bytes of the ciphertext, the decrypted data of the last 16 bytes will be checked.
    //If the value of the last byte is 16, the 16-byte string data will be discarded.
    //If the value of the last byte is smaller than 16, it indicates that the original text was padded during encryption.
    The padding will be discarded.
    for (int i = 0; i < szDstContent.Length; i++)
    {
        if (szDstContent[i] <= 16)
        {
            szDstContent[i] = 0;
        }
    }
    //The \0 at the end needs to be removed.
    return Encoding.UTF8.GetString(szDstContent).Replace("\0", "");
}
}
```

3.4 Security Service

Management and Configuration

Function	Request URI
Get or set advanced parameters of security management	GET or PUT <i>/ISAPI/Security/advanced?format=json</i>
Get or set security questions in batch	GET or PUT <i>/ISAPI/Security/questionConfiguration</i>
Search for security logs	POST <i>/ISAPI/ContentMgmt/security/logSearch</i>
Security Mode Level of Private Protocol	• Get configuration capability of security mode level of private protocol: GET <i>/ISAPI/Security/CommuMode/capabilities?format=json</i> • Get parameters of security mode level of private protocol: GET <i>/ISAPI/Security/CommuMode?format=json</i> • Set parameters of security mode level of private protocol: PUT <i>/ISAPI/Security/CommuMode?format=json</i>
Get capability of selecting certificate	GET <i>/ISAPI/Security/certificate/select/capabilities?format=json</i>
Get or set parameters of selecting certificate	GET or PUT <i>/ISAPI/Security/certificate/select/<functionName>?format=json</i>
Get or set certificate type parameters of web service	GET or PUT <i>/ISAPI/Security/webCertificate</i>

Network Certificate

Function	Request URI
Get CA (Certificate Authority) certificate capability	GET <i>/ISAPI/Security/deviceCertificate/capabilities?format=json</i>
Import network certificate to device	PUT <i>/ISAPI/Security/deviceCertificate</i>
Import CA (Certificate Authority) certificate to device	PUT <i>/ISAPI/Security/deviceCertificate?customID=</i>

Function	Request URI
Network Certificate Search	- Get capability of certificate search: GET <i>/ISAPI/Security/deviceCertificate/certificates/capabilities?format=json</i> - Search for certificate information in a batch: GET <i>/ISAPI/Security/deviceCertificate/certificates?format=json</i> - Search for information of a specific device certificate: GET <i>/ISAPI/Security/deviceCertificate/certificates/<customID>?format=json</i>
Delete network certificate	DELETE <i>/ISAPI/Security/deviceCertificate/certificates/<ID></i> or <i>/ISAPI/Security/deviceCertificate/certificates/<customID>?format=json</i>

Authentication Certificate

Function	Request URI
Get, generate, or delete certificate signature request	GET, PUT , or DELETE <i>/ISAPI/Security/serverCertificate/certSignReq</i>
Download authentication certificate	GET <i>/ISAPI/Security/serverCertificate/downloadCertSignReq</i>
Get, install, or delete authentication certificate	GET, PUT , or DELETE <i>/ISAPI/Security/serverCertificate/certificate</i>
Export device's self-signed certificate	GET <i>/ISAPI/Security/deviceCertificate/selfSignCert</i>
Authentication Certificate Status Search	- Get status of one authentication certificate: GET <i>/ISAPI/Security/deviceCertificate/certificates/<ID>/status?format=json</i> - Get status of all authentication certificates: GET <i>/ISAPI/Security/deviceCertificate/certificates/status?format=json</i>
Generate one authentication certificate	PUT <i>/ISAPI/Security/deviceCertificate/certificates/<ID>/recreate?format=json</i>
Generate all authentication certificates	PUT <i>/ISAPI/Security/deviceCertificate/certificates/recreate?format=json</i>

Client/Server Certificate

Function	Request URI
Get client/server certificate capability	GET <i>/ISAPI/Security/serverCertificate/capabilities?format=json</i>
Generate PKCS#10 signature request of client/server certificate	POST <i>/ISAPI/Security/serverCertificate/certSignReq?customID=</i>
Get or generate PKCS#10 signature request of client/server self-signed certificate	GET or PUT <i>/ISAPI/Security/serverCertificate/selfSignCert?customID=</i>
Get information of multiple client/server certificates in a batch	GET <i>/ISAPI/Security/serverCertificate/certificates?format=json</i>
Get or delete information of a specific client/server certificate	GET or DELETE <i>/ISAPI/Security/serverCertificate/certificates/<customID>?format=json</i>
Import client/server certificate to device	POST <i>/ISAPI/Security/serverCertificate/certificate?customID=</i>
Export client/server certificate	GET <i>/ISAPI/Security/serverCertificate/downloadCertSignReq?customID=</i>

Chapter 4 Login

Function	Request URI
Login by digest	GET <i>/ISAPI/Security/userCheck</i>
Lock for illegal login	GET or PUT <i>/ISAPI/Security/illegalLoginLock</i>
Get or set maximum failed login attempts	GET or PUT <i>/ISAPI/Security/loginLinkNum?format=json</i>

Chapter 5 Device Management

5.1 Device Activation

5.1.1 Directly Activate Device

For the connected devices, you must activate them to make sure the device password is reconfigured and the password string conforms to the security rule.

Steps

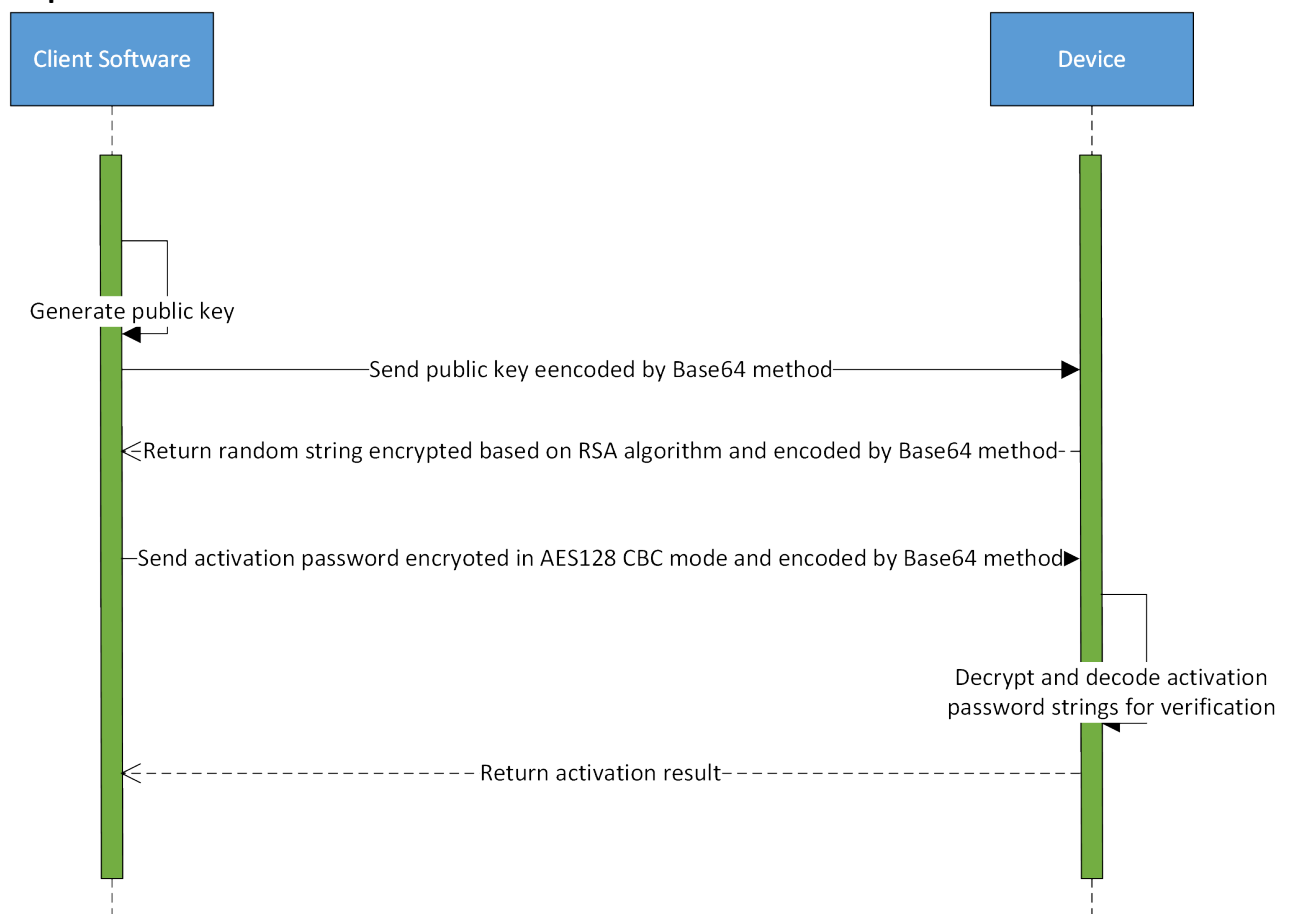


Figure 5-1 Direct Activation Sequence Diagram

1. Client software generates public and private key.



Note

The size of public key is 1024 bits (128 bytes). When transforming to the hexadecimal string, each byte corresponds to two characters (e.g., **127** → **FF**), so the public key is transformed to a 256-byte string finally.

2. Client software encodes the public key (a 256-byte string) by Base64 method and sends it to device.
 3. Device decodes the encoded public key by Base64 method and generates random strings.
 4. Device encrypts the random strings based on RSA algorithm and encodes the encrypted strings by Base64 method.
 5. Call ***/ISAPI/Security/challenge*** by POST method to get the encrypted and encoded random strings .
 6. Client software decodes the obtained random strings by Base64 method and decrypts the random strings by using private key.
 7. Client software encrypts the activation password with the random strings in AES128 CBC mode and encodes the password by Base64 method.
 8. Call ***/ISAPI/System/activate*** by PUT method to send the activation password strings to device.
-



Note

The sent activation password strings consist of first 16 characters of random strings and actual password.

Example

If the first 16 characters of random strings is "aaaabbbbccccdddd", and the actual password is "Abc12345", the encrypted activation password strings is "aaaabbbbccccddddAbc12345".

9. Device decodes the activation password by Base64 method and decrypts the password with the random string in AES128 CBC mode for verification.
10. Device starts activating and sends the activation result to client software.

5.1.2 Activate Network Camera via NVR

For network devices, except directly activating them, you can also activate them via NVR if they can be searched on the same network domain of NVR.

Steps

1. Call ***/ISAPI/ContentMgmt/InputProxy/search*** by GET method to search for the online network devices in the same network domain with the NVR.
2. **Optional:** Call ***/ISAPI/ContentMgmt/InputProxy/channels/activate/capabilities*** by GET method to get the activation capability of network devices for reference.
3. Call ***/ISAPI/ContentMgmt/InputProxy/channels/activate*** by PUT method to activate the searched online devices via NVR.

5.2 Reset Password by Setting Recovery Email

A recovery email is added or configured for resetting the password as required. The admin user can set a recovery email after activating the device, and then receive the verification code from the manufacturer via the recovery email to reset the device password.

Before You Start

Make sure you have activated the device.

Steps

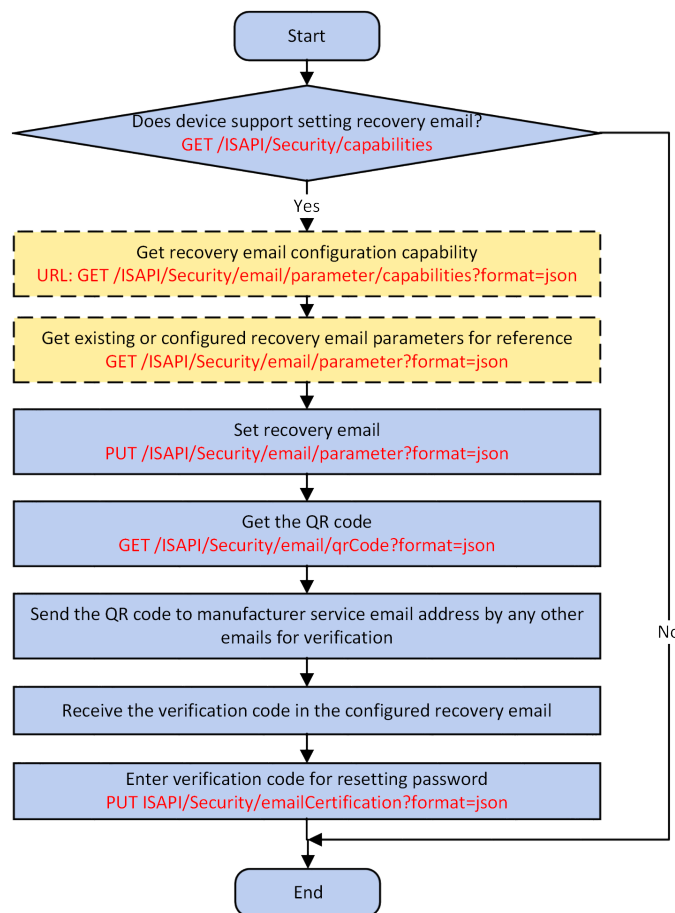


Figure 5-2 Programming Flow of Resetting Password by Setting Recovery Email

Note

You can reset password by other methods, such as answering security question, importing GUID files, and so on. But all methods should be supported by device, so you should call **/ISAPI/Security/extern/capabilities** by GET method to get the other security capability (***XML_externSecurityCap***) before choosing method to reset password.

1. Call **/ISAPI/Security/capabilities** by GET method to check if the device supports setting recovery email.

The security capability is returned in the message ***XML_SecurityCap***.

If the node **<isSupportSecurityEmail>** exists in the returned message and its value is "true", it indicates that setting recovery email is supported by the device, and you can continue to perform the following steps; otherwise, end this task.

2. **Optional:** Call ***/ISAPI/Security/email/parameter/capabilities?format=json*** by GET method to get recovery email configuration capability.

The recovery email configuration capability is returned in the message ***JSON_SecurityEmailCap*** by the output parameter pointer (***lpOutputParam***).

3. **Optional:** Call ***/ISAPI/Security/email/parameter?format=json*** by GET method to get the existing or configured recovery email parameters for reference.
4. Call ***/ISAPI/Security/email/parameter?format=json*** by PUT method to set the recovery email for the device.
5. Call ***/ISAPI/Security/email/qrCode?format=json*** by GET method to get the QR code of the recovery email.
6. Send the QR code to manufacturer service email address by any other emails for verification.
7. Receive the verification code in the configured recovery email.
8. Call ***/ISAPI/Security/emailCertification?format=json*** by PUT method to Enter the received verification code for resetting the device password.

5.3 Online Upgrade Device

For the devices that can connect to Guarding Vision Server, you can online upgrade their firmware via Guarding Vision Client, and get the upgrade progress. Besides, in the condition of bad network, you can also enable automatic download of upgrade package in the background to improve the upgrade speed.

Before You Start

Make sure you have activated the device.

Steps

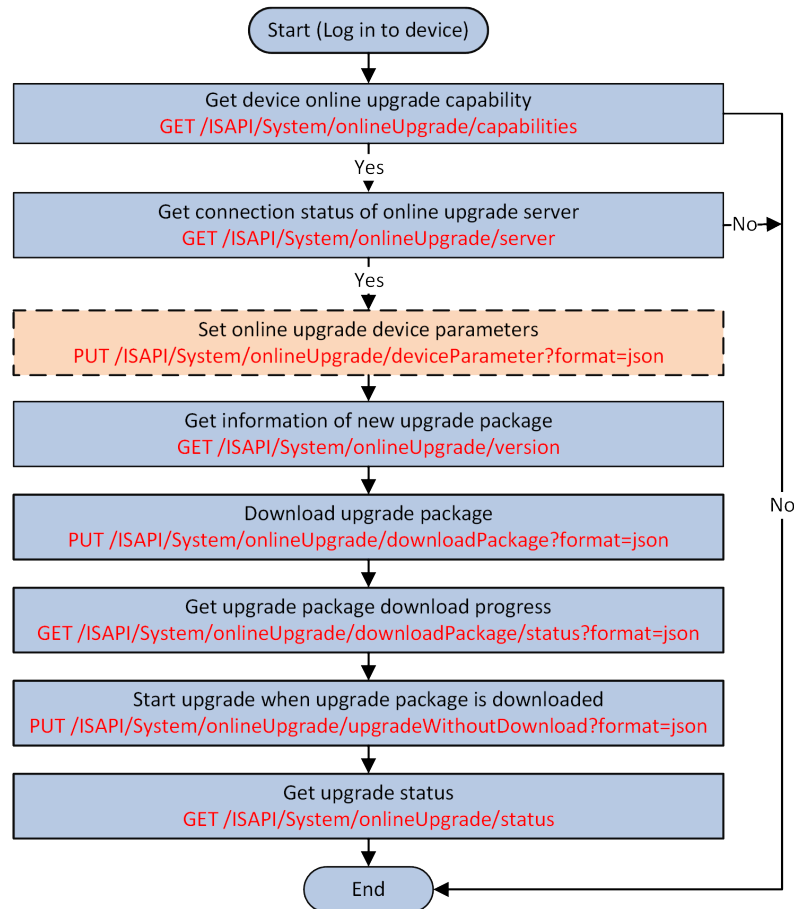


Figure 5-3 Programming Flow of Online Upgrade Device

1. Call `/ISAPI/System/onlineUpgrade/capabilities` by GET method to check if the device support online upgrade.
The online upgrade capability set is returned in *XML_OnlineUpgradeCap*.
2. Call `/ISAPI/System/onlineUpgrade/server` by GET method to get the connection status of online upgrade server.



Note

Only when the node `<connectStatus>` is returned and values "true", you can go on for next step. Otherwise, you should end this task.

The upgrade server connection status is returned in *XML_OnlineUpgradeServer*.

3. **Optional:** Call `/ISAPI/System/onlineUpgrade/deviceParameter?format=json` by PUT method to set the device online upgrade parameters.



Note

Before setting the device online upgrade parameters, you can get these parameters by calling ***/ISAPI/System/onlineUpgrade/deviceParameter?format=json*** via GET method.

4. Call ***/ISAPI/System/onlineUpgrade/version*** by GET method to get the new upgrade package information.
5. Call ***/ISAPI/System/onlineUpgrade/downloadPackage?format=json*** by PUT method to download the upgrade package.
6. **Optional:** During downloading the upgrade package, you can perform the following operations.

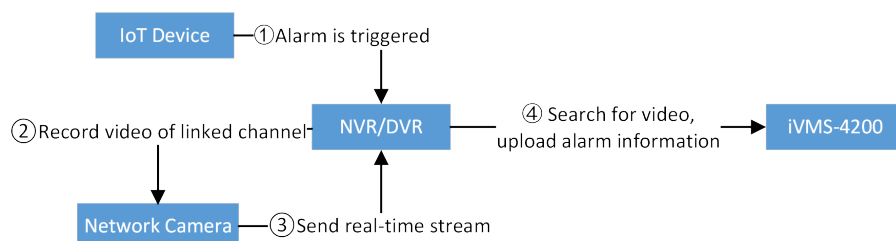
Pause Download	PUT <i>/ISAPI/System/onlineUpgrade/downloadPackage/pause?format=json</i>
Resume Download	PUT <i>/ISAPI/System/onlineUpgrade/downloadPackage/resume?format=json</i>
Cancel Download	DELETE <i>/ISAPI/System/onlineUpgrade/downloadPackage?format=json</i>
7. Call ***/ISAPI/System/onlineUpgrade/downloadPackage/status?format=json*** by GET method to get the upgrade package download progress.
8. Call ***/ISAPI/System/onlineUpgrade/upgradeWithoutDownload?format=json*** by PUT method to start upgrade when the upgrade package is downloaded.
9. Call ***/ISAPI/System/onlineUpgrade/status*** by GET method to get the upgrade status.

5.4 IoT Devices Access

The IoT devices here include access control terminal, security control panel, video intercom, analog camera RF, etc. The NVR/DVR supports accessing IoT devices for device management, status search, event/alarm linkage, and alarm receiving. When the IoT devices are added to NVR/DVR, the NVR/DVR can receive the alarm of IoT device, link the channel for recording, and receive the real-time stream when alarm is triggered.

This chapter mainly introduces the methods of adding IoT devices, and configuring event/alarm of IoT devices.

Application Scenario



5.4.1 Add IoT Devices to NVR/DVR

For convenient management of IoT device, such as status searching, event/alarm linkage, alarm receiving, and so on, you should add the IoT device to NVR/DVR first.

Before You Start

Make sure you have logged in to the device.

Steps

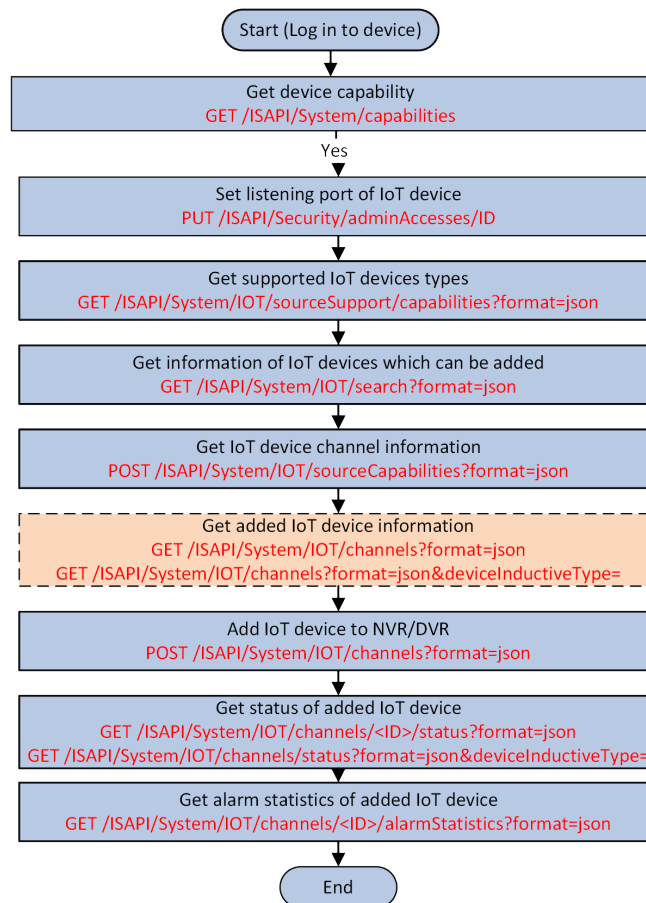


Figure 5-4 Programming Flow of Adding IoT Device to NVR/DVR

1. Call **/ISAPI/System/capabilities** by GET method to get the device capability for checking if the IoT device access function is supported via the node **<IOTCap>**.
The device capability **XML_DeviceCap** is returned.
2. Call **/ISAPI/Security/adminAccesses/<ID>** by PUT method to configure listening port before adding the OPTeX and Luminite IoT devices and set the value of **<protocol>** in **XML_AdminAccessProtocol** to "IOT".
3. Call **/ISAPI/System/IOT/sourceSupport/capabilities?format=json** by GET method to get the supported IoT devices types of device.
The supported IoT devices types are returned in **JSON_IOTSourceSupport**.

4. Call **`/ISAPI/System/IOT/search?format=json`** by GET method to get information of IoT devices which can be added.
The IoT devices which can be added is returned in **`JSON_IOTSourceList`**.
5. Call **`/ISAPI/System/IOT/sourceCapabilities?format=json`** by POST method to get the number of IoT device channels which can be added.
The supported source capability is returned in **`JSON_SourceCapabilities`**.
6. **Optional:** Get the information of added IoT devices.
 - Call **`/ISAPI/System/IOT/channels?format=json`** by GET method to get the information of added IoT devices.
 - Call **`/ISAPI/System/IOT/channels?format=json&deviceInductiveType=`** by GET method to get the information of added IoT devices according to inductive type
7. Call **`/ISAPI/System/IOT/channels?format=json`** by POST method to Add the IoT device to NVR/DVR.
8. Call **`/ISAPI/System/IOT/channels/<ID>/basicParam?format=json`** by PUT method to set the basic parameters of added IoT device according to channel ID.
9. **Optional:** Call **`/ISAPI/System/IOT/channels/<ID>?format=json`** by PUT method to Set the specified added IoT device according to channel ID.



Note

Before setting the specified IoT device, you can get the IoT device by calling **`/ISAPI/System/IOT/channels/<ID>?format=json`** via GET method.

10. **Optional:** Delete the added IoT device.
 - Call **`/ISAPI/System/IOT/channels/<ID>?format=json`** by DELETE method to delete the specified IoT channel.
 - Call **`/ISAPI/System/IOT/channels/<ID>/all?format=json`** by DELETE method to delete all channels (video channel and IoT channel) of added IoT device.
11. Get the status of added IoT device.
 - Call **`/ISAPI/System/IOT/channels/status?format=json`** by GET method to get the status of added IoT device.
 - Call **`/ISAPI/System/IOT/channels/status?format=json&deviceInductiveType=`** by GET method to get the status of added IoT device according to inductive type.
 - Call **`/ISAPI/System/IOT/channels/<ID>/status?format=json`** by GET method to get the status of specified channel of added IoT device.
12. Call **`/ISAPI/System/IOT/channels/<ID>/alarmStatistics?format=json`** by GET method to get the alarm statistics of added IoT device according to channel ID.
13. **Optional:** Call **`/ISAPI/System/accessDevice/associatedChannel?format=json`** by POST method to get the linked channel information of added IoT device.
14. **Optional:** Import or export the list of added IoT devices.

Import Call **`/ISAPI/System/IOT/channelConfig?format=json`** by PUT method to import the list of added IoT devices

Export Call ***/ISAPI/System/IOT/channelConfig?format=json*** by GET method to export the list of added IoT devices

Example

```
PUT /ISAPI/System/IOT/channelConfig?format=json
Accept: text/html, application/xhtml+xml,
Accept-Language: us-EN
Content-Type: File/ Opaque Data
User-Agent: Mozilla/5.0 (compatible; MSIE 9.0; Windows NT 6.1; WOW64; Trident/5.0)
Accept-Encoding: gzip, deflate
Host: 10.10.36.29:8080
Content-Length: 9907
Connection: Keep-Alive
Cache-Control: no-cache
Content-Disposition: form-data; name="XX";
Content-Length: 9907

.....JFIF.....` `.....C.....
..
..... $' "#..(7),01444.'9=82<.342...C. ....
```

5.4.2 Configure Alarm of IoT Device

When the IoT device is added to NVR/DVR, you can configure the alarm parameters such as arming schedule, linkage method, OSD, and so on for receiving the alarm of IoT device.

Before You Start

- Make sure you have logged in to the device.
- Make sure the IoT device is added to NVR/DVR.

Steps

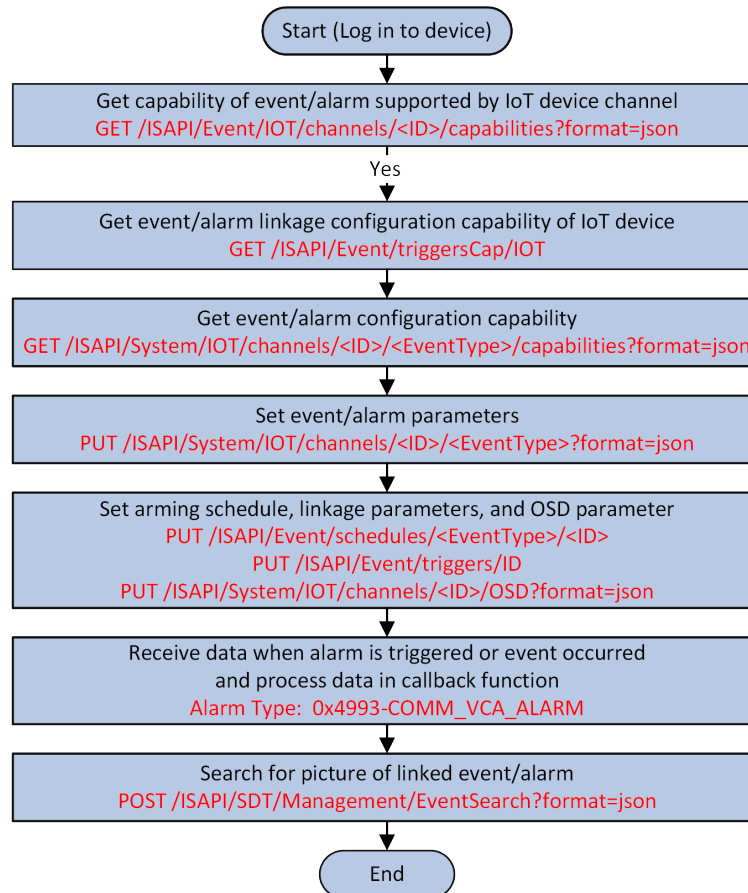


Figure 5-5 Programming Flow of Configuring Alarm of IoT Device

1. Call **`/ISAPI/Event/IOT/channels/<ID>/capabilities?format=json`** by GET method to get the IoT device channel capability.
The event capability supported by IoT device channel is returned in **`JSON_IOTChannelEventCap`**.
2. Call **`/ISAPI/Event/triggersCap/IOT`** by GET method to get the event/alarm linkage configuration capability of IoT device.
The capability is returned in **`XML_IOTTriggersCap`**.
3. Call **`/ISAPI/System/IOT/channels/<ID>/<EventType>/capabilities?format=json`** by GET method to get the event configuration capability.
The event configuration capability is returned in **`JSON_XXCap`**.
4. Call **`/ISAPI/System/IOT/channels/<ID>/<EventType>?format=json`** by PUT method to set the event/alarm parameters.
5. Call **`/ISAPI/Event/schedules/<EventType>/<ID>`** by PUT method to set the arming schedule
6. Call **`/ISAPI/Event/triggers/<event>/<ID>`** by PUT method to set the event/alarm linkage parameters.
7. Configure OSD.

- 1) Call **`/ISAPI/System/IOT/channels/<ID>/OSD/capabilities?format=json`** by GET method to get the OSD configuration capability.
 - 2) Call **`/ISAPI/System/IOT/channels/<ID>/OSD?format=json`** by GET method to Get the OSD parameters.
 - 3) Call **`/ISAPI/System/IOT/channels/<ID>/OSD?format=json`** by PUT method to set the OSD parameters.
8. Receive event/alarm information.
- Receive event/alarm in arming mode, see details in ***Receive Alarm/Event in Arming Mode*** .
 - Receive event/alarm in listening mode, see details in ***Receive Alarm/Event in Listening Mode*** .
9. Call **`/ISAPI/SDT/Management/EventSearch?format=json`** by POST method to search for the picture of linked event/alarm.



Note

Before searching for event, you can get the event search capability by calling **`/ISAPI/SDT/Management/EventSearch/capabilities?format=json`** via GET method.

10. **Optional:** Call **`/ISAPI/System/IOT/linkageChannels?format=json`** by POST method to Get linked channel of searched event.

Chapter 6 Device Configuration and Maintenance

6.1 Basic

Device Information Settings

Function	Request URI
Get device capability	GET <i>/ISAPI/System/deviceInfo/capabilities</i>
Get or set device information	GET or PUT <i>/ISAPI/System/deviceInfo</i>

Shut Down, Reboot and Restore to Default Settings

Function	Request URI
Shut down device	PUT <i>/ISAPI/System/shutdown?format=json</i>
Reboot device	PUT <i>/ISAPI/System/reboot</i>
Restore device to default settings	PUT <i>/ISAPI/System/factoryReset?mode=</i>

Upgrade

Function	Request URI
Get upgrade capability of peripherals' firmwares	GET <i>/ISAPI/System/AcsUpdate/capabilities</i>
Upgrade device's firmware	PUT or POST <i>/ISAPI/System/updateFirmware</i>
Upgrade slave device's firmware	PUT or POST <i>/ISAPI/System/updateFirmware?type=&id=</i>
Upgrade peripheral module's firmware	PUT or POST <i>/ISAPI/System/updateFirmware?type=&moduleAddress=</i>
Get device upgrading status	GET <i>/ISAPI/System/upgradeStatus</i>

Configuration File

Function	Request URI
Get capability of exporting files from device	GET <i>/ISAPI/System/fileExport/capabilities?format=json</i>
Securely export files from device	POST <i>/ISAPI/System/fileExport?format=json</i>

Function	Request URI
	 Note To check if exporting files from device is supported by the device, you can call GET <i>/ISAPI/System/capabilities</i> to get the device capability (<i>XML_DeviceCap</i>). If supports, the node <i><isSptFileExport></i> will be returned.

Algorithm

Function	Request URI
Get algorithm version information of device	GET <i>/ISAPI/System/algorithmsVersion</i>
Get capability of getting algorithm library information	GET <i>/ISAPI/System/algorithmsInfo/capabilities?format=json</i>
Get algorithm library information	GET <i>/ISAPI/System/algorithmsInfo?format=json</i>

6.1.1 Securely Export and Import Configuration Files

You can export configuration files of devices to back up basic device parameters (such as network parameters, stream parameters, etc.) timely, and the configuration files will be encrypted with the encryption key when being exported. If several devices adopt the same configuration, you can import configuration files to configure multiple devices in a batch. The devices will decrypt the encrypted configuration file using the encryption key which ensures that the configuration file cannot be used by unauthorized devices.

Steps

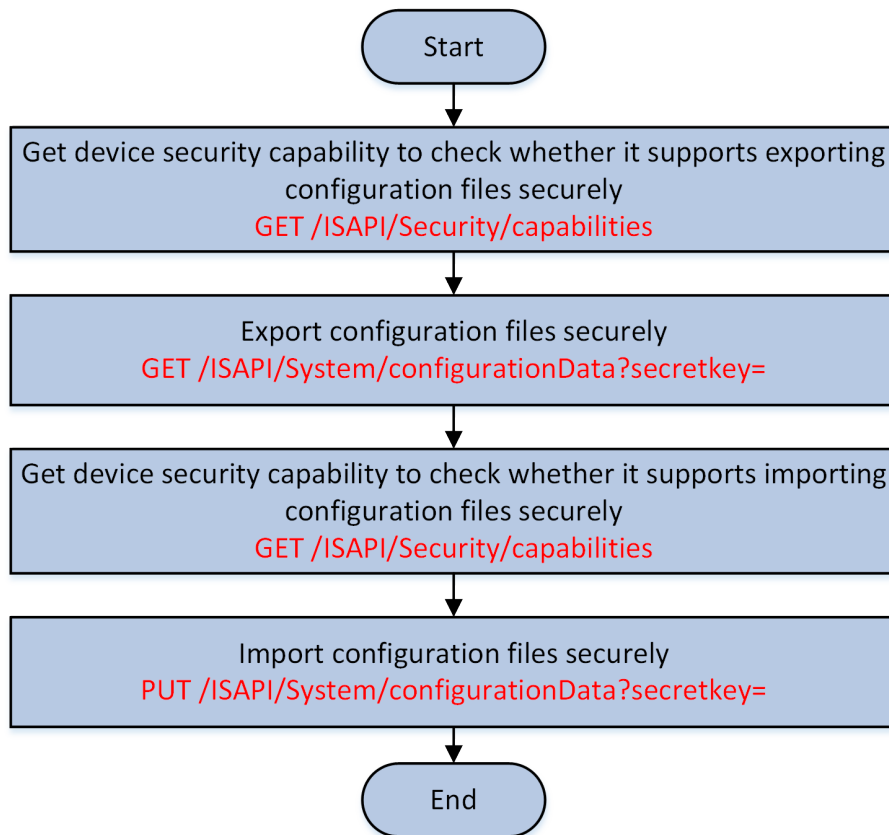


Figure 6-1 Programming Flow of Securely Exporting and Importing Configuration Files

1. Call **/ISAPI/Security/capabilities** by GET method to get the device security capability for checking whether the device supports securely exporting configuration files.
The device security capability is returned in the message ***XML_SecurityCap*** .
If this function is supported by the device, the node **<isSupportConfigFileExport>** will be returned and its value is "true", then you can perform the following steps.
Otherwise, please end this task.
2. Call **/ISAPI/System/configurationData?secretkey=** by GET method to export the configuration file from the device securely.

Note

The query parameter **secretKey** is the sensitive information that should be encrypted for transmission, and the parameter range to be encrypted is determined by the attributes "min" and "max" of the node **<cfgFileSecretKeyLenLimit>** in message ***XML_SecurityCap*** . For details about encrypting sensitive information, refer to ***Encrypt Sensitive Information*** .

3. Call **/ISAPI/Security/capabilities** by GET method to get the device security capability for checking whether the device supports securely importing configuration files.
The device security capability is returned in the message ***XML_SecurityCap*** .

If this function is supported by the device, the node **<isSupportConfigFileImport>** will be returned and its value is "true", then you can perform the following steps.

Otherwise, please end this task.

4. Call ***/ISAPI/System/configurationData?secretkey=*** by PUT method to import the configuration file to the device securely.



Note

- The query parameter **secretKey** is the sensitive information that should be encrypted for transmission, and the parameter range to be encrypted is determined by the attributes "min" and "max" of the field **<cfgFileSecretKeyLenLimit>** in the message ***XML_SecurityCap*** . For details about encrypting sensitive information, refer to ***Encrypt Sensitive Information*** .
 - The value of **secretKey** for importing the configuration file should be the same as that for exporting the file. Otherwise, importing will fail.
-

6.2 Log

Serial Port Log Redirection

Function	Request URI
Get configuration capability of serial port log redirection	GET <i>/ISAPI/System/serialLogCfg/capabilities?format=json</i>
Get or set parameters of serial port log redirection	GET or PUT <i>/ISAPI/System/serialLogCfg?format=json</i> Note To check whether the device supports configuring serial port log redirection, you can call <i>/ISAPI/System/capabilities</i> by GET method to get the device capability (<i>XML_DeviceCap</i>). If it supports, the node <isSptSerialLogCfg> will be returned and its value is "true".

Log Settings

Function	Request URI
Get log configuration capability	GET <i>/ISAPI/ContentMgmt/logConfig/capabilities</i>
Get or set log parameters	GET or PUT <i>/ISAPI/ContentMgmt/logConfig</i>

6.3 Status

Device Status

Function	Request URI
Get device status capability	GET <i>/ISAPI/System/workingstatus/capabilities?format=json</i>
Get all statuses of device	GET <i>/ISAPI/System/workingstatus?format=json</i>

Channel Status

Function	Request URI
Get status(es) of all channels or one channel	GET or POST <i>/ISAPI/System/workingstatus/chanStatus?format=json</i>
Get statuses of all channels by condition	POST <i>/ISAPI/System/Video/inputs/channels?format=json</i>
Get status of one channel by condition	POST <i>/ISAPI/System/Video/inputs/channels/<ID>?format=json</i>



Note

To check if getting channel status by condition is supported by device, you can call GET */ISAPI/System/capabilities* to get the device capability (*XML_DeviceCap*). If supports, the node *<isSupportFlexible>* will be returned.

Others

Function	Request URI
Get status(es) of all HDDs or one HDD	GET or POST <i>/ISAPI/System/workingstatus/hdStatus?format=json</i>
Get triggered I/O status	GET <i>/ISAPI/System/workingstatus/IOStatus?format=json</i>
Get CPU and memory status	GET <i>/ISAPI/System/status</i>
Upload status changes	Refer to Subscribe Alarm/Event in Arming Mode for integration process of uploading device status changes, and the node <i><type></i> in the request message <i>XML_SubscribeEvent</i> for the URIs (POST <i>/ISAPI/Event/notification/subscribeEvent</i> and PUT <i>/ISAPI/Event/notification/subscribeEvent/<ID></i>) should be set to "statusChanged".

The changes details can be obtained by calling */ISAPI/Event/notification/alertStream* via GET method.

 Note

To check if getting device status and uploading status changes are supported by device, you can call ***/ISAPI/System/capabilities*** by GET method to get the device capability (***XML_DeviceCap***). If supports, the nodes ***<isSupportSimpleDevStatus>***, ***<isSupportSimpleDevStatus>***, and ***<isSupportChangedUpload>*** will be returned and their values are "true".

6.4 Calibration

Accelerometer Zero Bias Calibration

Function	Request URI
Get device capability to check if device supports accelerometer zero bias calibration	GET <i>/ISAPI/System/capabilities</i> ; the device capability is returned in the message <i>XML_DeviceCap</i> . If supports, the node <i><isSupportZeroBiasCalibration></i> is returned in the message <i>XML_DeviceCap</i> and its value is "true".
Calibrate accelerometer zero bias	PUT <i>/ISAPI/System/zeroBiasCalibration/channels/<ID>?format=json</i>

Installation Angle Calibration

Function	Request URI
Get the device capability to check if device supports installation angle calibration	GET <i>/ISAPI/System/capabilities</i> ; the device capability is returned in the message <i>XML_DeviceCap</i> . If supports, the node <i><isSupportInstallationAngleCalibration></i> is returned in the message <i>XML_DeviceCap</i> and its value is "true".
Get capability of installation angle calibration	GET <i>/ISAPI/System/installationAngleCalibration/channels/<ID>/capabilities?format=json</i>
Get installation angle calibration status or calibrate installation angle	GET or PUT <i>/ISAPI/System/installationAngleCalibration/channels/<ID>?format=json</i>

Laser Optical Axis

Function	Request URI
Get configuration capability of laser optical axis	GET <i>/ISAPI/System/laserOpticalAxis/capabilities?format=json</i>
Get or set laser optical axis parameters	GET or PUT <i>/ISAPI/System/laserOpticalAxis?format=json</i>
Adjust laser optical axis	PUT <i>/ISAPI/System/laserOpticalAxis/goto?format=json</i>

6.5 Auto Power On or Off

Function	Request URI
Get capability of auto power on or off	GET <i>/ISAPI/System/autoSwitch/capabilities?format=json</i>
Get or set parameters of auto power on or off	GET or PUT <i>/ISAPI/System/autoSwitch?format=json</i>

6.6 Advanced

Automatic Maintenance

Function	Request URI
Get configuration capability of automatic maintenance	GET <i>/ISAPI/System/autoMaintenance/capabilities?format=json</i>
Get or set parameters of automatic maintenance	GET or PUT <i>/ISAPI/System/autoMaintenance?format=json</i>



Note

To check if the device supports automatic maintenance, you can call the GET */ISAPI/System/capabilities* to get the device capability (*XML_DeviceCap*). If supports, the node **<isSupportAutoMaintenance>** will be returned.

Device Diagnosis

Function	Request URI
Get device diagnosis capability	GET <i>/ISAPI/System/diagnosis/capabilities?format=json</i>
Diagnose device	POST <i>/ISAPI/System/diagnosis?format=json</i>



Note

To check if configuring device diagnosis is supported by the device, you can call GET */ISAPI/System/capabilities* to get the device capability (*XML_DeviceCap*). If supports, the node *<isSptDiagnosis>* will be returned.

Export Diagnose Information

Function	Request URI
Get capability of exporting device diagnose information	GET <i>/ISAPI/System/diagnosedData/parameter/capabilities</i>
Get or set parameters of exporting device diagnose information	GET or PUT <i>/ISAPI/System/diagnosedData/parameter</i>

Device Anti-Theft

Function	Request URI
Get device anti-theft configuration capability	GET <i>/ISAPI/System/guardAgainstTheft/capabilities</i>
Get or set device anti-theft parameters	GET or PUT <i>/ISAPI/System/guardAgainstTheft</i>



Note

To check if the anti-theft configuration is supported by the device, you can call GET */ISAPI/System/capabilities* to get the device capability (*XML_DeviceCap*). If this function is supported, the node *<isSupportGuardAgainstTheft>* will be returned and its value is "true".

Device Discovery

Function	Request URI
Get the configuration capability of device discovery mode	GET <i>/ISAPI/System/discoveryMode/capabilities</i>
Get or set the device discovery mode	GET or PUT <i>/ISAPI/System/discoveryMode</i>

Chapter 7 User Management

You can add, edit, and delete users to manage for logging in to the device via ISAPI protocol. And you can also assign permission to users according to actual needs.

Manage Users

Function	Request URI
Get the configuration capability of a specific user	GET <i>/ISAPI/Security/users/<ID>/capabilities</i>
Add one user	GET <i>/ISAPI/Security/users</i>
Get, edit, or delete all users	GET, PUT, or DELETE <i>/ISAPI/Security/users</i>
Get, edit, or delete one user	GET, PUT, or DELETE <i>/ISAPI/Security/users/<ID></i>
Get information of online users	GET <i>/ISAPI/Security/onlineUser</i>



Note

The function of managing users cannot be integrated via EZVIZ Open Platform.

User Permission

Function	Request URI
Get configuration capability of all users' permissions	GET <i>/ISAPI/Security/UserPermission/capabilities</i>
Get guest's permission capability	GET <i>/ISAPI/Security/UserPermission/viewerCap</i>
Get operator's permission capability	GET <i>/ISAPI/Security/UserPermission/operatorCap</i>
Get administrator's permission capability	GET <i>/ISAPI/Security/UserPermission/adminCap</i>
Get User Permission	• Get permissions of all users: GET <i>/ISAPI/Security/UserPermission</i> • Get permission of one user: GET <i>/ISAPI/Security/UserPermission/<ID></i> • Get local permission of one user: GET <i>/ISAPI/Security/UserPermission/<ID>/localPermission</i>

Function	Request URI
	- Get remote permission of one user: GET <i>/ISAPI/Security/UserPermission/<ID>/remotePermission</i> - Get permission of anonymous user: GET <i>/ISAPI/Security/UserPermission/anonymouslogin</i>
Assign Permission	- Assign permission to all users: PUT <i>/ISAPI/Security/UserPermission</i> - Assign permission to one user: PUT <i>/ISAPI/Security/UserPermission/<ID></i> - Assign local permission to one user: PUT <i>/ISAPI/Security/UserPermission/<ID>/localPermission</i> - Assign remote permission to one user: PUT <i>/ISAPI/Security/UserPermission/<ID>/remotePermission</i> - Assign permission to anonymous user: PUT <i>/ISAPI/Security/UserPermission/anonymouslogin</i>



Note

The user permission function cannot be integrated via EZVIZ Open Platform.

Lock and Unlock User

Function	Request URI
Get configuration capability of user locking and unlocking	GET <i>/ISAPI/System/userLock/config/capabilities?format=json</i> .
Get information of all locked users	GET <i>/ISAPI/System/userLock/lockedUsers?format=json</i>
Get user locking parameter	GET <i>/ISAPI/System/userLock/config?format=json</i>
Lock user	PUT <i>/ISAPI/System/userLock/config?format=json</i>
Unlock user	PUT <i>/ISAPI/System/userLock/unlockUser?format=json</i>

7.1 Configure Double Verification

Double verification helps to protect the critical video files of NVR/DVR by limiting playback and download. The basic concept is that two users should always be required to start playback and download. For example, when a normal user A (operator or guest) wants to play back the video of a channel which requires double verification, he/she should ask a double verification user to enter the correct user name and password for double verification.

Before You Start

Make sure you have logged in to the device.

Steps

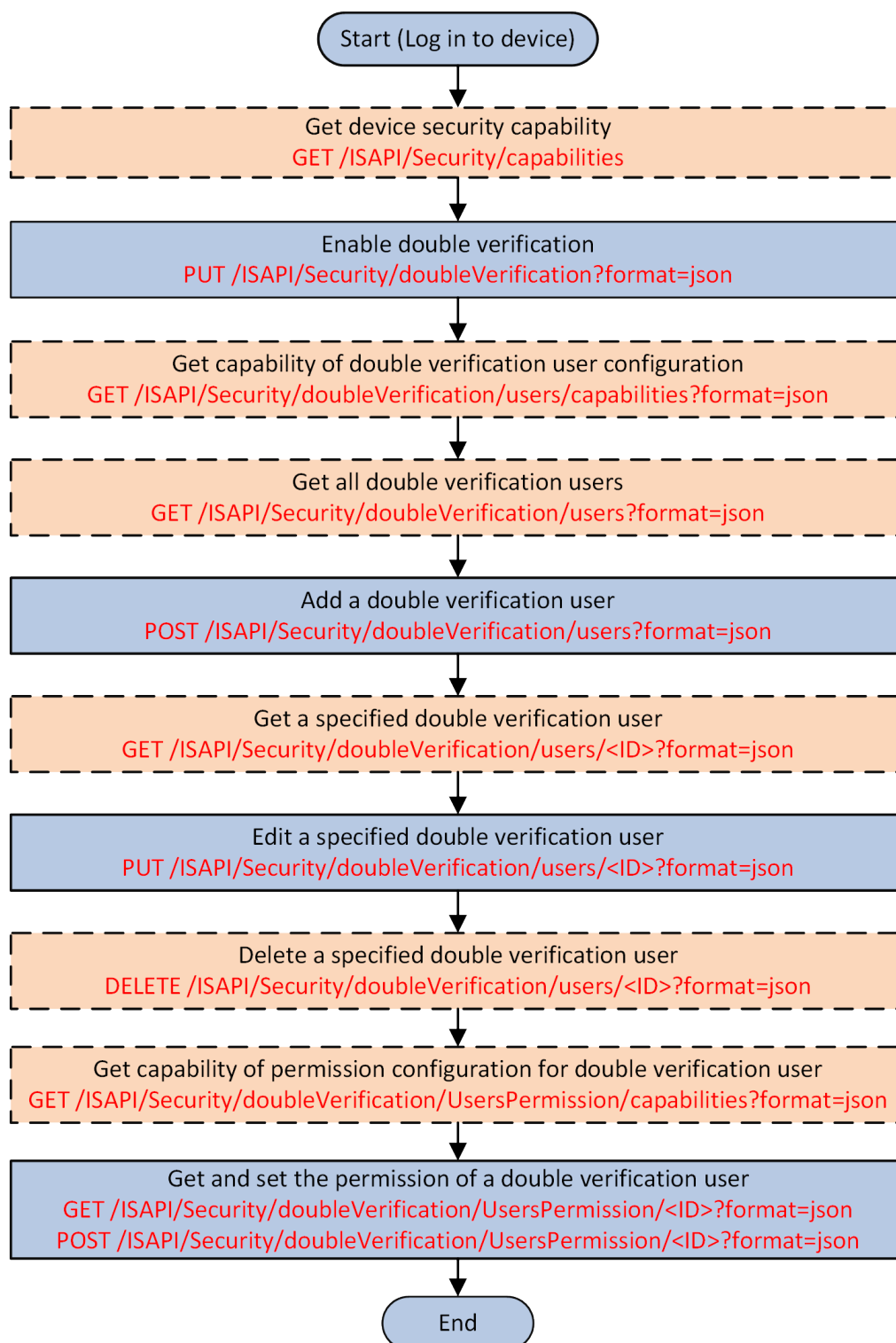


Figure 7-1 Programming Flow of Configuring Double Verification

Note

- Only the admin can configure double verification.
 - The admin is not required for double verification.
 - The double verification user name and password is only for double verification, and cannot be used for login.
-

1. **Optional:** Call ***/ISAPI/Security/capabilities*** by GET method to get the device security capability to check whether the double verification function is supported.

The security capability ***XML_SecurityCap*** is returned.

2. Call ***/ISAPI/Security/doubleVerification?format=json*** by PUT method to enable double verification
-

Note

Before enabling double verification, you can check whether double verification is enabled by calling ***/ISAPI/Security/doubleVerification?format=json*** via GET method.

3. **Optional:** Call ***/ISAPI/Security/doubleVerification/users/capabilities?format=json*** by GET method to get the capability of double verification user configuration.
 4. Call ***/ISAPI/Security/doubleVerification/users?format=json*** by POST method to add a double verification user.
-

Note

Before adding a double verification user, you can get all verification users by calling ***/ISAPI/Security/doubleVerification/users?format=json*** via GET method.

The ID of added double verification user is returned in ***JSON_id***.

5. Call ***/ISAPI/Security/doubleVerification/users/<ID>?format=json*** by PUT method to edit a specified double verification user.
-

Note

Before editing the double verification user, you can get the user information by calling ***/ISAPI/Security/doubleVerification/users/<ID>?format=json*** via GET method.

6. **Optional:** Call ***/ISAPI/Security/doubleVerification/users/<ID>?format=json*** by DELETE method to delete a specified double verification user.
 7. **Optional:** Call ***/ISAPI/Security/doubleVerification/UsersPermission/capabilities?format=json*** by GET method to get the capability of permission configuration for double verification users.
 8. Call ***/ISAPI/Security/doubleVerification/UsersPermission/<ID>?format=json*** by PUT method to set the permission of a specified double verification user.
-



Note

Before setting the permission of a double verification user, you can get the permission parameters by calling ***/ISAPI/Security/doubleVerification/UsersPermission/<ID>?format=json*** via GET method.

What to do next

Configure the user permission, determine whether double verification are required for them during playback and download.

Chapter 8 Video and Audio

This part provides the request URIs to implement the video and audio related functions, such as video search, live view, playback, two-way audio, control video in live view (e.g., focus and iris adjustment), and so on.

8.1 Search for Recording Start and End Time by Channel

You can check whether the channel has recorded videos and get the recording start and end time.

Steps

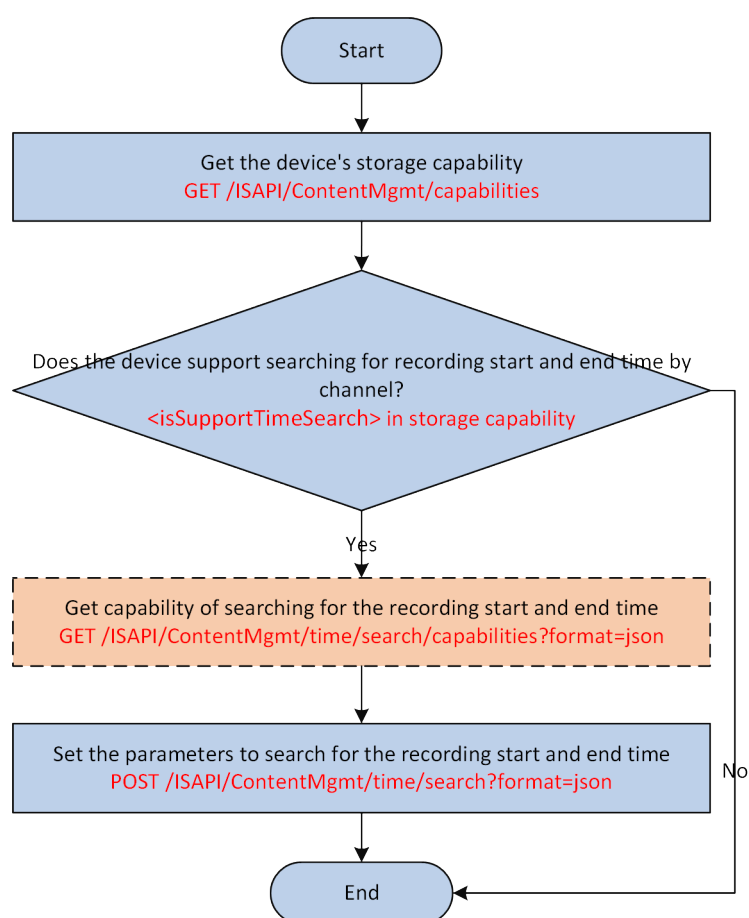


Figure 8-1 API Calling Flow of Searching for Recording Start and End Time by Channel

1. Call **/ISAPI/ContentMgmt/capabilities** by GET method to get the device's storage capability to check whether it supports searching for recording start and end time.

The device's storage capability is returned in the message **XML_RacmCap**.

If the node **<isSupportTimeSearch>** is returned in the message and its value is "true", it indicates that searching for recording start and end time is supported and you can continue to perform the following steps.

Otherwise, searching for recording start and end time is not supported, please end this task.

2. **Optional:** Call ***/ISAPI/ContentMgmt/time/search/capabilities?format=json*** by GET method to get the capability of searching for the recording start and end time for reference.

The capability is returned in the message ***JSON_Cap_TimeSearchCond***.

3. Call ***/ISAPI/ContentMgmt/time/search?format=json*** by POST method to set the parameters to search for the recording start and end time.

8.2 Live View and Playback

Live view helps to remotely get the real-time video or audio of monitoring area and know the status of human, vehicle, object, and so on. During live view, you can record and store the remarkable videos for remotely playback.

Live View

Table 8-1 Normal Operation

Function	Request URI
Start or stop live view	PLAY or TEARDOW

Table 8-2 Auto-Switch in Live View

Function	Request URI
Get configuration capability of live view auto-switch	GET <i>/ISAPI/System/Video/outputs/PreviewSwitch/capabilities</i>
Get configuration capability of live view auto-switch by video output	GET <i>/ISAPI/System/Video/outputs/PreviewSwitch/capabilities?videoOutType=</i>
Get or set parameters of live view auto-switch	GET or PUT <i>/ISAPI/System/Video/outputs/PreviewSwitch?groupID=&videoOutType=</i> GET <i>/ISAPI/System/Video/outputs/PreviewSwitch?groupID=&videoOutType=&previewFrameNo=</i>

Table 8-3 Control in Live View

Function	Request URI
Perform manual focus	PUT <i>/ISAPI/System/Video/inputs/channels/<ID>/focus</i>
Manually adjust iris	PUT <i>/ISAPI/System/Video/inputs/channels/<ID>/iris</i>

Playback

Function	Request URI
Start or stop playback	PLAY or TEARDOW <i>rtsp://<host>[:port]/ISAPI/Streaming/tracks/<ID>?starttime=&endtime=</i>
Refresh video index before playback for instant playback	PUT <i>/ISAPI/ContentMgmt/record/control/manualRefresh/channels/<ID></i>

Example

Sample Code for Starting and Stopping Live View

Start Live View:

DESCRIBE rtsp://10.17.133.46:554/ISAPI/streaming/channels/101 RTSP/1.0

CSeq: 0

Accept:application/sdp

User-Agent:NKPlayer-1.00.00.081112

RTSP/1.0 401 Unauthorized

CSeq: 0

WWW-Authenticate: Digest realm="4419b66d2485", nonce="47f172cffa4ce6e6942bbdcfa5931d0f", stale="FALSE"

Date: Tue, Mar 14 2017 10:42:35 GMT

DESCRIBE rtsp://10.17.133.46:554/ISAPI/streaming/channels/101 RTSP/1.0

CSeq:1

Accept:application/sdp

Authorization: Digest username="admin",

realm="4419b66d2485",

nonce="47f172cffa4ce6e6942bbdcfa5931d0f",

uri="rtsp://10.17.133.46:554/ISAPI/streaming/channels/101",

response="da5deb4033515f74ed0483d2da3f7fe0"

User-Agent:NKPlayer-1.00.00.081112

RTSP/1.0 200 OK

CSeq: 1

Content-Type: application/sdp

Content-Length: 604

v=0

o=- 1489488155469547 1489488155469547 IN IP4 10.17.133.46

s=Media Presentation

e=NONE

b=AS:5100

t=0 0

a=control:*

m=video 0 RTP/AVP 96

c=IN IP4 0.0.0.0

b=AS:5000

a=recvonly

a=x-dimensions:1280,720

a=control:trackID=1

[illegible]

```
Session:846509115;timeout=60
Rate-Control:yes
Scale:1.000
User-Agent:NKPlayer-1.00.00.081112

RTSP/1.0 200 OK
CSeq: 4
Session: 846509115
Scale: 1.0
RTP-Info: url=trackID=1;seq=55143,url=trackID=2;seq=38217
Date: Tue, Mar 14 2017 10:42:35 GMT

Stop Live View
TEARDOWN rtsp://10.17.133.46:554/ISAPI/streaming/channels/101 RTSP/1.0
CSeq: 5
Authorization: Digest username="admin",
realm="4419b66d2485",
nonce="47f172cffa4ce6e6942bbdcfa5931d0f",
uri="rtsp://10.17.133.46:554/ISAPI/streaming/channels/101",
response="31d5e172b87db528aa5b1dc07af3f530"
Session:846509115;timeout=60
User-Agent: NKPlayer-1.00.00.081112

RTSP/1.0 200 OK
CSeq: 5
Session: 846509115
Date: Tue, Mar 14 2017 10:42:36 GMT
```

Example

Sample Code for Starting and Stopping Playback

```
Enable:
DESCRIBE rtsp://10.17.133.46:554/ISAPI/streaming/tracks/101?
starttime=20170313T230652Z&endtime=20170314T025706Z RTSP/1.0
CSeq:0
Accept:application/sdp
User-Agent:NKPlayer-1.00.00.081112

RTSP/1.0 401 Unauthorized
CSeq: 0
WWW-Authenticate: Digest realm="4419b66d2485",
nonce="a0ecd9b1586ff9461f02f910035d0486",
stale="FALSE"
Date: Tue, Mar 14 2017 10:57:23 GMT

DESCRIBE rtsp://10.17.133.46:554/ISAPI/streaming/tracks/101?
starttime=20170313T230652Z&endtime=20170314T025706Z RTSP/1.0
CSeq:1
Accept:application/sdp
Authorization: Digest username="admin",
realm="4419b66d2485",
nonce="a0ecd9b1586ff9461f02f910035d0486",
```

```

uri="rtsp://10.17.133.46:554/ISAPI/streaming/tracks/101?
starttime=20170313T230652Z&endtime=20170314T025706Z",
response="1ebae505abbafd431fb433cec6e85189"
User-Agent:NKPlayer-1.00.00.081112

RTSP/1.0 200 OK
CSeq: 1
Content-Type: application/sdp
Content-Length: 604

v=0
o=- 1489489043122939 1489489043122939 IN IP4 10.17.133.46
s=Media Presentation
e=NONE
b=AS:5100
t=0 0
a=control:*
m=video 0 RTP/AVP 96
c=IN IP4 0.0.0.0
b=AS:5000
a=recvonly
a=x-dimensions:1280,720
a=control:trackID=1
a=rtpmap:96 H264/90000
a=fmtp:96 profile-level-id=420029; packetization-mode=1; sprop-parameter-
sets=Z2QAH6wXKgFAFuhAAAHCAABX5CE=,aP44sA==
m=audio 0 RTP/AVP 8
c=IN IP4 0.0.0.0
b=AS:50
a=recvonly
a=control:trackID=2
a=rtpmap:8 PCMA/8000
a=Media_header:MEDIAINFO=494D4B48010200000400000111710110401F000000FA00000000000000000000000000
000000000000;
a=appversion:1.0

SETUP rtsp://10.17.133.46:554/ISAPI/streaming/tracks/101?
starttime=20170313T230652Z&endtime=20170314T025706Z/trackID=1 RTSP/1.0
CSeq:2
Authorization: Digest username="admin",
realm="4419b66d2485",
nonce="a0ecd9b1586ff9461f02f910035d0486",
uri="rtsp://10.17.133.46:554/ISAPI/streaming/tracks/101?
starttime=20170313T230652Z&endtime=20170314T025706Z",
response="6a7a733b41908bd9549a4ac07c498509"
Transport:RTP/AVP/UDP;unicast;client_port=20056-20057;ssrc=0
User-Agent:NKPlayer-1.00.00.081112

RTSP/1.0 200 OK
CSeq: 2
Session: 2049381566;timeout=60
Transport: RTP/AVP/UDP;unicast;client_port=20056-20057;ssrc=7d3c07bc;server_port=8220-8221;mode="play"

```

Date: Tue, Mar 14 2017 10:57:23 GMT

SETUP rtsp://10.17.133.46:554/ISAPI/streaming/tracks/101?
starttime=20170313T230652Z&endtime=20170314T025706Z/trackID=2 RTSP/1.0
CSeq:3
Authorization: Digest username="admin",
realm="4419b66d2485",
nonce="a0ecd9b1586ff9461f02f910035d0486",
uri="rtsp://10.17.133.46:554/ISAPI/streaming/tracks/101?
starttime=20170313T230652Z&endtime=20170314T025706Z",
response="6a7a733b41908bd9549a4ac07c498509"
Session:2049381566;timeout=60
Transport:RTP/AVP/UDP;unicast;client_port=20058-20059;ssrc=0
User-Agent:NKPlayer-1.00.00.081112

RTSP/1.0 200 OK
CSeq: 3
Session: 2049381566;timeout=60
Transport: RTP/AVP/UDP;unicast;client_port=20058-20059;ssrc=06557745;server_port=8226-8227;mode="play"
Date: Tue, Mar 14 2017 10:57:23 GMT

PLAY rtsp://10.17.133.46:554/ISAPI/streaming/tracks/101?
starttime=20170313T230652Z&endtime=20170314T025706Z RTSP/1.0
CSeq:4
Authorization: Digest username="admin",
realm="4419b66d2485",
nonce="a0ecd9b1586ff9461f02f910035d0486",
uri="rtsp://10.17.133.46:554/ISAPI/streaming/tracks/101?
starttime=20170313T230652Z&endtime=20170314T025706Z",
response="fb986d385a7d839052ec4f0b2b70c631"
Session:2049381566;timeout=60
Range:clock=20170313T230652Z-20170314T025706Z
Rate-Control:yes
Scale:1.000
User-Agent:NKPlayer-1.00.00.081112

RTSP/1.0 200 OK
CSeq: 4
Session: 2049381566
Scale: 1.000
RTP-Info: url=trackID=1;seq=1,url=trackID=2;seq=1
Date: Tue, Mar 14 2017 10:57:23 GMT

Pause:
PAUSE rtsp://10.17.133.46:554/ISAPI/streaming/tracks/101?
starttime=20170313T230652Z&endtime=20170314T025706Z RTSP/1.0
CSeq: 5
Authorization: Digest username="admin",
realm="4419b66d2485",
nonce="a0ecd9b1586ff9461f02f910035d0486",
uri="rtsp://10.17.133.46:554/ISAPI/streaming/tracks/101?
starttime=20170313T230652Z&endtime=20170314T025706Z",

response="d229c4d2419df553212bc7d74bbfd432"

Session:2049381566;timeout=60

User-Agent: NKPlayer-1.00.00.081112

RTSP/1.0 200 OK

CSeq: 5

Session: 2049381566

Date: Tue, Mar 14 2017 10:57:24 GMT

Play in 2× Speed:

PLAY rtsp://10.17.133.46:554/ISAPI/streaming/tracks/101?

starttime=20170313T230652Z&endtime=20170314T025706Z RTSP/1.0

CSeq:6

Authorization: Digest username="admin",

realm="4419b66d2485",

nonce="a0ecd9b1586ff9461f02f910035d0486",

uri="rtsp://10.17.133.46:554/ISAPI/streaming/tracks/101?

starttime=20170313T230652Z&endtime=20170314T025706Z",

response="fb986d385a7d839052ec4f0b2b70c631"

Session:2049381566;timeout=60

Scale:2.000

User-Agent:NKPlayer-1.00.00.081112

RTSP/1.0 200 OK

CSeq: 6

Session: 2049381566

Scale: 2.000

RTP-Info: url=trackID=1;seq=1,url=trackID=2;seq=1

Date: Tue, Mar 14 2017 10:57:24 GMT

Pause:

PAUSE rtsp://10.17.133.46:554/ISAPI/streaming/tracks/101?

starttime=20170313T230652Z&endtime=20170314T025706Z RTSP/1.0

CSeq: 7

Authorization: Digest username="admin",

realm="4419b66d2485",

nonce="a0ecd9b1586ff9461f02f910035d0486",

uri="rtsp://10.17.133.46:554/ISAPI/streaming/tracks/101?

starttime=20170313T230652Z&endtime=20170314T025706Z",

response="d229c4d2419df553212bc7d74bbfd432"

Session:2049381566;timeout=60

User-Agent: NKPlayer-1.00.00.081112

RTSP/1.0 200 OK

CSeq: 7

Session: 2049381566

Date: Tue, Mar 14 2017 10:57:25 GMT

Play in 4× Speed:

PLAY rtsp://10.17.133.46:554/ISAPI/streaming/tracks/101?

starttime=20170313T230652Z&endtime=20170314T025706Z RTSP/1.0

CSeq:8

```
Authorization: Digest username="admin",
realm="4419b66d2485",
nonce="a0ecd9b1586ff9461f02f910035d0486",
uri="rtsp://10.17.133.46:554/ISAPI/streaming/tracks/101?
starttime=20170313T230652Z&endtime=20170314T025706Z",
response="fb986d385a7d839052ec4f0b2b70c631"
Session:2049381566;timeout=60
Scale:4.000
User-Agent:NKPlayer-1.00.00.081112

RTSP/1.0 200 OK
CSeq: 8
Session: 2049381566
Scale: 4.000
RTP-Info: url=trackID=1;seq=1,url=trackID=2;seq=1
Date: Tue, Mar 14 2017 10:57:25 GMT

Disable:
TEARDOWN rtsp://10.17.133.46:554/ISAPI/streaming/tracks/101?
starttime=20170313T230652Z&endtime=20170314T025706Z RTSP/1.0
CSeq: 9
Authorization: Digest username="admin",
realm="4419b66d2485",
nonce="a0ecd9b1586ff9461f02f910035d0486",
uri="rtsp://10.17.133.46:554/ISAPI/streaming/tracks/101?
starttime=20170313T230652Z&endtime=20170314T025706Z",
response="50c86f44780a497ee07622b70c050bcd"
Session:2049381566;timeout=60
User-Agent: NKPlayer-1.00.00.081112

RTSP/1.0 200 OK
CSeq: 9
Session: 2049381566
Date: Tue, Mar 14 2017 10:57:28 GMT
```

8.3 Start Two-Way Audio

Two-way audio function enables the voice talk of the camera. You can get not only the live video but also the real-time audio from the camera. If the device has multiple two-way audio channels, you can select the channel to start two-way audio.

Steps

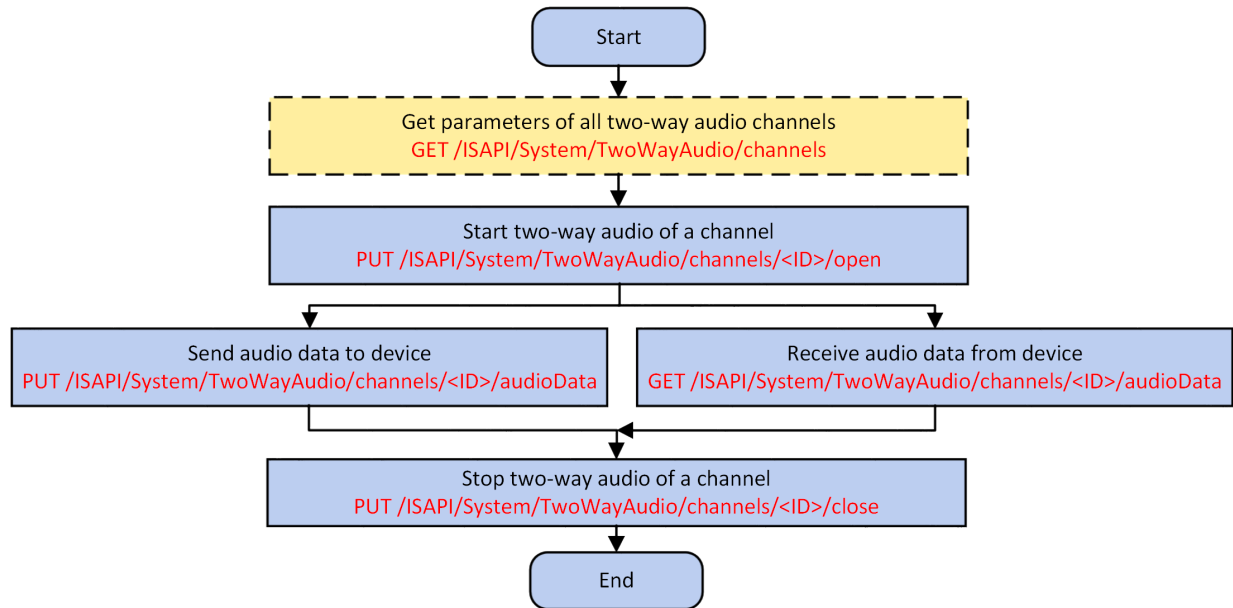


Figure 8-2 Programming Flow of Starting Two-Way Audio

1. **Optional:** Call `/ISAPI/System/TwoWayAudio/channels` by GET method to get parameters, including channel No., encoding mode, and so on, of all two-way audio channels.



Note

You can call `/ISAPI/System/TwoWayAudio/channels/capabilities` by GET method to get the audio configuration capability of all two-way audio channels for knowing the configuration details and notices.

2. Call `/ISAPI/System/TwoWayAudio/channels/<ID>/open` by PUT method to specify a channel of device to start the two-way audio.
3. Perform the following operations to transmit the audio data between platform and device.

Send Audio Data to Device	PUT <code>/ISAPI/System/TwoWayAudio/channels/<ID>/audioData</code>
----------------------------------	---

Receive Audio Data from Device	GET <code>/ISAPI/System/TwoWayAudio/channels/<ID>/audioData</code>
---------------------------------------	---

4. Call `/ISAPI/System/TwoWayAudio/channels/<ID>/close` by PUT method to stop two-way audio of the specific channel.

8.4 Stream Encoding

The following request URIs are used for configuring encoding parameters of video and audio stream by one channel or multiple channels.

Encode Stream of Multiple Channels

Function	Request URI
Get or set encoding parameters of multiple channels	GET or PUT <i>/ISAPI/Streaming/channels</i>
Get encoding status of multiple channels	GET <i>/ISAPI/Streaming/status</i>

Encode Stream of One Channel

Function	Request URI
Get encoding capability of one channel	GET <i>/ISAPI/Streaming/channels/<ID>/capabilities</i>
Get dynamic encoding capability of one channel	GET <i>/ISAPI/Streaming/channels/<ID>/dynamicCap</i> .
Get or set encoding parameters of one channel	GET or PUT <i>/ISAPI/Streaming/channels/<ID></i>
Get encoding status of one channel	GET <i>/ISAPI/Streaming/channels/<ID>/status</i>

Chapter 9 Capture and Recording

For some remarkable views in the live view, you can manually capture the views or record the video segments and save the captured pictures or videos to the configured storage. For recording, you can also configure recording schedule to record videos continuously or based on command or event.

Capture

Function	Request URI
Capture picture (available for front-end devices)	GET <i>/ISAPI/Streaming/channels/<ID>/picture</i>
Capture picture (available for rear-end devices)	GET <i>/ISAPI/ContentMgmt/StreamingProxy/channels/<ID>/Picture</i>
Get configuration capability of capturing pictures by schedule	GET <i>/ISAPI/Snapshot/channels/<ID>/capabilities</i>
Get or set parameters of multiple channels for capturing pictures by schedule	GET or PUT <i>/ISAPI/Snapshot/channels</i>
Get or set parameters of a specific channel for capturing pictures by schedule	GET or PUT <i>/ISAPI/Snapshot/channels/<ID></i>
Get capability of digital video channel proxy	GET <i>/ISAPI/ContentMgmt/StreamingProxy/channels/<ID>/capabilities</i>
Get capability of capturing picture in URL format	GET <i>/ISAPI/ContentMgmt/StreamingProxy/channels/<ID>/PictureByUrl/capabilities?format=json</i>
Capture picture in URL format	POST <i>/ISAPI/ContentMgmt/StreamingProxy/channels/<ID>/PictureByUrl?format=json</i>

Recording

Table 9-1 Normal Operation

Function	Request URI
Get video track type	GET <i>/ISAPI/ContentMgmt/record/profile</i>
Set storage for recorded files	PUT <i>/ISAPI/ContentMgmt/record/storageMounts</i>

Function	Request URI
	 Note To check if setting recording storage is supported, you can call the GET <i>/ISAPI/ContentMgmt/capabilities</i> to get the device storage management capability (<i>XML_RacmCap</i>).

Table 9-2 Recording Schedule

Function	Request URI
Get configuration capability of a recording schedule	GET <i>/ISAPI/ContentMgmt/record/tracks/<ID>/capabilities</i>
Get all recording schedules	GET <i>/ISAPI/ContentMgmt/record/tracks</i>
Add a recording schedule	POST <i>/ISAPI/ContentMgmt/record/tracks</i>
Set a recording schedule	PUT <i>/ISAPI/ContentMgmt/record/tracks/<ID></i>

Table 9-3 Manual Recording

Function	Request URI
Start manual recording	POST <i>/ISAPI/ContentMgmt/record/control/manual/start/tracks/<ID></i>
Stop manual recording	POST <i>/ISAPI/ContentMgmt/record/control/manual/stop/tracks/<ID></i>

Chapter 10 PTZ Control

PTZ control is to implement the panning, tilting, zoom functions of cameras, set presets for specific monitoring areas to fast switch the cameras as needed, and group the user-defined presets as a scanning track for patrol.

Basic PTZ Parameters

Function	Request URI
Get PTZ control capability	GET <i>/ISAPI/PTZCtrl/channels/<ID>/capabilities</i>
Get or set zoom/focus coordinate parameters	GET or PUT <i>/ISAPI/PTZCtrl/channels/<ID>/zoomFocus</i>
Maximum Tilt Angle	- Get configuration capability of maximum tilt angle: GET <i>/ISAPI/PTZCtrl/channels/<ID>/maxelevation/capabilities</i> - Get or set parameters of maximum tilt angle: GET or PUT <i>/ISAPI/PTZCtrl/channels/<ID>/maxelevation</i>
Get PTZ position information of current channel	GET <i>/ISAPI/PTZCtrl/channels/<ID>/save</i>
Set PTZ position information of current channel	PUT <i>/ISAPI/PTZCtrl/channels/<ID>/save?format=json</i>
Get PTZ status of a specific channel	GET <i>/ISAPI/PTZCtrl/channels/<ID>/status</i>
Preset Configuration	- Get all presets' parameters: GET <i>/ISAPI/PTZCtrl/channels/<ID>/presets</i> - Set all presets' parameters: PUT <i>/ISAPI/PTZCtrl/channels/<ID>/presets</i> - Added a preset: POST <i>/ISAPI/PTZCtrl/channels/<ID>/presets</i> - Delete all presets: DELETE <i>/ISAPI/PTZCtrl/channels/<ID>/presets</i> - Get a preset's parameters: GET <i>/ISAPI/PTZCtrl/channels/<ID>/presets/<ID></i> - Set a preset's parameters: PUT <i>/ISAPI/PTZCtrl/channels/<ID>/presets/<ID></i> - Delete a preset: DELETE <i>/ISAPI/PTZCtrl/channels/<ID>/presets/<ID></i>
Get parking configuration capability	GET <i>/ISAPI/PTZCtrl/channels/<ID>/parkaction/capabilities</i>
Get or set parking parameters	GET or PUT <i>/ISAPI/PTZCtrl/channels/<ID>/parkaction</i>

Other PTZ Parameters

Function	Request URI
Lens	- Initialize lens: PUT <i>/ISAPI/PTZCtrl/channels/<ID>/onepushfoucs/reset</i> - Get lens correction capability: GET <i>/ISAPI/PTZCtrl/channels/<ID>/lensCorrection/capabilities?format=json</i> - Get or set lens correction parameters: GET or PUT <i>/ISAPI/PTZCtrl/channels/<ID>/lensCorrection?format=json</i>  Note To check if the lens correction is supported by device, you can call the GET <i>/ISAPI/PTZCtrl/channels/<ID>/capabilities</i> to get the PTZ control capability (<i>XML_PTZChanelCap</i>). If supports, the node <i><isSupportLensCorrection></i> will be returned.
Get or set PTZ locking time duration	GET or PUT <i>/ISAPI/PTZCtrl/channels/<ID>/lockPTZ</i>  Note You can call the GET <i>/ISAPI/PTZCtrl/channels/<ID>/capabilities</i> to get the supported locking time duration (<i><lockTime></i>) from the PTZ control capability (<i>XML_PTZChanelCap</i>).
Get or set PTZ OSD parameters	GET or PUT <i>/ISAPI/PTZCtrl/channels/<ID>/PTZOSDDisplay</i>
Get or set moving object tracking status	GET or PUT <i>/ISAPI/PTZCtrl/channels/<ID>/moveAutoTracking?format=json</i>
Get configuration capability of tracking parameters	GET <i>/ISAPI/PTZCtrl/channels/<ID>/trackingRatio/capabilities?format=json</i>
Get or set tracking parameters	GET or PUT <i>/ISAPI/PTZCtrl/channels/<ID>/trackingRatio?format=json</i>

PTZ Auxiliaries Control

Function	Request URI
Get or set all PTZ auxiliaries status	GET or PUT <i>/ISAPI/PTZCtrl/channels/<ID>/auxcontrols</i>
Get or set the status of single PTZ auxiliary	GET or PUT <i>/ISAPI/PTZCtrl/channels/<ID>/auxcontrols/<ID></i>

E-PTZ Parameters

Function	Request URI
Get or set e-PTZ parameters	GET or PUT <i>/ISAPI/Image/channels/<ID>/EPTZ</i>
Get capability of switching e-PTZ mode	GET <i>/ISAPI/Image/channels/<ID>/EPTZ/mode/capabilities?format=json</i>
Get or set parameters of switching e-PTZ mode	GET or PUT <i>/ISAPI/Image/channels/<ID>/EPTZ/mode?format=json</i>

Gyroscope Calibration

Function	Request URI
Get gyroscope attitude calibration	GET <i>/ISAPI/PTZCtrl/channels/<ID>/gyroVerify/capabilities?format=json</i>
Set gyroscope attitude calibration parameters	PUT <i>/ISAPI/PTZCtrl/channels/<ID>/gyroVerify?format=json</i>

GPS Calibration

Function	Request URI
Get capability of verifying GPS calibration	GET <i>/ISAPI/System/GPSVerification/channels/<ID>/points/capabilities?format=json</i>
Get GPS calibration configuration capability	GET <i>/ISAPI/System/GPSCalibration/channels/<ID>/capabilities?format=json</i>
Configure a single GPS calibration point.	- Get parameters of a single GPS calibration point: GET <i>/ISAPI/System/GPSCalibration/channels/<ID>/points/<PID>?format=json</i> - Set a single GPS calibration point: PUT <i>/ISAPI/System/GPSCalibration/channels/<ID>/points/<PID>?format=json</i> - Delete a single GPS calibration point: DELETE <i>/ISAPI/System/GPSCalibration/channels/<ID>/points/<PID>?format=json</i>
Configure multiple GPS calibration points	- Get parameters of multiple GPS calibration points in a batch: GET <i>/ISAPI/System/GPSCalibration/channels/<ID>/points?format=json</i> - Set multiple GPS calibration points in a batch:

Function	Request URI
	PUT /ISAPI/System/GPSCalibration/channels/<ID>/points?format=json Delete multiple GPS calibration points in a batch DELETE /ISAPI/System/GPSCalibration/channels/<ID>/points?format=json
Enable GPS calibration	PUT /ISAPI/System/GPSPTZCalibration/channels/<ID>/calibration?format=json
Get capability of verifying GPS calibration	GET /ISAPI/System/GPSVerification/channels/<ID>/points/capabilities?format=json
Configure multiple verification points	- Add a verification points: POST /ISAPI/System/GPSVerification/channels/<ID>/points?format=json - Get information of multiple verification points: GET /ISAPI/System/GPSVerification/channels/<ID>/points?format=json - Set multiple verification points: PUT /ISAPI/System/GPSVerification/channels/<ID>/points?format=json - Delete multiple verification points: DELETE /ISAPI/System/GPSVerification/channels/<ID>/points?format=json
Configure one specified verification point	- Get information of one specified verification point: GET /ISAPI/System/GPSVerification/channels/<ID>/points/<ID>?format=json - Set information of one specified verification point: PUT /ISAPI/System/GPSVerification/channels/<ID>/points/<ID>?format=json - Delete one specified verification point: DELETE /ISAPI/System/GPSVerification/channels/<ID>/points/<ID>?format=json
Get GPS calibration verification results in a batch	GET /ISAPI/System/GPSVerification/channels/<ID>/points/resultInfo?format=json
Get or set parameters of displaying GPS calibration verification result	GET or PUT /ISAPI/System/GPSVerification/channels/<ID>/display?format=json

10.1 Control PTZ via Digital Channel

Function	Request URI
Get PTZ control capability of digital channels	GET <i>/ISAPI/ContentMgmt/PTZCtrlProxy/capabilities</i>
Get PTZ control capability of a digital channel	GET <i>/ISAPI/ContentMgmt/PTZCtrlProxy/channels/<ID>/capabilities</i>
Get or set parameters of all digital PTZ channels	GET or PUT <i>/ISAPI/ContentMgmt/PTZCtrlProxy/channels</i>
Get or set parameters of a digital PTZ channel	GET or PUT <i>/ISAPI/ContentMgmt/PTZCtrlProxy/channels/<ID></i>
Perform scheduled PTZ control via a digital channel	PUT <i>/ISAPI/ContentMgmt/PTZCtrlProxy/channels/<ID>/momentary</i>
Perform 3D positioning of a digital PTZ channel	PUT <i>/ISAPI/ContentMgmt/PTZCtrlProxy/channels/<ID>/position3D</i>
Manually control PTZ via a digital channel	PUT <i>/ISAPI/ContentMgmt/PTZCtrlProxy/channels/<ID>/continuous</i>
Control fan of a digital PTZ channel	PUT <i>/ISAPI/ContentMgmt/PTZCtrlProxy/channels/<ID>/autoPan</i>
Perform one-touch focus via a digital PTZ channel	PUT <i>/ISAPI/ContentMgmt/PTZCtrlProxy/channels/<ID>/onpushfoucs/start</i>
Get, set, or delete all presets of a digital PTZ channel	GET, PUT, or DELETE <i>/ISAPI/ContentMgmt/PTZCtrlProxy/channels/<ID>/presets</i>
Add a preset via a digital PTZ channel	POST <i>/ISAPI/ContentMgmt/PTZCtrlProxy/channels/<ID>/presets</i>
Get, set, or delete a preset of a digital PTZ channel	GET, PUT, or DELETE <i>/ISAPI/ContentMgmt/PTZCtrlProxy/channels/<ID>/presets/<ID></i>
Call a preset via a digital PTZ channel	PUT <i>/ISAPI/ContentMgmt/PTZCtrlProxy/channels/<ID>/presets/<ID>/goto</i>
Get, set, or delete all patrols of a digital PTZ channel	GET, PUT, or DELETE <i>/ISAPI/ContentMgmt/PTZCtrlProxy/channels/<ID>/patrols</i>
Add a patrol via a digital PTZ channel	POST <i>/ISAPI/ContentMgmt/PTZCtrlProxy/channels/<ID>/patrols</i>
Get, set, or delete a patrol of a digital PTZ channel	GET, PUT, or DELETE <i>/ISAPI/ContentMgmt/PTZCtrlProxy/channels/<ID>/patrols/<ID></i>

Function	Request URI
Get a patrol's status of a digital PTZ channel	GET <i>/ISAPI/ContentMgmt/PTZCtrlProxy/channels/<ID>/patrols/<ID>/status</i>
Start calling a patrol of a digital PTZ channel	PUT <i>/ISAPI/ContentMgmt/PTZCtrlProxy/channels/<ID>/patrols/<ID>/start</i>
Stop calling a patrol of a digital PTZ channel	PUT <i>/ISAPI/ContentMgmt/PTZCtrlProxy/channels/<ID>/patrols/<ID>/stop</i>
Pause calling a patrol of a digital PTZ channel	PUT <i>/ISAPI/ContentMgmt/PTZCtrlProxy/channels/<ID>/patrols/<ID>/pause</i>

10.2 Configure Master-Slave Tracking

The master-slave tracking system is formed by one fixed camera and one PTZ camera. The PTZ camera smartly tracks multiple moving targets, and the fixed camera offers the panoramic view. The master-slave tracking system provides detection and tracking of moving targets, and multiple targets can be tracked simultaneously. The system mainly applies to major industrial infrastructures requiring perimeter prevention, such as airport, prison, railway, museum, enterprise, high-end villas, etc.

Steps

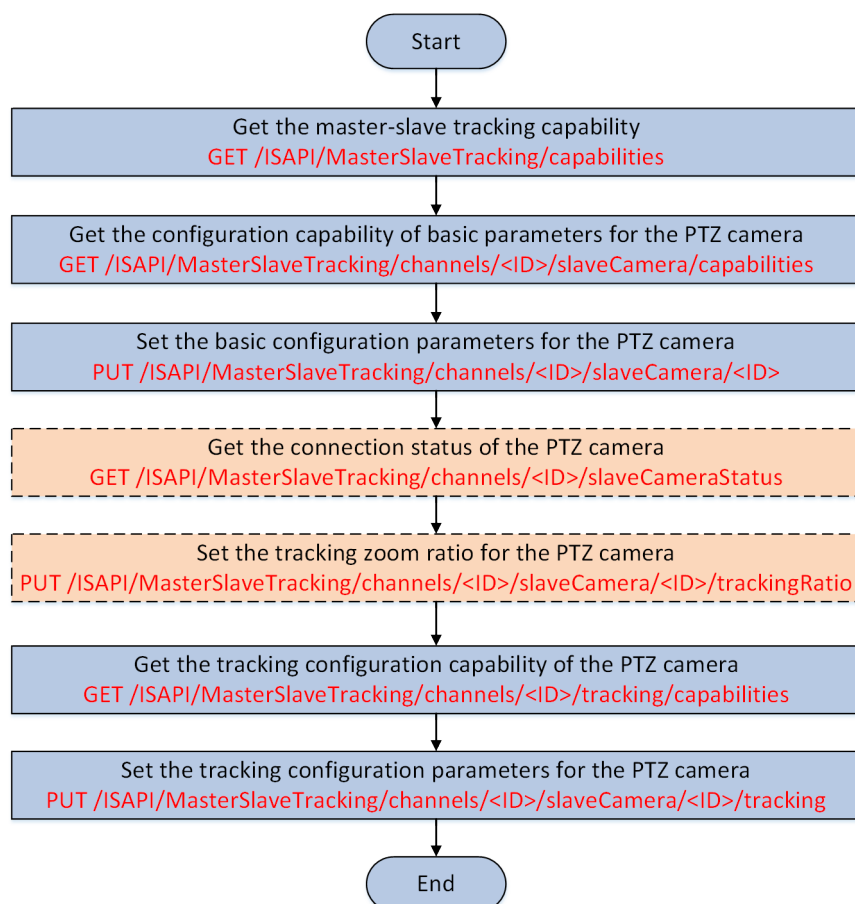


Figure 10-1 API Calling Flow of Configuring Master-Slave Tracking

1. Call **/ISAPI/MasterSlaveTracking/capabilities** by GET method to get the master-slave tracking capability to know the supported configurations.
2. Call **/ISAPI/MasterSlaveTracking/channels/<ID>/slaveCamera/capabilities** by GET method to get the configuration capability of basic parameters for the PTZ camera to know the basic parameters that can be configured.
3. Call **/ISAPI/MasterSlaveTracking/channels/<ID>/slaveCamera/<ID>** by PUT method to set the basic configuration parameters for the PTZ camera.
4. **Optional:** Call **/ISAPI/MasterSlaveTracking/channels/<ID>/slaveCameraStatus** by GET method to get the connection status of the PTZ camera.
5. **Optional:** Call **/ISAPI/MasterSlaveTracking/channels/<ID>/slaveCamera/<ID>/trackingRatio** by PUT method to set the tracking zoom ratio for the PTZ camera.
6. Call **/ISAPI/MasterSlaveTracking/channels/<ID>/tracking/capabilities** by GET method to get the tracking configuration capability of the PTZ camera to know the tracking parameters that can be configured.
7. Call **/ISAPI/MasterSlaveTracking/channels/<ID>/slaveCamera/<ID>/tracking** by PUT method to set the tracking configuration parameters for the PTZ camera.

10.3 Configure Panorama Tracking

Tracking Takeover and Linked Tracking Capture are two mutually exclusive modes for panorama tracking.

Linked Tracking Capture Parameters

Function	Request URI
Get capability of linked tracking capture configuration	GET <i>/ISAPI/MasterSlaveTracking/linkedTracking/capabilities?format=json</i>
Get or set linked tracking capture parameters	GET or PUT <i>/ISAPI/MasterSlaveTracking/linkedTracking?format=json</i>

Chapter 11 Storage Management

11.1 Configure Log Storage Mode

You can save the log files in default mode or custom mode. In the default mode, each HDD can save up to 64 MB log files, and the storage space will automatically be overwritten when the space usage reaches 64 MB. In the custom mode, it is available to specify HDDs to save log files.

Steps

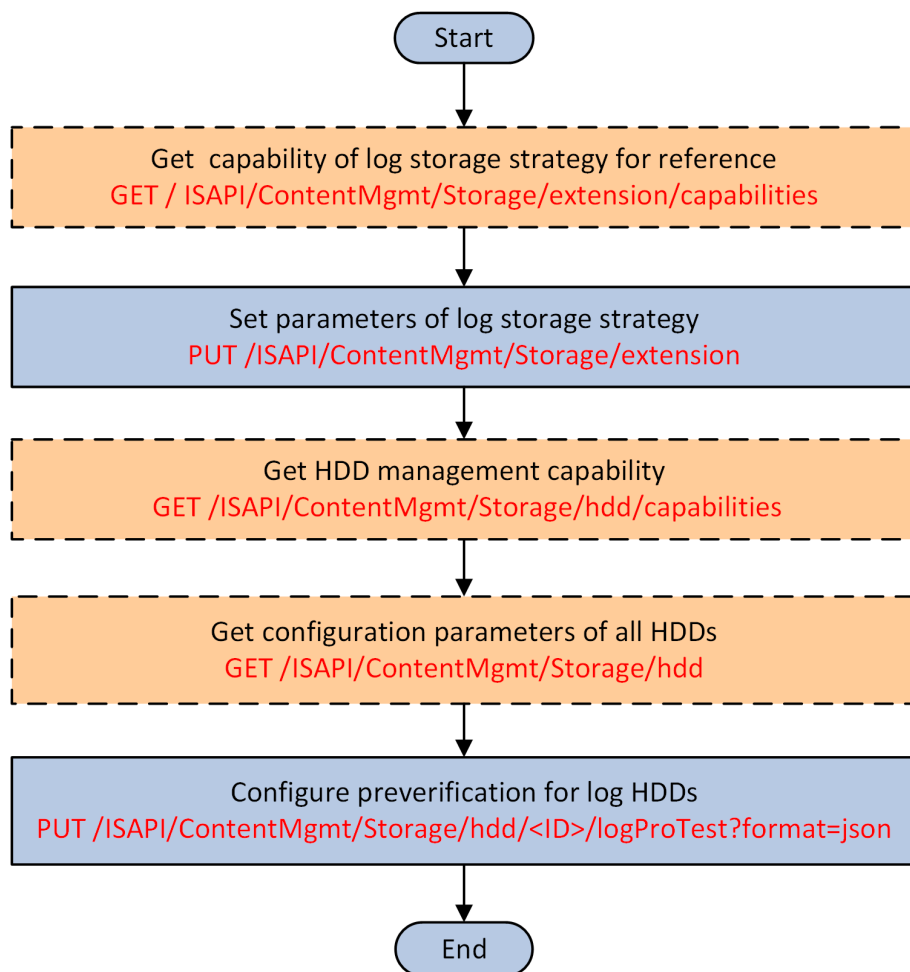


Figure 11-1 Programming Flow of Configuring Log Storage Mode

- Optional:** Call `/ISAPI/ContentMgmt/Storage/extension/capabilities` by GET method to get the capability of log storage strategy for reference.
The capability is returned in the message `XML_Cap_storageExtension`.
- Call `/ISAPI/ContentMgmt/Storage/extension` by PUT method, and set the node `<logStorageMode>` as "system" (system default mode) or "custom" (custom mode) in the message `XML_storageExtension`.

3. **Optional:** Call */ISAPI/ContentMgmt/Storage/hdd/capabilities* by GET method to get the HDD management capability.
The capability is returned in the message *XML_Cap_hddList* .
4. **Optional:** Call */ISAPI/ContentMgmt/Storage/hdd* by GET method to get configuration parameters of all HDDs.
The parameters are returned in the message *XML_hddList* .
5. Call */ISAPI/ContentMgmt/Storage/hdd/<ID>/logProTest?format=json* by PUT method to configure preverification for log HDDs

11.2 HDD Management

The HDD (Hard Disk Drive) is used for storage. Before storing information in HDD, you must format it, and you can also reallocate its space for better storage management.

HDD Configuration

Function	Request URI
Get HDD management capability	GET <i>/ISAPI/ContentMgmt/Storage/hdd/capabilities</i>
Get parameters of all HDDs	GET <i>/ISAPI/ContentMgmt/Storage/hdd</i>
Get or set parameters of one HDD	GET or PUT <i>/ISAPI/ContentMgmt/Storage/hdd/<ID></i>
Get parameters of all HDD quotas	GET <i>/ISAPI/ContentMgmt/Storage/quota</i>
Get parameters of an HDD quota	GET <i>/ISAPI/ContentMgmt/Storage/quota/<ID></i>
Set parameters of an HDD quota	PUT <i>/ISAPI/ContentMgmt/Storage/quota/<ID></i>
Verify HDD encryption password	PUT <i>/ISAPI/ContentMgmt/Storage/hdd/<ID>/encryptVerify?format=json</i>

HDD Formatting

Function	Request URI
Format all HDDs	PUT <i>/ISAPI/ContentMgmt/Storage/hdd/format</i>
Specify formatting type to format one HDD	PUT <i>/ISAPI/ContentMgmt/Storage/hdd/<ID>/format?formatType=</i>

Function	Request URI
Format multiple specified HDDs	PUT <i>/ISAPI/ContentMgmt/Storage/hdd/specifyHddFormat?format=json</i>
Format an encrypted HDD	PUT <i>/ISAPI/ContentMgmt/Storage/hdd/<ID>/encryptFormat?format=json</i>
Get formatting status of one HDD	GET <i>/ISAPI/ContentMgmt/Storage/hdd/<ID>/formatStatus</i>

HDD Clearing

Function	Request URI
Get HDD clearing capability	GET <i>/ISAPI/ContentMgmt/storage/hdd/clearingSpace/capabilities?format=json</i>
Get or set HDD clearing parameters	GET or PUT <i>/ISAPI/ContentMgmt/storage/hdd/clearingSpace?format=json</i>

HDD Checking

Function	Request URI
Get or set parameters for checking HDD status	GET or PUT <i>/ISAPI/ContentMgmt/Storage/hdd/SMARTTest/config</i>
Start checking HDD status	PUT <i>/ISAPI/ContentMgmt/Storage/hdd/<ID>/SMARTTest/start</i>
Get HDD checking status	GET <i>/ISAPI/ContentMgmt/Storage/hdd/<ID>/SMARTTest/status</i>
Start checking bad sectors of a HDD	PUT <i>/ISAPI/ContentMgmt/Storage/hdd/<ID>/BadSectorsTest/start</i>
Get bad sector checking status of a HDD	GET <i>/ISAPI/ContentMgmt/Storage/hdd/<ID>/BadSectorsTest/status</i>
Pause checking bad sectors of a HDD	PUT <i>/ISAPI/ContentMgmt/Storage/hdd/<ID>/BadSectorsTest/pause</i>
Resume checking bad sectors of a HDD	PUT <i>/ISAPI/ContentMgmt/Storage/hdd/<ID>/BadSectorsTest/resume</i>
Stop checking bad sectors of a HDD	PUT <i>/ISAPI/ContentMgmt/Storage/hdd/<ID>/BadSectorsTest/stop</i>

SHM (SkyHawk Health Management) Alarm Linkage

Function	Request URI
Get or set linkage parameters of HDD high temperature detection	GET or PUT <i>/ISAPI/Event/triggers/highHDTemperature</i>
Get or set linkage parameters of HDD low temperature detection	GET or PUT <i>/ISAPI/Event/triggers/lowHDTemperature</i>
Get or set linkage parameters of HDD impact detection	GET or PUT <i>/ISAPI/Event/triggers/hdImpact</i>
Get or set linkage parameters of HDD bad sector detection	GET or PUT <i>/ISAPI/Event/triggers/hdBadBlock</i>
Get or set linkage parameters of HDD severe fault detection	GET or PUT <i>/ISAPI/Event/triggers/severeHDFailure</i>

HDD Sync Status

Function	Request URI
Get HDD data sync status	GET <i>/ISAPI/ContentMgmt/Storage/hdd/<ID>/syncStatus?format=json</i>

11.3 Working Mode Configuration

Function	Request URI
Get configuration capability of current storage working mode	GET <i>/ISAPI/ContentMgmt/Storage/capabilities</i>
Get or set current working mode of storage.	GET or PUT <i>/ISAPI/ContentMgmt/storage</i>
Start disk data reconstruction	PUT <i>/ISAPI/ContentMgmt/Storage/dataReconstruction?format=json</i>
Get disk data reconstruction progress	GET <i>/ISAPI/ContentMgmt/Storage/dataReconstruction/progress?format=json</i>

11.4 SSD Management

The SSD (solid state driver) is used for storage. Here introduces the URIs to implement the functions of SSD file system format, file system upgrade, SSD firmware upgrade, SSD S.M.A.R.T (self-monitoring, analysis, and reporting technology) detection, and so on.

SSD File System

Function	Request URI
Format SSD file system	PUT <i>/ISAPI/System/SSDFileSystem/format?format=json</i>
Search for formatting progress of SSD file system	GET <i>/ISAPI/System/SSDFileSystem/format/status?format=json</i>
Get upgrade status of SSD file system	GET <i>/ISAPI/System/SSDFileSystem/upgrade?format=json</i>
Upgrade SSD file system	PUT <i>/ISAPI/System/SSDFileSystem/upgrade?format=json</i>
Search for upgrade progress of SSD file system	GET <i>/ISAPI/System/SSDFileSystem/upgrade/status?format=json</i>
Search for SSD storage quotas	GET <i>/ISAPI/System/SSDFileSystem/capacity?format=json</i>

SSD Configuration

Function	Request URI
Get SSD management capability	GET <i>/ISAPI/ContentMgmt/Storage/ssd/<ID>/capabilities?format=json</i>
Upgrade SSD firmware	POST <i>/ISAPI/System/SSDFileSystem/format/status?format=json</i>
Get SSD firmware upgrade progress	GET <i>/ISAPI/System/SSDFileSystem/upgrade?format=json</i>
Get a single SSD information	GET <i>/ISAPI/ContentMgmt/Storage/ssd/<ID></i>
Get all SSDs information	GET <i>/ISAPI/ContentMgmt/Storage/ssd</i>

SSD S.M.A.R.T. Detection

Function	Request URI
Run SSD S.M.A.R.T detection	PUT <i>/ISAPI/ContentMgmt/Storage/ssd/<ID>/SMARTTest/start</i>
Get SSD S.M.A.R.T detection status	GET <i>/ISAPI/ContentMgmt/Storage/ssd/<ID>/SMARTTest/status</i>

11.5 Others

Table 11-1 APIs about Additional Information

Function	API
Get storage capability of additional information	GET <i>/ISAPI/ContentMgmt/Storage/ExtraInfo/capabilities</i>
Get or set storage parameters of additional information	GET or PUT <i>/ISAPI/ContentMgmt/Storage/ExtraInfo</i>

Chapter 12 File Management

Table 12-1 Download Files

Function	Request URI
Get downloading capability	GET <i>/ISAPI/ContentMgmt/download/capabilities</i>
Download files	POST <i>/ISAPI/ContentMgmt/download</i>

12.1 Video Files

Search Video Files

Function	Request URI
Get search conditions	GET <i>/ISAPI/ContentMgmt/search/profile</i>
Search video files	POST <i>/ISAPI/ContentMgmt/search</i>  Note This URI is not only used to search video files, it can also be used to search any resources stored in the device.
Search video files by time	POST <i>/ISAPI/ContentMgmt/record/tracks/<ID>/dailyDistribution</i>
Get capability of searching video files by VCA event	GET <i>/ISAPI/ContentMgmt/SmartSearch/capabilities</i>
Search video files by VCA event	POST <i>/ISAPI/ContentMgmt/SmartSearch</i>

Lock or Unlock Video Files

Function	Request URI
Get video file locking and unlocking capability	GET <i>/ISAPI/ContentMgmt/record/control/locks/capabilities</i>
Lock or unlock video files by name	PUT <i>/ISAPI/ContentMgmt/record/control/locks/name</i>

Add Remarks to Video File

Function	Request URI
Get capability of adding remarks to video files	GET <i>/ISAPI/ContentMgmt/record/control/remark/capabilities?format=json</i>
Add remarks to video files	PUT <i>/ISAPI/ContentMgmt/record/control/remark?format=json</i>

Note

To check whether the device supports adding remarks to the video files, you can call */ISAPI/ContentMgmt/capabilities* by GET method to get the device storage capability (*XML_RacmCap*). If it supports, the node *<isSupportRemark>* will be returned and its value is "true".

Clear Stored Video Files

Function	Request URI
Get the capability of deleting files in a batch	GET <i>/ISAPI/ContentMgmt/record/control/cleanUp/capabilities?format=json</i>
Delete files in a batch	POST <i>/ISAPI/ContentMgmt/record/control/cleanUp?format=json</i>

Export Video or Picture Files to Device via USB

Function	Request URI
Get capability of exporting video or picture files to devices via USB	GET <i>/ISAPI/ContentMgmt/download/toUSB/capabilities?format=json</i>  Note To check whether this function is supported by the device, you can call <i>/ISAPI/ContentMgmt/download/capabilities</i> by GET method to get the downloading capability (<i>XML_DownloadAbility</i>). If it supports, the node <i><isSupportDownloadToUSB></i> will be returned and its value is "true".
Export video or picture files to devices via USB	POST <i>/ISAPI/ContentMgmt/download/toUSB?format=json</i>
Get progress of exporting video or picture files to devices via USB	GET <i>/ISAPI/ContentMgmt/download/toUSB/<taskId>/progress?format=json</i>

12.2 Audio Files

Custom Audio File Operations

Function	Request URI
Check whether custom audio operation is supported	Call <i>/ISAPI/Event/triggers/notifications/AudioAlarm/capabilities?format=json</i> by GET method. The capability is returned in <i>JSON_AudioAlarmCap</i> and if "true" is returned in the node <i>isSupportCustomAudio</i> , it indicates that custom audio operation is supported.
Download or delete custom audio file	Call <i>/ISAPI/Event/triggers/notifications/AudioAlarm/customAudio/<customAudioID>?format=json</i> by GET or DELETE method.
Get or edit custom audio file information	Call <i>/ISAPI/Event/triggers/notifications/AudioAlarm/customAudioInfo/<customAudioID>?format=json</i> by GET or PUT method.
Import custom audio file	Call <i>/ISAPI/Event/triggers/notifications/AudioAlarm/customAudio?format=json</i> by POST method.
Get encoding format information of custom audio file in a batch	Call <i>/ISAPI/Event/triggers/notifications/AudioAlarm/customAudioTypeInfo/channels?format=json</i> by GET method.

Audio Test

Function	Request URI
Check whether audio test is supported	Call <i>/ISAPI/Event/triggers/notifications/AudioAlarm/capabilities?format=json</i> by GET method. The capability is returned in <i>JSON_AudioAlarmCap</i> and if "true" is returned in the node <i>isSupportAudioTest</i> , it indicates that audio test is supported.
Audio test	Call <i>/ISAPI/Event/triggers/notifications/AudioAlarm/<audioID>/test?format=json</i> by PUT method.

12.3 Log Files

Table 12-2 Export Log to USB Flash Drive

Function	Request URI
Get the configuration capability of exporting log to USB flash drive	GET <i>/ISAPI/System/exportLogToUSB/capabilities?format=json</i>
Set parameters of exporting logs to USB flash drive	PUT <i>/ISAPI/System/exportLogToUSB/mode?format=json</i>
Get the status of exporting logs to USB flash drive	GET <i>/ISAPI/System/exportLogToUSB/status?format=json</i>

Table 12-3 Export Log to PC

Function	Request URI
Get the configuration capability of exporting logs to PC	GET <i>/ISAPI/System/exportUSBLog/capabilities?format=json</i>
Export logs to PC	PUT <i>/ISAPI/System/exportUSBLog?format=json</i>
Get the status of exporting logs to PC	GET <i>/ISAPI/System/exportUSBLog/status?format=json</i>

Table 12-4 Search and Export Log Files

Function	Request URI
Search for log files	Option1: POST <i>/ISAPI/ContentMgmt/logSearch</i> Option 2: POST <i>/ISAPI/ContentMgmt/logSearch?format=json</i>
Export log files	POST <i>/ISAPI/ContentMgmt/logSearch/dataPackage</i>  Note To check whether the device supports exporting log files, you can call <i>/ISAPI/ContentMgmt/capabilities</i> by GET method to get the storage capability (<i>XML_RacmCap</i>). If it supports, the node <i><isSupportLogDataPackage></i> is returned in the capability message and its value is "true".

Chapter 13 Alarm/Event

When the alarm is triggered or the event occurred, if you have configured alarm/event uploading parameters, you can receive and process the alarm/event information in the third-party platform or system by calling the request URLs of ISAPI. Two modes are available for receiving alarms, including arming mode and listening mode.

Arming Mode

When the alarm is triggered or event occurred, the third-party platform or system can send the request URL to the device for getting the alarm/event stream, and then the device uploads the response message with alarm/event information.

Listening Mode

When alarm is triggered or event occurred, the device uploads the alarm information automatically, and then the third-party platform or system can receives the alarm/event by configuring listening port of HTTP host server.

13.1 Configure Audio Detection Alarm

The audio detection function can detect the abnormal sound in the surveillance scene, such as the sudden increase or decrease of the sound intensity. If the alarm linkage is configured, some certain actions can be taken as response when the audio detection alarm is triggered.

Steps

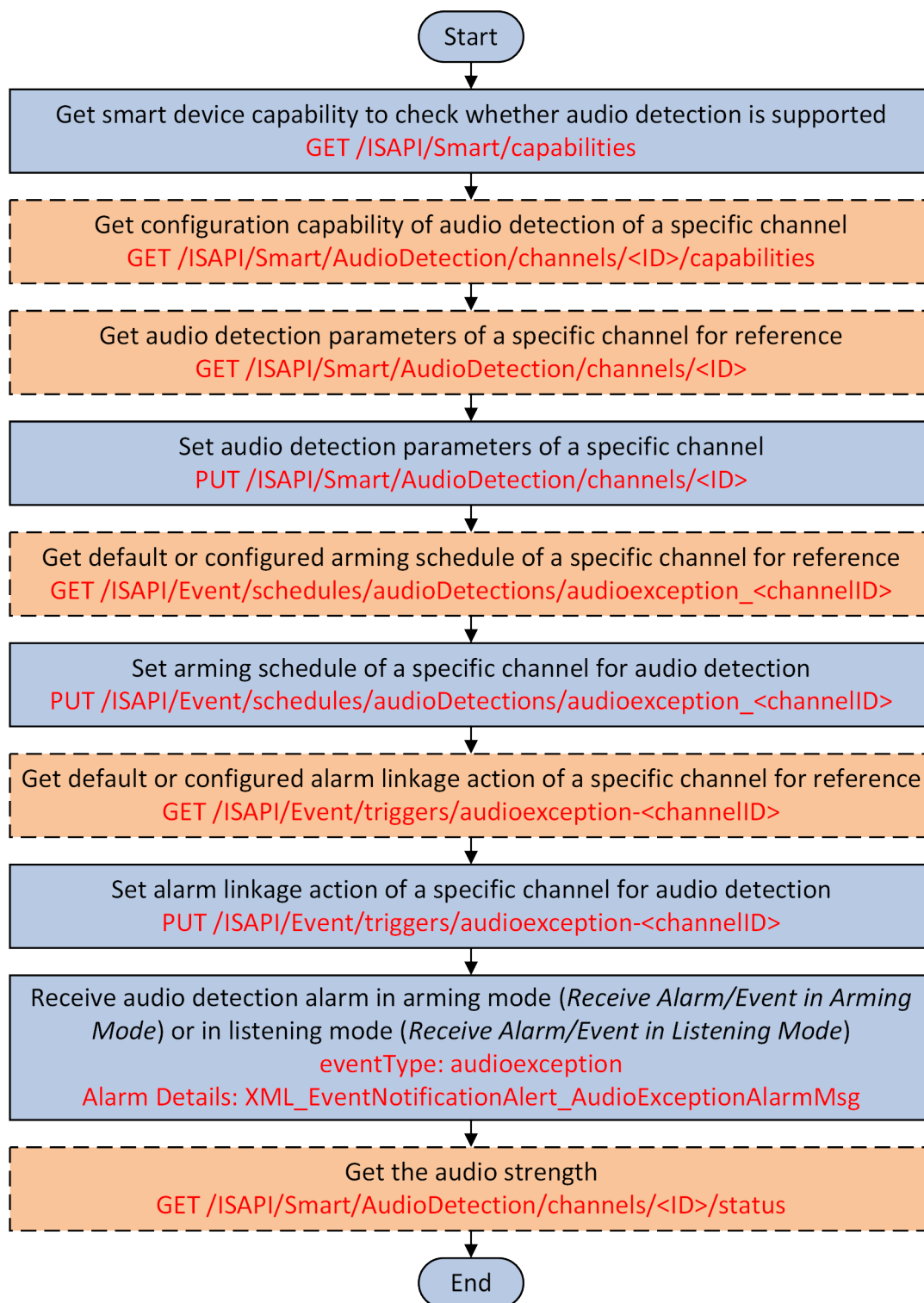


Figure 13-1 Programming Flow of Configuring Audio Detection Alarm

1. Call ***/ISAPI/Smart/capabilities*** by GET method to get the smart device capability and check whether the device supports audio detection.

The smart device capability is returned in the message ***XML_SmartCap*** .

If the node ***<isSupportAudioDetection>*** will be returned in the message and its value is true, it indicates that the audio detection is supported by the device, and then you can continue to perform the following operations. Otherwise, it indicates that the audio detection is not supported, please end this task.

2. **Optional:** Call ***/ISAPI/Smart/AudioDetection/channels/<ID>/capabilities*** by GET method to get the configuration capability of the audio detection of a specific channel for knowing about the configuration details or notices.
3. **Optional:** Call ***/ISAPI/Smart/AudioDetection/channels/<ID>*** by GET method to get the audio detection parameters of a specific channel for reference.
4. Call ***/ISAPI/Smart/AudioDetection/channels/<ID>*** by PUT method to set the audio detection parameters of a specific channel.
5. **Optional:** Call ***/ISAPI/Event/schedules/<EventType>/<ID>*** by GET method to get the default of configured arming schedule parameters of a specific channel for reference.



Note

The ***<EventType>*** in the request URI should be set to ***<audioDetections>*** and the ***<ID>*** in the request URI should be set in the format of ***<audioexception_<channelID>***, e.g., ***/ISAPI/Event/schedules/audioDetections/audioexception_101***.

6. Call ***/ISAPI/Event/schedules/<EventType>/<ID>*** by PUT method to set the arming schedule of a specific channel for audio detection.



Note

The ***<EventType>*** in the request URI should be set to ***<audioDetections>*** and the ***<ID>*** in the request URI should be set in the format of ***<audioexception_<channelID>***, e.g., ***/ISAPI/Event/schedules/audioDetections/audioexception_101***.

7. **Optional:** Call ***/ISAPI/Event/triggers/<eventType>-<ID>*** by GET method to get the default or configured alarm linkage actions of a specific channel for reference.



Note

The ***<eventType>*** in the request URI should be set to ***<audioexception>***, e.g., ***/ISAPI/Event/triggers/audioexception-101***.

8. Call ***/ISAPI/Event/triggers/<eventType>-<ID>*** by PUT method to set alarm linkage actions of a specific channel for audio detection.



Note

The ***<eventType>*** in the request URI should be set to ***<audioexception>***, e.g., ***/ISAPI/Event/triggers/audioexception-101***.

9. Receive the audio detection alarm in arming mode (refer to **Receive Alarm/Event in Arming Mode** for details) or in listening mode (refer to **Receive Alarm/Event in Listening Mode** for details) when the alarm is triggered.


Note

The audio detection alarm details are uploaded in the message **XML_EventNotificationAlert_AudioExceptionAlarmMsg** and the corresponding alarm type (**eventType**) is "audioexception".

10. **Optional:** Call **/ISAPI/Smart/AudioDetection/channels/<ID>/status** by GET method to get the audio strength.

13.2 Configure Exception Alarm

To monitor the device status, you can configure the exception alarm, such as supply voltage exception, PoE power exception, and so on. When the exception occurs, the configured linkage action will be triggered and the alarm information will be uploaded automatically.

Supply Voltage Exception

Function	Request URI
Get configuration capability of alarm linkage action	GET /ISAPI/Event/triggersCap
Get or set the linkage action of supply voltage exception alarm	GET or PUT /ISAPI/Event/triggers/<eventType>-<ID> , and the <ID> in the URI is "voltageinstable"
Receive supply voltage exception alarm in listening mode	Refer to Receive Alarm/Event in Listening Mode for details, the event type is "voltageinstable", and the alarm information is returned in message JSON_EventNotificationAlert_voltageinstable


Note

To check if the device supports supply voltage exception alarm, you can call **/ISAPI/Event/capabilities** by GET method to get the device event capability (**XML_EventCap**). If supports, the node **<isSupportVoltageInstable>** will be returned.

13.3 Configure Certificate Expiry Alarm

Generally, the device certificate is only valid in a specific period of time. You can configure the certificate expiry alarm to remind the user a few days in advance. When the certificate is expired, the alarm will be triggered and uploaded automatically.

Steps

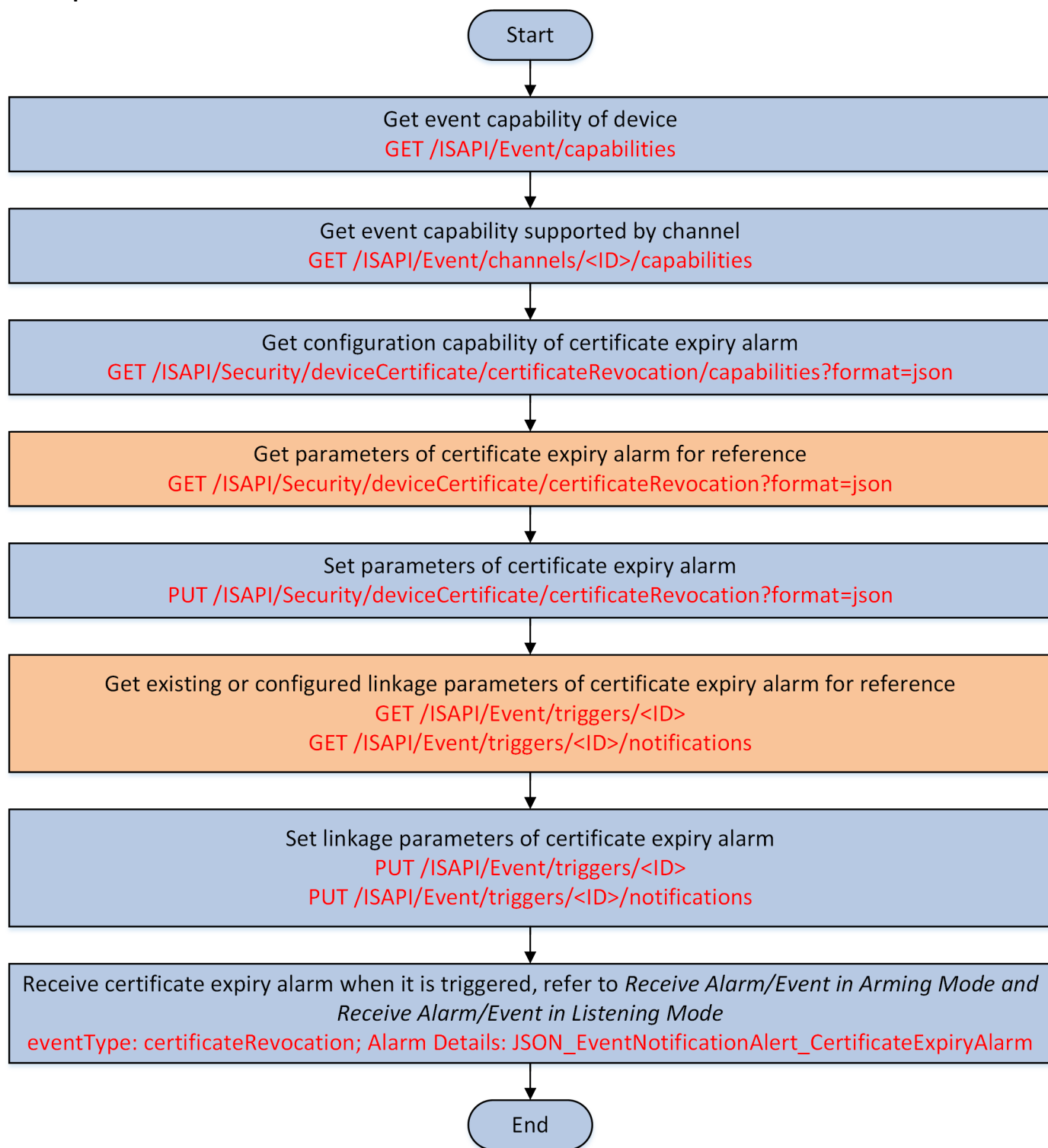


Figure 13-2 Programming Flow of Configuring Certificate Expiry Alarm

1. Call **`/ISAPI/Event/capabilities`** by GET method to get the event capability of the device for checking whether the device supports certificate expiry alarm.
The event capability of the device is returned in the message **`XML_EventCap`** . If the certificate expiry alarm is supported, the node **`<isSupportCertificateRevocation>`** will be returned and its value is "true", then you can perform the following steps. Otherwise, please end this task.
2. Call **`/ISAPI/Event/channels/<ID>/capabilities`** by GET method to get the event capability of the channel for checking whether the certificate expiry alarm is supported by the channel.
The event capability supported by the channel is returned in the message **`XML_ChannelEventCap`** . If the certificate expiry alarm is supported, the node **`<eventType>`** will be returned and it contains "certificateRevocation".
3. Call **`/ISAPI/Security/deviceCertificate/certificateRevocation/capabilities?format=json`** by GET method to get the configuration capability of certificate expiry alarm to know the supported parameters of certificate expiry alarm that can be configured.
4. **Optional:** Call **`/ISAPI/Security/deviceCertificate/certificateRevocation?format=json`** by GET method to get the parameters of certificate expiry alarm to get the existing or configured parameters for reference.
5. Call **`/ISAPI/Security/deviceCertificate/certificateRevocation?format=json`** by PUT method to set the parameters of certificate expiry alarm .
6. **Optional:** Call **`/ISAPI/Event/triggers/<eventType>-<ID>`** or **`/ISAPI/Event/triggers/<ID>/notifications`** by GET method to get the linkage parameters of certificate expiry alarm to get the existing or configured parameters for reference.



Note

The **`<ID>`** in the request URI refers to the channel No., and it should be set to the format "certificateRevocation-<channelID>".

7. Call **`/ISAPI/Event/triggers/<eventType>-<ID>`** or **`/ISAPI/Event/triggers/<ID>/notifications`** by PUT method to Set the linkage parameters of certificate expiry alarm.



Note

The **`<ID>`** in the request URI refers to the channel No., and it should be set to the format "certificateRevocation-<channelID>".

8. **Optional:** Receive the uploaded certificate expiry alarm in arming mode (see details in ***Receive Alarm/Event in Arming Mode***) or in listening mode (see details in ***Receive Alarm/Event in Listening Mode***) when the certificate is expired.

The certificate expiry alarm details are returned in the message **`JSON_EventNotificationAlert_CertificateExpiryAlarmMsg`** , and the corresponding alarm type (**`eventType`**) is "certificateRevocation".

13.4 Configure Network Traffic Pre-alarm

You can set or search for data plans, monitor data usage, and set a traffic limit. When the limit is exceeded, the alarm will be triggered and uploaded automatically.

Steps

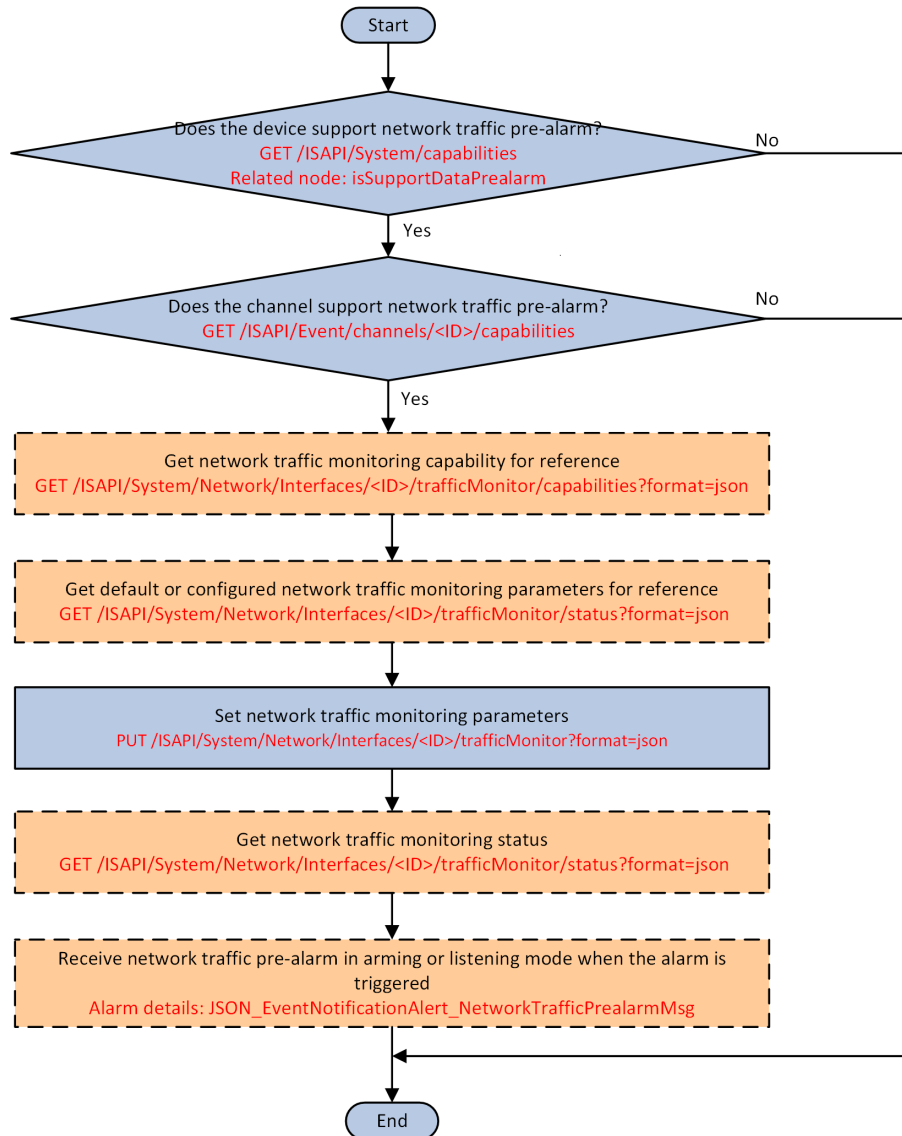


Figure 13-3 Programming Flow of Configuring Network Traffic Pre-alarm

1. Call **`/ISAPI/System/capabilities`** by GET to check whether the device supports network traffic pre-alarm.
2. Call **`/ISAPI/Event/channels/<ID>/capabilities`** by GET to check whether the channel supports network traffic pre-alarm.
3. **Optional:** Call **`/ISAPI/System/Network/WirelessDial/Interfaces/<ID>/trafficMonitor/capabilities?format=json`** by GET to get network traffic monitoring capability for reference.

The capability is returned in the message *JSON_TrafficMonitorCap* .

4. **Optional:** Call */ISAPI/System/Network/WirelessDial/Interfaces/<ID>/trafficMonitor?format=json* by GET to get default or configured network traffic monitoring parameters for reference.

The parameters are returned in the message *JSON_TrafficMonitor* .

5. Call */ISAPI/System/Network/WirelessDial/Interfaces/<ID>/trafficMonitor?format=json* by PUT to set parameters of network traffic monitoring.
6. **Optional:** Call */ISAPI/System/Network/WirelessDial/Interfaces/<ID>/trafficMonitor/status?format=json* by GET to get network traffic monitoring status.

The status is returned in the message *JSON_TrafficMonitorStatus* .

7. **Optional:** Receive the uploaded network traffic alarm in arming mode (see details in *Receive Alarm/Event in Arming Mode*) or in listening mode (see details in *Receive Alarm/Event in Listening Mode*).



Note

The traffic pre-alarm only supports the default linkage action (upload to center) and arming schedule (all-day schedule).

The network traffic alarm details are returned in the message *JSON_EventNotificationAlert_NetworkTrafficPrealarmMsg* , and the corresponding alarm type (*eventType*) is "dataPrealarm".

13.5 Configure Motion Detection Alarm

Motion detection detects the moving objects in the configured surveillance area, and a series of actions can be taken when the alarm is triggered. In order to detect the moving objects accurately and reduce the false alarm rate, detection target types (including human and vehicle), and detection region types (including region, grid, and ROI) are selectable for different motion detection environment.

Steps

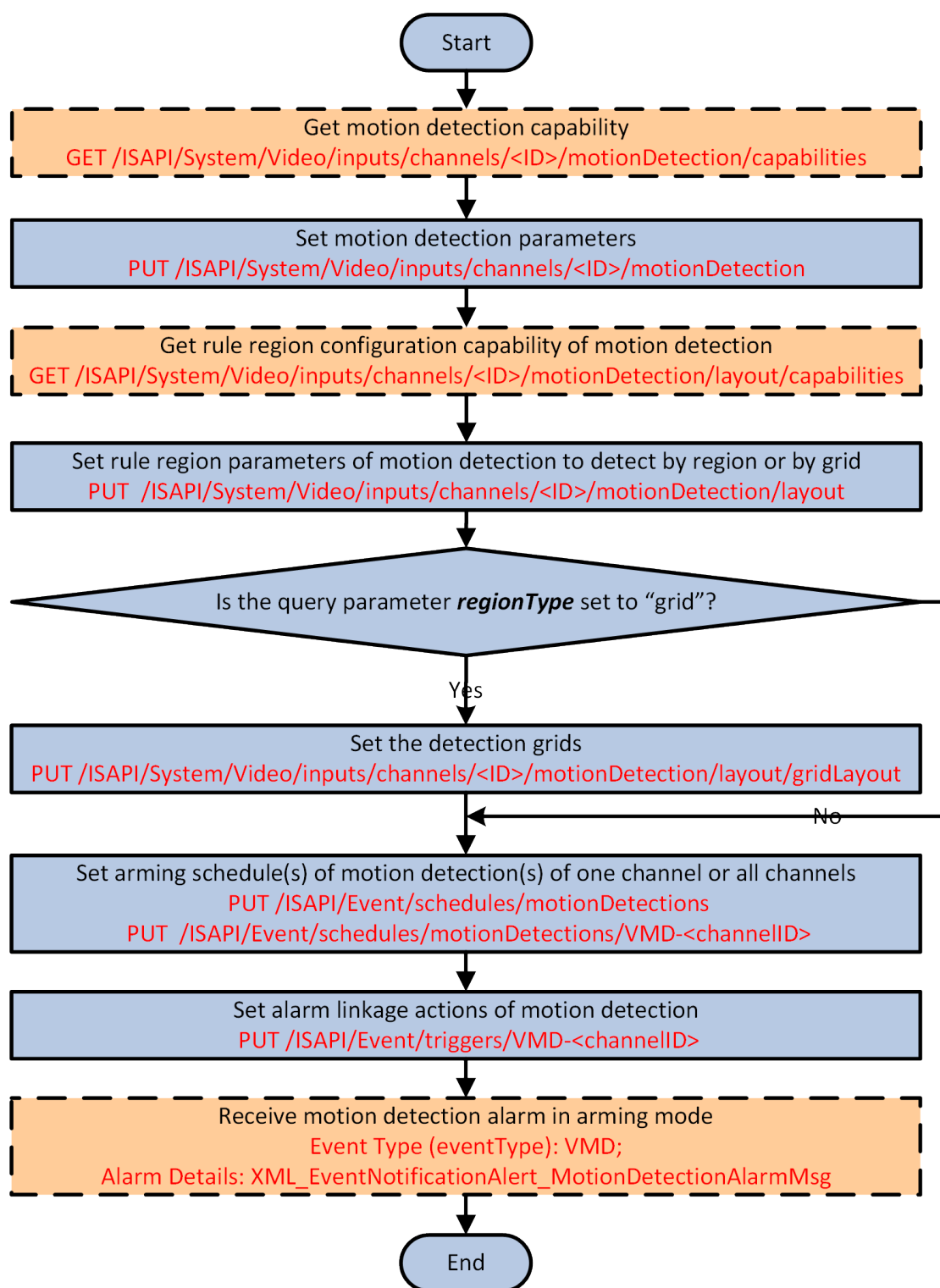


Figure 13-4 API Calling Flow of Configuring Motion Detection Alarm

1. Configure motion detection.

- 1) **Optional:** Call `/ISAPI/System/Video/inputs/channels/<ID>/motionDetection/capabilities` by GET method to get the motion detection capability for reference.
- 2) **Optional:** Call `/ISAPI/System/Video/inputs/channels/<ID>/motionDetection` by GET method to get default or configured motion detection parameters for reference.
- 3) Call `/ISAPI/System/Video/inputs/channels/<ID>/motionDetection` by PUT method to set motion detection parameters.
2. Configure rule region of motion detection to detect by region or by grid.
 - 1) **Optional:** Call `/ISAPI/System/Video/inputs/channels/<ID>/motionDetection/layout/capabilities` by GET method to get rule region configuration capability of motion detection for reference.
 - 2) **Optional:** Call `/ISAPI/System/Video/inputs/channels/<ID>/motionDetection/layout` by GET method to get default or configured rule region parameters of motion detection for reference.
 - 3) Call `/ISAPI/System/Video/inputs/channels/<ID>/motionDetection/layout` by PUT method to set rule region of motion detection to detect by region or by grid.
3. Check the value of query parameter **regionType** in the request URI of `/ISAPI/System/Video/inputs/channels/<ID>/motionDetection/layout/gridLayout` .
 - If the query parameter **regionType** in the request URI is set to "grid", call `/ISAPI/System/Video/inputs/channels/<ID>/motionDetection/layout/gridLayout` by PUT method to set the detection grids.
 - If the query parameter **regionType** in the request URI is set to "region", perform the following steps.
4. Configure arming schedule(s) of motion detection(s).
 - Call `/ISAPI/Event/schedules/motionDetections` by PUT method to set arming schedules of all channels' motion detection.
 - Call `/ISAPI/Event/schedules/motionDetections/<ID>` by PUT method and set the **<ID>** in the request URI to "VMD-**<channelID>**" (e.g., `/ISAPI/Event/schedules/motionDetections/VMD-101`) to set arming schedule of a channel's motion detection.
5. Call `/ISAPI/Event/triggers/<eventType>-<ID>` by PUT method and set the **<ID>** in the request URI to "VMD-**<channelID>**" (e.g., `/ISAPI/Event/triggers/VMD-101`) to set alarm linkage action of a channel's motion detection.
6. Call the flow of **Receive Alarm/Event in Arming Mode** to receive the motion detection alarm information if it is triggered and uploaded.

The motion detection alarm information with the event type (**eventType**) of "VMD" will be returned in the message **XML_EventNotificationAlert_MotionDetectionMsg** .

13.6 Configure Vibration Detection Alarm

Vibration detection function can be applied to the scenes in which the real-time vibration status of devices should be detected in case people damage the devices on purpose. You can configure the arming schedule and linkage actions of the vibration detection alarm. For those triggered alarms, you can also search for the related videos or pictures.

Steps

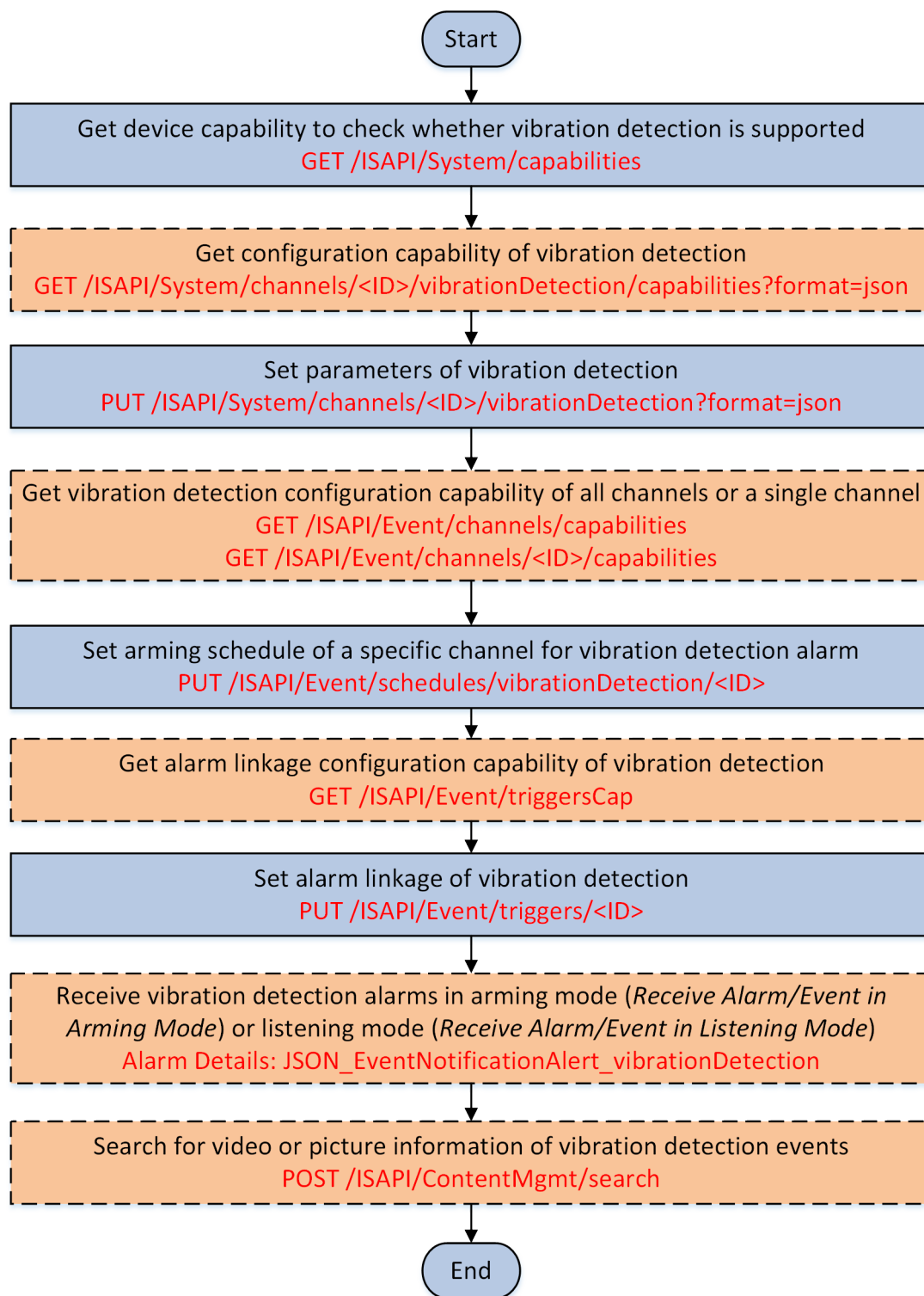


Figure 13-5 API Calling Flow of Configuring Vibration Detection Alarm

1. Call ***/ISAPI/System/capabilities*** by GET method to get the device capability and check whether vibration detection is supported by the device.

The device capability is returned in the message ***XML_DeviceCap***.

If the node ***<isSupportVibrationDetection>*** is returned in the message and its value is "true", it indicates that the vibration detection is supported, and then you can continue to perform the following steps.

Otherwise, the vibration detection is not supported, please end this task.

2. **Optional:** Call ***/ISAPI/System/channels/<ID>/vibrationDetection/capabilities?format=json*** by GET method to get configuration capability of vibration detection.
3. Call ***/ISAPI/System/channels/<ID>/vibrationDetection?format=json*** by PUT method to set parameters of vibration detection.
4. **Optional:** Call ***/ISAPI/Event/channels/capabilities*** or ***/ISAPI/Event/channels/<ID>/capabilities*** by GET method to get event configuration capability of all channels or a single channel for knowing the configuration details or notices.
5. Call ***/ISAPI/Event/schedules/vibrationDetection/<ID>*** by PUT method and set the ***<ID>*** in the format of "vibrationDetection-***<channelID>***", e.g., ***/ISAPI/Event/schedules/vibrationDetection/vibrationDetection-1***, to set arming schedule of a specific channel for vibration detection alarm.
6. **Optional:** Call ***/ISAPI/Event/triggersCap*** by GET method to get alarm linkage configuration capability of vibration detection.

The alarm linkage capability is returned in the message ***XML_EventTriggersCap***.

7. Call ***/ISAPI/Event/triggers/<eventType>-<ID>*** by PUT method and set the ***<ID>*** in the format of "vibrationDetection-***<channelID>***", e.g., ***/ISAPI/Event/triggers/vibrationDetection-1***, to set the alarm linkages.
8. **Optional:** Receive vibration detection alarm in arming mode (refer to ***Receive Alarm/Event in Arming Mode*** for details) or in listening mode (refer to ***Receive Alarm/Event in Listening Mode*** for details) when alarm is triggered.



Note

The vibration detection alarm details are uploaded in the message ***JSON_EventNotificationAlert_vibrationDetection*** and the corresponding alarm type (***eventType***) is "vibrationDetection".

9. **Optional:** Call ***/ISAPI/ContentMgmt/search*** by POST method to search for video or picture information of the vibration detection alarms.



Note

To check whether searching for video or picture information of vibration detection events is supported, you can call ***/ISAPI/ContentMgmt/capabilities*** by GET method to get the storage capability message ***XML_RacmCap*** and check whether the event type "vibrationDetection" exists in the nodes ***recordSearchType*** and ***pictureSearchType***.

13.7 Receive Alarm/Event in Arming Mode

When alarm is triggered or event occurred, and the alarm/event linkage is configured, you can send request message to device for getting the alarm/event stream, and then the device uploads the corresponding response message, which contains alarm/event information.

Before You Start

Make sure you have configured alarm/event and triggered the alarm/event. For configuring alarm/event parameters, refer to the some typical applications of alarm/event configuration.

Steps

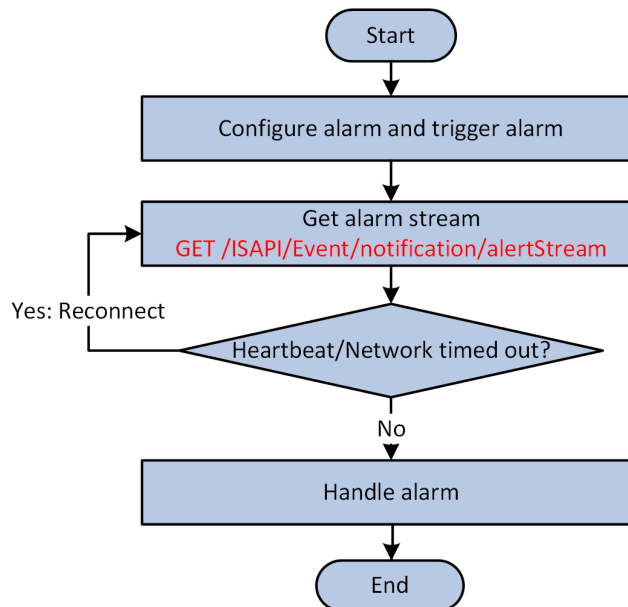


Figure 13-6 API Calling Flow of Receiving Alarm/Event in Arming Mode

1. Call `/ISAPI/Event/notification/alertStream` by GET to get the alarm/event stream.
2. Check if the heartbeat receiving timed out or network disconnected.
 - If the heartbeat keeps alive and the network still connected, perform the following step to continue.
 - If the heartbeat receiving timed out or network disconnected, perform the above step repeatedly until reconnected.
3. Receive and process the alarm/event information.

Example

Sample Code of Receiving Alarm/Event in Arming Mode (without Binary Picture Data)

```
GET /ISAPI/Event/notification/alertStream HTTP/1.1
Host: data_gateway_ip
Connection: Keep-Alive
```

```
HTTP/1.1 401 Unauthorized
Date: Sun, 01 Apr 2018 18:58:53 GMT
```

```
Server:
Content-Length: 178
Content-Type: text/html
Connection: keep-alive
Keep-Alive: timeout=10, max=99
WWW-Authenticate: Digest qop="auth",
realm="IP Camera(C2183)",
nonce="4e5468694e7a42694e7a4d364f4449354d7a6b354d54513d",
stale="FALSE"

GET /ISAPI/Event/notification/alertStream HTTP/1.1
Authorization: Digest username="admin",
realm="IP Camera(C2183)",
nonce="4e5468694e7a42694e7a4d364f4449354d7a6b354d54513d",
uri="/ISAPI/Event/notification/alertStream",
cnonce="3d183a245b8729121ae4ca3d41b90f18",
nc=00000001,
qop="auth",
response="f2e0728991bb031f83df557a8f185178"
Host: 10.6.165.192

HTTP/1.1 200 OK
MIME-Version: 1.0
Connection: close
Content-Type: multipart/mixed; boundary=<frontier>

--<frontier>
Content-Type: application/xml; charset="UTF-8"
Content-Length: text_length

<EventNotificationAlert/>
--<frontier>
```



Note

Some alarm data is in JSON format, so the **Content-Type** may be "application/xml" or "application/json".

13.8 Receive Alarm/Event in Listening Mode

When alarm is triggered or event occurred, and the alarm/event linkage is configured, the device uploads the alarm/event information automatically, you can receive the alarm/event by configuring the listening port of HTTP host server.

Before You Start

Make sure you have configured alarm/event and triggered the alarm/event. For configuring alarm/event parameters, refer to the some typical applications of alarm/event configuration.

Steps

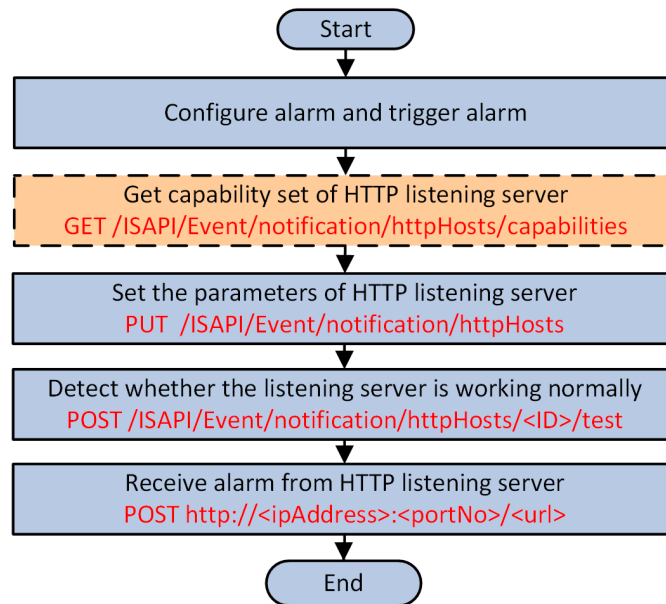


Figure 13-7 API Calling Flow of Receiving Alarm/Event in Listening Mode

1. **Optional:** Call `/ISAPI/Event/notification/httpHosts/capabilities` by GET method to get the capability of HTTP listening server.
2. Call `/ISAPI/Event/notification/httpHosts` by PUT method to set the parameters (including listening address and listening port) of HTTP listening server.



Note

Before setting the listening server, you'd better call the URI by GET method to get the default or configured parameters for reference.

3. Call `/ISAPI/Event/notification/httpHosts/<ID>/test` by POST method to check if the listening server is working normally.
4. Call `http://<ipAddress>:<portNo>/<url>` by POST method to receive the alarm/event information from the listening server.

Example

Sample Code of Receiving Alarm/Event in Listening Mode

- with Binary Picture Data

```

//Request
POST requestUrl HTTP/1.1
Host: data_gateway_ip:port
Accept-Language: en-US
Date: YourDate
Content-Type: multipart/form-data;boundary=<frontier>
Content-Length: body_length
Connection: keep-alive

--<frontier>
  
```

```
Content-Disposition: form-data; name="Event_Type"
Content-Type: text/xml
Content-Length: xml_length

<EventNotificationAlert/>
--<frontier>
Content-Disposition: form-data; name="Picture_Name"
Content-Length: image_length
Content-Type: image/jpeg

[binary picture data]
--<frontier>--

//Response
HTTP/1.1 HTTP statusCode
Date: YourDate
Connection: close
```

- without Binary Picture Data

```
//Request
POST requestUrl HTTP/1.1
Host: data_gateway_ip:port
Accept-Language: en-US
Date: YourDate
Content-Type: text/xml;
Content-Length: text_length
Connection: Close

<EventNotificationAlert/>

//Response
HTTP/1.1 HTTP statusCode
Date: YourDate
Connection: close
```

Note

- The **Host** is the HTTP server domain name or IP address and port No.
 - Some alarm data is in JSON format, so the **Content-Type** may be "text/xml" or "text/json".
-

13.9 Subscribe Alarm/Event in Arming Mode

For arming mode, the platform will connect to the devices automatically and send commands to the devices for uploading alarm/event information when the alarm is triggered or event occurred. To reduce the CPU and bandwidth usage of platform, and improve the device processing performance, the platform can subscribe alarm/event types to receive alarm/event information as required.

Before You Start

Configure alarm/event and trigger the alarm/event. For configuring alarm/event parameters, refer to the some typical applications of alarm/event configuration.

Steps

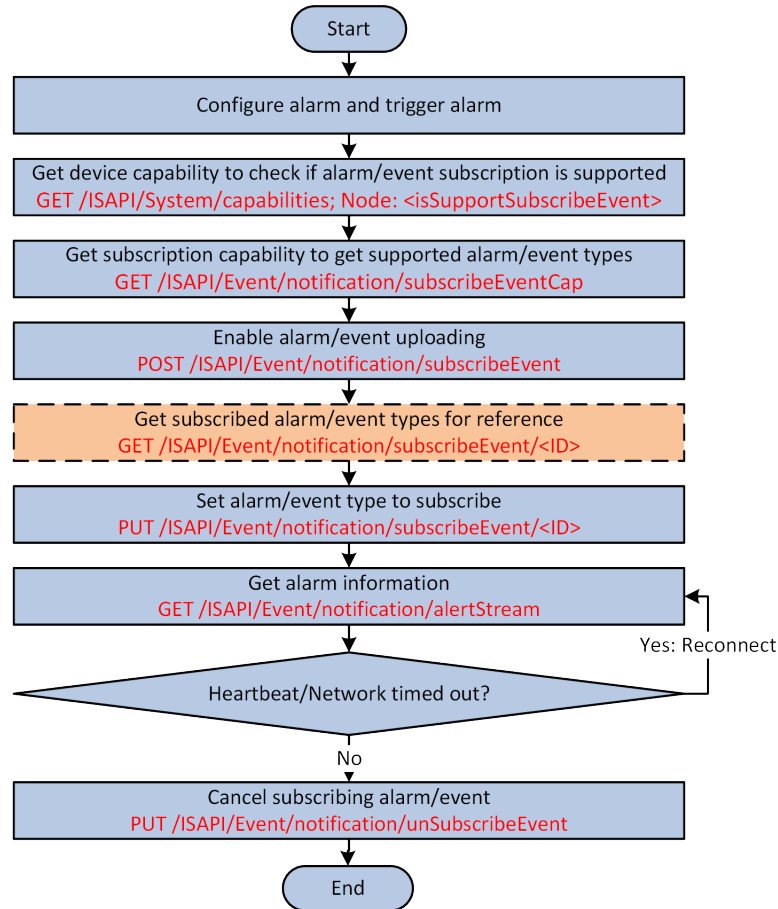


Figure 13-8 Programming Flow of Subscribing Alarm/Event in Arming Mode

1. Call `/ISAPI/System/capabilities` by GET method to get device capability to check whether the alarm/event subscription is supported.

The device capability is returned in the message *XML_DeviceCap*.

If the node `<isSupportSubscribeEvent>` is also returned in the message and its value is "true", it indicates that alarm/event subscription is supported by device, and you can continue to perform the following steps;

Otherwise, alarm/event subscription is not supported, please end the task.

2. Call `/ISAPI/Event/notification/subscribeEventCap` by GET method to get subscription capability to get supported alarm/event types.

The alarm/event subscription capability is returned in the message *XML_SubscribeEventCap*.

3. Call `/ISAPI/Event/notification/subscribeEvent` by POST method to enable alarm/event uploading.

Note

The sub node **<type>** of event type list (**EventList**) in the request message **XML_SubscribeEvent** can be set to different values (refer to *Supported Alarm/Event Types* for details) to subscribe different types of alarm/event information.

4. **Optional:** Call **/ISAPI/Event/notification/subscribeEvent/<ID>** by GET method to get subscribed alarm/event types for reference.
 5. Call **/ISAPI/Event/notification/subscribeEvent/<ID>** by PUT to set alarm/event type to subscribe.
-

Note

The sub node **<type>** of event type list (**EventList**) in the request message **XML_SubscribeEvent** can be set to different values (refer to *Supported Alarm/Event Types* for details) to subscribe different types of alarm/event information.

6. Call **/ISAPI/Event/notification/alertStream** by GET method to receive the alarm/event information.
7. Check if the heartbeat receiving timed out or network disconnected.
 - If the heartbeat keeps alive and the network still connected, perform the following step to continue.
 - If the heartbeat receiving timed out or network disconnected, perform the above step repeatedly until reconnected.
8. Call **/ISAPI/Event/notification/unSubscribeEvent** by PUT method to cancel subscribing alarm/event.

Chapter 14 Parameter Configuration

14.1 I/O Settings

I/O (Alarm Input and Output)

Function	Request URI
Get I/O configuration capability	GET <i>/ISAPI/System/IO/capabilities</i> .
Get alarm input and output information	GET <i>/ISAPI/System/IO</i>
Get I/O status	GET <i>/ISAPI/System/IO/status</i>

Alarm Input

Function	Request URI
Get configuration capability of alarm inputs	GET <i>/ISAPI/System/IO/inputs/capabilities</i> .
Get parameters of all alarm inputs	GET <i>/ISAPI/System/IO/inputs</i>
Get or set parameters of one alarm input	GET or PUT <i>/ISAPI/System/IO/inputs/<ID></i>
Get status of one alarm input	GET <i>/ISAPI/System/IO/inputs/<ID>/status</i>
Alarm Input Configuration of Digital Channel	 Note - To check if the alarm input configuration of digital channel is supported, you can call the GET <i>/ISAPI/ContentMgmt/capabilities</i> to get the device storage management capability (<i>XML_RacmCap</i>). - If this function is supported, the node <i><isSupportIOInputProxy></i> is returned and its value is "true". - Get or set alarm input parameters of all digital channels: GET or PUT <i>/ISAPI/ContentMgmt/IOProxy/inputs</i> - Added alarm input configuration of a specific digital channel: POST <i>/ISAPI/ContentMgmt/IOProxy/inputs</i> - Get or set alarm input parameters of a specific digital channel: GET or PUT <i>/ISAPI/ContentMgmt/IOProxy/inputs/<ID></i>

Function	Request URI
	- Delete alarm input configuration of a specific digital channel: DELETE <i>/ISAPI/ContentMgmt/IOProxy/inputs/<ID></i> - Get alarm input status of a specific digital channel: GET <i>/ISAPI/ContentMgmt/IOProxy/inputs/<ID>/status</i>

Alarm Output

Function	Request URI
Basic Alarm Output Parameters	- Get configuration capability of alarm outputs: GET <i>/ISAPI/System/IO/outputs/capabilities</i> - Get parameters of all alarm outputs: GET <i>/ISAPI/System/IO/outputs</i> - Get or set parameters of one alarm output: GET or PUT <i>/ISAPI/System/IO/outputs/<ID></i> - Get status of one alarm output: GET <i>/ISAPI/System/IO/outputs/<ID>/status</i>
Supplement Light Alarm Output	- Get configuration capability of supplement light alarm outputs: GET <i>/ISAPI/Event/triggers/notifications/whiteLightAlarm/capabilities?format=json</i> - Get or set parameters of supplement light alarm outputs: GET or PUT <i>/ISAPI/Event/triggers/notifications/whiteLightAlarm?format=json</i>
Audible Alarm Output	- Get configuration capability of audible alarm outputs: GET <i>/ISAPI/Event/triggers/notifications/AudioAlarm/capabilities?format=json</i> - Get or set parameters of audible alarm outputs: GET or PUT <i>/ISAPI/Event/triggers/notifications/AudioAlarm?format=json</i>
Manually trigger an alarm output	PUT <i>/ISAPI/System/IO/outputs/<ID>/trigger</i>
Alarm Output Configuration of Digital Channel	 Note To check if the alarm output configuration of digital channel is supported, you can call the GET <i>/ISAPI/ContentMgmt/capabilities</i> to get the device storage management capability (<i>XML_RacmCap</i>). If this function is supported, the node <i><isSupportIOOutputProxy></i> is returned and its value is "true".

Function	Request URI
	- Get or set alarm output parameters of all digital channels: GET or PUT <i>/ISAPI/ContentMgmt/IOProxy/outputs</i> - Added alarm output configuration of a specific digital channel: POST <i>/ISAPI/ContentMgmt/IOProxy/outputs</i> - Get or set alarm output parameters of a specific digital channel: GET or PUT <i>/ISAPI/ContentMgmt/IOProxy/outputs/<ID></i> - Delete alarm output configuration of a specific digital channel: DELETE <i>/ISAPI/ContentMgmt/IOProxy/outputs/<ID></i> - Get alarm output status of a specific digital channel: GET <i>/ISAPI/ContentMgmt/IOProxy/outputs/<ID>/status</i> - Trigger alarm output of a specific digital channel: PUT <i>/ISAPI/ContentMgmt/IOProxy/outputs/<ID>/trigger</i>  Note The alarm details are uploaded by the message <i>XML_EventNotificationAlert_AlarmInputAlarmMsg</i> .

14.2 Video/Image Settings

Basic Image Parameters

Function	Request URI
Get image configuration capability	GET <i>/ISAPI/Image/channels/<ID>/capabilities</i>
Get or set image parameters of all channels	GET or PUT <i>/ISAPI/Image/channels</i>
Get or set image parameters of one channel	GET or PUT <i>/ISAPI/Image/channels/<ID></i>
Get basic image parameters of all channels	GET <i>/ISAPI/Image/channels/<ID>/imageModes</i>
Get basic image parameters of one channel	GET <i>/ISAPI/Image/channels/<ID>/imageMode/<ID></i>
Get configuration parameters of all image modes of all channels	GET <i>/ISAPI/Image/channels/imageModes</i>

Function	Request URI
Get configuration parameters of all image modes of a specific channel	GET <i>/ISAPI/Image/channels/imageModes/<ID></i>
Reset image parameters	PUT <i>/ISAPI/Image/channels/<ID>/reset</i>
Restore to default image settings	PUT <i>/ISAPI/Image/channels/<ID>/restore</i>

Camera Video/Image Parameters

Function	Request URI
Get or set day/night mode parameters	GET or PUT <i>/ISAPI/Image/channels/<ID>/ISPMode</i>
Get or set image auto flip parameters	GET or PUT <i>/ISAPI/Image/channels/<ID>/ImageFlip</i>
Get or set wide dynamic range (WDR) parameters	GET or PUT <i>/ISAPI/Image/channels/<ID>/WDR</i>
Get or set backlight compensation (BLC) parameters	GET or PUT <i>/ISAPI/Image/channels/<ID>/BLC</i>
Get or set day/night auto switch parameters	GET or PUT <i>/ISAPI/Image/channels/<ID>/IrcutFilter</i>
Get or set 3-dimension digital noise reduction (3D DNR) parameters	GET or PUT <i>/ISAPI/Image/channels/<ID>/noiseReduce</i>
Get or set white balance (WB) parameters	GET or PUT <i>/ISAPI/Image/channels/<ID>/whiteBalance</i>
Get or set sharpness parameters	GET or PUT <i>/ISAPI/Image/channels/<ID>/sharpness</i>
Get or set defog parameters	GET or PUT <i>/ISAPI/Image/channels/<ID>/dehaze</i>
Get or set image standard	GET or PUT <i>/ISAPI/Image/channels/<ID>/powerLineFrequency</i>

Function	Request URI
Get or set image rotate mode parameters	GET or PUT <i>/ISAPI/Image/channels/<ID>/corridor</i>
Get or set video input mode parameters	GET or PUT <i>/ISAPI/Image/channels/<ID>/capturemode</i>

Table 14-1 Exposure

Function	Request URI
Get or set exposure parameters	GET or PUT <i>/ISAPI/Image/channels/<ID>/exposure</i>
Get or set shutter parameters in exposure mode	GET or PUT <i>/ISAPI/Image/channels/<ID>/shutter</i>
Get or set gain parameters in exposure mode	GET or PUT <i>/ISAPI/Image/channels/<ID>/gain</i>

Table 14-2 Image Adjustment

Function	Request URI
Get image adjustment capability	GET <i>/ISAPI/Image/channels/<ID>/color/capabilities</i>
Get or set image adjustment parameters	GET or PUT <i>/ISAPI/Image/channels/<ID>/color</i>

Other Video/Image Parameters

Table 14-3 Focus

Function	Request URI
Get focus configuration capability	GET <i>/ISAPI/Image/channels/<ID>/focusConfiguration/capabilities</i>
Get or set focus parameters	GET or PUT <i>/ISAPI/Image/channels/<ID>/focusConfiguration</i>
Perform regional focus	PUT <i>/ISAPI/Image/channels/<ID>/regionalFocus</i>  Note To check if the regional focus is supported by device, you can call the GET to get the image capability (<i>XML_ImageCap</i>). If supports, the node <isSupportRegionalFocus> will be returned in the capability.

Table 14-4 Target Coloration

Function	Request URI
Get image configuration capability of a specific channel	GET <i>/ISAPI/Image/channels/<ID>/capabilities</i>
Get or set the palettes parameters	GET or PUT <i>/ISAPI/Image/channels/<ID>/Palettes</i>

Table 14-5 On Screen Display (OSD)

Function	Request URI
Get or set OSD parameters	GET or PUT <i>/ISAPI/System/Video/inputs/channels/<ID>/overlays</i>  Note To check if the OSD language configuration is supported by device, you can call the GET / to get the video capability (<i>XML_VideoCap</i>). If supports, the node <i><OSDLanguage></i> will be returned.
Get text overlay configuration capability of a specific video digital input channel	GET <i>/ISAPI/ContentMgmt/InputProxy/channels/<ID>/video/overlays/capabilities</i>
Get or set text overlay parameters of a specific video digital input channel	GET or PUT <i>/ISAPI/ContentMgmt/InputProxy/channels/<ID>/video/overlays</i>
Get or set text language of OSD	GET <i>/ISAPI/System/Video/inputs/OSDLanguage</i>

Table 14-6 Stream Configuration for Displaying VCA Rules of Smart Events

Function	Request URI
Get capability of configuring stream for displaying VCA rules of smart events	GET <i>/ISAPI/Streaming/channels/<ID>/smartOverlap/capabilities?format=json</i>  Note To check if the stream configuration for displaying VCA rules of smart events is supported by the device, you can call the GET <i>/ISAPI/System/capabilities</i> to get the device capability (<i>XML_DeviceCap</i>). If this function is supported, the node

Function	Request URI
	<supportSmartOverlapChannles> will be returned and its value is "true".
Get or set stream parameters for displaying VCA rules of smart events	GET or PUT <i>/ISAPI/Streaming/channels/<ID>/smartOverlap?format=json</i>

Table 14-7 Target Enhancement

Function	Request URI
Get target enhancement capability	GET <i>/ISAPI/Image/channels/<ID>/targetEnhancement/capabilities</i>
Get or set target enhancement parameters	GET or PUT <i>/ISAPI/Image/channels/<ID>/targetEnhancement</i>

Table 14-8 Privacy Mask

Function	Request URI
Get configuration capability of privacy mask	GET <i>/ISAPI/System/Video/inputs/channels/<ID>/privacyMask/privacyMaskCap</i>
Get or set privacy mask parameters	GET or PUT <i>/ISAPI/System/Video/inputs/channels/<ID>/privacyMask</i>
Get or set multiple privacy mask regions	GET or PUT <i>/ISAPI/System/Video/inputs/channels/<ID>/privacyMask/regions</i>
Get or set one privacy mask region	GET or PUT <i>/ISAPI/System/Video/inputs/channels/<ID>/privacyMask/regions/<ID></i>

Table 14-9 Distortion Correction

Function	Request URI
Get configuration capability of distortion correction	GET <i>/ISAPI/Image/channels/<ID>/lensDistortionCorrection/capabilities</i>
Get or set distortion correction parameters	GET or PUT <i>/ISAPI/Image/channels/<ID>/lensDistortionCorrection</i>

Table 14-10 Target Cropping

Function	Request URI
Get configuration capability of target cropping	GET <i>/ISAPI/Streaming/channels/<ID>/regionClip/capabilities</i>
Get or set target cropping parameters	GET or PUT <i>/ISAPI/Streaming/channels/<ID>/regionClip</i>

Table 14-11 Temperature Range

Function	Request URI
Get temperature range configuration capability	GET <i>/ISAPI/Image/channels/<ID>/tempRange/capabilities</i>
Get or set temperature range	GET or PUT <i>/ISAPI/Image/channels/<ID>/tempRange</i>

Table 14-12 Mounting Scenario

Function	Request URI
Get configuration capability of mounting scenario mode	GET <i>/ISAPI/Image/channels/<ID>/mountingScenario/capabilities</i>
Get or set mounting scenario mode by channel	GET or PUT <i>/ISAPI/Image/channels/<ID>/mountingScenario</i>

14.3 Audio Settings

Basic Audio Configuration

Function	Request URI
Get audio capability	GET <i>/ISAPI/System/Audio/capabilities</i>
Get dynamic audio capability by channel	GET <i>/ISAPI/System/Audio/channels/<ID>/dynamicCap</i>
Get parameters of all audio channels	GET <i>/ISAPI/System/Audio/channels</i>
Get parameters of a specific audio channel	GET <i>/ISAPI/System/Audio/channels/<ID></i>

Audio Input Configuration

Function	Request URI
Get audio input capability of a specific channel	GET <i>/ISAPI/System/Audio/AudioIn/channels/<ID>/capabilities</i>
Get or set audio input parameters of a specific channel	GET or PUT <i>/ISAPI/System/Audio/AudioIn/channels/<ID></i>
Get audio input configuration capability of all channels	GET <i>/ISAPI/System/Audio/AudioIn/capabilities</i>
Get or set audio input parameters of all channels	GET or PUT <i>/ISAPI/System/Audio/AudioIn</i>

Audio Output Configuration

Function	Request URI
Get audio output capability of a specific channel	GET <i>/ISAPI/System/Audio/AudioOut/channels/<ID>/capabilities</i>
Get or set audio output parameters of a specific channel	GET or PUT <i>/ISAPI/System/Audio/AudioOut/channels/<ID></i>

14.4 Channel Settings

Basic Channel Parameters

Function	Request URI
Get all channels' attributes	GET <i>/ISAPI/AUXInfo/attributes/Channels</i>
Get one channel's attributes	GET <i>/ISAPI/AUXInfo/attributes/Channels/<ID></i>

Video Input Channel

Function	Request URI
Get capability of configuring video input channels in a batch	GET <i>/ISAPI/System/Video/inputs/channels/capabilities</i>
Get configuration capability by video input channel	GET <i>/ISAPI/System/Video/inputs/channels/<ID>/capabilities</i>

Function	Request URI
Get parameters of all video input channels	GET <i>/ISAPI/System/Video/inputs/channels?format=json</i>
Get parameters of one video input channel	<i>/ISAPI/System/Video/inputs/channels/<ID>?format=json</i>

Video Output Channel

Function	Request URI
Get parameters of all video outputs	GET <i>/ISAPI/System/Video/outputs</i>
Get configuration capability by video output channel	GET <i>/ISAPI/System/Video/outputs/channels/<ID>/capabilities</i>
Get parameters of all video output channels	GET <i>/ISAPI/System/Video/outputs/channels</i>
Get parameters of one video output channel	GET <i>/ISAPI/System/Video/outputs/channels/<ID></i>

Digital Channel



Note

To check if the digital channel management or configuration is supported, you can call the GET */ISAPI/ContentMgmt/capabilities* to get the device storage management capability (*XML_RacmCap*).

Function	Request URI
Get management capability of all digital channels	GET <i>/ISAPI/ContentMgmt/InputProxy/channels/capabilities</i>
Get management capability of one digital channel	GET <i>/ISAPI/ContentMgmt/InputProxy/channels/<ID>/capabilities</i>
Get supported number of digital channels	POST <i>/ISAPI/ContentMgmt/InputProxy/sourceCapability</i>
Get parameters of all digital channels	GET <i>/ISAPI/ContentMgmt/InputProxy/channels</i>
Get status of all digital channels	GET <i>/ISAPI/ContentMgmt/InputProxy/channels/status</i>

Function	Request URI
Get, add, or delete parameters of one digital channel	GET, POST, or PUT <i>/ISAPI/ContentMgmt/InputProxy/channels</i>
Get status of one digital channel	GET <i>/ISAPI/ContentMgmt/InputProxy/channels/<ID>/status</i>
Get or set control parameters of a digital channel	GET or PUT <i>/ISAPI/ContentMgmt/InputProxy/channels/<ID>/chanCtrl</i>
Reboot a digital channel	GET <i>/ISAPI/ContentMgmt/InputProxy/channels/<ID>/reboot</i>
Check if port mapping is required for connecting to a digital channel	POST <i>/ISAPI/ContentMgmt/InputProxy/channels/<ID>/portMapParam</i>
Get or set the device parameters of a digital channel	GET or PUT <i>/ISAPI/ContentMgmt/InputProxy/channels/<ID>/deviceInfo</i>
Get device parameters of all digital channels	POST <i>/ISAPI/ContentMgmt/InputProxy/deviceInfo</i>
Get firmware code list of the digital channel	GET <i>/ISAPI/System/firmwareCodeV2</i>

14.5 Peripherals Settings

Function	Request URI
Get peripherals configuration capability	GET <i>/ISAPI/System/externalDevice/capabilities</i>
Get configuration capability of built-in supplement light	GET <i>/ISAPI/Image/channels/<ID>/supplementLight/capabilities</i>
Get or set parameters of built-in supplement light	GET or PUT <i>/ISAPI/Image/channels/<ID>/supplementLight</i>  Note To check if configuring built-in supplement light is supported, you can call the GET <i>/ISAPI/Image/channels/<ID>/capabilities</i> to get the image channel capability (<i>XML_Cap_ImageChannel</i>). If supports, the node <i><SupplementLight></i> will be returned in the message.

14.6 Picture Storage Server Settings

Function	Request URI
Get picture storage server capability	GET <i>/ISAPI/System/PictureServer/capabilities?format=json</i>
Get picture storage server parameters	GET <i>/ISAPI/System/PictureServer?format=json</i>
Set picture storage server parameters	PUT <i>/ISAPI/System/PictureServer?format=json</i>

Chapter 15 System Configuration

Function	Request URI
Get Configuration Capability of Device Language	GET <i>/ISAPI/System/DeviceLanguage/capabilities</i>
Get or Set Device Language	GET or PUT <i>/ISAPI/System/DeviceLanguage</i>
Get USB Mode Configuration Capability	GET <i>/ISAPI/System/usb/capabilities?format=json</i>
Get or Set USB Mode	GET or PUT <i>/ISAPI/System/usb?format=json</i>

15.1 Network Settings

This chapter lists the request URIs for communication and network configurations of devices, including network interface settings, wireless network settings, network access settings, email settings, and so on.

Function	Request URI
Get network capability	GET <i>/ISAPI/System/Network/capabilities</i>
Import network certificate	GET <i>/ISAPI/Security/deviceCertificate</i>
Remotely get connection socket IP	GET <i>/ISAPI/System/Network/socketIP</i>  Note To check if getting socket IP is supported by device, you can call the GET <i>/ISAPI/System/Network/capabilities</i> to get the device capability (<i>XML_DeviceCap</i>). If supports, the node <i><isSupportGetLinkSocketIP></i> will be returned.
Network Self-Adaptive	- Get self-adaptive configuration capability: GET <i>/ISAPI/System/Network/adaption/capabilities?format=json</i> - Get self-adaptive parameters: GET <i>/ISAPI/System/Network/adaption?format=json&streamType=</i> - Set self-adaptive parameters: PUT <i>/ISAPI/System/Network/adaption?format=json&streamType=</i>

Function	Request URI
	 Note To check if network self-adaptive configuration is supported, you can get the network capability <i>XML_NetworkCap</i> by the <i>/ISAPI/System/Network/capabilities</i> . If supports, the node <i><Adaption></i> will be returned in the capability message.

15.1.1 Email

Function	Request URI
Get email configuration capability	GET <i>/ISAPI/System/Network/mailling/capabilities</i>
Get or set parameters of multiple emails	GET or PUT <i>/ISAPI/System/Network/mailling</i>
Get or set parameters of one email	GET or PUT <i>/ISAPI/System/Network/mailling/<ID></i>
Execute email test	POST <i>/ISAPI/System/Network/mailling/test</i>

15.1.2 Network Access

Network Access Test by Ping

Function	Request URI
Set ping test parameters	POST <i>/ISAPI/System/Network/pingtest</i>

Protocol Access

Function	Request URI
Get configuration capability of protocol access	GET <i>/ISAPI/Security/adminAccesses/capabilities</i>
Get or set access parameters of multiple protocols	GET or PUT <i>/ISAPI/Security/adminAccesses</i>
Get or set access parameters of one protocol	GET or PUT <i>/ISAPI/Security/adminAccesses/<ID></i>

Function	Request URI
Get 802.1 protocol access capability	GET <i>/ISAPI/System/Network/interfaces/<ID>/ieee802.1x/capabilities</i>
Get or set access parameters of 802.1 protocol	GET or PUT <i>/ISAPI/System/Network/interfaces/<ID>/ieee802.1x</i>

Hik-Connect Access

Function	Request URI
Get the Hik-Connect access configuration capability	GET <i>/ISAPI/System/Network/EZVIZ/capabilities</i>
Get or set access parameters of Hik-Connect	GET or PUT <i>/ISAPI/System/Network/EZVIZ</i>  Note To check if the device supports accessing to Hik-Connect, you can call the request URI: GET <i>/ISAPI/System/Network/capabilities</i> to get the network capability (<i>XML_NetworkCap</i>). If supports, the node <i><isSupportEhome></i> will be returned in the capability and its value is "true".
Edit verification code for Hik-Connect	PUT <i>/ISAPI/System/Network/EZVIZ/secretKey?format=json</i>  Note To check if the device supports editing verification code for Hik-Connect, you can call the request URI: GET <i>/ISAPI/System/Network/capabilities</i> to get the network capability sets <i>XML_NetworkCap</i> . If supports, the node <i><secretKey></i> will be returned.
Get Hik-Connect QR code	- GET <i>/ISAPI/System/Network/EZVIZ/QRCode?format=json</i> - GET <i>/ISAPI/System/Network/EZVIZ/QRCode</i>
Unbind devices from Hik-Connect account	PUT <i>/ISAPI/System/Network/EZVIZ/unbind?format=json</i>

EHome Server Access

Function	Request URI
Get configuration capability of EHome server access	GET <i>/ISAPI/System/Network/Ehome/capabilities</i>
Get or set access parameters of EHome server	GET or PUT <i>/ISAPI/System/Network/Ehome</i>

Function	Request URI
Get or set access parameters of EHome server according to the center ID	GET or PUT <i>/ISAPI/System/Network/Ehome?centerID=</i>
Get configuration capability of report uploading method	GET <i>/ISAPI/SecurityCP/ReportCenterCfg/capabilities?format=json</i>
Get or set parameters of report uploading method	GET or PUT <i>/ISAPI/SecurityCP/ReportCenterCfg/<ID>?format=json</i>

Note

To check if the device supports accessing to EHome server, you can call the request URI: GET */ISAPI/System/Network/capabilities* to get the network capability (*XML_NetworkCap*). If supports, the node *<isSupportEZVIZ>* will be returned in the capability and its value is "true".

SSH Server Access

Function	Request URI
Get the SSH parameters configuration capability	GET <i>/ISAPI/System/Network/ssh/capabilities</i>
Get or set access parameters of SSH server	GET or PUT <i>/ISAPI/System/Network/ssh</i>

Note

To check if the device supports accessing to SSH server, you can call the request URI: GET */ISAPI/System/Network/capabilities* to get the network capability (*XML_NetworkCap*). If supports, the node *<isSupportSSH>* will be returned in the capability and its value is "true".

SNMP Access

Function	Request URI
Get batch configuration parameters of SNMP	GET <i>/ISAPI/System/Network/SNMP</i>
Get advanced configuration parameters of SNMP	GET <i>/ISAPI/System/Network/SNMP/advanced</i>
Get all trap address parameters of SNMP in version 3	GET <i>/ISAPI/System/Network/SNMP/advanced/trapReceivers</i>
Set all trap address parameters of SNMP in version 3	PUT <i>/ISAPI/System/Network/SNMP/advanced/trapReceivers</i>

Function	Request URI
Add a new trap address of SNMP in version 3	POST <i>/ISAPI/System/Network/SNMP/advanced/trapReceivers</i>
Delete all trap addresses of SNMP in version 3	DELETE <i>/ISAPI/System/Network/SNMP/advanced/trapReceivers</i>
Get a single trap address's parameters of SNMP in version 3	GET <i>/ISAPI/System/Network/SNMP/advanced/trapReceivers/<ID></i>
Set parameters for a single trap address of SNMP in version 3	PUT <i>/ISAPI/System/Network/SNMP/advanced/trapReceivers/<ID></i>
Delete a single address of SNMP in version 3	DELETE <i>/ISAPI/System/Network/SNMP/advanced/trapReceivers/<ID></i>

15.1.3 Network Interface

Network Interface

Function	Request URI
Get configuration capability of multiple network interfaces	GET <i>/ISAPI/System/Network/interfaces/capabilities</i>
Get configuration capability of one network interface	GET <i>/ISAPI/System/Network/interfaces/<ID>/capabilities</i>
Get information of multiple network interfaces	GET <i>/ISAPI/System/Network/interfaces</i>
Get or set parameters of one network interface	GET or PUT <i>/ISAPI/System/Network/interfaces/<ID></i>
Get IP address of one network interface	GET or PUT <i>/ISAPI/System/Network/interfaces/<ID>/ipAddress</i>
Get connection configuration capability of a specific network interface	GET <i>/ISAPI/System/Network/interfaces/<ID>/link/capabilities</i>
Get or set connection parameters of a specific network interface	GET or PUT <i>/ISAPI/System/Network/interfaces/<ID>/link</i>

PoE

Function	Request URI
Get configuration capability of PoE port	GET <i>/ISAPI/System/Network/POE/capabilities?format=json</i>
Get or set PoE port parameters	GET or PUT <i>/ISAPI/System/Network/POE?format=json</i>

UPnP

Function	Request URI
Get the UPnP configuration capability	GET <i>/ISAPI/System/Network/UPnP/capabilities</i>
Get or set UPnP parameters	GET or PUT <i>/ISAPI/System/Network/UPnP</i>
Get or set parameters of all UPnP ports	GET or PUT <i>/ISAPI/System/Network/UPnP/ports</i>
Get or set parameters of one UPnP port	GET or PUT <i>/ISAPI/System/Network/UPnP/ports/<ID></i>
Get mapping statuses of all UPnP ports	GET <i>/ISAPI/System/Network/UPnP/ports/status</i>
Get mapping status of one UPnP port	GET <i>/ISAPI/System/Network/UPnP/ports/<ID>/status</i>

15.1.4 Wireless Network

Wi-Fi

Function	Request URI	
Get Wi-Fi configuration capability	GET <i>/ISAPI/System/Network/interfaces/<ID>/wireless/capabilities</i>	
Get or set Wi-Fi parameters of one network interface	GET or PUT <i>/ISAPI/System/Network/interfaces/<ID>/wireless</i>	
WPS (Wi-Fi Protected Setup)	Get WPS configuration capability	GET <i>/ISAPI/System/Network/WPS/capabilities</i>
	Get or set WPS parameters of a specific network interface	GET or PUT <i>/ISAPI/System/Network/interfaces/<ID>/WPS</i>

Function	Request URI	
	Set parameters of PBC (Push Button Configuration) automatic connection of WPS	PUT <i>/ISAPI/System/Network/interfaces/<ID>/WPS/AutoConnect</i>
	Get or set parameters of PIN (Personal Identification Number) code of wireless access point	GET or PUT <i>/ISAPI/System/Network/interfaces/<ID>/WPS/ApPinCode</i>
	Get or set PIN (Personal Identification Number) code of WPS of device	GET or PUT <i>/ISAPI/System/Network/interfaces/<ID>/WPS/devicePinCode</i>
	Update PIN (Personal Identification Number) code of WPS of device	PUT <i>/ISAPI/System/Network/interfaces/<ID>/WPS/devicePinCodeUpdate</i>

Wireless Hotspot

Function	Request URI
Get configuration capability of wireless hotspot	GET <i>/ISAPI/System/Network/interfaces/<ID>/wirelessServer/capabilities</i>
Get or set wireless hotspot parameters	GET or PUT <i>/ISAPI/System/Network/interfaces/<ID>/wirelessServer</i>
Get list of devices connected to hotspot	GET <i>/ISAPI/System/Network/interfaces/<ID>/wirelessServer/accessDeviceList</i>
Get information of a device connected to hotspot	GET <i>/ISAPI/System/Network/interfaces/<ID>/wireless/accessDeviceList/<ID></i>
Get capability of devices connected to hotspot	GET <i>/ISAPI/System/Network/interfaces/<ID>/wireless/accessDeviceList/capabilities</i>
Get hotspot list	GET <i>/ISAPI/System/Network/interfaces/<ID>/wireless/accessPointList</i>
Get one hotspot information	GET <i>/ISAPI/System/Network/interfaces/<ID>/wireless/accessPointList/<ID></i>

Wireless Dial

Function	Request URI
Get configuration capability of wireless dial	GET <i>/ISAPI/System/Network/WirelessDial/Interfaces/<ID>/dial/capabilities</i>
Get or set wireless dial parameters	GET or PUT <i>/ISAPI/System/Network/WirelessDial/Interfaces/<ID>/dial</i>
Get wireless dial schedule	GET <i>/ISAPI/System/Network/WirelessDial/Interfaces/<ID>/schedule</i>

Network Keepalive

Function	Request URI
Get network keepalive capability	GET <i>/ISAPI/System/Network/WirelessDial/Interfaces/<ID>/keepAlive/capabilities?format=json</i>
Get or set network keepalive parameters	GET or PUT <i>/ISAPI/System/Network/WirelessDial/Interfaces/<ID>/keepAlive?format=json</i> .
Test network keepalive configuration	POST <i>/ISAPI/System/Network/WirelessDial/Interfaces/<ID>/keepAlive/test?format=json</i>

Network Traffic Monitoring

Function	Request URI
Get network traffic monitoring capability	GET <i>/ISAPI/System/Network/WirelessDial/Interfaces/<ID>/trafficMonitor/capabilities?format=json</i> ;
Get or set network traffic monitoring parameters	GET or PUT <i>/ISAPI/System/Network/WirelessDial/Interfaces/<ID>/trafficMonitor?format=json</i> .
Get network traffic monitoring status	GET <i>/ISAPI/System/Network/WirelessDial/Interfaces/<ID>/trafficMonitor/status?format=json</i> .
Network traffic pre-alarm to be uploaded	Alarm Details: <i>JSON_EventNotificationAlert_NetworkTrafficPrealarmMsg</i>

SMS Configuration

Function	Request URI
Get dial information configuration capability	GET <i>/ISAPI/System/Network/WirelessDial/Interfaces/<ID>/messageConfig/messageConfigCap</i>):
Get or set parameters of dial information configuration	GET or PUT <i>/ISAPI/System/Network/WirelessDial/Interfaces/<ID>/messageConfig</i>
Get all dial whitelist configuration	GET <i>/ISAPI/System/Network/WirelessDial/Interfaces/<ID>/messageConfig/WhiteList/<ID></i>)
Get or set specific dial whitelist configuration	GET or PUT <i>/ISAPI/System/Network/WirelessDial/Interfaces/<ID>/messageConfig/WhiteList</i>

15.1.5 Network Service

Software Service

Function	Request URI
Get configuration capability of software service	GET <i>/ISAPI/System/Software/channels/<ID>/capabilities</i>
Get or set parameters of software service	GET or PUT <i>/ISAPI/System/Software/channels/<ID></i>

Session Initialization Protocol (SIP) Server

Function	Request URI
Get configuration capability of a specific SIP server	GET <i>/ISAPI/System/Network/SIP/<ID>/capabilities</i>
Get configuration capability of all SIP servers	GET <i>/ISAPI/System/Network/SIP/capabilities</i>
Get or set parameters of all SIP servers	GET or PUT <i>/ISAPI/System/Network/SIP</i>
Get or set parameters of one SIP server	GET or PUT <i>/ISAPI/System/Network/SIP/<ID></i>
Get configuration capability of SIP server	GET <i>/ISAPI/System/Network/SIP/<ID>/SIPInfo/capabilities</i>
Get or set information of one SIP server	GET or PUT <i>/ISAPI/System/Network/SIP/<ID>/SIPInfo</i>

Function	Request URI
Get or set information of all SIP servers	GET or PUT <i>/ISAPI/System/Network/SIP/<ID>/SIPInfo/multiInfo</i>
Get configuration capability of SIP intercom	GET <i>/ISAPI/System/Network/SIP/voiceTalkCtrl/capabilities?format=json</i>
Get or set SIP intercom parameters	GET or PUT <i>/ISAPI/System/Network/SIP/voiceTalkCtrl/<ID>?format=json</i>

File Transfer Protocol (FTP) Server

Function	Request URI
Get configuration capability of FTP server	GET <i>/ISAPI/System/Network/ftp/capabilities</i>
Get or set parameters of one FTP server	GET or PUT <i>/ISAPI/System/Network/ftp/<ID></i>
Get or set parameters of all FTP servers	GET or PUT <i>/ISAPI/System/Network/ftp</i>
Perform FTP server test	POST <i>/ISAPI/System/Network/ftp/test</i>

Dynamic Domain Name Server (DDNS)

Function	Request URI
Get DDNS configuration capability	GET <i>/ISAPI/System/Network/DDNS/capabilities</i>
Get or set parameters of all DDNSs	GET or PUT <i>/ISAPI/System/Network/DDNS</i>
Get or set parameters of one DDNS	GET or PUT <i>/ISAPI/System/Network/DDNS/<ID></i>
Get configuration capability of dynamic domain name	GET <i>/ISAPI/System/Network/interfaces/<ID>/dynamicHostName/capabilities?format=json</i>
Get or set dynamic domain name	GET or PUT <i>/ISAPI/System/Network/interfaces/<ID>/dynamicHostName?format=json</i>

Note

To check if setting dynamic domain name is supported, you can call the GET */ISAPI/System/Network/capabilities* to get the network capability (*XML_NetworkCap*). If supports, the node **<isSupportDynamicHostName>** is returned and its value is "true".

Log Server

Function	Request URI
Get log server configuration capability	GET <i>/ISAPI/System/logServer/capabilities</i>
Get or set log server parameters	GET or PUT <i>/ISAPI/System/logServer</i>
Perform log server test	POST <i>/ISAPI/System/logServer/test</i>

Static Routing

Function	Request URI
Get, set, or delete static routing parameters	GET, PUT, or DELETE <i>/ISAPI/System/Network/StaticRoute</i>

15.2 Time Settings

This chapter lists the request URIs for device time configuration, including setting time, setting time zone, and so on.

Function	Request URI
Get time configuration capability	GET <i>/ISAPI/System/time/capabilities</i>  Note To check if time configuration is supported, you can call the GET <i>/ISAPI/System/capabilities</i> to get the device capability (<i>XML_DeviceCap</i>). If supported, the node <code><isSupportTimeCap></code> will be returned.
Get or set time type	GET or PUT <i>/ISAPI/System/time/timeType?format=json</i>
Get or set time parameters	GET or PUT <i>/ISAPI/System/time</i>
Get or set time zone	GET or PUT <i>/ISAPI/System/time/timeZone</i>
Get or set local time parameters	GET or PUT <i>/ISAPI/System/time/localTime</i>
NTP Server	• Get NTP server capability: GET <i>/ISAPI/System/time/ntpServers/capabilities</i> • Get, set, or delete multiple NTP servers: GET, PUT, or DELETE <i>/ISAPI/System/time/ntpServers</i>

Function	Request URI
	- Get set, or delete one NTP server: GET, PUT, or DELETE <i>/ISAPI/System/time/ntpServers/<ID></i> - Add one NTP server: POST <i>/ISAPI/System/time/ntpServers</i> - Execute NTP server test: POST <i>/ISAPI/System/time/ntpServers/test</i>

15.3 System Diagnose

This chapter lists the request URIs for getting system diagnose information and the exporting status.

Function	Request URI
Get diagnose information	GET <i>/ISAPI/System/diagnosedData</i>
Get process of exporting diagnose information	GET <i>/ISAPI/System/diagnosedData/exportStatus</i>  Note To check if getting diagnose information is supported, you can call <i>/ISAPI/System/capabilities</i> by GET method to get the device capability (<i>XML_DeviceCap</i> >. If supports, the node <i><isSupportDiagnosedData></i> will be returned.

15.4 Data Replenishment

In the condition of disconnection between platform and NVR, the NVR saves the data of people counting, heat map, temperature, and vehicle. When the connection is resumed, the platform can get the specified data from NVR during disconnection period.

People Counting Data Replenishment

Function	Request URI
Get data replenishment capability of people counting	GET <i>/ISAPI/System/Video/inputs/channels/counting/collection/capabilities?format=json</i>  Note Before getting people counting data replenishment capability, you should call <i>/ISAPI/System/Video/capabilities</i> by GET method to check whether this function is supported by the

Function	Request URI
	device via the node <isSupportCountingCollection> in <i>XML_VideoCap</i> .
Perform data replenishment of people counting	POST <i>/ISAPI/System/Video/inputs/channels/counting/collection?format=json</i>

Heat Map Data Replenishment

Function	Request URI
Get data replenishment capability of heat map	GET <i>/ISAPI/System/Video/inputs/channels/heatMap/collection/capabilities?format=json</i>  Note Before getting people counting data replenishment capability, you should call <i>/ISAPI/System/Video/capabilities</i> by GET method to check whether this function is supported by the device via the node <isSupportHeatmapCollection> in <i>XML_VideoCap</i> .
Perform data replenishment of heat map	POST <i>/ISAPI/System/Video/inputs/channels/heatMap/collection?format=json</i>

Temperature Data Replenishment

Function	Request URI
Get data replenishment capability of temperature	GET <i>/ISAPI/Thermal/temperature/collection/capabilities?format=json</i>  Note Before getting temperature data replenishment capability, you should call <i>/ISAPI/Thermal/capabilities</i> by GET method to check whether this function is supported by the device via the node <isSupportTemperatureCollection> in <i>XML_ThermalCap</i> .
Perform temperature data replenishment	POST <i>/ISAPI/Thermal/temperature/collection?format=json</i>

Vehicle Detection Data Replenishment

Function	Request URI
Get intelligent management capability	GET <i>/ISAPI/SDT/Management/IntelligentSearch/capabilities?format=json</i>
Perform vehicle detection data replenishment	POST <i>/ISAPI/SDT/Management/IntelligentSearch?format=json</i>  Note Before searching for vehicle detection data, you should check whether the device supports this function. If it supports, the node <isSupportIntelligentSearch> will be returned in the intelligent management capability JSON_Cap_IntelliManagement (call <i>/ISAPI/SDT/Management/capabilities?format=json</i> by GET method to get) and its value is "true".

Appendix A. Request URIs

A.1 /ISAPI/AUXInfo

A.1.1 /ISAPI/AUXInfo/attributes/Channels

Get the attributes of channels.

Request URI Definition

Table A-1 GET /ISAPI/AUXInfo/attributes/Channels

Method	GET
Description	Get the attributes of channels.
Query	None.
Request	None.
Response	<i>XML_ChannelInfoList</i>

A.1.2 /ISAPI/AUXInfo/attributes/Channels/<ID>

Get the attributes of a specific channel according to channel ID.

Request URI Definition

Table A-2 GET /ISAPI/AUXInfo/attributes/Channels/<ID>

Method	GET
Description	Get the attributes of a specific channel according to channel ID.
Query	None.
Request	None.
Response	<i>XML_ChannelInfo</i>

Remarks

The <ID> in the request URI is the channel ID.

A.2 /ISAPI/ContentMgmt

A.2.1 /ISAPI/ContentMgmt/capabilities

Get storage capability.

Request URI Definition

Table A-3 GET /ISAPI/ContentMgmt/capabilities

Method	GET
Description	Get storage capability.
Query	None.
Request	None.
Response	Succeeded: <i>XML_RacmCap</i> Failed: <i>XML_ResponseStatus</i>

A.2.2 /ISAPI/ContentMgmt/download

Download the file via plug-in.

Request URI Definition

Table A-4 POST /ISAPI/ContentMgmt/download

Method	POST
Description	Download the file.
Query	None.
Request	<i>XML_downloadRequest</i>
Response	File

A.2.3 /ISAPI/ContentMgmt/download/capabilities

Get the downloading capability sets.

Request URI Definition

Table A-5 GET /ISAPI/ContentMgmt/download/capabilities

Method	GET
Description	Get the downloading capability sets.
Query	None.
Request	None.
Response	Succeeded: <i>XML_DownloadAbility</i> Failed: <i>XML_ResponseStatus</i>

A.2.4 /ISAPI/ContentMgmt/download/toUSB/<taskId>/progress?format=json

Get the progress of exporting files to devices via USB.

Request URI Definition

Table A-6 GET /ISAPI/ContentMgmt/download/toUSB/<taskId>/progress?format=json

Method	GET
Description	Get the progress of exporting files to devices via USB.
Query	format: determine the format of request or response message.
Request	None.
Response	Succeeded: <i>JSON_ProgressInfo</i> Failed: <i>JSON_ResponseStatus</i>

Remarks

The <taskId> in the request URI refers to the task ID, and it is returned in the message *JSON_ExporttoUSB_TaskInfo* after calling the request URI: POST */ISAPI/ContentMgmt/download/toUSB?format=json* .

A.2.5 /ISAPI/ContentMgmt/download/toUSB/capabilities?format=json

Get the capability of exporting files to devices via USB.

Request URI Definition

Table A-7 GET /ISAPI/ContentMgmt/download/toUSB/capabilities?format=json

Method	GET
Description	Get the capability of exporting files to devices via USB.
Query	format: determine the format of request or response message.
Request	None.
Response	Succeeded: <i>JSON_Cap_ExportInfo</i> Failed: <i>JSON_ResponseStatus</i>

A.2.6 /ISAPI/ContentMgmt/download/toUSB?format=json

Export files to devices via USB.

Request URI Definition

Table A-8 POST /ISAPI/ContentMgmt/download/toUSB?format=json

Method	POST
Description	Export files to devices via USB.
Query	format: determine the format of request or response message.
Request	<i>JSON_ExportInfo</i>
Response	Succeeded: <i>JSON_ExporttoUSB_TaskInfo</i> Failed: <i>JSON_ResponseStatus</i>

A.2.7 /ISAPI/ContentMgmt/InputProxy/channels

Operations about management of all digital channels.

Request URI Definition

Table A-9 GET /ISAPI/ContentMgmt/InputProxy/channels

Method	GET
Description	Get parameters of all digital channels.
Query	None.

Request	None.
Response	<i>XML_InputProxyChannelList</i>

Table A-10 PUT /ISAPI/ContentMgmt/InputProxy/channels

Method	PUT
Description	Set parameters of all digital channels.
Query	None.
Request	<i>XML_InputProxyChannelList</i>
Response	<i>XML_ResponseStatus</i>

Table A-11 POST /ISAPI/ContentMgmt/InputProxy/channels

Method	POST
Description	Add a digital channel.
Query	None.
Request	<i>XML_InputProxyChannel</i>
Response	<i>XML_ResponseStatus</i>

A.2.8 /ISAPI/ContentMgmt/InputProxy/channels/<ID>

Operations about management of a specific digital channel.

Request URI Definition

Table A-12 GET /ISAPI/ContentMgmt/InputProxy/channels/<ID>

Method	GET
Description	Get parameter of a specific digital channel.
Query	None.
Request	None.
Response	Succeeded: <i>XML_InputProxyChannel</i> Failed: <i>XML_ResponseStatus</i>

Table A-13 PUT /ISAPI/ContentMgmt/InputProxy/channels/<ID>

Method	PUT
Description	Set parameters of a specific digital channel.
Query	None.

Request	<i>XML_InputProxyChannel</i>
Response	<i>XML_ResponseStatus</i>

Table A-14 DELETE /ISAPI/ContentMgmt/InputProxy/channels/<ID>

Method	DELETE
Description	Delete a specific digital channel.
Query	None.
Request	None.
Response	<i>XML_ResponseStatus</i>

Remarks

The <ID> in the request URI refers to the digital channel ID.

A.2.9 /ISAPI/ContentMgmt/InputProxy/channels/<ID>/capabilities

Get management capability of a specific digital channel.

Request URI Definition**Table A-15 GET /ISAPI/ContentMgmt/InputProxy/channels/<ID>/capabilities**

Method	GET
Description	Get management capability of a specific digital channel.
Query	None.
Request	None.
Response	Succeeded: <i>XML_Cap_InputProxyChannel</i> Failed: <i>XML_ResponseStatus</i>

Remarks

The <ID> in the request URI refers to the digital channel ID.

A.2.10 /ISAPI/ContentMgmt/InputProxy/channels/<ID>/chanCtrl

Get or set control parameters of a specific channel.

Request URI Definition**Table A-16 GET /ISAPI/ContentMgmt/InputProxy/channels/<ID>/chanCtrl**

Method	GET
Description	Get control parameters of a specific channel.
Query	None.
Request	None.
Response	Succeeded: <i>XML_ChanCtrl</i> Failed: <i>XML_ResponseStatus</i>

Table A-17 PUT /ISAPI/ContentMgmt/InputProxy/channels/<ID>/chanCtrl

Method	PUT
Description	Set control parameters of a specific channel.
Query	None.
Request	<i>XML_ChanCtrl</i>
Response	<i>XML_ResponseStatus</i>

Remarks

The <ID> in the request URI refers to the channel ID.

A.2.11 /ISAPI/ContentMgmt/InputProxy/channels/<ID>/deviceInfo

Get or set the device parameters of a digital channel.

Request URI Definition**Table A-18 GET /ISAPI/ContentMgmt/InputProxy/channels/<ID>/deviceInfo**

Method	GET
Description	Get the device parameters of a digital channel.
Query	None.
Request	None.
Response	Succeeded: <i>XML_DeviceInfo</i> Failed: <i>XML_ResponseStatus</i>

Table A-19 PUT /ISAPI/ContentMgmt/InputProxy/channels/<ID>/deviceInfo

Method	PUT
Description	Set the device parameters of a digital channel.
Query	None.
Request	<i>XML_DeviceInfo</i>
Response	<i>XML_ResponseStatus</i>

Remarks

The <ID> in the request URI refers to channel ID.

A.2.12 /ISAPI/ContentMgmt/InputProxy/channels/<ID>/portMapParam

Check if port mapping is required for connecting to camera.

Request URI Definition

Table A-20 POST /ISAPI/ContentMgmt/InputProxy/channels/<ID>/portMapParam

Method	POST
Description	Check if port mapping is required for connecting to camera according to the IP address of terminal.
Query	None.
Request	<i>XML_PortMapParam</i>
Response	Succeeded: <i>XML_PortMapParamRet</i> Failed: <i>XML_ResponseStatus</i>

Remarks

The <ID> in the request URI refers to the channel ID.

A.2.13 /ISAPI/ContentMgmt/InputProxy/channels/<ID>/reboot

Reboot a specific channel.

Request URI Definition

Table A-21 GET /ISAPI/ContentMgmt/InputProxy/channels/<ID>/reboot

Method	GET
Description	Reboot a specific channel.

Query	None.
Request	None.
Response	<i>XML_ResponseStatus</i>

Remarks

The <ID> in the request URI refers to the channel ID.

A.2.14 /ISAPI/ContentMgmt/InputProxy/channels/<ID>/status

Get status of a specific digital channel.

Request URI Definition

Table A-22 GET /ISAPI/ContentMgmt/InputProxy/channels/<ID>/status

Method	GET
Description	Get status of a specific digital channel.
Query	None.
Request	None.
Response	<i>XML_InputProxyChannelStatus</i>

Remarks

The <ID> in the request URI refers to the digital channel ID.

A.2.15 /ISAPI/ContentMgmt/InputProxy/channels/<ID>/video/overlays

Get or set text overlay parameters of a specific video digital input channel.

Request URI Definition

Table A-23 GET /ISAPI/ContentMgmt/InputProxy/channels/<ID>/video/overlays

Method	GET
Description	Get text overlay parameters of a specific video digital input channel.
Query	None.
Request	None.
Response	Succeeded: <i>XML_Channel_VideoOverlay</i> Failed: <i>XML_ResponseStatus</i>

Table A-24 PUT /ISAPI/ContentMgmt/InputProxy/channels/<ID>/video/overlays

Method	PUT
Description	Set text overlay parameters of a specific video digital input channel.
Query	None.
Request	<i>XML_Channel_VideoOverlay</i>
Response	<i>XML_ResponseStatus</i>

Remarks

The <ID> in the request URI refers to the channel No.

A.2.16 /ISAPI/ContentMgmt/InputProxy/channels/<ID>/video/overlays/capabilities

Get the text overlay configuration capability of a specific video digital input channel.

Request URI Definition**Table A-25 GET /ISAPI/ContentMgmt/InputProxy/channels/<ID>/video/overlays/capabilities**

Method	GET
Description	Get the text overlay configuration capability of a specific video digital input channel.
Query	None.
Request	None.
Response	Succeeded: <i>XML_ChannelCap_VideoOverlay</i> Failed: <i>XML_ResponseStatus</i>

Remarks

The <ID> in the request URI refers to the channel No.

A.2.17 /ISAPI/ContentMgmt/InputProxy/channels/activate

Activate the network devices.

Request URI Definition

Table A-26 PUT /ISAPI/ContentMgmt/InputProxy/channels/activate

Method	PUT
Description	Activate the network devices.
Query	None
Request	<i>XML_VideoSourceActivation</i>
Response	<i>XML_ResponseStatus</i>

A.2.18 /ISAPI/ContentMgmt/InputProxy/channels/activate/capabilities

Get capability of activating network devices.

Request URI Definition

Table A-27 GET /ISAPI/ContentMgmt/InputProxy/channels/activate/capabilities

Method	GET
Description	Get capability of activating network devices.
Query	None
Request	None.
Response	Succeeded: <i>XML_VideoSourceActivationCapability</i> Failed: <i>XML_ResponseStatus</i>

A.2.19 /ISAPI/ContentMgmt/InputProxy/channels/capabilities

Get management capability of all digital channels.

Request URI Definition

Table A-28 GET /ISAPI/ContentMgmt/InputProxy/channels/capabilities

Method	GET
Description	Get management capability of all digital channels.
Query	None.
Request	None.
Response	<i>XML_Cap_InputProxyChannelListCap</i>

A.2.20 /ISAPI/ContentMgmt/InputProxy/channels/status

Get status of all digital channels.

Request URI Definition

Table A-29 GET /ISAPI/ContentMgmt/InputProxy/channels/status

Method	GET
Description	Get status of all digital channels.
Query	None.
Request	None.
Response	<i>XML_InputProxyChannelStatusList</i>

A.2.21 /ISAPI/ContentMgmt/InputProxy/deviceInfo

Get device parameters of all digital channels.

Request URI Definition

Table A-30 POST /ISAPI/ContentMgmt/InputProxy/deviceInfo

Method	POST
Description	Get device parameters of all digital channels.
Query	None.
Request	<i>XML_ProxyDevicesInfoCondition</i>
Response	Succeeded: <i>XML_ProxyDevicesInfoResult</i> Failed: <i>XML_ResponseStatus</i>

A.2.22 /ISAPI/ContentMgmt/InputProxy/search

Search for network cameras in the LAN (Local Area Network) that can be connected.

Request URI Definition

Table A-31 GET /ISAPI/ContentMgmt/InputProxy/search

Method	GET
Description	Search for network cameras in the LAN (Local Area Network) that can be connected.

Query	None.
Request	None.
Response	Succeeded: <i>XML_VideoSourceList</i> Failed: <i>XML_ResponseStatus</i>

A.2.23 /ISAPI/ContentMgmt/InputProxy/sourceCapability

Get supported number of digital channels.

Request URI Definition

Table A-32 POST /ISAPI/ContentMgmt/InputProxy/sourceCapability

Method	POST
Description	Get supported number of digital channels.
Query	None.
Request	<i>XML_sourceDescriptor</i>
Response	<i>XML_sourceCapability</i>

A.2.24 /ISAPI/ContentMgmt/IOProxy/inputs

Operations about the alarm input configuration of all digital channels.

Request URI Definition

Table A-33 GET /ISAPI/ContentMgmt/IOProxy/inputs

Method	GET
Description	Get the alarm input parameters of all digital channels.
Query	None.
Request	None.
Response	Succeeded: <i>XML_IOProxyInputPortList</i> Failed: <i>XML_ResponseStatus</i>

Table A-34 PUT /ISAPI/ContentMgmt/IOProxy/inputs

Method	PUT
Description	Set the alarm input parameters of all digital channels.
Query	None.

Request	<i>XML_IOProxyInputPortList</i>
Response	<i>XML_ResponseStatus</i>

Table A-35 POST /ISAPI/ContentMgmt/IOProxy/inputs

Method	POST
Description	Add the alarm input configuration of a specific digital channel.
Query	None.
Request	<i>XML_IOProxyInputPort</i>
Response	<i>XML_ResponseStatus</i>

A.2.25 /ISAPI/ContentMgmt/IOProxy/inputs/<ID>

Operations about the alarm input configuration of a specific digital channel.

Request URI Definition

Table A-36 GET /ISAPI/ContentMgmt/IOProxy/inputs/<ID>

Method	GET
Description	Get the alarm input parameters of a specific digital channel.
Query	None.
Request	None.
Response	Succeeded: <i>XML_IOProxyInputPort</i> Failed: <i>XML_ResponseStatus</i>

Table A-37 PUT /ISAPI/ContentMgmt/IOProxy/inputs/<ID>

Method	PUT
Description	Set the alarm input parameters of a specific digital channel.
Query	None.
Request	<i>XML_IOProxyInputPort</i>
Response	<i>XML_ResponseStatus</i>

Table A-38 DELETE /ISAPI/ContentMgmt/IOProxy/inputs/<ID>

Method	DELETE
Description	Delete the alarm input configuration of a specific digital channel.
Query	None.

Request	None.
Response	<i>XML_ResponseStatus</i>

Remarks

The <ID> in the request URI refers to the digital channel No.

A.2.26 /ISAPI/ContentMgmt/IOProxy/inputs/<ID>/status

Get the alarm input status of a specific digital channel.

Request URI Definition

Table A-39 GET /ISAPI/ContentMgmt/IOProxy/inputs/<ID>/status

Method	GET
Description	Get the alarm input status of a specific digital channel.
Query	None.
Request	None.
Response	Succeeded: <i>XML_IOPortStatus</i> Failed: <i>XML_ResponseStatus</i>

Remarks

The <ID> in the request URI refers to the digital channel No.

A.2.27 /ISAPI/ContentMgmt/IOProxy/outputs

Operations about the alarm output configuration of all digital channels.

Request URI Definition

Table A-40 GET /ISAPI/ContentMgmt/IOProxy/outputs

Method	GET
Description	Get the alarm output parameters of all digital channels.
Query	None.
Request	None.
Response	Succeeded: <i>XML_IOProxyOutputPortList</i> Failed: <i>XML_ResponseStatus</i>

Table A-41 PUT /ISAPI/ContentMgmt/IOProxy/outputs

Method	PUT
Description	Set the alarm output parameters of all digital channels.
Query	None.
Request	<i>XML_IOProxyOutputPortList</i>
Response	<i>XML_ResponseStatus</i>

Table A-42 POST /ISAPI/ContentMgmt/IOProxy/outputs

Method	POST
Description	Add the alarm output configuration of a specific digital channel.
Query	None.
Request	<i>XML_IOProxyOutputPort</i>
Response	<i>XML_ResponseStatus</i>

A.2.28 /ISAPI/ContentMgmt/IOProxy/outputs/<ID>

Operations about the alarm output configuration of a specific digital channel.

Request URI Definition

Table A-43 GET /ISAPI/ContentMgmt/IOProxy/outputs/<ID>

Method	GET
Description	Get the alarm output parameters of a specific digital channel.
Query	None.
Request	None.
Response	Succeeded: <i>XML_IOProxyOutputPort</i> Failed: <i>XML_ResponseStatus</i>

Table A-44 PUT /ISAPI/ContentMgmt/IOProxy/outputs/<ID>

Method	PUT
Description	Set the alarm output parameters of a specific digital channel.
Query	None.
Request	<i>XML_IOProxyOutputPort</i>
Response	<i>XML_ResponseStatus</i>

Table A-45 DELETE /ISAPI/ContentMgmt/IOProxy/outputs/<ID>

Method	DELETE
Description	Delete the alarm output configuration of a specific digital channel.
Query	None.
Request	None.
Response	<i>XML_ResponseStatus</i>

Remarks

The <ID> in the request URI refers to the digital channel No.

A.2.29 /ISAPI/ContentMgmt/IOProxy/outputs/<ID>/status

Get the alarm output status of a specific digital channel.

Request URI Definition
Table A-46 GET /ISAPI/ContentMgmt/IOProxy/outputs/<ID>/status

Method	GET
Description	Get the alarm output status of a specific digital channel.
Query	None.
Request	None.
Response	Succeeded: <i>XML_IOPortStatus</i> Failed: <i>XML_ResponseStatus</i>

Remarks

The <ID> in the request URI refers to the digital channel No.

A.2.30 /ISAPI/ContentMgmt/IOProxy/outputs/<ID>/trigger

Trigger alarm output of a specific digital channel.

Request URI Definition
Table A-47 PUT /ISAPI/ContentMgmt/IOProxy/outputs/<ID>/trigger

Method	PUT
Description	Trigger alarm output of a specific digital channel.
Query	None.

Request	<i>XML_IOPortData</i>
Response	<i>XML_ResponseStatus</i>

Remarks

The <ID> in the request URI refers to the digital channel No.

A.2.31 /ISAPI/ContentMgmt/logConfig

Get or set log parameters.

Request URI Definition

Table A-48 GET /ISAPI/ContentMgmt/logConfig

Method	GET
Description	Get log parameters.
Query	None.
Request	None.
Response	Succeeded: <i>XML_LogConfig</i> Failed: <i>XML_ResponseStatus</i>

Table A-49 PUT /ISAPI/ContentMgmt/logConfig

Method	PUT
Description	Set log parameters.
Query	None.
Request	<i>XML_LogConfig</i>
Response	<i>XML_ResponseStatus</i>

A.2.32 /ISAPI/ContentMgmt/logConfig/capabilities

Get log configuration capability.

Request URI Definition

Table A-50 GET /ISAPI/ContentMgmt/logConfig/capabilities

Method	GET
Description	Get log configuration capability.
Query	None.

Request	None.
Response	Succeeded: <i>XML_Cap_LogConfig</i> Failed: <i>XML_ResponseStatus</i>

A.2.33 /ISAPI/ContentMgmt/logSearch

Search for log files.

Request URI Definition

Table A-51 POST /ISAPI/ContentMgmt/logSearch

Method	POST
Description	Search for log files.
Query	None.
Request	<i>XML_Log_CMSearchDescription</i>
Response	Succeeded: <i>XML_Log_CMSearchResult</i> Failed: <i>XML_ResponseStatus</i>

A.2.34 /ISAPI/ContentMgmt/logSearch?format=json

Search log information.

Request URI Definition

Table A-52 POST /ISAPI/ContentMgmt/logSearch?format=json

Method	POST
Description	Search log information.
Query	format: determine the format of request or response message.
Request	<i>JSON_SearchCondition</i>
Response	Succeeded: <i>JSON_MatchList</i> Failed: <i>JSON_ResponseStatus</i>

A.2.35 /ISAPI/ContentMgmt/logSearch/dataPackage

Export device log files.

Request URI Definition

Table A-53 POST /ISAPI/ContentMgmt/logSearch/dataPackage

Method	POST
Description	Export device log files.
Query	None.
Request	<i>XML_CMSearchDataPackage</i>
Response	Succeeded: <i>XML_CMSearchDataPackageResult</i> Failed: <i>XML_ResponseStatus</i>

A.2.36 /ISAPI/ContentMgmt/PTZCtrlProxy/capabilities

Get PTZ control capability of digital channels.

Request URI Definition

Table A-54 /ISAPI/ContentMgmt/PTZCtrlProxy/capabilities

Method	GET
Description	Get PTZ control capability of digital channels.
Query	None
Request	None
Response	Succeeded: <i>XML_PTZCtrlCap</i> Failed: <i>XML_ResponseStatus</i>

A.2.37 /ISAPI/ContentMgmt/PTZCtrlProxy/channels

Get or set parameters of all digital PTZ channels.

Request URI Definition

Table A-55 GET /ISAPI/ContentMgmt/PTZCtrlProxy/channels

Method	GET
Description	Get parameters of all digital PTZ channels.
Query	None

Request	None
Response	Succeeded: <i>XML_PTZChannelList</i> Failed: <i>XML_ResponseStatus</i>

Table A-56 PUT /ISAPI/ContentMgmt/PTZCtrlProxy/channels

Method	PUT
Description	Set parameters of all digital PTZ channels.
Query	None
Request	<i>XML_PTZChannelList</i>
Response	<i>XML_ResponseStatus</i>

A.2.38 /ISAPI/ContentMgmt/PTZCtrlProxy/channels/<ID>

Get or set parameters of a digital PTZ channel.

Request URI Definition

Table A-57 GET /ISAPI/ContentMgmt/PTZCtrlProxy/channels/<ID>

Method	GET
Description	Get parameters of a digital PTZ channel.
Query	None
Request	None
Response	Succeeded: <i>XML_PTZChannel</i> Failed: <i>XML_ResponseStatus</i>

Table A-58 PUT /ISAPI/ContentMgmt/PTZCtrlProxy/channels/<ID>

Method	PUT
Description	Set parameters of a digital PTZ channel.
Query	None
Request	<i>XML_PTZChannel</i>
Response	<i>XML_ResponseStatus</i>

Remarks

The <ID> in the request URI refers to the digital PTZ channel ID.

A.2.39 /ISAPI/ContentMgmt/PTZCtrlProxy/channels/<ID>/autoPan

Control the fans.

Request URI Definition**Table A-59 PUT /ISAPI/ContentMgmt/PTZCtrlProxy/channels/<ID>/autoPan**

Method	PUT
Description	Control the fans.
Query	None
Request	<i>XML_autoPanData</i>
Response	<i>XML_ResponseStatus</i>

Remarks

The <ID> in the request URI refers to the digital channel No.

A.2.40 /ISAPI/ContentMgmt/PTZCtrlProxy/channels/<ID>/capabilities

Get the capability of a digital PTZ channel.

Request URI Definition**Table A-60 GET /ISAPI/ContentMgmt/PTZCtrlProxy/channels/<ID>/capabilities**

Method	GET
Description	Get the capability of a digital PTZ channel.
Query	None
Request	None
Response	Succeeded: <i>XML_PTZChanelCap</i> Failed: <i>XML_ResponseStatus</i>

Remarks

The <ID> in the request URI refers to the digital channel No.

A.2.41 /ISAPI/ContentMgmt/PTZCtrlProxy/channels/<ID>/continuous

Manually control PTZ.

Request URI Definition**Table A-61 PUT /ISAPI/ContentMgmt/PTZCtrlProxy/channels/<ID>/continuous**

Method	PUT
Description	Manually control PTZ.
Query	None
Request	<i>XML_Manual_PTZData</i>
Response	<i>XML_ResponseStatus</i>

Remarks

The <ID> in the request URI refers to the channel No.

A.2.42 /ISAPI/ContentMgmt/PTZCtrlProxy/channels/<ID>/momentary

Perform scheduled PTZ control via digital channel.

Request URI Definition**Table A-62 PUT /ISAPI/ContentMgmt/PTZCtrlProxy/channels/<ID>/momentary**

Method	PUT
Description	Perform scheduled PTZ control via digital channel.
Query	None
Request	<i>XML_Scheduled_PTZData</i>
Response	<i>XML_ResponseStatus</i>

Remarks

The <ID> in the request URI refers to the digital channel No.

A.2.43 /ISAPI/ContentMgmt/PTZCtrlProxy/channels/<ID>/onpushfoucs/start

Perform one-touch focus via a digital PTZ channel.

Request URI Definition**Table A-63 PUT /ISAPI/ContentMgmt/PTZCtrlProxy/channels/<ID>/onpushfoucs/start**

Method	PUT
Description	Perform one-touch focus via a digital PTZ channel.

Query	None
Request	None
Response	<i>XML_ResponseStatus</i>

Remarks

The <ID> in the request URI refers to the digital channel No.

A.2.44 /ISAPI/ContentMgmt/PTZCtrlProxy/channels/<ID>/patrols

Get, set, or delete all patrols of a digital PTZ channel, or add a patrol.

Request URI Definition

Table A-64 GET /ISAPI/ContentMgmt/PTZCtrlProxy/channels/<ID>/patrols

Method	GET
Description	Get all patrols of a digital PTZ channel.
Query	None
Request	None
Response	Succeeded: <i>XML_PTZPatrolList</i> Failed: <i>XML_ResponseStatus</i>

Table A-65 PUT /ISAPI/ContentMgmt/PTZCtrlProxy/channels/<ID>/patrols

Method	GET
Description	Set all patrols of a digital PTZ channel.
Query	None
Request	<i>XML_PTZPatrolList</i>
Response	<i>XML_ResponseStatus</i>

Table A-66 POST /ISAPI/ContentMgmt/PTZCtrlProxy/channels/<ID>/patrols

Method	POST
Description	Add a patrol.
Query	None
Request	<i>XML_PTZPatrol</i>
Response	<i>XML_ResponseStatus</i>

Table A-67 DELETE /ISAPI/ContentMgmt/PTZCtrlProxy/channels/<ID>/patrols

Method	DELETE
Description	Delete all patrols of a digital PTZ channel.
Query	None
Request	None
Response	<i>XML_ResponseStatus</i>

Remarks

The <ID> in the request URI refers to the digital channel No.

A.2.45 /ISAPI/ContentMgmt/PTZCtrlProxy/channels/<ID>/patrols/<ID>

Get, set, or delete a patrol of a digital PTZ channel.

Request URI Definition
Table A-68 GET /ISAPI/ContentMgmt/PTZCtrlProxy/channels/<ID>/patrols/<ID>

Method	GET
Description	Get a patrol of a digital PTZ channel.
Query	None
Request	None
Response	Succeeded: <i>XML_PTZPatrol</i> Failed: <i>XML_ResponseStatus</i>

Table A-69 PUT /ISAPI/ContentMgmt/PTZCtrlProxy/channels/<ID>/patrols/<ID>

Method	PUT
Description	Set a patrol of a digital PTZ channel.
Query	None
Request	<i>XML_PTZPatrol</i>
Response	<i>XML_ResponseStatus</i>

Table A-70 DELETE /ISAPI/ContentMgmt/PTZCtrlProxy/channels/<ID>/patrols/<ID>

Method	DELETE
Description	Delete a patrol of a digital PTZ channel.
Query	None

Request	None
Response	<i>XML_ResponseStatus</i>

Remarks

The first <ID> in the request URI refers to the digital channel No., and the second <ID> is the patrol No.

A.2.46 /ISAPI/ContentMgmt/PTZCtrlProxy/channels/<ID>/patrols/<ID>/pause

Pause calling a patrol of a digital PTZ channel.

Request URI Definition

Table A-71 PUT /ISAPI/ContentMgmt/PTZCtrlProxy/channels/<ID>/patrols/<ID>/pause

Method	PUT
Description	Pause calling a patrol of a digital PTZ channel.
Query	None
Request	None
Response	<i>XML_ResponseStatus</i>

Remarks

The first <ID> in the request URI refers to the digital channel No., and the second <ID> is the patrol No.

A.2.47 /ISAPI/ContentMgmt/PTZCtrlProxy/channels/<ID>/patrols/<ID>/start

Start calling a patrol of a digital PTZ channel.

Request URI Definition

Table A-72 PUT /ISAPI/ContentMgmt/PTZCtrlProxy/channels/<ID>/patrols/<ID>/start

Method	PUT
Description	Start calling a patrol of a digital PTZ channel.
Query	None
Request	None
Response	<i>XML_ResponseStatus</i>

Remarks

The first <ID> in the request URI refers to the digital channel No., and the second <ID> is the patrol No.

A.2.48 /ISAPI/ContentMgmt/PTZCtrlProxy/channels/<ID>/patrols/<ID>/status

Get a patrol's status of a digital PTZ channel.

Request URI Definition

Table A-73 GET /ISAPI/ContentMgmt/PTZCtrlProxy/channels/<ID>/patrols/<ID>/status

Method	GET
Description	Get a patrol's status of a digital PTZ channel.
Query	None
Request	None
Response	Succeeded: <i>XML_PTZPatrolStatus</i> Failed: <i>XML_ResponseStatus</i>

Remarks

The first <ID> in the request URI refers to the digital channel No., and the second <ID> is the patrol No.

A.2.49 /ISAPI/ContentMgmt/PTZCtrlProxy/channels/<ID>/patrols/<ID>/stop

Stop calling a patrol of a digital PTZ channel.

Request URI Definition

Table A-74 PUT /ISAPI/ContentMgmt/PTZCtrlProxy/channels/<ID>/patrols/<ID>/stop

Method	PUT
Description	Stop calling a patrol of a digital PTZ channel.
Query	None
Request	None
Response	<i>XML_ResponseStatus</i>

Remarks

The first <ID> in the request URI refers to the digital channel No., and the second <ID> is the patrol No.

A.2.50 /ISAPI/ContentMgmt/PTZCtrlProxy/channels/<ID>/position3D

Perform 3D positioning.

Request URI Definition

Table A-75 PUT /ISAPI/ContentMgmt/PTZCtrlProxy/channels/<ID>/position3D

Method	PUT
Description	Perform 3D positioning.
Query	None
Request	<i>XML_Position3D</i>
Response	<i>XML_ResponseStatus</i>

Remarks

The <ID> in the request URI refers to the digital channel No.

A.2.51 /ISAPI/ContentMgmt/PTZCtrlProxy/channels/<ID>/presets

Get, set or delete presets of a digital PTZ channel, or add a preset.

Request URI Definition

Table A-76 GET /ISAPI/ContentMgmt/PTZCtrlProxy/channels/<ID>/presets

Method	GET
Description	Get presets of a digital PTZ channel.
Query	None
Request	None
Response	Succeeded: <i>XML_PTZPresetList</i> Failed: <i>XML_ResponseStatus</i>

Table A-77 PUT /ISAPI/ContentMgmt/PTZCtrlProxy/channels/<ID>/presets

Method	PUT
Description	Set presets of a digital PTZ channel.
Query	None
Request	<i>XML_PTZPresetList</i>
Response	<i>XML_ResponseStatus</i>

Table A-78 POST /ISAPI/ContentMgmt/PTZCtrlProxy/channels/<ID>/presets

Method	POST
Description	Add a preset to a digital PTZ channel.
Query	None
Request	<i>XML_PTZPreset</i>
Response	<i>XML_ResponseStatus</i>

Table A-79 DELETE /ISAPI/ContentMgmt/PTZCtrlProxy/channels/<ID>/presets

Method	DELETE
Description	Delete presets of a digital PTZ channel.
Query	None
Request	None
Response	<i>XML_ResponseStatus</i>

Remarks

The <ID> in the request URI refers to the digital channel No.

A.2.52 /ISAPI/ContentMgmt/PTZCtrlProxy/channels/<ID>/presets/<ID>

Get, set, or delete a preset of a digital PTZ channel.

Request URI Definition

Table A-80 GET /ISAPI/ContentMgmt/PTZCtrlProxy/channels/<ID>/presets/<ID>

Method	GET
Description	Get a preset of a digital PTZ channel.
Query	None
Request	None
Response	Succeeded: <i>XML_PTZPreset</i> Failed: <i>XML_ResponseStatus</i>

Table A-81 PUT /ISAPI/ContentMgmt/PTZCtrlProxy/channels/<ID>/presets/<ID>

Method	PUT
Description	Set a preset of a digital PTZ channel.
Query	None

Request	<i>XML_PTZPreset</i>
Response	<i>XML_ResponseStatus</i>

Table A-82 DELETE /ISAPI/ContentMgmt/PTZCtrlProxy/channels/<ID>/presets/<ID>

Method	DELETE
Description	Delete a preset of a digital PTZ channel.
Query	None
Request	None
Response	<i>XML_ResponseStatus</i>

Remarks

The first <ID> in the request URI refers to the digital channel No., and the second <ID> indicates the preset No.

A.2.53 /ISAPI/ContentMgmt/PTZCtrlProxy/channels/<ID>/presets/<ID>/goto

Call a preset of a digital PTZ channel.

Request URI Definition**Table A-83 PUT /ISAPI/ContentMgmt/PTZCtrlProxy/channels/<ID>/presets/<ID>/goto**

Method	PUT
Description	Call a preset of a digital PTZ channel.
Query	None
Request	None
Response	<i>XML_ResponseStatus</i>

Remarks

The first <ID> in the request URI refers to the digital channel No., and the second <ID> indicates the preset No.

A.2.54 /ISAPI/ContentMgmt/record/control/cleanUp/capabilities?format=json

Get the capability of deleting saved files in a batch.

Request URI Definition

Table A-84 GET /ISAPI/ContentMgmt/record/control/cleanUp/capabilities?format=json

Method	GET
Description	Get the capability of deleting saved files in a batch.
Query	format : determine the format of request or response message.
Request	None.
Response	Succeeded: <i>JSON_Cap_CleanUpFileList</i> Failed: <i>JSON_ResponseStatus</i>

A.2.55 /ISAPI/ContentMgmt/record/control/cleanUp?format=json

Delete saved files in a batch.

Request URI Definition

Table A-85 POST /ISAPI/ContentMgmt/record/control/cleanUp?format=json

Method	POST
Description	Delete saved files in a batch.
Query	format : determine the format of request or response message.
Request	<i>JSON_CleanUpFileList</i>
Response	<i>JSON_ResponseStatus</i>

A.2.56 /ISAPI/ContentMgmt/record/control/locks/capabilities

Get video locking or unlocking capability.

Request URI Definition

Table A-86 GET /ISAPI/ContentMgmt/record/control/locks/capabilities

Method	GET
Description	Get video locking or unlocking capability.
Query	security : the version No. of encryption scheme. When security does not exist, it indicates that the data is not encrypted; when security is 1, it indicates that the nodes of sensitive information in the message are encrypted in AES128 CBC mode; when security is 2, it indicates

	that the nodes of sensitive information in the message are encrypted in AES256 CBC mode. iv : the initialization vector, and it is required when security is 1 or 2.
Request	None
Response	Succeeded: <i>XML_LockCap</i> Failed: <i>XML_ResponseStatus</i>

A.2.57 /ISAPI/ContentMgmt/record/control/locks/name

Lock or unlock videos by file name.

Request URI Definition

Table A-87 PUT /ISAPI/ContentMgmt/record/control/locks/name

Method	PUT
Description	Lock or unlock videos by file name.
Query	security : the version No. of encryption scheme. When security does not exist, it indicates that the data is not encrypted; when security is 1, it indicates that the nodes of sensitive information in the message are encrypted in AES128 CBC mode; when security is 2, it indicates that the nodes of sensitive information in the message are encrypted in AES256 CBC mode. iv : the initialization vector, and it is required when security is 1 or 2.
Request	<i>XML_LockByName</i>
Response	<i>XML_ResponseStatus</i>

A.2.58 /ISAPI/ContentMgmt/record/control/manual/start/tracks/<ID>

Start manual recording.

Request URI Definition

Table A-88 POST /ISAPI/ContentMgmt/record/control/manual/start/tracks/<ID>

Method	POST
Description	Start manual recording.
Query	None.

Request	None.
Response	<i>XML_ResponseStatus</i>

Remarks

The <ID> in the request URI refers to the recording channel ID.

A.2.59 /ISAPI/ContentMgmt/record/control/manual/stop/tracks/<ID>

Stop manual recording of a specific channel.

Request URI Definition

Table A-89 POST /ISAPI/ContentMgmt/record/control/manual/stop/tracks/<ID>

Method	POST
Description	Stop manual recording of a specific channel.
Query	None.
Request	None.
Response	<i>XML_ResponseStatus</i>

Remarks

The <ID> in the request URI refers to the recording channel ID.

A.2.60 /ISAPI/ContentMgmt/record/control/manualRefresh/channels/<ID>

Refresh the video mode manually before playback.

Request URI Definition

Table A-90 PUT /ISAPI/ContentMgmt/record/control/manualRefresh/channels/<ID>

Method	PUT
Description	Refresh the video mode manually before playback.
Query	None.
Request	None.
Response	<i>XML_ResponseStatus</i>

Remarks

The <ID> in the request URI refers to the channel No.

A.2.61 /ISAPI/ContentMgmt/record/control/remark/capabilities?format=json

Get capability of adding remarks for videos by video ID.

Request URI Definition

Table A-91 GET /ISAPI/ContentMgmt/record/control/remark/capabilities?format=json

Method	GET
Description	Get capability of adding remarks for videos by video ID.
Query	format : determine the format of request or response message.
Request	None
Response	Succeeded: <i>JSON_Cap_RemarkList</i> Failed: <i>XML_ResponseStatus</i>

A.2.62 /ISAPI/ContentMgmt/record/control/remark?format=json

Add remarks for videos by video ID.

Request URI Definition

Table A-92 PUT /ISAPI/ContentMgmt/record/control/remark?format=json

Method	PUT
Description	Add remarks for videos by video ID.
Query	format : determine the format of request or response message.
Request	<i>JSON_RemarkList</i>
Response	<i>XML_ResponseStatus</i>

A.2.63 /ISAPI/ContentMgmt/record/profile

Get video track type.

Request URI Definition

Table A-93 GET /ISAPI/ContentMgmt/record/profile

Method	GET
Description	Get video track type.
Query	None.

Request	None.
Response	Succeeded: <i>XML_CMSRecordProfile</i> Failed: <i>XML_ResponseStatus</i>

A.2.64 /ISAPI/ContentMgmt/record/storageMounts

Set recording storage parameters.

Request URI Definition

Table A-94 PUT /ISAPI/ContentMgmt/record/storageMounts

Method	PUT
Description	Set recording storage parameters, including root directory, size, and so on.
Query	None.
Request	<i>XML_MountList</i>
Response	<i>XML_ResponseStatus</i>

A.2.65 /ISAPI/ContentMgmt/record/tracks

Operations about recording schedule configuration.

Request URI Definition

Table A-95 GET /ISAPI/ContentMgmt/record/tracks

Method	GET
Description	Get all recording schedules.
Query	None.
Request	None.
Response	Succeeded: <i>XML_TrackList</i> Failed: <i>XML_ResponseStatus</i>

Table A-96 PUT /ISAPI/ContentMgmt/record/tracks

Method	PUT
Description	Set all recording schedules.
Query	None.

Request	<i>XML_TrackList</i>
Response	<i>XML_ResponseStatus</i>

Table A-97 POST /ISAPI/ContentMgmt/record/tracks

Method	POST
Description	Add a recording schedule.
Query	None.
Request	<i>XML_Track</i>
Response	<i>XML_ResponseStatus</i>

A.2.66 /ISAPI/ContentMgmt/record/tracks/<ID>

Operations about a recording schedule.

Request URI Definition

Table A-98 GET /ISAPI/ContentMgmt/record/tracks/<ID>

Method	GET
Description	Get a recording schedule.
Query	None.
Request	None.
Response	Succeeded: <i>XML_Track</i> Failed: <i>XML_ResponseStatus</i>

Table A-99 PUT /ISAPI/ContentMgmt/record/tracks/<ID>

Method	PUT
Description	Set a recording schedule.
Query	None.
Request	<i>XML_Track</i>
Response	<i>XML_ResponseStatus</i>

Table A-100 DELETE /ISAPI/ContentMgmt/record/tracks/<ID>

Method	DELETE
Description	Delete a recording schedule.
Query	None.

Request	None.
Response	<i>XML_ResponseStatus</i>

Remarks

The <ID> in the request URI refers to the recording schedule ID.

A.2.67 /ISAPI/ContentMgmt/record/tracks/<ID>/capabilities

Get the configuration capability of the recording schedule.

Request URI Definition

Table A-101 GET /ISAPI/ContentMgmt/record/tracks/<ID>/capabilities

Method	GET
Description	Get the configuration capability of the recording schedule.
Query	None.
Request	None.
Response	Succeeded: <i>XML_Cap_Track</i> Failed: <i>XML_ResponseStatus</i>

Remarks

The <ID> in the request URI refers to the recording schedule ID.

A.2.68 /ISAPI/ContentMgmt/record/tracks/<ID>/dailyDistribution

Search recorded video files by date.

Request URI Definition

Table A-102 POST /ISAPI/ContentMgmt/record/tracks/<ID>/dailyDistribution

Method	POST
Description	Search recorded video files by date.
Query	None.
Request	<i>XML_trackDailyParam</i>
Response	<i>XML_trackDailyDistribution</i>

Remarks

The <ID> in the request URI refers to the recording channel ID.

A.2.69 /ISAPI/ContentMgmt/search

Search for specified resources.

Request URI Definition

Table A-103 GET or POST /ISAPI/ContentMgmt/search

Method	GET or POST
Description	Search for specific resources.
Query	None.
Request	<i>XML_VideoPic_CMSearchDescription</i>
Response	Succeeded: <i>XML_VideoPic_CMSearchResult</i> Failed: <i>XML_ResponseStatus</i>

Example

Sample Code for Searching Video Files Stored in Device

```
POST /ISAPI/ContentMgmt/search HTTP/1.1
Host: 10.17.132.49
Content-Length: 493
Connection: Keep-Alive
Authorization: Digest username="admin",
realm="DS-2CD2F12FWD-IWS",
nonce="4e7a46474e305a454d5445365a4445314e6a51345a54413d",
uri="/ISAPI/ContentMgmt/search",
cnonce="ce22590094d2f2bb352fc3c4cd2a1ca3",
nc=00000019,
response="ad6f2c23636f25c6db5911a113375ea9",
qop="auth"

<?xml version="1.0" encoding="utf-8"?>
<CMSearchDescription>
  <searchID>C77384AD-66A0-0001-E7C2-1151F04F90B0</searchID>
  <trackIDList>
    <trackID>101</trackID>
  </trackIDList>
  <timeSpanList>
    <timeSpan>
      <startTime>2017-03-13T16:00:00Z</startTime>
      <endTime>2017-03-16T15:59:59Z</endTime>
    </timeSpan>
  </timeSpanList>
  <maxResults>40</maxResults>
  <searchResultPostion>0</searchResultPostion>
  <metadataList>
    <metadataDescriptor>//recordType.meta.std-cgi.com</metadataDescriptor>
  </metadataList>
```



```
</CMSearchDescription>
```

```
HTTP/1.1 200 OK
```

```
Date: Wed, 15 Mar 2017 09:40:02 GMT
```

```
Connection: close
```

```
Content-Length: 1649
```

```
Content-Type: application/xml
```

```
<?xml version="1.0" encoding="UTF-8"?>
```

```
<CMSearchResult version="2.0" xmlns="http://www.isapi.com/ver20/XMLSchema">
```

```
<searchID>{C77384AD-66A0-0001-E7C2-1151F04F90B0}</searchID>
```

```
<responseStatus>true</responseStatus>
```

```
<responseStatusStrg>OK</responseStatusStrg>
```

```
<numOfMatches>29</numOfMatches>
```

```
<matchList>
```

```
<searchMatchItem>
```

```
<sourceID>{0000000000-0000-0000-0000-000000000000}</sourceID>
```

```
<trackID>101</trackID>
```

```
<timeSpan>
```

```
<startTime>2017-03-14T10:32:01Z</startTime>
```

```
<endTime>2017-03-14T10:40:42Z</endTime>
```

```
</timeSpan>
```

```
<mediaSegmentDescriptor>
```

```
<contentType>video</contentType>
```

```
<codecType>H.264-BP</codecType>
```

```
<playbackURI>rtsp://10.17.132.49/Streaming/tracks/101/?
```

```
starttime=20170314T103201Z&endtime=20170314T104042Z&name=ch01_08000000016000000&size=260358144</playbackURI>
```

```
</mediaSegmentDescriptor>
```

```
<metadataMatches>
```

```
<metadataDescriptor>recordType.meta.isapi.com/timing</metadataDescriptor>
```

```
</metadataMatches>
```

```
</searchMatchItem>
```

```
<searchMatchItem>
```

```
<sourceID>{0000000000-0000-0000-0000-000000000000}</sourceID>
```

```
<trackID>101</trackID>
```

```
<timeSpan>
```

```
<startTime>2017-03-14T10:40:42Z</startTime>
```

```
<endTime>2017-03-14T10:53:14Z</endTime>
```

```
</timeSpan>
```

```
<mediaSegmentDescriptor>
```

```
<contentType>video</contentType>
```

```
<codecType>H.264-BP</codecType>
```

```
<playbackURI>rtsp://10.17.132.49/Streaming/tracks/101/?
```

```
starttime=20170314T104042Z&endtime=20170314T105314Z&name=ch01_08000000017000000&size=260603904</playbackURI>
```

```
</mediaSegmentDescriptor>
```

```
<metadataMatches>
```

```
<metadataDescriptor>recordType.meta.isapi.com/timing</metadataDescriptor>
```

```
</metadataMatches>
```

```
</searchMatchItem>
```

```
</matchList>  
</CMSearchResult>
```

A.2.70 /ISAPI/ContentMgmt/search/profile

Get video search conditions.

Request URI Definition

Table A-104 GET /ISAPI/ContentMgmt/search/profile

Method	GET
Description	Get video search conditions.
Query	None.
Request	None.
Response	Succeeded: <i>XML_CMSearchProfile</i> Failed: <i>XML_ResponseStatus</i>

A.2.71 /ISAPI/ContentMgmt/security/logSearch

Search for security logs.

Request URI Definition

Table A-105 POST /ISAPI/ContentMgmt/security/logSearch

Method	POST
Description	Search for security logs.
Query	None.
Request	<i>XML_VideoPic_CMSearchDescription</i>
Response	Succeeded: <i>XML_VideoPic_CMSearchResult</i> Failed: <i>XML_ResponseStatus</i>

A.2.72 /ISAPI/ContentMgmt/SmartSearch

Search for video files by VCA event.

Request URI Definition

Table A-106 POST /ISAPI/ContentMgmt/SmartSearch

Method	POST
Description	Search for video files by VCA event.
Query	None.
Request	<i>XML_SmartSearchDescription</i>
Response	<i>XML_SmartSearchResult</i>

A.2.73 /ISAPI/ContentMgmt/SmartSearch/capabilities

Get capability of searching videos by VCA event.

Request URI Definition

Table A-107 GET /ISAPI/ContentMgmt/SmartSearch/capabilities

Method	GET
Description	Get capability of searching videos by VCA event.
Query	None.
Request	None.
Response	Succeeded: <i>XML_Cap_SmartSearchDescription</i> Failed: <i>XML_ResponseStatus</i>

A.2.74 /ISAPI/ContentMgmt/storage

Get or set the current working mode of storage.

Request URI Definition

Table A-108 GET /ISAPI/ContentMgmt/storage

Method	GET
Description	Get the current working mode of storage.
Query	None.
Request	None.
Response	Succeeded: <i>XML_storage</i>

	Failed: <i>XML_ResponseStatus</i>
--	-----------------------------------

Table A-109 PUT /ISAPI/ContentMgmt/storage

Method	PUT
Description	Set the current working mode of storage.
Query	None.
Request	<i>XML_storage</i>
Response	<i>XML_ResponseStatus</i>

A.2.75 /ISAPI/ContentMgmt/Storage/capabilities

Get the configuration capability of current storage working mode.

Request URI Definition

Table A-110 GET /ISAPI/ContentMgmt/Storage/capabilities

Method	GET
Description	Get the configuration capability of current storage working mode.
Query	None.
Request	None.
Response	Succeeded: <i>XML_Cap_storage</i> Failed: <i>XML_ResponseStatus</i>

A.2.76 /ISAPI/ContentMgmt/Storage/dataReconstruction/progress?format=json

Get the disk data reconstruction progress.

Request URI Definition

Table A-111 GET /ISAPI/ContentMgmt/Storage/dataReconstruction/progress?format=json

Method	GET
Description	Get the disk data reconstruction progress.
Query	format: determine the format of request or response message.
Request	None.
Response	Succeeded: <i>JSON_Reconstruction_ProgressLists</i> Failed: <i>JSON_ResponseStatus</i>

A.2.77 /ISAPI/ContentMgmt/Storage/dataReconstruction?format=json

Start disk data reconstruction.

Request URI Definition

Table A-112 PUT /ISAPI/ContentMgmt/Storage/dataReconstruction?format=json

Method	PUT
Description	Start disk data reconstruction.
Query	format: determine the format of request or response message.
Request	<i>JSON_DataReconstruction</i>
Response	<i>JSON_ResponseStatus</i>

A.2.78 /ISAPI/ContentMgmt/Storage/extension

Get or set parameters of storage strategy.

Request URI Definition

Table A-113 GET /ISAPI/ContentMgmt/Storage/extension

Method	GET
Description	Get parameters of storage strategy.
Query	None.
Request	None.
Response	Succeeded: <i>XML_storageExtension</i> Failed: <i>XML_ResponseStatus</i>

Table A-114 PUT /ISAPI/ContentMgmt/Storage/extension

Method	PUT
Description	Set parameters of storage strategy.
Query	None.
Request	<i>XML_storageExtension</i>
Response	<i>XML_ResponseStatus</i>

A.2.79 /ISAPI/ContentMgmt/Storage/extension/capabilities

Get configuration capability of log storage.

Request URI Definition

Table A-115 GET /ISAPI/ContentMgmt/Storage/extension/capabilities

Method	GET
Description	Get configuration capability of log storage.
Query	None.
Request	None
Response	Succeeded: <i>XML_Cap_storageExtension</i> Failed: <i>JSON_ResponseStatus</i>

A.2.80 /ISAPI/ContentMgmt/Storage/ExtraInfo

Operations about the configuration for storing additional information.

Request URI Definition

Table A-116 GET /ISAPI/ContentMgmt/Storage/ExtraInfo

Method	GET
Description	Get the parameters for storing additional information.
Query	None.
Request	None.
Response	Succeeded: <i>XML_ExtraInfo</i> Failed: <i>XML_ResponseStatus</i>

Table A-117 PUT /ISAPI/ContentMgmt/Storage/ExtraInfo

Method	PUT
Description	Set the parameters for storing additional information.
Query	None.
Request	<i>XML_ExtraInfo</i>
Response	<i>XML_ResponseStatus</i>

A.2.81 /ISAPI/ContentMgmt/Storage/ExtraInfo/capabilities

Get the capability of storing additional information.

Request URI Definition

Table A-118 GET /ISAPI/ContentMgmt/Storage/ExtraInfo/capabilities

Method	GET
Description	Get the capability of storing additional information.
Query	None.
Request	None.
Response	Succeeded: <i>XML_Cap_ExtraInfo</i> Failed: <i>XML_ResponseStatus</i>

A.2.82 /ISAPI/ContentMgmt/Storage/hdd

Get configuration parameters of all HDDs.

Request URI Definition

Table A-119 GET /ISAPI/ContentMgmt/Storage/hdd

Method	GET
Description	Get configuration parameters of all HDDs.
Query	None.
Request	None.
Response	Succeeded: <i>XML_hddList</i> Failed: <i>XML_ResponseStatus</i>

A.2.83 /ISAPI/ContentMgmt/Storage/hdd/<ID>

Get and set parameters for a specific HDD.

Request URI Definition

Table A-120 GET /ISAPI/ContentMgmt/Storage/hdd/<ID>

Method	GET
Description	Get parameters for a specific HDD.

Query	None.
Request	None.
Response	Succeeded: <i>XML_hdd</i> Failed: <i>XML_ResponseStatus</i>

Table A-121 PUT /ISAPI/ContentMgmt/Storage/hdd/<ID>

Method	PUT
Description	Set parameters for a specific HDD.
Query	None.
Request	<i>XML_hdd</i>
Response	<i>XML_ResponseStatus</i>

Remarks

The <ID> in the request URI refers to the HDD ID.

A.2.84 /ISAPI/ContentMgmt/Storage/hdd/<ID>/BadSectorsTest/pause

Pause checking bad sectors of a HDD.

Request URI Definition**Table A-122 PUT /ISAPI/ContentMgmt/Storage/hdd/<ID>/BadSectorsTest/pause**

Method	PUT
Description	Pause checking bad sectors of a HDD.
Query	None.
Request	None.
Response	<i>XML_ResponseStatus</i>

Remarks

The <ID> in the request URI refers to the HDD No.

A.2.85 /ISAPI/ContentMgmt/Storage/hdd/<ID>/BadSectorsTest/resume

Resume checking dad sectors of a HDD.

Request URI Definition**Table A-123 PUT /ISAPI/ContentMgmt/Storage/hdd/<ID>/BadSectorsTest/resume**

Method	PUT
Description	Resume checking bad sectors of a HDD.
Query	None.
Request	None.
Response	<i>XML_ResponseStatus</i>

Remarks

The <ID> in the request URI refers to the HDD No.

A.2.86 /ISAPI/ContentMgmt/Storage/hdd/<ID>/BadSectorsTest/start

Start checking bad sectors of a HDD.

Request URI Definition**Table A-124 PUT /ISAPI/ContentMgmt/Storage/hdd/<ID>/BadSectorsTest/start**

Method	PUT
Description	Start checking bad sectors of a HDD.
Query	None.
Request	<i>XML_BadSectorsTest</i>
Response	<i>XML_ResponseStatus</i>

Remarks

The <ID> in the request URI refers to the HDD No.

A.2.87 /ISAPI/ContentMgmt/Storage/hdd/<ID>/BadSectorsTest/status

Get bad sector checking status of a HDD.

Request URI Definition**Table A-125 GET /ISAPI/ContentMgmt/Storage/hdd/<ID>/BadSectorsTest/status**

Method	GET
Description	Get bad sector checking status of a HDD.

Query	None.
Request	None.
Response	Succeeded: <i>XML_BadSectorsTestStatus</i> Failed: <i>XML_ResponseStatus</i>

Remarks

The <ID> in the request URI refers to the HDD No.

A.2.88 /ISAPI/ContentMgmt/Storage/hdd/<ID>/BadSectorsTest/stop

Stop checking bad sectors of a HDD.

Request URI Definition

Table A-126 PUT /ISAPI/ContentMgmt/Storage/hdd/<ID>/BadSectorsTest/stop

Method	PUT
Description	Stop checking bad sectors of a HDD.
Query	None.
Request	None.
Response	<i>XML_ResponseStatus</i>

Remarks

The <ID> in the request URI refers to the HDD No.

A.2.89 /ISAPI/ContentMgmt/Storage/hdd/<ID>/encryptFormat?format=json

Format an encrypted HDD.

Request URI Definition

Table A-127 PUT /ISAPI/ContentMgmt/Storage/hdd/<ID>/encryptFormat?format=json

Method	PUT
Description	Format an encrypted HDD.
Query	security : the version No. of encryption scheme. When security does not exist, it indicates that the data is not encrypted; when security is 1, it indicates that the nodes of sensitive information in the message are encrypted in AES128 CBC mode; when security is 2, it indicates

	that the nodes of sensitive information in the message are encrypted in AES256 CBC mode. iv : the initialization vector, and it is required when security is 1 or 2. format : determine the format of request or response message.
Request	<i>JSON_EncryptFormat</i>
Response	<i>JSON_ResponseStatus</i>

Remarks

The <ID> in the request URI refers to the HDD ID.

A.2.90 /ISAPI/ContentMgmt/Storage/hdd/<ID>/encryptVerify?format=json

Verify the HDD encryption password.

Request URI Definition

Table A-128 PUT /ISAPI/ContentMgmt/Storage/hdd/<ID>/encryptVerify?format=json

Method	PUT
Description	Verify the HDD encryption password.
Query	security : the version No. of encryption scheme. When security does not exist, it indicates that the data is not encrypted; when security is 1, it indicates that the nodes of sensitive information in the message are encrypted in AES128 CBC mode; when security is 2, it indicates that the nodes of sensitive information in the message are encrypted in AES256 CBC mode. iv : the initialization vector, and it is required when security is 1 or 2. format : determine the format of request or response message.
Request	<i>JSON_EncryptVerify</i>
Response	<i>JSON_ResponseStatus</i>

Remarks

The <ID> in the request URI refers to the HDD ID.

A.2.91 /ISAPI/ContentMgmt/Storage/hdd/<ID>/formatStatus

Get the formatting status of a specific HDD.

Request URI Definition**Table A-129 GET /ISAPI/ContentMgmt/Storage/hdd/<ID>/formatStatus**

Method	GET
Description	Get the formatting status of a specific HDD.
Query	None.
Request	None.
Response	Succeeded: <i>XML_formatStatus</i> Failed: <i>XML_ResponseStatus</i>

Remarks

The <ID> in the request URI refers to the HDD ID.

A.2.92 /ISAPI/ContentMgmt/Storage/hdd/<ID>/format?formatType=

Format a specific HDD according to the specified formatting type. This URI is only available for SD card.

Request URI Definition**Table A-130 PUT /ISAPI/ContentMgmt/Storage/hdd/<ID>/format?formatType=**

Method	PUT
Description	Format a specific HDD according to the specified formatting type.
Query	formatType : formatting type, the available values are "EXT4" and "FAT32". But if you adopt the default formatting type "FAT32", this query parameter is not required in the URI.
Request	None.
Response	<i>XML_ResponseStatus</i>

Remarks

The <ID> in the request URI refers to the HDD ID.

A.2.93 /ISAPI/ContentMgmt/Storage/hdd/<ID>/logProTest?format=json

Configure preverification for log HDDs.

Request URI Definition

Table A-131 PUT /ISAPI/ContentMgmt/Storage/hdd/<ID>/logProTest?format=json

Method	PUT
Description	Configure preverification for log HDD.
Query	format: determine the format of request or response message.
Request	None.
Response	Succeeded: <i>JSON_logProTestResult</i> Failed: <i>JSON_ResponseStatus</i>

Remarks

The <ID> in the request URI refers to the HDD No.

A.2.94 /ISAPI/ContentMgmt/Storage/hdd/<ID>/SMARTTest/start

Start checking HDD status.

Request URI Definition

Table A-132 PUT /ISAPI/ContentMgmt/Storage/hdd/<ID>/SMARTTest/start

Method	PUT
Description	Start checking HDD status.
Query	None.
Request	<i>XML_HDDSMARTTest</i>
Response	<i>XML_ResponseStatus</i>

Remarks

The <ID> in the request URI refers to the HDD No.

A.2.95 /ISAPI/ContentMgmt/Storage/hdd/<ID>/SMARTTest/status

Get HDD checking status.

Request URI Definition

Table A-133 GET /ISAPI/ContentMgmt/Storage/hdd/<ID>/SMARTTest/status

Method	GET
Description	Get HDD checking status.
Query	None.
Request	None.
Response	Succeeded: <i>XML_SMARTTestStatus</i> Failed: <i>XML_ResponseStatus</i>

Remarks

The <ID> in the request URI refers to the HDD No.

A.2.96 /ISAPI/ContentMgmt/Storage/hdd/capabilities

Get HDD management capability.

Request URI Definition

Table A-134 GET /ISAPI/ContentMgmt/Storage/hdd/capabilities

Method	GET
Description	Get HDD management capability.
Query	None.
Request	None.
Response	Succeeded: <i>XML_Cap_hddList</i> Failed: <i>XML_ResponseStatus</i>

A.2.97 /ISAPI/ContentMgmt/storage/hdd/clearingSpace/capabilities?format=json

Get configuration capability of clearing HDD.

Request URI Definition

Table A-135 GET /ISAPI/ContentMgmt/storage/hdd/clearingSpace/capabilities?format=json

Method	GET
Description	Get configuration capability of clearing HDD.

Query	format: determine the format of request or response message.
Request	None
Response	Succeeded: <i>JSON_Cap_ClearingSpaceConfig</i> Failed: <i>JSON_ResponseStatus</i>

A.2.98 /ISAPI/ContentMgmt/storage/hdd/clearingSpace?format=json

Get or set parameters of clearing HDD.

Request URI Definition

Table A-136 GET /ISAPI/ContentMgmt/storage/hdd/clearingSpace?format=json

Method	GET
Description	Get parameters of clearing HDD.
Query	format: determine the format of request or response message.
Request	None
Response	Succeeded: <i>JSON_ClearingSpaceConfig</i> Failed: <i>JSON_ResponseStatus</i>

Table A-137 PUT /ISAPI/ContentMgmt/storage/hdd/clearingSpace?format=json

Method	PUT
Description	Set parameters of clearing HDD.
Query	format: determine the format of request or response message.
Request	<i>JSON_ClearingSpaceConfig</i>
Response	<i>JSON_ResponseStatus</i>

A.2.99 /ISAPI/ContentMgmt/Storage/hdd/format

Format all HDDs.

Request URI Definition

Table A-138 PUT /ISAPI/ContentMgmt/Storage/hdd/format

Method	PUT
Description	Format all HDDs.
Query	None.

Request	None.
Response	<i>XML_ResponseStatus</i>

A.2.100 /ISAPI/ContentMgmt/Storage/hdd/SMARTTest/config

Operations about the configuration of HDD SMART status detection.

Request URI Definition

Table A-139 GET /ISAPI/ContentMgmt/Storage/hdd/SMARTTest/config

Method	GET
Description	Get the parameters of HDD SMART status detection.
Query	None.
Request	None.
Response	Succeeded: <i>XML_SMARTTestConfig</i> Failed: <i>XML_ResponseStatus</i>

Table A-140 PUT /ISAPI/ContentMgmt/Storage/hdd/SMARTTest/config

Method	PUT
Description	Set the parameters of HDD SMART status detection.
Query	None.
Request	<i>XML_SMARTTestConfig</i>
Response	<i>XML_ResponseStatus</i>

A.2.101 /ISAPI/ContentMgmt/Storage/hdd/<ID>/syncStatus?format=json

Get the HDD data sync status. Now only supports data sync from HDD to SSD.

Request URI Definition

Table A-141 GET /ISAPI/ContentMgmt/Storage/hdd/<ID>/syncStatus?format=json

Method	GET
Description	Get the HDD data sync status.
Query	format: determine the format of request or response message.
Request	None.
Response	Succeeded: <i>JSON_SyncStatus</i>

	Failed: <i>JSON_ResponseStatus</i>
--	------------------------------------

Remarks

- The <ID> in the request URI refers to the HDD ID.
- You can call this URI only when the value of sub node **status** (**hddList**> **hdd**>**status**) in XML_storage is "synching" (related URI: /ISAPI/ContentMgmt/Storage).

A.2.102 /ISAPI/ContentMgmt/Storage/hdd/specifyHddFormat?format=json

Specify multiple HDDs to be formatted.

Request URI Definition

Table A-142 PUT /ISAPI/ContentMgmt/Storage/hdd/specifyHddFormat?format=json

Method	PUT
Description	Specify multiple HDDs to format.
Query	format : determine the format of request or response message.
Request	<i>JSON_HddFormatList</i>
Response	<i>JSON_ResponseStatus</i>

A.2.103 /ISAPI/ContentMgmt/Storage/quota

Get parameters of all HDD quotas.

Request URI Definition

Table A-143 GET /ISAPI/ContentMgmt/Storage/quota

Method	GET
Description	Get parameters of all HDD quotas.
Query	None.
Request	None.
Response	Succeeded: <i>XML_diskQuotaList</i> Failed: <i>XML_ResponseStatus</i>

A.2.104 /ISAPI/ContentMgmt/Storage/quota/<ID>

Get parameters of an HDD quota.

Request URI Definition

Table A-144 GET /ISAPI/ContentMgmt/Storage/quota/<ID>

Method	GET
Description	Get parameters of an HDD quota.
Query	None.
Request	None.
Response	Succeeded: <i>XML_diskQuota</i> Failed: <i>XML_ResponseStatus</i>

Table A-145 PUT /ISAPI/ContentMgmt/Storage/quota/<ID>

Method	PUT
Description	Get parameters of an HDD quota.
Query	None.
Request	<i>XML_diskQuota</i>
Response	<i>XML_ResponseStatus</i>

A.2.105 /ISAPI/ContentMgmt/Storage/ssd

Get the information of all SSDs.

Request URI Definition

Table A-146 GET /ISAPI/ContentMgmt/Storage/ssd

Method	GET
Description	Get the information of all SSDs.
Query	None.
Request	None.
Response	Succeeded: <i>XML_ssdList</i> Failed: <i>XML_ResponseStatus</i>

A.2.106 /ISAPI/ContentMgmt/Storage/ssd/<ID>

Get the specified SSD information.

Request URI Definition

Table A-147 GET /ISAPI/ContentMgmt/Storage/ssd/<ID>

Method	GET
Description	Get the specified SSD information.
Query	None.
Request	None.
Response	Succeeded: <i>XML_ssd</i> Failed: <i>XML_ResponseStatus</i>

Remarks

The <ID> in the request URI refers to the SSD ID.

A.2.107 /ISAPI/ContentMgmt/Storage/ssd/<ID>/capabilities?format=json

Get the SSD management capability.

Request URI Definition

Table A-148 GET /ISAPI/ContentMgmt/Storage/ssd/<ID>/capabilities?format=json

Method	GET
Description	Get the SSD management capability.
Query	format: determine the format of request or response message.
Request	None.
Response	Succeeded: <i>JSON_Cap_SSD</i> Failed: <i>JSON_ResponseStatus</i>

Remarks

- The <ID> in the request URI refers to the SSD ID.
- If the node **isSupportSSDManagement** is returned in the message *XML_Cap_storage* (related URI: */ISAPI/ContentMgmt/Storage/capabilities*) and its value is true, it indicates that the device supports this function.

A.2.108 /ISAPI/ContentMgmt/Storage/ssd/<ID>/SMARTTest/status

Get the SSD S.M.A.R.T (self-monitoring, analysis, and reporting technology) detection status.

Request URI Definition

Table A-149 GET /ISAPI/ContentMgmt/Storage/ssd/<ID>/SMARTTest/status

Method	GET
Description	Get the SSD S.M.A.R.T (self-monitoring, analysis, and reporting technology) detection status.
Query	None.
Request	None.
Response	Succeeded: <i>XML_SSD_SmartTestStatus</i> Failed: <i>XML_ResponseStatus</i>

Remarks

The <ID> in the request URI refers to the SSD ID.

A.2.109 /ISAPI/ContentMgmt/Storage/ssd/<ID>/SMARTTest/start

Run SSD S.M.A.R.T (self-monitoring, analysis, and reporting technology) detection.

Request URI Definition

Table A-150 PUT /ISAPI/ContentMgmt/Storage/ssd/<ID>/SMARTTest/start

Method	PUT
Description	Run SSD S.M.A.R.T (self-monitoring, analysis, and reporting technology) detection.
Query	None.
Request	<i>XML_SSDSMARTTest</i>
Response	<i>XML_ResponseStatus</i>

Remarks

- The <ID> in the request URI refers to the SSD ID.
- If the node **isSupportSSDSMARTTest** is returned in the message ***XML_RacmCap*** (related URI: ***/ISAPI/ContentMgmt/capabilities***) and its value is true, it indicates that the device supports this function.

A.2.110 /ISAPI/ContentMgmt/Storage/ssd/<ID>/upgrade/status?format=json

Get the SSD firmware upgrade progress.

Request URI Definition

Table A-151 GET /ISAPI/ContentMgmt/Storage/ssd/<ID>/upgrade/status?format=json

Method	GET
Description	Get the SSD firmware upgrade progress.
Query	format: determine the format of request or response message.
Request	None.
Response	Succeeded: <i>JSON_SSD_UpgradeStatus</i> Failed: <i>JSON_ResponseStatus</i>

Remarks

The <ID> in the request URI refers to the SSD ID.

A.2.111 /ISAPI/ContentMgmt/Storage/ssd/<ID>/upgrade?format=json

Upgrade the SSD firmware.

Request URI Definition

Table A-152 POST /ISAPI/ContentMgmt/Storage/ssd/<ID>/upgrade?format=json

Method	POST
Description	Upgrade the SSD firmware.
Query	format: determine the format of request or response message.
Request	Binary data in form format
Response	<i>JSON_ResponseStatus</i>

Remarks

- The <ID> in the request URI refers to the SSD ID.
- If the node **isSupportSSDUpgrade** is returned in the message *JSON_Cap_SSD* (related URI: */ISAPI/ContentMgmt/Storage/ssd/<ID>/capabilities?format=json*) and its value is true, it indicates that the device supports this function.

A.2.112 /ISAPI/ContentMgmt/StreamingProxy/channels/<ID>/Picture

Capture pictures via digital channel.

Request URI Definition

Table A-153 GET /ISAPI/ContentMgmt/StreamingProxy/channels/<ID>/Picture

Method	GET
Description	Capture pictures via digital channel.
Query	None
Request	None
Response	Succeeded: Captured picture data, which is transmitted over HTTP. Failed: <i>XML_ResponseStatus</i>

Remarks

- The <ID> in the request URI refers to the digital channel ID.
- The supported picture resolutions include QCIF (176 × 144), CIF (352 × 288) , and D1 (704 × 576).

A.2.113 /ISAPI/ContentMgmt/StreamingProxy/channels/<ID>/capabilities

Get the capability of digital video channel proxy.

Request URI Definition

Table A-154 GET /ISAPI/ContentMgmt/StreamingProxy/channels/<ID>/capabilities

Method	GET
Description	Get the capability of digital video channel proxy.
Query	None.
Request	None.
Response	Succeeded: <i>XML_Cap_StreamingChannel</i> Failed: <i>XML_ResponseStatus</i>

Remarks

The <ID> in the request URI refers to streaming channel ID, which is equal to (channel No. × 100 + stream type), the stream type can be: 1 (main stream), 2 (sub-stream), and so on. For example, when **ID** is 101, it indicates the main stream of channel 1, when **ID** is 302, it indicates the sub-stream of channel 3.

A.2.114 /ISAPI/ContentMgmt/StreamingProxy/channels/<ID>/PictureByUrl/capabilities?format=json

Get the capability of capturing picture in URL format

Request URI Definition

Table A-155 GET /ISAPI/ContentMgmt/StreamingProxy/channels/<ID>/PictureByUrl/capabilities?format=json

Method	GET
Description	Get the capability of capturing picture in URL format
Query	format: determine the format of request or response message.
Request	None.
Response	Succeeded: <i>JSON_Cap_CaputreDescription</i> Failed: <i>JSON_ResponseStatus</i>

Remarks

The <ID> in the request URI refers to streaming channel ID, which is equal to (channel No. × 100 + stream type), the stream type can be: 1 (main stream), 2 (sub-stream), and so on. For example, when ID is 101, it indicates the main stream of channel 1, when ID is 302, it indicates the sub-stream of channel 3.

A.2.115 /ISAPI/ContentMgmt/StreamingProxy/channels/<ID>/PictureByUrl?format=json

Capture the picture in URL format.

Request URI Definition

Table A-156 POST /ISAPI/ContentMgmt/StreamingProxy/channels/<ID>/PictureByUrl?format=json

Method	POST
Description	Capture the picture in URL format.
Query	format: determine the format of request or response message.
Request	<i>JSON_CaputreDescription</i>
Response	Succeeded: <i>JSON_CaputreResult</i> Failed: <i>JSON_ResponseStatus</i>

Remarks

- The <ID> in the request URI refers to streaming channel ID, which is equal to (channel No. × 100 + stream type), the stream type can be: 1 (main stream), 2 (sub-stream), and so on. For example, when ID is 101, it indicates the main stream of channel 1, when ID is 302, it indicates the sub-stream of channel 3.
- If the node **isSupportPictureByUrl** is returned in the message ***XML_Cap_StreamingChannel*** (related URI: ***/ISAPI/ContentMgmt/StreamingProxy/channels/<ID>/capabilities***) and its value is true, it indicates that the device supports this function.

A.2.116 /ISAPI/ContentMgmt/time/search/capabilities?format=json

Get the capability of searching for recording start and end time.

Request URI Definition

Table A-157 GET /ISAPI/ContentMgmt/time/search/capabilities?format=json

Method	GET
Description	Get the capability of searching for recording start and end time.
Query	format : determine the format of request or response message.
Request	None.
Response	Succeeded: <i>JSON_Cap_TimeSearchCond</i> Failed: <i>JSON_ResponseStatus</i>

A.2.117 /ISAPI/ContentMgmt/time/search?format=json

Search for the recording start and end time.

Request URI Definition

Table A-158 POST /ISAPI/ContentMgmt/time/search?format=json

Method	POST
Description	Search for the recording start and end time.
Query	format : determine the format of request or response message.
Request	<i>JSON_TimeSearchCond</i>
Response	Succeeded: <i>JSON_TimeSearchResult</i> Failed: <i>JSON_ResponseStatus</i>

A.3 /ISAPI/Event

A.3.1 /ISAPI/Event/capabilities

Get the device event capability.

Request URI Definition

Table A-159 GET /ISAPI/Event/capabilities

Method	GET
Description	Get the device event capability set.
Query	None.
Request	None.
Response	Succeeded: <i>XML_EventCap</i> Failed: <i>XML_ResponseStatus</i>

A.3.2 /ISAPI/Event/channels/<ID>/capabilities

Get event capabilities supported by the channel.

Request URI Definition

Table A-160 GET /ISAPI/Event/channels/<ID>/capabilities

Method	GET
Description	Get event capabilities supported by the channel.
Query	None.
Request	None.
Response	Succeeded: <i>XML_ChannelEventCap</i> Failed: <i>XML_ResponseStatus</i>

Remarks

The <ID> in the URI refers to the channel ID.

A.3.3 /ISAPI/Event/channels/capabilities

Get the event capability of all channels.

Request URL Definition**Table A-161 GET /ISAPI/Event/channels/capabilities**

Method	GET
Description	Get the event capability of all channels.
Query	None.
Request	None.
Response	Succeeded: <i>XML_ChannelEventCapList</i> Failed: <i>XML_ResponseStatus</i>

A.3.4 /ISAPI/Event/IOT/channels/<ID>/capabilities?format=json

Get the event capabilities supported by IoT device channel.

Request URI Definition**Table A-162 GET /ISAPI/Event/IOT/channels/<ID>/capabilities?format=json**

Method	GET
Description	Get the event capabilities supported by IoT device channel.
Query	format: determine the format of request or response message.
Request	None.
Response	Succeeded: <i>JSON_IOTChannelEventCap</i> Filed: <i>JSON_ResponseStatus</i>

A.3.5 /ISAPI/Event/notification/alertStream

Get the uploaded heartbeat or alarm/event information.

Request URI Definition**Table A-163 GET /ISAPI/Event/notification/alertStream**

Method	GET
Description	Get the heartbeat or uploaded alarm/event information.
Query	None.

Request	None.
Response	Option 1: <i>XML_EventNotificationAlert_AlarmEventInfo</i> or <i>XML_EventNotificationAlert_HeartbeatInfo</i> Option 2: <i>JSON_EventNotificationAlert_Alarm/EventInfo</i>  Note The messages here only show the format of alarm/event information to be uploaded. For details, refer to the corresponding alarm/event configuration chapters.

Remarks

- After calling this URI, a persistent connection is set up between the device and the platform, and the alarm or event information will be uploaded from device continuously once the alarm is triggered or event occurred.
- You can check if the XML response message is the heartbeat information according to the nodes **<eventType>** and **<eventState>**. If the values of these two node are "videoloss" and "inactive", respectively, the returned message is the heartbeat information.

A.3.6 /ISAPI/Event/notification/httpHosts

Get or set parameters of all HTTP listening servers, add a HTTP listening server, and delete all HTTP listening servers.

Request URI Definition

Table A-164 GET /ISAPI/Event/notification/httpHosts

Method	GET
Description	Get parameters of all HTTP listening servers.
Query	security : the version No. of encryption scheme. When security does not exist, it indicates that the data is not encrypted; when security is 1, it indicates that the nodes of sensitive information in the message are encrypted in AES128 CBC mode; when security is 2, it indicates that the nodes of sensitive information in the message are encrypted in AES256 CBC mode.
Request	None
Response	Succeeded: <i>XML_HttpHostNotificationList</i> Failed: <i>XML_ResponseStatus</i>

Table A-165 PUT /ISAPI/Event/notification/httpHosts

Method	PUT
Description	Set parameters of all HTTP listening servers.
Query	security : the version No. of encryption scheme. When security does not exist, it indicates that the data is not encrypted; when security is 1, it indicates that the nodes of sensitive information in the message are encrypted in AES128 CBC mode; when security is 2, it indicates that the nodes of sensitive information in the message are encrypted in AES256 CBC mode.
Request	<i>XML_HttpHostNotificationList</i>
Response	<i>XML_ResponseStatus</i>

Table A-166 POST /ISAPI/Event/notification/httpHosts

Method	POST
Description	Add a HTTP listening server.
Query	security : the version No. of encryption scheme. When security does not exist, it indicates that the data is not encrypted; when security is 1, it indicates that the nodes of sensitive information in the message are encrypted in AES128 CBC mode; when security is 2, it indicates that the nodes of sensitive information in the message are encrypted in AES256 CBC mode.
Request	<i>XML_HttpHostNotification</i>
Response	<i>XML_ResponseStatus</i>

Table A-167 DELETE /ISAPI/Event/notification/httpHosts

Method	DELETE
Description	Delete all HTTP listening servers.
Query	None
Request	None
Response	<i>XML_ResponseStatus</i>

A.3.7 /ISAPI/Event/notification/httpHosts/<ID>

Get or set the parameters of a HTTP listening server, and delete a HTTP listening server.

Request URI Definition

Table A-168 GET /ISAPI/Event/notification/httpHosts/<ID>

Method	GET
Description	Get the parameters of a HTTP listening server.
Query	security : the version No. of encryption scheme. When security does not exist, it indicates that the data is not encrypted; when security is 1, it indicates that the nodes of sensitive information in the message are encrypted in AES128 CBC mode; when security is 2, it indicates that the nodes of sensitive information in the message are encrypted in AES256 CBC mode.
Request	None
Response	Succeeded: <i>XML_HttpHostNotification</i> Failed: <i>XML_ResponseStatus</i>

Table A-169 PUT /ISAPI/Event/notification/httpHosts/<ID>

Method	PUT
Description	Set the parameters of a HTTP listening server.
Query	security : the version No. of encryption scheme. When security does not exist, it indicates that the data is not encrypted; when security is 1, it indicates that the nodes of sensitive information in the message are encrypted in AES128 CBC mode; when security is 2, it indicates that the nodes of sensitive information in the message are encrypted in AES256 CBC mode.
Request	<i>XML_HttpHostNotification</i>
Response	<i>XML_ResponseStatus</i>

Table A-170 DELETE /ISAPI/Event/notification/httpHosts/<ID>

Method	DELETE
Description	Delete a HTTP listening server.
Query	security : the version No. of encryption scheme. When security does not exist, it indicates that the data is not encrypted; when security is 1, it indicates that the nodes of sensitive information in the message are encrypted in AES128 CBC mode; when security is 2, it indicates that the nodes of sensitive information in the message are encrypted in AES256 CBC mode.

Request	None
Response	<i>XML_ResponseStatus</i>

Remarks

The <ID> in the request URI refers to the HTTP listening server No.

A.3.8 /ISAPI/Event/notification/httpHosts/<ID>/test

Check whether the HTTP listening server is working normally.

Request URI Definition

Table A-171 POST /ISAPI/Event/notification/httpHosts/<ID>/test

Method	POST
Description	Check whether the HTTP listening server is working normally.
Query	None
Request	<i>XML_HttpHostNotification</i>
Response	Succeeded: <i>XML_HttpHostTestResult</i> Failed: <i>XML_ResponseStatus</i>

Remarks

The <ID> in the request URI refers to the HTTP listening server ID.

A.3.9 /ISAPI/Event/notification/httpHosts/capabilities

Get the configuration capabilities of all HTTP listening servers.

Request URI Definition

Table A-172 GET /ISAPI/Event/notification/httpHosts/capabilities

Method	GET
Description	Get the configuration capabilities of all HTTP listening servers.
Query	None
Request	None
Response	Succeeded: <i>XML_HttpHostNotificationCap</i> Failed: <i>XML_ResponseStatus</i>

A.3.10 /ISAPI/Event/notification/subscribeEvent

Subscribe events/alarms in arming mode.

Request URI Definition

Table A-173 POST /ISAPI/Event/notification/subscribeEvent

Method	POST
Description	Subscribe events/alarms in arming mode.
Query	None.
Request	<i>XML_SubscribeEvent</i>
Response	Succeeded: <i>XML_SubscribeEventResponse</i> or <i>XML_EventNotificationAlert_SubscriptionHeartbeat</i> and alarm/event details message Failed: <i>XML_ResponseStatus</i>

Remarks

The ***XML_EventNotificationAlert_SubscriptionHeartbeat*** and alarm/event details message is uploaded repeatedly. The default time interval of uploading heartbeat information is 30s.

A.3.11 /ISAPI/Event/notification/subscribeEvent/<ID>

Get or set alarm/event subscription parameters.

Request URI Definition

Table A-174 GET /ISAPI/Event/notification/subscribeEvent/<ID>

Method	GET
Description	Get alarm/event subscription parameters.
Query	None.
Request	None.
Response	Succeeded: <i>XML_SubscribeEvent</i> Failed: <i>XML_ResponseStatus</i>

Table A-175 PUT /ISAPI/Event/notification/subscribeEvent/<ID>

Method	PUT
Description	Set alarm/event subscription parameters.

Query	None.
Request	<i>XML_SubscribeEvent</i>
Response	<i>XML_ResponseStatus</i>

Remarks

The <ID> in the request URI refers to the subscription No. which is returned by the device. After the persistent connection for receiving events or alarms in arming mode is closed, the device will release the resource used by the <ID>.

A.3.12 /ISAPI/Event/notification/subscribeEventCap

Get event/alarm subscription capability.

Request URI Definition

Table A-176 GET /ISAPI/Event/notification/subscribeEventCap

Method	GET
Description	Get event/alarm subscription capability.
Query	None.
Request	None.
Response	Succeeded: <i>XML_SubscribeEventCap</i> Failed: <i>XML_ResponseStatus</i>

A.3.13 /ISAPI/Event/notification/unSubscribeEvent

Cancel subscribing alarm/event.

Request URI Definition

Table A-177 PUT /ISAPI/Event/notification/unSubscribeEvent

Method	PUT
Description	Cancel subscribing alarm/event.
Query	None.
Request	None.
Response	<i>XML_ResponseStatus</i>

A.3.14 /ISAPI/Event/schedules/motionDetections

Get or set arming schedule of motion detection of all channels.

Request URI Definition

Table A-178 GET /ISAPI/Event/schedules/motionDetections

Method	GET
Description	Get arming schedule of motion detection of all channels.
Query	None
Request	None
Response	Succeeded: <i>XML_MotionDetectionScheduleList</i> Failed: <i>XML_ResponseStatus</i>

Table A-179 PUT /ISAPI/Event/schedules/motionDetections

Method	PUT
Description	Set arming schedule of motion detection of all channels.
Query	None
Request	<i>XML_MotionDetectionScheduleList</i>
Response	<i>XML_ResponseStatus</i>

A.3.15 /ISAPI/Event/schedules/motionDetections/<ID>

Get or set the arming schedule of motion detection by channel.

Request URI Definition

Table A-180 GET /ISAPI/Event/schedules/motionDetections/<ID>

Method	GET
Description	Get the arming schedule of motion detection by channel.
Query	None
Request	None
Response	Succeeded: <i>XML_Schedule</i> Failed: <i>XML_ResponseStatus</i>

Table A-181 PUT /ISAPI/Event/schedules/motionDetections/<ID>

Method	PUT
Description	Set the arming schedule of motion detection by channel.
Query	None
Request	<i>XML_Schedule</i>
Response	<i>XML_ResponseStatus</i>

Remarks

The <ID> in the request URI refers to the ID of channel that triggered motion detection alarm. It consists of alarm type name ("VMD") and channel ID, its format is "VMD-<channelID>", e.g., "VMD-101".

A.3.16 /ISAPI/Event/schedules/vibrationDetection/<ID>

Get or set arming schedule of vibration detection by channel.

Request URI Definition

Table A-182 GET /ISAPI/Event/schedules/vibrationDetection/<ID>

Method	GET
Description	Get arming schedule of vibration detection by channel.
Query	None.
Request	None.
Response	Succeeded: <i>XML_Schedule</i> Failed: <i>XML_ResponseStatus</i>

Table A-183 PUT /ISAPI/Event/schedules/vibrationDetection/<ID>

Method	PUT
Description	Set arming schedule of vibration detection by channel.
Query	None.
Request	<i>XML_Schedule</i>
Response	<i>XML_ResponseStatus</i>

Remarks

The <ID> in the request URI should be set in the format of "vibrationDetection-<channelID>", e.g., /ISAPI/Event/schedules/vibrationDetection/vibrationDetection-101.

A.3.17 /ISAPI/Event/schedules/<EventType>/<ID>

Operations about arming schedule of specified event type.

Request URI Definition

Table A-184 GET /ISAPI/Event/schedules/<EventType>/<ID>

Method	GET
Description	Get the arming schedule of specified event type.
Query	None.
Request	None.
Response	Succeeded: <i>XML_Schedule</i> Failed: <i>XML_ResponseStatus</i>

Table A-185 PUT /ISAPI/Event/schedules/<EventType>/<ID>

Method	PUT
Description	Set the arming schedule of specified event type.
Query	None.
Request	<i>XML_Schedule</i>
Response	<i>XML_ResponseStatus</i>

Remarks

The <EventType> in the URI refers to the specified event type.

The <ID> in the URI refers to the ID of channel, which is triggered by alarm.

A.3.18 /ISAPI/Event/triggers/hdBadBlock

Operations about the linkage configuration of the HDD bad sector detection.

Request URI Definition

Table A-186 GET /ISAPI/Event/triggers/hdBadBlock

Method	GET
Description	Get the linkage parameters of the HDD bad sector detection.
Query	None.

Request	None.
Response	Succeeded: <i>XML_EventTrigger</i> Failed: <i>XML_ResponseStatus</i>

Table A-187 PUT /ISAPI/Event/triggers/hdBadBlock

Method	PUT
Description	Set the linkage parameters of the HDD bad sector detection.
Query	None.
Request	<i>XML_EventTrigger</i>
Response	<i>XML_ResponseStatus</i>

A.3.19 /ISAPI/Event/triggers/hdImpact

Operations about the linkage configuration of the HDD impact detection.

Request URI Definition

Table A-188 GET /ISAPI/Event/triggers/hdImpact

Method	GET
Description	Get the linkage parameters of the HDD impact detection.
Query	None.
Request	None.
Response	Succeeded: <i>XML_EventTrigger</i> Failed: <i>XML_ResponseStatus</i>

Table A-189 PUT /ISAPI/Event/triggers/hdImpact

Method	PUT
Description	Set the linkage parameters of the HDD impact detection.
Query	None.
Request	<i>XML_EventTrigger</i>
Response	<i>XML_ResponseStatus</i>

A.3.20 /ISAPI/Event/triggers/highHDTemperature

Operations about the linkage configuration of the HDD high temperature detection.

Request URI Definition

Table A-190 GET /ISAPI/Event/triggers/highHDTemperature

Method	GET
Description	Get the linkage parameters of the HDD high temperature detection.
Query	None.
Request	None.
Response	Succeeded: <i>XML_EventTrigger</i> Failed: <i>XML_ResponseStatus</i>

Table A-191 PUT /ISAPI/Event/triggers/highHDTemperature

Method	PUT
Description	Set the linkage parameters of the HDD high temperature detection.
Query	None.
Request	<i>XML_EventTrigger</i>
Response	<i>XML_ResponseStatus</i>

A.3.21 /ISAPI/Event/triggers/lowHDTemperature

Operations about the linkage configuration of the HDD low temperature detection.

Request URI Definition

Table A-192 GET /ISAPI/Event/triggers/lowHDTemperature

Method	GET
Description	Get the linkage parameters of the HDD low temperature detection.
Query	None.
Request	None.
Response	Succeeded: <i>XML_EventTrigger</i> Failed: <i>XML_ResponseStatus</i>

Table A-193 PUT /ISAPI/Event/triggers/lowHDTemperature

Method	PUT
Description	Set the linkage parameters of the HDD low temperature detection.
Query	None.

Request	<i>XML_EventTrigger</i>
Response	<i>XML_ResponseStatus</i>

A.3.22 /ISAPI/Event/triggers/severeHDFailure

Operations about the linkage configuration of the HDD severe fault detection.

Request URI Definition

Table A-194 GET /ISAPI/Event/triggers/severeHDFailure

Method	GET
Description	Get the linkage parameters of the HDD severe fault detection.
Query	None.
Request	None.
Response	Succeeded: <i>XML_EventTrigger</i> Failed: <i>XML_ResponseStatus</i>

Table A-195 PUT /ISAPI/Event/triggers/severeHDFailure

Method	PUT
Description	Set the linkage parameters of the HDD severe fault detection.
Query	None.
Request	<i>XML_EventTrigger</i>
Response	<i>XML_ResponseStatus</i>

A.3.23 /ISAPI/Event/triggers/<eventType>-<ID>

Get, set, or delete the alarm linkage action by channel.

Request URI Definition

Table A-196 GET /ISAPI/Event/triggers/<eventType>-<ID>

Method	GET
Description	Get the alarm linkage action by channel.
Query	None
Request	None
Response	Succeeded: <i>XML_EventTrigger</i>

	Failed: <i>XML_ResponseStatus</i>
--	-----------------------------------

Table A-197 PUT /ISAPI/Event/triggers/<eventType>-<ID>

Method	PUT
Description	Set the alarm linkage action by channel.
Query	None
Request	<i>XML_EventTrigger</i>
Response	<i>XML_ResponseStatus</i>

Table A-198 DELETE /ISAPI/Event/triggers/<eventType>-<ID>

Method	DELETE
Description	Delete the alarm linkage action by channel.
Query	None
Request	None
Response	<i>XML_ResponseStatus</i>

Remarks

The **<eventType>** in the request URI refers to the predefined event or alarm type name, and the **<ID>** is the No. of the event detection channel. For example, if the No. of the face capture channel is 101, the "**<eventType>-<ID>**" is "faceSnap-101".

A.3.24 /ISAPI/Event/triggers/<ID>/notifications

Operations about configurations of alarm/event linkage actions.

Request URI Definition**Table A-199 GET /ISAPI/Event/triggers/<ID>/notifications**

Method	GET
Description	Get the configuration parameters of alarm/event linkage actions.
Query	None.
Request	None.
Response	Succeeded: <i>XML_EventTriggerNotificationList</i> Failed: <i>XML_ResponseStatus</i>

Table A-200 PUT /ISAPI/Event/triggers/<ID>/notifications

Method	PUT
Description	Set the configuration parameters for alarm/event linkage actions.
Query	None.
Request	<i>XML_EventTriggerNotificationList</i>
Response	<i>XML_ResponseStatus</i>

Table A-201 DELETE /ISAPI/Event/triggers/<ID>/notifications

Method	DELETE
Description	Delete the configuration parameters of alarm/event linkage actions.
Query	None.
Request	None.
Response	<i>XML_ResponseStatus</i>

Remarks

The <ID> in the request URI refers to No. of alarm triggered channel.

A.3.25 /ISAPI/Event/triggersCap/IOT

Get the alarm linkage configuration capability of IoT device.

Request URI Definition

Table A-202 GET /ISAPI/Event/triggersCap/IOT

Method	GET
Description	Get the alarm linkage configuration capability of IoT device.
Query	None.
Request	None.
Response	<i>XML_IOTTriggersCap</i>

A.3.26 /ISAPI/Event/triggers/notifications/AudioAlarm/<audioID>/test?format=json

Audio test.

Request URI Definition

Table A-203 PUT /ISAPI/Event/triggers/notifications/AudioAlarm/<audioID>/test?format=json

Method	PUT
Description	Audio test.
Query	format: determine the format of request or response message.
Request	None.
Response	<i>JSON_ResponseStatus</i>

Remarks

The <audioID> in the URI refers to audio ID.

A.3.27 /ISAPI/Event/triggers/notifications/AudioAlarm/capabilities?format=json

Get audible warning configuration capability.

Request URI Definition

Table A-204 GET /ISAPI/Event/triggers/notifications/AudioAlarm/capabilities?format=json

Method	GET
Description	Get audible warning configuration capability.
Query	format: determine the format of request or response message.
Request	None.
Response	Succeeded: <i>JSON_AudioAlarmCap</i> Failed: <i>JSON_ResponseStatus</i>

A.3.28 /ISAPI/Event/triggers/notifications/AudioAlarm/customAudio/<customAudioID>?format=json

Download or delete custom alarm audio file.

Request URI Definition

Table A-205 GET /ISAPI/Event/triggers/notifications/AudioAlarm/customAudio/<customAudioID>?format=json

Method	GET
Description	Download custom alarm audio file.

Query	format: determine the format of request or response message.
Request	None.
Response	Succeeded: The custom alarm audio file information is returned. Failed: <i>JSON_ResponseStatus</i>

Table A-206 DELETE /ISAPI/Event/triggers/notifications/AudioAlarm/customAudio/<customAudioID>?format=json

Method	DELETE
Description	Delete custom audio file.
Query	format: determine the format of request or response message.
Request	None.
Response	<i>JSON_ResponseStatus</i>

Remarks

<customAudioID> in the URI refers to custom alarm audio ID and the value can be 1 to 3.

A.3.29 /ISAPI/Event/triggers/notifications/AudioAlarm/customAudio?format=json

Import custom alarm audio file.

Request URI Definition

Table A-207 POST /ISAPI/Event/triggers/notifications/AudioAlarm/customAudio?format=json

Method	POST
Description	Import custom alarm audio file.
Query	format: determine the format of request or response message. type: audio file type: url, bianry.
Request	<i>JSON_ImportCustomAudio</i>
Response	Succeeded: <i>JSON_CustomAudioID</i> Failed: <i>JSON_ResponseStatus</i>

A.3.30 /ISAPI/Event/triggers/notifications/AudioAlarm/customAudioInfo/<customAudioID>?format=json

Get or edit custom alarm audio file information.

Request URI Definition

Table A-208 GET /ISAPI/Event/triggers/notifications/AudioAlarm/customAudioInfo/<customAudioID>?format=json

Method	GET
Description	Get custom alarm audio file information.
Query	format: determine the format of request or response message.
Request	None.
Response	Succeeded: <i>JSON_CustomAudioInfo</i> Failed: <i>JSON_ResponseStatus</i>

Table A-209 PUT /ISAPI/Event/triggers/notifications/AudioAlarm/customAudioInfo/<customAudioID>?format=json

Method	PUT
Description	Edit custom audio file information.
Query	format: determine the format of request or response message.
Request	<i>JSON_CustomAudioInfo</i>
Response	<i>JSON_ResponseStatus</i>

Remarks

The <customAudioID> in the URI refers to custom alarm audio ID and the value can be 1 to 3.

A.3.31 /ISAPI/Event/triggers/notifications/AudioAlarm/customAudioInfo?format=json

Get the custom alarm audio file.

Request URI Definition

Table A-210 GET /ISAPI/Event/triggers/notifications/AudioAlarm/customAudioInfo?format=json

Method	GET
Description	Get the custom alarm audio file.
Query	format: determine the format of request or response message.
Request	None.
Response	Succeeded: <i>JSON_GetCustomAudio</i> Failed: <i>JSON_ResponseStatus</i>

A.3.32 /ISAPI/Event/triggers/notifications/AudioAlarm/customAudioTypeInfo/channels?format=json

Get encoding format information of custom audio file in a batch.

Request URI Definition

Table A-211 GET /ISAPI/Event/triggers/notifications/AudioAlarm/customAudioTypeInfo/channels?format=json

Method	GET
Description	Get encoding format information of custom audio file in a batch.
Query	format: determine the format of request or response message.
Request	None.
Response	Succeeded: <i>JSON_CustomAudioTypeInfoList</i> Failed: <i>JSON_ResponseStatus</i>

A.3.33 /ISAPI/Event/triggers/notifications/AudioAlarm?format=json

Get or set audible warning parameters.

Request URI Definition

Table A-212 GET /ISAPI/Event/triggers/notifications/AudioAlarm?format=json

Method	GET
Description	Get parameters of audible warning.
Query	format: determine the format of request or response message. alarmType: alarm type, including "behavior"-behavior analysis, "thermometry"-temperature measurement, "dynamicFire"-fire detection, "smokingMode"-smoke detection, "noMaskDetection"-no wearing mask detection.
Request	None.
Response	Succeeded: <i>JSON_AudioAlarm</i> Failed: <i>JSON_ResponseStatus</i>



Note

For example, if you want to get the audible warning parameters of thermometry, the request URI is: GET /ISAPI/Event/triggers/notifications/AudioAlarm?format=json&alarmType=thermometry

Table A-213 PUT /ISAPI/Event/triggers/notifications/AudioAlarm?format=json

Method	PUT
Description	Set parameters of audible warning.
Query	format : determine the format of request or response message.
Request	<i>JSON_AudioAlarm</i>
Response	<i>JSON_ResponseStatus</i>

A.3.34 /ISAPI/Event/triggers/notifications/whiteLightAlarm?format=json

Operations about configuration of supplement light alarm linkage.

Request URI Definition

Table A-214 GET /ISAPI/Event/triggers/notifications/whiteLightAlarm?format=json

Method	GET
Description	Get parameters of supplement light alarm linkage.
Query	format : determine the format of request or response message.
Request	None.
Response	Succeeded: <i>JSON_WhiteLightAlarm</i> Failed: <i>JSON_ResponseStatus</i>

Table A-215 PUT /ISAPI/Event/triggers/notifications/whiteLightAlarm?format=json

Method	PUT
Description	Set parameters of supplement light alarm linkage.
Query	format : determine the format of request or response message.
Request	<i>JSON_WhiteLightAlarm</i>
Response	<i>JSON_WhiteLightAlarm</i>

A.3.35 /ISAPI/Event/triggers/notifications/whiteLightAlarm/capabilities?format=json

Get the configuration capability of supplement light alarm linkage.

Request URI Definition

Table A-216 GET /ISAPI/Event/triggers/notifications/whiteLightAlarm/capabilities?format=json

Method	GET
Description	Get the configuration capability of supplement light alarm linkage.
Query	format : determine the format of request or response message.
Request	None.
Response	<i>JSON_WhiteLightAlarmCap</i>

A.3.36 /ISAPI/Event/triggersCap

Get alarm linkage capability.

Request URI Definition

Table A-217 GET /ISAPI/Event/triggersCap

Method	GET
Description	Get alarm linkage capability.
Query	None
Request	None
Response	Succeeded: <i>XML_EventTriggersCap</i> Failed: <i>XML_ResponseStatus</i>

A.3.37 http://<ipAddress>:<portNo>/<url>

HTTP listening sever sends alarm information to alarm center.

Request URL Definition

Table A-218 POST http://<ipAddress>:<portNo>/<url>

Method	POST
Description	HTTP listening sever sends alarm information to alarm center.
Query	None
Request	None
Response	Succeeded: <i>XML_EventNotificationAlert_AlarmEventInfo</i> or <i>JSON_EventNotificationAlert_Alarm/EventInfo</i>

	Failed: <i>XML_ResponseStatus</i>
--	--

Remarks

- The <**ipAddress**> in the request URL refers to the IP address or domain name of HTTP listening server, the <**portNo**> is the port No. of HTTP listening server, and the <**url**> represents the streaming URL, which is configured via the HTTP listening server.
- The default port No. is 80, so the request URL without port No. is also valid.

A.4 /ISAPI/Image

A.4.1 /ISAPI/Image/channels

Operations about image configuration of all channels.

Request URI Definition

Table A-219 GET /ISAPI/Image/channels

Method	GET
Description	Get image parameters of all channels.
Query	None.
Request	None.
Response	Succeeded: <i>XML_ImageChannellist</i> Failed: <i>XML_ResponseStatus</i>

Table A-220 PUT /ISAPI/Image/channels

Method	GET
Description	Set image parameters of all channels.
Query	None.
Request	<i>XML_ImageChannellist</i>
Response	<i>XML_ResponseStatus</i>

A.4.2 /ISAPI/Image/channels/<ID>

Operations about image configuration of a specific channel.

Request URI Definition**Table A-221 GET /ISAPI/Image/channels/<ID>**

Method	GET
Description	Get image parameters of a specific channel.
Query	None.
Request	None.
Response	Succeeded: <i>XML_ImageChannel</i> Failed: <i>XML_ResponseStatus</i>

Table A-222 PUT /ISAPI/Image/channels/<ID>

Method	PUT
Description	Set image parameters of a specific channel.
Query	None.
Request	<i>XML_ImageChannel</i>
Response	<i>XML_ResponseStatus</i>

Remarks

The <ID> in the request URI refers to the channel ID.

A.4.3 /ISAPI/Image/channels/<ID>/BLC

Operations about BLC (Blacklist Compensation) configuration of a specific channel.

Request URI Definition**Table A-223 GET /ISAPI/Image/channels/<ID>/BLC**

Method	GET
Description	Get BLC (Blacklist Compensation) parameters of a specific channel.
Query	None.
Request	None.
Response	Succeeded: <i>XML_BLC</i> Failed: <i>XML_ResponseStatus</i>

Table A-224 PUT /ISAPI/Image/channels/<ID>/BLC

Method	PUT
Description	Set BLC (Blacklist Compensation) parameters of a specific channel.
Query	None.
Request	<i>XML_BLC</i>
Response	<i>XML_ResponseStatus</i>

Remarks

The <ID> in the request URI refers to the channel ID.

A.4.4 /ISAPI/Image/channels/<ID>/capabilities

Get image configuration capability of a specific channel.

Request URI Definition**Table A-225 GET /ISAPI/Image/channels/<ID>/capabilities**

Method	GET
Description	Get image configuration capability of a specific channel.
Query	None.
Request	None.
Response	Succeeded: <i>XML_Cap_ImageChannel</i> Failed: <i>XML_ResponseStatus</i>

Remarks

The <ID> in the request URI refers to the channel No.

A.4.5 /ISAPI/Image/channels/<ID>/capturemode

Operations about video input mode configuration of a specific channel.

Request URI Definition**Table A-226 GET /ISAPI/Image/channels/<ID>/capturemode**

Method	GET
Description	Get video input modes of a specific channel.
Query	None.

Request	None.
Response	Succeeded: <i>XML_CaptureMode</i> Failed: <i>XML_ResponseStatus</i>

Table A-227 PUT /ISAPI/Image/channels/<ID>/capturemode

Method	PUT
Description	Set video input mode of a specific channel.
Query	None.
Request	<i>XML_CaptureMode</i>
Response	<i>XML_ResponseStatus</i>

Remarks

The <ID> in the request URI refers to the channel ID.

A.4.6 /ISAPI/Image/channels/<ID>/color

Operations about the image adjustment parameters of a specific channel.

Request URI Definition**Table A-228 GET /ISAPI/Image/channels/<ID>/color**

Method	GET
Description	Get the image adjustment parameters of a specific channel.
Query	None.
Request	None.
Response	Succeeded: <i>XML_Color</i> Failed: <i>XML_ResponseStatus</i>

Table A-229 PUT /ISAPI/Image/channels/<ID>/color

Method	PUT
Description	Set the image adjustment parameters of a specific channel.
Query	None.
Request	<i>XML_Color</i>
Response	<i>XML_ResponseStatus</i>

Remarks

The <ID> in the request URI refers to the channel No.

A.4.7 /ISAPI/Image/channels/<ID>/color/capabilities

Get the image adjustment capability of a specified channel.

Request URI Definition**Table A-230 GET /ISAPI/Image/channels/<ID>/color/capabilities**

Method	GET
Description	Get the image adjustment capability of a specified channel.
Query	None.
Request	None.
Response	<i>XML_Cap_Color</i>

Remarks

The <ID> in the request URI refers to the channel No.

A.4.8 /ISAPI/Image/channels/<ID>/corridor

Operations about image rotate mode configuration of a specific channel.

Request URI Definition**Table A-231 GET /ISAPI/Image/channels/<ID>/corridor**

Method	GET
Description	Get image rotate mode of a specific channel.
Query	None.
Request	None.
Response	Succeeded: <i>XML_corridor</i> Failed: <i>XML_ResponseStatus</i>

Table A-232 PUT /ISAPI/Image/channels/<ID>/corridor

Method	PUT
Description	Get image rotate mode of a specific channel.
Query	None.

Request	<i>XML_corridor</i>
Response	<i>XML_ResponseStatus</i>

Remarks

The <ID> in the request URI refers to the channel ID.

A.4.9 /ISAPI/Image/channels/<ID>/dehaze

Operations about defog mode configuration of a specific channel.

Request URI Definition**Table A-233 GET /ISAPI/Image/channels/<ID>/dehaze**

Method	GET
Description	Get defog mode parameters of a specific channel.
Query	None.
Request	None.
Response	Succeeded: <i>XML_De haze</i> Failed: <i>XML_ResponseStatus</i>

Table A-234 PUT /ISAPI/Image/channels/<ID>/dehaze

Method	PUT
Description	Set defog mode parameters of a specific channel.
Query	None.
Request	<i>XML_De haze</i>
Response	<i>XML_ResponseStatus</i>

Remarks

The <ID> in the request URI refers to the channel ID.

A.4.10 /ISAPI/Image/channels/<ID>/EPTZ

Get or set the e-PTZ configuration.

Request URI Definition**Table A-235 GET /ISAPI/Image/channels/<ID>/EPTZ**

Method	GET
Description	Get the e-PTZ parameters.
Query	None.
Request	None.
Response	Succeeded: <i>XML_EPTZ</i> Failed: <i>XML_ResponseStatus</i>

Table A-236 PUT /ISAPI/Image/channels/<ID>/EPTZ

Method	PUT
Description	Set the e-PTZ parameters.
Query	None.
Request	<i>XML_EPTZ</i>
Response	<i>XML_ResponseStatus</i>

Remarks

The <ID> in the request URI refers to the channel No.

A.4.11 /ISAPI/Image/channels/<ID>/EPTZ/mode/capabilities?format=json

Get the capability of switching e-PTZ mode.

Request URI Definition**Table A-237 GET /ISAPI/Image/channels/<ID>/EPTZ/mode/capabilities?format=json**

Method	GET
Description	Get the capability of switching e-PTZ mode.
Query	format: determine the format of request or response message.
Request	None.
Response	Succeeded: <i>JSON_EPTZModeCap</i> Failed: <i>JSON_ResponseStatus</i>

Remarks

The <ID> in the request URI refers to the channel No.

A.4.12 /ISAPI/Image/channels/<ID>/EPTZ/mode?format=json

Get or set the configuration for switching e-PTZ mode.

Request URI Definition

Table A-238 GET /ISAPI/Image/channels/<ID>/EPTZ/mode?format=json

Method	GET
Description	Get the parameters of switching e-PTZ mode.
Query	format : determine the format of request or response message.
Request	None.
Response	Succeeded: <i>JSON_EPTZMode</i> Failed: <i>JSON_ResponseStatus</i>

Table A-239 PUT /ISAPI/Image/channels/<ID>/EPTZ/mode?format=json

Method	PUT
Description	Set the parameters for switching e-PTZ mode.
Query	format : determine the format of request or response message.
Request	<i>JSON_EPTZMode</i>
Response	<i>JSON_ResponseStatus</i>

Remarks

The <ID> in the request URI refers to the channel No.

A.4.13 /ISAPI/Image/channels/<ID>/exposure

Operations about exposure mode configuration of a specific channel.

Request URI Definition

Table A-240 GET /ISAPI/Image/channels/<ID>/exposure

Method	GET
Description	Get the exposure mode of a specific channel.
Query	None.
Request	None.
Response	Succeeded: <i>XML_Exposure</i>

	Failed: <i>XML_ResponseStatus</i>
--	-----------------------------------

Table A-241 PUT /ISAPI/Image/channels/<ID>/exposure

Method	PUT
Description	Set the exposure mode of a specific channel.
Query	None.
Request	<i>XML_Exposure</i>
Response	<i>XML_ResponseStatus</i>

Remarks

The <ID> in the request URI refers to the channel ID.

A.4.14 /ISAPI/Image/channels/<ID>/focusConfiguration

Get or set the focus parameters of a specified channel.

Request URI Definition**Table A-242 GET /ISAPI/Image/channels/<ID>/focusConfiguration**

Method	GET
Description	Get the focus parameters of a specified channel.
Query	None.
Request	None.
Response	<i>XML_FocusConfiguration</i>

Table A-243 PUT /ISAPI/Image/channels/<ID>/focusConfiguration

Method	PUT
Description	Set the focus parameters of a specified channel.
Query	None.
Request	<i>XML_FocusConfiguration</i>
Response	<i>XML_ResponseStatus</i>

A.4.15 /ISAPI/Image/channels/<ID>/focusConfiguration/capabilities

Get the focus configuration capability.

Request URI Definition**Table A-244 GET /ISAPI/Image/channels/<ID>/focusConfiguration/capabilities**

Method	GET
Description	Get the focus configuration capability.
Query	None.
Request	None.
Response	Succeeded: <i>XML_Cap_FocusConfiguration</i> Failed: <i>XML_ResponseStatus</i>

Remarks

The <ID> in the request URI refers to the channel No.

A.4.16 /ISAPI/Image/channels/<ID>/gain

Operations about gain configuration in exposure of a specific channel.

Request URI Definition**Table A-245 GET /ISAPI/Image/channels/<ID>/gain**

Method	GET
Description	Get gain parameters in exposure of a specific channel.
Query	None.
Request	None.
Response	Succeeded: <i>XML_Gain</i> Failed: <i>XML_ResponseStatus</i>

Table A-246 PUT /ISAPI/Image/channels/<ID>/gain

Method	PUT
Description	Set gain parameters in exposure of a specific channel.
Query	None.
Request	<i>XML_Gain</i>
Response	<i>XML_ResponseStatus</i>

Remarks

- The <ID> in the request URI refers to the channel ID.
- PUT method is only valid when the node <ExposureType> in the message *XML_Exposure* is set to "GainFirst".

A.4.17 /ISAPI/Image/channels/<ID>/imageCap

Get the image capability of a specific channel.

Request URI Definition

Table A-247 GET /ISAPI/Image/channels/<ID>/imageCap

Method	GET
Description	Get the image capability of a specific channel.
Query	None.
Request	NULL.
Response	<i>XML_ImageCap</i>

A.4.18 /ISAPI/Image/channels/<ID>/ImageFlip

Operations about the image flipping status of a specific channel.

Request URI Definition

Table A-248 GET /ISAPI/Image/channels/<ID>/ImageFlip

Method	GET
Description	Get the image flipping status of a specific channel.
Query	None.
Request	None.
Response	Succeeded: <i>XML_ImageFlip</i> Failed: <i>XML_ResponseStatus</i>

Table A-249 PUT /ISAPI/Image/channels/<ID>/ImageFlip

Method	PUT
Description	Update the image flipping status of a specific channel.
Query	None.

Request	<i>XML_ImageFlip</i>
Response	<i>XML_ResponseStatus</i>

Remarks

The <ID> in the request URI refers to the channel No.

A.4.19 /ISAPI/Image/channels/<ID>/imageMode/<ID>

Get the default configuration parameters of a specific image mode of a specific channel.

Request URI Definition

Table A-250 GET /ISAPI/Image/channels/<ID>/imageMode/<ID>

Method	GET
Description	Get the default configuration parameters of a specific image mode of a specific channel.
Query	None.
Request	None.
Response	<i>XML_ImageMode</i>

Remarks

This function is not supported by the traffic camera.

A.4.20 /ISAPI/Image/channels/<ID>/imageModes

Get the default image mode parameters of a specific channel.

Request URI Definition

Table A-251 GET /ISAPI/Image/channels/<ID>/imageModes

Method	GET
Description	Get the default image mode parameters of a specific channel.
Query	None.
Request	None.
Response	<i>XML_ImageModeList</i>

Remarks

- This API is used to get default image mode configuration parameters. For different scenes, you are recommended to set different image mode parameters for the device.
- The image mode is not supported by the traffic camera.

A.4.21 /ISAPI/Image/channels/<ID>/IrcutFilter

Operations about day/night auto switch configuration of a specific channel.

Request URI Definition**Table A-252 GET /ISAPI/Image/channels/<ID>/IrcutFilter**

Method	GET
Description	Get day/night auto switch parameters of a specific channel.
Query	None.
Request	None.
Response	Succeeded: <i>XML_IrcutFilter</i> Failed: <i>XML_ResponseStatus</i>

Table A-253 PUT /ISAPI/Image/channels/<ID>/IrcutFilter

Method	PUT
Description	Set day/night auto switch parameters of a specific channel.
Query	None.
Request	<i>XML_IrcutFilter</i>
Response	<i>XML_ResponseStatus</i>

Remarks

The <ID> in the request URI refers to the channel ID.

A.4.22 /ISAPI/Image/channels/<ID>/ISPMode

Operations about the day/night mode configuration of a specific channel.

Request URI Definition

Table A-254 GET /ISAPI/Image/channels/<ID>/ISPMode

Method	GET
Description	Get the day/night mode parameters of a specific channel.
Query	None.
Request	None.
Response	Succeeded: <i>XML_ISPMode</i> Failed: <i>XML_ResponseStatus</i>

Table A-255 PUT /ISAPI/Image/channels/<ID>/ISPMode

Method	PUT
Description	Set the day/night mode parameters of a specific channel.
Query	None.
Request	<i>XML_ISPMode</i>
Response	<i>XML_ResponseStatus</i>

Remarks

The <ID> in the request URI refers to the channel No.

A.4.23 /ISAPI/Image/channels/<ID>/lensDistortionCorrection

Get or set parameters of image distortion correction.

Request URI Definition

Table A-256 GET /ISAPI/Image/channels/<ID>/lensDistortionCorrection

Method	GET
Description	Get parameters of image distortion correction.
Query	None.
Request	None.
Response	Succeeded: <i>XML_LensDistortionCorrection</i> Failed: <i>XML_ResponseStatus</i>

Table A-257 PUT /ISAPI/Image/channels/<ID>/lensDistortionCorrection

Method	PUT
Description	Set parameters of image distortion correction.
Query	None.
Request	<i>XML_LensDistortionCorrection</i>
Response	<i>XML_ResponseStatus</i>

Remarks

The <ID> in the request URI refers to the video input channel ID.

A.4.24 /ISAPI/Image/channels/<ID>/lensDistortionCorrection/capabilities

Get capability of image distortion correction.

Request URI Definition

Table A-258 GET /ISAPI/Image/channels/<ID>/lensDistortionCorrection/capabilities

Method	GET
Description	Get capability of image distortion correction.
Query	None.
Request	None.
Response	Succeeded: <i>XML_Cap_LensDistortionCorrection</i> Failed: <i>XML_ResponseStatus</i>

Remarks

The <ID> in the request URI refers to the video input channel ID.

A.4.25 /ISAPI/Image/channels/<ID>/mountingScenario

Operations about the mounting scenario mode of a camera.

Request URI Definition

Table A-259 GET /ISAPI/Image/channels/<ID>/mountingScenario

Method	GET
Description	Get the mounting scenario mode of a camera.
Query	None.

Request	None.
Response	<i>XML_MountingScenario</i>

Table A-260 PUT /ISAPI/Image/channels/<ID>/mountingScenario

Method	PUT
Description	Set the mounting scenario mode of a camera.
Query	None.
Request	<i>XML_MountingScenario</i>
Response	<i>XML_ResponseStatus</i>

Remarks

- The <ID> in the request URI refers to channel ID.
- The image parameters should be obtained again when mounting scenario mode changes.

A.4.26 /ISAPI/Image/channels/<ID>/mountingScenario/capabilities

Get the capability of configurations of mounting scenario mode.

Request URI Definition**Table A-261 GET /ISAPI/Image/channels/<ID>/mountingScenario/capabilities**

Method	GET
Description	Get the capability of configurations of mounting scenario mode.
Query	None.
Request	None.
Response	<i>XML_Cap_MountingScenario</i>

Remarks

The <ID> in the request URI refers to channel ID.

A.4.27 /ISAPI/Image/channels/<ID>/noiseReduce

Operations about 3D DNR (Digital Noise Reduction) configuration in auto mode of a specific channel.

Request URI Definition

Table A-262 GET /ISAPI/Image/channels/<ID>/noiseReduce

Method	GET
Description	Get 3D DNR (Digital Noise Reduction) parameters in auto mode of a specific channel.
Query	None.
Request	None.
Response	Succeeded: <i>XML_NoiseReduce</i> Failed: <i>XML_ResponseStatus</i>

Table A-263 PUT /ISAPI/Image/channels/<ID>/noiseReduce

Method	PUT
Description	Set 3D DNR (Digital Noise Reduction) parameters in auto mode of a specific channel.
Query	None.
Request	<i>XML_NoiseReduce</i>
Response	<i>XML_ResponseStatus</i>

Remarks

The <ID> in the request URI refers to the channel ID.

A.4.28 /ISAPI/Image/channels/<ID>/Palettes

Get or set the palettes parameters.

Request URI Definition

Table A-264 GET /ISAPI/Image/channels/<ID>/Palettes

Method	GET
Description	Get the palettes configuration parameters.
Query	None.
Request	None.
Response	<i>XML_Palettes</i>

Table A-265 PUT /ISAPI/Image/channels/<ID>/Palettes

Method	PUT
Description	Set the palettes parameters.
Query	None.
Request	None.
Response	<i>XML_ResponseStatus</i>

A.4.29 /ISAPI/Image/channels/<ID>/powerLineFrequency

Operations about image standard configuration of a specific channel.

Request URI Definition

Table A-266 GET /ISAPI/Image/channels/<ID>/powerLineFrequency

Method	GET
Description	Get image standard parameters of a specific channel.
Query	None.
Request	None.
Response	Succeeded: <i>XML_powerLineFrequency</i> Failed: <i>XML_ResponseStatus</i>

Table A-267 PUT /ISAPI/Image/channels/<ID>/powerLineFrequency

Method	PUT
Description	Set image standard parameters of a specific channel.
Query	None.
Request	<i>XML_powerLineFrequency</i>
Response	<i>XML_ResponseStatus</i>

Remarks

The <ID> in the request URI refers to the channel ID.

A.4.30 /ISAPI/Image/channels/<ID>/regionalFocus

Focus on a specific region during live view.

Request URI Definition

Table A-268 GET /ISAPI/Image/channels/<ID>/regionalFocus

Method	PUT
Description	Focus on a specific region during live view.
Query	None.
Request	<i>XML_RegionalFocus</i>
Response	<i>XML_ResponseStatus</i>

Remarks

The <ID> in the URI is the channel ID.

A.4.31 /ISAPI/Image/channels/<ID>/reset

Reset the image parameters of a specific channel.

Request URI Definition

Table A-269 PUT /ISAPI/Image/channels/<ID>/reset

Method	PUT
Description	Reset the image parameters of a specific channel.
Query	None.
Request	None.
Response	<i>XML_ResponseStatus</i>

Remarks

The <ID> in the request URI refers to the channel ID.

A.4.32 /ISAPI/Image/channels/<ID>/restore

Restore the image parameters of a specific channel to default settings.

Request URI Definition

Table A-270 PUT /ISAPI/Image/channels/<ID>/restore

Method	PUT
Description	Restore the image parameters of a specific channel to default settings.
Query	None.
Request	None.
Response	<i>XML_ResponseStatus</i>

Remarks

The <ID> in the request URI refers to the channel ID.

A.4.33 /ISAPI/Image/channels/<ID>/sharpness

Operations about the sharpness parameters of s specific channel.

Request URI Definition

Table A-271 GET /ISAPI/Image/channels/<ID>/sharpness

Method	GET
Description	Get the sharpness parameters of s specific channel.
Query	None.
Request	None.
Response	Succeeded: <i>XML_Sharpness</i> Failed: <i>XML_ResponseStatus</i>

Table A-272 PUT /ISAPI/Image/channels/<ID>/sharpness

Method	PUT
Description	Set the sharpness parameters of s specific channel.
Query	None.
Request	<i>XML_Sharpness</i>
Response	<i>XML_ResponseStatus</i>

Remarks

The <ID> in the request URI refers to the channel ID.

A.4.34 /ISAPI/Image/channels/<ID>/shutter

Operations about shutter configuration in exposure of a specific channel.

Request URI Definition

Table A-273 GET /ISAPI/Image/channels/<ID>/shutter

Method	GET
Description	Get shutter parameters in exposure of a specific channel.
Query	None.
Request	None.
Response	Succeeded: <i>XML_Shutter</i> Failed: <i>XML_ResponseStatus</i>

Table A-274 PUT /ISAPI/Image/channels/<ID>/shutter

Method	PUT
Description	Set shutter parameters in exposure of a specific channel.
Query	None.
Request	<i>XML_Shutter</i>
Response	<i>XML_ResponseStatus</i>

Remarks

- The <ID> in the request URI refers to the channel ID.
- PUT method is only valid when the node <ExposureType> in the message *XML_Exposure* is set to "ShutterFirst".

A.4.35 /ISAPI/Image/channels/<ID>/supplementLight

Get or set supplement light parameters by channel.

Request URI Definition

Table A-275 GET /ISAPI/Image/channels/<ID>/supplementLight

Method	GET
Description	Get supplement light parameters by channel.
Query	None.

Request	None.
Response	Succeeded: <i>XML_SupplementLight</i> Failed: <i>XML_ResponseStatus</i>

Table A-276 PUT /ISAPI/Image/channels/<ID>/supplementLight

Method	PUT
Description	Set supplement light parameters by channel.
Query	None.
Request	<i>XML_SupplementLight</i>
Response	<i>XML_ResponseStatus</i>

Remarks

The <ID> in the request URI refers to the channel ID.

A.4.36 /ISAPI/Image/channels/<ID>/supplementLight/capabilities

Get supplement light configuration capability by channel.

Request URI Definition

Table A-277 GET /ISAPI/Image/channels/<ID>/supplementLight/capabilities

Method	GET
Description	Get supplement light configuration capability by channel.
Query	None.
Request	None.
Response	Succeeded: <i>XML_Cap_SupplementLight</i> Failed: <i>XML_ResponseStatus</i>

Remarks

The <ID> in the request URI refers to the channel ID.

A.4.37 /ISAPI/Image/channels/<ID>/targetEnhancement

Get or set the target enhancement parameters.

Request URI Definition

Table A-278 GET /ISAPI/Image/channels/<ID>/targetEnhancement

Method	GET
Description	Get the target enhancement parameters.
Query	None.
Request	None.
Response	Succeeded: <i>XML_TargetEnhancement</i> Failed: <i>XML_ResponseStatus</i>

Table A-279 PUT /ISAPI/Image/channels/<ID>/targetEnhancement

Method	PUT
Description	Set the target enhancement parameters.
Query	None.
Request	<i>XML_TargetEnhancement</i>
Response	<i>XML_ResponseStatus</i>

Remarks

- This URI is used in combination with URI */ISAPI/Image/channels/<ID>* .
- The <ID> in the request URI refers to the channel No.

A.4.38 /ISAPI/Image/channels/<ID>/targetEnhancement/capabilities

Get the target enhancement capability.

Request URI Definition

Table A-280 GET /ISAPI/Image/channels/<ID>/targetEnhancement/capabilities

Method	GET
Description	Get the target enhancement capability.
Query	None.
Request	None.
Response	Succeeded: <i>XML_TargetEnhancementCap</i> Failed: <i>XML_ResponseStatus</i>

Remarks

The <ID> in the request URI refers to the channel No.

A.4.39 /ISAPI/Image/channels/<ID>/tempRange

Operation about temperature range parameters of the specified channel.

Request URI Definition**Table A-281 GET /ISAPI/Image/channels/<ID>/tempRange**

Method	GET
Description	Get the temperature range parameters.
Query	None.
Request	None.
Response	Succeeded: <i>XML_tempRange</i> Failed: <i>XML_ResponseStatus</i>

Table A-282 PUT /ISAPI/Image/channels/<ID>/tempRange

Method	PUT
Description	Set the temperature range parameters.
Query	None.
Request	None.
Response	Succeeded: <i>XML_tempRange</i> Failed: <i>XML_ResponseStatus</i>

Remarks

The <ID> in the request URI refers to the channel No.

A.4.40 /ISAPI/Image/channels/<ID>/tempRange/capabilities

Get the temperature range configuration capability of a specified channel.

Request URI Definition

Table A-283 GET

Method	GET
Description	Get the temperature range configuration capability of a specified channel.
Query	None.
Request	None.
Response	Succeeded: <i>XML_Cap_tempRange</i> Failed: <i>XML_ResponseStatus</i>

Remarks

The <ID> in the request URI refers to the channel No.

A.4.41 /ISAPI/Image/channels/<ID>/WDR

Operations about the WDR (Wide Dynamic Range) configuration of a specific channel.

Request URI Definition

Table A-284 GET /ISAPI/Image/channels/<ID>/WDR

Method	GET
Description	Get WDR (Wide Dynamic Range) parameters of a specific channel.
Query	None.
Request	None.
Response	Succeeded: <i>XML_WDR</i> Failed: <i>XML_ResponseStatus</i>

Table A-285 PUT /ISAPI/Image/channels/<ID>/WDR

Method	PUT
Description	Set WDR (Wide Dynamic Range) parameters of a specific channel.
Query	None.
Request	<i>XML_WDR</i>
Response	<i>XML_ResponseStatus</i>

Remarks

The <ID> in the request URI refers to the channel ID.

A.4.42 /ISAPI/Image/channels/<ID>/whiteBalance

Operations about the white balance parameters of s specific channel.

Request URI Definition**Table A-286 GET /ISAPI/Image/channels/<ID>/whiteBalance**

Method	GET
Description	Get the white balance parameters of s specific channel.
Query	None.
Request	None.
Response	Succeeded: <i>XML_WhiteBalance</i> Failed: <i>XML_ResponseStatus</i>

Table A-287 PUT /ISAPI/Image/channels/<ID>/whiteBalance

Method	PUT
Description	Set the white balance parameters of s specific channel.
Query	None.
Request	<i>XML_WhiteBalance</i>
Response	<i>XML_ResponseStatus</i>

Remarks

The <ID> in the request URI refers to the channel No.

A.4.43 /ISAPI/Image/channels/capabilities

Get the complete video and image configuration capability of all channels.

Request URI Definition**Table A-288 GET /ISAPI/Image/channels/capabilities**

Method	GET
Description	Get the complete video and image configuration capability of all channels.

Query	None.
Request	None.
Response	Succeeded: <i>XML_Cap_ImageChannel</i> Failed: <i>XML_ResponseStatus</i>

A.4.44 /ISAPI/Image/channels/imageModes

Get the configuration parameters of all image modes of all channels.

Request URI Definition

Table A-289 GET /ISAPI/Image/channels/imageModes

Method	GET
Description	Get the configuration parameters of all image modes of all channels.
Query	None.
Request	None.
Response	Succeeded: <i>XML_ImageModeList</i> Failed: <i>XML_ResponseStatus</i>

A.4.45 /ISAPI/Image/channels/imageModes/<ID>

Get the configuration parameters of all image modes of a specific channel.

Request URI Definition

Table A-290 GET /ISAPI/Image/channels/imageModes/<ID>

Method	GET
Description	Get the configuration parameters of all image modes of a specific channel.
Query	None.
Request	None.
Response	Succeeded: <i>XML_ImageMode</i> Failed: <i>XML_ResponseStatus</i>

Remarks

The <ID> in the request URI refers to the channel No. which is of type int.

A.5 /ISAPI/MasterSlaveTracking

A.5.1 /ISAPI/MasterSlaveTracking/capabilities

Get the master-slave tracking capability.

Request URI Definition

Table A-291 GET /ISAPI/MasterSlaveTracking/capabilities

Method	GET
Description	Get the master-slave tracking capability.
Query	None.
Request	None.
Response	<i>XML_MasterSlaveTrackingCap</i>

A.5.2 /ISAPI/MasterSlaveTracking/channels/<ID>/slaveCamera/<ID>

Operations about the basic configuration of the PTZ camera.

Request URI Definition

Table A-292 GET /ISAPI/MasterSlaveTracking/channels/<ID>/slaveCamera/<ID>

Method	GET
Description	Get the basic configuration parameters of the PTZ camera.
Query	None.
Request	None.
Response	<i>XML_SlaveCamera</i>

Table A-293 PUT /ISAPI/MasterSlaveTracking/channels/<ID>/slaveCamera/<ID>

Method	PUT
Description	Set the basic configuration parameters for the PTZ camera.
Query	None.
Request	<i>XML_SlaveCamera</i>
Response	<i>XML_ResponseStatus</i>

Remarks

The first <ID> in the request URI refers to the channel No. The second <ID> in the request URI refers to the PTZ camera ID which ranges from 1 to 4.

A.5.3 /ISAPI/MasterSlaveTracking/channels/<ID>/slaveCamera/<ID>/tracking

Operations about the tracking configuration of the PTZ camera.

Request URI Definition

Table A-294 GET /ISAPI/MasterSlaveTracking/channels/<ID>/slaveCamera/<ID>/tracking

Method	GET
Description	Get the tracking configuration parameters of the PTZ camera.
Query	None.
Request	None.
Response	<i>XML_Tracking</i>

Table A-295 PUT /ISAPI/MasterSlaveTracking/channels/<ID>/slaveCamera/<ID>/tracking

Method	PUT
Description	Set the tracking configuration parameters for the PTZ camera.
Query	None.
Request	<i>XML_Tracking</i>
Response	<i>XML_ResponseStatus</i>

Remarks

The first <ID> in the request URI refers to the channel No. The second <ID> in the request URI refers to the PTZ camera ID.

A.5.4 /ISAPI/MasterSlaveTracking/channels/<ID>/slaveCamera/<ID>/trackingRatio

Set the tracking zoom ratio for the PTZ camera.

Request URI Definition

Table A-296 PUT /ISAPI/MasterSlaveTracking/channels/<ID>/slaveCamera/<ID>/trackingRatio

Method	PUT
Description	Set the tracking zoom ratio for the PTZ camera.

Query	None.
Request	None.
Response	<i>XML_ResponseStatus</i>

A.5.5 /ISAPI/MasterSlaveTracking/channels/<ID>/slaveCamera/capabilities

Get the configuration capability of basic parameters for the PTZ camera.

Request URI Definition

Table A-297 GET /ISAPI/MasterSlaveTracking/channels/<ID>/slaveCamera/capabilities

Method	GET
Description	Get the configuration capability of basic parameters for the PTZ camera.
Query	None.
Request	None.
Response	<i>XML_Cap_SlaveCamera</i>

Remarks

The <ID> in the request URI refers to the channel No.

A.5.6 /ISAPI/MasterSlaveTracking/channels/<ID>/slaveCameraStatus

Get the connection status of the PTZ camera.

Request URI Definition

Table A-298 GET /ISAPI/MasterSlaveTracking/channels/<ID>/slaveCameraStatus

Method	GET
Description	Get the connection status of the PTZ camera.
Query	None.
Request	None.
Response	<i>XML_SlaveCameraStatus</i>

Remarks

The <ID> in the request URI refers to the PTZ camera ID.

A.5.7 /ISAPI/MasterSlaveTracking/channels/<ID>/tracking/capabilities

Get the tracking configuration capability of the PTZ camera.

Request URI Definition

Table A-299 GET /ISAPI/MasterSlaveTracking/channels/<ID>/tracking/capabilities

Method	GET
Description	Get the tracking configuration capability of the PTZ camera.
Query	None.
Request	None.
Response	<i>XML_Cap_Tracking</i>

A.5.8 /ISAPI/MasterSlaveTracking/linkedTracking/capabilities?format=json

Get capability of configuring advanced parameters of linked tracking capture.

Request URI Definition

Table A-300 GET /ISAPI/MasterSlaveTracking/linkedTracking/capabilities?format=json

Method	GET
Description	Get capability of configuring advanced parameters of linked tracking capture.
Query	format : determine the format of request or response message.
Request	None.
Response	Succeeded: <i>JSON_LinkedTrackingCap</i> Failed: <i>JSON_ResponseStatus</i>

A.5.9 /ISAPI/MasterSlaveTracking/linkedTracking?format=json

Get or set the advanced parameters of linked tracking capture.

Request URI Definition

Table A-301 GET /ISAPI/MasterSlaveTracking/linkedTracking?format=json

Method	GET
Description	Get the advanced parameters of linked tracking capture.

Query	format: determine the format of request or response message.
Request	None.
Response	Succeeded: <i>JSON_LinkedTracking</i> Failed: <i>JSON_ResponseStatus</i>

Table A-302 PUT /ISAPI/MasterSlaveTracking/linkedTracking?format=json

Method	PUT
Description	Set the advanced parameters of linked tracking capture.
Query	format: determine the format of request or response message.
Request	<i>JSON_LinkedTracking</i>
Response	<i>JSON_ResponseStatus</i>

A.6 /ISAPI/SDT

A.6.1 /ISAPI/SDT/Management/capabilities?format=json

Get the intelligent management capability of the device.

Request URI Definition

Table A-303 GET /ISAPI/SDT/Management/capabilities?format=json

Method	GET
Description	Get the intelligent management capability of the device to check the intelligent functions supported by the device.
Query	format: determine the format of request or response message.
Request	None.
Response	Succeeded: <i>JSON_Cap_IntelliManagement</i> Failed: <i>JSON_ResponseStatus</i>

A.6.2 /ISAPI/SDT/Management/EventSearch?format=json

Search for event.

Request URI Definition

Table A-304 POST /ISAPI/SDT/Management/EventSearch?format=json

Method	POST
Description	Search for event.
Query	format: determine the format of request or response message.
Request	<i>JSON_EventSearchCond</i>
Response	Succeeded: <i>JSON_EventSearchResult</i> Failed: <i>JSON_ResponseStatus</i>

A.6.3 /ISAPI/SDT/Management/EventSearch/capabilities?format=json

Get the event search capability.

Request URI Definition

Table A-305 GET /ISAPI/SDT/Management/EventSearch/capabilities?format=json

Method	GET
Description	Get the event search capability.
Query	format: determine the format of request or response message.
Request	None.
Response	<i>JSON_EventSearchCap</i>

A.6.4 /ISAPI/SDT/Management/IntelligentSearch/capabilities?format=json

Get intelligent search capability.

Request URI Definition

Table A-306 GET /ISAPI/SDT/Management/IntelligentSearch/capabilities?format=json

Method	GET
Description	Get intelligent search capability.
Query	format: determine the format of request or response message.
Request	None
Response	Succeeded: <i>JSON_IntelligentSearchCap</i> Failed: <i>JSON_ResponseStatus</i>

A.6.5 /ISAPI/SDT/Management/IntelligentSearch?format=json

Start intelligent search.

Request URI Definition

Table A-307 POST /ISAPI/SDT/Management/IntelligentSearch?format=json

Method	POST
Description	Start intelligent search.
Query	format : determine the format of request or response message.
Request	<i>JSON_IntelligentSearchCondition</i>
Response	Succeeded: <i>JSON_IntelligentSearchResult</i> Failed: <i>JSON_ResponseStatus</i>

A.7 /ISAPI/PTZCtrl

A.7.1 /ISAPI/PTZCtrl/channels/<ID>

Get or set the single PTZ control parameters.

Request URI Definition

Table A-308 GET /ISAPI/PTZCtrl/channels/<ID>

Method	GET
Description	Get the single PTZ control parameters.
Query	None
Request	None
Response	Succeeded: <i>XML_PTZChannel</i> Failed: <i>XML_ResponseStatus</i>

Table A-309 PUT /ISAPI/PTZCtrl/channels/<ID>

Method	PUT
Description	Set the single PTZ control parameters.
Query	None

Request	<i>XML_PTZChannel</i>
Response	<i>XML_ResponseStatus</i>

Remarks

The <ID> in the URI refers to the channel No.

A.7.2 /ISAPI/PTZCtrl/channels/<ID>/auxcontrols

Get or set all PTZ auxiliaries status.

Request URI Definition

Table A-310 GET /ISAPI/PTZCtrl/channels/<ID>/auxcontrols

Method	GET
Description	Get all PTZ auxiliaries status.
Query	None.
Request	None.
Response	Succeeded: <i>XML_PTZAuxList</i> Failed: <i>XML_ResponseStatus</i>

Table A-311 PUT /ISAPI/PTZCtrl/channels/<ID>/auxcontrols

Method	PUT
Description	Set all PTZ auxiliaries status.
Query	None.
Request	<i>XML_PTZAuxList</i>
Response	<i>XML_ResponseStatus</i>

Remarks

The <ID> in the URI refers to the channel No.

A.7.3 /ISAPI/PTZCtrl/channels/<ID>/auxcontrols/<ID>

Get or set the specified PTZ auxiliary status.

Request URI Definition

Table A-312 GET /ISAPI/PTZCtrl/channels/<ID>/auxcontrols/<ID>

Method	GET
Description	Get the specified PTZ auxiliary status.
Query	None.
Request	None.
Response	Succeeded: <i>XML_PTZAux</i> Failed: <i>XML_ResponseStatus</i>

Table A-313 PUT /ISAPI/PTZCtrl/channels/<ID>/auxcontrols/<ID>

Method	PUT
Description	Set the specified PTZ auxiliary status.
Query	None.
Request	<i>XML_PTZAux</i>
Response	<i>XML_ResponseStatus</i>

Remarks

- The first <ID> in the URI refers to the channel number.
- The second <ID> in refers to the auxiliary number, e.g., when there is only one wiper, the second <ID> is "1", when there are multiple wipers, the second <ID> varies according to the requirement.

A.7.4 /ISAPI/PTZCtrl/channels/<ID>/capabilities

Get the PTZ control capabilities.

Request URI Definition

Table A-314 GET /ISAPI/PTZCtrl/channels/ID/capabilities

Method	GET
Description	Get the PTZ control capabilities.
Query	None
Request	None.
Response	Succeeded: <i>XML_PTZChanelCap</i> Failed: <i>XML_ResponseStatus</i>

A.7.5 /ISAPI/PTZCtrl/channels/<ID>/EagleFocusing/auto/capabilities?format=json

Get the capability of rapid focus automatic calibration.

Request URI Definition

Table A-315 GET /ISAPI/PTZCtrl/channels/<ID>/EagleFocusing/auto/capabilities?format=json

Method	GET
Description	Get the capability of rapid focus automatic calibration.
Query	format : determine the format of request or response message.
Request	None.
Response	Succeeded: <i>JSON_AutoEagleFocusingCap</i> Failed: <i>JSON_ResponseStatus</i>

Remarks

The <ID> in the URI refers to the channel ID.

A.7.6 /ISAPI/PTZCtrl/channels/<ID>/EagleFocusing/auto?format=json

Automatically calibrate for rapid focus.

Request URI Definition

Table A-316 GET /ISAPI/PTZCtrl/channels/<ID>/EagleFocusing/auto?format=json

Method	GET
Description	Automatically calibrate for rapid focus.
Query	format : determine the format of request or response message.
Request	None.
Response	Succeeded: <i>JSON_AutoEagleFocusing</i> Failed: <i>JSON_ResponseStatus</i>

Remarks

The <ID> in the URI refers to the channel ID.

A.7.7 /ISAPI/PTZCtrl/channels/<ID>/EagleFocusing/capabilities

Get the rapid focus capability.

Request URI Definition**Table A-317 GET /ISAPI/PTZCtrl/channels/<ID>/EagleFocusing/capabilities**

Method	GET
Description	Get the rapid focus capability.
Query	None.
Request	None.
Response	Succeeded: <i>XML_Cap_EagleFocusing</i> Failed: <i>XML_ResponseStatus</i>

Remarks

The <ID> in the URI refers to the channel ID.

A.7.8 /ISAPI/PTZCtrl/channels/<ID>/GPS?format=json

Control the device turn to corresponding place according to applied GPS information.

Request URI Definition**Table A-318 PUT /ISAPI/PTZCtrl/channels/<ID>/GPS?format=json**

Method	PUT
Description	Control the device turn to corresponding place according to applied GPS information.
Query	format: determine the format of request or response message.
Request	<i>JSON_GPSInfo</i>
Response	<i>JSON_ResponseStatus</i>

Remarks

The <ID> in the URI refers to channel No.

A.7.9 /ISAPI/PTZCtrl/channels/<ID>/gyroVerify/capabilities?format=json

Get the gyroscope calibration capability.

Request URI Definition

Table A-319 GET /ISAPI/PTZCtrl/channels/<ID>/gyroVerify/ capabilities?format=json

Method	GET
Description	Get the gyroscope calibration capability.
Query	format: determine the format of request or response message.
Request	None.
Response	Succeeded: <i>JSON_GyroVerifyCap</i> Failed: <i>JSON_ResponseStatus</i>

Remarks

The <ID> in the URI refers to channel No.

A.7.10 /ISAPI/PTZCtrl/channels/<ID>/gyroVerify?format=json

Set the gyroscope calibration parameters.

Request URI Definition

Table A-320 PUT /ISAPI/PTZCtrl/channels/<ID>/gyroVerify?format=json

Method	PUT
Description	Set the gyroscope calibration parameters.
Query	format: determine the format of request or response message.
Request	<i>JSON_GyroVerify</i>
Response	<i>JSON_ResponseStatus</i>

Remarks

The <ID> in the URI refers to channel No.

A.7.11 /ISAPI/PTZCtrl/channels/<ID>/lensCorrection/capabilities?format=json

Get the lens correction capability.

Request URI Definition

Table A-321 GET /ISAPI/PTZCtrl/channels/<ID>/lensCorrection/capabilities?format=json

Method	GET
Description	Get the lens correction capability.
Query	format: determine the format of request or response message.
Request	None.
Response	<i>JSON_LensCorrectionCap</i>

A.7.12 /ISAPI/PTZCtrl/channels/<ID>/lensCorrection?format=json

Get or set the configuration parameters of lens correction.

Request URI Definition

Table A-322 GET /ISAPI/PTZCtrl/channels/<ID>/lensCorrection?format=json

Method	GET
Description	Get the configuration parameters of lens correction.
Query	format: determine the format of request or response message.
Request	None.
Response	Succeeded: <i>JSON_LensCorrection</i> Failed: <i>XML_ResponseStatus</i>

Table A-323 PUT /ISAPI/PTZCtrl/channels/<ID>/lensCorrection?format=json

Method	PUT
Description	Set the lens correction parameters.
Query	format: determine the format of request or response message.
Request	<i>JSON_LensCorrection</i>
Response	<i>XML_ResponseStatus</i>

A.7.13 /ISAPI/PTZCtrl/channels/<ID>/lockPTZ

Operations about locking PTZ of a specific channel.

Request URI Definition

Table A-324 GET /ISAPI/PTZCtrl/channels/<ID>/lockPTZ

Method	GET
Description	Get the remaining time before unlocking PTZ of a specific channel.
Query	None.
Request	None.
Response	<i>XML_LockPTZ</i>

Table A-325 PUT /ISAPI/PTZCtrl/channels/<ID>/lockPTZ

Method	PUT
Description	Lock PTZ of a specific channel.
Query	None.
Request	<i>XML_LockPTZ</i>
Response	<i>XML_ResponseStatus</i>

A.7.14 /ISAPI/PTZCtrl/channels/<ID>/maxelevation

Get or set the max. tilt-angle parameters.

Request URI Definition

Table A-326 GET /ISAPI/PTZCtrl/channels/<ID>/maxelevation

Method	GET
Description	Get the max. tilt-angle configuration parameters
Query	None.
Request	None.
Response	<i>XML_MaxElevation</i>

Table A-327 PUT /ISAPI/PTZCtrl/channels/<ID>/maxelevation

Method	PUT
Description	Set the max. tilt-angle
Query	None.
Request	<i>XML_MaxElevation</i>
Response	<i>XML_ResponseStatus</i>

Remarks

The <ID> in the URI indicates the channel ID.

A.7.15 /ISAPI/PTZCtrl/channels/<ID>/maxelevation/capabilities

Get the max. tilt-angle capability.

Request URI Definition

Table A-328 GET /ISAPI/PTZCtrl/channels/<ID>/maxelevation/capabilities

Method	GET
Description	Get the max. tilt-angle capability.
Query	None.
Request	None.
Response	<i>XML_Cap_MaxElevation</i>

Remarks

The <ID> in the URI indicates the channel ID.

A.7.16 /ISAPI/PTZCtrl/channels/<ID>/moveAutoTracking?format=json

Get or set moving object tracking status.

Request URI Definition

Table A-329 GET /ISAPI/PTZCtrl/channels/<ID>/moveAutoTracking?format=json

Method	GET
Description	Get moving object tracking status.
Query	format: determine the format of request or response message.
Request	None
Response	Succeeded: <i>JSON_MoveAutoTracking</i> Failed: <i>JSON_ResponseStatus</i>

Table A-330 PUT /ISAPI/PTZCtrl/channels/<ID>/moveAutoTracking?format=json

Method	PUT
Description	Set moving object tracking status.
Query	format: determine the format of request or response message.

Request	<i>JSON_MoveAutoTracking</i>
Response	<i>JSON_ResponseStatus</i>

Remarks

The <ID> in the request URI refers to the PTZ channel ID.

A.7.17 /ISAPI/PTZCtrl/channels/<ID>/onpushfoucs/reset

Initialize the lens.

Request URI Definition

Table A-331 PUT /ISAPI/PTZCtrl/channels/<ID>/onpushfoucs/reset

Method	PUT
Description	Initialize the lens.
Query	None.
Request	None.
Response	<i>XML_ResponseStatus</i>

Remarks

The <ID> in the request URI refers to the channel No.

A.7.18 /ISAPI/PTZCtrl/channels/<ID>/parkaction

Get or set the parking parameters.

Request URI Definition

Table A-332 GET /ISAPI/PTZCtrl/channels/<ID>/parkaction

Method	GET
Description	Get the parking parameters.
Query	None.
Request	None.
Response	Succeeded: <i>XML_ParkAction</i> Failed: <i>XML_ResponseStatus</i>

Table A-333 PUT /ISAPI/PTZCtrl/channels/<ID>/parkaction

Method	PUT
Description	Set the parking parameters.
Query	None.
Request	<i>XML_ParkAction</i>
Response	<i>XML_ResponseStatus</i>

A.7.19 /ISAPI/PTZCtrl/channels/<ID>/parkaction/capabilities

Get the parking configuration capability.

Request URI Definition

Table A-334 GET /ISAPI/PTZCtrl/channels/<ID>/parkaction/capabilities

Method	GET
Description	Get the parking configuration capability.
Query	None.
Request	None.
Response	Succeeded: <i>XML_Cap_ParkAction</i> Failed: <i>XML_ResponseStatus</i>

A.7.20 /ISAPI/PTZCtrl/channels/<ID>/presets

Operations about all presets' configurations.

Request URI Definition

Table A-335 GET /ISAPI/PTZCtrl/channels/<ID>/presets

Method	GET
Description	Get all presets' parameters.
Query	None.
Request	None.
Response	Succeeded: <i>XML_PTZPresetList</i> Failed: <i>XML_ResponseStatus</i>

Table A-336 PUT /ISAPI/PTZCtrl/channels/<ID>/presets

Method	PUT
Description	Set all presets' parameters.
Query	None.
Request	<i>XML_PTZPresetList</i>
Response	<i>XML_ResponseStatus</i>

Table A-337 POST /ISAPI/PTZCtrl/channels/<ID>/presets

Method	POST
Description	Add a preset.
Query	None.
Request	<i>XML_PTZPreset</i>
Response	<i>XML_ResponseStatus</i>

Table A-338 DELETE /ISAPI/PTZCtrl/channels/<ID>/presets

Method	DELETE
Description	Delete all presets.
Query	None.
Request	None.
Response	<i>XML_ResponseStatus</i>

Remarks

The <ID> in the URI is the camera ID.

A.7.21 /ISAPI/PTZCtrl/channels/<ID>/presets/<ID>

Get or set the a preset's parameters, or delete a preset.

Request URI Definition

Table A-339 GET /ISAPI/PTZCtrl/channels/<ID>/presets/<ID>

Method	GET
Description	Get a preset's parameters.
Query	None.

Request	None.
Response	Succeeded: <i>XML_PTZPreset</i> Failed: <i>XML_ResponseStatus</i>

Table A-340 PUT /ISAPI/PTZCtrl/channels/<ID>/presets/<ID>

Method	PUT
Description	Set a preset's parameters.
Query	None.
Request	<i>XML_PTZPreset</i>
Response	<i>XML_ResponseStatus</i>

Table A-341 DELETE /ISAPI/PTZCtrl/channels/<ID>/presets/<ID>

Method	DELETE
Description	Delete a preset.
Query	None.
Request	None.
Response	<i>XML_ResponseStatus</i>

A.7.22 /ISAPI/PTZCtrl/channels/<ID>/PTZOSDDisplay

Get or set the on-screen display parameters of the PTZ status.

Request URI Definition

Table A-342 GET /ISAPI/PTZCtrl/channels/<ID>/PTZOSDDisplay

Method	GET
Description	Get the on-screen display parameters of the PTZ status.
Query	None.
Request	None.
Response	Succeeded: <i>XML_PTZOSDDisplay</i> Failed: <i>XML_ResponseStatus</i>

Table A-343 PUT /ISAPI/PTZCtrl/channels/<ID>/PTZOSDDisplay

Method	PUT
Description	Set the on-screen display parameters of the PTZ status.

Query	None.
Request	<i>XML_PTZOSDDisplay</i>
Response	<i>XML_ResponseStatus</i>

Remarks

The <ID> in the request URI refers to the PTZ channel No.

A.7.23 /ISAPI/PTZCtrl/channels/<ID>/save

Get and save the PTZ position information of the current channel.

Request URI Definition

Table A-344 GET /ISAPI/PTZCtrl/channels/<ID>/save

Method	GET
Description	Get and save the PTZ position information (including panning, tilting, and zooming parameters) of the current channel.
Query	None.
Request	None.
Response	Succeeded: <i>XML_PTZStatus</i> Failed: <i>XML_ResponseStatus</i>

A.7.24 /ISAPI/PTZCtrl/channels/<ID>/save?format=json

Save the PTZ position information of the current channel.

Request URI Definition

Table A-345 PUT /ISAPI/PTZCtrl/channels/<ID>/save?format=json

Method	PUT
Description	Save the PTZ position information, including panning, tilting, and zooming parameters, of the current channel.
Query	format: determine the format of request or response message.
Request	None.
Response	<i>JSON_ResponseStatus</i>

Remarks

The <ID> in the request URI refers to the PTZ channel No.

A.7.25 /ISAPI/PTZCtrl/channels/<ID>/status

Get the PTZ status of a specific channel of the device.

Request URI Definition

Table A-346 GET /ISAPI/PTZCtrl/channels/<ID>/status

Method	GET
Description	Get the PTZ status of a specific channel of the device.
Query	None.
Request	None.
Response	Succeeded: <i>XML_PTZStatus</i> Failed: <i>XML_ResponseStatus</i>

A.7.26 /ISAPI/PTZCtrl/channels/<ID>/trackingRatio/capabilities?format=json

Get the configuration capability of tracking parameters.

Request URI Definition

Table A-347 GET /ISAPI/PTZCtrl/channels/<ID>/trackingRatio/capabilities?format=json

Method	GET
Description	Get the configuration capability of tracking parameters.
Query	format: determine the format of request or response message.
Request	None.
Response	Succeeded: <i>JSON_TrackingRatioCap</i> Failed: <i>JSON_ResponseStatus</i>

A.7.27 /ISAPI/PTZCtrl/channels/<ID>/trackingRatio?format=json

Get or set the tracking parameters.

Request URI Definition

Table A-348 GET /ISAPI/PTZCtrl/channels/<ID>/trackingRatio?format=json

Method	GET
Description	Get the tracking parameters.
Query	format : determine the format of request or response message.
Request	None.
Response	Succeeded: <i>JSON_TrackingRatio</i> Failed: <i>JSON_ResponseStatus</i>

Table A-349 PUT /ISAPI/PTZCtrl/channels/<ID>/trackingRatio?format=json

Method	PUT
Description	Set the tracking parameters.
Query	format : determine the format of request or response message.
Request	<i>JSON_TrackingRatio</i>
Response	<i>JSON_ResponseStatus</i>

A.7.28 /ISAPI/PTZCtrl/channels/<ID>/zoomFocus

Operations about zoom and focus coordinates configuration for zoom camera module.

Request URI Definition

Table A-350 GET /ISAPI/PTZCtrl/channels/<ID>/zoomFocus

Method	GET
Description	Get parameters of zoom and focus coordinates of zoom camera module.
Query	None.
Request	None.
Response	<i>XML_ZoomFocus</i>

Table A-351 PUT /ISAPI/PTZCtrl/channels/<ID>/zoomFocus

Method	PUT
Description	Set parameters of zoom and focus coordinates of zoom camera module.

Query	None.
Request	<i>XML_ZoomFocus</i>
Response	<i>XML_ResponseStatus</i>

Remarks

The <ID> in the request URI refers to the channel ID.

A.8 /ISAPI/Security

A.8.1 /ISAPI/Security/adminAccesses

Get or set parameters of all protocols supported by device.

Request URI Definition**Table A-352 GET /ISAPI/Security/adminAccesses**

Method	GET
Description	Get parameters of all protocols supported by device.
Query	None.
Request	None.
Response	Succeeded: <i>XML_AdminAccessProtocolList</i> Failed: <i>XML_ResponseStatus</i>

Table A-353 PUT /ISAPI/Security/adminAccesses

Method	PUT
Description	Set parameters of all protocols supported by device.
Query	None.
Request	<i>XML_AdminAccessProtocolList</i>
Response	<i>XML_ResponseStatus</i>

A.8.2 /ISAPI/Security/adminAccesses/<ID>

Get or set the parameters of a specific protocol that supported by device.

Request URI Definition

Table A-354 GET /ISAPI/Security/adminAccesses/<ID>

Method	GET
Description	Get the parameters of a specific protocol that supported by device.
Query	None.
Request	None.
Response	Succeeded: <i>XML_AdminAccessProtocol</i> Failed: <i>XML_ResponseStatus</i>

Table A-355 PUT /ISAPI/Security/adminAccesses/<ID>

Method	PUT
Description	Set the parameters of a specific protocol that supported by device.
Query	None.
Request	<i>XML_AdminAccessProtocol</i>
Response	<i>XML_ResponseStatus</i>

Remarks

The <ID> in the request URI refers to the protocol ID.

A.8.3 /ISAPI/Security/adminAccesses/capabilities

Get device protocol capability.

Request URI Definition

Table A-356 GET /ISAPI/Security/adminAccesses/capabilities

Method	GET
Description	Get device protocol capability.
Query	None.
Request	None.
Response	Succeeded: <i>XML_Cap_AdminAccessProtocolList</i> Failed: <i>XML_ResponseStatus</i>

A.8.4 /ISAPI/Security/advanced?format=json

Get or set advanced parameters of security.

Request URI Definition

Table A-357 GET /ISAPI/Security/advanced?format=json

Method	GET
Description	Get advanced configuration parameters of security.
Query	format : determine the format of request or response message.
Request	None.
Response	Succeeded: <i>JSON_SecurityAdvanced</i> Failed: <i>JSON_ResponseStatus</i>

Table A-358 PUT /ISAPI/Security/advanced?format=json

Method	PUT
Description	Set advanced parameters of security.
Query	format : determine the format of request or response message.
Request	<i>JSON_SecurityAdvanced</i>
Response	<i>JSON_ResponseStatus</i>

A.8.5 /ISAPI/Security/capabilities

Get the security capability of the device.

Request URI Definition

Table A-359 GET /ISAPI/Security/capabilities

Method	GET
Description	Get the security capability of the device.
Query	username : user name, string, it should be encrypted.
Request	None.
Response	Succeeded: <i>XML_SecurityCap</i> Failed: <i>XML_ResponseStatus</i>

A.8.6 /ISAPI/Security/certificate/select/<functionName>?format=json

Get or set the parameters of selecting the certificate.

Request URI Definition

Table A-360 GET /ISAPI/Security/certificate/select/<functionName>?format=json

Method	GET
Description	Get the parameters of selecting the certificate.
Query	format: determine the format of request or response message.
Request	None.
Response	Succeeded: <i>JSON_CertificateSelect</i> Failed: <i>JSON_ResponseStatus</i>

Table A-361 PUT /ISAPI/Security/certificate/select/<functionName>?format=json

Method	PUT
Description	Set the parameters of selecting the certificate.
Query	format: determine the format of request or response message.
Request	<i>JSON_CertificateSelect</i>
Response	<i>JSON_ResponseStatus</i>

Remarks

The <functionName> in the request URI refers to the function name.

A.8.7 /ISAPI/Security/certificate/select/capabilities?format=json

Get the capability of selecting the certificate.

Request URI Definition

Table A-362 GET /ISAPI/Security/certificate/select/capabilities?format=json

Method	GET
Description	Get the capability of selecting the certificate.
Query	format: determine the format of request or response message.
Request	None.
Response	Succeeded: <i>JSON_Cap_CertificateSelect</i>

	Failed: <i>JSON_ResponseStatus</i>
--	------------------------------------

A.8.8 /ISAPI/Security/challenge

Get the random challenge strings.

Request URI Definition

Table A-363 POST /ISAPI/Security/challenge

Method	POST
Description	Get the random challenge strings.
Query	None.
Request	<i>XML_PublicKey</i>
Response	<i>XML_Challenge</i>

A.8.9 /ISAPI/Security/CommuMode/capabilities?format=json

Get the configuration capability of the security mode level of the private protocol.

Request URI Definition

Table A-364 GET /ISAPI/Security/CommuMode/capabilities?format=json

Method	GET
Description	Get the configuration capability of the security mode level of the private protocol.
Query	format: determine the format of request or response message.
Request	None.
Response	Succeeded: <i>JSON_Cap_CommMode</i> Failed: <i>JSON_ResponseStatus</i>

A.8.10 /ISAPI/Security/CommuMode?format=json

Operations about the configuration of the security mode level of the private protocol.

Request URI Definition

Table A-365 GET /ISAPI/Security/CommuMode?format=json

Method	GET
Description	Get the parameters of the security mode level of the private protocol.
Query	format: determine the format of request or response message.
Request	None.
Response	Succeeded: <i>JSON_CommMode</i> Failed: <i>JSON_ResponseStatus</i>

Table A-366 PUT /ISAPI/Security/CommuMode?format=json

Method	PUT
Description	Set the parameters of the security mode level of the private protocol.
Query	format: determine the format of request or response message.
Request	<i>JSON_CommMode</i>
Response	<i>JSON_ResponseStatus</i>

A.8.11 /ISAPI/Security/deviceCertificate

Import network certificate to device.

Request URI Definition

Table A-367 PUT /ISAPI/Security/deviceCertificate

Method	PUT
Description	Import network certificate to device.
Query	type: certificate type. It can be set to "wpa" (upload WPA certificates), "ieee802.1x" (upload ieee802.1x certificates), "securityLog" (upload CA certificates of security logs) or "LogServer" (log server certificate); if no value is assigned to this parameter, it indicates that the WPA certificate will be imported.
Request	Certificate data (in binary format)
Response	<i>XML_ResponseStatus</i>

Remarks

The imported certificate type is determined by "Content-type" in binary certificate data, which contains "x-509-ca-cert", "x-509-client-cert", and "x-509-client-key".

A.8.12 /ISAPI/Security/deviceCertificate/capabilities?format=json

Get the CA (Certificate Authority) certificate capability.

Request URI Definition

Table A-368 GET /ISAPI/Security/deviceCertificate/capabilities?format=json

Method	GET
Description	Get the CA (Certificate Authority) certificate capability.
Query	format : determine the format of request or response message.
Request	None.
Response	Succeeded: <i>JSON_DeviceCertificateCap</i> Failed: <i>JSON_ResponseStatus</i>

A.8.13 /ISAPI/Security/deviceCertificate/certificateRevocation/capabilities?format=json

Get the configuration capability of the certificate expiry alarm.

Request URI Definition

Table A-369 GET /ISAPI/Security/deviceCertificate/certificateRevocation/capabilities?format=json

Method	GET
Description	Get the configuration capability of the certificate expiry alarm.
Query	format : determine the format of request or response message.
Request	None.
Response	Succeeded: <i>JSON_CertificateRevocationCap</i> Failed: <i>JSON_ResponseStatus</i>

A.8.14 /ISAPI/Security/deviceCertificate/certificateRevocation?format=json

Get or set the configuration of certificate expiry alarm.

Request URI Definition

Table A-370 GET /ISAPI/Security/deviceCertificate/certificateRevocation?format=json

Method	GET
Description	Get the parameters of certificate expiry alarm.
Query	format : determine the format of request or response message.
Request	None.
Response	Succeeded: <i>JSON_CertificateRevocation</i> Failed: <i>JSON_ResponseStatus</i>

Table A-371 PUT /ISAPI/Security/deviceCertificate/certificateRevocation?format=json

Method	PUT
Description	Set the parameters of certificate expiry alarm.
Query	format : determine the format of request or response message.
Request	<i>JSON_CertificateRevocation</i>
Response	<i>JSON_ResponseStatus</i>

A.8.15 /ISAPI/Security/deviceCertificate/certificates/<customID>?format=json

Get or delete the information of a specific device certificate.

Request URI Definition

Table A-372 GET /ISAPI/Security/deviceCertificate/certificates/<customID>?format=json

Method	GET
Description	Get the information of a specific device certificate.
Query	format : determine the format of request or response message.
Request	None.
Response	Succeeded: <i>JSON_DeviceCertificate</i> Failed: <i>JSON_ResponseStatus</i>

Table A-373 DELETE /ISAPI/Security/deviceCertificate/certificates/<customID>?format=json

Method	DELETE
Description	Delete the information of a specific device certificate.
Query	format: determine the format of request or response message.
Request	None.
Response	<i>JSON_ResponseStatus</i>

Remarks

The <customID> in the request URI refers to the custom certificate ID.

A.8.16 /ISAPI/Security/deviceCertificate/certificates/<ID>

Delete the certificate of a specific device.

Request URI Definition

Table A-374 DELETE /ISAPI/Security/deviceCertificate/certificates/<ID>

Method	DELETE
Description	Delete the certificate of a specific device.
Query	None.
Request	None.
Response	<i>XML_ResponseStatus</i>

Remarks

The <ID> in the request URI is the device ID returned when searching certificate information.

A.8.17 /ISAPI/Security/deviceCertificate/certificates/<ID>/recreate?format=json

Regenerate a specific abnormal certificate.

Request URI Definition

Table A-375 PUT /ISAPI/Security/deviceCertificate/certificates/<ID>/recreate?format=json

Method	PUT
Description	Regenerate a specific certificate for authentication client.
Query	format: determine the format of request or response message.

Request	None.
Response	<i>JSON_ResponseStatus</i>

Remarks

- The <ID> in the request URI refers to the certificate ID.
- This URI is only supported by HTTPS certificate, and it is available when the certificate exception is detected.

A.8.18 /ISAPI/Security/deviceCertificate/certificates/<ID>/status?format=json

Get the status of a specific certificate.

Request URI Definition

Table A-376 GET /ISAPI/Security/deviceCertificate/certificates/<ID>/status?format=json

Method	GET
Description	Get the status of a specific certificate.
Query	format: determine the format of request or response message.
Request	None.
Response	Succeeded: <i>JSON_DeviceCertificateStatus</i> Failed: <i>JSON_ResponseStatus</i>

Remarks

The <ID> in the request URI is the certificate ID, which is generated and returned by device.

A.8.19 /ISAPI/Security/deviceCertificate/certificates/capabilities?format=json

Get certificate search capability.

Request URI Definition

Table A-377 GET /ISAPI/Security/deviceCertificate/certificates/capabilities?format=json

Method	GET
Description	Get certificate search capability.
Query	format: determine the format of request or response message.
Request	None.
Response	Succeeded: <i>JSON_DeviceCertificatesCap</i>

	Failed: <i>JSON_ResponseStatus</i>
--	------------------------------------

A.8.20 /ISAPI/Security/deviceCertificate/certificates/recreate?format=json

Regenerate all abnormal certificates.

Request URI Definition

Table A-378 PUT /ISAPI/Security/deviceCertificate/certificates/recreate?format=json

Method	PUT
Description	Regenerate all abnormal certificates.
Query	format : determine the format of request or response message.
Request	None.
Response	<i>JSON_ResponseStatus</i>

Remarks

This URI is only supported by HTTPS certificate, and it is available when the certificate exception is detected.

A.8.21 /ISAPI/Security/deviceCertificate/certificates/status?format=json

Get status of all certificates.

Request URI Definition

Table A-379 GET /ISAPI/Security/deviceCertificate/certificates/status?format=json

Method	GET
Description	Get status of all certificates.
Query	format : determine the format of request or response message.
Request	None.
Response	Succeeded: <i>JSON_DeviceCertificateStatusList</i> Failed: <i>JSON_ResponseStatus</i>

A.8.22 /ISAPI/Security/deviceCertificate/certificates?format=json

Get device certificate information in a batch.

Request URI Definition

Table A-380 GET /ISAPI/Security/deviceCertificate/certificates?format=json

Method	GET
Description	Get device certificate information in a batch.
Query	format : determine the format of request or response message. type : certificate type, it can be set to "wpa" (get all WPA certificates), "ieee802.1x" (get all ieee802.1x certificates), and "securityLog" (get all CA certificates of security logs); if no value is assigned to type , all CA certificates will be searched and obtained.
Request	None.
Response	Succeeded: <i>JSON_DeviceCertificates</i> Failed: <i>JSON_ResponseStatus</i>

A.8.23 /ISAPI/Security/deviceCertificate/selfSignCert

Export the device's self-signed certificate.

Request URI Definition

Table A-381 GET /ISAPI/Security/deviceCertificate/selfSignCert

Method	GET
Description	Export the device's self-signed certificate.
Query	None.
Request	None.
Response	Succeeded: Binary data. Failed: <i>XML_ResponseStatus</i>

A.8.24 /ISAPI/Security/deviceCertificate?customID=

Import a CA (Certificate Authority) certificate to the device.

Request URI Definition**Table A-382 PUT /ISAPI/Security/deviceCertificate?customID=**

Method	PUT
Description	Import a CA (Certificate Authority) certificate to the device. The CA certificate is used for 802.1x (radius) with various authentication mechanism.
Query	customID : custom certificate ID, string, it consists of digits and characters.
Request	Certificate data (its format depends on the device).
Response	<i>XML_ResponseStatus</i>

Remarks

- When importing CA certificate by calling this URI, the function of CA certificate will not be distinguished. The CA certificate and its function will be bound together afterward.
- The imported certificate type is determined by "Content-type" in the certificate data, which contains "x-x509-ca-cert" (CA certificate (root certificate)), "x-x509-client-cert" (client certificate), and "x-x509-client-key" (client password).

A.8.25 /ISAPI/Security/doubleVerification/users/<ID>?format=json

Operations about a specified double verification user.

Request URI Definition**Table A-383 GET /ISAPI/Security/doubleVerification/users/<ID>?format=json**

Method	GET
Description	Get a double verification user.
Query	format : determine the format of request or response message. security : the version No. of encryption scheme. When security does not exist, it indicates that the data is not encrypted; when security is 1, it indicates that the nodes of sensitive information in the message are encrypted in AES128 CBC mode; when security is 2, it indicates that the nodes of sensitive information in the message are encrypted in AES256 CBC mode. iv : the initialization vector, and it is required when security is 1 or 2.
Request	None.
Response	<i>JSON_User</i>

Table A-384 PUT /ISAPI/Security/doubleVerification/users/<ID>?format=json

Method	PUT
Description	Set a double verification user.
Query	format: determine the format of request or response message. security: the version No. of encryption scheme. When security does not exist, it indicates that the data is not encrypted; when security is 1, it indicates that the nodes of sensitive information in the message are encrypted in AES128 CBC mode; when security is 2, it indicates that the nodes of sensitive information in the message are encrypted in AES256 CBC mode. iv: the initialization vector, and it is required when security is 1 or 2.
Request	<i>JSON_User</i>
Response	<i>JSON_ResponseStatus</i>

Table A-385 DELETE /ISAPI/Security/doubleVerification/users/<ID>?format=json

Method	DELETE
Description	Delete a double verification user.
Query	format: determine the format of request or response message. security: the version No. of encryption scheme. When security does not exist, it indicates that the data is not encrypted; when security is 1, it indicates that the nodes of sensitive information in the message are encrypted in AES128 CBC mode; when security is 2, it indicates that the nodes of sensitive information in the message are encrypted in AES256 CBC mode. iv: the initialization vector, and it is required when security is 1 or 2.
Request	None.
Response	<i>JSON_ResponseStatus</i>

Remarks

The <ID> in the URI indicates the user ID.

A.8.26 /ISAPI/Security/doubleVerification/users/capabilities?format=json

Get the double verification configuration capability.

Request URI Definition**Table A-386 GET /ISAPI/Security/doubleVerification/users/capabilities?format=json**

Method	GET
Description	Get the double verification configuration capability.
Query	format : determine the format of request or response message.
Request	None.
Response	<i>JSON_UserCap</i>

A.8.27 /ISAPI/Security/doubleVerification/users?format=json

Get all double verification users and add a double verification user.

Request URI Definition**Table A-387 GET /ISAPI/Security/doubleVerification/users?format=json**

Method	GET
Description	Get all double verification users.
Query	format : determine the format of request or response message. security : the version No. of encryption scheme. When security does not exist, it indicates that the data is not encrypted; when security is 1, it indicates that the nodes of sensitive information in the message are encrypted in AES128 CBC mode; when security is 2, it indicates that the nodes of sensitive information in the message are encrypted in AES256 CBC mode. iv : the initialization vector, and it is required when security is 1 or 2.
Request	None.
Response	<i>JSON_UserList</i>

Table A-388 POST /ISAPI/Security/doubleVerification/users?format=json

Method	POST
Description	Add a double verification user.
Query	format : determine the format of request or response message. security : the version No. of encryption scheme. When security does not exist, it indicates that the data is not encrypted; when security is 1, it indicates that the nodes of sensitive information in the message are encrypted in AES128 CBC mode; when security is 2, it indicates

	that the nodes of sensitive information in the message are encrypted in AES256 CBC mode. iv : the initialization vector, and it is required when security is 1 or 2.
Request	<i>JSON_User</i>
Response	Succeeded: <i>JSON_ResponseStatus</i> and <i>JSON_id</i> Failed: <i>JSON_ResponseStatus</i>

A.8.28 /ISAPI/Security/doubleVerification/UsersPermission/<ID>?format=json

Get and set the permission of double verification user.

Request URI Definition

Table A-389 GET /ISAPI/Security/doubleVerification/UsersPermission/<ID>?format=json

Method	GET
Description	Get the configuration parameters of double verification user permission.
Query	format : determine the format of request or response message.
Request	None.
Response	Succeeded: <i>JSON_UserPermission</i> Failed: <i>JSON_ResponseStatus</i>

Table A-390 PUT /ISAPI/Security/doubleVerification/UsersPermission/<ID>?format=json

Method	PUT
Description	Set the double verification user permission.
Query	format : determine the format of request or response message.
Request	<i>JSON_UserPermission</i>
Response	<i>JSON_ResponseStatus</i>

Remarks

The <ID> in the URI indicates the user ID.

A.8.29 /ISAPI/Security/doubleVerification/UsersPermission/capabilities?format=json

Get the capability of permission configuration for double verification user.

Request URI Definition

Table A-391 GET /ISAPI/Security/doubleVerification/UsersPermission/capabilities?format=json

Method	GET
Description	Get the capability of permission configuration for double verification user.
Query	format: determine the format of request or response message.
Request	None.
Response	Succeeded: <i>JSON_UserPermissionCap</i> Failed: <i>JSON_ResponseStatus</i>

A.8.30 /ISAPI/Security/doubleVerification?format=json

Get the capability of double verification enabling configuration.

Request URI Definition

Table A-392 GET /ISAPI/Security/doubleVerification?format=json

Method	GET
Description	Get the capability of double verification enabling configuration.
Query	format: determine the format of request or response message.
Request	None.
Response	<i>JSON_DoubleVerification</i>

Table A-393 PUT /ISAPI/Security/doubleVerification?format=json

Method	PUT
Description	Enable or disable the double verification.
Query	format: determine the format of request or response message.
Request	<i>JSON_DoubleVerification</i>
Response	<i>JSON_ResponseStatus</i>

A.8.31 /ISAPI/Security/email/parameter/capabilities?format=json

Get recovery email configuration capability (only available in LAN and for admin user).

Request URI Definition

Table A-394 GET /ISAPI/Security/email/parameter/capabilities?format=json

Method	GET
Description	Get recovery email configuration capability.
Query	format: determine the format of request or response message. security: the version No. of encryption scheme. When security does not exist, it indicates that the data is not encrypted; when security is 1, it indicates that the nodes of sensitive information in the message are encrypted in AES128 CBC mode; when security is 2, it indicates that the nodes of sensitive information in the message are encrypted in AES256 CBC mode. iv: the initialization vector, and it is required when security is 1 or 2.
Request	None.
Response	<i>JSON_SecurityEmailCap</i>

A.8.32 /ISAPI/Security/email/parameter?format=json

Operations about recovery email configuration (only available for LAN and for admin user).

Request URI Definition

Table A-395 GET /ISAPI/Security/email/parameter?format=json

Method	GET
Description	Get recovery email parameters.
Query	format: determine the format of request or response message. security: the version No. of encryption scheme. When security does not exist, it indicates that the data is not encrypted; when security is 1, it indicates that the nodes of sensitive information in the message are encrypted in AES128 CBC mode; when security is 2, it indicates that the nodes of sensitive information in the message are encrypted in AES256 CBC mode. iv: the initialization vector, and it is required when security is 1 or 2.
Request	None.
Response	<i>JSON_SecurityEmail</i>

Table A-396 PUT /ISAPI/Security/email/parameter?format=json

Method	PUT
Description	Set recovery email parameters.
Query	format: determine the format of request or response message. security: the version No. of encryption scheme. When security does not exist, it indicates that the data is not encrypted; when security is 1, it indicates that the nodes of sensitive information in the message are encrypted in AES128 CBC mode; when security is 2, it indicates that the nodes of sensitive information in the message are encrypted in AES256 CBC mode. iv: the initialization vector, and it is required when security is 1 or 2.
Request	<i>JSON_SecurityEmail</i>
Response	<i>JSON_ResponseStatus</i>

A.8.33 /ISAPI/Security/email/qrCode?format=json

Get the QR code of the configured recovery email (only available for LAN and for admin user).

Request URI Definition

Table A-397 GET /ISAPI/Security/email/qrCode?format=json

Method	GET
Description	Get the QR code of the configured recovery email.
Query	format: determine the format of request or response message. security: the version No. of encryption scheme. When security does not exist, it indicates that the data is not encrypted; when security is 1, it indicates that the nodes of sensitive information in the message are encrypted in AES128 CBC mode; when security is 2, it indicates that the nodes of sensitive information in the message are encrypted in AES256 CBC mode. iv: the initialization vector, and it is required when security is 1 or 2.
Request	None.
Response	<i>JSON_SecurityEmailQrCode</i>

A.8.34 /ISAPI/Security/emailCertification?format=json

Reset password by the verification code via recovery email (only available for LAN and for admin user).

Request URI Definition

Table A-398 PUT /ISAPI/Security/emailCertification?format=json

Method	PUT
Description	Reset password by the verification code via recovery email.
Query	format : determine the format of request or response message.
Request	<i>JSON_EmailCertification</i>
Response	<i>JSON_ResponseStatus</i>

Remarks

For the encryption of passwords and security answers, first transform them by UTF8, and then transcode them by BASE64, finally, encrypt them by AES128CB.

A.8.35 /ISAPI/Security/extern/capabilities

Get capability of other security configuration.

Request URI Definition

Table A-399 GET /ISAPI/Security/extern/capabilities

Method	GET
Description	Get capability of other security configuration.
Query	None.
Request	None.
Response	<i>XML_externSecurityCap</i>

A.8.36 /ISAPI/Security/illegalLoginLock

Get or set locking parameters of illegal login.

Request URI Definition

Table A-400 GET /ISAPI/Security/illegalLoginLock

Method	GET
Description	Get locking parameters of illegal login.
Query	None.
Request	None.
Response	Succeeded: <i>XML_IllegalLoginLock</i> Failed: <i>XML_ResponseStatus</i>

Table A-401 PUT /ISAPI/Security/illegalLoginLock

Method	PUT
Description	Set locking parameters of illegal login.
Query	None.
Request	<i>XML_IllegalLoginLock</i>
Response	<i>XML_ResponseStatus</i>

A.8.37 /ISAPI/Security/loginLinkNum?format=json

Get or set maximum number of logged in accounts.

Request URI Definition

Table A-402 GET /ISAPI/Security/loginLinkNum?format=json

Method	GET
Description	Get maximum number of logged in accounts.
Query	format: determine the format of request or response message.
Request	None.
Response	Succeeded: <i>JSON_LoginLinkNum</i> Failed: <i>JSON_ResponseStatus</i>

Table A-403 PUT /ISAPI/Security/loginLinkNum?format=json

Method	PUT
Description	Set maximum number of logged in accounts.
Query	format: determine the format of request or response message.

Request	<i>JSON_LoginLinkNum</i>
Response	<i>JSON_ResponseStatus</i>

A.8.38 /ISAPI/Security/onlineUser

Get the information of online users.

Request URI Definition

Table A-404 GET /ISAPI/Security/onlineUser

Method	GET
Description	Get the information of online users.
Query	security: the version No. of encryption scheme. When security does not exist, it indicates that the data is not encrypted; when security is 1, it indicates that the nodes of sensitive information in the message are encrypted in AES128 CBC mode; when security is 2, it indicates that the nodes of sensitive information in the message are encrypted in AES256 CBC mode. iv: the initialization vector, and it is required when security is 1 or 2.
Request	None.
Response	Succeeded: <i>XML_OnlineUserList</i> Failed: <i>XML_ResponseStatus</i>

A.8.39 /ISAPI/Security/questionConfiguration

Get or set device security questions.

Request URI Definition

Table A-405 GET /ISAPI/Security/questionConfiguration

Method	GET
Description	Get device security questions.
Query	None.
Request	None.
Response	Succeeded: <i>XML_SecurityQuestion</i> Failed: <i>XML_ResponseStatus</i>

Table A-406 PUT /ISAPI/Security/questionConfiguration

Method	PUT
Description	Set device security questions.
Query	security : the version No. of encryption scheme. When security does not exist, it indicates that the data is not encrypted; when security is 1, it indicates that the nodes of sensitive information in the message are encrypted in AES128 CBC mode; when security is 2, it indicates that the nodes of sensitive information in the message are encrypted in AES256 CBC mode. iv : the initialization vector, and it is required when security is 1 or 2.
Request	<i>XML_SecurityQuestion</i>
Response	<i>XML_ResponseStatus</i>

A.8.40 /ISAPI/Security/RTSPCertificate

Get or set parameters of devices' RTSP authentication mode.

Request URI Definition

Table A-407 GET /ISAPI/Security/RTSPCertificate

Method	GET
Description	Get parameters of devices' RTSP authentication mode.
Query	None.
Request	None.
Response	Succeeded: <i>XML_RTSPCertificate</i> Failed: <i>XML_ResponseStatus</i>

Table A-408 PUT /ISAPI/Security/RTSPCertificate

Method	PUT
Description	Set parameters of devices' RTSP authentication mode.
Query	None.
Request	<i>XML_RTSPCertificate</i>
Response	<i>XML_ResponseStatus</i>

A.8.41 /ISAPI/Security/RTSPCertificate/capabilities

Get the capability of devices' RTSP authentication mode.

Request URI Definition

Table A-409 GET /ISAPI/Security/RTSPCertificate/capabilities

Method	GET
Description	Get the capability of devices' RTSP authentication mode.
Query	None.
Request	None.
Response	Succeeded: <i>XML_RTSPCertificateCap</i> Failed: <i>XML_ResponseStatus</i>

A.8.42 /ISAPI/Security/serverCertificate/capabilities?format=json

Get the client/server certificate capability.

Request URI Definition

Table A-410 GET /ISAPI/Security/serverCertificate/capabilities?format=json

Method	GET
Description	Get the client/server certificate capability.
Query	format: determine the format of request or response message.
Request	None.
Response	Succeeded: <i>JSON_ServerCertificateCap</i> Failed: <i>JSON_ResponseStatus</i>

A.8.43 /ISAPI/Security/serverCertificate/certificate

Get, upload, and delete device certificate information.

Request URI Definition

Table A-411 GET /ISAPI/Security/serverCertificate/certificate

Method	GET
Description	Get device certificate information.

Query	type: certificate type, it can be set to "CCClientCertificate" (client certificate for CC authentication) or "LogServer" (log server certificate); if no value is assigned to type , it represents the HTTPS certificate.
Request	None.
Response	Succeeded: <i>XML_CertificateInfo</i> <i>XML_ResponseStatus</i>

Table A-412 PUT /ISAPI/Security/serverCertificate/certificate

Method	PUT
Description	Upload certificate to device.
Query	type: certificate type, it can be set to "CCClientCertificate" (client certificate for CC authentication) or "LogServer" (log server certificate); if no value is assigned to type , it represents the HTTPS certificate.
Request	Certificate data (binary)
Response	<i>XML_ResponseStatus</i>

Table A-413 DELETE /ISAPI/Security/serverCertificate/certificate

Method	DELETE
Description	Delete the installed certificate of device.
Query	type: certificate type, it can be set to "CCClientCertificate" (client certificate for CC authentication) or "LogServer" (log server certificate); if no value is assigned to type , it represents the HTTPS certificate.
Request	None.
Response	<i>XML_ResponseStatus</i>

A.8.44 /ISAPI/Security/serverCertificate/certificate?customID=

Import the client/server certificate to the device.

Request URI Definition

Table A-414 POST /ISAPI/Security/serverCertificate/certificate?customID=

Method	POST
Description	Import the client/server certificate to the device.

Query	security: the version No. of encryption scheme. When security does not exist, it indicates that the data is not encrypted; when security is 1, it indicates that the nodes of sensitive information in the message are encrypted in AES128 CBC mode; when security is 2, it indicates that the nodes of sensitive information in the message are encrypted in AES256 CBC mode. iv: the initialization vector, and it is required when security is 1 or 2. customID: custom certificate ID, string, it consists of digits and characters.
Request	<i>XML_CertificateReq_ImportCert</i> +certificate data.
Response	<i>XML_ResponseStatus</i>

Remarks

- When the node <certificateMode> in the request message *XML_CertificateReq_ImportCert* is set to "signingRequest", it means importing a self-signed certificate to the device.
- When the node <certificateMode> in the request message *XML_CertificateReq_ImportCert* is set to "privateKey", it means importing the certificate along with the private key by the following two steps.
 - Import the message *XML_CertificateReq_ImportCert* and the certificate file.



Note

The node <dataType> in the message *XML_CertificateReq_ImportCert* should be set to "certificate".

- Import the message *XML_CertificateReq_ImportCert* and the private key.



Note

- The node <dataType> in the message *XML_CertificateReq_ImportCert* should be set to "privateKey".
- This step depends on the first step. If exception occurs, the device will return error.

A.8.45 /ISAPI/Security/serverCertificate/certificates?format=json

Get the information of multiple client/server certificates in a batch.

Request URI Definition

Table A-415 GET /ISAPI/Security/serverCertificate/certificates?format=json

Method	GET
Description	Get the information of multiple client/server certificates in a batch.

Query	format: determine the format of request or response message.
Request	None.
Response	Succeeded: <i>JSON_ServerCertificates</i> Failed: <i>JSON_ResponseStatus</i>

A.8.46 /ISAPI/Security/serverCertificate/certificates/<customID>?format=json

Get or delete the information of a specific client/server certificate.

Request URI Definition

Table A-416 GET /ISAPI/Security/serverCertificate/certificates/<customID>?format=json

Method	GET
Description	Get the information of a specific client/server certificate.
Query	format: determine the format of request or response message.
Request	None.
Response	Succeeded: <i>JSON_DeviceCertificate</i> Failed: <i>JSON_ResponseStatus</i>

Table A-417 DELETE /ISAPI/Security/serverCertificate/certificates/<customID>?format=json

Method	DELETE
Description	Delete the information of a specific client/server certificate.
Query	format: determine the format of request or response message.
Request	None.
Response	<i>JSON_ResponseStatus</i>

Remarks

The <customID> in the request URI refers to the custom certificate ID.

A.8.47 /ISAPI/Security/serverCertificate/certSignReq

Get signature request information, generate, and delete signature request of device certificate.

Request URI Definition

Table A-418 GET /ISAPI/Security/serverCertificate/certSignReq

Method	GET
Description	Get signature request information of device certificate.
Query	type : certificate type, it can be set to "CCClientCertificate" (client certificate for CC authentication) or "LogServer" (log server certificate); if no value is assigned to type , it represents the HTTPS certificate.
Request	None.
Response	Succeeded: <i>XML_CertificateReqInfo</i> Failed: <i>XML_ResponseStatus</i>

Table A-419 PUT /ISAPI/Security/serverCertificate/certSignReq

Method	PUT
Description	Generate signature request information of device certificate.
Query	type : certificate type, it can be set to "CCClientCertificate" (client certificate for CC authentication) or "LogServer" (log server certificate); if no value is assigned to type , it represents the HTTPS certificate.
Request	<i>XML_CertificateReq</i>
Response	<i>XML_ResponseStatus</i>

Table A-420 DELETE /ISAPI/Security/serverCertificate/certSignReq

Method	DELETE
Description	Delete signature request of device certificate.
Query	type : certificate type, it can be set to "CCClientCertificate" (client certificate for CC authentication) or "LogServer" (log server certificate); if no value is assigned to type , it represents the HTTPS certificate.
Request	None.
Response	<i>XML_ResponseStatus</i>

A.8.48 /ISAPI/Security/serverCertificate/certSignReq?customID=

Create a PKCS#10 signature request of the client/server certificate.

Request URI Definition

Table A-421 POST /ISAPI/Security/serverCertificate/certSignReq?customID=

Method	POST
Description	Create a PKCS#10 signature request of the client/server certificate.
Query	customID : custom certificate ID, string, it consists of digits and characters.
Request	<i>XML_CertificateReq</i>
Response	Succeeded: <i>XML_CertificateResult</i> Failed: <i>XML_ResponseStatus</i>

A.8.49 /ISAPI/Security/serverCertificate/downloadCertSignReq

Download the certificate of authentication client.

Request URI Definition

Table A-422 GET /ISAPI/Security/serverCertificate/downloadCertSignReq

Method	GET
Description	Download the certificate of authenticated client.
Query	type : certificate type, it can be set to "CCClientCertificate" (client certificate for CC authentication) or "LogServer" (log server certificate); if no value is assigned to type , it represents the HTTPS certificate.
Request	None.
Response	Certificate data (binary)

A.8.50 /ISAPI/Security/serverCertificate/downloadCertSignReq?customID=

Export the client/server certificate.

Request URI Definition

Table A-423 GET /ISAPI/Security/serverCertificate/downloadCertSignReq?customID=

Method	GET
Description	Export the client/server certificate.

Query	customID: custom certificate ID, string, it consists of digits and characters.
Request	None.
Response	Certificate data.

Remarks

The returned certificate data should be either formatted exactly according to PKCS#10 standard or a PKCS#10 file encoded in PEM format.

A.8.51 /ISAPI/Security/serverCertificate/selfSignCert?customID=

Get or generate the PKCS#10 signature request of the client/server self-signed certificate.

Request URI Definition

Table A-424 GET /ISAPI/Security/serverCertificate/selfSignCert?customID=

Method	GET
Description	Get the PKCS#10 signature request of the client/server self-signed certificate.
Query	customID: custom certificate ID, string, it consists of digits and characters.
Request	None.
Response	Succeeded: <i>XML_CertificateReq</i> Failed: <i>XML_ResponseStatus</i>

Table A-425 PUT /ISAPI/Security/serverCertificate/selfSignCert?customID=

Method	PUT
Description	Generate the PKCS#10 signature request of the client/server self-signed certificate.
Query	customID: custom certificate ID, string, it consists of digits and characters.
Request	<i>XML_CertificateReq</i>
Response	<i>XML_ResponseStatus</i>

A.8.52 /ISAPI/Security/userCheck

Log in to the device by digest.

Request URI Definition

Table A-426 GET /ISAPI/Security/userCheck

Method	GET
Description	Log in to the device by digest. This URI is used to check whether the user name matches with the password.
Query	None.
Request	None.
Response	Succeeded: <i>XML_userCheck</i> Failed: <i>XML_ResponseStatus</i>

A.8.53 /ISAPI/Security/UserPermission

Operations about the user permission of the device.

Request URI Definition

Table A-427 GET /ISAPI/Security/UserPermission

Method	GET
Description	Get the user permission of the device.
Query	None.
Request	None.
Response	Succeeded: <i>XML_UserPermissionList</i> Failed: <i>XML_ResponseStatus</i>

Table A-428 PUT /ISAPI/Security/UserPermission

Method	PUT
Description	Set the user permission of the device.
Query	None.
Request	<i>XML_UserPermissionList</i>
Response	<i>XML_ResponseStatus</i>

A.8.54 /ISAPI/Security/UserPermission/<ID>

Operations about a specific user's permission.

Request URI Definition

Table A-429 GET /ISAPI/Security/UserPermission/<ID>

Method	GET
Description	Get a specific user's permission.
Query	None.
Request	None.
Response	Succeeded: <i>XML_UserPermission</i> Failed: <i>XML_ResponseStatus</i>

Table A-430 PUT /ISAPI/Security/UserPermission/<ID>

Method	PUT
Description	Set a specific user's permission.
Query	None.
Request	<i>XML_UserPermission</i>
Response	<i>XML_ResponseStatus</i>

A.8.55 /ISAPI/Security/UserPermission/<ID>/localPermission

Get and set the local permission of a specified user.

Request URI Definition

Table A-431 GET /ISAPI/Security/UserPermission/<ID>/localPermission

Method	GET
Description	Get the local permission of a specified user.
Query	None.
Request	None.
Response	<i>XML_localPermission</i>

Table A-432 PUT /ISAPI/Security/UserPermission/<ID>/localPermission

Method	PUT
Description	Set the local permission of a specified user.
Query	None.

Request	<i>XML_localPermission</i>
Response	<i>XML_ResponseStatus</i>

A.8.56 /ISAPI/Security/UserPermission/<ID>/remotePermission

Get and set the remote permission of a specified user.

Request URI Definition

Table A-433 GET /ISAPI/Security/UserPermission/<ID>/remotePermission

Method	GET
Description	Get the remote permission of a specified user.
Query	None.
Request	None.
Response	<i>XML_remotePermission</i>

Table A-434 PUT /ISAPI/Security/UserPermission/<ID>/remotePermission

Method	PUT
Description	Set the remote permission of a specified user.
Query	None.
Request	<i>XML_remotePermission</i>
Response	<i>XML_ResponseStatus</i>

A.8.57 /ISAPI/Security/UserPermission/adminCap

Get the user permission capability of the administrator.

Request URI Definition

Table A-435 GET /ISAPI/Security/UserPermission/adminCap

Method	GET
Description	Get the user permission capability of the administrator.
Query	None.
Request	None.
Response	Succeeded: <i>XML_UserPermissionCap</i> Failed: <i>XML_ResponseStatus</i>

Remarks

Only permissions that can be configured by the admin user should be returned.

A.8.58 /ISAPI/Security/UserPermission/anonymouslogin

Get or assign permission of the anonymous user.

Request URI Definition**Table A-436 GET /ISAPI/Security/UserPermission/anonymouslogin**

Method	GET
Description	Get permission of the anonymous user.
Query	None.
Request	None.
Response	Succeeded: <i>XML_anonymouslogin</i> Failed: <i>XML_ResponseStatus</i>

Table A-437 PUT /ISAPI/Security/UserPermission/anonymouslogin

Method	PUT
Description	Assign permission of the anonymous user.
Query	None.
Request	<i>XML_anonymouslogin</i>
Response	<i>XML_ResponseStatus</i>

A.8.59 /ISAPI/Security/UserPermission/capabilities

Get the configuration capability of all users' permissions.

Request URI Definition**Table A-438 GET /ISAPI/Security/UserPermission/capabilities**

Method	GET
Description	Get the configuration capability of all users' permissions.
Query	None.
Request	None.
Response	Succeeded: <i>XML_Cap_UserPermissionList</i>

	Failed: <i>XML_ResponseStatus</i>
--	-----------------------------------

A.8.60 /ISAPI/Security/UserPermission/operatorCap

Get the user permission capability of the operator.

Request URI Definition

Table A-439 GET /ISAPI/Security/UserPermission/operatorCap

Method	GET
Description	Get the user permission capability of the operator.
Query	None.
Request	None.
Response	Succeeded: <i>XML_UserPermissionCap</i> Failed: <i>XML_ResponseStatus</i>

Remarks

Only permissions that can be configured by the operator should be returned.

A.8.61 /ISAPI/Security/UserPermission/viewerCap

Get the user permission capability of the viewer.

Request URI Definition

Table A-440 GET /ISAPI/Security/UserPermission/viewerCap

Method	GET
Description	Get the user permission capability of the viewer.
Query	None.
Request	None.
Response	<i>XML_UserPermissionCap</i>

A.8.62 /ISAPI/Security/users

Get, set, or delete information of all users; add a new user.

Request URI Definition

Table A-441 GET /ISAPI/Security/users

Method	GET
Description	Get information of all users.
Query	security: the version No. of encryption scheme. When security does not exist, it indicates that the data is not encrypted; when security is 1, it indicates that the nodes of sensitive information in the message are encrypted in AES128 CBC mode; when security is 2, it indicates that the nodes of sensitive information in the message are encrypted in AES256 CBC mode. iv: the initialization vector, and it is required when security is 1 or 2.
Request	None
Response	Succeeded: <i>XML_UserList</i> Failed: <i>XML_ResponseStatus</i>

Table A-442 PUT /ISAPI/Security/users

Method	PUT
Description	Set information of all users.
Query	security: the version No. of encryption scheme. When security does not exist, it indicates that the data is not encrypted; when security is 1, it indicates that the nodes of sensitive information in the message are encrypted in AES128 CBC mode; when security is 2, it indicates that the nodes of sensitive information in the message are encrypted in AES256 CBC mode. iv: the initialization vector, and it is required when security is 1 or 2.
Request	<i>XML_UserList</i>
Response	<i>XML_ResponseStatus</i>

Table A-443 DELETE /ISAPI/Security/users

Method	DELETE
Description	Delete information of all users.
Query	None
Request	None
Response	<i>XML_ResponseStatus</i>

Table A-444 POST /ISAPI/Security/users

Method	POST
Description	Add a user.
Query	security: the version No. of encryption scheme. When security does not exist, it indicates that the data is not encrypted; when security is 1, it indicates that the nodes of sensitive information in the message are encrypted in AES128 CBC mode; when security is 2, it indicates that the nodes of sensitive information in the message are encrypted in AES256 CBC mode. iv: the initialization vector, and it is required when security is 1 or 2.
Request	<i>XML_User</i>
Response	<i>XML_ResponseStatus</i>

Remarks

- A default account "admin" must be provided with the administrator permission and it cannot be deleted.
- Passwords can only be uploaded and cannot be displayed when getting the user list of the device.
- The user ID should be returned when adding users to the user list.

Example

Sample Code for Adding User

```
POST /ISAPI/Security/users HTTP/1.1
Host: 10.17.132.49
Content-Length: 335
Connection: Keep-Alive
Authorization: Digest username="admin",
realm="DS-2CD2F12FWD-IWS",
nonce="4e6a4d774d6a63304f544936593255335a474d334f54673d",
uri="/ISAPI/Security/users",
cnonce="178e14d4977c835db891b5392b5d0a67",
nc=00000015,
response="d92518fb00cd2772cf212f5326b251be",
qop="auth"
```

```
<?xml version="1.0" encoding="UTF-8"?>
<User>
  <id>0</id>
  <userName>test</userName>
  <password>1qaz2wsx</password>
  <bondIpList>
    <bondIp>
      <id>1</id>
      <ipAddress>0.0.0.0</ipAddress>
```

```
<ipv6Address>::</ipv6Address>
</bondIp>
</bondIpList>
<macAddress></macAddress>
<userLevel>Operator</userLevel>
<attribute>
  <inherent>false</inherent>
</attribute>
</User>

HTTP/1.1 200 OK
Date: Tue, 14 Mar 2017 20:24:31 GMT
Server: App-webs/
Connection: close
Content-Length: 288
Content-Type: application/xml

<?xml version="1.0" encoding="UTF-8"?>
<ResponseStatus version="2.0" xmlns="http://www.isapi.com/ver20/XMLSchema">
  <requestURL>/ISAPI/Security/users</requestURL>
  <statusCode>1</statusCode>
  <statusString>OK</statusString>
  <id>2</id>
  <subStatusCode>ok</subStatusCode>
</ResponseStatus>
```

Example

Sample Code for Setting User Permission

```
PUT /ISAPI/Security/UserPermission/2 HTTP/1.1
Host: 10.17.132.49
Content-Length: 891
Connection: Keep-Alive
Authorization: Digest username="admin",
realm="DS-2CD2F12FWD-IWS",
nonce="4e6a4d774d6a63304f544936593255335a474d334f54673d",
uri="/ISAPI/Security/UserPermission/2",
cnonce="178e14d4977c835db891b5392b5d0a67",
nc=00000016,
response="f69a20db00b36e9bb46c26c06e9bae80",
qop="auth"
```

```
<?xml version="1.0" encoding="utf-8"?>
<UserPermission>
  <id>2</id>
  <userID>2</userID>
  <userType>operator</userType>
  <remotePermission>
    <parameterConfig>false</parameterConfig>
    <logOrStateCheck>true</logOrStateCheck>
    <upgrade>false</upgrade>
    <voiceTalk>true</voiceTalk>
```

```
<restartOrShutdown>false</restartOrShutdown>
<alarmOutOrUpload>false</alarmOutOrUpload>
<controlLocalOut>false</controlLocalOut>
<transparentChannel>false</transparentChannel>
<preview>true</preview>
<record>true</record>
<ptzControl>true</ptzControl>
<playBack>true</playBack>
<videoChannelPermissionList>
  <videoChannelPermission>
    <id>1</id>
    <preview>true</preview>
    <record>true</record>
    <playBack>true</playBack>
  </videoChannelPermission>
</videoChannelPermissionList>
<ptzChannelPermissionList>
  <ptzChannelPermission>
    <id>1</id>
    <ptzControl>true</ptzControl>
  </ptzChannelPermission>
</ptzChannelPermissionList>
</remotePermission>
</UserPermission>
```

HTTP/1.1 200 OK

Date: Tue, 14 Mar 2017 20:24:31 GMT

Server: App-webs/

Connection: close

Content-Length: 288

Content-Type: application/xml

```
<?xml version="1.0" encoding="UTF-8"?>
<ResponseStatus version="2.0" xmlns="http://www.isapi.com/ver20/XMLSchema">
  <requestURL>/ISAPI/Security/UserPermission/2</requestURL>
  <statusCode>1</statusCode>
  <statusString>OK</statusString>
  <subStatusCode>ok</subStatusCode>
</ResponseStatus>
```

A.8.63 /ISAPI/Security/users/<ID>

Get, set, or delete information of a specific user.

Request URI Definition

Table A-445 GET /ISAPI/Security/users/<ID>

Method	GET
Description	Get a specific user's configuration of the device.
Query	security: the version No. of encryption scheme. When security does not exist, it indicates that the data is not encrypted; when security is 1, it indicates that the nodes of sensitive information in the message are encrypted in AES128 CBC mode; when security is 2, it indicates that the nodes of sensitive information in the message are encrypted in AES256 CBC mode. iv: the initialization vector, and it is required when security is 1 or 2.
Request	None.
Response	Succeeded: <i>XML_User</i> Failed: <i>XML_ResponseStatus</i>

Table A-446 PUT /ISAPI/Security/users/<ID>

Method	PUT
Description	Get information of a specific user.
Query	security: the version No. of encryption scheme. When security does not exist, it indicates that the data is not encrypted; when security is 1, it indicates that the nodes of sensitive information in the message are encrypted in AES128 CBC mode; when security is 2, it indicates that the nodes of sensitive information in the message are encrypted in AES256 CBC mode. iv: the initialization vector, and it is required when security is 1 or 2.
Request	<i>XML_User</i>
Response	<i>XML_ResponseStatus</i>

Table A-447 DELETE /ISAPI/Security/users/<ID>

Method	DELETE
Description	Delete information of a specific user.
Query	security: the version No. of encryption scheme. When security does not exist, it indicates that the data is not encrypted; when security is 1, it indicates that the nodes of sensitive information in the message are encrypted in AES128 CBC mode; when security is 2, it indicates

	that the nodes of sensitive information in the message are encrypted in AES256 CBC mode. iv : the initialization vector, and it is required when security is 1 or 2. loginPassword : encrypted administrator password.
Request	None
Response	<i>XML_ResponseStatus</i>

Remarks

- The <ID> in the request URI refer to user ID.
- The administrator account cannot be deleted.

A.8.64 /ISAPI/Security/users/<ID>/capabilities

Get the configuration capability of a specific user.

Request URI Definition

Table A-448 GET /ISAPI/Security/users/<ID>/capabilities

Method	GET
Description	Get the configuration capability of a specific user.
Query	security : the version No. of encryption scheme. When security does not exist, it indicates that the data is not encrypted; when security is 1, it indicates that the nodes of sensitive information in the message are encrypted in AES128 CBC mode; when security is 2, it indicates that the nodes of sensitive information in the message are encrypted in AES256 CBC mode. iv : the initialization vector, and it is required when security is 1 or 2.
Request	None.
Response	Succeeded: <i>XML_Cap_User</i> Failed: <i>XML_ResponseStatus</i>

Remarks

The <ID> in the request URI refers to the user ID.

A.8.65 /ISAPI/Security/webCertificate

Get or set the certificate type parameters of web service.

Request URI Definition

Table A-449 GET /ISAPI/Security/webCertificate

Method	GET
Description	Get the certificate type parameters of web service.
Query	None.
Request	None.
Response	Succeeded: <i>XML_WebCertificate</i> Failed: <i>XML_ResponseStatus</i>

Table A-450 PUT /ISAPI/Security/webCertificate

Method	PUT
Description	Set the certificate type parameters of web service.
Query	None.
Request	<i>XML_WebCertificate</i>
Response	<i>XML_ResponseStatus</i>

A.8.66 /ISAPI/SecurityCP/ReportCenterCfg/capabilities?format=json

Get the configuration capability of the report uploading method.

Request URI Definition

Table A-451 GET /ISAPI/SecurityCP/ReportCenterCfg/capabilities?format=json

Method	GET
Description	Get the configuration capability of the report uploading method.
Query	format: determine the format of request or response message.
Request	None.
Response	Succeeded: <i>JSON_Cap_ReportCenterCfg</i> Failed: <i>JSON_ResponseStatus</i>

A.8.67 /ISAPI/SecurityCP/ReportCenterCfg/<ID>?format=json

Operations about the configuration of the report uploading method.

Request URI Definition

Table A-452 GET /ISAPI/SecurityCP/ReportCenterCfg/<ID>?format=json

Method	GET
Description	Get the parameters of the report uploading method.
Query	format : determine the format of request or response message.
Request	None.
Response	Succeeded: <i>JSON_ReportCenterCfg</i> Failed: <i>JSON_ResponseStatus</i>

Table A-453 PUT /ISAPI/SecurityCP/ReportCenterCfg/<ID>?format=json

Method	PUT
Description	Set the parameters of the report uploading method.
Query	format : determine the format of request or response message.
Request	<i>JSON_ReportCenterCfg</i>
Response	<i>JSON_ResponseStatus</i>

Remarks

The <ID> in the request URI refers to the center group No.

A.9 /ISAPI/Smart

A.9.1 /ISAPI/Smart/capabilities

Get the capability of smart devices.

Request URI Definition

Table A-454 GET /ISAPI/Smart/capabilities

Method	GET
Description	Get the capability of smart devices.
Query	None
Request	None
Response	Succeeded: <i>XML_SmartCap</i> Failed: <i>XML_ResponseStatus</i>

A.9.2 /ISAPI/Smart/AudioDetection/channels/<ID>

Get or set the audio detection parameters of a specific channel.

Request URI Definition

Table A-455 GET /ISAPI/Smart/AudioDetection/channels/<ID>

Method	GET
Description	Get the audio detection parameters of a specific channel.
Query	None.
Request	None.
Response	Succeeded: <i>XML_AudioDetection</i> Failed: <i>XML_ResponseStatus</i>

Table A-456 PUT /ISAPI/Smart/AudioDetection/channels/<ID>

Method	PUT
Description	Set the audio detection parameters of a specific channel.
Query	None.
Request	<i>XML_AudioDetection</i>
Response	<i>XML_ResponseStatus</i>

A.9.3 /ISAPI/Smart/AudioDetection/channels/<ID>/capabilities

Get the configuration capability of audio detection of a specific channel.

Request URI Definition

Table A-457 GET /ISAPI/Smart/AudioDetection/channels/<ID>/capabilities

Method	GET
Description	Get the configuration capability of audio detection of a specific channel.
Query	None.
Request	None.
Response	Succeeded: <i>XML_Cap_AudioDetection</i> Failed: <i>XML_ResponseStatus</i>

A.9.4 /ISAPI/Smart/AudioDetection/channels/<ID>/status

Get the audio strength.

Request URI Definition

Table A-458 GET /ISAPI/Smart/AudioDetection/channels/<ID>/status

Method	GET
Description	Get the audio strength.
Query	None.
Request	None.
Response	Succeeded: <i>XML_AudioStrengthStatus</i> Failed: <i>XML_ResponseStatus</i>

A.10 /ISAPI/Snapshot

A.10.1 /ISAPI/Snapshot/channels

Get or set the parameters of multiple channels for capturing pictures by schedule.

Request URL Definition

Table A-459 GET /ISAPI/Snapshot/channels

Method	GET
Description	Get the parameters of multiple channels for capturing pictures by schedule.
Query	None.
Request	None.
Response	Succeeded: <i>XML_SnapshotChannellist</i> Failed: <i>XML_ResponseStatus</i>

Table A-460 PUT /ISAPI/Snapshot/channels

Method	PUT
Description	Set the parameters of multiple channels for capturing pictures by schedule.
Query	None.

Request	<i>XML_SnapshotChannelList</i>
Response	<i>XML_ResponseStatus</i>

A.10.2 /ISAPI/Snapshot/channels/<ID>

Get or set the parameters of a specific channel for capturing pictures by schedule.

Request URL Definition

Table A-461 GET /ISAPI/Snapshot/channels/<ID>

Method	GET
Description	Get the parameters of a specific channel for capturing pictures by schedule.
Query	None.
Request	None.
Response	Succeeded: <i>XML_SnapshotChannel</i> Failed: <i>XML_ResponseStatus</i>

Table A-462 PUT /ISAPI/Snapshot/channels/<ID>

Method	PUT
Description	Set the parameters of a specific channel for capturing pictures by schedule.
Query	None.
Request	<i>XML_SnapshotChannel</i>
Response	<i>XML_ResponseStatus</i>

Remarks

The <ID> in the request URL refers to the channel No.

A.10.3 /ISAPI/Snapshot/channels/<ID>/capabilities

Get the configuration capability of capturing pictures by schedule.

Request URI Definition

Table A-463 GET /ISAPI/Snapshot/channels/<ID>/capabilities

Method	GET
Description	Get the configuration capability of capturing pictures by schedule.
Query	None.
Request	None.
Response	Succeeded: <i>XML_Cap_SnapshotChannel</i> Failed: <i>XML_ResponseStatus</i>

A.11 /ISAPI/Streaming

A.11.1 /ISAPI/Streaming/channels

Operations about the encoding configuration of multiple channels.

Request URI Definition

Table A-464 GET /ISAPI/Streaming/channels

Method	GET
Description	Get the encoding parameters of multiple channels.
Query	None.
Request	None.
Response	<i>XML_StreamingChannelList</i>

Table A-465 PUT /ISAPI/Streaming/channels

Method	PUT
Description	Set the encoding parameters of multiple channels.
Query	None.
Request	<i>XML_StreamingChannelList</i>
Response	<i>XML_ResponseStatus</i>

Table A-466 POST /ISAPI/Streaming/channels

Method	POST
Description	Add an encoding channel.

Query	None.
Request	<i>XML_StreamingChannel</i>
Response	<i>XML_ResponseStatus</i>

Table A-467 DELETE /ISAPI/Streaming/channels

Method	DELETE
Description	Delete all encoding channels.
Query	None.
Request	None.
Response	<i>XML_ResponseStatus</i>

A.11.2 /ISAPI/Streaming/channels/<ID>

Operations about the encoding configurations of a specific channel.

Request URI Definition

Table A-468 GET /ISAPI/Streaming/channels/<ID>

Method	GET
Description	Get the encoding parameters of a specific channel.
Query	None.
Request	None.
Response	Succeeded: <i>XML_StreamingChannel</i> Failed: <i>XML_ResponseStatus</i>

Table A-469 PUT /ISAPI/Streaming/channels/<ID>

Method	PUT
Description	Set the encoding parameters of a specific channel.
Query	None.
Request	<i>XML_StreamingChannel</i>
Response	<i>XML_ResponseStatus</i>

Table A-470 DELETE /ISAPI/Streaming/channels/<ID>

Method	DELETE
Description	Delete the encoding parameters of a specific channel.

Query	None
Request	None.
Response	<i>XML_ResponseStatus</i>

Remarks

The <ID> in the URI refers to the channel ID.

Example

Interaction Example of Getting Encoding Parameters of A Specific Channel

```
GET /ISAPI/Streaming/channels/444 HTTP/1.1
```

```
...
```

```
HTTP/1.1 200 OK
```

```
Content-Type: application/xml; charset="UTF-8"
```

```
Content-Length: ISAPI
```

```
<?xml version="1.0" encoding="UTF-8"?>
<StreamingChannel version="2.0" xmlns="http://www.isapi.org/ver20/XMLSchema">
  <id>444</id>
  <channelName>Input 1 MPEG-4 ASP</channelName>
  <enabled>true</enabled>
  <Transport>
    <rtspPortNo>554</rtspPortNo>
    <maxPacketSize>1446</maxPacketSize>
    <ControlProtocolList>
      <ControlProtocol>
        <streamingTransport>RTSP</streamingTransport>
      </ControlProtocol>
      <ControlProtocol>
        <streamingTransport>HTTP</streamingTransport>
      </ControlProtocol>
    </ControlProtocolList>
  </Transport>
  <Video>
    <enabled>true</enabled>
    <videoInputChannelID>2</videoInputChannelID>
    <videoCodecType>MPEG4</videoCodecType>
    <videoScanType>progressive</videoScanType>
    <videoResolutionWidth> 640</videoResolutionWidth>
    <videoResolutionHeight>480</videoResolutionHeight>
    <videoPositionX>0</videoPositionX>
    <videoPositionY>0</videoPositionY>
    <videoQualityControlType>CBR</videoQualityControlType>
    <constantBitRate>2000</constantBitRate>
    <maxFrameRate>2500</maxFrameRate>
    <keyFrameInterval>1000</keyFrameInterval>
    <rotationDegree>0</rotationDegree>
    <mirrorEnabled>false</mirrorEnabled>
    <snapShotImageType>JPEG</snapShotImageType>
  </Video>
```



```

<Audio>
  <enabled>>false</enabled>
  <audioInputChannelID>2</audioInputChannelID>
  <audioCompressionType> G.726</audioCompressionType>
  <audioBitRate>24</audioBitRate>
  <audioSamplingRate>8</audioSamplingRate>
</Audio>
</StreamingChannel>

```

A.11.3 /ISAPI/Streaming/channels/<ID>/capabilities

Get encoding capability of a specific channel.

Request URI Definition

Table A-471 GET /ISAPI/Streaming/channels/<ID>/capabilities

Method	GET
Description	Get the encoding capability of a specific channel.
Query	None.
Request	None.
Response	Succeeded: <i>XML_Cap_StreamingChannel</i> Failed: <i>XML_ResponseStatus</i>

Remarks

The <ID> in the URI refers to the channel ID.

Example

Interaction Example of Getting Encoding Capability of A Specific Channel

```

GET /ISAPI/Streaming/channels/444/capabilities HTTP/1.1
...
HTTP/1.1 200 OK
Content-Type: application/xml; charset="UTF-8"
Content-Length: ISAPI

<?xml version="1.0" encoding="UTF-8"?>
<StreamingChannel version="2.0" xmlns="http://www.isapi.org/ver20/XMLSchema">
  <id opt="111,222,333,444">444</id>
  <channelName min="0" max="64">Input 1 MPEG-4 ASP</channelName>
  <enabled opt="true,false" def="true">true</enabled>
  <Transport>
    <rtspPortNo min="0" max="65535" def="554">554</rtspPortNo>
    <maxPacketSize min="0" max="1500">1446</maxPacketSize>
    <audioPacketLength min="0" max="5000"/>
    <audioInboundPacketLength min="0" max="5000"/>
    <audioInboundPortNo min="0" max="65535"/>
  </Transport>

```

```
<videoSourcePortNo min="0" max="65535"/>
<audioSourcePortNo min="0" max="65535"/>
<ControlProtocolList>
  <ControlProtocol>
    <streamingTransport opt="RTSP/RTP,HTTP">RTSP</streamingTransport>
  </ControlProtocol>
  <ControlProtocol>
    <streamingTransport opt="RTSP/RTP,HTTP">HTTP</streamingTransport>
  </ControlProtocol>
</ControlProtocolList>
<Unicast>
  <enabled opt="true,false" def="false"/>
  <rtpTransportType opt="RTP/UDP,RTP/TCP"/>
</Unicast>
<Multicast>
  <enabled opt="true,false" def="false"/>
  <userTriggerThreshold/>
  <videoDestPortNo min="0" max="65535"/>
  <audioDestPortNo min="0" max="65535"/>
  <destIPAddress min="8" max="16"/>
  <destIPv6Address min="15" max="39"/>
  <ttl min="0" max="127" def="1"/>
</Multicast>
<Security>
  <enabled opt="true,false" def="false"/>
</Security>
</Transport>
<Video>
  <enabled opt="true,false">true</enabled>
  <videoInputChannelID opt="1,2,3,4">2</videoInputChannelID>
  <videoCodecType opt="MJPEG,MPEG4">MPEG4</videoCodecType>
  <videoScanType opt="interlaced,progressive">progressive</videoScanType>
  <videoResolutionWidth min="0" max="640">640</videoResolutionWidth>
  <videoResolutionHeight min="0" max="480">480</videoResolutionHeight>
  <videoPositionX min="0" max="640">0</videoPositionX>
  <videoPositionY min="0" max="480">0</videoPositionY>
  <videoQualityControlType opt="CBR,VBR">CBR</videoQualityControlType>
  <constantBitRate min="50" max="4000" dynamic="true">2000</constantBitRate>
  <maxFrameRate opt="2500,1250,625,312,156,78, 830" dynamic="true">2500</maxFrameRate>
  <keyFrameInterval min="0", max="10000">1000</keyFrameInterval>
  <rotationDegree opt="0,90,180,270" def="0">0</rotationDegree>
  <mirrorEnabled opt="true,false" def="false">false</mirrorEnabled>
  <snapshotImageType opt="JPEG" def="JPEG">JPEG</snapshotImageType>
</Video>
<Audio>
  <enabled opt="true,false" def="false">false</enabled>
  <audioInputChannelID opt="1,2,3,4">2</audioInputChannelID>
  <audioCompressionType opt="G.726,G.711ulaw" def="G.726">G.726</audioCompressionType>
  <audioBitRate opt="16,24,32,40" def="32" dynamic="true">24</audioBitRate>
  <audioSamplingRate opt="8" dynamic="true">8</audioSamplingRate>
  <audioResolution opt="3,4,5,6" dynamic="true"/>
```

```
</Audio>  
</StreamingChannel>
```

A.11.4 /ISAPI/Streaming/channels/<ID>/dynamicCap

Get the dynamic encoding capability of a specific channel.

Request URI Definition

Table A-472 GET /ISAPI/Streaming/channels/<ID>/dynamicCap

Method	GET
Description	Get the dynamic encoding capability of a specific channel.
Query	None.
Request	None.
Response	<i>XML_DynamicCap</i>

Remarks

The <ID> in the URI refers to the channel ID.

A.11.5 /ISAPI/Streaming/channels/<ID>/picture

Manually capture picture in the video stream of a specific channel.

Request URI Definition

Table A-473 GET /ISAPI/Streaming/channels/<ID>/picture

Method	GET
Description	Manually capture picture in the video stream of a specific channel.
Query	(Optional) videoResolutionWidth : width of captured picture resolution. (Optional) videoResolutionHeight : height of captured picture resolution. (Optional) snapShotImageType : captured picture format, only the JPEG format is supported.
Request	None.
Response	Picture data, which is transmitted on HTTP.

Remarks

- The <ID> in the request URI refers to the streaming channel ID, which is equal to (channel No. × 100 + 1). For example, if a camera only has one channel, its ID is 101.
- This URI only supports capturing picture in main stream, and only JPEG format is supported.

Example

Sample Code for Manually Capturing Picture

```
GET /ISAPI/Streaming/channels/1/picture HTTP/1.1
Host: 10.17.132.49
Connection: Keep-Alive
Authorization: Digest username="admin",
realm="DS-2CD2F12FWD-IWS",
nonce="4d305134517a55344d6a55365a4445324f544130597a453d",
uri="/ISAPI/Streaming/channels/1/picture",
cnonce="481bb1afa48b24e512778b084bfe4977",
nc=00000001,
response="e5670d5d9d40c522afabb11b5fa35268",
qop="auth"

HTTP/1.1 200 OK
Content-Type: image/jpeg; charset="UTF-8"
Content-Length:19991

.....JFIF.....C.
.
...
.. .....!... .)10.)-;3:J>36F7,- (Binary picture data)
```

A.11.6 /ISAPI/Streaming/channels/<ID>/regionClip

Get or set target cropping parameters by streaming channel.

Request URI Definition

Table A-474 GET /ISAPI/Streaming/channels/<ID>/regionClip

Method	GET
Description	Get target cropping parameters by streaming channel.
Query	None.
Request	None.
Response	Succeeded: <i>XML_RegionClip</i> Failed: <i>XML_ResponseStatus</i>

Table A-475 PUT /ISAPI/Streaming/channels/<ID>/regionClip

Method	PUT
Description	Set target cropping parameters by streaming channel.
Query	None.
Request	<i>XML_RegionClip</i>
Response	<i>XML_ResponseStatus</i>

Remarks

The <ID> in the request URI refers to streaming channel ID.

A.11.7 /ISAPI/Streaming/channels/<ID>/regionClip/capabilities

Get configuration capability of target cropping by streaming channel.

Request URI Definition**Table A-476 GET /ISAPI/Streaming/channels/<ID>/regionClip/capabilities**

Method	GET
Description	Get configuration capability of target cropping by streaming channel.
Query	None.
Request	None.
Response	Succeeded: <i>XML_Cap_RegionClip</i> Failed: <i>XML_ResponseStatus</i>

Remarks

The <ID> in the request URI refers to streaming channel ID.

A.11.8 /ISAPI/Streaming/channels/<ID>/smartOverlap/capabilities?format=json

Get the capability of configuring the stream for displaying VCA rules of smart events.

Request URI Definition**Table A-477 GET /ISAPI/Streaming/channels/<ID>/smartOverlap/capabilities?format=json**

Method	GET
Description	Get the capability of configuring the stream for displaying VCA rules of smart events.

Query	format: determine the format of request or response message.
Request	None.
Response	Succeeded: <i>JSON_SmartOverlapCap</i> Failed: <i>JSON_ResponseStatus</i>

Remarks

The <ID> in the request URI refers to the channel No.

A.11.9 /ISAPI/Streaming/channels/<ID>/smartOverlap?format=json

Get or set the stream parameters for displaying VCA rules of smart events.

Request URI Definition

Table A-478 GET /ISAPI/Streaming/channels/<ID>/smartOverlap?format=json

Method	GET
Description	Get the stream parameters for displaying VCA rules of smart events.
Query	format: determine the format of request or response message.
Request	None.
Response	Succeeded: <i>JSON_SmartOverlap</i> Failed: <i>JSON_ResponseStatus</i>

Table A-479 PUT /ISAPI/Streaming/channels/<ID>/smartOverlap?format=json

Method	PUT
Description	Set the stream parameters for displaying VCA rules of smart events.
Query	format: determine the format of request or response message.
Request	<i>JSON_SmartOverlap</i>
Response	<i>JSON_ResponseStatus</i>

Remarks

The <ID> in the request URI refers to the channel No.

A.11.10 /ISAPI/Streaming/channels/<ID>/status

Get the streaming or encoding status of a specific channel.

Request URI Definition

Table A-480 GET /ISAPI/Streaming/channels/<ID>/status

Method	GET
Description	Get the streaming or encoding status of a specific channel.
Query	None.
Request	None.
Response	Succeeded: <i>XML_StreamingSessionStatusList</i> Failed: <i>XML_ResponseStatus</i>

Remarks

The <ID> in the request URI refers to the streaming channel ID.

A.11.11 /ISAPI/Streaming/encryption/capabilities?format=json

Get stream encryption capability.

Request URI Definition

Table A-481 GET /ISAPI/Streaming/encryption/capabilities?format=json

Method	GET
Description	Get stream encryption capability.
Query	format: determine the format of request or response message.
Request	None.
Response	<i>JSON_EncryptionCap</i>

A.11.12 /ISAPI/Streaming/encryption/secretKey?format=json

Operations about the configuration of stream encryption key.

Request URI Definition

Table A-482 GET /ISAPI/Streaming/encryption/secretKey?format=json

Method	GET
Description	Get the configuration parameters of stream encryption key.
Query	format: determine the format of request or response message.

	security : the version No. of encryption scheme. When security does not exist, it indicates that the data is not encrypted; when security is 1, it indicates that the nodes of sensitive information in the message are encrypted in AES128 CBC mode; when security is 2, it indicates that the nodes of sensitive information in the message are encrypted in AES256 CBC mode. iv : the initialization vector, and it is required when security is 1 or 2.
Request	None.
Response	<i>JSON_secretKey</i>

Table A-483 PUT /ISAPI/Streaming/encryption/secretKey?format=json

Method	GET
Description	Set stream encryption key.
Query	format : determine the format of request or response message. security : the version No. of encryption scheme. When security does not exist, it indicates that the data is not encrypted; when security is 1, it indicates that the nodes of sensitive information in the message are encrypted in AES128 CBC mode; when security is 2, it indicates that the nodes of sensitive information in the message are encrypted in AES256 CBC mode. iv : the initialization vector, and it is required when security is 1 or 2.
Request	<i>JSON_secretKey</i>
Response	<i>JSON_ResponseStatus</i>

A.11.13 /ISAPI/Streaming/encryption?format=json

Operations about the configuration of stream encryption.

Request URI Definition

Table A-484 GET /ISAPI/Streaming/encryption?format=json

Method	GET
Description	Get the configuration parameters of stream encryption.
Query	format : determine the format of request or response message.
Request	None.
Response	<i>JSON_EnableEncryption</i>

Table A-485 PUT /ISAPI/Streaming/encryption?format=json

Method	GET
Description	Enable stream encryption.
Query	format : determine the format of request or response message.
Request	<i>JSON_EnableEncryption</i>
Response	<i>JSON_ResponseStatus</i>

A.11.14 /ISAPI/Streaming/status

Get the streaming or encoding status of all channels.

Request URI Definition

Table A-486 GET /ISAPI/Streaming/status

Method	GET
Description	Get the streaming or encoding status of all channels.
Query	None.
Request	None.
Response	Succeeded: <i>XML_StreamingStatus</i> Failed: <i>XML_ResponseStatus</i>

A.11.15 rtsp://<host>[:port]/ISAPI/Streaming/channels/<ID>

Get basic parameter or session information for live view, or start and stop live view.

Request URL Definition

Table A-487 DESCRIBE rtsp://<host>[:port]/ISAPI/Streaming/channels/<ID>

Method	DESCRIBE
Description	Get basic parameters for live view.
Query	None.
Request	None.
Response	None.

Table A-488 SETUP rtsp://<host>[:port]/ISAPI/Streaming/channels/<ID>

Method	SETUP
Description	Get session information for live view.
Query	None.
Request	None.
Response	None.

Table A-489 PLAY rtsp://<host>[:port]/ISAPI/Streaming/channels/<ID>

Method	PLAY
Description	Start live view.
Query	None.
Request	None.
Response	None.

Table A-490 TEARDOW rtsp://<host>[:port]/ISAPI/Streaming/channels/<ID>

Method	TEARDOW
Description	Stop live view.
Query	None.
Request	None.
Response	None.

Remarks

The <ID> in the URL is defined as (channel No.)*100+(stream type value).

channel No.

For analog channel, the No. starts from 1, and for digital channel, the No. starts from the last analog channel No.

E.g., if the device has 16 analog channels and 16 digital channels, the analog channel No. is between 1 and 16, and the digital channel No. is between 17 and 32.

stream type value

The stream type values contain 1, 2, and 3. 1-main stream, 2-sub-stream, 3-third stream.

E.g., if ID is 101, it indicates the main stream of channel No.1; if ID is 102, it indicates the sub-stream of channel No. 2; if ID is 1601, it indicates the main stream of channel 16;...

A.11.16 rtsp://<host>[:port]/ISAPI/Streaming/channels/<ID>?npq=

Operations about multicast based on network protocol quality (NPQ).

Request URL Definition

Table A-491 DESCRIBE rtsp://<host>[:port]/ISAPI/Streaming/channels/<ID>?npq=

Method	DESCRIBE
Description	Request for getting basic parameters for multicast based on NPQ.
Query	npq : Network Protocol Quality (NPQ), this query parameter is used to enable streaming according to NPQ and by FEC (Forward Error Correction) method if its value is set to "fec".
Request	None.
Response	None.

Table A-492 SETUP rtsp://<host>[:port]/ISAPI/Streaming/channels/<ID>?npq=

Method	SETUP
Description	Request for getting session information for multicast based on NPQ.
Query	npq : Network Protocol Quality (NPQ), this query parameter is used to enable streaming according to NPQ and by FEC (Forward Error Correction) method if its value is set to "fec".
Request	None.
Response	None.

Table A-493 PLAY rtsp://<host>[:port]/ISAPI/Streaming/channels/<ID>?npq=

Method	PLAY
Description	Start multicast based on NPQ.
Query	npq : Network Protocol Quality (NPQ), this query parameter is used to enable streaming according to NPQ and by FEC (Forward Error Correction) method if its value is set to "fec".
Request	None.
Response	None.

Table A-494 TEARDOW rtsp://<host>[:port]/ISAPI/Streaming/channels/<ID>?npq=

Method	TEARDOW
Description	Stop multicast based on NPQ.

Query	npq : Network Protocol Quality (NPQ), this query parameter is used to enable streaming according to NPQ and by FEC (Forward Error Correction) method if its value is set to "fec".
Request	None.
Response	None.

Remarks

The <ID> in the request URL is the value of (channel No. × 100 + stream type value), e.g., if the channel No. is 1, and the stream type value is 1, the ID equals to 101.

- For channel No., if the device contains analog and digital channels, the analog channel No. starts from 1, and the digital channel No. starts from the next No. of the end analog channel. For example, if the device contains 16 analog channels and 16 digital channels, the analog channel No. is between 1 and 16, while the digital channel No. is between 17 and 32.
- For stream type, only two values are available, that is, 1 (main stream) and 2 (sub-stream).

A.11.17 rtsp://<host>[:port]/ISAPI/Streaming/tracks/<ID>?starttime=&endtime=

Operations about playback of a specific channel.

Request URL Definition

Table A-495 DESCRIBE rtsp://<host>[:port]/ISAPI/Streaming/tracks/<ID>?starttime=&endtime=

Method	DESCRIBE
Description	Request for getting basic parameters for playback.
Query	starttime : start time of the target stream, whose format is based on the ISO 8601 standard. (Optional) endtime : end time of the target stream, whose format is based on the ISO 8601 standard. If this field is empty, the stream will be continuously obtained until the session is terminated or paused manually.
Request	None.
Response	None.

Table A-496 SETUP rtsp://<host>[:port]/ISAPI/Streaming/tracks/<ID>?starttime=&endtime=

Method	SETUP
Description	Request for getting session information for playback.
Query	starttime : start time of the target stream, whose format is based on the ISO 8601 standard.

	(Optional) endtime : end time of the target stream, whose format is based on the ISO 8601 standard. If this field is empty, the stream will be continuously obtained until the session is terminated or paused manually.
Request	None.
Response	None.

Table A-497 PLAY rtsp://<host>[:port]/ISAPI/Streaming/tracks/<ID>?starttime=&endtime=

Method	PLAY
Description	Start playback.
Query	starttime : start time of the target stream, whose format is based on the ISO 8601 standard. (Optional) endtime : end time of the target stream, whose format is based on the ISO 8601 standard. If this field is empty, the stream will be continuously obtained until the session is terminated or paused manually.
Request	None.
Response	None.

Table A-498 TEARDOW rtsp://<host>[:port]/ISAPI/Streaming/tracks/<ID>?starttime=&endtime=

Method	TEARDOW
Description	Stop playback.
Query	starttime : start time of the target stream, whose format is based on the ISO 8601 standard. (Optional) endtime : end time of the target stream, whose format is based on the ISO 8601 standard. If this field is empty, the stream will be continuously obtained until the session is terminated or paused manually.
Request	None.
Response	None.

Remarks

The <ID> in the URL is defined as (channel No.)*100+(stream type value).

channel No.

For analog channel, the No. starts from 1, and for digital channel, the No. starts from the last analog channel No.

E.g., if the device has 16 analog channels and 16 digital channels, the analog channel No. is between 1 and 16, and the digital channel No. is between 17 and 32.

stream type value

The stream type values contain 1, 2, and 3. 1-main stream, 2-sub-stream, 3-third stream.

E.g., if **ID** is 101, it indicates the main stream of channel No.1; if **ID** is 102, it indicates the sub-stream of channel No. 2; if **ID** is 1601, it indicates the main stream of channel 16;...

A.12 /ISAPI/System

A.12.1 /ISAPI/System/accessDevice/associatedChannel?format=json

Get the linked channel information of added IoT device.

Request URI Definition

Table A-499 GET /ISAPI/System/accessDevice/associatedChannel?format=json

Method	GET
Description	Get the linked channel information of added IoT device.
Query	format : determine the format of request or response message.
Request	<i>JSON_IOT_ChannelInfoList</i>
Response	<i>JSON_AssociatedChannelList</i>

A.12.2 /ISAPI/System/AcsUpdate/capabilities

Get the capability of upgrading device (slave access control device, peripheral module, etc.).

Request URI Definition

Table A-500 GET /ISAPI/System/AcsUpdate/capabilities

Method	GET
Description	Get the capability of upgrading device (slave access control device, peripheral module, etc.).
Query	None.
Request	None.
Response	Succeeded: <i>XML_Cap_AcsUpdate</i> Failed: <i>XML_ResponseStatus</i>

A.12.3 /ISAPI/System/activate

Device activation.

Request URI Definition

Table A-501 PUT /ISAPI/System/activate

Method	PUT
Description	Send activation password to activate the device.
Query	None.
Request	<i>XML_ActivateInfo</i>
Response	<i>XML_ResponseStatus</i>

A.12.4 /ISAPI/System/algorithmsInfo/capabilities?format=json

Get the capability of getting the algorithm library information.

Request URI Definition

Table A-502 GET /ISAPI/System/algorithmsInfo/capabilities?format=json

Method	GET
Description	Get the capability of getting the algorithm library information.
Query	format: determine the format of request or response message.
Request	None.
Response	Succeeded: <i>JSON_AlgorithmsInfoCap</i> Failed: <i>JSON_ResponseStatus</i>

A.12.5 /ISAPI/System/algorithmsInfo?format=json

Get the algorithm library information.

Request URI Definition

Table A-503 GET /ISAPI/System/algorithmsInfo?format=json

Method	GET
Description	Get the algorithm library information.
Query	format: determine the format of request or response message.

Request	None.
Response	Succeeded: <i>JSON_AlgorithmsInfo</i> Failed: <i>JSON_ResponseStatus</i>

A.12.6 /ISAPI/System/algorithmsVersion

Get the version information of algorithm library.

Request URI Definition

Table A-504 GET /ISAPI/System/algorithmsVersion

Method	GET
Description	Get the version information of algorithm library.
Query	None
Request	None
Response	Succeeded: <i>XML_AlgorithmsVersion</i> Failed: <i>XML_ResponseStatus</i>

A.12.7 /ISAPI/System/Audio/AudioIn

Get or set audio input parameters of all channels.

Request URI Definition

Table A-505 GET /ISAPI/System/Audio/AudioIn

Method	GET
Description	Get audio input parameters of all channels.
Query	None.
Request	None.
Response	Succeeded: <i>XML_AudioInConfig</i> Failed: <i>XML_ResponseStatus</i>

Table A-506 PUT /ISAPI/System/Audio/AudioIn

Method	PUT
Description	Set audio input parameters of all channels.
Query	None.

Request	<i>XML_AudioInConfig</i>
Response	<i>XML_ResponseStatus</i>

A.12.8 /ISAPI/System/Audio/AudioIn/capabilities

Get the audio input configuration capability of all channels.

Request URI Definition

Table A-507 GET /ISAPI/System/Audio/AudioIn/capabilities

Method	GET
Description	Get the audio input configuration capability of all channels.
Query	None.
Request	None.
Response	Succeeded: <i>XML_AudioInConfigCap</i> Failed: <i>XML_ResponseStatus</i>

A.12.9 /ISAPI/System/Audio/AudioIn/channels/<ID>

Get or set audio input parameters of a specific channel.

Request URI Definition

Table A-508 GET /ISAPI/System/Audio/AudioIn/channels/<ID>

Method	GET
Description	Get audio input parameters of a specific channel.
Query	None.
Request	None.
Response	Succeeded: <i>XML_AudioIn</i> Failed: <i>XML_ResponseStatus</i>

Table A-509 PUT /ISAPI/System/Audio/AudioIn/channels/<ID>

Method	PUT
Description	Set audio input parameters of a specific channel.
Query	None.

Request	<i>XML_AudioIn</i>
Response	<i>XML_ResponseStatus</i>

Remarks

The <ID> in the request URI refers to the channel No.

A.12.10 /ISAPI/System/Audio/AudioIn/channels/<ID>/capabilities

Get the audio input capability of a specific channel.

Request URI Definition

Table A-510 GET /ISAPI/System/Audio/AudioIn/channels/<ID>/capabilities

Method	GET
Description	Get the audio input capability of a specific channel.
Query	None.
Request	None.
Response	Succeeded: <i>XML_AudioInCap</i> Failed: <i>XML_ResponseStatus</i>

Remarks

The <ID> in the request URI refers to the channel No.

A.12.11 /ISAPI/System/Audio/AudioOut/channels/<ID>

Get or set audio output parameters of a specific channel.

Request URI Definition

Table A-511 GET /ISAPI/System/Audio/AudioOut/channels/<ID>

Method	GET
Description	Get audio output parameters of a specific channel.
Query	None.
Request	None.
Response	Succeeded: <i>XML_AudioOut</i> Failed: <i>XML_ResponseStatus</i>

Table A-512 PUT /ISAPI/System/Audio/AudioOut/channels/<ID>

Method	PUT
Description	Set audio output parameters of a specific channel.
Query	None.
Request	<i>XML_AudioOut</i>
Response	<i>XML_ResponseStatus</i>

Remarks

The <ID> in the request URI refers to the channel No.

A.12.12 /ISAPI/System/Audio/AudioOut/channels/<ID>/capabilities

Get the audio output capability of a specific channel.

Request URI Definition**Table A-513 GET /ISAPI/System/Audio/AudioOut/channels/<ID>/capabilities**

Method	GET
Description	Get the audio output capability of a specific channel.
Query	None.
Request	None.
Response	Succeeded: <i>XML_AudioOutCap</i> Failed: <i>XML_ResponseStatus</i>

Remarks

The <ID> in the request URI refers to the channel No.

A.12.13 /ISAPI/System/Audio/capabilities

Get the audio capability.

Request URI Definition**Table A-514 GET /ISAPI/System/Audio/capabilities**

Method	GET
Description	Get the audio capability.
Query	None.

Request	None.
Response	Succeeded: <i>XML_AudioCap</i> Failed: <i>XML_ResponseStatus</i>

A.12.14 /ISAPI/System/Audio/channels

Get the configuration of all audio channels of the device.

Request URI Definition

Table A-515 GET /ISAPI/System/Audio/channels

Method	GET
Description	Get the configuration of all audio channels of the device.
Query	None.
Request	None.
Response	Succeeded: <i>XML_AudioChannelList</i> Failed: <i>XML_ResponseStatus</i>

A.12.15 /ISAPI/System/Audio/channels/<ID>

Get the configuration of a specific audio channel of the device.

Request URI Definition

Table A-516 GET /ISAPI/System/Audio/channels/<ID>

Method	GET
Description	Get the configuration of a specific audio channel of the device.
Query	None.
Request	None.
Response	Succeeded: <i>XML_AudioChannel</i> Failed: <i>XML_ResponseStatus</i>

A.12.16 /ISAPI/System/Audio/channels/<ID>/dynamicCap

Get dynamic audio capability by channel.

Request URI Definition

Table A-517 GET /ISAPI/System/Audio/channels/<ID>/dynamicCap

Method	GET
Description	Get dynamic audio capability by channel.
Query	None.
Request	<i>XML_AudioDescriptor</i>
Response	Succeeded: <i>XML_DynamicCap</i> Failed: <i>XML_ResponseStatus</i>

Remarks

The <ID> in the request URI refers to the audio input channel ID.

A.12.17 /ISAPI/System/autoMaintenance/capabilities?format=json

Get the capability of automatic maintenance configuration.

Request URI Definition

Table A-518 GET /ISAPI/System/autoMaintenance/capabilities?format=json

Method	GET
Description	Get the capability of automatic maintenance configuration.
Query	format: determine the format of request or response message.
Request	None.
Response	<i>JSON_AutoMaintenanceCap</i>

A.12.18 /ISAPI/System/autoMaintenance?format=json

Get automatic maintenance configuration parameters.

Request URI Definition

Table A-519 GET /ISAPI/System/autoMaintenance?format=json

Method	GET
Description	Get automatic maintenance configuration parameters.
Query	format: determine the format of request or response message.

Request	None.
Response	<i>JSON_AutoMaintenance</i>

Table A-520 PUT /ISAPI/System/autoMaintenance?format=json

Method	PUT
Description	Set automatic maintenance parameters.
Query	format: determine the format of request or response message.
Request	<i>JSON_AutoMaintenance</i>
Response	<i>JSON_ResponseStatus</i>

A.12.19 /ISAPI/System/autoSwitch/capabilities?format=json

Get configuration capability of auto power on or off.

Request URI Definition

Table A-521 GET /ISAPI/System/autoSwitch/capabilities?format=json

Method	GET
Description	Get configuration capability of auto power on or off.
Query	format: determine the format of request or response message.
Request	None.
Response	Succeeded: <i>JSON_AutoSwitchCap</i> Failed: <i>JSON_ResponseStatus</i>

A.12.20 /ISAPI/System/autoSwitch?format=json

Get or set parameters of auto power on or off.

Request URI Definition

Table A-522 GET /ISAPI/System/autoSwitch?format=json

Method	GET
Description	Get parameters of auto power on or off.
Query	format: determine the format of request or response message.
Request	None.
Response	Succeeded: <i>JSON_AutoSwitch</i>

	Failed: <i>JSON_ResponseStatus</i>
--	------------------------------------

Table A-523 PUT /ISAPI/System/autoSwitch?format=json

Method	PUT
Description	Set parameters of auto power on or off.
Query	None.
Request	<i>JSON_AutoSwitch</i>
Response	<i>JSON_ResponseStatus</i>

A.12.21 /ISAPI/System/Bluetooth/capabilities

Get the bluetooth configuration capability.

Request URI Definition

Table A-524 GET /ISAPI/System/Bluetooth/capabilities

Method	GET
Description	Get the bluetooth configuration capability.
Query	None.
Request	None.
Response	Succeeded: <i>XML_BluetoothCap</i> Failed: <i>XML_ResponseStatus</i>

A.12.22 /ISAPI/System/Bluetooth/ibeaconParam

Operations about the ibeacon bluetooth configuration.

Request URI Definition

Table A-525 GET /ISAPI/System/Bluetooth/ibeaconParam

Method	GET
Description	Get the ibeacon bluetooth parameters.
Query	None.
Request	None.
Response	Succeeded: <i>XML_IbeaconParam</i> Failed: <i>XML_ResponseStatus</i>

Table A-526 PUT /ISAPI/System/Bluetooth/ibeaconParam

Method	PUT
Description	Set the ibeacon bluetooth parameters.
Query	None.
Request	<i>XML_IbeaconParam</i>
Response	<i>XML_ResponseStatus</i>

A.12.23 /ISAPI/System/Bluetooth/ibeaconParam/capabilities

Get the ibeacon bluetooth configuration capability.

Request URI Definition

Table A-527 GET /ISAPI/System/Bluetooth/ibeaconParam/capabilities

Method	GET
Description	Get the ibeacon bluetooth configuration capability.
Query	None.
Request	None.
Response	Succeeded: <i>XML_Cap_IbeaconParam</i> Failed: <i>XML_ResponseStatus</i>

A.12.24 /ISAPI/System/capabilities

Get device capability.

Request URI Definition

Table A-528 GET /ISAPI/System/capabilities

Method	GET
Description	Get device capability.
Query	None
Request	None.
Response	Succeeded: <i>XML_DeviceCap</i> Failed: <i>XML_ResponseStatus</i>

A.12.25 /ISAPI/System/channels/<ID>/vibrationDetection/capabilities?format=json

Get the capability of vibration detection.

Request URI Definition**Table A-529 GET /ISAPI/System/channels/<ID>/vibrationDetection/capabilities?format=json**

Method	GET
Description	Get the capability of vibration detection.
Query	format: determine the format of request or response message.
Request	None.
Response	Succeeded: <i>JSON_VibrationDetectionCap</i> Failed: <i>JSON_ResponseStatus</i>

Remarks

The <ID> in the URI refers to the channel ID.

A.12.26 /ISAPI/System/channels/<ID>/vibrationDetection?format=json

Get or set the vibration detection parameters.

Request URI Definition**Table A-530 GET /ISAPI/System/channels/<ID>/vibrationDetection?format=json**

Method	GET
Description	Get vibration detection parameters.
Query	format: determine the format of request or response message.
Request	None.
Response	Succeeded: <i>JSON_VibrationDetection</i> Failed: <i>JSON_ResponseStatus</i>

Table A-531 PUT /ISAPI/System/channels/<ID>/vibrationDetection?format=json

Method	PUT
Description	Set vibration detection parameters.
Query	format: determine the format of request or response message.

Request	<i>JSON_VibrationDetection</i>
Response	<i>JSON_ResponseStatus</i>

Remarks

The <ID> in the URI refers to the channel ID.

A.12.27 /ISAPI/System/configurationData?secretkey=

Import or export configuration files securely.

Request URI Definition

Table A-532 GET /ISAPI/System/configurationData?secretkey=

Method	GET
Description	Export the configuration files securely.
Query	security: the version No. of encryption scheme. When security does not exist, it indicates that the data is not encrypted; when security is 1, it indicates that the nodes of sensitive information in the message are encrypted in AES128 CBC mode; when security is 2, it indicates that the nodes of sensitive information in the message are encrypted in AES256 CBC mode. iv: the initialization vector, and it is required when security is 1 or 2. secretkey: the verification key, it is provided by the upper-layer. It should be encrypted for exporting and recorded for importing.
Request	None.
Response	Opaque data (binary data).

Table A-533 PUT or POST /ISAPI/System/configurationData?secretkey=

Method	PUT, POST
Description	Import the configuration files securely.
Query	security: the version No. of encryption scheme. When security does not exist, it indicates that the data is not encrypted; when security is 1, it indicates that the nodes of sensitive information in the message are encrypted in AES128 CBC mode; when security is 2, it indicates that the nodes of sensitive information in the message are encrypted in AES256 CBC mode. iv: the initialization vector, and it is required when security is 1 or 2.

	secretkey : the verification key, it must be the same as the exported one and should be encrypted for importing.
Request	Opaque data (binary data for PUT method, data in form format for POST method).
Response	<i>XML_ResponseStatus</i>

Remarks

The device may reboot after importing the configuration file.

A.12.28 /ISAPI/System/configurationData?type=

Import or export device configuration files safely.

Request URI Definition

Table A-534 GET /ISAPI/System/configurationData?type=

Method	GET
Description	Export device configuration files safely.
Query	security: the version No. of encryption scheme. When security does not exist, it indicates that the data is not encrypted; when security is 1, it indicates that the nodes of sensitive information in the message are encrypted in AES128 CBC mode; when security is 2, it indicates that the nodes of sensitive information in the message are encrypted in AES256 CBC mode. iv: the initialization vector, and it is required when security is 1 or 2. secretkey: the verification key, which is provided by the upper-layer and should be encrypted for importing. type: the type of configuration parameters to be exported or imported, which contains "OSD", "MODE", "IMAGE" and "IOOUT" types. For "OSD" type, the OSD (On Screen Display) parameters will be imported or exported; for "MODE" type, the application mode parameters will be imported or exported; for "IMAGE" type, the image parameters will be imported or exported; for "IOOUT" type, the IO output parameters will be imported or exported. Multiple types can be selected and they should be separated by commas, e.g., /ISAPI/System/configurationData?type=OSD,MODE,IMAGE,IOOUT. If no type is selected, it is recommended to return failure response message or set type to NULL.

Request	None.
Response	Succeeded: Opaque data. Failed: <i>XML_ResponseStatus</i>

Table A-535 PUT or POST /ISAPI/System/configurationData?type=

Method	PUT, POST
Description	Import device configuration files safely.
Query	security: the version No. of encryption scheme. When security does not exist, it indicates that the data is not encrypted; when security is 1, it indicates that the nodes of sensitive information in the message are encrypted in AES128 CBC mode; when security is 2, it indicates that the nodes of sensitive information in the message are encrypted in AES256 CBC mode. iv: the initialization vector, and it is required when security is 1 or 2. secretkey: the verification key, which is provided by the upper-layer and should be encrypted for importing. type: the type of configuration parameters to be exported or imported, which contains "OSD", "MODE", "IMAGE" and "IOOUT" types. For "OSD" type, the OSD (On Screen Display) parameters will be imported or exported; for "MODE" type, the application mode parameters will be imported or exported; for "IMAGE" type, the image parameters will be imported or exported; for "IOOUT" type, the IO output parameters will be imported or exported. Multiple types can be selected and they should be separated by commas, e.g., /ISAPI/System/configurationData?type=OSD,MODE,IMAGE,IOOUT. If no type is selected, it is recommended to return failure response message or set type to NULL.
Request	Opaque data.
Response	<i>XML_ResponseStatus</i>

Remarks

- This URI will import or export all configuration parameters by default.
- The device needs to reboot for the imported parameters to take effect.

A.12.29 /ISAPI/System/deviceInfo

Operations about the device information.

Request URI Definition

Table A-536 GET /ISAPI/System/deviceInfo

Method	GET
Description	Get the device information.
Query	None
Request	None.
Response	Succeeded: <i>XML_DeviceInfo</i> Failed: <i>XML_ResponseStatus</i>

Table A-537 PUT /ISAPI/System/deviceInfo

Method	PUT
Description	Set the device information.
Query	None
Request	<i>XML_DeviceInfo</i>
Response	<i>XML_ResponseStatus</i>

A.12.30 /ISAPI/System/deviceInfo/capabilities

Get the device information configuration capability.

Request URI Definition

Table A-538 GET /ISAPI/System/deviceInfo/capabilities

Method	GET
Description	Get the device information configuration capability.
Query	None.
Request	None.
Response	Succeeded: <i>XML_Cap_DeviceInfo</i> Failed: <i>XML_ResponseStatus</i>

A.12.31 /ISAPI/System/DeviceLanguage

Operations about device language configuration.

Request URI Definition

Table A-539 GET /ISAPI/System/DeviceLanguage

Method	GET
Description	Get device language configuration parameters.
Query	None.
Request	None.
Response	Succeeded: <i>XML_DeviceLanguage</i> Failed: <i>XML_ResponseStatus</i>

Table A-540 PUT /ISAPI/System/DeviceLanguage

Method	PUT
Description	Set device language parameters.
Query	None.
Request	<i>XML_DeviceLanguage</i>
Response	<i>XML_ResponseStatus</i>

A.12.32 /ISAPI/System/DeviceLanguage/capabilities

Get the capability of device language configuration.

Request URI Definition

Table A-541 GET /ISAPI/System/DeviceLanguage/capabilities

Method	GET
Description	Get the capability of device language configuration.
Query	None.
Request	None.
Response	Succeeded: <i>XML_Cap_DeviceLanguage</i> Failed: <i>XML_ResponseStatus</i>

A.12.33 /ISAPI/System/diagnosedData

Get device diagnosis information.

Request URI Definition

Table A-542 GET /ISAPI/System/diagnosedData

Method	GET
Description	Get device diagnosis information.
Query	None.
Request	None.
Response	Non-transparent data

Remarks

Before getting the device diagnosis information, you should get the information export status via URI GET */ISAPI/System/diagnosedData/exportStatus* , only when the node <status> in *XML_ExportStatus* is "unexport", you can call this URI to get the device diagnosis information.

A.12.34 /ISAPI/System/diagnosedData/exportStatus

Get the export status of device diagnosis information.

Request URI Definition

Table A-543 GET /ISAPI/System/diagnosedData/exportStatus

Method	GET
Description	Get the export status of device diagnosis information.
Query	None.
Request	None.
Response	<i>XML_ExportStatus</i>

A.12.35 /ISAPI/System/diagnosedData/parameter

Get or set the parameters of exporting device diagnose information.

Request URI Definition

Table A-544 GET /ISAPI/System/diagnosedData/parameter

Method	GET
Description	Get the parameters of exporting device diagnose information.
Query	None.

Request	None.
Response	Succeeded: <i>XML_DiagnosedDataParameter</i> Failed: <i>XML_ResponseStatus</i>

Table A-545 PUT /ISAPI/System/diagnosedData/parameter

Method	PUT
Description	Set the parameters of exporting device diagnose information.
Query	None.
Request	<i>XML_DiagnosedDataParameter</i>
Response	<i>XML_ResponseStatus</i>

Remarks

The speed dome does not support getting parameters of exporting diagnose information.

A.12.36 /ISAPI/System/diagnosedData/parameter/capabilities

Get the capability of exporting device diagnose information.

Request URI Definition**Table A-546 GET /ISAPI/System/diagnosedData/parameter/capabilities**

Method	GET
Description	Get the capability of exporting device diagnose information.
Query	None.
Request	None.
Response	Succeeded: <i>XML_Cap_DiagnosedDataParameter</i> Failed: <i>XML_ResponseStatus</i>

A.12.37 /ISAPI/System/diagnosis/capabilities?format=json

Get the device diagnosis capability.

Request URI Definition**Table A-547 GET /ISAPI/System/diagnosis/capabilities?format=json**

Method	GET
Description	Get the device diagnosis capability.

Query	format : determine the format of request or response message.
Request	None.
Response	Succeeded: <i>JSON_DiagnosisCondCap</i> Failed: <i>JSON_ResponseStatus</i>

A.12.38 /ISAPI/System/diagnosis?format=json

Diagnose the device.

Request URI Definition

Table A-548 POST /ISAPI/System/diagnosis?format=json

Method	POST
Description	Diagnose the device.
Query	format : determine the format of request or response message.
Request	<i>JSON_DiagnosisCond</i>
Response	<i>JSON_DiagnosisResult</i>

A.12.39 /ISAPI/System/discoveryMode

Get or set the device discovery mode.

Request URI Definition

Table A-549 GET /ISAPI/System/discoveryMode

Method	GET
Description	Get the device discovery mode.
Query	None.
Request	None.
Response	Succeeded: <i>XML_DiscoveryMode</i> Failed: <i>XML_ResponseStatus</i>

Table A-550 PUT /ISAPI/System/discoveryMode

Method	PUT
Description	Set the device discovery mode.
Query	None.

Request	<i>XML_DiscoveryMode</i>
Response	<i>XML_ResponseStatus</i>

A.12.40 /ISAPI/System/discoveryMode/capabilities

Get the configuration capability of device discovery mode.

Request URI Definition

Table A-551 GET /ISAPI/System/discoveryMode/capabilities

Method	GET
Description	Get the configuration capability of device discovery mode.
Query	None.
Request	None.
Response	Succeeded: <i>XML_Cap_DiscoveryMode</i> Failed: <i>XML_ResponseStatus</i>

A.12.41 /ISAPI/System/doubleLensParking/capabilities

Get the capability of dual-lens parking space camera.

Request URI Definition

Table A-552 GET /ISAPI/System/doubleLensParking/capabilities

Method	GET
Description	Get the capability of dual-lens parking space camera.
Query	None.
Request	None.
Response	Succeeded: <i>XML_DoubleLensParkingCap</i> Failed: <i>XML_ResponseStatus</i>

A.12.42 /ISAPI/System/dumpData

Get the dump file data of the device.

Request URI Definition

Table A-553 GET /ISAPI/System/dumpData

Method	GET
Description	Get the dump file data of the device.
Query	None.
Request	None.
Response	Opaque data.

Remarks

The dump file data of the device may be in binary format or other format.

A.12.43 /ISAPI/System/exportLogToUSB/capabilities?format=json

Get the configuration capability of exporting log to USB flash drive.

Request URI Definition

Table A-554 GET /ISAPI/System/exportLogToUSB/capabilities?format=json

Method	GET
Description	Get the configuration capability of exporting log to USB flash drive.
Query	format: determine the format of request or response message.
Request	None.
Response	Succeeded: <i>JSON_ExportLogToUSBCap</i> Failed: <i>JSON_ResponseStatus</i>

A.12.44 /ISAPI/System/exportLogToUSB/mode?format=json

Set parameters of exporting logs to USB flash drive.

Request URI Definition

Table A-555 PUT /ISAPI/System/exportLogToUSB/mode?format=json

Method	PUT
Description	Set parameters of exporting logs to USB flash drive.
Query	format: determine the format of request or response message.

Request	<i>JSON_ExportLogToUSB</i>
Response	<i>JSON_ResponseStatus</i>

A.12.45 /ISAPI/System/exportLogToUSB/status?format=json

Get the status of exporting logs to USB flash drive.

Table A-556 GET /ISAPI/System/exportLogToUSB/status?format=json

Method	GET
Description	Get the status of exporting logs to USB flash drive.
Query	format: determine the format of request or response message.
Request	None.
Response	Succeeded: <i>JSON_ExportLogToUSBStatus</i> Failed: <i>JSON_ResponseStatus</i>

A.12.46 /ISAPI/System/exportUSBLog/capabilities?format=json

Get the configuration capability of exporting logs to PC.

Request URI Definition

Table A-557 GET /ISAPI/System/exportUSBLog/capabilities?format=json

Method	GET
Description	Get the configuration capability of exporting logs to PC.
Query	format: determine the format of request or response message.
Request	None.
Response	Succeeded: <i>JSON_ExportUSBLogCap</i> Failed: <i>JSON_ResponseStatus</i>

A.12.47 /ISAPI/System/exportUSBLog/status?format=json

Get the status of exporting logs to PC.

Request URI Definition

Table A-558 GET /ISAPI/System/exportUSBLog/status?format=json

Method	GET
Description	Get the status of exporting logs to PC.
Query	format: determine the format of request or response message.
Request	None.
Response	Succeeded: <i>JSON_ExportUSBLogStatus</i> Failed: <i>JSON_ResponseStatus</i>

A.12.48 /ISAPI/System/exportUSBLog?format=json

Export logs from USB flash drive to PC.

Table A-559 PUT /ISAPI/System/exportUSBLog?format=json

Method	PUT
Description	Export logs from USB flash drive to PC.
Query	format: determine the format of request or response message.
Request	<i>JSON_ExportUSBLog</i>
Response	<i>JSON_ResponseStatus</i>

A.12.49 /ISAPI/System/externalDevice/capabilities

Get peripherals configuration capability.

Request URI Definition

Table A-560 GET /ISAPI/System/externalDevice/capabilities

Method	GET
Description	Get peripherals configuration capability.
Query	None.
Request	None.
Response	<i>XML_Cap_ExternalDevice</i>

A.12.50 /ISAPI/System/factoryReset?mode=

Restore the device to default settings.

Request URI Definition

Table A-561 PUT /ISAPI/System/factoryReset?mode=

Method	PUT
Description	Restore the device to default settings.
Query	mode: The recovery mode, which contains "full", "basic", and "part" mode. The default mode is "full", and all the device parameters will be restored to default settings; for "basic" mode, the device parameters, except network and user parameters, will be restored to default settings; for "part" mode, only a part of default settings will be restored.
Request	None.
Response	<i>XML_ResponseStatus</i>

Remarks

- For access control devices, this URI is only supported by facial recognition terminal, and the "full" and "basic" recovery modes are not supported by passthrough method.
- For facial recognition terminal, if the recovery mode is set to "part", i.e., /ISAPI/System/factoryReset?mode=part, the settings of card, face, event, fingerprint, schedule template, network, user name, and password will be reserved without being restored.

A.12.51 /ISAPI/System/fileExport/capabilities?format=json

Get the capability of exporting files from the device.

Request URI Definition

Table A-562 GET /ISAPI/System/fileExport/capabilities?format=json

Method	GET
Description	Get the capability of exporting files from the device.
Query	format: determine the format of request or response message.
Request	None.
Response	Succeeded: <i>JSON_FileExportCap</i> Failed: <i>JSON_ResponseStatus</i>

A.12.52 /ISAPI/System/fileExport?format=json

Export files from the device.

Request URI Definition

Table A-563 POST /ISAPI/System/fileExport?format=json

Method	POST
Description	Export files from the device.
Query	format : determine the format of request or response message.
Request	<i>JSON_FileExport</i>
Response	<i>JSON_FileExportResult</i>

A.12.53 /ISAPI/System/firmwareCodeV2

Get firmware code list.

Request URI Definition

Table A-564 GET /ISAPI/System/firmwareCodeV2

Method	GET
Description	Get firmware code list.
Query	None.
Request	None.
Response	Succeeded: <i>XML_FirmwareCodeList</i> Failed: <i>XML_ResponseStatus</i>

A.12.54 /ISAPI/System/GPSCalibration/channels/<ID>/capabilities?format=json

Get GPS calibration configuration capability.

Request URI Definition

Table A-565 GET /ISAPI/System/GPSCalibration/channels/<ID>/capabilities?format=json

Method	GET
Description	Get GPS calibration configuration capability.
Query	format : determine the format of request or response message.

Request	None.
Response	Succeeded: <i>JSON_GPSCalibrationCap</i> Failed: <i>JSON_ResponseStatus</i>

Remarks

The <ID> in the request URI refers to the video channel ID.

A.12.55 /ISAPI/System/GPSCalibration/channels/<ID>/points/<PID>?format=json

Get, set or delete configuration parameters of a single GPS calibration point.

Request URI Definition

Table A-566 GET /ISAPI/System/GPSCalibration/channels/<ID>/points/<PID>?format=json

Method	GET
Description	Get the configuration parameters of a single GPS calibration point.
Query	format: determine the format of request or response message.
Request	None.
Response	Succeeded: <i>JSON_GPSCalibration</i> Failed: <i>JSON_ResponseStatus</i>

Table A-567 PUT /ISAPI/System/GPSCalibration/channels/<ID>/points/<PID>?format=json

Method	PUT
Description	Set the configuration parameters of a single GPS calibration point.
Query	format: determine the format of request or response message.
Request	<i>JSON_GPSCalibration</i>
Response	<i>JSON_ResponseStatus</i>

Table A-568 DELETE /ISAPI/System/GPSCalibration/channels/<ID>/points/<PID>?format=json

Method	DELETE
Description	Delete the configuration parameters of a single GPS calibration point.
Query	format: determine the format of request or response message.
Request	None.
Response	<i>JSON_ResponseStatus</i>

Remarks

The first <ID> in the request URI refers to the video channel ID and the second <ID> refers to calibration point ID.

A.12.56 /ISAPI/System/GPSCalibration/channels/<ID>/points/status?format=json

Get GPS calibration status.

Request URI Definition

Table A-569 GET /ISAPI/System/GPSCalibration/channels/<ID>/points/status?format=json

Method	GET
Description	Get GPS calibration status.
Query	format: determine the format of request or response message.
Request	None.
Response	Succeeded: <i>JSON_GPSCalibrationStatus</i> Failed: <i>JSON_ResponseStatus</i>

Remarks

The <ID> in the request URI refers to the video channel ID.

A.12.57 /ISAPI/System/GPSCalibration/channels/<ID>/points?format=json

Get, set or delete configuration parameters of multiple GPS calibration points in a batch.

Request URI Definition

Table A-570 GET /ISAPI/System/GPSCalibration/channels/<ID>/points?format=json

Method	GET
Description	Get configuration parameters of multiple GPS calibration points in a batch.
Query	format: determine the format of request or response message.
Request	None.
Response	Succeeded: <i>JSON_GPSCalibrationList</i> Failed: <i>JSON_ResponseStatus</i>

Table A-571 PUT /ISAPI/System/GPSCalibration/channels/<ID>/points?format=json

Method	PUT
Description	Set configuration parameters of multiple GPS calibration points in a batch.
Query	format: determine the format of request or response message.
Request	<i>JSON_GPSCalibrationList</i>
Response	<i>JSON_ResponseStatus</i>

Table A-572 DELETE /ISAPI/System/GPSCalibration/channels/<ID>/points?format=json

Method	DELETE
Description	Delete configuration parameters of multiple GPS calibration points in a batch.
Query	format: determine the format of request or response message.
Request	None.
Response	<i>JSON_ResponseStatus</i>

Remarks

The <ID> in the request URI refers to the video channel ID.

A.12.58 /ISAPI/System/GPSPTZCalibration/channels/<ID>/calibration?format=json

Enable GPS calibration.

Request URI Definition

Table A-573 PUT /ISAPI/System/GPSPTZCalibration/channels/<ID>/calibration?format=json

Method	PUT
Description	Enable GPS calibration.
Query	format: determine the format of request or response message.
Request	None.
Response	<i>JSON_ResponseStatus</i>

Remarks

The <ID> in the request URI refers to the video channel ID.

A.12.59 /ISAPI/System/GPSVerification/channels/<ID>/display?format=json

Get or set the parameters of displaying GPS calibration verification result.

Request URI Definition**Table A-574 GET /ISAPI/System/GPSVerification/channels/<ID>/display?format=json**

Method	GET
Description	Get the parameters of displaying GPS calibration verification result.
Query	format: determine the format of request or response message.
Request	None.
Response	Succeeded: <i>JSON_VerificationDisplay</i> Failed: <i>JSON_ResponseStatus</i>

Table A-575 PUT /ISAPI/System/GPSVerification/channels/<ID>/display?format=json

Method	PUT
Description	Set the parameters of displaying GPS calibration verification result.
Query	format: determine the format of request or response message.
Request	<i>JSON_VerificationDisplay</i>
Response	<i>JSON_ResponseStatus</i>

Remarks

The <ID> in the URI refers to channel No.

A.12.60 /ISAPI/System/GPSVerification/channels/<ID>/points/<ID>?format=json

Get, set, or delete a specified verification point.

Request URI Definition**Table A-576 GET /ISAPI/System/GPSVerification/channels/<ID>/points/<ID>?format=json**

Method	GET
Description	Get the parameters of a specified verification point.
Query	format: determine the format of request or response message.
Request	None.
Response	Succeeded: <i>JSON_VerificationPoint</i>

	Failed: <i>JSON_ResponseStatus</i>
--	------------------------------------

Table A-577 PUT /ISAPI/System/GPSVerification/channels/<ID>/points/<ID>?format=json

Method	PUT
Description	Set the parameters of a specified verification point.
Query	format: determine the format of request or response message.
Request	<i>JSON_VerificationPoint</i>
Response	<i>JSON_ResponseStatus</i>

Table A-578 DELETE /ISAPI/System/GPSVerification/channels/<ID>/points/<ID>?format=json

Method	DELETE
Description	Delete a specified verification point.
Query	format: determine the format of request or response message.
Request	None.
Response	<i>JSON_ResponseStatus</i>

Remarks

The first <ID> in the URI refers to channel No., and second <ID> refers to verification point No.

A.12.61 /ISAPI/System/GPSVerification/channels/<ID>/points/capabilities?format=json

Get the capability of verifying GPS calibration.

Request URI Definition**Table A-579 GET /ISAPI/System/GPSVerification/channels/<ID>/points/capabilities?format=json**

Method	GET
Description	Get the capability of verifying GPS calibration.
Query	format: determine the format of request or response message.
Request	None.
Response	Succeeded: <i>JSON_VerificationPointList</i> Failed: <i>JSON_ResponseStatus</i>

Remarks

The <ID> in the URI refers to channel No.

A.12.62 /ISAPI/System/GPSVerification/channels/<ID>/points/resultInfo?format=json

Get the verification results of GPS calibration in a batch according to channel No.

Request URI Definition

Table A-580 GET /ISAPI/System/GPSVerification/channels/<ID>/points/resultInfo?format=json

Method	GET
Description	Get the verification results of GPS calibration in a batch according to channel No.
Query	format: determine the format of request or response message.
Request	None.
Response	Succeeded: <i>JSON_GPSVerification_ResultInfoList</i> Failed: <i>JSON_ResponseStatus</i>

Remarks

The <ID> in the URI refers to channel No.

A.12.63 /ISAPI/System/GPSVerification/channels/<ID>/points?format=json

Get, set, or delete multiple verification points in a batch, or add a verification point.

Request URI Definition

Table A-581 GET /ISAPI/System/GPSVerification/channels/<ID>/points?format=json

Method	GET
Description	Get the parameters of verification points in a batch according to channel No.
Query	format: determine the format of request or response message.
Request	None.
Response	Succeeded: <i>JSON_VerificationPointList</i> Failed: <i>JSON_ResponseStatus</i>

Table A-582 PUT /ISAPI/System/GPSVerification/channels/<ID>/points?format=json

Method	PUT
Description	Set the parameters of verification points in a batch according to channel No.
Query	format: determine the format of request or response message.
Request	<i>JSON_VerificationPointList</i>
Response	<i>JSON_ResponseStatus</i>

Table A-583 DELETE /ISAPI/System/GPSVerification/channels/<ID>/points?format=json

Method	DELETE
Description	Delete all verification points in a batch according to channel No.
Query	format: determine the format of request or response message.
Request	None.
Response	<i>JSON_ResponseStatus</i>

Table A-584 POST /ISAPI/System/GPSVerification/channels/<ID>/points?format=json

Method	POST
Description	Add a verification point according to channel No.
Query	format: determine the format of request or response message.
Request	<i>JSON_VerificationPoint</i>
Response	Succeeded: <i>JSON_GPSVerification_ResultInfo</i> Failed: <i>JSON_ResponseStatus</i>

Remarks

The <ID> in the URI refers to channel No.

A.12.64 /ISAPI/System/guardAgainstTheft

Get or set device anti-theft parameters.

Request URI Definition

Table A-585 GET /ISAPI/System/guardAgainstTheft

Method	GET
Description	Get the device anti-theft parameters.

Query	None.
Request	None.
Response	Succeeded: <i>XML_GuardAgainstTheft</i> Failed: <i>XML_ResponseStatus</i>

Table A-586 PUT /ISAPI/System/guardAgainstTheft

Method	PUT
Description	Set the device anti-theft parameters.
Query	None.
Request	<i>XML_GuardAgainstTheft</i>
Response	<i>XML_ResponseStatus</i>

A.12.65 /ISAPI/System/guardAgainstTheft/capabilities

Get the device anti-theft configuration capability.

Request URI Definition

Table A-587 GET /ISAPI/System/guardAgainstTheft/capabilities

Method	GET
Description	Get the device anti-theft configuration capability.
Query	None.
Request	None.
Response	Succeeded: <i>XML_Cap_GuardAgainstTheft</i> Failed: <i>XML_ResponseStatus</i>

A.12.66 /ISAPI/System/guideConfig/<guideEvent>/capabilities?format=json

Get the capability of quick setup instruction for specified event.

Request URI Definition

Table A-588 GET /ISAPI/System/guideConfig/<guideEvent>/capabilities?format=json

Method	GET
Description	Get the capability of quick setup instruction for specified event.
Query	format: determine the format of request or response message.

Request	None.
Response	Succeeded: <i>JSON_GuideConfigCap</i> Failed: <i>JSON_ResponseStatus</i>

Remarks

The <**guideEvent**> in the URI refers to the event, which supports quick setup, and now only "linkageCapture" (capture by linkage) is supported, i.e., the URI is: /ISAPI/System/guideConfig/linkageCapture/capabilities?format=json

A.12.67 /ISAPI/System/guideConfig/<guideEvent>?format=json

Get the parameters of quick setup instruction for specified event.

Request URI Definition

Table A-589 GET /ISAPI/System/guideConfig/<guideEvent>?format=json

Method	GET
Description	Get the parameters of quick setup instruction for specified event.
Query	format : determine the format of request or response message.
Request	None.
Response	Succeeded: <i>JSON_GuideConfig</i> Failed: <i>JSON_ResponseStatus</i>

Remarks

The <**guideEvent**> in the URI refers to the event, which supports quick setup, and now only "linkageCapture" (capture by linkage) is supported, i.e., the URI is: /ISAPI/System/guideConfig/linkageCapture?format=json

A.12.68 /ISAPI/System/Hardware

Operations about the device hardware configuration.

Request URI Definition

Table A-590 GET /ISAPI/System/Hardware

Method	GET
Description	Get the device hardware configuration parameters.
Query	None.

Request	None.
Response	Succeeded: <i>XML_HardwareService</i> Failed: <i>XML_ResponseStatus</i>

Table A-591 PUT /ISAPI/System/Hardware

Method	PUT
Description	Set the device hardware parameters.
Query	None.
Request	<i>XML_HardwareService</i>
Response	<i>XML_ResponseStatus</i>

A.12.69 /ISAPI/System/Hardware/defog

Operations about the defogging control configuration.

Request URI Definition

Table A-592 GET /ISAPI/System/Hardware/defog

Method	GET
Description	Get the defogging control configuration parameters.
Query	None.
Request	None.
Response	Succeeded: <i>XML_Defog</i> Failed: <i>XML_ResponseStatus</i>

Table A-593 PUT /ISAPI/System/Hardware/defog

Method	PUT
Description	Set the defogging control parameters.
Query	None.
Request	<i>XML_Defog</i>
Response	<i>XML_ResponseStatus</i>

A.12.70 /ISAPI/System/installationAngleCalibration/channels/<ID>/capabilities?format=json

Get the capability of installation angle calibration.

Request URI Definition

Table A-594 GET /ISAPI/System/installationAngleCalibration/channels/<ID>/capabilities?format=json

Method	GET
Description	Get the capability of installation angle calibration.
Query	format: determine the format of request or response message.
Request	None.
Response	Succeeded: <i>JSON_Cap_CalibrationStatus</i> Failed: <i>JSON_ResponseStatus</i>

Remarks

The <ID> in the request URI refers to the channel No.

A.12.71 /ISAPI/System/installationAngleCalibration/channels/<ID>?format=json

Get the installation angle calibration status or calibrate the installation angle.

Request URI Definition

Table A-595 GET /ISAPI/System/installationAngleCalibration/channels/<ID>?format=json

Method	GET
Description	Get the installation angle calibration status.
Query	format: determine the format of request or response message.
Request	None.
Response	Succeeded: <i>JSON_CalibrationStatus</i> Failed: <i>JSON_ResponseStatus</i>

Table A-596 PUT /ISAPI/System/installationAngleCalibration/channels/<ID>?format=json

Method	PUT
Description	Calibrate the installation angle.
Query	format: determine the format of request or response message.

Request	None.
Response	<i>JSON_ResponseStatus</i>

Remarks

The <ID> in the request URI refers to the channel No.

A.12.72 /ISAPI/System/IO

Get the alarm input and output information.

Request URI Definition

Table A-597 GET /ISAPI/System/IO

Method	GET
Description	Get the alarm input and output information.
Query	None.
Request	None.
Response	Succeeded: <i>XML_IOPortList</i> Failed: <i>XML_ResponseStatus</i>

A.12.73 /ISAPI/System/IO/capabilities

Get alarm input and output configuration capability.

Request URI Definition

Table A-598 GET /ISAPI/System/IO/capabilities

Method	GET
Description	Get alarm input and output configuration capability.
Query	None.
Request	None.
Response	<i>XML_IOCcap</i>

A.12.74 /ISAPI/System/IO/inputs

Get the information of all alarm input ports.

Request URI Definition**Table A-599 Get the information of all alarm input ports.**

Method	GET
Description	Get the information of all alarm input ports.
Query	None.
Request	None.
Response	<i>XML_IOInputPortList</i>

A.12.75 /ISAPI/System/IO/inputs/<ID>

Get or set an alarm input port.

Request URI Definition**Table A-600 GET /ISAPI/System/IO/inputs/<ID>**

Method	GET
Description	Get an alarm input's information.
Query	None.
Request	None.
Response	<i>XML_IOInputPort</i>

Table A-601 PUT /ISAPI/System/IO/inputs/<ID>

Method	PUT
Description	Set an alarm input's information.
Query	None.
Request	<i>XML_IOInputPort</i>
Response	<i>XML_ResponseStatus</i>

A.12.76 /ISAPI/System/IO/inputs/<ID>/status

Get the status of a specific alarm input.

Request URI Definition

Table A-602 GET /ISAPI/System/IO/inputs/<ID>/status

Method	GET
Description	Get the status of a specific alarm input.
Query	None.
Request	None.
Response	Succeeded: <i>XML_IOPortStatus</i> Failed: <i>XML_ResponseStatus</i>

Remarks

The <ID> in the request URI refers to the alarm input ID.

A.12.77 /ISAPI/System/IO/inputs/capabilities

Get alarm input configuration capability.

Request URI Definition

Table A-603 GET /ISAPI/System/IO/inputs/capabilities

Method	GET
Description	Get the alarm input configuration capability.
Query	None.
Request	None.
Response	<i>XML_Cap_IOInputPortList</i>

A.12.78 /ISAPI/System/IO/outputs

Get the information of all I/O output ports.

Request URI Definition

Table A-604 GET /ISAPI/System/IO/outputs

Method	GET
Description	Get the information of all I/O output ports.
Query	None.

Request	None.
Response	<i>XML_IOOutputPortList</i>

A.12.79 /ISAPI/System/IO/outputs/capabilities

Get the I/O output capability.

Request URI Definition

Table A-605 GET /ISAPI/System/IO/outputs/capabilities

Method	GET
Description	Get the I/O output capability.
Query	None.
Request	None.
Response	<i>XML_Cap_IOOutputPortList</i>

A.12.80 /ISAPI/System/IO/outputs/<ID>

Get or set the information of specified I/O output port.

Table A-606 GET /ISAPI/System/IO/outputs/<ID>

Method	GET
Description	Get the information of specified I/O output port.
Query	None.
Request	None.
Response	<i>XML_IOOutputPort</i>

Table A-607 PUT /ISAPI/System/IO/outputs/<ID>

Method	PUT
Description	Set the information of specified alarm output port.
Query	<i>XML_IOOutputPort</i>
Request	None.
Response	<i>XML_ResponseStatus</i>

A.12.81 /ISAPI/System/IO/outputs/<ID>/status

Get status of a specific alarm output.

Request URI Definition

Table A-608 GET /ISAPI/System/IO/outputs/<ID>/status

Method	GET
Description	Get status of a specific alarm output.
Query	None.
Request	None.
Response	Succeeded: <i>XML_IOPortStatus</i> Failed: <i>XML_ResponseStatus</i>

Remarks

The <ID> in the request URI refers to the alarm output ID.

A.12.82 /ISAPI/System/IO/outputs/<ID>/trigger

Manually trigger a specific alarm output.

Request URI Definition

Table A-609 PUT /ISAPI/System/IO/outputs/<ID>/trigger

Method	PUT
Description	Manually trigger a specific alarm output.
Query	none.
Request	<i>XML_IOPortData</i>
Response	<i>XML_ResponseStatus</i>

Remarks

The <ID> in the request URI refers to the alarm output ID.

A.12.83 /ISAPI/System/IO/status

Get alarm input and output status.

Request URI Definition**Table A-610 GET /ISAPI/System/IO/status**

Method	GET
Description	Get alarm input and output status.
Query	None.
Request	None.
Response	Succeeded: <i>XML_IOPortStatusList</i> Failed: <i>XML_ResponseStatus</i>

A.12.84 /ISAPI/System/IOT/channelConfig?format=json

Import or export the list of added IoT devices.

Request URI Definition**Table A-611 GET /ISAPI/System/IOT/channelConfig?format=json**

Method	GET
Description	Export the list of added IoT devices.
Query	format : determine the format of request or response message. security : the version No. of encryption scheme. When security does not exist, it indicates that the data is not encrypted; when security is 1, it indicates that the nodes of sensitive information in the message are encrypted in AES128 CBC mode; when security is 2, it indicates that the nodes of sensitive information in the message are encrypted in AES256 CBC mode. iv : the initialization vector, and it is required when security is 1 or 2.
Request	None.
Response	Succeeded: Opaque Data Failed: <i>JSON_ResponseStatus</i>

Table A-612 PUT /ISAPI/System/IOT/channelConfig?format=json

Method	PUT
Description	Import the list of added IoT devices.
Query	format : determine the format of request or response message.

	security : the version No. of encryption scheme. When security does not exist, it indicates that the data is not encrypted; when security is 1, it indicates that the nodes of sensitive information in the message are encrypted in AES128 CBC mode; when security is 2, it indicates that the nodes of sensitive information in the message are encrypted in AES256 CBC mode. iv : the initialization vector, and it is required when security is 1 or 2.
Request	Opaque Data
Response	<i>JSON_ResponseStatus</i> <i>JSON_IOT_ErrorList</i>

Remarks

The Opaque Data should be encrypted.

A.12.85 /ISAPI/System/IOT/channels/<ID>/<EventType>/capabilities?format=json

Get the event configuration capability of IoT device.

Request URI Definition

Table A-613 GET /ISAPI/System/IOT/channels/<ID>/<EventType>/capabilities?format=json

Method	GET
Description	Get the event configuration capability of IoT device.
Query	format : determine the format of request or response message.
Request	None.
Response	Succeeded: <i>JSON_XXCap</i> Failed: <i>JSON_ResponseStatus</i>

A.12.86 /ISAPI/System/IOT/channels/<ID>/<EventType>?format=json

Get or set the event/alarm parameters.

Request URI Definition

Table A-614 GET /ISAPI/System/IOT/channels/<ID>/<EventType>?format=json

Method	GET
Description	Get the event/alarm parameters.
Query	format : determine the format of request or response message.

Request	None.
Response	Succeeded: <i>JSON_XX</i> Failed: <i>JSON_ResponseStatus</i>

Table A-615 PUT /ISAPI/System/IOT/channels/<ID>/<EventType>?format=json

Method	PUT
Description	Set the event/alarm parameters.
Query	format: determine the format of request or response message.
Request	<i>JSON_XX</i>
Response	<i>JSON_ResponseStatus</i>

A.12.87 /ISAPI/System/IOT/channels/<ID>/alarmStatistics?format=json

Get the alarm statistics of added IoT device according to channel ID.

Request URI Definition

Table A-616 GET /ISAPI/System/IOT/channels/<ID>/alarmStatistics?format=json

Method	GET
Description	Get the alarm statistics of added IoT device according to channel ID.
Query	format: determine the format of request or response message.
Request	None.
Response	Succeeded: <i>JSON_AlarmStatistics</i> Failed: <i>JSON_ResponseStatus</i>

A.12.88 /ISAPI/System/IOT/channels/<ID>/all?format=json

Delete all channels (video channel and IoT channel) of added IoT device

Request URI Definition

Table A-617 DELETE /ISAPI/System/IOT/channels/<ID>/all?format=json

Method	DELETE
Description	Delete all channels (video channel and IoT channel) of added IoT device
Query	format: determine the format of request or response message.

Request	None.
Response	<i>JSON_ResponseStatus</i>

A.12.89 /ISAPI/System/IOT/channels/<ID>/basicParam?format=json

Set the basic parameters of added IoT device according to channel ID.

Request URI Definition

Table A-618 PUT /ISAPI/System/IOT/channels/<ID>/basicParam?format=json

Method	PUT
Description	Set the basic parameters of added IoT device according to channel ID.
Query	format: determine the format of request or response message.
Request	<i>JSON_BasicParam</i>
Response	<i>JSON_ResponseStatus</i>

A.12.90 /ISAPI/System/IOT/channels/<ID>/OSD/capabilities?format=json

Get the OSD configuration capability of specified IoT device channel.

Request URI Definition

Table A-619 GET /ISAPI/System/IOT/channels/<ID>/OSD/capabilities?format=json

Method	GET
Description	Get the OSD configuration capability of specified IoT device channel.
Query	format: determine the format of request or response message.
Request	None.
Response	Succeeded: <i>JSON_OSDCap</i> Failed: <i>JSON_ResponseStatus</i>

A.12.91 /ISAPI/System/IOT/channels/<ID>/OSD?format=json

Get or set the OSD parameters.

Request URI Definition

Table A-620 GET /ISAPI/System/IOT/channels/<ID>/OSD?format=json

Method	GET
Description	Get the OSD parameters.
Query	format: determine the format of request or response message.
Request	None.
Response	Succeeded: <i>JSON_OSD</i> Failed: <i>JSON_ResponseStatus</i>

Table A-621 PUT /ISAPI/System/IOT/channels/<ID>/OSD?format=json

Method	PUT
Description	Set the OSD parameters.
Query	format: determine the format of request or response message.
Request	<i>JSON_OSD</i>
Response	<i>JSON_ResponseStatus</i>

Remarks

The <ID> in the URI is defied as the channel ID.

A.12.92 /ISAPI/System/IOT/channels/<ID>/status?format=json

Get the status of specified channel of added IoT device.

Request URI Definition

Table A-622 GET /ISAPI/System/IOT/channels/<ID>/status?format=json

Method	GET
Description	Get the status of specified channel of added IoT device.
Query	format: determine the format of request or response message. security: the version No. of encryption scheme. When security does not exist, it indicates that the data is not encrypted; when security is 1, it indicates that the nodes of sensitive information in the message are encrypted in AES128 CBC mode; when security is 2, it indicates that the nodes of sensitive information in the message are encrypted in AES256 CBC mode. iv: the initialization vector, and it is required when security is 1 or 2.

Request	None.
Response	<i>JSON_IOTChannelStatus</i>

Remarks

The <ID> in the URI is defined as channel ID.

A.12.93 /ISAPI/System/IOT/channels/<ID>?format=json

Operations about specified added IoT device.

Request URI Definition

Table A-623 GET /ISAPI/System/IOT/channels/<ID>?format=json

Method	GET
Description	Get the information of added IoT device according to channel ID.
Query	format: determine the format of request or response message. security: the version No. of encryption scheme. When security does not exist, it indicates that the data is not encrypted; when security is 1, it indicates that the nodes of sensitive information in the message are encrypted in AES128 CBC mode; when security is 2, it indicates that the nodes of sensitive information in the message are encrypted in AES256 CBC mode. iv: the initialization vector, and it is required when security is 1 or 2.
Request	None.
Response	Succeeded: <i>JSON_IOTChannel</i> Failed: <i>JSON_ResponseStatus</i>

Table A-624 PUT /ISAPI/System/IOT/channels/<ID>?format=json

Method	PUT
Description	Set the added IoT device according to channel ID.
Query	format: determine the format of request or response message. security: the version No. of encryption scheme. When security does not exist, it indicates that the data is not encrypted; when security is 1, it indicates that the nodes of sensitive information in the message are encrypted in AES128 CBC mode; when security is 2, it indicates that the nodes of sensitive information in the message are encrypted in AES256 CBC mode. iv: the initialization vector, and it is required when security is 1 or 2.

Request	<i>JSON_IOTChannel</i>
Response	<i>JSON_ResponseStatus</i>

Table A-625 DELETE /ISAPI/System/IOT/channels/<ID>?format=json

Method	DELETE
Description	Delete the specified IoT device according to channel ID.
Query	format: determine the format of request or response message.
Request	None.
Response	<i>JSON_ResponseStatus</i>

Remarks

The <ID> in the URI is defined as channel ID.

A.12.94 /ISAPI/System/IOT/channels/status?format=json

Get the status of added IoT device.

Request URI Definition

Table A-626 GET /ISAPI/System/IOT/channels/status?format=json

Method	GET
Description	Get the status of added IoT device.
Query	format: determine the format of request or response message. security: the version No. of encryption scheme. When security does not exist, it indicates that the data is not encrypted; when security is 1, it indicates that the nodes of sensitive information in the message are encrypted in AES128 CBC mode; when security is 2, it indicates that the nodes of sensitive information in the message are encrypted in AES256 CBC mode. iv: the initialization vector, and it is required when security is 1 or 2.
Request	None.
Response	Succeeded: <i>JSON_IOTChannelStatusList</i> Failed: <i>JSON_ResponseStatus</i>

A.12.95 /ISAPI/System/IOT/channels/status?format=json&deviceInductiveType=

Get the status of added IoT device according to inductive type.

Request URI Definition

Table A-627 GET /ISAPI/System/IOT/channels/status?format=json&deviceInductiveType=

Method	GET
Description	Get the status of added IoT device according to inductive type.
Query	format: determine the format of request or response message. security: the version No. of encryption scheme. When security does not exist, it indicates that the data is not encrypted; when security is 1, it indicates that the nodes of sensitive information in the message are encrypted in AES128 CBC mode; when security is 2, it indicates that the nodes of sensitive information in the message are encrypted in AES256 CBC mode. iv: the initialization vector, and it is required when security is 1 or 2. deviceInductiveType: device inductive type, its value can be obtained by URI: GET <i>/ISAPI/System/IOT/sourceSupport/capabilities?format=json</i> , now there are two available values: "inductiveType1" and "inductiveType2"
Request	None.
Response	<i>JSON_IOTChannelStatusList</i>

A.12.96 /ISAPI/System/IOT/channels?format=json

Operations about IoT devices.

Request URI Definition

Table A-628 GET /ISAPI/System/IOT/channels?format=json

Method	GET
Description	Get the information of added IoT devices.
Query	format: determine the format of request or response message. security: the version No. of encryption scheme. When security does not exist, it indicates that the data is not encrypted; when security is 1, it indicates that the nodes of sensitive information in the message are encrypted in AES128 CBC mode; when security is 2, it indicates that the nodes of sensitive information in the message are encrypted in AES256 CBC mode. iv: the initialization vector, and it is required when security is 1 or 2.

Request	None.
Response	<i>JSON_IOTChannelList</i>

Table A-629 POST /ISAPI/System/IOT/channels?format=json

Method	POST
Description	Add the IoT devices to NVR/DVR.
Query	format : determine the format of request or response message. security : the version No. of encryption scheme. When security does not exist, it indicates that the data is not encrypted; when security is 1, it indicates that the nodes of sensitive information in the message are encrypted in AES128 CBC mode; when security is 2, it indicates that the nodes of sensitive information in the message are encrypted in AES256 CBC mode. iv : the initialization vector, and it is required when security is 1 or 2.
Request	<i>JSON_IOTChannel</i>
Response	Succeeded: <i>JSON_IOT_ChannelInfo</i> Failed: <i>JSON_ResponseStatus</i>

A.12.97 /ISAPI/System/IOT/channels?format=json&deviceInductiveType=

Get the informaiton of added IoT devices according to device inductive type.

Request URI Definition

Table A-630 GET /ISAPI/System/IOT/channels?format=json&deviceInductiveType=

Method	GET
Description	Get the added IoT devices according to device inductive type.
Query	format : determine the format of request or response message. deviceInductiveType : device inductive type, its value can be obtained by URI: GET <i>/ISAPI/System/IOT/sourceSupport/capabilities?format=json</i> , now there are two available values: "inductiveType1" and "inductiveType2" security : the version No. of encryption scheme. When security does not exist, it indicates that the data is not encrypted; when security is 1, it indicates that the nodes of sensitive information in the message are encrypted in AES128 CBC mode; when security is 2, it indicates that the nodes of sensitive information in the message are encrypted in AES256 CBC mode.

	iv : the initialization vector, and it is required when security is 1 or 2.
Request	None.
Response	<i>JSON_IOTChannelList</i>

A.12.98 /ISAPI/System/IOT/linkageChannels?format=json

Get the linked channel of searched event.

Request URI Definition

Table A-631 POST /ISAPI/System/IOT/linkageChannels?format=json

Method	POST
Description	Get the linked channel of searched event.
Query	format : determine the format of request or response message.
Request	<i>JSON_LinkageChansCond</i>
Response	Succeeded: <i>JSON_Result</i> Failed: <i>JSON_ResponseStatus</i>

A.12.99 /ISAPI/System/IOT/search?format=json

Get the IoT devices which can be added.

Request URI Definition

Table A-632 GET /ISAPI/System/IOT/search?format=json

Method	GET
Description	Get the IoT devices which can be added.
Query	format : determine the format of request or response message. security : the version No. of encryption scheme. When security does not exist, it indicates that the data is not encrypted; when security is 1, it indicates that the nodes of sensitive information in the message are encrypted in AES128 CBC mode; when security is 2, it indicates that the nodes of sensitive information in the message are encrypted in AES256 CBC mode. iv : the initialization vector, and it is required when security is 1 or 2.
Request	None.
Response	<i>JSON_IOTSourceList</i>

A.12.100 /ISAPI/System/IOT/sourceCapabilities?format=json

Get the number of IoT device channels.

Request URI Definition**Table A-633 GET /ISAPI/System/IOT/sourceCapabilities?format=json**

Method	GET
Description	Get the number of IoT device channels.
Query	format: determine the format of request or response message. security: the version No. of encryption scheme. When security does not exist, it indicates that the data is not encrypted; when security is 1, it indicates that the nodes of sensitive information in the message are encrypted in AES128 CBC mode; when security is 2, it indicates that the nodes of sensitive information in the message are encrypted in AES256 CBC mode. iv: the initialization vector, and it is required when security is 1 or 2.
Request	<i>JSON_IOTSourceDescription</i>
Response	<i>JSON_SourceCapabilities</i>

A.12.101 /ISAPI/System/IOT/sourceSupport/capabilities?format=json

Get supported protocol type of IoT devices.

Request URI Definition**Table A-634 GET /ISAPI/System/IOT/sourceSupport/capabilities?format=json**

Method	GET
Description	Get supported protocol type of IoT devices.
Query	format: determine the format of request or response message.
Request	None.
Response	<i>JSON_IOTSourceSupport</i>

A.12.102 /ISAPI/System/laserOpticalAxis/capabilities?format=json

Get the configuration capability of laser optical axis.

Request URI Definition

Table A-635 GET /ISAPI/System/laserOpticalAxis/capabilities?format=json

Method	GET
Description	Get the configuration capability of laser optical axis.
Query	format: determine the format of request or response message.
Request	None.
Response	Succeeded: <i>JSON_LaserOpticalAxisCap</i> Failed: <i>JSON_ResponseStatus</i>

A.12.103 /ISAPI/System/laserOpticalAxis/goto?format=json

Adjust the laser optical axis.

Request URI Definition

Table A-636 PUT /ISAPI/System/laserOpticalAxis/goto?format=json

Method	PUT
Description	Adjust the laser optical axis.
Query	format: determine the format of request or response message.
Request	<i>JSON_LaserOpticalAxis_direction</i>
Response	<i>JSON_ResponseStatus</i>

A.12.104 /ISAPI/System/laserOpticalAxis?format=json

Get or set laser optical axis parameters.

Request URI Definition

Table A-637 GET /ISAPI/System/laserOpticalAxis?format=json

Method	GET
Description	Get the laser optical axis parameters.
Query	format: determine the format of request or response message.
Request	None.
Response	Succeeded: <i>JSON_LaserOpticalAxis</i>

	Failed: <i>JSON_ResponseStatus</i>
--	------------------------------------

Table A-638 PUT /ISAPI/System/laserOpticalAxis?format=json

Method	PUT
Description	Configure the laser optical axis.
Query	format : determine the format of request or response message.
Request	<i>JSON_LaserOpticalAxis</i>
Response	<i>JSON_ResponseStatus</i>

A.12.105 /ISAPI/System/logServer

Get or set log server parameters.

Request URI Definition

Table A-639 GET /ISAPI/System/logServer

Method	GET
Description	Get log server parameters.
Query	None.
Request	None.
Response	Succeeded: <i>XML_LogServer</i> Failed: <i>XML_ResponseStatus</i>

Table A-640 PUT /ISAPI/System/logServer

Method	PUT
Description	Set log server parameters.
Query	None.
Request	<i>XML_LogServer</i>
Response	<i>XML_ResponseStatus</i>

A.12.106 /ISAPI/System/logServer/capabilities

Get log server configuration capability.

Request URI Definition

Table A-641 GET /ISAPI/System/logServer/capabilities

Method	GET
Description	Get log server configuration capability.
Query	None.
Request	None.
Response	Succeeded: <i>XML_LogServerCap</i> Failed: <i>XML_ResponseStatus</i>

A.12.107 /ISAPI/System/logServer/test

Perform security log server test.

Request URI Definition

Table A-642 POST /ISAPI/System/logServer/test

Method	POST
Description	Perform security log server test.
Query	None.
Request	<i>XML_LogServerTestDescription</i>
Response	<i>XML_ResponseStatus</i>

A.12.108 /ISAPI/System/Network/adaption/capabilities?format=json

Get network self-adaptive configuration capability.

Request URI Definition

Table A-643 GET /ISAPI/System/Network/adaption/capabilities?format=json

Method	GET
Description	Get network self-adaptive configuration capability.
Query	format: determine the format of request or response message.
Request	None.
Response	Succeeded: <i>JSON_AdaptionCap</i>

	Failed: <i>JSON_ResponseStatus</i>
--	------------------------------------

A.12.109 /ISAPI/System/Network/adaption?format=json&streamType=

Operations about network self-adaptive configuration.

Request URI Definition

Table A-644 GET /ISAPI/System/Network/adaption?format=json&streamType=

Method	GET
Description	Get network self-adaptive parameters.
Query	format: determine the format of request or response message. streamType: stream types, integer, the following values are available: 0-main stream, 1-sub-stream, 2-third stream, 3-virtual stream, 4-stream 5, 5-stream 6, 7-stream 7, 8-stream 8, ..., and so on.
Request	None.
Response	Succeeded: <i>JSON_Adaption</i> Failed: <i>JSON_ResponseStatus</i>

Table A-645 PUT /ISAPI/System/Network/adaption?format=json&streamType=

Method	GET
Description	Set network self-adaptive parameters.
Query	format: determine the format of request or response message. streamType: stream types, integer, the following values are available: 0-main stream, 1-sub-stream, 2-third stream, 3-virtual stream, 4-stream 5, 5-stream 6, 7-stream 7, 8-stream 8, ..., and so on.
Request	<i>JSON_Adaption</i>
Response	<i>XML_ResponseStatus</i>

A.12.110 /ISAPI/System/Network/Bond

Get all network bonding configuration parameters.

Request URI Definition

Table A-646 GET /ISAPI/System/Network/Bond

Method	GET
Description	Get all network bonding configuration parameters.
Query	None.
Request	None.
Response	Succeeded: <i>XML_BondList</i> Failed: <i>XML_ResponseStatus</i>

A.12.111 /ISAPI/System/Network/Bond/<ID>

Operations about the bonding configuration of a specific network interface.

Request URI Definition

Table A-647 GET /ISAPI/System/Network/Bond/<ID>

Method	GET
Description	Get the bonding parameters of a specific network interface.
Query	None.
Request	None.
Response	Succeeded: <i>XML_Bond</i> Failed: <i>XML_ResponseStatus</i>

Table A-648 PUT /ISAPI/System/Network/Bond/<ID>

Method	PUT
Description	Set the bonding parameters of a specific network interface.
Query	None.
Request	<i>XML_Bond</i>
Response	<i>XML_ResponseStatus</i>

A.12.112 /ISAPI/System/Network/capabilities

Get the network capability.

Request URI Definition

Table A-649 GET /ISAPI/System/Network/capabilities

Method	GET
Description	Get the network capability.
Query	None.
Request	None.
Response	Succeeded: <i>XML_NetworkCap</i> Failed: <i>XML_ResponseStatus</i>

A.12.113 /ISAPI/System/Network/channels/<ID>/buletooth/status

Operations about the bluetooth status configuration of a specific channel.

Request URI Definition

Table A-650 GET /ISAPI/System/Network/channels/<ID>/buletooth/status

Method	GET
Description	Get the bluetooth status parameters of a specific channel.
Query	None.
Request	None.
Response	Succeeded: <i>XML_BluetoothStatus</i> Failed: <i>XML_ResponseStatus</i>

Table A-651 PUT /ISAPI/System/Network/channels/<ID>/buletooth/status

Method	PUT
Description	Set the bluetooth status parameters of a specific channel.
Query	None.
Request	<i>XML_BluetoothStatus</i>
Response	<i>XML_ResponseStatus</i>

Remarks

The <ID> in the request URI refers to the channel No.

A.12.114 /ISAPI/System/Network/DDNS

Operations about all DDNS configuration parameters.

Request URI Definition

Table A-652 GET /ISAPI/System/Network/DDNS

Method	GET
Description	Get all DDNS configuration parameters.
Query	None.
Request	None.
Response	Succeeded: <i>XML_DDNSList</i> Failed: <i>XML_ResponseStatus</i>

Table A-653 PUT /ISAPI/System/Network/DDNS

Method	PUT
Description	Set all DDNS configuration parameters.
Query	None.
Request	<i>XML_DDNSList</i>
Response	<i>XML_ResponseStatus</i>

A.12.115 /ISAPI/System/Network/DDNS/capabilities

Get DDNS (Dynamic Domain Name Server) configuration capability.

Request URI Definition

Table A-654 GET /ISAPI/System/Network/DDNS/capabilities

Method	GET
Description	Get DDNS (Dynamic Domain Name Server) configuration capability.
Query	None.
Request	None.
Response	Succeeded: <i>XML_DDNSList</i> Failed: <i>XML_ResponseStatus</i>

A.12.116 /ISAPI/System/Network/DDNS/<ID>

Operations about the configuration of a specific DDNS.

Request URI Definition

Table A-655 GET /ISAPI/System/Network/DDNS/<ID>

Method	GET
Description	Get the configuration parameters of a specific DDNS.
Query	None.
Request	None.
Response	Succeeded: <i>XML_DDNS</i> Failed: <i>XML_ResponseStatus</i>

Table A-656 PUT /ISAPI/System/Network/DDNS/<ID>

Method	PUT
Description	Set the parameters of a specific DDNS.
Query	None.
Request	<i>XML_DDNS</i>
Response	<i>XML_ResponseStatus</i>

A.12.117 /ISAPI/System/Network/DDNS/<ID>/capabilities

Get the configuration capability of a specific DDNS.

Request URI Definition

Table A-657 GET /ISAPI/System/Network/DDNS/<ID>/capabilities

Method	GET
Description	Get the configuration capability of a specific DDNS.
Query	None.
Request	None.
Response	Succeeded: <i>XML_Cap_DDNS</i> Failed: <i>XML_ResponseStatus</i>

A.12.118 /ISAPI/System/Network/Ehome

Operations about the EHome server access configuration.

Request URI Definition

Table A-658 GET /ISAPI/System/Network/Ehome

Method	GET
Description	Get the EHome server access parameters.
Query	None.
Request	None.
Response	Succeeded: <i>XML_Ehome</i> Failed: <i>XML_ResponseStatus</i>

Table A-659 PUT /ISAPI/System/Network/Ehome

Method	PUT
Description	Set the EHome server access parameters.
Query	None.
Request	<i>XML_Ehome</i>
Response	<i>XML_ResponseStatus</i>

A.12.119 /ISAPI/System/Network/Ehome/capabilities

Get the configuration capability of EHome server access.

Request URI Definition

Table A-660 GET /ISAPI/System/Network/Ehome/capabilities

Method	GET
Description	Get the configuration capability of EHome server access.
Query	None.
Request	None.
Response	Succeeded: <i>XML_Cap_EHome</i> Failed: <i>XML_ResponseStatus</i>

A.12.120 /ISAPI/System/Network/Ehome?centerID=

Operations about the EHome server access configuration.

Request URI Definition**Table A-661 GET /ISAPI/System/Network/Ehome?centerID=**

Method	GET
Description	Get the EHome server access parameters.
Query	security : the version No. of encryption scheme. When security does not exist, it indicates that the data is not encrypted; when security is 1, it indicates that the nodes of sensitive information in the message are encrypted in AES128 CBC mode; when security is 2, it indicates that the nodes of sensitive information in the message are encrypted in AES256 CBC mode. iv : the initialization vector, and it is required when security is 1 or 2. centerID : EHome center ID, and if there is only one center, centerID is 1.
Request	None.
Response	Succeeded: <i>XML_Ehome</i> Failed: <i>XML_ResponseStatus</i>

Table A-662 PUT /ISAPI/System/Network/Ehome?centerID=

Method	PUT
Description	Set the EHome server access parameters.
Query	security : the version No. of encryption scheme. When security does not exist, it indicates that the data is not encrypted; when security is 1, it indicates that the nodes of sensitive information in the message are encrypted in AES128 CBC mode; when security is 2, it indicates that the nodes of sensitive information in the message are encrypted in AES256 CBC mode. iv : the initialization vector, and it is required when security is 1 or 2. centerID : EHome center ID, and if there is only one center, centerID is 1.
Request	<i>XML_Ehome</i>
Response	<i>XML_ResponseStatus</i>

A.12.121 /ISAPI/System/Network/EZVIZ

Operations about Hik-Connect access configurations.

Request URI Definition

Table A-663 GET /ISAPI/System/Network/EZVIZ

Method	GET
Description	Get Hik-Connect parameters.
Query	security : the version No. of encryption scheme. When security does not exist, it indicates that the data is not encrypted; when security is 1, it indicates that the nodes of sensitive information in the message are encrypted in AES128 CBC mode; when security is 2, it indicates that the nodes of sensitive information in the message are encrypted in AES256 CBC mode. iv : the initialization vector, and it is required when security is 1 or 2.
Request	None.
Response	<i>XML_EZVIZ</i>

Table A-664 PUT /ISAPI/System/Network/EZVIZ

Method	PUT
Description	Set Hik-Connect parameters.
Query	security : the version No. of encryption scheme. When security does not exist, it indicates that the data is not encrypted; when security is 1, it indicates that the nodes of sensitive information in the message are encrypted in AES128 CBC mode; when security is 2, it indicates that the nodes of sensitive information in the message are encrypted in AES256 CBC mode. iv : the initialization vector, and it is required when security is 1 or 2.
Request	<i>XML_EZVIZ</i>
Response	<i>XML_ResponseStatus</i>

A.12.122 /ISAPI/System/Network/EZVIZ/QRCode

Get the Hik-Connect QR code of the device.

Request URI Definition**Table A-665 GET /ISAPI/System/Network/EZVIZ/QRCode**

Method	GET
Description	Get the Hik-Connect QR code of the device.
Query	None.
Request	None.
Response	Succeeded: Binary data (QR code in binary format) Failed: <i>XML_ResponseStatus</i>

A.12.123 /ISAPI/System/Network/EZVIZ/QRCode?format=json

Get the Hik-Connect QR code.

Request URI Definition**Table A-666 GET /ISAPI/System/Network/EZVIZ/QRCode?format=json**

Method	GET
Description	Get the Hik-Connect QR code.
Query	security : the version No. of encryption scheme. When security does not exist, it indicates that the data is not encrypted; when security is 1, it indicates that the nodes of sensitive information in the message are encrypted in AES128 CBC mode; when security is 2, it indicates that the nodes of sensitive information in the message are encrypted in AES256 CBC mode. iv : the initialization vector, and it is required when security is 1 or 2. format : determine the format of request or response message.
Request	None.
Response	Succeeded: <i>JSON_EZVIZQRCode</i> Failed: <i>JSON_ResponseStatus</i>

Remarks

- You can use smart phone to scan the QR code to download Hik-Connect app or use Hik-Connect to scan the QR code to bind the device.
- If the node **isSupportEZVIZQRCode** is returned in the message ***XML_Cap_EZVIZ*** (related URI: ***/ISAPI/System/Network/EZVIZ/capabilities***) and the value is true, it indicates that the device supports this function.

A.12.124 /ISAPI/System/Network/EZVIZ/capabilities

Get the Hik-Connect access configuration capability.

Request URI Definition

Table A-667 GET /ISAPI/System/Network/EZVIZ/capabilities

Method	GET
Description	Get the Hik-Connect access configuration capability.
Query	None.
Request	None.
Response	Succeeded: <i>XML_Cap_EZVIZ</i> Failed: <i>XML_ResponseStatus</i>

A.12.125 /ISAPI/System/Network/EZVIZ/secretKey?format=json

Edit the verification code for Hik-Connect.

Request URI Definition

Table A-668 PUT /ISAPI/System/Network/EZVIZ/secretKey?format=json

Method	PUT
Description	Edit the verification code for Hik-Connect.
Query	security: the version No. of encryption scheme. When security does not exist, it indicates that the data is not encrypted; when security is 1, it indicates that the nodes of sensitive information in the message are encrypted in AES128 CBC mode; when security is 2, it indicates that the nodes of sensitive information in the message are encrypted in AES256 CBC mode. iv: the initialization vector, and it is required when security is 1 or 2. format: determine the format of request or response message.
Request	<i>JSON_EZVIZSecretKey</i>
Response	<i>JSON_ResponseStatus</i>

A.12.126 /ISAPI/System/Network/EZVIZ/unbind?format=json

Unbind devices from Hik-Connect account.

Request URI Definition**Table A-669 PUT /ISAPI/System/Network/EZVIZ/unbind?format=json**

Method	PUT
Description	Unbind devices from Hik-Connect account.
Query	format : determine the format of request or response message. security : the version No. of encryption scheme. When security does not exist, it indicates that the data is not encrypted; when security is 1, it indicates that the nodes of sensitive information in the message are encrypted in AES128 CBC mode; when security is 2, it indicates that the nodes of sensitive information in the message are encrypted in AES256 CBC mode. iv : the initialization vector, and it is required when security is 1 or 2.
Request	<i>JSON_EZVIZUnbind</i>
Response	<i>JSON_ResponseStatus</i>

Remarks

This URI can only be used in the same network segment in a LAN by admin users.

A.12.127 /ISAPI/System/Network/ftp

Operations about all FTP configurations.

Request URI Definition**Table A-670 GET /ISAPI/System/Network/ftp**

Method	GET
Description	Get all FTP parameters.
Query	None.
Request	None.
Response	Succeeded: <i>XML_FTPNotificationList</i> Failed: <i>XML_ResponseStatus</i>

Table A-671 PUT /ISAPI/System/Network/ftp

Method	PUT
Description	Set all FTP parameters.
Query	None.

Request	<i>XML_FTPNotificationList</i>
Response	<i>XML_ResponseStatus</i>

A.12.128 /ISAPI/System/Network/ftp/<ID>

Operations about the configuration of a specific FTP.

Request URI Definition

Table A-672 GET /ISAPI/System/Network/ftp/<ID>

Method	GET
Description	Get the parameters of a specific FTP.
Query	None.
Request	None.
Response	Succeeded: <i>XML_FTPNotification</i> Failed: <i>XML_ResponseStatus</i>

Table A-673 PUT /ISAPI/System/Network/ftp/<ID>

Method	PUT
Description	Set the parameters of a specific FTP.
Query	None.
Request	<i>XML_FTPNotification</i>
Response	<i>XML_ResponseStatus</i>

A.12.129 /ISAPI/System/Network/ftp/capabilities

Get the FTP capability.

Request URI Definition

Table A-674 GET /ISAPI/System/Network/ftp/capabilities

Method	GET
Description	Get the FTP capability.
Query	None.
Request	None.
Response	Succeeded: <i>XML_Cap_FTPNotificationList</i>

	Failed: <i>XML_ResponseStatus</i>
--	--

A.12.130 /ISAPI/System/Network/ftp/test

Perform test to check if the FTP server is available.

Request URI Definition

Table A-675 POST /ISAPI/System/Network/ftp/test

Method	POST
Description	Perform test to check if the FTP server is available.
Query	None.
Request	<i>XML_FTPTestDescription</i>
Response	Succeeded: <i>XML_FTPTestResult</i> Failed: <i>XML_ResponseStatus</i>

A.12.131 /ISAPI/System/Network/interfaces

Get the information of multiple network interfaces.

Request URI Definition

Table A-676 GET /ISAPI/System/Network/interfaces

Method	GET
Description	Get the information of multiple network interfaces.
Query	None.
Request	None.
Response	Succeeded: <i>XML_NetworkInterfaceList</i> Failed: <i>XML_ResponseStatus</i>

Example

Sample Code of Getting Information of Multiple Network Interfaces

```
GET /ISAPI/System/Network/interfaces HTTP/1.1
```

```
...
```

```
HTTP/1.1 200 OK
```

```
Content-Type: application/xml; charset="UTF-8"
```

```
Content-Length: xxx
```

```
<?xml version="1.0" encoding="UTF-8"?>
```

```
<NetworkInterfaceList version="2.0" xmlns="http://www.isapi.org/ver20/XMLSchema">
  <NetworkInterface>
    <id>1</id>
    <IPAddress>
      <ipVersion>v4</ipVersion>
      <addressingType>static</addressingType>
      <ipAddress>172.6.64.7</ipAddress>
      <subnetMask>255.255.255.0</subnetMask>
      <DefaultGateway>
        <ipAddress>172.6.64.1</ipAddress>
      </DefaultGateway>
      <PrimaryDNS>
        <ipAddress>192.0.0.200</ipAddress>
      </PrimaryDNS>
    </IPAddress>
    <Discovery>
      <UPnP>
        <enabled>true</enabled>
      </UPnP>
      <Zeroconf>
        <enabled>true</enabled>
      </Zeroconf>
    </Discovery>
    <Link>
      <MACAddress> 00:40:48:4C:7F:F2</MACAddress>
      <autoNegotiation>true</autoNegotiation>
      <speed>1000</speed>
      <duplex>full</duplex>
      <MTU>1500</MTU>
    </Link>
  </NetworkInterface>
</NetworkInterfaceList>
```

A.12.132 /ISAPI/System/Network/interfaces/<ID>

Operations about the configuration of a specific network interface.

Request URI Definitions

Table A-677 GET /ISAPI/System/Network/interfaces/<ID>

Method	GET
Description	Get the information of a specific network interface.
Query	None.
Request	None.
Response	Succeeded: <i>XML_NetworkInterface</i> Failed: <i>XML_ResponseStatus</i>

PUT /ISAPI/System/Network/interfaces/<ID>

Method	PUT
Description	Set a specific network interface.
Query	None.
Request	<i>XML_NetworkInterface</i>
Response	<i>XML_ResponseStatus</i>

A.12.133 /ISAPI/System/Network/interfaces/<ID>/capabilities

Get the capability of a specific network interface.

Request URI Definition**Table A-678 GET /ISAPI/System/Network/interfaces/<ID>/capabilities**

Method	GET
Description	Get the capability of a specific network interface.
Query	None.
Request	None.
Response	Succeeded: <i>XML_Cap_NetworkInterface</i> Failed: <i>XML_ResponseStatus</i>

A.12.134 /ISAPI/System/Network/interfaces/<ID>/discovery

Operations about the network discovery configuration.

Request URI Definition**Table A-679 GET /ISAPI/System/Network/interfaces/<ID>/discovery**

Method	GET
Description	Get the network discovery configuration parameters.
Query	None.
Request	None.
Response	Succeeded: <i>XML_Discovery</i> Failed: <i>XML_ResponseStatus</i>

Table A-680 PUT /ISAPI/System/Network/interfaces/<ID>/discovery

Method	PUT
Description	Set the network discovery configuration parameters.
Query	None.
Request	<i>XML_Discovery</i>
Response	<i>XML_ResponseStatus</i>

A.12.135 /ISAPI/System/Network/interfaces/<ID>/discovery/capabilities

Get the configuration capability of network discovery.

Request URI Definition

Table A-681 GET /ISAPI/System/Network/interfaces/<ID>/discovery/capabilities

Method	GET
Description	Get the configuration capability of network discovery.
Query	None.
Request	None.
Response	Succeeded: <i>XML_Cap_Discovery</i> Failed: <i>XML_ResponseStatus</i>

A.12.136 /ISAPI/System/Network/interfaces/<ID>/dynamicHostName?format=json

Get or set dynamic domain name.

Request URI Definition

Table A-682 GET /ISAPI/System/Network/interfaces/<ID>/dynamicHostName?format=json

Method	GET
Description	Get dynamic domain name.
Query	format: determine the format of request or response message.
Request	None.
Response	Succeeded: <i>JSON_DynamicHostName</i> Failed: <i>JSON_ResponseStatus</i>

Table A-683 PUT /ISAPI/System/Network/interfaces/<ID>/dynamicHostName?format=json

Method	PUT
Description	Set dynamic domain name.
Query	format: determine the format of request or response message.
Request	<i>JSON_DynamicHostName</i>
Response	<i>JSON_ResponseStatus</i>

Remarks

The <ID> in the request URI refers to the network interface ID.

A.12.137 /ISAPI/System/Network/interfaces/<ID>/dynamicHostName/capabilities?format=json

Get configuration capability of dynamic domain name.

Request URI Definition**Table A-684 GET /ISAPI/System/Network/interfaces/<ID>/dynamicHostName/capabilities?format=json**

Method	GET
Description	Get configuration capability of dynamic domain name.
Query	format: determine the format of request or response message.
Request	None.
Response	Succeeded: <i>JSON_DynamicHostNameCap</i> Failed: <i>JSON_ResponseStatus</i>

Remarks

The <ID> in the request URI refers to the network interface ID.

A.12.138 /ISAPI/System/Network/interfaces/<ID>/ieee802.1x

Operations about configuration of IEEE 802.1x protocol access.

Request URI Definition

Table A-685 GET /ISAPI/System/Network/interfaces/<ID>/ieee802.1x

Method	GET
Description	Get parameters of IEEE 802.1x protocol access.
Query	None.
Request	None.
Response	Succeeded: <i>XML_IEEE802_1x</i> Failed: <i>XML_ResponseStatus</i>

Table A-686 PUT /ISAPI/System/Network/interfaces/<ID>/ieee802.1x

Method	PUT
Description	Set parameters of IEEE 802.1x protocol access.
Query	None.
Request	<i>XML_IEEE802_1x</i>
Response	<i>XML_ResponseStatus</i>

Remarks

The <ID> in the request URI refers to the network interface ID.

A.12.139 /ISAPI/System/Network/interfaces/<ID>/ieee802.1x/capabilities

Get configuration capability of IEEE802.1x protocol access.

Request URI Definition

Table A-687 GET /ISAPI/System/Network/interfaces/<ID>/ieee802.1x/capabilities

Method	GET
Description	Get configuration capability of IEEE802.1x protocol access.
Query	None.
Request	None.
Response	<i>XML_Cap_IEEE802_1x</i>

A.12.140 /ISAPI/System/Network/interfaces/<ID>/ipAddress

Operations about IP address configuration of a specific network interface.

Request URI Definition

Table A-688 GET /ISAPI/System/Network/interfaces/<ID>/ipAddress

Method	GET
Description	Get the IP address of a specific network interface.
Query	None.
Request	None.
Response	Succeeded: <i>XML_IPAddress</i> Failed: <i>XML_ResponseStatus</i>

Table A-689 PUT /ISAPI/System/Network/interfaces/<ID>/ipAddress

Method	PUT
Description	Set IP address of a specific network interface.
Query	None.
Request	<i>XML_IPAddress</i>
Response	<i>XML_ResponseStatus</i>

Example

Sample Code of Setting IP Address of A Specific Network Interface

```
PUT /ISAPI/System/Network/interfaces/1/ipAddress HTTP/1.1
...
Content-Type: application/xml; charset="UTF-8"
Content-Length: xxx

<?xml version="1.0" encoding="UTF-8"?>
<IPAddress version="2.0" xmlns="http://www.isapi.org/ver20/XMLSchema">
  <ipVersion>v4</ipVersion>
  <addressingType>static</addressingType>
  <ipAddress>172.6.64.16</ipAddress>
  <subnetMask>255.255.255.0</subnetMask>
  <DefaultGateway>
    <ipAddress>172.6.64.1</ipAddress>
  </DefaultGateway>
  <PrimaryDNS>
    <ipAddress>192.0.0.200</ipAddress>
  </PrimaryDNS>
</IPAddress>

HTTP/1.1 200 OK
...
Content-Type: application/xml; charset="UTF-8"
Content-Length:xxx
```



```
<?xml version="1.0" encoding="UTF-8"?>
<ResponseStatus version="2.0" xmlns="http://www.isapi.org/ver20/XMLSchema">
  <requestURL>/Network/interfaces/1/ipAddress</requestURL>
  <statusCode>1</statusCode>
  <statusString>OK</statusString>
</ResponseStatus>
```

A.12.141 /ISAPI/System/Network/interfaces/<ID>/ipAddress/capabilities

Get the IP address configuration capability of a specific network interface.

Request URI Definition

Table A-690 GET /ISAPI/System/Network/interfaces/<ID>/ipAddress/capabilities

Method	GET
Description	Get the IP address configuration capability of a specific network interface.
Query	None.
Request	None.
Response	Succeeded: <i>XML_Cap_IpAddress</i> Failed: <i>XML_ResponseStatus</i>

A.12.142 /ISAPI/System/Network/interfaces/<ID>/link

Operations about the connection configuration of a specific network interface.

Request URI Definition

Table A-691 GET /ISAPI/System/Network/interfaces/<ID>/link

Method	GET
Description	Get the connection parameters of a specific network interface.
Query	None.
Request	None.
Response	Succeeded: <i>XML_Link</i> Failed: <i>XML_ResponseStatus</i>

Table A-692 PUT /ISAPI/System/Network/interfaces/<ID>/link

Method	PUT
Description	Set the connection parameters of a specific network interface.
Query	None.
Request	<i>XML_Link</i>
Response	<i>XML_ResponseStatus</i>

A.12.143 /ISAPI/System/Network/interfaces/<ID>/link/capabilities

Get the connection configuration capability of a specific network interface.

Request URI Definition

Table A-693 GET /ISAPI/System/Network/interfaces/<ID>/link/capabilities

Method	GET
Description	Get the connection configuration capability of a specific network interface.
Query	None.
Request	None.
Response	Succeeded: <i>XML_Cap_Link</i> Failed: <i>XML_ResponseStatus</i>

A.12.144 /ISAPI/System/Network/interfaces/<ID>/NetworkMode

Operations about the working mode configuration of the network connection.

Request URI Definition

Table A-694 GET /ISAPI/System/Network/interfaces/<ID>/NetworkMode

Method	GET
Description	Get the working mode parameters of the network connection.
Query	None.
Request	None.
Response	Succeeded: <i>XML_NetWorkMode</i> Failed: <i>XML_ResponseStatus</i>

Table A-695 PUT /ISAPI/System/Network/interfaces/<ID>/NetworkMode

Method	PUT
Description	Set the working mode parameters of the network connection.
Query	None.
Request	<i>XML_NetWorkMode</i>
Response	<i>XML_ResponseStatus</i>

Remarks

The <ID> in the request URI refers to the network interface ID.

A.12.145 /ISAPI/System/Network/interfaces/<ID>/wireless

Operations about Wi-Fi configuration of a specific network interface.

Request URI Definition

Table A-696 GET /ISAPI/System/Network/interfaces/<ID>/wireless

Method	GET
Description	Get the Wi-Fi parameters of a specific network interface.
Query	security : the version No. of encryption scheme. When security does not exist, it indicates that the data is not encrypted; when security is 1, it indicates that the nodes of sensitive information in the message are encrypted in AES128 CBC mode; when security is 2, it indicates that the nodes of sensitive information in the message are encrypted in AES256 CBC mode. iv : the initialization vector, and it is required when security is 1 or 2.
Request	None.
Response	Succeeded: <i>XML_Wireless</i> Failed: <i>XML_ResponseStatus</i>

Table A-697 PUT /ISAPI/System/Network/interfaces/<ID>/wireless

Method	PUT
Description	Set Wi-Fi parameters of a specific network interface.
Query	security : the version No. of encryption scheme. When security does not exist, it indicates that the data is not encrypted; when security is 1, it indicates that the nodes of sensitive information in the message are encrypted in AES128 CBC mode; when security is 2, it indicates

	that the nodes of sensitive information in the message are encrypted in AES256 CBC mode. iv : the initialization vector, and it is required when security is 1 or 2.
Request	<i>XML_Wireless</i>
Response	<i>XML_ResponseStatus</i>

A.12.146 /ISAPI/System/Network/interfaces/<ID>/wireless/accessDeviceList/<ID>

Get the information of a specific device connected to the hotspot.

Request URI Definition

Table A-698 GET /ISAPI/System/Network/interfaces/<ID>/wireless/accessDeviceList/<ID>

Method	GET
Description	Get the information of a specific device connected to the hotspot.
Query	None.
Request	None.
Response	Succeeded: <i>XML_accessDevice</i> Failed: <i>XML_ResponseStatus</i>

Remarks

The first <ID> in the URI refers to the network interface ID, and the second <ID> refers to the device ID.

A.12.147 /ISAPI/System/Network/interfaces/<ID>/wireless/accessDeviceList/capabilities

Get the capability set of the device connected to the hotspot.

Request URI Definition

Table A-699 GET /ISAPI/System/Network/interfaces/<ID>/wireless/accessDeviceList/capabilities

Method	GET
Description	Get the capability set of the device connected to the hotspot.
Query	None.
Request	None.
Response	Succeeded: <i>XML_Cap_accessDeviceList</i>

	Failed: <i>XML_ResponseStatus</i>
--	-----------------------------------

Remarks

The <ID> in the URI refers to the network interface ID.

A.12.148 /ISAPI/System/Network/interfaces/<ID>/wireless/accessPointList

Get Wi-Fi hotspot list.

Request URI Definition

Table A-700 GET /ISAPI/System/Network/interfaces/<ID>/wireless/accessPointList

Method	GET
Description	Get Wi-Fi hotspot list.
Query	None.
Request	None.
Response	Succeeded: <i>XML_accessPointList</i> Failed: <i>XML_ResponseStatus</i>

A.12.149 /ISAPI/System/Network/interfaces/<ID>/wireless/accessPointList/<ID>

Get a specific Wi-Fi hotspot.

Request URI Definition

Table A-701 GET /ISAPI/System/Network/interfaces/<ID>/wireless/accessPointList/<ID>

Method	GET
Description	Get a specific Wi-Fi hotspot.
Query	None.
Request	None.
Response	Succeeded: <i>XML_accessPoint</i> Failed: <i>XML_ResponseStatus</i>

A.12.150 /ISAPI/System/Network/interfaces/<ID>/wireless/capabilities

Get the Wi-Fi configuration capability.

Request URI Definition

Method	GET
Description	Get the Wi-Fi configuration capability.
Query	security : the version No. of encryption scheme. When security does not exist, it indicates that the data is not encrypted; when security is 1, it indicates that the nodes of sensitive information in the message are encrypted in AES128 CBC mode; when security is 2, it indicates that the nodes of sensitive information in the message are encrypted in AES256 CBC mode. iv : the initialization vector, and it is required when security is 1 or 2.
Request	None.
Response	Succeeded: <i>XML_Cap_Wireless</i> Failed: <i>XML_ResponseStatus</i>

A.12.151 /ISAPI/System/Network/interfaces/<ID>/wirelessServer

Operations about wireless hotspot configuration.

Request URI Definitions**Table A-702 GET /ISAPI/System/Network/interfaces/<ID>/wirelessServer**

Method	GET
Description	Get wireless hotspot parameters.
Query	None.
Request	None.
Response	<i>XML_WirelessServer</i>

Table A-703 PUT /ISAPI/System/Network/interfaces/<ID>/wirelessServer

Method	PUT
Description	Set wireless hotspot parameters.
Query	None.
Request	<i>XML_WirelessServer</i>
Response	<i>XML_ResponseStatus</i>

Remarks

The <ID> in the URI refers to the network interface ID.

A.12.152 /ISAPI/System/Network/interfaces/<ID>/wirelessServer/accessDeviceList

Get list of devices that are connected to the hotspot.

Request URI Definition

Table A-704 GET /ISAPI/System/Network/interfaces/<ID>/wirelessServer/accessDeviceList

Method	GET
Description	Get list of devices that are connected to the hotspot.
Query	None.
Request	None.
Response	Succeeded: <i>XML_accessDeviceList</i> Failed: <i>XML_ResponseStatus</i>

Remarks

The <ID> in the URI refers to the network interface ID.

A.12.153 /ISAPI/System/Network/interfaces/<ID>/wirelessServer/capabilities

Get configuration capability of wireless hotspot.

Request URI Definition

Table A-705 GET /ISAPI/System/Network/interfaces/<ID>/wirelessServer/capabilities

Method	GET
Description	Get configuration capability of wireless hotspot.
Query	None.
Request	None.
Response	<i>XML_Cap_WirelessServer</i>

A.12.154 /ISAPI/System/Network/interfaces/<ID>/wirelessServerStatus

Operations about the wireless hotspot status configuration.

Request URI Definition

Table A-706 GET /ISAPI/System/Network/interfaces/<ID>/wirelessServerStatus

Method	GET
Description	Get the wireless hotspot status parameters.
Query	None.
Request	None.
Response	Succeeded: <i>XML_WirelessServerStatus</i> Failed: <i>XML_ResponseStatus</i>

Table A-707 PUT /ISAPI/System/Network/interfaces/<ID>/wirelessServerStatus

Method	PUT
Description	Set the wireless hotspot status parameters.
Query	None.
Request	<i>XML_WirelessServerStatus</i>
Response	<i>XML_ResponseStatus</i>

Remarks

The <ID> in the request URI refers to the network interface ID.

A.12.155 /ISAPI/System/Network/interfaces/<ID>/wirelessServerStatus/capabilities

Get the wireless hotspot status capability.

Request URI Definition

Table A-708 GET /ISAPI/System/Network/interfaces/<ID>/wirelessServerStatus/capabilities

Method	GET
Description	Get the wireless hotspot status capability.
Query	None.
Request	None.
Response	Succeeded: <i>XML_Cap_WirelessServerStatus</i> Failed: <i>XML_ResponseStatus</i>

Remarks

The <ID> in the request URI refers to the network interface ID.

A.12.156 /ISAPI/System/Network/interfaces/<ID>/wirelessStatus

Operations about the configuration of wireless network connection status.

Request URI Definition**Table A-709 GET /ISAPI/System/Network/interfaces/<ID>/wirelessStatus**

Method	GET
Description	Get the parameters of wireless network connection status.
Query	None.
Request	None.
Response	Succeeded: <i>XML_WirelessStatus</i> Failed: <i>XML_ResponseStatus</i>

Table A-710 PUT /ISAPI/System/Network/interfaces/<ID>/wirelessStatus

Method	PUT
Description	Set the parameters of wireless network connection status.
Query	None.
Request	<i>XML_WirelessStatus</i>
Response	<i>XML_ResponseStatus</i>

Remarks

The <ID> in the request URI refers to the network interface ID.

A.12.157 /ISAPI/System/Network/interfaces/<ID>/wirelessStatus/capabilities

Get the configuration capability of wireless network connection status.

Request URI Definition**Table A-711 GET /ISAPI/System/Network/interfaces/<ID>/wirelessStatus/capabilities**

Method	GET
Description	Get the configuration capability of wireless network connection status.

Query	None.
Request	None.
Response	Succeeded: <i>XML_Cap_WirelessStatus</i> Failed: <i>XML_ResponseStatus</i>

Remarks

The <ID> in the request URI refers to the network interface ID.

A.12.158 /ISAPI/System/Network/interfaces/<ID>/WPS

Get or set the WPS (Wi-Fi Protected Setup) parameters of a specific network interface.

Request URI Definition

Table A-712 GET /ISAPI/System/Network/interfaces/<ID>/WPS

Method	GET
Description	Get the WPS (Wi-Fi Protected Setup) parameters of a specific network interface.
Query	None.
Request	None.
Response	Succeeded: <i>XML_WPS</i> Failed: <i>XML_ResponseStatus</i>

Table A-713 PUT /ISAPI/System/Network/interfaces/<ID>/WPS

Method	PUT
Description	Set the WPS (Wi-Fi Protected Setup) parameters of a specific network interface.
Query	None.
Request	<i>XML_WPS</i>
Response	<i>XML_ResponseStatus</i>

Remarks

The <ID> in the request URI refers to the network interface ID.

A.12.159 /ISAPI/System/Network/interfaces/<ID>/WPS/ApPinCode

Operations about the configuration of PIN (Personal Identification Number) code of the wireless access point.

Request URI Definition

Table A-714 GET /ISAPI/System/Network/interfaces/<ID>/WPS/ApPinCode

Method	GET
Description	Get the parameters of PIN (Personal Identification Number) code of the wireless access point.
Query	None.
Request	None.
Response	Succeeded: <i>XML_WpsApPincode</i> Failed: <i>XML_ResponseStatus</i>

Table A-715 PUT /ISAPI/System/Network/interfaces/<ID>/WPS/ApPinCode

Method	PUT
Description	Set the parameters of PIN (Personal Identification Number) code of the wireless access point.
Query	None.
Request	<i>XML_WpsApPincode</i>
Response	<i>XML_ResponseStatus</i>

Remarks

The <ID> in the request URI refers to the network interface ID, and it is set to 2 by default.

A.12.160 /ISAPI/System/Network/interfaces/<ID>/WPS/AutoConnect

Set the configuration of PBC (Push Button Configuration) automatic connection of WPS (Wi-Fi Protected Setup).

Request URI Definition

Table A-716 PUT /ISAPI/System/Network/interfaces/<ID>/WPS/AutoConnect

Method	PUT
Description	Set the parameters of PBC (Push Button Configuration) automatic connection of WPS (Wi-Fi Protected Setup).

Query	None.
Request	<i>XML_WPS</i>
Response	<i>XML_ResponseStatus</i>

Remarks

The <ID> in the request URI refers to the network interface ID, and it is set to 2 by default.

A.12.161 /ISAPI/System/Network/interfaces/<ID>/WPS/devicePinCode

Get the PIN (Personal Identification Number) code of WPS (Wi-Fi Protected Setup) of the device.

Request URI Definition

Table A-717 GET /ISAPI/System/Network/interfaces/<ID>/WPS/devicePinCode

Method	GET
Description	Get the PIN (Personal Identification Number) code of WPS (Wi-Fi Protected Setup) of the device.
Query	None.
Request	None.
Response	PIN code string.

A.12.162 /ISAPI/System/Network/interfaces/<ID>/WPS/devicePinCodeUpdate

Update the PIN (Personal Identification Number) code of WPS (Wi-Fi Protected Setup) of the device.

Request URI Definition

Table A-718 PUT /ISAPI/System/Network/interfaces/<ID>/WPS/devicePinCodeUpdate

Method	PUT
Description	Update the PIN (Personal Identification Number) code of WPS (Wi-Fi Protected Setup) of the device.
Query	None.
Request	None.
Response	PIN code string.

Remarks

The <ID> in the request URI refers to the network interface ID, and it is set to 2 by default.

A.12.163 /ISAPI/System/Network/interfaces/capabilities

Get the configuration capability of multiple network interfaces.

Request URI Definition

Table A-719 GET /ISAPI/System/Network/interfaces/capabilities

Method	GET
Description	Get the configuration capability of multiple network interfaces.
Query	None.
Request	None.
Response	Succeeded: <i>XML_Cap_NetworkInterfaceList</i> Failed: <i>XML_ResponseStatus</i>

A.12.164 /ISAPI/System/Network/ipFilter

Operations about IP address filter configuration.

Request URI Definition

Table A-720 GET /ISAPI/System/Network/ipFilter

Method	GET
Description	Get the IP filter configuration parameters.
Query	None.
Request	None.
Response	Succeeded: <i>XML_IPFilter</i> Failed: <i>XML_ResponseStatus</i>

Table A-721 PUT /ISAPI/System/Network/ipFilter

Method	PUT
Description	Set the IP address filter.
Query	None.

Request	<i>XML_IPFilter</i>
Response	<i>XML_ResponseStatus</i>

A.12.165 /ISAPI/System/Network/ipFilter/capabilities

Get IP address filter configuration capability.

Request URI Definition

Table A-722 GET /ISAPI/System/Network/ipFilter/capabilities

Method	GET
Description	Get the IP address configuration capability.
Query	None.
Request	None.
Response	Succeeded: <i>XML_Cap_IPFilter</i> Failed: <i>XML_ResponseStatus</i>

A.12.166 /ISAPI/System/Network/MACFilter

Operations about MAC address filter configuration.

Request URI Definition

Table A-723 GET /ISAPI/System/Network/MACFilter

Method	GET
Description	Get the MAC address filter configuration parameters.
Query	None.
Request	None.
Response	<i>XML_MACFilter</i>

Table A-724 PUT /ISAPI/System/Network/MACFilter

Method	PUT
Description	Set the MAC address filter.
Query	None.
Request	<i>XML_MACFilter</i>
Response	<i>XML_ResponseStatus</i>

A.12.167 /ISAPI/System/Network/MACFilter/capabilities

Get the MAC address filter configuration capability.

Request URI Definition

Table A-725 GET /ISAPI/System/Network/MACFilter/capabilities

Method	GET
Description	Get the MAC address filter configuration capability.
Query	None.
Request	None.
Response	<i>XML_Cap_MACFilter</i>

A.12.168 /ISAPI/System/Network/mailing

Operations about email configuration.

Request URI Definition

Table A-726 GET /ISAPI/System/Network/mailing

Method	GET
Description	Get email configuration parameters.
Query	None.
Request	None.
Response	<i>XML_mailingList</i>

Table A-727 PUT /ISAPI/System/Network/mailing

Method	PUT
Description	Set email parameters.
Query	None.
Request	<i>XML_mailingList</i>
Response	<i>XML_ResponseStatus</i>

A.12.169 /ISAPI/System/Network/mailing/<ID>

Operations about the configuration of a specific email.

Request URI Definition

Table A-728 GET /ISAPI/System/Network/mailing/<ID>

Method	GET
Description	Get the configuration parameters of a specific email.
Query	None.
Request	None.
Response	<i>XML_mailing</i>

Table A-729 PUT /ISAPI/System/Network/mailing/<ID>

Method	PUT
Description	Set the parameters of a specific email.
Query	None.
Request	<i>XML_mailing</i>
Response	<i>XML_ResponseStatus</i>

Remarks

The <ID> in the URI refers to the email ID.

A.12.170 /ISAPI/System/Network/mailing/capabilities

Get email configuration capability.

Request URI Definition

Table A-730 GET /ISAPI/System/Network/mailing/capabilities

Method	GET
Description	Get email configuration capability.
Query	None.
Request	None.
Response	<i>XML_Cap_mailingList</i>

A.12.171 /ISAPI/System/Network/mailing/test

Execute the email test to check if the email server and address are available.

Request URI Definition

Table A-731 POST /ISAPI/System/Network/mailing/test

Method	POST
Description	Execute the email test to check if the email server and address are available.
Query	None.
Request	<i>XML_mailingTestDescription</i>
Response	<i>XML_mailingTestResult</i>

A.12.172 /ISAPI/System/Network/pingtest

Start a ping test and get the test result.

Request URI Definition

Table A-732 POST /ISAPI/System/Network/pingtest

Method	POST
Description	Start a ping test and get the test result.
Query	None.
Request	<i>XML_pingTestDescription</i>
Response	<i>XML_pingTestResult</i>

A.12.173 /ISAPI/System/Network/POE/capabilities?format=json

Get capability of PoE port parameters configuration.

Request URI Definition

Table A-733 GET /ISAPI/System/Network/POE/capabilities?format=json

Method	GET
Description	Get the capability of PoE port parameters configuration.
Query	format: determine the format of request or response message.
Request	None.
Response	<i>JSON_Cap_POE</i>

A.12.174 /ISAPI/System/Network/POE?format=json

Get or set the PoE port configuration parameters.

Request URI Definition

Table A-734 GET /ISAPI/System/Network/POE?format=json

Method	GET
Description	Get PoE port configuration parameters.
Query	format : determine the format of request or response message.
Request	None.
Response	<i>JSON_POE</i>

Table A-735 PUT /ISAPI/System/Network/POE?format=json

Method	PUT
Description	Set PoE port configuration parameters.
Query	format : determine the format of request or response message.
Request	<i>JSON_POE</i>
Response	<i>JSON_ResponseStatus</i>

A.12.175 /ISAPI/System/Network/resourceStatistics?format=json

Get the information of network resource statistics.

Request URI Definition

Table A-736 GET /ISAPI/System/Network/resourceStatistics?format=json

Method	GET
Description	Get the information of network resource statistics.
Query	None.
Request	None.
Response	Succeeded: <i>JSON_resourceStatistics</i> Failed: <i>JSON_ResponseStatus</i>

A.12.176 /ISAPI/System/Network/SIP

Operations about the configuration of a specific SIP (Session Initiation Protocol) server.

Request URI Definition

Table A-737 GET /ISAPI/System/Network/SIP

Method	GET
Description	Get the parameters of a specific SIP (Session Initiation Protocol) server.
Query	None.
Request	None.
Response	Succeeded: <i>XML_SIPServerList</i> Failed: <i>XML_ResponseStatus</i>

Table A-738 PUT /ISAPI/System/Network/SIP

Method	PUT
Description	Set the parameters of a specific SIP (Session Initiation Protocol) server.
Query	None.
Request	<i>XML_SIPServerList</i>
Response	<i>XML_ResponseStatus</i>

A.12.177 /ISAPI/System/Network/SIP/<ID>

Operations about the configuration of a specific SIP (Session Initiation Protocol) server.

Request URI Definition

Table A-739 GET /ISAPI/System/Network/SIP/<ID>

Method	GET
Description	Get the parameters of a specific SIP (Session Initiation Protocol) server.
Query	None.
Request	None.
Response	Succeeded: <i>XML_SIPServer</i>

	Failed: <i>XML_ResponseStatus</i>
--	-----------------------------------

Table A-740 PUT /ISAPI/System/Network/SIP/<ID>

Method	PUT
Description	Set the parameters of a specific SIP (Session Initiation Protocol) server.
Query	None.
Request	<i>XML_SIPServer</i>
Response	<i>XML_ResponseStatus</i>

A.12.178 /ISAPI/System/Network/SIP/<ID>/capabilities

Get the configuration capability of a specific SIP (Session Initiation Protocol) server.

Request URI Definition

Table A-741 GET /ISAPI/System/Network/SIP/<ID>/capabilities

Method	GET
Description	Get the configuration capability of a specific SIP (Session Initiation Protocol) server.
Query	None.
Request	None.
Response	Succeeded: <i>XML_Cap_SIPServer</i> Failed: <i>XML_ResponseStatus</i>

A.12.179 /ISAPI/System/Network/SIP/<ID>/SIPInfo

Operations about the configuration of a piece of SIP (Session Initiation Protocol) information.

Request URI Definition

Table A-742 GET /ISAPI/System/Network/SIP/<ID>/SIPInfo

Method	GET
Description	Get the parameters of a piece of SIP (Session Initiation Protocol) information.
Query	None.

Request	None.
Response	Succeeded: <i>XML_SIPInfo</i> Failed: <i>XML_ResponseStatus</i>

Table A-743 PUT /ISAPI/System/Network/SIP/<ID>/SIPInfo

Method	PUT
Description	Set the parameters of a piece of SIP (Session Initiation Protocol) information.
Query	None.
Request	<i>XML_SIPInfo</i>
Response	<i>XML_ResponseStatus</i>

A.12.180 /ISAPI/System/Network/SIP/<ID>/SIPInfo/capabilities

Get the configuration capability of a specific piece of SIP (Session Initiation Protocol) information.

Request URI Definition

Table A-744 GET /ISAPI/System/Network/SIP/<ID>/SIPInfo/capabilities

Method	GET
Description	Get the configuration capability of a specific piece of SIP (Session Initiation Protocol) information.
Query	None.
Request	None.
Response	Succeeded: <i>XML_Cap_SIPInfo</i> Failed: <i>XML_ResponseStatus</i>

A.12.181 /ISAPI/System/Network/SIP/<ID>/SIPInfo/multiInfo

Get or set the information list of SIP servers.

Request URI Definition

Table A-745 GET /ISAPI/System/Network/SIP/<ID>/SIPInfo/multiInfo

Method	GET
Description	Get the information list of SIP servers.

Query	None.
Request	None.
Response	Succeeded: <i>XML_SIPInfoList</i> Failed: <i>XML_ResponseStatus</i>

Table A-746 PUT /ISAPI/System/Network/SIP/<ID>/SIPInfo/multiInfo

Method	PUT
Description	Set the information list of SIP servers.
Query	None.
Request	<i>XML_SIPInfoList</i>
Response	<i>XML_ResponseStatus</i>

Remarks

The <ID> in the request URI refers to the SIP server ID.

A.12.182 /ISAPI/System/Network/SIP/capabilities

Get the configuration capability of all SIP (Session Initiation Protocol) servers.

Request URL Definition

Table A-747 GET /ISAPI/System/Network/SIP/capabilities

Method	GET
Description	Get the configuration capability of all SIP (Session Initiation Protocol) servers.
Query	None.
Request	None.
Response	Succeeded: <i>XML_SIPServerListCap</i> Failed: <i>XML_ResponseStatus</i>

A.12.183 /ISAPI/System/Network/SIP/voiceTalkCtrl/<ID>?format=json

Get or set SIP (Session Initiation Protocol) intercom parameters.

Request URI Definition

Table A-748 GET /ISAPI/System/Network/SIP/voiceTalkCtrl/<ID>?format=json

Method	GET
Description	Get SIP (Session Initiation Protocol) intercom parameters.
Query	format: determine the format of request or response message.
Request	None.
Response	Succeeded: <i>JSON_voiceTalkCtrl</i> Failed: <i>JSON_ResponseStatus</i>

Table A-749 PUT /ISAPI/System/Network/SIP/voiceTalkCtrl/<ID>?format=json

Method	PUT
Description	Set SIP (Session Initiation Protocol) intercom parameters.
Query	format: determine the format of request or response message.
Request	<i>JSON_voiceTalkCtrl</i>
Response	<i>JSON_ResponseStatus</i>

Remarks

The <ID> in the request URI is 1 by default.

A.12.184 /ISAPI/System/Network/SIP/voiceTalkCtrl/capabilities?format=json

Get the configuration capability of SIP (Session Initiation Protocol) intercom.

Request URI Definition

Table A-750 GET /ISAPI/System/Network/SIP/voiceTalkCtrl/capabilities?format=json

Method	GET
Description	Get the configuration capability of SIP (Session Initiation Protocol) intercom.
Query	format: determine the format of request or response message.
Request	None.
Response	Succeeded: <i>JSON_voiceTalkCtrlCap</i> Failed: <i>JSON_ResponseStatus</i>

A.12.185 /ISAPI/System/Network/SNMP

Get or set batch configuration parameters of SNMP (Simple Network Management Protocol).

Request URI Definition

Table A-751 GET /ISAPI/System/Network/SNMP

Method	GET
Description	Get batch configuration parameters of SNMP.
Query	security : the version No. of encryption scheme. When security does not exist, it indicates that the data is not encrypted; when security is 1, it indicates that the nodes of sensitive information in the message are encrypted in AES128 CBC mode; when security is 2, it indicates that the nodes of sensitive information in the message are encrypted in AES256 CBC mode. iv : the initialization vector, and it is required when security is 1 or 2.
Request	None.
Response	Succeeded: <i>XML_SNMP</i> Failed: <i>XML_ResponseStatus</i>

Table A-752 PUT /ISAPI/System/Network/SNMP

Method	PUT
Description	Set batch configuration parameters of SNMP.
Query	security : the version No. of encryption scheme. When security does not exist, it indicates that the data is not encrypted; when security is 1, it indicates that the nodes of sensitive information in the message are encrypted in AES128 CBC mode; when security is 2, it indicates that the nodes of sensitive information in the message are encrypted in AES256 CBC mode. iv : the initialization vector, and it is required when security is 1 or 2.
Request	<i>XML_SNMP</i>
Response	<i>XML_ResponseStatus</i>

A.12.186 /ISAPI/System/Network/SNMP/advanced

Get or set advanced parameters of SNMP (Simple Network Management Protocol).

Request URI Definition

Table A-753 GET /ISAPI/System/Network/SNMP/advanced

Method	GET
Description	Get advanced parameters of SNMP.
Query	None.
Request	None.
Response	Succeeded: <i>XML_SNMPSAdvanced</i> Failed: <i>XML_ResponseStatus</i>

Table A-754 PUT /ISAPI/System/Network/SNMP/advanced

Method	PUT
Description	Set advanced parameters of SNMP.
Query	None.
Request	<i>XML_SNMPSAdvanced</i>
Response	<i>XML_ResponseStatus</i>

A.12.187 /ISAPI/System/Network/SNMP/advanced/trapReceivers

Get, set, delete parameters of all trap address, or add a new trap address of SNMP in version 3.

Request URI Definition

Table A-755 GET /ISAPI/System/Network/SNMP/advanced/trapReceivers

Method	GET
Description	Get all trap address parameters of SNMP in version 3.
Query	None.
Request	None.
Response	Succeeded: <i>XML_SNMPTrapReceiverList</i> Failed: <i>XML_ResponseStatus</i>

Table A-756 PUT /ISAPI/System/Network/SNMP/advanced/trapReceivers

Method	PUT
Description	Set parameters for all trap addresses of SNMP in version 3.
Query	None.

Request	<i>XML_SNMPTrapReceiverList</i>
Response	<i>XML_ResponseStatus</i>

Table A-757 DELETE /ISAPI/System/Network/SNMP/advanced/trapReceivers

Method	DELETE
Description	Delete all trap address parameters of SNMP in version 3.
Query	None.
Request	None.
Response	<i>XML_ResponseStatus</i>

Table A-758 POST /ISAPI/System/Network/SNMP/advanced/trapReceivers

Method	POST
Description	Add a new trap address of SNMP.
Query	None.
Request	<i>XML_SNMPTrapReceiver</i>
Response	<i>XML_ResponseStatus</i>

A.12.188 /ISAPI/System/Network/SNMP/advanced/trapReceivers/<ID>

Get, set or delete parameters for a single trap address.

Request URI Definition

Table A-759 GET /ISAPI/System/Network/SNMP/advanced/trapReceivers/<ID>

Method	GET
Description	Get trap address parameters of SNMP in version 3.
Query	None.
Request	None.
Response	Succeeded: <i>XML_SNMPTrapReceiver</i> Failed: <i>XML_ResponseStatus</i>

Table A-760 PUT /ISAPI/System/Network/SNMP/advanced/trapReceivers/<ID>

Method	PUT
Description	Set parameters for all trap addresses of SNMP in version 3.
Query	None.

Request	<i>XML_SNMPTrapReceiver</i>
Response	<i>XML_ResponseStatus</i>

Table A-761 DELETE /ISAPI/System/Network/SNMP/advanced/trapReceivers/<ID>

Method	DELETE
Description	Delete a trap address of SNMP in version 3.
Query	None.
Request	None.
Response	<i>XML_ResponseStatus</i>

A.12.189 /ISAPI/System/Network/socketIP

Get the socket IP of current connection.

Request URI Definition

Table A-762 GET /ISAPI/System/Network/socketIP

Method	GET
Description	Get the socket IP of current connection.
Query	None.
Request	None.
Response	<i>XML_SocketIP</i>

A.12.190 /ISAPI/System/Network/ssh

Operations about SSH server configurations.

Request URI Definition

Table A-763 GET /ISAPI/System/Network/ssh

Method	GET
Description	Get SSH server parameters.
Query	None.
Request	None.
Response	<i>XML_SSH</i>

Table A-764 PUT /ISAPI/System/Network/ssh

Method	PUT
Description	Set SSH server parameters.
Query	None.
Request	<i>XML_SSH</i>
Response	<i>XML_ResponseStatus</i>

A.12.191 /ISAPI/System/Network/ssh/capabilities

Get the SSH parameters configuration capability.

Request URL Definition

Table A-765 GET /ISAPI/System/Network/ssh/capabilities

Method	GET
Description	Get the SSH parameters configuration capability.
Query	None.
Request	None.
Response	Succeeded: <i>XML_Cap_SSH</i> Failed: <i>XML_ResponseStatus</i>

A.12.192 /ISAPI/System/Network/StaticRoute

Get, set, or delete the parameters of static route.

Request URI Definition

Table A-766 GET /ISAPI/System/Network/StaticRoute

Method	GET
Description	Get the parameters of static route.
Query	None.
Request	None.
Response	Succeeded: <i>XML_staticRouteList</i> Failed: <i>XML_ResponseStatus</i>

Table A-767 PUT /ISAPI/System/Network/StaticRoute

Method	PUT
Description	Set the parameters of static route.
Query	None.
Request	<i>XML_staticRouteList</i>
Response	<i>XML_ResponseStatus</i>

Table A-768 DELETE /ISAPI/System/Network/StaticRoute

Method	DELETE
Description	Delete the parameters of static route.
Query	None.
Request	None.
Response	<i>XML_ResponseStatus</i>

A.12.193 /ISAPI/System/Network/telnetd

Operations about the telnetd service configuration.

Request URI Definition

Table A-769 GET /ISAPI/System/Network/telnetd

Method	GET
Description	Get the telnetd service parameters.
Query	None.
Request	None.
Response	Succeeded: <i>XML_Telnetd</i> Failed: <i>XML_ResponseStatus</i>

Table A-770 PUT /ISAPI/System/Network/telnetd

Method	PUT
Description	Set the telnetd service parameters.
Query	None.
Request	<i>XML_Telnetd</i>
Response	<i>XML_ResponseStatus</i>

A.12.194 /ISAPI/System/Network/telnetd/capabilities

Get the telnetd service capability.

Request URI Definition

Table A-771 GET /ISAPI/System/Network/telnetd/capabilities

Method	GET
Description	Get the telnetd service capability.
Query	None.
Request	None.
Response	Succeeded: <i>XML_Cap_Telnetd</i> Failed: <i>XML_ResponseStatus</i>

A.12.195 /ISAPI/System/Network/UPnP

Operations about UPnP configuration.

Request URI Definition

Table A-772 GET /ISAPI/System/Network/UPnP

Method	GET
Description	Get UPnP parameters.
Query	None.
Request	None.
Response	<i>XML_UPnP</i>

Table A-773 PUT /ISAPI/System/Network/UPnP

Method	PUT
Description	Set UPnP parameters.
Query	None.
Request	<i>XML_UPnP</i>
Response	<i>XML_ResponseStatus</i>

A.12.196 /ISAPI/System/Network/UPnP/capabilities

Get the UPnP configuration capability.

Request URI Definition

Table A-774 GET /ISAPI/System/Network/UPnP/capabilities

Method	GET
Description	Get the UPnP configuration capability.
Query	None.
Request	None.
Response	Succeeded: <i>XML_Cap_UPnPCap</i> Failed: <i>XML_ResponseStatus</i>

A.12.197 /ISAPI/System/Network/UPnP/ports

Operations about configuration of all UPnP ports.

Request URI Definition

Table A-775 GET /ISAPI/System/Network/UPnP/ports

Method	GET
Description	Get the parameters of all UPnP ports.
Query	None.
Request	None.
Response	<i>XML_ports</i>

Table A-776 PUT /ISAPI/System/Network/UPnP/ports

Method	PUT
Description	Set the parameters of all UPnP ports.
Query	None.
Request	<i>XML_ports</i>
Response	<i>XML_ResponseStatus</i>

A.12.198 /ISAPI/System/Network/UPnP/ports/status

Get the mapping statuses of all UPnP ports.

Request URI Definition**Table A-777 GET /ISAPI/System/Network/UPnP/ports/status**

Method	GET
Description	Get the mapping statuses of all UPnP ports.
Query	None.
Request	None.
Response	<i>XML_portsStatus</i>

A.12.199 /ISAPI/System/Network/UPnP/ports/<ID>

Operations about configuration of a specific UPnP port.

Request URI Definition**Table A-778 GET /ISAPI/System/Network/UPnP/ports/<ID>**

Method	GET
Description	Get the parameters of a specific UPnP port.
Query	None.
Request	None.
Response	Succeeded: <i>XML_port</i> Failed: <i>XML_ResponseStatus</i>

Table A-779 PUT /ISAPI/System/Network/UPnP/ports/<ID>

Method	PUT
Description	Set the parameters of a specific UPnP port.
Query	None.
Request	<i>XML_port</i>
Response	<i>XML_ResponseStatus</i>

Remarks

The <ID> in the request URI refers to UPnP port ID.

A.12.200 /ISAPI/System/Network/UPnP/ports/<ID>/status

Get mapping status of a specific UPnP port.

Request URI Definition

Table A-780 GET /ISAPI/System/Network/UPnP/ports/<ID>/status

Method	GET
Description	Get mapping status of a specific UPnP port.
Query	None.
Request	None.
Response	Succeeded: <i>XML_portStatus</i> Failed: <i>XML_ResponseStatus</i>

Remarks

The <ID> in the request URI refers to UPnP port ID.

A.12.201 /ISAPI/System/Network/verificationCodeCheck

Verify the encryption key or verification code.

Request URI Definition

Table A-781 POST /ISAPI/System/Network/verificationCodeCheck

Method	POST
Description	Verify the encryption key or verification code.
Query	None.
Request	<i>XML_CheckInfo</i>
Response	<i>XML_ResponseStatus</i>

A.12.202 /ISAPI/System/Network/WirelessDial/Interfaces

Get or set parameters for all dial network cards.

Request URI Definition

Table A-782 GET /ISAPI/System/Network/WirelessDial/Interfaces

Method	GET
Description	Get parameters for all dial network cards.
Query	None.
Request	None.
Response	Succeeded: <i>XML_WirelessDialInterface</i> Failed: <i>XML_ResponseStatus</i>

Table A-783 PUT /ISAPI/System/Network/WirelessDial/Interfaces

Method	PUT
Description	Set parameters for all dial network cards.
Query	None.
Request	<i>XML_WirelessDialInterface</i>
Response	<i>XML_ResponseStatus</i>

A.12.203 /ISAPI/System/Network/WirelessDial/Interfaces/<ID>

Get or set parameters for a single dial network card.

Request URI Definition

Table A-784 GET /ISAPI/System/Network/WirelessDial/Interfaces/<ID>

Method	GET
Description	Get parameters for a single dial network card.
Query	None.
Request	None.
Response	Succeeded: <i>XML_WirelessDialInterfaceList</i> Failed: <i>XML_ResponseStatus</i>

Table A-785 PUT /ISAPI/System/Network/WirelessDial/Interfaces/<ID>

Method	PUT
Description	Set the wireless dial-up parameters.
Query	None.

Request	<i>XML_WirelessDialInterfaceList</i>
Response	<i>XML_ResponseStatus</i>

Remarks

The <ID> in the request URI refers to the network interface ID.

A.12.204 /ISAPI/System/Network/WirelessDial/Interfaces/<ID>/dial

Operations about the wireless dial-up configuration.

Request URI Definition

Table A-786 GET /ISAPI/System/Network/WirelessDial/Interfaces/<ID>/dial

Method	GET
Description	Get the configuration parameters of wireless dial-up.
Query	security: the version No. of encryption scheme. When security does not exist, it indicates that the data is not encrypted; when security is 1, it indicates that the nodes of sensitive information in the message are encrypted in AES128 CBC mode; when security is 2, it indicates that the nodes of sensitive information in the message are encrypted in AES256 CBC mode. iv: the initialization vector, and it is required when security is 1 or 2.
Request	None.
Response	Succeeded: <i>XML_Dial</i> Failed: <i>XML_ResponseStatus</i>

Table A-787 PUT /ISAPI/System/Network/WirelessDial/Interfaces/<ID>/dial

Method	PUT
Description	Set the wireless dial-up parameters.
Query	security: the version No. of encryption scheme. When security does not exist, it indicates that the data is not encrypted; when security is 1, it indicates that the nodes of sensitive information in the message are encrypted in AES128 CBC mode; when security is 2, it indicates that the nodes of sensitive information in the message are encrypted in AES256 CBC mode. iv: the initialization vector, and it is required when security is 1 or 2.
Request	<i>XML_Dial</i>
Response	<i>XML_ResponseStatus</i>

A.12.205 /ISAPI/System/Network/WirelessDial/Interfaces/<ID>/dial/capabilities

Get the wireless dial-up capability.

Request URI Definition**Table A-788 GET /ISAPI/System/Network/WirelessDial/Interfaces/<ID>/dial/capabilities**

Method	GET
Description	Get the wireless dial-up capability.
Query	security : the version No. of encryption scheme. When security does not exist, it indicates that the data is not encrypted; when security is 1, it indicates that the nodes of sensitive information in the message are encrypted in AES128 CBC mode; when security is 2, it indicates that the nodes of sensitive information in the message are encrypted in AES256 CBC mode. iv : the initialization vector, and it is required when security is 1 or 2.
Request	None.
Response	Succeeded: <i>XML_Cap_Dial</i> Failed: <i>XML_ResponseStatus</i>

A.12.206 /ISAPI/System/Network/WirelessDial/Interfaces/<ID>/dialstatus

Operations about the configuration of wireless dial-up connection status.

Request URI Definition**Table A-789 GET /ISAPI/System/Network/WirelessDial/Interfaces/<ID>/dialstatus**

Method	GET
Description	Get the parameters of wireless dial-up connection status.
Query	None.
Request	None.
Response	Succeeded: <i>XML_Dialstatus</i> Failed: <i>XML_ResponseStatus</i>

Table A-790 PUT /ISAPI/System/Network/WirelessDial/Interfaces/<ID>/dialstatus

Method	PUT
Description	Set the parameters of wireless dial-up connection status.

Query	None.
Request	<i>XML_DialStatus</i>
Response	<i>XML_ResponseStatus</i>

Remarks

The <ID> in the request URI refers to the network interface ID.

A.12.207 /ISAPI/System/Network/WirelessDial/Interfaces/<ID>/keepAlive/capabilities?format=json

Get the configuration capability of network keepalive.

Request URI Definition

Table A-791 GET /ISAPI/System/Network/WirelessDial/Interfaces/<ID>/keepAlive/capabilities?format=json

Method	GET
Description	Get the configuration capability of network keepalive.
Query	format: determine the format of request or response message.
Request	None.
Response	Succeeded: <i>JSON_KeepAliveCap</i> Failed: <i>JSON_ResponseStatus</i>

Remarks

The <ID> in the request URI refers to the network interface ID.

A.12.208 /ISAPI/System/Network/WirelessDial/Interfaces/<ID>/keepAlive/test?format=json

Test network keepalive configuration.

Request URI Definition

Table A-792 POST /ISAPI/System/Network/WirelessDial/Interfaces/<ID>/keepAlive/test?format=json

Method	POST
Description	Test network keepalive configuration.
Query	format: determine the format of request or response message.

Request	<i>JSON_KeepAlive</i>
Response	<i>JSON_TestResult</i>

Remarks

The <ID> in the request URI refers to the network interface ID.

A.12.209 /ISAPI/System/Network/WirelessDial/Interfaces/<ID>/keepAlive?format=json

Set network keepalive parameters.

Request URI Definition

Table A-793 PUT /ISAPI/System/Network/WirelessDial/Interfaces/<ID>/keepAlive?format=json

Method	PUT
Description	Set network keepalive parameters.
Query	format : determine the format of request or response message.
Request	<i>JSON_KeepAlive</i>
Response	<i>JSON_ResponseStatus</i>

Remarks

The <ID> in the request URI refers to the network interface ID.

A.12.210 /ISAPI/System/Network/WirelessDial/Interfaces/<ID>/messageConfig

Get or set dial-up information parameters.

Request URI Definition

Table A-794 GET /ISAPI/System/Network/WirelessDial/Interfaces/<ID>/messageConfig

Method	GET
Description	Get dial-up information parameters.
Query	None.
Request	None.
Response	Succeeded: <i>XML_messageConfig</i> Failed: <i>XML_ResponseStatus</i>

Table A-795 PUT /ISAPI/System/Network/WirelessDial/Interfaces/<ID>/messageConfig

Method	PUT
Description	Set dial-up information parameters.
Query	None.
Request	<i>XML_messageConfig</i>
Response	<i>XML_ResponseStatus</i>

Remarks

The <ID> in the request URI refers to the network interface ID.

A.12.211 /ISAPI/System/Network/WirelessDial/Interfaces/<ID>/messageConfig/messageConfigCap

Get dial-up information capability.

Request URI Definition

Table A-796 GET /ISAPI/System/Network/WirelessDial/Interfaces/<ID>/messageConfig/messageConfigCap

Method	GET
Description	Get dial-up information capability.
Query	None.
Request	None.
Response	Succeeded: <i>XML_messageConfigCap</i> Failed: <i>XML_ResponseStatus</i>

Remarks

The <ID> in the request URI refers to the network interface ID.

A.12.212 /ISAPI/System/Network/WirelessDial/Interfaces/<ID>/messageConfig/WhiteList

Get or set all whitelist configuration.

Request URI Definition**Table A-797 GET /ISAPI/System/Network/WirelessDial/Interfaces/<ID>/messageConfig/WhiteList**

Method	GET
Description	Get all whitelist configuration.
Query	None.
Request	None.
Response	Succeeded: <i>XML_SMSWhiteList</i> Failed: <i>XML_ResponseStatus</i>

Table A-798 PUT /ISAPI/System/Network/WirelessDial/Interfaces/<ID>/messageConfig/WhiteList

Method	PUT
Description	Set all whitelist configuration.
Query	None.
Request	<i>XML_SMSWhiteList</i>
Response	<i>XML_ResponseStatus</i>

Remarks

The <ID> in the request URI refers to the network interface ID.

A.12.213 /ISAPI/System/Network/WirelessDial/Interfaces/<ID>/messageConfig/WhiteList/<ID>

Get or set specific whitelist configuration.

Request URI Definition**Table A-799 GET /ISAPI/System/Network/WirelessDial/Interfaces/<ID>/messageConfig/WhiteList/<ID>**

Method	GET
Description	Get specific whitelist configuration.
Query	security : the version No. of encryption scheme. When security does not exist, it indicates that the data is not encrypted; when security is 1, it indicates that the nodes of sensitive information in the message are encrypted in AES128 CBC mode; when security is 2, it indicates

	that the nodes of sensitive information in the message are encrypted in AES256 CBC mode. None. iv : the initialization vector, and it is required when security is 1 or 2.
Request	None.
Response	Succeeded: <i>XML_ListMember</i> Failed: <i>XML_ResponseStatus</i>

Table A-800 PUT /ISAPI/System/Network/WirelessDial/Interfaces/<ID>/messageConfig/WhiteList/<ID>

Method	PUT
Description	Set specific whitelist configuration.
Query	security : the version No. of encryption scheme. When security does not exist, it indicates that the data is not encrypted; when security is 1, it indicates that the nodes of sensitive information in the message are encrypted in AES128 CBC mode; when security is 2, it indicates that the nodes of sensitive information in the message are encrypted in AES256 CBC mode. None. iv : the initialization vector, and it is required when security is 1 or 2.
Request	<i>XML_ListMember</i>
Response	<i>XML_ResponseStatus</i>

Remarks

The <ID> in the request URI refers to the network interface ID.

A.12.214 /ISAPI/System/Network/WirelessDial/Interfaces/<ID>/schedule

Get or set dial-up schedule parameters.

Request URI Definition

Table A-801 GET /ISAPI/System/Network/WirelessDial/Interfaces/<ID>/schedule

Method	GET
Description	Get dial-up schedule parameters.
Query	None.
Request	None.
Response	Succeeded: <i>XML_Schedule</i> Failed: <i>XML_ResponseStatus</i>

Table A-802 PUT /ISAPI/System/Network/WirelessDial/Interfaces/<ID>/schedule

Method	PUT
Description	Set dial-up schedule parameters.
Query	None.
Request	<i>XML_Schedule</i>
Response	<i>XML_ResponseStatus</i>

Remarks

The <ID> in the request URI refers to the network interface ID.

A.12.215 /ISAPI/System/Network/WirelessDial/Interfaces/<ID>/trafficMonitor/capabilities?format=json

Get traffic monitoring configuration capability.

Request URI Definition

Table A-803 GET /ISAPI/System/Network/WirelessDial/Interfaces/<ID>/trafficMonitor/capabilities?format=json

Method	GET
Description	Get traffic monitoring configuration capability.
Query	format: determine the format of request or response message.
Request	None.
Response	Succeeded: <i>JSON_TrafficMonitorCap</i> Failed: <i>JSON_ResponseStatus</i>

A.12.216 /ISAPI/System/Network/WirelessDial/Interfaces/<ID>/trafficMonitor/status?format=json

Get traffic monitoring status.

Request URI Definition

Table A-804 GET /ISAPI/System/Network/WirelessDial/Interfaces/<ID>/trafficMonitor/status?format=json

Method	GET
Description	Get traffic monitoring status.

Query	format: determine the format of request or response message.
Request	None.
Response	Succeeded: <i>JSON_TrafficMonitorStatus</i> Failed: <i>JSON_ResponseStatus</i>

A.12.217 /ISAPI/System/Network/WirelessDial/Interfaces/<ID>/trafficMonitor?format=json

Get traffic monitoring parameters.

Request URI Definition

Table A-805 GET /ISAPI/System/Network/WirelessDial/Interfaces/<ID>/trafficMonitor?format=json

Method	GET
Description	Get traffic monitoring parameters.
Query	format: determine the format of request or response message.
Request	None.
Response	Succeeded: <i>JSON_TrafficMonitor</i> Failed: <i>JSON_ResponseStatus</i>

Table A-806 PUT /ISAPI/System/Network/WirelessDial/Interfaces/<ID>/trafficMonitor?format=json

Method	PUT
Description	Set traffic monitoring parameters.
Query	None.
Request	<i>JSON_TrafficMonitor</i>
Response	<i>JSON_ResponseStatus</i>

Remarks

The <ID> in the request URI refers to the network interface ID.

A.12.218 /ISAPI/System/Network/WPS/capabilities

Get the WPS (Wi-Fi Protected Setup) configuration capability.

Request URI Definition**Table A-807 GET /ISAPI/System/Network/WPS/capabilities**

Method	GET
Description	Get the WPS (Wi-Fi Protected Setup) configuration capability.
Query	None.
Request	None.
Response	Succeeded: <i>XML_Cap_WPS</i> Failed: <i>XML_ResponseStatus</i>

A.12.219 /ISAPI/System/onlineUpgrade/capabilities

Get the device online upgrade capability.

Request URI Definition**Table A-808 GET /ISAPI/System/onlineUpgrade/capabilities**

Method	GET
Description	Get the device online upgrade capability.
Query	None.
Request	None.
Response	<i>XML_OnlineUpgradeCap</i>

A.12.220 /ISAPI/System/onlineUpgrade/deviceParameter?format=json

Get and set the device online upgrade parameters.

Request URI Definition**Table A-809 GET /ISAPI/System/onlineUpgrade/deviceParameter?format=json**

Method	GET
Description	Get the device online upgrade parameters.
Query	format: determine the format of request or response message.
Request	None.
Response	Succeeded: <i>JSON_OnlineUpgradeParameter</i> Failed: <i>JSON_ResponseStatus</i>

Table A-810 PUT /ISAPI/System/onlineUpgrade/deviceParameter?format=json

Method	PUT
Description	Set the device online upgrade parameters.
Query	format: determine the format of request or response message.
Request	<i>JSON_OnlineUpgradeParameter</i>
Response	<i>JSON_ResponseStatus</i>

A.12.221 /ISAPI/System/onlineUpgrade/downloadPackage/pause?format=json

Pause the upgrade package download.

Request URI Definition

Table A-811 PUT /ISAPI/System/onlineUpgrade/downloadPackage/pause?format=json

Method	PUT
Description	Pause upgrade package download.
Query	format: determine the format of request or response message.
Request	None.
Response	<i>JSON_ResponseStatus</i>

A.12.222 /ISAPI/System/onlineUpgrade/downloadPackage/resume?format=json

Resume the upgrade package download.

Request URI Definition

Table A-812 PUT /ISAPI/System/onlineUpgrade/downloadPackage/resume?format=json

Method	PUT
Description	Resume the upgrade package download.
Query	format: determine the format of request or response message.
Request	None.
Response	<i>JSON_ResponseStatus</i>

A.12.223 /ISAPI/System/onlineUpgrade/downloadPackage/status?format=json

Get the upgrade package download progress.

Request URI Definition

Table A-813 GET /ISAPI/System/onlineUpgrade/downloadPackage/status?format=json

Method	GET
Description	Get the upgrade package download progress.
Query	format: determine the format of request or response message.
Request	None.
Response	Succeeded: <i>JSON_DownloadPackageStatus</i> Failed: <i>JSON_ResponseStatus</i>

A.12.224 /ISAPI/System/onlineUpgrade/downloadPackage?format=json

Start or cancel downloading upgrade package to device.

Request URI Definition

Table A-814 PUT /ISAPI/System/onlineUpgrade/downloadPackage?format=json

Method	PUT
Description	Start to download upgrade package to device.
Query	format: determine the format of request or response message.
Request	None.
Response	<i>JSON_ResponseStatus</i>

Table A-815 DELETE /ISAPI/System/onlineUpgrade/downloadPackage?format=json

Method	DELETE
Description	Cancel the upgrade package download.
Query	format: determine the format of request or response message.
Request	None.
Response	<i>JSON_ResponseStatus</i>

A.12.225 /ISAPI/System/onlineUpgrade/ignoreCurrentVersion?format=json

Ignore the current upgrade package version.

Request URI Definition

Table A-816 PUT /ISAPI/System/onlineUpgrade/ignoreCurrentVersion?format=json

Method	PUT
Description	Ignore the current upgrade package version.
Query	format: determine the format of request or response message.
Request	None.
Response	<i>JSON_ResponseStatus</i>

Remarks

If the current upgrade package version is ignored, it will not be found in further detection.

A.12.226 /ISAPI/System/onlineUpgrade/server

Get the online upgrade server status.

Request URI Definition

Table A-817 GET /ISAPI/System/onlineUpgrade/server

Method	GET
Description	Get the online upgrade server status.
Query	None.
Request	None.
Response	<i>XML_OnlineUpgradeServer</i>

A.12.227 /ISAPI/System/onlineUpgrade/status

Get the online upgrade progress of device.

Request URI Definition

Table A-818 GET /ISAPI/System/onlineUpgrade/status

Method	GET
Description	Get the online upgrade progress of device.
Query	None.
Request	None.
Response	<i>XML_OnlineUpgradeStatus</i>

A.12.228 /ISAPI/System/onlineUpgrade/upgradeWithoutDownload?format=json

Start device upgrade with existed upgrade package.

Request URI Definition

Table A-819 PUT /ISAPI/System/onlineUpgrade/upgradeWithoutDownload?format=json

Method	PUT
Description	Start upgrade with existed upgrade package.
Query	format: determine the format of request or response message.
Request	None.
Response	<i>JSON_ResponseStatus</i>

A.12.229 /ISAPI/System/onlineUpgrade/version

Get the information of new upgrade package.

Request URI Definition

Table A-820 GET /ISAPI/System/onlineUpgrade/version

Method	GET
Description	Get the information of new upgrade package.
Query	check
Request	None.
Response	<i>XML_OnlineUpgradeVersion</i>

A.12.230 /ISAPI/System/PictureServer/capabilities?format=json

Get the picture storage server capability.

Request URI Definition

Table A-821 GET /ISAPI/System/PictureServer/capabilities?format=json

Method	GET
Description	Get the picture storage server capability.
Query	format: determine the format of request or response message.

Request	None.
Response	Succeeded: <i>JSON_Cap_PictureServerInformation</i> Failed: <i>JSON_ResponseStatus</i>

A.12.231 /ISAPI/System/PictureServer?format=json

Operations about the picture storage server configuration parameters.

Request URI Definition

Table A-822 GET /ISAPI/System/PictureServer?format=json

Method	GET
Description	Get the picture storage server parameters.
Query	format: determine the format of request or response message. security: the version No. of encryption scheme. When security does not exist, it indicates that the data is not encrypted; when security is 1, it indicates that the nodes of sensitive information in the message are encrypted in AES128 CBC mode; when security is 2, it indicates that the nodes of sensitive information in the message are encrypted in AES256 CBC mode. iv: the initialization vector, and it is required when security is 1 or 2.
Request	None.
Response	Succeeded: <i>JSON_PictureServerInformation</i> Failed: <i>JSON_ResponseStatus</i>

Table A-823 PUT /ISAPI/System/PictureServer?format=json

Method	PUT
Description	Set the picture storage server parameters.
Query	format: determine the format of request or response message. security: the version No. of encryption scheme. When security does not exist, it indicates that the data is not encrypted; when security is 1, it indicates that the nodes of sensitive information in the message are encrypted in AES128 CBC mode; when security is 2, it indicates that the nodes of sensitive information in the message are encrypted in AES256 CBC mode. iv: the initialization vector, and it is required when security is 1 or 2.

Request	<i>JSON_PictureServerInformation</i>
Response	<i>JSON_ResponseStatus</i>

A.12.232 /ISAPI/System/reboot

Reboot device.

Request URI Definition

Table A-824 PUT /ISAPI/System/reboot

Method	PUT
Description	Reboot device.
Query	None.
Request	None.
Response	<i>XML_ResponseStatus</i>

A.12.233 /ISAPI/System/Serial/authentication/capabilities?format=json

Get capability of serial authentication service.

Request URI Definition

Table A-825 GET /ISAPI/System/Serial/authentication/capabilities?format=json

Method	GET
Description	Get capability of serial authentication service.
Query	format: determine the format of request or response message.
Request	None.
Response	Succeeded: <i>JSON_AuthenticationCap</i> Failed: <i>JSON_ResponseStatus</i>

A.12.234 /ISAPI/System/Serial/authentication?format=json

Get or set parameters of serial authentication service.

Request URI Definition

Table A-826 GET /ISAPI/System/Serial/authentication?format=json

Method	GET
Description	Get parameters of serial authentication service.
Query	format : determine the format of request or response message.
Request	None.
Response	Succeeded: <i>JSON_Authentication</i> Failed: <i>JSON_ResponseStatus</i>

Table A-827 PUT /ISAPI/System/Serial/authentication?format=json

Method	PUT
Description	Set parameters of serial authentication service.
Query	format : determine the format of request or response message.
Request	<i>JSON_Authentication</i>
Response	<i>JSON_ResponseStatus</i>

A.12.235 /ISAPI/System/Serial/capabilities

Get the serial port capability of the device.

Request URI Definition

Table A-828 GET /ISAPI/System/Serial/capabilities

Method	GET
Description	Get the serial port capability of the device.
Query	None.
Request	None.
Response	Succeeded: <i>XML_SerialCap</i> Failed: <i>XML_ResponseStatus</i>

A.12.236 /ISAPI/System/Serial/ports

Get the list of serial ports supported by the device.

Request URI Definition

Table A-829 GET /ISAPI/System/Serial/ports

Method	GET
Description	Get the list of serial ports supported by the device.
Query	None.
Request	None.
Response	Succeeded: <i>XML_SerialPorList</i> Failed: <i>XML_ResponseStatus</i>

A.12.237 /ISAPI/System/Serial/ports/<ID>

Operations about the configuration of a specific serial port supported by the device.

Request URI Definition

Table A-830 GET /ISAPI/System/Serial/ports/<ID>

Method	GET
Description	Get the parameters of a specific port supported by the device.
Query	None.
Request	None.
Response	Succeeded: <i>XML_SerialPort</i> Failed: <i>XML_ResponseStatus</i>

Table A-831 PUT /ISAPI/System/Serial/ports/<ID>

Method	PUT
Description	Set the parameters of a specific serial port supported by the device.
Query	None.
Request	<i>XML_SerialPort</i>
Response	<i>XML_ResponseStatus</i>

A.12.238 /ISAPI/System/Serial/ports/<ID>/capabilities

Get the capability of a specific serial port.

Request URI Definition

Table A-832 GET /ISAPI/System/Serial/ports/<ID>/capabilities

Method	GET
Description	Get the capability of a specific serial port.
Query	None.
Request	None.
Response	Succeeded: <i>XML_Cap_SerialPort</i> Failed: <i>XML_ResponseStatus</i>

A.12.239 /ISAPI/System/Serial/ports/capabilities

Get the capability of all serial ports.

Request URI Definition

Table A-833 GET /ISAPI/System/Serial/ports/capabilities

Method	GET
Description	Get the capability of all serial ports.
Query	None.
Request	None.
Response	Succeeded: <i>XML_Cap_SerialPortList</i> Failed: <i>XML_ResponseStatus</i>

A.12.240 /ISAPI/System/Serial/ports/command

Send a command to a serial port.

Request URI Definition

Table A-834 PUT /ISAPI/System/Serial/ports/command

Method	PUT
Description	Send a command to a serial port.
Query	None.

Request	<i>XML_SerialCommand</i> or query string with raw command in binary format
Response	<i>XML_ResponseStatus</i>

Remarks

- If the device is an encoder that converts analog signals to digital signals and it is connected to analog cameras with PTZ enabled, the encoder will forward the command to the appropriate serial port according to the **<chainNo>** node of the XML message or the query string of the HTTP request.
- If the device is a digital camera with PTZ enabled, the camera will forward the PTZ command to the corresponding serial port.
- The serial port command can be encapsulated in the **<command>** node of the XML message, and the command should be encoded in hexadecimal notation; or the serial port command can be directly uploaded as payload of the HTTP request, and in this case the **Content-Type** should be set to "application/octet-stream", and the query string **chainNo** should be set.

Example

Sample Code of Command in XML Format

```
PUT /ISAPI/System/Serial/ports/999/command HTTP/1.1
Content-Type: application/xml; charset="UTF-8"
Content-Length: xxx

<?xml version="1.0" encoding="UTF-8"?>
<SerialCommand>
  <chainNo>0</chainNo>
  <command>ab45be8778cd</command>
</SerialCommand>
```

Example

Sample Code of Command Using Query String

```
/ISAPI/System/Serial/ports/999/command?chainNo=1 HTTP/1.1
Content-Type: application/octet-stream
Content-Length: xxx
(...Raw data of the command should be inserted here...)
```

A.12.241 /ISAPI/System/Serial/ports/command/capabilities

Get the serial port command capability.

Request URI Definition

Table A-835 GET /ISAPI/System/Serial/ports/command/capabilities

Method	GET
Description	Get the serial port command capability.
Query	None.
Request	None.
Response	Succeeded: <i>XML_Cap_SerialCommand</i> Failed: <i>XML_ResponseStatus</i>

A.12.242 /ISAPI/System/serialLogCfg/capabilities?format=json

Get the configuration capability of serial port log redirection.

Request URI Definition

Table A-836 GET /ISAPI/System/serialLogCfg/capabilities?format=json

Method	GET
Description	Get the configuration capability of serial port log redirection.
Query	format: determine the format of request or response message.
Request	None.
Response	Succeeded: <i>JSON_SerialLogCfgCap</i> Failed: <i>JSON_ResponseStatus</i>

A.12.243 /ISAPI/System/serialLogCfg?format=json

Operations about the redirection configuration of serial port log.

Request URI Definition

Table A-837 GET /ISAPI/System/serialLogCfg?format=json

Method	GET
Description	Get the redirection parameters of serial port log.
Query	format: determine the format of request or response message.

Request	None.
Response	Succeeded: <i>JSON_SerialLogCfg</i> Failed: <i>JSON_ResponseStatus</i>

Table A-838 PUT /ISAPI/System/serialLogCfg?format=json

Method	PUT
Description	Set the redirection parameters of serial port log.
Query	format: determine the format of request or response message.
Request	<i>JSON_SerialLogCfg</i>
Response	<i>JSON_ResponseStatus</i>

A.12.244 /ISAPI/System/shutdown?format=json

Shut down the device.

Request URI Definition

Table A-839 PUT /ISAPI/System/shutdown?format=json

Method	PUT
Description	Shut down the device.
Query	format: determine the format of request or response message.
Request	None.
Response	<i>JSON_ResponseStatus</i>

A.12.245 /ISAPI/System/Software/channels/<ID>

Operation about software service configuration.

Request URI Definition

Table A-840 GET /ISAPI/System/Software/channels/<ID>

Method	GET
Description	Get software service parameters.
Query	None.
Request	None.
Response	<i>XML_SoftwareService</i>

Table A-841 PUT /ISAPI/System/Software/channels/<ID>

Method	PUT
Description	Set software service parameters.
Query	None.
Request	<i>XML_SoftwareService</i>
Response	<i>XML_ResponseStatus</i>

Remarks

The <ID> in the request URI refers to the channel No.

A.12.246 /ISAPI/System/Software/channels/<ID>/capabilities

Get the configuration capability of software service.

Request URI Definition
Table A-842 GET /ISAPI/System/Software/channels/<ID>/capabilities

Method	GET
Description	Get the configuration capability of software service.
Query	None.
Request	None.
Response	<i>XML_Cap_SoftwareService</i>

Remarks

The <ID> in the request URI refers to the channel No.

A.12.247 /ISAPI/System/SSDFileSystem/capacity?format=json

Search for SSD storage quotas.

Request URI Definition
Table A-843 GET /ISAPI/System/SSDFileSystem/capacity?format=json

Method	GET
Description	Search for SSD storage quotas.
Query	format: determine the format of request or response message.

Request	None.
Response	Succeeded: <i>JSON_SSDCapacity</i> Failed: <i>JSON_ResponseStatus</i>

Remarks

If the node **isSupportSSDFileSystemCapacity** is returned in the message *XML_DeviceCap* (related URI: */ISAPI/System/capabilities*) and its value is true, it indicates that the device supports this function.

A.12.248 /ISAPI/System/SSDFileSystem/format?format=json

Format SSD file system.

Request URI Definition

Table A-844 PUT /ISAPI/System/SSDFileSystem/format?format=json

Method	PUT
Description	Format SSD file system.
Query	format: determine the format of request or response message.
Request	None.
Response	<i>JSON_ResponseStatus</i>

Remarks

Formatting SSD will erase its data, including those in face picture library, and cancel the alarms linked to the library. The device will restart after formatting.

A.12.249 /ISAPI/System/SSDFileSystem/format/status?format=json

Search for the progress of SSD file system formatting.

Request URI Definition

Table A-845 GET /ISAPI/System/SSDFileSystem/format/status?format=json

Method	GET
Description	Search for the progress of SSD file system formatting.
Query	format: determine the format of request or response message.
Request	None.
Response	Succeeded: <i>JSON_SSDFormatStatus</i>

	Failed: <i>JSON_ResponseStatus</i>
--	------------------------------------

A.12.250 /ISAPI/System/SSDFileSystem/upgrade?format=json

Get the upgrade status of SSD file system or upgrade the SSD file system. The device will automatically reboot after upgrade.

Request URI Definition

Table A-846 GET /ISAPI/System/SSDFileSystem/upgrade?format=json

Method	GET
Description	Get the upgrade status of SSD file system.
Query	format: determine the format of request or response message.
Request	None.
Response	Succeeded: <i>JSON_SSDUpgrade</i> Failed: <i>JSON_ResponseStatus</i>

Table A-847 PUT /ISAPI/System/SSDFileSystem/upgrade?format=json

Method	PUT
Description	Upgrade the SSD file system.
Query	format: determine the format of request or response message.
Request	None.
Response	<i>JSON_ResponseStatus</i>

A.12.251 /ISAPI/System/SSDFileSystem/upgrade/status?format=json

Search for the upgrade progress of SSD file system.

Request URI Definition

Table A-848 GET /ISAPI/System/SSDFileSystem/upgrade/status?format=json

Method	GET
Description	Search for the upgrade progress of SSD file system.
Query	format: determine the format of request or response message.
Request	None.
Response	Succeeded: <i>JSON_SSDUpgradeStatus</i>

	Failed: <i>JSON_ResponseStatus</i>
--	------------------------------------

A.12.252 /ISAPI/System/status

Get device status, e.g., CPU, memory, and so on.

Request URI Definition

Table A-849 GET /ISAPI/System/status

Method	GET
Description	Get device status, e.g., CPU, memory, and so on.
Query	None.
Request	None.
Response	Succeeded: <i>XML_DeviceStatus</i> Failed: <i>XML_ResponseStatus</i>

A.12.253 /ISAPI/System/syncSignalOutput/<ID>

Operations about the configuration of synchronous signal output.

Request URI Definition

Table A-850 GET /ISAPI/System/syncSignalOutput/<ID>

Method	GET
Description	Get the parameters of synchronous signal output.
Query	None.
Request	None.
Response	Succeeded: <i>XML_SyncSignalOutputList</i> Failed: <i>XML_ResponseStatus</i>

Table A-851 PUT /ISAPI/System/syncSignalOutput/<ID>

Method	PUT
Description	Set the parameters of synchronous signal output.
Query	None.
Request	<i>XML_SyncSignalOutputList</i>
Response	<i>XML_ResponseStatus</i>

A.12.254 /ISAPI/System/syncSignalOutput/<ID>/capabilities

Get the configuration capability of synchronous signal output.

Request URI Definition

Table A-852 GET /ISAPI/System/syncSignalOutput/<ID>/capabilities

Method	GET
Description	Get the configuration capability of synchronous signal output.
Query	None.
Request	None.
Response	Succeeded: <i>XML_Cap_SyncSignalOutputList</i> Failed: <i>XML_ResponseStatus</i>

A.12.255 /ISAPI/System/time

Get or set device time parameters.

Request URI Definition

Table A-853 GET /ISAPI/System/time

Method	GET
Description	Get device time parameters.
Query	None
Request	None
Response	Succeeded: <i>XML_Time</i> Failed: <i>XML_ResponseStatus</i>

Table A-854 PUT /ISAPI/System/time

Method	PUT
Description	Set device time parameters.
Query	None
Request	<i>XML_Time</i>
Response	<i>XML_ResponseStatus</i>

A.12.256 /ISAPI/System/time/capabilities

Get the time capability of the device.

Request URI Definition

Table A-855 GET /ISAPI/System/time/capabilities

Method	GET
Description	Get the time capability of the device.
Query	None.
Request	None.
Response	Succeeded: <i>XML_Cap_Time</i> Failed: <i>XML_ResponseStatus</i>

A.12.257 /ISAPI/System/time/localTime

Operations about the configuration of device local time.

Request URI Definition

Table A-856 GET /ISAPI/System/time/localTime

Method	GET
Description	Get the configuration parameters of device local time.
Query	None.
Request	None.
Response	String with date and time information (ISO8601 format, e.g., 2018-02-10T17:30:08)

Table A-857 PUT /ISAPI/System/time/localTime

Method	PUT
Description	Set device local time.
Query	None.
Request	String with date and time information (ISO8601 format, e.g., 2018-02-10T17:30:08)
Response	<i>XML_ResponseStatus</i>

A.12.258 /ISAPI/System/time/ntpServers

Get or set parameters of all NTP servers, add a NTP server, or delete all NTP servers.

Request URI Definition

Table A-858 GET /ISAPI/System/time/ntpServers

Method	GET
Description	Get parameters of all NTP servers.
Query	None
Request	None
Response	Succeeded: <i>XML_NTPServerList</i> Failed: <i>XML_ResponseStatus</i>

Table A-859 PUT /ISAPI/System/time/ntpServers

Method	PUT
Description	Set parameters of all NTP servers.
Query	None
Request	<i>XML_NTPServerList</i>
Response	<i>XML_ResponseStatus</i>

Table A-860 POST /ISAPI/System/time/ntpServers

Method	POST
Description	Add a NTP server.
Query	None
Request	<i>XML_NTPServer</i>
Response	<i>XML_ResponseStatus</i>

Table A-861 DELETE /ISAPI/System/time/ntpServers

Method	DELETE
Description	Delete all NTP servers.
Query	None
Request	None
Response	<i>XML_ResponseStatus</i>

A.12.259 /ISAPI/System/time/ntpServers/<ID>

Get or set parameters of a NTP server, or delete a NTP server.

Request URI Definition**Table A-862 GET /ISAPI/System/time/ntpServers/<ID>**

Method	GET
Description	Get the parameters of a NTP server.
Query	None
Request	None
Response	Succeeded: <i>XML_NTPServer</i> Failed: <i>XML_ResponseStatus</i>

Table A-863 PUT /ISAPI/System/time/ntpServers/<ID>

Method	PUT
Description	Set the parameters of a NTP server.
Query	None
Request	<i>XML_NTPServer</i>
Response	<i>XML_ResponseStatus</i>

Table A-864 DELETE /ISAPI/System/time/ntpServers/<ID>

Method	DELETE
Description	Delete a NTP server.
Query	None
Request	None
Response	<i>XML_ResponseStatus</i>

A.12.260 /ISAPI/System/time/ntpServers/<ID>/capabilities

Get the configuration capability of a specific NTP server.

Request URI Definition

Table A-865 GET /ISAPI/System/time/ntpServers/<ID>/capabilities

Method	GET
Description	Get the configuration capability of a specific NTP server.
Query	None.
Request	None.
Response	Succeeded: <i>XML_Cap_NTPServer</i> Failed: <i>XML_ResponseStatus</i>

A.12.261 /ISAPI/System/time/ntpServers/capabilities

Get the configuration capability of all NTP servers.

Request URI Definition

Table A-866 GET /ISAPI/System/time/ntpServers/capabilities

Method	GET
Description	Get the configuration capability of all NTP servers.
Query	None
Request	None
Response	Succeeded: <i>XML_Cap_NTPServerList</i> Failed: <i>XML_ResponseStatus</i>

A.12.262 /ISAPI/System/time/ntpServers/test

Execute a test to check if the NTP server is available.

Request URI Definition

Table A-867 POST /ISAPI/System/time/ntpServers/test

Method	POST
Description	Execute a test to check if the NTP server is available.
Query	None

Request	<i>XML_NTPTestDescription</i>
Response	Succeeded: <i>XML_NTPTestResult</i> Failed: <i>XML_ResponseStatus</i>

A.12.263 /ISAPI/System/time/timeType?format=json

Operations about device time type configuration.

Request URI Definition

Table A-868 GET /ISAPI/System/time/timeType?format=json

Method	GET
Description	Get the device time type.
Query	format: determine the format of request or response message.
Request	None.
Response	<i>JSON_TimeType</i>

Table A-869 PUT /ISAPI/System/time/timeType?format=json

Method	PUT
Description	Set the device time type.
Query	format: determine the format of request or response message.
Request	<i>JSON_TimeType</i>
Response	<i>JSON_ResponseStatus</i>

A.12.264 /ISAPI/System/time/timeZone

Operations about the configuration of device time zone.

Request URI Definition

Table A-870 GET /ISAPI/System/time/timeZone

Method	GET
Description	Get the device time zone parameters.
Query	None.
Request	None.
Response	String containing time zone information

Table A-871 PUT /ISAPI/System/time/timeZone

Method	PUT
Description	Set the device time zone parameters.
Query	None.
Request	String containing time zone information
Response	<i>XML_ResponseStatus</i>

A.12.265 /ISAPI/System/TwoWayAudio/channels

Get audio parameters of all two-way audio channels.

Request URI Definition

Table A-872 GET /ISAPI/System/TwoWayAudio/channels

Method	GET
Description	Get parameters of all two-way audio channels.
Query	None.
Request	None.
Response	Succeeded: <i>XML_TwoWayAudioChannelList</i> Failed: <i>XML_ResponseStatus</i>

Example

Sample Code for Getting Parameters of All Two-Way Audio Channels

```
GET /ISAPI/System/TwoWayAudio/channels HTTP/1.1
Host: 10.17.132.49
Connection: Keep-Alive
Authorization: Digest username="admin",
realm="DS-2CD2F12FWD-IWS",
nonce="4e4446464e6a6333516a63365a4445304f47526a4e32553d",
uri="/ISAPI/System/TwoWayAudio/channels",
cnonce="145ef1bc3ab456be09918d39b77e78ae",
nc=00000021,
response="d10bab23689c60e95241230fff3181b0",
qop="auth"
```

```
HTTP/1.1 200 OK
Date: Wed, 15 Mar 2017 09:25:29 GMT
Server: App-webs/
Connection: close
Content-Length: 467
Content-Type: application/xml
```

```
<?xml version="1.0" encoding="UTF-8"?>
<TwoWayAudioChannelList version="2.0" xmlns="http://www.isapi.com/ver20/XMLSchema">
  <TwoWayAudioChannel version="2.0" xmlns="http://www.isapi.com/ver20/XMLSchema">
    <id>1</id>
    <enabled>>false</enabled>
    <audioCompressionType>G.711ulaw</audioCompressionType>
    <audioInputType>MicIn</audioInputType>
    <speakerVolume>50</speakerVolume>
    <noisereduce>>false</noisereduce>
  </TwoWayAudioChannel>
</TwoWayAudioChannelList>
```

A.12.266 /ISAPI/System/TwoWayAudio/channels/<ID>

Operations about the configuration of a specific two-way audio channel.

Request URI Definition

Table A-873 GET /ISAPI/System/TwoWayAudio/channels/<ID>

Method	GET
Description	Get the parameters of a specific two-way audio channel.
Query	None.
Request	None.
Response	Succeeded: <i>XML_TwoWayAudioChannel</i> Failed: <i>XML_ResponseStatus</i>

Table A-874 PUT /ISAPI/System/TwoWayAudio/channels/<ID>

Method	PUT
Description	Set the parameters of a specific two-way audio channel.
Query	None.
Request	<i>XML_TwoWayAudioChannel</i>
Response	<i>XML_ResponseStatus</i>

A.12.267 /ISAPI/System/TwoWayAudio/channels/<ID>/audioData

Receive or send audio data from or to a specific two-way audio channel.

Request URI Definition

Table A-875 GET /ISAPI/System/TwoWayAudio/channels/<ID>/audioData

Method	GET
Description	Receive audio data from a specific two-way audio channel.
Query	sessionId : communication session ID, which is required when the two-way audio is started between multiple channels and a channel; otherwise, the session ID is not required.
Request	None.
Response	Succeeded: audio data Failed: <i>XML_ResponseStatus</i>

Table A-876 PUT /ISAPI/System/TwoWayAudio/channels/<ID>/audioData

Method	PUT
Description	Send audio data to a specific two-way audio channel.
Query	sessionId : communication session ID, which is required when the two-way audio is started between multiple channels and a channel; otherwise, the session ID is not required.
Request	Audio data.
Response	<i>XML_ResponseStatus</i>

Remarks

The <ID> in the request URI refers to the two-way audio channel ID.

Example

Sample Code for Sending Audio Data to Device

```
PUT /ISAPI/System/TwoWayAudio/channels/1/audioData HTTP/1.1
HOST: 10.17.132.49
Authorization: Digest username="admin",
realm="DS-2CD2F12FWD-IWS",
nonce="4e3055314e6a64434e7a59365a4445304f545668596a6b3d",
uri="/ISAPI/System/TwoWayAudio/channels/1/audioData",
response="6a03c7e85c17a35bae9ee6f3cfd254b"
Connection: keep-alive
Content-Length: 0
Content-Type: application/octet-stream

HTTP/1.1 200 OK

..U....U.....U.....U..... (Binary audio data)
```

Example

Sample Code for Receiving Audio Data from Device

```
GET /ISAPI/System/TwoWayAudio/channels/1/audioData HTTP/1.1
HOST: 10.17.132.49
Authorization: Digest username="admin",
realm="DS-2CD2F12FWD-IWS",
nonce="4e3055314e6a64434e7a59365a4445304f5456685a44673d",
uri="/ISAPI/System/TwoWayAudio/channels/1/audioData",
response="7ea0ec7bd102b3c4cb225cbf33adb8f8"
Connection: keep-alive
```

```
HTTP/1.1 200 OK
Content-Type: audio/basic
```

```
Q.....U..dzR..\.FfN...q[....z..q.....` (Binary audio data)
```

A.12.268 /ISAPI/System/TwoWayAudio/channels/<ID>/capabilities

Get the capability of a specific two-way audio channel.

Request URI Definition

Table A-877 GET /ISAPI/System/TwoWayAudio/channels/<ID>/capabilities

Method	GET
Description	Get the capability of a specific two-way audio channel.
Query	None.
Request	None.
Response	Succeeded: <i>XML_TwoWayAudioChannelCap</i> Failed: <i>XML_ResponseStatus</i>

A.12.269 /ISAPI/System/TwoWayAudio/channels/<ID>/close

Stop two-way audio of a specific channel.

Request URI Definition

Table A-878 PUT /ISAPI/System/TwoWayAudio/channels/<ID>/close

Method	PUT
Description	Stop two-way audio of a specific channel.
Query	None.

Request	None.
Response	<i>XML_ResponseStatus</i>

Remarks

The <ID> in the request URI refers to the two-way audio channel ID.

Example

Sample Code for Stopping Two-Way Audio

```
PUT /ISAPI/System/TwoWayAudio/channels/1/close HTTP/1.1
HOST: 10.17.132.49
Authorization: Digest username="admin",
realm="DS-2CD2F12FWD-IWS",
nonce="4e54457a4d7a67354d6a6b365a4445304f5463794d32553d",
uri="/ISAPI/System/TwoWayAudio/channels/1/close",
response="7d5acdc3ccf7c080400802800d7471e3"
Content-Length: 0

HTTP/1.1 200 OK
Date: Wed, 15 Mar 2017 09:25:35 GMT
Server: App-webs/
Connection: close
Content-Length: 298
Content-Type: application/xml

<?xml version="1.0" encoding="UTF-8"?>
<ResponseStatus version="2.0" xmlns="http://www.isapi.com/ver20/XMLSchema">
  <requestURL>/ISAPI/System/TwoWayAudio/channels/1/close</requestURL>
  <statusCode>1</statusCode>
  <statusString>OK</statusString>
  <subStatusCode>ok</subStatusCode>
</ResponseStatus>
```

A.12.270 /ISAPI/System/TwoWayAudio/channels/<ID>/open

Start two-way audio of a specific channel.

Request URI Definition

Table A-879 PUT /ISAPI/System/TwoWayAudio/channels/<ID>/open

Method	PUT
Description	Start two-way audio of a specific channel.
Query	None.
Request	None.
Response	Succeeded: <i>XML_TwoWayAudioSession</i>

Failed: ***XML_ResponseStatus***

Remarks

The <ID> in the request URI refers to the two-way audio channel ID.

Example

Sample Code for Start Two-Way Audio

```
PUT /ISAPI/System/TwoWayAudio/channels/1/open HTTP/1.1
HOST: 10.17.132.49
Authorization: Digest username="admin",
realm="DS-2CD2F12FWD-IWS",
nonce="4e3055314e6a64434e7a59365a4445304f5456685957453d",
uri="/ISAPI/System/TwoWayAudio/channels/1/open",
response="368dda22535b9783bdccafc3b2ded29a"
Content-Length: 0

HTTP/1.1 200 OK
Date: Wed, 15 Mar 2017 09:25:29 GMT
Server: App-webs/
Connection: close
Content-Length: 183
Content-Type: application/xml

<?xml version="1.0" encoding="UTF-8"?>
<TwoWayAudioSession version="2.0" xmlns="http://www.isapi.com/ver20/XMLSchema">
  <sessionId>2093716360</sessionId>
</TwoWayAudioSession>
```

A.12.271 /ISAPI/System/TwoWayAudio/channels/capabilities

Get the audio configuration capability of all two-way audio channels.

Request URI Definition

Table A-880 GET /ISAPI/System/TwoWayAudio/channels/capabilities

Method	GET
Description	Get the audio configuration capability of all two-way audio channels.
Query	None.
Request	None.
Response	Succeeded: <i>XML_Cap_TwoWayAudioChannelList</i> Failed: <i>XML_ResponseStatus</i>

A.12.272 /ISAPI/System/unitConfig/capabilities?format=json

Get the capability of unit unified configuration.

Request URI Definition

Table A-881 GET /ISAPI/System/unitConfig/capabilities?format=json

Method	GET
Description	Get the capability of unit unified configuration.
Query	format : determine the format of request or response message.
Request	None.
Response	Succeeded: <i>JSON_unitConfigCap</i> Failed: <i>JSON_ResponseStatus</i>

A.12.273 /ISAPI/System/unitConfig?format=json

Get and set the unit unified configuration parameters.

Request URI Definition

Table A-882 GET /ISAPI/System/unitConfig?format=json

Method	GET
Description	Get the unit unified configuration parameters.
Query	format : determine the format of request or response message.
Request	None.
Response	Succeeded: <i>JSON_unitConfig</i> Failed: <i>JSON_ResponseStatus</i>

Table A-883 PUT /ISAPI/System/unitConfig?format=json

Method	PUT
Description	Set the unit unified configuration parameters.
Query	format : determine the format of request or response message.
Request	<i>JSON_unitConfig</i>
Response	<i>JSON_ResponseStatus</i>

A.12.274 /ISAPI/System/updateFirmware

Upgrade the device firmware.

Request URI Definition

Table A-884 PUT or POST /ISAPI/System/updateFirmware

Method	PUT, POST
Description	Upgrade the device firmware.
Query	None.
Request	Opaque data (binary data for PUT method, data in form format for POST method).
Response	<i>XML_ResponseStatus</i>

Remarks

After upgrading the device firmware, the device will automatically reboot according to the response message. If upgrading failed, the following reasons may be returned: "upgrading"-the device is upgrading, "badFlash"-flash error, "badLanguage"-language mismatches.

Example

The URL Interaction in Form Format

```
/ISAPI/System/updateFirmware
Accept: text/html, application/xhtml+xml,
Accept-Language: zh-CN
Content-Type: multipart/form-data; boundary=-----7e13971310878
User-Agent: Mozilla/5.0 (compatible; MSIE 9.0; Windows NT 6.1; WOW64; Trident/5.0)
Accept-Encoding: gzip, deflate
Host: 10.10.36.29:8080
Content-Length: 9907
Connection: Keep-Alive
Cache-Control: no-cache

-----7e13971310878
Content-Disposition: form-data; name="updateFile";
Content-Type: application/octet-stream
Content-Length: 9907

.....`.....C.....
..
.....$. ' ",#.(7),01444.'9=82<.342...C. ....
-----7e13971310878--
```

A.12.275 /ISAPI/System/updateFirmware?type=&id=

Upgrade the slave device's firmware.

Request URI Definition**Table A-885 PUT or POST /ISAPI/System/updateFirmware?type=&id=**

Method	PUT, POST
Description	Upgrade the slave device's firmware.
Query	type: device or module type, string type, it can be set to one of the following values: "cardReader"-card reader, "FPModule"-fingerprint module, "securityModule"-secure door control unit, "extendModule"- (IO) extension module, "channelController"-lane controller, "IRModule"-IR module, "lampModule"-indicator module, "elevatorController"-slave elevator controller, and "FPAlgorithmProgram"-fingerprint algorithm program of the card reader, "uboot". id: slave device No., e.g., /ISAPI/System/updateFirmware?type=cardReader&id=1 refers to upgrading firmware of card reader 1.
Request	Opaque data (binary data for PUT method, data in form format for POST method).
Response	<i>XML_ResponseStatus</i>

Remarks

After upgrading the slave device's firmware, the device will automatically reboot according to the response message. If upgrading failed, the following reasons may be returned: "upgrading"-the device is upgrading, "badFlash"-flash error, "badLanguage"-language mismatches.

A.12.276 /ISAPI/System/updateFirmware?type=&moduleAddress=

Upgrade the peripheral module's firmware.

Request URI Definition**Table A-886 PUT or POST /ISAPI/System/updateFirmware?type=&moduleAddress=**

Method	PUT, POST
Description	Upgrade the peripheral module's firmware.

Query	type: device or module type, string type, it can be set to one of the following values: "keypad", "wirelessRecv"-wireless receiving module, "wiredZone"-wired zone module. moduleAddress: module address, e.g., /ISAPI/System/updateFirmware?type=wirelessRecv&moduleAddress=1 refers to upgrading firmware of the wireless receiving module with the module address 1.
Request	Opaque data (binary data for PUT method, data in form format for POST method).
Response	<i>XML_ResponseStatus</i>

Remarks

After upgrading the peripheral module's firmware, the device will automatically reboot according to the response message. If upgrading failed, the following reasons may be returned: "upgrading"-the device is upgrading, "badFlash"-flash error, "badLanguage"-language mismatches.

A.12.277 /ISAPI/System/upgradeStatus

Get the device upgrading status and progress.

Request URI Definition

Table A-887 GET /ISAPI/System/upgradeStatus

Method	GET
Description	Get the device upgrading status and progress.
Query	None.
Request	None.
Response	Succeeded: <i>XML_upgradeStatus</i> Failed: <i>XML_ResponseStatus</i>

A.12.278 /ISAPI/System/usb/capabilities?format=json

Get the USB parameters configuration capability.

Request URI Definition

Table A-888 GET /ISAPI/System/usb/capabilities?format=json

Method	GET
Description	Get the USB parameters configuration capability.
Query	format : determine the format of request or response message.
Request	None.
Response	Succeeded: <i>JSON_USBCap</i> Failed: <i>JSON_ResponseStatus</i>

A.12.279 /ISAPI/System/usb?format=json

Get or set the USB parameters.

Request URI Definition

Table A-889 GET /ISAPI/System/usb?format=json

Method	GET
Description	Get the USB parameters.
Query	format : determine the format of request or response message.
Request	None.
Response	Succeeded: <i>JSON_USB</i> Failed: <i>JSON_ResponseStatus</i>

Table A-890 PUT /ISAPI/System/usb?format=json

Method	PUT
Description	Set the USB parameters.
Query	format : determine the format of request or response message.
Request	<i>JSON_USB</i>
Response	<i>JSON_ResponseStatus</i>

A.12.280 /ISAPI/System/userLock/config/capabilities?format=json

Get the configuration capability of locking IP address.

Request URI Definition

Table A-891 GET /ISAPI/System/userLock/config/capabilities?format=json

Method	GET
Description	Get the configuration capability of locking IP address.
Query	format: determine the format of request or response message.
Request	None.
Response	Succeeded: <i>JSON_LockCfgCap</i> Failed: <i>JSON_ResponseStatus</i>

A.12.281 /ISAPI/System/userLock/config?format=json

Operations about IP address locking configuration.

Request URI Definition

Table A-892 GET /ISAPI/System/userLock/config?format=json

Method	GET
Description	Get the parameters of IP address locking.
Query	format: determine the format of request or response message.
Request	None.
Response	Succeeded: <i>JSON_LockCfg</i> Failed: <i>JSON_ResponseStatus</i>

Table A-893 PUT /ISAPI/System/userLock/config?format=json

Method	PUT
Description	Set the parameters of IP address locking.
Query	format: determine the format of request or response message.
Request	<i>JSON_LockCfg</i>
Response	<i>JSON_ResponseStatus</i>

A.12.282 /ISAPI/System/userLock/lockedUsers?format=json

Get all locked IP addresses.

Request URI Definition

Table A-894 GET /ISAPI/System/userLock/lockedUsers?format=json

Method	GET
Description	Get all locked IP addresses.
Query	format : determine the format of request or response message.
Request	None.
Response	Succeeded: <i>JSON_List_IPAddress</i> Failed: <i>JSON_ResponseStatus</i>

A.12.283 /ISAPI/System/userLock/unlockUser?format=json

Unlock the IP address.

Request URI Definition

Table A-895 PUT /ISAPI/System/userLock/unlockUser?format=json

Method	PUT
Description	Unlock the IP address.
Query	format : determine the format of request or response message.
Request	<i>JSON_opertype</i>
Response	<i>JSON_ResponseStatus</i>

A.12.284 /ISAPI/System/Video/capabilities

Get video channel capability.

Request URI Definition

Table A-896 GET /ISAPI/System/Video/capabilities

Method	GET
Description	Get video channel capability.
Query	None
Request	None
Response	Succeeded: <i>XML_VideoCap</i>

	Failed: <i>XML_ResponseStatus</i>
--	--

A.12.285 /ISAPI/System/Video/inputs

Get the video input configuration of the device.

Request URI Definition

Table A-897 GET /ISAPI/System/Video/inputs

Method	GET
Description	Get the video input configuration parameters of the device.
Query	None.
Request	None.
Response	Succeeded: <i>XML_VideoInput</i> Failed: <i>XML_ResponseStatus</i>

A.12.286 /ISAPI/System/Video/inputs/channels

Get the configuration of all video input channels of the device.

Request URI Definition

Table A-898 GET /ISAPI/System/Video/inputs/channels

Method	GET
Description	Get the configuration parameters of all video input channels of the device.
Query	None.
Request	None.
Response	Succeeded: <i>XML_VideoInputChannelList</i> Failed: <i>XML_ResponseStatus</i>

A.12.287 /ISAPI/System/Video/inputs/channels/<ID>

Operations about the configuration of a specific video input channel of the device.

Request URI Definition**Table A-899 GET /ISAPI/System/Video/inputs/channels/<ID>**

Method	GET
Description	Get the parameters of a specific video input channel of the device.
Query	None.
Request	None.
Response	Succeeded: <i>XML_VideoInputChannel</i> Failed: <i>XML_ResponseStatus</i>

Table A-900 PUT /ISAPI/System/Video/inputs/channels/<ID>

Method	PUT
Description	Set the parameters of a specific video input channel of the device.
Query	None.
Request	<i>XML_VideoInputChannel</i>
Response	<i>XML_ResponseStatus</i>

See Also

The <ID> in the request URI refers to the channel No.

A.12.288 /ISAPI/System/Video/inputs/channels?format=json

Get the information of all channels according to the request message.

Request URI Definition**Table A-901 POST /ISAPI/System/Video/inputs/channels?format=json**

Method	POST
Description	Get the information of all channels according to the request message.
Query	format: determine the format of request or response message.
Request	<i>JSON_InputParam</i>
Response	Succeeded: <i>JSON_ChannelInfoList</i> Failed: <i>JSON_ResponseStatus</i>

A.12.289 /ISAPI/System/Video/inputs/channels/<ID>/focus

Perform manual focus on a specific channel.

Request URI Definition**Table A-902 PUT /ISAPI/System/Video/inputs/channels/<ID>/focus**

Method	PUT
Description	Perform manual focus on a specific channel.
Query	None.
Request	<i>XML_FocusData</i>
Response	<i>XML_ResponseStatus</i>

Remarks

The <ID> in the request URI refers to video input channel ID.

A.12.290 /ISAPI/System/Video/inputs/channels/<ID>?format=json

Get the information of a channel according to the request message.

Request URI Definition**Table A-903 POST /ISAPI/System/Video/inputs/channels/<ID>?format=json**

Method	POST
Description	Get the information of a channel according to the request message.
Query	format: determine the format of request or response message.
Request	
Response	Succeeded: <i>JSON_ChannellInfo</i> Failed: <i>JSON_ResponseStatus</i>

Remarks

The <ID> in the request URI refers to the channel No.

A.12.291 /ISAPI/System/Video/inputs/channels/<ID>/capabilities

Get the configuration capability of a video input channel.

Request URI Definition

Table A-904 GET /ISAPI/System/Video/inputs/channels/<ID>/capabilities

Method	GET
Description	Get the configuration capability of a video input channel.
Query	None.
Request	None.
Response	Succeeded: <i>XML_Cap_VideoInputChannel</i> Failed: <i>XML_ResponseStatus</i>

Remarks

The <ID> in the request URI refers to video input channel ID.

A.12.292 /ISAPI/System/Video/inputs/channels/<ID>/iris

Manually adjust the iris of a specific video input channel.

Request URI Definition

Table A-905 PUT /ISAPI/System/Video/inputs/channels/<ID>/iris

Method	PUT
Description	Manually adjust the iris of a specific video input channel.
Query	None.
Request	<i>XML_IrisData</i>
Response	<i>XML_ResponseStatus</i>

Remarks

The <ID> in the request URI refers to video input channel ID.

A.12.293 /ISAPI/System/Video/inputs/channels/<ID>/motionDetection

Get or set the motion detection parameters of a specified video input channel.

Request URI Definition**Table A-906 GET /ISAPI/System/Video/inputs/channels/<ID>/motionDetection**

Method	GET
Description	Get the motion detection parameters of a specified video input channel.
Query	None.
Request	None.
Response	Succeeded: <i>XML_MotionDetection</i> Failed: <i>XML_ResponseStatus</i>

Table A-907 PUT /ISAPI/System/Video/inputs/channels/<ID>/motionDetection

Method	PUT
Description	Set the motion detection parameters of a specified video input channel.
Query	None.
Request	<i>XML_MotionDetection</i>
Response	<i>XML_ResponseStatus</i>

Remarks

The <ID> in the request URI refers to the video input channel No.

A.12.294 /ISAPI/System/Video/inputs/channels/<ID>/motionDetection/capabilities

Get the motion detection capability of a specified video input channel.

Request URI Definition**Table A-908 GET /ISAPI/System/Video/inputs/channels/<ID>/motionDetection/capabilities**

Method	GET
Description	Get the motion detection capability of a specified video input channel.
Query	None.
Request	None.
Response	Succeeded: <i>XML_Cap_MotionDetection</i>

	Failed: <i>XML_ResponseStatus</i>
--	-----------------------------------

Remarks

The <ID> in the request URI refers to the video input channel No.

A.12.295 /ISAPI/System/Video/inputs/channels/<ID>/motionDetection/layout

Get or set rule region parameters of motion detection.

Request URI Definition**Table A-909 GET /ISAPI/System/Video/inputs/channels/<ID>/motionDetection/layout**

Method	GET
Description	Get rule region parameters of motion detection.
Query	regionType : Detect by region or by grid, string, the available value is "region" or "grid".
Request	None
Response	Succeeded: <i>XML_MotionDetectionLayout</i> Failed: <i>XML_ResponseStatus</i>

Table A-910 PUT /ISAPI/System/Video/inputs/channels/<ID>/motionDetection/layout

Method	PUT
Description	Set rule region parameters of motion detection.
Query	regionType : Detect by region or by grid, string, the available value is "region" or "grid".
Request	<i>XML_MotionDetectionLayout</i>
Response	<i>XML_ResponseStatus</i>

Remarks

The <ID> in the URI refers to the video input channel ID.

A.12.296 /ISAPI/System/Video/inputs/channels/<ID>/motionDetection/layout/capabilities

Get rule region configuration capability of motion detection.

Request URI Definition

Table A-911 GET /ISAPI/System/Video/inputs/channels/<ID>/motionDetection/layout/capabilities

Method	GET
Description	Get rule region configuration capability of motion detection.
Query	None
Request	None
Response	Succeeded: <i>XML_Cap_MotionDetectionLayout</i> Failed: <i>XML_ResponseStatus</i>

Remarks

The <ID> in the URI refers to the video input channel ID.

A.12.297 /ISAPI/System/Video/inputs/channels/<ID>/motionDetection/layout/gridLayout

Get or set grid layout parameters of motion detection.

Request URI Definition

Table A-912 GET /ISAPI/System/Video/inputs/channels/<ID>/motionDetection/layout/gridLayout

Method	GET
Description	Get grid layout parameters of motion detection.
Query	regionType : Detect by region or by grid, string, the available value is "region" or "grid".
Request	None
Response	Succeeded: <i>XML_MotionDetectionGridLayout</i> Failed: <i>XML_ResponseStatus</i>

Table A-913 PUT /ISAPI/System/Video/inputs/channels/<ID>/motionDetection/layout/gridLayout

Method	PUT
Description	Set grid layout parameters of motion detection.
Query	regionType : Detect by region or by grid, string, the available value is "region" or "grid".

Request	<i>XML_MotionDetectionGridLayout</i>
Response	<i>XML_ResponseStatus</i>

Remarks

The <ID> in the URI refers to the video input channel ID.

A.12.298 /ISAPI/System/Video/inputs/channels/<ID>/overlays

Operations about OSD configuration of a specific video input channel.

Request URI Definition**Table A-914 GET /ISAPI/System/Video/inputs/channels/<ID>/overlays**

Method	GET
Description	Get OSD configuration parameters of a specific video input channel.
Query	None.
Request	None.
Response	Succeeded: <i>XML_VideoOverlay</i> Failed: <i>XML_ResponseStatus</i>

Table A-915 PUT /ISAPI/System/Video/inputs/channels/<ID>/overlays

Method	PUT
Description	Set OSD parameters of a specific video input channel.
Query	None.
Request	<i>XML_VideoOverlay</i>
Response	<i>XML_ResponseStatus</i>

Remarks

The <ID> in the request URI refers to the video input channel ID.

Example

Sample Code for Getting OSD Parameters

```
GET /ISAPI/System/Video/inputs/channels/1/overlays HTTP/1.1
Host: 10.17.132.49
Connection: Keep-Alive
Authorization: Digest username="admin",
realm="DS-2CD2F12FWD-IWS",
nonce="4d3046464f454e425254633659325534595749784f546b3d",
uri="/ISAPI/System/Video/inputs/channels/1/overlays",
cnonce="03e455047789a833deaae75e44381653",
```

```
nc=00000017,  
response="ae141a01b415c1b9ab50ddea7ff467c3",  
qop="auth"
```

```
HTTP/1.1 200 OK  
Date: Tue, 14 Mar 2017 20:38:40 GMT  
Server: App-webs/  
Connection: close  
Content-Length: 1087  
Content-Type: application/xml
```

```
<?xml version="1.0" encoding="UTF-8"?>  
<VideoOverlay version="2.0" xmlns="http://www.isapi.com/ver20/XMLSchema">  
  <normalizedScreenSize>  
    <normalizedScreenWidth>704</normalizedScreenWidth>  
    <normalizedScreenHeight>576</normalizedScreenHeight>  
  </normalizedScreenSize>  
  <attribute>  
    <transparent>false</transparent>  
    <flashing>false</flashing>  
  </attribute>  
  <fontSize>64*64</fontSize>  
  <TextOverlayList size="1">  
    <TextOverlay>  
      <id>1</id>  
      <enabled>true</enabled>  
      <positionX>0</positionX>  
      <positionY>576</positionY>  
      <displayText>222</displayText>  
    </TextOverlay>  
  </TextOverlayList>  
  <DateTimeOverlay>  
    <enabled>true</enabled>  
    <positionX>0</positionX>  
    <positionY>544</positionY>  
    <dateStyle>MM-DD-YYYY</dateStyle>  
    <timeStyle>24hour</timeStyle>  
    <displayWeek>true</displayWeek>  
  </DateTimeOverlay>  
  <channelNameOverlay version="2.0" xmlns="http://www.isapi.com/ver20/XMLSchema">  
    <enabled>true</enabled>  
    <positionX>512</positionX>  
    <positionY>64</positionY>  
  </channelNameOverlay>  
  <frontColorMode>auto</frontColorMode>  
  <frontColor>000000</frontColor>  
</VideoOverlay>
```


A.12.299 /ISAPI/System/Video/inputs/channels/<ID>/overlays/BatteryPowerOverlay

Operations about the battery overlay configuration of a specific video input channel.

Request URI Definition

Table A-916 GET /ISAPI/System/Video/inputs/channels/<ID>/overlays/BatteryPowerOverlay

Method	GET
Description	Get the battery overlay parameters of a specific video input channel.
Query	None.
Request	None.
Response	Succeeded: <i>XML_BatteryPowerOverlay</i> Failed: <i>XML_ResponseStatus</i>

Table A-917 PUT /ISAPI/System/Video/inputs/channels/<ID>/overlays/BatteryPowerOverlay

Method	PUT
Description	Set the battery overlay parameters of a specific video input channel.
Query	None.
Request	<i>XML_BatteryPowerOverlay</i>
Response	<i>XML_ResponseStatus</i>

Remarks

The <ID> in the request URI refers to the video input channel No.

A.12.300 /ISAPI/System/Video/inputs/channels/<ID>/overlays/BatteryPowerOverlay/capabilities

Get the configuration capability of battery overlay of a specific video input channel.

Request URI Definition

Table A-918 GET /ISAPI/System/Video/inputs/channels/<ID>/overlays/BatteryPowerOverlay/capabilities

Method	GET
Description	Get the configuration capability of battery overlay of a specific video input channel.

Query	None.
Request	None.
Response	Succeeded: <i>XML_Cap_BatteryPowerOverlay</i> Failed: <i>XML_ResponseStatus</i>

A.12.301 /ISAPI/System/Video/inputs/channels/<ID>/overlays/capabilities

Get the OSD capability of a specific video input channel.

Request URI Definition

Table A-919 GET /ISAPI/System/Video/inputs/channels/<ID>/overlays/capabilities

Method	GET
Description	Get the OSD capability of a specific video input channel.
Query	None.
Request	None.
Response	Succeeded: <i>XML_Cap_VideoOverlay</i> Failed: <i>XML_ResponseStatus</i>

A.12.302 /ISAPI/System/Video/inputs/channels/<ID>/overlays/channelNameOverlay

Operations about the configuration of channel name overlay of a specific video input channel.

Request URI Definition

Table A-920 GET /ISAPI/System/Video/inputs/channels/<ID>/overlays/channelNameOverlay

Method	GET
Description	Get the parameters of channel name overlay of a specific video input channel.
Query	None.
Request	None.
Response	Succeeded: <i>XML_channelNameOverlay</i> Failed: <i>XML_ResponseStatus</i>

Table A-921 PUT /ISAPI/System/Video/inputs/channels/<ID>/overlays/channelNameOverlay

Method	PUT
Description	Set the parameters of channel name overlay of a specific video input channel.
Query	None.
Request	<i>XML_channelNameOverlay</i>
Response	<i>XML_ResponseStatus</i>

Remarks

The <ID> in the request URI refers to the video input channel No.

A.12.303 /ISAPI/System/Video/inputs/channels/<ID>/overlays/dateTimeOverlay

Operations about the configuration of date and time information overlay of a specific video input channel.

Request URI Definition**Table A-922 GET /ISAPI/System/Video/inputs/channels/<ID>/overlays/dateTimeOverlay**

Method	GET
Description	Get the parameters of date and time information overlay of a specific video input channel.
Query	None.
Request	None.
Response	Succeeded: Failed: <i>XML_ResponseStatus</i>

Table A-923 PUT /ISAPI/System/Video/inputs/channels/<ID>/overlays/dateTimeOverlay

Method	PUT
Description	Set the parameters of date and time information overlay of a specific video input channel.
Query	None.
Request	
Response	<i>XML_ResponseStatus</i>

Remarks

The <ID> in the request URI refers to the video input channel No.

A.12.304 /ISAPI/System/Video/inputs/channels/<ID>/overlays/dateTimeOverlay/capabilities

Get the capability of date and time information overlay of a specific video input channel.

Request URI Definition

Table A-924 GET /ISAPI/System/Video/inputs/channels/<ID>/overlays/dateTimeOverlay/capabilities

Method	GET
Description	Get the capability of date and time information overlay of a specific video input channel.
Query	None.
Request	None.
Response	Succeeded: <i>XML_Cap_DatetimeOverlay</i> Failed: <i>XML_ResponseStatus</i>

Remarks

The <ID> in the request URI refers to the video input channel No.

A.12.305 /ISAPI/System/Video/inputs/channels/<ID>/overlays/text

Operations about the text overlay configuration of a specific video input channel.

Request URI Definition

Table A-925 GET /ISAPI/System/Video/inputs/channels/<ID>/overlays/text

Method	GET
Description	Get the text overlay parameters of a specific video input channel.
Query	None.
Request	None.
Response	Succeeded: <i>XML_TextOverlayList</i> Failed: <i>XML_ResponseStatus</i>

Table A-926 PUT /ISAPI/System/Video/inputs/channels/<ID>/overlays/text

Method	PUT
Description	Set the text overlay parameters of a specific video input channel.
Query	None.
Request	<i>XML_TextOverlayList</i>
Response	<i>XML_ResponseStatus</i>

Table A-927 POST /ISAPI/System/Video/inputs/channels/<ID>/overlays/text

Method	POST
Description	Add a text to overlay on the video for a specific video input channel.
Query	None.
Request	<i>XML_TextOverlay</i>
Response	<i>XML_ResponseStatus</i>

Table A-928 DELETE /ISAPI/System/Video/inputs/channels/<ID>/overlays/text

Method	DELETE
Description	Delete the text overlay parameters of a specific video input channel.
Query	None.
Request	None.
Response	<i>XML_ResponseStatus</i>

A.12.306 /ISAPI/System/Video/inputs/channels/<ID>/overlays/text/<ID>

Operations about the configuration of a specific piece of text overlay information of a specific video input channel.

Request URI Definition

Table A-929 GET /ISAPI/System/Video/inputs/channels/<ID>/overlays/text/<ID>

Method	GET
Description	Get the parameters of a specific piece of text overlay information of a specific video input channel.
Query	None.

Request	None.
Response	Succeeded: <i>XML_TextOverlay</i> Failed: <i>XML_ResponseStatus</i>

Table A-930 PUT /ISAPI/System/Video/inputs/channels/<ID>/overlays/text/<ID>

Method	PUT
Description	Set the parameters of a specific piece of text overlay information of a specific video input channel.
Query	None.
Request	<i>XML_TextOverlay</i>
Response	<i>XML_ResponseStatus</i>

Table A-931 DELETE /ISAPI/System/Video/inputs/channels/<ID>/overlays/text/<ID>

Method	DELETE
Description	Delete the parameters of a specific piece of text overlay information of a specific video input channel.
Query	None.
Request	None.
Response	<i>XML_ResponseStatus</i>

A.12.307 /ISAPI/System/Video/inputs/channels/<ID>/privacyMask

Get or set privacy mask parameters.

Request URI Definition

Table A-932 GET /ISAPI/System/Video/inputs/channels/<ID>/privacyMask

Method	GET
Description	Get or set privacy mask parameters.
Query	None.
Request	None.
Response	Succeeded: <i>XML_PrivacyMask</i> Failed: <i>XML_ResponseStatus</i>

Remarks

The <ID> in the request URI refers to the video input channel ID.

A.12.308 /ISAPI/System/Video/inputs/channels/<ID>/privacyMask/privacyMaskCap

Get configuration capability of privacy mask.

Request URI Definition**Table A-933 GET /ISAPI/System/Video/inputs/channels/<ID>/privacyMask/privacyMaskCap**

Method	GET
Description	Get configuration capability of privacy mask.
Query	None.
Request	None.
Response	Succeeded: <i>XML_PrivacyMaskCap</i> Failed: <i>XML_ResponseStatus</i>

Remarks

The <ID> in the request URI refers to the video input channel ID.

A.12.309 /ISAPI/System/Video/inputs/channels/<ID>/privacyMask/regions

Get, set, or delete parameters of privacy mask regions, and add a region.

Request URI Definition**Table A-934 GET /ISAPI/System/Video/inputs/channels/<ID>/privacyMask/regions**

Method	GET
Description	Get parameters of privacy mask regions.
Query	None.
Request	None.
Response	Succeeded: <i>XML_PrivacyMaskRegionList</i> Failed: <i>XML_ResponseStatus</i>

Table A-935 PUT /ISAPI/System/Video/inputs/channels/<ID>/privacyMask/regions

Method	PUT
Description	Set parameters of privacy mask regions.
Query	None.

Request	<i>XML_PrivacyMaskRegionList</i>
Response	<i>XML_ResponseStatus</i>

Table A-936 DELETE /ISAPI/System/Video/inputs/channels/<ID>/privacyMask/regions

Method	DELETE
Description	Delete parameters of privacy mask regions.
Query	None.
Request	None.
Response	<i>XML_ResponseStatus</i>

Table A-937 POST /ISAPI/System/Video/inputs/channels/<ID>/privacyMask/regions

Method	POST
Description	Add a privacy mask region.
Query	None.
Request	<i>XML_PrivacyMaskRegion</i>
Response	<i>XML_ResponseStatus</i>

Remarks

The <ID> in the request URI refers to the video input channel ID.

A.12.310 /ISAPI/System/Video/inputs/channels/<ID>/privacyMask/regions/<ID>

Get, set, or delete parameters of a privacy mask region.

Request URI Definition**Table A-938 GET /ISAPI/System/Video/inputs/channels/<ID>/privacyMask/regions/<ID>**

Method	GET
Description	Get parameters of a privacy mask region.
Query	None.
Request	None.
Response	Succeeded: <i>XML_PrivacyMaskRegion</i> Failed: <i>XML_ResponseStatus</i>

Table A-939 PUT /ISAPI/System/Video/inputs/channels/<ID>/privacyMask/regions/<ID>

Method	PUT
Description	Set parameters of a privacy mask region.
Query	None.
Request	<i>XML_PrivacyMaskRegion</i>
Response	<i>XML_ResponseStatus</i>

Table A-940 DELETE /ISAPI/System/Video/inputs/channels/<ID>/privacyMask/regions/<ID>

Method	DELETE
Description	Delete parameters of a privacy mask region.
Query	None.
Request	None.
Response	<i>XML_ResponseStatus</i>

Remarks

The first <ID> in the request URI refers to the video input channel ID; and second <ID> is the privacy mask region ID.

A.12.311 /ISAPI/System/Video/inputs/channels/<ID>/VCAResource

Operation about configuration of intelligent resources switch.

Request URI Definition

Table A-941 GET /ISAPI/System/Video/inputs/channels/<ID>/VCAResource

Method	GET
Description	Get parameters of intelligent resources switch.
Query	None.
Request	None.
Response	Succeeded: <i>XML_VCAResource</i> Failed: <i>XML_ResponseStatus</i>

Table A-942 PUT /ISAPI/System/Video/inputs/channels/<ID>/VCAResource

Method	PUT
Description	Set parameters of intelligent resources switch.

Query	None.
Request	<i>XML_VCAResource</i>
Response	<i>XML_ResponseStatus</i>

Remarks

- The <ID> in the request URI refers to the video input channel ID.
- The intelligent resources switch supports three capture mode, i.e., target capture, face capture, and perimeter capture.
 - For target capture mode, the panoramic camera captures the human body and vehicle pictures, the speed dome tracks and capture the human body for human body or face recognition and comparison. If matched, the thumbnails and original pictures of human body and face will be uploaded; otherwise, only the thumbnail and original picture of face will be uploaded.
 - For face capture mode, the panoramic camera transmits the videos without any intelligent analysis, the speed dome captures face picture in up to 4 scenes.
 - For perimeter capture, the panoramic camera captures vehicle pictures and performs HMS +DGS detection, and it supports detecting and tracking the small target within 40 m; while the speed dome tracks and captures the detected target, and outputs thumbnails and original pictures of human body and face.

A.12.312 /ISAPI/System/Video/inputs/channels/<ID>/VCAResource/capabilities

Get capability of intelligent resources switch.

Request URI Definition

Table A-943 GET /ISAPI/System/Video/inputs/channels/<ID>/VCAResource/capabilities

Method	GET
Description	Get capability of intelligent resources switch.
Query	None.
Request	None.
Response	Succeeded: <i>XML_Cap_VCAResource</i> Failed: <i>XML_ResponseStatus</i>

A.12.313 /ISAPI/System/Video/inputs/channels/capabilities

Get the capability of configuring video input channels in a batch.

Request URI Definition

Table A-944 GET /ISAPI/System/Video/inputs/channels/capabilities

Method	GET
Description	Get the capability of configuring video input channels in a batch.
Query	None.
Request	None.
Response	Succeeded: <i>XML_VideoInputChannelsCap</i> Failed: <i>XML_ResponseStatus</i>

A.12.314 /ISAPI/System/Video/inputs/channels/counting/collection/capabilities?format=json

Get the capability of people counting data replenishment.

Request URI Definition

Table A-945 GET /ISAPI/System/Video/inputs/channels/counting/collection/capabilities?format=json

Method	GET
Description	Get the capability of people counting data replenishment.
Query	format: determine the format of request or response message.
Request	None.
Response	<i>JSON_PeopleCounting_CollectionDescriptionCap</i>

A.12.315 /ISAPI/System/Video/inputs/channels/counting/collection?format=json

Perform people counting replenishment.

Request URI Definition

Table A-946 POST /ISAPI/System/Video/inputs/channels/counting/collection?format=json

Method	POST
Description	Perform people counting replenishment.
Query	format: determine the format of request or response message.

Request	<i>JSON_PeopleCounting_CollectionDescription</i>
Response	<i>JSON_PeopleCounting_CollectionResult</i>

A.12.316 /ISAPI/System/Video/inputs/channels/heatMap/collection/capabilities?format=json

Get the capability of heat map data replenishment.

Request URI Definition

Table A-947 GET /ISAPI/System/Video/inputs/channels/heatMap/collection/capabilities?format=json

Method	GET
Description	Get the capability of heat map data replenishment.
Query	format: determine the format of request or response message.
Request	None.
Response	<i>JSON_HeatMap_CollectionDescriptionCap</i>

A.12.317 /ISAPI/System/Video/inputs/channels/heatMap/collection?format=json

Perform heat map data replenishment.

Request URI Definition

Table A-948 POST /ISAPI/System/Video/inputs/channels/heatMap/collection?format=json

Method	POST
Description	Perform heat map data replenishment.
Query	format: determine the format of request or response message.
Request	<i>JSON_HeatMap_CollectionDescription</i>
Response	<i>JSON_HeatMap_CollectionResult</i>

A.12.318 /ISAPI/System/Video/inputs/OSDLanguage

Operations about OSD language configuration.

Request URI Definition

Table A-949 GET /ISAPI/System/Video/inputs/OSDLanguag

Method	GET
Description	Get OSD language parameters.
Query	None.
Request	None.
Response	<i>XML_Language</i>

Table A-950 PUT /ISAPI/System/Video/inputs/OSDLanguag

Method	PUT
Description	Set OSD language parameters.
Query	None.
Request	<i>XML_Language</i>
Response	<i>XML_ResponseStatus</i>

A.12.319 /ISAPI/System/Video/outputs

Get configuration parameters of video outputs.

Request URI Definition

Table A-951 GET /ISAPI/System/Video/outputs

Method	GET
Description	Get configuration parameters of all video outputs.
Query	None.
Request	None.
Response	Succeeded: <i>XML_VideoOutput</i> Failed: <i>XML_ResponseStatus</i>

A.12.320 /ISAPI/System/Video/outputs/channels

Get configuration parameters of multiple video outputs.

Request URI Definition

Table A-952 GET /ISAPI/System/Video/outputs/channels

Method	GET
Description	Get configuration parameters of multiple video outputs.
Query	None.
Request	None.
Response	Succeeded: <i>XML_VideoOutputChannelList</i> Failed: <i>XML_ResponseStatus</i>

A.12.321 /ISAPI/System/Video/outputs/channels/<ID>

Get or set parameters of a video output.

Request URI Definition

Table A-953 GET /ISAPI/System/Video/outputs/channels/<ID>

Method	GET
Description	Get parameters of a video output.
Query	None.
Request	None.
Response	Succeeded: <i>XML_VideoOutputChannel</i> Failed: <i>XML_ResponseStatus</i>

Table A-954 PUT /ISAPI/System/Video/outputs/channels/<ID>

Method	PUT
Description	Set parameters of a video output.
Query	None.
Request	<i>XML_VideoOutputChannel</i>
Response	<i>XML_ResponseStatus</i>

Remarks

The <ID> in the request URI refers to the video output ID.

A.12.322 /ISAPI/System/Video/outputs/channels/<ID>/capabilities

Get video output configuration capability.

Request URI Definition

Table A-955 GET /ISAPI/System/Video/outputs/channels/<ID>/capabilities

Method	GET
Description	Get video output configuration capability.
Query	None.
Request	None.
Response	Succeeded: <i>XML_Cap_VideoOutputChannel</i> Failed: <i>XML_ResponseStatus</i>

Remarks

The <ID> in the request URI refers to the video output ID.

A.12.323 /ISAPI/System/Video/outputs/PreviewSwitch/capabilities

Get capability of auto-switch configuration during live view.

Request URI Definition

Table A-956 GET /ISAPI/System/Video/outputs/PreviewSwitch/capabilities

Method	GET
Description	Get capability of auto-switch configuration during live view.
Query	None.
Request	None.
Response	Succeeded: <i>XML_Cap_PreviewSwitch</i> Failed: <i>XML_ResponseStatus</i>

A.12.324 /ISAPI/System/Video/outputs/PreviewSwitch/capabilities? videoOutType=

Get capability of auto-switch configuration during live view by video output.

Request URI Definition**Table A-957 GET /ISAPI/System/Video/outputs/PreviewSwitch/capabilities?videoOutType=**

Method	GET
Description	Get capability of auto-switch configuration during live view by video output.
Query	videoOutType: video output type, string type, it can be the following values: CVBS, HDMI, VGA, AUXOutput1, AUXOutput2, AUXOutput3, AUXOutput4, extendHDMI1, extendHDMI2, extendHDMI3, extendHDMI4, noSameSourceVGA1, noSameSourceVGA2, noSameSourceVGA3, noSameSourceVGA4, sameSourceVGA, mainLCD, AUX-VGA, AUX-LCD, and zeroChannel. If there are more than two outputs are homologous, you can combine these outputs by "/", e.g., if HDM1 and VGA1 are homologous outputs, the value of videoOutType equals to HDM1/VGA1.
Request	None.
Response	Succeeded: <i>XML_PreviewSwitchVideoOutCap</i> Failed: <i>XML_ResponseStatus</i>

A.12.325 /ISAPI/System/Video/outputs/PreviewSwitch?groupID=&videoOutType=

Operations about auto-switch configuration in live view.

Request URI Definition**Table A-958 GET /ISAPI/System/Video/outputs/PreviewSwitch?groupID=&videoOutType=**

Method	GET
Description	Get configuration parameters of auto-switch in live view.
Query	groupNo: group No., integer type, which is used to group the windows for configuration, and up to 64 windows can be added to a group. videoOutType: video output type, string type, it can be the following values: CVBS, HDMI, VGA, AUXOutput1, AUXOutput2, AUXOutput3, AUXOutput4, extendHDMI1, extendHDMI2, extendHDMI3, extendHDMI4, noSameSourceVGA1, noSameSourceVGA2, noSameSourceVGA3, noSameSourceVGA4, sameSourceVGA, mainLCD, AUX-VGA, AUX-LCD, and zeroChannel. If there are more than two outputs are homologous, you can combine these outputs

	by "/", e.g., if HDM1 and VGA1 are homologous outputs, the value of videoOutType equals to HDM1/VGA1.
Request	None.
Response	Succeeded: <i>XML_PreviewSwitch</i> Failed: <i>XML_ResponseStatus</i>

Table A-959 PUT /ISAPI/System/Video/outputs/PreviewSwitch?groupID=&videoOutType=

Method	PUT
Description	Set parameters for auto-switch in live view.
Query	groupNo : group No., integer type, which is used to group the windows for configuration, and up to 64 windows can be added to a group. videoOutType : video output type, string type, it can be the following values: CVBS, HDMI, VGA, AUXOutput1, AUXOutput2, AUXOutput3, AUXOutput4, extendHDMI1, extendHDMI2, extendHDMI3, extendHDMI4, noSameSourceVGA1, noSameSourceVGA2, noSameSourceVGA3, noSameSourceVGA4, sameSourceVGA, mainLCD, AUX-VGA, AUX-LCD, and zeroChannel. If there are more than two outputs are homologous, you can combine these outputs by "/", e.g., if HDM1 and VGA1 are homologous outputs, the value of videoOutType equals to HDM1/VGA1.
Request	<i>XML_PreviewSwitch</i>
Response	<i>XML_ResponseStatus</i>

A.12.326 /ISAPI/System/Video/outputs/PreviewSwitch?groupID=&videoOutType=&previewFrameNo=

Get configuration parameters of auto-switch in live view.

Request URI Definition

Table A-960 GET /ISAPI/System/Video/outputs/PreviewSwitch?groupID=&videoOutType=&previewFrameNo=

Method	GET
Description	Get configuration parameters of auto-switch in live view.
Query	groupNo : group No., integer type, which is used to group the windows for configuration, and up to 64 windows can be added to a group.

	videoOutType: video output type, string type, it can be the following values: CVBS, HDMI, VGA, AUXOutput1, AUXOutput2, AUXOutput3, AUXOutput4, extendHDMI1, extendHDMI2, extendHDMI3, extendHDMI4, noSameSourceVGA1, noSameSourceVGA2, noSameSourceVGA3, noSameSourceVGA4, sameSourceVGA, mainLCD, AUX-VGA, AUX-LCD, and zeroChannel. If there are more than two outputs are homologous, you can combine these outputs by "/", e.g., if HDM1 and VGA1 are homologous outputs, the value of videoOutType equals to HDM1/VGA1. (Optional) previewFrameNo: number of live view windows, integer string type, and it can be the following values: 1, 4, 6, 8, 9, 25, 32, 36, auto1 (custom window division 1), auto2 (custom window division 2), auto3 (custom window division 3), and auto4 (custom window division 4).
Request	None.
Response	Succeeded: <i>XML_PreviewSwitch</i> Failed: <i>XML_ResponseStatus</i>

A.12.327 /ISAPI/System/wiresharkToUSB/capabilities?format=json

Get the configuration capability of exporting captured packets to USB flash drive.

Request URI Definition

Table A-961 GET /ISAPI/System/wiresharkToUSB/capabilities?format=json

Method	GET
Description	Get the configuration capability of exporting captured packets to USB flash drive.
Query	format: determine the format of request or response message.
Request	None.
Response	Succeeded: <i>JSON_Cap_WiresharkToUSB</i> Failed: <i>JSON_ResponseStatus</i>

A.12.328 /ISAPI/System/wiresharkToUSB/mode?format=json

Set parameters of exporting captured packets to USB flash drive.

Request URI Definition**Table A-962 PUT /ISAPI/System/wiresharkToUSB/mode?format=json**

Method	PUT
Description	Set parameters of exporting captured packets to USB flash drive.
Query	format: determine the format of request or response message.
Request	<i>JSON_WiresharkToUSB</i>
Response	<i>JSON_ResponseStatus</i>

A.12.329 /ISAPI/System/wiresharkToUSB/status?format=json

Get the status of exporting captured packets to USB flash drive.

Request URI Definition**Table A-963 GET /ISAPI/System/wiresharkToUSB/status?format=json**

Method	GET
Description	Get the status of exporting captured packets to USB flash drive.
Query	format: determine the format of request or response message.
Request	None.
Response	Succeeded: <i>JSON_WiresharkToUSBStatus</i> Failed: <i>JSON_ResponseStatus</i>

A.12.330 /ISAPI/System/workingstatus/hdStatus?format=json

Get HDD status.

Request URI Definition**Table A-964 GET /ISAPI/System/workingstatus/hdStatus?format=json**

Method	GET
Description	Get all HDDs' status.
Query	format: determine the format of request or response message.
Request	None.
Response	Succeeded: <i>JSON_HDStatus</i>

	Failed: <i>JSON_ResponseStatus</i>
--	------------------------------------

Table A-965 POST /ISAPI/System/workingstatus/hdStatus?format=json

Method	POST
Description	Get a specified HDD's status.
Query	format: determine the format of request or response message.
Request	<i>JSON_HDCond</i>
Response	Succeeded: <i>JSON_HDStatus</i> Failed: <i>JSON_ResponseStatus</i>

A.12.331 /ISAPI/System/workingstatus/chanStatus?format=json

Get channel status.

Request URI Definition

Table A-966 GET /ISAPI/System/workingstatus/chanStatus?format=json

Method	GET
Description	Get all channels' status.
Query	format: determine the format of request or response message.
Request	None.
Response	Succeeded: <i>JSON_ChanStatus</i> Failed: <i>JSON_ResponseStatus</i>

Table A-967 POST /ISAPI/System/workingstatus/chanStatus?format=json

Method	POST
Description	Get a specified channel's status.
Query	format: determine the format of request or response message.
Request	<i>JSON_ChanCond</i>
Response	Succeeded: <i>JSON_ChanStatus</i> Failed: <i>JSON_ResponseStatus</i>

A.12.332 /ISAPI/System/workingstatus/capabilities?format=json

Get capability of getting all statuses of device.

Request URI Definition

Table A-968 GET /ISAPI/System/workingstatus/capabilities?format=json

Method	GET
Description	Get capability of getting all statuses of device.
Query	format: determine the format of request or response message.
Request	None.
Response	Succeeded: <i>JSON_Cap_WorkingStatus</i> Failed: <i>JSON_ResponseStatus</i>

A.12.333 /ISAPI/System/workingstatus?format=json

Get device working status.

Request URI Definition

Table A-969 GET /ISAPI/System/workingstatus?format=json

Method	GET
Description	Get device working status.
Query	format: determine the format of request or response message.
Request	None.
Response	Succeeded: <i>JSON_WorkingStatus</i> Failed: <i>JSON_ResponseStatus</i>

Remarks

Up to 20 bytes are allowed for a device's working status; up to 100 bytes are allowed for a channel's status, a HDD's status, and an IO's status; up to 30 bytes are allowed for a two-way audio channel's status.

A.12.334 /ISAPI/System/workingstatus/IOStatus?format=json

Get device IO status.

Request URI Definition

Table A-970 GET /ISAPI/System/workingstatus/IOStatus?format=json

Method	GET
Description	Get device IO status.
Query	format: determine the format of request or response message.
Request	None.
Response	Succeeded: <i>JSON_IOStatus</i> Failed: <i>JSON_ResponseStatus</i>

A.12.335 /ISAPI/System/zeroBiasCalibration/channels/<ID>?format=json

Calibrate the accelerometer bias.

Request URI Definition

Table A-971 PUT /ISAPI/System/zeroBiasCalibration/channels/<ID>?format=json

Method	PUT
Description	Calibrate the accelerometer bias.
Query	format: determine the format of request or response message.
Request	None.
Response	<i>JSON_ResponseStatus</i>

Remarks

The <ID> in the request URI refers to the channel No.

A.13 /ISAPI/Thermal

A.13.1 /ISAPI/Thermal/capabilities

Get thermal capability.

Request URI Definition

Table A-972 GET /ISAPI/Thermal/capabilities

Method	GET
Description	Get the thermal capability.
Query	None.
Request	None.
Response	<i>XML_ThermalCap</i>

A.13.2 /ISAPI/Thermal/temperature/collection/capabilities?format=json

Get the capability of temperature data replenishment.

Request URI Definition

Table A-973 GET /ISAPI/Thermal/temperature/collection/capabilities?format=json

Method	GET
Description	Get the capability of temperature data replenishment.
Query	format: determine the format of request or response message.
Request	None.
Response	<i>JSON_Temperature_CollectionDescriptionCap</i>

A.13.3 /ISAPI/Thermal/temperature/collection?format=json

Perform the temperature data replenishment.

Request URI Definition

Table A-974 GET /ISAPI/Thermal/temperature/collection?format=json

Method	POST
Description	Perform the temperature data replenishment.
Query	format: determine the format of request or response message.
Request	<i>JSON_CollectionDescription</i>
Response	<i>JSON_CollectionResult</i>

Appendix B. Appendixes

B.1 Request and Response Messages

B.1.1 JSON Messages

JSON_Adaption

Adaption message in JSON format

```
{
  "Adaption": {
    "enable": ,
    /*required, boolean type, whether to enable network self-adaptive function: true-no, false-yes; this function is bound
    with IntraRefresh encoding strategy*/
    "policy": ,
    /*required,self-adaptive strategy, 0-self-adaptive (adjust the resolution and bit rate, the resolution value should be
    smaller than or equal to the configured value, but the frame rate is constant), 1-resolution in priority (adjust bit rate,
    resolution and frame rate are constant), 2-error correction and retransmission (resolution and frame rate are
    constant)*/
    "percentage":
    /*this node is required when policy is "2", error correction percentage, unit: %, which is between 0 and 100; lager
    percentage represents more corrected data, more higher requirement for bandwidth, and better effect*/
  }
}
```

JSON_AdaptionCap

AdaptionCap message in JSON format

```
{
  "AdaptionCap": {
    "enable": "true,false",
    /*required, boolean type, whether to enable network self-adaptive function: true-no, false-yes; this function is bound
    with IntraRefresh encoding strategy*/
    "policy": {
      /*required,self-adaptive strategy, 0-self-adaptive (adjust the resolution and bit rate, the resolution value should be
      smaller than or equal to the configured value, but the frame rate is constant), 1-resolution in priority (adjust bit rate,
      resolution and frame rate are constant), 2-error correction and retransmission (resolution and frame rate are
      constant)*/
      "@opt": "0,1,2"
    },
    "percentage": {
      /*this node is required when policy is "2", error correction percentage, unit: %, which is between 0 and 100; lager
      percentage represents more corrected data, more higher requirement for bandwidth, and better effect*/
    }
  }
}
```



```
"@min": 0,  
"@max": 100,  
}  
}  
}
```

JSON_AlarmStatistics

AlarmStatistics message in JSON format

```
{  
  "AlarmStatistics":{  
    /*optional, alarm statistics of added IoT devices*/  
    "totalNumber": "",  
    /*required, the total number of alarms, integer*/  
    "time": "",  
    /*required, statistic time, integer, unit: second*/  
  }  
}
```

JSON_AlgorithmsInfo

AlgorithmsInfo message in JSON format

```
{  
  "AlgorithmsInfo":{  
    "type": "",  
    /*required, string, algorithm library type, e.g, "face"*/  
    "name": "",  
    /*required, string, algorithm library name which is used to distinguish different algorithms, the maximum size is 256 bytes*/  
    "versionInfo":{  
      /*required, algorithm library version information*/  
      "major": ,  
      /*required, integer, major version of the algorithm library*/  
      "minor": ,  
      /*required, integer, minor version of the algorithm library*/  
      "revision":  
      /*required, integer, revision version of the algorithm library*/  
    },  
    "platform": "",  
    /*required, string, platform where the algorithm library runs*/  
    "os": "",  
    /*required, string, operating system adaptive to the algorithm library*/  
    "accuracy": "",  
    /*optional, string, computational accuracy of the algorithm library*/  
    "encryption": "",  
    /*required, string, encryption method*/  
    "buildTime": "",  
  }  
}
```

```
/*required, string, compiling date in the format "buildyyyymmdd"*/
  "versionProperties":""
/*required, string, version attribute: "alpha", "beta", "release", "custom"*/
}
}
```

JSON_AlgorithmsInfoCap

AlgorithmsInfoCap capability message in JSON format

```
{
  "AlgorithmsInfoCap":{
    "type":{
/*required, string, algorithm library type, e.g., "face"*/
      "@opt":["face"]
    },
    "maxSize": ,
/*required, integer, supported number of algorithm libraries*/
    "name":{
/*required, string, algorithm library name which is used to distinguish different algorithms, the maximum size is 256 bytes*/
      "@min": ,
      "@max":
    },
    "versionInfo":{
/*required, algorithm library version information*/
      "major":{
/*required, integer, major version of the algorithm library*/
        "@min": ,
        "@max":
      },
      "minor":{
/*required, integer, minor version of the algorithm library*/
        "@min": ,
        "@max":
      },
      "revision":{
/*required, integer, revision version of the algorithm library*/
        "@min": ,
        "@max":
      }
    },
    "platform":{
/*required, string, platform where the algorithm library runs*/
      "@min": ,
      "@max":
    },
    "accuracy":{
/*optional, string, computational accuracy of the algorithm library*/
      "@min": ,
      "@max":
    }
  }
}
```

```
,
"os":{
/*required, string, operating system adaptive to the algorithm library*/
"@min": ,
"@max":
},
"encryption":{
/*required, string, encryption method*/
"@min": ,
"@max":
},
"buildTime":{
/*required, string, compiling date in the format "buildyyyymmdd"*/
"@min": ,
"@max":
},
"versionProperties":{
/*required, string, version attribute: "alpha", "beta", "release", "custom"*/
"@min": ,
"@max":
}
}
}
```

JSON_AssociatedChannelList

AssociatedChannelList message in JSON format

```
{
  "AssociatedChannelList": [{
/*optional, list of linked channels*/
    "channel": "",
/*required, channel No., integer*/
    "type": "",
/*required, channel type: "video"-video channel, "IoT"-IoT channel, string*/
    "associatedVideoChannels": "",
/*optional, linked video channel No., array, and the unit is integer*/
    "associatedIoTChannels": ""
/*optional, linked IoT channel No., array, and the unit is integer*/
  }]
}
```

JSON_AudioAlarm

AudioAlarm message in JSON format

```
{
  "AudioAlarm":{
    "audioID": ,
```

```

/*required, alarm sound types, 1-siren, 2-"Warning, this is a restricted area", 3-"Warning, this is a restricted area,
please keep away", 4-"Warning, this is a no-parking zone", 5-"Warning, this is a no-parking zone, please keep away",
6-"Attention please. The area is under surveillance", 7-"Welcome, Please notice that the area is under surveillance",
8-"Welcome", 9-"Danger! Please keep away", 10-siren + "Danger, please keep away", 11-Audio Warning, 12-Beep
Sound, 13-custom alarm sound 1, 14-custom alarm sound 2, 15-custom alarm sound 3*/
"audioVolume": ,
/*required, int, volume, which is between 1 and 100*/
"alarmTimes": ,
/*required, int, times of alarm, which is between 1 and 50*/
"TimeRangeList": [{
/*optional, alarm output schedule list*/
"week": ,
"TimeRange": [{
"id": ,
/*required, int, ID of time period for each day*/
"beginTime": "",
/*required, string, start time in ISO8601 format*/
"endTime": ""
/*required, string, end time in ISO8601 format*/
}}
}}
"audioClass": "",
/*optional, string, read-only, sound type, "alertAudio"-alarm sound (get the supported alarm sound types from the
node AlertAudioTypeListCap), "promptAudio"-prompt sound, "customAudio"-custom sound (configure it according to
customAudioID); if this node is not returned, you should get the supported sound types from the node
audioTypeListCap*/
"alertAudioID": ,
/*dependent, int, read-only, alarm sound ID, this node is valid only when audioClass is alertAudio, 1-"Siren",
2-"Warning, this is a restricted area", 3-"Warning, this is a restricted area, please keep away", 4-"Warning, this is a no-
parking zone", 5-"Warning, this is a no-parking zone, please keep away", 6-"Attention please. The area is under
surveillance", 7-"Welcome, Please notice that the area is under surveillance", 8-"Welcome", 9-"Danger! Please keep
away", 10-"(Siren)&Danger, please keep away", 11-Audio Warning*/
"customAudioID": ,
/*dependent, int, read-only, custom sound ID, this node will be returned when audioClass is "customAudio". The
alarm sound file can only be uploaded via Hik-Connect in specified format. If there is no value returned in this node,
the custom sound (customAudio) will not be returned,either*/
"alarmType": "",
/*optional, string, alarm type: "behavior"-behavior analysis, "thermometry"-thermometry, "dynamicFire"-fire
detection, "smokingMode"-smoke detection, "noMaskDetection"-no wearing mask detection*/
}
}

```

Remarks

1. The user can configure the sound file via the node **audioID** only, and when the **audioID** is specified, its value will be automatically synchronized to the nodes of **audioClass**, **alertAudioID**, and **customAudioID**, which are read-only.
2. When the value of **audioID** is from 1 to 11, the corresponding value of **audioClass** is "alertAudio", and the value of **alertAudioID** is also from 1 to 11; when the value of **audioID** is 12, the corresponding value of **audioClass** is "promptAudio"; when the value of **audioID** is from

13 to 15, the corresponding value of **audioClass** is "customAudio", and the value of **customAudioID** is from 1 to 3.

JSON_AudioAlarmCap

JSON message about audible warning configuration capability

```
{
  "AudioAlarmCap":{
    "audioTypeListCap":{{
      /*required, alarm sound types, 1-siren, 2-"Warning, this is a restricted area", 3-"Waring, this is a restricted area,
      please keep away", 4-"Warning, this is a no-parking zone", 5-"Warning, this is a no-parking zone, please keep away",
      6-"Attention please. The area is under surveillance", 7-"Welcome, Please notice that the area is under surveillance",
      8-"Welcome", 9-"Danger! Please keep away", 10-siren + "Danger, please keep away", 11-Audio Warning, 12-Beep
      Sound, 13-custom alarm sound 1, 14-custom alarm sound 2, 15-custom alarm sound 3, 16-AI custom audio file*/
      "audioID": 1,
      "audioDescription":"siren"
    },
    {
      "audioID": 2,
      "audioDescription":"Warning, this is a restricted area"
    },
    {
      "audioID": 3,
      "audioDescription":"Waring, this is a restricted area, please keep away"
    }
  }},
  "audioVolume":{
    /*required, int, volume, which is between 1 ad 100*/
    "@min": 1,
    "@max": 100,
    "@def": 100
  },
  "alarmTimes":{
    /*required, int, times of audible warning, which is between 1 and 50*/
    "@min": 1,
    "@max": 50,
    "@def": 5
  },
  "TimeRangeCap":{
    /*optional, audible warning schedule capability*/
    "week":{
      /*required, int, days of the week: 1-Monday, 2-Tuesday, 3-Wednesday, 4-Thursday, 5-Friday,6-Saturday, 7-Sunday*/
      "@opt":"1, 2, 3, 4, 5, 6, 7"
    },
    "id":{
      /*required, int, ID of time period for each day*/
      "@maxSize": 8
    },
    "beginTime":{
      /*required, string, start time in ISO8601 format*/
      "@min":"00:00",
```

```

    "@max": "24:00"
  },
  "endTime": {
    /*required, string, end time in ISO8601 format*/
    "@min": "00:00",
    "@max": "24:00"
  }
}
"audioClass": {
  /*optional, string, read-only, sound type, "alertAudio"-alarm sound (get the supported alarm sound types from the
  node AlertAudioTypeListCap), "promptAudio"-prompt sound, "customAudio"-custom sound (configure it according to
  customAudioID); if this node is not returned, you should get the supported sound types from the node
  audioTypeListCap*/
  "@opt": "alertAudio, promptAudio, customAudio",
  "@def": "alertAudio"
},
"AlertAudioTypeListCap": [{
  /*dependent, read-only, list of supported alarm sound types, this node will be returned when audioClass is
  "alertAudio": 1-"Siren", 2-"Warning, this is a restricted area", 3-"Waring, this is a restricted area, please keep away",
  4-"Warning, this is a no-parking zone", 5-"Warning, this is a no-parking zone, please keep away", 6-"Attention please.
  The area is under surveillance", 7-"Welcome, Please notice that the area is under surveillance", 8-"Welcome",
  9-"Danger! Please keep away", 10-"(Siren)&Danger, please keep away, 11-Audio Warning"*/
  "alertAudioID": 1,
  "alertAudioDescription": "Siren"
}, {
  "alertAudioID": 2,
  "alertAudioDescription": "Warning, this is a restricted area"
}, {
  "alertAudioID": 3,
  "alertAudioDescription": "Waring, this is a restricted area, please keep away"
}
],
"customAudioID": {
  /*dependent, int, read-only, custom sound ID, this node will be returned when audioClass is "customAudio". The
  alarm sound file can only be uploaded via Hik-Connect in specified format. If there is no value returned in this node,
  the custom sound (customAudio) will not be returned,either*/
  "@min": 1,
  "@max": 3,
  "@def": 1
},
"alarmType": {
  /*optional, string, alarm type: "behavior"-behavior analysis, "thermometry"-thermometry, "dynamicFire"-fire
  detection, "smokingMode"-smoke detection*/
  "@opt": "behavior,thermometry,dynamicFire,smokingMode",
  "@def": "thermometry"
}
}
"AlarmBindAudioList": [{
  /*optional, alarm type and the list of audio related to alarm type*/
  "alarmType": "behavior",
  /*optional, string, alarm type: "behavior"-behavior analysis, "thermometry"-temperature measurement,
  "dynamicFire"-fire detection, "smokingMode"-smoke detection, "noMaskDetection"-no wearing mask detection*/
  "audioID": {

```

```
/*optional, string, the audible warning type ID corresponding to alarm type, behavior analysis corresponds to 1-12,
thermometry corresponds to 13, fire detection corresponds to 14, smoke detection corresponds to 15*/
"@opt": "1,2,3,4,5,6,7,8,9,10,11,12"
}
}}
"isSupportCustomAudio": true,
/*optional, boolean, whether the device supports uploading, downloading, and editing custom alarm sound file*/
"isSupportAudioTest": true,
/*optional, boolean, whether the device supports sound file test*/
"customAudioName": {
/*optional, string, name of custom sound file, the maximum size is 24 bytes*/
"@min": 1,
"@max": 24
}
"isSupportCustomAudioFormatInfo": true,
/*optional, boolean, whether the device supports getting the custom audio file format*/
"isSupportCustomAudioURL": true,
/*optional, boolean, whether the device supports importing the custom audio file in URL format; if this node is not
returned, the device dose not support the function*/
"isSupportCustomAudioFiles": true
/*optional, boolean, whether the device supports importing the custom audio files in a batch; if this node is not
returned, the device dose not support the function*/
}
}
```

Remarks

1. The user can configure the sound file via the node **audioID** only, and when the **audioID** is specified, its value will be automatically synchronized to the nodes of **audioClass**, **alertAudioID**, and **customAudioID**, which are read-only.
2. When the value of **audioID** is from 1 to 11, the corresponding value of **audioClass** is "alertAudio", and the value of **alertAudioID** is also from 1 to 11; when the value of **audioID** is 12, the corresponding value of **audioClass** is "promptAudio"; when the value of **audioID** is from 13 to 15, the corresponding value of **audioClass** is "customAudio", and the value of **customAudioID** is from 1 to 3.

JSON_Authentication

JSON message about parameters of serial authentication service

```
{
  "Authentication":{
    "enabled":,
/*required, boolean, whether to enable serial authentication service*/
    "serviceClosurePeriod":
/*dependent, integer, service life, unit: day; it is required when enabled is "false"*/
  }
}
```

JSON_AuthenticationCap

JSON message about serial authentication service

```
{
  "AuthenticationCap":{
    "enabled":{
      /*required, boolean, whether to enable serial authentication service*/
      "@opt":[
        true,
        false
      ]
    },
    "serviceClosurePeriod":{
      /*dependent, integer, service life, unit: day; it is required when enabled is "false"*/
      "@min":1,
      "@max":30,
      "@def":7
    }
  }
}
```

JSON_AutoEagleFocusing

AutoEagleFocusing message in JSON format

```
{
  "AutoEagleFocusing": {
    /*required, auto calibration of rapid focus*/
    "EagleFocusingRegion": [{
      /*optional, string, rapid focus region*/
      "sid": "",
      /*optional, string, scene ID*/
      "spotNum": ,
      /*optional, integer, the number of calibration points*/
      "type": "",
      /*optional, string, region type now only supports "line"*/
      "Region": [{
        /*optional, region list*/
        "x": ,
        "y":
      }],
      "AbsoluteHigh": {
        /*optional, PTZ position*/
        "elevation": ,
        /*optional, xs:integer, tilt angle, range: [-900,2700]*/
        "azimuth": ,
        /*optional, xs:integer, azimuth, range: [0,3600]*/
        "absoluteZoom": 33
      }
    }
  }
}
```



```
/*optional, xs:integer, zoom, range: [1,1000]*/
}
}}
}
}
```

JSON_AutoEagleFocusingCap

AutoEagleFocusingCap message in JSON format

```
{
  "AutoEagleFocusingCap": {
    /*required, auto calibration capability of rapid focus*/
    "EagleFocusingRegion": {
      "sid": {
        /*optional, string, scene ID*/
        "@min": 1,
        "@max": 2
      },
      "spotNum": {
        /*optional, integer, the number of calibration points*/
        "@min": 1,
        "@max": 64
      },
      "type": {
        /*optional, string, region type now only supports "line"*/
        "@opt": "line"
      },
      "RegionCap": {
        "minSize": 3,
        /*required, integer, the minimum number of region edges*/
        "maxSize": 10,
        /*required, integer, the maximum number of region edges*/
        "x": {
          /*required, float, X-coordinate, value range: 0.000 to 1*/
          "@min": 0.000,
          "@max": 1.000,
          "#text": 0.120
        },
        "y": {
          /*required, float, Y-coordinate, value range: 0.000 to 1*/
          "@min": 0.000,
          "@max": 1.000,
          "#text": 0.120
        }
      },
      "AbsoluteHighCap": {
        "elevation": {
          /*optional, xs:integer, tilt angle, range: [-900,2700]*/
          "@min": -900,
          "@max": 2700
        }
      }
    }
  }
}
```

```
    },
    "azimuth": {
/*optional, xs:integer, azimuth, range: [0,3600]*/
        "@min": 0,
        "@max": 3600
    },
    "absoluteZoom": {
/*optional, xs:integer, zoom, range: [1,1000]*/
        "@min": 1,
        "@max": 1000
    }
}
}
```

JSON_AutoGotoCfg

AutoGotoCfg message in JSON format

```
{
  "AutoGotoCfg":{
    "enable": ,
/*optional, boolean type, whether to enable*/
    "autoRecoveryTime":
/*optional, integer type, automatically restored time after the PTZ control is disabled, unit: second*/
  }
}
```

JSON_AutoGotoCfgCap

AutoGotoCfgCap message in JSON format

```
{
  "AutoGotoCfgCap":{
    "enable":"true,false",
/*optional, whether to enable*/
    "autoRecoveryTime":{
/*optional, automatically restored time after the PTZ control is disabled, unit: second*/
      "@min": ,
      "@max":
    }
  }
}
```

JSON_AutoMaintenance

AutoMaintenance message in JSON format

```
{
  "AutoMaintenance":{
    "enabled": true,
    /*required, boolean, enable or not*/
    "dayOfWeek": "",
    /*required, integer, day of the week, ranges from 1 to 7, 1-Monday, 2-Tuesday, and so on*/
    "rebootTime": "",
    /*required, reboot time, corrects to minute, it is 0 by default, ISO8601 format, string*/
  }
}
```

JSON_AutoMaintenanceCap

AutoMaintenanceCap message in JSON format

```
{
  "AutoMaintenanceCap":{
    "enabled": "true,false",
    /*required, boolean, enable or not*/
    "dayOfWeek":{
      /*required, integer, day of the week, ranges from 1 to 7, 1-Monday, 2-Tuesday, and so on*/
      "@min": 1,
      "@max": 7,
      "#text": 1
    },
  }
}
```

JSON_AutoSwitch

AutoSwitch message in JSON format

```
{
  "AutoSwitch":{
    "enabled":true,
    /*required, boolean, whether to enable power on or off*/
    "openTime":"10:00:00",
    /*required, string, power on time, which is accurate to minute with the time format of ISO8601*/
    "closeTime":"10:00:00"
  }
}
```

JSON_AutoSwitchCap

AutoSwitchCap message in JSON format

```
{
  "AutoSwitchCap":{
    "enabled":"true,false"
    /*required, boolean, whether to enable auto power on or off*/
  }
}
```

JSON_BasicParam

BasicParam message in JSON format

```
{
  "BasicParam":{
    /*required, basic parameters of added IoT device*/
    "channelName": ""
    /*required, channel name, string, the max. length is 32*/
  }
}
```

JSON_CalibrationStatus

CalibrationStatus message in JSON format

```
{
  "CalibrationStatus":{
    /*required, string type, installation angle calibration status: "Calibrated"-calibrated, "Uncalibrated"-uncalibrated*/
  }
}
```

JSON_Cap_CalibrationStatus

CalibrationStatus message in JSON format

```
{
  "CalibrationStatus":{
    /*required, string type, installation angle calibration status: "Calibrated"-calibrated, "Uncalibrated"-uncalibrated*/
    "@opt":"Calibrated,Uncalibrated"
  }
}
```

JSON_Cap_CaputreDescription

JSON message about capability of capturing picture in URL format

```
{
  "CaputreDescription":{
    "imageType":{
      /*required, string, picture type: "JPEG"*/
      "@opt":["JPEG"]
    },
    "URLType":{
      /*required, string, URL type: "cloudURL"*/
      "@opt":["cloudURL"]
    }
  }
}
```

JSON_Cap_CertificateSelect

CertificateSelect capability message in JSON format

```
{
  "CertificateSelect":{
    "Function":{
      /*required, function list*/
      "functionName":"HTTPS",
      /*required, function name, it should be set to one of the following values:
      "HTTPS","WebSockets","SDK_OVER_TLS","SRTP","securityLog","ieee802.1x"*/
      "certificateType":["client", "server", "CA"]
      /*required, certificate type required by this function, it should be set to one or multiple values of the following
      options: "client","server","CA"*/
    },
    "clientCertificate":{
      /*optional, optional client certificate types*/
      "@opt":["name1"]
    },
    "serverCertificate":{
      /*optional, optional server certificate types*/
      "@opt":["name2"]
    },
    "CACertificate":{
      /*optional, optional CA certificate types*/
      "@opt":["name3"]
    },
    "SRTPSecurityAlgorithm":{
      /*optional, string, optional encryption algorithms*/
      "@opt":["AES128", "AES256"]
    }
  }
}
```

```
}  
}
```

JSON_Cap_CleanUpFileList

Message about getting the capability of deleting saved files in a batch in JSON format

```
{  
  "CleanUpFileList": {  
    "@size": ,  
    /*required, int, the maximum number of files that can be deleted in a single time*/  
    "trackId":  
    /*required, boolean, No. of supported files, read-only*/  
  }  
}
```

JSON_Cap_ClearingSpaceConfig

JSON message about configuration capability of HDD clearing

```
{  
  "ClearingSpaceConfig":{  
    /*when the free space of HDD reaches the configured value of insufficientSpace, and if the value of clearingOriginalData is true, the configured HDD space size will be cleared*/  
    "insufficientSpace":{  
      /*required, int, HDD free space, which is used as the threshold to start clearing*/  
      "@min":0,  
      /*minimum value of the range*/  
      "@max":100  
      /*maximum value of the range*/  
    },  
    "clearingOriginalData":{  
      /*required, boolean, whether to enable clearing the original data*/  
      "opt":"true,false"  
    },  
    "clearingSpace":{  
      /*required, int, space size to clear*/  
      "@min":0,  
      /*minimum value of the range*/  
      "@max":100  
      /*maximum value of the range*/  
    }  
  }  
}
```

JSON_Cap_CommuMode

CommuMode capability message in JSON format

```
{
  "CommuMode":{
    "mode":{
      /*required, string type, security mode level of the private protocol: "compatibilityMode"-compatibility mode,
      "safeMode"-security mode*/
      "@opt":"compatibilityMode,safeMode"
    }
  }
}
```

JSON_Cap_ExportInfo

ExportInfo capability message in JSON format

```
{
  "ExportInfo":{
    "mediaType":{
      /*required, string, exported data type: "video", "audio", "metadata", "text", "mixed", "other". It should be the same
      as the search type. For pictures, this node should be set to "metadata"*/
      "@opt":["video", "audio", "metadata", "text", "mixed", "other"]
    },
    "playbackURLList":{
      /*required, array, file path*/
      "@size":1
    }
  }
}
```

JSON_Cap_IntelliManagement

JSON message about capability of intelligent management

```
{
  "requestURL": "",
  "statusCode": ,
  "statusString": "",
  "subStatusCode": "",
  "errorCode": ,
  "errorMsg": "",
  /*see the description of this node and the above nodes in the message of JSON_ResponseStatus*/
  "isSupportTask": true,
  /*optional, boolean type, whether it supports task: "true"-yes, "false"-no*/
  "isSupportCtrlCenter": true,
```

```
/*optional, boolean type, whether it supports control center: "true"-yes, "false"-no*/
"isSupportIntelligentSearch":true,
/*optional, boolean type, whether it supports intelligent search: "true"-yes, "false"-no*/
"isSupportExecuteControl":true,
/*optional, boolean type, whether it supports arming: "true"-yes, "false"-no*/
"isSupportDataStatistics":true,
/*optional, boolean type, whether it supports statistics: "true"-yes, "false"-no*/
"isSupportIntelligentStorage":true,
/*optional, boolean type, whether it supports intelligent storage: "true"-yes, "false"-no*/
"isSupportServer":true,
/*optional, boolean type, whether it supports service configuration: "true"-yes, "false"-no*/
"isSupportAlgorithmModel":true,
/*optional, boolean type, whether it supports algorithm model configuration: "true"-yes, "false"-no*/
"isSupportCluster":true,
/*optional, boolean type, whether it supports cluster: "true"-yes, "false"-no*/
"isSupportSearchResultDisplay":true,
/*optional, boolean type, whether it supports display configuration: "true"-yes, "false"-no*/
"isSupportTranscodeAnalysisUnit":true,
/*optional, boolean type, whether it supports transcoding analysis unit: "true"-yes, "false"-no*/
"isSupportFactoryReset":true,
/*optional, boolean type, whether it supports restoring to factory settings: "true"-yes, "false"-no*/
"isSupportUPS":true,
/*optional, boolean type, whether it supports UPS: "true"-yes, "false"-no*/
"isSupportCity":true,
/*optional, boolean type, whether it supports city: "true"-yes, "false"-no*/
"isSupportClusterEnableStatus":true,
/*optional, boolean type, whether it supports enabling cluster status: "true"-yes, "false"-no*/
"isSupportSensitivity":true,
/*optional, boolean type, whether it supports sensitivity configuration: "true"-yes, "false"-no*/
"isSupportConfidence":true,
/*optional, boolean type, whether it supports confidence configuration: "true"-yes, "false"-no*/
"isSupportEventSearch":true,
/*optional, boolean, whether it supports event search: "true"-yes, "false"-no*/
"isSupportAudioFileUpload":true,
/*optional, boolean type, whether it supports importing audio file: "true"-yes, "false"-no*/
"isSupportIntelligentMode":true,
/*optional, boolean type, whether it supports intelligent mode configuration: "true"-yes, "false"-no*/
"saveCapturedDataMaxDays":90,
/*optional, int*, the captured pictures can be saved in local HDD up to 90 days by default, when the number of days
reaches the limit, the records will be overwritten by day./
"isSupportAlgorithmTraining": true,
/*optional, boolean, whether it supports algorithm training, "true"-yes, "false"-no*/
"isSupportDataResync":true,
/*optional, boolean, whether it supports data resynchronization: "true"-yes, "false"-no*/
"isSupportCertificateImport":true,
/*optional, boolean, whether it supports importing software license certificate: "true"-yes, "false"-no*/
"isSupportFingerprint":true,
/*optional, boolean, whether it supports exporting hardware fingerprint: "true"-yes, "false"-no*/
"isSupportAuthorization":true,
/*optional, boolean, whether it supports authorization configuration of the algorithm center: "true"-yes, "false"-no*/
"isSupportLicensingDongle":true,
/*optional, boolean, whether it supports dongle: "true"-yes, "false"-no*/
```



```
"isSupportAlgorithmUpdate":true,
/*optional, boolean, whether it supports updating algorithm: "true"-yes, "false"-no*/
"isSupportModellingMode":true,
/*optional, boolean, whether it supports modeling mode configuration*/
"isSupportListLoading":true,
/*optional, boolean, whether it supports list library loading configuration*/
"isSupportTasksStatistics": true,
/*optional, boolean, whether it supports analysis task statistics*/
"isSupportTargetsStatistics": true,
/*optional, boolean, whether it supports analysis target statistics*/
"isSupportAuthAddress":true,
/*optional, boolean, whether it supports authorizing address*/
"isSupportCertificateStatus": true,
/*optional, boolean, whether it supports getting status of software authorization certificate*/
"certificateImportProductID": 555,
/*optional, int, product ID of software authorization, corresponding URI: /ISAPI/SDT/Management/certificateImport/
<productID>?format=json and /ISAPI/SDT/Management/fingerprint/<productID>?format=json*/
"algorithmAuthProductID": 556,
/*optional, int, product ID of algorithm authorization*/
"isSupportAnalysisResult":true,
/*optional, boolean, whether it supports searching for and counting the number of behavior analysis events: "true"-
yes, "false"-no*/
"isShowCenterComputing":true
/*optional, boolean, whether to display the configuration interface of algorithm resources: "true"-yes, "false"-no. The
configuration interface of algorithm resources will be displayed only when this field is "true" in the single smart
mode*/
"isSupportVehicleRegion": true,
/*optional, boolean, whether it supports vehicle region configuration*/
"isSupportStorageParam":true,
/*optional, boolean, whether it supports storage parameters configuration*/
"isSupportFingerprintByName": true,
/*optional, boolean, whether it supports exporting hardware fingerprint*/
"isSupportAnalysisResultPerimeter": true,
/*optional, boolean, whether it supports perimeter protection (including line crossing detection, intrusion detection,
region entrance detection, and region exiting detection); if this node is not returned, it indicates that the events are
not classified as perimeter protection and behavior analysis*/
}
```

See Also

JSON_ResponseStatus

JSON_Cap_PictureServerInformation

PictureServerInformation capability message in JSON format

```
{
  "PictureServerInformation":{
    "pictureServerType":{
      /*required, string type, picture storage server type: "tomact", "VRB", "cloudStorage"-cloud storage, "KMS"*/
      "@opt":"tomact,VRB,cloudStorage,KMS",
```

```
"#text":""
},
"addressingFormatType":{
/*required, string type, format type of the picture storage server address: "ipaddress"-IP address (default),
"hostname"-host name*/
"@opt":"ipaddress,hostname",
"#text":""
},
"hostName":{
/*string type, domain name of the picture storage server, the string length is between 0 and 64. This field is valid
when addressingFormatType is "hostname"*/
"@min":0,
"@max":64,
"#text":""
},
"ipv4Address":{
/*string type, IPv4 address of the picture storage server, the string length is between 0 and 64. This field is valid when
addressingFormatType is "ipaddress"*/
"@min":0,
"@max":64,
"#text":""
},
"ipv6Address":{
/*string type, IPv6 address of the picture storage server, the string length is between 0 and 128. This field is valid
when addressingFormatType is "ipaddress"*/
"@min":0,
"@max":128,
"#text":""
},
"portNo":{
/*required, integer type, port No. of the picture storage server, which is between 1024 and 65535*/
"@min":1024,
"@max":65535,
"#text":
},
"underlyingProtocol":{
/*optional, string, bottom-level protocol of the picture storage server: "HTTP", "HTTPS". This field is valid when
pictureServerType contains "cloudStorage"*/
"@opt":["http","https"]
},
"cloudStorage":{
/*parameters of the cloud storage server, which is valid when pictureServerType is "cloudStorage"*/
"cloudManageHttpPort":{
/*required, integer type, HTTP port No. for central management of the cloud storage server, which is between 1024
and 65535*/
"@min":1024,
"@max":65535,
"#text":
},
"cloudTransDataPort":{
/*required, integer type, data transmission port No. of the cloud storage server, which is between 1024 and 65535*/
"@min":1024,
```

```
"@max":65535,
"#text":
},
"cloudCmdPort":{
/*required, integer type, signaling port No. of the cloud storage server, which is between 1024 and 65535*/
"@min":1024,
"@max":65535,
"#text":
},
"cloudHeartBeatPort":{
/*required, integer type, heartbeat port No. of the cloud storage server, which is between 1024 and 65535*/
"@min":1024,
"@max":65535,
"#text":
},
"cloudStorageHttpPort":{
/*required, integer type, HTTP port No. of the cloud storage server, which is between 1024 and 65535*/
"@min":1024,
"@max":65535,
"#text":
},
"cloudUsername":{
/*required, string type, user name of the cloud storage server, the string length is between 0 and 32*/
"@min":0,
"@max":32,
"#text":""
},
"cloudPassword":{
/*required, string type, password of the cloud storage server, the string length is between 0 and 32*/
"@min":0,
"@max":32,
"#text":""
},
"cloudPoolId":{
/*required, integer type, cloud storage pool ID, which is between 1 and 4294967295. If this field is not configured by
the upper-level, this field will be set to 1 by default*/
"@min":1,
"@max":4294967295,
"#text":
},
"cloudPoolIdEx":{
/*optional, string, cloud storage pool ID, this node is valid when cloud storage pool ID of type string is supported*/
"@min":0,
"@max":0,
"#text":""
},
"cloudProtocolVersion":{
/*required, string type, protocol version of the cloud storage server, the string length is between 0 and 32*/
"@min":0,
"@max":32,
"#text":""
},
},
```

```
"cloudAccessKey":{
/*string type, cloud storage server access_key, the string length is between 0 and 64. This field is valid when
cloudProtocolVersion is "V2.0"*/
  "@min":0,
  "@max":64
},
"cloudSecretKey":{
/*string type, cloud storage server secret_key, the string length is between 0 and 64. This field is valid when
cloudProtocolVersion is "V2.0"*/
  "@min":0,
  "@max":64
}
}
}
```

JSON_Cap_POE

POE capability message in JSON format

```
{
  "POE": {
    "ipVersion": {
/*optional, supported IP address type, string type*/
      "@opt": "ipV4,ipV6",
      "#text": "ipV4"
    }
  }
}
```

JSON_Cap_RemarkList

JSON message about capability of adding remarks for videos by video ID

```
{
  "RemarkList": {
    "@size": 100,
/*maximum number of remarks can be added for once, the default value is 100*/
    "trackId": ,
/*whether it supports file No.*/
    "remarks": ,
/*whether it supports adding remarks for file*/
    "recordName":{
/*optional, string, file name of video, audio, or picture*/
      "@min":1,
      "@max":100
    },
    "recorderCode":{
/*optional, string, ID of device that collects data, which should be encrypted*/
```

```
"@min":1,
"@max":100
},
"policeCode":{
/*optional, string, ID of police that collects the data, which should be encrypted*/
"@min":1,
"@max":100
}
}
}
```

JSON_Cap_ReportCenterCfg

ReportCenterCfg capability message in JSON format

```
{
  "ReportCenterCfg":{
    "CenterID":{
/*center group No.*/
"@min": ,
"@max":
    },
    "enable":"true,false",
/*optional, boolean type, whether to enable uploading report*/
    "ChanAlarmMode":{
/*alarm channel of the center group*/
      "maxSize":1,
      "id":{
/*optional, integer type, channel ID: 1-main channel, 2-backup channel 1, 3-backup channel 2, 4-backup channel 3*/
"@min":1,
"@max":2
      },
      "chanAlarmMode":{
/*optional, string type, alarm channel mode: "T1"-T1 channel, "T2"-T2 channel, "N1"-N1 channel, "N2"-N2 channel,
"G1"-G1 channel, "G2"-G2 channel, "N3"-N3 channel, "N4"-N4 channel*/
"@opt":"T1,T2,N1,N2,G1,G2,N3,N4"
      }
    }
  }
}
```

JSON_Cap_SSD

JSON message about SSD management capability

```
{
  "SSD":{
    "isSupportSSDUpgrade":true
/*optional, boolean, whether the device supports SSD firmware upgrade (related URI: /ISAPI/ContentMgmt/
```

```
storage/ssd/<ID>/upgrade?format=json)*/
}
}
```

JSON_Cap_TimeSearchCond

Capability message about searching for recording start and end time in JSON format

```
{
  "TimeSearchCond": {
    "@size": 1024,
    /*the maximum number of search conditions that can be set*/
    "channelID": {
      /*required, int, channel No.*/
      "@min": 1,
      "@max": 256
    },
    /*value range of the channel No.*/
    },
    "streamType": {
      /*optional, string, stream type, "main,sub,adaptive"*/
      "@opt": [
        "main",
        "sub",
        "adaptive"
      ]
    }
  },
  "TimeSearchResult": {
    "@size": 1024,
    /*the maximum number of results that can be returned*/
    "channelID": {
      /*required, int, channel No.*/
      "@min": 1,
      "@max": 256
    },
    /*value range of the channel No.*/
    },
    "streamType": {
      /*optional, string, stream type, "main,sub,adaptive"*/
      "@opt": [
        "main",
        "sub",
        "adaptive"
      ]
    },
    "hasRecord": {
      /*required, boolean, whether the video exists, startTime and endTime are valid only when the value of hasRecord is
      "true"*/
      "@opt": [
        true,
        false
      ]
    }
  }
}
```

```
    },
    "startTime": {
/*optional, string, recording start time, e.g., 2010-01-01T00:00:00+08:00, this field is required when the value of
hasRecord is "true"*/
        "@min": 0,
        "@max": 64
    },
    "endTime": {
/*optional, string, recording end time, e.g., 2010-01-01T00:00:00+08:00, this field is required when the value of
hasRecord is "true"*/
        "@min": 0,
        "@max": 64
    }
}
}
```

JSON_Cap_WiresharkToUSB

JSON message about configuration capability of exporting captured packets to USB flash drive

```
{
  "WiresharkToUSB": {
    "switch": "",
    "operateCmd": "",
    "DeviceCondList": [
      {
        "DeviceCond": {
          "id": ,
          "devType": "",
          "devID": ,
          "subSys": ,
        }
      }
    ]
  }
}
```

JSON_Cap_WorkingStatus

WorkingStatus capability message in JSON format

```
{
  "WorkingStatusCap":{
    "devStatus":{
      "@opt": "0,1,2"
/*required, integer type, value of device status: 0-normal, 1-CPU usage, higher than 85%, 2-hardware error (e.g., serial
port exception)*/
    },
    "ChanStatus":{
```

```
"chanNo": "",
/*required, integer type, channel No., which starts from 1*/
"enable": "",
/*integer type, it is valid for analog channel only, 0-disable, 1-enable*/
"online": "",
/*required, integer type, online status: 0-offline, 1-online*/
"record":{
  "@opt": "0,1,2",
/*integer type, recording status: 0-recording, 1-recording exception (HDD exception), 2-recording exception (network camera offline), 3-recording exception (other reason)*/
},
"signal": "",
/*required, integer type, signal status: 0-normal, 1-signal loss*/
"linkNum": "",
/*required, integer type, number of software clients connected to this channel*/
"bitRate": ""
/*required, integer type, channel bit rate, unit: Kpbs*/
},
"HDStatus":{
  "hdNo": "",
/*required, integer type, HDD No., which starts from 1*/
"enable": "",
"status":{
  "@opt": "0,1,2",
/*required, integer type, HDD status: 0-activate, 1-sleep, 2-exception, 3-sleepy HDD error, 4-unformatted, 5-disconnected (for network HDD), 6-formatting*/
},
"volume": "",
/*required, integer type, HDD capacity, unit: MB*/
"linkNum": "",
/*integer type, *number of connections*/
"freeSpace": "1"
/*required, integer type, free space, unit: MD*/
}
}
}
```

JSON_CaputreDescription

JSON message about parameters of capturing picture in URL format

```
{
  "CaputreDescription":{
    "imageType":"JPEG",
/*required, string, picture type: "JPEG"*/
    "URLType":"cloudURL",
/*required, string, URL type: "cloudURL"-cloud storage or third party storage, "localURL"-device local storage*/
/*When the message is transmitted via ISUP protocol, the "URLType" should be "cloudURL"*/
  }
}
```


JSON_CaputreResult

JSON message about picture capture result

```
{
  "CaputreResult":{
    "URLType":"cloudURL",
    /*required, string, URL type: "cloudURL"-cloud storage or third party storage, "localURL"-device local storage*/
    "imageURL":""
  }
}
```

JSON_CertificateRevocation

CertificateRevocation message in JSON format

```
{
  "CertificateRevocation":{
    "enabled": ,
    /*required, boolean, whether to enable certificate expiry alarm*/
    "expireAlarmTime": ,
    /*required, integer, number of days to prompt in advance before the certificate is expired*/
    "detecteTime": "",
    /*required, time, detection time of certificate expiry alarm*/
    "intervalTime":
    /*required, integer, alarm interval, unit: day*/
  }
}
```

JSON_CertificateRevocationCap

CertificateRevocationCap capability message in JSON format

```
{
  "CertificateRevocationCap":{
    "enabled":{
    /*required, boolean, whether to enable certificate expiry alarm*/
      "@opt":[true, false]
    },
    "expireAlarmTime":{
    /*required, integer, number of days to prompt in advance before the certificate is expired*/
      "@min":1,
      "@max":30,
      "@def":7
    },
    "detecteTime":{
    /*required, time, detection time of certificate expiry alarm*/

```

```
"@min":"00:00",
"@max":"24:00",
"@def":"10:00"
},
"intervalTime":{
/*required, integer, alarm interval, unit: day*/
"@min":1,
"@max":30,
"@def":1
}
}
}
```

JSON_CertificateSelect

CertificateSelect message in JSON format

```
{
  "CertificateSelect":{
/*you can choose one field among clientCertificate, serverCertificate, and CACertificate to configure as needed*/
    "clientCertificate": "",
/*optional, string, client certificate type*/
    "serverCertificate": "",
/*optional, string, server certificate type*/
    "CACertificate": "",
/*optional, string, CA certificate type*/
    "SRTPSecurityAlgorithm": ""
/*optional, string, encryption algorithm*/
  }
}
```

JSON_ChanCond

ChanCond message in JSON format

```
{
  "ChanCond":{
    "chanNo": [...,...]
/*specify channel No. to get the status*/
  }
}
```

JSON_ChanStatus

ChanStatus message in JSON format

```
{
  "ChanStatus":[{
    "chanNo": ,
    /*required, integer type, channel No., which starts from 1*/
    "enable": ,
    /*integer type, it is valid for analog channel only, 0-disable, 1-enable*/
    "online": ,
    /*required, integer type, online status: 0-offline, 1-online*/
    "record": ,
    /*required, whether the device is recording, 0-no, 1-yes*/
    "recordStatus": ,
    /*integer type, recording status: 0-recording, 1-recording exception (HDD exception), 2-recording exception (network
    camera offline), 3-recording exception (other reason)*/
    "signal": ,
    /*required, integer type, signal status: 0-normal, 1-signal loss*/
    "linkNum": ,
    /*required, integer type, number of software clients connected to this channel*/
    "bitRate":
    /*required, integer type, channel bit rate, unit: Kpbs*/
  },
  {
    "chanNo": ,
    /*required, integer type, channel No., which starts from 1*/
    "enable": ,
    /*integer type, it is valid for analog channel only, 0-disable, 1-enable*/
    "online": ,
    /*required, integer type, online status: 0-offline, 1-online*/
    "record": ,
    /*required, whether the device is recording, 0-no, 1-yes*/
    "recordStatus": ,
    /*integer type, recording status: 0-recording, 1-recording exception (HDD exception), 2-recording exception (network
    camera offline), 3-recording exception (other reason)*/
    "signal": ,
    /*required, integer type, signal status: 0-normal, 1-signal loss*/
    "linkNum": ,
    /*required, integer type, number of software clients connected to this channel*/
    "bitRate":
    /*required, integer type, channel bit rate, unit: Kpbs*/
  }]
}
```

JSON_ChannelInfoList

ChannelInfoList message in JSON format

```
{
  "ChannelInfoList":{
    "ChannelInfo":[{
      "chanNo": ,
      /*required, integer type, channel No.*/
```

```
"enable": ,
/*integer type, it is valid for analog channel only, 0-disable, 1-enable*/
"name": "",
/*string type, channel name*/
"online": ,
/*integer type, online status: 0-offline, 1-online*/
"linknum":
}}
}
}
```

JSON_ChannelInfo

ChannelInfo message in JSON format

```
{
  "ChannelInfo": [{
    "chanNo": ,
/*required, integer type, channel No.*/
    "enable": ,
/*integer type, it is valid for analog channel only, 0-disable, 1-enable*/
    "name": "",
/*string type, channel name*/
    "online": ,
/*integer type, online status: 0-offline, 1-online*/
    "linknum": ,
/*number of live view channels*/
  }]
}
```

JSON_ChangedStatus

ChangedStatus message in JSON format

```
{
  ChangedStatus: {
    "seq": ,
/*32-byte integer type, alarm No., for alarm acknowledge*/
    "devStatus": ,
/*integer type, value of device status: 0-normal, 1-CPU usage, higher than 85%, 2-hardware error (e.g., serial port exception)*/
    "devCapHash": "";
/*global Hash value of device capability, which is generated by MD5, and encrypted by Base64; for NVR, the Hash value of network camera should be the calculation source*/
    "zeroChanEnable": ,
/*boolean type, whether the channel-zero is enabled: true-yes, false-no*/
    "cycleRecordEnable": true,
/*boolean type, whether to enable recording, true-yes, false-no*/
    "chanChanged": [...,...,...]
  }
}
```

```
/*integer type, channel status changes, e.g., delete, edit, add, the Hash value of this channel changed*/
  "hdChanged": [...,...,...]
/*integer type, HDD status changes, e.g., delete, edit, add*/
  "ChanStatus": [{
    "chanNo": ,
/*integer type, channel No., read-only*/
    "enable": 1,
/*integer type, it is valid for analog channel only, 0-disable, 1-enable*/
    "online": ,
/*required, integer type, online status: 0-offline, 1-online*/
    "recordStatus": ,
/*integer type, recording status: 0-recording, 1-recording exception (HDD exception), 2-recording exception (network
camera offline), 3-recording exception (other reason)*/
    "signal": ,
/*required, integer type, signal status: 0-normal, 1-signal loss*/
    "arming": ,
/*integer type, network camera arming status: 0-armed, 1-arming failed*/
  }],
  "HDStatus": [{
    "hdNo": ,
/*required, integer type, HDD No., which starts from 1*/
    "status": ,
/*required, integer type, HDD status: 0-activate, 1-sleep, 2-exception, 3-sleepy HDD error, 4-unformatted, 5-
disconnected (for network HDD), 6-formatting*/
  }],
  "CapChanged": [{
    "name": "",
/*required, root node of changed capability set*/
    "chan": ,
/*channel No., it is required when transmitting channel capability; it is optional when transmitting device capability*/
    "url": "/ISAPI/Thermal/channels/1/fireDetection/capabilities",
/*capability set URL, it is required for capability set based on ISAPI protocol*/
  }],
  {
    "name": "",
/*required, capability set name, which is usually the root node*/
    "chan": 1,
/*channel No., it is required when transmitting channel capability; it is optional when transmitting device capability*/
    "abilityType": 0x700,
/*capability type, it is required if the capability is not based on ISAPI protocol*/
  }]
}
```

JSON_CleanUpFileList

Message about deleting saved files in a batch in JSON format

```
{
  "CleanUpFileList": [{
```

```
"trackId": ""
/*required, string, file No.*/
}
]
}
```

JSON_ClearingSpaceConfig

JSON message about HDD clearing parameters

```
{
  "ClearingSpaceConfig":{
    /*when the free space of HDD reaches the configured value of insufficientSpace, and if the value of clearingOriginalData is true, the configured HDD space size will be cleared*/
    "insufficientSpace": ,
    /*required, int, HDD free space, which is used as the threshold to start clearing*/
    "clearingOriginalData": ,
    /*required, boolean, whether to enable clearing the original data*/
    "clearingSpace":
    /*required, int, space size to clear*/
  }
}
```

JSON_CommuMode

CommuMode message in JSON format

```
{
  "CommuMode":{
    "mode":""
    /*required, string type, security mode level of the private protocol: "compatibilityMode"-compatibility mode, "safeMode"-security mode*/
  }
}
```

JSON_CustomAudioID

JSON message about custom alarm sound ID

```
{
  "customAudioID":
  /*required, string, custom alarm sound ID, the value can be 1 to 3*/
}
```

JSON_CustomAudioInfo

JSON message about custom alarm sound file information

```
{
  "CustomAudioInfo": {
    /*required, related information about custom alarm sound file*/
    "customAudioID": ,
    /*required, int, custom alarm sound file ID, the value can be 1 to 3*/
    "customAudioName": ""
  }
}
```

JSON_CustomAudioTypeInfoList

JSON message about encoding format of custom audio file

```
{
  "CustomAudioTypeInfoList": [{
    /*encoding format list of custom audio file*/
    "CustomAudioTypeInfo": {
      /*encoding format of custom audio file*/
      "channelID": 1,
      /*required, channel ID*/
      "audioType": ["OPUS", "G726"],
      /*required, string, audio encoding format type: OPUS, G726; now supports analog channels only*/
      "bitRate": 64,
      /*required, int, bit rate, unit: kbps*/
      "samplingRate": 48,
      /*optional, int, sampling rate, unit: k*/
      "samplingBits": 10,
      /*optional, int, sampling bits, unit: bits*/
      "AudioChannelNum": 1,
      /*optional, int, the number of audio channels*/
      "AudioFileMaxCapacity": 10
    }
  ]
}
```

JSON_DataReconstruction

JSON message about disk data reconstruction parameters

```
{
  "DataReconstruction": {
    "dataType": "record",
  }
}
```

```
/*required, string, reconstruction data type: "all", "record"*/
  "diskID": [1,2],
/*optional, array of integers, disk ID for single disk mode*/
  "IRaid": [1,2]
/*optional, array of integers, RAID ID for RAID mode*/
}
}
```

JSON_DeviceCertificate

DeviceCertificate message in JSON format

```
{
  "DeviceCertificate":{
    "CertificateInfo":{
      "issuerDN": "",
/*required, string, read-only, name of the authority that issued the certificate*/
      "subjectDN": "",
/*required, string, read-only, certificate holder name*/
      "startDate": "",
/*required, date, read-only, start date of the certificate validity period, it is accurate to day*/
      "endDate": "",
/*required, date, read-only, end date of the certificate validity period, it is accurate to day*/
      "type": "",
/*optional, string, read-only, certificate type: "https", "websockets", "SDKoverTLS", "SRTP", "securityLog",
"ieee802.1x". Multiple types should be separated by commas*/
      "status": "",
/*optional, string, read-only, certificate status: "normal", "expired", "exceptional"*/
      "customID": ""
/*optional, string, read-only, custom certificate ID, it is created when the user imports the certificate and it consists of
digits and characters*/
    }
  }
}
```

JSON_DeviceCertificateCap

DeviceCertificateCap capability message in JSON format

```
{
  "DeviceCertificateCap":{
    "customID":{
/*required, string, custom certificate ID, it is created when the user imports the certificate and it consists of digits and
characters*/
      "@min":1,
      "@max":64,
      "status":{
/*optional, string, certificate status: "normal", "expired", "exceptional"*/
        "@opt":["normal", "expired", "exceptional"]
      }
    }
  }
}
```



```
}  
}  
}  
}
```

JSON_DeviceCertificates

DeviceCertificates message in JSON format

```
{  
  "DeviceCertificates":{  
    "CertificateInfoList":{  
      "CertificateInfo": [{  
        "id": ,  
        /*required, integer32, read-only, certificate ID, which starts from 1*/  
        "issuerDN": "",  
        /*required, string, read-only, name of the authority that issued the certificate*/  
        "subjectDN": "",  
        /*required, string, read-only, certificate holder name*/  
        "startDate": "",  
        /*required, read-only, start date of certificate expiry period, it is accurate to day*/  
        "endDate": "",  
        /*required, read-only, end date of certificate expiry period, it is accurate to day*/  
        "type": "",  
        /*optional, read-only, certificate type: "wpa", "securityLog", "ieee802.1x", "HTTPS", "WebSocketS", "SDK_OVER_TLS",  
        "SRTP". Multiple types can be separated by commas*/  
        "status": "",  
        /*optional, string, read-only, certificate status: "normal", "expired", "exceptional"*/  
        "customID": ""  
        /*optional, string, read-only, custom certificate ID, it is created when the user imports the certificate and it consists of  
        digits and characters*/  
      }]  
    }  
  }  
}
```

JSON_DeviceCertificatesCap

DeviceCertificatesCap message in JSON format

```
{  
  "DeviceCertificatesCap":{  
    "CertificateInfoList":{  
      "@size": 64,  
      /*required, integer32 type, read-only*/  
      "CertificateInfo": [{  
        "id": {  
          /*required, integer32 type, read-only, certificate ID, which starts from 1*/  
          "@min":1,  

```

```
"@max":64,
"#text":
}
"issuerDN": "",
/*required, string type, read-only, certificate issue*/
"subjectDN": "",
/*required, string type, read-only, certificate holder name*/
"startDate": "",
/*required, read-only, start date of certificate expiry period, it accurate to day*/
"endDate": ""
/*required, read-only, end date of certificate expiry period, it accurate to day*/
"type": {
/*optional, read-only, certificate type*/
"@opt": "wpa, securityLog, ieee802.1x"
}
}}
}
}
}
```

JSON_DeviceCertificateStatus

DeviceCertificateStatus message in JSON format

```
{
  "DeviceCertificateStatus": {
    /*required, status of device certificate*/
    "id": "",
    /*required, read-only, string type, certificate ID, which is generated and returned by device*/
    "status": ""
    /*required, string type, certificate status: normal, abnormal*/
  }
}
```

JSON_DeviceCertificateStatusList

DeviceCertificateStatusList message in JSON format

```
{
  "DeviceCertificateStatusList": [{
    "DeviceCertificateStatus": {
      /*required, status of device certificate*/
      "id": "",
      /*required, read-only, string type, certificate ID, which is generated and returned by device*/
      "status": ""
      /*required, string type, certificate status: normal, abnormal*/
    }
  ]
}
```

```
}}  
}
```

JSON_DiagnosisCond

DiagnosisCond message in JSON format

```
{  
  "DiagnosisCond":{  
    "command": "",  
    /*required, string type, diagnosis command*/  
    "selfdefineCmd": ""  
    /*string type, custom diagnosis command, it is valid when command is set to "selfdefine"*/  
  }  
}
```

JSON_DiagnosisCondCap

DiagnosisCondCap capability message in JSON format

```
{  
  "DiagnosisCondCap":{  
    /*diagnosis condition capability*/  
    "command":{  
      /*required, supported diagnosis command*/  
      "@opt":[  
        "system",  
        /*diagnose system*/  
        "alarm",  
        /*diagnose alarm*/  
        "peripheral",  
        /*diagnose peripheral*/  
        "wifi",  
        /*diagnose Wi-Fi network*/  
        "cloud",  
        /*diagnose cloud*/  
        "GPRS/3G/4G",  
        /*diagnose mobile network*/  
        "IPC",  
        /*diagnose network camera*/  
        "ARC",  
        /*diagnose alarm center*/  
        "selfdefine"  
        /*custom diagnosis command*/  
      ],  
    },  
    "selfdefineCmd":{  
      /*length range of the custom diagnosis command, it is valid when "selfdefine" is included in command*/  
      "@min": ,  
    }  
  }  
}
```

```
/*required, integer type, the minimum length of the custom diagnosis command*/
"@max":
/*required, integer type, the maximum length of the custom diagnosis command*/
}
}
}
```

JSON_DiagnosisResult

DiagnosisResult message in JSON format

```
{
  "DiagnosisResult":{
    "result":""
/*required, string type, diagnosis result*/
  }
}
```

JSON_DoubleVerification

DoubleVerification message in JSON format

```
{
  "DoubleVerification":{
    "enable":""
/*optional, whether enables double verification, boolean*/
  }
}
```

JSON_DownloadPackageStatus

DownloadPackageStatus message in JSON format

```
{
  "DownloadPackageStatus":{
/*required, upgrade package download status*/
    "status": "notDownload,downloading,pause,finish,incorrectPackage,hdOperationFailed",
/*required, string, download status: "notDownload"-not download, "downloading"-downloading, "pause"-paused,
"finish"-completed, "incorrectPackage"-incorrect upgrade package format, "hdOperationFailed"-HD operation failed*/
    "total": "",
/*optional, float, upgrade package total size, unit: MB, corrects to one decimal place*/
    "remain": "",
/*optional, float, remaining space, unit: MB, corrects to one decimal place*/
    "speed": ""
/*optional, float, download speed, unit: KB/s, corrects to two decimal places*/
    "remainTime": "",
/*optional, integer, estimated remaining time, unit: s*/
  }
}
```

```
"progress": 0,  
/*required, integer, progress, ranges from 0 to 100*/  
}  
}
```

JSON_DynamicHostName

DynamicHostName message in JSON format

```
{  
  "DynamicHostName": {  
    /*required, configuration parameters of dynamic domain name*/  
    "enabled": ,  
    /*required, boolean type, whether to enable dynamic domain name*/  
    "hostName": ""  
    /*required, string type, domain name*/  
  }  
}
```

JSON_DynamicHostNameCap

DynamicHostNameCap message in JSON format

```
{  
  "DynamicHostNameCap": {  
    /*required, configuration capability of dynamic domain name*/  
    "enabled": "true,false",  
    /*required, boolean type, whether to enable dynamic domain name*/  
    "hostName": {  
      /*required, string type, domain name*/  
      "@min": 0,  
      "@max": 64  
    }  
  }  
}
```

JSON_EmailCertification

Enter a short description of your reference here (optional).

Enter the syntax information of your reference here (optional).

Enter the actual information in this section (optional).

Example

Enter an example to illustrate your reference here (optional).

JSON_EnableEncryption

EnableEncryption message in JSON format

```
{
  "enable": true,
  /*required, boolean type, whether to enable stream encryption, true=yes, false=no*/
}
```

JSON_EncryptFormat

EncryptFormat message in JSON format

```
{
  "EncryptFormat":{
    "password":"","
    /*required, string, password, it will be encrypted according to the encryption vector*/
    "encryptFormatType":""
    /*optional, string, HDD formatting type: "FAT32", "EXT4"*/
  }
}
```

JSON_EncryptionCap

Stream encryption capability in JSON format.

```
{
  "enable": "true,false",
  /*required, boolean type, whether to enable stream encryption, true=yes, false=no*/
  "secretKey":{
    /*required, string type, encryption key*/
    "@min": 6,
    "@max": 12,
    "#text": "user"
  }
}
```

JSON_EncryptVerify

EncryptVerify message in JSON format

```
{
  "EncryptVerify":{
    "password":""
    /*required, string, password, it will be encrypted according to the encryption vector*/
  }
}
```

```
}  
}
```

JSON_EPTZMode

EPTZMode message in JSON format

```
{  
  "EPTZMode":{  
    "mode":"","  
    /*required, modes supported by e-PTZ: "cruise"-patrol (PTZ supports preset, patrol, zoom, and eight directions),  
    "autoTrack"-automatic tracking*/  
    "AutoTrack":{  
    /*dependency, auto-tracking mode configuration*/  
      "sensitivity": ,  
    /*optional, integer, sensitivity*/  
      "detectionTarget":["human", "vehicle"],  
    /*detection target*/  
      "ReginonList":[{  
        "id": ,  
    /*required, rule ID*/  
        "Region":[{  
    /*required, rule area, the number of edges is between 3 and 10*/  
      "x": ,  
    /*required, float, X-coordinate, range: [0.000,1]*/  
      "y":  
    /*required, float, Y-coordinate, range: [0.000,1]*/  
      }]  
    },  
    "FilterSize":{  
    /*optional, size filter of rectangle*/  
      "MaxTargetRect":{  
        "height": ,  
    /*required, float, height, range: [0.000,1.000], the value is accurate to three decimal places*/  
        "width": ,  
    /*required, float, width, range: [0.000,1.000], the value is accurate to three decimal places*/  
        "x": ,  
    /*required, float, X-coordinate, range: [0.000,1.000], the value is accurate to three decimal places. The upper-left  
corner is the origin*/  
        "y":  
    /*required, float, Y-coordinate, range: [0.000,1.000], the value is accurate to three decimal places. The upper-left  
corner is the origin*/  
      },  
      "MinTargetRect":{  
        "height": ,  
    /*required, float, height, range: [0.000,1.000], the value is accurate to three decimal places*/  
        "width": ,  
    /*required, float, width, range: [0.000,1.000], the value is accurate to three decimal places*/  
        "x": ,  
    /*required, float, X-coordinate, range: [0.000,1.000], the value is accurate to three decimal places. The upper-left  
corner is the origin*/
```

```
    "y":  
    /*required, float, Y-coordinate, range: [0.000,1.000], the value is accurate to three decimal places. The upper-left  
    corner is the origin*/  
    }  
  }  
}  
}
```

JSON_EPTZModeCap

EPTZModeCap capability message in JSON format

```
{  
  "EPTZModeCap":{  
    "mode":{  
    /*required, modes supported by e-PTZ: "cruise"-patrol (PTZ supports preset, patrol, zoom, and eight directions),  
    "autoTrack"-automatic tracking*/  
      "@opt":["cruise", "autoTrack"]  
    },  
    "AutoTrackCap":{  
    /*optional, this node is returned when mode contains "autoTrack"*/  
      "sensitivity":{  
    /*optional, integer, sensitivity*/  
        "@min":0,  
        "@max":100  
      },  
      "detectionTarget":{  
    /*detection target*/  
        "@opt":["human", "vehicle", "all"]  
      },  
      "ReginonList":{  
        "id":{  
    /*required, rule ID*/  
          "@min":1,  
          "@max":4  
        },  
        "Region":{  
    /*required, rule area, the number of edges is between 3 and 10*/  
          "number":{  
    /*required, supported number of edges*/  
            "@min":3,  
            "@max":10  
          },  
          "x":{  
    /*required, float, X-coordinate, range: [0.000,1]*/  
            "@min":0,  
            "@max":1  
          },  
          "y":{  
    /*required, float, Y-coordinate, range: [0.000,1]*/
```



```
"@min":0,
"@max":1
}
}
},
"FilterSize":{
/*optional, size filter of rectangle*/
"MaxTargetRect":{
"height":{
/*required, float, height, range: [0.000,1.000], the value is accurate to three decimal places*/
"@min":0,
"@max":1
},
"width":{
/*required, float, width, range: [0.000,1.000], the value is accurate to three decimal places*/
"@min":0,
"@max":1
},
"x":{
/*required, float, X-coordinate, range: [0.000,1.000], the value is accurate to three decimal places. The upper-left
corner is the origin*/
"@min":0,
"@max":1
},
"y":{
/*required, float, Y-coordinate, range: [0.000,1.000], the value is accurate to three decimal places. The upper-left
corner is the origin*/
"@min":0,
"@max":1
}
},
"MinTargetRect":{
"height":{
/*required, float, height, range: [0.000,1.000], the value is accurate to three decimal places*/
"@min":0,
"@max":1
},
"width":{
/*required, float, width, range: [0.000,1.000], the value is accurate to three decimal places*/
"@min":0,
"@max":1
},
"x":{
/*required, float, X-coordinate, range: [0.000,1.000], the value is accurate to three decimal places. The upper-left
corner is the origin*/
"@min":0,
"@max":1
},
"y":{
/*required, float, Y-coordinate, range: [0.000,1.000], the value is accurate to three decimal places. The upper-left
corner is the origin*/
"@min":0,
```

```
    "@max":1
  }
}
}
}
}
```

JSON_EventNotificationAlert_Alarm/EventInfo

EventNotificationAlert message with alarm or event information in JSON format.

```
{
  "ipAddress": "",
  /*required, device IPv4 address , string, the maximum size is 32 bytes*/
  "ipv6Address": "",
  /*optional, device IPv6 address, string, the maximum size is 128 bytes*/
  "portNo": ,
  /*optional, device port No., integer32*/
  "protocol": "",
  /*optional, protocol type, "HTTP, HTTPS", string, the maximum size is 32 bytes*/
  "macAddress": "",
  /*optional, MAC address, string, the maximum size is 32 bytes, e.g., 01:17:24:45:D9:F4*/
  "channelID": "",
  /*optional, device channel No., integer32*/
  "dateTime": "",
  /*optional, string, alarm/event triggered or occurred time based on ISO8601, the maximum size is 32 bytes, e.g.,
  2009-11-14T15:27Z*/
  "activePostCount": "",
  /*required, alarm/event frequency, integer32*/
  "eventType": "",
  /*required, alarm/event type, "captureResult, faceCapture,...", string, the maximum size is 128 bytes*/
  "eventState": "",
  /*required, string, the maximum size is 32 bytes, durative alarm/event status: "active"-valid, "inactive"-invalid*/
  "eventDescription": "",
  /*required, event description, string, the maximum size is 128 bytes*/
  "deviceId": "",
  /*string type, device ID*/
  "uuid": "",
  /*string type, event UUID, which is used to uniquely identify an event, the standard UUID format is xxxxxxxx-xxxx-xxxx-
  xxxx-xxxxxxxxxxxx*/
  ...
  /*optional, for different alarm/event types, the nodes are different, see the message examples in different
  applications*/
}
```

JSON_EventNotificationAlert_CertificateExpiryAlarmMsg

The certificate expiry alarm details are uploaded in JSON format of EventNotificationAlert message, here shows an example.

```
Content-Type: multipart/form-data; boundary=MIME_boundary
--MIME_boundary
Content-Type: application/json
Content-Length:

{
  "ipAddress":172.6.64.7,
  "ipv6Address": "",
  "portNo":80,
  "protocol":"HTTP",
  "macAddress":"01:17:24:45:D9:F4",
  "channelID":1,
  "dateTime":"2018-11-29T15:32:55+08:00",
  "activePostCount":1,
  "eventType":"certificateRevocation",
  "eventState":"active",
  "eventDescription":"Certificate Revocation Detection",
  /*refer to the message JSON_EventNotificationAlert_Alarm/EventInfo for description details of the above nodes*/
  "channelName":"ABC",
  /*required, channel name (camera name)*/
  "deviceId":"test0123",
  /*optional, device ID, it is also the PUID and it must be returned when passing through ISAPI event by ISUP*/
  "Result":[{"
    "customID": "",
    /*required, string, custom certificate ID*/
    "certificateType":"CA",
    /*required, string, certificate type: "CA"-CA certificate, "client/server"-client or server certificate*/
    "type":["securityLog"],
    /*optional, function type: "HTTPS", "WebSockets", "SDK_OVER_TLS", "SRTP", "securityLog", "ieee802.1x"*/
    "status":"expiring",
    /*optional, certificate status: "expiring"-expiring, "expired"-expired, "exceptional"-exceptional*/
    "startDate":"2010-04-17",
    /*required, date, start date of the certificate expiry date, it is accurate to day*/
    "endDate":"2010-04-17"
  }]
}
```

See Also

JSON_EventNotificationAlert_Alarm/EventInfo

JSON_EventNotificationAlert_HDDBadSectorEventMsg

The event details of the HDD bad sector detection are uploaded in JSON format of EventNotificationAlert message, here shows an example.

```
{
  "ipAddress": "",
  /*required, string type, IPv4 address of the alarm device, the maximum length is 32 bytes*/
  "ipv6Address": "",
  /*optional, string type, IPv6 address of the alarm device, the maximum length is 128 bytes*/
  "portNo": ,
  /*optional, integer32 type, port No. of the alarm device*/
  "protocolType": "",
  /*optional, string type, protocol type: "HTTP", "HTTPS", the maximum length is 32 bytes*/
  "macAddress": "",
  /*optional, string type, MAC address, the maximum length is 32 bytes*/
  "channelID": ,
  /*optional, integer32 type, device channel No. that triggers alarms*/
  "dateTime": "",
  /*required, string type, alarm triggering time in ISO8601 time format, e.g., 2018-03-13T19:42:27+08:00, the maximum
length is 32 bytes*/
  "activePostCount": ,
  /*required, integer32 type, times that the same alarm has been uploaded*/
  "eventType": "",
  /*required, triggered event type: "hdBadBlock"-HDD bad sector detection event*/
  "eventState": "",
  /*required, string type, event triggering status: "active"-triggered, "inactive"-not triggered (heartbeat data), the
maximum length is 32 bytes*/
  "eventDescription": "",
  /*required, string type, event description: "HDD bad sector detection event", the maximum length is 128 bytes*/
  "deviceId": "",
  /*optional, device ID, which is the PUID and should be returned for ISUP alarms, e.g., "test0123"*/
  "HDBadBlock": {
    "diskNo":
  /*required, integer32 type, HDD No.*/
  }
}
```

JSON_EventNotificationAlert_HDDHighTemperatureEventMsg

The event details of the HDD high temperature detection are uploaded in JSON format of EventNotificationAlert message, here shows an example.

```
{
  "ipAddress": "",
  /*required, string type, IPv4 address of the alarm device, the maximum length is 32 bytes*/
  "ipv6Address": "",
  /*optional, string type, IPv6 address of the alarm device, the maximum length is 128 bytes*/
  "portNo": ,
```

```
/*optional, integer32 type, port No. of the alarm device*/
"protocolType": "",
/*optional, string type, protocol type: "HTTP", "HTTPS", the maximum length is 32 bytes*/
"macAddress": "",
/*optional, string type, MAC address, the maximum length is 32 bytes*/
"channelID": ,
/*optional, integer32 type, device channel No. that triggers alarms*/
"dateTime": "",
/*required, string type, alarm triggering time in ISO8601 time format, e.g., 2018-03-13T19:42:27+08:00, the maximum
length is 32 bytes*/
"activePostCount": ,
/*required, integer32 type, times that the same alarm has been uploaded*/
"eventType": "",
/*required, triggered event type: "highHDDTemperature"-HDD high temperature detection event*/
"eventState": "",
/*required, string type, event triggering status: "active"-triggered, "inactive"-not triggered (heartbeat data), the
maximum length is 32 bytes*/
"eventDescription": "",
/*required, string type, event description: "HDD high temperature detection event", the maximum length is 128
bytes*/
"deviceId": "",
/*optional, device ID, which is the PUID and should be returned for ISUP alarms, e.g., "test0123"*/
"HighHDDTemperature": {
  "diskNo":
/*required, integer32 type, HDD No.*/
}
}
```

JSON_EventNotificationAlert_HDDImpactEventMsg

The event details of the HDD impact detection are uploaded in JSON format of EventNotificationAlert message, here shows an example.

```
{
  "ipAddress": "",
  /*required, string type, IPv4 address of the alarm device, the maximum length is 32 bytes*/
  "ipv6Address": "",
  /*optional, string type, IPv6 address of the alarm device, the maximum length is 128 bytes*/
  "portNo": ,
  /*optional, integer32 type, port No. of the alarm device*/
  "protocolType": "",
  /*optional, string type, protocol type: "HTTP", "HTTPS", the maximum length is 32 bytes*/
  "macAddress": "",
  /*optional, string type, MAC address, the maximum length is 32 bytes*/
  "channelID": ,
  /*optional, integer32 type, device channel No. that triggers alarms*/
  "dateTime": "",
  /*required, string type, alarm triggering time in ISO8601 time format, e.g., 2018-03-13T19:42:27+08:00, the maximum
length is 32 bytes*/
  "activePostCount": ,
}
```

```
/*required, integer32 type, times that the same alarm has been uploaded*/
"eventType": "",
/*required, triggered event type: "hdImpact"-HDD impact detection event*/
"eventState": "",
/*required, string type, event triggering status: "active"-triggered, "inactive"-not triggered (heartbeat data), the
maximum length is 32 bytes*/
"eventDescription": "",
/*required, string type, event description: "HDD impact detection event", the maximum length is 128 bytes*/
"deviceId": "",
/*optional, device ID, which is the PUID and should be returned for ISUP alarms, e.g., "test0123"*/
"HDImpact": {
  "diskNo":
/*required, integer32 type, HDD No.*/
}
}
```

JSON_EventNotificationAlert_HDDLowTemperatureEventMsg

The event details of the HDD low temperature detection are uploaded in JSON format of EventNotificationAlert message, here shows an example.

```
{
  "ipAddress": "",
  /*required, string type, IPv4 address of the alarm device, the maximum length is 32 bytes*/
  "ipv6Address": "",
  /*optional, string type, IPv6 address of the alarm device, the maximum length is 128 bytes*/
  "portNo": ,
  /*optional, integer32 type, port No. of the alarm device*/
  "protocolType": "",
  /*optional, string type, protocol type: "HTTP", "HTTPS", the maximum length is 32 bytes*/
  "macAddress": "",
  /*optional, string type, MAC address, the maximum length is 32 bytes*/
  "channelID": ,
  /*optional, integer32 type, device channel No. that triggers alarms*/
  "dateTime": "",
  /*required, string type, alarm triggering time in ISO8601 time format, e.g., 2018-03-13T19:42:27+08:00, the maximum
length is 32 bytes*/
  "activePostCount": ,
  /*required, integer32 type, times that the same alarm has been uploaded*/
  "eventType": "",
  /*required, triggered event type: "lowHDTemperature"-HDD low temperature detection event*/
  "eventState": "",
  /*required, string type, event triggering status: "active"-triggered, "inactive"-not triggered (heartbeat data), the
maximum length is 32 bytes*/
  "eventDescription": "",
  /*required, string type, event description: "HDD low temperature detection event", the maximum length is 128
bytes*/
  "deviceId": "",
  /*optional, device ID, which is the PUID and should be returned for ISUP alarms, e.g., "test0123"*/
  "LowHDTemperature": {
```

```
"diskNo":  
/*required, integer32 type, HDD No.*/  
}  
}
```

JSON_EventNotificationAlert_HDDSevereFaultEventMsg

The event details of the HDD severe fault detection are uploaded in JSON format of EventNotificationAlert message, here shows an example.

```
{  
  "ipAddress": "",  
  /*required, string type, IPv4 address of the alarm device, the maximum length is 32 bytes*/  
  "ipv6Address": "",  
  /*optional, string type, IPv6 address of the alarm device, the maximum length is 128 bytes*/  
  "portNo": ,  
  /*optional, integer32 type, port No. of the alarm device*/  
  "protocolType": "",  
  /*optional, string type, protocol type: "HTTP", "HTTPS", the maximum length is 32 bytes*/  
  "macAddress": "",  
  /*optional, string type, MAC address, the maximum length is 32 bytes*/  
  "channelID": ,  
  /*optional, integer32 type, device channel No. that triggers alarms*/  
  "dateTime": "",  
  /*required, string type, alarm triggering time in ISO8601 time format, e.g., 2018-03-13T19:42:27+08:00, the maximum  
length is 32 bytes*/  
  "activePostCount": ,  
  /*required, integer32 type, times that the same alarm has been uploaded*/  
  "eventType": "",  
  /*required, triggered event type: "severeHDFailure"-HDD severe fault detection event*/  
  "eventState": "",  
  /*required, string type, event triggering status: "active"-triggered, "inactive"-not triggered (heartbeat data), the  
maximum length is 32 bytes*/  
  "eventDescription": "",  
  /*required, string type, event description: "HDD severe fault detection event", the maximum length is 128 bytes*/  
  "deviceId": "",  
  /*optional, device ID, which is the PUID and should be returned for ISUP alarms, e.g., "test0123"*/  
  "SevereHDFailure": {  
    "diskNo":  
    /*required, integer32 type, HDD No.*/  
  }  
}
```

JSON_EventNotificationAlert_NetworkTrafficPrealarmMsg

JSON message about details of network traffic pre-alarm.

```
{  
  "ipAddress": "",
```

```
/*required, string, IPv4 address of security control device, the maximum size is 32 bytes*/
"ipv4Address": "",
/*optional, string, IPv6 address of security control device, the maximum size is 128 bytes*/
"portNo": ,
/*optional, integer32, port No. of security control device*/
"protocolType": "",
/*optional, string, protocol types: "HTTP,HTTPS,Ehome", the maximum size is 32 bytes*/
"macAddress": "",
/*optional, string, MAC address, the maximum size is 32 bytes*/
"channelID": ,
/*optional, integer32, device channel No. of the device that triggers alarm*/
"dateTime": "",
/*required, string, time of alarm trigger with the format of ISO8601, the maximum size is 32 bytes*/
"activePostCount": ,
/*required, integer32, number of times that an alarm has been uploaded*/
"eventType": "",
/*required, string, event type, here it should be set to "dataPrealarm"; the maximum size is 128 bytes*/
"eventState": "",
/*required, event status: "active"-triggered alarm, "inactive"-not triggered alarm, the maximum size is 32 bytes*/
"eventDescription": "",
/*required, string, event description*/
"deviceId": "",
/*optional, PUID, e.g., test0123*/
"DataPrealarm":{
/*req, traffic pre-alarm event*/
"interfaceType": "",
/*required, string, network card type: "4g,wiredAndWireless"*/
"interfaceID": ,
/*required, integer, network card ID*/
"packageType": "",
/*required, string, data plan type: "day"-daily plan, "month"-monthly plan, "year"-annual plan*/
"trafficValue": ,
/*required, float, data value in the plan, which should be accurate to three decimal places, unit: MB*/
"usedTrafficValue":
/*required, float, data usage, which should be accurate to three decimal places, unit: MB*/
}
}
```

JSON_EventNotificationAlert_vibrationDetection

Message about alarm details of vibration detection in JSON format

```
{
  "ipAddress": "",
  /*required, string, IPv4 address of alarm device, the maximum size is 32 bytes*/
  "ipv4Address": "",
  /*optional, string, IPv6 address of alarm device, the maximum size is 128 bytes*/
  "portNo": ,
  /*optional, integer32, port No. of alarm device*/
  "protocol": "",
  /*optional, string, protocol type: "HTTP, HTTPS, EHome", it should be the same as the protocolType in /ISAPI/Event/
```



```
notification/httpHosts/<ID>, and the maximum size is 32 bytes. This node should be set to "HTTP" when the JSON
message is transmitted by SDK; it should be set to "EHome" when the JSON message is transmitted by ISUP; it should
be set to "HTTPS" when the transmission protocol is HTTPS*/
"macAddress": "",
/*optional, string, MAC address, the maximum size is 32 bytes*/
"channelID": ,
/*optional, integer32, channel ID of the device that triggers alarm*/
"dateTime": "",
/*required, string, alarm triggering time in ISO8601 format, the maximum size is 32 bytes*/
"activePostCount": ,
/*required, integer32, times of the same alarm being uploaded*/
"eventType": "",
/*required, string, event type that triggers alarm, here it should be set to "vibrationDetection" (vibration detection),
the maximum size is 128 bytes*/
"eventState": "",
/*required, string, event triggering status: "active"-triggered, "inactive"-not triggered (heartbeat data), the maximum
size is 32 bytes*/
"eventDescription": "",
/*required, string, event description (e.g., vibration detection), the maximum size is 128 bytes*/
"channelName": " ",
/*optional, string, channel name (camera name), which should be the same as the channelName in /ISAPI/Streaming/
channels/<ID>*/
"deviceId": " ",
/*optional, device ID which is also the PUID, which should be returned in ISUP alarm and should be the same as the
deviceId in /ISAPI/System/Network/Ehome*/
"VibrationDetection": {
  "sensitivity":
/*optional, int, sensitivity, value range: [0-100]*/
}
}
```

JSON_EventNotificationAlert_voltageinstable

The supply voltage exception alarm is uploaded in the JSON format of EventNotificationAlert message.

```
{
  "ipAddress": "",
/*required, device IPv4 address , string, the maximum size is 32 bytes*/
  "ipv6Address": "",
/*optional, device IPv6 address, string, the maximum size is 128 bytes*/
  "portNo": ,
/*optional, device port No., integer32*/
  "protocol": "",
/*optional, protocol type, "HTTP, HTTPS", string, the maximum size is 32 bytes*/
  "macAddress": "",
/*optional, MAC address, string, the maximum size is 32 bytes, e.g., 01:17:24:45:D9:F4*/
  "channelID": "",
/*optional, device channel No., integer32*/
  "dateTime": "",
```

```
/*optional, string, alarm/event triggered or occurred time based on ISO8601, the maximum size is 32 bytes, e.g.,
2009-11-14T15:27Z*/
"activePostCount":,
/*required, alarm/event frequency, integer32*/
"eventType": "voltageinstable",
/*required, alarm/event type, string, the maximum size is 128 bytes*/
"eventState": "",
/*required, string, the maximum size is 32 bytes, durative alarm/event status: "active"-valid, "inactive"-invalid*/
"eventDescription": "",
/*required, event description, string, the maximum size is 128 bytes, "overvoltage"-high supply voltage,
"undervoltage"-low supply voltage*/
"voltageValue": ""
/*optional, supply voltage value, float type, unit: V, corrects to one decimal place*/
}
```

JSON_EventSearchCap

EventSearchCap message in JSON format

```
{
  "startTime": "2004-05-03T17:30:08Z",
  /*required, start time, ISO8601_time, string*/
  "endTime": "2004-05-03T17:30:08Z",
  /*required, end time, ISO8601_time, string*/
  "resultMaxNum": 100,
  /*required, supported maximum number of searching, int*/
  "channelIDLen":{
    /*optional, channel ID length*/
    "@min":,
    "@max":
  },
  "eventType":{
    /*event type: studentStoodUp-student stands up, accessController-access controller, videoIntercom-video intercom,
    OPTEX-OPTEX security control panel, Luminite-Luminite security control panel, GJD-GJD security control panel,
    cameraDetector-detector, securityControlPanel-security control panel, multiple selections are separated by comma,
    string*/
    "@opt": "studentStoodUp, accessController,videoIntercom, OPTEX,Luminite,GJD, cameraDetector,
securityControlPanel",
    "#text": "studentStoodUp"
  },
  "type":{
    /*optional, channel type: video-video channel, IOT-IOT channel, if this node does not exist, it indicates video channel,
    string*/
    "@opt": "video,IOT",
    "#text": "video"
  },
  "AccessController":{
    /*optional*/
    "eventType":{
      /*optional, event type, if this node does not exist, it indicates searching all supported events, string*/
      "@opt": "",
```

```
"#text": ""
},
"nameLen":{
/*optional, name length*/
"@min": ,
"@max":
},
"cardNoLen":{
/*optional, card No. length*/
"@min": ,
"@max":
},
},
"VideoIntercom":{
/*optional*/
"eventType":{
/*optional, optional, event type, if this node does not exist, it indicates searching all supported events, string*/
"@opt": "",
"#text": ""
},
"nameLen":{
/*optional, name length*/
"@min": ,
"@max":
},
"cardNoLen":{
/*optional, card No. length*/
"@min": ,
"@max":
},
},
"OPTEX":{
/*optional*/
"eventType":{
/*optional, optional, event type, if this node does not exist, it indicates searching all supported events, string*/
"@opt": "",
"#text": ""
},
},
"Luminite":{
/*optional*/
"eventType":{
/*optional, optional, event type, if this node does not exist, it indicates searching all supported events, string*/
"@opt": "",
"#text": ""
},
},
"GJD":{
/*optional*/
"eventType":{
/*optional, optional, event type, if this node does not exist, it indicates searching all supported events, string*/
"@opt": "",
```

```
"#text": ""
},
},
"CameraDetector":{
/*optional*/
"eventType":{
/*optional, optional, event type, if this node does not exist, it indicates searching all supported events, string*/
"@opt": "",
"#text": ""
},
},
"SecurityControlPanel":{
/*optional, security control panel event*/
"eventType":{
/*optional, optional, event type, if this node does not exist, it indicates searching all supported events, string*/
"@opt": ""
},
"zoneNo":{
/*required, zone No., integer*/
"size": 1,
/*required, the maximum number of items, integer*/
"@min": 1,
"@max": 16,
"#text": 1
},
}
}
```

JSON_EventSearchCond

EventSearchCond message in JSON format

```
{
"searchID": "",
/*required, string, unique record search ID.*/
"searchResultPosition": "",
/*required, initial position of search result in the list, integer, When there are multiple records, and cannot get all
records in one time searching, you can search the records followed specified position for next search*/
"maxResults": "",
/*required, the max. number of results for current search, int*/
"startTime": "",
/*required, start time, ISO8601_time, string*/
"endTime": "",
/*required, end time, ISO8601_time, string*/
"choiceChannel": [{
/*optional*/
"channelID": "",
/*optional, channel ID,string*/
}],
"eventType": "",
/*required, event type: studentStoodUp-student stands up, accessController-access controller, videoIntercom-video
```

```
intercom, OPTEX-OPTEX security control panel, Luminite-Luminite security control panel, GJD-GJD security control
panel, cameraDetector-detector, securityControlPanel-security control panel, multiple selections are separated by
comma, string*/
"type": "video,IOT",
/*optional, channel type: video-video channel, IOT-IOT channel, if this node does not exist, it indicates video channel,
string*/
"AccessController":{
  "eventType": "",
/*optional, IoT event type, if this node does not exist, it indicates searching all supported events, string*/
  "name": "",
/*optional, name*/
  "cardNo": "",
/*optional, card No.*/
},
"VideoIntercom":{
  "eventType": "",
/*optional, IoT event type, if this node does not exist, it indicates searching all supported events, string*/
  "name": "",
/*optional, name*/
  "cardNo": "",
/*optional, card No.*/
},
"OPTEX":{
  "eventType": "",
/*optional, IoT event type, if this node does not exist, it indicates searching all supported events, string*/
},
"Luminite":{
  "eventType": "",
/*optional, IoT event type, if this node does not exist, it indicates searching all supported events, string*/
},
"GJD":{
  "eventType": "",
/*optional, IoT event type, if this node does not exist, it indicates searching all supported events, string*/
},
"CameraDetector":{
  "eventType": "",
/*optional, IoT event type, if this node does not exist, it indicates searching all supported events, string*/
}
"SecurityControlPanel":{
  "eventType": "",
/*optional, IoT event type, if this node does not exist, it indicates searching all supported events, string*/
  "zoneNo": [1,2]
/*optional, zone No., it is valid when eventType values "zone", array, the sub type is integer*/
}
}
```

JSON_EventSearchResult

EventSearchResult message in JSON format

```
{
  "responseStatusStrg": "",
  /*optional, searching status: OK- Searching ended, NO MATCHES-No matched data found, MORE-Search again for
  more results, string, the max. length is 32, {dep if errcode == 1 && errMsg == ok}*/
  "numOfMatches": "",
  /*optional, returned number of results for current search, integer32, {dep if errcode == 1 && errMsg == ok}*/
  "totalMatches": "",
  /*optional, total number of matched results, integer32, {dep if errcode == 1 && errMsg == ok}*/
  "eventInfo": [{
    "captureTime": "",
    /*required, capture time, ISO8601_time, string*/
    "channelID": "",
    /*optional, camera ID, string*/
    "channelName": "",
    /*optional, camera name*/
    "picUrl": "",
    /*optional, picture URL, string*/
    "subPicUrl": "",
    /*optional, thumbnail URL, string */
    "eventType": "",
    /*required, event type: studentStoodU-student stands up, videoIntercom-video intercom, string*/
    "studentStoodUp":{
    /*optional, student stand up event information*/
      "studentsStoodUp_number": ""
    }
    /* optional, xs:integer, the number of students, who are standing up*/
  },
  "AccessController":{
    "eventType ": "",
    /*required, event type, string*/
    "name": "",
    /*optional, name*/
    "cardNo": "",
    /*optional, card No.*/
    "cardType": "",
    /*optional, card type: invalid-invalid, ordinary-normal card, disabled-card for the disabled,blacklist-card in blacklist,
    patrol-patrol card, super-super card, guest-visitor card, remove-removed card*/
    "videoChannelInfo": [{
    /*optional, video channel information*/
      "videoChannel": "",
    /*required, video channel No., integer*/
      "picUrl": "",
    /*optional, picture URL, string*/
    }]
  },
  "VideoIntercom":{
    "eventType ": "",
    /*required, event type, string*/
    "name": "",
    /*optional, name*/
    "cardNo": "",
    /*optional, card No.*/
    "cardType": "invalid,ordinary,disabled,blacklist,patrol,super,guest,remove",
```

```
/*optional, card type: invalid-invalid, ordinary-normal card, disabled-card for the disabled,blacklist-card in blacklist,
patrol-patrol card, super-super card, guest-visitor card, remove-removed card*/
  "videoChannelInfo": [{
/*optional, video channel information*/
    "videoChannel": "",
/*required, video channel No., integer*/
    "picUrl": "",
/*optional, picture URL, string*/
  }]
},
"OPTEx":{
  "eventType ": "",
/*required, event type, string*/
  "videoChannelInfo": [{
/*optional, video channel information*/
    "videoChannel": "",
/*required, video channel No., integer*/
    "picUrl": "",
/*optional, picture URL, string*/
  }]
},
"Luminite":{
  "eventType ": "",
/*required, event type, string*/
  "videoChannelInfo": [{
/*optional, video channel information*/
    "videoChannel": "",
/*required, video channel No., integer*/
    "picUrl": "",
/*optional, picture URL, string*/
  }]
},
"GJD":{
  "eventType ": "",
/*required, event type, string*/
  "videoChannelInfo": [{
/*optional, video channel information*/
    "videoChannel": "",
/*required, video channel No., integer*/
    "picUrl": "",
/*optional, picture URL, string*/
  }]
},
"CameraDetector":{
  "eventType ": "",
/*required, event type, string*/
  "videoChannelInfo": [{
/*optional, video channel information*/
    "videoChannel": "",
/*required, video channel No., integer*/
    "picUrl": "",
/*optional, picture URL, string*/
```

```
    }}
  },
  "SecurityControlPanel":{
    "eventType ": "",
    /*required, event type, string*/
    "videoChannelInfo": [{
    /*optional, video channel information*/
      "videoChannel": "",
    /*required, video channel No., integer*/
      "picUrl": "",
    /*optional, picture URL, string*/
    }]
    "zoneNo": "",
    /*optional, zone No., it is valid when eventType values "zone", integer*/
    "code":1103,
    /*optional, CID event No., it is valid when eventType values "zone", integer*/
    "zoneName": ""
  }
  }}
}
```

JSON_ExportInfo

ExportInfo message in JSON format

```
{
  "ExportInfo":{
    "mediaType": "",
    /*required, string, exported data type: "video", "audio", "metadata", "text", "mixed", "other". It should be the same
    as the search type. For pictures, this node should be set to "metadata"*/
    "playbackURLList":
    /*required, array, file path*/
  }
}
```

JSON_ExportLogToUSB

JSON message about parameters of exporting logs to USB flash drive

```
{
  "ExportLogToUSB": {
    "switch": "",
    "DeviceCondList": [
      {
        "DeviceCond": {
          "id": ,
          "devType": "",
          "devID": ,
```



```
"subSys": ,  
  }  
}  
]  
}  
}
```

JSON_ExportLogToUSBCap

JSON message about capability of exporting logs to USB flash drive

```
{  
  "ExportLogToUSBCap": {  
    "switch": {  
      "@opt": [  
        "stop",  
        "start"  
      ]  
    },  
    "DeviceCondList": [  
      {  
        "DeviceCond": {  
          "id": {  
            "@min": 1,  
            "@max": ,  
          },  
          "devType": {  
            "@opt": [  
              "mainDev",  
              "subDev"  
            ]  
          },  
          "devID": {  
            "@min": ,  
            "@max":  
          },  
          "subSys": {  
            "@min": ,  
            "@max":  
          },  
        },  
      ]  
    },  
  ]  
}
```

JSON_ExportLogToUSBStatus

JSON message about status of exporting logs to USB flash drive

```
{
  "ExportLogToUSBStatus":{
    "DeviceLogExportList":[{
      "DeviceLogExport":{
        "devType": "",
        "devID": ,
        "subSys": ,
        "switch": "",
        "exportStatus": "",
        "exceptionMsg": ""
      }
    }]
  }
}
```

JSON_ExporttoUSB_TaskInfo

TaskInfo message in JSON format

```
{
  "TaskInfo":{
    "taskId": ""
  }
  /*required, string, task ID*/
}
```

JSON_ExportUSBLog

JSON message exporting logs from USB flash drive to PC

```
{
  "ExportUSBLog": {
    "switch": "",
    /*required, string, exporting switch: "start"-start exporting, "stop"-stop exporting*/
    "exportType": "",
    /*required, string, exporting mode: "byTime"-by time, "byAll"-all files*/
    "startTime": "",
    /*optional, string, start time in ISO 8601 format, which should be accurate to millisecond, value range: [25,29], e.g.,
    "1970-01-01T00:00:00.000+08:00"*/
    "stopTime": ""
    /*optional, string, end time in ISO 8601 format, which should be accurate to millisecond, value range: [25,29], e.g.,
    "1970-01-01T00:00:00.000+08:0"*/
  }
}
```

```
}  
}
```

JSON_ExportUSBLogCap

JSON message about configuration capability of exporting logs to PC.

```
{  
  "ExportUSBLogCap": {  
    "switch": {  
      "@opt": ["stop", "start"]  
    }, /*required, string, export switch, start-start exporting, stop-stop exporting*/  
    "exportType": {  
      "@opt": ["byTime", "byAll"]  
    }, /*required, string, exporting mode: "byTime"-by time, "byAll"-all files*/  
    "startTime": {  
      "@min": ,  
      "@max":  
    }, /*optional, string, start time in ISO 8601 format, which should accurate to millisecond, range: [25,29], e.g.,:  
    "1970-01-01T00:00:00.000+08:00"*/  
    "stopTime": {  
      "@min": ,  
      "@max":  
    }, /*optional, string, end time in ISO 8601 format, which should accurate to millisecond, range: [25,29], e.g.,:  
    "1970-01-01T00:00:00.000+08:00"*/  
  }  
}
```

JSON_ExportUSBLogStatus

JSON message about status of exporting logs to PC

```
{  
  "ExportUSBLogStatus": {  
    "percent": , /*required, integer, exporting progress, range: [0-100]*/  
    "filePath": "" /*optional, string, exporting path of log files, which is valid only when percent is "100"*/  
  }  
}
```

JSON_EZVIZQRCode

JSON message about Hik-Connect QR code

```
{  
  "EZVIZQRCode": {  
    "QRCode": ""  
  }  
} /*required, string, Hik-Connect QR code string, the maximum length is 1024 bytes*/
```

```
}  
}
```

JSON_EZVIZSecretKey

EZVIZSecretKey message in JSON format

```
{  
  "EZVIZSecretKey":{  
    /*required, verification code for Hik-Connect, string, sensitive information should be encrypted, the string length  
    ranges from 0 to 64. E.g., 34020000002000000001*/  
    "secretKey": ""  
  }  
}
```

JSON_EZVIZUnbind

JSON message about parameters for unbinding devices from Hik-Connect account

```
{  
  "EZVIZUnbind":{  
    /*required, condition parameters for unbinding devices from Hik-Connect account*/  
    "password":""  
    /*required, string, write-only, login password of the admin user. This node is sensitive information that needs to be  
    encrypted*/  
  }  
}
```

JSON_FileExport

JSON message about the configuration parameters of exporting files from the device

```
{  
  "FileExport":{  
    "fileType": "",  
    /*required, string, file type: "serialLog"-serial port log, "coreDump"-core dump log, "debug"-debug log, "selfDefine"-  
    custom file type to be exported, "errorInfo"-error information*/  
    "selfDefineCmd": "",  
    /*string, custom file type to be exported, it is valid when fileType is set to "selfDefine"*/  
    "devType": "",  
    /*optional, string, device type: "mainDev" -main device(default), "subDev"-partition*/  
    "subSys": ""  
    /*optional, string, partition No., this node is valid when the value of devType is "subDev"*/  
  }  
}
```

JSON_FileExportCap

JSON message about the capability of exporting files from the device

```
{
  "FileExportCap":{
    "fileType":{
      "@opt":["serialLog","coreDump","debug","selfDefine"]
      /*required, string, file type: "serialLog"-serial port log, "coreDump"-core dump log, "debug"-debug log, "selfDefine"-
      custom file type to be exported, "errorInfo"-error information*/
    },
    "selfDefineCmd":{
      /*string, custom file type to be exported, it is valid when fileType is set to "selfDefine"*/
      "@min": ,
      /*required, int, the minimum length of the custom file type to be exported*/
      "@max":
      /*required, int, the maximum length of the custom file type to be exported*/
    }
  }
}
```

JSON_FileExportResult

FileExportResult message in JSON format

```
{
  "FileExportResult":{
    "url":""
    /*required, string type, file URL, e.g., "http://ip:port//ISAPI/ContentMgmt/logSearch/dataPackage/data?identifier"*/
  }
}
```

JSON_GetCustomAudio

JSON Message about getting the custom alarm audio file

```
{
  "CustomAudioInfo": {
    "CustomAudioList":[{
      /*required, the file list of custom alarm sound*/
      "customAudioID": ,
      /*required, int, custom alarm sound ID, the value can be 1 to 3*/
      "customAudioName": ""
      /*required, string, name of the custom alarm sound, the maximum size is 24 bytes*/
      "AudioType":"AI"
      /*optional, string, audio file type: AI-AI audio file*/
    }]
  }
}
```

```
}  
}
```

JSON_GPSCalibration

Message about configuration parameters of a single GPS calibration point in JSON format

```
{  
  "GPSCalibration":{  
    "id": ,  
    /*optional, int, calibration point ID, which is used for parsing in upper layer and cannot be configured*/  
    "calibrated":,  
    /*optional, boolean, calibration status*/  
    "longitude":,  
    /*required, float, longitude, accurate to 6 decimal places*/  
    "latitude":,  
    /*required, float, latitude, accurate to 6 decimal places*/  
    "x":,  
    /*required, float, x-coordinate, accurate to 6 decimal places*/  
    "y":  
    /*required, float, y-coordinate, accurate to 6 decimal places*/  
  }  
}
```

JSON_GPSCalibrationCap

Message about GPS calibration capability in JSON format

```
{  
  "GPSCalibrationCap":{  
    /*required, GPS calibration configuration capability*/  
    "enabled": "true,false",  
    /*required, boolean, whether the channel supports GPS calibration configuration*/  
    "id": {  
    /*required, int, calibration No.*/  
      "@min":1,  
      "@max":64,  
      "@text":1  
    },  
    "Longitude":{  
    /*required, float, longitude, accurate to 6 decimal places*/  
      "@min":-180.000000,  
      "@max":180.000000,  
      "@text":56.000000  
    },  
    "Latitude":{  
    /*required, float, latitude, accurate to 6 decimal places*/  
      "@min":-90.000000,  
      "@max":90.000000,  
    }  
  }  
}
```

```
"@text":32.000000
},
"x":{
/*required, float, x-coordinate, accurate to 6 decimal places*/
"@min":0.000000,
"@max":1.000000,
"@text":1.000000
},
"y":{
/*required, float, y-coordinate, accurate to 6 decimal places*/
"@min":0.000000,
"@max":1.000000,
"@text":1.000000
},
"showEnabled": "true,false",
/*optional, boolean, whether to enable calibration stream overlay, if it is set to "true", calibration identification will be
displayed on the live view screen*/
"calibratated":"true,false",
/*optional, boolean, calibration status*/
"isSupportCalibrationStatus":true
/*optional, boolean, whether it supports getting all the calibration status (related URI: /ISAPI/System/
GPSCalibratation/channels/<ID>/points/status?format=json)*/
}
}
```

JSON_GPSCalibratationList

Message about configuration parameters of multiple GPS calibration points in JSON format

```
{
  "GPSCalibratationList":[{
    "id": ,
/*required, int, calibration point ID*/
    "calibratated":,
/*optional, boolean, calibration status*/
    "longitude":,
/*required, float, longitude, accurate to 6 decimal places*/
    "latitude":,
/*required, float, latitude, accurate to 6 decimal places*/
    "x":,
/*required, float, x-coordinate, accurate to 6 decimal places*/
    "y":
/*required, float, y-coordinate, accurate to 6 decimal places*/
  }]
}
```

JSON_GPSCalibrationStatus

Message about GPS calibration status in JSON format

```
{
  "GPSCalibrationStatus": {
    /*required, GPS calibration status*/
    "status": ""
    /*required, string, GPS calibration status, calib-calibrated already,noCalib-Not been calibrated yet*/
  }
}
```

JSON_GuideConfig

GuideConfig message in JSON format

```
{
  "GuideConfig": {
    "GuideStep": [{
      /*required, configuration steps of quick setup instruction*/
      "stepIndex": ,
      /*required, integer, the maximum supported index*/
      "StepFunction": {
        /*required, string, instruction functions*/
        "channelID": ,
        /*optional, integer, channel number*/
        "type": ""
        /*required, "masterSlaveTracking" (linkage calibration, corresponding URL: /ISAPI/MasterSlaveTracking/channels/<ID>/
        slaveCameraCalibrating/capabilities), "mixedTargetDetection" (multi-target type detection, corresponding URL : /ISAPI/
        Intelligent/channels/<ID>/mixedTargetDetection/capabilities?format=json), eagleFocusing (rapid focus, corresponding
        URL: /ISAPI/PTZCtrl/channels/<ID>/EagleFocusing/capabilities)*/
      },
      "status": "unfinished"
      /*required, string, instruction status, whether the single configuration has completed*/
    }],
    "guideStatus": "unfinished"
    /*required, string, instruction overall status, whether all configurations have completed*/
  }
}
```

JSON_GuideConfigCap

GuideConfigCap message in JSON format

```
{
  "GuideConfigCap": {
    "GuideStep": {
      /*required, configuration steps of quick setup instruction*/
    }
  }
}
```



```
"stepIndexNum": ,
/*required, integer, the maximum supported index*/
"StepFunction": {
/*required, string, instruction functions*/
"channels": ,
/*optional, integer, channel number*/
"type": {
"@opt": ["masterSlaveTracking", "mixedTargetDetection", "eagleFocusing"]
/*required, "masterSlaveTracking" (linkage calibration, corresponding URL: /ISAPI/MasterSlaveTracking/channels/<ID>/
slaveCameraCalibrating/capabilities), "mixedTargetDetection" (multi-target type detection, corresponding URL : /ISAPI/
Intelligent/channels/<ID>/mixedTargetDetection/capabilities?format=json), eagleFocusing (rapid focus, corresponding
URL: /ISAPI/PTZCtrl/channels/<ID>/EagleFocusing/capabilities)*/
},
"guideStatus": {
/*required, string, instruction status, whether the instruction configuration has completed*/
"@opt": ["finished", "unfinished"]
}
},
"MasterSlaveTrackingCap": {
/*optional, dep:StepFunction, linkage calibration capability*/
"enabled": {
/*required, boolean, whether to enable, the default value is "true"*/
"@opt": "true,false",
"@def": "true"
},
"CalibratingMode": {
/*required, string, calibration mode*/
"@opt": "auto",
"@def": "auto"
}
},
"MixedTargetDetectionCap": {
/*optional, dep:StepFunction, capability of multi-target type detection*/
"enabled": {
/*required, boolean, whether to enable, the default value is "true"*/
"@opt": "true,false",
"@def": "true"
},
"RuleInfoCap": {
/*required, rule information*/
"maxSize": 1,
/*required, integer, number of supported rules*/
"ruleID": {
/*required, integer, rule ID*/
"@min": 1,
"@max": 1
},
"RegionCap": {
/*required, rule region capability*/
"minSize": 3,
/*required, integer, the minimum region edges*/
"maxSize": 10,
```

```
/*required, integer, the maximum region edges*/
"x": {
/*required, float, X-coordinate, range: from 0.000 to 1*/
"@min": 0.000,
"@max": 1.000,
"#text": 0.120
},
"y": {
/*required, float, Y-coordinate, range: from 0.000 to 1*/
"@min": 0.000,
"@max": 1.000,
"#text": 0.120
}
}
},
"EagleFocusingCap": {
/*optional, dep:StepFunction, rapid focus capability*/
"focusMode": {
/*required, string, focus mode*/
"@opt": "auto",
"@def": "auto"
},
"controlEnabled": {
/*required, boolean, whether to enable rapid focus, corresponding URL: /ISAPI/PTZCtrl/channels/<ID>/EagleFocusing/
control/capabilities*/
"@opt": "true,false",
"@def": "true"
}
}
}
}
```

JSON_HeatMap_CollectionDescription

CollectionDescription message in JSON format

```
{
  "CollectionDescription":{
    "searchID": "",
    /*required, string, unique search ID, the content of searchID remains unchanged when search condition remains
    unchanged.*/
    "searchResultPosition": "",
    /*required, initial position of search result list, integer type. When there are multiple records, and cannot get all
    records in one time searching, you can search the records followed specified position for next search*/
    "maxResults": "",
    /*required, number of matched records per search, integer type*/
    "timeSpanList": [{
    /*required, time interval list*/
    "startTime": "",
```

```
/*required, start time, ISO8601_time, string*/
    "endTime": "",
/*required, end time, ISO8601_time, string*/
    },
    "channels": "",
/*optional, sub type, array, integer32, related channel No., array. If the channel information is not specified, it
indicates searching all channels*/
    "statisticsType": "PDC",
/*required, string, statistic type: OLD-heat map, PDC-people counting, DURATION-people dwell time, INTERSECTION-
people counting of intersection*/
    }
}
```

JSON_HeatMap_CollectionDescriptionCap

CollectionDescriptionCap message in JSON format

```
{
  "CollectionDescriptionCap":{
    "timeRange": "",
/*required, supported time range for replenishment, unit: day, integer*/
    "timeSpanMaxNum": "",
/*required, the maximum number of time buckets supported for one time searching, integer*/
    "resultMaxNum": "",
/*required, the maximum number of items can be searched, integer*/
    "channelMaxNum": "",
/*required, the maximum number of channels supported for one time search, integer*/
    "statisticsType":{
/*required, string, statistics type: OLD-heat map, PDC-people counting statistics, DURATION-people staying time
duration, INTERSECTION-people counting of intersection*/
      "@opt": "OLD,PDC,DURATION,INTERSECTION"
    }
  }
}
```

JSON_HeatMap_CollectionResult

CollectionResult message in JSON format

```
{
  "CollectionResult":{
    "responseStatusStrg": "",
/*required, searching status: OK- Searching ended, NO MATCHES-No matched data found, MORE-Search again for
more results, string, the max. length is 32, {dep if errcode == 1 && errMsg == ok}*/
    "numOfMatches": "",
/*required, returned number of results for current search, integer32, {dep if errcode == 1 && errMsg == ok}*/
    "totalMatches": "",
/*required, total number of matched results, integer32, {dep if errcode == 1 && errMsg == ok}*/
    "targets": [{
```

```

/*optional, people gathering result*/
    "startTime": "",
/*required, start time*/
    "endTime": "",
/*required, end time*/
    "channel": "",
/*required, integer, channel No.*/
    "statisticsType": "",
/*required, string, statistic type: OLD-heat map, PDC-people counting, DURATION-people staying time, INTERSECTION-
people counting of intersection*/
    "HeatMapValue": [{
        "heatmapDataType": "",
/*optional, string, heat map data type, it is valid only when statisticsType values "PDC". "PDC_stayNum"-the number
of people dwell in the image, "PDC_leaveNum"-the number of exiting people in the image*/
        "maxHeatMapValue": ,
/*required, integer, the max. heat value, 4-byte*/
        "minHeatMapValue": ,
/*required, integer, the min. heat value, 4-byte*/
        "timeHeatMapValue": ,
/*required, integer, the average heat value, 4-byte*/
        "lineValue": ,
/*required, integer, row value, 4-byte*/
        "columnValue": ,
/*required, integer, column value, 4-byte*/
        "curNumber": ,
/*optional, integer, the number of current people, it is valid only when statisticsType values "PDC"*/
        "leaveNumber": ,
/*optional, integer, the number of exiting people, it is valid only when statisticsType values "PDC"*/
        "totalTime": ,
/*optional, integer, total dwell time, it is valid only when statisticsType values "DURATION"*/
    }],
    "heatmapURL": "",
/*optional, string, URL of pixel data for heat map image, it is valid when statisticsType values "OLD", "DURATION" or
"INTERSECTION"*/
    "PDC_stayURL": "",
/*optional, string, URL of pixel data for dwell people in heat map image, it is valid when statisticsType values "PDC"*/
    "PDC_leaveURL": "",
/*optional, string, URL of pixel data for exiting people, it is valid when statisticsType values "PDC"*/
    "arrayUnitType": ""
/*optional, string, matrix unit data type (each pixel data type of matrix information): "byte"-a byte, "short"-2 bytes,
"int"-four bytes*/
    }]
}
}

```

JSON_HDCond

HDCond message in JSON format

```

{
  "HDCond":{

```

```
"hdNo": [...,...]
/*specify the channel No. to get the corresponding HDD status*/
}
}
```

JSON_HddFormatList

HddFormatList message in JSON format

```
{
  "HddFormatList": [{
    /*required, HDD list to be formatted*/
    "HddFormat": {
      /*required, initialization parameter of a HDD*/
      "id": ,
      /*required, string type, ID of HDDs that specified to be formatted*/
      "formatType": ""
      /*optional, string type, formatting type: FAT32, EXT4; this node is only available for SD card; if this node does not exist,
      the formatting type is the default type "FAT32"*/
    }
  ]
}
```

JSON_HDStatus

HDStatus message in JSON format

```
{
  "HDStatus": [{
    "hdNo": ,
    /*required, integer type, HDD No., which starts from 1*/
    "status": ,
    /*required, integer type, HDD status: 0-activate, 1-sleep, 2-exception, 3-sleepy HDD error, 4-unformatted, 5-
    disconnected (for network HDD), 6-formatting*/
    "volume": ,
    /*required, integer type, HDD capacity, unit: MB*/
    "freeSpace":
    /*required, integer type, free space, unit: MD*/
  ]
}
```

JSON_id

id message in JSON format

```
{
  "id": "",
```

```
/*optional, integer, user ID, it returns when double verification user is added*/  
}
```

JSON_ImportCustomAudio

JSON message about information of custom alarm audio file to be imported

Import Audio File in URL format

```
{  
  "CustomAudioInfo": {  
    /*required, related information about custom alarm audio file*/  
    "customAudioName": "",  
    /*required, string, custom alarm audio file name, the maximum size is 24 bytes*/  
    "channels":[1,2],  
    /*optional, integer, ID of the channel, which the audio file should be sent to; the voice coding format of each channel  
    should be the same; if this node is empty or not configured, the audio file will be sent to all channels*/  
    "AudioType":"AI",  
    /*optional, string, audio file type: AI-AI audio file*/  
    "isCover":false  
    /*optional, boolean, whether to cover the audio file with same name: true=yes, false=no (default)*/  
  }  
}
```

Import Audio File in Binary Format

```
Content-Type: multipart/form-data; boundary=-----7e13971310878  
-----7e13971310878  
Content-Disposition: form-data; name="CustomAudioInfo";  
Content-Type: text/json  
  
{  
  "CustomAudioInfo": {  
    /*required, related information about custom alarm audio file*/  
    "customAudioName": "",  
    /*required, string, custom alarm audio file name, the maximum size is 24 bytes*/  
    "AudioType":"AI",  
    /*optional, string, audio file type: AI-AI audio file*/  
    "isCover":false  
    /*optional, boolean, whether to cover the audio file with same name: true=yes, false=no (default)*/  
  }  
}  
-----7e13971310878  
Content-Disposition: form-data; name="audioData";  
Content-Type: audio/wav  
  
Opaque Data, binary audio file  
-----7e13971310878--
```

JSON_InputParam

InputParam message in JSON format

```
{
  "InputParam":{
    "condType":["name", "enable", "online", "linknum"]
    /*enter channel name, enable or disable channel, online status, and number of connections*/
  }
}
```

JSON_IntelligentSearchCap

JSON message about intelligent search capability

```
{
  "requestURL": "",
  /*optional, string, request URL*/
  "statusCode": 1,
  /*required, int, status code*/
  "statusString": "",
  /*required, string, status description*/
  "subStatusCode": "",
  /*required, string, sub status code*/
  "errorCode": 1,
  /*optional, int, error code, this node is required when statusCode is not 1 and it corresponds to subStatusCode*/
  "errorMsg": "ok",
  /*optional, string, error details, this node is required, when statusCode is not 1 and it can be error details about a specific parameter*/
  "startTime": "2004-05-03T17:30:08+08:00",
  /*required, string, start time in ISO8601 time format. This node only indicates that the device supports this field*/
  "endTime": "2004-05-03T17:30:08+08:00",
  /*required, string, end time in ISO8601 time format. This node only indicates that the device supports this field*/
  "resultMaxNum": 100,
  /*required, int, supported maximum number of results that can be searched*/
  "channelIDLen": {
    /*optional, camera ID length*/
    "min": 1,
    "max": 32
  },
  "channelNameLen": {
    /*optional, camera name length*/
    "min": 1,
    "max": 32
  },
  "streamType": "realtime,historyvideo,localvideo",
  /*optional, string, streaming type, "realtime,historyvideo,localvideo"*/
  "targetRect": {
    /*required, target frame*/

```

```
"height":{
  "min":1,
  "max":100
},
"width":{
  "min":1,
  "max":100
},
"x":{
  "min":1,
  "max":100
},
"y":{
  "min":1,
  "max":100
}
},
"picUrlLen":{
/*required, URL length of the large picture*/
  "min":1,
  "max":128
},
"subPicUrlLen": {
/*required, length of the thumbnail URL*/
  "min": 1,
  "max": 1
},
"targetID": {
/*optional, string, linkage ID between face and human body*/
  "min": 1,
  "max": 1
},
"targetType": "vehicle",
/*string, search object type*/
"targetSize": "",
/*optional, string, target size*/
"direction": "",
/*optional, string, target direction*/
"speed": "",
/*optional, string, target speed*/
"ageGroup": "",
/*optional, string, age*/
"gender": "",
/*optional, string, gender, string, gender*/
"glass": "",
/*optional, string, wear glasses or not*/
"bag": "",
/*optional, string, whether with bag or not*/
"hat": "",
/*optional, string, whether with hat or not*/
"mask": "",
/*optional, string, whether with mask or not*/
```



```

"jacket": "",
/*optional, string, jacket type*/
"trousersType": "",
/*optional, string, trousers type*/
"hairStyle": "",
/*optional, string, hair style*/
"jacketColor": "",
/*optional, string, jacket color*/
"trousersColor": "",
/*optional, string, trousers color*/
"ride": "",
/*optional, string, ride a bike or not*/
"things": "",
/*optional, string, whether with handbag or not*/
"cyclingType": "",
/*optional, string, vehicle type*/
"cyclingPersonNumber": "",
/*optional, string, number of passengers*/
"isSupportLicense": true,
/*required, boolean type, whether it supports license plate number*/
"plateType": "unknown,92TypeCivil,arm,upDownMilitay,92TypeArm,leftRightMilitay,
02TypePersonalized,yellowTwoLine,
04NewMilitay,embassy,oneLineArm,twoLineArm,yellow1225FarmVehicle,green1325FarmVehicle,yellow1325FarmVehi
cle,motorola,coach,tempTravl,trailer,consulate,hongKongMacao,tempEntry,civilAviation,newEnergy",
/*optional, string, license plate type: license plate type: "unknown", "92TypeCivil"-92-style civil vehicle, "arm"-police
vehicle, "upDownMilitay"-military vehicle (top-bottom type), "92TypeArm"-92-style police vehicle, "leftRightMilitay"-
military vehicle (left-right type), "02TypePersonalized"-02-style customized vehicle, "yellowTwoLine"-yellow two-line
rear license plate, "04NewMilitay"-04-style new military vehicle, "embassy"-embassy vehicle, "oneLineArm"-new
armed police vehicle (one-line), "twoLineArm"-new armed police vehicle (two-line), "yellow1225FarmVehicle"-yellow
agricultural vehicle with 1225 structure, "green1325FarmVehicle"-green agricultural vehicle with 1325 structure,
"yellow1325FarmVehicle"-yellow agricultural vehicle with 1325 structure, "motorola"-motorcycle, "coach"-driver-
training vehicle, "tempTravl"-vehicle with temporary license plate, "trailer"-trailer, "consulate"-consular vehicle,
"hongKongMacao"-vehicle entering and leaving Hong Kong/Macao, "tempEntry"-temporary entry vehicle,
"civilAviation"-civil aviation license plate, "newEnergy"-new energy license plate. Multiple types should be separated
by commas*/

"vehicleColor": "white,silver,gray,black,red,deepBlue,blue,yellow,green,brown,pink,purple,deepGray,cyan,orange,unkn
own",
/*optional, string, vehicle color: "unknown", "white", "silver"-silvery, "gray", "black", "red", "deepBlue"-dark blue,
"blue", "yellow", "green", "brown", "pink", "purple", "deepGray"-dark gray, "cyan", "orange". Multiple colors should
be separated by commas*/
"plateColor": "white,yellow,blue,black,green,civilAviationBlack,civilAviationGreen,other",
/*optional, string, license plate color: "white", "yellow", "blue", "black", "green", "civilAviationBlack"-civil aviation
black, "civilAviationGreen"-civil aviation green, "0xff" or "other"-other color, "unknown"-unrecognized. Multiple colors
should be separated by commas*/

"vehicleType": "largeBus,truck,vehicle,van,buggy,pedestrian,twoWheelVehicle,threeWheelVehicle,SUVMPV,mediumBus
,motorVehicle,nonmotorVehicle,smallCar,miniCar,pickupTruck,unknown",
/*optional, string, vehicle type: "largeBus"-large-sized bus, "truck"-truck, "vehicle"-salon car, "van"-minivan, "buggy"-
light truck, "pedestrian", "twoWheelVehicle"-two wheeler, "threeWheelVehicle"-tricycle, "SUVMPV"-SUV/MPV,
"mediumBus"-middle-sized bus, "motorVehicle"-motor vehicle (it will be transformed to "vehicle" (salon car) in the
platform), "nonmotorVehicle"-non-motor vehicle (it will be transformed to "threeWheelVehicle" (tricycle) in the

```

platform), "smallCar"-small sedan (it will be transformed to "vehicle" (salon car) in the platform), "miniCar"-mini sedan (it will be transformed to "vehicle" (salon car) in the platform), "pickupTruck"-pick-up truck, "unknown".

Multiple types should be separated by commas*/

```
"isSupportVehicleLogo":true,
/*optional, boolean type, whether it supports vehicle parent brand*/
"isSupportVehicleSubLogo":true,
/*optional, boolean type, whether it supports vehicle sub brand*/
"isSupportVehicleModel":true,
/*optional, boolean type, whether it supports vehicle model year*/
"pilotSafebelt":"","
/*optional, string type, whether the driver buckled up*/
"pilotSunvisor":"","
/*optional, string type, whether the driver's sun visor is pulled down*/
"vicePilotSafebelt":"","
/*optional, string type, whether the front passenger buckled up*/
"vicePilotSunvisor":"","
/*optional, string type, whether the front passenger's sun visor is pulled down*/
"uphone":"","
/*optional, string type, whether the person is making a call*/
"dangmark":"","
/*optional, string, whether it is dangerous goods vehicle*/
"envprosign":"","
/*optional, string, whether to display yellow label vehicle*/
"muckTruck":"","
/*optional, string, whether it is a dump truck*/
"vehicleState":"","
/*optional, string, license plate status*/
"pdvs":"","
/*optional, string, whether there are people sticking out of the sunroof*/
"vehicleHead":"","
/*optional, string, license plate recognition direction*/
"pendant":"","
/*optional, string, whether there is window hangings detected*/
"temporaryLicense":"","
/*optional, string, whether it is a temporary license plate*/
"tricycleCanopy":"","
/*optional, string, whether there is a hood on the tricycle*/
"decoration":"","
/*optional, string, whether there is decoration detected*/
"tissueBox":"","
/*optional, string, whether there is tissue box detected*/
"card":"","
/*optional, string, whether it is with cards*/
"cardType":"","
/*optional, string, card type*/
"copilot":"","
/*optional, string, whether there is a person in the co-driver room*/
"frontChild":"","
/*optional, string, whether the co-driver is with baby in arm*/
"sunroof":"","
/*optional, string, whether it is with a sunroof*/
"luggageRack":"","
```

```
/*optional, string, whether it is with a luggage rack*/
"vehicleSprayPainted": "",
/*optional, string, whether the vehicle is painted*/
"spareTire": "",
/*optional, string, whether it is with a spare tire*/
"coverPlate": "",
/*optional, string, whether the heavy truck is covered with back cover*/
"label": "",
/*optional, string, whether there is inspection label detected*/
"plateRecogRegionIDNum":
/*optional, number of IDs of the license plate recognition region*/ {
  "min": 1,
  "max": 1
},
"discuss": {
/*optional, discussion*/
  "confidence": 99.99990,
/*optional, float, confidence*/
  "value": ""
/*optional, discussion type, "unknown", "no", "yes"*/
},
"channelMaxNum":1,
/*required, integer type, maximum number of channels that support simultaneous search*/
"supportSearchByUTC":true
/*optional, boolean type, whether it supports searching by UTC*/
"linkFaceBodyId": {
  "@min": 0,
  "@max": 64
},
/*optional, linkage ID between human face and body, the maximum length is 64 bytes*/
"sortType": {
/*optional, string, sorting type: similarity-by similarity(by default), time-by time, the maximum length is 32 bytes*/
  "@opt":["similarity", "time"]
},
"plateCategory": {
/*optional, additional information of license plate*/
  "@min": 0,
  "@max": 0
}
}
```

JSON_IntelligentSearchCondition

JSON message about intelligent search conditions

```
{
  "searchID": "",
/*required, string, search ID. The content of searchID remains unchanged if the search conditions remain unchanged.
This node is used to check the same search. When the device performance is limited, you can search asynchronously
by applying conditions with the same searchID several times and getting the search progress*/
  "searchResultPosition": ,
```

```
/*required, int, the start position of the search result in the result list. When there are multiple records and you
cannot get all search results at a time, you can search for the records after the specified position next time*/
"maxResults": ,
/*required, int, maximum number of search results obtained this time*/
"startTime": "",
/*required, string, start time in ISO8601 time format*/
"endTime": "",
/*required, string, end time in ISO8601 time format*/
"choiceChannel": [{
/*optional*/
"channelID": "",
/*optional, string, camera No.*/
"plateRecogRegionID":
/*optional, int, license plate recognition area No. If this node is not configured, it refers to all areas*/
}],
"targetType": "",
/*optional, string, search object type. If this node is not returned, it indicates no limit*/
"targetSize": "",
/*optional, string, target size*/
"direction": "",
/*optional, string, target direction*/
"speed": "",
/*optional, string, target speed*/
"sortType": "",
/*optional, string, sorting type: "similarity"-by similarity(by default), "time"-by time, the maximum length is 32 bytes*/
"humanInfo": {
/*dependent, this node is valid only when the value of targetType is human*/
"ageGroup": "",
/*optional, string, age*/
"gender": "",
/*optional, string, gender, string, gender*/
"glass": "",
/*optional, string, wear glasses or not*/
"bag": "",
/*optional, string, whether with bag or not*/
"hat": "",
/*optional, string, whether with hat or not*/
"mask": "",
/*optional, string, whether with mask or not*/
"jacketType": "",
/*optional, string, jacket type*/
"trousersType": "",
/*optional, string, trousers type*/
"hairStyle": "",
/*optional, string, hair style*/
"jacketColor": "",
/*optional, string, jacket color, "white,black"*/
"trousersColor": "",
/*optional, string, trousers color, "white,black"*/
"ride": "",
/*optional, string, ride a bike or not*/
"things": "",
```

```
/*optional, string, whether with handbag or not*/
  "cyclingType":"","
/*optional, string, vehicle type*/
  "cyclingPersonNumber":"","
/*optional, string, number of passengers*/
  "hID":"","
/*required, string, person ID, the maximum length is 32 bytes*/
  "name":"","
/*required, string, name of the person, the maximum length is 32 bytes*/
  "linkFaceBodyId":"","
/*optional, linkage ID between human face and body, the maximum length is 64 bytes*/
},
"vehicleInfo ":{
/*this node is valid when targetType is "vehicle"*/
  "license":"","
/*optional, string, license plate number*/
  "plateCategory":"","
/*optional, additional information of license plate*/
  "plateType":"","
/*optional, string, license plate type: license plate type: "unknown", "92TypeCivil"-92-style civil vehicle, "arm"-police
vehicle, "upDownMilitay"-military vehicle (top-bottom type), "92TypeArm"-92-style police vehicle, "leftRightMilitay"-
military vehicle (left-right type), "02TypePersonalized"-02-style customized vehicle, "yellowTwoLine"-yellow two-line
rear license plate, "04NewMilitay"-04-style new military vehicle, "embassy"-embassy vehicle, "oneLineArm"-new
armed police vehicle (one-line), "twoLineArm"-new armed police vehicle (two-line), "yellow1225FarmVehicle"-yellow
agricultural vehicle with 1225 structure, "green1325FarmVehicle"-green agricultural vehicle with 1325 structure,
"yellow1325FarmVehicle"-yellow agricultural vehicle with 1325 structure, "motorola"-motorcycle, "coach"-driver-
training vehicle, "tempTravl"-vehicle with temporary license plate, "trailer"-trailer, "consulate"-consular vehicle,
"hongKongMacao"-vehicle entering and leaving Hong Kong/Macao, "tempEntry"-temporary entry vehicle,
"civilAviation"-civil aviation license plate, "newEnergy"-new energy license plate. If this node is not returned, it
indicates no limit*/
  "vehicleColor":"","
/*optional, string, vehicle color: "unknown", "white", "silver"-silvery, "gray", "black", "red", "deepBlue"-dark blue,
"blue", "yellow", "green", "brown", "pink", "purple", "deepGray"-dark gray, "cyan", "orange". Multiple colors should
be separated by commas. If this node is not returned, it indicates no limit*/
  "vehicleType":"","
/*optional, string, vehicle type: "largeBus"-large-sized bus, "truck"-truck, "vehicle"-salon car, "van"-minivan, "buggy"-
light truck, "pedestrian", "twoWheelVehicle"-two wheeler, "threeWheelVehicle"-tricycle, "SUVMPV"-SUV/MPV,
"mediumBus"-middle-sized bus, "unknown". If this node is not returned, it indicates no limit*/
  "vehicleLogo": ,
/*optional, int, vehicle parent brand. If this node is not returned, it indicates no limit*/
  "vehicleSubLogo": ,
/*optional, int, vehicle sub brand. If this node is not returned, it indicates no limit*/
  "vehicleModel": ,
/*optional, int, vehicle model year. If this node is not returned, it indicates no limit*/
  "plateColor":"","
/*optional, string, license plate color: "white", "yellow", "blue", "black", "green", "civilAviationBlack"-civil aviation
black, "civilAviationGreen"-civil aviation green, "0xff" or "other"-other color. If this node is not returned, it indicates no
limit*/
  "pilotSafebelt":"","
/*optional, string, whether the driver buckled up. If this node is not returned, it indicates no limit*/
  "pilotSunvisor":"","
/*optional, string, whether the driver's sun visor is pulled down. If this node is not returned, it indicates no limit*/
```

```
"vicePilotSafebelt": "",
/*optional, string, whether the front passenger buckled up*/
"vicePilotSunvisor": "",
/*optional, string, whether the front passenger's sun visor is pulled down*/
"uphone": "",
/*optional, string, whether the person is making a call*/
"dangmark": "",
/*optional, string, whether it is dangerous goods vehicle*/
"vehicleState": "",
/*optional, string, license plate status*/
"pdvs": "",
/*optional, string, whether there are people sticking out of the sunroof*/
"vehicleHead": "",
/*optional, string, license plate recognition direction*/
"pendant": "",
/*optional, string, whether there is window hangings detected*/
"temporaryLicense": "",
/*optional, string, whether it is a temporary license plate*/
"tricycleCanopy": "",
/*optional, string, whether there is a hood on the tricycle*/
"decoration": "",
/*optional, string, whether there is decoration detected*/
"tissueBox": "",
/*optional, string, whether there is tissue box detected*/
"card": "",
/*optional, string, whether it is with cards*/
"cardType": "",
/*optional, string, card type*/
"copilot": "",
/*optional, string, whether there is a person in the co-driver room*/
"frontChild": "",
/*optional, string, whether the co-driver is with baby in arm*/
"muckTruck": "",
/*optional, string, whether it is a dump truck*/
"sunroof": "",
/*optional, string, whether it is with a sunroof*/
"luggageRack": "",
/*optional, string, whether it is with a luggage rack*/
"vehicleSprayPainted": "",
/*optional, string, whether the vehicle is painted*/
"spareTire": "",
/*optional, string, whether it is with a spare tire*/
"coverPlate": "",
/*optional, string, whether the heavy truck is covered with back cover*/
"label": "",
/*optional, string, whether there is inspection label detected*/
},
"twoWheelVehicle": {
/*dependent, this node is valid only when the value of targetType is "twoWheelVehicle"*/
"ageGroup": "",
/*optional, string, whether to display age group*/
"gender": "",
```

```
/*optional, string, whether to display gender*/
  "glass": "",
/*optional, string, whether to display glasses*/
  "bag": "",
/*optional, string, whether to display back bag*/
  "hat": "",
/*optional, string, whether to display hat*/
  "mask": "",
/*optional, string, whether to display mask*/
  "jacketType": "",
/*optional, string, whether to display jacket type*/
  "hairStyle": "",
/*optional, string, whether to display hairstyle*/
  "jacketColor": "",
/*optional, string, whether to display jacket color*/
  "cyclingType": "",
/*optional, string, whether to display cycling type*/
  "cyclingPersonNumber": ""
/*optional, string, whether to display number of people cycling*/
},
"threeWheelVehicle": {
/*dependent, this node is valid only when the value of targetType is "threeWheelVehicle"*/
  "ageGroup": "",
/*optional, string, whether to display age group*/
  "gender": "",
/*optional, string, whether to display gender*/
  "glass": "",
/*optional, string, whether to display glasses*/
  "bag": "",
/*optional, string, whether to display back bag*/
  "hat": "",
/*optional, string, whether to display hat*/
  "mask": "",
/*optional, string, whether to display mask*/
  "jacketType": "",
/*optional, string, whether to display jacket type*/
  "hairStyle": "",
/*optional, string, whether to display hairstyle*/
  "jacketColor": "",
/*optional, string, whether to display jacket color*/
  "cyclingPersonNumber": ""
/*optional, string, whether to display number of people cycling*/
}
}
```

JSON_IntelligentSearchResult

JSON message about intelligent search results

```
{
  "requestURL": "",
```

```
/*required, string type, request URL*/
"statusCode": ,
/*required, int type, status code*/
"statusString": "",
/*required, string type, status description*/
"subStatusCode": "",
/*required, string type, sub status code*/
"errorCode": ,
/*optional, int type, error code, this node is required when statusCode is not 1 and it corresponds to subStatusCode*/
"errorMsg": "",
/*optional, string type, error details, this node is required, when statusCode is not 1 and it can be error details about a
specific parameter*/
"responseStatusStr": "",
/*required, string type, searching status: "OK"-searching completed, "NO MATCH"-no matched results, "MORE"-
searching for more results. The maximum length is 32 bytes. This node is valid when errorCode is 1 and errorMsg is
"ok"*/
"numOfMatches": ,
/*required, integer32, number of results returned this time. This node is valid when errorCode is 1 and errorMsg is
"ok"*/
"totalMatches": ,
/*required, integer32, total number of matched results. This node is valid when errorCode is 1 and errorMsg is "ok"*/
"progress": ,
/*optional, integer32 type, VCA search progress*/
"targetInfo": {
/*this node is valid when progress is 100*/
"captureTime": "",
/*required, string type, capture time in ISO8601 time format*/
"channelID": "",
/*optional, string type, camera No.*/
"channelName": "",
/*optional, string type, camera name*/
"streamType": "",
/*optional, string type, streaming type: "realtime,historyvideo,localvideo"*/
"targetRect": {
/*required, target frame*/
"height": ,
"width": ,
"x": ,
"y": ,
},
"picUrl": "",
/*optional, string type, large picture URL*/
"subPicUrl": "",
/*optional, string, thumbnail URL*/
"targetType": "",
/*optional, string type, search object type*/
"targetSize": "",
/*optional, string, target size*/
"direction": "",
/*optional, string, target direction*/
"speed": "",
/*optional, string, target speed*/
```



```
"modelKey": "",
/* optional, string, mark of human body which is used to raise up the efficiency of searching for human body by
picture*/
  "AbsoluteHigh": {
/* optional, PTZ position*/
    "elevation": ,
/* optional, int, tilting parameter, the value is between -900 and 2700*/
    "azimuth": ,
/* optional, int, panning parameter, the value is between 0 and 3600*/
    "absoluteZoom":
/* optional, int, zooming parameter, the value is between 1 and 1000*/
  },
  "humanInfo":{
/* dependent, this node is valid only when the value of targetType is human*/
    "ageGroup": "",
/* optional, string, age*/
    "gender": "",
/* optional, string, gender, string, gender*/
    "glass": "",
/* optional, string, wear glasses or not*/
    "bag": "",
/* optional, string, whether with bag or not*/
    "hat": "",
/* optional, string, whether with hat or not*/
    "mask": "",
/* optional, string, whether with mask or not*/
    "jacketType": "",
/* optional, string, jacket type*/
    "trousersType": "",
/* optional, string, trousers type*/
    "hairStyle": "",
/* optional, string, hair style*/
    "jacketColor": "",
/* optional, string, jacket color*/
    "trousersColor": "",
/* optional, string, trousers coloe*/
    "ride": "",
/* optional, string, ride a bike or not*/
    "things": "",
/* optional, string, whether with handbag or not*/
    "cyclingType": "",
/* optional, string, vehicle type*/
    "cyclingPersonNumber": "",
/* optional, string, number of passengers*/
    "posture":{
/* optional, posture*/
      "confidence":,
/* optional, float, confidence*/
      "value": ""
    },
/* optional, string, posture type*/
  },
  "listening":{
```

```
/*optional, listening*/
  "confidence";
/*optional, float, listening*/
  "value":""
/*optional, string, listening type*/
  },
  "reading":{
/*optional, reading*/
  "confidence";
/*optional, float, confidence*/
  "value":""
/*optional, string, reading type*/
  },
  "writing":{
/*optional, writing*/
  "confidence";
/*optional, float, confidence*/
  "value":""
/*optional, string, writing type*/
  },
  "raiseHand":{
/*optional, raise hand*/
  "confidence";
/*optional, float, confidence*/
  "value":""
/*optional, string, hand raising type*/
  },
  "lyingOnTable":{
/*optional, lying on the table*/
  "confidence";
/*optional, float, confidence*/
  "value":""
/*optional, string, types of lying on the table*/
  },
  "playPhone":{
/*optional, playing phone*/
  "confidence";
/*optional, float, confidence*/
  "value":""
/*optional, string , types of playing phone*/
  },
  "orientation":{
/*optional, orientation*/
  "confidence";
/*optional, float, confidence*/
  "value":""
/*optional, string, types of orientation*/
  },
  "discuss":{
/*optional, discussion*/
  "confidence";
/*optional, float, confidence*/
```

```
    "value": ""
  /*optional, string, types of discussion: unknown, no discussion, have discussion*/
  },
  "studentScene": "",
  /*optional, string, school scene: "takeLesson", "answer", "discuss"*/
  "studentNum": ,
  /*optional, int, number of students*/
  "linkFaceBodyId": ""
  /*optional, linkage ID between human body and face, the maximum length is 64 bytes*/
  },
  "vehicleInfo": {
  /*this node is valid when targetType is "vehicle"*/
    "license": "",
  /*optional, string type, license plate number*/
    "plateCategory": "",
  /*optional, additional information of license plate*/
    "plateType": "",
  /*optional, string type, license plate type: license plate type: "unknown", "92TypeCivil"-92-style civil vehicle, "arm"-
  police vehicle, "upDownMilitary"-military vehicle (top-bottom type), "92TypeArm"-92-style police vehicle,
  "leftRightMilitary"-military vehicle (left-right type), "02TypePersonalized"-02-style customized vehicle, "yellowTwoLine"-
  yellow two-line rear license plate, "04NewMilitary"-04-style new military vehicle, "embassy"-embassy vehicle,
  "oneLineArm"-new armed police vehicle (one-line), "twoLineArm"-new armed police vehicle (two-line),
  "yellow1225FarmVehicle"-yellow agricultural vehicle with 1225 structure, "green1325FarmVehicle"-green agricultural
  vehicle with 1325 structure, "yellow1325FarmVehicle"-yellow agricultural vehicle with 1325 structure, "motorola"-
  motorcycle, "coach"-driver-training vehicle, "tempTravel"-vehicle with temporary license plate, "trailer"-trailer,
  "consulate"-consular vehicle, "hongKongMacao"-vehicle entering and leaving Hong Kong/Macao, "tempEntry"-
  temporary entry vehicle, "civilAviation"-civil aviation license plate, "newEnergy"-new energy license plate*/
    "vehicleColor": "",
  /*optional, string type, vehicle color: "unknown", "white", "silver"-silvery, "gray", "black", "red", "deepBlue"-dark
  blue, "blue", "yellow", "green", "brown", "pink", "purple", "deepGray"-dark gray, "cyan", "orange". Multiple colors
  should be separated by commas*/
    "vehicleType": "",
  /*optional, string type, vehicle type: "largeBus"-large-sized bus, "truck"-truck, "vehicle"-salon car, "van"-minivan,
  "buggy"-light truck, "pedestrian", "twoWheelVehicle"-two wheeler, "threeWheelVehicle"-tricycle, "SUVMPV"-SUV/
  MPV, "mediumBus"-middle-sized bus, "unknown"*/
    "vehicleLogo": ,
  /*optional, int type, vehicle parent brand*/
    "vehicleSubLogo": ,
  /*optional, int type, vehicle sub brand*/
    "vehicleModel": ,
  /*optional, int type, vehicle model year*/
    "plateColor": "",
  /*optional, string type, license plate color: "white", "yellow", "blue", "black", "green", "civilAviationBlack"-civil
  aviation black, "civilAviationGreen"-civil aviation green, "0xff" or "other"-other color.*/
    "pilotSafebelt": "",
  /*optional, string type, whether the driver buckled up*/
    "pilotSunvisor": "",
  /*optional, string type, whether the driver's sun visor is pulled down*/
    "vicePilotSafebelt": "",
  /*optional, string type, whether the front passenger buckled up*/
    "vicePilotSunvisor": "",
  /*optional, string type, whether the front passenger's sun visor is pulled down*/
```

```
"uphone":"","  
/*optional, string type, whether the person is making a call*/  
"dangmark":"","  
/*optional, string, whether it is dangerous goods vehicle*/  
"vehicleState":"","  
/*optional, string, license plate status*/  
"pdvs":"","  
/*optional, string, whether there are people sticking out of the sunroof*/  
"vehicleHead":"","  
/*optional, string, license plate recognition direction*/  
"pendant":"","  
/*optional, string, whether there is window hangings detected*/  
"temporaryLicense":"","  
/*optional, string, whether it is a temporary license plate*/  
"tricycleCanopy":"","  
/*optional, string, whether there is a hood on the tricycle*/  
"decoration":"","  
/*optional, string, whether there is decoration detected*/  
"tissueBox":"","  
/*optional, string, whether there is tissue box detected*/  
"card":"","  
/*optional, string, whether it is with cards*/  
"cardNum": ,  
/*optional, int, number of cards*/  
"cardType": [{  
/*optional, array, card type*/  
"value": ""  
/*optional, string, value of different card types*/  
}],  
"copilot":"","  
/*optional, string, whether there is a person in the co-driver room*/  
"frontChild":"","  
/*optional, string, whether the co-driver is with baby in arm*/  
"muckTruck":"","  
/*optional, string, whether it is a dump truck*/  
"sunroof":"","  
/*optional, string, whether it is with a sunroof*/  
"luggageRack":"","  
/*optional, string, whether it is with a luggage rack*/  
"vehicleSprayPainted":"","  
/*optional, string, whether the vehicle is painted*/  
"spareTire":"","  
/*optional, string type, whether it is with a spare tire*/  
"coverPlate":"","  
/*optional, string, whether the heavy truck is covered with back cover*/  
"label":"","  
/*optional, string, whether there is inspection label detected*/  
"labelNum": ,  
/*optional, int, number of labels*/  
"labelPosition": [{  
/*optional, label position*/  
"height": ,
```

```
/*required, float, height*/
  "width": ,
/*required, float, width*/
  "x": ,
/*required, float, X-coordinate*/
  "y":
/*required, float, Y-coordinate*/
  }},
  "region": "",
/*optional, string type, region that the license plate belongs to: "EU"-Europe, "ER"-Russian region, "EUandCIS"-Europe
and Russia, "ME"-the Middle East, "All"-all regions, "other"*/
  "country": ""
/*optional, string type, country that the license plate belongs to: "EU"-Europe, "ER"-Russian region, "EUandCIS"-
Europe and Russia, "ME"-the Middle East, "All"-all regions, "other"*/
  },
  "twoWheelVehicle": {
/*dependent, this node is valid only when the value of targetType is "twoWheelVehicle"*/
    "ageGroup": "",
/*optional, string, whether to display age group*/
    "gender": "",
/*optional, string, whether to display gender*/
    "glass": "",
/*optional, string, whether to display glasses*/
    "bag": "",
/*optional, string, whether to display back bag*/
    "hat": "",
/*optional, string, whether to display hat*/
    "mask": "",
/*optional, string, whether to display mask*/
    "jacketType": "",
/*optional, string, whether to display jacket type*/
    "hairStyle": "",
/*optional, string, whether to display hairstyle*/
    "jacketColor": "",
/*optional, string, whether to display jacket color*/
    "cyclingType": "",
/*optional, string, whether to display cycling type*/
    "cyclingPersonNumber": ""
/*optional, string, whether to display number of people cycling*/
  },
  "threeWheelVehicle": {
/*dependent, this node is valid only when the value of targetType is "threeWheelVehicle"*/
    "ageGroup": "",
/*optional, string, whether to display age group*/
    "gender": "",
/*optional, string, whether to display gender*/
    "glass": "",
/*optional, string, whether to display glasses*/
    "bag": "",
/*optional, string, whether to display back bag*/
    "hat": "",
/*optional, string, whether to display hat*/
```

```
"mask": "",
/*optional, string, whether to display mask*/
"jacketType": "",
/*optional, string, whether to display jacket type*/
"hairStyle": "",
/*optional, string, whether to display hairstyle*/
"jacketColor": "",
/*optional, string, whether to display jacket color*/
"cyclingType": "",
/*optional, string, whether to display cycling type*/
"cyclingPersonNumber": ""
/*optional, string, whether to display number of people cycling*/
},
"humanID": "",
/*optional, string, face picture library ID*/
"humanURL": "",
/*optional, string, face picture URL*/
"name": "",
/*optional, string, name of the student*/
"ClassRegions": [{
/*optional, coordinates of the people in the classroom*/
  "x": ,
  "y":
}]
}]
}
```

JSON_IOStatus

JSON message about alarm input and output status

```
{
  "IOStatus": {
    "IOInTrig": [...,...,...],
    /*required, int, triggered alarm input No., e.g., 3201-alarm input No.1 of digital channel No.32*/
    "IOOutTrig": [...,...,...]
  }
}
```

JSON_IOT_ChannelInfo

ChannelInfo message in JSON format

```
{
  "ChannelInfo":{
    "channel": "",
    /*required, channel No., it should be returned when the channel is added, integer*/
  }
}
```

```
}  
}
```

JSON_IOT_ChannelInfoList

ChannelInfoList message in JSON format

```
{  
  "ChannelInfoList": [{  
    /*optional, channel information, if this node does not exist, it indicates all channels, array*/  
    "channel": "",  
    /*required, channel No., integer*/  
    "type": ""  
    /*required, channel type: "video"-video channel, "IOT"-IoT channel, string*/  
  }]  
}
```

JSON_IOT_ErrorList

ErrorList message in JSON format

```
{  
  "ErrorList": [{  
    /*dep, detailed error information, it is valid when subStatusCode values "badParameters"*/  
    "errorRowNo": "",  
    /*required, error row No., integer*/  
    "errorType": "",  
    /*required, error type, string, channelNoInvalid-invalid channel No., channelNoConflict-conflicted channel No.,  
    channel IP/Domain invalid-invalid channel IP or domain name, channel IP/Domain conflict-conflicted channel IP or  
    domain name, "channel IP conflict with local IP"-channel IP is conflicted with local IP, protocolError-incorrect protocol,  
    adminPortError-incorrect admin port, channelError-incorrect channel No., UserNameInvalid-invalid user name,  
    passwordInvalid-invalid password, transProtocolError-incorrect transmission protocol, inductiveTypeInvalid-invalid  
    inductive type*/  
  }]  
}
```

JSON_IOTChannel

IOTChannel message in JSON format

```
{  
  "IOTChannel":{  
    /*optional, added IoT devices information*/  
    "channel": "",  
    /*optional, channel No., if this node does not exist, it indicates any free channel, integer*/  
    "IOTProtocolType": "",  
    /*required, IOT protocol type: "HIKVISION"-Hikvision device, "OPTEX"-OPTEX security control panel, "Luminite"-
```

```
Luminite security control panel, "GJD"-GJD security control panel, "SIA-CID"-SIA-CID protocol, string, the max. length is 32*/
"addressingFormatType": "",
/*optional, address type, "ipaddress"-ip address, "hostname"-host name, string, the max. length is 32*/
"hostName": "",
/*optional, host name, it is required when addressingFormatType values "hostname", string, the max. length is 64*/
"ipAddress": "",
/*optional, device IPV4 address, it is required when addressingFormatType values "ipaddress", string, the max. length is 32*/
"ipv6Address": "",
/*optional, device IPV6 address, it is required when addressingFormatType values "ipaddress", string, the max. length is 128*/
"protocol": "",
/*optional, protocol type: "tcp"-TCP protocol, "udp"-UDP protocol, "multicast"-Multicast protocol, string, the max. length is 32*/
"portNo": "",
/*optional, device port No., integer*/
"userName": "",
/*optional, user name, string, the sensitive information is encrypted, the max. length is 32*/
"password": "",
/*read-only, password, string, the sensitive information is encrypted, the max. length is 16. It will be applied only when configuring, and it is not returned when getting information*/
"deviceChannel": "",
/*required, added IoT channel No., integer*/
"videoChannels": [1, 2],
/*optional, added video channel No., including the cameras and IoT devices, array, and the array unit is integer*/
"addAll": "",
/*optional, whether adds all channels (including video and IoT channel) of device, boolean*/
"inductiveDeviceType": "",
/*optional, inductive device type, "inductiveType1"-inductive type 1, "inductiveType2"-inductive type 2. When getting added channel information according inductive type, if this node does not exist, all added device channels will be returned, string*/
"zoneNum": ""
/*optional, the number of added zones, it is valid when IOTProtocolType values "SIA-CID", integer*/
}
}
```

Remarks



Note

For different IoT devices, the inputted parameters in IOTChannel message are different.

Example

IOTChannel Message of IoT Devices via Private Protocol

```
"IOTChannel":{
  "IOTProtocolType": "HIKVISION",
  "addressingFormatType": "ipaddress",
  "ipAddress": "10.11.12.13",
  "protocol": "tcp",
  "portNo": 80,
```



```
"userName": "admin",
"password": "12345",
"deviceChannel": 1,
"videoChannels": [1, 2]
}
```

Example

IOTChannel Message of IoT Devices via SIA-CID Protocol

```
"IOTChannel":{
  "IOTProtocolType": "HIKVISION",
  "addressingFormatType": "ipaddress",
  "ipAddress": "10.11.12.13",
  "protocol": "tcp",
  "portNo": 80,
  "userName": "admin",
  "password": "12345",
  "deviceChannel": 1,
  "videoChannels": [1, 2],
  "zoneNum": 1
}
```

Example

IOTChannel Message of OPTEX IoT Devices

```
"IOTChannel":{
  "IOTProtocolType": "OPTEX",
  "addressingFormatType": "ipaddress",
  "ipAddress": "10.11.12.13",
  "protocol": "tcp",
  "deviceChannel": 1
}
```

Example

IOTChannel Message of Luminite IoT Devices

```
"IOTChannel":{
  "IOTProtocolType": "Luminite",
  "addressingFormatType": "ipaddress",
  "ipAddress": "10.11.12.13",
  "protocol": "tcp",
  "portNo": 80,
  "password": "12345",
  "deviceChannel": 1
}
```

Example

IOTChannel Message of GJD IoT Devices

```
"IOTChannel":{
  "IOTProtocolType": "GJD",
  "addressingFormatType": "ipaddress",
  "ipAddress": "10.11.12.13",
}
```

```
"deviceChannel": 1
}
```

JSON_IOTChannelEventCap

IOTChannelEventCap message in JSON format

```
{
  "IOTChannelEventCap":{
    /*required, event capability*/
    "AccessController":{
      /*optional, access controller event*/
      "eventType":{
        /*required, event type, array, the sub type is string*/
        "@opt": ["authenticated", "openingDoor"]
      },
    },
    "VideoIntercom":{
      /*optional, video intercom event*/
      "eventType":{
        /*required, event type, string*/
        "@opt": ["tampering", "duressAlarm"]
      },
    },
    "GJD":{
      /*optional, GJD security control panel event*/
      "eventType":{
        /*required, event type, string*/
        "@opt": ["pirDetection", "tampering"]
      },
    },
    "Luminite":{
      /*optional, Luminite security control panel*/
      "eventType":{
        /*required, event type, string*/
        "@opt": ["pirDetection", "pirTamper"]
      },
    },
    "OPTEx":{
      /*optional, OPTEx security control panel*/
      "eventType":{
        /*required, event type, string*/
        "@opt": ["pcFn", "pcFr"]
      },
    },
    "cameraDetector":{
      /*optional, detector event*/
      "eventType":{
        /*required, event type, string*/
        "@opt": ["wirelessTemperature", "wirelessPIR"]
      },
    },
  },
}
```

```
{,
  "SecurityControlPanel":{
/*optional, security control panel*/
  "eventType":{
/*required, event type, string*/
    "@opt": ["zone","host"]
  },
  "zoneNum":{
/*required, the number of zones, integer*/
    "@min": 1,
    "@max": 16,
    "#text": 1
  },
}
}
```

JSON_IOTChannelList

IOTChannelList message in JSON format

```
{
  "IOTChannelList": [{
    "IOTChannel"
/*optional, added IoT devices information, see details in
    JSON_IOTChannel
*/
  }]
}
```

See Also

JSON_IOTChannel

JSON_IOTChannelStatus

IOTChannelStatus message in JSON format

```
{
  "IOTChannelStatus":{
/*optional, added IoT device status*/
    "IOTChannel":{
/*optional, added IoT device information, see details in
        JSON_IOTChannel
*/
    },
    "channelName": "",
/*required, channel name, string, type, the max. length is 32*/
    "onlineStatus": "",
/*required, online status, boolean*/
    "channelDetectResult": "connecting,networkUnreachable,accessAbnormal,normal,gaurdFailed",
  },
}
```

```
/*optional, channel detection status: "connecting"-connecting, "networkUnreachable"-unreachable network,
"accessAbnormal"-abnormal access, "normal"-normal, "guardFailed"-arming failed, string*/
  "SecurityStatus":{
/*optional, security status*/
    "passwordStatus": ""
/*optional, password status: notActivated-device is inactivated, risk-risky password, weak-weak password, medium-
medium password, strong-strong password, invalid-invalid status, string*/
  },
  "DeviceInfo":{
/*optional, device information*/
    "deviceName": "",
/*optional, device name, string, the max. length is 32*/
    "deviceType": "",
/*required, device type: "accessController"-access controller, "videoIntercom"-video intercom, "OPTEX"-OPTEX
security control panel, Luminite-Luminite security control panel, GJD-GJD security control panel, securityControlPanel-
security control panel, string*/
    "model": "",
/*optional, device model, string, the max. length is 128*/
    "firmwareVersion": "",
/*optional, firmware version, string, the max. length is 128*/
    "firmwareReleasedDate": ""
/*optional, firmware compiled date, string, the max. length is 128*/
  }
}
}
```

JSON_IOTChannelStatusList

IOTChannelStatusList message in JSON format

```
{
  "IOTChannelStatusList": [{
    "IOTChannelStatus":
/*optional, added IoT device status, see details in JSON_IOTChannelStatus*/
  }]
}
```

See Also

JSON_IOTChannelStatus

JSON_IOTSourceDescription

IOTSourceDescription message in JSON format

```
{
  "IOTSourceDescription":{
/*optional, protocol information*/
    "IOTProtocolType": "",
/*required, IOT protocol type: "HIKVISION"-Hikvision device, "OPTEX"-OPTEX security control panel, "Luminite"-
```

```
Luminite security control panel, "GJD"-GJD security control panel, "SIA-CID"-SIA-CID protocol, string, the max. length is 32*/
"addressingFormatType": "",
/*optional, address type, "ipaddress"-ip address, "hostname"-host name, string, the max. length is 32*/
"hostName": "",
/*optional, host name, it is required when addressingFormatType values "hostname", string, the max. length is 64*/
"ipAddress": "",
/*optional, device IPV4 address, it is required when addressingFormatType values "ipaddress", string, the max. length is 32*/
"ipv6Address": "",
/*optional, device IPV6 address, it is required when addressingFormatType values "ipaddress", string, the max. length is 128*/
"protocol": "",
/*optional, protocol type: "tcp"-TCP protocol, "udp"-UDP protocol, "multicast"-Multicast protocol, string, the max. length is 32*/
"portNo": "",
/*optional, device port No., integer*/
"userName": "",
/*optional, user name, string, the sensitive information is encrypted, the max. length is 32*/
"password": ""
/*optional, password, string, the sensitive information is encrypted, the max. length is 16*/
}
}
```

JSON_IOTSourceList

IOTSourceList message in JSON format

```
{
  "IOTSourceList": [{
    /*optional, source information, array*/
    "IOTProtocolType": "",
    /*required, IOT protocol type: "HIKVISION"-Hikvision device, "OPTEX"-OPTEX security control panel, "Luminite"-Luminite security control panel, "GJD"-GJD security control panel, "SIA-CID"-SIA-CID protocol, string, the max. length is 32*/
    "addressingFormatType": "",
    /*optional, address type, "ipaddress"-ip address, "hostname"-host name, string, the max. length is 32*/
    "hostName": "",
    /*optional, host name, it is required when addressingFormatType values "hostname", string, the max. length is 64*/
    "ipAddress": "",
    /*optional, device IPV4 address, it is required when addressingFormatType values "ipaddress", string, the max. length is 32*/
    "ipv6Address": "",
    /*optional, device IPV6 address, it is required when addressingFormatType values "ipaddress", string, the max. length is 128*/
    "portNo": "",
    /*optional, device port No., integer*/
    "userName": "",
    /*optional, user name, string, sensitive information will be encrypted, the max. length is 32*/
    "password": "",
    /*optional, password, string, sensitive information will be encrypted, the max. length is 16*/
  }]
}
```

```
"activeStatus": "",
/*required, active or not, boolean*/
"deviceType": "",
/*optional, string, devive type: accessController-access controller, videoIntercom-video intercom, OPTEX-OPTEX
security control panel, Luminite-Luminite security control panel, GJD-GJD security control panel, securityControlPanel-
security control panel*/
"macAddress": "",
/*optional, device MAC address, string, the max. length is 48*/
"serialNumber": "",
/*optional, device serial No., string, the max. length is 48*/
"firmwareVersion": "",
/*optional, firmware version (including compiled date), string, the max. length is 128*/
}}
}
```

JSON_IOTSourceSupport

IOTSourceSupport message in JSON format

```
{
  "IOTSourceSupport":{
    /*required, supported source information*/
    "sourceNum": "",
    /*required, the number of protocols, integer*/
    "IOTSourceDescriptions": [{
    /*optional, protocol informaiton, array*/
      "IOTProtocolType": "",
      /*required, IOT protocol type: "HIKVISION"-Hikvision device, "OPTEX"-OPTEX security control panel, "Luminite"-
      Luminite security control panel, "GJD"-GJD security control panel, "SIA-CID"-SIA-CID protocol, string, the max. length is
      32*/
      "addressingFormatType":{
        /*optional, address type, "ipaddress"-ip address, "hostname"-host name, string, the max. length is 32*/
        "@opt": "ipaddress,hostname",
        "#text": "ipaddress"
      },
      "protocol":{
        /*optional, protocol type, "tcp, udp, muticast", string, the max. length is 32*/
        "@opt": "tcp,udp,multicast",
        "#text": "tcp"
      },
      "portNo":{
        /*optional, device port No., integer*/
        "@min": 0,
        "@max": 65535,
        "#text": 80
      },
      "userName ":{
        /*optional, user name, string, the max. length is 32*/
        "@min": 0,
        "@max": 32,
        "#text": "admin"
      }
    }
  ]
}
```

```
,
  "password":{
/*optional, password, string, the max. length is 16*/
    "@min": 0,
    "@max": 16,
    "#text": "12345"
  },
  "maxVideoChannelNum": "",
/*optional, the max. number of supported video channel, integer*/
  "zoneNum":{
/*optional, number of connected zones, integer, it is valid when the IOTProtocolType values "SIA-CID"*/
    "@min": 1,
    "@max": 256,
    "#text": 1
  },
},
}],
  "DeviceInductiveType":{
/*optional, device inductive type*/
    "inductiveType1": ["accessController", "videoIntercom"],
/*optional, array, inductive type 1: accessController-access controller, videoIntercom-video intercom, OPTEX-OPTEX
security control panel, Luminite-Luminite security control panel, GJD-GJD security control panel, securityControlPanel-
security control panel, the sub type is string*/
    "inductiveType2": ["OPTEX", "Luminite", "GJD", "securityControlPanel"]
/*optional, array, inductive type 2: accessController-access controller, videoIntercom-video intercom, OPTEX-OPTEX
security control panel, Luminite-Luminite security control panel, GJD-GJD security control panel,securityControlPanel-
security control panel, the sub type is string*/
  }
}
}
```

JSON_GPSInfo

JSON message about GPS information

```
{
  "GPSInfo":{
/*required, actual GPS information*/
    "longitude": ,
/*required, float, longitude, range: [-180.000000,180.000000], corrects to 6 decimal places*/
    "latitude":
/*required, float latitude, range: [-90.000000,90.000000], corrects to 6 decimal places*/
  }
}
```

JSON_GPSVerification_ResultInfo

JSON message about calibration verification result information

```
{
  "ResultInfo":{
    /*required, calibration verification result information*/
    "id": ,
    /*required, int, verification point ID; PanoVu series camera supports up to 16 points, and speed dome supports up to 64 points*/
    "longitude": ,
    /*required, float, longitude, corrects to 6 decimal places, range:[-180.000000,180.000000]*/
    "latitude":
    /*required, float, latitude, corrects to 6 decimal places, range: [-90.000000,90.000000]*/
  }
}
```

JSON_GPSVerification_ResultInfoList

JSON message about calibration verification results

```
{
  "ResultInfoList":[{
    /*required, verification results information*/
    "id": ,
    /*required, int, verification point ID; PanoVu series camera supports up to 16 points, and speed dome supports up to 64 points*/
    "longitude": ,
    /*required, float, longitude which is calculated based on the pan and tilt, corrects to 6 decimal places, range: [-180.000000,180.000000]*/
    "latitude":
    /*required, float, latitude which is calculated based on the pan and tilt, corrects to 6 decimal places, range: [-90.000000,90.000000]*/
  }]
}
```

JSON_GyroVerify

JSON message about gyroscope calibration parameters

```
{
  "GyroVerify": {
    /*required, gyroscope calibration*/
    "enabled":
    /*required, whether to enable gyroscope calibration: true=yes, false=no*/
  }
}
```


JSON_GyroVerifyCap

JSON message about gyroscope calibration capability.

```
{
  "GyroVerifyCap":{
    "enabled":{
      /*required, whether to enable gyroscope calibration*/
      "@opt": [true,false],
      "@def":false
    }
  }
}
```

JSON_KeepAlive

KeepAlive message in JSON format

```
{
  "KeepAlive":{
    "enabled": "",
    /*required, boolean, whether to enable network keepalive*/
    "interval": "",
    /*required, integer, heartbeat interval, unit: second*/
    "addressingFormatType": "",
    /*required, string, address types: "ipaddress"-IP address, "hostname"-host name, the max. length is 32*/
    "hostName": "",
    /*optional, string, host name, which is required when addressingFormatType is "hostname", the max. length is 64*/
    "ipAddress": "",
    /*optional, string, IPv4 address, which is required when addressingFormatType is "ipaddress", the max. length is 32*/
    "ipv6Address": "",
    /*optional, string, IPv6 address, which is valid when addressingFormatType is "ipaddress", the max. length is 128*/
  }
}
```

JSON_KeepAliveCap

KeepAliveCap message in JSON format

```
{
  "KeepAliveCap":{
    "enabled": "true,false",
    /*required, boolean, whether to enable network keepalive*/
    "interval":{
      "@min": 60,
      "@max": 300,
    }
  }
}
```

```
"@def": 300
},
/*required, integer, heart beat interval which is 300s by default, unit: second*/
"addressingFormatType":{
  "@opt": ["ipaddress", "hostname"],
},
/*required, string, address types: "ipaddress"-IP address, "hostname"-host name, the max. length is 32*/
"supportServerTest": true,
/*optional, boolean, whether to support server test*/
}
}
```

JSON_LaserOpticalAxis

JSON message about laser optical axis parameters.

```
{
  "LaserOpticalAxis": {
    "adjustmentEnabled": ,
    /*required, boolean, whether it enables adjusting laser optical axis*/
    "sensitivity":
    /*required, int, sensitivity*/
  }
}
```

JSON_LaserOpticalAxis_direction

JSON message about direction of adjusting laser optical axis

```
{
  "direction": ""
  /*required, string, direction of adjusting laser optical axis: "up"-up, "down"-down, "left"-left, "right"-right*/
}
```

JSON_LaserOpticalAxisCap

JSON message about capability of configuring laser optical axis

```
{
  "LaserOpticalAxisCap": {
    "adjustmentEnabled": {
    /*required, boolean, whether it enables adjusting laser optical axis*/
      "@opt": [true,false]
    },
    "sensitivity": {
    /*required, int, sensitivity*/
      "@min": 1,

```

```
"@max": 100
}
"direction": {
/*required, string, direction of adjusting laser optical axis: "up"-up, "down"-down, "left"-left, "right"-right*/
"@opt":["up","down","left","right"],
"#text":"up"
}
}
}
```

JSON_LensCorrection

LensCorrection in JSON format

```
{
  "LensCorrection":{
    "enabled": "",
/*required, enable or not*/
    "TimeTaskList": [{
      "TimeTaskBlock":{
        "dayOfWeek": "",
/*required, integer32, ISO8601 weekday number, 1=Monday*/
        "TimeTaskRange": [{
          "taskID": "",
/*required, integer32, task ID, starts from 1*/
          "beginTime": "",
/*required, begin time, accurate to second, string type*/
          "endTime": ""
/*required, end time, accurate to second, string type*/
        }]
      }
    }]
  }
}
```

JSON_LensCorrectionCap

LensCorrectionCap message in JSON format

```
{
  "LensCorrectionCap":{
    "enabled": "true,false",
/*required, enable or not*/
    "TimeTaskList":{
      "maxSize": ,
      "TimeTaskBlock":{
        "dayOfWeek": {
/*required, integer32, ISO8601 weekday number, 1=Monday*/
          "@min": "",

```

```
    "@max": "",
  },
  "TimeTaskRange": {
    "maxSize": ,
    "taskID": {
      /*required, integer32, task ID, starts from 1*/
      "@min": ,
      "@max": ,
    },
    "beginTime": "01:00:00",
    /*required, begin time, corrects to second, string*/
    "endTime": "02:00:00"
  }
  /*required, end time, corrects to second, string*/
}
}
}
}
```

JSON_LinkageChansCond

LinkageChansCond message in JSON format

```
{
  "LinkageChansCond":{
    "eventType": "",
    /*required, string, for access controller or video intercom, the EventType == AcsEvent_major_minor, for security control panel, the EventType == AlarmHost_zoneNo*/
    "monitorId": "",
    /*dep, string, camera ID, strlen=[32,64]. For access controller or video intercom, the format is: device serial No._DOOR_door No., for security control panel, the format is: device serial No._ALARM_security control panel No.*/
    "unlockType": ""
    /*optional, unlock type, string, password-unlock by password, hijacking-hijacking unlock, card-unlock by swiping card, householder-unlock by householder, centerplatform-unlock by center platform, bluetooth-bluetooth unlock, qrcode-unlock by QR code, face-unlock by face, fingerprint-unlock by fingerprint*/
  }
}
```

JSON_LinkedTracking

JSON message about advanced parameters of linked tracking capture

```
{
  "LinkedTracking":{
    /*required, advanced parameters of linked tracking capture*/
    "trackingTakeoverEnabled": ,
    /*optional, boolean, whether to enable tracking takeover, it is disabled by default; the function is supported only when you select VCA resource as Smart Event*/
    "linkedTrackingCaptureEnabled": ,
  }
}
```

```
/*optional, boolean, whether to enable linked tracking capture, it is disabled by default; the function is supported only
when you select VCA resource as Smart Event*/
  "captureThreshold": ,
/*optional, int, capture threshold, it refers for the quality of face to trigger capture and alarm, range: [0,100], default
value: 20*/
  "Exposure":{
/*optional, face exposure*/
  "exposureEnabled": ,
/*optional, boolean, whether to enable face exposure, it is disabled by default*/
  "brightRef": ,
/*optional, int, reference brightness of a face in the face exposure mode, range: [0,100]*/
  "expDurationTime":
/*optional, int, minimum duration, it is the extra time the device keeps the face exposure lever after the face
disappears from the scene; range: [0,60], unit: minute, default value: 5m*/
  },
  "RemoveDuplicate":{
/*optional, remove duplicates of captured face pictures*/
  "enabled": ,
/*optional, whether to enable removing duplicates of captured face pictures*/
  "threshold": ,
/*optional, int, threshold of removing duplicates, when the picture similarity is larger than the value, the picture is
duplicated*/
  "faceScore": ,
/*optional, int, the face grading threshold for removing duplicates, only when the face grading is larger than the
configured threshold, the face picture will be compared for removing duplicates, in case that low quality face pictures
influence the comparison effect*/
  "updateTime":
/*optional, int, face modal data updating time interval (the time interval between the modal data being added into
the library and being deleted)*/
  }
}
}
```

JSON_LinkedTrackingCap

JSON message about configuration capability of advanced parameters of linked tracking capture

```
{
  "LinkedTrackingCap":{
/*required, advanced parameters capability of linked tracking capture*/
  "trackingTakeoverEnabled":{
/*optional, boolean, whether to enable tracking takeover, it is disabled by default; the function is supported only
when you select VCA resource as Smart Event*/
    "@opt": [true, false],
    "@def": false
  },
  "linkedTrackingCaptureEnabled":{
/*optional, boolean, whether to enable linked tracking capture, it is disabled by default; the function is supported only
when you select VCA resource as Smart Event*/
    "@opt": [true, false],
    "@def": false
  }
}
```

```
,
  "captureThreshold":{
/*optional, int, capture threshold, it refers for the quality of face to trigger capture and alarm, range: [0,100], default
value: 20*/
    "@min":0,
    "@max":100,
    "@def":20
  },
  "ExposureCap":{
/*optional, face exposure capability*/
    "exposureEnabled":{
/*optional, boolean, whether to enable face exposure, it is disabled by default*/
      "@opt": [true, false],
      "@def":false
    },
    "brightRef":{
/*optional, int, reference brightness of a face in the face exposure mode, range: [0,100]*/
      "@min":0,
      "@max":100,
      "@def":50
    },
    "expDurationTime":{
/*optional, int, minimum duration, it is the extra time the device keeps the face exposure lever after the face
disappears from the scene; range: [0,60], unit: minute, default value: 5m*/
      "@min":1,
      "@max":60,
      "@def":5
    }
  },
  "RemoveDuplicateCap":{
/*optional, capability of removing duplicates of captured face pictures*/
    "enabled":{
/*optional, whether to enable removing duplicates of captured face pictures*/
      "@opt": [true, false],
      "@def":false
    },
    "threshold":{
/*optional, int, threshold of removing duplicates, when the picture similarity is larger than the value, the picture is
duplicated*/
      "@min":70,
      "@max":100,
      "@def":88
    },
    "faceScore":{
/*optional, int, the face grading threshold for removing duplicates, only when the face grading is larger than the
configured threshold, the face picture will be compared for removing duplicates, in case that low quality face pictures
influence the comparison effect*/
      "@min":20,
      "@max":60,
      "@def":20
    },
    "updateTime":{
```

```
/*optional, int, face modal data updating time interval (the time interval between the modal data being added into the library and being deleted)*/
  "@min":0,
  "@max":300,
  "@def":30
}
}
}
}
```

Remarks

Tracking takeover and linked tracking capture are two mutually exclusive modes for panorama tracking, they cannot be enabled at the same time.

JSON_List_IPAddress

List_IPAddress message in JSON format

```
{
  "List":[{"
    "IPAddress":{"
      "ipVersion":"","
/*required, string type, version information of IP address: "v4"-IPv4, "v6"-IPv6*/
      "ipAddress":""
/*required, string type, IP address*/
    }
  }]
}
```

JSON_LockCfg

LockCfg message in JSON format

```
{
  "LockCfg":{"
    "maxTryTimes": ,
/*optional, integer, the maximum attempts of logging in by the IP address*/
    "lockedTime":
/*optional, integer, time range of locking the IP address, unit: second*/
  }
}
```

JSON_LockCfgCap

LockCfgCap capability message in JSON format

```
{
  "LockCfgCap":{
    "maxTryTimes":{
      /*optional, the maximum attempts of logging in by the IP address*/
      "@opt":"5,...",
    },
    "lockedTime":{
      /*optional, time range of locking the IP address, unit: second*/
      "@min": ,
      "@max":
    }
  }
}
```

JSON_LoginLinkNum

LoginLinkNum message in JSON format

```
{
  "LoginLinkNum": {
    "maxLinkNum": ,
    /*required, integer type, maximum number of logged in accounts*/
    "adminPassword":""
    /*required, string type, confirm the password of admin user, this node is valid only when setting maximum number (PUT), otherwise, this node does not exits*/
  }
}
```

JSON_logProTestResult

JSON message about preverification results of logging HDD.

```
{
  "requestURL": "",
  /*optional, string type, request URL*/
  "statusCode": ,
  /*required, integer type, status code*/
  "statusString": "",
  /*required, string type, status description*/
  "subStatusCode": "",
  /*required, string type, sub status code*/
  "errorCode": ,
  /*optional, integer type, error code, which corresponds to subStatusCode, this field is required when statusCode is not 1. The returned value is the transformed decimal number*/
  "errorMsg": "",
  /*optional, string type, error details, this field is required when statusCode is not 1*/
  "logProTestResult":{
    /*preverification results*/
  }
}
```



```
"enough": ,
/*required, boolean, whether there is enough storage space in the HDD: "true"-enough, "false"-not enough*/
"CoverDate":{
/*dependent, it is returned when the field enough is "false"*/
  "beginTime": "",
/*required, string, start time of HDD coverage, which should in ISO 8601 format,the maximum size is 32 bytes*/
  "endTime": "",
/*required, string, end time of HDD coverage, which should in ISO 8601 format,the maximum size is 32 bytes*/
  "coverSpace":
/*required, integer, size of the covered HDD, unit: GB*/
}
}
}
```

JSON_MatchList

JSON Message about log search results

```
{
  "requestURL": "",
/*optional, string, request URL*/
  "statusCode": ,
/*required, int, status code*/
  "statusString": "",
/*required, string, status description*/
  "subStatusCode": "",
/*required, string, sub status code*/
  "errorCode": ,
/*optional, int, error code, which corresponds to the subStatusCode; this node is required when the value of
statusCode is not 1*/
  "errorMsg": "",
/*optional, string, error description, which corresponds to errorCode*/
  "searchID": "",
/*required, string, search ID, which is used to check whether the current search requester is the same as the previous
one. If they are the same, the search record will be stored in the device to speed up the next search, the maximum
length is 32 bytes*/
  "responseStatusStr": "",
/*required, string, search status: "OK"-search completed, "FAILED"-search failed, "MORE"-more data waiting to be
searched, "PARAM ERROR"-incorrect parameters, "INVALID TIME"-invalid time, "TIMEOUT"-timeout, "NO MATCHES"-no
matched data, the maximum length is 32 bytes*/
  "numOfMatches": ,
/*required, integer32, number of matched data records*/
  "matchList": [{
/*optional, array, list of matched data records*/
    "searchMatchItem": {
/*optional, single matched record*/
      "logDescriptor": {
/*required, log description*/
        "metaID": "",
/*required, string, log type string, log type includes "Alarm"-alarm log, "Exception"-exception log, "Operation"-
operation log, and "Information"-additional information; e.g., "log.std-cgi.com/Alarm", the maximum length is 64
```

```
bytes*/
  "startTime": "",
/*required, string, log time, ISO8601 time format, the maximum length is 32 bytes*/
  "localId": "",
/*optional, string, local channel, different logs are corresponding to different local channel types, the maximum length
is 32 bytes*/
  "paraType": "",
/*optional, string, function modules that generate the logs: "videoout, image, encode, network, alarm, exception,
decoder, rs232, preview, security, datetime, frametype"; the maximum length is 32 bytes*/
  "userName": "",
/*optional, string, user name that performs current search task, the maximum length is 32 bytes*/
  "ipAddress": "",
/*optional, string, device IP address, the maximum length is 32 bytes*/
  "logInfo": ""
/*optional, string, log information, the maximum length is 32 bytes*/
  "operationType":,
/*optional, operation type, it is valid only when metaId is set to "Operation": 0-all, 1-software started, 2-software
stopped, 3-configuration page entered, 4-parameters edited, 5-file search, 6-device connected, 7-information
collection started, 8-information collection ended, 9-devices disconnected, 10-password changed, 11-disk space
checked, 12-network status checked*/
  "remarks": ""
/*optional, operation remarks*/
}
}
}}
```

JSON_MoveAutoTracking

JSON message about status of moving object tracking

```
{
  "MoveAutoTracking":{
    "enabled":
/*required, boolean, whether to enable moving object tracking*/
  }
}
```

JSON_OnlineUpgradeParameter

OnlineUpgradeParameter message in JSON format

```
{
  "OnlineUpgradeParameter":{
/*required, online upgrade parameters*/
    "autoDownloadPackage": "",
/*required, whether enables automatic download of upgrade package, boolean, it values "false" by default*/
    "timingUpgrade": "",
/*optional, whether enable scheduled upgrade, boolean, it values "false" by default*/
```

```
"upgradeTime": "",
/*optional, upgrade time, corrects to minute, and second is 0 by default, ISO8601 format, string type, it is valid when
timingUpgrade values "true"*/
}
}
```

JSON_operType

operType message in JSON format

```
{
  "operType": "",
  /*required, string type, operation type: "unlock"-unlock one or multiple IP addresses, "unlockAll"-unlock all IP
  addresses*/
  "List": [{
    "IPAddress": {
      "ipVersion": "",
      /*required, string, IP address version information: "v4"-IPv4, "v6"-IPv6. This node is valid only when openType is
      "unlock"*/
      "ipAddress": ""
    }
    /*required, string, IP address, this node is valid only when openType is "unlock"*/
  }
  ]
}
```

JSON_OSD

OSD message in JSON format

```
{
  "OSD": {
    /*required, OSD information*/
    "overlayItem": "",
    /*required, string, overlaid item, supports multiple items, each item is separated by comma; deviceName-device
    name, cardNo-card number, eventName-event name, name-name, identityCard-ID card, privacyInfo-private
    information*/
    "characterEncoding": "",
    /*required, character encoding, string*/
    "fontSize": "",
    /*required, string, font size: big-large, medium-medium, small-small*/
    "fontColor": "",
    /*required, font color, hexBinary*/
    "overlayType": "",
    /*required, overlay type: "flip, scroll", string*/
    "displayTime": "",
    /*required, display time, unit; second, integer*/
    "PrivacyInfo": [
      /*optional, private information list, it is required when the value of overlayItem contains "privacyInfo", array*/
      "content": ""
    ]
  }
}
```

```
/*required, private information, string, the max. length is 16*/
  },
  "normalizedScreenSize":{
    /*required, read-only, normalized coordinates*/
    "normalizedScreenWidth": "",
    /*required, normalized width, integer*/
    "normalizedScreenHeight": ""
  },
  "Rect":{
    /*required, OSD position*/
    "height": "",
    /*required, height, float*/
    "width": "",
    /*required, width, float*/
    "x": "",
    /*required, X-coordinate, float*/
    "y": "",
    /*required, Y-coordinate, float*/
  }
}
```

JSON_OSDCap

OSDCap message in JSON format

```
{
  "OSDCap":{
    /*required, OSD information*/
    "overlayItem":{
      /*required, string, overlaid item, supports multiple items, each item is separated by comma; deviceName-device
      name, cardNo-card number, eventName-event name, name-name, identityCard-ID card, privacyInfo-private
      information*/
      "@opt": "deviceName,cardNo,eventName,name,identityCard,privacyInfo",
      "#text": "deviceName"
    },
    "characterEncoding":{
      /*required, character encoding, string*/
      "@opt": "GB2312,Latin-1",
      "#text": "GB2312"
    },
    "fontSize":{
      /*required, string, font size: big-large, medium-medium, small-small*/
      "@opt": "big,medium,small",
      "#text": "big"
    },
    "fontColor":{
      /*required, font color, hexBinary*/
      "@opt": "FFFFFF",
    },
  }
}
```

```
"overlayType":{
/*required, overlay type: "flip, scroll", string*/
"@opt": "flip,scroll",
"#text": "flip"
},
"displayTime":{
/*required, display time, unit; second, integer*/
"@min": 1,
"@max": 120,
"#text": 15
},
"PrivacyInfo": [{
/*optional, private information list, it is required when the value of overlayItem contains "privacyInfo", array*/
"size": 3,
/*required, the max. number of items, integer*/
"content":{
/*required, private information, string, the max. length is 16*/
"@min": 1,
"@max": 16,
"#text": "12345"
},
}],
"Rect":{
/*required, OSD position*/
"height":{
/*required, height, float*/
"@min": 0.000,
"@max": 1.000,
"#text": 0.311
},
"width":{
/*required, width, float*/
"@min": 0.000,
"@max": 1.000,
"#text": 0.311
},
"x":{
/*required, X-coordinate, float*/
"@min": 0.000,
"@max": 1.000,
"#text": 0.311
},
"y":{
/*required, Y-coordinate, float*/
"@min": 0.000,
"@max": 1.000,
"#text": 0.311
},
}
}
}
```

JSON_PeopleCounting_CollectionDescription

CollectionDescription message in JSON format

```
{
  "CollectionDescription":{
    "searchID": "",
    /*required, string, unique search ID, the content of searchID remains unchanged when search condition remains
    unchanged.*/
    "searchResultPosition": "",
    /*required, initial position of search result list, integer type. When there are multiple records, and cannot get all
    records in one time searching, you can search the records followed specified position for next search*/
    "maxResults": "",
    /*required, number of matched records per search, integer type*/
    "timeSpanList": [{
    /*required, time interval list*/
      "startTime": "",
      /*required, start time, ISO8601_time, string*/
      "endTime": "",
      /*required, end time, ISO8601_time, string*/
    }],
    "channels": "",
    /*optional, sub type, array, integer32, related channel No., array. If the channel information is not specified, it
    indicates searching all channels*/
  }
}
```

JSON_PeopleCounting_CollectionDescriptionCap

CollectionDescriptionCap message in JSON format.

```
{
  "CollectionDescriptionCap":{
    "timeRange": "",
    /*required, supported time range for people counting replenishment, unit: day, integer*/
    "timeSpanMaxNum": "",
    /*required, the maximum number of time buckets supported for one time search integer*/
    "resultMaxNum": "",
    /*required, the maximum number of items can be searched, integer*/
    "channelMaxNum": ""
  }
}
```

JSON_PeopleCounting_CollectionResult

CollectionResult message in JSON format

```
{
  "CollectionResult":{
    "responseStatusStrg": "",
    /*required, searching status: OK- Searching ended, NO MATCHES-No matched data found, MORE-Search again for
    more results, string, the max. length is 32, {dep if errcode == 1 && errMsg == ok}*/
    "numOfMatches": "",
    /*required, returned number of results for current search, integer32, {dep if errcode == 1 && errMsg == ok}*/
    "totalMatches": "",
    /*required, total number of matched results, integer32, {dep if errcode == 1 && errMsg == ok}*/
    "targets": [{
      /*optional, people gathering result*/
      "startTime": "",
      /*required, start time of people gathering*/
      "endTime": "",
      /*required, end time of people gathering*/
      "channel": "",
      /*required, integer, channel No.*/
      "enter": "",
      /*required, integer, the number of people entered*/
      "exit": "",
      /*required, integer, the number of people exited*/
      "pass": "",
      /*required, integer, the number of people passed by*/
    }]
  }
}
```

JSON_PictureServerInformation

PictureServerInformation message in JSON format

```
{
  "PictureServerInformation":{
    "pictureServerType": "",
    /*required, string type, picture storage server type: "tomact,VRB,cloudStorage,KMS"*/
    "addressingFormatType": "",
    /*required, string type, format type of the picture storage server address: "ipaddress"-IP address (default),
    "hostname"-host name*/
    "hostName": "",
    /*string type, domain name of the picture storage server, the string length is between 0 and 64. This field is valid
    when addressingFormatType is "hostname"*/
    "ipv4Address": "",
    /*string type, IPv4 address of the picture storage server, the string length is between 0 and 64. This field is valid when
    addressingFormatType is "ipaddress"*/
    "ipv6Address": "",
    /*string type, IPv6 address of the picture storage server, the string length is between 0 and 128. This field is valid
    when addressingFormatType is "ipaddress"*/
    "portNo": ,
    /*required, integer type, port No. of the picture storage server, which is between 1024 and 65535*/
    "underlyingProtocol": "",
  }
}
```

```
/*optional, string, bottom-level protocol of the picture storage server: "HTTP", "HTTPS". This field is valid when
pictureServerType contains "cloudStorage". If this field does not exist, the default bottom-level protocol is HTTP*/
"cloudStorage":{
/*parameters of the cloud storage server, which is valid when pictureServerType is "cloudStorage"*/
"cloudManageHttpPort": ,
/*required, integer type, HTTP port No. for central management of the cloud storage server, which is between 1024
and 65535*/
"cloudTransDataPort": ,
/*required, integer type, data transmission port No. of the cloud storage server, which is between 1024 and 65535.
This field is not supported by access control devices*/
"cloudCmdPort": ,
/*required, integer type, signaling port No. of the cloud storage server, which is between 1024 and 65535*/
"cloudHeartBeatPort": ,
/*required, integer type, heartbeat port No. of the cloud storage server, which is between 1024 and 65535. This field
is not supported by access control devices*/
"cloudStorageHttpPort": ,
/*required, integer type, HTTP port No. of the cloud storage server, which is between 1024 and 65535. This field is not
supported by access control devices*/
"cloudUsername": "",
/*required, string type, user name of the cloud storage server, the string length is between 0 and 32. This field is not
supported by access control devices*/
"cloudPassword": "",
/*required, string type, password of the cloud storage server, the string length is between 0 and 32. This field is not
supported by access control devices*/
"cloudPoolId": ,
/*required, integer type, cloud storage pool ID, which is between 1 and 4294967295. If this field is not configured by
the upper-level, this field will be set to 1 by default*/
"cloudPoolIdEx": "",
/*optional, string type, cloud storage pool ID, this node is valid when cloud storage pool ID of type string (cloud
storage protocol in version 3.0) is supported*/
"cloudProtocolVersion": "",
/*required, string type, protocol version of the cloud storage server, the string length is between 0 and 32*/
"cloudAccessKey": "",
/*string type, cloud storage server access_key, the string length is between 0 and 64. This field is valid when
cloudProtocolVersion is "V2.0"*/
"cloudSecretKey": ""
/*string type, cloud storage server secret_key, the string length is between 0 and 64. This field is valid when
cloudProtocolVersion is "V2.0"*/
}
}
}
```

JSON_POE

POE message in JSON format

```
{
"POE": {
"@opt": "ipV4,ipV6",
/*optional, IPv4 address, string type, the max. length is 32 bytes*/
"#text": "ipV4"
```



```
/*optional, IPv6 address, string type, the max. length is 128*/  
}  
}  
}
```

JSON_ProgressInfo

ProgressInfo message in JSON format

```
{  
  "ProgressInfo":{  
    /*required, progress information*/  
    "status": "",  
    /*required, string, exporting status: "exporting", "failed", "success"*/  
    "progress": ""  
    /*required, string, exporting progress, format example: 2/101 (there are 101 files in total to be exported, and 2 files  
    have been exported)*/  
  }  
}
```

JSON_Reconstruction_ProgressLists

JSON message about disk data reconstruction progress

```
{  
  "ProgressLists": [{  
    "diskID": 1,  
    /*required, int, disk ID*/  
    "progress": 100,  
    /*required, int, disk data reconstruction progress*/  
    "status": 1  
    /*required, int, status: 0-reconstructing, 1-device exception (applying for memory failed/opening files failed/deleting  
    files failed), 2-abnormal disconnection, 3-reconstructed*/  
  }]  
}
```

JSON_RemarkList

JSON message about remark information list of videos

```
{  
  "RemarkList": [{  
    "trackId": "",  
    /*required, file No.*/  
    "remarks": "",  
    /*required, string, file remarks, which should be encrypted*/  
    "recordName": "",  
  }]  
}
```

```
/*optional, string, file name of video, audio, or picture*/
"recorderCode":"","
/*optional, string, ID of device that collects data, which should be encrypted*/
"policeCode":"","
/*optional, string, ID of police that collects data, which should be encrypted*/
}}
}
```

JSON_ReportCenterCfg

ReportCenterCfg message in JSON format

```
{
  "ReportCenterCfg":{
    "enable": ,
    /*optional, boolean type, whether to enable uploading report*/
    "ChanAlarmMode":[{
      /*alarm channel of the center group*/
      "id": ,
      /*optional, integer type, channel ID: 1-main channel, 2-backup channel 1, 3-backup channel 2, 4-backup channel 3*/
      "chanAlarmMode":""
    }]
  }
}
```

JSON_ResponseStatus

JSON message about response status

```
{
  "requestURL":"","
  /*optional, string, request URL*/
  "statusCode": ,
  /*required, int, status code*/
  "statusString":"","
  /*required, string, status description*/
  "subStatusCode":"","
  /*required, string, sub status code*/
  "errorCode": ,
  /*optional, int, error code, which corresponds to subStatusCode, this field is required when statusCode is not 1. The
  returned value is the transformed decimal number*/
  "errorMsg":"","
  /*optional, string, error details, this field is required when statusCode is not 1*/
}
```



Note

See **Response Codes of Text Protocol** for details about the status codes, sub status codes, error codes, and error descriptions.

JSON_resourceStatistics

resourceStatistics message in JSON format

```
{
  "inputBandwidth": ,
  /*required, input bandwidth, unit: kbps, float type, corrects to one decimal*/
  "outputBandwidth": ,
  /*required, output bandwidth, unit: kbps, float type, corrects to one decimal*/
  "StreamingInformation": [{
    /*optional, stream information, array*/
    "moduleType": "",
    /*required, module type: "remoteStreaming"-remote streaming, "remotePlayBack"-remote playback, string type, the
    maximum length is 32 bits*/
    "ipAddress": "",
    /*required, device IPv4 address, string type, the maximum length is 32 bits*/
    "ipv6Address": "",
    /*optional, device IPv6 address, string type, the maximum length is 128 bits*/
    "bandwidth":
  /*required, bandwidth, unit: kbps, float type, corrects to one decimal*/
  }],
  "inputPictureBandwidth": ,
  /*optional, picture input bandwidth (including all interfaces of inputting binary picture data), unit: kbps, float type,
  corrects to one decimal*/
  "outputPictureBandwidth": ,
  /*optional, picture output bandwidth (including all interfaces of returning binary picture data), unit: kbps, float type,
  corrects to one decimal*/
}
```

JSON_Result

Result message in JSON format

```
{
  "Result":{
    "channels": [1, 2],
    /*optional, integer, linked video channel No.*/
    "IoTChannels": [1, 2]
    /*optional, integer, linked IoT channel NO.*/
  }
}
```

JSON_SSD_UpgradeStatus

JSON message about SSD firmware upgrade progress

```
{
  "UpgradeStatus": {
    "upgrading": false,
    /*required, read-only, boolean, whether the SSD firmware is upgrading*/
    "percent":100
    /*required, read-only, int, SSD firmware upgrade progress, range: [0,100]; this node is valid only when the value of
    upgrading is true*/
  }
}
```

JSON_SearchCondition

JSON message about condition information of log search

```
{
  "searchID": "",
  /*required, string, Search ID, which is used to check whether the current search requester is the same as the previous
  one. If they are the same, the search record will be stored in the device to speed up the next search, the maximum
  length is 32 bytes*/
  "timeSpanList": {
    /*required, list of time periods*/
    "timeSpan": {
      /*required, time periods*/
      "startTime": "",
      /*required, string, start time of search, ISO8601 time format, the maximum length is 32 bytes*/
      "endTime": ""
      /*required, string, end time of search, ISO8601 time format, the maximum length is 32 bytes*/
    }
  },
  "metaID": "",
  /*required, string, log type string, log type includes "Alarm"-alarm log, "Exception"-exception log, "Operation"-
  operation log, and "Information"-additional information; e.g., "log.std-cgi.com/Alarm", the maximum length is 64
  bytes*/
  "searchResultPostion": ,
  /*required, integer32, the start position of search result in result list*/
  "maxResults": ,
  /*required, integer32, the maximum number of records supported in this search*/
  "wildcard": "",
  /*optional, string, wildcard, which is used for fuzzy search, the maximum length is 48 bytes*/
  "operationType": ,
  /*optional, operation type, it is valid only when metaID is set to "Operation": 0-all, 1-software started, 2-software
  stopped, 3-configuration page entered, 4-parameters edited, 5-file search, 6-device connected, 7-information
  collection started, 8-information collection ended, 9-devices disconnected, 10-password changed, 11-disk space
  checked, 12-network status checked*/
}
```

JSON_secretKey

secretKey message in JSON format

```
{
  "secretKey": "",
  /*required, string type, encryption key*/
}
```

JSON_SecurityAdvanced

SecurityAdvanced message in JSON format

```
{
  "SecurityAdvanced ": {
    "securityEnhanced": ,
    /*optional, boolean type, whether to enable security reinforcement: true*/
    "noOperationEnabled": ,
    /*optional, boolean type, whether to enable control timeout: true*/
    "noOperationTime": ,
    /*optional, integer type, control timeout, value range: [1,60], default value: 15, unit: minute*/
    "digestStatus": "SHA256"
    /*read-only, string type, digest algorithm: SHA256, MD5*/
  }
}
```

JSON_SecurityEmail

SecurityEmail message in JSON format

```
{
  "SecurityEmail": {
    "SecurityInformation": [{
      /*required, recovery email information*/
      "emailAddress": "",
      /*required, string type, email address, the sensitive information should be encrypted*/
    }],
  },
}
```

JSON_SecurityEmailCap

SecurityEmailCap message in JSON format

```
{
  "SecurityEmailCap":
```

```
/*required, recovery email configuration capability*/
{
  "emailAddressNum": ,
/*required, number of recovery emails*/
  "emailAddress":
/*required, string type, recovery email information*/
  {
    "@min": 1,
    "@max": 128,
  },
}
}
```

JSON_SecurityEmailQrCode

SecurityEmailQrCode message in JSON format

```
{
  "SecurityEmailQrCode":
/*optional, QR code of recovery email*/
  {
    "SecurityInformation": [{
/*required, security information, the sensitive information should be encrypted*/
      "emailAddress": "",
/*required, string type, email address, the sensitive information should be encrypted*/
      "qrCode": "",
/*required, string type, QR code, the maximum size is 1024 bytes, the sensitive information should be encrypted*/
    }],
    "serverEmailAddress": "",
/*optional, string type, manufacturer service email address, the sensitive information should be encrypted*/
    "isDisclaimerDisplay":
/*optional, boolean type, whether to display disclaimer*/
  },
}
```

JSON_SerialLogCfg

SerialLogCfg message in JSON format

```
{
  "SerialLogCfg":{
    "enabled":
/*required, boolean type, whether to enable serial port log redirection: "true"-yes, "false"-no*/
  }
}
```

JSON_SerialLogCfgCap

SerialLogCfgCap capability message in JSON format

```
{
  "SerialLogCfgCap":{
    "enabled":"true, false"
    /*required, boolean type, whether to enable serial port log redirection: "true"-yes, "false"-no*/
  }
}
```

JSON_ServerCertificateCap

ServerCertificateCap capability message in JSON format

```
{
  "ServerCertificateCap":{
    "customID":{
      /*required, string, custom certificate ID, it is created when the user imports the certificate and it consists of digits and characters*/
      "@min":1,
      "@max":64
    },
    "status":{
      /*optional, string, certificate status: "normal", "expired", "exceptional"*/
      "@opt":["normal", "expired", "exceptional"]
    },
    "RSAKeyLength":{
      /*optional, integer, RSA key length*/
      "@opt":[1024, 2048]
    },
    "countryName":{
      /*optional, string, country*/
      "@min":1,
      "@max":32
    },
    "commonName":{
      /*optional, string, domain name or IP address*/
      "@min":1,
      "@max":32
    },
    "validity":{
      /*optional, integer, valid days*/
      "@min":1
    },
    "passwd":{
      /*optional, string, password*/
      "@min":1,
      "@max":64
    }
  }
}
```

```
,
  "stateOrProvinceName":{
/*optional, string, province or state*/
    "@min":1,
    "@max":32
  },
  "localityName":{
/*optional, string, region*/
    "@min":1,
    "@max":32
  },
  "organizationName":{
/*optional, string, organization*/
    "@min":1,
    "@max":32
  },
  "organizationUnitName":{
/*optional, string, organization unit*/
    "@min":1,
    "@max":32
  },
  "email":{
/*optional, string, email*/
    "@min":1,
    "@max":32
  }
}
}
```

JSON_ServerCertificates

ServerCertificates message in JSON format

```
{
  "ServerCertificates":{
    "CertificateInfo":[{
      "issuerDN":"","
/*required, string, read-only, name of the authority that issued the certificate*/
      "subjectDN":"","
/*required, string, read-only, certificate holder name*/
      "startDate":"","
/*required, date, read-only, start date of the certificate validity period, it is accurate to day*/
      "endDate":"","
/*required, date, read-only, end date of the certificate validity period, it is accurate to day*/
      "type":"","
/*optional, string, read-only, certificate type: "HTTPS", "WebSocketS", "SDK_OVER_TLS", "SRTP", "securityLog",
"ieee802.1x". Multiple types should be separated by commas*/
      "status":"","
/*optional, string, read-only, certificate status: "normal", "expired", "exceptional"*/
      "customID":"","
/*optional, string, read-only, custom certificate ID, it is created when the user imports the certificate and it consists of
```



```
digits and characters*/
    }}
  }
}
```

JSON_SmartOverlap

SmartOverlap message in JSON format

```
{
  "SmartOverlap":{
    "enabled": ,
    /*required, boolean, whether to enable the stream*/
    "streamType": ,
    /*required, integer, stream type: 1-main stream, 2-sub-stream, 3-third stream*/
    "bkgImageOverlapEnabled":
    /*required, boolean, whether to overlay rule frame and target frame on the background picture*/
  }
}
```

JSON_SmartOverlapCap

SmartOverlapCap capability message in JSON format

```
{
  "SmartOverlapCap":{
    "enabled":{
    /*required, boolean, whether to enable the stream*/
      "@opt":[ true, false]
    },
    "streamType":{
    /*required, integer, stream type: 0-disable, 1-main stream, 2-sub-stream, 3-third stream*/
      "@opt":[2]
    },
    "bkgImageOverlapEnabled":{
    /*required, boolean, whether to overlay rule frame and target frame on the background picture*/
      "@opt":[true, false]
    },
    "supportEventType":{
    /*required, string, supported event type: "fieldDetection"-intrusion, "attendedBaggage"-object removal,
    "unattendedBaggage"-unattended baggage, "regionExiting"-region exiting, "regionEntrance"-region entrance,
    "lineDetection"-line crossing*/
      "@opt":["fieldDetection", "attendedBaggage", "unattendedBaggage", "regionExiting", "regionEntrance",
    "lineDetection"]
    }
  }
}
```

JSON_SourceCapabilities

SourceCapabilities message in JSON format

```
{
  "SourceCapabilities":{
    /*required, the supported source capabilities*/
    "IOTChannelNum": "",
    /*required, the number of IoT channels,integer*/
    "videoChannelNum": "",
    /*optional, the number of video channels, integer*/
    "accessChannels": [{
      /*optional, array, added channel No.*/
      "channel": "",
      /*required, channel No., integer*/
      "type": "",
      /*required, channel type: "video"-video channel, "IOT"-IoT channel, string*/
    }]
  }
}
```

JSON_SSDCapacity

JSON message about SSD storage quotas

```
{
  "SSDCapacity":{
    "total":70.5,
    /*required, float, SSD total capacity, unit: GB*/
    "systemReserved":10.5,
    /*required, float, reserved capacity, unit: GB*/
    "free":10.5,
    /*required, float, remained capacity, unit: GB*/
    "faceLib":10.5,
    /*optional, float, face picture library capacity, unit: GB*/
    "strangerLib":10.5,
    /*optional, float, strangers library capacity, unit: GB*/
    "faceFrequencyLib":10.5,
    /*optional, float, capacity of people frequency statistics library, unit: GB*/
    "alarmCache":10.5,
    /*optional, float, alarm cache capacity, unit: GB*/
    "faceIndex":10.5,
    /*optional, float, face picture index capacity, unit: GB*/
  }
}
```

JSON_SSDFormatStatus

JSON message about progress of SSD file system formatting

```
{
  "SSDFormatStatus": {
    "formatting": "false",
    /*required, read-only, boolean, whether the device is formatting*/
    "percent":100
    /*required, read-only, int, device formatting progress, range: [0,100]*/
  }
}
```

JSON_SSDUpgrade

JSON message about upgrade status of SSD file system

```
{
  "SSDUpgrade": {
    "upgrading": "false"
    /*required, read-only, boolean, whether the device has been upgraded*/
  }
}
```

JSON_SSDUpgradeStatus

JSON message about progress of SSD file system upgrade

```
{
  "SSDUpgradeStatus": {
    "upgrading": "false",
    /*required, read-only, boolean, whether the SSD file system is upgrading*/
    "percent":100
    /*required, read-only, int, device upgrade progress, range: [0,100]*/
  }
}
```

JSON_SyncStatus

JSON message about HDD data sync status

```
{
  "SyncStatus":{
    "synching":true,
    /*required, boolean, whether the HDD is in sync*/
    "percent":90
  }
}
```

```
/*optional, int, sync progress, range: [0,100]%*/  
}  
}
```

JSON_CollectionDescription

CollectionDescription message in JSON format

```
{  
  "CollectionDescription":{  
    "searchID": "",  
    /*required, string, unique search ID, the content of searchID remains unchanged when search condition remains  
    unchanged.*/  
    "searchResultPosition": "",  
    /*required, initial position of search result list, integer type. When there are multiple records, and cannot get all  
    records in one time searching, you can search the records followed specified position for next search*/  
    "maxResults": "",  
    /*required, number of matched records per search, integer type*/  
    "timeSpanList": [{  
    /*required, time interval list*/  
      "startTime": "",  
      /*required, start time, ISO8601_time, string*/  
      "endTime": "",  
      /*required, end time, ISO8601_time, string*/  
    }],  
    "channels": "",  
    /*optional, sub type, array, integer32, related channel No., array. If the channel information is not specified, it  
    indicates searching all channels*/  
    "alarmLevel": "TMA",  
    /*required, string, alarm level: TMA--thermometry alarm, TMPA-thermometry pre-alarm*/  
  }  
}
```

JSON_Temperature_CollectionDescriptionCap

CollectionDescriptionCap message in JSON format

```
{  
  "CollectionDescriptionCap":{  
    "timeRange": 15,  
    /*required, supported time range for replenishment, unit: day, integer */  
    "timeSpanMaxNum": 1,  
    /*required, the maximum number of time buckets supported for one time search, integer*/  
    "resultMaxNum": 100,  
    /*required, the maximum number of items can be searched, integer*/  
    "channelMaxNum": 1,  
    /*required, the maximum number of channels supported for one time search, integer*/  
    "alarmLevel":{  
    /*required, string, alarm level: TMA--thermometry alarm, TMPA-thermometry pre-alarm*/  
    }
```

```
"@opt": "TMA,TMPA"
}
}
}
```

JSON_CollectionResult

CollectionResult message in JSON format

```
{
  "CollectionResult":{
    "responseStatusStrg": "",
    /*required, searching status: OK- Searching ended, NO MATCHES-No matched data found, MORE-Search again for
    more results, string, the max. length is 32, {dep if errcode == 1 && errMsg == ok}*/
    "numOfMatches": "",
    /*required, returned number of results for current search, integer32, {dep if errcode == 1 && errMsg == ok}*/
    "totalMatches": "",
    /*required, total number of matched results, integer32, {dep if errcode == 1 && errMsg == ok}*/
    "targets": [{
    /*optional, statistics result*/
      "time": "2017-08-25T11:34:59+08:00",
    /*required, statistic time*/
      "channel": "",
    /*required, integer, channel No.*/
      "Region": [{
    /*required, float, X-coordinate, ranges from 0.000 to 1*/
      "x": "",
    /*required, float, Y-coordinate, ranges from 0.000 to 1*/
      "y": ""
      }],
      "thermometryUnit": "",
    /*required, string, temperature unit: celsius, fahrenheit, kelvin*/
      "ruleTemperature": ,
    /*required, float, themometry rule*/
      "currTemperature": ,
    /*required, float, current temperature*/
      "ruleCalibType": "",
    /*required, string, rule calibration type: point,line,region*/
      "ruleType": "",
    /*required, string, rule: "highest temp is higher than"-the maximum temperature is higher than, "lowest temp is
    higher than"-the minimum temperature is higher than, "average temp is higher than"-the average temperature is
    higher than, "temp diff is higher than"-temperature difference is higher than, "highest temp is lower than"-the
    maximum temperature is lower than, "lowest temp is lower than"-the minimum temperature is lower than, "average
    temp is lower than"-the average temperature is lower than, "temp diff is lower than"-temperature difference is lower
    than*/
      "MaximumTemperaturePoint":{
    /*optional, the maximum temperature point*/
      "x": "",
    /*required, float, X-coordinate, ranges from 0.000 to 1*/
      "y": ""
    /*required, float, Y-coordinate, ranges from 0.000 to 1*/
      }
    }
  ]
}
```

```
    },
    "AbsoluteHigh":{
/*optional, absolute height*/
    "elevation": "",
/*required, float,  elevation*/
    "azimuth": "",
/*required, float,  azimuth*/
    "absoluteZoom": ""
/*required, float,  absolute zoom*/
    },
    "ruleID": "",
/*required, integer,  rule ID*/
    "presetNo": "",
/*required, integer,  preset ID*/
    "visibleLightURL": "",
/*optional, string,  visible light picture URL*/
    "thermalURL": "",
/*optional, string,  thermal picture URL*/
    "thermalInfoURL": "",
/*optional, string,  thermal additional information URL*/
    }}
  }
}
```

JSON_TestResult

TestResult message in JSON format

```
{
  "TestResult":{
    "status":"ok",
/*required, string, server status: "ok"-normal, "connectServerFail"-server connection failed
  }
}
```

JSON_TimeSearchCond

Message about recording time search condition in JSON format

```
{
  "TimeSearchCond": {
    "channelID": ,
/*required, int, channel No.*/
    "streamType": ""
/*required, string, stream type, "main,sub,adaptive"*/
  }
}
```

JSON_TimeSearchResult

Message about recording time search result in JSON format

```
{
  "TimeSearchResult": {
    "channelID": ,
    /*required, int, channel No.*/
    "streamType": "",
    /*required, string, stream type, "main,sub,adaptive"*/
    "hasRecord": ,
    /*required, boolean, whether video exists, startTime and endTime are valid only when the value of hasRecord is
    "true"*/
    "startTime": "",
    /*optional, string, recording start time, e.g., 2010-01-01T00:00:00+08:00, this field is required when the value of
    hasRecord is "true"*/
    "endTime": "",
    /*optional, string, recording end time, e.g., 2010-01-01T00:00:00+08:00, this field is required when the value of
    hasRecord is "true"*/
  }
}
```

JSON_TimeType

TimeType message in JSON format

```
{
  "TimeType":{
    "type": ""
  }
  /*required, value: local, UTC*/
}
```

JSON_TrackingRatio

JSON message about tracking parameters

```
{
  "TrackingRatio":{
    /*required, tracking parameters of the speed dome*/
    "coefficient": ,
    /*required, int, tracking zoom ratio, it is between 1 and 100 and the default value is 50. The tracking zoom ratio is the
    ratio of the target to the image (unit: percentage). The radar will send the target's distance to the application during
    tracking. For human bodies, the ratio of a person's height to the image will be calculated assuming that the person is
    1.7 meters tall; for vehicles, the ratio of the vehicle's width to the image will be calculated assuming that the vehicle is
    2 meters wide*/
    "durationTime": ,
    /*required, int, tracking duration, it is between 5 and 60 and the default value is 20, unit: second*/
  }
}
```

```
"whiteLight":  
/*optional, boolean, whether to support the flashing light*/  
"trackingTakeoverEnabled":  
/*required, boolean, whether to enable tracking takeover*/  
}  
}
```

JSON_TrackingRatioCap

JSON message about the configuration capability of tracking parameters

```
{  
  "TrackingRatioCap":{  
    /*required, configuration capability of the speed dome's tracking parameters*/  
    "coefficient":{  
      /*required, int, tracking zoom ratio, it is between 1 and 100 and the default value is 50. The tracking zoom ratio is the  
      ratio of the target to the image (unit: percentage). The radar will send the target's distance to the application during  
      tracking. For a human body, the ratio of its height to the image will be calculated assuming that the person is 1.7  
      meters tall; for a vehicle, the ratio of its width to the image will be calculated assuming that the vehicle is 2 meters  
      wide*/  
      "@min":1,  
      "@max":100,  
      "@def":50  
    },  
    "durationTime":{  
      /*required, int, tracking duration, it is between 5 and 60 and the default value is 20, unit: second*/  
      "@min":5,  
      "@max":60,  
      "@def":20  
    },  
    "whiteLight":{  
      /*optional, boolean, whether to support the flashing light*/  
      "@opt":[true, false]  
    }  
    "trackingTakeoverEnabled": {  
      /*optional, boolean, whether to enable tracking takeover; by default, it is false*/  
      "@opt": [true, false]  
    }  
  }  
}
```

JSON_TrafficMonitor

TrafficMonitor message in JSON format

```
{  
  "TrafficMonitor":{  
    "enabled":,  
    /*required, boolean, whether to enable traffic monitoring*/  
  }  
}
```



```
"packageType":"","
/*required, string, data plan types: "day"-daily plan, "month"-monthly plan, "year"-annual plan */
"trafficValue";
/*required, float, data value, which should be accurate to three decimal places, unit:MB */
"overrunOperation":"","
/*optional, string, operations when the data exceeds: "brokenNetworkAndAlert"-turn off the cellular network and
alert users, "alert"-notify users */
  "alertType":[
    "beep",
    "email",
    "SMS"
  ],
/*required, string, notification methods: "beep"-audio warning, "email"-send email, "SMS"-send message, "center"-
upload to center*/
  "alertValue";
/*required, integer, alarm value(%)*/
  "startDay":
/*optional, integer, the start day of the monthly plan, which is valid only when packageTypeis "month"*/
}
}
```

JSON_TrafficMonitorCap

TrafficMonitorCap message in JSON format

```
{
  "TrafficMonitorCap":{
    "enabled":"","
/*required, boolean, whether to enable traffic monitoring*/
    "packageType":{
      "@opt":[
        "day",
        "month",
        "year"
      ]
    },
/*required, string, data plan types: "day"-daily plan, "month"-monthly plan, "year"-annual plan*/
    "overrunOperation":{
      "@opt":[
        "brokenNetworkAndAlert",
        "alert"
      ]
    },
/*optional, string, operations after the data exceeds: "brokenNetworkAndAlert"-turn off the cellular network and alert
users, "alert"-notify users */
  },
  "alertType":{
    "@opt":[
      "beep",
      "email",
      "SMS",
      "center"
    ]
  }
}
```

```
]
},
/*required, string, notification methods: "beep"-audio warning, "email"-send email, "SMS"-send message, "center"-
upload to center*/
  "alertValue":{
    "@min":,
    "@max":
  },
/*required, integer, alarm value(%)*/
  "supportTrafficMonitorStatus":
/*optional, boolean, whether it supports getting the current network data usage*/
}
}
```

JSON_TrafficMonitorStatus

TrafficMonitorStatus message in JSON format

```
{
  "TrafficMonitorStatus":{
    "remainTrafficValue":,
/*required, float, remaining data, which should be accurate to three decimal places, unit:MB*/
    "trafficValue":,
/*required, float, data value, which should be accurate to three decimal places, unit:MB*/
    "todayUsedTrafficValue":
/*required, float, today's data usage, which should be accurate to three decimal places, unit:MB*/
    "todayUsedUpstreamTrafficValue":,
/*optional, float, today's upstream data usage, which should be accurate to three decimal places, unit:MB*/
    "todayUsedDownstreamTrafficValue": ,
/*optional, float, today's downstream data usage, which should be accurate to three decimal places, unit:MB*/
    "monthUsedTrafficValue": ,
/*optional, float, data usage of this month, which should be accurate to three decimal places, unit:MB*/
    "monthUsedUpstreamTrafficValue": ,
/*optional, float, upstream data usage of this month, which should be accurate to three decimal places, unit:MB*/
    "monthUsedDownstreamTrafficValue": ,
/*optional, float, downstream data usage of this month, which should be accurate to three decimal places, unit:MB*/
    "yearUsedTrafficValue": ,
/*optional, float, data usage of this year, which should be accurate to three decimal places, unit:MB*/
    "yearUsedUpstreamTrafficValue": ,
/*optional, float, upstream data usage of this year, which should be accurate to three decimal places, unit:MB*/
    "yearUsedDownstreamTrafficValue": ,
/*optional, float, downstream data usage of this year, which should be accurate to three decimal places, unit:MB*/
    "enabled": ,
/*optional, boolean, whether traffic monitoring is enabled*/
    "packageType": "",
/*optional, string, data plan types: "day"-daily plan, "month"-monthly plan, "year"-annual plan*/
    "alertValue":
/*optional, integer, alarm value(%)*/
  }
}
```

JSON_unitConfig

unitConfig message in JSON format

```
{
  "enabled": "",
  /*required, enable unit unifying configuration or not, boolean type*/
  "temperatureRange": "",
  /*optional, string type, temperature unit: degreeCentigrade-Centigrade (°C), degreeFahrenheit-Fahrenheit (°F) ,
  degreeKelvin-Kelvin (K), the default value is "degreeCentigrade"*/
  "distanceUnit": ""
  /*optional, string type, distance unit: centimeter, meter, feet, the default value is "centimeter"*/
}
```

Remarks

- When the **enabled** values "false", it indicates the unit unifying function is disabled, the unit can be configured in thermomery basic settings, and the configured unit has no influence on the unit in system settings.
- When the **enabled** values "false", the thermomery unit configuration in applied URL remains unchanged and takes effect; while when the **enabled** values "true" and the thermometry unit is different with the system unit, the thermometry unit configuration in applied URL takes no effect and the error code will be returned.

JSON_unitConfigCap

unitConfigCap message in JSON format

```
{
  "enabled":"true,false",
  /*required, enable unit unified configuration or not, boolean type*/
  "temperatureRange":{
    /*optional, string type, temperature unit: degreeCentigrade-Centigrade (°C), degreeFahrenheit-Fahrenheit (°F) ,
    degreeKelvin-Kelvin (K), the default value is "degreeCentigrade"*/
    "@opt":"degreeCentigrade,degreeFahrenheit,degreeKelvin"
  },
  "distanceUnit":{
    /*optional, string type, distance unit: centimeter, meter, feet, the default value is "centimeter"*/
    "@opt":"centimeter,meter,feet"
  }
}
```

JSON_USB

JSON message about USB parameters

```
{
  "USB": {
    "usbMode": ""
  /*optional, string, USB mode: "UVC", "NCM"*/
  }
}
```

JSON_USBCap

JSON message about USB parameters configuration capability

```
{
  "USBCap": {
    "usbMode": {
  /*optional, string, USB mode: "UVC", "NCM"*/
      "@opt":["UVC", "NCM"],
      "#text":"UVC"
    }
  }
}
```

JSON_User

User message in JSON format

```
{
  "User":{
    "userName": "",
  /*required, string, double verification user name*/
    "password": ""
  /*required, string, double verification password*/
  }
}
```

JSON_UserCap

UserCap message in JSON format

```
{
  "UserCap":{
    "userMaxNumber": "",
  /*required, the maximum number of supported double verification users*/
    "userNameLen":{
  /*required, the length of double verification user name*/
      "@min": "",
      "@max": "",
    },
  }
```

```
"passwordLen":{
/*required, the length of double verification user password*/
  "@min": ,
  "@max": ,
}
}
}
```

JSON_UserList

UserList message in JSON format

```
{
  "UserList": [{
    "id": ,
/*required, integer, user ID*/
    "userName":"","
/*required, string, double verification user name*/
  }]
}
```

JSON_UserPermission

UserPermission message in JSON format

```
{
  "UserPermission":{
    "localPlayback":{
/*optional, local playback permission*/
      "enable": "",
/*required, enables permission or not, boolean*/
      "channelPermission": [{
/*optional, channel permission*/
        "channelID": ,
/*required, channel ID, integer*/
        "enable": ""
/*required, enables permission or not, boolean*/
      }]
    },
    "localBackup":{
/*optional, local backup permission*/
      "enable": "",
/*required, enables permission or not, boolean*/
      "channelPermission": [{
/*optional, channel permission*/
        "channelID": ,
/*required, channel ID, integer*/
        "enable": ""
/*required, enables permission or not, boolean*/
      }]
    }
  }
}
```

```
    }}
  },
  "remotePlayback":{
/*optional, remote playback and download permission*/
    "enable": "",
/*required, enables permission or not, boolean*/
    "channelPermission": [{
/*optional, channel permission*/
      "channelID": ,
/*required, channel ID, integer*/
      "enable": ""
/*required, enables permission or not, boolean*/
    }]
  }
}
```

JSON_UserPermissionCap

UserPermissionCap message in JSON format

```
{
  "UserPermissionCap":{
    "localPlayback":{
/*required, local playback permission*/
      "channelID":{
/*required, the channel which supports configuring local playback permission*/
        "@min": "",
        "@max": ,
      },
      "enable": "true,false"
/*required, enables permission or not, boolean*/
    },
    "localBackup":{
/*required, local backup permission*/
      "channelID":{
/*required, the channel which supports configuring local backup permission*/
        "@min": "",
        "@max": ,
      },
      "enable": "true,false"
/*required, enables permission or not, boolean*/
    },
    "remotePlayback":{
/*required, remote playback permission*/
      "channelID":{
/*required, the channel which supports configuring remote playback permission*/
        "@min": "",
        "@max": "",
        "#text": ""
      },

```

```
"enable": "true,false"
/*required, enables permission or not, boolean*/
}
}
}
```

JSON_VerificationDisplay

JSON message about parameters of calibration verification result display

```
{
  "VerificationDisplay": {
    /*required, display calibration verification result*/
    "enabled":
    /*required, boolean, whether to display the calculated point position and verification point position on video: true-yes*/
  }
}
```

JSON_VerificationPoint

JSON message about calibration verification point information

```
{
  "VerificationPoint":[
    /*required, calibration verification point*/
    "pan": ,
    /*optional, float, pan, range: [0.00,360.00]*/
    "tilt": ,
    /*optional, float, tilt, range: [-90.00,90.00]*/
    "zoom": ,
    /*optional, float, zoom, range: [0.00,90.00]*/
    "longitude": ,
    /*required, float, longitude which is calculated based on the pan and tilt, corrects to 6 decimal places, range: [-180.000000,180.000000]*/
    "latitude": ,
    /*required, float, latitude which is calculated based on the pan and tilt, corrects to 6 decimal places, range: [-90.000000,90.000000]*/
    "x": ,
    /*optional, float, x-coordinate, [0.000,1.000]*/
    "y":
    /*optional, float, y-coordinate, [0.000,1.000]*/
  ]
}
```

JSON_VerificationPointList

JSON message about verification points information

```
{
  "VerificationPointList":[{
    /*required, calibration verification points*/
    "id": ,
    /*required, int, verification point ID; PanoVu series camera supports up to 16 points, and speed dome supports up to 64 points*/
    "pan": ,
    /*optional, float, pan, range: [0.00,360.00]*/
    "tilt": ,
    /*optional, float, tilt, range: [-90.00,90.00]*/
    "zoom": ,
    /*optional, float, zoom, range: [0.00,90.00]*/
    "longitude": ,
    /*required, float, longitude, corrects to 6 decimal places, range:[-180.000000,180.000000]*/
    "latitude": ,
    /*required, float, latitude, corrects to 6 decimal places, range: [-90.000000,90.000000]*/
    "x": ,
    /*optional, float, x-coordinate, [0.000,1.000]*/
    "y":
    /*optional, float, y-coordinate, [0.000,1.000]*/
  }]
}
```

JSON_VerificationPointsCap

JSON message about capability of verify calibration

```
{
  "VerificationPointsCap":{
    /*required, calibration verification capability*/
    "id":{
      /*required, int, verification point ID; PanoVu series camera supports up to 16 points, and speed dome supports up to 64 points*/
      "@min":1,
      "@max":64
    },
    "pan":{
      /*optional, float, pan*/
      "@min":0,
      "@max":360
    },
    "tilt":{
      /*optional, float, tilt*/
      "@min":-90,
      "@max":90
    }
  }
}
```



```
,
  "zoom":{
/*optional, float, zoom*/
    "@min":0,
    "@max":90
  },
  "longitude":{
/*required, float, longitude, corrects to 6 decimal places*/
    "@min":-180,
    "@max":180
  },
  "latitude":{
/*required, float, latitude, corrects to 6 decimal places*/
    "@min":-90,
    "@max":90
  },
  "x":{
/*optional, float, x-coordinate*/
    "@min":0,
    "@max":1
  },
  "y":{
/*optional, float, y-coordinate*/
    "@min":0,
    "@max":1
  },
  "isSupportDisplay":
/*optional, boolean, whether device supports display the verification result (related URI: /ISAPI/System/
GPSVerification/channels/<ID>/display?format=json)*/
  }
}
```

JSON_VibrationDetection

Vibration detection message in JSON format

```
{
  "VibrationDetection": {
/*vibration detection*/
    "enabled": ,
/*required, boolean, whether to enable vibration detection, "true"-enable, "false"-disable*/
    "sensitivity":
/*required, int, sensitivity, value range: [0-100], by default: 50*/
  }
}
```

JSON_VibrationDetectionCap

Message about vibration detection capability in JSON format

```
{
  "VibrationDetectionCap": {
    /*vibration detection capability*/
    "enabled": {
      /*required, boolean, whether to enable vibration detection,"true"-enable, "false"-disable*/
      "@opt": [true, false]
    },
    "sensitivity": {
      /*required, int, sensitivity, value range: [0-100]*/
      "@min": 0,
      "@max": 100,
      "@def": 50
    }
  }
}
```

JSON_voiceTalkCtrl

JSON message about SIP (Session Initiation Protocol) intercom parameters

```
{
  "voiceTalkCtrl":{
    "enable":true
    /*required, boolean, whether to enable SIP intercom: true=yes, false=no*/
  }
}
```

JSON_voiceTalkCtrlCap

JSON message about the configuration capability of SIP (Session Initiation Protocol) intercom

```
{
  "voiceTalkCtrlCap":{
    "enable":[true,false]
    /*required, boolean, whether to enable SIP intercom: true=yes, false=no*/
  }
}
```

JSON_WhiteLightAlarm

WhiteLightAlarm message in JSON format

```
{
  "WhiteLightAlarm":{
    "durationTime": ,
    /*required, int, flashing duration of supplement light, which is between 1 and 60 s*/
    "frequency": "",
    /*required, string, flashing frequency of supplement light: "high"-flashing 0.5s per second, "medium"-flashing 0.8s per
    1.6s, "low"-flashing 1s per 2s*/
    "brightness": ,
    /*optional, int, supplement light brightness, which is between 1 and 100*/
    "TimeRangeList": [{
    /*optional, alarm output schedule list*/
      "week": ,
    /*required, int, days of the week: 1-Monday, 2-Tuesday, 3-Wednesday, 4-Thursday, 5-Friday, 6-Saturday, 7-Sunday*/
      "TimeRange": [{
        "id": ,
    /*required, int, ID of time period for each day*/
        "beginTime": "",
    /*required, string, start time in ISO8601 format*/
        "endTime": ""
    /*required, string, end time in ISO8601 format*/
      }]
    }]
  }
}
```

JSON_WhiteLightAlarmCap

WhiteLightAlarmCap message in JSON format

```
{
  "WhiteLightAlarmCap":{
    "durationTime":{
    /*required, int, flashing duration of supplement light, which is between 1 and 60 s*/
      "@min": 1,
      "@max": 60,
      "@def": 15
    },
    "frequency":{
    /*required, string, flashing frequency of supplement light: "high"-flashing 0.5s per second, "medium"-flashing 0.8s per
    1.6s, "low"-flashing 1s per 2s*/
      "@opt": "high,medium,low,normallyOn",
      "@def": "high"
    },
    "brightness":{
    /*optional, int, supplement light brightness, which is between 1 and 100*/
      "@min": 1,
      "@max": 100,
      "@def": 50
    },
    "TimeRangeCap":{
```

```
/*optional, alarm output schedule capability*/
  "week":{
/*required, int, days of the week: 1-Monday, 2-Tuesday, 3-Wednesday, 4-Thursday, 5-Friday,6-Saturday, 7-Sunday*/
    "@opt":"1,2,3,4,5,6,7"
  },
  "id":{
/*required, int, ID of time period for each day*/
    "@maxSize": 8
  },
  "beginTime":{
/*required, string, start time in ISO8601 format*/
    "@min":"00:00",
    "@max":"24:00"
  },
  "endTime":{
/*required, string, end time in ISO8601 format*/
    "@min":"00:00",
    "@max":"24:00"
  }
  },
  "whiteLightMode":{
/*optional, string, supplement light mode: "keepOn"-always on, "flashing"-flashing*/
    "@opt":"keepOn,flashing",
    "@def":"keepOn"
  }
}
}
```

JSON_WiresharkToUSB

JSON message about parameters of exporting captured packets to USB flash drive

```
{
  "WiresharkToUSB": {
    "switch": "",
    "operateCmd": "",
    "DeviceCondList": [
      {
        "DeviceCond": {
          "id": ,
          "devType": "",
          "devID": ,
          "subSys": ,
        }
      }
    ]
  }
}
```

JSON_WiresharkToUSBStatus

JSON message about status of exporting captured packets to USB flash drive

```
{
  "WiresharkToUSBStatus":{
    "WiresharkExportList":[{
      "WiresharkExport":{
        "devType": "",
        "devID": ,
        "subSys": ,
        "switch": "",
        "operateCmd": "",
        "exportStatus": "",
        "exceptionMsg": ""
      }
    }]
  }
}
```

JSON_WorkingStatus

WorkingStatus message in JSON format

```
{
  "WorkingStatus":{
    "devStatus": ,
    /*required, integer type, value of device status: 0-normal, 1-CPU usage, higher than 85%, 2-hardware error (e.g., serial
    port exception)*/
    "ChanStatus":[{
      "chanNo": ,
      /*required, integer type, channel No., which starts from 1*/
      "enable": ,
      /*integer type, it is valid for analog channel only, 0-disable, 1-enable*/
      "online": ,
      /*required, integer type, online status: 0-offline, 1-online*/
      "record": ,
      /*required, whether the device is recording, 0-no, 1-yes*/
      "recordStatus": ,
      /*integer type, recording status: 0-recording, 1-recording exception (HDD exception), 2-recording exception (network
      camera offline), 3-recording exception (other reason)*/
      "signal": ,
      /*required, integer type, signal status: 0-normal, 1-signal loss*/
      "linkNum": ,
      /*required, integer type, number of software clients connected to this channel*/
      "bitRate": ,
      /*required, integer type, channel bit rate, unit: Kpbs*/
    }],
    "HDStatus":{
```

```
"hdNo": ,
/*required, integer type, HDD No., which starts from 1*/
"status": ,
/*required, integer type, HDD status: 0-activate, 1-sleep, 2-exception, 3-sleepy HDD error, 4-unformatted, 5-
disconnected (for network HDD), 6-formatting*/
"volume": ,
/*required, integer type, HDD capacity, unit: MB*/
"freeSpace":
/*required, integer type, free space, unit: MD*/
}},
"IOStatus":{
  "IOInTrig":[...,...,...],
/*integer type, triggered alarm input No., two places and below: analog alarm input No., two places and above: digital
alarm input No.; and the low 2-bit is the alarm input No., the 3-bit or above is the digital channel No., e.g., 3201-alarm
input No.1 of digital channel No.32*/
  "IOOutTrig":[...,...,...]
/*integer type, triggered alarm output No., two places and below: analog alarm output No., two places and above:
digital alarm output No.; and the lower 2 bits are the alarm output No., the 3-bit or above is the digital channel No.,
e.g., 3201-alarm output No.1 of digital channel No.32*/
}
}
}
```

JSON_XX

XX message in JSON format

```
{
  "XX":{
/*required, event/alarm configuration information*/
    "enabled": "",
/*required, enable or not, boolean*/
    "zoneName": ""
/*optional, zone name, it is valid only when event type is zone, string*/
  }
}
```

Remarks

The XX in the parameter "XX" corresponds to detailed event type. E.g., if the event type is humanRecognition, then the returned node is "HumanRecognition".

JSON_XXCap

XXCap message in JSON format

```
{
  "XXCap":{
/*required, event configuration information*/
```

```
"enabled": "true,false",
/*required, enable or not, boolean*/
"zoneName":{
/*optional, zone name, it is valid only when event type is zone, string*/
"@min": 1,
"@max": 64,
},
}
}
```

Remarks

The XX in the parameter "**XXCap**" corresponds to detailed event type. E.g., if the event type is humanRecognition, then the returned node is "HumanRecognitionCap".

B.1.2 XML Messages

XML_accessDevice

accessDevice message in XML format

```
<accessDevice version="2.0" xmlns="http://www.isapi.org/ver20/XMLSchema">
  <id><!--req, xs:integer--></id>
  <MACAddress><!--opt, xs:string--></MACAddress>
  <ipV4Address><!--dep, xs:string--></ipV4Address>
  <accessTime><!--req, xs:time, ISO8601 data--></accessTime>
</accessDevice>
```

XML_accessDeviceList

accessDeviceList message in XML format

```
<accessDeviceList version="2.0" xmlns="http://www.isapi.org/ver20/XMLSchema">
  <accessDevice/><!--see details in the message of XML_accessDevice-->
</accessDeviceList>
```

See Also

XML_accessDevice

XML_accessPoint

accessPoint message in XML format

```
<accessPoint version="2.0" xmlns="http://www.isapi.org/ver20/XMLSchema">
  <id><!--req, xs:integer--></id>
  <networkMode>
```

```
<!--opt, xs:string, "infrastructure, adhoc"-->
</networkMode>
<channel><!--opt, xs:string, "1 to 14, auto"--></channel>
<ssid><!--req, xs:string--></ssid>
<speed><!--opt, xs:integer, unit: Mbps--></speed>
<signalStrength><!--opt, xs:integer, signal strength, which is between 0 and 100--></signalStrength>
<securityMode>
  <!--req, xs:string, "disable,WEP,WPA-personal,WPA2-personal,WPA-RADIUS,WPA-enterprise,WPA2-enterprise"-->
</securityMode>
<connected><!--opt, xs:boolean--></connected>
<connecting><!--opt, xs:boolean, whether the device is connecting to the Wi-Fi--></connecting>
</accessPoint>
```

XML_accessPointList

accessPointList message in XML format

```
<accessPointList version="2.0" xmlns="http://www.isapi.org/ver20/XMLSchema">
  <accessPoint/><!--see details in the message of XML_accessPoint>
</accessPointList>
```

See Also

XML_accessPoint

XML_ActivateInfo

ActivateInfo message in XML format.

```
<ActivateInfo version="2.0" xmlns="http://www.isapi.org/ver20/XMLSchema">
  <password><!--req, xs:string--></password>
</ActivateInfo>
```

XML_AdminAccessProtocol

AdminAccessProtocol message in XML format

```
<AdminAccessProtocol version="2.0" xmlns="http://www.isapi.org/ver20/XMLSchema">
  <id><!--required, xs:string, ID--></id>
  <enabled><!--optional, xs:boolean, "true,false"--></enabled>
  <protocol><!--required, xs:string, protocol name:
  "HTTP,HTTPS,RTSP,DEV_MANAGE,IOT,WebSocket,WebSocketS,SDK_OVER_TLS,TLS1_1Enable,TLS1_2Enable,SRTP,MATRI
  X_GATEWAY,Bonjour"--></protocol>
  <portNo><!--required, xs:integer--></portNo>
  <redirectToHttps>
    <!--opt, xs: boolean, whether to automatically go to HTTPS when connecting to HTTP port and HTTPS is enabled-->
  </redirectToHttps>
  <streamOverTls>
```



```
<!--opt, xs: boolean, whether to enable TLS link encryption when the streaming mode is SDK_OVER_TLS, true=yes, false=no-->
</streamOverTls>
<TLS1_1Enable><!--dep, xs:boolean--></TLS1_1Enable>
<TLS1_2Enable><!--dep, xs:boolean--></TLS1_2Enable>
</AdminAccessProtocol>
```

Remarks

"TLS1_1Enable" and "TLS1_2Enable" are not available for port configuration.

XML_AdminAccessProtocolList

AdminAccessProtocolList message in XML format

```
<AdminAccessProtocolList version="2.0" xmlns="http://www.isapi.org/ver20/XMLSchema">
  <AdminAccessProtocol/><!--see details in the message of XML_AdminAccessProtocol-->
</AdminAccessProtocolList>
```

See Also

XML_AdminAccessProtocol

XML_AlgorithmsVersion

Message about version information of algorithm library in XML format

```
<AlgorithmsVersion version="2.0" xmlns="http://www.isapi.org/ver20/XMLSchema">
  <AlgorithmsVersionInfoList><!--optional-->
    <AlgorithmsVersionInfo>
      <majorVersion><!--required, xs:integer, major version No.--></majorVersion>
      <minorVersion><!--required, xs:integer, minor version No.--></minorVersion>
      <revisionNumber><!--required, xs:integer, revision No.--></revisionNumber>
      <buildNumber><!--required, xs:integer, build No.--></buildNumber>
      <versionYear><!--required, xs:integer, date: year--></versionYear>
      <versionMonth><!--required, xs:integer, date: month--></versionMonth>
      <versionDay><!--required, xs:integer, date: day--></versionDay>
      <name>
        <!--required, xs:string, algorithm library name, "faceProperties"-face attributes, "behaviorAnalysis"-behavior
        analysis, "humanAttribute"-human body attributes, "faceSnap"-face capture, "faceRecognition"-face recognition,
        "faceContrast"-face picture comparison, "peopleCounting"-people counting, "faceScore"-face picture score,
        "personQueueDetection"-queue management, "depthMap"-depth map, "safetyHelmetDetection"-hard hat detection,
        "streetBehavior"-street behavior, "workBehavior"-work behavior, "cityManagement"-city management,
        "fireEscapeDetection"-fire engine access detection, "takingElevatorDetection"-elevator detection-->
        <name>
          </AlgorithmsVersionInfo>
        </AlgorithmsVersionInfoList>
      </AlgorithmsVersion>
```

Example

AlgorithmsVersion Message Example

```
<AlgorithmsVersion version="2.0" xmlns="http://www.isapi.com/ver20/XMLSchema">
  <AlgorithmsVersionInfoList>
    <AlgorithmsVersionInfo>
      <majorVersion>1</majorVersion>
      <minorVersion>0</minorVersion>
      <revisionNumber>1</revisionNumber>
      <versionYear>17</versionYear>
      <versionMonth>12</versionMonth>
      <versionDay>12</versionDay>
      <name>faceSnap</name>
    </AlgorithmsVersionInfo>
    <AlgorithmsVersionInfo>
      <majorVersion>1</majorVersion>
      <minorVersion>0</minorVersion>
      <revisionNumber>0</revisionNumber>
      <versionYear>17</versionYear>
      <versionMonth>10</versionMonth>
      <versionDay>31</versionDay>
      <name>faceScore</name>
    </AlgorithmsVersionInfo>
    <AlgorithmsVersionInfo>
      <majorVersion>3</majorVersion>
      <minorVersion>4</minorVersion>
      <revisionNumber>1</revisionNumber>
      <versionYear>17</versionYear>
      <versionMonth>12</versionMonth>
      <versionDay>25</versionDay>
      <name>faceContrast</name>
    </AlgorithmsVersionInfo>
  </AlgorithmsVersionInfoList>
</AlgorithmsVersion>
```

XML_anonymouslogin

XML message about parameters of managing anonymous user permission

```
<?xml version="1.0" encoding="utf-8"?>
<anonymouslogin version="2.0" xmlns="http://www.isapi.org/ver20/XMLSchema">
  <enabled><!--required, xs:boolean--></enabled>
</anonymouslogin>
```

XML_AudioCap

AudioCap capability message in XML format

```
<AudioCap version="2.0" xmlns="http://www.isapi.org/ver20/XMLSchema">
  <audioInputNums><!--req, xs:integer--></audioInputNums>
  <audioOutputNums><!--req, xs:integer--></audioOutputNums>
</AudioCap>
```

XML_AudioChannel

AudioChannel message in XML format

```
<AudioChannel version="2.0" xmlns="http://www.isapi.org/ver20/XMLSchema">
  <id><!--required, xs:string, audio channel index--></id>
  <enabled><!--required, xs:boolean, whether to enable the audio channel--></enabled>
  <type opt="audioIn,audioOut"><!--optional, xs:string, audio channel type: "audioIn"-audio input, "audioOut"-audio
output--></type>
  <name><!--optional, xs:string, audio channel name--></name>
</AudioChannel>
```

XML_AudioChannelList

AudioChannelList message in XML format

```
<AudioChannelList version="2.0" xmlns="http://www.isapi.org/ver20/XMLSchema">
  <AudioChannel/><!--opt, see details in the message of XML_AudioChannel-->
</AudioChannelList>
```

See Also

XML_AudioChannel

XML_AudioDetection

XML message about audio detection parameters

```
<AudioDetection version="2.0" xmlns="http://www.isapi.org/ver20/XMLSchema">
  <id><!--required, xs:string, ID--></id>
  <audioInputException>
    <enabled><!--required, xs:boolean--></enabled>
  </audioInputException>
  <soundIntensityMutation><!--optional-->
    <enabled><!--required, xs:boolean--></enabled>
    <sensitivityLevel><!--required, xs:integer--></sensitivityLevel>
    <mutationThreshold><!--required, xs:integer--></mutationThreshold>
  </soundIntensityMutation>
  <SteepFall><!--optional-->
    <enabled><!--required, xs:boolean--></enabled>
    <sensitivityLevel><!--required, xs:integer--></sensitivityLevel>
  </SteepFall>
```

```
<AudioLoss><!--optional-->
  <enabled><!--required, xs:boolean--></enabled>
  <sensitivityLevel><!--optional, xs:integer, sensitivity, it is between 1 and 100 and the default value is 50--></
sensitivityLevel>
</AudioLoss>
</AudioDetection>
```

XML_AudioDescriptor

AudioDescriptor message in XML format

```
<AudioDescriptor version="2.0" xmlns="http://www.isapi.org/ver20/XMLSchema">
  <audioCompressionType>
    <!--opt, xs:string, audio encoding type: "G.711alaw,G.711ulaw,G.726,G.729,G.729a,G.
729b,PCM,MP3,AC3,AAC,ADPCM,MP2L2"-->
  </audioCompressionType>
  <audioSamplingRate><!--opt, xs:float, kHz--></audioSamplingRate>
</AudioDescriptor>
```

XML_AudioIn

AudioIn message in XML format

```
<AudioIn version="2.0" xmlns="http://www.isapi.org/ver20/XMLSchema">
  <id>
    <!--req, xs:string-->
  </id>
  <MixAudioIn>
    <!--req, ro-->
    <enabled>
      <!--req, xs:Boolean-->
    </enabled>
    <audioInputType>
      <!--opt, xs:string-->
    </audioInputType>
    <highPassFilter>
      <!--req, xs:integer, unit: Hz-->
    </highPassFilter>
    <noiseMargin>
      <!--req, xs:integer, unit: dB-->
    </noiseMargin>
    <AutoLimitWave>
      <!--dep-->
    <FBCEnable>
      <!--req, xs:Boolean-->
    </FBCEnable>
    <mode>
      <!--req, xs:string-->
    </mode>
```

```
<filterQValue>
  <!--req, xs:string-->
</filterQValue>
<staticFilterNum>
  <!--req, xs:integer-->
</staticFilterNum>
</AutoLimitWave>
</MixAudioIn>
<AudioInVolumelist>
  <AudioInVlome>
    <type>
      <!--req, xs:string, "audioInput,audioEncode"-->
    </type>
    <volume>
      <!--req, xs:integer-->
    </volume>
  </AudioInVlome>
</AudioInVolumelist>
</AudioIn>
```

XML_AudioInCap

AudioInCap capability message in XML format

```
<AudioInCap version="2.0" xmlns="http://www.isapi.org/ver20/XMLSchema">
  <id>
    <!--req, xs:string-->
  </id>
  <MixAudioIn>
    <!--req, ro-->
    <enabled opt="true,false">
      <!--req, xs:Boolean-->
    </enabled>
    <audioInputType opt="micIn, lineIn">
      <!--opt, xs:string-->
    </audioInputType>
    <highPassFilter
      opt="0,8,16,24,31,39,47,55,63,71,79,87,94,102,110,118,126,134,142,150,157,165,173,181,189,197,205,213,220,228,
      236,244,252,260,268,276,283,291,299,307,315,323,331,339,346,354,362,370,378,386,394,402,409,417,425,433,441,
      449,457,465,472,480,488,496,504,512,520,528,535,543,551,559,567,575,583,591,598,606,614,622,630,638,646,654,
      661,669,677,685,693,701,709,717,724,732,740,748,756,764,772,780,787,795,803,811,819,827,835,843,850,858,866,
      874,882,890,898,906,913,921,929,937,945,953,961,969,976,984,992,1000" default="30">
      <!--req, xs:integer, unit: Hz-->
    </highPassFilter>
    <noiseMargin
      opt="1000,660,657,653,650,647,643,640,637,633,-630,627,623,620,617,613,610,607,603,600,597,593,590,587,583,5
      80,577,573,570,567,563,560,557,553,550,547,543,540,537,533,530,527,523,520,517,513,510,507,503,500,497,493,4
      90,487,483,480,477,473,470,467,463,460,457,453,450,447,443,440,437,433,430,427,423,420,417,413,410,407,403,4
      00,397,393,390,387,383,380,377,373,370,367,363,360,357,353,350,347,343,340,337,333,330,327,323,320,317,313,3
      10,307,303,300,297,293,290,287,283,280,277,273,270,267,263,260,257,253,250,247,243,240" default="30">
      <!--req, xs:integer, unit: dB-->
    </noiseMargin>
  </MixAudioIn>
</AudioInCap>
```

```
</noiseMargin>
<AutoLimitWave>
  <!--dep, audioInputType-->
  <FBCEnable opt="true,false">
    <!--req, xs:Boolean-->
  </FBCEnable>
  <mode opt="fast,general,slow" default="general">
    <!--req, xs:string-->
  </mode>
  <filterQValue opt="40,10" default="40">
    <!--req, xs:integer-->
  </filterQValue>
  <staticFilterNum min="0" max="12" default="0">
    <!--req, xs:integer-->
  </staticFilterNum>
</AutoLimitWave>
</MixAudioIn>
<AudioInVolumelist>
  <AudioInVlome>
    <type>
      <!--req, xs:string, "audioOutput,audioEncode"-->
    </type>
    <volume min="0" max="127" defalut="50">
      <!--req, xs:integer-->
    </volume>
  </AudioInVlome>
</AudioInVolumelist>
</AudioInCap>
```

XML_AudioInConfig

XML message about audio input parameters of all channels

```
<AudioInConfig version="2.0" xmlns="http://www.isapi.org/ver20/XMLSchema">
  <FBCEnable><!--required, xs:boolean, whether to enable FBC: true, false--></FBCEnable>
  <AECEnable><!--optional, xs:boolean, whether to enable AEC (Acoustic Echo Cancellation): true, false--></
AECEnable>
</AudioInConfig>
```

XML_AudioInConfigCap

XML message about the audio input configuration capability of all channels

```
<AudioInConfigCap version="2.0" xmlns="http://www.isapi.org/ver20/XMLSchema">
  <FBCEnable opt="true,false"><!--required, xs:boolean--></FBCEnable>
  <AECEnable opt="true,false"><!--optional, xs:boolean, whether it supports AEC (Acoustic Echo Cancellation): true,
false--></AECEnable>
</AudioInConfigCap>
```

XML_AudioOut

AudioOut message in XML format

```
<AudioOut version="2.0" xmlns="http://www.isapi.org/ver20/XMLSchema">
  <id><!--req, xs:string--></id>
  <MixAudioOut>
    <enabled><!--req, xs:Boolean--></enabled>
    <modulatorEnbale><!--req, xs:Boolean--></modulatorEnbale>
    <postFilter><!--req, xs:Boolean--></postFilter>
    <limitPressure><!--req, xs:Boolean--></limitPressure>
    <modulatorValue><!--req, xs:integer--></modulatorValue>
    <triggerTime><!--req, xs:integer, unit: ms--></triggerTime>
    <freeTime><!--req, xs:integer, unit: ms--></freeTime>
    <compressThreshold><!--req, xs:integer--></compressThreshold>
    <compressMode><!--req, xs:string--></compressMode>
    <compressRate><!--req, xs:integer--></compressRate>
    <recoveryGain><!--req, xs:integer--></recoveryGain>
    <outputGain><!--req, xs:integer --></outputGain>
  </MixAudioOut>
  <AudioOutVolumelist>
    <AudioOutVlome>
      <type><!--req, xs:string, "audioOutput,audioDecode"--></type>
      <volume><!--req, xs:integer--></volume>
    </AudioOutVlome>
  </AudioOutVolumelist>
</AudioOut>
```

XML_AudioOutCap

AudioOutCap capability message in XML format

```
<AudioOutCap xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0" >
  <id>
    <!--req, xs:string-->
  </id>
  <MixAudioOut>
    <enabled opt="true,false" >
      <!--req, xs:Boolean-->
    </enabled>
    <modulatorEnbale opt="true,false" >
      <!--req, xs:Boolean-->
    </modulatorEnbale>
    <postFilter opt="false,true" default="true" >
      <!--req, xs:Boolean-->
    </postFilter>
    <limitPressure opt="false,true" default="true" >
      <!--req, xs:Boolean-->
    </limitPressure>
```

```
<modulatorValue min="0" max="127" default="10" >
  <!--req, xs:integer-->
</modulatorValue>
<triggerTime
opt="1,1,1,1,1,1,1,1,1,1,1,1,1,1,1,1,2,2,2,2,2,2,2,2,2,2,3,3,3,3,3,3,4,4,4,4,4,4,5,5,5,5,6,6,6,7,7,7,8,8,8,9,9,10,
10,11,11,12,12,13,14,15,15,16,17,18,19,20,21,23,24,25,27,29,30,32,34,36,38,41,43,46,49,52,55,59,63,67,72,76,82,87,
93,10,17,114,123,132,141,152,163,175,188,23,218,235,254,274,296,320,347,375,47,441,479,520,566,616,671,731,79
8,872" default="4" >
  <!--req, xs:integer, unit: ms-->
</triggerTime>
<freeTime
opt="98,11,15,18,112,116,120,124,129,133,138,143,148,154,159,165,171,177,184,191,198,26,213,222,230,239,248,2
58,268,278,289,31,313,325,338,352,366,381,397,413,430,448,467,486,57,528,551,575,60,626,653,682,712,743,776,8
11,848,887,927,970,1014,1062,1111,1163,1218,1276,1337,142,1470,1541,1617,1697,1781,1870,1964,2063,2168,227
9,2397,2521,2653,2792,2940,3097,3263,3439,3626,3825,4035,4259,4497,4750,5020,536,5611,5936,6282,6651,7045,
7465,7914,8393,895,9452,10037,10664,11335,12053,12824,13650,14537,15489,16511,17610,18793,20065,21435,22
911,2452,26219,28073,30075,32239,34580,37114,39859,42834,46062" default="42" >
  <!--req, xs:integer, unit: ms-->
</freeTime>
<compressThreshold opt="1,2,4,8" defalut="2" >
  <!--req, xs:integer-->
</compressThreshold>
<compressMode opt="soft,hard" default="soft" >
  <!--req, xs:string-->
</compressMode>
<compressRate min="0" max="127" >
  <!--req, xs:integer-->
</compressRate>
<recoveryGain opt="1,2,4,8" defalut="2" >
  <!--req, xs:integer-->
</recoveryGain>
<outputGain
opt="100,421,361,325,30,281,265,252,240,230,221,212,25,198,192,186,180,175,170,165,161,156,152,148,145,141,1
38,134,131,128,125,122,120,117,114,112,110,17,15,13,10,98,96,94,92,90,88,86,85,83,81,79,78,76,74,73,71,70,68,67,
65,64,62,61,60,58,57,56,54,53,52,51,49,48,47,46,45,43,42,41,40,39,38,37,36,35,34,33,32,31,30,29,28,27,26,25,24,23,
23,22,21,20,19,18,17,17,16,15,14,13,12,12,11,10,9,9,8,7,6,6,5,4,3,3,2,1,1,0" >
  <!--req, xs:integer-->
</outputGain>
</MixAudioOut>
<AudioOutVolumelist>
  <AudioOutVlome>
    <type>
      <!--req, xs:string, "audioOutput,audioEncode"-->
    </type>
    <volume min="0" max="127" defalut="50" >
      <!--req, xs:integer-->
    </volume>
  </AudioOutVlome>
</AudioOutVolumelist>
</AudioOutCap>
```


XML_AudioStrengthStatus

XML message about the audio strength parameters

```
<AudioStrengthStatus version="2.0" xmlns="http://www.isapi.org/ver20/XMLSchema">
  <id><!--required, xs:string--></id>
  <audioStrength><!--read-only, required, xs:integer--></audioStrength>
</AudioStrengthStatus>
```

XML_autoPanData

XML message about fan control parameters

```
<?xml version="1.0" encoding="utf-8"?>
<autoPanData version="2.0" xmlns="http://www.isapi.org/ver20/XMLSchema">
  <autoPan><!--required, xs:integer, range: [0,100]--></autoPan>
</autoPanData>
```

XML_BadSectorsTest

BadSectorsTest message in XML format

```
<BadSectorsTest version="2.0" xmlns="http://www.isapi.org/ver20/XMLSchema">
  <testType><!--opt, xs: string, "full,metadata"--></testType>
</BadSectorsTest>
```

XML_BadSectorsTestStatus

BadSectorsTestStatus message in XML format

```
<BadSectorsTestStatus version="2.0" xmlns="http://www.isapi.org/ver20/XMLSchema">
  <diskID><!--req, xs: string--></diskID>
  <MaskAreaList><!--req-->
    <MaskArea><!--list-->
      <maskAreaID><!--req, xs: integer--></maskAreaID>
      <startLBA><!--opt,xs:integer--></startLBA>
      <endLBA><!--opt, xs: integer--></endLBA>
    </MaskArea>
  </MaskAreaList>
  <BlockAreaTestStatus><!--opt-->
    <testType><!--opt, xs: string, "full,metadata"--></testType>
    <testStatus><!--req, xs: string, "none,running,pause,complete,exceed,abort"--></testStatus>
    <firstBlock><!--opt, xs: integer--></firstBlock>
    <lastBlock><!--opt, xs: integer--></lastBlock>
    <currentBlock><!--opt, xs: integer--></currentBlock>
  <BadSectorsList><!--opt-->
```

```
<BadSectors><!--list-->
  <id><!--req, xs: integer--></id>
  <block><!--opt, xs: integer--></block>
</BadSectors>
</BadSectorsList>
</BlockAreaTestStatus>
</BadSectorsTestStatus>
```

XML_BatteryPowerOverlay

BatteryPowerOverlay message in XML format

```
<BatteryPowerOverlay version="2.0" xmlns="http://www.isapi.org/ver20/XMLSchema">
  <enabled><!--req, xs:boolean--></enabled>
  <positionX><!--req, xs:integer, X-coordinate--></positionX>
  <positionY><!--req, xs:integer, Y-coordinate--></positionY>
</BatteryPowerOverlay>
```

XML_BLC

BLC message in XML format

```
<BLC version="2.0" xmlns="http://www.isapi.com/ver20/XMLSchema">
  <enabled/><!--req, xs: boolean-->
  <BLCMode/><!--opt, xs: string, BLC mode: "UP, DOWN, LEFT, RIGHT, CENTER, MULTI-AREA, Region"-->
  <BLCLevel><!--opt, xs: integer--></BLCLevel>
  <BLCRegionList><!--dep-->
    <BLCRegion>
      <id><!--req, xs: integer--></id>
      <RegionCoordinatesList/>
    </BLCRegion>
  </BLCRegionList>
</BLC>
```

XML_BluetoothCap

BluetoothCap capability message in XML format

```
<BluetoothCap version="2.0" xmlns="http://www.isapi.org/ver20/XMLSchema">
  <isSupportIbeacon>
    <!--opt, xs:boolean, whether to support configuring ibeacon bluetooth parameters-->
  </isSupportIbeacon>
</BluetoothCap>
```

XML_BluetoothStatus

BluetoothStatus message in XML format

```
<BluetoothStatus version="2.0" xmlns="http://www.isapi.org/ver20/XMLSchema">
  <connectStatus><!--opt, xs:boolean--></connectStatus>
  <connectDeviceName><!--opt, xs:string--></connectDeviceName>
</BluetoothStatus>
```

XML_Bond

Bond message in XML format

```
<Bond version="2.0" xmlns="urn:selfextension:ISAPIext-ver10-xsd">
  <id>
    <!--req, xs:integer-->
  </id>
  <enabled>
    <!--req, xs:boolean-->
  </enabled>
  <workMode>
    <!--req, xs:string, working mode: "balance-rr"-round-robin, "active-backup"-->
  </workMode>
  <primaryIf>
    <!--req, xs:string, default route ID-->
  </primaryIf>
  <slaveIfList>
    <!--req-->
    <ethernetIfId>
      <!--req, xs:string, ID-->
    </ethernetIfId>
  </slaveIfList>
  <IPAddress>
    <ipVersion>
      <!--req, xs:string, "v4,v6,dual"-->
    </ipVersion>
    <addressingType>
      <!--req, xs:string, "static,dynamic,apipa"-->
    </addressingType>
    <ipAddress>
      <!--dep, xs:string-->
    </ipAddress>
    <subnetMask>
      <!--dep, xs:string, subnet mask for IPv4 address-->
    </subnetMask>
    <ipv6Address>
      <!--dep, xs:string-->
    </ipv6Address>
    <bitMask>
```

```
<!--dep, xs:integer, bitmask for IPv6 address-->
</bitMask>
<DefaultGateway>
  <!--dep-->
  <ipAddress>
    <!--dep, xs:string-->
  </ipAddress>
  <ipv6Address>
    <!--dep, xs:string-->
  </ipv6Address>
</DefaultGateway>
<PrimaryDNS>
  <!--dep-->
  <ipAddress>
    <!--dep, xs:string-->
  </ipAddress>
  <ipv6Address>
    <!--dep, xs:string-->
  </ipv6Address>
</PrimaryDNS>
<SecondaryDNS>
  <!--dep-->
  <ipAddress>
    <!--dep, xs:string-->
  </ipAddress>
  <ipv6Address>
    <!--dep, xs:string-->
  </ipv6Address>
</SecondaryDNS>
</IPAddress>
</Bond>
```

XML_BondList

BondList message in XML format

```
<BondList version="2.0" xmlns="urn:selfextension:ISAPIext-ver10-xsd">
  <Bond/><!--see details in the message of XML_Bond-->
</BondList>
```

See Also

XML_Bond

XML_Cap_accessDeviceList

accessDeviceList capability message in XML format

```
<accessDeviceList version="2.0" xmlns="http://www.isapi.org/ver20/XMLSchema">
  <accessDevice size="4">
```

```
<id><!--req, xs:integer--></id>
<MACAddress><!--opt, xs:string--></MACAddress>
<ipV4Address><!--opt, xs:string--></ipV4Address>
<accessTime><!--req, xs:time, ISO8601 data--></accessTime>
</accessDevice>
</accessDeviceList>
```

XML_Cap_AcsUpdate

AcsUpdate capability message in XML format

```
<AcsUpdate version="2.0" xmlns="http://www.isapi.org/ver20/XMLSchema">
  <type
opt="cardReader,FPMModule,securityModule,extendModule,channelController,IRModule,lampModule,elevatorControll
er,FPAAlgorithmProgram,uboot,keypad,wirelessRecv,wiredZone">
    <!--opt, upgrading type: "cardReader"-card reader, "FPMModule"-fingerprint module, "securityModule"-secure door
control unit, "extendModule"-(IO) extension module, "channelController"-lane controller, "IRModule"-IR module,
"lampModule"-indicator module, "elevatorController"-slave elevator controller, "FPAAlgorithmProgram"-fingerprint
algorithm program of the card reader, "uboot", "keypad", "wirelessRecv"-wireless receiving module, "wiredZone"-
wired zone module-->
  </type>
  <cardReaderNo min="" max="">
    <!--opt, range of No. of the card reader, this node is valid only when <type> contains "cardReader"-->
  </cardReaderNo>
  <FPMModuleNo min="" max="">
    <!--opt, range of No. of the fingerprint module, this node is valid only when <type> contains "FPMModule"-->
  </FPMModuleNo>
  <securityModuleNo min="" max="">
    <!--opt, range of No. of the secure door control module, this node is valid only when <type> contains
"securityModule"-->
  </securityModuleNo>
  <extendModuleNo min="" max="">
    <!--opt, range of No. of the (IO) extension module, this node is valid only when <type> contains "extendModule"-->
  </extendModuleNo>
  <channelControllerNo min="" max="">
    <!--opt, range of No. of the lane controller, this node is valid only when <type> contains "channelController"-->
  </channelControllerNo>
  <IRModuleNo min="" max="">
    <!--opt, range of No. of the IR module, this node is valid only when <type> contains "IRModule"-->
  </IRModuleNo>
  <lampModuleNo min="" max="">
    <!--opt, range of No. of the indicator module, this node is valid only when <type> contains "lampModule"-->
  </lampModuleNo>
  <elevatorControllerNo min="" max="">
    <!--opt,range of No. of the slave elevator controller, this node is valid only when <type> contains
"elevatorController"-->
  </elevatorControllerNo>
  <FPAAlgorithmProgramNo min="" max="">
    <!--opt, range of No. of the fingerprint algorithm program of the card reader, this node is valid only when <type>
contains "FPAAlgorithmProgram"-->
  </FPAAlgorithmProgramNo>
```

```
<keypadAddress opt="1,3,5">
  <!--opt, range of keypad module address, this node is valid when <type> contains "keypad"-->
</keypadAddress>
<wirelessRecvAddress opt="1,3,5">
  <!--opt, range of wireless receiving module address, this node is valid when <type> contains "wirelessRecv"-->
</wirelessRecvAddress>
<wiredZoneAddress opt="1,3,5">
  <!--opt, range of wired zone module address, this node is valid when <type> contains "wiredZone"-->
</wiredZoneAddress>
</AcsUpdate>
```

XML_Cap_AdminAccessProtocolList

AdminAccessProtocol capability message in XML format

```
<AdminAccessProtocolList version="2.0" xmlns="http://www.isapi.com/ver20/XMLSchema">
  <AdminAccessProtocol><!--multiple <AdminAccessProtocol> nodes are allowed-->
    <id><!--req, xs: string, ID--></id>
    <enabled opt="true,false"><!--opt, xs: boolean--></enabled>
    <protocol
opt="HTTP,HTTPS,SDK,RTSP,DEV_MANAGE,WebSocket,WebSocketS,SDK_OVER_TLS,SRTP,MATRIX_GATEWAY,IOT,Bonjour"
    <!--req, xs: string-->
  </protocol>
  <portNo min="2000" max="65535" default="8443"><!--req, xs:integer --></portNo>
  <redirectToHttps opt="true,false">
    <!--opt, xs: boolean, whether to automatically go to HTTPS when connecting to HTTP port and HTTPS is enabled.
This node is valid when protocol is "HTTPS"-->
  </redirectToHttps>
  <streamOverTls opt="true,false">
    <!--opt, xs: boolean, whether to enable TLS link encryption when the streaming mode is SDK_OVER_TLS, true=yes,
false-no-->
  </streamOverTls>
  <TLS1_1Enable opt="true,false"><!--dep, xs: boolean, whether to enable TLS version 1.1, it is valid when protocol is
"HTTPS", "true"-yes, "false"-no--></TLS1_1Enable>
  <TLS1_2Enable opt="true,false"><!--dep, xs: boolean, whether to enable TLS version 1.2, it is valid when protocol is
"HTTPS", "true"-yes, "false"-no--></TLS1_2Enable>
</AdminAccessProtocol>
  <TLS1_0Enable opt="true,false">
    <!--dep, xs: boolean, whether to enable TLS version 1.0, it is valid when protocol is "HTTPS", "true"-yes, "false"-no-->
  </TLS1_0Enable>
  <TLS1_1Enable opt="true,false">
    <!--dep, xs: boolean, whether to enable TLS version 1.1, it is valid when protocol is "HTTPS", "true"-yes, "false"-no-->
  </TLS1_1Enable>
  <TLS1_2Enable opt="true,false">
    <!--dep, xs: boolean, whether to enable TLS version 1.2, it is valid when protocol is "HTTPS", "true"-yes, "false"-no-->
  </TLS1_2Enable>
</AdminAccessProtocolList>
```

XML_Cap_AudioDetection

XML message about the configuration capability of audio detection

```
<AudioDetection version="2.0" xmlns="http://www.isapi.org/ver20/XMLSchema">
  <id><!--required, xs:string, ID--></id>
  <audioInputException>
    <enabled opt="true,false"><!--required, xs:boolean--></enabled>
  </audioInputException>
  <soundIntensityMutation>
    <enabled opt="true,false"><!--required, xs:boolean--></enabled>
    <sensitivityLevel min="1" max="100"><!--required, xs:integer--></sensitivityLevel>
    <mutationThreshold min="1" max="100"><!--required, xs:integer--></mutationThreshold>
  </soundIntensityMutation>
  <SteepFall><!--optional-->
    <enabled opt="true,false"><!--required, xs:boolean--></enabled>
    <sensitivityLevel min="1" max="100"><!--required, xs:integer--></sensitivityLevel>
  </SteepFall>
  <mutexAbility opt="PDC"/><!--optional, read-only, xs:string, "PDC"-->
  <isSupportMultiScene><!--optional, xs:boolean--></isSupportMultiScene>
  <AudioLoss><!--opt-->
    <enabled opt="true,false"><!--required, xs:boolean--></enabled>
    <sensitivityLevel min="" max="" def=""><!--optional, xs:integer, sensitivity, it is between 1 and 100 and its default value is 50--></sensitivityLevel>
  </AudioLoss>
</AudioDetection>
```

XML_Cap_BatteryPowerOverlay

BatteryPowerOverlay capability message in XML format

```
<BatteryPowerOverlay version="2.0" xmlns="http://www.isapi.org/ver20/XMLSchema">
  <enabled>
    <!--req, xs:boolean-->
  </enabled>
  <positionX>
    <!--req, xs:integer, X-coordinate-->
  </positionX>
  <positionY>
    <!--req, xs:integer, Y-coordinate-->
  </positionY>
</BatteryPowerOverlay>
```

XML_Cap_Color

Color capability message in XML format

```
<?xml version="1.0" encoding="utf-8"?>
<Color version="2.0" xmlns="http://www.isapi.org/ver20/XMLSchema">
  <brightnessLevel min="" max=""><!--opt,xs:integer, brightness--></brightnessLevel>
  <contrastLevel min="" max=""><!--opt,xs:integer, contrast--></contrastLevel>
  <saturationLevel min="" max=""><!--opt,xs:integer, saturation--></saturationLevel>
  <hueLevel min="" max=""><!--opt, xs:integer, hue--></hueLevel>
  <grayScale><!--opt, gray scale-->
    <grayScaleMode><!--opt,xs:string, gray scale mode: "indoor,outdoor"--></grayScaleMode>
  </grayScale>
  <nightMode opt="true,false">
    <!--opt, xs:boolean, enable night mode, when its value is "true", the saturation can be adjusted, otherwise, the
saturation cannot be adjusted-->
  </nightMode>
</Color>
```

XML_Cap_DatetimeOverlay

DatetimeOverlay capability message in XML format

```
<DateTimeOverlay version="2.0" xmlns="http://www.isapi.org/ver20/XMLSchema">
  <enabled>
    <!--req, xs:boolean, whether to enable date and time overlay-->
  </enabled>
  <positionX min="0" max="1000">
    <!--req, xs:integer, X-coordinate-->
  </positionX>
  <positionY min="0" max="1000">
    <!--req, xs:integer, Y-coordinate-->
  </positionY>
  <dateStyle>
    <!--opt, xs:string, date format: "YYYY-MM-DD, MM-DD-YYYY, DD-MM-YYYY, CHR-YYYY-MM-DD, CHR-MM-DD-YYYY,
CHR-DD-MM-YYYY"-->
  </dateStyle>
  <timeStyle>
    <!--opt, xs:string, "12hour, 24hour"-->
  </timeStyle>
  <displayWeek>
    <!--opt, xs:boolean, whether to display week information-->
  </displayWeek>
  <displayMilliSecond>
    <!--opt, xs:boolean, whether to display millisecond-->
  </displayMilliSecond>
</DateTimeOverlay>
```

XML_Cap_DDNS

DDNS capability message in XML format


```
<DDNS version="2.0" xmlns="http://www.isapi.org/ver20/XMLSchema">
  <id>
    <!--req, xs:string-->
  </id>
  <enabled>
    <!--req, xs:boolean, "true,false"-->
  </enabled>
  <ethernetIfId opt="1,2,3...">
    <!--opt, xs:string, ID-->
  </ethernetIfId>
  <provider opt="IPServer,DynDNS,PeanutHall,NO-IP,HiDDNS">
    <!--req, xs:string-->
  </provider>
  <serverAddress>
    <addressingFormatType opt="ipaddress,hostname">
      <!--req, xs:string-->
    </addressingFormatType>
    <hostname min="0" max="260">
      <!--dep, xs:string-->
    </hostname>
    <ipAddress min="0" max="64">
      <!--dep, xs:string-->
    </ipAddress>
    <ipv6Address min="0" max="64">
      <!--dep, xs:string-->
    </ipv6Address>
  </serverAddress>
  <portNo min="0" max="65535">
    <!--opt, xs:integer-->
  </portNo>
  <deviceDomainName min="0" max="64">
    <!--dep, xs:string-->
  </deviceDomainName>
  <username min="0" max="32">
    <!--dep, xs:string-->
  </userName>
  <password min="0" max="16">
    <!--wo, dep, xs:string-->
  </password>
</DDNS>
```

XML_Cap_DeviceInfo

DeviceInfo capability message in XML format

```
<DeviceInfo version="2.0" xmlns="http://www.isapi.org/ver20/XMLSchema">
  <deviceName min="" max=""><!--req, xs:string, device name--></deviceName>
  <DockStation>
    <!--opt, dock station configuration-->
  </Platform>
```

```
<!--opt, platform configuration-->
<type opt="none, 9533, 8618, ISAPI"><!--req, xs:string, platform type--></type>
<ip><!--opt, xs:string, IP address --></ip>
<port><!--opt, xs:integer, communication port--></port>
<userName><!--req, xs:string, user name, which is used for the dock station to log in to platform--></userName>
<password><!--req, xs:string, password, which is used for the dock station to log in to platform, it should be
encrypted--></password>
</Platform>
<centralStorageBackupEnabled opt="true, false"><!--opt, xs:boolean, whether to enable central storage backup--></
centralStorageBackupEnabled>
</DockStation>
<languageType opt="chinese,english,spanish,portuguese,italian,french,russian,german,polish"/><!--opt, xs:string-->
<deviceId min="0" max="128"><!--ro, req, xs:string, uuid--></deviceId>
<deviceDescription min="0" max="16"><!--opt, xs:string--></deviceDescription>
<deviceLocation opt="STD-CGI, hangzhou"><!--opt, xs:string--></deviceLocation>
<systemContact opt="STD-CGI, Hikvision.China"><!--opt, req, xs:string--></systemContact>
<model min="0" max="64"><!--ro, req, xs:string--></model>
<serialNumber min="0" max="48"><!--ro, req, xs:string--></serialNumber>
<macAddress min="0" max="64"><!--ro, req, xs:string--></macAddress>
<firmwareVersion min="0" max="64"><!--ro, req, xs:string--></firmwareVersion>
<firmwareReleasedDate min="0" max="64"><!--ro, opt, xs:string--></firmwareReleasedDate>
<bootVersion min="0" max="16"><!--ro, opt, xs:string--></bootVersion>
<bootReleasedDate min="0" max="16"><!--ro, opt, xs:string--></bootReleasedDate>
<hardwareVersion min="0" max="16"><!--ro, opt, xs:string--></hardwareVersion>
<encoderVersion min="0" max="64"><!--ro, opt, xs:string--></encoderVersion>
<encoderReleasedDate min="0" max="64"><!--ro, opt, xs:string--></encoderReleasedDate>
<decoderVersion min="0" max="64"><!--ro, opt, xs:string--></decoderVersion>
<decoderReleasedDate min="0" max="64"><!--ro, opt, xs:string--></decoderReleasedDate>
<deviceType opt="IPCamera, IPDome, DVR, HybirdNVR, NVR, DVS, IPZoom"><!--ro, req, xs:string--></deviceType>
<telecontrolID min="1" max="255"><!--opt, xs:integer, "1-255"--></telecontrolID>
<supportBeep><!--opt, xs:boolean: "true,false"--></supportBeep>
<firmwareVersionInfo><!--ro, opt, xs:string--></firmwareVersionInfo>
<subChannelEnabled><!--opt, xs:boolean: "true,false"--></subChannelEnabled>
<thrChannelEnabled><!--opt, xs:boolean: "true,false"--></thrChannelEnabled>
<actualFloorNum><!--req, xs:integer, "1-128"--></actualFloorNum>
<radarVersion><!--opt, xs:string, radar version--></radarVersion>
<powerOnMode opt="button,adapter" def="button"><!--optional, xs:string, device startup mode: "button"-press
button to power on (default), "adapter"-connect adapter to power on--></powerOnMode>
<webVersion><!--optional, read-only, xs:string, web version No., it is the current value by default--></webVersion>
<deviceRFProgramVersion><!--optional, read-only, xs:string, version No. of the device's RF (Radio Frequency)
program, it is the current value by default--></deviceRFProgramVersion>
<securityModuleSerialNo><!--optional, read-only, xs:string, serial No. of the security module, it is the current value
by default--></securityModuleSerialNo>
<securityModuleVersion><!--optional, read-only, xs:string, version No. of the security module, it is the current value
by default--></securityModuleVersion>
<securityChipVersion><!--optional, read-only, xs:string, version No. of the security chip, it is the current value
by default--></securityChipVersion>
<securityModuleKeyVersion><!--optional, read-only, xs:string, version No. of the security module key, it is the current
value by default--></securityModuleKeyVersion>
<UIDLampRecognition><!--optional, information of the UID lamp recognition device-->
  <enabled opt="true,false"><!--optional, xs:boolean, whether to enable--></enabled>
</UIDLampRecognition>
```

```
<bootTime><!--optional, xs:string, read-only, system booting time, ISO 8601 format; the maximum length is 32 bytes--></bootTime>
</DeviceInfo>
```

XML_Cap_DefaultParam

DefaultParam capability message in XML format

```
<DefaultParam version="1.0" xmlns="http://www.isapi.org/ver20/XMLSchema">
  <DialNum min="1" max="32"><!--opt, xs:string--></DialNum>
  <Username min="1" max="32"><!--opt, xs:string--></Username>
  <Password min="1" max="32"><!--opt, xs:string--></Password>
  <APNname min="1" max="32"><!--opt, xs:string--></APNname>
  <VerifyProto opt="auto,CHAP,PAP"><!--req, xs:string, verification protocol: "auto, CHAP, PAP"--></VerifyProto>
</DefaultParam>
```

XML_Cap_DeviceLanguage

DeviceLanguage capability message in XML format

```
<DeviceLanguage version="2.0" xmlns="http://www.isapi.org/ver20/XMLSchema">
  <language
    opt="SimChinese,TraChinese,English,Russian,Bulgarian,Hungarian,Greek,German,Italian,Czech,Slovakia,French,Polish,D
    utch,Portuguese,Spanish,Romanian,Turkish,Japanese,Danish,Swedish,Norwegian,Finnish,Korean,Thai,Estonia,Vietname
    se,Hebrew,Latvian,Arabic,Sovenian,Croatian,Lithuanian,Serbian,BrazilianPortuguese,Indonesian,Ukrainian">
    <!--req, xs:string, language supported by the device: "SimChinese"-simplified Chinese, "TraChinese"-traditional
    Chinese, "English", "Russian", "Bulgarian", "Hungarian", "Greek", "German", "Italian", "Czech", "Slovakia", "French",
    "Polish", "Dutch", "Portuguese", "Spanish", "Romanian", "Turkish", "Japanese", "Danish", "Swedish", "Norwegian",
    "Finnish", "Korean", "Thai", "Estonia", "Vietnamese", "Hebrew", "Latvian", "Arabic", "Sovenian"-Slovenian, "Croatian",
    "Lithuanian", "Serbian", "BrazilianPortuguese"-Brazilian Portuguese, "Indonesian", "Ukrainian"-->
  </language>
</DeviceLanguage>
```

XML_Cap_DiagnosedDataParameter

XML message about capability of exporting diagnose information

```
<?xml version="1.0" encoding="utf-8"?>
<DiagnosedDataParameter version="2.0" xmlns="http://www.isapi.org/ver20/XMLSchema">
  <enabled opt="true,false">
    <!--required, xs:boolean, whether to enable exporting diagnose information: true=yes, false=no-->
  </enabled>
  <HardwareInfo opt="true,false">
    <!--optional, xs:boolean,opt="true,false", whether to enable exporting hardware information, such as running
    information of motor, fan, heater, compressor, supplement light, wiper, and so on-->
  </HardwareInfo>
  <SoftwareInfo opt="true,false">
```

```
<!--optional, xs:boolean, whether to enable exporting software information, such as software status, running log,
system information-->
</SoftwareInfo>
<logInfo opt="true,false">
  <!--optional, xs:boolean, whether to enable exporting startup log--></logInfo>
</DiagnosedDataParameter>
```

XML_Cap_Dial

Dial capability message in XML format

```
<Dial version="2.0" xmlns="http://www.isapi.org/ver20/XMLSchema">
  <enabled opt="true,false" def="false"><!--req, xs:boolean--></enabled>
  <DialMethod opt="auto,manual">
    <!--req, xs:string, "auto, manual", set it to manual when accessing the private network, and you can configure dial-
up parameters(including access number, user name, password, APN, and verification protocol-->
  </DialMethod>
  <SwitchMethod opt="auto,4GFirst,3GFirst,manualto2G,manualto3G,manualto4G,cableFirst">
    <!--req, xs:string, "auto,4GFirst,3GFirst, manualto2G, manualto3G,manualto4G,cableFirst"-->
  </SwitchMethod>
  <OfflineTime min="30" max="65535"><!--opt, xs:integer, unit: second--></OfflineTime>
  <UIMCardNum min="1" max="32"><!--opt, xs:string--></UIMCardNum>
  <DialNum min="1" max="32"><!--opt, xs:string--></DialNum>
  <Username min="1" max="32"><!--opt, xs:string--></Username>
  <Password min="1" max="32"><!--opt, xs:string--></Password>
  <APNname min="1" max="32"><!--opt, xs:string--></APNname>
  <SIMNum min="" max=""><!--opt, xs:string, SIM card No. (mobile phone number)--></SIMNum>
  <MTU min="100" max="1500"><!--opt, xs: integer--></MTU>
  <VerifyProto opt="auto,CHAP,PAP"><!--req, xs:string, verification protocol: "auto, CHAP, PAP"--></VerifyProto>
  <DefaultParam/><!--opt, default parameters, see details in the message of XML_Cap_DefaultParam-->
  <netAPN min="" max=""><!--opt, xs:string, APN configuration for the private network-->
  <Flow><!--opt, flow configuration-->
    <limitEnabled opt="true,false"><!--opt, xs:boolean, whether to enable flow limitation--></limitEnabled>
    <consumeFlow opt="true,false"><!--opt, xs:boolean, whether supports displaying flow usage, unit: MB--></
consumeFlow>
    <threshold min="" max=""><!--opt, threshold of flow, unit: MB--></threshold>
  </Flow>
  <enabled4G opt="true,false" def="true"><!-- opt, xs:boolean, whether to enable 4G--></enabled4G>
  <isSupportNetworkKeepAlive><!--opt, xs:boolean, whether it supports network keepalive --></
isSupportNetworkKeepAlive>
  <enabledDNS opt="true,false" def="true"><!-- opt, xs:boolean, whether to enable DNS manual configuration--></
enabledDNS>
  <pinCode min="" max=""/><!--optional, write-only, xs:string, PIN code-->
  <ISPName min="" max=""><!--optional, xs:string, ISP (Internet Service Provider) name--></ISPName>
  <IMEI min="" max=""><!--optional, xs:string, IMEI code--></IMEI>
  <ICCID><!--optional, xs:string, ICCID code--></ICCID>
  <netType opt="2G,3G,4G"><!--optional, xs:string, network type--></netType>
</Dial>
```

See Also

XML_Cap_DefaultParam

XML_Cap_Discovery

Discovery capability message in XML format

```
<Discovery version="2.0" xmlns="http://www.isapi.org/ver20/XMLSchema">
  <UPnP><!--req-->
    <enabled><!--req, xs:boolean, "true, false"--></enalbed>
  </UPnP>
  <ZeroConf><!--opt, channel-zero configuration-->
    <enabled><!--req, xs:boolean, "true, false"--></enabled>
  </ZeroConf>
</Discovery>
```

XML_Cap_DiscoveryMode

XML message about terminal SADP configuration capability

```
<DiscoveryMode xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0" >
  <ONVIFDiscoveryMode>
    <!--optional, xs:string, "discoverable,nonDiscoverable-->
  </ONVIFDiscoveryMode>
  <ISDiscoveryMode>
    <!-- optional, xs:string, "discoverable,nonDiscoverable-->
  </ISDiscoveryMode>
  <PcapMode>
    <!--optional, xs:string, "discoverable,nonDiscoverable-->
  </PcapMode>
  <SADP>
    <!--optional, xs:string,"discoverable,nonDiscoverable-->
  </SADP>
</DiscoveryMode>
```

XML_Cap_EagleFocusing

EagleFocusing capability message in XML format

```
<?xml version="1.0" encoding="utf-8"?>
<EagleFocusing version="2.0" xmlns="http://www.isapi.org/ver20/XMLSchema">
  <id><!--req,xs:string--></id>
  <enabled opt="true,false"><!--req, xs:boolean, "true"-start calibration, "false"-stop calibration--></enabled>
  <normalizedScreenSize><!--req,ro-->
    <normalizedScreenWidth><!--req,ro,xs:integer--></normalizedScreenWidth>
    <normalizedScreenHeight><!--req,ro,xs:integer--></normalizedScreenHeight>
```

```

</normalizedScreenSize>
<EagleFocusingRegionList><!--opt-->
  <EagleFocusingRegion><!--list-->
    <id><!--req, xs:string--></id>
    <sid><!--opt,xs:string, scene ID--></sid>
    <rate><!--opt,xs:integer, rate--></rate>
    <spotNum><!--opt,xs:integer, the number of calibration points--></spotNum>
    <type><!--req,xs:string,"line", region type, now only supports "line"--></type>
    <Region><!--dep, depends on type-->
      <RegionCoordinatesList size="4"><!--opt-->
        <RegionCoordinates><!--list-->
          <positionX><!--req,xs:integer,coordinate--></positionX>
          <positionY><!--req,xs:integer,coordinate--></positionY>
        </RegionCoordinates>
      </RegionCoordinatesList>
    </Region>
  </EagleFocusingRegion>
</EagleFocusingRegionList>
<sceneNum min="" max=""><!--opt,xs:integer, the maximum number of supported scenes--></sceneNum>
<isSupportSceneGoto opt="true,false"><!--opt,xs:boolean, whether device supports turning to specified scene--></
isSupportSceneGoto>
  <rate min="" max=""><!--opt,xs:integer, rate--></rate>
  <spotNum min="" max=""><!--opt,xs:integer, the number of calibration points on line--></spotNum>
  <onlyReadParam opt="rate,spotNum"><!--opt,xs:string,"rate,spotNum"--></onlyReadParam>
  <isSupportAuto opt="true,false"><!--req,xs:boolean, whether device supports automatic calibration of rapid focus--
></isSupportAuto>
</EagleFocusing>

```

XML_Cap_EHome

EHome capability message in XML format

```

<Ehome version="2.0" xmlns="http://www.isapi.org/ver20/XMLSchema">
  <id min="1" max="1"><!--opt, xs:string, EHome center ID--></id>
  <GPRSAddressingFormatType opt="ipaddress"><!--opt, xs:string, GPRS address type. If this node is not returned, the
address type will be determined by addressingFormatType which indicates that the address types configured by
different networks do not need to be distinguished--></GPRSAddressingFormatType>
  <addressingFormatType opt="ipaddress,hostname"><!--req, xs:string--></addressingFormatType>
  <hostName min="" max=""><!--dep, xs:string, domain name--></hostName>
  <ipAddress min="" max=""><!--dep, xs:string--></ipAddress>
  <ipv6Address min="" max=""><!--dep, xs:string--></ipv6Address>
  <portNo min="" max=""><!--opt, xs:integer--></portNo>
  <deviceId min="" max=""><!--req, xs:string--></deviceId>
  <registerStatus min="" max=""><!--ro, xs:boolean--></registerStatus>
  <key min="" max=""><!--opt, xs:string, encryption key of EHome version 5.0--></key>
  <version min="" max=""><!--ro, xs:string--></version>
  <netWork opt="0,1,2,3,4"/><!--opt,xs:integer, 0-make no sense, 1-automatic, 2-wired network preferred, 3-wired
network, 4-3G/4G/GPRS-->
  <voiceDeviceType opt="bluetooth,client,local"><!--opt, xs:string, two-way audio device type: "bluetooth"-bluetooth
device, "client", "local"--></voiceDeviceType>
  <protocolVersion opt="v2.0,v2.6,v4.0,v5.0"><!--opt, xs:string, protocol version: "v2.0,v2.6,v4.0,v5.0". If this node is

```

set to "v2.0", the device can only use protocol v2.0 to register; if this node is set to "v2.6", "v4.0" or "v5.0", the device will firstly use this protocol to register. If this node is not returned, the protocol version will be determined by `<version>--></protocolVersion>`
`</Ehome>`

XML_Cap_ExternalDevice

XML message about external device configuration capability

```
<?xml version="1.0" encoding="utf-8"?>
<ExternalDevice version="2.0" xmlns="http://www.isapi.org/ver20/XMLSchema">
  <SupplementLight/><!--opt, whether device supports configuring supplement light, see XML_Cap_SupplementLight
for details-->
  <THScreen/><!--opt, capability of peripheral screen configuration, corresponding url: /ISAPI/System/externalDevice/
THScreen-->
  <isSupportSupplementLightChannelSetting>
    <!--opt, xs:boolean, whether device supports configuring peripheral supplement light of specified channel,
corresponding url: /ISAPI/System/channel/<ID>/externalDevice/supplementLight-->
  </isSupportSupplementLightChannelSetting>
</ExternalDevice>
```

See Also

XML_Cap_SupplementLight

XML_Cap_ExtrInfo

ExtrInfo capability message in XML format

```
<ExtrInfo version="2.0" xmlns="http://www.isapi.org/ver20/XMLSchema">
  <diskMode opt="singleDisk,RAID"><!--req, xs:string, disk mode: "singleDisk"-single disk, "RAID". The device needs to
reboot after changing the disk mode--></diskMode>
</ExtrInfo>
```

XML_Cap_EZVIZ

XML message about the Hik-Connect access configuration capability

```
<?xml version="1.0" encoding="utf-8"?>
<EZVIZ version="2.0" xmlns="http://www.isapi.org/ver20/XMLSchema">
  <!--redirect: whether to enable redirecting server address-->
  <!--verificationCode: only admin users are allowed to edit the verification code-->
  <enabled opt="true,false"><!--required, xs:boolean--></enabled>
  <registerStatus opt="true,false"><!--read-only, optional, xs:boolean--></registerStatus>
  <redirect opt="true,false"><!--optional, xs:boolean--></redirect>
  <serverAddress><!--optional-->
    <addressingFormatType min="" max=""><!--required, xs:string, "ipaddress,hostname"--></addressingFormatType>
    <hostName min="" max=""><!--dependent, xs:string--></hostName>
```

```
<ipAddress min="" max=""><!--dependent, xs:string--></ipAddress>
<ipv6Address min="" max=""><!--dependent, xs:string--></ipv6Address>
</serverAddress>
<verificationCode min="" max=""><!--optional, xs:string--></verificationCode>
<offlineStatus opt="secretKeyInvalid"><!--read-only, dependent, xs:string, device offline status: "secretKeyInvalid"-
invalid verification code, it is valid when the value of registerStatus is "false"--></offlineStatus>
<streamEncryptEnabled opt="true,false"><!--optional, xs:boolean, whether to enable stream encryption--></
streamEncryptEnabled>
<isSupportUpISAPI><!--optional, xs:boolean, whether the device supports uplink transmitting ISAPI message-->true</
isSupportUpISAPI>
<isSupportDownISAPI><!--optional, xs:boolean, whether the device supports downlink transmitting ISAPI message--
>true</isSupportDownISAPI>
<isSupportEZVIZQRCode><!--optional, xs:boolean, whether the device supports getting the Hik-Connect QR code
(related URI: /ISAPI/System/Network/EZVIZ/QRCode?format=json)--></isSupportEZVIZQRCode>
</EZVIZ>
```

XML_Cap_FocusConfiguration

FocusConfiguration capability message in XML format

```
<FocusConfiguration>
<focusStyle opt="AUTO,MANUAL,SEMIAUTOMATIC"><!--req, xs:string--></focusStyle>
<focusLimited opt="50,100,300,600,1000,2000,5000,10000,15000" def="1000"><!--opt, xs:integer--></focusLimited>
<focusPosition/><!--dep, xs:integer, depends on FocusStyle-->
<focusSpeed><!--opt, xs:integer--></focusSpeed>
<focusSensitivity min="0" max="2" def="1"><!--opt, xs:integer, sensitivity of focus, ranging from 0 to 2. It is valid when
the focus mode is automatic or semi-automatic--></focusSensitivity>
<temperatureChangeAdaptEnabled opt="true,false" --><!--opt, xs:boolean--></temperatureChangeAdaptEnabled>
<relativeFocusPos min="0" max="4000" def=""><!--opt, xs:integer--></relativeFocusPos>
<highTemperaturePriority opt="true,false"><!--opt, xs:boolean, enable high temperature priority mode--></
highTemperaturePriority>
</FocusConfiguration>
```

XML_Cap_FTPNotification

FTPNotification capability message in XML format

```
<FTPNotification version="2.0" xmlns="http://www.isapi.org/ver20/XMLSchema">
<id>
<!--req, xs:string, FTP ID-->
</id>
<enabled>
<!--req, xs:boolean, whether to enable the FTP server-->
</enabled>
<useSSL>
<!--opt, xs:boolean-->
</useSSL>
<addressingFormatType opt="ipaddress,hostname">
<!--req, xs:string, this node can only be set to "ipaddress"-->
```



```
</addressingFormatType>
<hostName>
  <!--dep, xs:string-->
</hostName>
<ipAddress>
  <!--dep, xs:string-->
</ipAddress>
<ipv6Address>
  <!--dep, xs:string-->
</ipv6Address>
<portNo>
  <!--opt, xs:integer, FTP port No.-->
</portNo>
<userName>
  <!--req, xs:string, user name-->
</userName>
<password>
  <!--wo, xs:string, password-->
</password>
<passiveModeEnabled>
  <!--opt, xs:boolean-->
</passiveModeEnabled>
<annoyftp>
  <!--opt, xs:boolean-->
</annoyftp>
<uploadPicture>
  <!--opt, xs:boolean-->
</uploadPicture>
<uploadVideoClip>
  <!--opt, xs:boolean-->
</uploadVideoClip>
<uploadPath>
  <!--req-->
  <pathDepth>
    <!--req, xs:integer, directory levels, up to 16 levels can be supported-->
  </pathDepth>
  <topDirNameRule opt="none, devName, devId, devIp, positionInfo, time_month, time_date, illegalType, direction,
site, chanName, chanId, lanId, customize ,time, buildUnitNo">
    <!--dep, xs:string, parent directory name: "none", "devName"-device name, "devId"-device ID, "devIp"-device IP
address, "positionInfo"-camera 1, "time_month"-usage date (YYYY-MM), "time_date"-usage date (YYYY-MM-DD),
"illegalType"-violation type, "direction"-direction, "site"-place, "chanName"-channel name, "chanId"-channel No.,
"lanId"-lane No., "customize"-custom, "time", "buildUnitNo"-building No. and unit No. This node is set to NULL by
default-->
  </topDirNameRule>
  <topDirName min="" max="">
    <!--dep, xs:string, custom string for parent directory, the maximum string length is 32 bytes. This node is valid
when <topDirNameRule> is "customize". If attributes of "min" and "max" are not obtained after parsing, the default
value will be returned without error message-->
    <topDirName/><!--dep, xs:string, custom string for parent directory, the maximum string length is 32 bytes. This
node is valid when <topDirNameRule> is "customize"-->
    <subDirNameRule opt="none, devName, devId, devIp, positionInfo, time_month, time_date, illegalType, direction,
site, chanName, chanId, lanId, customize ,time, buildUnitNo">
```

```
<!--dep, xs:string, child directory name-->
</subDirNameRule>
<subDirName min="" max="">
  <!--dep, xs:string, custom string for child directory, the maximum string length is 32 bytes. This node is valid when
<subDirNameRule> is "customize". If attributes of "min" and "max" are not obtained after parsing, the default value
will be returned without error message-->
  <subDirName/><!--dep, xs:string, custom string for child directory, the maximum string length is 32 bytes. This node
is valid when <subDirNameRule> is "customize"-->
  <threeDirNameRule opt="none, devName, devId, devIp, positionInfo, time_month, time_date, illegalType, direction,
site, chanName, chanId, lanId, customize ,time, buildUnitNo">
    <!--dep, xs:string, the third directory name-->
    </threeDirNameRule>
    <threeDirName/><!--req, xs:string, custom string for the third directory, the maximum string length is 32 bytes. This
node is valid when <threeDirNameRule> is "customize"-->
    <fourDirNameRule opt="none, devName, devId, devIp, positionInfo, time_month, time_date, illegalType, direction,
site, chanName, chanId, lanId, customize ,time, buildUnitNo">
      <!--dep, xs:string, the fourth directory name-->
      </fourDirNameRule>
      <fourDirName/><!--req, xs:string, custom string for the fourth directory, the maximum string length is 32 bytes. This
node is valid when <fourDirNameRule> is "customize"-->
    </uploadPath>
    <picArchivingInterval min="" max="">
      <!--opt, xs:integer, the value is between 1 and 30, 0-close-->
    </picArchivingInterval>
    <picNameRuleType opt="default,prefix">
      <!--opt, xs:string-->
    </picNameRuleType>
    <picNamePrefix min="0" max="32">
      <!--dep, xs:string-->
    </picNamePrefix>
    <ftpPicNameRuleType opt="videoIntercom,ITC">
      <!--req, xs:string, type of FTP picture name rule: "videoIntercom"-rule used by video intercom products, "ITC"-rule
used by traffic cameras-->
    </ftpPicNameRuleType>
    <FTPPicNameRule>
      <!--dep, picture name rule of a specific FTP-->
      <ItemList/><!--req, see details in the message of XML_Cap_ItemList-->
      <delimiter>
        <!--req, xs:string, delimiter, which is a single character and the default value is "_"-->
      </delimiter>
      <customStr min="1" max="128">
        <!--req, xs:string, custom string-->
      </customStr>
    </FTPPicNameRule>
    <upDataType opt="0,1,2">
      <!-- opt, xs:integer, picture uploading type: 0-all, 1-checkpoint, 2-violation. When only one FTP server is enabled,
this node can only be set to 0. When two FTP servers are both enabled, you should set 1 for one FTP server and set 2
for another FTP server, which means that two FTP servers cannot be set to the same type-->
    </upDataType>
    <uploadPlateEnable>
      <!--opt, xs:boolean, whether to enable uploading license plate thumbnail-->
    </uploadPlateEnable>
```

```
<site min="1" max="128">
  <!--req, xs:string, place, the maximum string length is 128 bytes-->
</site>
<roadNum min="1" max="32">
  <!--req, xs:string, intersection No., the maximum string length is 32 bytes-->
</roadNum>
<instrumentNum min="1" max="32">
  <!--req, xs:string, device No., the maximum string length is 32 bytes-->
</instrumentNum>
<direction min="1" max="32">
  <!--req, xs:string, direction No., the maximum string length is 32 bytes-->
</direction>
<directionDesc min="1" max="32">
  <!--req, xs:string, direction description, the maximum string length is 32 bytes-->
</directionDesc>
<monitoringInfo1 min="1" max="44">
  <!--req, xs:string, camera 1 information, the maximum string length is 44 bytes-->
</monitoringInfo1>
<uploadAttachedInfomation>
  <!--req, xs:boolean, whether to upload additional information-->
</uploadAttachedInfomation>
<BrokenNetHttp><!--opt, whether it supports ANR (automatic network replenishment)-->
  <enabled opt="true,false"><!--opt, xs:boolean, whether to enable ANR (automatic network replenishment)--></
enabled>
  <supportEventType opt="personQueueCounting"/><!--opt, xs:string, event types supporting FTP ANR:
"personQueueCounting"-person queue counting detection, "personQueueTime"-person queue time detection,
"personQueueRealTimeData"-upload real-time data of person queue counting detection, "faceCapture"-face capture
and recognition, "fieldDetection"-intrusion, "attendedBaggage"-object removal, "unattendedBaggage"-unattended
baggage, "regionExiting"-region exiting, "regionEntrance"-region entrance, "lineDetection"-line crossing detection-->
</BrokenNetHttp>
</FTPNotification>
```

See Also

XML_Cap_ItemList

XML_Cap_FTPNotificationList

FTPNotificationList capability message in XML format

```
<FTPNotificationList version="2.0" xmlns="http://www.isapi.org/ver20/XMLSchema">
  <FTPNotification/><!--opt, see details in the message of XML_Cap_FTPNotification-->
</FTPNotificationList>
```

See Also

XML_Cap_FTPNotification

XML_Cap_GuardAgainstTheft

GuardAgainstTheft capability message in XML format

```
<GuardAgainstTheft version="2.0" xmlns="http://www.isapi.org/ver20/XMLSchema">
  <enabled opt="true,false"><!--required, xs:boolean, whether it supports enabling device anti-theft--></enabled>
  <SMSEnabled opt="true,false"><!--required, xs:boolean, whether it supports enabling SMS--></SMSEnabled>
  <phoneNum min="0" max="32"><!--dependency, xs:string, phone No.--></phoneNum>
  <longitudeLatitudeEnabled opt="true,false"><!--dependency, xs:boolean, whether it supports enabling longitude and
latitude--></longitudeLatitudeEnabled>
  <siteLocationEnabled opt="true,false"><!--dependency, xs:boolean, whether it supports enabling site location--></
siteLocationEnabled>
  <siteLocationInfo min="0" max="128"><!--dependency, xs:string, site location information--></siteLocationInfo>
  <timeEnabled opt="true,false"><!--dependency, xs:boolean, whether it supports enabling time--></timeEnabled>
  <customInfo min="0" max="128"><!--dependency, xs:string, custom information--></customInfo>
</GuardAgainstTheft>
```

XML_Cap_HardwareService

HardwareService capability message in XML format

```
<HardwareService version="2.0" xmlns="http://www.isapi.org/ver20/XMLSchema">
  <lrLightSwitch><!--opt-->
    <mode opt="open,close"><!--req, xs:string --></mode>
  </lrLightSwitch>
  <ABF><!--opt-->
    <enabled><!--req, xs:boolean --></enabled>
  </ABF>
  <LED><!--opt-->
    <enabled><!--req, xs:boolean --></enabled>
  </LED>
  <Defog><!--opt-->
    <enabled><!--req, xs:boolean --></enabled>
  </Defog>
  <SupplementLight><!--opt-->
    <enabled><!--req, xs:boolean --></enabled>
    <isSupportFireLaserLight opt="true,false"><!--opt,xs:boolean--></isSupportFireLaserLight>
    <isSupportSupplementLightWord opt="true,false"><!--opt,xs:boolean--></isSupportSupplementLightWord>
    <captureWithSupplimentLightEnabled opt="true,false">
      <!--opt, xs:boolean, enable snapshot supplement light or not-->
    </captureWithSupplimentLightEnabled>
  </SupplementLight>
  <Deicing><!--opt-->
    <enabled><!--req, xs:boolean --></enabled>
  </Deicing>
  <ManualDeicing><!--opt-->
    <enabled><!--req, xs:boolean --></enabled>
  </ManualDeicing>
  <mutexAbility opt="laserLight, deicing">
```

```
<!--req, mutual exclusion ability, the laser light and deicing (including manual deicing and automatic deicing) are mutual exclusive-->
</mutexAbility>
<HighTemperatureProtection><!--opt-->
  <enabled><!--req, xs:boolean --></enabled>
  <temperatureType opt="90,100,110" def="90"><!--dep, xs:string,unit:°C --></temperatureType>
</HighTemperatureProtection>
</HardwareService>
```

XML_Cap_hddList

hddList capability message in XML format

```
<hddList version="2.0" xmlns="http://www.isapi.org/ver20/XMLSchema" size="">
  <hdd><!--list-->
    <id><!--read-only, required, xs: string; ID--></id>
    <hddName><!--read-only, required, xs: string--></hddName>
    <hddPath><!--read-only, optional, xs: string--></hddPath>
    <hddType opt="IDE,SATA,eSATA,NFS,iSCSI,Virtual Disk"><!--read-only, required, xs: string--></hddType>
    <status
      opt="ok,unformatted,error,idle,mismatch,offline,smartFailed,reparing,formatting,notexist,unRecordHostFormatted">
      <!--read-only, required, xs: string, unRecordHostFormatted--unformatted in education sharing system-->
    </status>
    <capacity><!--read-only, required, xs: float, unit: MB--></capacity>
    <freeSpace><!--read-only, required, xs: float, unit: MB--></freeSpace>
    <property opt="RW,read-only,Redund"><!--required, xs: string, HDD properties--></property>
    <group><!--optional, xs: string; HDD group ID--></group>
    <reservedDayCfg min="0" max="1000"><!--optional, xs:integer, video saving duration, it is between 0 and 1000,
unit: day--></reservedTimeCfg>
    <DataModeList><!--optional, read-only, current HDD allocation mode-->
      <DataMode>
        <type opt="recordStorage,pictureCloudStorage,fileStorage">
          <!--required, xs: string, storage application type, recordStorage-video storage, pictureCloudStorage-picture to be
saved in cloud storage, fileStorage-file-storage->
        </type>
        <occupancyRate><!--required, xs: integer, HDD usage, range: [0,100]--></occupancyRate>
      </DataMode>
    </DataModeList>
    <formatType opt="FAT32,EXT4" def="FAT32">
      <!--read-only, optional, xs: string, formatting type, this node is only available for SD card; if this node does not
exist, the default formatting type is FAT32-->
    </formatType>
    <Encryption>
      <passwordLen min="6" max="64"/>
      <encryptionStatus opt="unencrypted,encrypted,verifyFailed"><!--read-only, optional, xs:string, encryption status:
"unencrypted", "encrypted", "verifyFailed"-verification failed--></encryptionStatus>
      <encryptFormatType opt="FAT32,EXT4"><!--read-only, optional, xs:string--></encryptFormatType>
    </Encryption>
    <isSupportLogStorage><!--optional, xs:boolean, whether it can be used as a log HDD--></isSupportLogStorage>
```

```
</hdd>
</hddList>
```

XML_Cap_IbeaconParam

IbeaconParam capability message in XML format

```
<IbeaconParam version="2.0" xmlns="http://www.isapi.org/ver20/XMLSchema">
  <UUID min="" max="">
    <!--req, xs:string, parking lot ID, which supports using digits, letters and hyphen as the tag-->
  </UUID>
  <major min="" max="">
    <!--req, xs:integer, number of floors in the parking lot, which cannot be 0x00 for all-->
  </major>
  <minor min="" max="">
    <!--req, xs:integer, number of parking spaces on each floor, which cannot be 0x00 for all-->
  </minor>
  <sendPower min="" max="">
    <!--req, xs:integer, transmitted power: 01-0 dbm, 02- -6 dbm, 03- -23 dbm-->
  </sendPower>
  <frequency min="" max="">
    <!--req, xs:integer, transmitted frequency, which is the broadcast time interval, unit: 625 μs. The broadcast time interval is between 32 (20 ms) to 8000 (5s), and the default value is 160 (100 ms)-->
  </frequency>
  <measurePower min="" max="">
    <!--req, xs:integer, rated power, which is the RSSI (Received Signal Strength Indication) in the position of 1 meter away, unit: dbm-->
  </measurePower>
</IbeaconParam>
```

XML_Cap_IEEE802_1x

IEEE802_1x capability message in XML format

```
<IEEE802_1x version="2.0" xmlns="http://www.isapi.org/ver20/XMLSchema">
  <enabled><!--req, xs: boolean--></enabled>
  <authenticationProtocolType opt="EAP-TLS,EAP-TTLS,EAP-PEAP,EAP-LEAP,EAP-FAST,EAP-MD5">
    <!--req, xs: string-->
  </authenticationProtocolType>
  <innerTTLSAuthenticationMethod opt="MS-CHAP,MS-CHAPv2,PAP,EAP-MD5">
    <!--dep, xs: string, this node is required when <authenticationProtocolType> is "EAP-TLS"-->
  </innerTTLSAuthenticationMethod>
  <innerEAPProtocolType opt="EAP-POTP,MS-CHAPv2">
    <!--dep, xs: string, this node is required when <authenticationProtocolType> is "EAP-PEAP" or "EAP-FAST"-->
  </innerEAPProtocolType>
  <validateServerEnabled><!--dep, xs: boolean--></validateServerEnabled>
  <userName><!--dep, xs: string--></userName>
  <password><!--dep, xs: string--></password>
  <anonymousID><!--opt, xs: string--></anonymousID>
```

```
<autoPACProvisioningEnabled>
  <!--dep, xs: boolean, this node is required when <authenticationProtocolType> is "EAP-FAST"-->
</autoPACProvisioningEnabled>
<Extensions><!--opt-->
  <EAPOLVersion><!--opt, xs: string, "1, 2"--></EAPOLVersion>
</Extensions>
</IEEE802_1x>
```

XML_Cap_ImageChannel

XML message about image channel capability

```
<ImageChannel version="2.0" xmlns="http://www.isapi.org/ver20/XMLSchema">
  <id><!--req, xs:integer--></id>
  <enabled><!--req, xs:boolean--></enabled>
  <videoInputID><!--req, xs:integer--></videoInputID>
  <FocusConfiguration/><!--opt, refer to the message XML_Cap_FocusConfiguration for details-->
  <LensInitialization/><!--opt-->
  <ImageFlip/><!--opt-->
  <ImageFreeze/><!--opt-->
  <proportionalpan/><!--opt-->
  <WDR>
    <mode opt="close,open,auto"><!--req, xs: string, WDR mode: "open,close,auto"--></mode>
    <WDRLevel max="100" min="0"><!--opt, xs: integer--></WDRLevel>
    <WDRContrastLevel max="" min="0"><!--opt, xs:integer--></WDRContrastLevel>
    < WDRLevel1 max="" min="0"><!--opt,xs:integer-->< /WDRLevel1>
  </WDR>
  <BLC/><!--opt-->
  <NoiseReduce/><!--opt-->
  <ImageEnhancement/><!--opt-->
  <SlowShutter/><!--opt-->
  <DSS/><!--opt-->
  <WhiteBlance/><!--opt-->
  <Exposure/><!--opt, exposure configuration parameters, refer to the message XML_Exposure for details-->
  <DayNightGate/><!--opt-->
  <BrightEnhance/><!--opt-->
  <Sharpness>
    <SharpnessLevel max="100" min="0"><!--req, xs:integer--></SharpnessLevel>
  </Sharpness>
  <gammaCorrection/><!--opt-->
  <powerLineFrequency/><!--opt-->
  <Color><!--opt, image adjustment capability-->
    <brightnessLevel max="100" min="0"><!--opt, xs:integer, brightness--></brightnessLevel>
    <contrastLevel max="100" min="0"><!--opt,xs:integer, contrast--></contrastLevel>
    <saturationLevel max="100" min="0"><!--opt, xs:integer, saturation--></saturationLevel>
    <grayScale>
      <grayScaleMode opt="outdoor,indoor"><!--opt,xs:string, gray scale mode: "indoor,outdoor"--></grayScaleMode>
    </grayScale>
  </Color>
  <IrcutFilter/><!--opt, configuration parameters of day/night auto-switch, refer to the message XML_IrcutFilter for details-->
```

```

<Scene/><!--opt-->
<EPTZ/><!--opt-->
<EIS/><!--opt-->
<HLC/><!--opt-->
<ZoomLimit/><!--opt-->
<corridor/><!--opt-->
<Dehaze/><!--opt-->
<ImageMode opt="standard, indoor, outdoor, dimLight"/><!--opt, xs:string-->
<enableImageLossDetection><!--opt, boolean--></enableImageLossDetection>
<CaptureMode/><!--opt-->
<IrLight/><!--opt-->
<LensDistortionCorrection/><!--opt-->
<SupplementLight/><!--opt, supplement light configuration capability, refer to the message
XML_Cap_SupplementLight for details-->
<OpticalDehaze/><!--opt-->
<ManualRanging/><!--opt-->
<OIS/><!--opt-->
<isSupportlaserSpotManual><!--opt, boolean--></isSupportlaserSpotManual>
<isSupportLaserSpotAdjustment><!--opt, boolean--></isSupportLaserSpotAdjustment>
<DigitalZoom>
  <ZoomRatio opt="1x,2x,4x,8x,16x,32x"><!--req, xs:string--></ZoomRatio>
</DigitalZoom>
<Palettes>
  <mode
opt="WhiteHot,BlackHot,Fusion1,Rainbow,Fusion2,Ironbow1,Ironbow2,Sepia,Color1,Color2,IceFire,Rain,RedHot,Green
Hot,DeepBlue,Color3"><!--
opt,xs:string,"WhiteHot,BlackHot,Fusion1,Rainbow,Fusion2,Ironbow1,Ironbow2,Sepia,Color1,Color2,IceFire,Rain,RedH
ot,GreenHot,DeepBlue,Color3"--></mode>
  <ColorateTarget><!--dep,mode="WhiteHot","target coloration"-->
    <ColorateTargetModeList>
      <ColorateTargetMode><!--list-->
        <id><!--req,xs:integer "index,which starts from 1"--></id>
        <mode opt="colorateHotAreae,colorateIntervalArea,colorateColdArea"><!--
req,xs:string,"colorateHotAreae,colorateIntervalArea,colorateColdArea"--></mode>
        <enabled opt="true,false"><!--req,xs:boolean,"true,false"--></enabled>
        <TemperatureLimit><!--req "temperature range"-->
          <minTemperature><!--dep <mode> is colorateHotAreae/colorateIntervalArea,xs:float--></minTemperature>
          <maxTemperature><!--dep <mode> is colorateColdArea/colorateIntervalArea,xs:float--></maxTemperature>
        </TemperatureLimit>
        <Color><!--req,"area color"-->
          <R><!--req,xs:integer--></R>
          <G><!--req,xs:integer--></G>
          <B><!--req,xs:integer--></B>
        </Color>
      </ColorateTargetMode>
    </ColorateTargetModeList>
  </ColorateTarget>
  <supportColorateTargetMode opt="WhiteHot,BlackHot"><!--opt,xs:string, palette modes that support target
coloration--></supportColorateTargetMode>
</Palettes>
<ExposureSync opt="true,false" def="false">
  <enabled><!--req, xs:boolean--></enabled>

```



```

</ExposureSync>
<BrightnessSuddenChangeSuppressionCap/><!--opt-->
<isSupportIcr><!--opt, boolean--></isSupportIcr>
<isSupportMultishut><!--opt, boolean--></isSupportMultishut>
<isSupportPlateBright><!--opt, boolean--></isSupportPlateBright>
<isSupportJPEGParam><!--opt, boolean--></isSupportJPEGParam>
<isSupportDarkEnhance><!--opt, boolean--></isSupportDarkEnhance>
<isSupportHdr><!--opt, boolean--></isSupportHdr>
<isSupportLse><!--opt, boolean--></isSupportLse>
<isSupportMce><!--opt, boolean--></isSupportMce>
<isSupportScve><!--opt, boolean--></isSupportScve>
<isSupportSectionCtrl><!--opt, boolean--></isSupportSectionCtrl>
<isSupportAutoContrast><!--opt, boolean--></isSupportAutoContrast>
<isSupportGrayRange><!--opt, boolean--></isSupportGrayRange>
<isSupportLSEDetail><!--opt, boolean--></isSupportLSEDetail>
<isSupportCapture><!--opt, boolean--></isSupportCapture>
<isSupportBrightEnhance><!--opt, boolean--></isSupportBrightEnhance>
<isSupportRecord><!--opt, xs:boolean--></isSupportRecord>
<isSupportDefog><!--opt, xs:boolean--></isSupportDefog>
<isSupportGeneral><!--opt, xs:boolean, whether to display general configuration page on the interface--></
isSupportGeneral>
<isSupportIa><!--opt, xs:boolean, whether to display video configuration page on the interface--></isSupportIa>
<NoiseReduce2D>
  <noiseReduce2DEnable><!--req, xs:boolean--></noiseReduce2DEnable>
  <noiseReduce2DLevel min="0" max="100"><!--dep,xs:integer--></noiseReduce2DLevel>
</NoiseReduce2D>
<Shutter>
  <ShutterLevel min="100" max="40000"><!--req, xs:integer--></ShutterLevel>
</Shutter>
<Gain>
  <GainLevel min="0" max="100"><!--req, xs:integer--></GainLevel>
</Gain>
<TempRange/><!--opt, temperature range capability, see XML_Cap_tempRange for details-->
</ImageChannel>

```

Remarks

The nodes **<DigitalZoom>** and **<ExposureSync>** are not supported by thermographic automation thermal camera (DS-2TA03-15SVI, DS2TA06-25SVI).

XML_Cap_InputProxyChannel

InputProxyChannel capability message in XML format

```

<InputProxyChannel version="1.0" xmlns="http://www.isapi.com/ver20/XMLSchema">
  <id min="" max=""></id>
  <name min="" max=""></name>
  <sourceInputPortDescriptor>
    <adminProtocol opt="HIKVISION,SONY,ISAPI,ONVIF"></adminProtocol>
    <addressingFormatType opt="ipaddress,hostname"></addressingFormatType>
    <hostName min="" max=""></hostName>
  </sourceInputPortDescriptor>
</InputProxyChannel>

```

```
<ipAddress></ipAddress>
<ipv6Address></ipv6Address>
<adminPortNo min="2000" max="65535"></adminPortNo>
<srcInputPort min="" max=""></srcInputPort>
<userName min="" max=""><!--user name, which should be encrypted--></userName>
<password min="" max=""><!--password, which should be encrypted--></password>
<streamType opt="auto,tcp,udp"></streamType>
<deviceId min="" max=""></deviceId>
</sourceInputPortDescriptor>
<enableAnr opt="true,false"></enableAnr>
<NVRInfo>
  <ipAddressNVR>
    <!--opt, xs:string, IP address of NVR-->
  </ipAddressNVR>
  <portNVR min="2000" max="65535">
    <!--opt, xs:integer, port No. of NVR-->
  </portNVR>
  <ipcChannelNo min="" max="">
    <!--opt, xs:integer, channel No. of the network camera in NVR-->
  </ipcChannelNo>
</NVRInfo>
</InputProxyChannel>
```

XML_Cap_IOInputPortList

IOInputPortList capability message in XML format

```
<IOInputPortList version="2.0" xmlns="http://www.isapi.org/ver20/XMLSchema">
  <IOInputPort><!--list-->
    <id><!--required, xs:string, IO port No.--></id>
    <enabled opt="true,false"><!--required, xs:boolean, whether to enable--></enabled>
    <triggering opt="high,low"><!--required, xs:string, triggering mode: "high"-high level, "low"-low level--></triggering>
    <name min="" max=""><!--required, xs:string, IO port name--></name>
    <inputType opt="switch,semaphore"><!--optional, xs:string, input mode: "switch", "semaphore"--></inputType>
    <CombinationAlarmCap><!--optional, composite alarm capability-->
      <channel min="" max=""><!--required, xs:integer, channel No. range--></channel>
      <EventTypeList size=""><!--required, event type list, the attribute "size" indicates the maximum number of
supported events for composition-->
        <eventType><!--list, xs:string, event type: "VMD"-motion detection, "shelteralarm"-video tampering detection,
"facedetection"-face detection, "fielddetection"-intrusion detection, "linedetection"-line crossing detection,
"regionEntrance"-region entrance detection, "regionExitings"-region exiting detection, "loitering"-loitering detection,
"group"-people gathering detection, "rapidMove"-fast moving detection, "parking"-parking detection,
"unattendedBaggage"-unattended baggage detection, "attendedBaggage"-object removal detection, "ANPR",
"audioexception"-audio exception detection--></eventType>
      </EventTypeList>
    </CombinationAlarmCap>
    <supportIOInputType opt="local,digitalChannel"/><!--optional, supported IO port type: "local"-local IO port,
"digitalChannel"-digital channel IO-->
  </IOInputPort>
</IOInputPortList>
```

XML_Cap_IOOutputPortList

IOOutputPortList capability message in XML format

```
<IOOutputPortList version="2.0" xmlns="http://www.isapi.org/ver20/XMLSchema">
  <IOOutputPort>
    <id min="" max=""><!-- req, xs:integer, "2" --></id>
    <PowerOnState><!--req, output port configuration parameters when the device is powered on>
      <defaultState opt="high,low">
        <!--ro, req, xs:string,default output port signal when it is not triggered-->
      </defaultState>
      <outputState opt="high,low,pulse">
        <!--ro, req, xs:string, output port signal when it is being triggered-->
      </outputState>
      <pulseDuration min="" max="">
        <!--dep, xs:integer, duration of a output port signal when it is being triggered, it is valid when outputState is
"pulse", unit: milliseconds-->
      </pulseDuration>
    </PowerOnState>
    <name><!--opt, xs:string--></name>
    <IOUseType opt="disable,electricLock,custom"><!--opt, xs:string--></IOUseType>
    <normalStatus opt="open, close"><!--opt, xs:string,normal status: open-remain open, close-remain closed--></
normalStatus>
  </IOOutputPort>
</IOOutputPortList>
```

XML_Cap_IpAddress

IpAddress capability message in XML format

```
<IpAddress version="2.0" xmlns="http://www.isapi.org/ver20/XMLSchema">
  <ipVersion opt="IPv4,IPv6,dual">
    <!--req, xs:string-->
  </ipVersion>
  <addressingType opt="static,dynamic,apipa">
    <!--req, xs:string-->
  </addressingType>
  <ipAddress min="0" max="40">
    <!--dep, xs:string-->
  </ipAddress>
  <subnetMask min="0" max="15">
    <!--dep, xs:string-->
  </subnetMask>
  <ipv6Address min="0" max="40">
    <!--dep, xs:string-->
  </ipv6Address>
  <bitMask ask min="0" max="128">
    <!--dep, xs:integer-->
  </bitMask>
```

```
<DefaultGateway>
  <!--dep-->
  <ipAddress min="0" max="40">
    <!--dep, xs:string-->
  </ipAddress>
  <ipv6Address min="0" max="40">
    <!--dep, xs:string-->
  </ipv6Address>
</DefaultGateway>
<PrimaryDNS>
  <!--dep-->
  <ipAddress min="0" max="40">
    <!--dep, xs:string-->
  </ipAddress>
  <ipv6Address min="0" max="40">
    <!--dep, xs:string-->
  </ipv6Address>
</PrimaryDNS>
<SecondaryDNS>
  <!--dep-->
  <ipAddress min="0" max="40">
    <!--dep, xs:string-->
  </ipAddress>
  <ipv6Address min="0" max="40">
    <!--dep, xs:string-->
  </ipv6Address>
</SecondaryDNS>
<Ipv6Mode><!--opt-->
  <ipV6AddressingType opt="router,ra,manual,dhcp">
    <!--dep, xs:string-->
  </ipV6AddressingType>
  <ipv6AddressList>
    <v6Address>
      <id min="0" max="255">
        <!--dep, xs:string-->
      </id>
      <type opt="router,ra,manual,dhcp">
        <!--dep, xs:string-->
      </type>
      <address min="0" max="40">
        <!--dep, xs:string-->
      </address>
      <bitmask min="0" max="255">
        <!--dep, xs:integer-->
      </bitMask>
    </v6Address>
  </ipv6AddressList>
</Ipv6Mode>
</IpAddress>
```

XML_Cap_IPFilter

IPFilter capability message in XML format

```
<IPFilter version="2.0" xmlns="http://www.isapi.org/ver20/XMLSchema">
  <enabled opt="true, false"><!--req, xs:boolean--></enabled>
  <permissionType opt="deny, allow">
    <!--req, xs:string, IP address filter status: "deny, allow". If this node is configured, it will overwrite value of all
  <permissionType> in <IPFilterAddress>-->
  </permissionType>
  <IPFilterAddressList size="32"><!--opt, the attribute "size" refers to the maximum number of IP addresses supported
by the device. If the attribute "size" is not returned, the default supported number of IP addresses is 48-->
    <IPFilterAddress>
      <id min="" max=""><!--req, xs:string, ID--></id>
      <permissionType opt="deny, allow"><!--dep, ro, xs:string--></permissionType>
      <addressFilterType opt="mask, range">
        <!--dep, ro, xs:string, filter type: "mask"-single address, "range"-address range-->
      </addressFilterType>
      <AddressRange><!--dep, this node is valid when <addressFilterType> contains "range"-->
        <startIPAddress min="" max=""><!--dep, xs:string--></startIPAddress>
        <endIPAddress min="" max=""><!--dep, xs:string--></endIPAddress>
        <startIPv6Address min="" max=""><!--dep, xs:string--></startIPv6Address>
        <endIPv6Address min="" max=""><!--dep, xs:string--></endIPv6Address>
      </AddressRange>
      <AddressMask><!--dep, it is valid when <addressFilterType> contains "mask"-->
        <ipAddress min="" max=""><!--dep, xs:string--></ipAddress>
        <ipv6Address min="" max=""><!--dep, xs:string--></ipv6Address>
        <bitMask min="" max=""><!--opt, xs:string--></bitMask>
        <bitMaskIPv6 min="" max=""><!--req, xs:string, IPv6 prefix length--></bitMaskIPv6>
      </AddressMask>
      <describeMsg min="" max=""><!--opt, xs:string--></describeMsg>
    </IPFilterAddress>
  </IPFilterAddressList>
</IPFilter>
```

XML_Cap_ItemList

ItemList capability message in XML format

```
<ItemList size="15">
  <Item>
    <itemID min="1" max="15">
      <!--req, xs:string, item ID, which is between 1 and 15-->
    </itemID>
    <itemOrder
opt="none,devIp,time,buildUnitNo,outDoorDevNo,unlockType,devName,deviceNo,channelName,channelNo,
plateNo,plateColor,laneNo,carSpeed,
positionInfo1,pictureNo,CarNo,speedLimit,illegalCode,siteNo,directionNo,carColor,platePosition,carType,illegalType,cu
stom">
```

```
<!--req, xs:string, name element: "none", "devIp"-device IP address, "time", "buildUnitNo"-building No. and unit No., "outDoorDevNo"-door station, "unlockType"-unlocking type, "devName"-device name, "deviceNo"-device No., "channelName"-channel name, "channelNo"-channel No., "plateNo"-license plate number, "plateColor"-license plate color, "laneNo"-lane No., "carSpeed"-vehicle speed, "positionInfo1"-camera 1, "pictureNo"-picture No., "CarNo"-vehicle No., "speedLimit"-speed limit, "illegalCode"-violation code, "siteNo"-intersection No., "directionNo"-direction No., "carColor"-vehicle color, "platePosition"-license plate coordinates, "carType"-vehicle type, "illegalType"-violation type, "custom"-->
</itemOrder>
<itemCustomStr min="1" max="32">
  <!--req, xs:string, element custom string, which is between 1 and 32, unit: bytes. This node is valid only when
<itemOrder> is "custom". Currently traffic cameras only support one custom name-->
</itemCustomStr>
</Item>
</ItemList>
```

XML_Cap_LensDistortionCorrection

LensDistortionCorrection capability message in XML format

```
<LensDistortionCorrection version="2.0" xmlns="http://www.isapi.org/ver20/XMLSchema">
  <enabled><!--req, xs: boolean--></enabled>
  <level opt="level1,level2,level3,custom">
    <!--opt, xs: string, distortion correction level: level 1, level 2, level 3, custom-->
  </level>
  <accurateLevel min="0" max="100">
    <!--opt, xs: integer, correction strength, value range: [0,100]-->
  </accurateLevel>
  <zoomedInDistantViewLevel min="0" max="100">
    <!--opt, xs: integer, zooming in range, value range: [0-100], this node is valid when distortion correction is enabled-->
  </zoomedInDistantViewLevel>
  <horizontalFOV min="0" max="100"><!--opt, xs: integer, horizontal FOV, value range: [0,100]--></horizontalFOV>
  <verticalFOV min="0" max="100"><!--opt, xs: integer, vertical FOV, value range: [0,100]--></verticalFOV>
</LensDistortionCorrection>
```

XML_Cap_Link

Link capability message in XML format.

```
<Link version="2.0" xmlns="http://www.isapi.org/ver20/XMLSchema">
  <MACAddress><!--req, xs:string--></MACAddress>
  <autoNegotiation><!--req, xs:boolean, "true, false"--></autoNegotiation>
  <speed><!--req, xs:integer, "10.100, 1000"--></speed>
  <duplex><!--req, xs:string, "half, full"--></enalbed>
  <MTU><!--req, xs:integer, "500-1500"--></MTU>
</Link>
```

XML_Cap_LogConfig

LogConfig capability message in XML format.

```
<LogConfig version="2.0" xmlns="http://www.isapi.org/ver20/XMLSchema">
  <enabled><!--required, boolean type, whether to enable log--></enabled>
  <level opt="none,debug,info,error,fault,all">
    <!--required, string type, log types, multiple type can be selected, and each type should be separated by comma-->
  </level>
</LogConfig>
```

XML_Cap_MACFilter

MACFilter capability message in XML format

```
<MACFilter version="2.0" xmlns="http://www.isapi.org/ver20/XMLSchema">
  <enabled opt="true,false"><!--req, xs:boolean--></enabled>
  <permissionType opt="deny, allow"><!-- req, xs:string,"deny, allow"--></permissionType>
  <MACFilterAddressList size="">
    <MACFilterAddress>
      <id><!--req, xs:string;id--></id>
      <MACAddress max=""><!--req, xs:string--></MACAddress>
    </MACFilterAddress>
  </MACFilterAddressList>
</MACFilter>
```

XML_Cap_mailingList

mailingList capability message in XML format.

```
<mailingList version="2.0" xmlns="http://www.isapi.org/ver20/XMLSchema">
  <mailing><!--opt,xs:string--></mailing>
</mailingList>
```

XML_Cap_MaxElevation

MaxElevation capability message in XML format

```
<MaxElevation version="1.0" xmlns="http://www.isapi.org/ver20/XMLSchema">
  <mElevation min="-20" max="0"><!--req, xs:integer, the lower limit of max. tilt-angle--> </mElevation>
  <mVerElevation><!--req, xs:integer, the upper limit of max. tilt-angle, normally it is 90°--></mVerElevation>
</MaxElevation>
```

XML_Cap_MotionDetection

XML message about motion detection capability

```
<MotionDetection version="2.0" xmlns="http://www.isapi.org/ver20/XMLSchema">
  <enabled opt="true,false"><!--required, xs:boolean, whether to enable motion detection--></enabled>
  <enableHighlight opt="true,false"><!--optional, xs:boolean, whether to enable highlight--></enableHighlight>
  <samplingInterval><!--optional, xs:integer, number of frames--></samplingInterval>
  <startTriggerTime><!--optional, xs:integer, unit: millisecond--></startTriggerTime>
  <endTriggerTime><!--opt, xs:integer, unit: millisecond--></endTriggerTime>
  <regionType opt="grid,roi,none, region"><!--required, read-only, xs:string, region type: "grid", "roi", "none",
"region"--></regionType>
  <Grid><!--dependency-->
    <rowGranularity><!--required, read-only, xs:integer--></rowGranularity>
    <columnGranularity><!--required, read-only, xs:integer--></columnGranularity>
  </Grid>
  <ROI><!--dependency-->
    <normalizedScreenWidth><!--required, read-only, xs:integer--></normalizedScreenWidth>
    <normalizedScreenHeight><!--required, read-only, xs:integer--></normalizedScreenHeight>
  </ROI>
  <MotionDetectionLayout><!--required, rule region configuration of motion detection-->
    <sensitivityLevel min="0" max="100" step="20"><!--required, xs:integer, sensitivity--></sensitivityLevel>
    <layout>
      <gridMap>
        <!--dependent, xs:string, the image is divided in to N × N grids, each grid can be enabled (value: 1) or disabled
(value 0) motion detection, each four horizontal grids consist of a hexadecimal number, e.g., the corresponding
hexadecimal number of "0011" is 3-->
        </gridMap>
        <roiMap><!--dependent, xs:string--></roiMap>
        <RegionList size="4"><!--this node is available when the regionType is set to "region"-->
          <Region><!--list-->
            <id><!--required, xs:string--></id>
            <RegionCoordinatesList><!--opt-->
              <RegionCoordinates><!--list-->
                <positionX><!--required, xs:integer; X-coordinate--></positionX>
                <positionY><!--required, xs:integer; Y-coordinate--></positionY>
              </RegionCoordinates>
            </RegionCoordinatesList>
          </Region>
        </RegionList>
      </layout>
      <targetType opt="human,vehicle"><!--optional, xs:string, target type, "human", "vehicle"--></targetType>
    </MotionDetectionLayout>
    <enableWithMoving opt="true,false"><!--optional, xs:boolean--></enableWithMoving>
  </MotionDetection>
```

XML_Cap_MotionDetectionLayout

XML message about rule region parameters of motion detection


```
<?xml version="1.0" encoding="utf-8"?>
<MotionDetectionLayout version="2.0" xmlns="http://www.isapi.org/ver20/XMLSchema">
  <sensitivityLevel min="0" max="100" step="20"><!--required, xs:integer, sensitivity--></sensitivityLevel>
  <layout>
    <gridMap>
      <!--dependent, xs:string, the image is divided in to N x N grids, each grid can be enabled (value: 1) or disabled
      (value 0) motion detection, each four horizontal grids consist of a hexadecimal number, e.g., the corresponding
      hexadecimal number of "0011" is 3-->
    </gridMap>
    <roiMap><!--dependent, xs:string--></roiMap>
    <RegionList size="4"><!--this node is available when the regionType is set to "region"-->
      <Region><!--list-->
        <id><!--required, xs:string--></id>
        <RegionCoordinatesList><!--opt-->
          <RegionCoordinates><!--list-->
            <positionX><!--required, xs:integer; X-coordinate--></positionX>
            <positionY><!--required, xs:integer; Y-coordinate--></positionY>
          </RegionCoordinates>
        </RegionCoordinatesList>
      </Region>
    </RegionList>
  </layout>
  <targetType opt="human,vehicle"><!--optional, xs:string, target type, "human", "vehicle"--></targetType>
</MotionDetectionLayout>
```

XML_Cap_MountingScenario

MountingScenario capability message in XML format

```
<MountingScenario version="2.0" xmlns="http://www.isapi.org/ver20/XMLSchema">
  <mode
  opt="indoor,outdoor,day,night,morning,nightfall,mode1,mode2,mode3,mode4,street,lowillumination,custom1,custom
  2,normal,road,faceSnap,highway,road2,backlight,frontlight">
    <!--req, xs:string, "lowillumination"-low illumination, "backlight"-back light, "frontlight"-front light, "faceSnap"-face
    picture capture-->
  </mode>
</MountingScenario>
```

XML_Cap_NetworkInterface

NetworkInterface capability message in XML format

```
<NetworkInterface version="2.0" xmlns="http://www.isapi.org/ver20/XMLSchema">
  <id><!--req, xs:string--></id>
  <IPAddress><!--req-->
  <Wireless><!--opt-->
  <Discovery><!--opt-->
  <Link><!--opt-->
```

```
<defaultConnection><!--opt, xs:boolean--></defaultConnection>
<macAddress min="" max=""><!--opt, xs:string--></macAddress>
<EthernetPortList size="4"><!--opt, network interface information-->
  <EthernetPort><!--opt -->
    <id min="1",max="4"><!--req, xs: integer--></id>
    <MACAddress><!--req, xs:string--></MACAddress>
    <status opt="connected, disconnect"><!--opt, xs:string--></status>
    <speed><!--req, xs:integer, "10, 100, 1000, 10000"--></speed>
  </EthernetPort>
</EthernetPortList>
</NetworkInterface>
```

XML_Cap_NetworkInterfaceList

XML message about configuration capability of multiple network interfaces

```
<?xml version="1.0" encoding="utf-8"?>
<NetworkInterfaceList version="2.0" xmlns="http://www.isapi.org/ver20/XMLSchema">
  <NetworkInterface><!--optional-->
    <id><!--required, xs:string--></id>
    <IPAddress/><!--required-->
    <Wireless/><!--optional-->
    <Discovery/><!--optional-->
    <Link/><!--optional-->
    <defaultConnection><!--optional, xs:boolean--></defaultConnection>
    <macAddress min="" max=""><!--optional, xs:string--></macAddress>
    <EthernetPortList size="4"><!--optional, information about network interfaces-->
      <EthernetPort><!--optional-->
        <id min="1" max="4"><!--required, xs:integer--></id>
        <MACAddress> <!--required, xs:string--></MACAddress>
        <status opt="connected, disconnect"><!--optional, xs:string--></status>
        <speed> <!--required, xs:integer, "10,100,1000,10000"--></speed>
      </EthernetPort>
    </EthernetPortList>
    <interfaceName min="" max=""><!--optional, xs:string, name of the network interfaces--></interfaceName>
  </NetworkInterface>
</NetworkInterfaceList>
```

XML_Cap_NTPServer

NTPServer capability message in XML format

```
<NTPServer version="2.0" xmlns="http://www.isapi.org/ver20/XMLSchema">
  <id min="1" max=""><!--req, xs:string, ID--></id>
  <addressingFormatType opt="ipaddress,hostname"><!--req, xs:string, which field will be used to locate the NTP
server: "ipaddress, hostname"--></addressingFormatType>
  <hostname min="0" max="64"><!--dep, xs:string--></hostName>
  <ipAddress min="0" max="64"><!--dep, xs:string--></ipAddress>
  <ipv6Address min="0" max="64"><!--dep, xs:string--></ipv6Address>
```

```
<portNo min="1" max="65535"><!--opt, xs:integer--></portNo>
<synchronizeInterval min="0" max="65535"><!--opt, xs:integer, NTP time synchronization interval, unit: minute--></synchronizeInterval>
</NTPServer>
```

XML_Cap_NTPServerList

XML message about the configuration capability of all NTP servers

```
<NTPServerList version="2.0" xmlns="http://www.isapi.org/ver20/XMLSchema">
  <NTPServer><!--list, see details in the message XML_Cap_NTPServer--></NTPServer>
</NTPServerList>
```

Remarks

XML_Cap_NTPServer

XML_Cap_ParkAction

ParkAction capability message in XML format

```
<ParkAction version="2.0" xmlns="http://www.isapi.org/ver20/XMLSchema">
  <enabled><!--required, xs:boolean, whether to enable parking--></enabled>
  <Parktime min="" max=""><!--required, xs:integer, parking time, unit: second--></Parktime>
  <Action><!--parking actions-->
    <ActionType opt="atuoscan,framescan,randomscan,panoramascan,patrol,pattern,preset,areaScan"><!--required, xs:string, action type: "atuoscan,framescan,randomscan,panoramascan,patrol,pattern,preset,areaScan"--></ActionType>
    <ActionNum min="0" max="255"><!--required, xs:integer, parking No. This node will be used when <ActionType> is "patrol", "pattern", or "preset". For other parking action types, this node should be set to 0--></ActionNum>
  </Action>
</ParkAction>
```

XML_Cap_PreviewSwitch

PreviewSwitch message in XML format

```
<PreviewSwitch version="2.0" xmlns="http://www.isapi.org/ver20/XMLSchema">
  <groupNo min="0" max="15"><!--opt, xs: integer, group No.--></groupNo>
  <videoOutType
opt="CVBS,HDMI,VGA,AUXOutput1,AUXOutput2,AUXOutput3,AUXOutput4,extendHDMI1,extendHDMI2,extendHDMI3,extendHDMI4,noSameSourceVGA1,noSameSourceVGA2,noSameSourceVGA3,noSameSourceVGA4,sameSourceVGA,mainLCD,AUX-VGA,AUX-LCD,zeroChannel">
    <!--opt, xs: string, video output types, if there are more than two outputs are homologous, you can combine these outputs by "/" to be returned, e.g., if HDM1 and VGA1 are homologous outputs, "HDM1/VGA1" will be returned-->
  </videoOutType>
  <DisplayWindowList size="64">
    <DisplayWindow>
```

```
<id><!--req, xs: string, window No., which equals to id × groupNo--></id>
<displayChannelNo min="0" max="64"><!--req, xs: string, 0-not display, 1 to 64-displayed channel No.--></displayChannelNo>
</DisplayWindow>
</DisplayWindowList>
<previewFrameNo opt="1,4,6,8,9,25,32,36,auto1,auto2,auto3,auto4">
  <!--req, xs: string, number of live view windows, auto1-custom window division 1, auto2-custome window division 2, auto3-custome window division 3, auto4-custome window division 4-->
</previewFrameNo>
<sound><!--req, xs: boolean, whether to turn on audio during live view, true-yes--></sound>
<switchTime opt="0,5,10,20,30,60,120,300"><!--req, xs: string, switching interval, unit: s--></switchTime>
<sameSource>
  <!--dep, xs: boolean, whether the output is homologous, true-yes, this node is valid only when VideoOutType is "noSameSourceVGA1", "noSameSourceVGA2", "noSameSourceVGA3", and "noSameSourceVGA4"-->
</sameSource>
<supportGetByPreviewNum>
  <!--opt, whether supports setting number of live view windows in the URL, true-yes-->
</supportGetByPreviewNum>
<supportGetByVideoOutType>
  <!--opt, xs: boolean, whether supports getting capability by output-->
</supportGetByVideoOutType>
</PreviewSwitch>
```

XML_Cap_RegionClip

RegionClip capability message in XML format

```
<RegionClip version="2.0" xmlns="http://www.std-cgi.org/ver20/XMLSchema">
  <id><!--req, xs: string--></id>
  <enabled><!--req, xs: boolean--></enabled>
  <normalizedScreenSize><!--req, read-only-->
    <normalizedScreenWidth><!--req, read-only, xs: integer--></normalizedScreenWidth>
    <normalizedScreenHeight><!--req, read-only, xs: integer--></normalizedScreenHeight>
  </normalizedScreenSize>
  <regionType opt="rectangle,convexPolygon,concavePolygon"><!--req, ro, xs:string--></regionType>
  <videoResolutionWidth opt="704"></videoResolutionWidth>
  <videoResolutionHeight opt="576"></videoResolutionHeight>
  <ClipRegionList>
    <ClipRegion>
      <RegionCoordinatesList size="1">
        <RegionCoordinates><!--req-->
          <positionX><!--req, xs: integer; x-coordinate--></positionX>
          <positionY><!--req, xs: integer; y-coordinate--></positionY>
        </RegionCoordinates>
      </RegionCoordinatesList>
    </ClipRegion>
  </ClipRegionList>
</RegionClip>
```

XML_Cap_RuleInfo

Capability message of setting behavior analysis in XML format.

```
<RuleInfo version="2.0" xmlns="http://www.isapi.com/ver20/XMLSchema">
  <ruleId opt="1,2,3,4,5,6,7,8"><!--required, xs:string--></ruleId>
  <ruleName min="" max=""><!--required, xs:string--></ruleName>
  <combinedRuleId opt="15,16"><!--required, xs:string--></combinedRuleId>
  <enabled opt="true,false"><!--required, xs:boolean--></enabled>
  <ruleType opt="region,line"><!--required, xs:string--></ruleType>
  <LineDetectionParam>
    <detectionTarget opt="all,human,vehicle, human_vehicle "/>
    <!--detection target, node with underline indicates that it supports multiple selections, e.g., human_vehicle
    indicates human and vehicle-->
    <directionSensitivity opt="left-right,right-left,any" def="any"><!--required, xs:string--></directionSensitivity>
    <isSupportHumanMisinfoFilter>
      <!--optional, xs:boolean, whether supports false alarm filter of human: true-supports, false or no return-not
      support-->
    </isSupportHumanMisinfoFilter>
    <isSupportVehicleMisinfoFilter>
      <!--optional, xs:boolean, whether supports false alarm filter of vehicle: true-supports, false or no return-not
      support-->
    </isSupportVehicleMisinfoFilter>
    <sensitivity min="1" max="100" default="50"><!--required, xs:integer, sensitivity, from 1 to 100--></sensitivity>
  </LineDetectionParam>
  <FieldDetectionParam>
    <durationTime min="1" max="100" def="5"><!--required,xs:integer--></durationTime>
    <sensitivityLevel min="1" max="100"><!--required, xs:integer, sensitivity from 0 to 100--></sensitivityLevel>
    <detectionTarget opt="all,human,vehicle, human_vehicle "/>
    <!--detection target, node with underline indicates that it supports multiple selections, e.g., human_vehicle
    indicates human and vehicle-->
    <isSupportHumanMisinfoFilter>
      <!--optional, xs:boolean, whether supports false alarm filter of human: true-supports, false or no return-not
      support-->
    </isSupportHumanMisinfoFilter>
    <isSupportVehicleMisinfoFilter>
      <!--optional, xs:boolean, whether supports false alarm filter of vehicle: true-supports, false or no return-not
      support-->
    </isSupportVehicleMisinfoFilter>
  </FieldDetectionParam>
  <RegionEntranceParam>
    <detectionTarget opt="all,human,vehicle, human_vehicle "/>
    <!--detection target, node with underline indicates that it supports multiple selections, e.g., human_vehicle
    indicates human and vehicle-->
    <isSupportHumanMisinfoFilter>
      <!--optional, xs:boolean, whether supports false alarm filter of human: true-supports, false or no return-not
      support-->
    </isSupportHumanMisinfoFilter>
    <isSupportVehicleMisinfoFilter>
      <!--optional, xs:boolean, whether supports false alarm filter of vehicle: true-supports, false or no return-not
      support-->
    </isSupportVehicleMisinfoFilter>
  </RegionEntranceParam>
</RuleInfo>
```

```

</isSupportVehicleMisinfoFilter>
</RegionEntranceParam>
<RegionExitingParam>
  <detectTarget opt="all,human,vehicle, human_vehicle "/><!-- detection target, node with underline indicates
that it supports multiple selections, e.g., human_vehicle indicates human and vehicle-->
  <isSupportHumanMisinfoFilter>
    <!--optional, xs:boolean, whether supports false alarm filter of human, true-supports, false or no return-not
support-->
  </isSupportHumanMisinfoFilter>
  <isSupportVehicleMisinfoFilter>
    <!--optional, xs:boolean, whether supports false alarm filter of vehicle, true-supports, false or no return-not
support-->
  </isSupportVehicleMisinfoFilter>
</RegionExitingParam>
<PeopleNumChangeParam>
  <peopleNumThreshold min="" max="" default=""><!--optional, xs:integer--></peopleNumThreshold>
  <detectMode opt="greaterthan,lessThan,equal,notEqual"><!--optional, xs:string--></detectMode>
  <noneStateEffectiveEnabled><!--optional,xs:boolean--></noneStateEffectiveEnabled>
  <durationTime min="" max="" default=""><!--optional, xs:integer--></durationTime>
</PeopleNumChangeParam>
<SpacingChangeParam>
  <spacingThreshold min="" max="" default=""><!--optional, xs: integer--></spacingThreshold>
  <detectMode><!--optional,xs:string, "greaterthan,lessThan"--></detectMode>
  <durationTime min="" max="" default=""><!--optional, xs:integer--></durationTime>
</SpacingChangeParam>
<ViolentMotionParam>
  <sensitivity min="" max="" default=""><!--optional, xs:integer--></sensitivity>
</ViolentMotionParam>
<LeavePositionParam>
  <leaveDelay min="" max="" default=""><!--optional, xs:integer--></leaveDelay>
  <onPosition min="" max="" default=""><!--optional, xs:integer--></onPosition>
</LeavePositionParam>
<FailDownParam>
  <durationTime min="" max="" default=""><!--optional, xs:integer--></durationTime>
  <sensitivity min="" max="" def=""><!--optional, xs:integer, sensitivity--></sensitivity>
  <heightThreshold min="" max="" default=""><!--optional, xs:integer--></heightThreshold>
  <stackTarget opt="true,false"><!--optional, xs:boolean, whether to overlay alarm target frame--></stackTarget>
  <stackRule opt="true,false"><!--optional, xs:boolean, whether to overlay alarm rule frame--></stackRule>
</FailDownParam>
<RunningParam><!--running detection-->
  <speed min="" max="" default=""><!--optional, xs:integer--></speed>
  <durationTime min="" max="" def=""><!--optional, xs:integer, duration time--></durationTime>
  <sensitivity min="" max="" def=""><!--optional, xs:integer, sensitivity--></sensitivity>
  <stackTarget opt="true,false"><!--optional, xs:boolean, whether to overlay alarm target frame--></stackTarget>
  <stackRule opt="true,false"><!--optional, xs:boolean, whether to overlay alarm rule frame--></stackRule>
  <mode opt="single,many"><!--optional, xs:string, running mode, "single"-single person running, "many"-multiple
persons running--></mode>
</RunningParam>
<RetentionParam><!--dependent-->
  <durationTime min="" max="" default="">
    <!--required, xs:integer, duration time, from 60 seconds to 3600 seconds, default value: 1800 seconds-->
  </RetentionParam>

```

```

<ParkingParam>
  <durationTime min="5" max="100"><!-- required,xs:integer--> </durationTime>
</ParkingParam>
<eventType
opt="none,lineDetection,fieldDetection,regionEntrance,regionExiting,parking,peopleNumChange,spacingChange,violentMotion,leavePosition,failDown,running,retentionParam,playCellphone">
  <!--required, xs:string-->
</eventType>
<SizeFilter>
  <enabled opt="true,false"><!--required, xs:boolean--></enabled>
  <mode opt="pixels,actualSize"><!--required, xs:string, filter mode: "pixels"-filter according to pixel, "actualSize"-
filter according to actual size--></mode>
  <ObjectSizeList size="4">
    <ObjectSize><!--only return the default value of supported detection type-->
      <detectionTarget><!--optional,xs:string, detection target, "all"-all, "human"-human, "vehicle"-vehicle,
"human_vehicle"-human and vehicle--></detectionTarget>
      <MaxObjectSize><!--maximum size:float-->
        <positionX><!--required, xs:integer;coordinate--></positionX>
        <positionY><!--required, xs:integer;coordinate--></positionY>
        <width><!--required, xs:integer, default width value: when detectionTarget values "all", it is 500, "human"-200,
"vehicle"-500, "human_vehicle"-500, unit: cm--></width>
        <height><!--required, xs:integer, default height value: when detectionTarget values "all", it is 100, "human"-300,
"vehicle"-100, "human_vehicle"-100, unit: cm --></height>
      </MaxObjectSize>
      <MinObjectSize><!--minimum size:float-->
        <positionX><!--required, xs:integer;coordinate--></positionX>
        <positionY><!--required, xs:integer;coordinate--></positionY>
        <width><!--required, xs:integer, default width value: when detectionTarget values "all", it is 60, "human"-60,
"vehicle"-180, "human_vehicle"-60, unit: cm--></width>
        <height><!--required, xs:integer, default height value: when detectionTarget values "all", it is 10, "human"-10,
"vehicle"-140, "human_vehicle"-10, unit: cm--></height>
      </MinObjectSize>
    </ObjectSize>
  </ObjectSizeList>
</SizeFilter>
<RuleRegion>
  <RegionCoordinatesList size="4">
    <RegionCoordinates>
      <positionX min="" max=""><!--required, xs:integer--></positionX>
      <positionY min="" max=""><!--required, xs:integer--></positionY>
    </RegionCoordinates>
  </RegionCoordinatesList>
</RuleRegion>
<backgroundSuppression opt="open,close,selfAdapt"><!--optional, xs:string, background suppression--></
backgroundSuppression>
<PlayCellphoneParam><!--optional, playing mobile phone-->
  <durationTime min="1" max="3600" def=""><!--optional, xs:integer, duration time--></durationTime>
  <stackTarget opt="true,false"><!--optional, xs:boolean, whether to overlay alarm target frame--></stackTarget>
  <stackRule opt="true,false"><!--optional, xs:boolean, whether to overlay alarm rule frame--></stackRule>
</PlayCellphoneParam>
<streetBehavior opt="running,group,violentMotion,failDown">
  <!--optional, xs:string, street behavior capability: "running"-fast moving, "group"-people gathering, "violentMotion"-

```

```
violent motion, "failDown"-failing down-->
</streetBehavior>
<workBehavior opt="playCellphone,peopleNumChange,leavePosition,retention">
  <!--optional, xs:string, working behavior capability: "playCellphone"-playing mobile phone, "peopleNumChange"-
number of persons changed, "leavePosition"-absent, "retention"-loitering-->
</workBehavior>
<isSupportMultiRules><!--optional, xs:boolean, whether it supports multiple rules for one event; by default, only one
rule can be configured for one event--></isSupportMultiRules>
<isSupportSchedulesEventType><!--optional, xs:boolean, whether to configure arming schedule by event type; by
default, it is configured by rule ID--></isSupportSchedulesEventType>
<isSupportTriggerEventType><!--optional, xs:boolean, whether to configure alarm linkage by event type; by default, it
is configured by rule ID--></isSupportTriggerEventType>
<analysisMode opt="realTimeMode,autoSwitchMode"><!--optional, xs:string, analysis mode, the default mode is
"realTimeMode"--></analysisMode>
</RuleInfo>
```

Remarks

When the **mode** values "actualSize", the input value of **positionX** and **positionY** is 0.

XML_Cap_SalveCamera

SalveCamera capability message in XML format

```
<SlaveCamera version="2.0" xmlns="http://www.std-cgi.org/ver20/XMLSchema">
  <SlaveCameraInfoList>
    <SlaveCameraInfo>
      <id min="1" max="4">
        <!--req, xs:integer-->
      </id>
      <serverAddress>
        <addressingFormatType opt="ipaddress,hostname">
          <!--req, xs:string, "ipaddress,hostname"-->
        </addressingFormatType>
        <hostName max="">
          <!--dep, xs:string-->
        </hostName>
        <ipAddress>
          <!--dep, xs:string-->
        </ipAddress>
        <ipv6Address>
          <!--dep, xs:string-->
        </ipv6Address>
      </serverAddress>
      <userName max="">
        <!--req, xs:string-->
      </userName>
      <passWord max="">
        <!--req, xs:string-->
      </passWord>
      <portNo min="" max="">
```



```
<!--req, xs:integer-->
</portNo>
<loginStatus opt="login,logout">
  <!--req, xs:string-->
</loginStatus>
</SlaveCameraInfo>
</SlaveCameraInfoList>
</SlaveCamera>
```

XML_Cap_SerialCommand

SerialCommand capability message in XML format

```
<SerialCommand version="2.0" xmlns="http://www.isapi.org/ver20/XMLSchema">
  <chainNo min="1" max="16">
    <!--opt, xs:integer-->
  </chainNo>
  <command>
    <!--req, xs:hexBinary-->
  </command>
</SerialCommand>
```

XML_Cap_SerialPort

SerialPort capability message in XML format

```
<SerialPort version="2.0" xmlns="http://www.isapi.org/ver20/XMLSchema">
  <id min="0" max="255">
    <!--req, xs:string, serial port ID-->
  </id>
  <enabled>
    <!--req, xs:boolean, whether to enable the serial port: "true, false"-->
  </enabled>
  <serialNumber opt="1,2,3,4,5">
    <!--opt, xs:integer, serial port No., which is between 1 and 5-->
  </serialNumber>
  <serialPortType opt="RS485,RS422,RS232">
    <!--req, xs:string, serial port type: "RS485", "RS422", "RS232"-->
  </serialPortType>
  <duplexMode opt="half,full">
    <!--req, xs:string, duplex mode of the serial port: "half", "full"-->
  </duplexMode>
  <direction opt="monodirectional,bidirectional">
    <!--req, xs:string-->
  </direction>
  <baudRate range="600,1200,2400,4800,9600,19200,38400,57600,76800,115200">
    <!--req, xs:integer-->
  </baudRate>
  <dataBits min="5" max="8">
```

```
<!--req, xs:integer-->
</dataBits>
<parityType opt="none,even,odd,mark,space">
  <!--req, xs:string-->
</parityType>
<stopBits range="1,2">
  <!--req, xs:string, stop bit: "1,1.5,2"-->
</stopBits>
<workMode opt="console,transparent,narrowband,audiomixer,stairsControl,cardReader,disabled,custom">
  <!--dep, xs:string, working mode: "console","transparent","narrowband","audiomixer","stairsControl"-elevator
control,"cardReader"-card reader,"disabled","custom". This node is required only when <serialPortType> is set to
"RS232"-->
</workMode>
<flowCtrl opt="none,software,hardware">
  <!--req, xs:string-->
</flowCtrl>
<rs485WorkMode opt="Led, CaptureTrigger">
  <!--opt, xs:string, working mode of RS-485 serial port, which is used for LED display or triggering transmission of
captured pictures. This node is valid only when <serialPortType> is "RS485"-->
</rs485WorkMode>
<audiomixerProtocolType opt="HIKVISION">
  <!--opt, xs:string, "HIKVISION"-->
</audiomixerProtocolType>
<deviceName min="0" max="32"><!--dependent, xs:string, this node exists only when <serialPortType> is "RS485"-->
</deviceName>
<deviceType min="" max=""><!--dependent, xs:integer, this node exists only when <serialPortType> is "RS485"-->
</deviceType>
<deviceProtocol min="" max=""><!--dependent, xs:integer, this node exists only when <serialPortType> is "RS485"-->
</deviceProtocol>
<mode opt="readerMode,clientMode,externMode,accessControlHost, disabled "><!--dependent, xs:string, working
mode, this node exists only when <serialPortType> is "RS485"-->
</mode>
</SerialPort>
```

XML_Cap_SIPInfo

SIPInfo capability message in XML format

```
<SIPInfo version="2.0" xmlns="http://www.isapi.org/ver20/XMLSchema">
  <videoID opt="1,2,3">
    <!--req, xs:string-->
  </videoID>
  <AlarmInList>
    <AlarmIn>
      <id min="0" max="8">
        <!--req, xs:integer-->
      </id>
      <alarmInID min="0" max="64">
        <!--req, xs:string-->
      </alarmInID>
    </AlarmIn>
  </AlarmInList>
</SIPInfo>
```

```
</AlarmInList>
</SIPInfo>
```

XML_Cap_SIPServer

SIPServer capability message in XML format

```
<SIPServer version="2.0" xmlns="http://www.isapi.org/ver20/XMLSchema">
  <id min="1" max="">
    <!--req, xs:integer, the maximum ID depends on the number of channels-->
  </id>
  <localPort min="0" max="65535">
    <!--req, xs:integer-->
  </localPort>
  <streamID min="1" max="255">
    <!--req, xs:integer, stream type: 1-main stream, 2-sub-stream, 3-third stream-->
  </streamID>
  <Standard>
    <!--opt-->
    <registerStatus>
      <!--ro, req, xs:boolean, registration status: "false"-unregistered, "true"-registered-->
    </registerStatus>
    <enabled >
      <!--req, xs:string, "true"-log in, "false"-log out-->
    </enabled>
    <registrar min="0" max="128">
      <!--req, xs:string-->
    </registrar>
    <registrarPort min="1024" max="65535">
      <!--req, xs:integer-->
    </registrarPort>
    <proxy min="0" max="128">
      <!--req, xs:string-->
    </proxy>
    <proxyPort min="0" max="65535">
      <!--req, xs:integer-->
    </proxyPort>
    <displayName min="0" max="64">
      <!--req, xs:string-->
    </displayName>
    <username min="0" max="64">
      <!--req, xs:string-->
    </userName>
    <authID min="0" max="64">
      <!--req, xs:string-->
    </authID>
    <password min="0" max="32">
      <!--wo, req, xs:string-->
    </password>
    <expires min="0" max="100000">
      <!--req, xs:integer-->
```

```
</expires>
</Standard>
<GB28181>
  <!--opt-->
    <registerStatus>
      <!--req, xs:boolean, "true,false"-->
    </registerStatus>
    <enabled>
      <!--req, xs:boolean, "true,false"-->
    </enabled>
    <registrar min="0" max="128">
      <!--req, xs:string-->
    </registrar>
    <registrarPort min="1024" max="65535">
      <!--req, xs:integer-->
    </registrarPort>
    <serverId min="0" max="64">
      <!--req, xs:string-->
    </serverId>
    <serverDomain min="0" max="128">
      <!--req, xs:string-->
    </serverDomain>
    <username min="0" max="64">
      <!--req, xs:string-->
    </userName>
    <authID min="0" max="64">
      <!--req, xs:string-->
    </authID>
    <password min="0" max="32">
      <!--wo, req, xs:string-->
    </password>
    <expires min="30" max="100000">
      <!--req, xs:integer-->
    </expires>
    <liveTime min="30" max="100000">
      <!--req, xs:integer-->
    </liveTime>
    <heartbeatTime min="0" max="255">
      <!--req, xs:integer-->
    </heartbeatTime>
    <heartbeatCount min="3" max="255">
      <!--req, xs:integer-->
    </heartbeatCount>
  </GB28181>
</SIPServer>
```

XML_Cap_SmartSearchDescription

SmartSearchDescription capability message in XML format

```
<SmartSearchDescription version="2.0" xmlns="http://www.isapi.org/ver20/XMLSchema">
  <searchID>
    <!--req, xs:string, search ID, which is used to confirm the upper-level platform or system. If the platform or system
are same during two times of search, the search history will be recorded in the memory for next fast search-->
  </searchID>
  <searchResultPosition><!--req, xs: integer--></searchResultPosition>
  <maxResults><!--req, xs: integer--></maxResults>
  <trackID><!--req, xs: integer, channel ID, e.g., 101-main stream if first channel--></trackID>
  <startTime><!--req, xs: datetime, start time of validity period, e.g., 2013-06-10T 12:00:00Z--></startTime>
  <endTime><!--req, xs: datetime, end time of validity period, e.g., 2013-06-10T 13:00:00Z--></endTime>
  <type opt="motionDetection, traversePlaneDetection,FieldDetection, faceDetection">
    <!--req, xs: string; VCA event type-->
  </type>
  <MotionDetection><!--dep, it is valid when type is "motionDetection"-->
    <Grid>
      <rowGranularity></rowGranularity>
      <columnGranularity></columnGranularity>
    </Grid>
    <MotionDetectionLayout>
      <layout>
        <gridMap></gridMap>
      </layout>
    </MotionDetectionLayout>
    <sensitivity min="1", max="100"><!--opt, xs: integer--></sensitivity>
  </MotionDetection>
  <TraversePlane><!--dep, it is valid when type is "traversePlaneDetection"-->
    <TraversePlaneParam>
      <LineList>
        <Line>
          <StartPoint>
            <x><!--req, xs: float--></x>
            <y><!--req, xs: float--></y>
          </StartPoint>
          <EndPoint>
            <x><!--req, xs: float--></x>
            <y><!--req, xs: float--></y>
          </EndPoint>
        </Line>
      </LineList>
      <crossDirection opt="bothDirection, leftToRight, rightToLeft"><!--opt, xs: string, crossing direction--
    </crossDirection>
    <sensitivity min="1", max="100"><!-- opt, xs: integer--></sensitivity>
    <planeHeight><!--opt, xs: integer--></planeHeight>
    <detectionTarget opt="all, human, vehicle"><!--req, xs: string--><detectionTarget>
  </TraversePlaneParam>
  <dwPreTime><!--opt, xs: integer, pre-recording time--></dwPreTime>
  <dwDelayTime><!--opt, xs: integer,post-recording time--></dwDelayTime>
  <PTZPos>
    <P><!--opt, xs: integer--></P>
    <T><!--opt, xs: integer--></T>
    <Z><!--opt, xs: integer--></Z>
  </PTZPos>
```

```
<advancedType opt="humanFeature, objectFeature"><!--opt, xs: string--></advancedType>
<advance><!--dep, it is valid when type is "motionDetection"-->
  <HumanFeature>
    <age min="0" max="100"><!--opt, xs: integer--></age>
    <sex opt="male,female"><!--opt, xs: string--></sex>
    <eyeGlass><!--opt, xs: boolean--></eyeGlass>
  </HumanFeature><!--dep, it is valid when type is "humanFeature"-->
  <ObjectFeature>
    <colorRatel><!--opt, xs: integer--></colorRatel>
    <R><!--opt, xs: integer--></R>
    <G><!--opt, xs: integer--></G>
    <B><!--opt, xs: integer--></B>
  </ObjectFeature><!--dep, it is valid when type is "objectFeature"-->
</advance>
</TraversePlane>
<FieldDetection><!--dep, it is valid when type is "FieldDetection"-->
  <normalizedScreenSize>
    <normalizedScreenWidth><!--opt, xs: integer--></normalizedScreenWidth>
    <normalizedScreenHeight><!--opt, xs: integer--></normalizedScreenHeight>
  </normalizedScreenSize>
  <param>
    <region>
      <pointList>
        <point>
          <x><!--opt, xs: integer--></x>
          <y><!--opt, xs: integer--></y>
        </point>
      </pointList>
    </region>
  </param>
  <duration min="1", max="100"><!--opt, xs: integer--></duration>
  <sensitivity min="1", max="100"><!--opt, xs: integer--></sensitivity>
  <rate min="1", max="100"><!--opt, xs: integer--></rate>
  <detectionTarget opt="all, human, vehicle"><!--req, xs: string--><detectionTarget>
  <dwPreTime><!--opt, xs: integer, pre-recording time--></dwPreTime>
  <dwDelayTime><!--opt, xs: integer, post-recording time--></dwDelayTime>
  <PTZPos>
    <P><!--opt, xs: integer--></P>
    <T><!--opt, xs: integer--></T>
    <Z><!--opt, xs: integer--></Z>
  </PTZPos>
  <advancedType opt="humanFeature, objectFeature"><!--opt, xs: string--></advancedType>
  <advance><!--dep, it is valid when type is "motionDetection"-->
    <HumanFeature>
      <age min="0" max="100"><!--opt, xs: integer--></age>
      <sex opt="male,female"><!--opt, xs: string--></sex>
      <eyeGlass><!--opt, xs: boolean--></eyeGlass>
    </HumanFeature><!--dep, it is valid when type is "humanFeature"-->
    <ObjectFeature>
      <colorRatel><!--opt, xs: integer--></colorRatel>
      <R></R>
      <G></G>
```

```

    <B></B>
  </ObjectFeature><!--dep, it is valid when type is "objectFeature"-->
</advance>
</FieldDetection>
<FaceDetection><!--dep, it is valid when type is "faceDetection"-->
  <param>
    <region>
      <pointList>
        <point>
          <x/>
          <y/>
        </point>
      </pointList>
    </region>
  </param>
  <duration min="1", max="100"><!--opt, xs: integer--></duration>
  <sensitivity min="1", max="100"><!--opt, xs: integer--></sensitivity>
  <rate min="1", max="100"><!--opt, xs: integer--></rate>
  <detectionTarget opt="all, human, vehicle"><!--req, xs: string--><detectionTarget>
  <dwPreTime><!--opt, xs: integer, pre-recording time--></dwPreTime>
  <dwDelayTime><!--opt, xs: integer, post-recording time--></dwDelayTime>
  <PTZPos>
    <P><!--opt, xs: integer--></P>
    <T><!--opt, xs: integer--></T>
    <Z><!--opt, xs: integer--></Z>
  </PTZPos>
  <advancedType opt="humanFeature, objectFeature"><!--opt, xs: string--></advancedType>
  <advance><!--dep, it is valid when type is "motionDetection"-->
  <HumanFeature>
    <colorRatel><!--opt, xs: integer--></colorRatel>
    <age min="0" max="100"><!--opt, xs: integer--></age>
    <sex opt="male,female"><!--opt, xs: string--></sex>
    <eyeGlass><!--opt, xs: boolean--></eyeGlass>
  </HumanFeature> <!--dep, it is valid when type is "humanFeature"-->
  <ObjectFeature>
    <colorRatel><!--opt, xs: integer--></colorRatel>
    <R></R>
    <G></G>
    <B></B>
  </ObjectFeature><!--dep, it is valid when type is "objectFeature"-->
</advance>
</FaceDetection>
</SmartSearchDescription>

```

XML_Cap_SnapshotChannel

XML message about configuration capability of capturing pictures by schedule

```

<SnapshotChannel version="2.0" xmlns="http://www.isapi.org/ver20/XMLSchema">
  <id opt="1"><!--required, xs:string, ID--></id>
  <videoInputChannelID opt="1"><!--required, xs:string, video input channel ID--></videoInputChannelID>

```

```

<timingCapture>
  <enabled opt="true,false"><!--required, xs:boolean--></enabled>
  <supportSchedule opt="true,false"><!--required, xs:boolean--></supportSchedule>
  <compress>
    <pictureCodecType opt="JPEG"><!--required, xs:string, picture format: "JPEG,BMP,GIF,PNG"--></pictureCodecType>
    <pictureWidth opt="1280"><!--required, xs:integer--></pictureWidth>
    <pictureHeight opt="720"><!--required, xs:integer--></pictureHeight>
    <quality opt="40,60,80"><!--optional, xs:integer, quality in percentage, the value is between 0 and 100--></quality>
    <captureInterval min="1000" max="604800000"><!--optional, xs:integer, capture interval, unit: milliseconds--></
captureInterval>
  <captureNumber min="1" max="120" def="4"><!--optional, xs:integer--></captureNumber>
</compress>
</timingCapture>
<eventCapture><!--optional-->
  <enabled opt="true,false"><!--required, xs:boolean--></enabled>
  <supportSchedule opt="false"><!--optional, ro,xs:boolean--></supportSchedule>
  <compress>
    <pictureCodecType opt="JPEG"><!--required, xs:string, picture format: "JPEG,BMP,GIF,PNG"--></pictureCodecType>
    <pictureWidth opt="1280"><!--required, xs:integer--></pictureWidth>
    <pictureHeight opt="720"><!--required, xs:integer--></pictureHeight>
    <quality opt="40,60,80"><!--optional, xs:integer, quality in percentage, the value is between 0 and 100--></quality>
    <captureInterval min="1000" max="604800000"><!--optional, xs:integer, capture interval, unit: milliseconds--></
captureInterval>
  <captureNumber min="1" max="120" def="4"><!--optional, xs:integer--></captureNumber>
</compress>
</eventCapture>
<PromptDescription>
  <prompt1><!--optional, xs:string, this node will be returned only when it is supported. If this node is not supported,
the upper node will still exist--></prompt1>
</PromptDescription>
<StreamInformationList>
  <StreamInformation><!--list-->
    <streamType opt="mainstream,substream,stream3,stream4,stream5"><!--optional, xs:string--></streamType>
    <timingCapture>
      <enabled opt="true,false"><!--required, xs:boolean--></enabled>
      <supportSchedule opt="true,false"><!--required, xs:boolean--></supportSchedule>
      <compress>
        <pictureCodecType opt="JPEG"><!--required, xs:string, picture format: "JPEG,BMP,GIF,PNG"--></
pictureCodecType>
        <pictureWidth opt="1280"><!--required, xs:integer--></pictureWidth>
        <pictureHeight opt="720"><!--required, xs:integer--></pictureHeight>
        <quality opt="40,60,80"><!--optional, xs:integer, quality in percentage, the value is between 0 and 100--></
quality>
        <captureInterval min="1000" max="604800000"><!--optional, xs:integer, captuer inteval, unit: milliseconds--></
captureInterval>
        <captureNumber min="1" max="120" def="4"><!--optional, xs:integer--></captureNumber>
      </compress>
    </timingCapture>
    <eventCapture><!--optional-->
      <enabled opt="true,false"><!--required, xs:boolean--></enabled>
      <supportSchedule opt="false"><!--optional, read-only, xs:boolean--></supportSchedule>
      <compress>

```



```
<pictureCodecType opt="JPEG"><!--required, xs:string, picture format: "JPEG,BMP,GIF,PNG"--></pictureCodecType>
<pictureWidth opt="1280"><!--required, xs:integer--></pictureWidth>
<pictureHeight opt="720"><!--required, xs:integer--></pictureHeight>
<quality opt="40,60,80"><!--optional, xs:integer, quality in percentage, the value is between 0 and 100--></quality>
<captureInterval min="1000" max="604800000"><!--optional, xs:integer, capture interval, unit: milliseconds--></captureInterval>
<captureNumber min="1" max="120" def="4"><!--optional, xs:integer--></captureNumber>
</compress>
</eventCapture>
<PromptDescription>
  <prompt1><!--optional, xs:string, this node will be returned only when it is supported. If this node is not supported, the upper node will still exist--></prompt1>
</PromptDescription>
</StreamInformation>
</StreamInformationList>
</SnapshotChannel>
```

XML_Cap_SoftwareService

SoftwareService capability message in XML format

```
<SoftwareService version="2.0" xmlns="http://www.isapi.org/ver20/XMLSchema">
  <ThirdStream><!--opt, third stream configuration-->
    <enabled opt="true,false"><!--req, xs: boolean, whether to enable third stream configuration--></enabled>
    <thirdStreamPrompt opt="prompt5">
      <!--req, xs: string, set audible prompt for third stream, prompt5-"When third stream is enabled, functions such as intrusion detection, line crossing detection, region entrance detection, region exiting detection, face detection, HDMI, H.264+ and H265+, are not supported. New settings will take effect after a reboot."-->
    </thirdStreamPrompt>
  </ThirdStream>
  <PanoramaDisplay><!--opt, display the image range on panorama view-->
    <enabled opt="true,false" default="false">
      <!--req, xs: boolean, whether to enable image range display on panorama view-->
    </enabled>
  </PanoramaDisplay>
  <MotionDetect>
    <enabled opt="true,false"><!--whether to enable motion detection--></enabled>
    <MotionDetectPrompt opt="prompt1">
      <!--req, xs: string, motion detection audible prompt, prompt1-"When motion detection is enabled, functions such as intrusion detection, line crossing detection, region entrance detection, and region exiting detection, are not supported. New settings will take effect after a reboot."-->
    </MotionDetectPrompt>
  </MotionDetect>
</SoftwareService>
```

XML_Cap_SSH

SSH capability message in XML format

```
<?xml version="1.0" encoding="utf-8"?>
<SSH version="2.0" xmlns="http://www.isapi.org/ver20/XMLSchema">
  <enabled opt="true,false"><!--req, xs:boolean--></enabled>
  <isSupportSSHTest><!--opt, xs:boolean, whether supports SSH passthrough command test--></isSupportSSHTest>
</SSH>
```

XML_Cap_storage

XML message about configuration capability of storage working mode

```
<?xml version="1.0" encoding="utf-8"?>
<storage version="2.0" xmlns="http://www.isapi.org/ver20/XMLSchema">
  <hddList size=""><!--optional, the contents are same to the message XML_hddList (related URI: /ISAPI/ContentMgmt/Storage/hdd)--></hddList>
  <nasList size=""><!--optional, the contents are same to the message XML_nasList (related URI: /ISAPI/ContentMgmt/Storage/nas)--></nasList>
  <workMode opt="group,quota,extract"><!--optional, xs:string, working mode: "group", "quota", "extract"--></workMode>
  <supportDataReconstruction opt="all,record">
    <!--optional, xs:string, whether the device supports disk data reconstruction, and the supported data type for reconstruction: "all", "record"-->
  </supportDataReconstruction>
  <isSupportSSDManagement>
    <!--optional, xs:string, whether the device supports SSD management (related URI: /ISAPI/ContentMgmt/storage/ssd/<ID>/capabilities?format=json)-->
  </isSupportSSDManagement>
  <ssdList size=""><!--optional, the contents are same to the message XML_ssdList (related URI: /ISAPI/ContentMgmt/Storage/ssd)--></ssdList>
</storage>
```

XML_Cap_storageExtension

XML message about storage strategy capability.

```
<storageExtension xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0" >
  <LoopEnable,opt="true,false">
    <!--optional, xs:boolean, whether to enable overwriting-->
  </LoopEnable>
  <enableDormant>
    <!--optional, xs:boolean-->
  </enableDormant>
  <packDuration max="300" min="1">
    <!--optional, xs:integer, unit: minute-->
  </packDuration>
```

```
<logStorageMode opt="system,custom">
  <!--optional, xs:string, log storage mode: "system"-system default mode, "custom"-custom mode-->
</logStorageMode>
<logStorageId>
  <!--dependent, xs:integer, disk ID (saving logs in a disk) or array number (saving logs in disk arrays), which are valid
only when logStorageMode is "custom"; -->
</logStorageId>
<logStorageCycle min="0" max="90" def="90">
  <!--dependent, xs:integer, log storage period, which is valid only when logStorageMode is "custom"-->
</logStorageCycle>
</storageExtension>
```

XML_Cap_StreamingChannel

StreamingChannel capability message in XML format

```
<StreamingChannel version="2.0" xmlns="http://www.isapi.org/ver20/XMLSchema">
  <id opt="111,222,333,444"><!--req, xs:string, ID--></id>
  <channelName min="0" max="64"><!--req, xs:string, channel name--></channelName>
  <enabled opt="true,false" def="true"><!--req, xs:boolean--></enabled>
  <Transport><!--req-->
    <rtspPortNo min="0" max="65535" def="554">554</rtspPortNo>
    <maxPacketSize min="0" max="1500"><!--optional, xs:integer--></maxPacketSize>
    <audioPacketLength min="0" max="5000"/>
    <audioInboundPacketLength min="0" max="5000"/><!--optional, xs:integer-->
    <audioInboundPortNo min="0" max="65535"/><!--optional, xs:integer-->
    <videoSourcePortNo min="0" max="65535"/><!--optional, xs:integer-->
    <audioSourcePortNo min="0" max="65535"/><!--optional, xs:integer-->
    <ControlProtocolList><!--req-->
      <ControlProtocol><!--list-->
        <streamingTransport opt="RTSP/RTP,HTTP"><!--req, xs:string, "HTTP,RTSP,SHTTP,SRTP"--></streamingTransport>
      </ControlProtocol>
    </ControlProtocolList>
  </Transport>
  <Unicast><!--opt-->
    <enabled opt="true,false" def="false"/><!--req, xs:boolean-->
    <rtpTransportType opt="RTP/UDP,RTP/TCP"/><!--optional, xs:string, "RTP/UDP,RTP/TCP"-->
  </Unicast>
  <Multicast><!--opt-->
    <enabled opt="true,false" def="false"/><!--req, xs:boolean-->
    <userTriggerThreshold/><!--optional, xs:integer-->
    <videoDestPortNo min="1" max="65535" def="8860"/><!--optional, xs:integer-->
    <audioDestPortNo min="1" max="65535" def="8860"/><!--optional, xs:integer-->
    <destIPAddress min="8" max="16"/><!--dep, xs:string-->
    <destIPv6Address min="15" max="39"/><!--dep, xs:string-->
    <tTl min="0" max="127" def="1"/><!--optional, xs:integer-->
    <activeMulticastEnabled>
      <!--optional, xs: boolean, "true, false", whether to enable active multicast, which is mutually exclusive with
passive multicast-->
    </activeMulticastEnabled>
    <packagingFormat opt="RTP,TS,ES,PS"><!--optional, xs: string, encapsulation format--></packagingFormat>
    <FecInfo><!--opt-->
```

```

    <fecRatio min="0" max="100">
      <!--req, read-only, xs: integer, extra bandwidth occupation ratio of forward error correction (FEC) data, the value
is between 0 and 100, the default value is 0-->
    </fecRatio>
    <fecDestPortNo>
      <!--optional, xs: integer, port No. of FEC multicast, the default port No. is specified by device-->
    </fecDestPortNo>
  </FecInfo>
</Multicast>
<Security><!--opt-->
  <enabled opt="true,false" def="false"/><!--req, xs:boolean-->
  <certificateType opt="digest,digest/baisc" def="digest"><!--req, xs:string--></certificateType>
  <SecurityAlgorithm><!--dependency, this node is valid when certificateType is "digest" or "digest/basic"-->
    <algorithmType opt="MD5,SHA256,MD5/SHA256"><!--optional, xs:string, algorithm type: "MD5", "SHA256",
"MD5/SHA256"--></algorithmType>
  </SecurityAlgorithm>
</Security>
<SRTPMulticast><!--opt-->
  <SRTPVideoDestPortNo min="" max=""><!--optional, xs:integer--></SRTPVideoDestPortNo>
  <SRTPAudioDestPortNo min="" max=""><!--optional, xs:integer--></SRTPAudioDestPortNo>
</SRTPMulticast>
</Transport>
<Video>
  <enabled opt="true,false">true</enabled>
  <videoInputChannelID opt="1,2,3,4">2</videoInputChannelID>
  <videoCodecType opt="MJPEG,MPEG4">MPEG4</videoCodecType>
  <videoScanType opt="interlaced,progressive">progressive</videoScanType>
  <videoResolutionWidth min="0" max="640">640</videoResolutionWidth>
  <videoResolutionHeight min="0" max="480">480</videoResolutionHeight>
  <videoDiffResolutionList><!--optional, list, if this node exists, it indicates that the resolution depends on the node
<b>videoCodecType</b>, and <b>videoResolutionWidth</b> and <b>videoResolutionHeight</b> are invalid. For forward
compatibility, the maximum set of <b>videoCodecType</b>, <b>videoResolutionWidth</b>, and <b>videoResolutionHeight</b>-->
    <videoDiffResolution>
      <videoCodecType>
        <!--req, xs:string, "MPEG4,MJPEG,3GP,H.264,HK.264,MPNG,SVAC"-->
      </videoCodecType>
      <videoResolutionWidth><!--req, xs:integer--></videoResolutionWidth>
      <videoResolutionHeight><!--req, xs:integer--></videoResolutionHeight>
    </videoDiffResolution>
  </videoDiffResolutionList>
  <videoPositionX min="0" max="640">0</videoPositionX>
  <videoPositionY min="0" max="480">0</videoPositionY>
  <videoQualityControlType opt="CBR,VBR">CBR</videoQualityControlType>
  <constantBitRate min="50" max="4000" dynamic="true">2000</constantBitRate>
  <maxFrameRate opt="2500,1250,625,312,156,78, 830">2500</maxFrameRate>
  <keyFrameInterval min="0", max="10000">1000</keyFrameInterval>
  <rotationDegree opt="0,90,180,270" def="0">0</rotationDegree>
  <mirrorEnabled opt="true,false" def="false">false</mirrorEnabled>
  <snapshotImageType opt="JPEG" def="JPEG">JPEG</snapshotImageType>
  <IntelligentInfoDisplayMethod opt="player,non-player"><!--optional, xs:string, method of displaying intelligent
information: "player", "non-player"--></IntelligentInfoDisplayMethod>
  <minimumResolutionSupportedBySmartCode>

```

```
<!--optional, xs:string, the minimum resolution supported by smart coding (smart264 and smart265), e.g.,
"640*512". If this node is returned, it indicates the smart code of current device (H7, H5) is limited by the minimum
resolution, and the upper layer can get the supported minimum resolution for smart coding of current device. For the
resolution lower than the minimum resolution does not support smart coding. If this node is not returned, it indicates
the smart coding of current device (H3) is not limited by minimum resolution-->
</minimumResolutionSupportedBySmartCode>
</Video>
<Audio>
  <enabled opt="true,false" def="false">false</enabled>
  <audioInputChannelID opt="1,2,3,4">2</audioInputChannelID>
  <audioCompressionType opt="G.726,G.711ulaw" def="G.726">G.726</audioCompressionType>
  <audioBitRate opt="16,24,32,40" def="32" dynamic="true">24</audioBitRate>
  <audioSamplingRate opt="8" dynamic="true">8</audioSamplingRate>
  <audioResolution opt="3,4,5,6" dynamic="true"/>
</Audio>
<isSupportDynamicCapWithCondition><!--optional, xs:boolean, whether to support dynamic capability with
conditions--></isSupportDynamicCapWithCondition>
<enableCABAC><!--optional, xs:boolean, whether it supports compressing stream to improve performance--></
enableCABAC>
<subStreamRecStatus><!--optional, xs:boolean--></subStreamRecStatus>
<isSupportRefreshFrame><!--optional, xs:boolean, whether it supports refreshing frames when Smart264 is enabled--
></isSupportRefreshFrame>
<isSupportBareDataOverlay><!--optional, xs:boolean--></isSupportBareDataOverlay>
<isSupportRTCPCfg><!--optional, xs:boolean--></isSupportRTCPCfg>
<customStreamEnable><!--optional, xs:boolean, whether the stream is custom stream: "true"-yes, this node is not
returned-no--></customStreamEnable>
<isSupportPictureByUrl>
  <!--optional, xs: boolean, whether the device supports capture picture in URL format (related URI: /ISAPI/
ContentMgmt/StreamingProxy/channels/<ID>/PictureByUrl/capabilities?format=json)-->
</isSupportPictureByUrl>
</StreamingChannel>
```

XML_Cap_SupplementLight

SupplementLight capability message in XML format

```
<SupplementLight><!--opt-->
  <mode opt="schedule,off,on,auto"><!--opt, xs: string, adjustment mode of supplement light: "schedule,off,on,auto"--
></mode>
  <Schedule>
    <TimeRange><!--req-->
      <beginTime><!--req, xs: time, ISO8601 time--></beginTime>
      <endTime><!-- req, xs: time, ISO8601 time--></endTime>
    </TimeRange>
  </Schedule>
  <brightnessLimit min="0" max="100">
    <!--dep, xs: integer, brightness of supplement light, which is between 0 and 100; the brightness can be adjust when
<mode> is set to "on"-->
  </brightnessLimit>
  <supplementLightMode opt="mixed,whiteLight,close">
    <!--opt, xs: string, illumination mode: "mixed"-hybrid, "whitelight"-white light, "close"-disabled-->
```

```

</supplementLightMode>
<irLightBrightness min="0" max="100">
  <!--dep, xs: integer, brightness of IR supplement light, which is between 0 and 100; this node is valid only when
<supplementLightMode> is set to "mixed" and <mixedLightBrightnessRegulatMode> is set to "manual"-->
</irLightBrightness>
<mixedLightBrightnessRegulatMode opt="manual,auto">
  <!--dep, xs: string, brightness adjustment mode of hybrid supplement light; this node is valid only when
<supplementLightMode> is set to "mixed"-->
</mixedLightBrightnessRegulatMode>
<mixedModeSupportType opt="irLight,whiteLight">
  <!--dep, xs: string, light type of hybrid illumination mode, "irLight"-IR light, "whiteLight"-white light-->
</mixedModeSupportType>
<highIrLightBrightness min="0" max="100">
  <!--dep, xs: integer, brightness of far IR light, which is between 0 and 100; this node cannot be configured together
with node <irLightBrightness>-->
</highIrLightBrightness>
<highWhiteLightBrightness min="0" max="100">
  <!--dep, xs: integer, brightness of far white light, which is between 0 and 100; this node cannot be configured
together with node <brightnessLimit>-->
</highWhiteLightBrightness>
<lowIrLightBrightness min="0" max="100">
  <!--dep, xs: integer, brightness of near IR light, which is between 0 and 100; this node cannot be configured together
with node <irLightBrightness>-->
</lowIrLightBrightness>
<lowWhiteLightBrightness min="0" max="100">
  <!--dep, xs: integer, brightness of near white light, which is between 0 and 100; this node cannot be configured
together with node <brightnessLimit>-->
</lowWhiteLightBrightness>
<whiteLightBrightness min="0" max="100"><!--dep, xs: integer, white light brightness, which is between 0 and 100--
></whiteLightBrightness>
<irLightbrightnessLimit min="0" max="100"><!--dep, xs:integer, IR light brightness limit, range: [0,100]; this node is
valid when the value of mode "auto"--></irLightbrightnessLimit>
<whiteLightbrightnessLimit min="0" max="100"><!--dep, xs:integer, white light brightness limit, range: [0,100]; this
node is valid when the value of mode "auto"--></whiteLightbrightnessLimit>
<unscheduledBrightness min="" max=""><!--optional, xs:integer, brightness in the unscheduled time, by default it is
0 (the supplement light is disabled) and the value is between 0 and 100--></unscheduledBrightness>
</SupplementLight>

```

XML_Cap_SyncSignalOutputList

SyncSignalOutputList capability message in XML format

```

<SyncSignalOutputList version="2.0" xmlns="http://www.isapi.org/ver20/XMLSchema">
  <IOOutNum min="3" max="8">
    <!--req, xs:string, number of IO outputs-->
  </IOOutNum>
  <SyncSignalOutput>
    <id min="1" max="8">
      <!--req, xs:integer, ID, which is between 1 and 8-->
    </id>
    <IOWorkMode opt="flashLight,polarizer,continuousLight,flashLightSwitch">

```

```

    <!--req, xs:string, IO output mode: "flashLight"-strobe light control mode, "polarizer"-polarizer control mode,
    "continuousLight"-solid light control mode, "flashLightSwitch"-strobe light switching mode (used to switch between
    the strobe supplement light mode and IR mode). When this node is set to "polarizer", only <outputStatus>,
    <detectBrightnessEnable>, <brightnessThreshold>, <flashEnabled>, <startHour>, <startMinute>, <endHour>, and
    <endMinute> are valid, and the <outputStatus> can only be set to "high" or "low". When this node is set to
    "continuousLight", only <detectBrightnessEnable>, <brightnessThreshold>, <flashEnabled>, <startHour>,
    <startMinute>, <endHour>, and <endMinute> are valid-->
    </IOWorkMode>
    <defaultStatus opt="hige,low">
        <!--req, xs:string, effective status-->
    </defaultStatus>
    <outputStatus opt="high,low,pulse">
        <!--req, xs:string, default status-->
    </outputStatus>
    <aheadTime min="" max="">
        <!--req, xs:integer, pre-output time-->
    </aheadTime>
    <timeDelay min="" max="">
        <!--req, xs:integer, output duration-->
    </timeDelay>
    <freqMultiplyulti min="1" max="15">
        <!--req, xs:integer, frequency multiplication, which is between 1 and 15-->
    </freqMultiplyulti>
    <dutyRate min="0" max="40">
        <!--req, xs:integer, duty ratio, which is between 0 and 40-->
    </dutyRate>
    <postFlashEnable>
        <!--req, xs:boolean-->
    </postFlashEnable>
    <illegalFlashEnable>
        <!--req, xs:boolean-->
    </illegalFlashEnable>
    <videoFlashEnable>
        <!--req, xs:boolean-->
    </videoFlashEnable>
    <detectBrightnessEnable>
        <!--req, xs:boolean-->
    </detectBrightnessEnable>
    <brightnessThreshold min="0" max="100">
        <!--dep, xs:integer, brightness threshold, which is between 0 and 100-->
    </brightnessThreshold>
    <flashEnabled>
        <!--req, xs:boolean-->
    </flashEnabled>
    <startHour min="" max="">
        <!--dep, xs:integer-->
    </startHour>
    <startMinute min="" max="">
        <!--dep, xs:integer-->
    </startMinute>
    <endHour min="" max="">
        <!--dep, xs:integer-->

```

```
</endHour>
<endMinute min="" max="">
  <!--dep, xs:integer-->
</endMinute>
<plateBrightness>
  <!--req, xs:boolean-->
</plateBrightness>
<incrBrightEnable>
  <!--opt, xs:boolean, whether to enable brightness enhancement mode (for solid light mode)-->
</incrBrightEnable>
<incrBrightTime min="0" max="10000">
  <!--req, xs:integer, brightness enhancement duration, which is between 0 and 10000, unit: millisecond. This node is
valid only when <incrBrightEnable> is "true"-->
</incrBrightTime>
<incrBrightPercent min="0" max="100">
  <!--req, xs:integer, percentage of brightness enhancement, which is between 0 and 100. This node is valid only
when <incrBrightEnable> is "true"-->
</incrBrightPercent>
<brightness min="0" max="100">
  <!--dep, xs:integer, solid light brightness (for solid light mode), which is between 0 and 100-->
</brightness>
<delayCaptureTime min="1" max="1000">
  <!--dep, xs:integer, delayed capture time, which is between 1 and 1000, unit: millisecond. This node is valid only
when <incrBrightEnable> is "true"-->
</delayCaptureTime>
<manualBrightnessEnable>
  <!--req, xs:boolean, whether to enable adjusting brightness manually-->
</manualBrightnessEnable>
<manualBrightness>
  <!--dep, xs:integer, brightness adjusted manually, which is between 0 and 100-->
</manualBrightness>
</SyncSignalOutput>
</SyncSignalOutputList>
```

XML_Cap_Telnetd

Telnetd capability message in XML format

```
<Telnetd version="2.0" xmlns="http://www.isapi.org/ver20/XMLSchema">
  <enabled>
    <!--req, xs:boolean, whether to enable telnet: "true, false"-->
  </enabled>
  <radarMsgEnabled>
    <!--req, xs:boolean, whether to enable uploading radar debugging information: "true, false"-->
  </radarMsgEnabled>
  <illegalLoginLock>
    <!--req, xs:boolean, "true, false"-->
  </illegalLoginLock>
</Telnetd>
```


XML_Cap_tempRange

tempRange capability message in XML format

```
<?xml version="1.0" encoding="utf-8"?>
<TempRange version="2.0" xmlns="http://www.isapi.org/ver20/XMLSchema">
  <mode opt="automatic,manual">
    <!--req, xs:string, temperature range mode: "automatic"-automatic mode, "manual"-manual mode-->
  </mode>
  <temperatureUpperLimit min="-20" max="550" def="150">
    <!--dep, xs:float, maximum temperature, corrects to one decimal, the minimum value is "-20", the maximum value is "550", unit: Celsius degree; its value should be larger than temperatureLowerLimit; it is valid when the value of mode is "manual"-->
  </temperatureUpperLimit>
  <temperatureLowerLimit min="-20" max="550" def="0">
    <!--dep, xs:float, minimum temperature, corrects to one decimal, the minimum value is "-20", the maximum value is "550", unit: Celsius degree; its value should be smaller than temperatureUpperLimit; it is valid when the value of mode is "manual"-->
  </temperatureLowerLimit>
</TempRange>
```

XML_Cap_Time

Time capability message in XML format

```
<Time version="2.0" xmlns="http://www.isapi.org/ver20/XMLSchema">
  <timeMode opt="NTP,manual,satellite,timecorrect,platform"><!--required, xs:string--></timeMode>
  <localTime min="0" max="256">
    <!--dependent, xs:datetime, this node is required when <timemode> is "manual" or "local"-->
  </localTime>
  <timeZone min="0" max="256">
    <!--dependent, xs:string, POSIX time zone string, this node is required when <timemode> is "manual", "local" or "NTP"-->
  </timeZone>
  <satelliteInterval min="" max="">
    <!--dependent, xs:integer, minutes, time synchronization interval of locating by satellite, this node is valid only when <timemode> is "satellite"-->
  </satelliteInterval>
  <timeType opt="local, UTC">
    <!--optional, xs: string, time type, it can be local, or UTC, or local and UTC. If this node is not returned, it indicates that device does not support getting or setting time type-->
  </timeType>
  <platformType opt="EZVIZ">
    <!--dependent, xs: string, platform type: "EZVIZ"-Hik-Connect; it is valid only when the value of timeMode is "platform"-->
  </platformType>
</Time>
```

XML_Cap_Track

Track capability message in XML format

```
<Track version="2.0" xmlns="http://www.isapi.org/ver20/XMLSchema">
  <id><!--xs:integer, the value of <id> and <Channel> is the same. For example, 201 refers to the main stream of
channel 1, 202 refers to the sub-stream of channel 2, and so on--></id>
  <Channel><!--req, xs:integer--></Channel>
  <Enable><!--req, xs:boolean--></Enable>
  <Description><!--req, xs:string, e.g., trackType=standard,sourceTag=AXIS210a,contentType=video,codecType=MPEG4-
SP,resolution=640x480,frameRate=20 fps,bitrate=6000 kbps--></Description>
  <TrackGUID><!--req, xs:string, GUID generated by the client, e.g., A01AAAAA-BBBB-CCCC-DDDD-033595353625--></
TrackGUID>
  <Size><!--opt, xs:integer--></Size>
  <Duration min="0" max="750"><!--opt, xs:string, video expiry date, e.g., P10DT15H indicates that the video will
expire after 10 days and 15 hours--></Duration>
  <DefaultRecordingMode opt="CMR,MOTION,ALARM,EDR,ALARMANDMOTION,AllEvent,POS"><!--req, xs:string,
default video type--></DefaultRecordingMode>
  <LoopEnable><!--opt, xs:boolean, whether to support recurrently overwriting--></LoopEnable>
  <SrcDescriptor><!--video source description-->
    <SrcGUID><!--req, xs:string, GUID generated by the device, e.g., E800A543-9D53-4520-8BB8-9509062C692D--></
SrcGUID>
    <SrcChannel><!--req, xs:integer, source channel--></SrcChannel>
    <StreamHint><!--req, xs:string, e.g., "video, mp4, 640x480, 20 fps, 6000 kbps"--></StreamHint>
    <SrcDriver><!--req, xs:string, stream executable driver name, e.g., RTP/RTSP--></SrcDriver>
    <SrcType><!--opt, xs:string, source type, e.g., "mp4", "video"--></SrcType>
    <SrcUrl opt="rtsp://localhost/PSIA/Streaming/channels/101,rtsp://localhost/PSIA/Streaming/channels/102,rtsp://
localhost/PSIA/Streaming/channels/103,,rtsp://localhost/PSIA/Streaming/channels/107"><!--req, xs:string, e.g.,
rtsp://10.3.2.26/mpeg4/media.amp--></SrcUrl>
    <SrcUrlMethods><!--req, xs:string, methods supported by the source: "DESCRIBE,SETUP,PLAY,TEARDOWN"--></
SrcUrlMethods>
    <SrcLogin><!--req, xs:string, login source password--></SrcLogin>
  </SrcDescriptor>
  <TrackSchedule><!--recording schedule-->
    <ScheduleBlockList>
      <ScheduleBlock><!--list, recording schedule list-->
        <ScheduleBlockGUID><!--xs:string, schedule GUID, e.g., ABC12345-CDEF-4520-8BB8-7135789C8790--></
ScheduleBlockGUID>
        <ScheduleBlockType><!--req, xs:string--></ScheduleBlockType>
        <ScheduleAction><!--list-->
          <id><!--req, xs:integer--></id>
          <ScheduleActionStartTime><!--start time of the schedule-->
            <DayOfWeek><!--req, xs:string, "Monday,Tuesday,Wednesday,Thursday,Friday,Saturday,Sunday"--></
DayOfWeek><!--inclusive-->
            <TimeOfDay><!--xs:time, 00:00:00--></TimeOfDay>
          </ScheduleActionStartTime>
          <ScheduleActionEndTime><!--end time of the schedule-->
            <DayOfWeek><!--req, xs:string, "Monday,Tuesday,Wednesday,Thursday,Friday,Saturday,Sunday"--></
DayOfWeek><!--exclusive-->
            <TimeOfDay><!--xs:time, 08:00:00--></TimeOfDay>
          </ScheduleActionEndTime>
        </ScheduleAction>
      </ScheduleBlock>
    </ScheduleBlockList>
  </TrackSchedule>
</Track>
```

```
<ScheduleDSTEnable><!--req, xs:boolean, whether to enable DST (daylight saving time)--></ScheduleDSTEnable>
<Description><!--req, xs:string, PreMorning (midnight to 8 a.m., local time)--></Description>
<Actions><!--alarm or motion detection that triggers recording-->
  <Record><!--opt, xs:boolean--></Record>
  <Log><!--opt, xs:boolean--></Log>
  <SaveImg><!--opt, xs:boolean--></SaveImg>
  <ActionRecordingMode><!--req, xs:string, recording schedule type:
"CMR,EDR,ALARM,MOTION,ALARMANDMOTION,COMMAND,SMART"--></ActionRecordingMode>
  <PreRecordTimeSeconds><!--opt, xs:integer, pre-record time, it is between 0 and 10 and the default value is 5,
unit: second--></PreRecordTimeSeconds>
  <PostRecordTimeSeconds><!--opt, xs:integer, post-record time, it is between 0 and 20 and the default value is
5, unit: second--></PostRecordTimeSeconds>
</Actions>
</ScheduleAction>
</ScheduleBlock>
</ScheduleBlockList>
</TrackSchedule>
<CustomExtensionList>
  <CustomExtension>
    <CustomExtensionName><!--opt, xs:string, example: www.isapi.com/RaCM/trackExt/ver10--></
CustomExtensionName>
    <enableSchedule><!--opt, xs:boolean, whether to enable recording schedule configuration--></enableSchedule>
    <SaveAudio><!--opt, xs:boolean, whether to enable recording audio--></SaveAudio>
    <PreRecordTimeSeconds><!--opt, xs:integer, pre-record time, unit: second--></PreRecordTimeSeconds>
    <PostRecordTimeSeconds><!--opt, xs:integer, post-record time, unit: second--></PostRecordTimeSeconds>
    <HolidaySchedule>
      <ScheduleBlock>
        <ScheduleBlockGUID><!--req, xs:string, example: 00000000-0000-0000-0000-000000000000--></
ScheduleBlockGUID>
        <ScheduleBlockType><!--opt, xs:string, e.g., www.isapi.com/racm/schedule/ver10--></ScheduleBlockType>
      </ScheduleBlock>
    </HolidaySchedule>
  </CustomExtension>
</CustomExtensionList>
<IntelligentRecord><!--opt, xs:boolean, whether to enable VCA recording function: 0-no, 1-yes--></IntelligentRecord>
<delayTime opt="0,3,4,5,10,30,60,120,300"><!--opt, xs:integer, capture delay time, unit: second--></delayTime>
<durationEnabled opt="true,false"><!--opt, xs:boolean, whether to enable video expiry time. If this function is not
supported, this node will not be returned. If this function is supported, the video expiry date will be set by the node
<Duration>--></durationEnabled>
</Track>
```

XML_Cap_Tracking

Tracking capability message in XML format

```
<Tracking version="2.0" xmlns="http://www.std-cgi.org/ver20/XMLSchema">
  <enabled>
    <!--req, xs:boolean-->
  </enabled>
  <mode opt="auto>manual">
    <!--dep, xs:string, "auto>manual"-->
  </mode>
</Tracking>
```

```
</mode>
<trackingTime min="0" max="60" def="5">
  <!--dep, xs:integer, tracking time-->
</trackingTime>
<normalizedScreenSize>
  <!--req, ro-->
  <normalizedScreenWidth>
    <!--req, ro, xs:integer-->
  </normalizedScreenWidth>
  <normalizedScreenHeight>
    <!--req, ro, xs:integer-->
  </normalizedScreenHeight>
</normalizedScreenSize>
<ManualRegionCoordinatesList min="4" size="4">
  <!--dep, tracking area-->
  <RegionCoordinates>
    <!--opt-->
    <positionX>
      <!--req, xs:integer, X-coordinate-->
    </positionX>
    <positionY>
      <!--req, xs:integer, Y-coordinate-->
    </positionY>
  </RegionCoordinates>
</ManualRegionCoordinatesList>
<isNotSupportTrackingDisplay>
  <!--req, xs:boolean, whether to support displaying master-slave tracking on the interface: "true"-yes, "false"-no-->
</isNotSupportTrackingDisplay>
<Schedule>
  <!--opt,dep, schedule, this node is valid only when enabled is "true". By default the schedule is from 00:00 to 24:00
if no schedule is configured-->
  <TimeRange >
    <beginTime>
      <!--UTC time, e.g.: "2018-03-13T19:42:27+08:00"-->
    </beginTime>
    <endTime>
      <!--UTC time, e.g.: "2018-03-13T19:42:27+08:00"--->
    </endTime>
  </TimeRange >
</Schedule >
</Tracking>
```

XML_Cap_TwoWayAudioChannelList

XML message about the audio configuration capability of all two-way audio channels

```
<TwoWayAudioChannelList version="2.0" xmlns="http://www.isapi.org/ver20/XMLSchema">
  <TwoWayAudioChannel><!--list-->
    <id><!--required, xs:string, audio channel index--></id>
    <enabled><!--required, xs:boolean, whether to enable--></enabled>
    <audioCompressionType opt="G.711alaw,G.711ulaw,G.726,G.729,G.729a,G.
```

```

729b,PCM,MP3,AC3,AAC,ADPCM,MP2L2"><!--required, xs:string, encoding type of audio output: "G.711alaw,G.
711ulaw,G.726,G.729,G.729a,G.729b,PCM,MP3,AC3,AAC,ADPCM,MP2L2"--></audioCompressionType>
<audioInboundCompressionType opt="G.711alaw,G.711ulaw,G.726,G.729,G.729a,G.
729b,PCM,MP3,AC3,AAC,ADPCM"><!--optional, xs:string, encoding type of audio input: "G.711alaw,G.711ulaw,G.
726,G.729,G.729a,G.729b,PCM,MP3,AC3,AAC,ADPCM"--></audioInboundCompressionType>
<speakerVolume min="" max=""><!--optional, xs:integer, input volume, value range: [1,100]--></speakerVolume>
<microphoneVolume min="" max=""><!--optional, xs:integer, output volume, value range: [1,100]--></
microphoneVolume>
<noisereduce opt="true,false"><!--optional, xs:boolean, whether to enable--></noisereduce>
<audioBitRate min="" max=""><!--optional, xs:integer, audio frame rate, unit: kbs--></audioBitRate>
<audioInputType opt="MicIn,LineIn"><!--optional, xs:string, audio input: "MicIn,LineIn"--></audioInputType>
<associateVideoInputs><!--optional, linked audio input-->
<enabled opt="true,false"><!--required, xs:boolean, whether to enable--></enabled>
<videoInputChannelList><!--required, list of video input channels-->
<videoInputChannelID min="" max=""><!--list, xs:string, video input channel index--></videoInputChannelID>
</videoInputChannelList>
</associateVideoInputs>
<audioSamplingRate min="" max=""><!--optional, xs:float, audio sampling rate, unit: kHz--></audioSamplingRate>
<lineOutForbidden opt="true,false"><!--optional, read-only, xs:boolean, whether the audio output is not supported:
false or this node is not returned-the device supports audio output, true-the device does not support audio output--
></lineOutForbidden>
<muteDuringPanTilt opt="true,false"><!--dependent, xs:boolean, whether to mute during moving--></
muteDuringPanTilt>
</TwoWayAudioChannel>
</TwoWayAudioChannelList>

```

XML_Cap_UPnPcap

XML message about UPnP configuration capability

```

<?xml version="1.0" encoding="utf-8"?>
<UPnPcap version="2.0" xmlns="http://www.isapi.org/ver20/XMLSchema">
<enabled opt="true,false"/><!--required, xs:boolean-->
<isSupportPortsCfg opt="true,false"/><!--optional, xs:boolean-->
<ShortcutNatInfo><!--optional, configuration parameters about shortcut NAT port mapping-->
<enabled opt="true,false"/><!--optional, xs:boolean-->
<natRouterLanAddr><!--optional, IP information of local mapping-->
<ipVersion opt="v4,v6,dual"><!--required, xs:string--></ipVersion>
<ipAddress min="" max=""><!--dependent, xs:string--></ipAddress>
<ipv6Address min="" max=""><!--dependent, xs:string--></ipv6Address>
</natRouterLanAddr>
<natRouterWanAddr><!--optional, IP information of external mapping-->
<ipVersion opt="v4,v6,dual"><!--required, xs:string--></ipVersion>
<ipAddress min="" max=""><!--dependent, xs:string--></ipAddress>
<ipv6Address min="" max=""><!--dependent, xs:string--></ipv6Address>
</natRouterWanAddr>
<beginPort min="" max=""><!--optional, xs:integer, port No. when mapping starts, <b>beginPort</b> and <b>endPort</b>
specify the range of the port No. that supports mapping--></beginPort>
<endPort min="" max=""><!--optional, xs:integer, port No. when mapping ends, <b>beginPort</b> and <b>endPort</b> specify
the range of the port No. that supports mapping--></endPort>

```

```
</ShortcutNatInfo>  
</UPnPcap>
```

XML_Cap_User

XML message about user configuration capability

```
<User xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0" >  
  <id>  
    <!--required, xs:integer, user ID, value range: [1,16]-->  
  </id>  
  <userName>  
    <!--required, xs:string -->  
  </userName>  
  <password>  
    <!--required, xs:string, read-only-->  
  </password>  
  <bondIpAddressList>  
    <bondIpAddress>  
      <id>  
        <!--required, xs:integer -->  
      </id>  
      <ipAddress>  
        <!--dependent, xs:string -->  
      </ipAddress>  
      <ipv6Address>  
        <!--dependent, xs:string -->  
      </ipv6Address>  
    </bondIpAddress>  
  </bondIpAddressList>  
  <bondMacAddressList>  
    <bondMacAddress>  
      <id>  
        <!--required, xs:integer -->  
      </id>  
      <macAddress>  
        <!--optional, xs:string -->  
      </macAddress>  
    </bondMacAddress>  
  </bondMacAddressList>  
  <userLevel>  
    <!--optional, xs:string, user type: "Administrator", "Operator", "Viewer"-->  
  </userLevel>  
  <attribute>  
    <!--optional-->  
    <inherent>  
      <!--xs:boolean -->  
    </inherent>  
  </attribute>  
  <belongUserId>  
    <!--optional, xs:integer, upper-level user ID, "0"-there is no upper-level user-->
```

```
</belongUserId>
<belongOrgNo>
  <!--optional, xs:integer, organization No.-->
</belongOrgNo>
</User>
```

XML_Cap_UserPermissionList

XML message about the configuration capability of all users' permissions

```
<UserPermissionList version="2.0" xmlns="http://www.isapi.org/ver20/XMLSchema">
  <UserPermission><!--required-->
    <id min="" max=""/><!--required, xs:string-->
    <userId min="" max=""/><!--required, xs:string, user ID-->
    <userType optional="admin,operator,viewer"/><!--required, xs:string, user type: "admin, operator, viewer"-->
    <localPermission><!--optional-->
      <backup optional="true,false"/><!--optional, xs:boolean, whether it supports local backup. If this node is returned,
it indicates that the device supports this function. You can configure whether to assign this permission to users-->
      <record optional="true,false"/><!--optional, xs:boolean, whether it supports local manual operation-->
      <playBack optional="true,false"/><!--optional, xs:boolean, local playback-->
      <preview optional="true,false"/><!--optional, xs:boolean, local live view-->
      <videoChannelPermissionList><!--optional-->
        <videoChannelPermission> <!--optional-->
          <id min="" max=""/><!--required, xs:string-->
          <playBack optional="true,false"/><!--optional, xs:boolean-->
          <preview optional="true,false"/><!--optional, xs:boolean-->
          <record optional="true,false"/><!--optional, xs:boolean-->
          <backup optional="true,false"/><!--optional, xs:boolean-->
        </videoChannelPermission>
      </videoChannelPermissionList>
      <ptzControl optional="true,false"/><!--optional, xs:boolean, local PTZ control-->
      <ptzChannelPermissionList> <!--optional-->
        <ptzChannelPermission> <!--required-->
          <id min="" max=""/><!--required, xs:string-->
          <ptzControl optional="true,false"/><!--optional, xs:boolean, local PTZ control-->
        </ptzChannelPermission>
      </ptzChannelPermissionList>
      <logOrStateCheck optional="true,false"/><!--optional, xs:boolean, whether it supports view logs locally-->
      <parameterConfig optional="true,false"/><!--optional, xs:boolean, whether it supports setting parameters locally-->
      <restartOrShutdown optional="true,false"/><!--optional, xs:boolean, whether it supports shutting down and
rebooting the device locally-->
      <upgrade optional="true,false"/><!--optional, xs:boolean, whether it supports upgrading and formatting locally-->
      <manageChannel optional="true,false"/><!--optional, xs:boolean, whether it supports managing channels locally-->
    </localPermission>
    <remotePermission><!--optional-->
      <record optional="true,false"/><!--optional, xs:boolean, whether it supports recording remotely and manually-->
      <playBack optional="true,false"/><!--optional, xs:boolean, whether it supports remote playback-->
      <preview optional="true,false"/><!--optional, xs:boolean, whether it supports remote live view-->
      <videoChannelPermissionList><!--optional -->
        <videoChannelPermission><!--optional -->
          <id min="" max=""/><!--required, xs:string !-->
```

```

    <preview optional="true,false"/><!--optional, xs:boolean, whether it supports remote live view-->
    <playBack optional="true,false"/><!--optional, xs:boolean, whether it supports remote playback-->
    <record optional="true,false"/><!--optional, xs:boolean, whether it supports recording remotely and manually-->
  </videoChannelPermission>
</videoChannelPermissionList>
  <ptzControl optional="true,false"/><!--optional, xs:boolean, whether it supports remote PTZ control-->
  <ptzChannelPermissionList><!--optional -->
  <ptzChannelPermission><!--optional -->
    <id><!--required, this ID must correspond to the PTZ ID in /ISAPI/PTZCtrl/channels/<ID>--></id>
    <ptzControl optional="true,false"/><!--optional, xs:boolean, whether it supports controlling the PTZ remotely-->
  </ptzChannelPermission>
</ptzChannelPermissionList>
  <logOrStateCheck optional="true,false"/><!--optional, xs:boolean, whether it supports viewing logs remotely-->
  <parameterConfig optional="true,false"/><!--optional, xs:boolean, whether it supports setting parameters
remotely-->
    <restartOrShutdown optional="true,false"/><!--optional, xs:boolean, whether it supports shutting down and
rebooting remotely-->
    <upgrade optional="true,false"/><!--optional, xs:boolean, whether it supports upgrading and formatting remotely--
>
    <voiceTalk optional="true,false"/><!--optional, xs:boolean, whether it supports remote two-way audio-->
    <transParentChannel optional="true,false"/><!--optional, xs:boolean, whether it supports controlling the serial
port remotely-->
    <contorlLocalOut optional="true,false"/><!--optional, xs:boolean, whether it supports controlling the local output
remotely-->
    <manageChannel optional="true,false"/><!--optional, xs:boolean, whether it supports managing channels
remotely-->
    <alarmOutOrUpload optional="true,false"/><!--optional, xs:boolean, whether it supports request for uploading
alarms and alarm output remotely-->
    <IRAID>
      <rapidConfiguration optional="true,false"/><!--optional, xs:boolean, whether it supports configuring IRAID
permissions rapidly-->
      <reset optional="true,false"/><!--optional, xs:boolean, whether it supports configuring IRAID to reset
configuration permissions-->
    </IRAID>
  </remotePermission>
  <passwordValidity min="" max=""><!--required, xs:string, valid duration of the password, value range: [0, 365],
unit: day. Only the admin user can edit this parameter-->
</UserPermission>
</UserPermissionList>

```

XML_Cap_VCResource

Capability message of intelligent resources switch in XML format

```

<VCResource version="2.0" xmlns="http://www.isapi.org/ver20/XMLSchema">
  <type
opt="basicBehavior,fullBehavior,facesnapBehavior,facesnap,TFS,smartVehicleDetection,smartHVTDetection,smart,judic
ial,smart264AndRoadDetection,smart264AndFaceDetection,smart264AndHeatMap,smartVehicleIllegalParkingDetectio
n,smartIntelligentMonitor,smartTrafficDataCollection, roadDetection,humanRecognition, perimeterCapture,
vehicleDetection,HVTDetection,mixedTargetDetection,trackingCaptureMode,nonTrackingCaptureMode,close,faceHum
anModelingContrast,cityManagement,teacherBehavior,

```



```
12MPLiveView,personQueueDetection,verticalPeopleCounting,safetyHelmet,faceCounting">
  <!--required, xs: string, intelligent resource types, "trackingCaptureMode"-panoramic capture mode,
"nonTrackingCaptureMode"-single target capture mode, "faceHumanModelingContrast"-resources of face/human
body+face modeling+face picture comparison, "verticalPeopleCounting"-vertical people counting, "safetyHelmet"-
hard hat detection, "faceCounting"-face counting, "personArming"-person arming-->
</type>
<PromptList><!--optional, prompt for switching intelligent resources-->
  <Prompt><!--list-->
    <resourceType opt="roadDetection">
      <!--optional, xs: string, intelligent resource to prompt, currently, only "roadDetection" is supported-->
    </resourceType>
    <resourcePrompt opt="smartCode">
      <!--optional, xs: string, prompt contents-->
    </resourcePrompt>
  </Prompt>
</PromptList>
</VCAResource>
```

XML_Cap_VideoInputChannel

VideoInputChannel capability message in XML format

```
<VideoInputChannel version="2.0" xmlns="http://www.isapi.org/ver20/XMLSchema">
  <id><!--req, xs: string--></id>
  <inputPort><!--req, xs: string--></inputPort>
  <videoInputEnabled><!--opt, xs: boolean--></videoInputEnabled>
  <name><!--opt, xs: string--></name>
  <videoFormat opt="PAL,NTSC"><!--opt, xs: string, video standards--></videoFormat>
  <portType opt="SDI,OPT,VGA,HDMI,YPbPr"><!--opt, xs: string, port type--></portType>
  <resDesc><!--opt, xs: string--></resDesc>
</VideoInputChannel>
```

XML_Cap_VideoOutputChannel

VideoOutputChannel capability message in XML format

```
<VideoOutputChannel version="2.0" xmlns="http://www.isapi.org/ver20/XMLSchema">
  <id><!--req, xs: string; ID--></id>
  <type opt="VGA,CVBS,HDMI,Spot,SDI,LCD"><!--req, xs: string, video output type--></type>
  <menu><!--dep, read-only-->
    <mirrorMenu><!--req, xs: boolean--></mirrorMenu>
  </menu>
  <mode opt="close,clip,scale,open,SDI_1080P25,...,HDMI_1080P,HDMI_720P,HDMI_2160P">
    <!--opt, xs: string, video output mode-->
  </mode>
  <resolution opt="1920*1080/60HZ,1280*720/50HZ,..."><!--opt, xs: string, video resolution--></resolution>
</VideoOutputChannel>
```

XML_Cap_VideoOverlay

VideoOverlay capability message in XML format

```
<VideoOverlay version="2.0" xmlns="http://www.isapi.org/ver20/XMLSchema">
  <normalizedScreenSize>
    <!--req-->
    <normalizedScreenWidth>
      <!--ro, req, xs:integer-->
    </normalizedScreenWidth>
    <normalizedScreenHeight>
      <!--ro, req, xs:integer-->
    </normalizedScreenHeight>
  </normalizedScreenSize>
  <attribute>
    <!--opt-->
    <transparent>
      <!--req, xs:boolean-->
    </transparent>
    <flashing>
      <!--req, xs:boolean-->
    </flashing>
  </attribute>
  <TextOverlayList/><!--opt, see details in the message of XML_TextOverlayList-->
  <DateTimeOverlay/><!--opt, see details in the message of XML_DateTimeOverlay-->
  <channelNameOverlay/><!--opt, see details in the message of XML_channelNameOverlay-->
  <fontSize>
    <!--opt, xs:string, "adaptive,16*16,32*32,48*48,64*64,80*80,96*96,112*112,128*128"-->
  </fontSize>
  <frontColorMode>
    <!--opt, string, "auto,customize"-->
  </frontColorMode>
  <frontColor>
    <!--dep, xs:hexBinary;color-->
  </frontColor>
  <BatteryPowerOverlay/><!--opt, see details in the message of XML_BatteryPowerOverlay-->
  <alignment opt="customize,alignRight,alignLeft">
    <!--opt, xs:string, "customize,alignRight,alignLeft"-->
  </alignment>
  <publicSecurity>
    <!--req, xs:boolean-->
  </publicSecurity>
  <DeviceStatusDisplay><!--opt-->
    <batteryCapacityDisplay>
      <!--opt, xs:boolean-->
    </batteryCapacityDisplay>
    <chargingStatusDisplay>
      <!--opt, xs:boolean-->
    </chargingStatusDisplay>
    <bluetoothStatusDisplay>
      <!--opt, xs:boolean-->
    </bluetoothStatusDisplay>
  </DeviceStatusDisplay>
</VideoOverlay>
```

```
</bluetoothStatusDisplay>
<dialStatusDisplay>
  <!--opt, xs:boolean-->
</dialStatusDisplay>
</DeviceStatusDisplay>
<boundary>
  <!--opt, xs:integer-->
</boundary>
</VideoOverlay>
```

See Also

XML_TextOverlayList

XML_DateTimeOverlay

XML_channelNameOverlay

XML_BatteryPowerOverlay

XML_Cap_Wireless

Wireless capability message in XML format

```
<Wireless version="2.0" xmlns="http://www.isapi.org/ver20/XMLSchema">
  <enabled><!--req, xs:boolean--></enabled>
  <wirelessNetworkMode opt="infrastructure,adhoc"><!--opt, xs:string--></wirelessNetworkMode>
  <channel opt="1,2,3,4,5,6,7,8,9,10,11,12,13,14,auto"><!--opt, xs:string--></channel>
  <ssid min="" max=""><!--opt, xs:string--></ssid>
  <wmmEnabled><!--opt, xs:boolean--></wmmEnabled>
  <WirelessSecurity><!--opt-->
    <securityMode opt="disable,WEP,WPA-personal,WPA2-personal,WPA-RADIUS,WPA-enterprise,WPA2-
enterprise,WPA/WPA2">
      <!--opt, xs:string-->
    </securityMode>
    <WEP><!--dep, this node depends on <securityMode>-->
      <authenticationType opt="open,sharedkey,auto"><!--req, xs:string--></authenticationType>
      <defaultTransmitKeyIndex min="" max=""><!--req, xs:integer--></defaultTransmitKeyIndex>
      <wepKeyLength opt="64,128"><!--opt, xs:integer--></wepKeyLength>
      <EncryptionKeyList>
        <encryptionKey>
          <!--req, xs:hexBinary, WEP encryption key in hexadecimal format-->
        </encryptionKey>
      </EncryptionKeyList>
    </WEP>
    <WPA><!--dep, depends on <securityMode>-->
      <algorithmType opt="TKIP,AES,TKIP/AES"><!--req, xs:string--></algorithmType>
      <sharedKey><!-- req, xs:string, pre-shared key used in WPA--></sharedKey>
      <wpaKeyLength min="8" max="64"><!--req, xs: integer--></wpaKeyLength>
    </WPA>
    <support64bitKey opt="WPA-personal, WPA2-personal"/><!--opt, xs:string--></support64bitKey>
  </WirelessSecurity>
  <AccessPointList>
    <InterfaceDisplay><!--opt-->
```

```
<TableElementList>
  <TableElement>
    <!--opt,xs:string,opt="SSID,workingMode,securityMode,channel,signalStrength, speed,connectionStatus"-->
  </TableElement>
</TableElementList>
</InterfaceDisplay>
</AccessPointList>
<isSupportConnectStatus>
  <!-- opt, xs:boolean, whether supports Wi-Fi connection status-->
</isSupportConnectStatus>
<workScene opt="computerRoom,monitorTerminal"><!--opt, xs:string, working scenario--></workScene>
<protocol opt="802.11ac"><!--req,xs:string, protocol mode--></protocol>
<hideSsid><!--opt, xs:boolean--></hideSsid>
<ChannelConfig><!--opt-->
  <transmitPower opt="9,12,15,18,21,24,27"><!--opt, xs:integer, unit: dBm--><transmitPower>
  <ChannelCountryList><!--opt>
    <Country>
      <countryID><!--req, xs:integer, country code--></countryID>
    <ChannelList>
      <Node>
        <width opt="auto,20,40,80"><!--req, xs:string, bandwidth, unit: MHz--></width>
        <channel><!--req,dep, xs:string, unit: MHz></channel>
      </Node>
    </ChannelList>
  </Country>
</ChannelCountryList>
</ChannelConfig>
<isSupportNullSsid>
  <!--opt, xs: boolean, whether supports setting SSID to null, if supports, return "true", otherwise, this node will not
be returned. If the SSID is empty, the device will filter the SSID configuration when configuring Wi-Fi-->-->
</isSupportNullSsid>
</Wireless>
```

XML_Cap_WirelessServer

WirelessServer capability message in XML format

```
<WirelessServer version="2.0" xmlns="http://www.isapi.org/ver20/XMLSchema">
  <wifiApEnabled opt="true,false"><!--opt,xs: boolean,--></wifiApEnabled>
  <broadcastEnabled opt="true,false"><!--opt,xs: boolean--></broadcastEnabled>
  <wlanShareEnabled opt="true,false"><!--opt,xs: boolean--></wlanShareEnabled>
  <ssid min="" max=""><!--opt, xs: string--></ssid>
  <WirelessSecurity><!--req-->
    <securityMode opt="disable,WEP,WPA-personal,WPA2-personal,WPA-RADIUS,WPA-enterprise,WPA2-enterprise">
      <!--opt, xs: string-->
    </securityMode>
    <WEP><!--dep, depends on <securityMode>-->
      <authenticationType opt="open,sharedkey,auto">
        <!--req, xs: string-->
      </authenticationType>
      <defaultTransmitKeyIndex min="" max="">
```

```
<!--req, xs: integer-->
</defaultTransmitKeyIndex>
<wepKeyLength opt="64,128"><!--opt, xs: integer--></wepKeyLength>
<EncryptionKeyList size="">
  <encryptionKey>
    <!--req, xs: hexBinary, WEP encryption key in hexadecimal format-->
  </encryptionKey>
</EncryptionKeyList>
</WEP>
<WPA><!--dep, depends on <securityMode-->
  <algorithmType opt="TKIP,AES,TKIP/AES">
    <!--req, xs: string, "TKIP,AES,TKIP/AES"-->
  </algorithmType>
  <sharedKey><!--opt, xs: string, shared key used in WPA--></sharedKey>
  <wpaKeyLength min="" max=""><!-- opt, xs: integer, the key length is between 8 and 63--></wpaKeyLength>
  <defaultPassword><!--opt, xs: boolean--></defaultPassword>
</WPA>
</WirelessSecurity>
<DHCPEnabled opt="true,false"><!--opt, xs: boolean--></DHCPEnabled>
<ipVersion opt="v4,v6"><!--opt, xs:string--></ipVersion>
<HostIpAddress><!--opt-->
  <ipAddress><!--dep, xs:string--></ipAddress>
  <ipv6Address><!--dep, xs:string--></ipv6Address>
</HostIpAddress>
<IPMask><!--opt-->
  <subnetMask><!--dep, xs:string, subnet mask for IPv4 address--></subnetMask>
  <bitMask><!--dep, xs:integer, bitmask IPv6 address--></bitMask>
</IPMask>
<AddressPool><!--opt-->
  <startIPv4Address><!--dep, xs:string--></startIPv4Address>
  <endIPv4Address><!--dep, xs:string--></endIPv4Address>
  <startIPv6Address><!-- dep, xs:string--></startIPv6Address>
  <endIPv6Address><!--dep, xs:string--></endIPv6Address>
</AddressPool>
<DNSAddressList size="2"><!--opt-->
  <DNSAddress><!--opt>
    <id><!--opt,xs:string,start from 1--></id>
    <ipAddress><!--dep, xs: string--></ipAddress>
    <ipv6Address><!--dep, xs: string--></ipv6Address>
  </DNSAddress>
</DNSAddressList>
<GatewayAddress>
  <ipAddress><!--dep, xs:string--></ipAddress>
  <ipv6Address><!--dep, xs:string--></ipv6Address>
</GatewayAddress>
<wifiApModeType opt="true,false,auto">
  <!--opt, xs: string, current wireless access point (AP) mode, if this node is returned, it indicates that the device
  supports auto mode-->
</wifiApModeType>
</WirelessServer>
```

Remarks

For Client supports auto AP, the node <<**wifiApModeType**> is valid; for Client does not supports auto AP, the node <**wifiApEnabled**> is valid; the values of these two nodes will effect each other, and when <**wifiApModeType**> equals to "auto", the value of <**wifiApEnabled**> is "true".

XML_Cap_WirelessServerStatus

WirelessServerStatus capability message in XML format

```
<WirelessServerStatus version="1.0" xmlns="http://www.isapi.org/ver20/XMLSchema">
  <wifiApEnabled>
    <!--dep, xs:boolean-->
  </wifiApEnabled>
  <linkDevices min="0" max="20">
    <!--dep, xs:integer, maximum number of linked devices-->
  </linkDevices>
  <DeviceInfoList size="20">
    <!--opt-->
    <DeviceInfo>
      <!--opt-->
      <IpAddress>
        <!--req-->
        <ipAddress>
          <!--dep, xs:string-->
        </ipAddress>
        <ipv6Address>
          <!--dep, xs:string-->
        </ipv6Address>
      </IpAddress>
      <hostName>
        <!--opt, xs:string-->
      </hostName>
      <macAddress>
        <!--opt, xs:string-->
      </macAddress>
    </DeviceInfo>
  </DeviceInfoList>
</WirelessServerStatus>
```

XML_Cap_WirelessStatus

WirelessStatus capability message in XML format

```
<WirelessStatus version="1.0" xmlns="http://www.isapi.org/ver20/XMLSchema">
  <enabled>
    <!--dep, xs:boolean-->
  </enabled>
  <connectionEnabled>
```

```
<!--dep, xs:boolean-->
</connectionEnabled>
<IpAddress>
  <!--req-->
  <IpAddress>
    <!--dep, xs:string-->
  </IpAddress>
  <ipv6Address>
    <!--dep, xs:string-->
  </ipv6Address>
</IpAddress>
<SubnetMask>
  <!--req-->
  <IpAddress>
    <!--dep, xs:string-->
  </IpAddress>
  <ipv6Address>
    <!--dep, xs:string-->
  </ipv6Address>
</SubnetMask>
<RouterAddress>
  <!--req-->
  <IpAddress>
    <!--dep, xs:string-->
  </IpAddress>
  <ipv6Address>
    <!--dep, xs:string-->
  </ipv6Address>
</RouterAddress>
<DNSAddressList size="2">
  <!--opt-->
  <DNSAddress>
    <!--opt-->
    <id>
      <!--opt, xs:string, DNS address ID, which starts from 1-->
    </id>
    <IpAddress>
      <!--dep, xs:string-->
    </IpAddress>
    <ipv6Address>
      <!--dep, xs:string-->
    </ipv6Address>
  </DNSAddress>
</DNSAddressList>
</WirelessStatus>
```

XML_Cap_WPS

WPS capability message in XML format

```
<WPS version="2.0" xmlns="http://www.isapi.org/ver20/XMLSchema">
  <enable><!--required, xs:boolean, whether to enable WPS--></enable>
</WPS>
```

XML_CaptureMode

CaptureMode message in XML format

```
<CaptureMode version="2.0" xmlns="http://www.isapi.org/ver20/XMLSchema">
  <mode>
    <!--req, xs:string, video input mode: 1-640×480@30fps, 2-4CIF@30fps, 3-720P@25fps, 4-720P@30fps,
    5-720P@60fps, 6-1280×960@15fps, 7-1280×960@25fps, 8-1280×960@30fps, 9-1280×1024@30fps,
    10-1600×900@15fps, 11-1600×1200@15fps, 12-1080P@15fps, 13-1080P@25fps, 14-1080P@30fps,
    15-1080P@50fps, 16-1080P@60fps, 17-2048×1536@15fps, 18-2048×1536@20fps, 19-2048×1536@24fps,
    20-2048×1536@25fps, 21-2048×1536@30fps, 22-2560×2048@25fps, 23-2560×2048@30fps, 24-2560×1920@7.5fps,
    25-3072×2048@30fps, 26-1944×1212@24fps, 27-1944×1212@25fps-->
  </mode>
</CaptureMode>
```

XML_CertificateInfo

CertificateInfo message in XML format

```
<CertificateInfo version="1.0" xmlns="http://www.isapi.org/ver20/XMLSchema">
  <id><!--req, xs:string--></id>
  <version><!--opt, xs:string--></version>
  <IssuerDN><!--req, isapi:DN-->
    <countryName><!--req, xs:string--></countryName>
    <stateOrProvinceName><!--opt, xs:string--></stateOrProvinceName>
    <localityName><!--opt, xs:string--></localityName>
    <organizationName><!--opt, xs:string--></organizationName>
    <organizationUnitName><!--opt, xs:string--></organizationUnitName>
    <commonName><!--req, xs:string--></commonName>
    <email><!--opt, xs:string--></email>
  </IssuerDN>
  <SubjectDN><!--req, isapi:DN-->
    <countryName><!--req, xs:string--></countryName>
    <stateOrProvinceName><!--opt, xs:string--></stateOrProvinceName>
    <localityName><!--opt, xs:string--></localityName>
    <organizationName><!--opt, xs:string--></organizationName>
    <organizationUnitName><!--opt, xs:string--></organizationUnitName>
    <commonName><!--req, xs:string--></commonName>
    <email><!--opt, xs:string--></email>
  </SubjectDN>
  <signatureAlgorithm><!--req, xs:string, signature algorithm: RSA_3, RSA_F4--></signatureAlgorithm>
  <keyAlgorithm><!--opt, xs:string--></keyAlgorithm>
  <startDate><!--req, xs:time, in ISO8601 time format--></startDate>
  <endDate><!--req, xs:time, in ISO8601 time format--></endDate>
```



```
<serialNumber><!--req, xs: string, uuid--></serialNumber>
</CertificateInfo>
```

XML_CertificateReq

CertificateReq message in XML format

```
<CertificateReq version="2.0" xmlns="http://www.isapi.org/ver20/XMLSchema">
  <id><!--req, xs: string--></id>
  <SubjectDN><!--req, isapi:DN-->
    <countryName><!--req, xs: string--></countryName>
    <stateOrProvinceName><!--opt, xs: string--></stateOrProvinceName>
    <localityName><!--opt, xs: string--></localityName>
    <organizationName><!--opt, xs: string--></organizationName>
    <organizationUnitName><!--opt, xs: string--></organizationUnitName>
    <commonName><!--req, xs: string --></commonName>
    <email><!--opt, xs: string--></email>
  </SubjectDN>
  <validity><!--opt, xs: integer, expiry date, which is between 1 and 5000--></validity>
  <passwd><!--opt, xs: string, password to protect private key--></passwd>
  <RSAKeyLength><!--opt, xs: integer--></RSAKeyLength>
</CertificateReq>
```

XML_CertificateReq_ImportCert

CertificateReq message in XML format

```
<CertificateReq version="1.0" xmlns="http://www.isapi.org/ver20/XMLSchema">
  <certificateMode><!--opt, xs:string, importing method: "signingRequest"-signature request (import self-signed certificate), "privateKey"-private key--></certificateMode>
  <privateKeyMode><!--dep, xs:string, private key mode: "seperateKey", "PKCS#12"--></privateKeyMode>
  <seperateKeyPassword><!--dep, xs:string, password in seperateKey mode, it should be encrypted--></seperateKeyPassword>
  <PKCSPassword><!--dep, xs:string, password in PKCS#12 mode, it should be encrypted--></PKCSPassword>
  <dataType><!--dep, xs:string, data type: "certificate", "privateKey"-private key--></dataType>
</CertificateReq>
```

XML_CertificateReqInfo

CertificateReqInfo message in XML format

```
<CertificateReqInfo version="2.0" xmlns="http://www.isapi.org/ver20/XMLSchema">
  <id><!--req, xs: string--></id>
  <SubjectDN><!--req, isapi:DN-->
    <countryName><!--req, xs: string--></countryName>
    <stateOrProvinceName><!--opt, xs: string--></stateOrProvinceName>
    <localityName><!--opt, xs: string--></localityName>
```

```
<organizationName><!--opt, xs: string--></organizationName>
<organizationUnitName><!--opt, xs: string--></organizationUnitName>
<commonName><!--req, xs: string --></commonName>
<email><!--opt, xs: string--></email>
</SubjectDN>
<version><!--opt, xs: string--></version>
<validity><!--opt, xs:int, expiry date, which is between 1 and 5000--></validity>
<keyAlgorithm><!--opt, xs: string--></keyAlgorithm>
<passwd><!--opt, xs: string, password to protect private key--></passwd>
<RSAKeyLength><!--opt, xs: integer--></RSAKeyLength>
</CertificateReqInfo>
```

XML_CertificateResult

CertificateResult message in XML format

```
<CertificateResult version="1.0" xmlns="http://www.isapi.org/ver20/XMLSchema">
  <formatType><!--req, xs:string, certificate format, e.g., "PEM"--></formatType>
  <detailInfo><!--req, xs:string, certificate details--></detailInfo>
</CertificateResult>
```

XML_Challenge

Challenge message in XML format.

```
<Challenge version="2.0" xmlns="http://www.isapi.org/ver20/XMLSchema">
  <key><!--req, xs:string--></key>
</Challenge>
```

XML_ChanCtrl

ChanCtrl message (for traffic) in XML format

```
<ChanCtrl version="2.0" xmlns="http://www.isapi.org/ver20/XMLSchema">
  <streamGetType><!--streaming mode: 0-live view mode (default), 1-data receiving mode--></streamGetType>
</ChanCtrl>
```

XML_Channel_VideoOverlay

Message about the text overlay parameters of a specific video digital input channel in XML format.

```
<VideoOverlay version="2.0" xmlns="http://www.isapi.org/ver20/XMLSchema">
  <normalizedScreenSize><!--required-->
    <normalizedScreenWidth><!--required, read-only, xs:integer, normalized width--></normalizedScreenWidth>
    <normalizedScreenHeight><!--required, read-only, xs:integer, normalized height--></normalizedScreenHeight>
  </normalizedScreenSize>
```

```

<attribute><!--optional-->
  <transparent><!--required, xs:boolean, whether it is transparent--></transparent>
  <flashing><!--required, xs:boolean, whether it is flickering--></flashing>
</attribute>
<TextOverlayList><!--optional-->
  <TextOverlay><!--list-->
    <id><!--required, xs:string, text No.--></id>
    <enabled><!--required, xs:boolean, whether to enable--></enabled>
    <positionX><!--required, xs:integer, X-coordinate--></positionX>
    <positionY><!--required, xs:integer, Y-coordinate--></positionY>
    <displayText><!--required, xs:string, overlaid text--></displayText>
  </TextOverlay>
</TextOverlayList>
<DateTimeOverlay><!--optional-->
  <enabled><!--required, xs:boolean, whether to display the date--> </enabled>
  <positionX><!--required, xs:integer, X-coordinate--> </positionX>
  <positionY><!--required, xs:integer, Y-coordinate--> </positionY>
  <dateStyle><!--optional, xs:string, date format: "YYYY-MM-DD", "MM-DD-YYYY", "DD-MM-YYYY", "CHR-YYYY-MM-DD", "CHR-MM-DD-YYYY", "CHR-DD-MM-YYYY", "CHR-YYYY/MM/DD", "CHR-MM/DD/YYYY", "CHR-DD/MM/YYYY"--> </dateStyle>
  <timeStyle><!--optional, xs:string, time format: "12hour", "24hour"--> </timeStyle>
  <displayWeek><!--optional, xs:boolean, whether to display the day of the week--> </displayWeek>
</DateTimeOverlay>
<channelNameOverlay><!--optional-->
  <enabled><!--required, xs:boolean, whether to display the channel name--> </enabled>
  <positionX><!--required, xs:integer, X-coordinate--> </positionX>
  <positionY><!--required, xs:integer, Y-coordinate--> </positionY>
</channelNameOverlay>
<fontSize><!--optional, xs:string, font size: "adaptive", "16*16", "32*32", "48*48", "64*64"--></fontSize>
<frontColorMode><!--optional, string, font color mode: "auto", "customize"--></frontColorMode>
<frontColor><!--dependency, xs:hexBinary, font color--></frontColor>
<BatteryPowerOverlay><!--optional, battery power overlay-->
  <enabled><!--required, xs:boolean, whether to display the overlaid battery power--></enabled>
  <positionX><!--required, xs:integer, X-coordinate--> </positionX>
  <positionY><!--required, xs:integer, Y-coordinate--> </positionY>
</BatteryPowerOverlay>
<alignment><!--optional, xs:string, alignment mode: "customize"-custom, "alignRight"-align to right, "alignLeft"-align to left--></alignment>
<publicSecurity><!--required, xs:boolean, security surveillance--></publicSecurity>
</VideoOverlay>

```

XML_ChannelCap_VideoOverlay

Message about the text overlay configuration capability of a specific video digital input channel in XML format.

```

<VideoOverlay version="2.0" xmlns="http://www.isapi.org/ver20/XMLSchema">
  <normalizedScreenSize><!--required-->
    <normalizedScreenWidth><!--required, read-only, xs:integer, normalized width--></normalizedScreenWidth>
    <normalizedScreenHeight><!--required, read-only, xs:integer, normalized height--></normalizedScreenHeight>

```

```

</normalizedScreenSize>
<attribute><!--optional-->
  <transparent opt="true,false"><!--required, xs:boolean, whether it is transparent--></transparent>
  <flashing opt="true,false"><!--required, xs:boolean, whether it is flickering--></flashing>
</attribute>
<TextOverlayList size="8"><!--optional-->
  <TextOverlay><!--list-->
    <id><!--required, xs:string, text No.--></id>
    <enabled><!--required, xs:boolean, whether to enable--></enabled>
    <positionX><!--required, xs:integer, X-coordinate--></positionX>
    <positionY><!--required, xs:integer, Y-coordinate--></positionY>
    <displayText><!--required, xs:string, overlaid text--></displayText>
    <isPersistentText><!--optional, xs:boolean, whether the OSD overlaid text is the same as the originally configured
one after the device reboots--></isPersistentText>
  </TextOverlay>
</TextOverlayList>
<DateTimeOverlay>
  <enabled opt="true,false"><!--required, xs:boolean, whether to display the date--></enabled>
  <positionX><!--required, xs:integer, X-coordinate--></positionX>
  <positionY><!--required, xs:integer, Y-coordinate--></positionY>
  <dateStyle opt="YYYY-MM-DD,MM-DD-YYYY,DD-MM-YYYY,CHR-YYYY-MM-DD,CHR-MM-DD-YYYY,CHR-DD-MM-
YYYY,CHR-YYYY/MM/DD,CHR-MM/DD/YYYY,CHR-DD/MM/YYYY"><!--optional, xs:string, date format--></dateStyle>
  <timeStyle opt="12hour,24hour"><!--optional, xs:string, time format--></timeStyle>
  <displayWeek opt="true,false"><!--optional, xs:boolean, whether to display the day of the week--></displayWeek>
</DateTimeOverlay>
<channelNameOverlay>
  <enabled><!--required, xs:boolean, whether to display the channel name--></enabled>
  <positionX><!--required, xs:integer, X-coordinate--></positionX>
  <positionY><!--required, xs:integer, Y-coordinate--></positionY>
</channelNameOverlay>
<fontSize opt="16*16,32*32,48*48,64*64,adaptive"><!--optional, xs:string, font size, unit: pixel--></fontSize>
<frontColorMode opt="auto,customize"><!--optional, string, font color mode--></frontColorMode>
<frontColor><!--dependency, xs:hexBinary, font color--></frontColor>
<BatteryPowerOverlay/><!--optional, battery power overlay-->
<DeviceStatusDisplay><!--optional, device status display-->
  <batteryCapacityDisplay><!--optional, xs:boolean, whether to display battery capacity--></batteryCapacityDisplay>
  <chargingStatusDisplay><!--optional, xs:boolean, whether to display charging status--></chargingStatusDisplay>
  <bluetoothStatusDisplay><!--optional, xs:boolean, whether to display bluetooth status--></bluetoothStatusDisplay>
  <dialStatusDisplay><!--optional, xs:boolean, whether to display dial-up status--></dialStatusDisplay>
</DeviceStatusDisplay>
<alignment opt="customize,alignRight,alignLeft,GB,allRight,allLeft"><!--optional, xs:string, alignment mode:
"customize"-custom, "alignRight"-align to right, "alignLeft"-align to left, "allRight"-align all to right (align all OSD texts
to the right, including characters, time, title, and so on), "allLeft"-align all to left (align all OSD texts to the left,
including characters, time, title, and so on)--></alignment>
  <lowRightTextNumber min="1" max="6"/><!--optional, number of characters in the lower-left corner-->
  <lowLeftTextNumber min="7" max="8"/><!--optional, number of characters in the lower-right corner-->
  <boundary min="0" max="2"><!--optional, xs:integer, boundary--></boundary>
  <TextOverlaySize min="0" max="0"><!--optional, xs:integer, number of overlaid texts--></TextOverlaySize>
  <FloorNumOverlay><!--optional-->
    <enabled><!--required, xs:boolean, whether to enable displaying the overlaid floor No.--></enabled>
    <positionX><!--required, xs:integer, X-coordinate--></positionX>
    <positionY><!--required, xs:integer, Y-coordinate--></positionY>

```

```

</FloorNumOverlay>
<upDownboundary><!--optional, xs:integer, minimum size of the upper or lower boundary, unit: number of
characters, range: [0,2] (the value can be 0, 1, or 2)--></upDownboundary>
<leftRightboundary><!--optional, xs:integer, minimum size of the left or right boundary, unit: number of characters,
range: [0,2] (the value can be 0, 1, or 2)--></leftRightboundary>
<Angle><!--optional, device angle-->
  <TiltAngle><!--optional, tilting angle-->
    <enabled opt="true,false"><!--required, xs:boolean, whether to enable: true, false (default)--></enabled>
    <positionX min="0" max="1000"><!--required, xs:integer,X-coordinate--></positionX>
    <positionY min="0" max="1000"><!--required,xs:integer, Y-coordinate--></positionY>
  </TiltAngle>
</Angle>
</VideoOverlay>

```

XML_ChannelEventCap

Message about event capability of a single channel in XML format

```

<ChannelEventCap version="2.0" xmlns="http://www.isapi.org/ver20/XMLSchema">
  <eventType opt=""/>
  <!--required, xs:string, event types supported by the channel-->
  <shieldEventType opt="behavior,faceSnap,humanRecognition,faceCapture,targetCapture"/>
  <!--required, xs:string, event types that support area shield-->
  <channelID><!--optional, xs:integer, current device channel No.--></channelID>
  <id><!--required, xs:integer--></id>
</ChannelEventCap>

```

Remarks

The values of node **<eventType>** are shown in the table below.

eventType	Description
VMD (motionDetection)	Motion detection
Shelteralarm (tamperDetection)	Video tampering alarm
videoLoss	Video loss
ROI	Region of interest
facedetection (faceDetection)	Face detection
intelliTrace	Auto-track
fielddetection (fieldDetection)	Intrusion detection
defocus (defocusDetection)	Defocus detection
audioexception (audioDetection)	Audio exception detection
scenechangedetection (sceneChangeDetection)	Sudden scene change detection

eventType	Description
linedetection (lineDetection)	Line-crossing detection
regionEntrance	Region entrance detection
regionExiting	Region exiting detection
loitering	Loitering detection
group	People gathering detection
rapidMove	Fast moving detection
parking	Parking detection
unattendedBaggage	Unattended baggage
attendedBaggage	Object removal detection
peopleDetection	Human detection
storageDetection	HDD health detection
behavior	Behavior analysis
faceCapture(faceSnap)	Face capture
ANPR(vehicleDetection)	Vehicle detection
fireDetection	Fire detection
shipsDetection	Ship detection
PIR	PIR alarm
targetCapture (humanRecognition)	Target capture
alarmResult (faceContrast)	Face comparison
framesPeopleCounting	People counting in single frame
hide	Tampering detection
audioLoss	Audio loss
cameraAngleAnomaly	Image exception
violentMotion	Violent motion
trial	Tailing
humanEnter	Human entrance
operateOverTime	Operation timed out
stickUp	Sticking scrip
installScanner	Installing scanner

eventType	Description
faceDetect	ATM face detection
temperature (temperatureDetection)	Temperature difference detection
PeopleCounting	People counting
personQueueDetection	People queuing-up detection
heatmap	Heat map
mixedTargetDetection	Multi-target-type detection
MTD_face	Supported face attributes of multi-target-type detection (MTD)
MTD_human	Supported human attributes of multi-target-type detection (MTD)
MTD_vehicle	Supported motor vehicle attributes of multi-target-type detection (MTD)
MTD_nonMotor	Supported non-motor vehicle attributes of multi-target-type detection (MTD)
faceSnapModeling	Face capture modeling
HVTVehicleDetection	Mixed-traffic detection
PictureCaptureComparision	N:1 face comparison
IO	Sensor alarm
smokeDetection	Smoke detection
smokeAndFireDetection	Smoke and fire detection
diskfull	HDD is full
diskerror	HDD error
nicbroken	Network disconnected
ipconflict	IP address conflicted
illaccess	Illegal access
personDensityDetection	People density detection
vehicleControl	Upload alarm of vehicle in whitelist and blacklist
vehicleRcogResult	Vehicle secondary recognition
cardMatch	Authentication
overSpeed	Overspeed alarm

eventType	Description
highTempAlarm	High temperature alarm
abnormalAcceleration	Abnormal accelerated speed alarm
failDown	People falling down
leavePosition	Absence detection
peopleNumChange	The number of people changed
retention	Overstay detection
running	Running
thermometry	Temperature measurement
heatmapPDC	Heat map people counting
heatmapDuration	Heat map people staying duration
intersectionAnalysis	Intersection analysis
AID_abandonedObject	Thrown object in traffic events
AID_pedestrian	Pedestrian detection in traffic events
AID_congestion	Congestion detection in traffic events
AID_roadBlock	Roadblock detection in traffic events
AID_construction	Construction detection in traffic events
AID_trafficAccident	Traffic accident detection in traffic events
AID_fogDetection	Fog detection in traffic events
TFS_illegalParking	Illegal parking of enforcement events
TFS_wrongDirection	Wrong-way driving of enforcement events
TFS_crossLane	Driving on lane line of enforcement events
TFS_laneChange	Lane change of enforcement events
TFS_vehicleExist	Vehicle existing of enforcement events
TFS_turnRound	Turning round of enforcement events
TFS_parallelParking	Parallel parking of enforcement events
TPS	Traffic data collection configuration
luma	Brightness exception
chroma	Color cast detection
snow	Snow noise

eventType	Description
streak	Stripe noise
freeze	Image freeze
sigLose	Signal loss
clarity	Clarity exception
jitter	Image flutter
block	Video tampering
flowers	Blurred screen detection
noise	Image noise
ghost	Abnormal light spot detection
purple	Image purple edge detection
ICR	ICR exception detection
protectiveFilm	Protective film unremoved
certificateRevocation	Certificate is expired
dataPreAlarm	Traffic pre-alarm
vibrationDetection	Vibration detection
radarFieldDetection	Radar intrusion detection
radarLineDetection	Radar line crossing detection
personArmingTrack	Person arming
manualPersonArmingTrack	Manual person arming
fireEscapeDetection	Fire engine access detection
takingElevatorDetection	Elevator detection

The value in the bracket is the old version, and considering the compatibility, both new and old value will be returned.

XML_ChannelEventCapList

Message about event capability of all channels in XML format

```
<ChannelEventCapList version="2.0" xmlns="http://www.isapi.org/ver20/XMLSchema">
  <ChannelEventCap/>
</ChannelEventCapList>
```

See Also

XML_ChannelEventCap

XML_ChannelInfo

ChannelInfo message in XML format.

```
<ChannelInfo version="2.0" xmlns="http://www.isapi.org/ver20/XMLSchema">
  <id><!--required, xs:integer, channel number--></id>
  <EagleEye>
    <!--optional, the channel supports the functions of PanoVu series camera-->
    <StreamFusion>
      <!--optional, "panoramic + PTZ camera streaming channel, it is not returned when not support-->
      <InsertChanNo><!--optional, xs:integer, access channel number--></InsertChanNo>
    </StreamFusion>
    <PanoramicMetaData>
      <!--optional, panoramic camera streaming channel, it is not returned when not support-->
      <InsertChanNo><!--optional, xs:integer, access channel number--></InsertChanNo>
    </PanoramicMetaData>
  </EagleEye>
  <FishEye>
    <!--optional, the channel supports the functions of fisheye camera-->
    <FishEyeStream>
      <!--optional, fisheye camera stream, it is used for software decoding of play library, it is not returned when not support-->
      <InsertChanNo><!--optional, xs:integer, access channel number--></InsertChanNo>
    </FishEyeStream>
  </FishEye>
  <Thermal>
    <!--optional, the channel supports the functions of thermal products-->
    <TemperatureMetaData>
      <!--optional, temperature measurement metadata, it is not returned when not support-->
      <InsertChanNo><!--optional, xs:integer, access channel number--></InsertChanNo>
    </TemperatureMetaData>
    <ThermometryDetection>
      <!--optional, temperature measurement detection, it is not returned when not support-->
      <InsertChanNo><!--optional, xs:integer, access channel number--></InsertChanNo>
    </ThermometryDetection>
    <ThermometryDiffDetection>
      <!--optional, temperature difference detection, it is not returned when not support-->
      <InsertChanNo><!--optional, xs:integer, access channel number--></InsertChanNo>
    </ThermometryDiffDetection>
    <FireDetection>
      <!--optional, fire detection, it is not returned when not support-->
      <InsertChanNo><!--optional, xs:integer, access channel number--></InsertChanNo>
    </FireDetection>
    <SmokeDetection>
      <!--optional, smoke detection, it is not returned when not support-->
      <InsertChanNo><!--optional, xs:integer, access channel number--></InsertChanNo>
    </SmokeDetection>
  </Thermal>
</ChannelInfo>
```

```
<ShipsDetection>
  <!--optional, ship detection, it is not returned when not support-->
  <InsertChanNo><!--optional, xs:integer, access channel number--></InsertChanNo>
</ShipsDetection>
<IntelligentRuleDisplay>
  <!--optional, intelligent rule display parameters of thermal application, it is not returned when not support-->
  <InsertChanNo><!--optional, xs:integer, access channel number--></InsertChanNo>
</IntelligentRuleDisplay>
<IntelligentBehaviorRule>
  <!--optional, intelligent behavior rule configuration, it is not returned when not support-->
  <InsertChanNo><!--optional, xs:integer, access channel number--></InsertChanNo>
</IntelligentBehaviorRule>
<DPC>
  <!--optional, defective pixel correction, it is not returned when not support-->
  <InsertChanNo><!--optional, xs:integer, access channel number--></InsertChanNo>
</DPC>
<Metadata>
  <!--optional, metadata extraction, it is not returned when not support-->
  <InsertChanNo><!--optional, xs:integer, access channel number--></InsertChanNo>
</Metadata>
<ThermometryShieldMask>
  <!--optional, temperature measurement shielded area, it is not returned when not support-->
  <InsertChanNo><!--optional, xs:integer, access channel number--></InsertChanNo>
</ThermometryShieldMask>
<FaceThermometry>
  <!--optional, face thermography, it is not returned when not support-->
  <InsertChanNo><!--optional, xs:integer, access channel number--></InsertChanNo>
</FaceThermometry>
<BlackBody>
  <!--optional, black body, it is not returned when not support-->
  <InsertChanNo><!--optional, xs:integer, access channel number--></InsertChanNo>
</BlackBody>
<BodyTemperatureCompensation>
  <!--optional, body temperature compensation, it is not returned when not support-->
  <InsertChanNo><!--optional, xs:integer, access channel number--></InsertChanNo>
</BodyTemperatureCompensation>
<LensCorrection>
  <!--optional, lens correction, when this node is not returned, it indicates not support-->
  <InsertChanNo><!--optional, xs:integer, access channel number--></InsertChanNo>
</LensCorrection>
<BurningPrevention>
  <!--optional, burning prevention, when this node is not returned, it indicates not support-->
  <InsertChanNo><!--optional, xs:integer, access channel number--></InsertChanNo>
</BurningPrevention>
<FaceSnapThermometry>
  <!--optional, upload the captured face picture with temperature information, it is not returned when this function
is not supported-->
  <InsertChanNo><!--optional, xs:integer, access channel number--></InsertChanNo>
</FaceSnapThermometry>
</Thermal>
<isSupportChangedUpload/><!--optional, xs: boolean, whether supports uploading status changes-->
<PanoramaCamera><!--optional, PTZ camera channel-->
```

```
<PTZCtrl>
  <InsertChanNo><!--optional, xs:integer, access channel number--></InsertChanNo>
</PTZCtrl>
</PanoramaCamera>
<GlobalCamera><!--optional, Panoramic camera channe-->
  <InsertChanNo><!--optional, xs:integer, access channel number--></InsertChanNo>
</GlobalCamera>
</ChannelInfo>
```

XML_ChannelInfoList

ChannelInfoList message in XML format

```
<ChannelInfoList>version="2.0" xmlns="http://www.isapi.org/ver20/XMLSchema">
  <ChannelInfo/><!--see details in the message of XML_ChannelInfo-->
</ChannelInfoList>
```

See Also

XML_ChannelInfo

XML_channelNameOverlay

channelNameOverlay message in XML format

```
<channelNameOverlay version="2.0" xmlns="http://www.isapi.org/ver20/XMLSchema">
  <enabled>
    <!--req, xs:boolean, whether to enable channel name overlay-->
  </enabled>
  <name>
    <!--req, xs:string, channel name-->
  </name>
  <positionX>
    <!--req, xs:integer, X-coordinate-->
  </positionX>
  <positionY>
    <!--req, xs:integer, Y-coordinate-->
  </positionY>
</channelNameOverlay>
```

XML_CheckInfo

CheckInfo message in XML format

```
<CheckInfo version="2.0" xmlns="http://www.isapi.org/ver20/XMLSchema">
  <randomString><!--req, xs: string, random string--></randomString>
  <verificationCode>
    <!--opt, xs: string, verification code to be sent, which will be encrypted by sha256, and then convert the
```

hexadecimal code to a string, finally it will be encoded by Base64. The format is first 32-byte of **randomString** + actual verification code-->
</verificationCode>
</CheckInfo>

XML_CMSearchDataPackage

CMSearchDataPackage message in XML format.

```
<CMSearchDataPackage version="2.0" xmlns="http://www.isapi.org/ver20/XMLSchema">
  <metald>
    <!--required, string type, metadata ID, its format is log.std-cgi.com+/major type+/minor type-->
  </metald>
  <timeSpanList>
    <timeSpan>
      <startTime><!--required, start time with ISO8601 time format, e.g., 2017-08-02T00:00:00Z--></startTime>
      <endTime><!--required, end time with ISO8601 time format, e.g., 2017-08-02T00:00:00Z--></endTime>
    </timeSpan>
  </timeSpanList>
</CMSearchDataPackage>
```

XML_CMSearchDataPackageResult

CMSearchDataPackageResult message in XML format.

```
<CMSearchDataPackageResult version="2.0" xmlns="http://www.isapi.org/ver20/XMLSchema">
  <url>
    <!--required, string type, file URL, e.g., http://ip:port//ISAPI/ContentMgmt/logSearch/dataPackage/data?xx-->
  </url>
</CMSearchDataPackageResult>
```

XML_CMSearchProfile

CMSearchProfile message in XML format

```
<CMSearchProfile version="1.0" xmlns="ISAPillianxce.org:resourcedescription">
  <searchProfile><!--req, xs: string, video search conditions--></searchProfile>
  <textSearch>
    <!--req, xs: string, text type to be searched, such as recording information text and metadata text-->
  </textSearch>
  <maxSearchTimespans>
    <!--req, xs: integer, maximum time period of results returned in each search-->
  </maxSearchTimespans>
  <maxSearchTracks><!--req, xs: integer, maximum tracks can be searched--></maxSearchtracks>
  <maxSearchSources>
    <!--req, xs: integer, maximum recording channels can be searched-->
  </maxSearchSources>
```

```
<maxSearchMetadatas>
  <!--req, xs: integer, maximum number of metadata can be searched-->
</maxSearchMetadatas>
<maxSearchMatchResults>
  <!--req, xs: integer, maximum search results can be returned-->
</maxSearchMatchResults>
<maxSearchTimeout><!--req, xs: integer, maximum search timeout--></maxSearchTimeout>
<maxConcurrentSearches>
  <!--req, xs: integer, maximum concurrent search operations-->
</maxConcurrentSearches>
</CMSearchProfile>
```

XML_VideoPic_CMSearchResult

XML message about search results of videos and pictures

```
<CMSearchResult version="2.0" xmlns="http://www.isapi.com/ver20/XMLSchema">
  <searchID><!--req, xs:string, search ID--></searchID>
  <responseStatus><!--req, xs:string--></responseStatus>
  <responseStatusStrg><!--req, xs:string--></responseStatusStrg>
  <numOfMatches><!--opt, xs:integer--></numOfMatches>
  <matchList>
    <searchMatchItem>
      <sourceID></sourceID>
      <trackID></trackID>
      <timeSpan>
        <startTime></startTime>
        <endTime></endTime>
      </timeSpan>
      <mediaSegmentDescriptor>
        <contentType><!--req, xs:string, file type, corresponding to search condition--></contentType>
        <codecType><!--req, xs:string, video encoding format--></codecType>
        <rateType><!--opt, xs:string, video bit rate,4Mbps--></rateType>
        <playbackURI><!--opt, xs:string, video file URL, "name=" and "size=" are file name and file size respectively, e.g.,
rtsp://10.65.130.168/Streaming/tracks/201/?
starttime=20190213T091134Z&amp;endtime=20190213T092116Z&amp;name=01010000093000100&amp;size=7323
35288--></playbackURI>
        <lockStatus><!--opt, xs:string, status of video file: lock-locked file, unlock-unlocked file--></lockStatus>
        <remainLockTime><!--dep, xs:integer, unit:s, remaining locking time, which is returned only when lockStatus is
set to lock--></remainLockTime>
        <name><!--opt, xs:string, file name--></name>
        <size><!--opt, xs:integer, file size--></size>
        <mediaID><!--opt, xs:integer, file ID--></mediaID>
        <remark><!--opt, xs:string, file tag--></remark>
        <GPSInfo><!--opt, xs:string, file GPS information--></GPSInfo>
        <recorderCode><!--opt, xs:string, body camera No.--></recorderCode>
        <uploadState><!--opt, xs:string, status: uploaded, nonUploaded, uploadFailed, uploading--></uploadState>
        <uploadTime><!--dep, xs:datetime, time of completing video uploading, e.g., "2004-05-03T17:30:08+08:00"--></
uploadTime>
        <policeCode><!--opt, xs:string, No. of the policeman who recorded the video--></policeCode>
        <policeName><!--opt, xs:string, name of the policeman who recorded the video--></policeName>
```

```
<shootingTime><!--opt, xs:datetime, captured or recorded time, e.g., "2004-05-03T17:30:08+08:00"--></shootingTime>
<FileTime><!--opt, xs:integer, time duration of the video file, unit:min--></FileTime>
<remarks><!--opt, xs:string, file remarks--></remarks>
<cloudStorageType><!--opt, xs:string, storage server type: 0-cloud storage, 1-cvr, 2-pstor, which should be consistent with cloudStorageType in /ISAPI/ContentMgmt/channels/<ID>/cloudStorage/<ID>--></cloudStorageType>
<fileUrl><!--opt, xs:string, file URL--></fileUrl>
<thumbnailUrl><!--opt, xs:string, thumbnail URL of picture or video--></thumbnailUrl>
<encryption>
  <enabled><!--opt, xs:boolean, whether to encrypt the file--></enabled>
  <encryptionkey><!--opt, xs:string, encryption key, which is returned only when enabled is set to true--></encryptionkey>
  <encryptionLevel><!--opt, xs:integer, encryption level: 1-AES128, 2-AES256, which is returned only when enabled is set to true--></encryptionLevel>
</encryption>
</mediaSegmentDescriptor>
<metadataMatches>
  <metadataDescriptor><!--searching conditions:
"evidence,illegalParking,crosslane,vehicleexist,lanechange,wrongdirection,congestion,turnround,pedestrian,constructi
on,roadBlock,abandonedObject,trafficAccident,fogDetection"--></metadataDescriptor>
</metadataMatches>
</searchMatchItem>
</matchList>
</CMSearchResult>
```

XML_CMSRecordProfile

CMSRecordProfile message in XML format

```
<CMRecordProfile version="2.0" xmlns="http://www.isapi.org/ver20/XMLSchema">
  <trackType><!--opt, xs: string, video track type--></trackType>
</CMRecordProfile>
```

XML_Color

XML message about color adjustment parameters

```
<Color version="2.0" xmlns="http://www.isapi.org/ver20/XMLSchema">
  <brightnessLevel><!--opt, xs:integer, brightness--></brightnessLevel>
  <contrastLevel><!--opt,xs:integer, contrast--></contrastLevel>
  <saturationLevel><!--opt,xs:integer, saturation--></saturationLevel>
  <hueLevel><!--opt, xs:integer, hue--></hueLevel>
  <grayScale>
    <grayScaleMode><!--opt,xs:string, gray scale mode: "indoor,outdoor"--></grayScaleMode>
  </grayScale>
  <nightMode>
    <!--opt, xs:boolean, enable night mode, when its value is "true", the saturation can be adjusted, otherwise, the saturation cannot be adjusted-->
  </nightMode>
</Color>
```

```
</nightMode>
</Color>
```

XML_corridor

corridor message in XML format

```
<corridor version="2.0" xmlns="http://www.std-cgi.org/ver20/XMLSchema">
  <enabled><!--req, xs: boolean, whether to enable rotate mode--></enabled>
  <mirror><!--opt, xs: string, mirror mode: "on,off"--></mirror>
</corridor>
```

XML_DateTimeOverlay

DateTimeOverlay message in XML format

```
<DateTimeOverlay version="2.0" xmlns="http://www.isapi.org/ver20/XMLSchema">
  <enabled>
    <!--req, xs:boolean, whether to enable date and time overlay-->
  </enabled>
  <positionX>
    <!--req, xs:integer, X-coordinate-->
  </positionX>
  <positionY>
    <!--req, xs:integer, Y-coordinate-->
  </positionY>
  <dateStyle>
    <!--opt, xs:string, date format: "YYYY-MM-DD, MM-DD-YYYY, DD-MM-YYYY, CHR-YYYY-MM-DD, CHR-MM-DD-YYYY,
CHR-DD-MM-YYYY"-->
  </dateStyle>
  <timeStyle>
    <!--opt, xs:string, "12hour, 24hour"-->
  </timeStyle>
  <displayWeek>
    <!--opt, xs:boolean, whether to display week information-->
  </displayWeek>
  <displayMilliSecond>
    <!--opt, xs:boolean, whether to display millisecond-->
  </displayMilliSecond>
</DateTimeOverlay>
```

XML_DDNS

DDNS message in XML format

```
<DDNS version="2.0" xmlns="http://www.isapi.org/ver20/XMLSchema">
  <id>
```



```
<!--req, xs:string-->
</id>
<enabled>
  <!--req, xs:boolean, "true,false"-->
</enabled>
<ethernetIfId>
  <!--opt, xs:string, ID. This node is used to link the DDNS to a network interface which is used by the DDNS client to
register-->
</ethernetIfId>
<provider>
  <!--req, xs:string, "IPServer, DynDNS, PeanutHall, NO-IP, HiDDNS..."-->
</provider>
<serverAddress><!--DDNS address. This node is required when <provider> is "IPServer"-->
  <addressingFormatType>
    <!--req, xs:string, method used to locate the NTP server: "ipaddress, hostname"-->
  </addressingFormatType>
  <hostname>
    <!--dep, xs:string-->
  </hostname>
  <ipAddress>
    <!--dep, xs:string-->
  </ipAddress>
  <ipv6Address>
    <!--dep, xs:string-->
  </ipv6Address>
</serverAddress>
<portNo>
  <!--opt, xs:integer, port No-->
</portNo>
<deviceDomainName>
  <!--dep, xs:string, domain name of the device-->
</deviceDomainName>
<username>
  <!--dep, xs:string-->
</userName>
<password>
  <!--wo, dep, xs:string-->
</password>
</DDNS>
```

XML_DDNSList

DDNSList message in XML format

```
<DDNSList version="2.0" xmlns="http://www.isapi.org/ver20/XMLSchema">
  <DDNS/><!--req, see details in the message of XML_DDNS-->
</DDNSList>
```

See Also

XML_DDNS

XML_DefaultParam

DefaultParam message in XML format

```
<DefaultParam version="1.0" xmlns="http://www.isapi.org/ver20/XMLSchema">
  <DialNum><!--opt, xs:string--></DialNum>
  <Username><!--opt, xs:string--></Username>
  <Password><!--opt, xs:string--></Password>
  <APNname><!--opt, xs:string--></APNname>
  <VerifyProto><!--req, xs:string, "auto, CHAP, PAP"--></VerifyProto>
</DefaultParam>
```

XML_Defog

Defog message in XML format

```
<Defog version="2.0" xmlns="http://www.isapi.org/ver20/XMLSchema">
  <enabled>
    <!--req, xs:boolean-->
  </enabled>
</Defog>
```

XML_De haze

Dehaze message in XML format

```
<Dehaze version="2.0" xmlns="http://www.isapi.org/ver20/XMLSchema">
  <DehazeMode><!--opt, xs: string, "open,close,auto"-->
  <DehazeLevel> <!--opt, xs: integer--></DehazeLevel>
</Dehaze>
```

XML_DeviceCap

XML message about device capability

```
<DeviceCap version="2.0" xmlns="http://www.isapi.org/ver20/XMLSchema">
  <SysCap><!--optional-->
    <isSupportDst><!--optional, xs: boolean, whether it supports daylight saving time--></isSupportDst>
    <NetworkCap><!--optional, xs: boolean, network capability-->
    <IOCap><!--optional, IO capability-->
    <SerialCap><!--optional, serial port capability-->
    <VideoCap><!--optional, video capability, see details in the message of XML_VideoCap-->
    <AudioCap><!--optional, audio capability-->
    <isSupportHoliday><!--optional, xs:boolean--></isSupportHoliday>
  <RebootConfigurationCap>
    <Genetec><!--optional, xs:boolean--></Genetec>
```

```
<ONVIF><!--optional, xs:boolean--></ONVIF>
<RTSP><!--optional, xs:boolean--></RTSP>
<HTTP><!--optional, xs:boolean--></HTTP>
<SADP>
  <ISDiscoveryMode><!--optional, xs:boolean--></ISDiscoveryMode>
  <PcapMode><!--optional, xs:boolean--></PcapMode>
</SADP>
<IPCAddStatus><!--optional, xs:boolean--></IPCAddStatus>
</RebootConfigurationCap>
<isSupportExternalDevice><!--optional, xs:boolean--></isSupportExternalDevice>
<isSupportChangedUpload>
  <!--optional, xs: boolean, whether it supports uploading status changes-->
</isSupportChangedUpload>
<isSupportGettingWorkingStatus>
  <!--optional, xs:boolean, whether it supports getting device status-->
</isSupportGettingWorkingStatus>
<isSupportGettingChannelInfoByCondition>
  <!--optional, xs:boolean-->
</isSupportGettingChannelInfoByCondition>
<isSupportDiagnosedDataParameter>
  <!--optional, xs:boolean-->
</isSupportDiagnosedDataParameter>
<isSupportSimpleDevStatus>
  <!--optional, xs: boolean, whether it supports getting device working status-->
</isSupportSimpleDevStatus>
<isSupportFlexible>
  <!--optional, xs: boolean, whether it supports getting channel status by condition-->
</isSupportFlexible>
<isSupportPTZChannels>
  <!--optional, xs:boolean, whether it supports returning PTZ channel (which is different from the video channel)-->
</isSupportPTZChannels>
<isSupportSubscribeEvent>
  <!--optional, xs:boolean, whether it supports alarm or event subscription: "true,false"-->
</isSupportSubscribeEvent>
<isSupportDiagnosedData>
  <!--optional, xs:boolean, "true,false", whether it supports diagnosis data-->
</isSupportDiagnosedData>
<isSupportTimeCap>
  <!--optional, xs:boolean, whether it supports time capability-->
</isSupportTimeCap>
<isSupportThermalStreamData>
  <!--optional, xs:boolean, whether it supports uploading thermal stream data in real-time. If it is supported, the
returned value is "true"; otherwise, this node will not be returned-->
</isSupportThermalStreamData>
<isSupportPostUpdateFirmware>
  <!--optional,xs:boolean,"true,false", whether it supports upgrading the firmware-->
</isSupportPostUpdateFirmware>
<isSupportPostConfigData>
  <!--optional, xs:boolean,"true,false", whether it supports importing or exporting the configuration file-->
</isSupportPostConfigData>
<isSupportUserLock>
  <!--optional, xs:boolean,"true,false", whether it supports locking user-->
```

```
</isSupportUserLock>
<isSupportModuleLock><!--optional, xs:boolean, whether it supports locking the module: "true,false"--></
isSupportModuleLock>
<isSupportSoundCfg><!--optional, xs:boolean--></isSupportSoundCfg>
<isSupportMetadata>
  <!--optional, xs:boolean, if it is supported, return "true", otherwise, this node will not be returned-->
</isSupportMetadata>
<isSupportShutdown><!--optional, xs:boolean, whether it supports shutdown configuration--></
isSupportShutdown>
  <supportSmartOverlapChannles opt="1"/><!--optional, xs:boolean, whether it supports stream configuration of
smart events. If this function is supported, this node and the corresponding channel ID will be returned; otherwise,
this node will not be returned-->
</SysCap>
<voicetalkNums><!--optional, xs:integer, the number of two-way audio channels--></voicetalkNums>
<isSupportSnapshot><!--optional, xs:boolean, whether it supports capture: "true, false"--></isSupportSnapshot>
<SecurityCap/><!--optional, security capability-->
<EventCap/><!--optional, event capability-->
<ITCCap><!--opt--></ITCCap>
<ImageCap/><!--optional, image capability-->
<RacmCap/><!--optional, storage capability-->
<PTZCtrlCap>
  <isSupportPatrols><!--optional, xs:boolean--></isSupportPatrols>
</PTZCtrlCap>
<SmartCap/><!--optional, intelligent capability-->
<isSupportEhome><!--optional, xs:boolean--></isSupportEhome>
<isSupportStreamingEncrypt><!--optional, xs:boolean--></isSupportStreamingEncrypt>
<TestCap>
  <isSupportEmailTest><!--optional, xs:boolean--></isSupportEmailTest>
</TestCap>
<ThermalCap/><!--optional, temperature measurement capability-->
<WLAlarmCap/><!--optional, wireless alarm capability-->
<SecurityCPCapabilities/><!--optional, security control panel capability-->
<isSupportGIS>
  <!--optional, xs:boolean, whether it supports GIS capability-->
</isSupportGIS>
<isSupportCompass>
  <!--optional, xs:boolean-->
</isSupportCompass>
<isSupportRoadInfoOverlays>
  <!--optional, xs:boolean-->
</isSupportRoadInfoOverlays>
<isSupportFaceCaptureStatistics>
  <!--optional, xs:boolean-->
</isSupportFaceCaptureStatistics>
<isSupportExternalDevice>
  <!--optional, xs:boolean-->
</isSupportExternalDevice>
<isSupportElectronicsEnlarge>
  <!--optional, xs:boolean, whether it supports digital zoom-->
</isSupportElectronicsEnlarge>
<isSupportRemoveStorage>
  <!--optional, xs:boolean-->
```

```
</isSupportRemoveStorage>
<isSupportCloud>
  <!--optional, xs:boolean-->
</isSupportCloud>
<isSupportRecordHost>
  <!--optional, xs:boolean-->
</isSupportRecordHost>
<isSupportEagleEye>
  <!--optional, xs:boolean, whether it supports PanoVu series camera-->
</isSupportEagleEye>
<isSupportPanorama>
  <!--optional, xs:boolean, whether it supports panorama-->
</isSupportPanorama>
<isSupportFirmwareVersionInfo>
  <!--optional, xs:boolean, whether it supports displaying firmware version information-->
</isSupportFirmwareVersionInfo>
<isSupportExternalWirelessServer>
  <!--optional, xs: boolean-->
</isSupportExternalWirelessServer>
<isSupportSetupCalibration>
  <!--optional, xs:boolean, whether it supports setting calibration-->
</isSupportSetupCalibration>
<isSupportGetmutexFuncErrMsg>
  <!--optional, xs:boolean, whether it supports getting mutex information-->
</isSupportGetmutexFuncErrMsg>
<isSupportTokenAuthenticate><!--optional, xs:boolean--></isSupportTokenAuthenticate>
<isSupportStreamDualVCA><!--optional, xs:boolean--></isSupportStreamDualVCA>
<isSupportlaserSpotManual>
  <!--optional, boolean, whether it supports laser spot configuration-->
</isSupportlaserSpotManual>
<isSupportRTMP><!--optional, xs:boolean--></isSupportRTMP>
<isSupportTraffic><!--optional, xs:boolean--></isSupportTraffic>
<isSupportLaserSpotAdjustment>
  <!--optional, boolean, whether it supports adjusting laser spot size-->
</isSupportLaserSpotAdjustment>
<VideoIntercomCap/><!--optional, video intercom capability-->
<isSupportSafetyCabin>
  <!--optional, xs:boolean-->
</isSupportSafetyCabin>
<isSupportPEA>
  <!--optional, xs:boolean, whether it supports one-touch security control panel capability-->
</isSupportPEA>
<isSupportCurrentLock>
  <!--optional, xs:boolean, whether it supports locking current configuration-->
</isSupportCurrentLock>
<isSupportGuardAgainstTheft>
  <!--optional, xs:boolean, whether it supports device anti-theft configuration-->
</isSupportGuardAgainstTheft>
<isSupportPicInfoOverlap>
  <!--optional, xs:boolean, whether it supports picture information overlay-->
</isSupportPicInfoOverlap>
<isSupportPlay>
```

```
<!--optional, xs: boolean, whether it supports live view: "true,false"-->
</isSupportPlay>
<isSupportPlayback>
  <!--optional, xs: boolean, whether it supports playback: "true,false"-->
</isSupportPlayback>
<UHFRFIDReader>
  <!--optional, supported capability of UHF RFID card reader-->
  <isSupportBasicInformation>
    <!--optional, xs:boolean, whether it supports basic parameters of UHF RFID card reader-->
  </isSupportBasicInformation>
  <isSupportHardDiskStorageTest>
    <!--optional, xs:boolean, whether it supports hard disk storage test of UHF RFID card reader-->
  </isSupportHardDiskStorageTest>
</UHFRFIDReader>
<isSupportIntelligentStructureAnalysis>
  <!--optional, xs:boolean, whether it supports structured VCA-->
</isSupportIntelligentStructureAnalysis>
<isSupportIntelligentAnalysisEngines>
  <!--optional, xs:boolean, whether it supports VCA engine configuration-->
</isSupportIntelligentAnalysisEngines>
<PreviewDisplayNum>
  <!--optional, xs:integer, the number of live view windows, which is the number of simultaneous live view windows
controlled by the device. Limited by the performance of DeepinMind series network video recorder, currently only live
view of a network camera is supported, and playback is not supported-->
</PreviewDisplayNum>
<isSupportBoard opt="true,false">
  <!--optional, xs:boolean, whether it supports protocol related to sub-board-->
</isSupportBoard>
<ResourceSwitch>
  <workMode opt="4KPreview,educationRecord">
    <!--req, xs:string, device working mode : "4KPreview"-4K live view mode, "educationRecord"-education recording
mode-->
  </workMode>
</ResourceSwitch>
<isSupportCustomStream><!--optional, xs:boolean--></isSupportCustomStream>
<isSupportTriggerCapCheck>
  <!--optional, xs:boolean, whether it supports verifying capability of alarm linkage actions-->
</isSupportTriggerCapCheck>
<isSupportActiveMulticast>
  <!--optional, xs: boolean, whether it supports active multicast-->
</isSupportActiveMulticast>
<isSupportChannelEventCap>
  <!--optional, xs:boolean, whether it supports getting event capability by channel-->
</isSupportChannelEventCap>
<isSupportPictureServer>
  <!-- opt, xs:boolean, whether it supports picture storage server-->
</isSupportPictureServer>
<isSupportVideoCompositeAlarm>
  <!--optional, xs:boolean, whether it supports video double check alarm-->
</isSupportVideoCompositeAlarm>
<isSupportSensorCalibrating>
  <!--optional, xs:boolean, whether it supports double sensor calibration-->
```

```
</isSupportSensorCalibrating>
<isSupportChannelEventListCap>
  <!--optional, xs:boolean, whether it supports getting event capability of all channels-->
</isSupportChannelEventListCap>
<VCAResourceChannelsCap>
  <!--optional, whether it supports independently switching to another VCA resource by channel-->
  <ChannelsList>
    <channelsID>
      <!--req, xs:integer, channel No. supported by the device-->
    </channelsID>
  </ChannelsList>
</VCAResourceChannelsCap>
<SensorCap/><!--optional, intelligent cabinet capability-->
<isSupportSecurityCP/>
  <!--optional, xs:boolean, whether it supports the applications of security control panel: "true, false"-->
</isSupportSecurityCP>
<isSupportClientProxyWEB>
  <!--optional, xs:boolean, whether it supports the function that the client proxy passes through the remote web
configuration: "true"-->
</isSupportClientProxyWEB>
<WEBLocation>
  <!--optional, string type, web page location: "local"-local device, "remote"-remote location. If this node is not
returned, the web page will be in the local device by default-->
</WEBLocation>
<isSupportTime/>
  <!--optional, xs:boolean, "true, false", whether it supports time configuration-->
</isSupportTime>
<isSupportTimeZone/>
  <!--optional, xs:boolean, "true, false", whether it supports daylight saving time (DST) configuration-->
</isSupportTimeZone>
<isSupportMixedTargetDetection>
  <!--optional, xs:boolean, "true, false", whether it supports multi-target-type detection-->
</isSupportMixedTargetDetection>
<isSupportFaceContrastMode>
  <!--optional, xs:boolean, whether it supports face picture comparison mode-->
</isSupportFaceContrastMode>
<isSupportPictureCaptureComparision>
  <!--optional, xs:boolean, whether it supports face picture N:1 comparison between face pictures captured by the
camera and imported face pictures-->
</isSupportPictureCaptureComparision>
<isSupportGPSCalibratation>
  <!--optional, xs:boolean, whether it supports GPS calibration capability-->
</isSupportGPSCalibratation>
<isSupportChannelFullEventListCap>
  <!--optional, xs:boolean, whether it supports getting event list capability of all channels-->
</isSupportChannelFullEventListCap>
<isSupportAUXInfoCap>
  <!--optional, xs:boolean, whether it supports getting property capability of all channels-->
</isSupportAUXInfoCap>
<isSupportCalibrationFile>
  <!--optional, xs:boolean, whether it supports importing calibration file-->
</isSupportCalibrationFile>
```

```
<isSupportDisplayTrajectory>
  <!--optional, xs:boolean, whether it supports displaying trajectory-->
</isSupportDisplayTrajectory>
<maximumSuperPositionTime opt="5,10,20,30">
  <!--dep,xs:integer, the maximum time of trajectory displaying, unit: second, it is valid only when displaying
trajectory is supported-->
</maximumSuperPositionTime>
<isSupportUnitConfig>
  <!--optional, xs:boolean, whether it supports unit configuration-->
</isSupportUnitConfig>
<isSupportAutoMaintenance>
  <!--optional, xs:boolean, whether it supports automatic maintenance. When this node exists and values "true", it
indicates support-->
</isSupportAutoMaintenance>
<isSupportGetLinkSocketIP>
  <!--optional, xs: boolean, "true,false", whether it supports getting the SocketIP of current connection-->
</isSupportGetLinkSocketIP>
<isSupportIntelligentSearch>
  <!--optional, xs:boolean, whether it supports intelligent search-->
</isSupportIntelligentSearch>
<IOTCap><!--optional, xs:boolean, IoT device access capability-->
  <supportChannelNum>
    <!--req, xs:integer, number of supported channels of IoT device-->
  </supportChannelNum>
  <startChannelNo>
    <!--optional, xs:integer, initial channel ID, if this node is not inputted, it indicates that the initial channel ID is 1-->
  </startChannelNo>
  <isSupportlinkageChannelsSearch>
    <!--optional, boolean, returns "true" if support, returns "false" if not support-->
  </isSupportlinkageChannelsSearch>
</IOTCap>
<isSupportEncryption>
  <!--optional, xs: boolean, stream encryption capability-->
</isSupportEncryption>
<AIDEventSupport opt="abandonedObject, pedestrian, congestion, roadBlock, construction, trafficAccident,
fogDetection, wrongDirection, illegalParking, SSharpDriving, lowSpeed, dragRacing">
  <!--optional, xs:string, supported traffic incident type: "abandonedObject"-objects dropped down, "pedestrian"-
pedestrian, "congestion"-congestion, "roadBlock"-roadblock, "construction"-construction, "trafficAccident"-traffic
accident, "fogDetection"-fog, "wrongDirection"-wrong-way driving, "illegalParking"-illegal parking, "SSharpDriving"-
slalom driving, "lowSpeed"-driving in low speed, "dragRacing"-street racing-->
</AIDEventSupport>
<TFSEventSupport
opt="illegalParking ,wrongDirection,crossLane,laneChange,vehicleExist,turnRound,parallelParking,notKeepDistance,not
SlowZebraCrossing,overtakeRightSide,lowSpeed,dragRacing,changeLaneContinuously,SSharpDriving,largeVehicleOccup
yLine,jamCrossLine">
  <!--optional, xs:string, supported enforcement event type: "illegalParking"-illegal parking, "wrongDirection"-wrong-
way driving, "crossLane"-driving on the lane line, "laneChange"-illegal lane change, "vehicleExist"-motor vehicle on
non-motor vehicle lane, "turnRound"-illegal U-turn, "parallelParking"-parallel parking, "notKeepDistance"-not keeping
vehicle distance, "notSlowZebraCrossing"-not slowing down at zebra crossing, "overtakeRightSide"-overtaking on the
right, "lowSpeed"-driving in low speed, "dragRacing"-street racing, "changeLaneContinuously"-continuous lane
change, "SSharpDriving"-slalom driving, "largeVehicleOccupLine"-lane occupation by large-sized vehicle,
"jamCrossLine"-queue jumping-->
```



```
</TFEventSupport>
<isVehicleStatisticsSupport>
  <!--optional, xs: boolean, whether it supports setting parameters for traffic data collection-->
</isVehicleStatisticsSupport>
<isSupportIntersectionAnalysis>
  <!--optional, xs: boolean, whether it supports intersection analysis-->
</isSupportIntersectionAnalysis>
<supportRemoteCtrl opt="up,down,left,right,enter,menu,num,power,esc,edit,F1,.prev,rec,play,stop,notSupport"/><!--
whether it supports remote control-->
<isSptDiagnosis>
  <!--optional, xs:boolean, whether it supports device diagnosis: "true", "false"-->
</isSptDiagnosis>
<isSptSerialLogCfg>
  <!--optional, xs:boolean, whether it supports configuring serial port log redirection: "true", "false"-->
</isSptSerialLogCfg>
<isSptFileExport>
  <!--optional, xs:boolean, whether it supports exporting files from the device: "true", "false"-->
</isSptFileExport>
<isSptCertificationStandard>
  <!--optional, xs:boolean, whether it supports configuring authentication standard for security control panel: "true",
"false"-->
</isSptCertificationStandard>
<isSptKeypadLock>
  <!--optional, xs:boolean, whether it supports locking keypad: "true", "false"-->
</isSptKeypadLock>
<MixedTargetDetection><!--optional, whether the device supports recognizing specific target among mixed targets-->
  <isSupportFaceRecognition><!--optional, xs:boolean, whether it supports face recognition--></
isSupportFaceRecognition>
  <isSupportHumanRecognition><!--optional, xs:boolean, whether it supports human body recognition--></
isSupportHumanRecognition>
  <isSupportVehicleRecognition><!--optional, xs:boolean, whether it supports vehicle recognition--></
isSupportVehicleRecognition>
</MixedTargetDetection>
<isSupportDiscoveryMode><!--optional, xs:boolean--></isSupportDiscoveryMode>
<streamEncryptionType>
  <!--dep, xs:string, stream encryption type: "RTP/TLS", "SRTP/UDP", "SRTP/MULTICAST". This node is valid when
<isSupportEncryption> is "true", and the device can support one or more stream encryption types-->
</streamEncryptionType>
<isSupportLms><!--optional, xs:boolean, whether it supports laser--></isSupportLms>
<isSupportLCDScreen><!--optional, xs:boolean, whether it supports LCD screen--></isSupportLCDScreen>
<isSupportBluetooth><!--optional, xs:boolean, whether it supports bluetooth--></isSupportBluetooth>
<isSupportAcsUpdate>
  <!--optional, whether it supports upgrading slave access control devices or peripheral modules: "true"-yes, this
node is not returned-no-->
</isSupportAcsUpdate>
<isSupportAccessControlCap>
  <!--optional, whether it supports access control capability: "true"-yes, this node is not returned-no-->
</isSupportAccessControlCap>
<isSupportIDCardInfoEvent><!--optional, whether it supports ID card swiping event: "true"-yes. This node will not be
returned if this function is not supported--></isSupportIDCardInfoEvent>
<OpenPlatformCap><!--optional, embedded open platform capability, refer to the message XML_OpenPlatformCap
for details-->
```

```
<isSupportInstallationAngleCalibration>
  <!--optional, xs:boolean, whether it supports installation angle calibration-->
</isSupportInstallationAngleCalibration>
<isSupportZeroBiasCalibration>
  <!--optional, xs:boolean, whether it supports zero bias calibration-->
</isSupportZeroBiasCalibration>
<isSupportDevStatus><!--optional, xs:boolean, whether device supports getting device status--></
isSupportDevStatus>
  <isSupportRadar><!--optional, xs:boolean, whether it supports the security radar--></isSupportRadar>
  <isSupportRadarChannels><!--optional, xs:boolean, whether it supports getting radar channels--></
isSupportRadarChannels>
  <radarIPDForm><!--optional, xs:string, radar form: "single"-single radar, "double_diagonal"-two radars forming an
180° diagonal, "double_vertical"-two radars forming a 90° vertical angle--></radarIPDForm>
  <isSupportRadarFieldDetection><!--optional, xs:boolean, whether it supports intrusion detection (radar)--></
isSupportRadarFieldDetection>
  <isSupportRadarLineDetection><!--optional, xs:boolean, whether it supports line crossing detection (radar)--></
isSupportRadarLineDetection>
  <mixedTargetDetectionWebNoDisplay><!--optional, xs:boolean, whether to enable not displaying multi-target-type
recognition--></mixedTargetDetectionWebNoDisplay>
  <SHMCap><!--opt-->
  <isSupportHighHDDTemperature><!--optional, xs:boolean, whether it supports HDD high temperature detection--></
isSupportHighHDDTemperature>
  <isSupportLowHDDTemperature><!--optional, xs:boolean, whether it supports HDD low temperature detection--></
isSupportLowHDDTemperature>
  <isSupportHDImpact><!--optional, xs:boolean, whether it supports HDD impact detection--></isSupportHDImpact>
  <isSupportHDBadBlock><!--optional, xs:boolean, whether it supports HDD bad sector detection--></
isSupportHDBadBlock>
  <isSupportSevereHDFailure><!--optional, xs:boolean, whether it supports HDD severe fault detection--></
isSupportSevereHDFailure>
  </SHMCap>
  <isSupportBVCorrect><!--optional, xs:boolean, whether it supports configuring camera correction parameters--></
isSupportBVCorrect>
  <guideEventSupport opt="linkageCapture">
    <!--optional,xs:string, events which support quick setup by instruction, "linkageCapture"-capture by linkage-->
  </guideEventSupport>
  <isSupportAutoSwitch><!--optional, xs:boolean, whether it supports auto switch--> true</isSupportAutoSwitch>
  <isSupportDataPrealarm><!--optional,xs:boolean, whether it supports traffic pre-alarm event--></
isSupportDataPrealarm>
  <supportGISEvent opt="AID,TPS,ANPR,mixedTargetDetection">
    <!--optional, xs:string, event types that support GIS information access: AID (corresponding SDK event:
COMM_ALARM_AID_V41), TPS (corresponding SDK event: COMM_ALARM_TPS_REAL_TIME), ANPR (corresponding
SDK event: COMM_ITS_PLATE_RESULT), mixedTargetDetection-mixed targets detection-->
  </supportGISEvent>
  <isSupportIntelligentMode><!--optional, xs:boolean, whether it supports intelligent scene switch (related URI:/ISAPI/
System/IntelligentSceneSwitch?format=json)--></isSupportIntelligentMode>
  <isSupportCertificateCaptureEvent><!--optional, xs:boolean, whether it supports certificate capture and comparison
events: true=yes. If this function is not supported, this node will not be returned--></
isSupportCertificateCaptureEvent>
  <isSupportAlgorithmsInfo><!--optional, xs:boolean, whether it supports getting the algorithm library version
information: true=yes. If this function is not supported, this node will not be returned--></isSupportAlgorithmsInfo>
  <isSupportVibrationDetection><!--optional, xs:boolean, whether it supports vibration detection--></
isSupportVibrationDetection>
```

```
<isSupportFaceTemperatureMeasurementEvent><!--optional, xs:boolean, whether it supports uploading face
thermography events (eventType: "FaceTemperatureMeasurementEvent")--></
isSupportFaceTemperatureMeasurementEvent>
<isSupportQRCodeEvent><!--optional, xs:boolean, whether it supports uploading QR code events (eventType:
"QRCodeEvent")--></isSupportQRCodeEvent>
<isSupportPersonArmingTrack><!--optional, xs:boolean, whether device supports person arming (related URI: /ISAPI/
Intelligent/channels/<ID>/personArmingTrack/capabilities?format=json)--></isSupportPersonArmingTrack>
<isSupportManualPersonArmingTrack><!--optional, xs:boolean, whether device supports manual person arming
(related URI: /ISAPI/Intelligent/channels/<ID>/manualPersonArmingTrack?format=json)--></
isSupportManualPersonArmingTrack>
<isSupportGPSCalibrationMode><!--optional, xs:boolean, whether device supports GPS calibration (related URI: /
ISAPI/System/GPSCalibration/channels/<ID>/mode?format=json)--></isSupportGPSCalibrationMode>
<isSupportGPSVerification><!--optional, xs:boolean, whether device supports GPS verification (related URI: /ISAPI/
System/GPSVerification/channels/<ID>/points?format=json)--></isSupportGPSVerification>
<isSupportHBDLib><!--optional, xs:boolean, whether device supports human body picture library (related URI: /ISAPI/
Intelligent/HBDLib/capabilities?format=json)--></isSupportHBDLib>
<isSupportFireEscapeDetection><!--optional, xs:boolean, whether the device supports fire engine access detection
(related URI: /ISAPI/Intelligent/channels/<ID>/fireEscapeDetection/capabilities?format=json)--></
isSupportFireEscapeDetection>
<isSupportTakingElevatorDetection><!--optional, xs:boolean, whether the device supports elevator detection
(related URI: /ISAPI/Intelligent/channels/<ID>/takingElevatorDetection/capabilities?format=json)--></
isSupportTakingElevatorDetection>
<isSupportSSDFileSystemUpgrade><!--optional, xs:boolean, whether the device supports SSD file system upgrade
(related URI: /ISAPI/System/SSDFileSystem/upgrade?format=json)--></isSupportSSDFileSystemUpgrade>
<isSupportSSDFileSystemFormat><!--optional, xs:boolean, whether the device supports SSD file system formatting
(related URI: /ISAPI/System/SSDFileSystem/format?format=json)--></isSupportSSDFileSystemFormat>
<isSupportSSDFileSystemCapacity><!--optional, xs:boolean, whether the device supports getting space distribution
information of SSD file system (related URI: /ISAPI/System/SSDFileSystem/capacity?format=json)--></
isSupportSSDFileSystemCapacity>
<isSupportAIOpenPlatform><!--optional, xs:boolean, whether the device supports AI open platform capabilities; if
supports, this node will be returned and its value is true; if not, this node will not be returned--></
isSupportAIOpenPlatform>
<isSupportPictureDownloadError><!--optional, xs:boolean, whether the device supports reporting picture download
failure--></isSupportPictureDownloadError>
</DeviceCap>
```

XML_DeviceInfo

XML message about device information

```
<DeviceInfo version="2.0" xmlns="http://www.isapi.org/ver20/XMLSchema">
  <deviceName><!--required, xs:string--></deviceName>
  <deviceId><!--required, read-only, xs:string, uuid--></deviceId>
  <deviceDescription>
    <!--optional, xs:string, description about the device defined in RFC1213. For network camera, this node is set to
    "IPCamera"; for network speed dome, this node is set to "IPDome"; for DVR or DVS, this node is set to "DVR" or
    "DVS"-->
  </deviceDescription>
  <deviceLocation><!--optional, xs:string, actual location of the device--></deviceLocation>
  <deviceStatus><!--optional, read-only, xs:string, device status: "normal", "abnormal"-->
  <DetailAbnormalStatus>
```

```
<!--dependent, error status details, it is valid only when deviceStatus is "abnormal"-->
<hardDiskFull>
  <!--optional, read-only, xs: boolean, whether the error of "HDD full" occurred: "true"-yes,"false"-no-->
</hardDiskFull>
<hardDiskError>
  <!--optional, read-only, xs:boolean, whether the error of "HDD error" occurred: "true"-yes,"false"-no-->
</hardDiskError>
<ethernetBroken>
  <!--optional, read-only, xs: boolean, whether the error of "network disconnected" occurred: "true"-yes,"false"-no-->
>
</ethernetBroken>
<ipaddrConflict>
  <!--optional, read-only, xs: boolean, whether the error of "IP address conflicted" occurred: "true"-yes,"false"-no-->
</ipaddrConflict>
<illegalAccess>
  <!--optional, read-only, xs: boolean, whether the error of "illegal login" occurred: "true"-yes,"false"-no-->
</illegalAccess>
<recordError>
  <!--optional, read-only, xs: boolean, whether the error of "recording exception" occurred: "true"-yes,"false"-no-->
</recordError>
<raidLogicDiskError>
  <!--optional, read-only, xs: boolean, whether the error of "RAID exception" occurred: "true"-yes,"false"-no-->
</raidLogicDiskError>
<spareWorkDeviceError>
  <!--optional, read-only, xs: boolean, whether the error of "working device exception" occurred: "true"-yes,"false"-no-->
</spareWorkDeviceError>
</DetailAbnormalStatus>
<systemContact><!--optional, xs:string, contact information of the device--></systemContact>
<model><!--required, read-only, xs:string--></model>
<serialNumber><!--required, read-only, xs:string--></serialNumber>
<macAddress><!--required, read-only, xs:string--></macAddress>
<firmwareVersion><!--required, read-only, xs:string--></firmwareVersion>
<firmwareReleasedDate><!--optional, read-only, xs:string--></firmwareReleasedDate>
<bootVersion><!--optional, read-only, xs:string--></bootVersion>
<bootReleasedDate><!--optional, read-only, xs:string--></bootReleasedDate>
<hardwareVersion><!--optional, read-only, xs:string--></hardwareVersion>
<encoderVersion><!--optional, read-only, xs:string--></encoderVersion>
<encoderReleasedDate><!--optional, read-only, xs:string--></encoderReleasedDate>
<decoderVersion><!--optional, read-only, xs:string--></decoderVersion>
<decoderReleasedDate><!--optional, read-only, xs:string--></decoderReleasedDate>
<softwareVersion><!--optional, read-only, xs:string, software version--></softwareVersion>
<capacity><!--optional, read-only, xs:integer, unit: MB, device capacity--></capacity>
<usedCapacity><!--optional, read-only, xs:integer, unit: MB, capacity usage--></usedCapacity>
<deviceType>
  <!--required, read-only, xs:string,device type: "IPCamera", "IPDome", "DVR", "HybirdNVR", "NVR", "DVS", "IPZoom",
  "CVR", "Radar", "PerimeterRadar"-perimeter radar, "ACS", "PHA"-Axiom hybrid security control panel-->
</deviceType>
<telecontrolID><!--optional, xs:integer, keyfob control ID, the value is between 1 and 255--></telecontrolID>
<supportBeep><!--optional, xs:boolean--></supportBeep>
<supportVideoLoss><!--optional, xs:boolean, whether it supports video loss detection--></supportVideoLoss>
<firmwareVersionInfo><!--optional, read-only, xs:string, firmware version information--></firmwareVersionInfo>
```

```
<actualFloorNum>
  <!--required, xs: integer, actual number of floors, which is between 1 and 128-->
</actualFloorNum>
<subChannelEnabled><!--optional, xs:boolean, whether to support sub-stream live view: "true"-yes, "false"-no--></subChannelEnabled>
<thrChannelEnabled><!--optional, xs:boolean, whether to support third stream live view: "true"-yes, "false"-no--></thrChannelEnabled>
<radarVersion><!--optional, xs:string, radar version--></radarVersion>
<cameraModuleVersion><!--read-only, xs:string, camera module version--></cameraModuleVersion>
<mainversion><!--optional, xs:integer, main version No. which is between 1 and 255--></mainversion>
<subversion><!--optional, xs:integer, sub version No. which is between 1 and 255--></subversion>
<upgradeversion><!--optional, xs:integer, upgraded version No. which is between 1 and 255--></upgradeversion>
<customizeversion><!--optional, xs:integer, customized version No. which is between 1 and 255--></customizeversion>
<companyName><!--optional, xs:string, the manufacturing company's abbreviation--></companyName>
<copyright><!--optional, xs:string, copyright information--></copyright>
<systemName><!--optional, xs:string, storage system name: "storageManagement"-storage management system, "distributedStorageManagement"-distributed storage management system--></systemName>
<systemStatus><!--optional, xs:string, system status: "configured"-configured, "unConfigured"-not configured--></systemStatus>
<isLeaderDevice><!--optional, xs:boolean, whether it is the corresponding device of the resource IP address--></isLeaderDevice>
<clusterVersion><!--dependent, xs:string, system cluster version. This node is valid when <isLeaderDevice> is true--></clusterVersion>
<manufacturer><!--optional, xs:string, manufacturer information: "hikvision"-Hikvision devices; for neutral devices, this node should be empty--></manufacturer>
<customizedInfo><!--optional, xs:string, order No. of the customization project. For baseline devices, this node is empty; for custom devices, the order No. of the customization project will be returned by this node--></customizedInfo>
<localZoneNum><!--optional, xs:integer, number of local zones--></localZoneNum>
<alarmOutNum><!--optional, xs:integer, number of alarm outputs--></alarmOutNum>
<distanceResolution><!--optional, xs:float, resolution of distance, unit: meter--></distanceResolution>
<angleResolution><!--optional, xs:float, resolution of angle, unit: degree--></angleResolution>
<speedResolution><!--optional, xs:float, resolution of speed, unit: m/s--></speedResolution>
<detectDistance><!--optional, xs:float, detection distance, unit: meter--></detectDistance>
<languageType opt="chinese,english,spanish,portuguese,italian,french,russian,turkish,greek,czech"><!--optional, xs:string--></languageType>
<relayNum><!--optional, xs:integer, number of local relays--></relayNum>
<electroLockNum><!--optional, xs:integer, number of local electronic locks--></electroLockNum>
<RS485Num><!--optional, xs:integer, number of local RS-485--></RS485Num>
<powerOnMode><!--optional, xs:string, device startup mode: "button"-press button to power on (default), "adapter"-connect adapter to power on--></powerOnMode>
<DockStation>
  <!--optional, dock station configuration-->
  <Platform>
    <!--optional, platform configuration-->
    <type><!--required, xs:string, platform type: none, 9533, 8618, ISAPI--></type>
    <ip><!--optional, xs:string, IP address --></ip>
    <port><!--optional, xs:integer, communication port--></port>
    <userName><!--required, xs:string, user name, which is used for the dock station to log in to platform--></userName>
    <password><!--required, xs:string, password, which is used for the dock station to log in to platform, it should be
```

```
encrypted--></password>
</Platform>
<centralStorageBackupEnabled><!--optional, xs:boolean, whether to enable central storage backup--></
centralStorageBackupEnabled>
</DockStation>
<webVersion><!--optional, read-only, xs:string, web version No.--></webVersion>
<deviceRFProgramVersion><!--optional, read-only, xs:string, version No. of the device's RF (Radio Frequency)
program--></deviceRFProgramVersion>
<securityModuleSerialNo><!--optional, read-only, xs:string, serial No. of the security module--></
securityModuleSerialNo>
<securityModuleVersion><!--optional, read-only, xs:string, version No. of the security module--></
securityModuleVersion>
<securityChipVersion><!--optional, read-only, xs:string, version No. of the security chip--></securityChipVersion>
<securityModuleKeyVersion><!--optional, read-only, xs:string, version No. of the security module key--></
securityModuleKeyVersion>
<UIDLampRecognition><!--optional, information of the UID lamp recognition device-->
<enabled><!--optional, xs:boolean, whether to enable--></enabled>
</UIDLampRecognition>
<bootTime><!--optional, xs:string, read-only, system booting time, ISO 8601 format; the maximum length is 32 bytes--
></bootTime>
</DeviceInfo>
```

XML_DeviceLanguage

DeviceLanguage message in XML format

```
<DeviceLanguage version="2.0" xmlns="http://www.isapi.org/ver20/XMLSchema">
<language>
<!--required, xs:string, language supported by the device: "SimChinese"-simplified Chinese, "TraChinese"-traditional
Chinese, "English", "Russian", "Bulgarian", "Hungarian", "Greek", "German", "Italian", "Czech", "Slovakia", "French",
"Polish", "Dutch", "Portuguese", "Spanish", "Romanian", "Turkish", "Japanese", "Danish", "Swedish", "Norwegian",
"Finnish", "Korean", "Thai", "Estonia", "Vietnamese", "Hebrew", "Latvian", "Arabic", "Sovenian"-Slovenian, "Croatian",
"Lithuanian", "Serbian", "BrazilianPortuguese"-Brazilian Portuguese, "Indonesian", "Ukrainian"-->
</language>
</DeviceLanguage>
```

XML_DeviceStatus

DeviceStatus message in XML format

```
<DeviceStatus version="2.0" xmlns="http://www.isapi.org/ver20/XMLSchema">
<currentDeviceTime><!--opt, xs: datetime--></currentDeviceTime>
<deviceUpTime><!--opt, xs: integer, seconds--></deviceUpTime>
<TemperatureList><!--opt-->
<Temperature>
<tempSensorDescription><!--req, xs: string--></tempSensorDescription>
<temperature><!--req, xs: float--></temperature>
</Temperature>
</TemperatureList>
```

```
<FanList><!--opt-->
  <Fan>
    <fanDescription><!--req, xs: string--></fanDescription>
    <speed><!--req, xs: integer--></speed>
  </Fan>
</FanList>
<PressureList><!--opt-->
  <Pressure>
    <pressureSensorDescription><!--req, xs: string--></pressureSensorDescription>
    <pressure><!--req, xs: integer--></pressure>
  </Pressure>
</PressureList>
<TamperList><!--opt-->
  <Tamper><tamperSensorDescription><!--req, xs: string--></tamperSensorDescription>
    <tamper><!--req, xs: boolean--> </tamper>
  </Tamper>
</TamperList>
<CPUList><!--opt-->
  <CPU>
    <cpuDescription><!--req, xs: string--></cpuDescription>
    <cpuUtilization><!--req, xs: integer, percentage, which is between 0 and 100--></cpuUtilization>
  </CPU>
</CPUList>
<MemoryList><!--opt-->
  <Memory>
    <memoryDescription><!--req, xs: string--></memoryDescription>
    <memoryUsage><!--req, xs: float, unit: MB--></memoryUsage>
    <memoryAvailable><!--req, xs: float, unit: MB--></memoryAvailable>
  </Memory>
</MemoryList>
<openFileHandles><!--opt, xs: integer--></openFileHandles>
<CameraList><!--opt-->
  <Camera>
    <zoomReverseTimes><!--req, xs: integer--></zoomReverseTimes>
    <zoomTotalSteps><!--req, xs: integer--></zoomTotalSteps>
    <focusReverseTimes><!--req, xs: integer--></focusReverseTimes>
    <focusTotalSteps><!--req, xs: integer--></focusTotalSteps>
    <irisShiftTimes><!--req, xs: integer--></irisShiftTimes>
    <irisTotalSteps><!--req, xs: integer--></irisTotalSteps>
    <icrShiftTimes><!--req, xs: integer--></icrShiftTimes>
    <icrTotalSteps><!--req, xs: integer--></icrTotalSteps>
    <lensIntirTimes><!--req, xs: integer--></lensIntirTimes>
    <cameraRunTotalTime><!--req, xs: integer--></cameraRunTotalTime>
  </Camera>
</CameraList>
<DomeInfoList><!--opt-->
  <DomeInfo>
    <domeRunTotalTime><!--opt, xs: integer--></domeRunTotalTime >
    <runTimeUnderNegativetwenty><!--opt, xs: integer--></runTimeUnderNegativetwenty>
    <runTimeBetweenNttwentyPforty><!--opt, xs: integer--></runTimeBetweenNttwentyPforty>
    <runtimeOverPositiveforty><!--opt, xs: integer--></runtimeOverPositiveforty>
    <panTotalRounds><!--opt, xs: integer--></panTotalRounds>
```

```
<tiltTotalRounds><!--opt, xs: integer--></tiltTotalRounds>
<heatState><!--opt, xs: integer--></heatState>
<fanState><!--opt, xs: integer--></fanState>
</DomelInfo>
</DomelInfoList>
<deviceStatus><!--req, xs: string, device working status: "normal, abnormal 等"--></deviceStatus>
<dialSignalStrength>
  <!--optional, xs:integer, 4G signal strength, it is between 1 and 3-->
</dialSignalStrength>
<USBStatusList>
  <USBStatus>
    <!--list-->
    <id>
      <!--required, xs:integer-->
    </id>
    <state>
      <!--optional, xs:string, USB status: "connected", "fullCapacity", "deviceException", "disconnected"-->
    </state>
  </USBStatus>
</USBStatusList>
<WifiStatusList>
  <WifiStatus>
    <!--list-->
    <id>
      <!--required, xs:integer-->
    </id>
    <state>
      <!--optional, xs:string, Wi-Fi status: "disabled"-Wi-Fi is disabled, "enable"-Wi-Fi is enabled, "disconnected"-Wi-Fi
is disconnected, "enableHotspot"-Wi-Fi hot spot is enabled-->
    </state>
  </WifiStatus>
</WifiStatusList>
<AlertStreamServerList><!--opt, user of arming device-->
  <AlertStreamServer><!--list-->
    <id><!--opt, xs:integer--></id>
    <protocolType><!--opt, xs:string, "SDK,ISAPI", arming protocol type--></protocolType>
    <ipAddress><!--dep, xs:string, IP address of platform, server, NVR/CVR, and so on--></ipAddress>
  </AlertStreamServer>
</AlertStreamServerList>
</DeviceStatus>
```

XML_DiagnosedDataParameter

XML message about parameters of exporting diagnose information

```
<?xml version="1.0" encoding="utf-8"?>
<DiagnosedDataParameter version="2.0" xmlns="http://www.isapi.org/ver20/XMLSchema">
  <enabled>
    <!--required, xs:boolean, whether to enable exporting diagnose information: true=yes, false=no-->
  </enabled>
  <HardwareInfo>
```



```
<!--optional, xs:boolean, opt="true,false", whether to enable exporting hardware information, such as running
information of motor, fan, heater, compressor, supplement light, wiper, and so on-->
</HardwareInfo>
<SoftwareInfo>
  <!--optional, xs:boolean, opt="true,false", whether to enable exporting software information, such as software
status, running log, system information-->
</SoftwareInfo>
<logInfo>
  <!--optional, xs:boolean, opt="true,false", whether to enable exporting startup log--></logInfo>
</DiagnosedDataParameter>
```

XML_Dial

Dial message in XML format

```
<Dial version="1.0" xmlns="http://www.isapi.org/ver20/XMLSchema">
  <enabled><!--req, xs:boolean--></enabled>
  <DialMethod><!--req, xs:string, "auto, manual"--></DialMethod>
  <SwitchMethod>
    <!--req, xs:string, "auto,4GFirst,3GFirst, manualto2G, manualto3G, manualto4G"-->
  </SwitchMethod>
  <OfflineTime><!--opt, xs:integer--></OfflineTime>
  <UIMCardNum><!--opt, xs:string--></UIMCardNum>
  <DialNum><!--opt, xs:string--></DialNum>
  <Username><!--opt, xs:string--></Username>
  <Password><!--opt, xs:string--></Password>
  <APNname><!--opt, xs:string--></APNname>
  <SIMNum><!--opt, xs:string, SIM card No. (mobile phone number)--></SIMNum>
  <MTU><!--opt, xs: integer--></MTU>
  <VerifyProto><!--req, xs:string, "auto, CHAP, PAP"--></VerifyProto>
  <DefaultParam/><!--opt, default parameters, see details in the message of XML_DefaultParam-->
  <netAPN><!--opt, xs:string, APN configuration of the private network--></netAPN>
  <Flow><!--opt, network traffic flow configuration-->
    <limitEnabled><!--opt, xs:boolean, whether to enable traffic flow limitation--></limitEnabled>
    <consumeFlow><!--opt, xs:float, unit: MB--></consumeFlow>
    <threshold><!--opt, xs:integer, unit: MB--></threshold>
  </Flow>
  <pinCode><!--opt, wo, xs:string--></pinCode>
  <enabled4G><!-- opt, xs:boolean, whether to enable 4G--></enabled4G>
  <enabledDNS><!-- opt, xs:boolean, whether to enable DNS manual configuration--></enabledDNS>
  <ISPName><!--optional, xs:string, ISP (Internet Service Provider) name--></ISPName>
  <IMEINo><!--optional, xs:string, IMEI code--></IMEINo>
  <ICCID><!--optional, xs:string, ICCID code--></ICCID>
  <netType><!--optional, xs:string, network type--></netType>
</Dial>
```

See Also

XML_DefaultParam

XML_Dialstatus

Dialstatus message in XML format

```
<Dialstatus version="2.0" xmlns="http://www.isapi.org/ver20/XMLSchema">
  <RealtimeMode>
    <!-- ro, xs:string, real time mode,
"CDMA1x,EVDO,HYBRID,GSM,GPRS,EDGE,WCDMA,HSDPA,HSUPA,HSPA,TDSCDMA,TD-LTE,FDD-LTE,LTE,UNKNOWN"-->
  </RealtimeMode>
  <UIMInfo>
    <!-- ro, xs:string, UIM cord status: "UNKNOWN"-unknown, "VALID"-valid, "NOVALID"-invalid, "ROAM"-roam,
"NOEXIST"-not exist, "WRONGPINCODE"-PIN code error,"NONETWORKSERVICE"-connectionless network service,
"ONLYSUPPORTEMERGENCYCALL"-the current network only supports emergency calls, "PINCODELOCKED"—PIN code
locked-->
  </UIMInfo>
  <SignalStrength><!-- ro, xs: integer, signal strength--></SignalStrength>
  <Dialstat><!-- ro, xs:string ,dial-up status: "disconnect"-disconnected, "dialing"-dialing, "success"-dialed, "shmError"-
dialing exception, "certificationFailure"-authentication failed, "invalidDialingCharacters"-invalid dialing characters,
"networkTimeout"-network timeout, "wirelessModemError"-wireless module error,"SIMError"-SIM card error,
"networkError"-network error, "unknown"-unknown error--></Dialstat>
  <IpAddress> <!-- req -->
    <IpAddress><!-- dep, xs:string--></IpAddress>
    <ipv6Address><!-- dep, xs:string --></ipv6Address>
  </IpAddress>
  <SubnetMask><!-- req -->
    <IpAddress><!-- dep, xs:string--></IpAddress>
    <ipv6Address><!-- dep, xs:string--></ipv6Address>
  </SubnetMask>
  <Gateway><!-- req -->
    <IpAddress><!-- dep, xs:string--></IpAddress>
    <ipv6Address><!-- dep, xs:string--></ipv6Address>
  </Gateway>
  <DNSAddress> <!-- req -->
    <IpAddress><!-- dep, xs:string--></IpAddress>
    <ipv6Address><!-- dep, xs:string--></ipv6Address>
  </DNSAddress>
  <subDNSAddress><!-- opt, sub DNS address-->
    <IpAddress><!-- dep, xs:string --></IpAddress>
    <ipv6Address><!-- dep, xs:string--></ipv6Address>
  </subDNSAddress>
  <operator><!-- ro, opt,xs:string,operator--></operator>
  <enableMessageConfig><!--opt, xs:boolean,whether to enable SMS service--></enableMessageConfig>
  <registrationStatus><!--ro,opt, xs:boolean, registration status--></registrationStatus>
</Dialstatus>
```

XML_Discovery

Discovery message in XML format

```
<Discovery version="2.0" xmlns="http://www.isapi.org/ver20/XMLSchema">
  <UPnP><!--req-->
    <enabled><!--req, xs:boolean, whether to enable UPnP: "true, false"--></enalbed>
  </UPnP>
  <ZeroConf><!--opt, channel-zero configuration-->
    <enabled><!--req, xs:boolean, "true, false"--></enabled>
  </ZeroConf>
</Discovery>
```

XML_DiscoveryMode

XML message about terminal SADP configuration

```
<DiscoveryMode xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0" >
  <ONVIFDiscoveryMode>
    <!--optional, xs:string, "discoverable,nonDiscoverable-->
  </ONVIFDiscoveryMode>
  <ISDiscoveryMode>
    <!-- optional, xs:string, "discoverable,nonDiscoverable-->
  </ISDiscoveryMode>
  <PcapMode>
    <!--optional, xs:string, "discoverable,nonDiscoverable-->
  </PcapMode>
  <SADP>
    <!--optional, xs:string, "discoverable,nonDiscoverable-->
  </SADP>
</DiscoveryMode>
```

XML_diskQuota

diskQuota message in XML format

```
<diskQuota version="2.0" xmlns="http://www.isapi.org/ver20/XMLSchema">
  <id><!--req, xs: integer, channel ID--></id>
  <useVideoQuota><!--read-only, xs: integer, quota for video files, unit: MB--></useVideoQuota>
  <usePictureQuota><!--read-only, xs: integer, quota for picture files, unit: MB--></usePictureQuota>
  <totalDiskVolume><!--read-only, xs: integer, total disk capacity, unit: MB--></totalDiskVolume>
  <videoQuota><!--req, xs: integer, unit: MB--></videoQuota>
  <pictureQuota><!--opt, xs: integer, unit: MB--></pictureQuota>
  <type><!--opt, xs: string, "volume, ratio, absent-ratio, otherwise-volume"--></type>
  <videoQuotaRatio><!--dep, xs: integer, video quota ratio (%): [0,100]--></videoQuotaRatio>
  <pictureQuotaRatio><!--dep, xs: integer, picture quota ratio (%): [0,100]--></pictureQuotaRatio>
  <totalVideoVolume><!--read-only, xs: integer, unit: MB--></totalVideoVolume>
  <totalPictureVolume><!--ro, xs: integer, unit: MB--></totalPictureVolume>
  <freeVideoQuota><!--ro, xs: integer, free space for video files, unit: MB--></freeVideoQuota>
  <freePictureQuota><!--ro, xs: integer, free space for picture files, unit: MB--></freePictureQuota>
  <reservedTimeCfg><!--optional, xs:integer, video saving time, value range: [0,1000], unit: day--></reservedTimeCfg>
</diskQuota>
```

XML_diskQuotaList

XML message about parameters of all HDD quotas

```
<diskQuotaList version="2.0" xmlns="http://www.isapi.org/ver20/XMLSchema">
  <diskQuota><!--optional, see details in the message XML_diskQuota--></diskQuota>
</diskQuotaList>
```

See Also

XML_diskQuota

XML_DoubleLensParkingCap

DoubleLensParkingCap capability message in XML format

```
<DoubleLensParkingCap version="2.0" xmlns="http://www.isapi.org/ver20/XMLSchema">
  <cameraType opt = "masterCamera, slaveCamera">
    <!--req, xs:string, dual-lens camera type: "masterCamera"-fixed dual-lens camera, "slaveCamera"-PTZ dual-lens
camera-->
  </cameraType>
</DoubleLensParkingCap>
```

XML_DownloadAbility

DownloadAbility message in XML format

```
<DownloadAbility version="2.0" xmlns="http://www.isapi.org/ver20/XMLSchema">
  <isSupportDownloadbyTime>
    <!--opt, xs:boolean, whether it supports download by time-->
  </isSupportDownloadbyTime>
  <isSupportDownloadbyFileName>
    <!--opt, xs:boolean, whether it supports download by file name -->
  </isSupportDownloadbyFileName>
  <isSupportDownloadToUSB>
    <!--opt, xs:boolean, whether it supports exporting files to devices via USB-->
  </isSupportDownloadToUSB>
</DownloadAbility>
```

XML_downloadRequest

XML message about downloading request

```
<downloadRequest version="1.0" xmlns="http://www.isapi.org/ver20/XMLSchema">
  <playbackURI>
    <!--required, xs:string, playback URL, returned by the search service. It contains the information of file name and
size, e.g., <playbackURI>rtsp://IpAddress/Streaming/tracks/101?starttime=2016-07-18
```

```
00:00:00&endtime=2016-07-18 00:25:04Z&name=00000000721000000&size=1065437356</playbackURI-->
</playbackURI>
<downloadCourseFile>
  <!--optional, xs:boolean, whether to download the course files-->
</downloadCourseFile>
</downloadRequest>
```

Remarks

When download file by name, the **playbackURI** inputs the file start time, end time, file name, and size; when download file by time, the **playbackURI** inputs the start time and end time.

XML_DynamicCap

DynamicCap capability message in XML format.

```
<DynamicCap version="2.0" xmlns="http://www.isapi.org/ver20/XMLSchema">
  <ResolutionAvailableDescriptorList><!--req-->
    <ResolutionAvailableDescriptor><!--req-->
      <videoResolutionWidth>
        <!--required, xs:integer-->
      </videoResolutionWidth>
      <videoResolutionHeight>
        <!--required, xs:integer-->
      </videoResolutionHeight>
      <supportedFrameRate>
        <!--required, xs:string, frame rate supported by this resolution, and its value is multiplied by 100 to be returned,
e.g., if the returned values are "2000, 2200, 2500", the actual frame rates are "20, 22, 25"-->
      </supportedFrameRate>
      <supportedBitRate min="" max=""/>
      <!--optional, xs:integer,in kbps-->
    </ResolutionAvailableDescriptor>
  </ResolutionAvailableDescriptorList>
  <CodecParamDescriptorList>
    <CodecParamDescriptor>
      <videoCodecType><!-- required, xs: string, "MPEG4,MJPEG,3GP,H.264,HK.264"--></videoCodecType>
      <isSupportProfile><!--dependent, xs: boolean--></isSupportProfile>
      <CBRCap><!--constant bit rate-->
        <isSupportSmooth><!--dependent, xs: boolean--></isSupportSmooth>
      </CBRCap>
      <VBRCap><!--variable bit rate-->
        <isSupportSmooth><!--dependent, xs: boolean--></isSupportSmooth>
      </VBRCap>
      <isSupportSVC><!--optional, xs: boolean--></isSupportSVC>
      <isSupportCABAC><!--optional, xs: boolean--></isSupportCABAC>
      <SmartCodecCap><!--optional-->
        <readOnlyParams opt="keyFrameInterval,Profile,SVC,fixedQuality">
          <!--optional, ro, xs: string, read-only options: I frame interval, encoding complexity, SVC, picture quality-->
        </readOnlyParams>
      </SmartCodecCap>
    </CodecParamDescriptor>
  </CodecParamDescriptorList>
  <BitrateType>
    <Constant><!--optional, constant bit rate-->
```

```

    <support opt="videoBitrate">
      <!--optional, xs: string, "averageVideoBitrate"-average bit rate, "videoBitrate"-upper-limit of bit rate-->
    </support>
    <hiddenAbility opt="averageVideoBitrate">
      <!--optional, xs:string, "averageVideoBitrate"-average bit rate, "videoBitrate"-upper-limit of bit rate-->
    </hiddenAbility>
  </Constant>
  <Variable><!--optional, variable bit rate-->
    <support opt="averageVideoBitrate">
      <!--optional, xs: string, "averageVideoBitrate"-average bit rate, "videoBitrate"-upper-limit of bit rate-->
    </support>
    <readOnlyAbility opt="videoBitrate">
      <!--optional, xs: string, "averageVideoBitrate"-average bit rate, "videoBitrate"-upper-limit of bit rate-->
    </readOnlyAbility>
  </Variable>
</BitrateType>
  <vbrAverageDefault><!--dependent, xs: integer, recommended average bit rate, unit: Kbps--></
vbrAverageDefault>
  <smart264EnabledPrompt opt="prompt1,prompt2,prompt3, prompt6, prompt25">
    <!--optional, wo, xs: string, audible prompt for Smart264 enabled-->
  </smart264EnabledPrompt>
  <smart265EnabledPrompt opt="prompt1,prompt2, prompt3, prompt25">
    <!--optional, wo, xs: string, audible prompt for Smart265 enabled-->
  </smart265EnabledPrompt>
</SmartCodecCap>
</CodecParamDescriptor>
</CodecParamDescriptorList>
<AudioDescriptorList>
  <AudioDescriptor>
    <audioCompressionType>
      <!--required, xs: string, audio encoding type: "G.711alaw,G.711ulaw,G.726,G.729,G.729a,G.
729b,PCM,MP3,AC3,AAC,ADPCM,MP2L2"-->
    </audioCompressionType>
    <audioItemList>
      <audioItem>
        <audioSamplingRate default=""><!--optional, xs: string--></audioSamplingRate>
        <audioBitRate opt=""><!--dependent, xs: integer--></audioBitRate>
        <noiseReduce opt="true,false"><!--required, xs: string--></noiseReduce>
      <audioItem>
    </audioItemList>
  </AudioDescriptor>
</AudioDescriptorList>
</DynamicCap>

```

XML_Ehome

Ehome message in XML format

```

<Ehome version="2.0" xmlns="http://www.isapi.org/ver20/XMLSchema">
  <enabled><!--opt, xs: boolean--></enabled>
  <addressingFormatType>

```

```
<!--req, xs: string, address format type: "ipaddress"-IP address, "hostname"-host name-->
</addressingFormatType>
<hostName><!--dep, xs: string, host name, this node is valid only when <addressingFormatType> is "hostname"--></hostName>
<ipAddress><!--dep, xs: string, IP address, this node is valid only when <addressingFormatType> is "ipaddress"--></ipAddress>
<ipv6Address><!--dep, xs: string, IPv6 address, this node is valid only when <addressingFormatType> is "ipaddress"--></ipv6Address>
<portNo><!--opt, xs: integer, port No.--></portNo>
<deviceId><!--req, xs: string, device ID--></deviceId>
<registerStatus><!--ro, xs: boolean, registration status--></registerStatus>
<key><!--opt, xs: string, encryption key of ISUP version 5.0, which is required for ISUP version 50--></key>
<version><!--ro, xs: string, version No.--></version>
<netWork><!--opt, xs:integer, 0-make no sense, 1-automatic, 2-wired network preferred, 3-wired network, 4-3G/4G/GPRS--></netWork>
<voiceDeviceType><!--opt, xs:string, two-way audio device type: "bluetooth"-bluetooth device, "client", "local"--></voiceDeviceType>
<protocolVersion><!--opt, xs:string, protocol version: "v2.0,v2.6,v4.0,v5.0". If this node is set to "v2.0", the device can only use protocol v2.0 to register; if this node is set to "v2.6", "v4.0" or "v5.0", the device will firstly use this protocol to register. If this node is not returned, the protocol version will be determined by <version--></protocolVersion>
</Ehome>
```

XML_EPTZ

EPTZ message in XML format

```
<EPTZ version="2.0" xmlns="http://www.isapi.org/ver20/XMLSchema">
  <enabled><!--req, xs:string, whether to enable e-PTZ: "true"-yes, "false"-no--></enabled>
  <streamType><!--opt, xs:integer, stream type: 1-main stream, 2-sub-stream, 3-third stream, 4-fourth stream--></streamType>
</EPTZ>
```

XML_EventCap

EventCap capability message in XML format

```
<EventCap version="2.0" xmlns="http://www.isapi.org/ver20/XMLSchema">
  <isSupportHDFull><!--optional, xs:boolean, "true"-support, "false"-not support--></isSupportHDFull>
  <isSupportHDError><!--optional, xs:boolean, "true"-support, "false"-not support--></isSupportHDError>
  <isSupportNicBroken><!--optional, xs:boolean, "true"-support, "false"-not support--></isSupportNicBroken>
  <isSupportIpConflict><!--optional, xs:boolean, "true"-support, "false"-not support--></isSupportIpConflict>
  <isSupportIllaAccess><!--optional, xs:boolean, "true"-support, "false"-not support--></isSupportIllaAccess>
  <isSupportViException><!--optional, xs:boolean, "true"-support, "false"-not support--></isSupportViException>
  <isSupportViMismatch><!--optional, xs:boolean, "true"-support, "false"-not support--></isSupportViMismatch>
  <isSupportRecordException><!--optional, xs:boolean, "true"-support, "false"-not support--></isSupportRecordException>
  <isSupportTriggerFocus><!--optional, xs:boolean, "true"-support, "false"-not support--></isSupportTriggerFocus>
  <isSupportMotionDetection><!--optional, xs:boolean, "true"-support, "false"-not support--></isSupportMotionDetection>
```

```
<isSupportVideoLoss><!--optional, xs:boolean, "true"-support, "false"-not support--></isSupportVideoLoss>
<isSupportTamperDetection><!--optional, xs:boolean, "true"-support, "false"-not support--></
isSupportTamperDetection>
<isSupportStudentsStoodUp><!--optional, xs:boolean, "true"-support, "false"-not support--></
isSupportStudentsStoodUp>
<isSupportFramesPeopleCounting><!--optional, xs:boolean, "true"-support, "false"-not support--></
isSupportFramesPeopleCounting>
<isSupportRaidException><!--optional, xs:boolean, "true"-support, "false"-not support--></isSupportRaidException>
<isSupportSpareException><!--optional, xs:boolean, "true"-support, "false"-not support--></
isSupportSpareException>
<isSupportPoePowerException><!--optional, xs:boolean, "true"-support, "false"-not support--></
isSupportPoePowerException>
<isSupportRegionEntrance><!--optional, xs:boolean, "true"-support, "false"-not support--></
isSupportRegionEntrance>
<isSupportRegionExiting><!--optional, xs:boolean, "true"-support, "false"-not support--></isSupportRegionExiting>
<isSupportLoitering><!--optional, xs:boolean, "true"-support, "false"-not support--></isSupportLoitering>
<isSupportGroup><!--optional, xs:boolean, "true"-support, "false"-not support--></isSupportGroup>
<isSupportRapidMove><!--optional, xs:boolean, "true"-support, "false"-not support--></isSupportRapidMove>
<isSupportFireDetection><!--optional, xs:boolean, "true"-support, "false"-not support--></isSupportFireDetection>
<isSupportIllegalParking><!--optional, xs:boolean, whether it supports illegal parking detection: "true"-support,
"false"-not support--></isSupportIllegalParking>
<isSupportUnattendedBaggage><!--optional, xs:boolean --></isSupportUnattendedBaggage>
<isSupportAttendedBaggage><!--optional, xs:boolean, "true"-support, "false"-not support--></
isSupportAttendedBaggage>
<isSupportHumanAttribute><!--optional, xs:boolean, "true"-support, "false"-not support--></
isSupportHumanAttribute>
<isSupportFaceContrast><!--optional, xs:boolean, "true"-support, "false"-not support--></isSupportFaceContrast>
<isSupportFaceLib><!--optional, xs:boolean, "true"-support, "false"-not support--></isSupportFaceLib>
<isSupportWhiteListFaceContrast><!--optional, xs:boolean, "true"-support, "false"-not support--></
isSupportWhiteListFaceContrast>
<isSupportBlackListFaceContrast><!--optional, xs:boolean, whether it supports blacklist face comparison: "true"-
support, "false"-not support--></isSupportBlackListFaceContrast>
<isSupportHumanRecognition><!--optional, xs:boolean, "true"-support, "false"-not support--></
isSupportHumanRecognition>
<isSupportFaceSnap><!--optional, xs:boolean, "true"-support, "false"-not support--></isSupportFaceSnap>
<isSupportPersonDensityDetection><!--optional, xs:boolean, "true"-support, "false"-not support--></
isSupportPersonDensityDetection>
<isSupportMixedTargetDetection><!--optional, xs:boolean, whether it supports multi-target-type detection alarm:
"true"-support, "false"-not support--></isSupportMixedTargetDetection>
<isSupportPedestrian><!--optional, xs:boolean, whether it supports pedestrian detection: "true"-support, "false"-not
support--></isSupportPedestrian>
<isSupportTrafficAccident><!--optional, xs:boolean, whether it supports traffic accident detection: "true"-support,
"false"-not support--></isSupportTrafficAccident>
<isSupportConstruction><!--optional, xs:boolean, whether it supports construction detection: "true"-support, "false"-
not support--></isSupportConstruction>
<isSupportRoadBlock><!--optional, xs:boolean, whether it supports roadblock detection: "true"-support, "false"-not
support--></isSupportRoadBlock>
<isSupportAbandonedObject><!--optional, xs:boolean, whether it supports thrown object detection: "true"-support,
"false"-not support--></isSupportAbandonedObject>
<isSupportParallelParking><!--optional, xs:boolean, whether it supports parallel parking detection: "true"-support,
"false"-not support--></isSupportParallelParking>
<isSupportParkingState><!--optional, xs:boolean, whether it supports parking space status detection: "true"-support,
```



```
"false"-not support, currently this node is not supported--></isSupportParkingState>
  <isSupportCongestion><!--optional, xs:boolean, whether it supports congestion detection: "true"-support, "false"-
not support--></isSupportCongestion>
  <isSupportVehicleStatistics><!--optional, xs:boolean, whether it supports data collection: "true"-support, "false"-not
support--></isSupportVehicleStatistics>
  <isSupportWrongDirection><!--optional, xs:boolean, whether it supports wrong-way driving detection: "true"-
support, "false"-not support--></isSupportWrongDirection>
  <isSupportTrunRound><!--optional, xs:boolean, whether it supports U-turning detection: "true"-support, "false"-not
support--></isSupportTrunRound>
  <isSupportCrossLane><!--optional, xs:boolean, whether it supports driving on the lane line detection: "true"-support,
"false"-not support--></isSupportCrossLane>
  <isSupportLaneChange><!--optional, xs:boolean, whether it supports illegal lane change detection: "true"-support,
"false"-not support--></isSupportLaneChange>
  <isSupportVehicleExist><!--optional, xs:boolean, whether it supports motor vehicle on non-motor vehicle lane
detection: "true"-support, "false"-not support--></isSupportVehicleExist>
  <isSupportFogDetection><!--optional, xs:boolean, whether it supports fog detection: "true"-support, "false"-not
support--></isSupportFogDetection>
  <isSupportIntersectionAnalysis><!--optional, xs: boolean, whether it supports configuring intersection analysis alarm:
"true"-support, "false"-not support--></isSupportIntersectionAnalysis>
  <isSupportVoltageInstable><!--optional,xs:boolean, whether it supports supply voltage exception alarm: "true"-
support, "false"-not support--></isSupportVoltageInstable>
  <isSupportSafetyHelmetDetection><!--optional, xs:boolean, whether it supports hard hat detection: "true"-support,
"false"-not support--></isSupportSafetyHelmetDetection>
  <isSupportCertificateRevocation><!--optional, xs:boolean, whether it supports certificate expiry alarm--></
isSupportCertificateRevocation>
  <isSupportNoMaskDetection><!--optional, xs:boolean, whether device supports no wearing mask detection--></
isSupportNoMaskDetection>
  <isSupportTMPA><!--optional, xs:boolean, whether device supports temperature measurement pre-alarm--></
isSupportTMPA>
  <RuleScheduleCap><!--optional, capability of setting arming schedule by rule-->
    <isSupportCityManagement>
      <!--optional, xs:boolean, whether the device supports setting arming schedule by rule for intelligent city
management; if supports, the value is true, otherwise, this node will not be returned-->
    </isSupportCityManagement>
  </RuleScheduleCap>
</EventCap>
```

XML_EventNotificationAlert_AlarmEventInfo

EventNotificationAlert message with alarm/event information in XML format.

```
<EventNotificationAlert version="2.0" xmlns="http://www.isapi.org/ver20/XMLSchema">
  <ipAddress><!--dep, xs:string, device IPv4 address--></ipAddress>
  <ipv6Address><!--dep, xs:string, device IPv6 address--></ipv6Address>
  <portNo><!--opt, xs:integer, device port number--></portNo>
  <protocol><!--opt, xs:string, protocol type for uploading alarm/event information, "HTTP,HTTPS"--></protocol>
  <macAddress><!--opt, xs:string, MAC address--></macAddress>
  <channelID><!--dep, xs:string, device channel No., starts from 1--></channelID>
  <dateTime><!--req, alarm/event triggered or occurred time, format: 2017-07-19T10:06:41+08:00--></dateTime>
  <activePostCount><!--req, xs:integer, alarm/event frequency, starts from 1--></activePostCount>
  <eventType><!--req, xs:string, alarm/event type, "peopleCounting, ANPR,..."--></eventType>
```

```
<eventState>
  <!--req, xs:string, durative alarm/event status: "active"-valid, "inactive"-invalid, e.g., when a moving target is
  detected,
    the alarm/event information will be uploaded continuously unit the status is set to "inactive"-->
</eventState>
<eventDescription><!--req, xs:string, alarm/event description--></eventDescription>
<...><!--opt, for different alarm/event types, the nodes are different, see the message examples in different
applications--></...>
</EventNotificationAlert>
```

XML_EventNotificationAlert_AlarmInputAlarmMsg

XML Message about Alarm Input Alarm Details

```
<?xml version="1.0" encoding="utf-8"?>
<EventNotificationAlert version="2.0" xmlns="http://www.isapi.org/ver20/XMLSchema">
  <ipAddress><!--dependent, xs:string, IPv4 address--></ipAddress>
  <ipv6Address><!--dependent, xs:string, IPv6 address--></ipv6Address>
  <portNo><!--optional, xs:integer, device's communication port No.--></portNo>
  <protocolType><!--optional, xs:string, transport protocol type: HTTP, HTTPS, EHome--></protocolType>
  <macAddress><!--optional, xs:string, MAC address--></macAddress>
  <dynChannelID><!--optional, xs:string, digital channel No.--></dynChannelID>
  <channelID><!--dependent, xs:string, channel No.--></channelID>
  <dateTime>
    <!--required, xs:datetime, event occurred time, which is in IOS 8601 format, e.g., 2017-04-22T15:39:01+08:00-->
  </dateTime>
  <activePostCount><!--required, xs:integer, event occurred times--></activePostCount>
  <eventType><!--required, xs:string, event types, here it should be set to "IO"--></eventType>
  <eventState><!--required, xs:string, event status (for persistent event): active, inactive--></eventState>
  <eventDescription><!--required, xs:string, event description--></eventDescription>
  <inputIOPortID><!--optional, xs:integer, local alarm input ID--></inputIOPortID>
  <dynInputIOPortID><!--optional, xs:integer, dynamic alarm input ID--></dynInputIOPortID>
  <channelName><!--optional, xs:string, channel name--></channelName>
  <deviceId><!--optional, xs:string, device ID--></deviceId>
  <Extensions>
    <serialNumber><!--required, xs:string, serial No.--></serialNumber>
    <eventPush><!--required, xs:string, event or alarm notification details--></eventPush>
  </Extensions>
</EventNotificationAlert>
```

XML_EventNotificationAlert_AudioExceptionAlarmMsg

XML Message about Audio Exception Alarm Details

```
<?xml version="1.0" encoding="utf-8"?>
<EventNotificationAlert version="2.0" xmlns="http://www.isapi.org/ver20/XMLSchema">
  <ipAddress><!--dependent, xs:string, IPv4 address--></ipAddress>
  <ipv6Address><!--dependent, xs:string, IPv6 address--></ipv6Address>
  <portNo><!--optional, xs:integer, device's communication port No.--></portNo>
```

```
<protocolType><!--optional, xs:string, transport protocol type: HTTP, HTTPS, EHome--></protocolType>
<macAddress><!--optional, xs:string, MAC address--></macAddress>
<dynChannelID><!--optional, xs:string, digital channel No.--></dynChannelID>
<channelID><!--dependent, xs:string, channel No.--></channelID>
<dateTime>
  <!--required, xs:datetime, event occurred time, which is in IOS 8601 format, e.g., 2017-04-22T15:39:01+08:00-->
</dateTime>
<activePostCount><!--required, xs:integer, event occurred times--></activePostCount>
<eventType><!--required, xs:string, event types, here it should be set to "audioexception"--></eventType>
<eventState><!--required, xs:string, event status (for persistent event): active, inactive--></eventState>
<eventDescription><!--required, xs:string, event description--></eventDescription>
<channelName><!--optional, xs:string, channel name--></channelName>
<deviceId><!--optional, xs:string, device ID--></deviceId>
<Extensions>
  <serialNumber><!--required, xs:string, serial No.--></serialNumber>
  <eventPush><!--required, xs:string, event or alarm notification details--></eventPush>
</Extensions>
</EventNotificationAlert>
```

XML_EventNotificationAlert_DefocusDetectionAlarmMsg

XML Message about Alarm Details of Defocus Detection

```
<?xml version="1.0" encoding="utf-8"?>
<EventNotificationAlert version="2.0" xmlns="http://www.isapi.org/ver20/XMLSchema">
  <ipAddress><!--dependent, xs:string, IPv4 address--></ipAddress>
  <ipv6Address><!--dependent, xs:string, IPv6 address--></ipv6Address>
  <portNo><!--optional, xs:integer, device's communication port No.--></portNo>
  <protocolType><!--optional, xs:string, transport protocol type: HTTP, HTTPS, EHome--></protocolType>
  <macAddress><!--optional, xs:string, MAC address--></macAddress>
  <dynChannelID><!--optional, xs:string, digital channel No.--></dynChannelID>
  <channelID><!--dependent, xs:string, channel No.--></channelID>
  <dateTime>
    <!--required, xs:datetime, event occurred time, which is in IOS 8601 format, e.g., 2017-04-22T15:39:01+08:00-->
  </dateTime>
  <activePostCount><!--required, xs:integer, event occurred times--></activePostCount>
  <eventType><!--required, xs:string, event types, here it should be set to "defocus"--></eventType>
  <eventState><!--required, xs:string, event status (for persistent event): active, inactive--></eventState>
  <eventDescription><!--required, xs:string, event description--></eventDescription>
  <channelName><!--optional, xs:string, channel name--></channelName>
  <deviceId><!--optional, xs:string, device ID--></deviceId>
  <Extensions>
    <serialNumber><!--required, xs:string, serial No.--></serialNumber>
    <eventPush><!--required, xs:string, event or event notification details--></eventPush>
  </Extensions>
</EventNotificationAlert>
```

XML_EventNotificationAlert_HDDErrorAlarmMsg

XML message about HDD error alarm details

```
<?xml version="1.0" encoding="utf-8"?>
<EventNotificationAlert version="2.0" xmlns="http://www.isapi.org/ver20/XMLSchema">
  <ipAddress><!--dependent, xs:string, IPv4 address--></ipAddress>
  <ipv6Address><!--dependent, xs:string, IPv6 address--></ipv6Address>
  <portNo><!--optional, xs:integer, device's communication port No.--></portNo>
  <protocolType><!--optional, xs:string, transport protocol type: HTTP, HTTPS, EHome--></protocolType>
  <macAddress><!--optional, xs:string, MAC address--></macAddress>
  <dynChannelID><!--optional, xs:string, digital channel No.--></dynChannelID>
  <channelID><!--dependent, xs:string, channel No.--></channelID>
  <dateTime>
    <!--required, xs:datetime, event occurred time, which is in IOS 8601 format, e.g., 2017-04-22T15:39:01+08:00-->
  >
  </dateTime>
  <activePostCount><!--required, xs:integer, event occurred times--></activePostCount>
  <eventType><!--required, xs:string, event types, here it should be set to "diskerror"--></eventType>
  <eventState><!--required, xs:string, event status (for persistent event): active, inactive--></eventState>
  <eventDescription><!--required, xs:string, event description--></eventDescription>
  <channelName><!--optional, xs:string, channel name--></channelName>
  <deviceId><!--optional, xs:string, device ID--></deviceId>
  <diskNo><!--optional, xs:integer, HDD No.--></diskNo>
  <HDDList><!--dependent, xs:integer, HDD list, this node is valid when <eventType> is "diskfull, diskerror"-->
    <HDD>
      <id><!--required, xs:integer, index No., which starts from 1--></id>
      <diskNumber><!--required, xs:integer, HDD ID--></diskNumber>
    </HDD>
  </HDDList>
</EventNotificationAlert>
```

XML_EventNotificationAlert_HeartbeatInfo

EventNotificationAlert message with heartbeat information (when there is no alarm is triggered) in XML format

```
<EventNotificationAlert version="2.0" xmlns="http://www.isapi.org/ver20/XMLSchema">
  <ipv6Address><!--dep, xs:string, device IPv6 address--></ipv6Address>
  <portNo><!--opt, xs:integer, device port number--></portNo>
  <protocol><!--opt, xs:string, protocol type for uploading alarm/event information, "HTTP,HTTPS"--></protocol>
  <macAddress><!--opt, xs:string, MAC address--></macAddress>
  <channelID><!--dep, xs:string, device channel No., starts from 1--></channelID>
  <dateTime><!--req, heartbeat uploaded time, format: 2017-07-19T10:06:41+08:00--></dateTime>
  <activePostCount><!--req, xs:integer, heartbeat frequency, starts from 1--></activePostCount>
  <eventType><!--req, xs:string, for heartbeat, it is "videoloss"--></eventType>
  <eventState>
    <!--req, xs:string, for heartbeat, it is "inactive"-->
  </eventState>
```

```
<eventDescription><!--req, xs: string, description--></eventDescription>
</EventNotificationAlert>
```

Remarks

- For network camera or network speed dome with the version 5.5.0 and lower, the heartbeat frequency is 300 ms per heartbeat.
- For network camera or network speed dome with the version 5.5.0 and higher, the heartbeat frequency is 10 s per heartbeat. If no heartbeat received for continuous 30 s, it indicates that the heartbeat is timed out.

Example

Message Example of Heartbeat

```
<EventNotificationAlert version="2.0" xmlns="http://www.isapi.com/ver20/XMLSchema">
  <ipAddress>10.17.133.46</ipAddress>
  <portNo>80</portNo>
  <protocol>HTTP</protocol>
  <macAddress>44:19:b6:6d:24:85</macAddress>
  <channelID>1</channelID>
  <dateTime>2017-05-04T11:20:02+08:00</dateTime>
  <activePostCount>0</activePostCount>
  <eventType>videoloss</eventType>
  <eventState>inactive</eventState>
  <eventDescription>videoloss alarm</eventDescription>
</EventNotificationAlert>
```

XML_EventNotificationAlert_IllegalLoginAlarmMsg

XML message about illegal login alarm details

```
<?xml version="1.0" encoding="utf-8"?>
<EventNotificationAlert version="2.0" xmlns="http://www.isapi.org/ver20/XMLSchema">
  <ipAddress><!--dependent, xs:string, IPv4 address--></ipAddress>
  <ipv6Address><!--dependent, xs:string, IPv6 address--></ipv6Address>
  <portNo><!--optional, xs:integer, device's communication port No.--></portNo>
  <protocolType><!--optional, xs:string, transport protocol type: HTTP, HTTPS, EHome--></protocolType>
  <macAddress><!--optional, xs:string, MAC address--></macAddress>
  <dynChannelID><!--optional, xs:string, digital channel No.--></dynChannelID>
  <channelID><!--dependent, xs:string, channel No.--></channelID>
  <dateTime>
    <!--required, xs:datetime, event occurred time, which is in IOS 8601 format, e.g., 2017-04-22T15:39:01+08:00-->
  >
  </dateTime>
  <activePostCount><!--required, xs:integer, event occurred times--></activePostCount>
  <eventType><!--required, xs:string, event types, here it should be set to "illaccess"--></eventType>
  <eventState><!--required, xs:string, event status (for persistent event): active, inactive--></eventState>
  <eventDescription><!--required, xs:string, event description--></eventDescription>
  <channelName><!--optional, xs:string, channel name--></channelName>
```

```
<deviceId><!--optional, xs:string, device ID--></deviceId>
</EventNotificationAlert>
```

XML_EventNotificationAlert_MotionDetectionMsg

XML Message about Details of Motion Detection Event

```
<?xml version="1.0" encoding="utf-8"?>
<EventNotificationAlert version="2.0" xmlns="http://www.isapi.org/ver20/XMLSchema">
  <ipAddress><!--dependent, xs:string, IPv4 address--></ipAddress>
  <ipv6Address><!--dependent, xs:string, IPv6 address--></ipv6Address>
  <portNo><!--optional, xs:integer, device's communication port No.--></portNo>
  <protocolType><!--optional, xs:string, transport protocol type: HTTP, HTTPS, EHome--></protocolType>
  <macAddress><!--optional, xs:string, MAC address--></macAddress>
  <dynChannelID><!--optional, xs:string, digital channel No.--></dynChannelID>
  <channelID><!--dependent, xs:string, channel No.--></channelID>
  <dateTime>
    <!--required, xs:datetime, event occurred time, which is in IOS 8601 format, e.g., 2017-04-22T15:39:01+08:00-->
  </dateTime>
  <activePostCount><!--required, xs:integer, event occurred times--></activePostCount>
  <eventType><!--required, xs:string, event types, here it should be set to "VMD"--></eventType>
  <eventState><!--required, xs:string, event status (for persistent event): active, inactive--></eventState>
  <eventDescription><!--required, xs:string, event description--></eventDescription>
  <DetectionRegionList><!--optional, detection region parameter list (for expert mode only)-->
    <DetectionRegionEntry><!--list, parameters of a detection region (for expert mode only)-->
      <regionID><!--required, xs:string, detection region ID--></regionID>
      <sensitivityLevel><!--required, xs:integer, sensitivity level, value range: [0,100]--></sensitivityLevel>
      <RegionCoordinatesList><!--required, detection region coordinates-->
        <RegionCoordinates>
          <positionX><!--required, xs:integer, x-coordinate--></positionX>
          <positionY><!--required, xs:integer, y-coordinate--></positionY>
        </RegionCoordinates>
      </RegionCoordinatesList>
    </DetectionRegionEntry>
  </DetectionRegionList>
  <channelName><!--optional, xs:string, channel name--></channelName>
  <deviceId><!--optional, xs:string, device ID--></deviceId>
  <Extensions>
    <serialNumber><!--required, xs:string, serial No.--></serialNumber>
    <eventPush><!--required, xs:string, event notification details--></eventPush>
  </Extensions>
</EventNotificationAlert>
```

XML_EventNotificationAlert_PIRAlarmMsg

XML Message about PIR (Passive Infrared) Alarm Details

```
<?xml version="1.0" encoding="utf-8"?>
<EventNotificationAlert version="2.0" xmlns="http://www.isapi.org/ver20/XMLSchema">
```

```
<ipAddress><!--dependent, xs:string, IPv4 address--></ipAddress>
<ipv6Address><!--dependent, xs:string, IPv6 address--></ipv6Address>
<portNo><!--optional, xs:integer, device's communication port No.--></portNo>
<protocolType><!--optional, xs:string, transport protocol type: HTTP, HTTPS, EHome--></protocolType>
<macAddress><!--optional, xs:string, MAC address--></macAddress>
<dynChannelID><!--optional, xs:string, digital channel No.--></dynChannelID>
<channelID><!--dependent, xs:string, channel No.--></channelID>
<dateTime>
  <!--required, xs:datetime, event occurred time, which is in IOS 8601 format, e.g., 2017-04-22T15:39:01+08:00-->
</dateTime>
<activePostCount><!--required, xs:integer, event occurred times--></activePostCount>
<eventType><!--required, xs:string, event types, here it should be set to "PIR"--></eventType>
<eventState><!--required, xs:string, event status (for persistent event): active, inactive--></eventState>
<eventDescription><!--required, xs:string, event description--></eventDescription>
<channelName><!--optional, xs:string, channel name--></channelName>
<deviceId><!--optional, xs:string, device ID--></deviceId>
</EventNotificationAlert>
```

XML_EventNotificationAlert_SceneChangeDetectionAlarmMsg

XML Message about Alarm Details of Scene Change Detection

```
<?xml version="1.0" encoding="utf-8"?>
<EventNotificationAlert version="2.0" xmlns="http://www.isapi.org/ver20/XMLSchema">
  <ipAddress><!--dependent, xs:string, IPv4 address--></ipAddress>
  <ipv6Address><!--dependent, xs:string, IPv6 address--></ipv6Address>
  <portNo><!--optional, xs:integer, device's communication port No.--></portNo>
  <protocolType><!--optional, xs:string, transport protocol type: HTTP, HTTPS, EHome--></protocolType>
  <macAddress><!--optional, xs:string, MAC address--></macAddress>
  <dynChannelID><!--optional, xs:string, digital channel No.--></dynChannelID>
  <channelID><!--dependent, xs:string, channel No.--></channelID>
  <dateTime>
    <!--required, xs:datetime, event occurred time, which is in IOS 8601 format, e.g., 2017-04-22T15:39:01+08:00-->
  </dateTime>
  <activePostCount><!--required, xs:integer, event occurred times--></activePostCount>
  <eventType><!--required, xs:string, event types, here it should be set to "scenechangedetection"--></eventType>
  <eventState><!--required, xs:string, event status (for persistent event): active, inactive--></eventState>
  <eventDescription><!--required, xs:string, event description--></eventDescription>
  <channelName><!--optional, xs:string, channel name--></channelName>
  <deviceId><!--optional, xs:string, deviceID--></deviceId>
  <Extensions>
    <serialNumber><!--required, xs:string, serial No.--></serialNumber>
    <eventPush><!--required, xs:string, event notification details--></eventPush>
  </Extensions>
</EventNotificationAlert>
```

XML_EventNotificationAlert_SubscriptionHeartbeat

Heartbeat information message returned when subscribing alarm/event, and it is in XML format.

```
<EventNotificationAlert version="2.0" xmlns="http://www.isapi.com/ver20/XMLSchema">
  <ipAddress>10.17.133.46</ipAddress>
  <portNo>80</portNo>
  <protocol>HTTP</protocol>
  <macAddress>44:19:b6:6d:24:85</macAddress>
  <channelID>1</channelID>
  <dateTime>2017-05-04T11:20:02+08:00</dateTime>
  <activePostCount>0</activePostCount>
  <eventType>heartBeat</eventType>
  <eventState>active</eventState>
  <eventDescription>heartBeat</eventDescription>
</EventNotificationAlert>
```

See Also

XML_EventNotificationAlert_AlarmEventInfo

XML_EventNotificationAlert_VideoLossAlarmMsg

XML Message about Video Loss Alarm Details

```
<?xml version="1.0" encoding="utf-8"?>
<EventNotificationAlert version="2.0" xmlns="http://www.isapi.org/ver20/XMLSchema">
  <ipAddress><!--dependent, xs:string, IPv4 address--></ipAddress>
  <ipv6Address><!--dependent, xs:string, IPv6 address--></ipv6Address>
  <portNo><!--optional, xs:integer, device's communication port No.--></portNo>
  <protocolType><!--optional, xs:string, transport protocol type: HTTP, HTTPS, EHome--></protocolType>
  <macAddress><!--optional, xs:string, MAC address--></macAddress>
  <dynChannelID><!--optional, xs:string, digital channel No.--></dynChannelID>
  <channelID><!--dependent, xs:string, channel No.--></channelID>
  <dateTime>
    <!--required, xs:datetime, event occurred time, which is in IOS 8601 format, e.g., 2017-04-22T15:39:01+08:00-->
  </dateTime>
  <activePostCount><!--required, xs:integer, event occurred times--></activePostCount>
  <eventType><!--required, xs:string, event types, here it should be set to "videoloss"--></eventType>
  <eventState><!--required, xs:string, event status (for persistent event): active, inactive--></eventState>
  <eventDescription><!--required, xs:string, event description--></eventDescription>
  <channelName><!--optional, xs:string, channel name--></channelName>
  <deviceId><!--optional, xs:string, device ID--></deviceId>
  <Extensions>
    <serialNumber><!--required, xs:string, serial No.--></serialNumber>
    <eventPush><!--required, xs:string, event notification details--></eventPush>
  </Extensions>
</EventNotificationAlert>
```


XML_EventNotificationAlert_VideoTamperingAlarmMsg

XML Message about Video Tampering Alarm Details

```
<?xml version="1.0" encoding="utf-8"?>
<EventNotificationAlert version="2.0" xmlns="http://www.isapi.org/ver20/XMLSchema">
  <ipAddress><!--dependent, xs:string, IPv4 address--></ipAddress>
  <ipv6Address><!--dependent, xs:string, IPv6 address--></ipv6Address>
  <portNo><!--optional, xs:integer, device's communication port No.--></portNo>
  <protocolType><!--optional, xs:string, transport protocol type: HTTP, HTTPS, EHome--></protocolType>
  <macAddress><!--optional, xs:string, MAC address--></macAddress>
  <dynChannelID><!--optional, xs:string, digital channel No.--></dynChannelID>
  <channelID><!--dependent, xs:string, channel No.--></channelID>
  <dateTime>
    <!--required, xs:datetime, event occurred time, which is in IOS 8601 format, e.g., 2017-04-22T15:39:01+08:00-->
  </dateTime>
  <activePostCount><!--required, xs:integer, event occurred times--></activePostCount>
  <eventType><!--required, xs:string, event types, here it should be set to "shelteralarm"--></eventType>
  <eventState><!--required, xs:string, event status (for persistent event): active, inactive--></eventState>
  <eventDescription><!--required, xs:string, event description--></eventDescription>
  <channelName><!--optional, xs:string, channel name--></channelName>
  <deviceId><!--optional, xs:string, device ID--></deviceId>
  <Extensions>
    <serialNumber><!--required, xs:string, serial No.--></serialNumber>
    <eventPush><!--required, xs:string, event notification details--></eventPush>
  </Extensions>
</EventNotificationAlert>
```

XML_EventTrigger

Linkage parameter message in XML format

```
<EventTrigger version="2.0" xmlns="http://www.isapi.org/ver20/XMLSchema">
  <id><!--required, xs:string, ID--></id>
  <eventType>
    <!--required, xs:string, see details in the "Remarks" below-->
  </eventType>
  <eventDescription><!--optional, xs:string--></eventDescription>
  <inputIOPortID><!--dependent, xs:string, alarm input ID--></inputIOPortID>
  <dynInputIOPortID><!--dependent, xs:string, dynamic alarm input ID--></dynInputPortID>
  <videoInputChannelID>
    <!--dependent, xs:string, video input channel ID, it is valid when <eventType> is "VMD, videoloss, tamperdetection, regionEntrance, regionExiting, loitering, group, rapidMove, parking, unattendedBaggage, attendedBaggage"-->
  </videoInputChannelID>
  <dynVideoInputChannelID><!--dependent, xs:string, dynamic video input channel ID--></dynVideoInputChannelID>
  <intervalBetweenEvents><!--optional, xs:integer, event time interval, unit: second--></intervalBetweenEvents>
  <WLSensorID><!--dependent, xs:string, ID--></WLSensorID>
  <EventTriggerNotificationList/><!--optional, alarm/event linkage actions, see details in the message of
```

```
XML_EventTriggerNotificationList-->
</EventTrigger>
```

Remarks

The node **<eventType>** can be the following values: IO, VMD, videoloss, raidfailure, recordingfailure, badvideo, POS, analytics, fanfailure, overheat, tamperdetection, diskfull, diskerror, nicbroken, ipconflict, illaccess, videomismatch, resolutionmismatch, radifailure, PIR, WLSensor, spareException, poePowerException, heatmap, counting, linedetection, fielddetection, regionEntrance, regionExiting, loitering, group,rapidMove, parking, unattendedBaggage, attendedBaggage, HUMANATTRIBUTE, blacklist, whitelist, peopleDetection, allVehicleList, otherVehicleList, vehicledetection, storageDetection, shipsDetection, humanAttribute, faceContrast, blacklistFaceContrast, whitelistFaceContrast, faceSnap, faceLib, personDensityDetection, personQueueDetecton, mixedTargetDetection, HVTVehicleDetection, illegalParking, pedestrian, trafficAccident, construction, roadblock, abandonedObject, parallelParking, parkingState, congestion, intersectionAnalysis, heatMap, thermometry, shipsFlowDetection, dredgerDetection, reverseEntrance, luma, highHDTemperature, lowHDTemperature, hdImpact, hdBadBlock, SevereHDFailure, safetyHelmetDetection, vibrationDetection, HBDLib,TMPA,faceThermometry,noMaskDetection

See Also

XML_EventTriggerNotificationList

XML_EventTriggerCapType

XML message about capability of alarm linkage action types

```
<EventTriggerCapType version="2.0" xmlns="http://www.isapi.org/ver20/XMLSchema">
  <isSupportCenter><!--optional, xs:boolean--></isSupportCenter>
  <isSupportRecord><!--optional, xs:boolean--></isSupportRecord>
  <isSupportMonitorAlarm><!--optional, xs:boolean--></isSupportMonitorAlarm>
  <isSupportBeep><!--optional, xs: boolean, whether it supports audible warning--></isSupportBeep>
  <isSupportIO><!--optional, xs:boolean--></isSupportIO>
  <isSupportFTP><!--optional, xs:boolean--></isSupportFTP>
  <isSupportEmail><!--optional, xs:boolean--></isSupEmail>
  <isSupportLightAudioAlarm><!--optional, xs:boolean--></isSupportLightAudioAlarm>
  <isSupportFocus><!--optional, xs:boolean--></isSupportFocus>
  <isSupportPTZ><!--optional, xs:boolean--></isSupportPTZ>
  <maxPresetActionNum>
    <!--dependent, xs:integer, it is valid only when <isSupportPTZ> is "true"-->
  </maxPresetActionNum>
  <maxPatrolActionNum>
    <!--dependent, xs:integer, it is valid only when <isSupportPTZ> is "true"-->
  </maxPatrolActionNum>
  <maxPatternActionNum>
    <!--dependent, xs:integer, it is valid only when <isSupportPTZ> is "true"-->
  </maxPatternActionNum>
  <isSupportTrack><!--optional, xs:boolean, whether it supports PTZ linked tracking--></isSupportTrack>
```

```

<isSupportWhiteLight>
  <!--optional, xs: boolean, whether it supports supplement light alarm linkage-->
</isSupportWhiteLight>
<isSupportCloud><!--optional, xs:boolean, whether it supports upload to the cloud--></isSupportCloud>
<targetNotificationInterval max="1000" min="0" default="30"><!--xs:integer, range: [0, 1000], the default value is 30,
unit: seconds, this node is valid for <MotionDetectionTriggerCap> and <TamperDetectionTriggerCap> and this node is
valid when <isSupportPTZ> is "true"--></targetNotificationInterval>
<direction opt="both,forward,reverse"><!--xs:string, triggering direction, this node is valid for the node
<BlackListTriggerCap>, <WhiteListTriggerCap>, and <VehicleDetectionTriggerCap>--></direction>
<presetDurationTime min="" max=""><!--dependent, xs:integer--></presetDurationTime>
<isSupportSMS><!--optional, xs:boolean, whether to support SMS (Short Message Service)--></isSupportSMS>
<maxCellphoneNum><!--dependent, xs:integer, the maximum number of cellphones, which is node is valid only
when <isSupportSMS> is "true"--></maxCellphoneNum>
<isSupportOSD><!--optional, xs:boolean--></isSupportOSD>
<isSupportAudio><!--optional, xs:boolean, whether it supports setting audio alarm independently. If this node is set
to "true", audio alarm and buzzer alarm can be linked separately, and the lineage method is audio--></isSupportAudio>
<AudioAction><!--dependent, this node is valid when <isSupportBeep> is "true" or <isSupportAudio> is "true"-->
  <audioTypeList>
    <audioType><!--list-->
      <audioID><!--required, xs:integer, alarm sound type--></audioID>
      <audioDescription><!--required, xs:string, alarm sound description, it should correspond to the alarm sound type--
--></audioDescription>
    </audioType>
  </audioTypeList>
  <alarmTimes opt="0,1,2,3,4,5,6,7,8,9,255"><!--required, xs:integer, alarm times, it is between 0 and 9, 255-
continuous alarm, unit: time--></alarmTimes>
</AudioAction>
<isSupportSMS><!--optional, xs:boolean --></isSupportSMS>
<maxCellphoneNum><!--dependent, if <isSupportSMS> is true, xs:integer--></maxCellphoneNum>
<isNotSupportCenterModify><!--optional, xs:boolean, whether editing configuration parameters of the surveillance
center is not supported: "true"-yes (configuration parameters of the surveillance center cannot be edited), "false" or
this node is not returned-no (configuration parameters of the surveillance center can be edited)--></
isNotSupportCenterModify>
<isSupportMessageConfig>
  <!--optional, xs:boolean, whether it supports SMS configuration, if supports, set cellphoneNumber to null-->
</isSupportMessageConfig>
<isSupportAnalogOutput><!--optional, xs:boolean, whether it supports IO output of linkage analog channel--></
isSupportAnalogOutput>
<isSupportIOOutputUnify><!--optional, xs:boolean, whether it supports configuration of IO output--></
isSupportIOOutputUnify>
<isSupportFaceContrast><!--optional, xs:boolean, whether it supports face picture comparison linkage--></
isSupportFaceContrast>
<isSupportSiren><!--optional, xs:boolean, whether it supports siren linkage--></isSupportSiren>
<isSupportOutput><!--optional, xs:boolean, whether it supports relay linkage--></isSupportOutput>
</EventTriggerCapType>

```

XML_EventTriggerNotification

Event linkage notification message in XML format

```
<EventTriggerNotification><!--opt-->
  <id><!--required, xs:string, device ID--></id>
  <notificationMethod>
    <!--required, xs:string, linkage actions, opt="email,IM,IO,syslog,HTTP,FTP,beep,ptz,record, monitorAlarm, center,
    LightAudioAlarm,focus,trace,cloud,SMS,whiteLight,audio,whiteLight,faceContrast,siren,output"-->
  </notificationMethod>
  <notificationRecurrence>
    <!--optional, xs:string, "beginning,beginningandend,recurring"-->
  </notificationRecurrence>
  <notificationInterval><!--dependent, xs:integer, unit: millisecond--></notificationInterval>
  <outputIOPortID><!--dependent, xs:string, video output No., it is required only when notificationMethod is "IO"--></
outputIOPortID>
  <dynOutputIOPortID><!--dependent, xs:string, dynamic video output No., it is required only when
notificationMethod is "IO"--></dynOutputIOPortID>
  <videoInputID><!--dependent, xs:string, video input No., it is required only when notificationMethod is "record"--></
videoInputID>
  <dynVideoInputID><!--dependent, xs:string, dynamic video input No., it is required only when notificationMethod is
"record"--></dynVideoInputID>
  <ptzAction><!--dependent, it is required only when notificationMethod is "ptz"-->
    <ptzChannelID><!--required, xs:string, PTZ channel ID--></ptzChannelID>
    <actionName><!--required, xs:string, PTZ control type: "preset", "pattern", "patrol"--></actionName>
    <actionNum><!--dependent, xs:integer></actionNum>
  </ptzAction>
  <WhiteLightAction><!--dependent, white light linkage parameters, this node is valid when notificationMethod is
"whiteLight"-->
    <whiteLightDurationTime><!--required, xs:integer, white light flashing duration, it is between 1 and 60, unit:
second--></whiteLightDurationTime>
  </WhiteLightAction>
  <cellphoneNumber><!--dependent, xs:string, min="0" max="11",cellphone number--></cellphoneNumber-->
</EventTriggerNotification>
```

XML_EventTriggerNotificationList

EventTriggerNotificationList message in XML format

```
<EventTriggerNotificationList version="2.0" xmlns="http://www.isapi.org/ver20/XMLSchema">
  <EventTriggerNotification/><!--opt, see details in the message of XML_EventTriggerNotification-->
</EventTriggerNotificationList>
```

See Also

XML_EventTriggerNotification

XML_EventTriggersCap

XML message about linkage capabilities of different alarm categories

```
<EventTriggersCap version="2.0" xmlns="http://www.isapi.org/ver20/XMLSchema">
  <DiskfullTriggerCap><!--optional, xs: EventTriggerCapType--></DiskfullTriggerCap>
```

```
<DiskerrorTriggerCap><!--optional, xs: EventTriggerCapType--></DiskerrorTriggerCap>
<NicbrokenTriggerCap><!--optional, xs: EventTriggerCapType--></NicbrokenTriggerCap>
<IpconflictTriggerCap><!--optional, xs: EventTriggerCapType--></IpconflictTriggerCap>
<IllaccesTriggerCap><!--optional, xs: EventTriggerCapType--></IllaccesTriggerCap>
<BadvideoTriggerCap><!--optional, xs: EventTriggerCapType--></BadvideoTriggerCap>
<VideomismatchTriggerCap><!--optional, xs: EventTriggerCapType--></VideomismatchTriggerCap>
<IOTriggerCap><!--optional, xs: EventTriggerCapType--></IOTriggerCap>
<LineDetectTriggerCap><!--optional, xs: EventTriggerCapType--></LineDetectTriggerCap>
<RegionEntranceTriggerCap><!--optional, xs: EventTriggerCapType--></RegionEntranceTriggerCap>
<RegionExitingTriggerCap><!--optional, xs: EventTriggerCapType--></RegionExitingTriggerCap>
<LoiteringTriggerCap><!--optional, xs: EventTriggerCapType--></LoiteringTriggerCap>
<GroupDetectionTriggerCap><!--optional, xs: EventTriggerCapType--></GroupDetectionTriggerCap>
<RapidMoveTriggerCap><!--optional, xs: EventTriggerCapType--></RapidMoveTriggerCap>
<ParkingTriggerCap><!--optional, xs: EventTriggerCapType--></ParkingTriggerCap>
<UnattendedBaggageTriggerCap><!--optional, xs: EventTriggerCapType--></UnattendedBaggageTriggerCap>
<AttendedBaggageTriggerCap><!--optional, xs: EventTriggerCapType--></AttendedBaggageTriggerCap>
<FireDetectionTriggerCap><!--optional, xs: EventTriggerCapType--></FireDetectionTriggerCap>
<FireDetectionCap><!--optional, xs: EventTriggerCapType--></FireDetectionCap>
<StorageDetectionTriggerCap><!--optional, xs: EventTriggerCapType--></StorageDetectionTriggerCap>
<ShipsDetectionTriggerCap><!--optional, xs: EventTriggerCapType--></ShipsDetectionTriggerCap>
<ThermometryCap><!--optional, xs: EventTriggerCapType--></ThermometryCap>
<VandalProofTriggerCap><!--optional, xs: EventTriggerCapType--></VandalProofTriggerCap>
<BlackListTriggerCap><!--optional, xs: EventTriggerCapType, configuration capability of blacklist arming linkage--></
BlackListTriggerCap>
<WhiteListTriggerCap><!--optional, xs: EventTriggerCapType, configuration capability of whitelist arming linkage--></
WhiteListTriggerCap>
<AllVehicleListTriggerCap><!--optional,xs:EventTriggerCapType, configuration capability of other list arming linkage--
></AllVehicleListTriggerCap>
<OtherVehicleListTriggerCap><!--optional,xs:EventTriggerCapType--></OtherVehicleListTriggerCap>
<PeopleDetectionTriggerCap><!--optional,xs:EventTriggerCapType--></PeopleDetectionTriggerCap>
<PIRAAlarmCap><!--optional, xs: EventTriggerCapType--></PIRAAlarmCap>
<TamperDetectionTriggerCap><!--optional, xs: EventTriggerCapType--></TamperDetectionTriggerCap>
<DefocusDetectionTriggerCap><!--optional, xs: EventTriggerCapType--></DefocusDetectionTriggerCap>
<FaceDetectionTriggerCap><!--optional, xs: EventTriggerCapType--></FaceDetectionTriggerCap>
<SceneChangeDetectionTriggerCap><!--optional, xs: EventTriggerCapType--></SceneChangeDetectionTriggerCap>
<VandalProofAlarmCap><!--optional, xs: EventTriggerCapType--></VandalProofAlarmCap>
<JudgmentTriggerCap><!--optional, xs: EventTriggerCapType--></JudgmentTriggerCap>
<FightingTriggerCap><!--optional, xs: EventTriggerCapType--></FightingTriggerCap>
<RisingTriggerCap><!--optional, xs: EventTriggerCapType--></RisingTriggerCap>
<DozingTriggerCap><!--optional, xs: EventTriggerCapType--></DozingTriggerCap>
<CountingTriggerCap><!--optional, xs: EventTriggerCapType--></CountingTriggerCap>
<VideoLossTriggerCap><!--optional, xs: EventTriggerCapType--></VideoLossTriggerCap>
<HideTriggerCap><!--optional, xs:EventTriggerCapType--></HideTriggerCap>
<AlarmInTriggerCap><!--optional, xs: EventTriggerCapType--></AlarmInTriggerCap>
<VehicleDetectionTriggerCap><!--optional, xs: EventTriggerCapType--></VehicleDetectionTriggerCap>
<AudioExceptionCap><!--optional, xs: EventTriggerCapType--></AudioExceptionCap>
<FiledDetectTriggerCap><!--optional, xs: EventTriggerCapType--></FiledDetectTriggerCap>
<MotionDetectionTriggerCap><!--optional, xs: EventTriggerCapType--></MotionDetectionTriggerCap>
<TemperatureCap><!--optional, xs: EventTriggerCapType--></TemperatureCap>
<IntelligentTriggerCap><!--optional, xs: EventTriggerCapType--></IntelligentTriggerCap>
<FaceContrastTriggerCap><!--optional, xs: EventTriggerCapType, face picture comparison alarm linkage--></
FaceContrastTriggerCap>
```

```
<PersonDensityDetectionTriggerCap><!--optional, xs: EventTriggerCapType--></PersonDensityDetectionTriggerCap>
<PersonQueueDetectionTriggerCap><!--optional, xs: EventTriggerCapType, queue management alarm linkage--></
PersonQueueDetectionTriggerCap>
<WhiteListFaceContrastTriggerCap><!--optional, xs: EventTriggerCapType--></WhiteListFaceContrastTriggerCap>
<HumanRecognitionTriggerCap><!--optional,xs: EventTriggerCapType--></HumanRecognitionTriggerCap>
<FaceSnapTriggerCap><!--optional, xs: EventTriggerCapType--></FaceSnapTriggerCap>
<isSupportWhiteLightAction>
  <!--dependent, xs: boolean, see details in EventTriggerCapType, it is valid when isSupportWhiteLight is "true"-->
</isSupportWhiteLightAction>
<isSupportAudioAction>
  <!--dependent, xs: boolean, see details in EventTriggerCapType, it is valid when isSupportBeep is "true"-->
</isSupportAudioAction>
<HFPDTriggerCap><!--optional, xs: EventTriggerCapType--></HFPDTriggerCap>
<MixedTargetDetectionCap><!--optional, xs: EventTriggerCapType--></MixedTargetDetectionCap>
<HVTVehicleDetectionTriggerCap><!--optional, xs: EventTriggerCapType--></HVTVehicleDetectionTriggerCap>
<VCATriggerCap><!--optional, xs: EventTriggerCapType--></VCATriggerCap>
<PIRCap><!--optional, xs: EventTriggerCapType--></PIRCap>
<IllegalParkingTriggerCap><!--optional, xs: EventTriggerCapType, whether it supports illegal parking detection--></
IllegalParkingTriggerCap>
<PedestrianTriggerCap><!--optional, xs: EventTriggerCapType, whether it supports pedestrian detection--></
PedestrianTriggerCap>
<TrafficAccidentTriggerCap><!--optional, xs: EventTriggerCapType, whether it supports traffic accident detection--></
TrafficAccidentTriggerCap>
<ConstructionTriggerCap><!--optional, xs: EventTriggerCapType, whether it supports construction detection--></
ConstructionTriggerCap>
<RoadBlockTriggerCap><!--optional, xs: EventTriggerCapType, whether it supports roadblock detection--></
RoadBlockTriggerCap>
<AbandonedObjectTriggerCap><!--optional, xs: EventTriggerCapType, whether it supports objects dropped down
detection--></AbandonedObjectTriggerCap>
<ParallelParkingTriggerCap><!--optional, xs: EventTriggerCapType, whether it supports parallel parking detection--></
ParallelParkingTriggerCap>
<ParkingStateTriggerCap><!--optional, xs: EventTriggerCapType, whether it supports parking space status detection,
currently this node is not supported--></ParkingStateTriggerCap>
<CongestionTriggerCap><!--optional, xs: EventTriggerCapType, whether it supports congestion detection--></
CongestionTriggerCap>
<IntersectionAnalysisCap><!--optional, xs: EventTriggerCapType, whether it supports intersection analysis--></
IntersectionAnalysisCap>
<ShipsFlowDetectionTriggerCap><!--optional,xs:EventTriggerCapType, ship flow detection--></
ShipsFlowDetectionTriggerCap>
<dredgerDetectionTriggerCap><!--optional,xs:EventTriggerCapType, dredger detection--></
dredgerDetectionTriggerCap>
<voltageInstableTriggerCap><!--optional,xs:EventTriggerCapType, supply voltage exception--></
voltageInstableTriggerCap>
<HighHDDTemperatureTriggerCap><!--optional, xs:EventTriggerCapType, HDD high temperature detection--></
HighHDDTemperatureTriggerCap>
<LowHDDTemperatureTriggerCap><!--optional, xs:EventTriggerCapType, HDD low temperature detection--></
LowHDDTemperatureTriggerCap>
<HDImpactTriggerCap><!--optional, xs:EventTriggerCapType, HDD impact detection--></HDImpactTriggerCap>
<HDBadBlockTriggerCap><!--optional, xs:EventTriggerCapType, HDD bad sector detection--></
HDBadBlockTriggerCap>
<SevereHDFailureTriggerCap><!--optional, xs:EventTriggerCapType, HDD severe fault detection--></
SevereHDFailureTriggerCap>
```

```
<HUMANATTRIBUTECap><!--optional, xs:EventTriggerCapType--></HUMANATTRIBUTECap>
<HumanAttributeTriggerCap><!--optional, xs:EventTriggerCapType, human body attribute--></
HumanAttributeTriggerCap>
  <BlackListFaceContrastTriggerCap><!--optional, xs:EventTriggerCapType, alarm linkage capability of blacklist face
comparison--></BlackListFaceContrastTriggerCap>
  <FaceLibTriggerCap><!--optional, xs:EventTriggerCapType--></FaceLibTriggerCap>
  <SafetyHelmetDetectionTriggerCap><!--optional, xs:EventTriggerCapType, alarm linkage capability of hard hat
detection--></SafetyHelmetDetectionTriggerCap>
  <VibrationDetectionTriggerCap><!--optional, xs:EventTriggerCapType, alarm linkage capability of vibration detection--
></VibrationDetectionTriggerCap>
  <RadarLineDetectionTriggerCap><!--optional, xs:EventTriggerCapType, alarm linkage capability of radar line crossing
detection--></RadarLineDetectionTriggerCap>
  <RadarFieldDetectionTriggerCap><!--optional, xs:EventTriggerCapType, alarm linkage capability of radar intrusion
detection--></RadarFieldDetectionTriggerCap>
  <HBDLibTriggerCap><!--optional, xs:EventTriggerCapType, alarm linkage capability of human body picture library--></
HBDLibTriggerCap>
  <FaceThermometryCap><!--optional, xs:EventTriggerCapType--></FaceThermometryCap>
  <NoMaskDetectionTriggerCap><!--optional, xs:EventTriggerCapType, alarm linkage capability of no wearing mask
detection--></NoMaskDetectionTriggerCap>
  <TMPATriggerCap><!--optional, xs:EventTriggerCapType, alarm linkage capability of temperature measurement pre-
alarm--></TMPATriggerCap>
  <FireEscapeDetectionTriggerCap><!--optional, xs:EventTriggerCapType, alarm linkage capability of fire engine access
detection--></FireEscapeDetectionTriggerCap>
  <TakingElevatorDetectionTriggerCap><!--optional, xs:EventTriggerCapType, alarm linkage capability of elevator
detection--></TakingElevatorDetectionTriggerCap>
  <RuleTriggerCap><!--optional, linkage capability of rule triggered alarm -->
    <isSupportCityManagement>
      <!--optional, xs:boolean, whether the city management supports setting linkage actions by area; if supports, the
value is true, otherwise, this node will not be returned-->
    </isSupportCityManagement>
  </RuleTriggerCap>
</EventTriggersCap>
```

See Also

XML_EventTriggerCapType

XML_ExportStatus

ExportStatus message in XML format

```
<ExportStatus version="2.0" xmlns="http://www.isapi.org/ver20/XMLSchema">
  <status><!--opt, xs:string, export status: "export"-exporting, "unexport"-not export--></status>
  <message><!--dep, xs:string, it is valid when status is "export"--></message>
</ExportStatus>
```

XML_Exposure

Exposure message in XML format

```
<Exposure version="2.0" xmlns="http://www.isapi.org/ver20/XMLSchema">
  <ExposureType/><!--req, xs:string, "auto, IrisFirst, ShutterFirst, gainFirst, manual, plris, T5280-PQ1, T5289-PQ1,
T1140-PQ1, T2712-PQ1, HV1250P-MPIR, plris-General"-->
  <autoIrisLevel>
    <!--dep, xs:integer-->
  </autoIrisLevel>
  <OverexposeSuppress><!--opt-->
    <enabled>
      <!--req, xs:boolean-->
    </enabled>
    <Type>
      <!--dep, xs:string, this node depends on <enabled>, "AUTO,MANUAL"-->
    </Type>
    <DistanceLevel>
      <!--dep, xs: integer, this node depends on <Type>-->
    </DistanceLevel>
    <shortIRDistanceLevel>
      <!--dep, xs: integer, this node depends on <Type>-->
    </shortIRDistanceLevel>
    <longIRDistanceLevel>
      <!--dep, xs: integer, this node depends on <Type>-->
    </longIRDistanceLevel>
  </OverexposeSuppress>
  <plris><!--opt-->
    <plrisType/><!--dep, xs:string, this node depends on <ExposureType>, "AUTO,MANUAL"-->
    <IrisLevel>
      <!--dep, xs:integer, this node depends on <plrisType>-->
    </IrisLevel>
  </plris>
  <PlrisGeneral><!--dep, it is available when <ExposureType> is set to "plris-General"-->
    <irisLevel>
      <!--opt, iris level of general lens, which is between 1 and 100, the default level is 50-->
    </irisLevel>
  </PlrisGeneral>
  <exposureLevel>
    <!--opt, xs:integer , this node depends on <ExposureType>-->
  </exposureLevel>
  <faceExposure><!--opt, face exposure-->
    <enabled opt="true,false">
      <!--opt, xs:boolean, whether to enable face exposure-->
    </enabled>
    <sensitivity min="0" max="100">
      <!--opt, xs:integer, sensitivity-->
    </sensitivity>
  </faceExposure>
</Exposure>
```

XML_externSecurityCap

externSecurityCap message in XML format


```
<externSecurityCap>
  <RestAdminPassWord>
    <isSupportWithSecurityQuestion>
      <!--opt, xs: boolean, whether supports resetting password by answering security questions, "true,false" -->
    </isSupportWithSecurityQuestion>
    <isSupportWithGUIDFileData>
      <!-- opt, xs: boolean, whether supports resetting password by importing GUID file, "true,false" -->
    </isSupportWithGUIDFileData>
    <isSupportWithSecurityEmail>
      <!-- opt, xs: boolean, whether supports resetting password by setting recovery email, "true,false" -->
    </isSupportWithSecurityEmail>
  </RestAdminPassWord>
  <SecurityLimits><!--opt-->
    <LoginPasswordLenLimit min="1" max="16">
      <!--opt, minimum and maximum lengths of login password-->
    </LoginPasswordLenLimit>
    <SecurityAnswerLenLimit min="1" max="128">
      <!--opt, minimum and maximum answer length of security questions-->
    </SecurityAnswerLenLimit>
  </SecurityLimits>
</externSecurityCap>
```

XML_ExtraInfo

ExtraInfo message in XML format

```
<ExtraInfo version="2.0" xmlns="http://www.isapi.org/ver20/XMLSchema">
  <diskMode><!--req, xs:string, disk mode: "singleDisk"-single disk, "RAID". The device needs to reboot after changing the disk mode--></diskMode>
</ExtraInfo>
```

XML_EZVIZ

XML message about the Hik-Connect access configuration parameters

```
<EZVIZ version="2.0" xmlns="http://www.isapi.org/ver20/XMLSchema">
  <enabled><!--required, xs:boolean--></enabled>
  <registerStatus><!--read-only, optional, xs:boolean--></registerStatus>
  <redirect><!--optional, xs:boolean, whether to allow the device to redirect the server address--></redirect>
  <serverAddress><!--optional-->
    <addressingFormatType>
      <!--required, xs:string, "ipaddress,hostname"-->
    </addressingFormatType>
    <hostName><!--dependent, xs:string--></hostName>
    <ipAddress><!--dependent, xs:string--></ipAddress>
    <ipv6Address><!--dependent, xs:string--></ipv6Address>
  </serverAddress>
  <verificationCode>
```

```
<!--optional, xs:string, verification code. Only the "admin" user can edit the verification code using the control. The
verification code will be displayed for other users. If this node is returned, setting verification code is supported-->
</verificationCode>
<offlineStatus>
  <!--ro, dependent, xs:string, it is valid when registerStatus values "false", device offline status,
opt="secretKeyInvalid"-invalid verification code-->
</offlineStatus>
<enabledTiming>
  <!--optional, xs:boolean, whether to enable Hik-Connect timing: true=yes, false=no (default); this node is valid only
when the value of parameter <platformType> in XML_Time is "EZVIZ" (related URI: /ISAPI/System/time)-->
</enabledTiming>
<version><!--optional, xs:string, read-only, version information, foramt: ""Vx.x.xbuildYYMMDD--></version>
<operateCode><!--optional, xs:string, operation code for binding device--></operateCode>
<netWork>
  <!--optional, xs:integer, 0-null ; 1-automatic ; 2-wired network priority ; 3-wired network(by default), 4-3G/4G/
GPRS-->
</netWork>
<convergenceCloudEnabled><!--optional, xs:boolean--></convergenceCloudEnabled>
<streamEncryptEnabled><!--optional, xs:boolean, whether to enable stream encryption, true=yes (default), false-
no--></streamEncryptEnabled>
<upISAPIStatus><!--read-only, optional, xs:string, status of uplink transmitting ISAPI message: "open", "close"--></
upISAPIStatus>
<bindStatus><!--optional, xs:string, read-only, Hik-Connect account binding status of the current device: "bind"-
bound, "unbind"-unbound. When "bind" is returned, you can unbind the device from Hik-Connect account by calling
the URI /ISAPI/System/Network/EZVIZ/unbind--></bindStatus>
</EZVIZ>
```

XML_FirmwareCodeList

Message about firmware code list in XML format

```
<?xml version="1.0" encoding="utf-8"?>
<FirmwareCodeList version="2.0" xmlns="http://www.isapi.org/ver20/XMLSchema">
  <FirmwareCode><!--list-->
    <index><!--required, xs:integer, index No., which starts from 1--></index>
    <code><!--required, xs:string, recognition code--></code>
    <version><!--required, xs:string, firmware version--></version>
    <channel><!--optional, xs:integer, IP channel No.--></channel>
    <devTypeName><!--optional, xs:string, device type of the corresponding device of the IP channel--></
devTypeName>
  </FirmwareCode>
</FirmwareCodeList>
```

XML_FocusConfiguration

FocusConfiguration message in XML format

```
<FocusConfiguration version="2.0" xmlns="http://www.isapi.org/ver20/XMLSchema">
  <focusStyle><!--req, xs:string, focus type, "AUTO, MANUAL, SEMIAUTOMATIC"-->
```

```
<focusLimited/><!--opt, xs:integer, the minimum focus length, unit: cm-->
<focusPosition/><!--dep, depends on "FocusStyle", xs:integer-->
<focusSpeed><!--opt, xs:integer--></focusSpeed>
<focusSensitivity>
  <!--opt, xs:integer, focus sensitivity, ranges from 0 to 2, it is valid when "focusStyle" is "MANUAL" or
"SEMIAUTOMATIC"-->
</focusSensitivity>
<temperatureChangeAdaptEnabled><!--opt, xs:boolean, "true,false"--></temperatureChangeAdaptEnabled>
<relativeFocusPos>
  <!--opt, xs:integer, relative focus value, it is valid when "focusStyle" is "MANUAL" or "SEMIAUTOMATIC"-->
</relativeFocusPos>
<highTemperaturePriority><!--opt,xs:boolean, whether to enable high temperature priority mode--></
highTemperaturePriority>
</FocusConfiguration>
```

XML_FocusData

FocusData message in XML format

```
<FocusData version="2.0" xmlns="http://www.isapi.org/ver20/XMLSchema">
  <focus>
    <!--req, xs: integer, a vector, whose value is the percentage of maximum focus speed: negative value-focus near;
positive value-focus far-->
  </focus>
</FocusData>
```

XML_formatStatus

formatStatus message in XML format

```
<formatStatus version="1.0" xmlns="http://www.isapi.org/ver20/XMLSchema">
  <formatting><!--ro, req, xs: boolean--></formatting>
  <percent><!--ro, req, xs: integer, formatting progress percentage, which is between 0 and 100--></percent>
</formatStatus>
```

XML_FTPNotification

FTPNotification message in XML format

```
<FTPNotification version="2.0" xmlns="http://www.isapi.org/ver20/XMLSchema">
  <id>
    <!--req, xs:string, FTP ID-->
  </id>
  <enabled>
    <!--req, xs:boolean, whether to enable the FTP server-->
  </enabled>
  <useSSL>
```

```
<!--opt, xs:boolean-->
</useSSL>
<addressingFormatType opt="ipaddress,hostname">
  <!--req, xs:string, this node can only be set to "ipaddress"-->
</addressingFormatType>
<hostName>
  <!--dep, xs:string-->
</hostName>
<ipAddress>
  <!--dep, xs:string-->
</ipAddress>
<ipv6Address>
  <!--dep, xs:string-->
</ipv6Address>
<portNo>
  <!--opt, xs:integer, FTP port No.-->
</portNo>
<userName>
  <!--req, xs:string, user name-->
</userName>
<password>
  <!--wo, xs:string, password-->
</password>
<passiveModeEnabled>
  <!--opt, xs:boolean-->
</passiveModeEnabled>
<annoyftp>
  <!--opt, xs:boolean-->
</annoyftp>
<uploadPicture>
  <!--opt, xs:boolean-->
</uploadPicture>
<uploadVideoClip>
  <!--opt, xs:boolean-->
</uploadVideoClip>
<uploadPath>
  <!--req-->
  <pathDepth>
    <!--req, xs:integer, directory levels, up to 16 levels can be supported-->
  </pathDepth>
  <topDirNameRule>
    <!--dep, xs:string, parent directory name: "none", "devName"-device name, "devId"-device ID, "devIp"-device IP
address, "positionInfo"-camera 1, "time_month"-usage date (YYYY-MM), "time_date"-usage date (YYYY-MM-DD),
"illegalType"-violation type, "direction"-direction, "site"-place, "chanName"-channel name, "chanId"-channel No.,
"laneId"-lane No., "customize"-custom, "time", "buildUnitNo"-building No. and unit No. This node is set to NULL by
default-->
  </topDirNameRule>
  <topDirName/><!--dep, xs:string, custom string for parent directory, the maximum string length is 32 bytes. This
node is valid when <topDirNameRule> is "customize"-->
  <subDirNameRule>
    <!--dep, xs:string, child directory name: "none", "devName"-device name, "devId"-device ID, "devIp"-device IP
address, "positionInfo"-camera 1, "time_month"-usage date (YYYY-MM), "time_date"-usage date (YYYY-MM-DD),
```

"illegalType"-violation type, "direction"-direction, "site"-place, "chanName"-channel name, "chanId"-channel No., "laneId"-lane No., "customize"-custom, "time", "buildUnitNo"-building No. and unit No. This node is set to NULL by default-->

```
</subDirNameRule>
```

<subDirName/><!--dep, xs:string, custom string for child directory, the maximum string length is 32 bytes. This node is valid when <subDirNameRule> is "customize"-->

```
<threeDirNameRule>
```

<!--dep, xs:string, the third directory name: "none", "devName"-device name, "devId"-device ID, "devIp"-device IP address, "positionInfo"-camera 1, "time_month"-usage date (YYYY-MM), "time_date"-usage date (YYYY-MM-DD), "illegalType"-violation type, "direction"-direction, "site"-place, "chanName"-channel name, "chanId"-channel No., "laneId"-lane No., "customize"-custom, "time", "buildUnitNo"-building No. and unit No. This node is set to NULL by default-->

```
</threeDirNameRule>
```

<threeDirName/><!--req, xs:string, custom string for the third directory, the maximum string length is 32 bytes. This node is valid when <threeDirNameRule> is "customize"-->

```
<fourDirNameRule>
```

<!--dep, xs:string, the fourth directory name: "none", "devName"-device name, "devId"-device ID, "devIp"-device IP address, "positionInfo"-camera 1, "time_month"-usage date (YYYY-MM), "time_date"-usage date (YYYY-MM-DD), "illegalType"-violation type, "direction"-direction, "site"-place, "chanName"-channel name, "chanId"-channel No., "laneId"-lane No., "customize"-custom, "time", "buildUnitNo"-building No. and unit No. This node is set to NULL by default-->

```
</fourDirNameRule>
```

<fourDirName/><!--req, xs:string, custom string for the fourth directory, the maximum string length is 32 bytes. This node is valid when <fourDirNameRule> is "customize"-->

```
</uploadPath>
```

```
<picArchivingInterval>
```

<!--opt, xs:integer, the value is between 1 and 30, 0-close-->

```
</picArchivingInterval>
```

```
<picNameRuleType>
```

<!--opt, xs:string, picture name rule type: "default, prefix"-->

```
</picNameRuleType>
```

```
<picNamePrefix>
```

<!--dep, xs:string, prefix length of the picture name, which is between 0 and 32-->

```
</picNamePrefix>
```

```
<ftpPicNameRuleType>
```

<!--req, xs:string, type of FTP picture name rule: "videoIntercom"-rule used by video intercom products, "ITC"-rule used by traffic cameras-->

```
</ftpPicNameRuleType>
```

```
<FTPPicNameRule>
```

<!--dep, picture name rule of a specific FTP-->

```
<ItemList/><!--req, see details in the message of XML_ItemList-->
```

```
<delimiter>
```

<!--req, xs:string, delimiter, which is a single character and the default value is "_"-->

```
</delimiter>
```

```
<customStr>
```

<!--req, xs:string, custom string, its length is between 1 and 128-->

```
</customStr>
```

```
</FTPPicNameRule>
```

```
<upDataType>
```

<!-- opt, xs:integer, picture uploading type: 0-all, 1-checkpoint, 2-violation. When only one FTP server is enabled, this node can only be set to 0. When two FTP servers are both enabled, you should set 1 for one FTP server and set 2 for another FTP server, which means that two FTP servers cannot be set to the same type-->

```
</upDataType>
<uploadPlateEnable>
  <!--opt, xs:boolean, whether to enable uploading license plate thumbnail-->
</uploadPlateEnable>
<site>
  <!--req, xs:string, place, the maximum string length is 128 bytes-->
</site>
<roadNum>
  <!--req, xs:string, intersection No., the maximum string length is 32 bytes-->
</roadNum>
<instrumentNum>
  <!--req, xs:string, device No., the maximum string length is 32 bytes-->
</instrumentNum>
<direction>
  <!--req, xs:string, direction No., the maximum string length is 32 bytes-->
</direction>
<directionDesc>
  <!--req, xs:string, direction description, the maximum string length is 32 bytes-->
</directionDesc>
<monitoringInfo1>
  <!--req, xs:string, camera 1 information, the maximum string length is 44 bytes-->
</monitoringInfo1>
<uploadAttachedInfomation>
  <!--req, xs:boolean, whether to upload additional information-->
</uploadAttachedInfomation>
<brokenNetHttp><!--opt, xs:boolean, whether to enable ANR (automatic network replenishment)--></brokenNetHttp>
</FTPNotification>
```

See Also

XML_ItemList

XML_FTPNotificationList

FTPNotificationList message in XML format

```
<FTPNotificationList version="2.0" xmlns="http://www.isapi.org/ver20/XMLSchema">
  <FTPNotification/><!--opt, see details in the message of XML_FTPNotification-->
</FTPNotificationList>
```

See Also

XML_FTPNotification

XML_FTPTestDescription

FTPTestDescription message in XML format

```
<FTPTestDescription version="2.0" xmlns="http://www.isapi.org/ver20/XMLSchema">
  <useSSL><!--opt, xs: boolean--></useSSL>
  <addressingFormatType>
    <!--req, xs: string, "ipaddress,hostname" -->
  </addressingFormatType>
  <hostName><!--dep, xs: string--></hostName>
  <ipAddress><!--dep, xs: string--></ipAddress>
  <ipv6Address><!--dep, xs: string--></ipv6Address>
  <portNo><!--opt, xs: integer--></portNo>
  <userName><!--req, xs: string--></userName>
  <password><!--wo, xs: string--></password>
  <passiveModeEnabled><!--opt, xs: boolean--></passiveModeEnabled>
  <annoyftp><!--opt, xs: boolean--></annoyftp>
  <uploadPath><!--req-->
    <pathDepth><!--req, xs: integer, the value is from 0 to 2--></pathDepth>
    <topDirNameRule>
      <!--dep, xs: string, "devName, devId, devIp, customize"-->
    </topDirNameRule>
    <topDirName/><!--dep, xs: string-->
    <subDirNameRule>
      <!--dep, xs: string, "chanName, chanId, customize"-->
    </subDirNameRule>
    <subDirName/><!--dep, xs: string-->
  </uploadPath>
</FTPTestDescription>
```

XML_FTPTestResult

FTPTestResult message in XML format

```
<FTPTestResult version="2.0" xmlns="http://www.isapi.org/ver20/XMLSchema">
  <errorDescription><!--req, xs: string--></errorDescription>
</FTPTestResult>
```

XML_Gain

Gain message in XML format

```
<Gain version="1.0" xmlns="http://www.isapi.org/ver20/XMLSchema">
  <GainLevel/><!--dep, xs:integer, this node is valid when <ExposureType> in the message XML_Exposure is
  "GainFisrt"-->
  <GainWindow><!--opt-->
    <RegionCoordinatesList><!--opt-->
      <RegionCoordinates><!--opt-->
        <positionX><!--req, xs: integer; x-coordinate--></positionX>
        <positionY><!--req, xs: integer; y-coordinate--></positionY>
      </RegionCoordinates>
    </RegionCoordinatesList>
  </GainWindow>
</Gain>
```

```
</GainWindow>  
</Gain>
```

See Also

XML_Exposure

XML_GuardAgainstTheft

GuardAgainstTheft message in XML format

```
<GuardAgainstTheft version="2.0" xmlns="http://www.isapi.org/ver20/XMLSchema">  
  <enabled><!--required, xs:boolean, whether to support enabling device anti-theft--></enabled>  
  <SMSEnabled><!--required, xs:boolean, whether to support enabling SMS--></SMSEnabled>  
  <phoneNum><!--dependency, xs:string, phone No.--></phoneNum>  
  <longitudeLatitudeEnabled><!--dependency, xs:boolean, whether to support enabling longitude and latitude--></longitudeLatitudeEnabled>  
  <siteLocationEnabled><!--optional, xs:boolean, whether to support enabling site location--></siteLocationEnabled>  
  <siteLocationInfo><!--dependency, xs:string, site location information--></siteLocationInfo>  
  <timeEnabled><!--optional, xs:boolean, whether to support enabling time--></timeEnabled>  
  <customInfo><!--optional, xs:string, custom information--></customInfo>  
</GuardAgainstTheft>
```

XML_HardwareService

HardwareService message in XML format

```
<HardwareService version="2.0" xmlns="http://www.isapi.org/ver20/XMLSchema">  
  <IrLightSwitch><!--opt-->  
    <mode><!--req, xs:string, "open,close"--></mode>  
  </IrLightSwitch>  
  <ABF><!--opt-->  
    <enabled><!--req, xs:boolean--></enabled>  
  </ABF>  
  <LED><!--opt-->  
    <enabled><!--req, xs:boolean--></enabled>  
  </LED>  
  <Defog>  
    <!--opt -->  
    <enabled>  
      <!--req, xs:boolean-->  
    </enabled>  
  </Defog>  
  <SupplementLight>  
    <!--opt-->  
    <enabled>  
      <!--req, xs:boolean-->  
    </enabled>  
  <isSupportFireLaserLight>  
    <!--opt, xs:boolean, "true, false"-->
```



```
</isSupportFireLaserLight>
<isSupportSupplementLightWord>
  <!--opt, xs:boolean, "true, false", whether to support displaying supplement light prompt-->
</isSupportSupplementLightWord>
<captureWithSupplimentLightEnabled>
  <!--opt, xs:boolean, "true,false", enable snapshot supplement light or not-->
</captureWithSupplimentLightEnabled>
</SupplementLight>
<Deicing><!--opt-->
  <enabled><!--req, xs:boolean--></enabled>
</Deicing>
<ManualDeicing><!--opt-->
  <enabled><!--req, xs:boolean--></enabled>
</ManualDeicing>
<mutexAbility>
  <!--req, "laserLight,deicing", mutual exclusion ability, the laser light and deicing (including manual deicing and
automatic deicing) are mutual exclusive-->
</mutexAbility>
<VisibleMovementPower><!--opt-->
  <enabled><!--req, xs:boolean--></enabled>
</VisibleMovementPower>
<ThermalMovementPower><!--opt-->
  <enabled><!--req, xs:boolean--></enabled>
</ThermalMovementPower>
<PtzPower><!--opt-->
  <enabled><!--req, xs:boolean--></enabled>
</PtzPower>
<powerSavingControl><!--opt, xs:string, power saving strategy: "sleepMode"-sleeping mode, lowConsumptionMode-
低功耗模式 --></powerSavingControl>
<HighTemperatureProtection><!--opt-->
  <enabled><!--req, xs:boolean --></enabled>
  <temperatureType><!--dep, xs:string,unit:°C, "90,100,110" --></temperatureType>
</HighTemperatureProtection>
</HardwareService>
```

XML_hdd

XML message about HDD parameters

```
<hdd version="2.0" xmlns="http://www.isapi.org/ver20/XMLSchema">
  <id><!--read only, req, xs: string; ID--></id>
  <hddName><!--read only, req, xs: string--></hddName>
  <hddPath><!--read only, opt, xs: string--></hddPath>
  <hddType><!--read only, req, xs: string, "IDE, SATA, eSATA, NFS, iSCSI, Virtual Disk"--></hddType>
  <status>
    <!--required, read only, xs: string, HDD status:
"ok,unformatted,error,idle,mismatch,offline,smartFailed,reparing,formatting,notexist,unRecordHostFormatted,synching
,syncError,unloaded". "unRecordHostFormatted"--unformatted in education sharing system, "synching"-synchronizing,
"syncError"-synchronization exception-->
  </status>
  <capacity><!--read only, req, xs: float, unit: MB--></capacity>
```

```
<freeSpace><!--read only, req, xs: float, unit: MB--></freeSpace>
<property><!--required, xs: string, HDD properties, "RW,RO,Redund"--></property>
<group><!--optional, xs: string; HDD group ID--></group>
<reservedDayCfg><!--optional, xs:integer, video saving duration, it is between 0 and 1000, unit: day--></
reservedTimeCfg>
<DataModeList><!--opt, ro, current HDD allocation mode-->
  <DataMode>
    <type><!--required, xs: string, storage application type: "recordStorage"-video storage, "pictureCloudStorage"-
picture to be saved in cloud storage, "fileStorage"-file storage--></type>
    <occupancyRate><!--required, xs: integer, HDD usage, range: [0,100]--></occupancyRate>
  </DataMode>
</DataModeList>
<formatType>
  <!--optional, read only, xs: string, formatting type: FAT32 (default) and EXT4, this node is only available for SD card; if
this node does not exist, the default formatting type is FAT32-->
</formatType>
<encryptionStatus><!--optional, read only, xs:string, encryption status: "unencrypted", "encrypted", "verifyFailed"-
verification failed--></encryptionStatus>
</hdd>
```

XML_hddList

XML message about HDD list parameters

```
<hddList xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0" >
  <hdd xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0" >
    <id>
      <!--read-only, required, xs:string, HDD ID-->
    </id>
    <hddName>
      <!--read-only, required, xs:string-->
    </hddName>
    <hddPath>
      <!--read-only, optional, xs:string-->
    </hddPath>
    <hddType>
      <!--read-only, required, xs:string, "IDE,SATA,eSATA,NFS,iSCSI,Virtual Disk,NTFS",etc-->
    </hddType>
    <status>
      <!--read-only, required,
xs:string, "ok,unformatted,error,idle,mismatch,offline,smartFailed,repairing,formatting,notexist,unRecordHostFormatted
,maintaining,maintainingPercent,synching,syncError,unloaded"-->
    </status>
    <capacity>
      <!--read-only, required, xs:float,in MB-->
    </capacity>
    <freeSpace>
      <!--read-only, required, xs:float,in MB-->
    </freeSpace>
    <property>
      <!--required, xs:string "RW,read-only, Redund"-->
    </property>
  </hdd>
</hddList>
```

```
</property>
<group>
  <!--optional, xs:string; id-->
</group>
<DataModeList>
  <!--optional,ro-->
  <DataMode>
    <!--list-->
    <type>
      <!--required, xs:string,opt="recordStorage,pictureCloudStorage,fileStorage"-->
    </type>
    <occupancyRate>
      <!--required, xs:integer [0,100]-->
    </occupancyRate>
  </DataMode>
</DataModeList>
<formatType>
  <!--read-only, optional, xs:string; "FAT32,EXT4"-->
</formatType>
<ipAddress>
  <!--optional, xs:string-->
</ipAddress>
<ipv6Address>
  <!--optional, xs:string -->
</ipv6Address>
<formatType>
  <!--read-only, optional, xs:string; "FAT32,EXT4"-->
</formatType>
<encryptionStatus>
  <!--read-only, optional, xs:string, encryption status: "unencrypted"-unencrypted, "encrypted"-encrypted,
"verifyFailed"-verification failed-->
</encryptionStatus>
<logStorageEnable>
  <!--read-only, optional, xs:boolean, whether it is a log HDD-->
</logStorageEnable>
</hdd>
</hddList>
```

XML_HDDSMARTTest

HDDSMARTTest message in XML format

```
<HDDSMARTTest version="2.0" xmlns="http://www.isapi.org/ver20/XMLSchema">
  <testType><!--opt, xs: string, checking type: short,expanded,conveyance--></testType>
</HDDSMARTTest>
```

XML_Hello

Hello message in XML format

```
<?xml version="1.0" encoding="UTF-8"?>
<Hello>
  <Types>hello</Types>
  <!--device type value-->
  <DeviceType>1234</DeviceType>
  <!--device type description-->
  <DeviceDescription>DS-2CD4024F</DeviceDescription>
  <!--device serial No.-->
  <DeviceSN>DS-2CD4024F20130925CCCH435107838</DeviceSN>
  <!--command port number of private protocol-->
  <CommandPort>8000</CommandPort>
  <!--http port number-->
  <HttpPort>80</HttpPort>
  <!--MAC address-->
  <MAC>44-19-b7-10-98-32</MAC>
  <!--IPv4 address-->
  <IPv4Address>172.9.3.106</IPv4Address>
  <!--IPv4 subnet mask-->
  <IPv4SubnetMask>255.255.255.0</IPv4SubnetMask>
  <!--IPv4 gateway-->
  <IPv4Gateway>172.9.3.1</IPv4Gateway>
  <!--IPv6 address-->
  <IPv6Address>2000:1:2:3:4619:b6ff:fe01:dd9d</IPv6Address>
  <!--IPv6 gateway-->
  <IPv6Gateway>2000:1:2:3:4619:b6ff:fe01:1</IPv6Gateway>
  <!--size of IPv6 subnet mask-->
  <IPv6MaskLen>64</IPv6MaskLen>
  <!--enable/disable DHCP, true: enable, false: disable-->
  <DHCP>>false</DHCP>
  <!--total number of analog channels-->
  <AnalogChannelNum>32</AnalogChannelNum>
  <!--total number of digital channels-->
  <DigitalChannelNum>32</DigitalChannelNum>
  <!--device software version information, not encrypt-->
  <SoftwareVersion>V4.1.0 build130126</SoftwareVersion>
  <!--DSP version information, not encrypt-->
  <DSPVersion>V4.0 build130111</DSPVersion>
  <!--device started time-->
  <BootTime>2014-01-06 11:39:00</BootTime>
  <!--true: OEM device, false: baseline device-->
  <OEMCode>true</OEMCode>
  <!--OEM manufacturer information, optional-->
  <OEMInfo>Axis</OEMInfo>
  <!--device software version information, it is encrypted in AES128/AES256 CBC mode with fixed key-->
  <SoftwareVersionEncrypt>V4.1.0 build130126</SoftwareVersionEncrypt>
  <!--DSP version, it is encrypted in AES128/AES256 CBC mode with fixed key-->
  <DSPVersionEncrypt>V4.0 build130111</DSPVersionEncrypt>
```

```
<!--OEM manufacturer information, it is encrypted in AES128/AES256 CBC mode with fixed key, optional-->
<OEMInfoEncrypt>Axis</OEMInfoEncrypt>
<!--true: encrypted device, false: normal device>
<Encrypt>true</Encrypt>
<!--security code generated by specific algorithm (reserved), which is to prevent piracy-->
<SafeCode>123456</SafeCode>
<!--support upgrading reset password or not, true: yes, false: no>
<ResetAbility>true</ResetAbility>
<!--number of HDDs>
<DiskNumber>1</DiskNumber>
<!--true: activated, false: inactivated>
<Activated>true</Activated>
<!--support resetting password or not, true: yes, false: no>
<PasswordResetAbility>true</PasswordResetAbility>
<!--support synchronizing password of network camera or not, true: yes, false: no>
<SyncIPCPassWord>true</SyncIPCPassWord>
<!--support password reset mode 2 or not, true: yes, false: no>
<PasswordResetModeSecond>true</PasswordResetModeSecond>
<!--OEMCode details-->
<DetailOEMCode>10101</DetailOEMCode>
<!--true: EZVIZ device, false: baseline device-->
<EZVIZCode>true</EZVIZCode>
<!--support locking device or not, true: yes, false: no-->
<DeviceLock>true</DeviceLock>
</Hello>
```

XML_HttpHostNotification

XML message about parameters of a HTTP listening server

```
<HttpHostNotification version="2.0" xmlns="http://www.isapi.org/ver20/XMLSchema">
  <id><!--required, xs:string, ID--></id>
  <url><!--required, xs:string, the absolute path, e.g., http://<ipAddress>:<portNo>/<uri>--></url>
  <protocolType><!--required, xs:string, "HTTP,HTTPS,EHome"--></protocolType>
  <parameterFormatType><!--required, xs:string, alarm/event information format, "XML,JSON"--></
parameterFormatType>
  <addressingFormatType><!--required, xs:string, "ipaddress,hostname"--></addressingFormatType>
  <hostName><!--dependent, xs:string--></hostName>
  <ipAddress><!--dependent, xs:string--></ipAddress>
  <ipv6Address><!--dependent, xs:string--></ipv6Address>
  <portNo><!--optional, xs:integer--></portNo>
  <userName><!--dependent, xs:string--></userName>
  <password><!--dependent, xs:string--></password>
  <httpAuthenticationMethod><!--required, xs:string, "MD5digest,none"--></httpAuthenticationMethod>
  <ANPR><!--optional-->
    <detectionUpLoadPicturesType>
      <!--optional, xs:string, types of alarm picture to be uploaded: "all, licensePlatePicture, detectionPicture"-->
    </detectionUpLoadPicturesType>
  </ANPR>
  <eventType optional="AID,TFS,TPS"><!--required, xs:string--></eventType>
  <uploadImagesDataType>
```

```
<!--optional, xs:string, "URL", "binary" (default), for cloud storage, only "URL" is supported-->
</uploadImagesDataType>
<eventMode><!--optional, xs:string, "all,list"--></eventMode>
<EventList><!--dependent, it is valid only when eventMode is "list"-->
  <Event><!--required-->
    <type><!--required, xs:string--></type>
  </Event>
</EventList>
<channels><!--optional, xs:string, "1,2,3,4..."--></channels>
<SubscribeEvent/><!--optional, event subscription parameters, see details in the message of XML_SubscribeEvent-->
</HttpHostNotification>
```

XML_HttpHostNotificationCap

XML message about capability of HTTP listening server

```
<?xml version="1.0" encoding="utf-8"?>
<HttpHostNotificationCap version="2.0" xmlns="http://www.isapi.org/ver20/XMLSchema">
  <hostNumber>2</hostNumber>
  <urlLen max=""/>
  <protocolType opt="HTTP,HTTPS,EHome"/>
  <parameterFormatType opt="XML,querystring,JSON"/>
  <addressingFormatType opt="ipaddress,hostname"/>
  <ipAddress opt="ipv4,ipv6"/>
  <portNo min=""" max=""/>
  <userNameLen min=""" max=""/>
  <passwordLen min=""" max=""/>
  <httpAuthenticationMethod opt="MD5digest,none"/>
  <Extensions>
    <intervalBetweenEvents min=""" max=""/>
  </Extensions>
  <uploadImagesDataType opt="URL,binary"/>
  <ANPR><!--optional-->
    <detectionUpLoadPicturesType opt="all,licensePlatePicture,detectionPicture..."/><!--optional, xs:string, types of
alarm pictures to be uploaded-->
    <alarmHttpPushProtocol opt="baseline,custom"/>
  </ANPR>
  <httpBroken opt="true,false" def="true" ><!--optional, xs:boolean, whether to enable global ANR: true, false--></
httpBroken>
  <SubscribeEventCap>
    <heartbeat min=""" max=""/><!--optional, heartbeat time interval, unit: second-->
    <channelMode opt="all,list"/><!--required, all-subscribe events of all channels, list-subscribe event by channel-->
    <eventMode opt="all,list"/><!--required, event subscription mode: all-subscribe all events of all channels, list-
subscribe events by type, channel, and target-->
    <!--if the values of the two nodes channelMode and eventMode are both "all", it indicates that the device does not
support subscribe events by type and channel-->
    <EventList><!--dependent, alarm uploading mode, this node is valid only when eventMode is "list"-->
      <Event><!--required-->
        <type><!--required, xs:string, event types--></type>
        <pictureURLType opt="binary,localURL,cloudStorageURL" def=""/>
        <!--optional, xs:string, transmission format of alarm picture: "binary"-picture binary data, "localURL"-picture URL
```

```
from local device, "cloudStorageURL"-picture URL from cloud storage-->
  </Event>
</EventList>
<pictureURLType opt="binary,localURL,cloudStorageURL" def=""/>
  <!--optional, xs:string, transmission format of all alarm pictures: "binary"-picture binary data (default for camera),
  "localURL"-picture URL from local device (default for NVR/DVR), "cloudStorageURL"-picture URL from cloud storage;
  this node is in highest priority-->
  <ChangedUploadSub>
    <interval/><!--optional, xs:integer, the life cycle of arming GUID, unit: second, the default life cycle is 5 minutes; if
    the reconnection is not started during the life cycle, a new GUID will be generated-->
    <StatusSub>
      <all/><!--optional, xs:boolean, whether to subscribe all-->
      <channel/><!--optional, xs:boolean, subscribe channel status, this node is not required when the node all is "true"-->
    >
      <hd/><!--optional, xs:boolean, subscribe the HDD status, this node is not required when the node all is "true"-->
      <capability/><!--optional, xs:boolean, subscribe the capability changed status, this node is not required when the
      node all is "true"-->
    </StatusSub>
  </ChangedUploadSub>
</SubscribeEventCap>
</HttpHostNotificationCap>
```

XML_HttpHostNotificationList

HttpHostNotificationList message in XML format

```
<HttpHostNotificationList version="2.0" xmlns="http://www.isapi.org/ver20/XMLSchema">
  <HttpHostNotification>
    <id><!--req, xs:string, ID--></id>
    <url><!--req, xs:string--></url>
    <protocolType><!--req, xs:string, "HTTP,HTTPS"--></protocolType>
    <parameterFormatType><!--req, xs:string, alarm/event information format, "XML,JSON"--></parameterFormatType>
    <addressingFormatType><!--req, xs:string, "ipaddress,hostname"--></addressingFormatType>
    <hostName><!--dep, xs:string--></hostName>
    <ipAddress><!--dep, xs:string--></ipAddress>
    <ipv6Address><!--dep, xs:string--></ipv6Address>
    <portNo><!--opt, xs:integer--></portNo>
    <userName><!--dep, xs:string--></userName>
    <password><!--dep, xs:string--></password>
    <httpAuthenticationMethod><!--req, xs:string, "MD5digest,none"--></httpAuthenticationMethod>
    <uploadImagesDataType>
      <!--opt, xs:string, "URL", "binary" (default), for cloud storage, only "URL" is supported-->
    </uploadImagesDataType>
    <eventMode><!--opt, xs:string, "all,list"--></eventMode>
    <EventList><!--dep, it is valid only when eventMode is "list"-->
      <Event><!--req-->
        <type><!--req, xs:string--></type>
      </Event>
    </EventList>
    <channels><!--opt, xs:string, "1,2,3,4..."--></channels>
```

```
</HttpHostNotification>
</HttpHostNotificationList>
```

Example

HttpHostNotificationList Message Example

```
<HttpHostNotificationList version="2.0" xmlns="http://www.isapi.org/ver20/XMLSchema">
  <HttpHostNotification>
    <id>1</id>
    <url></url>
    <protocolType>HTTP</protocolType>
    <parameterFormatType>XML</parameterFormatType>
    <addressingFormatType>ipaddress</addressingFormatType>
    <ipAddress>0.0.0.0</ipAddress>
    <portNo>80</portNo>
    <userName></userName>
    <httpAuthenticationMethod>none</httpAuthenticationMethod>
  </HttpHostNotification>
</HttpHostNotificationList>
```

XML_HttpHostTestResult

HttpHostTestResult message in XML format.

```
<HttpHostTestResult version="2.0" xmlns="http://www.isapi.org/ver20/XMLSchema">
  <errorDescription>
    <!--req, xs:string-->
  </errorDescription>
</HttpHostTestResult>
```

XML_IbeaconParam

IbeaconParam message in XML format

```
<IbeaconParam xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <UUID min="" max="">
    <!--req, xs:string, parking lot ID, which supports using digits, letters and hyphen as the tag-->
  </UUID>
  <major min="" max="">
    <!--req, xs:integer, number of floors in the parking lot, which cannot be 0x00 for all-->
  </major>
  <minor min="" max="">
    <!--req, xs:integer, number of parking spaces in each floor, which cannot be 0x00 for all-->
  </minor>
  <sendPower min="" max="">
    <!--req, xs:integer, transmitted power: 01-0 dbm, 02- -6 dbm, 03- -23 dbm-->
  </sendPower>
  <frequency min="" max="">
    <!--req, xs:integer, transmitted frequency, which is the broadcast time interval, unit: 625 μs. The broadcast time
```


interval is between 32 (20 ms) to 8000 (5s), and the default value is 160 (100 ms)-->

```
</frequency>
<measurePower min="" max="">
  <!--req, xs:integer, rated power, which is the RSSI (Received Signal Strength Indication) in the position of one meter
away, unit: dbm-->
</measurePower>
</lbeaconParam>
```

XML_IEEE802_1x

IEEE802_1x message in XML format

```
<IEEE802_1x version="2.0" xmlns="http://www.isapi.org/ver20/XMLSchema">
  <enabled><!--req, xs:boolean--></enabled>
  <authenticationProtocolType>
    <!--req, xs:string, "EAP-TLS,EAP-TTLS,EAP-PEAP,EAP-LEAP,EAP-FAST,EAP-MD5"-->
  </authenticationProtocolType>
  <innerTTLSAuthenticationMethod>
    <!--dep, xs:string, "MS-CHAP,MS-CHAPv2,PAP,EAP-MD5", this node is required when <authenticationProtocolType>
is "EAP-TLS"-->
  </innerTTLSAuthenticationMethod>
  <innerEAPProtocolType>
    <!--dep, xs:string, "EAP-POTP,MS-CHAPv2", this node is required when <authenticationProtocolType> is "EAP-
PEAP" or "EAP-FAST"-->
  </innerEAPProtocolType>
  <validateServerEnabled><!--dep, xs:boolean--></validateServerEnabled>
  <userName><!--dep, xs:string--></userName>
  <password><!--dep, xs:string--></password>
  <anonymousID><!--opt, xs:string--></anonymousID>
  <autoPACProvisioningEnabled>
    <!--dep, xs:boolean, this node is required when <authenticationProtocolType> is "EAP-FAST"-->
  </autoPACProvisioningEnabled>
  <Extensions><!--opt-->
    <EAPOLVersion><!--opt, xs:string, "1, 2"--></EAPOLVersion>
  </Extensions>
</IEEE802_1x>
```

Remarks

- For EAP-LEAP or EAP-MD5, the parameters **userName**, **password**, and **Extensions** can be configured.
- For EAP-TLS, the parameters **userName**, **password**, and **Extensions** can be configured, and the certificates with different types (i.e., CA, user certificate, key) can be uploaded.

XML_IllegalLoginLock

IllegalLoginLock message in XML format

```
<IllegalLoginLock version="2.0" xmlns="http://www.isapi.org/ver20/XMLSchema">
  <enabled>true</enabled>
  <maxIllegalLoginTimes>
    <!--opt, xs: integer, maximum illegal login attempts, whose value is between 3 and 20, and the default value is 5-->
  </maxIllegalLoginTimes>
  <maxIllegalLoginLockTime><!--optional, xs:integer, lock duration when maximum illegal login attempts reached the
upper limit, range:[0,120], unit: minute--></maxIllegalLoginLockTime>
</IllegalLoginLock>
```

XML_ImageCap

ImageCap message in XML format

```
<ImageCap version="2.0" xmlns="http://www.isapi.com/ver20/XMLSchema">
  <isSupportRegionalExposure><!--opt, xs:boolean--></isSupportRegionalExposure>
  <isSupportRegionalFocus><!--opt, xs:boolean--></isSupportRegionalFocus>
</ImageCap>
```

XML_ImageChannel

ImageChannel message in XML format

```
<ImageChannel version="2.0" xmlns="http://www.isapi.org/ver20/XMLSchema">
  <id><!--req, xs: integer--></id>
  <enabled><!--req, xs: boolean--></enabled>
  <videoInputID><!--req, xs: integer--></videoInputID>
  <Defog/><!--opt, electronic defogging parameters, see details in the message of XML_Defog-->
  <NoiseReduce2D/><!--opt, 2D noise reduction parameters, see details in the message of XML_NoiseReduce2D-->
  <Focusconfiguration/><!--opt-->
  <LensInitialization/><!--opt-->
  <ImageFlip/><!--opt, image automatic flipping parameters, see details in the message of XML_ImageFlip-->
  <ImageFreeze/><!--opt-->
  <proportionalpan/><!--opt-->
  <WDR/><!--opt, WDR parameters, refer to the message XML_WDR for details-->
  <BLC/><!--opt, BLC parameters, refer to the message XML_BLC for details-->
  <NoiseReduce/><!--opt, 3D DNR parameters, see details in the message of XML_NoiseReduce-->
  <ImageEnhancement/><!--opt, image enhancement parameters, see details in the message of
XML_ImageEnhancement-->
  <DSS/><!--opt, low illumination electronic shutter parameters in exposure, see details in the message of XML_DSS-->
  <WhiteBalance/><!--opt, WB parameters, see details in the message of XML_WhiteBalance-->
  <Exposure/><!--opt, exposure parameters, see details in the message of XML_Exposure-->
  <Sharpness/><!--opt, sharpness parameters, see details in the message of XML_Sharpness-->
  <gammaCorrection/><!--opt, gamma correction parameters, see details in the message of XML_gammaCorrection-->
  <powerLineFrequency/><!--opt, image standard parameters, refer to the message XML_powerLineFrequency for
details-->
  <Color/><!--opt, image color parameters, see details in the message of XML_Color-->
  <IrcutFilter/><!--opt, day/night auto switch parameters, see details in the message of XML_IrxutFilter-->
  <ImageModeList/><!--opt, default image mode parameters, see details in the message of XML_ImageModeList-->
```

```
<BrightEnhance/><!--opt, brightness enhancement parameters, see details in the message of XML_BrightEnhance-->
<ISPMode/><!--opt, day/night mode parameters, see details in the message of XML_ISPMode-->
<Shutter/><!--opt, shutter parameters in exposure, see details in the message of XML_Shutter-->
<Gain/><!--opt, gain parameters, see details in the message of XML_Gain-->
<ImageIcrE/><!--opt, IR-cut filter parameters, see details in the message of XML_ImageIcrE-->
<ImageMultishut/><!--opt, multi-shutter parameters, see details in the message of XML_ImageMultishut-->
<PlateBright/><!--opt, license plate brightness compensation parameters, see details in the message of
XML_PlateBright-->
<JPEGParam/><!--opt, JPEG picture size parameters, see details in the message of XML_JPEGParam-->
<DarkEnhance/><!--opt, dark space enhancement parameters, see details in the message of XML_DarkEnhance-->
<Hdr/><!--opt, WDR (Wide Dynamic Range) parameters, see details in the message of XML_Hdr-->
<LSE/><!--opt, contrast enhancement parameters, see details in the message of XML_LSE-->
<MCE/><!--opt, memory color enhancement parameters, see details in the message of XML_MCE-->
<Svce/><!--opt, part contrast parameters, see details in the message of XML_Svce-->
<SectionCtrl/><!--opt, configuration parameters of picture exposure control by video segment, see details in the
message of XML_SectionCtrl-->
<AutoContrast/><!--opt, automatic contrast parameters, see details in the message of XML_AutoContrast-->
<GrayRange/><!--opt, grayscale range parameters, see details in the message of XML_GrayRange-->
<LSEDetail/><!--opt, contrast enhancement parameters, see details in the message of XML_LSE-->
<ITCImageSnap/><!--opt, captured picture parameters, see details in the message of XML_ITCImageSnap-->
<ImageRecord/><!--opt, image parameters in the recorded video, see details in the message of XML_ImageRecord-->
<Scene/><!--opt-->
<EPTZ/><!--opt-->
<EIS/><!--opt-->
<HLC/><!--opt-->
<ZoomLimit/><!--opt-->
<corridor/><!--opt, image rotate mode parameters, refer to the message XML_corridor for details-->
<Dehaze/><!--opt, defog mode parameters, refer to the message XML_De haze for details-->
<ImageMode/><!--opt, xs: string, image mode: "standard, indoor, outdoor, dimLight"-->
<enableImageLossDetection/><!--opt, xs: boolean--></enableImageLossDetection>
<CaptureMode/><!--opt, video input mode parameters, refer to the message XML_CaptureMode for details-->
<IrLight/><!--opt-->
<LensDistortionCorrection/><!--opt-->
<ExposureSync/><!--opt, exposure synchronization settings-->
<BrightnessSuddenChangeSuppression/><!--opt-->
<TempRange/><!--opt, temperature range, refer to the message XML_tempRange for details-->
<SupplementLight/><!-- opt -->
</ImageChannel>
```

XML_ImageChannellist

ImageChannellist message in XML format

```
<ImageChannellist version="2.0" xmlns="http://www.isapi.org/ver20/XMLSchema">
  <ImageChannel/><!--opt-->
</ImageChannellist>
```

See Also

XML_ImageChannel

XML_ImageFlip

ImageFlip message in XML format

```
<ImageFlip version="2.0" xmlns="http://www.isapi.org/ver20/XMLSchema">
  <enabled/><!--req, xs:boolean-->
  <ImageFlipStyle/><!--opt, xs:string, "LEFTRIGHT, UPDOWN, CENTER, AUTO", it can be enabled only when the value is
  "true"-->
  <flipAngle><!--opt, xs:string, "90, 180, 270"--></flipAngle>
</ImageFlip>
```

XML_ImageMode

ImageMode message in XML format

```
<ImageMode version="2.0" xmlns="http://www.isapi.org/ver20/XMLSchema">
  <type><!--req, xs:string, "standard, indoor, outdoor, dimLight"--></type>
  <recommendation><!--req, ro-->
    <brightnessLevel><!--opt, xs:integer, ranging from 0 to 100--></brightnessLevel>
    <contrastLevel><!--opt, xs:integer, ranging from 0 to 100--></contrastLevel>
    <sharpnessLevel><!--opt, xs:integer, ranging from 0 to 100--></sharpnessLevel>
    <saturationLevel><!--opt, xs:integer, ranging from 0 to 100--></saturationLevel>
    <hueLevel><!--opt, xs:integer, ranging from 0 to 100--></hueLevel>
    <deNoiseLevel><!--opt, xs:integer, ranging from 0 to 100--></deNoiseLevel>
  </recommendation>
</ImageMode>
```

XML_ImageModeList

ImageModeList message in XML format

```
<ImageModeList version="2.0" xmlns="http://www.isapi.org/ver20/XMLSchema">
  <ImageMode/><!--opt, see details in the message of XML_ImageMode-->
</ImageModeList>
```

See Also

XML_ImageMode

XML_InputProxyChannel

InputProxyChannel message in XML format

```
<InputProxyChannel version="2.0" xmlns="http://www.isapi.org/ver20/XMLSchema">
  <id><!--req, xs:string, starts from 1--></id>
  <name><!--opt, xs:string--></name>
  <sourceInputPortDescriptor><!--req-->
```

```
<adminProtocol><!--req, xs:string, "HIKVISION,SONY,ISAPI,ONVIF,..."--></adminProtocol>
<addressingFormatType><!--req, xs:string, "ipaddress,hostname"--></addressingFormatType>
<hostName><!--dep, xs:string, domain name--></hostName>
<ipAddress><!--dep, xs:string, IP address--></ipAddress>
<ipv6Address><!--dep, xs:string, IPv6 address--></ipv6Address>
<managePortNo><!--req, xs:integer--></managePortNo>
<srcInputPort><!--req, xs:string, channel No.--></srcInputPort>
<userName><!--req, xs:string, user name, which should be encrypted--></userName>
<password><!--req, wo, xs:string, password, which should be encrypted--></password>
<streamType><!--opt, xs:string, opt="auto,tcp,udp"--></streamType>
<deviceId><!--dep, xs:string--></deviceId>
<deviceTypeName><!--ro, opt, xs:string, device type name--></deviceTypeName>
<serialNumber><!--ro, opt, xs:string, device serial No.--></serialNumber>
<firmwareVersion><!--ro, opt, xs:string, firmware version--></firmwareVersion>
<firmwareCode><!--ro, opt, xs:string, firmware code--></firmwareCode>
</sourceInputPortDescriptor>
<enableAnr>
  <!--opt, xs:boolean, whether enables ANR funtion-->
</enableAnr>
<NVRInfo>
  <ipAddressNVR>
    <!--opt, xs:string, IP address of NVR-->
  </ipAddressNVR>
  <portNVR>
    <!--opt, xs:integer, port No. of NVR-->
  </portNVR>
  <ipcChannelNo>
    <!--opt, xs:integer, channel No. of the network camera in NVR-->
  </ipcChannelNo>
</NVRInfo>
</InputProxyChannel>
```

XML_InputProxyChannelList

InputProxyChannelList message in XML format

```
<InputProxyChannelList version="2.0" xmlns="http://www.isapi.org/ver20/XMLSchema">
  <InputProxyChannel/><!--opt, see details in
    XML_InputProxyChannel
  -->
</InputProxyChannelList>
```

XML_Cap_InputProxyChannelListCap

InputProxyChannelListCap capability message in XML format

```
<InputProxyChannelListCap version="2.0" xmlns="http://www.isapi.org/ver20/XMLSchema">
  <InputProxyChannel>
    <id min="" max=""/><!--req,xs:string,starts from 1-->
```

```
<name min="" max=""/><!--opt, xs:string-->
<sourceInputPortDescriptor><!--req-->
  <adminProtocol opt="HIKVISION,SONY,ISAPI,ONVIF,..."/><!--req, xs:string-->
  <addressingFormatType opt="ipaddress,hostname"/><!--req, xs:string-->
  <hostName min="" max=""/><!--dep, xs:string, domain name-->
  <ipAddress min="" max=""/><!--dep, xs:string, IP address-->
  <ipv6Address min="" max=""/><!--dep, xs:string, IPv6 address-->
  <managePortNo min="" max=""/><!--req, xs:integer, port number-->
  <srcInputPort min="" max=""/><!--req, xs:string, channel No.-->
  <userName min="" max=""/><!--req, xs:string-->
  <password min="" max=""/><!--req, wo, xs:string-->
  <streamType opt="auto,tcp,udp"/><!--opt, xs:string-->
  <deviceId min="" max=""/><!--dep, xs:string-->
</sourceInputPortDescriptor>
</InputProxyChannel>
</InputProxyChannelListCap>
```

XML_InputProxyChannelStatus

InputProxyChannelStatus message in XML format

```
<InputProxyChannelStatus version="1.0" xmlns="http://www.isapi.org/ver20/XMLSchema">
  <id><!--req, xs:string--></id>
  <sourceInputPortDescriptor/><!--req-->
  <online><!--req, xs:boolean, whether the camera is online--></online>
  <streamingProxyChannelIdList><!--req-->
    <streamingProxyChannelId>
      <!--req, xs:string, stream channel No., e.g., 101-main stream of channel 1, 102-sub-stream of channel 1-->
    </streamingProxyChannelId>
  </streamingProxyChannelIdList>
  <chanDetectResult>
    <!--opt, xs:string, network camera status: "connect"-connected, "overSysBandwidth"-insufficient bandwidth,
    "domainError"-incorrect domain name, "ipcStreamFail"-getting stream failed, "connecting", "chacnNoError"-incorrect
    channel No., "cipAddrConflictWithDev": IP address is conflicted with device address, "ipAddrConflicWithIpc"-IP
    address conflicted, "errorUserNameOrPasswd"-incorrect user name or password, "netUnreachable"-invalid network
    address, "unknownError"-unknown error, "notExist"-does not exist, "ipcStreamTypeNotSupport"-the stream
    transmission mode is not supported, "ipcResolutionNotSupport"-the resolution of network camera is not supported-->
  </chanDetectResult>
</InputProxyChannelStatus>
```

XML_InputProxyChannelStatusList

InputProxyChannelStatusList message in XML format

```
<InputProxyChannelStatusList version="2.0" xmlns="http://www.isapi.org/ver20/XMLSchema">
  <InputProxyChannelStatus/><!--opt, see details in XML_InputProxyChannelStatus-->
</InputProxyChannelStatusList>
```

See Also

XML_InputProxyChannelStatus

XML_IntelliCap

XML message about intelligent capability

```
<IntelliCap version="2.0" xmlns="http://www.isapi.org/ver20/XMLSchema">
  <isFaceSupport><!--optional, xs:boolean, whether it supports face detection--></isFaceSupport>
  <isBehaviorSupport><!--optional, xs:boolean, whether it supports behavior analysis--></isBehaviorSupport>
  <isLineDetectionSupport><!--optional, xs:boolean, whether it supports line crossing detection--></
isLineDetectionSupport>
  <isFieldDetectionSupport><!--optional, xs:boolean, whether it supports intrusion detection--></
isFieldDetectionSupport>
  <isRegionEntranceSupport><!--optional, xs:boolean, whether it supports region entrance detection--></
isRegionEntranceSupport>
  <isRegionExitingSupport><!--optional, xs:boolean, whether it supports region exiting detection--></
isRegionExitingSupport>
  <isLoiteringSupport><!--optional, xs:boolean, whether it supports loitering detection--></isLoiteringSupport>
  <isGroupSupport><!--optional, xs:boolean, whether it supports people gathering detection--></isGroupSupport>
  <isRapidMoveSupport><!--optional, xs:boolean, whether it supports fast moving detection--></
isRapidMoveSupport>
  <isParkingSupport><!--optional, xs:boolean, whether it supports parking detection--></isParkingSupport>
  <isUnattendedBaggageSupport><!--optional, xs:boolean, whether it supports unattended baggage detection--></
isUnattendedBaggageSupport>
  <isAttendedBaggageSupport><!--optional, xs:boolean, whether it supports object removal detection--></
isAttendedBaggageSupport>
  <isTeacherSupport><!--optional, xs:boolean, whether it supports teacher's behavior detection--></
isTeacherSupport>
  <isStudentSupport><!--optional, xs:boolean, whether it supports student's behavior detection--></
isStudentSupport>
  <isCombinedSupport><!--optional, xs:boolean, whether it supports combined detection--></isCombinedSupport>
  <isTrafficSupport><!--optional, xs:boolean, whether it supports traffic detection--></isTrafficSupport>
  <RestoreLib>
    <libName opt="studentsStoodUp,peopleCounting,shipDetection"><!--optional, xs:string,--></libName>
  </RestoreLib>
  <RestoreDefParamForbid>
    <!--optional, xs:boolean "true, false", forbid restoring algorithm library to default. When the node does not exist, it
indicates restoring to default is not supported. When the node exists and its value is "ture", it indicates support, when
the value is "false", it indicates not support-->
  </RestoreDefParamForbid>
  <RestoreAlgLibParam>
    <!--optional, the capabilities of restoring algorithm library parameters according to algorithm library name-->
    <libName opt="faceSnap,HMS,behaviorAnalysis"><!--optional, xs:string--></libName>
  </RestoreAlgLibParam>
  <isFaceCaptureStatisticsSupport><!--whether it supports face picture statistics--></isFaceCaptureStatisticsSupport>
  <isSupportPersonQueueDetection><!--whether it supports queue management--></
isSupportPersonQueueDetection>
  <isSupportIntersectionAnalysis>
    <!--optional, xs: boolean, whether it supports intersection analysis-->
```

```
</isSupportIntersectionAnalysis>
<mixedTargetDetectionWithoutAttribute><!--optional, xs: boolean--></mixedTargetDetectionWithoutAttribute>
<isSupportUploadFacePictureByForm><!--optional, xs:boolean, whether it supports uploading face pictures by form-->
</isSupportUploadFacePictureByForm>
<isSupportUploadFacePictureByUrl><!--optional, xs: boolean, whether it supports uploading face pictures by URL-->
</isSupportUploadFacePictureByUrl>
<isSupportUploadHumanPictureByForm><!--optional, xs:boolean, whether it supports uploading human pictures in
form--></isSupportUploadHumanPictureByForm>
<isSupportMisinfoFilterStatisticalMode><!--optional, xs:boolean, whether it supports statistics of filtering face
comparison--></isSupportMisinfoFilterStatisticalMode>
<isSupportFaceScore><!--optional, xs:boolean, whether it supports face grading configuration (camera)--></
isSupportFaceScore>
<HumanRecognitionModeSearchCap>
  <searchTargetsNumMax><!--optional, xs:integer, maximum number of sample pictures that can be imported for
searching by picture--></searchTargetsNumMax>
  <HumanMode>
    <searchCond opt="age_group,gender,jacet_color,glass,bag,ride,unlimit"/><!--setting multiple search conditions is
supported-->
    <similarity min="0.0" max="100.0"/><!--optional, xs:float, similarity, range: [0.0,100.0]-->
  </HumanMode>
  <HumanInfo><!--optional, person attribute information, it is required when searching for picture by attributes-->
    <age_group opt="child,infant,kid,middle,middleAged,old,prime,teenager,unknown,young">
      <!--optional, xs:string, age group: "child,infant,kid,middle,middleAged,old,prime,teenager,unknown,young"-->
    </age_group>
    <gender opt="female,male,unknown"><!--optional, xs:string, gender: "female,male,unknown"--></gender>
    <jacet_color
opt="black,black,blue,brown,cyan,deepBlue,deepGray,gray,green,mixture,orange,pink,purple,red,silver,unknown,whit
e,yellow">
      <!--optional, xs:string, tops color-->
    </jacet_color>
    <glass opt="yes,no,unknown,sunglasses"><!--optional, xs:string, whether the person is wearing glasses--></glass>
    <bag opt="yes,no,unknown"><!--optional, xs:string, whether the person is carrying a bag--></bag>
    <ride opt="yes,no,unknown"><!--optional, xs:string, whether the person is riding--></ride>
  </HumanInfo>
  <positive opt="true,false"/><!--optional, xs:boolean, whether it is false human body recognition alarm: "true"-yes,
"false"-no-->
  <eventType opt="unlimit,humanRecognition">
    <!--optional, xs:string, event type: "unlimit"-no limit, "humanRecognition"-human body detection alarm-->
  </eventType>
  <isSupportMultiChannelSearch>
    <!--optional, xs:boolean, whether it supports multi-channel search-->
  </isSupportMultiChannelSearch>
  <isSupportTotalSearchResult>
    <!--optional, xs:boolean, whether it supports limiting number of results that can be obtained after a single search-->
  </isSupportTotalSearchResult>
</HumanRecognitionModeSearchCap>
<VehicleRecognitionModeSearchCap>
  <searchTargetsNumMax><!--optional, xs:integer, maximum number of sample pictures that can be imported for
searching by picture--></searchTargetsNumMax>
  <eventType opt="unlimit,vehicleBlackList,vehicleWhiteList"/><!--xs:string, event type: "unlimit"-no limit,
"vehicleBlackList"-vehicle blacklist, "vehicleWhiteList"-vehicle whitelist-->
```



```
<VehicleMode>
  <searchCond opt="licensePlate,vehicleLogo,vehicleSubLogoRecog,vehicleType,vehicleColor,unlimit"/>
  <similarity min="0.0" max="100.0"/><!--optional, xs:float, similarity, range: [0.0,100.0]-->
</VehicleMode>
<isSupportMultiChannelSearch>
  <!--optional, xs:boolean, whether it supports multi-channel search-->
</isSupportMultiChannelSearch>
<isSupportTotalSearchResult>
  <!--optional, xs:boolean, whether it supports limiting number of results that can be obtained after a single search-->
>
  </isSupportTotalSearchResult>
</VehicleRecognitionModeSearchCap>
<FaceContrastAnalyzeCap><!--capability of manual comparison and analysis of face pictures-->
  <isSupportTrigger opt="false,true"><!--optional, xs:boolean--></isSupportTrigger>
  <SearchCondition><!--optional-->
    <startTime><!--required, xs:time, ISO8601 time--></startTime>
    <endTime><!--required, xs:time, ISO8601 time--></endTime>
  </SearchCondition>
</FaceContrastAnalyzeCap>
<FramesPeopleCountingCap><!--capability of people counting statistics in a single frame-->
  <Statistics>
    <SearchCondition>
      <startTime><!--required, xs:time,ISO8601 time--></startTime>
      <endTime><!--required, xs:time,ISO8601 time--></endTime>
    </SearchCondition>
  </Statistics>
  <MaskRegion><!--optional, shielded region-->
    <maxRegionNum><!--optional, xs:integer, number of regions--></maxRegionNum>
    <Region>
      <vertexNum min="3" max="10"><!--optional, xs:integer, number of region vertexes--></vertexNum>
    </Region>
  </MaskRegion>
</FramesPeopleCountingCap>
<FaceContrastPersonInfoExtend><!--configuration capability of face comparison tag-->
  <personInfoCap>
    <maxPersonInfo min="0" max="4">
      <!--optional, xs:integer, maximum number of person tags-->
    </maxPersonInfo>
  </personInfoCap>
  <personInfoFDlibCap>
    <maxPersonInfo min="0" max="4">
      <!--optional, xs:integer, maximum number of person tags-->
    </maxPersonInfo>
  </personInfoFDlibCap>
</FaceContrastPersonInfoExtend>
<isSupportSafetyHelmetDetection>
  <!-- opt, xs:boolean, whether it supports hard hat detection-->
</isSupportSafetyHelmetDetection>
<isSupportDistanceRecognition><!--optional, xs:boolean, whether supports distance measurement--></
isSupportDistanceRecognition>
  <isSupportATM><!--optional, xs:boolean, ATM intelligent configuraiton, corresponds to URI: /ISAPI/Intelligent/ATM/
capabilities?format=json--></isSupportATM>
```

```
<isSupportFaceTemperature><!--optional, xs:boolean, whether the device supports face thermography, corresponds
to URI: /ISAPI/Intelligent/faceTemperature/capabilities?format=json--></isSupportFaceTemperature>
<isSupportFireEscapeDetectionSearch>
  <!--optional, xs:boolean, whether the device supports searching for fire engine access detection events of multiple
channels (related URI: URI/ISAPI/Intelligent/fireEscapeDetection/search/capabilities?format=json)-->
</isSupportFireEscapeDetectionSearch>
<isSupportTakingElevatorDetectionSearch>
  <!--optional, xs:boolean, whether the device supports searching for elevator detection events of multiple channels
(related URI: /ISAPI/Intelligent/takingElevatorDetection/search/capabilities?format=json)-->
</isSupportTakingElevatorDetectionSearch>
<isSupportCityManagementSearch>
  <!--optional, xs:boolean, whether the device supports searching for city management events of multiple channels
(related URI: /ISAPI/Intelligent/cityManagement/search/capabilities?format=json)-->
</isSupportCityManagementSearch>
<isSupportSafetyHelmetDetectionSearch>
  <!--optional, xs:boolean, whether the device supports searching for hard hat detection events of multiple channels
(related URI: /ISAPI/Intelligent/safetyHelmetDetection/search/capabilities?format=json)-->
</isSupportSafetyHelmetDetectionSearch>
</IntelliCap>
```

XML_IOCap

IOCap capability message in XML format

```
<IOCap version="2.0" xmlns="http://www.isapi.org/ver20/XMLSchema">
  <IOInputPortNums>
    <!--opt, xs:integer-->
  </IOInputPortNums>
  <IOOutputPortNums>
    <!--opt, xs:integer-->
  </IOOutputPortNums>
  <isSupportStrobeLamp>
    <!--opt, xs:integer-->
  </isSupportStrobeLamp>
  <SoftIOInputPortNums>
    <!--opt, xs:integer-->
  </SoftIOInputPortNums>
  <isSupportIOOutputAdvanceParameter>
    <!--opt, xs:boolean, whether supports configuration of alarm input advanced parameters-->
  </isSupportIOOutputAdvanceParameter>
  <isSupportCombinationAlarm>
    <!--opt, xs:boolean, whether supports composite alarm-->
  </isSupportCombinationAlarm>
</IOCap>
```

XML_IOInputPort

IOInputPort message in XML format

```
<IOInputPort version="2.0" xmlns="http://www.isapi.org/ver20/XMLSchema">
  <id><!--required, xs:integer, IO port No.--></id>
  <enabled><!--required, xs:boolean, whether to enable: "true,false"--></enabled>
  <IODescriptor><!--optional, I/O port description-->
    <proxyProtocol>
      <!--required, string, access protocol, "HIKVISION, AXIS, PANASONIC, BOSCH, PELCO, SONY..."-->
    </proxyProtocol>
    <userName><!--required, string, user name--></userName>
    <addressingFormatType>
      <!--required, string, address type: "ipaddress,hostname"-->
    </addressingFormatType>
    <hostName><!--dep, xs:string--></hostName>
    <ipAddress><!--dep, xs:string--></ipAddress>
    <ipv6Address><!--dep, xs:string--></ipv6Address>
    <managePortNo><!--required, integer, manage port--></managePortNo>
    <innerIOPortID><!--required, integer, I/O port--></innerIOPortID>
  </IODescriptor>
  <triggering><!--required, xs:string, condition for triggering alarm output: "high"-high level, "low"-low level. Both the
high level and the low level can trigger continuously--></triggering>
  <name><!--optional, xs:string--></name>
  <IOUseType>
    <!--optional, xs:string, IO using method: "disable"-unused, "openDoor"-open the door, "doorStatus"-door status,
"custom"-custom-->
  </IOUseType>
  <inputType>
    <!--optional, xs:string, input mode: "switch", "semaphore"-->
  </inputType>
  <CombinationAlarm><!--optional, composite alarm information list-->
    <channel><!--required, xs:integer, channel No.--></channel>
    <EventTypeList><!--required, event type list-->
      <eventType><!--list, xs:string, event type: "VMD"-motion detection, "shelteralarm"-video tampering detection,
"facedetection"-face detection, "fielddetection"-intrusion detection, "linedetection"-line crossing detection,
"regionEntrance"-region entrance detection, "regionExitings"-region exiting detection, "loitering"-loitering detection,
"group"-people gathering detection, "rapidMove"-fast moving detection, "parking"-parking detection,
"unattendedBaggage"-unattended baggage detection, "attendedBaggage"-object removal detection, "ANPR",
"audioexception"-audio exception detection--></eventType>
    </EventTypeList>
  </CombinationAlarm>
</IOInputPort>
```

XML_IOInputPortList

IOInputPortList message in XML format

```
<IOInputPortList version="2.0" xmlns="http://www.isapi.org/ver20/XMLSchema">
  <IOInputPort/><!--opt, alarm input, see details in XML_IOInputPort-->
</IOInputPortList>
```

See Also

XML_IOInputPort

XML_IOOutputPort

JSON message about alarm output information

```
<IOOutputPort version="2.0" xmlns="http://www.isapi.org/ver20/XMLSchema">
  <id><!--required, xs:integer, "2"--></id>
  <PowerOnState>
    <!--required, output port configuration parameters when the device is powered on-->
    <defaultState>
      <!--read-only, required, xs:string, default output port signal when it is not triggered, "high,low"-->
    </defaultState>
    <outputState>
      <!--read-only, required, output port signal when it is being triggered, xs:string, "high,low,pulse"-->
    </outputState>
    <pulseDuration>
      <!--dependent, xs:integer, duration of a output port signal when it is being triggered, it is valid when outputState is
"pulse", unit: milliseconds -->
    </pulseDuration>
  </PowerOnState>
  <name><!--optional, xs:string--></name>
  <IOUseType><!--optional, xs:string, "disable,electricLock,custom"--></IOUseType>
  <normalStatus><!--optional, xs:string, normal status: open-remain open, close-remain closed--></normalStatus>
</IOOutputPort>
```

XML_IOOutputPortList

XML message about alarm output list

```
<IOOutputPortList version="2.0" xmlns="http://www.isapi.org/ver20/XMLSchema">
  <IOOutputPort/><!--optional, see details in XML_IOOutputPort-->
</IOOutputPort>
```

See Also

XML_IOOutputPort

XML_IOPortData

XML message about triggering parameters of alarm output

```
<IOPortData xmlns="http://www.isapi.org/ver20/XMLSchema">
  <outputState><!--required, xs:string, output level: "high, low"--></outputState>
</IOPortData>
```

XML_IOPortList

XML message about the alarm input and output information list.

```
<IOPortList version="2.0" xmlns="http://www.isapi.org/ver20/XMLSchema">
  <IOInputPortList/><!--optional, see details in XML_IOInputPortList-->
  <IOOutputPortList/><!--optional, see details in XML_IOOutputPortList-->
</IOPortList>
```

XML_IOPortStatus

XML message about alarm output status

```
<IOPortStatus version="2.0" xmlns="http://www.isapi.org/ver20/XMLSchema">
  <ioPortID><!--required, xs: integer, I/O No.: 1, 2--></ioPortID>
  <ioPortType><!--required, xs: string, I/O type: "input", "output"--></ioPortType>
  <ioState><!--required, xs: string, I/O status: "active", "inactive"--></ioState>
</IOPortStatus>
```

XML_IOPortStatusList

IOPortStatusList message in XML format

```
<IOPortStatusList version="2.0" xmlns="http://www.isapi.org/ver20/XMLSchema">
  <IOPortStatus><!--req-->
    <ioPortID><!--req, xs: integer, I/O No.: 1, 2--></ioPortID>
    <ioPortType><!--req, xs: string, I/O type: input, output--></ioPortType>
    <ioState><!-- req, xs: string, I/O status: active, inactive--></ioState>
  </IOPortStatus>
</IOPortStatusList>
```

XML_IOProxyInputPort

IOProxyInputPort message in XML format

```
<IOProxyInputPort version="2.0" xmlns="http://www.isapi.org/ver20/XMLSchema">
  <id><!--req, xs:string--></id>
  <enabled><!--req, xs:boolean--></enabled>
  <IODescriptor><!--req, xs:string, description about the IO port connected to the front-end device-->
    <proxyProtocol><!--req, xs:string, "HIKVISION, AXIS, PANASONIC, BOSCH, PELCO, SONY, ..."--></proxyProtocol>
    <userName><!--req, wo, xs:string --></userName>
    <password><!--req, wo, xs:string --></password>
    <addressingFormatType><!--dep, req, xs:enumeration, "ipaddress, hostname, ..."--></addressingFormatType>
    <hostName><!--dep, xs:string--></hostName>
    <ipAddress><!--dep, xs:string--></ipAddress>
    <ipv6Address><!--dep, xs:string--></ipv6Address>
```

```
<managePortNo><!--req, xs:integer--></managePortNo>
<innerIOPortID><!--req, xs:string, ID--></innerIOPortID>
</IODescriptor>
<triggering><!--req, xs:string, "high, low, rising, falling"--></triggering>
<name><!--opt, xs:string--></name>
<CombinationAlarm><!--opt, information list of the combined alarm-->
  <channel><!--req, xs:integer, channel No.--></channel>
  <EventTypeList><!--req, event type list-->
    <eventType>
      <!--list, xs:string, event type-->
    </eventType>
  </EventTypeList>
</CombinationAlarm>
</IOProxyInputPort>
```

XML_IOProxyInputPortList

IOProxyInputPortList message in XML format

```
<IOProxyInputPortList version="2.0" xmlns="http://www.isapi.org/ver20/XMLSchema">
  <IOProxyInputPort/><!--opt, input configuration of one digital channel, see details in the message of
XML_IOProxyInputPort-->
</IOProxyInputPortList>
```

See Also

XML_IOProxyInputPort

XML_IOProxyOutputPort

IOProxyOutputPort message in XML format

```
<IOProxyOutputPort version="2.0" xmlns="http://www.isapi.org/ver20/XMLSchema">
  <id><!--req, xs:string--></id>
  <IODescriptor/><!--req-->
  <PowerOnState><!--req-->
    <defaultState><!--req, xs:string, "high, low"--></defaultState>
    <outputState><!--req, xs:string, "high, low, pulse"--></outputState>
    <pulseDuration><!--dep, xs:integer, unit: milliseconds--></pulseDuration>
  </PowerOnState>
  <name><!--opt, xs:string--></name>
</IOProxyOutputPort>
```

XML_IOProxyOutputPortList

IOProxyOutputPortList message in XML format

```
<IOProxyOutputPortList version="2.0" xmlns="http://www.isapi.org/ver20/XMLSchema">
  <IOProxyOutputPort/><!--opt, output configuration of one digital channel, see details in the message of
XML_IOProxyOutputPort-->
</IOProxyOutputPortList>
```

See Also

XML_IOProxyOutputPort

XML_IOTTriggersCap

IOTTriggersCap message in XML format

```
<?xml version="1.0" encoding="utf-8"?>
<IOTTriggersCap version="2.0" xmlns="http://www.isapi.org/ver20/XMLSchema">
  <AccessController><!--opt, security control panel-->
    <XXTriggerCap><!--opt,xs: EventTriggerCapType--></XXTriggerCap>
  </AccessController>
  <VideoIntercom><!--opt, video intercom-->
    <XXTriggerCap><!--opt,xs: EventTriggerCapType --></XXTriggerCap>
  </VideoIntercom>
  <GJD><!--opt, GJD security control panel-->
    <XXTriggerCap><!--opt,xs: EventTriggerCapType--></XXTriggerCap>
  </GJD>
  <Luminite><!--opt, Luminite security control panel-->
    <XXTriggerCap><!--opt,xs: EventTriggerCapType--></XXTriggerCap>
  </Luminite>
  <OPTEx><!--opt, OPTEx security control panel-->
    <XXTriggerCap><!--opt,xs: EventTriggerCapType--></XXTriggerCap>
  </OPTEx>
  <CameraDetector><!--opt, detector-->
    <XXTriggerCap><!--opt,xs: EventTriggerCapType--></XXTriggerCap>
  </CameraDetector>
</IOTTriggersCap>
```

Remarks

The XX in the node **<XXTriggerCap>** corresponds to detailed event type. E.g., if the event type is humanRecognition, then the returned node is <HumanRecognitionTriggerCap>.

See Also

XML_EventTriggerCapType

XML_IPAddress

IPAddress message in XML format

```
<IPAddress version="2.0" xmlns="http://www.isapi.org/ver20/XMLSchema">
  <ipVersion><!--req, xs:string, "v4,v6,dual"--></ipVersion>
```

```
<addressingType><!--req, xs:string, "static,dynamic,api"--></addressingType>
<ipAddress><!--dep, xs:string, ipv4 address--></ipAddress>
<subnetMask><!--dep, xs:string, subnet mask for IPv4 address--></subnetMask>
<ipv6Address><!--dep, xs:string--></ipv6Address>
<bitMask><!--dep, xs:integer, bitmask IPv6 address--></bitMask>
<DefaultGateway><!--dep-->
  <ipAddress><!--dep, xs:string--></ipAddress>
  <ipv6Address><!--dep, xs:string--></ipv6Address>
</DefaultGateway>
<PrimaryDNS><!--dep-->
  <ipAddress><!--dep, xs:string--></ipAddress>
  <ipv6Address><!--dep, xs:string--></ipv6Address>
</PrimaryDNS>
<SecondaryDNS><!--dep-->
  <ipAddress><!--dep, xs:string--></ipAddress>
  <ipv6Address><!--dep, xs:string--></ipv6Address>
</SecondaryDNS>
<Ipv6Mode><!--opt-->
  <ipV6AddressingType><!--dep, xs:string, "router,ra,manual,dhcp"--></ipV6AddressingType>
  <ipv6AddressList>
    <v6Address>
      <id><!--dep, xs:string; id--></id>
      <type><!--dep, xs:string, "router,ra,manual,dhcp"--></type>
      <address><!--dep, xs:string--></address>
      <bitMask><!--dep, xs:integer--></bitMask>
    </v6Address>
  </ipv6AddressList>
</Ipv6Mode>
</IPAddress>
```

XML_IPFilter

IPFilter message in XML format

```
<IPFilter version="2.0" xmlns="http://www.isapi.org/ver20/XMLSchema">
  <enabled><!--req, xs:boolean--></enabled>
  <permissionType><!--opt, xs:string, "deny,allow"--></permissionType>
  <IPFilterAddressList size = "32"/><!--opt, the character size indicates the max. number of supported IP address. See
XML_IPFilterAddressList for details-->
</IPFilter>
```

See Also

XML_IPFilterAddressList

XML_IPFilterAddress

IPFilterAddress message in XML format


```
<IPFilterAddress version="2.0" xmlns="http://www.isapi.org/ver20/XMLSchema">
  <id><!--req, xs:string;id--></id>
  <permissionType><!--dep, xs:string, "deny,allow" --></permissionType>
  <addressFilterType><!--ro, xs:string, "mask, range"--></addressFilterType>
  <AddressRange><!--dep, it is valid when <addressFilterType> is "range"-->
    <startIPAddress><!--dep, xs:string--></startIPAddress>
    <endIPAddress><!--dep, xs:string--></endIPAddress>
    <startIPv6Address><!--dep, xs:string--></startIPv6Address>
    <endIPv6Address><!--dep, xs:string--></endIPv6Address>
  </AddressRange>
  <AddressMask><!--dep, it is valid when <addressFilterType> is "mask"-->
    <ipAddress><!--dep, xs:string--></ipAddress>
    <ipv6Address><!--dep, xs:string--></ipv6Address>
    <bitMask><!--opt, xs:string--></bitMask>
  </AddressMask>
</IPFilterAddress>
```

XML_IPFilterAddressList

IPFilterAddressList message in XML format

```
<IPFilterAddressList version="2.0" xmlns="http://www.isapi.org/ver20/XMLSchema">
  <IPFilterAddress/><!--opt, see XML_IPFilterAddress for details-->
</IPFilterAddressList>
```

See Also

XML_IPFilterAddress

XML_IrcutFilter

IrcutFilter message in XML format

```
<IrcutFilter version="2.0" xmlns="http://www.isapi.org/ver20/XMLSchema">
  <IrcutFilterType>
    <!--opt, xs: string, day/night auto switch mode: "auto, day, night, schedule, eventTrigger, darkFighterX,
darkFighterXAuto, darkFighterXSchedule"-->
  </IrcutFilterType>
  <dayToNightFilterLevel>
    <!--opt, xs: string, level of switching day to night: "low, normal, high"-->
  </dayToNightFilterLevel>
  <dayToNightFilterTime>
    <!--opt, xs: integer, time interval of switching day to night-->
  </dayToNightFilterTime>
  <nightToDayFilterLevel>
    <!--opt, xs: string, level of switching night to day: "low, normal, high"-->
  </nightToDayFilterLevel>
  <nightToDayFilterTime>
    <!--opt, xs: integer, time interval of switching night to day-->
```

```
</nightToDayFilterTime>
<Schedule><!--dep-->
  <scheduleType><!--req, xs: string, schedule type: "day,night"></scheduleType>
  <TimeRange><!--req, time period on schedule-->
    <beginTime><!--req, xs: time, which is in ISO8601 time format--></beginTime>
    <endTime><!--req, xs:time, which is in ISO8601 time format--></endTime>
  </TimeRange>
</Schedule>
<EventTrigger><!--dep-->
  <eventType><!--req, xs: string, event type: "IO,VMD"></eventType>
  <IrcutFilterAction><!--req, xs:string, "day,night"></IrcutFilterAction >
</EventTrigger>
</IrcutFilter>
```

XML_IrisData

IrisData message in XML format

```
<IrisData version="2.0" xmlns="http://www.isapi.org/ver20/XMLSchema">
  <iris>
    <!--req, xs: integer, a vector, whose value is the percentage of the maximum iris adjusting speed: negative value-
close iris; positive value-open iris-->
  </iris>
</IrisData>
```

XML_ISPMode

ISPMode message in XML format

```
<ISPMode version="2.0" xmlns="http://www.isapi.org/ver20/XMLSchema">
  <mode><!--opt, xs:string, "auto,schedule"--></mode>
  <Schedule><!--dep-->
    <scheduleType><!--req, xs:string, "day,night"--></scheduleType>
    <TimeRange><!--req-->
      <beginTime><!--req, xs:time, ISO8601 time--></beginTime>
      <endTime><!--req, xs:time, ISO8601 time--></endTime>
    </TimeRange>
  </Schedule>
</ISPMode>
```

XML_ItemList

ItemList message in XML format

```
<ItemList>
  <Item>
    <itemID>
```

```
<!--req, xs:string, item ID, which is between 1 and 15-->
</itemID>
<itemOrder>
  <!--req, xs:string, name element: "none", "devIp"-device IP address, "time", "buildUnitNo"-building No. and unit
  No., "outDoorDevNo"-door station, "unlockType"-unlocking type, "devName"-device name, "deviceNo"-device No.,
  "channelName"-channel name, "channelNo"-channel No., "plateNo"-license plate number, "plateColor"-license plate
  color, "laneNo"-lane No., "carSpeed"-vehicle speed, "positionInfo1"-camera 1, "pictureNo"-picture No., "CarNo"-
  vehicle No., "speedLimit"-speed limit, "illegalCode"-violation code, "siteNo"-intersection No., "directionNo"-direction
  No., "carColor"-vehicle color, "platePosition"-license plate coordinates, "carType"-vehicle type, "illegalType"-violation
  type, "custom"-->
  </itemOrder>
  <itemCustomStr>
    <!--req, xs:string, element custom string, which is between 1 and 32, unit: bytes. This node is valid only when
    <itemOrder> is "custom". Currently traffic cameras only support one custom name-->
    </itemCustomStr>
  </Item>
</ItemList>
```

XML_Language

Language message in XML format

```
<Language version="2.0" xmlns="http://www.isapi.org/ver20/XMLSchema">
  <type><!--req, xs: string, "GBK,EUC-KR", def="GBK"--></type>
</Language>
```

XML_LensDistortionCorrection

LensDistortionCorrection message in XML format

```
<PrivacyMaskRegion version="2.0" xmlns="http://www.isapi.org/ver20/XMLSchema">
  <id><!--req, xs: integer--></id>
  <enabled><!--req, xs: boolean--></enabled>
  <RegionCoordinatesList><!--req-->
    <RegionCoordinates><!--list-->
      <positionX><!--req, xs: integer; coordinate--></positionX>
      <positionY><!--req, xs: integer; coordinate--></positionY>
    </RegionCoordinates>
  </RegionCoordinatesList>
  <privacymaskName><!--opt, xs: string--></privacymaskName>
  <maskType>
    <!--opt, xs: string, "gray,red,yellow,blue,orange,green,transparent,half-transparent,mosaic"-->
  </maskType>
  <zoomdoorlimit><!--opt, xs: integer, the value is between 10 and 1000--></zoomdoorlimit>
</PrivacyMaskRegion>
```

XML_Link

Link message in XML format

```
<Link version="2.0" xmlns="http://www.isapi.org/ver20/XMLSchema">
  <MACAddress><!--req, xs:string--></MACAddress>
  <autoNegotiation><!--req, xs:boolean--></autoNegotiation>
  <speed><!--req, xs:integer, "10, 100, 1000"--></speed>
  <duplex><!--req, xs:string, "half, full"--></duplex>
  <MTU><!--req, xs:integer--></MTU>
</Link>
```

XML_ListMember

ListMember message in XML format

```
<ListMember version="2.0" xmlns="http://www.isapi.org/ver20/XMLSchema">
  <id><!-- req, xs:string--></id>
  <phoneNumber><!-- req, xs:string, cellphone number, which is sensitive information that needs to be encrypted--></phoneNumber>
  <SupportEntryList><!-- req -->
    <SupportEntry>
      <entry><!--req,xs:string,control types: "SMSAlarm"-message linkage, "SMSCtrl"-SMS control, "CallCtrl"-calling control--></entry>
      <enabled><!-- opt, xs:boolean,"true,false" --></enabled>
    </SupportEntry>
  </SupportEntryList>
  <SMSAlarmTypeList><!-- dep -->
    <SMSAlarmType>
      <type><!-- req, xs:string, "diskfull, diskerror, nicbroken, ipconflict, illaccess, AlarmInErr, tamper, vmd, wireless, pir, callhelp, AudioDetection, scenechangeDetection, defocusDetection, facedetection, LineDetection, FieldDetection, regionEntrance, regionExiting, loitering, group, rapidMove, parking, unattendedBaggage, attendedBaggage, dataPrealarm, all", event type that triggers message alarms, "all"-determine the event type that triggers message alarms based on event linkage actions, related URI (/ISAPI/Event/triggers/<ID>) --></type>
      <enabled><!-- opt, xs:boolean,"true,false" --></enabled>
    </SMSAlarmType>
  </SMSAlarmTypeList>
  <SMSCtrlTypeList>
    <SMSCtrlType>
      <type><!-- req,xs:string, message operations: "messageReboot"-reboot message, "dial"-enable/disable dial-up, "getDialStatus"-get dial-up status, "SMSAlarm"- receive/ reject message linkage--></type>
      <enabled><!-- opt, xs:boolean,"true,false"--></enabled>
    </SMSCtrlType>
  </SMSCtrlTypeList>
  <name><!-- opt, xs:string,contact name, which is sensitive information that needs to be encrypted--></name>
</ListMember>
```

XML_localPermission

localPermission message in XML format

```
<localPermission version="2.0" xmlns="http://www.isapi.org/ver20/XMLSchema">
  <backup><!--opt, xs:boolean--></backup>
  <record><!--opt, xs:boolean--></record>
  <playBack><!--opt, xs:boolean--></playBack>
  <preview><!--opt, xs:boolean--></preview>
  <videoChannelPermissionList><!--opt-->
    <videoChannelPermission><!--opt-->
      <id><!--req, corresponds to the video input channel ID--></id>
      <playBack><!--opt, xs:boolean--></playBack>
      <preview><!--opt, xs:boolean--></preview>
      <record><!--opt, xs:boolean--></record>
      <backup><!--opt, xs:boolean--></backup>
      <playBackDoubleVerification>
        <!--opt, xs:boolean, whether supports secondary authentication for playback-->
      </playBackDoubleVerification>
      <backupDoubleVerification>
        <!--opt, xs:boolean, whether supports secondary authentication for backup-->
      </backupDoubleVerification>
    </videoChannelPermission>
  </videoChannelPermissionList>
  <ptzControl>
    <!--req, xs:boolean-->
  </ptzControl>
  <ptzChannelPermissionList><!--opt-->
    <ptzChannelPermission><!--req-->
      <id><!--req, corresponds to PTZ channel ID--></id>
      <ptzControl><!--opt, xs:boolean--></ptzControl>
    </ptzChannelPermission>
  </ptzChannelPermissionList>
  <logOrStateCheck><!--opt, xs:boolean--></logOrStateCheck>
  <parameterConfig><!--opt, xs:boolean--></parameterConfig>
  <restartOrShutdown><!--opt, xs:boolean--></restartOrShutdown>
  <upgrade><!--opt, xs:boolean--></upgrade>
</localPermission>
```

XML_localPermissionCap

localPermissionCap capability message in XML format

```
<localPermissionCap version="2.0" xmlns="http://www.isapi.org/ver20/XMLSchema">
  <backup><!--opt, xs:boolean--></backup>
  <record><!--opt, xs:boolean--></record>
  <playBack><!--opt, xs:boolean--></playBack>
  <preview><!--opt, xs:boolean--></preview>
  <videoChannelPermissionList><!--opt-->
```

```
<videoChannelPermission><!--opt-->
  <id><!--req, corresponds to the video input channel ID--></id>
  <playBack><!--opt, xs:boolean--></playBack>
  <preview><!--opt, xs:boolean--></preview>
  <record><!--opt, xs:boolean--></record>
  <backup><!--opt, xs:boolean--></backup>
  <playBackDoubleVerification>
    <!--opt, xs:boolean, whether supports secondary authentication for playback-->
  </playBackDoubleVerification>
  <backupDoubleVerification>
    <!--opt, xs:boolean, whether supports secondary authentication for backup-->
  </backupDoubleVerification>
</videoChannelPermission>
</videoChannelPermissionList>
<ptzControl>
  <!--req, xs:boolean-->
</ptzControl>
<ptzChannelPermissionList><!--opt-->
  <ptzChannelPermission><!--req-->
    <id><!--req, corresponds to PTZ channel ID--></id>
    <ptzControl><!--opt, xs:boolean--></ptzControl>
  </ptzChannelPermission>
</ptzChannelPermissionList>
<logOrStateCheck><!--opt, xs:boolean--></logOrStateCheck>
<parameterConfig><!--opt, xs:boolean--></parameterConfig>
<restartOrShutdown><!--opt, xs:boolean--></restartOrShutdown>
<upgrade><!--opt, xs:boolean--></upgrade>
</localPermissionCap>
```

XML_LockByName

XML message about parameters of locking and unlocking video by file name

```
<?xml version="1.0" encoding="utf-8"?>
<LockByName version="2.0" xmlns="http://www.isapi.org/ver20/XMLSchema">
  <TrackId><!--required, xs:string, channel ID, e.g., 101 indicates the main stream of channel No.1--></TrackId>
  <recordName><!--required, xs:string, file name, the name length is between 1 and 100--></recordName>
  <command><!--required, xs:string, operation command: "lock" and "unlock"--></command>
  <duration>
    <!--dependent, xs:integer, locking duration, unit: second; if the value is 0, it indicates that the video will be
    permanently locked, this node is valid only when command is set to "lock"-->
  </duration>
  <recorderCode><!--optional, xs:string, body camera No.--></recorderCode>
  <policeCode><!--optional, xs:string, No. of police that collects data, it should be encrypted--></policeCode>
</LockByName>
```

XML_LockCap

XML message about video locking and unlocking capability

```
<?xml version="1.0" encoding="utf-8"?>
<LockCap version="2.0" xmlns="http://www.isapi.org/ver20/XMLSchema">
  <LockByName><!--optional, lock or unlock video by file name-->
    <TrackId><!--required, xs:string--></TrackId>
    <recordName min="1" max="100"><!--required, xs:string, video file name--></recordName>
    <command opt="lock,unlock"><!--required, xs:string, operation command: "lock" and "unlock"--></command>
    <duration min="" max=""><!--dependent, xs:integer, locking duration, unit: second--></duration>
    <recorderCode min="1" max="100"><!--optional, xs:string, body camera No.--></recorderCode>
    <policeCode min="1" max="100">
      <!--optional, xs:string, No. of police that collects data, it should be encrypted-->
    </policeCode>
  </LockByName>
  <LockByTime><!--optional, lock or unlock video by time-->
    <TrackId><!--required, xs:string--></TrackId>
    <recordType opt="ALL, MANUAL, CMR, MOTION, ALARM, EDR, ALARMANDMOTION, COMMAND, SMART">
      <!--required, xs:string-->
    </recordType>
    <startDateTime><!--dependent, xs:time, ISO8601 time format, e.g., 2004-05-03T17:30:08+08:00--></startDateTime>
    <endDateTime><!--dependent, xs:time, ISO8601 time format, e.g., 2004-05-03T17:30:08+08:00--></endDateTime>
    <command opt="lock,unlock"><!--required, xs:string--></command>
    <duration min="" max=""><!--dependent, xs:integer--></duration>
  </LockByTime>
</LockCap>
```

XML_LockPTZ

LockPTZ message in XML format

```
<LockPTZ version="2.0" xmlns="http://www.isapi.org/ver20/XMLSchema">
  <lockTime>
    <!--req, xs:integer, PTZ is unlocked when this node is set to 0, unit: second-->
  </lockTime>
</LockPTZ>
```

XML_LogConfig

LogConfig message in XML format.

```
<LogConfig version="2.0" xmlns="http://www.isapi.org/ver20/XMLSchema">
  <enabled><!--required, boolean type, whether to enable log--></enabled>
  <level>
    <!--required, string type, log types: "none,debug,info,error,fault,all", multiple type can be selected, and each type
    should be separated by comma-->
  </level>
</LogConfig>
```

```
</level>
</LogConfig>
```

XML_LogServer

XML message about log server information

```
<LogServer version="2.0" xmlns="http://www.std-cgi.org/ver20/XMLSchema">
  <enabled><!--required, xs: boolean, opt="true,false"--></enabled>
  <addressingFormatType>
    <!--required, xs: string, "ipaddress,hostname"-->
  </addressingFormatType>
  <hostName><!--dependent, xs: string--></hostName>
  <ipAddress><!--dependent, xs: string--></ipAddress>
  <ipv6Address><!--dependent, xs: string--></ipv6Address>
  <portNo><!--optional, xs: integer--></portNo>
  <transmissionEncryption>
    <!--optional, xs:boolean, whether to enable transmission encryption: "true"-yes, "false"-no (default). If this field is
not supported, the default encryption method is TLS-->
  </transmissionEncryption>
  <checkCertificate>
    <!--optional, xs:boolean,whether to enable checking certificates-->
  </checkCertificate>
  <uploadInterval>
    <!--optional, xs:integer, log uploading interval, which is 1 hour by default, range: [1,24], unit: hour-->
  </uploadInterval>
</LogServer>
```

XML_LogServerCap

XML message about log server capability

```
<LogServerCap version="2.0" xmlns="http://www.std-cgi.org/ver20/XMLSchema">
  <enabled opt="true,false"></enabled>
  <addressingFormatType opt="ipaddress,hostname">
    <!--required, xs:string,"ipaddress,hostname"-->
  </addressingFormatType>
  <hostName min="" max="">
    <!--dependent, xs:string-->
  </hostName>
  <ipAddress min="" max=""><!--dep, xs:string--></ipAddress>
  <ipv6Address min="" max=""><!--dep, xs:string--></ipv6Address>
  <portNo min="" max=""><!--opt, xs:integer--></portNo>
  <transmissionEncryption opt="true,false">
    <!--optional, xs:boolean, whether to enable transmission encryption: "true"-yes, "false"-no (default). If this field is
not supported, the default encryption method is TLS-->
  </transmissionEncryption>
  <checkCertificate opt="true,false">
    <!--optional, xs:boolean,whether to enable checking certificates-->
  </checkCertificate>
</LogServerCap>
```



```
</checkCertificate>
<uploadInterval min="1" max="24" def="1">
  <!--optional, xs:integer, log uploading interval, which is 1 hour by default, range: [1,24], unit: hour-->
</uploadInterval>
</LogServerCap>
```

XML_LogServerTestDescription

LogServerTestDescription message in XML format

```
<LogServerTestDescription>
  <addressingFormatType>
    <!--required, xs:string, "ipaddress,hostname"-->
  </addressingFormatType>
  <hostName><!--dependent, xs:string--></hostName>
  <ipAddress><!--dependent, xs:string--></ipAddress>
  <ipv6Address><!--dependent, xs:string--></ipv6Address>
  <portNo><!--optional, xs:integer--></portNo>
  <transmissionEncryption><!--optional, xs:boolean, whether to enable transmission encryption, and it is disabled by
  default. If this field is not supported, the default encryption method used by the device is TLS--></
transmissionEncryption>
  <checkCertificate><!--optional, xs:boolean, whether to enable checking certificate--></checkCertificate>
  <uploadInterval><!--optional, xs:integer, log upload interval, which is 1 hour by default, range: [1,24], unit: hour--></
uploadInterval>
</LogServerTestDescription>
```

XML_Log_CMSearchDescription

XML message about log search condition

```
<CMSearchDescription version="2.0" xmlns="http://www.isapi.org/ver20/XMLSchema">
  <searchID>
    <!--required, xs:string, search ID, e.g., {812F04E0-4089-11A3-9A0C-0305E82C2906}-->
  </searchID>
  <trackIDList>
    <trackID>
      <!--required, xs:integer, ID, XX01-search for the main stream video of channel XX, XX03-search for the pictures of
  channel XX-->
    </trackID>
  </trackIDList>
  <metaId>
    <!--required, xs:string, format: log.std-cgi.com/<majorType>/<minorType>; <majorType> represents the major log
  type and <minorType> is the minor log type, see details in Log Type for ISAPI-->
  </metaId>
  <timeSpanList>
    <timeSpan>
      <startTime><!--required, xs:time, ISO8601 time, start time of search, e.g.: 2017-08-02T00:00:00Z--></startTime>
      <endTime><!--required, xs:time, ISO8601 time, end time of search, e.g.: 2017-08-02T23:59:59Z--></endTime>
    </timeSpan>
  </timeSpanList>
</CMSearchDescription>
```

```
</timeSpanList>
<searchResultPostion><!--required, xs:integer, start position of search--></searchResultPostion>
<maxResults><!--optional, xs:integer, maximum number of returned results--></maxResults>
</CMSearchDescription>
```

See Also

Log Types for ISAPI

XML_Log_CMSearchResult

XML message about log search results

```
<CMSearchResult version="2.0" xmlns="http://www.isapi.com/ver20/XMLSchema">
  <searchID><!--required, xs:string, search ID--></searchID>
  <responseStatus><!--required, xs:string, search status--></responseStatus>
  <responseStatusStrg><!--required, xs:string, search status string: Ok, NO MATCHES, More--></responseStatusStrg>
  <totalMatches><!--optional, xs:integer, total number of matched records--></totalMatches>
  <numOfMatches><!--optional, xs:integer, number of returned records--></numOfMatches>
  <matchList>
    <searchMatchItem>
      <logDescriptor>
        <metald><!--required, xs:string, format: log.std-cgi.com/<majorType>/<minorType>; <majorType> represents the
major log type and <minorType> is the minor log type, see details in Log Type for ISAPI--></metald>
        <StartDateTime><!--required, xs:time, ISO8601 time--></StartDateTime>
        <localID><!--optional, xs:string, channel No.--></localID>
        <paraType><!--optional, xs:string--></paraType>
        <userName><!--optional, xs:string--></userName>
        <logInfo><!--dependent, log details-->
          <OpenDoorRecord>
            <type><!--required, xs: string, unlocking type: password, hijack (unlock under duress), card, resident, center--
--></type>
          </OpenDoorRecord>
          <VisAlarmRecord>
            <type>
              <!--required, xs: string, alarm category: zone (zone alarm), dismantle (tampering alarm), hijack (duress alarm),
passwordErr (wrong password alarm), doorNotOpen-(door closed alarm), doorNotClose (door open alarm), SOS,
callReq (call request alarm), smartLockHijackFingerPrint (fingerprint duress alarm), smartLockHijackPassword
(password duress alarm), smartLockBreaking (forced-open door alarm), smartLockBeLocked (door locked alarm),
smartLockLowBattery (low battery alarm)-->
            </type>
          </VisAlarmRecord>
        </logInfo>
        <ipAddress><!--optional, xs:string--></ipAddress>
        <object><!--optional, xs:string, "network", "keypad", "remoteCtrl"-keyfob, "card"--></object>
        <params><!--optional, xs:string, parameters, such as zone No. and so on--></params>
        <seq><!--optional, xs:string, serial No.--></seq>
        <additionInformation><!--optional, xs:string, additional information--></additionInformation>
      </logDescriptor>
    </searchMatchItem>
```

```
</matchList>
</CMSearchResult>
```

See Also

Log Types for ISAPI

XML_MACFilter

MACFilter message in XML format

```
<MACFilter version="2.0" xmlns="http://www.isapi.org/ver20/XMLSchema">
  <enabled><!--req, xs:boolean--></enabled>
  <permissionType><!--req, xs:string,"deny, allow"--></permissionType>
  <MACFilterAddressList>
    <MACFilterAddress>
      <id><!--req, xs:string, id--></id>
      <MACAddress><!--req, xs:string--></MACAddress>
    </MACFilterAddress>
  </MACFilterAddressList>
</MACFilter>
```

XML_mailing

mailing message in XML format.

```
<mailing version="2.0" xmlns="http://www.isapi.org/ver20/XMLSchema">
  <id><!--req, xs:string--></id>
  <enabled><!--opt, xs:boolean--></enabled>
  <sender><!--req-->
    <name><!--req, xs:string--></name>
    <emailAddress><!--req, xs:string--></emailAddress>
    <smtp><!--req-->
      <enableAuthorization><!--req, xs:boolean--></enableAuthorization>
      <enableSSL><!--opt, xs:boolean--></enableSSL>
      <addressingFormatType>
        <!--req, xs:string,"ipaddress,hostname"-->
      </addressingFormatType>
      <hostName><!--dep, xs:string--></hostName>
      <ipAddress><!--dep, xs:string--></ipAddress>
      <ipv6Address><!--dep, xs:string--></ipv6Address>
      <portNo><!--opt, xs:integer--></portNo>
      <accountName><!--dep, xs:string--></accountName>
      <password><!--dep, xs:string--></password>
      <enableTLS><!--opt, xs:boolean--></enableTLS>
      <startTLS><!--dep, xs:boolean--></startTLS>
    </smtp>
  </sender>
  <receiverList><!--req-->
    <receiver><!--req-->
```

```
<id><!--req, xs:string--></id>
<name><!--req, xs:string--></name>
<emailAddress><!--req,xs:string--></emailAddress>
</receiver>
</receiverList>
<attachment><!--opt-->
  <snapshot><!--opt-->
    <enabled><!--req, xs:boolean--></enabled>
    <interval><!--req, xs:integer, unit: second--></interval>
  </snapshot>
</attachment>
</mailing>
```

XML_mailingList

mailingList message in XML format

```
<mailingList version="2.0" xmlns="http://www.isapi.org/ver20/XMLSchema">
  <mailing><!--opt,xs:string--></mailing>
</mailingList>
```

XML_mailingTestResult

mailingTestResult message in XML format

```
<mailingTestResult version="2.0" xmlns="http://www.isapi.org/ver20/XMLSchema">
  <errorDescription><!--req, xs:string--></errorDescription>
</mailingTestResult>
```

XML_mailingTestDescription

mailingTestDescription message in XML format.

```
<mailingTestDescription version="2.0" xmlns="http://www.isapi.org/ver20/XMLSchema">
  <sendName><!--opt, xs:string--></sendName>
  <sendEmailAddress><!--req, xs:string--></sendEmailAddress>
  <addressingFormatType>
    <!--req, xs:string, "ipaddress,hostname"-->
  </addressingFormatType>
  <hostName><!--dep, xs:string--></hostName>
  <ipAddress><!--dep, xs:string--></ipAddress>
  <ipv6Address><!--dep, xs:string--></ipv6Address>
  <portNo><!--req, xs:integer--></portNo>
  <enableSSL><!--opt, xs:boolean--></enableSSL>
  <enableAuthorization><!--req, xs:boolean--></enableAuthorization>
  <accountName><!--dep, xs:string--></accountName>
  <password><!--dep, xs:string--></password>
```

```
<receiverList><!--req-->
  <receiver><!--req-->
    <id><!--req, xs:string--></id>
    <name><!--req, xs:string--></name>
    <emailAddress><!--req, xs:string--></emailAddress>
  </receiver>
</receiverList>
</mailingTestDescription>
```

XML_MasterSlaveTrackingCap

MasterSlaveTrackingCap capability message in XML format

```
<MasterSlaveTrackingCap version="2.0" xmlns="http://www.isapi.org/ver20/XMLSchema">
  <isSupportSlaveCameraCfg>
    <!--optional, xs:boolean, whether to support PTZ camera configuration-->
  </isSupportSlaveCameraCfg>
  <isSupportSlaveCameraStatus>
    <!--optional, xs:boolean, whether to support PTZ camera status configuration-->
  </isSupportSlaveCameraStatus>
  <isSupportSlaveCameraTrackingRatio>
    <!--optional, xs:boolean, whether to support PTZ camera zoom ratio configuration-->
  </isSupportSlaveCameraTrackingRatio>
  <isSupportSlaveCameraTracking>
    <!--optional, xs:boolean, whether to support PTZ camera tracking-->
  </isSupportSlaveCameraTracking>
  <isSupportLinkedTracking>
    <!--optional, xs:boolean, whether device supports advanced parameters configuration of linked tracking (related
URI: /ISAPI/MasterSlaveTracking/linkedTracking?format=json)-->
  </isSupportLinkedTracking>
</MasterSlaveTrackingCap>
```

XML_MaxElevation

MaxElevation message in XML format

```
<MaxElevation version="1.0" xmlns="http://www.isapi.org/ver20/XMLSchema">
  <mElevation><!--req, xs:integer, the lower limit of max. tilt-angle--></mElevation>
</MaxElevation>
```

XML_messageConfig

messageConfig message in XML format

```
<messageConfig version="1.0" xmlns="http://www.isapi.org/ver20/XMLSchema">
  <enabled/> <!-- req, xs:boolean,"true,false" -->
```

```
<SMSWhiteList/>
</messageConfig>
```

XML_messageConfigCap

messageConfigCap message in XML format

```
<messageConfigCap version="2.0" xmlns="http://www.isapi.org/ver20/XMLSchema">
  <supportEntry opt="SMSAlarm,SMSCtrl,CallCtrl">
    <!--req,xs:string,control types: "SMSAlarm"-message linkage, "SMSCtrl"-SMS control, "CallCtrl"-calling control-->
  </supportEntry>
  <SMSAlarmType
    opt="diskfull,diskerror,nicbroken,ipconflict,illaccess,AlarmInErr,tamper,vmd,wireless,pir,callhelp,AudioDetection,scene
    changeDetection,defocusDetection,facedetection,LineDetection,FieldDetection,regionEntrance,regionExiting,loitering,
    group,rapidMove,parking,unattendedBaggage,attendedBaggage,dataPrealarm,all"/>
    <!--req, event type that triggers message alarms, "all"-determine the event type that triggers message alarms based
    on event linkage actions related URL: /ISAPI/Event/triggers/<ID>-->
  <SMSCtrlType opt="messageReboot,dial,getDialStatus,SMSAlarm"/>
    <!-- req,xs:string, message operations: "messageReboot"-reboot message, "dial"-enable/disable dial-up,
    "getDialStatus"-get dial-up status, "SMSAlarm"- receive/ reject message linkage-->
  <name min="0" max="32"><!--opt, xs:string, contact name, which is sensitive information that needs to be
    encrypted--></name>
  <maxListMemberNum><!--opt, xs:integer,max. number of SMS contacts, which is 8 by default--></
    maxListMemberNum>
</messageConfigCap>
```

XML_MotionDetection

XML message about motion detection parameters

```
<MotionDetection version="2.0" xmlns="http://www.isapi.org/ver20/XMLSchema">
  <enabled><!--required, xs:boolean, whether to enable motion detection--></enabled>
  <enableHighlight><!--optional, xs:boolean, whether to enable highlight--></enableHighlight>
  <samplingInterval><!--optional, xs:integer, number of frames--></samplingInterval>
  <startTriggerTime><!--optional, xs:integer, unit: millisecond--></startTriggerTime>
  <endTriggerTime><!--opt, xs:integer, unit: millisecond--></endTriggerTime>
  <regionType><!--required, read-only, xs:string, region type: "grid", "roi", "none", "region"--></regionType>
  <Grid><!--dependency-->
    <rowGranularity><!--required, read-only, xs:integer--></rowGranularity>
    <columnGranularity><!--required, read-only, xs:integer--></columnGranularity>
  </Grid>
  <ROI><!--dependency-->
    <normalizedScreenWidth><!--required, read-only, xs:integer--></normalizedScreenWidth>
    <normalizedScreenHeight><!--required, read-only, xs:integer--></normalizedScreenHeight>
  </ROI>
  <MotionDetectionLayout><!--required, rule region configuration of motion detection-->
    <sensitivityLevel><!--required, xs:integer, sensitivity, which is between 0 and 100--></sensitivityLevel>
    <layout>
      <gridMap>
```

```
<!--dependent, xs:string, the image is divided in to N x N grids, each grid can be enabled (value: 1) or disabled (value 0) motion detection, each four horizontal grids consist of a hexadecimal number, e.g., the corresponding hexadecimal number of "0011" is 3-->
</gridMap>
<roiMap><!--dependent,xs:string--></roiMap>
<RegionList size="4"><!--this node is available when the regionType is set to "region"-->
  <Region><!--list-->
    <id><!--required, xs:string--></id>
    <RegionCoordinatesList><!--opt-->
      <RegionCoordinates><!--list-->
        <positionX><!--required, xs:integer; X-coordinate--></positionX>
        <positionY><!--required, xs:integer; Y-coordinate--></positionY>
      </RegionCoordinates>
    </RegionCoordinatesList>
  </Region>
</RegionList>
</layout>
<targetType opt="human,vehicle"><!--optional, xs:string, target type, "human", "vehicle"--></targetType>
</MotionDetectionLayout>
<enableWithMoving><!--optional, xs:boolean--></enableWithMoving>
</MotionDetection>
```

XML_MotionDetectionGridLayout

XML message about grid layout parameters of motion detection

```
<?xml version="1.0" encoding="utf-8"?>
<MotionDetectionGridLayout version="2.0" xmlns="http://www.isapi.org/ver20/XMLSchema">
  <sensitivityLevel><!--required, xs:integer, sensitivity, which is between 0 and 100--></sensitivityLevel>
  <layout>
    <gridMap>
      <!--dependent, xs:string, the image is divided in to N x N grids, each grid can be enabled (value: 1) or disabled (value 0) motion detection, each four horizontal grids consist of a hexadecimal number, e.g., the corresponding hexadecimal number of "0011" is 3-->
    </gridMap>
    <roiMap><!--dep,xs:string--></roiMap>
    <RegionList size="4"><!--this node is available when the regionType is set to "region"-->
      <Region><!--list-->
        <id><!--required, xs:string--></id>
        <RegionCoordinatesList><!--opt-->
          <RegionCoordinates><!--list-->
            <positionX><!--required, xs:integer; X-coordinate--></positionX>
            <positionY><!--required, xs:integer; Y-coordinate--></positionY>
          </RegionCoordinates>
        </RegionCoordinatesList>
      </Region>
    </RegionList>
  </layout>
  <targetType opt="human,vehicle"><!--optional, xs:string, target type, "human", "vehicle"--></targetType>
</MotionDetectionGridLayout>
```

XML_MotionDetectionLayout

XML message about rule region parameters of motion detection

```
<?xml version="1.0" encoding="utf-8"?>
<MotionDetectionLayout version="2.0" xmlns="http://www.isapi.org/ver20/XMLSchema">
  <sensitivityLevel><!--required, xs:integer, sensitivity, which is between 0 and 100--></sensitivityLevel>
  <layout>
    <gridMap>
      <!--dependent, xs:string, the image is divided in to N x N grids, each grid can be enabled (value: 1) or disabled
      (value 0) motion detection, each four horizontal grids consist of a hexadecimal number, e.g., the corresponding
      hexadecimal number of "0011" is 3-->
    </gridMap>
    <roiMap><!--dependent,xs:string--></roiMap>
    <RegionList size="4"><!--this node is available when the regionType is set to "region"-->
      <Region><!--list-->
        <id><!--required, xs:string--></id>
        <RegionCoordinatesList><!--opt-->
          <RegionCoordinates><!--list-->
            <positionX><!--required, xs:integer; X-coordinate--></positionX>
            <positionY><!--required, xs:integer; Y-coordinate--></positionY>
          </RegionCoordinates>
        </RegionCoordinatesList>
      </Region>
    </RegionList>
  </layout>
  <targetType opt="human,vehicle"><!--optional, xs:string, target type, "human", "vehicle"--></targetType>
</MotionDetectionLayout>
```

XML_MotionDetectionScheduleList

XML message about arming schedules of motion detection

```
<?xml version="1.0" encoding="utf-8"?>
<MotionDetectionScheduleList version="2.0" xmlns="http://www.isapi.org/ver20/XMLSchema">
  <Schedule/><!--see details in the message XML_Schedule-->
</MotionDetectionScheduleList>
```

See Also

XML_Schedule

XML_MountingScenario

MountingScenario message in XML format

```
<MountingScenario version="2.0" xmlns="http://www.isapi.org/ver20/XMLSchema">
  <mode>
```



```
<!--req, xs:string,
"indoor,outdoor,day,night,morning,nightfall,mode1,mode2,mode3,mode4,highway,road2,faceSnap,backlight,frontlight
", "lowIllumination"-low illumination, "backlight"-back light, "frontlight"-front light, "faceSnap"-face picture capture-->
</mode>
</MountingScenario>
```

XML_MountList

MountList message in XML format

```
<MountList version="1.0" xmlns="http://www.isapi.com/ver20/XMLSchema">
  <Mount>
    <id><!--req, xs: integer, storage node ID--></id>
    <path><!--req, xs: string, saving path--></path>
    <dir><!--req, xs: string--></dir>
    <size><!--req, xs: string,, storage size--></size>
    <descr><!--req, xs: string, storage description--></descr>
  </Mount>
</MountList>
```

XML_Manual_PTZData

XML message about manual PTZ control parameters

```
<?xml version="1.0" encoding="utf-8"?>
<PTZData version="2.0" xmlns="http://www.isapi.org/ver20/XMLSchema">
  <pan><!--optional, xs:integer, panning range: [-100,100]--></pan>
  <tilt><!--optional, xs:integer, tilting range: [-100,100]--></tilt>
  <zoom><!--optional, xs:integer, zooming range: [-100,100]--></zoom>
</PTZData>
```

XML_NetworkCap

NetworkCap capability message in XML format

```
<NetworkCap version="2.0" xmlns="http://www.isapi.org/ver20/XMLSchema">
  <isSupportWireless>
    <!--req, xs:boolean, whether it supports accessing via wireless network, "true"-yes, "false"-no-->
  </isSupportWireless>
  <isSupportWAN><!--optional, xs:boolean, whether it supports WAN--></isSupportWAN>
  <isSupportPPPoE>
    <!--req, xs:boolean, whether it supports PPPoE, "true"-yes, "false"-no-->
  </isSupportPPPoE>
  <isSupportBond>
    <!--req, xs:boolean, whether it supports NIC bonding, "true"-yes, "false"-no-->
  </isSupportBond>
  <isSupport802_1x>
```

```
<!--req, xs:boolean, whether it supports 802_1x-->
</isSupport802_1x>
<isSupportNtp>
  <!--optional, xs:boolean, whether it supports NTP-->
</ isSupportNtp>
<isSupportFtp>
  <!--optional, xs:boolean, whether it supports FTP-->
</isSupportFtp>
<isSupportUpnp>
  <!--optional, xs:boolean, whether it supports UPnP-->
</isSupportUpnp>
<isSupportPNP>
  <!--optional, xs:boolean, whether it supports PnP-->
</isSupportPNP>
<isSupportDdns>
  <!--optional, xs:boolean, whether it supports DNS-->
</isSupportDdns>
<isSupportHttps>
  <!--optional, xs:boolean, whether it supports HTTPS-->
</isSupportHttps>
<SnmpCap><!--optional, whether it supports SNMP-->
  <isSupport><!--req, xs:boolean--></isSupport>
</SnmpCap>
<isSupportExtNetCfg>
  <!--optional, xs:boolean, whether it supports configuring extended network parameters-->
</isSupportExtNetCfg>
<isSupportIPFilter>
  <!--optional, xs:boolean, whether it supports filtering IP address-->
</isSupportIPFilter>
<isSupportSSH opt="true"><!--optional, xs:boolean, whether it supports SSH--></isSupportSSH>
<isSupportNetPreviewStrategy><!--optional, xs:boolean--></isSupportNetPreviewStrategy>
<isSupportEZVIZ>
  <!--optional, xs:boolean, whether it supports Hik-Connect-->
</isSupportEZVIZ>
<isSupportEhome>
  <!--optional, xs:boolean, whether it supports EHome (ISUP) protocol-->
</isSupportEhome>
<isSupportWirelessServer>
  <!--optional, xs:boolean, whether it supports Wi-Fi hotspot-->
</isSupportWirelessServer>
<isSupportWPS><!--optional, xs:boolean, whether it supports (WPS) Wi-Fi Protected Setup--></isSupportWPS>
<isWirelessMutexWithWirelessServer><!--optional, xs:boolean, whether it supports configuring the wireless access
to be mutually exclusive with the wireless service--></isWirelessMutexWithWirelessServer>
<isSupportWirelessDial>
  <!--optional, xs:boolean, whether it supports wireless dial configuration-->
</isSupportWirelessDial>
<WPS><!--optional, Wi-Fi Protected Setup configuration-->
  <NetworkInterfaceList size="2">
    <NetworkInterface>
      <id><!--req, xs:string, NIC ID--></id>
      <enabled><!--req, xs:boolean, whether the NIC is enabled--></enabled>
      <isSupportAutoConnect><!--optional, xs:boolean--></isSupportAutoConnect>
```

```
<isSupportDevicePinCode>
  <!--optional, xs:boolean, whether it supports device PIN code-->
</isSupportDevicePinCode>
<isSupportDevicePinCodeUpdate>
  <!--optional, xs:boolean, whether it supports updating device PIN code-->
</isSupportDevicePinCodeUpdate>
<ApPinCode><!--opt-->
  <ssid min="" max="">
    <!--optional, xs:string, maximum and minimum SSID length that can be returned by device-->
  </ssid>
  <pinCode min="" max="">
    <!--optional, xs:string, maximum and minimum PIN code length that can be returned by device-->
  </pinCode>
</ApPinCode>
</NetworkInterface>
</NetworkInterfaceList>
</WPS>
<isSupportMACFilter>
  <!--optional, xs:boolean, whether it supports filtering MAC address-->
</isSupportMACFilter>
<verificationCode max="">
  <!--optional, xs:string, the maximum length of verificationCode that can be returned by device-->
</verificationCode>
<WPSCap><!--opt-->
  <isSupport><!--req, xs: boolean--></isSupport>
  <isSupportAutoConnect><!--req, xs: boolean--></isSupportAutoConnect>
</WPSCap>
<NetWorkMode>
  <workMode><!--optional, xs:string, network mode: "close,wifi,wifiAp"--></workMode>
</NetWorkMode>
<VerificationCodeModification><!--optional, xs:string, whether the verification code can be edited by the admin
user-->
  <verificationCodeType opt="normal,empty"></verificationCodeType>
  <isSupportDeclarationURL><!--optional, xs:boolean, whether it supports URL declared by the service--></
isSupportDeclarationURL>
  <isSupportPrivacyPolicyURL><!--optional, xs:boolean, whether it supports the privacy policy URL--></
isSupportPrivacyPolicyURL>
  <verificationCodeModify opt="true,false">
    <!--optional, whether the verification code is edited: "true"-yes, "false"-no, no return-not support-->
  </verificationCodeModify>
  <Hyperlinks><!--opt-->
    <declarationURL><!--optional, xs:string--></declarationURL>
    <privacyPolicyURL><!--optional, xs:string--></privacyPolicyURL>
  </Hyperlinks>
  <isSupportVerificationCodeCheck>
    <!--optional, xs: boolean, whether it supports verifying and configuring the verification code, true=yes, if this node
is not returned or the value of the returned node is false, it indicates that not support-->
  </isSupportVerificationCodeCheck>
  <isSupportOldVerificationCode><!--optional, xs:boolean, whether it supports old EZVIZ password configuration. The
old password contains six uppercase letters--></isSupportOldVerificationCode>
</VerificationCodeModification>
<EZVIZSecretKey>
```

```

<!--optional, whether it supports capability of editing verification code for Hik-Connect-->
<offlineStatus>
  <!--ro,dep,xs:string; it is valid when registerStatus values "false", device offline status, opt="secretKeyInvalid"-
invalid verification code-->
</offlineStatus>
  <secretKey min="0" max="64"><!--optional, xs:string, verification code for Hik-Connect--></secretKey>
</EZVIZSecretKey>
<isSupportplatformAccess><!--optional, xs:boolean, capability of accessing the platform, whether it supports filtering
IP addresses that access to the platform--></isSupportplatformAccess>
<isSupportIntegrate><!--optional, xs:boolean--></isSupportIntegrate>
<isSupportIntelligentBoost><!--optional, xs:boolean, whether it supports bandwidth adaption--></
isSupportIntelligentBoost>
  <isSupportWebSocket><!--optional, xs:boolean--></isSupportWebSocket>
  <isSupportWebSocketS><!--optional, xs:boolean--></isSupportWebSocketS>
  <isSupportHostName>
    <!--optional, xs:boolean, whether it supports host name, related URI: /ISAPI/System/Network/interfaces/<ID>/
hostName/capabilities?format=json-->
  </isSupportHostName>
  <isSupportResourceStatistics><!--optional, xs:boolean, whether supports network resource information--></
isSupportResourceStatistics>
  <isSupportBandwidthLimit><!--optional, xs:boolean--></isSupportBandwidthLimit>
  <isSupportPOEPortsDisableServer><!--optional, xs:boolean--></isSupportPOEPortsDisableServer>
  <isSupportPOEConfiguration><!--optional, xs:boolean--></isSupportPOEConfiguration>
  <Adaption>
    <streamType opt="0,1,2,3,4,5,7,8,9,10">
      <!--stream types that support network self-adaptive during live view: 0-main stream, 1-sub-stream, 2-third stream,
3-virtual stream, 4-stream 5, 5-stream 6, 7-stream 7, 8-stream 8, ..., and so on-->
    </streamType>
    <isSupportPlayback><!--optional, xs: boolean, whether it supports self-adaptive during playback--></
isSupportPlayback>
  </Adaption>
  <isSupportVideolmgDB><!--optional, xs:boolean, whether it supports configuring the image and video library--></
isSupportVideolmgDB>
  <isSupportEventDataOverWebSocket opt="true,false"><!--optional, xs:boolean, whether it supports uploading
events via WebSocket--></isSupportEventDataOverWebSocket>
  <isSupportDynamicHostName><!--optional, xs:boolean, whether it supports configuring dynamic domain name--></
isSupportDynamicHostName>
  <isSupportWifiProbe><!--optional, xs:boolean, whether it supports Wi-Fi probe configuration--></
isSupportWifiProbe>
  <isSupportRFIDData><!--optional, xs:boolean, whether it supports configuration of RFID data collection--></
isSupportRFIDData>
  <isSupportwifiProbeSSID><!--optional, xs:boolean, whether it supports SSID configuration of Wi-Fi probe--></
isSupportwifiProbeSSID>
  <isSupportPOEPortsDisableAdaptiveServer><!--optional, xs:boolean--></isSupportPOEPortsDisableAdaptiveServer>
  <isSupportGetLinkSocketIP><!--optional, xs:boolean, whether it supports get the current linked SocketIP--></
isSupportGetLinkSocketIP>
  <isSupportTrafficMonitor><!--optional, xs:boolean, whether it supports traffic monitoring of 4G network card--> </
isSupportTrafficMonitor>
  <isSupportDMSAuthInfo><!--optional, xs:boolean, whether it supports DMS authentication information
management--></isSupportDMSAuthInfo>
  <isSupportWiredandWirelessTrafficMonitor><!--optional, xs:boolean, whether it supports monitoring traffic of wired
NIC and wireless NIC--></isSupportWiredandWirelessTrafficMonitor>

```

```
<isSupport4G><!--optional, xs:boolean, whether it supports 4G module--></isSupport4G>
<isSupport4GConfig><!--optional, xs:boolean, whether it supports 4G configuration--></isSupport4GConfig>
<isSupportSipTransmit><!--optional, xs:boolean, whether it supports configuring SIP transmission parameters--></
isSupportSipTransmit>
<isSupportWifiCascade><!--optional, xs:boolean, whether it supports enabling automatic cascading of device Wi-Fi,
related URI: /ISAPI/System/Network/wifiCascade?format=json--></isSupportWifiCascade>
<isSupportRelativeInfo><!--optional, xs:boolean, whether it supports getting information of connected devices,
related URI: /ISAPI/System/Network/wifiCascade/relativeInfo?format=json--> </isSupportRelativeInfo>
<isSupportEZVIZUnbind><!--optional, xs:boolean, whether it supports unbinding devices from the Hik-Connect
account, related URI: /ISAPI/System/Network/EZVIZ/unbind?format=json--></isSupportEZVIZUnbind>
<isSupportEZVIZQRcode><!--optional, xs:boolean, whether it supports getting the Hik-Connect QR code, related URI: /
ISAPI/System/Network/EZVIZ/QRcode?format=json--></isSupportEZVIZQRcode>
<isSupportEZVIZTiming><!--optional, xs:boolean, whether the device supports timing of Hik-Connect (related URI: /
ISAPI/System/Network/EZVIZ)--></isSupportEZVIZTiming>
</NetworkCap>
```

XML_NetworkInterface

NetworkInterface message in XML format

```
<NetworkInterface version="2.0" xmlns="http://www.isapi.org/ver20/XMLSchema">
  <id><!--req, xs:string--></id>
  <IPAddress/><!--req-->
  <Wireless/><!--opt-->
  <Discovery/><!--opt-->
  <Link/><!--opt-->
  <defaultConnection><!--opt, xs:boolean, default network connection, this node is required when the device has
multiple network interfaces--></defaultConnection>
  <ActiveMulticast>
    <enabled><!--req, xs:boolean--></enabled>
    <streamID opt="main"><!--req, xs:string--></streamID>
    <ipV4Address><!--opt, xs:string--></ipV4Address>
    <ipV6Address><!--opt, xs:string--></ipV6Address>
    <port min="" max=""><!--opt, xs:integer--></port>
  </ActiveMulticast>
  <macAddress min="" max=""><!--opt, xs:string--></macAddress>
  <EthernetPortList size="4"><!--opt, network interface information-->
    <EthernetPort><!--opt-->
      <id><!--req, xs: integer, min="1",max="4"--></id>
      <MACAddress><!--req, xs:string--></MACAddress>
      <status><!--opt, xs:string; opt="connected, disconnect"--></status>
      <speed><!--req, xs:integer, "10, 100, 1000,10000"--></speed>
    </EthernetPort>
  </EthernetPortList>
</NetworkInterface>
```

XML_NetworkInterfaceList

NetworkInterfaceList message in XML format

```
<NetworkInterfaceList version="2.0" xmlns="http://www.isapi.org/ver20/XMLSchema">
  <NetworkInterface/><!--opt, see details in the message of XML_NetworkInterface-->
</NetworkInterfaceList>
```

See Also

XML_NetworkInterface

XML_NetWorkMode

NetWorkMode message in XML format

```
<NetWorkMode version="2.0" xmlns="http://www.isapi.org/ver20/XMLSchema">
  <workMode><!--opt, xs:string, working mode: "close", "wifi"-Wi-Fi configuration, "wifiAp"-Wi-Fi access point (Wi-Fi
server) configuration--></workMode>
</NetWorkModeParam>
```

XML_NoiseReduce

NoiseReduce message in XML format

```
<NoiseReduce version="2.0" xmlns="http://www.isapi.org/ver20/XMLSchema">
  <mode><!--req, xs: string, 3D DNR mode: "close, general, advanced"--></mode>
  <GeneralMode><!--dep, this node is valid only when <mode> is "general"-->
    <generalLevel><!--req, xs: integer--></generalLevel>
  </GeneralMode>
  <AdvancedMode><!--dep, this node is valid only when <mode> is "advanced"-->
    <FrameNoiseReduceLevel><!--req, xs: integer--></FrameNoiseReduceLevel>
    <InterFrameNoiseReduceLevel><!--req, xs: integer--></InterFrameNoiseReduceLevel>
  </AdvancedMode>
</NoiseReduce>
```

XML_NTPServer

NTPServer message in XML format

```
<NTPServer version="2.0" xmlns="http://www.isapi.org/ver20/XMLSchema">
  <id><!--req, xs:string, NTP server ID--></id>
  <addressingFormatType>
    <!--req, xs:string, which field will be used to locate the NTP server: "ipaddress, hostname"-->
  </addressingFormatType>
  <hostName><!--dep, xs:string--></hostName>
  <ipAddress><!--dep, xs:string--></ipAddress>
```

```
<ipv6Address><!--dep, xs:string--></ipv6Address>
<portNo><!--opt, xs:integer--></portNo>
<synchronizeInterval>
  <!--opt, xs:integer, NTP time synchronization interval, unit: minute-->
</synchronizeInterval>
</NTPServer>
```

XML_NTPServerList

NTPServerList message in XML format

```
<NTPServerList version="2.0" xmlns="http://www.isapi.org/ver20/XMLSchema">
  <NTPServer/><!--opt, see details in the message of XML_NTPServer-->
</NTPServerList>
```

See Also

XML_NTPServer

XML_NTPTestDescription

NTPTestDescription message in XML format

```
<NTPTestDescription version="2.0" xmlns="http://www.isapi.org/ver20/XMLSchema">
  <addressingFormatType>
    <!--req, xs:string, "ipaddress,hostname"-->
  </addressingFormatType>
  <hostName><!--dep, xs:string--></hostName>
  <ipAddress><!--dep, xs:string--></ipAddress>
  <ipv6Address><!--dep, xs:string--></ipv6Address>
  <portNo><!--req, xs:integer--></portNo>
</NTPTestDescription>
```

XML_NTPTestResult

NTPTestResult message in XML format

```
<NTPTestResult version="2.0" xmlns="http://www.isapi.org/ver20/XMLSchema">
  <errorDescription><!--req, xs:string--></errorDescription>
</NTPTestResult>
```

XML_OnlineUpgradeCap

OnlineUpgradeCap message in XML format

```
<OnlineUpgradeCap version="2.0" xmlns="http://www.isapi.org/ver20/XMLSchema">
  <firmwareNum max="" />
  <!--req, the number of online upgrade packages, including full package and incremental package. Full package is
  used for upgrading whole firmware, while incremental package is used for upgrade certain unit, such as openssl
  library-->
  <firmwareCode max="" />
  <!--req, the maximum length of firmware code-->
  <firmwareVersion max="" />
  <!--req, the maximum length of version-->
  <firmwareCodeNumOnce max="" />
  <!--req, the maximum number of firmware codes can be obtained each time-->
  <upgradePercent min="" max="" />
  <!--req-->
  <Version>
    <!--req, upgrade package version information-->
    <newVersion max="" />
    <!--req-->
    <changeLog max="" />
    <!--req-->
  </Version>
  <DeviceParameter>
    <!--opt, online upgrade parameters-->
    <isSupportAutoDownloadPackage>
      <!--opt,xs:boolean,"true,false", whether supports automatic download of upgrade package-->
    </isSupportAutoDownloadPackage>
    <notSupportAutoUpgrade>
      <!--opt,xs:boolean,"true,false", whether not support automatic download of upgrade package and automatic
      upgrade-->
    </notSupportAutoUpgrade>
    <isSupportTimingUpgrade>
      <!--opt,xs:boolean,"true,false", whether supports scheduled upgrade-->
    </isSupportTimingUpgrade>
  </DeviceParameter>
  <ManualDownloadPackage>
    <!--opt, manually download upgrade package-->
    <supportOperation opt="start, cancel, pause, resume" />
    <!--opt, supported operations: "start, cancel, pause, resume"-->
  </ManualDownloadPackage>
  <isSupportIgnoreCurrentVersion>
    <!--opt, xs:boolean, "true,false", whether supports ignoring current version-->
  </isSupportIgnoreCurrentVersion>
</OnlineUpgradeCap>
```

XML_OnlineUpgradeStatus

OnlineUpgradeStatus message in XML format

```
<OnlineUpgradeStatus version="2.0" xmlns="http://www.isapi.org/ver20/XMLSchema">
  <status>
    <!--ro, req, xs:string,
```



```
"notUpgrade,upgrading,successful,languageMismatch,writeFlashError,packageTypeMismatch,packageVersionMismatch,netUnreachable,unknownError"-->
</status>
<percent><!-- ro, req, xs:integer "0-100" --></percent>
</OnlineUpgradeStatus>
```

XML_OnlineUpgradeVersion

OnlineUpgradeVersion message in XML format

```
<OnlineUpgradeVersion version="2.0" xmlns="http://www.isapi.org/ver20/XMLSchema">
  <newVersionAvailable><!--ro, req, xs:boolean, whether there is new verion of upgrade package--></newVersionAvailable>
  <newVersion><!--ro, dep, xs:string, new version No.--></newVersion>
  <changeLog><!--ro, dep, xs:string, update content of new version--></changeLog>
</OnlineUpgradeVersion>
```

XML_OnlineUpgradeServer

OnlineUpgradeServer message in XML format

```
<OnlineUpgradeServer version="2.0" xmlns="http://www.isapi.org/ver20/XMLSchema">
  <connectStatus><!--ro, req xs:boolean, online upgrade server connection status--></connectStatus>
</OnlineUpgradeServer>
```

XML_OnlineUserList

XML message about information of online users

```
<OnlineUserList version="2.0" xmlns="http://www.isapi.org/ver20/XMLSchema">
  <OnlineUser>
    <id><!--required, xs:string--></id>
    <name><!--optional, xs:string--></name>
    <type><!--optional, xs:string, "admin,operator,viewer"--></type>
    <loginTime><!--optional, xs:time, ISO 8601 time--></loginTime>
    <clientAddress>
      <ipAddress><!--optional, xs:string--></ipAddress>
      <ipv6Address><!--optional, xs:string--></ipv6Address>
    </clientAddress>
  </OnlineUser>
</OnlineUserList>
```

XML_Palettes

Palettes message in XML format

```
<Palettes version="2.0" xmlns="http://www.isapi.org/ver20/XMLSchema">
  <mode><!--opt, xs:string,
  "WhiteHot,BlackHot,Fusion1,Rainbow,Fusion2,Ironbow1,Ironbow2,Sepia,Color1,Color2,IceFire,Rain,RedHot,GreenHot,
  DeepBlue,Color3"--></mode>
  <ColorateTarget><!--dep, colorate target, it is valid when mode is "WhiteHot"-->
    <ColorateTargetModelList>
      <ColorateTargetMode>
        <id><!--req, xs:integer, serial number, starts from 1--></id>
        <mode>
          <!--req, xs:string, colorate target mode; colorateHotAreae-colorate area with temperature high than configured
          threshold, colorateIntervalArea-colorate area with temperature between the configured threshold, colorateColdArea-
          colorate area with temperature lower than the configured threshold-->
        </mode>
        <enabled><!--req, xs:bool, "true,false"--></enabled>
        <TemperatureLimit><!--req, temperature limit-->
          <minTemperature><!--dep, it is valid when mode is "colorateHotAreae"/"colorateIntervalArea", xs: float--></
minTemperature>
          <maxTemperature><!--dep, it is valid when mode is "colorateColdArea"/"colorateIntervalArea", xs: float--></
maxTemperature>
        </TemperatureLimit>
        <Color><!--req, area color-->
          <R><!--req, xs:integer--></R>
          <G><!--req, xs:integer--></G>
          <B><!--req, xs:integer--></B>
        </Color>
      </ColorateTargetMode>
    </ColorateTargetModelList>
  </ColorateTarget>
</Palettes>
```

XML_ParkAction

ParkAction message in XML format

```
<ParkAction version="2.0" xmlns="http://www.isapi.org/ver20/XMLSchema">
  <enabled><!--required, xs: boolean, whether to enable parking--></enabled>
  <Parktime><!--req, xs:integer, parking time, unit: second--></Parktime>
  <Action><!--parking actions-->
    <ActionType opt="autoscan,framescan,randomscan,panoramascan,patrol,pattern,preset,areaScan"><!--required,
xs:string, action type: "atuoscan,framescan,randomscan,panoramascan,patrol,pattern,preset,areaScan"--></
ActionType>
    <ActionNum><!--required, xs:integer, parking No. This node will be used when <ActionType> is "patrol", "pattern",
or "preset". For other parking action types, this node should be set to 0--></ActionNum>
  </Action>
</ParkAction>
```

XML_ParkingParam

XML message about rule parameters of parking detection

```
<ParkingParam><!--dep-->
  <durationTime>
    <!--req, xs:integer, duration time, from 5 seconds to 100 seconds, default value: 5s-->
  </durationTime>
</ParkingParam>
```

XML_pingTestDescription

pingTestDescription message in XML format

```
<pingTestDescription version="2.0" xmlns="http://www.isapi.org/ver20/XMLSchema">
  <ipAddress><!--dep, xs:string--></ipAddress>
</pingTestDescription>
```

XML_pingTestResult

pingTestResult message in XML format

```
<pingTestResult version="2.0" xmlns="http://www.isapi.org/ver20/XMLSchema">
  <status><!--req, xs:string, "used, not used"--></status>
</pingTestResult>
```

XML_Probe

Probe message in XML format.

```
<?xml version="1.0" encoding="UTF-8"?>
<Probe>
  <!--the UUID will be returned in the device response message for matching, if not matched, it will not be handled.-->
  <Uuid>8d2091bc-1dd2-11b2-807b-8ce748cf9334</Uuid>
  <Types>inquiry</Types>
</Probe>
```

XML_ProxyDevicesInfoCondition

Message about search condition of device parameters (all digital channels) in XML format

```
<ProxyDevicesInfoCondition version="2.0" xmlns="http://www.isapi.org/ver20/XMLSchema">
  <searchResultPosition><!--required, xs:integer--></searchResultPosition>
```

```
<maxResults><!--required, xs:integer--></maxResults>
</ProxyDevicesInfoCondition>
```

XML_ProxyDevicesInfoResult

Message about search result of device parameters (all digital channels) in XML format

```
<ProxyDevicesInfoResult version="2.0" xmlns="http://www.isapi.org/ver20/XMLSchema">
  <responseStatus><!--required, xs:string, "true,false", search status--></responseStatus>
  <responseStatusStrg><!--required, xs:string, "OK,NO MATCHES,MORE", search status description--></
responseStatusStrg>
  <numOfMatches><!--required, xs:integer, number of returned records--></numOfMatches>
  <totalMatches><!--required, xs:integer, number of matched records--></totalMatches>
  <DeviceInfoList><!--optional, list of task information-->
    <!--refer to XML_DeviceInfo-->
    <DeviceInfo>
      <deviceName><!--required, xs:string--></deviceName>
      <deviceId><!--required, read-only, xs:string, uuid--></deviceId>
      <deviceDescription><!--optional, xs:string--></deviceDescription>
      <deviceLocation><!--optional, xs:string--></deviceLocation>
      <deviceStatus><!--optional, read-only, xs:string,opt="normal,abnormal"--></deviceStatus>
      <DetailAbnormalStatus><!--dependent, this field is valid only when deviceStatus is "abnormal"-->
        <hardDiskFull><!--optional, read-only, xs:boolean, "true,false", whether disk is full--></hardDiskFull>
        <hardDiskError><!--optional, read-only, xs:boolean, "true,false", whether disk has error--></hardDiskError>
        <ethernetBroken><!--optional, read-only, xs:boolean, "true,false", whether it is disconnected--></
ethernetBroken>
        <ipaddrConflict><!--optional, read-only, xs:boolean, "true,false", whether IP address conflicts--></ipaddrConflict>
        <illegalAccess><!--optional, read-only, xs:boolean, "true,false", whether it is illegal access--></illegalAccess>
        <recordError><!--read-only, optional, xs:boolean, "true,false", whether video exception occurred--></recordError>
        <raidLogicDiskError><!--read-only, optional, xs:boolean, "true,false", whether raid logic disk exception occurred--
></raidLogicDiskError>
        <spareWorkDeviceError><!--read-only, optional, xs:boolean, "true,false", whether spare device exception
occurred--></spareWorkDeviceError>
      </DetailAbnormalStatus>
      <systemContact><!--optional, xs:string--></systemContact>
      <model><!--required, read-only, xs:string--></model>
      <serialNumber><!--required, read-only, xs:string--></serialNumber>
      <macAddress><!--required, read-only, xs:string--></macAddress>
      <firmwareVersion><!--required, read-only, xs:string--></firmwareVersion>
      <firmwareReleasedDate><!--optional, read-only, xs:string--></firmwareReleasedDate>
      <encoderVersion><!--optional, read-only, xs:string--></encoderVersion>
      <encoderReleasedDate><!--optional, read-only, xs:string--></encoderReleasedDate>
      <bootVersion><!--optional, read-only, xs:string--></bootVersion>
      <bootReleasedDate><!--optional, read-only, xs:string--></bootReleasedDate>
      <hardwareVersion><!--optional, read-only, xs:string--></hardwareVersion>
      <decoderVersion><!--optional, read-only, xs:string--></decoderVersion>
      <decoderReleasedDate><!--optional, read-only, xs:string--></decoderReleasedDate>
      <softwareVersion><!--optional, read-only, xs:string, software version--></softwareVersion>
      <capacity><!--optional, read-only, xs:integer, device capacity (MB)--></capacity>
      <usedCapacity><!--optional, read-only, xs:integer, used capacity (MB)--></usedCapacity>
      <deviceType><!--required, read-only, xs:string, "IPCamera,IPDome,DVR,HybirdNVR,NVR,DVS,IPZoom,CVR"--></
```

```
deviceType>
  <telecontrolID><!--optional, xs:integer, "1-255"--></telecontrolID>
  <supportBeep><!--optional, xs:boolean--></supportBeep>
  <supportVideoLoss><!--optional, xs:boolean--></supportVideoLoss>
  <firmwareVersionInfo><!--read-only, optional, xs:string--></firmwareVersionInfo>
  <actualFloorNum><!--optional, xs:integer, "1-128"--></actualFloorNum>
  <radarVersion><!--optional, xs:string--></radarVersion>
  <cameraModuleVersion><!--read-only, xs:string--></cameraModuleVersion>
  <mainversion><!--optional, xs:integer, "1-255"--></mainversion>
  <subversion><!--optional, xs:integer, "1-255"--></subversion>
  <upgradeversion><!--optional, xs:integer, "1-255"--></upgradeversion>
  <customizeversion><!--optional, xs:integer, "1-255"--></customizeversion>
  <companyName><!--optional, xs:string, manufacturer name--></companyName>
  <copyright><!--optional, xs:string, copyright information--></copyright>
  <systemName><!--optional, xs:string, storage system, "storageManagement,distributedStorageManagement",
"storageManagement"-storage management system, "distributedStorageManagement"-distributed storage
management system--></systemName>
  <systemStatus><!--optional, system status, xs:string, "configured,unConfigured", "configured"-system has been
configured, "unConfigured"-system has not been configured--></systemStatus>
  <isLeaderDevice><!--optional, xs:boolean, whether it is the corresponding device of the resource IP--></
isLeaderDevice>
  <clusterVersion><!--dependent, xs:string, system cluster version, this field is valid only when isLeaderDevice is
"true"--></clusterVersion>
  <manufacturer><!--optional, xs:string,manufacturer information used for differentiating OEM devices--></
manufacturer>
  <customizedInfo><!--optional, xs:string, custom project information--></customizedInfo>
  <channel><!--optional, xs:integer, when IPC device information is obtained via NVR, this node refers to the channel
of the NVR--></channel>
</DeviceInfo>
</DeviceInfoList>
</ProxyDevicesInfoResult>
```

See Also

XML_DeviceInfo

XML_port

port message in XML format

```
<port version="2.0" xmlns="http://www.isapi.org/ver20/XMLSchema">
  <id/><!--req, xs: string, ID-->
  <enabled/><!--req, xs: boolean-->
  <internalPort/><!--req, xs: string, "http,admin,rtsp,https,WebSocket,WebSocket,SDK_OVER_TLS,SRTP..."-->
  <externalPort/><!--req, xs:integer-->
</port>
```

XML_portStatus

portStatus message in XML format

```
<portStatus version="2.0" xmlns="http://www.isapi.org/ver20/XMLSchema">
  <id/><!--req, xs: string, ID-->
  <enabled/><!--req-->
  <internalPort/><!--req, xs: string, "http,admin,rtsp,https,WebSocket,WebSocket,SDK_OVER_TLS,SRTP..."-->
  <externalPort/><!--req, xs: integer-->
  <status/><!--req, xs: string, "inactive, active, conflict, ..."-->
</portStatus>
```

XML_ports

ports message in XML format

```
<ports version="2.0" xmlns="http://www.isapi.org/ver20/XMLSchema">
  <enabled/><!--req-->
  <mapmode><!--req, xs: string, "auto,manual"--></mapmode>
  <natRouterLanAddr><!--opt-->
    <ipVersion><!--req, xs: string, "v4,v6,dual"--></ipVersion>
    <ipAddress><!--dep, xs: string--></ipAddress>
    <ipv6Address><!--dep, xs: string--></ipv6Address>
  </natRouterLanAddr>
  <portList><!--req-->
    <port/><!--see details in the message XML_port-->
  </portList>
  <natType><!--req, xs: string, "manual, auto"--></natType>
</ports>
```

See Also

XML_port

XML_PortMapParam

PortMapParam message in XML format

```
<PortMapParam version="2.0" xmlns="http://www.isapi.org/ver20/XMLSchema">
  <userip><!--IP address of user terminal, or IP address of PoE0 or Eth0--></userip>
</PortMapParam>
```

XML_PortMapParamRet

PortMapParamRet message in XML format

```
<PortMapParamRet version="2.0" xmlns="http://www.isapi.org/ver20/XMLSchema">
  <mapResult><!--whether port mapping is required: 0-no, 1-yes, 2-unknown--></mapResult>
  <MapPortGop>
    <MapPort>
      <mapPortIdx><!--index No. of mapping ports--></mapPortIdx>
      <mapPortName><!--mapping port name--></mapPortName>
      <mapPortValue><!--mapping port number--></mapPortValue>
    </MapPort>
  </MapPortGop>
</PortMapParamRet>
```

XML_portsStatus

portsStatus message in XML format.

```
<portsStatus version="2.0" xmlns="http://www.isapi.org/ver20/XMLSchema">
  <enabled/><!--req-->
  <natRouterLanAddr><!--req-->
    <ipVersion><!--req, xs: string, "v4,v6,dual"--></ipVersion>
    <ipAddress><!--dep, xs: string--></ipAddress>
    <ipv6Address><!--dep, xs: string--></ipv6Address>
  </natRouterLanAddr>
  <natRouterWanAddr><!--req-->
    <ipVersion><!--req, xs: string, "v4,v6,dual"--></ipVersion>
    <ipAddress><!--dep, xs: string--></ipAddress>
    <ipv6Address><!--dep, xs: string--></ipv6Address>
  </natRouterWanAddr>
  <portStatusList><!--req-->
    <portStatus/><!--req, see details in the message XML_portStatus-->
  </portStatusList>
</portsStatus>
```

See Also

XML_portStatus

XML_Position3D

XML message about 3D positioning parameters

```
<?xml version="1.0" encoding="utf-8"?>
<Position3D version="2.0" xmlns="http://www.isapi.org/ver20/XMLSchema">
  <StartPoint>
    <positionX><!--required, xs:integer--></positionX>
    <positionY><!--required, xs:integer--></positionY>
  </StartPoint>
  <EndPoint>
    <positionX><!--required, xs:integer--></positionX>
    <positionY><!--required, xs:integer--></positionY>
```

```
</EndPoint>
</Position3D>
```

XML_powerLineFrequency

powerLineFrequency message in XML format

```
<powerLineFrequency version="2.0" xmlns="http://www.isapi.org/ver20/XMLSchema">
  <powerLineFrequencyMode/><!--opt, xs: string, standard mode: "50hz, 60hz"-->
</powerLineFrequency>
```

XML_PreviewSwitch

PreviewSwitch message in XML format

```
<PreviewSwitch version="2.0" xmlns="http://www.isapi.org/ver20/XMLSchema">
  <DisplayWindowList size="64">
    <!--req, if the number of channels is larger or equal to 64, 64 channels should be applied by a group for one time;
    otherwise, apply the maximum number of channels-->
    <DisplayWindow>
      <id><!--req, xs: string, actual window No., which equals to id × groupNo--></id>
      <displayChannelNo><!--req, xs: string, 0-not display, 1 to 64-displayed channel No., min="0" max="64"--></
displayChannelNo>
    </DisplayWindow>
  </DisplayWindowList>
  <previewFrameNo>
    <!--req, xs: string, number of live view windows: 1, 4, 6, 8, 9, 25, 32, 36, auto1-custom window division 1, auto2-
custome window division 2, auto3-custome window division 3, auto4-custome window division 4-->
  </previewFrameNo>
  <sound><!--req, xs: boolean, whether to turn on audio during live view: true=yes, false=no--></sound>
  <switchTime><!--req, xs: string, switching interval: 0, 5, 10, 20, 30, 60, 120, and 300, unit:s--></switchTime>
  <sameSource>
    <!--dep, xs: boolean, whether the output is homologous: true=yes, false=no, this node is valid only when
VideoOutType is "noSameSourceVGA1", "noSameSourceVGA2", "noSameSourceVGA3", and "noSameSourceVGA4"-->
  </sameSource>
</PreviewSwitch>
```

XML_PreviewSwitchVideoOutCap

PreviewSwitchVideoOutCap message in XML format

```
<PreviewSwitchVideoOutCap version="2.0" xmlns="http://www.isapi.org/ver20/XMLSchema">
  <DisplayWindowList size="64">
    <DisplayWindow>
      <id><!--req, xs: string, window No., which equals to id × groupNo--></id>
      <displayChannelNo min="0" max="64"><!--req, xs: string, 0-not display, 1 to 64-displayed channel No.--></
displayChannelNo>
```



```
</DisplayWindow>
</DisplayWindowList>
<previewFrameNo opt="1,4,6,8,9,25,32,36,auto1,auto2,auto3,auto4">
  <!--req, xs: string, number of live view windows, auto1-custom window division 1, auto2-custome window division
2, auto3-custome window division 3, auto4-custome window division 4-->
</previewFrameNo>
<sound><!--req, xs: boolean, whether to turn on audio during live view: true=yes, false=no--></sound>
<switchTime opt="0,5,10,20,30,60,120,300"><!--req, xs: string, switching interval, unit: s--></switchTime>
<sameSource>
  <!--dep, xs: boolean, whether the output is homologous: true=yes, false=no, this node is valid only when
VideoOutType is "noSameSourceVGA1", "noSameSourceVGA2", "noSameSourceVGA3", and "noSameSourceVGA4"-->
</sameSource>
<supportGetByPreviewNum>
  <!--opt, whether supports setting number of live view windows in the URL, true=yes-->
</supportGetByPreviewNum>
</PreviewSwitchVideoOutCap>
```

XML_PrivacyMask

PrivacyMask message in XML format

```
<PrivacyMask version="2.0" xmlns="http://www.isapi.org/ver20/XMLSchema">
  <enabled><!--req, xs: boolean--></enabled>
  <normalizedScreenSize><!--opt-->
    <normalizedScreenWidth><!--req, xs: integer--></normalizedScreenWidth>
    <normalizedScreenHeight><!--req, xs: integer--></normalizedScreenHeight>
  </normalizedScreenSize>
  <PrivacyMaskRegionList size=8/><!--opt-->
    <regionType><!--opt, xs: string, "quadrilateral"--></regionType>
  </PrivacyMask>
```

XML_PrivacyMaskCap

PrivacyMaskCap message in XML format

```
<PrivacyMaskCap version="2.0" xmlns="http://www.isapi.org/ver20/XMLSchema">
  <maskTypeDescriptor>
    <!--opt, xs: string, privacy mask type: "gray,red,yellow,blue,orange,green,transparent,half-transparent,mosaic"-->
  </maskTypeDescriptor>
  <minZoomdoorlimit><!--opt, xs: integer></minZoomdoorlimit>
  <maxZoomdoorlimit><!--opt, xs: integer></maxZoomdoorlimit>
  <videoPrivacyType opt="privacyMask,privacyCover">
    <!--opt, xs: string, "privacyMask"-video tampering, "privacyCover"-privacy mask-->
  </videoPrivacyType>
</PrivacyMaskCap>
```

XML_PrivacyMaskRegion

PrivacyMaskRegion message in XML format

```
<PrivacyMaskRegion version="2.0" xmlns="http://www.isapi.org/ver20/XMLSchema">
  <id><!--req, xs: integer--></id>
  <enabled><!--req, xs: boolean--></enabled>
  <RegionCoordinatesList><!--req-->
    <RegionCoordinates><!--req-->
      <positionX><!--req, xs: integer; coordinate--></positionX>
      <positionY><!--req, xs: integer; coordinate--></positionY>
    </RegionCoordinates>
  </RegionCoordinatesList>
  <privacymaskName><!--opt, xs: string--></privacymaskName>
  <maskType>
    <!--opt, xs:string "gray,red,yellow,blue,orange,green,transparent,half-transparent,mosaic,black"-->
  </maskType>
  <zoomdoorlimit><!--opt, xs: integer, the value is between 10 and 1000--></zoomdoorlimit>
</PrivacyMaskRegion>
```

XML_PrivacyMaskRegionList

PrivacyMaskRegionList message in XML format

```
<PrivacyMaskRegionList version="2.0" xmlns="http://www.isapi.org/ver20/XMLSchema">
  <PrivacyMaskRegion/><!--opt, refer to the message XML_PrivacyMaskRegion for details-->
</PrivacyMaskRegionList>
```

See Also

XML_PrivacyMaskRegion

XML_PTZAux

PTZAux message in XML format

```
<?xml version="1.0" encoding="utf-8"?>
<PTZAux version="2.0" xmlns="http://www.isapi.org/ver20/XMLSchema">
  <id><!--req,xs:string,id--></id>
  <type><!--req, ro, xs:string,"LIGHT,WIPER,FAN,HEATER",auxiliary type: light, wiper, fan, heater--></type>
  <status><!--req, xs:string,"on,off", auxiliary status: turned on, turned off--></status>
</PTZAux>
```

XML_PTZAuxList

PTZAuxList message in XML format

```
<?xml version="1.0" encoding="utf-8"?>
<PTZAuxList version="2.0" xmlns="http://www.isapi.org/ver20/XMLSchema">
  <PTZAux><!--list-->
    <id><!--req, xs:string, id--></id>
    <type><!--req, ro, xs:string, "LIGHT,WIPER,FAN,HEATER", auxiliary type: light, wiper, fan, heater--></type>
    <status><!--req, xs:string, "on,off", auxiliary status: turned on, turned off--></status>
  </PTZAux>
</PTZAuxList>
```

XML_PTZChannel

PTZChannel message in XML format

```
<PTZChanel version="2.0" xmlns="http://www.isapi.org/ver20/XMLSchema">
  <id><!--req, xs:integer--></id>
  <enabled><!--ro, req, xs:boolean--></enabled>
  <serialNumber><!--req, xs:integer--></serialNumber>
  <videoInputID><!--req, xs:integer--></videoInputID>
  <panMaxSpeed><!--ro, opt, xs:integer, degrees/sec--></panMaxSpeed>
  <tiltMaxSpeed><!--ro, opt, xs:integer, degrees/sec--></tiltMaxSpeed>
  <presetSpeed><!--opt, xs:integer, 1..8--></presetSpeed>
  <autoPatrolSpeed><!--opt, xs:integer, 0..100--></autoPatrolSpeed>
  <keyBoardControlSpeed><!--opt, xs:integer, 0..100--></keyBoardControlSpeed>
  <controlProtocol><!--opt, xs:string, "pelco-d,modbus-RTU,modbus-ASCII"--></controlProtocol>
  <controlAddress><!--opt-->
    <enabled><!--req, xs:boolean--></enabled>
    <Address><!--opt, xs:string, 1-255--></Address>
  </controlAddress>
  <defaultPresetID><!--opt, xs:string, id--></defaultPresetID>
  <PTZRs485Para><!--opt-->
    <baudRate><!--req, xs:integer--></baudRate>
    <dataBits><!--req, xs:integer--></dataBits>
    <parityType><!--req, xs:string, "none, even, odd, mark, space"--></parityType>
    <stopBits><!--req, xs:string, "1, 1.5, 2"--></stopBits>
    <flowCtrl><!--req, xs:string, "none, software, hardware"--></flowCtrl>
  </PTZRs485Para>
  <manualControlSpeed>
    <!--opt, xs:string, "pedestrian, nonMotorVehicle, motorVehicle, selfadaptive, compatible"-->
  </manualControlSpeed>
  <panSupport><!--optional, xs:boolean, whether it supports panning--></panSupport>
  <tiltSupport><!--optional, xs:boolean, whether it supports tilting--></tiltSupport>
  <zoomSupport><!--optional, xs:boolean, whether it supports zooming--></zoomSupport>
</PTZChannel>
```

XML_PTZChanelCap

XML message about PTZ channel capability

```
<PTZChanelCap version="2.0" xmlns="http://www.isapi.org/ver20/XMLSchema">
  <AbsolutePanTiltPositionSpace><!--optional-->
    <XRange/><!--required-->
    <YRange/><!--required-->
  </AbsolutePanTiltPositionSpace>
  <AbsoluteZoomPositionSpace><!--optional-->
    <ZRange/><!--required-->
  </AbsoluteZoomPositionSpace>
  <RelativePanTiltSpace><!--optional-->
    <XRange/><!--required-->
    <YRange/><!--required-->
  </RelativePanTiltSpace>
  <RelativeZoomSpace><!--optional-->
    <ZRange/><!--required-->
  </RelativeZoomSpace>
  <ContinuousPanTiltSpace><!--optional-->
    <XRange/><!--required-->
    <YRange/><!--required-->
  </ContinuousPanTiltSpace>
  <ContinuousZoomSpace><!--optional-->
    <ZRange/><!--required-->
  </ContinuousZoomSpace>
  <MomentaryPanTiltSpace><!--optional-->
    <XRange/><!--required-->
    <YRange/><!--required-->
  </MomentaryPanTiltSpace>
  <MomentaryZoomSpace><!--optional-->
    <ZRange/><!--required-->
  </MomentaryZoomSpace>
  <homePostionSupport><!--required, xs:boolean--></homePostionSupport>
  <maxPresetNum>
    <!--required, xs:integer, max. supported preset number-->
  </maxPresetNum>
  <maxPatrolNum>
    <!--required,xs:integer, max. supported patrol number-->
  </maxPatrolNum>
  <maxPatternNum>
    <!--required,xs:integer, max. supported pattern number-->
  </maxPatternNum>
  <maxLimitesNum>
    <!--required,xs:integer, max. supported limit number-->
  </maxLimitesNum>
  <maxTimeTaskNum>
    <!--required,xs:integer, max. supported timing task number-->
  </maxTimeTaskNum>
  <serialNumber min="1" max="4">
    <!--configuration capability of RS-485 serial port supported by current channel-->
  </serialNumber>
  <controlProtocol>
    <!--optional, xs:string,"pelco-d,modbus-RTU,modbus-ASCII", supported PTZ control protocol-->
  </controlProtocol>
  <controlAddress>
```

```
<!--optional, xs:string, 0-255, address-->
</controlAddress>
<PTZRs485Para>
  <!--optional, PTZ RS485 parameters capability-->
  <baudRate>
    <!--required, xs:integer, baud rate-->
  </baudRate>
  <dataBits>
    <!--required, xs:integer, data bit-->
  </dataBits>
  <parityType>
    <!--required, xs:string, "none,even,odd,mark,space", verification type-->
  </parityType>
  <stopBits>
    <!--required, xs:string, "1,1.5,2", stop bit-->
  </stopBits>
  <flowCtrl>
    <!--required, xs:string, "none, software, hardware", stream control type-->
  </flowCtrl>
</PTZRs485Para>
<PresetNameCap>
  <!--optional, preset name capability-->
  <presetNameSupport>
    <!--optional,xs:boolean, whether to support preset name?-->
  </presetNameSupport>
  <maxPresetNameLen>
    <!--dependent,xs:integer, preset name length-->
  </maxPresetNameLen>
  <specialNo/>
  <!--dependent, special preset-->
</PresetNameCap>
<isSupportPosition3D>
  <!--optional, xs:boolean, whether to support 3D position-->
</isSupportPosition3D>
<isSupportManualTrack>
  <!--optional,xs:boolean, whether to support manual tracking position(NET_DVR_PTZ_MANUALTRACE)?-->
</isSupportManualTrack>
<manualControlSpeed opt="compatible,pedestrian,nonMotorVehicle,motorVehicle,selfadaptive">
  <!--optional,xs:string, manual control speed: "compatible"-compatible mode, "pedestrian"-pedestrian,
  "nonMotorVehicle"-non-motor vehicle, "motorVehicle"-motor vehicle, "selfadaptive"-self adaptive-->
</manualControlSpeed>
<isSupportPtzlimiteds>
  <!--optional,xs:boolean, whether to support PTZ limitation-->
</isSupportPtzlimiteds>
<ParkAction/><!--see details in the message XML_ParkAction-->
<TimeTaskList/><!--see details in the message XML_TimeTaskList-->
<Thermometry>
  <maxThermometryPresetNum>
    <!--optional, xs:integer-->
  </maxThermometryPresetNum>
</Thermometry>
<isSupportPtzEagleFocusing>
```

```
<!--optional, xs:boolean-->
</isSupportPtzEagleFocusing>
<TrackingRatio>
  <!--optional, tracking zoom ratio-->
  <coefficient min="1" max="10" default="5">
    <!--optional, xs:integer, zoom ratio-->
  </coefficient>
</TrackingRatio>
<TrackInitPosition>
  <!--optional, tracking initial position-->
  <slaveCameraID>
    <!--optional, xs:integer, slave camera ID, 1..4 -->
  </slaveCameraID>
</TrackInitPosition>
<isSupportAbsoluteEx><!--optional,xs:boolean, whether to support extended capability for PTZ absolute position--></
isSupportAbsoluteEx>
<isSupportCruise><!--optional,xs:boolean, whether to support auto-switch mode--></isSupportCruise>
<isSupportAreaScan><!--optional,xs:boolean, whether to support area scan--></isSupportAreaScan>
<isSupportFaceSnap3D><!--optional, xs: boolean--></isSupportFaceSnap3D>
<ManualPTZLockCap><!--optional, PTZ manual locking capability-->
  <lockStatus opt="lock,unlock"><!--optional, xs:string, PTZ locking status, the default status is locked--></lockStatus>
  <unLockTime><!--optional, xs:integer, remaining unlocking time duration (the automatic locking time duration is
300 s in the unlocked status*--></unLockTime>
</ManualPTZLockCap>
<isSupportOnepushSynchronizeFOV>
  <!--optional, xs:boolean, whether supports one-touch synchronize FOV, return true for supports, and no return
indicates not support-->
</isSupportOnepushSynchronizeFOV>
<isSupportLensCorrection>
  <!--optional, xs:boolean, whether supports lens correction, return true for supports, and no return indicates not
support-->
</isSupportLensCorrection>
<isSupportPTZTrackStatus>
  <!--optional, xs:boolean, whether to support getting PTZ tracking linkage status-->
</isSupportPTZTrackStatus>
<ptzctrlSupportSerialNumber opt="1,3"><!--optional, xs:string, RS-485 serial port range supported in PTZ control
mode--></ptzctrlSupportSerialNumber>
<LFPositionCap><!--optional, capability of calibrating position for smart linkage-->
  <elevation><!--optional, xs:integer, range: [-900,2700]--></elevation>
  <azimuth><!--optional, xs:integer, range: [0,3600]--></azimuth>
  <absoluteZoom><!--optional, xs:integer, range: [1,1000]--></absoluteZoom>
</LFPositionCap>
<pqrsZoom min="" max=""/><!--optional, xs: integer, zooming coordinates of Sony zoom camera module-->
<mnstFocus min="" max=""/><!--optional, xs: integer, focus coordinates of Sony zoom camera module-->
<isSupportPTZSave>
  <!--optional, xs: boolean, whether supports saving the current PTZ position information-->
</isSupportPTZSave>
<isSupportPTZSaveGet>
  <!--optional, xs:boolean, whether to support saving the PTZ position information of the current channel by GET
method-->
</isSupportPTZSaveGet>
<isSupportAutoGotoCfg>
```

```
<!--optional, xs: boolean, whether supports automatically restoring to saved PTZ position: "true"-yes, this node is
not returned-no-->
<isSupportAutoGotoCfg>
<lockTime><!--optional, xs:integer, PTZ locking time, unit: second--></lockTime>
<isSupportPTZCtrlStatus><!--optional, xs:boolean, whether it supports getting the absolute PTZ position of the
current channel: "true"-yes, "false"-no. If this node is not returned, it does not necessarily indicate that the device
does not support this function--></isSupportPTZCtrlStatus>
<isSupportInitializationCtrl><!--optional, xs:boolean, whether it supports electric PTZ self-checking and control--></
isSupportInitializationCtrl>
<isSupportComponentCtrl><!--optional, xs:boolean, whether it supports PTZ component control--></
isSupportComponentCtrl>
<DoubleMirror><!--optional-->
  <enabled opt="true,false" def="false"><!--required, xs:boolean--></enabled>
</DoubleMirror>
<MoveAutoTracking><!--optional, moving object tracking-->
  <enabled opt="true,false"><!--required, xs:boolean, whether to enable this function--></enabled>
</MoveAutoTracking>
<isSupportTrackingRatioCfg><!--optional, xs:boolean, whether it supports configuring tracking parameters--></
isSupportTrackingRatioCfg >
<isSupportGPSCtrl>
  <!--optional, xs:boolean, whether current channel supports controlling pan and tilt by GPS, if supports, this node is
returned and the value is true; if not, this node is not returned (related URI: /ISAPI/PTZCtrl/channels/<ID>/GPS?
format=json)-->
</isSupportGPSCtrl>
<isSupportGyroVerify>
  <!--optional, xs:boolean, whether device supports gyroscope attitude calibration (related URI: /ISAPI/PTZCtrl/
channels/<ID>/gyroVerify?format=json)-->
</isSupportGyroVerify>
</PTZChanelCap>
```

See Also

XML_ParkAction

XML_TimeTaskList

XML_PTZChannelList

XML message about parameters of multiple digital PTZ channels

```
<?xml version="1.0" encoding="utf-8"?>
<PTZChannelList version="2.0" xmlns="http://www.isapi.org/ver20/XMLSchema">
  <PTZChannel><!--list-->
    <id><!--required, xs:integer--></id>
    <enabled><!--required, xs:boolean, read-only--></enabled>
    <serialNumber>
      <!--required, xs:integer, read-only, serial port No. of channel; if the value is 0, it indicates that no serial port is
linked-->
    </serialNumber>
    <videoInputID><!--required, xs:integer--></videoInputID>
    <panMaxSpeed><!--ro,optional, xs:integer, panning speed, unit: degree/second--></panMaxSpeed>
    <tiltMaxSpeed><!--ro,optional, xs:integer, tilting speed, unit: degree/second--></tiltMaxSpeed>
```

```
<presetSpeed><!--optional, xs:integer, preset speed, range: [1,8]--></presetSpeed>
<autoPatrolSpeed><!--optional, xs:integer, patrol speed, range: [0,100]--></autoPatrolSpeed>
<keyBoardControlSpeed><!--optional, xs:integer, PTZ control speed, range: [0,100]--></keyBoardControlSpeed>
<controlProtocol><!--optional, xs:string, control protocol type: pelco-d--></controlProtocol>
<controlAddress><!--optional-->
  <enabled><!--required, xs:boolean--></enabled>
  <Address><!--optional, xs:string, range: [1,255]--></Address>
</controlAddress>
<defaultPresetID><!--optional, xs:string, default preset No.--></defaultPresetID>
<PTZRs485Para><!--optional-->
  <baudRate><!--required, xs:integer--></baudRate>
  <dataBits><!--required, xs:integer--></dataBits>
  <parityType><!--required, xs:string, parity type: none, even, odd, mark, space--></parityType>
  <stopBits><!--required, xs:string, stop bit: 1, 1.5, 2--></stopBits>
  <flowCtrl><!--required, xs:string, encoding or decoding type: none, software, hardware--></flowCtrl>
</PTZRs485Para>
<manualControlSpeed>
  <!--optional, xs:string, manual control speed type: pedestrian, nonMotorVehicle, motorVehicle, selfadaptive,
compatible-->
</manualControlSpeed>
</PTZChannel>
</PTZChannelList>
```

XML_PTZCtrlCap

XML message about PTZ control capability of digital channels

```
<?xml version="1.0" encoding="utf-8"?>
<PTZCtrlCap version="2.0" xmlns="http://www.isapi.org/ver20/XMLSchema">
  <maxPTZChannleNum><!--optional, xs:integer, maximum number of PTZ channels--></maxPTZChannleNum>
  <isSupportPTZWithPreview>
    <!--optional, xs:boolean, whether the device supports controlling PTZ during live view-->
  </isSupportPTZWithPreview>
</PTZCtrlCap>
```

XML_PTZOSDDisplay

PTZOSDDisplay message in XML format

```
<PTZOSDDisplay version="2.0" xmlns="http://www.isapi.org/ver20/XMLSchema">
  <zoomlable><!--req, xs:string, "2sec, 5sec, 10sec, alwaysclose, alwaysopen"--></zoomlable>
  <azimuth><!--req, xs:string, "2sec, 5sec, 10sec, alwaysclose, alwaysopen"--></azimuth>
  <presetlable><!--req, xs:string, "2sec, 5sec, 10sec, alwaysclose, alwaysopen"--></presetlable>
  <actionStatusDisplayEnabled><!--opt, xs:boolean, display status or not--></actionStatusDisplayEnabled>
</PTZOSDDisplay>
```


XML_PTZPatrol

XML message about parameters of a patrol

```
<?xml version="1.0" encoding="utf-8"?>
<PTZPatrol version="2.0" xmlns="http://www.isapi.org/ver20/XMLSchema">
  <enabled><!--required, xs:boolean--></enabled>
  <id><!--required, xs:string, ID--></id>
  <patrolName><!--required, xs:string--></patrolName>
  <resumeType><!--optional, xs:string, type: relative, absolute--></resumeType>
  <PatrolSequenceList><!--required-->
    <PatrolSequence><!--required, list-->
      <presetID><!--required, xs:integer, preset No. in the patrol--></presetID>
      <seqSpeed><!--required, xs:integer, patrol speed--></seqSpeed>
      <delay><!--required, xs:integer, unit: millisecond--></delay>
    </PatrolSequence>
  </PatrolSequenceList>
</PTZPatrol>
```

XML_PTZPatrolList

XML message about patrol parameter list

```
<?xml version="1.0" encoding="utf-8"?>
<PTZPatrolList version="2.0" xmlns="http://www.isapi.org/ver20/XMLSchema">
  <PTZPatrol><!--list-->
    <enabled><!--required, xs:boolean--></enabled>
    <id><!--required, xs:string, ID--></id>
    <patrolName><!--required, xs:string--></patrolName>
    <resumeType><!--optional, xs:string, type: relative, absolute--></resumeType>
    <PatrolSequenceList><!--required-->
      <PatrolSequence><!--required, list-->
        <presetID><!--required, xs:integer, preset No. in the patrol--></presetID>
        <seqSpeed><!--required, xs:integer, patrol speed--></seqSpeed>
        <delay><!--required, xs:integer, unit: millisecond--></delay>
      </PatrolSequence>
    </PatrolSequenceList>
  </PTZPatrol>
</PTZPatrolList>
```

XML_PTZPatrolStatus

XML message about patrol status

```
<?xml version="1.0" encoding="utf-8"?>
<PTZPatrolStatus version="2.0" xmlns="http://www.isapi.org/ver20/XMLSchema">
  <patrolID><!--required, xs:string, ID--></patrolID>
```

```
<patrolStatus><!--required, xs:string, patrol status: running, stopped, paused--></patrolStatus>
</PTZPatrolStatus>
```

XML_PTZPreset

XML message about preset parameters

```
<PTZPreset version="2.0" xmlns="http://www.isapi.org/ver20/XMLSchema">
  <enabled><!--required, xs:boolean, whether to enable preset configuration--></enabled>
  <id><!--required, xs:string, preset No.--></id>
  <presetName><!--required, xs:string, preset name--></presetName>
  <AbsoluteHigh>
    <elevation><!--optional, xs:integer, tilting parameter, the value is between -900 and 2700--></elevation>
    <azimuth><!--optional, xs:integer, panning parameter, the value is between 0 and 3600--></azimuth>
    <absoluteZoom><!--optional, xs:integer, zooming parameter, the value is between 1 and 1000--></absoluteZoom>
  </AbsoluteHigh>
</PTZPreset>
```

XML_PTZPresetList

XML message about preset information list

```
<PTZPresetList version="2.0" xmlns="http://www.isapi.org/ver20/XMLSchema">
  <PTZPreset/><!--optional, see details in XML_PTZPreset-->
</PTZPresetList>
```

See Also

XML_PTZPreset

XML_PTZStatus

PTZStatus message in XML format

```
<PTZStatus version="2.0" xmlns="http://www.isapi.org/ver20/XMLSchema">
  <AbsoluteHigh><!--high-accuracy positioning which is accurate to one decimal place-->
    <elevation>
      <!--opt, xs:integer, tilting parameter, the value is between -900 and 2700, which corresponds to the range from
-90.0 degree to 270.0 degree vertically-->
    </elevation>
    <azimuth>
      <!--opt, xs:integer, panning parameter, the value is between 0 and 3600, which corresponds to the range from 0.0
degree to 360.0 degree horizontally-->
    </azimuth>
    <absoluteZoom>
      <!--opt, xs:integer, zooming parameter, the value is between 0 and 1000, which corresponds to the focus range
from 0.0 to 100.0-->
    </absoluteZoom>
```

```
</AbsoluteHigh>
</PTZStatus>
```

XML_PublicKey

PublicKey message in XML format.

```
<PublicKey version="2.0" xmlns="http://www.isapi.org/ver20/XMLSchema">
  <key><!--req, xs:string--></key>
</PublicKey>
```

XML_RacmCap

XML message about device storage capability

```
<?xml version="1.0" encoding="utf-8"?>
<RacmCap version="1.0" xmlns="http://www.isapi.org/ver20/XMLSchema">
  <isSupportZeroChan>
    <!--optional, xs:boolean, whether it supports channel-zero, "true"-yes, "false"-no-->
  </isSupportZeroChan>
  <inputProxyNums><!--optional, xs:integer, number of supported digital channels--></inputProxyNums>
  <eSATANums><!--optional, xs:integer, number of supported eSATAs--></eSATANums>
  <miniSASNums><!--optional, xs:integer, number of supported miniSATAs--></miniSASNums>
  <nasNums><!--optional, xs:integer, number of supported NASs--></nasNums>
  <ipSanNums><!--optional, xs:integer, number of supported IPSANs--></ipSanNums>
  <isSupportRaid>
    <!--optional, xs:boolean, whether it supports RAID, "true"-yes, "false"-no-->
  </isSupportRaid>
  <isSupportExtHdCfg>
    <!--optional, xs:boolean, whether it supports HDD advanced management, "true"-yes, "false"-no-->
  </isSupportExtHdCfg>
  <isSupportTransCode><!--optional, xs:boolean, whether it supports auto-switch, "true"-yes, "false"-no--></
isSupportTransCode>
  <isSupportIpclImport>
    <!--optional, xs:boolean, whether it supports importing configuration files to network camera, "true"-yes, "false"-
no-->
  </isSupportIpclImport>
  <NasMountType><!--optional-->
    <isNFSSupportAuthentication/><!--optional, xs:boolean, whether it supports NFS authentication, "true"-yes, "false"-
no-->
    <isCIFSSupportAuthentication/><!--optional, xs:boolean, whether it supports CIFS authentication, "true"-yes, "false"-
no-->
  </NasMountType>
  <isSupportIpcStreamType/><!--optional, xs:boolean-->
  <isSupportIOInputProxy/><!--optional, xs:boolean, whether it supports dynamic alarm input channel: "true"-yes,
"false"-no-->
  <isSupportIOOutputProxy/><!--optional, xs:boolean, whether it supports dynamic alarm output channel: "true"-yes,
"false"-no-->
  <isSupportPTZRs485Proxy/><!--optional, xs:boolean, whether it supports dynamic PTZ485 channel, "true"-yes,
```

```
"false"-no-->
<isSupportSrcIDSearch/><!--optional, xs:boolean, whether it supports search by stream ID, "true"-yes, "false"-no-->
<isSupportReversePlayback/><!--optional, xs:boolean, whether it supports reverse playback, "true"-yes, "false"-no-->
<isSupportSMARTTest/><!--optional, xs:boolean, whether it supports HDD checking, "true"-yes, "false"-no-->
<isSupportDownloadByTime/><!--optional, xs:boolean, whether it supports downloading by time, "true"-yes, "false"-no-->
<pictureSearchType
opt="AllEvent,CMR,MOTION,ALARM,EDR,ALARMANDMOTION,Command,pir,wlsensor,callhelp,facedetection,FieldDetection,scenechangedetection,LineDetection,regionEntrance,regionExiting,loitering,group,rapidMove,parking,unattendedBaggage,attendedBaggage,vehicleDetection>manual>manualSnapshot,playSnapshot,allPic,evidence,illegalParking,crosslane,vehicleexist,lanchange,wrongdirection,congestion,turnround,parallelParking,pedestrian,construction,roadblock,abandonedObject,trafficAccident,fogDetection, personQueueCounting,personQueueTime,mixedTargetDetection,safetyHelmet,accessController,videoInterCom,GJD,Luminite,OPTEX,securityControlPanel,playCellphone,vibrationDetection,ATMPanel,ATMSurround,ATMFace,ATMSafetyCabin"/>
<!--optional, xs:string, picture search conditions-->
<recordSearchType
opt="AllEvent,CMR,MOTION,ALARM,EDR,ALARMANDMOTION,Command,pir,wlsensor,callhelp,facedetection,FieldDetection,scenechangedetection,LineDetection,regionEntrance,regionExiting,loitering,group,rapidMove,parking,unattendedBaggage,attendedBaggage,vehicleDetection>manual>manualSnapshot,playSnapshot,AudioDetection,playCellphone,ruining,failDown,violentMotion,retention,allPerimeterEvent,allBehaviorEvent,vibrationDetection,ATMPanel,ATMSurround,ATMFace,ATMSafetyCabin"/>
<!--optional, xs:string, video search conditions-->
<isSupportActivatelpc/><!--optional, xs:boolean, whether it supports activating network camera, "true"-yes, "false"-no-->
<isSupportCheckIpcSecurity><!--optional-->
<isSupportCheckPassword/><!--optional, xs:boolean, whether it supports password verification, "true"-yes, "false"-no-->
</isSupportCheckIpcSecurity>
<isSupportMainAndSubRecord/>
<!--optional, xs:boolean, whether it supports recording in main stream or sub-stream, "true"-yes, "false"-no-->
<isSupportSyncIpcPassword>
<!--optional, xs:boolean, whether it supports synchronizing network camera's password, "true"-yes, "false"-no-->
</isSupportSyncIpcPassword>
<isSupportTransferIpc>
<!--optional, xs:boolean, whether it supports network camera passthrough function, "true"-yes, "false"-no-->
</isSupportTransferIpc>
<isSupportPOS><!--optional, xs:boolean, whether it supports POS, "true"-yes, "false"-no--></isSupportPOS>
<isSupportPassBackBasicCfg>
<!--optional, xs:boolean, whether it supports the ANR basic function of CVR, "true"-yes, "false"-no-->
</isSupportPassBackBasicCfg>
<PassBackTaskConfig><!--optional, task management function of ANR-->
<isSupportAddTask><!--optional, xs:boolean, whether it supports adding task, "true"-yes, "false"-no--></isSupportAddTask>
<isSupportSearchTask><!--optional, xs:boolean, whether it supports task search, "true"-yes, "false"-no--></isSupportSearchTask>
<isSupportControlTask><!--optional, xs:boolean, whether it supports task control, "true"-yes, "false"-no--></isSupportControlTask>
<isSupportDeleteTask><!--optional, xs:boolean, whether it supports deleting task, "true"-yes, "false"-no--></isSupportDeleteTask>
</PassBackTaskConfig>
<PassBackPlanConfig><!--optional, task plan of ANR-->
<isSupportAddPlan><!--optional, xs:boolean, whether it supports adding plan, "true"-yes, "false"-no--></isSupportAddPlan>
```

```
isSupportAddPlan>
  <isSupportSearchPlan><!--optional, xs:boolean, whether it supports plan search, "true"-yes, "false"-no--></
isSupportSearchPlan>
  <isSupportDeletePlan><!--optional, xs:boolean, whether it supports deleting plan, "true"-yes, "false"-no--></
isSupportDeletePlan>
</PassBackTaskConfig>
<IRAIDCap/><!--optional, network RAID-->
<isSupportStorageExtraInfo>
  <!--optional, xs:boolean, whether it supports configuring storage additional information, "true"-yes, "false"-no-->
</isSupportStorageExtraInfo>
<isSupportRecordStatus><!--optional, xs:boolean--></isSupportRecordStatus>
<supportAIDTFSType opt="illegalParking,wrongdirection crosslane,vehicleexist,lanexchange, turnround, evidence"/>
  <!--optional, xs:string, event picture search conditions supported by both AID and TFS-->
<isSupportRacmChannelsCap><!--optional, xs:boolean--></isSupportRacmChannelsCap>
<LockCap/><!--optional, lock or unlock video-->
<isSupportForamtAll><!--optional, xs:boolean, "true,false"--></isSupportForamtAll>
<isSupportExtendCabinetCfg>
  <!--optional, xs:boolean, whether it supports enclosure configuration-->
</isSupportExtendCabinetCfg>
<diskGroupNums><!--optional, xs:integer, number of supported HDD groups--></diskGroupNums>
<isSupportCountingSearchByUTC><!--optional, xs:boolean, whether it supports searching people counting results by
UTC time--></isSupportCountingSearchByUTC>
<isSupportPlaybackReverseByUTC><!--optional, xs:boolean, whether it supports reverse playback by UTC time--></
isSupportPlaybackReverseByUTC>
<isSupportWebPrivatePlaybackByUTC>
  <!--optional, xs: boolean, whether it supports playback based on Web private protocol-->
</isSupportWebPrivatePlaybackByUTC >
<isSupportFindCommonFileByUTC>
  <!--optional, xs: boolean, whether it supports extending the time zone for searching files-->
</isSupportFindCommonFileByUTC>
<isSupportFindEventFileByUTC><!--optional, xs: boolean, whether it supports extending the time zone for searching
files by event-->
</isSupportFindEventFileByUTC >
<isSupportSmartSearchRecordByUTC>
  <!--optional, xs: boolean, whether it supports extending the time zone for VCA search-->
</isSupportSmartSearchRecordByUTC>
<isSupportMRDSearchByTimeZone>
  <!--optional, xs: boolean, whether it supports extending the time zone for searching files by calendar-->
</isSupportMRDSearchByTimeZone>
<isSupportSearchRecordLabelByUTC>
  <!--optional, xs: boolean, whether it supports extending the time zone for searching video tags-->
</isSupportSearchRecordLabelByUTC>
<isSupportSearchPictureByUTC>
  <!--optional, xs: boolean, whether it supports extending the time zone for searching pictures-->
</isSupportSearchPictureByUTC>
<isSupportSmartSearchPictureByUTC>
  <!--optional, xs: boolean, whether it supports extending the time zone for searching pictures with smart
information-->
</isSupportSmartSearchPictureByUTC>
<isSupportFindLogByUTC>
  <!--optional, xs: boolean, whether it supports extending the time zone for searching log-->
</isSupportFindLogByUTC>
```

```
<isSupportUploadRecordByUTC>
  <!--optional, xs: boolean, whether it supports extending the time zone for uploading files to cloud storage-->
</isSupportUploadRecordByUTC>
<isSupportPlaybackByUTC>
  <!--optional, xs: boolean, whether it supports extending the time zone for playback by time and locating by time-->
</isSupportPlaybackByUTC>
<SecurityLog>
  <isSupportSecurityLog><!--optional, boolean, whether it supports security log--></isSupportSecurityLog>
  <isSupportLogServer><!--optional, boolean, whether it supports log server configuration--></isSupportLogServer>
  <isSupportLogServerTest><!--optional, xs: boolean, whether it supports log server test--></isSupportLogServerTest>
  <SecurityLogTypeList><!--required, xs: list, supported log type list-->
    <SecurityLogType>
      <primaryType><!--required, xs: string, major type--></primaryType>
      <secondaryType optional=""><!--required, xs: string, minor type--></secondaryType>
    </SecurityLogType>
  </SecurityLogTypeList>
</SecurityLog>
<iSptInputProxyChanCap>
  <!--optional, xs:boolean, whether it supports getting the capability of the digital channel-->
</iSptInputProxyChanCap>
<isSupportLogDataPackage>
  <!--optional, xs:boolean, whether it supports exporting logs: "true, false"-->
</isSupportLogDataPackage>
<logSearchTimeSpanNums>
  <!--optional, xs:integer, supported number of time periods for log search-->
</logSearchTimeSpanNums>
<isSupportManualRecord>
  <!--optional, xs:boolean, whether the device supports manual recording-->
</isSupportManualRecord>
<isSupportRemark><!--optional, xs:boolean, whether it supports file remarks--></isSupportRemark>
<FileUpload><!--optional, upload file-->
  <enabled><!--required, xs:boolean, whether it supports enabling file uploading--></enabled>
</FileUpload>
<PoliceInfoUpload><!--optional, upload police information-->
  <enabled><!--required, xs:boolean, whether it supports enabling police information uploading--></enabled>
</PoliceInfoUpload>
<PoliceInfo>
  <policeID><!--optional, xs:boolean, whether it supports police ID--></policeID>
  <policeCode><!--optional, xs:boolean, whether it supports police No.--></policeCode>
  <policeName><!--optional, xs:boolean, whether it supports police name--></policeName>
  <password><!--optional, xs:boolean, whether it supports police password--></password>
  <deviceID><!--optional, xs:boolean, whether it supports device ID--></deviceID>
</PoliceInfo>
<behaviorEventPicSearch
opt="allBehaviorEvent,running,group,violentMotion,failDown,playCellphone,peopleNumChange,leavePosition,retentio
n,sleepOnduty">
  <!--optional, xs:string, behavior analysis events supported by picture search, which is used for web display,
"allBehaviorEvent"-all events in behaviorEventPicSearch-->
</behaviorEventPicSearch>
<perimeterEventPicSearch
opt="allPerimeterEvent,linedetection,fielddetection,regionEntrance,regionExiting,loitering">
  <!--optional, xs:string, perimeter protection events that support picture search, which is used for web display,
```

```
"allPerimeterEvent"-all events in perimeterEventPicSearch-->
</perimeterEventPicSearch>
<isSupportAssignChannelID><!--optional, xs:boolean, whether it supports specify channel No. when adding IPC--></isSupportAssignChannelID>
<isSupportAssignStreamID><!--optional, xs:boolean, whether it supports specify stream ID of the channel when adding IPC--></isSupportAssignStreamID>
<isSupportTimeSearch><!--optional, xs:boolean, whether it supports searching for recording start and end time by channel--></isSupportTimeSearch>
<CloudStorageServerCap><!--optional-->
  <isSupportCloudStorageParameter><!--optional, xs:boolean, whether it supports configuring cloud storage parameters--></isSupportCloudStorageParameter>
  <isSupportCloudStoragePool><!--optional, xs:boolean, whether it supports configuring cloud storage pool parameters--></isSupportCloudStoragePool>
</CloudStorageServerCap>
<CMSearchCount/><!--optional, xs:boolean, whether it supports searching for file quantity of dock station-->
<isSupportSSDSMARTTest>
  <!--optional, xs:boolean, whether the device supports SSD S.M.A.R.T detection (related URI: /ISAPI/ContentMgmt/Storage/ssd/<ID>/SMARTTest/start)-->
</isSupportSSDSMARTTest>
</RacmCap>
```

XML_RegionalFocus

RegionalFocus message in XML format

```
<RegionalFocus version="2.0" xmlns="http://www.isapi.com/ver20/XMLSchema">
  <StartPoint><!--0-1000-->
    <positionX><!--req, xs:integer--></positionX>
    <positionY><!--req, xs:integer--></positionY>
  </StartPoint>
  <EndPoint><!--0-1000-->
    <positionX><!--req, xs:integer--></positionX>
    <positionY><!--req, xs:integer--></positionY>
  </EndPoint>
</RegionalFocus>
```

XML_RegionClip

RegionClip message in XML format

```
<RegionClip version="2.0" xmlns="http://www.std-cgi.org/ver20/XMLSchema">
  <id><!--req, xs: string--></id>
  <enabled><!--req, xs: boolean--></enabled>
  <normalizedScreenSize><!--req, read-only-->
    <normalizedScreenWidth><!--req, read-only, xs: integer--></normalizedScreenWidth>
    <normalizedScreenHeight><!--req, read-only, xs: integer--></normalizedScreenHeight>
  </normalizedScreenSize>
  <regionType><!--req, ro, xs:string, opt="rectangle,convexPolygon,concavePolygon"--></regionType>
  <videoResolutionWidth><!--opt="704"--></videoResolutionWidth>
```

```
<videoResolutionHeight><!--opt="576"--></videoResolutionHeight>
<ClipRegionList>
  <ClipRegion>
    <RegionCoordinatesList>
      <RegionCoordinates><!--req, size=4-->
        <positionX><!--req, xs: integer; x-coordinate--></positionX>
        <positionY><!--req, xs: integer; y-coordinate--></positionY>
      </RegionCoordinates>
    </RegionCoordinatesList>
  </ClipRegion>
</ClipRegionList>
</RegionClip>
```

XML_remotePermission

remotePermission message in XML format

```
<remotePermission version="2.0" xmlns="http://www.isapi.org/ver20/XMLSchema">
  <record><!--opt, xs: boolean--></record>
  <boolean><!--opt, xs: boolean--></boolean>
  <preview><!--opt, xs: boolean--></preview>
  <videoChannelPermissionList><!--opt-->
    <videoChannelPermission><!--opt-->
      <id><!--req, corresponds to the video input channel ID--></id>
      <preview><!--opt, xs: boolean--></preview>
      <palyBack><!--opt, xs: boolean--></palyBack>
      <record><!--opt, xs:boolean--></record>
      <playBackDoubleVerification>
        <!--opt, xs:boolean, whether supports secondary authentication for playback and download-->
      </playBackDoubleVerification>
    </videoChannelPermission>
  </videoChannelPermissionList>
  <ptzControl><!--opt, xs: boolean--></ptzControl>
  <ptzChannelPermissionList><!--opt-->
    <ptzChannelPermission><!--opt-->
      <id><!--req, corresponds to PTZ channel ID--></id>
      <ptzControl><!--opt, xs: boolean--></ptzControl>
    </ptzChannelPermission>
  </ptzChannelPermissionList>
  <logOrStateCheck><!--opt, xs: boolean, permission to view log and status--></logOrStateCheck>
  <parameterConfig><!--opt, xs: boolean, parameter configuration permission--></parameterConfig>
  <restartOrShutdown><!--opt, xs: boolean, permission to reboot and shutdown--></restartOrShutdown>
  <upgrade><!--opt, xs: boolean, permission to upgrade--></upgrade>
  <voiceTalk><!--opt, xs: boolean--></voiceTalk>
  <transParentChannel><!--opt, xs: boolean--></transParentChannel>
  <contorlLocalOut><!--opt, xs: boolean--></contorlLocalOut>
  <alarmOutOrUpload><!--opt, xs: boolean, permission to upload and output alarm--></alarmOutOrUpload>
  <factoryReset><!--opt, xs:boolean, restore default parameters--></factoryReset>
  <arm><!--opt, xs:boolean, arm--></arm>
  <disarm><!--opt, xs:boolean, disarm--></disarm>
  <accessControl><!--opt, xs:boolean, access control--></accessControl>
```



```
<subSysOrZoneArm><!--opt, xs:boolean, partition or zone arming permission--></subSysOrZoneArm>
<subSysOrZoneDisarm><!--opt, xs:boolean, partition or zone disarming permission--></subSysOrZoneDisarm>
<subSysOrZoneClearArm><!--opt, xs:boolean, permission to clear partition or zone alarms--></
subSysOrZoneClearArm>
<zoneBypass><!--opt, xs:boolean, permission to bypass zone--></zoneBypass>
<zoneBypassRecover><!--opt, xs:boolean, permission to recover bypassing zone--></zoneBypassRecover>
<IRAID>
  <rapidConfiguration><!--opt, xs:boolean--></rapidConfiguration>
  <reset><!--opt, xs:boolean--></reset>
</IRAID>
<subSystemList><!--opt, partitions linked to the operator when adding an operator or setting operator parameters-->
  <subSystem><!--opt, xs:integer, partition linked to the operator. When adding an operator or setting operator
parameters, one or more partitions can be linked to an operator. An operator should be linked to at least one partition
and up to four partitions--></subSystem>
</subSystemList>
</remotePermission>
```

XML_remotePermissionCap

remotePermissionCap capability message in XML format

```
<remotePermissionCap version="2.0" xmlns="http://www.isapi.org/ver20/XMLSchema">
  <record><!--opt, xs: boolean--></record>
  <booleant><!--opt, xs: boolean--></booleant>
  <preview><!--opt, xs: boolean--></preview>
  <videoChannelPermissionList><!--opt-->
    <videoChannelPermission><!--opt-->
      <id><!--req, corresponds to the video input channel ID--></id>
      <preview><!--opt, xs: boolean--></preview>
      <palyBack><!--opt, xs: boolean--></palyBack>
      <record><!--opt, xs:boolean--></record>
      <playBackDoubleVerification>
        <!--opt, xs:boolean, whether supports secondary authentication for playback and download-->
      </playBackDoubleVerification>
    </videoChannelPermission>
  </videoChannelPermissionList>
  <ptzControl><!--opt, xs: boolean--></ptzControl>
  <ptzChannelPermissionList><!--opt-->
    <ptzChannelPermission><!--opt-->
      <id><!--req, corresponds to PTZ channel ID--></id>
      <ptzControl><!--opt, xs: boolean--></ptzControl>
    </ptzChannelPermission>
  </ptzChannelPermissionList>
  <logOrStateCheck><!--opt, xs: boolean, permission to view log and status--></logOrStateCheck>
  <parameterConfig><!--opt, xs: boolean, parameter configuration permission--></parameterConfig>
  <restartOrShutdown><!--opt, xs: boolean, permission to reboot and shutdown--></restartOrShutdown>
  <upgrade><!--opt, xs: boolean, permission to upgrade--></upgrade>
  <voiceTalk><!--opt, xs: boolean--></voiceTalk>
  <transParentChannel><!--opt, xs: boolean--></transParentChannel>
  <contorlLocalOut><!--opt, xs: boolean--></contorlLocalOut>
  <alarmOutOrUpload><!--opt, xs: boolean, permission to upload and output alarm--></alarmOutOrUpload>
```

```
<factoryReset><!--opt, xs:boolean, restore default parameters--></factoryReset>
<arm><!--opt, xs:boolean, arm--></arm>
<disarm><!--opt, xs:boolean, disarm--></disarm>
<accessControl><!--opt, xs:boolean, access control--></accessControl>
<subSysOrZoneArm><!--opt, xs:boolean, partition or zone arming permission--></subSysOrZoneArm>
<subSysOrZoneDisarm><!--opt, xs:boolean, partition or zone disarming permission--></subSysOrZoneDisarm>
<subSysOrZoneClearArm><!--opt, xs:boolean, permission to clear partition or zone alarms--></
subSysOrZoneClearArm>
<zoneBypass><!--opt, xs:boolean, permission to bypass zone--></zoneBypass>
<zoneBypassRecover><!--opt, xs:boolean, permission to recover bypassing zone--></zoneBypassRecover>
<IRAID>
  <rapidConfiguration><!--opt, xs:boolean--></rapidConfiguration>
  <reset><!--opt, xs:boolean--></reset>
</IRAID>
<subSystemNo min="" max=""><!--opt, xs:integer, range of partition No. When adding an operator or setting
operator parameters, one or more partitions can be linked to the operator--></subSystemNo>
  <subSystem min="1" max="3"><!--opt, xs:integer, number of partitions that can be linked to the operator. The
attributes min="1" and max="3" are sample settings, which indicates that at least one partition and up to three
partitions can be linked. When adding an operator or setting operator parameters, one or more partitions can be
linked to an operator. Actually, an operator should be linked to at least one partition and up to four partitions--></
subSystemNo>
</remotePermissionCap>
```

XML_ResponseStatus

XML message about response status

```
<ResponseStatus version="2.0" xmlns="http://www.std-cgi.org/ver20/XMLSchema">
  <requestURL>
    <!--required, read-only, xs:string, request URL-->
  </requestURL>
  <statusCode>
    <!--required, read-only, xs:integer, status code: 0,1-OK, 2-Device Busy, 3-Device Error, 4-Invalid Operation, 5-Invalid
XML Format, 6-Invalid XML Content, 7-Reboot Required, 9-Additional Error-->
  </statusCode>
  <statusString>
    <!--required, read-only, xs:string, status description: OK, Device Busy, Device Error, Invalid Operation, Invalid XML
Format, Invalid XML Content, Reboot, Additional Error-->
  </statusString>
  <subStatusCode>
    <!--required, read-only, xs:string, describe the error reason in detail-->
  </subStatusCode>
</ResponseStatus>
```



See ***Response Codes of Text Protocol*** for details about sub status codes and corresponding error codes.

XML_ResponseStatus_AuthenticationFailed

ResponseStatus message in XML format for failed authentication.

```
<ResponseStatus version="1.0" xmlns="http://www.std-cgi.org/ver20/XMLSchema">
  <requestURL><!-- req, ro,xs:string --></requestURL>
  <statusCode><!-- req, ro,xs:integer --></statusCode>
  <statusString><!-- req, ro,xs:string --></statusString>
  <subStatusCode><!-- req, ro,xs:string --></subStatusCode>
  <lockStatus><!-- opt, ro,xs:string , "unlock,locked", locking status--></lockStatus>
  <retryTimes><!-- opt, ro,xs:integer, remaining authentication attempts--></retryTimes>
  <resLockTime><!-- opt, ro,xs:integer, remaining locking time, unit: second--></resLockTime>
</ResponseStatus>
```

XML_RTSPCertificate

XML message about parameters of RTSP authentication mode

```
<RTSPCertificate xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0" >
  <certificateType>
    <!--required, xs:string, authentication mode: digest (default), digest/basic-->
  </certificateType>
  <SecurityAlgorithm>
    <!--dependent, authentication algorithm, it is valid when certificateType is "digest" or "digest/basic"-->
    <algorithmType>
      <!--optional, xs:string, algorithm types: MD5, SHA256, MD5/SHA256-->
    </algorithmType>
  </SecurityAlgorithm>
</RTSPCertificate>
```

XML_RTSPCertificateCap

XML message about capability of RTSP authentication mode

```
<RTSPCertificateCap xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0" >
  <certificateType def="digest" opt="digest,digest/basic" >
    <!--required, xs:string-->
  </certificateType>
  <SecurityAlgorithm>
    <!--depend, authentication algorithm, it is valid when certificateType is "digest" or "digest/basic"-->
    <algorithmType opt="MD5,SHA256,MD5/SHA256" >
      <!--optional, xs:string, algorithm type-->
    </algorithmType>
  </SecurityAlgorithm>
</RTSPCertificateCap>
```

XML_Schedule

Schedule message in XML format

```
<Schedule version="2.0" xmlns="http://www.isapi.org/ver20/XMLSchema">
  <id><!--required, xs:string, ID--></id>
  <eventType>
    <!--optional, xs:string, alarm/event types, see details in the "Remarks" below-->
  </eventType>
  <inputIOPortID><!--read-only, dependent, xs:string, alarm input No.--></inputIOPortID>
  <outputIOPortID><!--read-only, dependent, xs:string, alarm output No.--></inputIOPortID>
  <videoInputChannelID><!--read-only, dependent, xs:string, video input channel ID--></videoInputChannelID>
  <TimeBlockList size="8"><!--required-->
    <TimeBlock><!--list-->
      <dayOfWeek>
        <!--optional, xs:integer, day of the week based on ISO8601, "1"=Monday, ....-->
      </dayOfWeek>
      <TimeRange><!--required-->
        <beginTime><!--required, xs:time, ISO8601 time--></beginTime>
        <endTime><!--required, xs:time, ISO8601 time--></endTime>
      </TimeRange>
      <CustomExtension>
        <vehicleDetectSceneID>
          <!--required, xs:interger-->
        </vehicleDetectSceneID>
      </CustomExtension>
    </TimeBlock>
  </TimeBlockList>
  <HolidayBlockList><!--optional-->
    <TimeBlock><!--list-->
      <TimeRange><!--required-->
        <beginTime><!--required, xs:time, ISO8601 time--></beginTime>
        <endTime><!--required, xs:time, ISO8601 time--></endTime>
      </TimeRange>
    </TimeBlock>
  </HolidayBlockList>
</Schedule>
```

Remarks

The node **<eventType>** can be set to the following values: IO, VMD, videoloss, PIR, linedetection, fielddetection, audioexception, facedetection, regionEntrance, regionExiting, loitering, group, rapidMove, parking, unattendedBaggage, attendedBaggage, storageDetection, shipsDetection, HUMANATTRIBUTE, humanAttribute, faceContrast, faceSnap, faceLib, whiteListFaceContrast, personDensityDetection, personQueueDetection, mixedTargetDetection, fireDetection, illegalParking, pedestrian, trafficAccident, construction, roadblock, abandonedObject, parallelParking, parkingState, congestion, intersectionAnalysis, heatMap, reverseEntrance, vehicledetect, safetyHelmetDetection, vibrationDetection, TMPA, faceThermometry, HBDLib.

XML_Scheduled_PTZData

XML message about parameters of scheduled PTZ control

```
<?xml version="1.0" encoding="utf-8"?>
<PTZData version="2.0" xmlns="http://www.isapi.org/ver20/XMLSchema">
  <pan><!--optional, xs:integer, panning range: [-100,100]--></pan>
  <tilt><!--optional, xs:integer, tilting range: [-100,100]--></tilt>
  <zoom><!--optional, xs:integer, zooming range: [-100,100]--></zoom>
  <Momentary>
    <duration><!--optional, xs:integer, unit: millisecond--></duration>
  </Momentary>
</PTZData>
```

XML_SecurityCap

SecurityCap capability message in XML format

```
<SecurityCap version="2.0" xmlns="http://www.isapi.org/ver20/XMLSchema">
  <supportUserNums><!--optional, xs:integer, number of supported users--></supportUserNums>
  <userBondIpNums><!--optional, xs:integer, number of bound IP addresses supported by the user--></
userBondIpNums>
  <userBondMacNums><!--optional, xs:integer, number of bound MAC addresses supported by the user--></
userBondMacNums>
  <isSupCertificate><!--optional, xs:boolean, whether it supports certification: "true", "false"--></isSupCertificate>
  <issupIllegalLoginLock><!--optional, xs: boolean, whether it supports locking illegal login: "true,false"--
><issupIllegalLoginLock>
  <isSupportOnlineUser><!--optional, xs: boolean, "true,false"--><isSupportOnlineUser>
  <isSupportAnonymous><!--optional, xs: boolean, "true,false"--><isSupportAnonymous>
  <isSupportStreamEncryption><!--optional, xs:boolean, whether it supports stream encryption: "true", "false"--></
isSupportStreamEncryption>
  <securityVersion opt="1,2"/><!--optional, xs:interger, encryption capability, each version contains encryption
algorithm and node range to be encrypted. "1"-encrypt by AES128, "2"-encrypt by AES256-->
  <keyIterateNum>
    <!--dependent xs:integer, iteration times, this node depends on securityVersion, and the value is usually between
100 and 1000-->
  </keyIterateNum>
  <isSupportUserCheck>
    <!--dependent optional, xs:boolean, whether it supports verifying password when editing/adding/deleting user
parameters, this node depends on securityVersion-->
  </isSupportUserCheck>
  <isSupportGUIDFileDataExport><!--optional, xs:boolean, "true,false"--></isSupportGUIDFileDataExport>
  <isSupportSecurityQuestionConfig><!--optional, xs:boolean, "true,false"--></isSupportSecurityQuestionConfig>
  <supportSecurityNode opt="wirelessServer,wirelessDial"><!--optional, xs:string, "true,false"--></
supportSecurityNode>
  <isSupportGetOnlineUserListSC><!--optional, xs:boolean, "true,false"--></isSupportGetOnlineUserListSC>
  <SecurityLimits><!--optional-->
    <LoginPasswordLenLimit min="1" max="16"><!--opt--></LoginPasswordLenLimit>
    <SecurityAnswerLenLimit min="1" max="128"><!--opt--></SecurityAnswerLenLimit>
```

```
</SecurityLimits>
<RSAKeyLength opt="512,1024,2048" def=2048/><!--optional, RSA key length of the HTTPS certificate-->
<isSupportONVIFUserManagement><!--optional, xs:boolean, whether it supports user management by ONVIF
protocol: "true", "false"--><isSupportONVIFUserManagement>
<WebCertificateCap><!--opt-->
  <CertificateType opt="basic, digest, digest/basic"><!--required, xs:string--></CertificateType>
  <SecurityAlgorithm><!--dependent, this node is valid when certificateType is "digest" or "digest/basic"-->
    <algorithmType opt="MD5,SHA256,MD5/SHA256"><!--optional, xs:string, algorithm type: "MD5", "SHA256",
"MD5/SHA256"--></algorithmType>
  </SecurityAlgorithm>
</WebCertificateCap>
<isSupportConfigFileImport>
  <!--optional, xs: boolean, whether it supports importing configuration files securely: "true"-yes. If it is not
supported, this node will not be returned-->
</isSupportConfigFileImport>
<isSupportConfigFileExport>
  <!-- optional, xs:boolean, whether it supports exporting configuration files securely: "true"-yes. If it is not
supported, this node will not be returned-->
</isSupportConfigFileExport>
<cfgFileSecretKeyLenLimit min="0" max="16">
  <!--optional, length limit of the configuration files' encryption key-->
</cfgFileSecretKeyLenLimit>
<isSupportDeviceCertificatesManagement>
  <!--optional, xs: boolean, whether it supports device certificate management: true=yes, if not support, this node will
not be returned-->
</isSupportDeviceCertificatesManagement>
<supportIPCActivatePassword><!--optional, xs:boolean, whether it supports configuring password for activating the
network camera--></supportIPCActivatePassword>
<isIrreversible>
  <!--dependent optional, xs:boolean, whether it supports irreversible password storage. If this node does not exist,
irreversible password storage is not supported-->
</isIrreversible>
<salt><!--optional, xs:string, salt of user name--></salt>
<keypadPassword min="1" max=""/><!--optional, xs:string, keypad password length. If different types of users have
different keypad password length, this node only indicates the administrator's keypad password length, otherwise this
node indicates that all types of users have the same keypad password length-->
<installerKeypadPassword min="1" max=""/><!--optional, xs:string, installer's keypad password length-->
<operatorKeypadPassword min="1" max=""/><!--optional, xs:string, operator's keypad password length-->
<userOperateType opt="1,2,3">
  <!--optional, xs:string, user operation type: "1"-network user, "2"-keypad user, "3"-network user and keypad user-->
</userOperateType>
<isSupportOnvifInfo><!--optional, xs:boolean, whether it supports the API of getting ONVIF protocol information. If
this node is returned and its value is "true", in indicates supporting this function; if this node is not returned, it
indicates that this function is not supported--></isSupportOnvifInfo>
<isSupportPictureURICertificate opt="true,false"><!--optional, xs:boolean, whether it supports configuring picture
URL authentication. If this node is not returned, it indicates that the default picture authentication method of the
device is digest--></isSupportPictureURICertificate>
<isSupportUnloggedUserPermissionConfig opt="true,false"><!--optional, xs:boolean, whether it supports configuring
permissions for users that have not logged in--></isSupportUnloggedUserPermissionConfig>
<isSupportUserNamePasswordCheckUpgrade><!--optional, xs:boolean, whether it supports upgrading user name
and password verification. If this node is not returned, it indicates that this function is not supported--></
isSupportUserNamePasswordCheckUpgrade>
```

```

<isSupportDeviceSelfSignCertExport><!--optional, xs:boolean--></isSupportDeviceSelfSignCertExport>
<isSupportSecurityEmail><!--optional, xs:boolean, whether it supports configuring security E-mail address. If this
node is not returned, it indicates that this function is not supported--></isSupportSecurityEmail>
<isSupportRTSPCertificate opt="true,false"><!--optional, xs:boolean--></isSupportRTSPCertificate>
<isSptUserEnabled><!--optional, xs:boolean, whether it supports configuration of enabling user: "true"-yes, "false"-
no--></isSptUserEnabled>
<isSptAdminCap><!--optional, xs:boolean, whether it supports getting administrator permission capability: "true"-
yes, "false"-no--></isSptAdminCap>
<DoubleVerificationCap><!--optional, double verification capability-->
  <isSupportUsersConfig>
    <!--optional, xs: boolean, "true,false", whether it supports configuring double verification user-->
  </isSupportUsersConfig>
  <isSupportUsersPermissionConfig>
    <!--optional, xs: boolean, "true,false", whether it supports configuring the permission of double verification user--
  >
  </isSupportUsersPermissionConfig>
</DoubleVerificationCap>
<isSupportCertificateCustomID><!--optional, xs:Boolean, whether it supports certificate configuration with custom
ID--></isSupportCertificateCustomID>
<maxIllegalLoginTimes min="3" max="20"def="5">
  <!--dependentxs:integer,"maximum illegal login attempts" -->
</maxIllegalLoginTimes>
<SecurityAdvanced>
  <securityEnhanced> <!--optional,xs:boolean,security reinforcement--> </securityEnhanced>
  <noOperationEnabled><!--required, xs:Boolean, whether to enable the timeout of no operation--></
noOperationEnabled>
  <noOperationTime min="1" max="60"def="15"><!--required, xs:integer,timeout of no operation, unit: minute--></
noOperationTime>
  <isSupportDigestStatus><!--optional, Boolean—digest status></isSupportDigestStatus>
</SecurityAdvanced>
<LoginLinkNum><maxLinkNum min="1" max="128"def="50"><!--required, xs:integer,maximum number of logged in
accounts--></maxLinkNum><LoginLinkNum>
  <isSupportCCClientCertificate><!--optional,Boolean—whether to enable the CC client certificate></
isSupportCCClientCertificate>
  <passwordValidity min="0" max="365"><!--optional, xs:string, password validity period, unit: day. If this node is
returned, it indicates supporting configuring password validity period. The password validity period can only be edited
by the administrator user--></passwordValidity>
  <maxIllegalLoginLockTime min="1" max="120" def="30"><!--optional, xs:integer, the lock duration when maximum
illegal login attempts reached the upper limit; unit: minute--></maxIllegalLoginLockTime>
</SecurityCap>

```

XML_SecurityQuestion

SecurityQuestion message in XML format

```

<SecurityQuestion version="2.0" xmlns="http://www.isapi.org/ver20/XMLSchema">
  <QuestionList>
    <Question><!--the number of security question must be 3-->
      <id><!--required, xs: integer, question ID, which corresponds to that of device--></id>
      <answer>
        <!--write-only, xs: string, answer of the security question, it will not be returned when getting question; when

```

```
setting question, this node will be encrypted (BASE64-->AES128CBE) for transmission-->
  </answer>
  <mark>
    <!--required, read-only, xs: boolean, whether the configured security question is marked-->
  </mark>
</Question>
</QuestionList>
<password>
  <!--write-only, xs: string, password that encrypted by CBC, this node will not be returned when getting question; but
it is required when setting question-->
</password>
</SecurityQuestion>
```

See Also

XML_SerialCap

Serial port capability message in XML format

```
<SerialCap version="2.0" xmlns="http://www.isapi.org/ver20/XMLSchema">
  <rs485PortNums min="1" max="5" def="1">
    <!--optional, xs: integer-->
  </rs485PortNums>
  <supportRS23Cconfig>
    <!--optional, xs: boolean-->
  </supportRS23Cconfig>
  <rs422PortNums opt="0">
    <!--optional, xs: integer-->
  </rs422PortNums>
  <rs232PortNums opt="1">
    <!--optional, xs: integer, range of RS-232 serial port No. supported by the device-->
  </rs232PortNums>
  <rs485WorkMode opt="Led, CaptureTrigger">
    <!--optional, xs: string, range of RS-485 serial port No. supported by the device-->
  </rs485WorkMode>
  <isSupportAuthenticationService>
    <!--optional, xs:boolean, whether the device supports serial port authentication-->
  </isSupportAuthenticationService>
</SerialCap>
```

XML_SerialCommand

SerialCommand message in XML format

```
<SerialCommand version="1.0" xmlns="urn:psalliance-org">
  <chainNo>
    <!--opt, xs:string-->
  </chainNo>
  <command>
```



```
<!--req, xs:hexBinary-->
</command>
</SerialCommand>
```

XML_SerialPorList

SerialPorList message in XML format

```
<SerialPortList version="2.0" xmlns="http://www.isapi.org/ver20/XMLSchema">
  <SerialPort/><!--opt, see details in the message of XML_SerialPort-->
</SerialPortList>
```

See Also

XML_SerialPort

XML_SerialPort

SerialPort message in XML format

```
<SerialPort version="2.0" xmlns="http://www.isapi.org/ver20/XMLSchema">
  <id>
    <!--req, xs:string, serial port ID-->
  </id>
  <enabled>
    <!--req, xs:boolean, whether to enable the serial port: "true, false"-->
  </enabled>
  <serialPortType>
    <!--req, xs:string, serial port type: "RS485", "RS422", "RS232"-->
  </serialPortType>
  <duplexMode>
    <!--req, xs:string, duplex mode of the serial port: "half", "full"-->
  </duplexMode>
  <direction>
    <!--req, xs:string, "monodirectional, bdirectional"-->
  </direction>
  <baudRate>
    <!--req, xs:integer-->
  </baudRate>
  <dataBits>
    <!--req, xs:integer-->
  </dataBits>
  <parityType>
    <!--req, xs:string, parity type: "none,even,odd,mark,space"-->
  </parityType>
  <stopBits>
    <!--req, xs:string, stop bit: "1,1.5,2"-->
  </stopBits>
  <workMode>
    <!--dep, xs:string, working mode: "console", "transparent", "audiomixer", "stairsControl"-elevator
```

```
control,"cardReader"-card reader,"disabled","custom". This node is required only when <serialPortType> is set to
"RS232"-->
</workMode>
<flowCtrl>
  <!--req, xs:string, "none, software, hardware"-->
</flowCtrl>
<rs485WorkMode>
  <!--opt, xs:string, working mode of RS-485 serial port, which is used for LED display or triggering transmission of
captured pictures: "Led, CaptureTrigger". This node is valid only when <serialPortType> is "RS485"-->
</rs485WorkMode>
<audiomixerProtocolType>
  <!--opt, xs:string, "HIKVISION". This node is required only when <workMode> is "audiomixer"-->
</audiomixerProtocolType>
<deviceName><!--deq, xs:string, device name, the maximum length is 32 bytes, this node is valid only when
<serialPortType> is "RS485"--></deviceName>
<deviceType><!--deq, xs:integer, device type, this node is valid only when <serialPortType> is "RS485"--></
deviceType>
<deviceProtocol><!--deq, xs:integer, device protocol, this node is valid only when <serialPortType> is "RS485"--></
deviceProtocol>
<mode><!--deq, xs:string, working mode: "readerMode,clientMode,externMode,accessControlHost,disabled", this
node is valid only when <serialPortType> is "RS485"--></mode>
<outputDataType><!--dep, xs:string, output data type: "cardNo,employeeNo", this node is valid when <mode>is
"accessControlHost"--></outputDataType>
</SerialPort>
```

XML_Cap_SerialPortList

XML message about the capability of all serial ports

```
<?xml version="1.0" encoding="utf-8"?>
<SerialPortList version="2.0" xmlns="http://www.hikvision.com/ver20/XMLSchema">
  <SerialPort>
    <id><!--required, xs:integer, serial port ID--></id>
    <enabled opt="true,false"><!--optional, xs:boolean, whether to enable--></enabled>
    <serialNumber opt="1,2,3,4,5"><!--optional, xs:integer, serial port No., value range: [1,5]--></serialNumber>
    <serialPortType opt="RS-485,RS-422,RS-232"><!--optional, xs:string, serial port type: "RS-485", "RS-422", "RS-232"--
    ></serialPortType>
    <duplexMode opt="half,full"><!--optional, xs:string, duplex mode of the serial port: "half" (half-duplex mode), "full"
(full-duplex mode)--></duplexMode>
    <direction opt="monodirectional,bidirectional"><!--optional, xs:string, communication direction of the serial port:
"monodirectional"-unidirectional, "bidirectional"-bidirectional--></direction>
    <baudRate opt="2400,4800,9600,19200,38400,57600,115200"><!--optional, xs:integer, baud rate--></baudRate>
    <dataBits opt="7,8"><!--optional, xs:integer, data bit. The data bits follow the start bit and they indicate the valid
information during communication. The number of data bits is determined by both communication parties and
generally it is 7 or 8--></dataBits>
    <parityType opt="none,even,odd,mark,space"><!--optional, xs:string, communication parity type of the serial port:
"none"-no parity bit, "even"-even parity, "odd"-odd parity, "mark"-the parity bit is always 1, "space"-the parity bit is
always 0--></parityType>
    <stopBits opt="1,1.5,2"><!--optional, xs:string, number of stop bits--></stopBits>
    <workMode opt="console,transparent,audiomixer,ptzctrl"><!--optional, xs:string, working mode--></workMode>
    <flowCtrl opt="none,software,hardware"><!--optional, xs:string, flow control of the serial port: "none"-disable,
```

```
"software"-by software, "hardware"-by hardware--></flowCtrl>
<audiomixerProtocolType opt="HIKVISION,DS-ZH101S"><!--optional, xs:string, protocol type of the audio mixer--></audiomixerProtocolType>
<isVariable opt="true,false"><!--optional, xs:boolean, whether it is a variable serial port--></isVariable>
<PTZInfo><!--dependent, this node is valid when the value of <b>workMode</b> contains "ptzCtrl"-->
  <bindChannelNo min="" max=""><!--required, xs:integer, linked channel No.--></bindChannelNo>
  <PTZProtocolList><!--required-->
    <PTZProtocol><!--optional-->
      <id><!--required, xs:integer, protocol index No.--></id>
      <protoDesc><!--required, xs:string, protocol description--></protoDesc>
    </PTZProtocol>
  </PTZProtocolList>
</PTZInfo>
<ScreenCtrlInfo><!--dependent, this node is valid when <b>workMode</b> contains "screenCtrl"-->
  <ScreenCtrlProtoList><!--required-->
    <ScreenCtrlProto><!--optional-->
      <id><!--required, xs:integer, protocol index No.--></id>
      <protoDesc><!--required, xs:string, protocol description--></protoDesc>
    </ScreenCtrlProto>
  </ScreenCtrlProtoList>
</ScreenCtrlInfo>
<MatrixCtrlInfo><!--dependent, this node is valid when <b>workMode</b> contains "matrix"-->
  <MatrixCtrlProtoList><!--required-->
    <MatrixCtrlProto><!--optional-->
      <id><!--required, xs:integer, protocol index No.--></id>
      <protoDesc><!--required, xs:string, protocol description--></protoDesc>
    </MatrixCtrlProto>
  </MatrixCtrlProtoList>
</MatrixCtrlInfo>
<KeyBoardCtrlInfo><!--dependent, this node is valid when <b>workMode</b> contains "keyboard"-->
  <KeyBoardCtrlProtoList><!--required-->
    <KeyBoardCtrlProto><!--optional-->
      <id><!--required, xs:integer, protocol index No.--></id>
      <protoDesc><!--required, xs:string, protocol description--></protoDesc>
    </KeyBoardCtrlProto>
  </KeyBoardCtrlProtoList>
</KeyBoardCtrlInfo>
<deviceName min="0" max="32"><!--dependent, xs:string, device name--></deviceName>
<deviceType min="" max=""><!--dependent, xs:integer, device type--></deviceType>
<deviceProtocol min="" max=""><!--dependent, xs:integer, device protocol--></deviceProtocol>
<mode opt="readerMode,clientMode,externMode,stairsControl,accessControlHost,disabled,custom"><!--dependent, xs:string, working mode--></mode>
  <outputDataType opt="cardNo,employeeNo"><!--dependent, xs:string, output data type. This node is valid when <b>mode</b> is accessControlHost--></outputDataType>
</SerialPort>
</SerialPortList>
```

XML_SIPInfo

SIPInfo message in XML format

```
<SIPInfo version="2.0" xmlns="http://www.isapi.org/ver20/XMLSchema">
  <videoID>
    <!--req, xs:string-->
  </videoID>
  <AlarmInList>
    <AlarmIn>
      <id>
        <!--req, xs:string-->
      </id>
      <alarmInID>
        <!--req, xs:string-->
      </alarmInID>
    </AlarmIn>
  </AlarmInList>
</SIPInfo>
```

XML_SIPInfoList

SIPInfoList message in XML format

```
<SIPInfoList version="2.0" xmlns="http://www.isapi.org/ver20/XMLSchema">
  <SIPInfo>
    <videoID><!--req, xs:string--></videoID>
    <AlarmInList>
      <AlarmIn>
        <id><!--req, xs:string--></id>
        <alarmInID><!--req, xs:string--></alarmInID>
      </AlarmIn>
    </AlarmInList>
    <VideoInputList><!--opt-->
      <VideoInput>
        <id><!--req, xs:string--></id>
        <videoInputID><!--req, xs:string--></videoInputID>
      </VideoInput>
    </VideoInputList>
    <AudioOutputList><!--opt,ID list of audio outputs-->
      <AudioOutput>
        <id><!--req, xs:string--></id>
        <audioOutputID><!--req, xs:string--></audioOutputID>
      </AudioOutput>
    </AudioOutputList>
    <platformNo><!--opt, xs:integer--></platformNo>
  </SIPInfo>
</SIPInfoList>
```

XML_SIPServer

SIPServer message in XML format

```
<SIPServer version="2.0" xmlns="http://www.isapi.org/ver20/XMLSchema">
  <id><!--req, xs:integer, the maximum ID depends on the number of channels--></id>
  <localPort><!--req, xs:integer, local port No., which is between 1 and 65535--></localPort>
  <streamID><!--req, xs:integer, stream type: 1-main stream, 2-sub-stream, 3-third stream--></streamID>
  <Standard><!--opt-->
    <registerStatus><!--ro, req, xs:boolean, registration status: "false"-unregistered, "true"-registered--></
registerStatus>
    <enabled><!--req, xs:string, "true"-log in, "false"-log out--></enabled>
    <registrar><!--req, xs:string--></registrar>
    <registrarPort><!--req, xs:integer--></registrarPort>
    <proxy><!--req, xs:string--></proxy>
    <proxyPort><!--req, xs:integer--></proxyPort>
    <displayName><!--req, xs:string--></displayName>
    <userName><!--req, xs:string--></userName>
    <authID><!--req, xs:string--></authID>
    <password><!--wo, req, xs:string--></password>
    <expires><!--req, xs:integer--></expires>
    <serverAddressType><!--optional, xs:string, server address type: "IPV4,domainName"--></serverAddressType>
    <domainName><!--optional, xs:string, domain name, this node is valid only when <serverAddressType> is
"domainName"--></domainName>
    <networkType><!--optional, xs:string, network type: "wireless Network,wired Network"--></networkType>
    <calledTargetUserName><!--optional, xs:string, called user name--></calledTargetUserName>
  </Standard>
  <GB28181><!--opt-->
    <registerStatus><!--req, xs:boolean, "true,false"--></registerStatus>
    <enabled><!--req, xs:boolean, "true,false"--></enabled>
    <registrar><!--req, xs:string--></registrar>
    <registrarPort><!--req, xs:integer--></registrarPort>
    <serverId><!--req, xs:string--></serverId>
    <serverDomain><!--req, xs:string--></serverDomain>
    <userName><!--req, xs:string--></userName>
    <authID><!--req, xs:string--></authID>
    <password><!--wo, req, xs:string--></password>
    <expires><!--req, xs:integer--></expires>
    <liveTime><!--req, xs:integer--></liveTime>
    <heartbeatTime><!--req, xs:integer--></heartbeatTime>
    <heartbeatCount><!--req, xs:integer--></heartbeatCount>
  </GB28181>
</SIPServer>
```

XML_SIPServerList

SIPServerList message in XML format

```
<SIPServerList version="2.0" xmlns="http://www.isapi.org/ver20/XMLSchema">
  <SIPServer/><!--opt, see details in the message of XML_SIPServer-->
</SIPServerList>
```

See Also

XML_SIPServer

XML_SIPServerListCap

SIPServerListCap capability message in XML format

```
<SIPServerListCap size="" version="2.0" xmlns="http://www.isapi.org/ver20/XMLSchema">
  <SIPServer><!--optional-->
    <id min="" max=""><!--required, xs:string--></id>
    <localPort min="" max=""><!--required, xs:integer--></localPort>
    <streamID opt="1,2"><!--required, xs:integer, stream type ID: 1-main stream, 2-sub stream--></streamID>
    <Standard><!--optional-->
      <registerStatus opt="true,false"><!--read-only, required, xs:boolean, registration status: "false"-unregistered,
"true"-registered--></registerStatus>
      <enabled opt="true,false"><!--required, xs:boolean--></enabled>
      <registrar min="" max=""><!--required, xs:string--></registrar>
      <registrarPort min="" max=""><!--required, xs:integer--></registrarPort>
      <proxy min="" max=""><!--required, xs:string--></proxy>
      <proxyPort min="" max=""><!--required, xs:integer--></proxyPort>
      <displayName min="" max=""><!--required, xs:string--></displayName>
      <userName min="" max=""><!--required, xs:string--></userName>
      <authID min="" max=""><!--required, xs:string--></authID>
      <password min="" max=""><!--write-only, required, xs:string--></password>
      <expires min="" max=""><!--required, xs:integer--></expires>
      <voiceTalk opt="true,false"><!--optional, xs:boolean, whether to enable SIP two-way audio--></voiceTalk>
      <isSptSIPVoiceTalk opt="true,false"><!--optional, xs:boolean, whether it supports SIP intercom, related URI: /ISAPI/
System/Network/SIP/voiceTalkCtrl/capabilities?format=json --></isSptSIPVoiceTalk>
      <serverAddressType><!--optional, xs:string, server address type: "IPv4,domainName"--></serverAddressType>
      <domainName min="" max=""><!--optional, xs:string, domain name, this node is valid only when
<serverAddressType> is "domainName"--></domainName>
      <networkType><!--optional, xs:string, network type: "wireless Network,wired Network"--></networkType>
      <calledTargetUserName min="" max=""><!--optional, xs:string, called user name--></calledTargetUserName>
    </Standard>
    <GB28181><!--optional-->
      <registerStatus opt="true,false"><!--read-only, required, xs:boolean, registration status: "false"-unregistered,
"true"-registered--></registerStatus>
      <enabled opt="true,false"><!--required, xs:boolean--></enabled>
      <registrar min="" max=""><!--required, xs:string--></registrar>
      <registrarPort min="" max=""><!--required, xs:integer--></registrarPort>
      <serverId min="" max=""><!--required, xs:string--></serverId>
      <serverDomain min="" max=""><!--required, xs:integer--></serverDomain>
      <userName min="" max=""><!--required, xs:string--></userName>
      <authID min="" max=""><!--required, xs:string--></authID>
      <password min="" max=""><!--write-only, required, xs:string--></password>
      <expires min="" max=""><!--required, xs:integer--></expires>
      <liveTime min="" max=""><!--required, xs:integer--></liveTime>
      <heartbeatTime min="" max=""><!--required, xs:integer--></heartbeatTime>
      <heartbeatCount min="" max=""><!--required, xs:integer--></heartbeatCount>
      <transportType opt="UDP,TCP,TLS"><!--optional, xs:string, "UDP,TCP,TLS"--></transportType>
```

```
<tcpConnectType opt="active,passive"><!--xs:string, dependency, this node is valid when transportType is "TCP"--></tcpConnectType>
<registerInterval min="" max=""><!--optional, xs:integer, registration interval, unit:second--></registerInterval>
</GB28181>
</SIPServer>
</SIPServerListCap>
```

XML_Sharpness

Sharpness message in XML format

```
<Sharpness version="2.0" xmlns="http://www.isapi.org/ver20/XMLSchema">
  <SharpnessMode><!--opt, xs:string, "manual", "auto"--></SharpnessMode>
  <SharpnessLevel><!--req, xs:integer--></SharpnessLevel>
</Sharpness>
```

XML_Shutter

Shutter message in XML format

```
<Shutter version="1.0" xmlns="http://www.isapi.org/ver20/XMLSchema">
  <ShutterLevel>
    <!--dep, xs: string, shutter level: "1/1, 1/2, 1/3, 1/6, 1/12, 1/25, 1/50, 1/75, 1/100, 1/120, 1/125, 1/150, 1/175,
    1/215, 1/225, 1/300, 1/400, 1/425, 1/600, 1/1000, 1/1250, 1/1750, 1/2500, 1/3500, 1/6000, 1/10000", this node is
    valid when <ExposureType> in the message XML_Exposure is "ShutterFisrt"-->
    <ShutterLevel>
  </Shutter>
```

See Also

XML_Exposure

XML_SlaveCamera

SlaveCamera message in XML format

```
<SlaveCameraInfo version="2.0" xmlns="http://www.std-cgi.org/ver20/XMLSchema">
  <id>
    <!--req, xs:string-->
  </id>
  <serverAddress>
    <addressingFormatType>
      <!--req, xs:string, "ipaddress,hostname"-->
    </addressingFormatType>
    <hostName>
      <!--dep, xs:string-->
    </hostName>
    <ipAddress>
```

```
<!--dep, xs:string-->
</ipAddress>
<ipv6Address>
  <!--dep, xs:string-->
</ipv6Address>
</serverAddress>
<userName>
  <!--req, xs:string-->
</userName>
<passWord>
  <!--wo, xs:string-->
</passWord>
<portNo>
  <!--req, xs:integer-->
</portNo>
<loginStatus>
  <!--req, xs:string, "login,logout"-->
</loginStatus>
</SlaveCameraInfo>
```

XML_SlaveCameraStatus

SlaveCameraStatus message in XML format

```
<SlaveCameraStatus version="2.0" xmlns="http://www.std-cgi.org/ver20/XMLSchema">
  <SlaveCameraLinkStatusList>
    <!--req-->
    <SlaveCameraLinkStatus>
      <id>
        <!--req, xs:string-->
      </id>
      <linkStatus/>
      <!--ro, xs:string, connection status: "online,offline"-->
    </SlaveCameraLinkStatus>
  </SlaveCameraLinkStatusList>
</SlaveCameraStatus>
```

XML_SmartCap

SmartCap capability message in XML format

```
<SmartCap version="2.0" xmlns="http://www.isapi.org/ver20/XMLSchema">
  <isSupportROI><!--opt, xs:boolean, whether to support ROI (Region of Interest)--></isSupportROI>
  <isSupportFaceDetect><!--opt, xs:boolean, whether to support face detection--></isSupportFaceDetect>
  <isSupportIntelliTrace><!--opt, xs:boolean--></isSupportIntelliTrace>
  <isSupportFieldDetection><!--opt, xs:boolean, whether to support region detection--></isSupportFieldDetection>
  <isSupportDefocusDetection><!--opt, xs:boolean--></isSupportDefocusDetection>
  <isSupportAudioDetection><!--opt, xs:boolean--></isSupportAudioDetection>
  <isSupportSceneChangeDetection><!--opt, xs:boolean, whether to support scene change detection--></
```



```
isSupportSceneChangeDetection>
<isSupportLineDetection><!--opt, xs:boolean--></isSupportLineDetection>
<isSupportRegionEntrance><!--opt, xs:boolean--></isSupportRegionEntrance>
<isSupportRegionExiting><!--opt, xs:boolean--></isSupportRegionExiting>
<isSupportLoitering><!--opt, xs:boolean--></isSupportLoitering>
<isSupportGroup><!--opt, xs:boolean--></isSupportGroup>
<isSupportRapidMove><!--opt, xs:boolean--></isSupportRapidMove>
<isSupportParking><!--opt, xs:boolean--></isSupportParking>
<isSupportUnattendedBaggage><!--opt, xs:boolean--></isSupportUnattendedBaggage>
<isSupportAttendedBaggage><!--opt, xs:boolean--></isSupportAttendedBaggage>
<isSupportPeopleDetection><!--opt, xs:boolean--></isSupportPeopleDetection>
<isSupportStorageDetection><!--opt, xs:boolean--></isSupportStorageDetection>
<isSupportShipsDetection><!--opt, xs:boolean--></isSupportShipsDetection>
<isSupportSmartCalibration><!--opt, xs:boolean--></isSupportSmartCalibration>
<isSupportShield><!--opt, xs:boolean, whether to support shielded area--></isSupportShield>
<isSupportAlgVersion><!--opt, xs:boolean, whether to support algorithm library version--></isSupportAlgVersion>
<isSupportVideoOverlap><!--opt, xs:boolean, whether to support text overlay--></isSupportVideoOverlap>
<isSupportParkingState><!--opt, xs:boolean, whether to support parking space status detection--></
isSupportParkingState>
<isSupportChannelResource><!--opt, xs:boolean--></ isSupportChannelResource>
<isSupportAnalysisUnitSwitch opt="true,false"><!--opt, xs:boolean, whether to support analysis unit switch--></
isSupportAnalysisUnitSwitch>
<isSupportHFPD><!--opt, xs:boolean, whether to support frequently appeared person detection--></isSupportHFPD>
<isSupportImageROI><!--opt, xs:boolean, whether to support smartJpeg (image ROI (Region of Interest))--></
isSupportImageROI>
</SmartCap>
```

XML_SmartSearchDescription

SmartSearchDescription message in XML format

```
<SmartSearchDescription version="2.0" xmlns="http://www.isapi.org/ver20/XMLSchema">
  <searchID>
    <!--req, xs:string, search ID, which is used to confirm the upper-level platform or system. If the platform or system
are same during two times of search, the search history will be recorded in the memory for next fast search-->
  </searchID>
  <searchResultPosition><!--req, xs: integer--></searchResultPosition>
  <maxResults><!--req, xs: integer--></maxResults>
  <trackID><!--req, xs: integer, channel ID, e.g., 101-main stream if first channel--></trackID>
  <startTime><!--req, xs: datetime, start time of validity period, e.g., 2013-06-10T 12:00:00Z--></startTime>
  <endTime><!--req, xs: datetime, end time of validity period, e.g., 2013-06-10T 13:00:00Z--></endTime>
  <type>
    <!--req, xs: string; VCA event type: "motionDetection, traversePlaneDetection,FieldDetection, faceDetection"-->
  </type>
  <MotionDetection><!--dep, it is valid when type is "motionDetection"-->
    <Grid>
      <rowGranularity></rowGranularity>
      <columnGranularity></columnGranularity>
    </Grid>
    <MotionDetectionLayout>
      <layout>
```

```

    <gridMap></gridMap>
  </layout>
</MotionDetectionLayout>
<sensitivity><!--opt, xs: integer, value range: [1,100]--></sensitivity>
</MotionDetection>
<TraversePlane><!--dep, it is valid when type is "traversePlaneDetection"-->
  <TraversePlaneParam>
    <LineList>
      <Line>
        <StartPoint>
          <x><!--req, xs: float--></x>
          <y><!--req, xs: float--></y>
        </StartPoint>
        <EndPoint>
          <x><!--req, xs: float--></x>
          <y><!--req, xs: float--></y>
        </EndPoint>
      </Line>
    </LineList>
    <crossDirection><!--opt, xs: string, crossing direction: "bothDirection, leftToRight, rightToLeft"--><crossDirection>
    <sensitivity><!--opt, xs: integer, value range: [1,100]--></sensitivity>
    <planeHeight><!--opt, xs: integer--></planeHeight>
    <detectionTarget><!--req, xs: string, detection target type: "all, human, vehicle"--><detectionTarget>
  </TraversePlaneParam>
  <dwPreTime><!--opt, xs: integer, pre-recording time--></dwPreTime>
  <dwDelayTime><!--opt, xs: integer, post-recording time--></dwDelayTime>
  <PTZPos>
    <P><!--opt, xs: integer--></P>
    <T><!--opt, xs: integer--></T>
    <Z><!--opt, xs: integer--></Z>
  </PTZPos>
  <advancedType><!--opt, xs: string, target sub type: "humanFeature, objectFeature"--></advancedType>
  <advance><!--dep, it is valid when type is "motionDetection"-->
    <HumanFeature>
      <age><!--opt, xs: integer, value range: [0,100]--></age>
      <sex><!--opt, xs: string, gender: "male,female"--></sex>
      <eyeGlass><!--opt, xs: boolean--></eyeGlass>
    </HumanFeature><!--dep, it is valid when advancedType is "humanFeature"-->
    <ObjectFeature>
      <colorRatel><!--opt, xs: integer--></colorRatel>
      <R><!--opt, xs: integer--></R>
      <G><!--opt, xs: integer--></G>
      <B><!--opt, xs: integer--></B>
    </ObjectFeature><!--dep, it is valid when advancedType is "objectFeature"-->
  </advance>
</TraversePlane>
<FieldDetection><!--dep, it is valid when type is "FieldDetection"-->
  <normalizedScreenSize>
    <normalizedScreenWidth><!--opt, xs: integer--></normalizedScreenWidth>
    <normalizedScreenHeight><!--opt, xs: integer--></normalizedScreenHeight>
  </normalizedScreenSize>
  <param>

```

```
<region>
  <pointList>
    <point>
      <x><!--opt, xs: integer--></x>
      <y><!--opt, xs: integer--></y>
    </point>
  </pointList>
</region>
</param>
<duration><!--opt, xs: integer, value range: [1,100]--></duration>
<sensitivity><!--opt, xs: integer, value range: [1,100]--></sensitivity>
<rate><!--opt, xs: integer, value range: [1,100]--></rate>
<detectionTarget><!--req, xs: string, detection target type: "all, human, vehicle"--><detectionTarget>
<dwPreTime><!--opt, xs: integer, pre-recording time--></dwPreTime>
<dwDelayTime><!--opt, xs: integer, post-recording time--></dwDelayTime>
<PTZPos>
  <P><!--opt, xs: integer--></P>
  <T><!--opt, xs: integer--></T>
  <Z><!--opt, xs: integer--></Z>
</PTZPos>
<advancedType><!--opt, xs: string, target sub type: "humanFeature, objectFeature"--></advancedType>
<advance><!--dep, it is valid when type is "motionDetection"-->
  <HumanFeature>
    <age><!--opt, xs: integer, value range: [0,100]--></age>
    <sex><!--opt, xs: string, gender: "male,female"--></sex>
    <eyeGlass><!--opt, xs: boolean--></eyeGlass>
  </HumanFeature><!--dep, it is valid when advancedType is "humanFeature"-->
  <ObjectFeature>
    <colorRatel><!--opt, xs: integer--></colorRatel>
    <R><!--opt, xs: integer--></R>
    <G><!--opt, xs: integer--></G>
    <B><!--opt, xs: integer--></B>
  </ObjectFeature><!--dep, it is valid when advancedType is "objectFeature"-->
</advance>
</FieldDetection>
<FaceDetection><!--dep, it is valid when type is "faceDetection"-->
  <param>
    <region>
      <pointList>
        <point>
          <x/>
          <y/>
        </point>
      </pointList>
    </region>
  </param>
  <duration><!--opt, xs: integer, value range: [1,100]--></duration>
  <sensitivity><!--opt, xs: integer, value range: [1,100]--></sensitivity>
  <rate><!--opt, xs: integer, value range: [1,100]--></rate>
  <detectionTarget><!--req, xs: string, detection target type: "all, human, vehicle"--><detectionTarget>
  <dwPreTime><!--opt, xs: integer, pre-recording time--></dwPreTime>
  <dwDelayTime><!--opt, xs: integer, post-recording time--></dwDelayTime>
```

```
<PTZPos>
  <P><!--opt, xs: integer--></P>
  <T><!--opt, xs: integer--></T>
  <Z><!--opt, xs: integer--></Z>
</PTZPos>
<advancedType><!--opt, xs: string, target sub type: "humanFeature, objectFeature"--></advancedType>
<advance><!--dep, it is valid when type is "motionDetection"-->
  <HumanFeature>
    <age><!--opt, xs: integer, value range: [0,100]--></age>
    <sex><!--opt, xs: string, gender: "male,female"--></sex>
    <eyeGlass><!--opt, xs: boolean--></eyeGlass>
  </HumanFeature><!--dep, it is valid when advancedType is "humanFeature"-->
  <ObjectFeature>
    <colorRatel><!--opt, xs: integer--></colorRatel>
    <R><!--opt, xs: integer--></R>
    <G><!--opt, xs: integer--></G>
    <B><!--opt, xs: integer--></B>
  </ObjectFeature><!--dep, it is valid when advancedType is "objectFeature"-->
</advance>
</FaceDetection>
</SmartSearchDescription>
```

XML_SmartSearchResult

SmartSearchResult message in XML format

```
<SmartSearchResult version="2.0" xmlns="http://www.isapi.org/ver20/XMLSchema">
  <searchID>
    <!--req, xs:string, search ID, which is used to confirm the upper-level platform or system. If the platform or system
are same during two times of search, the search history will be recorded in the memory for next fast search-->
  </searchID>
  <responseStatus><!--req, xs: string, "true,false"--></responseStatus>
  <responseStatusStrg><!--req, xs: string, "OK,NO MATCHES,MORE"--></responseStatusStrg>
  <numOfMatches><!--req, xs: integer--></numOfMatches>
  <totalMatches><!--req, xs: integer--></totalMatches>
  <matchList>
    <searchMatchItem><!--list-->
      <id><!--req, xs: integer--></id>
      <timeSpan>
        <startTime><!--req, xs: string, start time of validity period, e.g., 2013-06-10T 12:00:00Z--></startTime>
        <endTime><!--req,xs:string, end time of validity period, e.g., 2013-06-10T 12:00:00Z--></endTime>
      </timeSpan>
    </searchMatchItem>
  </matchList>
</SmartSearchResult>
```

XML_SMARTTestConfig

SMARTTestConfig message in XML format

```
<SMARTTestConfig version="2.0" xmlns="http://www.isapi.org/ver20/XMLSchema">
  <enable opt="true,false"><!--req, xs:string, whether to continue using the HDD when the HDD SMART detection
failed--></enable>
</SMARTTestConfig>
```

XML_SMARTTestStatus

SMARTTestStatus message in XML format

```
<SMARTTestStatus version="2.0" xmlns="http://www.isapi.org/ver20/XMLSchema">
  <id><!--req, xs: string--></id>
  <temprature><!--req, xs: integer, unit: °C--></temprature>
  <powerOnDay><!--req, xs: integer, unit: day--></powerOnDay>
  <selfEvaluatingStatus><!--req, xs: string, "ok, error"--></selfEvaluatingStatus>
  <allEvaluatingStatus><!--req, xs: string, "unctional,badsectors,fault"--></allEvaluatingStatus>
  <selfTestPercent><!--req, xs: integer--></selfTestPercent>
  <selfTestStatus>
    <!--req, xs: string,
"ok,aborted,interrupted,failed,unkown,electronic_element_error,servo_error,read_failed,progress,not_tested,not_rec
ognized"-->
  </selfTestStatus>
  <testType><!--req, xs: string, "short,expanded,conveyance"--></testType>
  <TestResultList><!--req, list, the maximum size is 30 bytes-->
    <TestResult><!--list-->
      <attributeID><!--req, xs: string--></attributeID>
      <attributeName><!--opt, xs: string--></attributeName>
      <status><!--req, xs: string, "ok,illegal"--></status>
      <flags><!--req, xs: integer--></flags>
      <thresholds><!--req, xs: integer--></thresholds>
      <value><!--req, xs: string--></value>
      <worst><!--req, xs: integer--></worst>
      <rawValue><!--req, xs: integer--></rawValue>
    </TestResult>
  </TestResultList>
</SMARTTestStatus>
```

XML_SMSWhiteList

SMSWhiteList message in XML format

```
<SMSWhiteList version="1.0" xmlns="http://www.isapi.org/ver20/XMLSchema" >
  <ListMember/>
</SMSWhiteList>
```

XML_SnapshotChannel

SnapshotChannel message in XML format

```
<SnapshotChannel version="2.0" xmlns="http://www.isapi.org/ver20/XMLSchema">
  <id><!--required, xs:string, ID--></id>
  <videoInputChannelID><!--required, xs:string, video input channel ID--></videoInputChannelID>
  <timingCapture><!--optional-->
    <enabled><!--required, xs:boolean, whether to enable capturing pictures by schedule--></enabled>
    <supportSchedule><!--optional, read-only, xs:boolean--></supportSchedule>
    <compress>
      <pictureCodecType><!--required, xs:string, picture format: "JPEG", "BMP", "GIF", "PNG"--></pictureCodecType>
      <pictureWidth><!--required, xs:integer, picture width--></pictureWidth>
      <pictureHeight><!--required, xs:integer, picture height--></pictureHeight>
      <quality><!--optional, xs:integer, picture quality (in percentage), it is between 1 and 100--></quality>
      <captureInterval><!--optional, xs:integer, capture interval, unit: millisecond--></captureInterval>
      <captureNumber><!--optional, xs:integer--></captureNumber>
    </compress>
  </timingCapture>
  <eventCapture><!--optional-->
    <enabled><!--required, xs:boolean--></enabled>
    <supportSchedule><!--optional, read-only, xs:boolean--></supportSchedule>
    <compress>
      <pictureCodecType><!--required, xs:string, picture format: "JPEG", "BMP", "GIF", "PNG"--></pictureCodecType>
      <pictureWidth><!--required, xs:integer, picture width--></pictureWidth>
      <pictureHeight><!--required, xs:integer, picture height--></pictureHeight>
      <quality><!--optional, xs:integer, picture quality (in percentage), it is between 0 and 100--></quality>
      <captureInterval><!--optional, xs:integer, capture interval, unit: millisecond--></captureInterval>
      <captureNumber><!--optional, xs:integer--></captureNumber>
    </compress>
  </eventCapture>
</SnapshotChannel>
```

XML_SnapshotChannelList

SnapshotChannelList message in XML format

```
<SnapshotChannelList version="2.0" xmlns="http://www.isapi.org/ver20/XMLSchema">
  <SnapshotChannel><!--optional, see details in the message XML_SnapshotChannel--></SnapshotChannel>
</SnapshotChannelList>
```

See Also

XML_SnapshotChannel

XML_SNMP

XML message about SNMP configuration

```
<SNMP version="2.0" xmlns="http://www.isapi.org/ver20/XMLSchema">
  <SNMPv1c/><!--dependent -->
  <SNMPv2c/><!--dependent-->
  <SNMPAdvanced/><!--dependent, advanced parameters-->
  <listenPort> <!--optional, xs:integer, SNMP port--></listenPort>
</SNMP>
```

XML_SNMPSAdvanced

XML message about SNMP advanced parameter configuration

```
<SNMPAdvanced xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0" >
  <localEngineID>
    <!--required, xs:hexBinary, local device engine-->
  </localEngineID>
  <authenticationNotificationEnabled>
    <!--optional, xs:boolean, notification of failure to enable authentication-->
  </authenticationNotificationEnabled>
  <SNMPUserList/>
    <!--optional -->
  <SNMPNotificationFilterList/>
    <!--optional, filtering table based on OIDS-->
  <notificationEnabled>
    <!--optional, xs:boolean -->
  </notificationEnabled>
  <SNMPNotificationReceiverList/>
    <!--optional-->
  <enabled>
    <!--required, xs:boolean -->
  </enabled>
  <SNMPTrapReceiverList/>
    <!--optional, all trap address information of SNMP in version 3-->
</SNMPAdvanced>
```

XML_SNMPTTrapReceiver

XML message about parameters of single trap address of SNMP

```
<SNMPTTrapReceiver xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0" >
  <id>
    <!--required, xs:string, ID -->
  </id>
  <ReceiverAddress>
    <!--required -->
  <addressingFormatType>
    <!--required, xs:string, "ipaddress,hostname"-->
  </addressingFormatType>
  <hostName>
```

```
<!--dependent, xs:string -->
</hostName>
<ipAddress>
  <!--dependent, xs:string -->
</ipAddress>
<ipv6Address>
  <!--dependent, xs:string -->
</ipv6Address>
<portNo>
  <!--optional, xs:integer -->
</portNo>
</ReceiverAddress>
<notificationType>
  <!--required, xs:string, "trap,inform" -->
</notificationType>
<communityString>
  <!--optional, xs:string -->
</communityString>
</SNMPTrapReceiver>
```

XML_SNMPTrapReceiverList

XML message about trap address parameters of SNMP

```
<SNMPTrapReceiverList version="2.0" xmlns="http://www.isapi.org/ver20/XMLSchema">
  <SNMPTrapReceiver/> <!--optional-->
</SNMPTrapReceiverList>
```

XML_SoftwareService

SoftwareService message in XML format

```
<SoftwareService version="2.0" xmlns="http://www.isapi.org/ver20/XMLSchema">
  <ThirdStream><!--opt, third stream configuration-->
    <enabled><!--req, xs: boolean, "true,false", whether to enable third stream configuration--></enabled>
  </ThirdStream>
  <PanoramaDisplay><!--opt, display the image range on panorama view-->
    <enabled>
      <!--req, xs: boolean, "true,false", whether to enable image range display on panorama view-->
    </enabled>
  </PanoramaDisplay>
  <MotionDetect>
    <enabled><!--whether to enable motion detetion--></enabled>
  </MotionDetect>
</SoftwareService>
```


XML_SocketIP

SocketIP message in XML format

```
<SocketIP version="2.0" xmlns="http://www.isapi.org/ver20/XMLSchema">
  <ipAddress>
    <!--opt, xs:string-->
  </ipAddress>
  <ipv6Address>
    <!--opt, xs:string-->
  </ipv6Address>
</SocketIP>
```

XML_sourceCapability

sourceCapability message in XML format

```
<sourceCapability version="2.0" xmlns="http://www.isapi.org/ver20/XMLSchema">
  <videoInputNums><!--req, xs:integer--></videoInputNums>
  <audioInputNums><!--opt, xs:integer--></audioInputNums>
</sourceCapability>
```

XML_sourceDescriptor

sourceDescriptor message in XML format

```
<sourceDescriptor version="2.0" xmlns="http://www.isapi.org/ver20/XMLSchema">
  <adminProtocol><!--req, xs:string, "HIKVISION,SONY,ISAPI,ONVIF,..."--></adminProtocol>
  <addressingFormatType><!--req,xs:string, opt="ipaddress,hostname"--></addressingFormatType>
  <hostName><!--dep,xs:string, domain name--></hostName>
  <ipAddress><!--dep,xs:string, IP address--></ipAddress>
  <ipv6Address><!--dep,xs:string, IPv6 address--></ipv6Address>
  <managePortNo><!--req,xs:integer, port number--></managePortNo>
  <userName><!--req,xs:string--></userName>
  <password><!--req,wo,xs:string--></password>
</sourceDescriptor>
```

XML_ssd

JSON message about a SSD information

```
<?xml version="1.0" encoding="utf-8"?>
<ssd version="2.0" xmlns="http://www.isapi.org/ver20/XMLSchema">
  <id><!--required, read-only, xs:string, SSD ID--></id>
  <ssidName><!--required, read-only, xs:string, SSD name--></ssidName>
```

```
<version><!--optional, xs:string, SSD firmware version--></version>
</ssd>
```

XML_ssdList

JSON message about information of all SSDs

```
<?xml version="1.0" encoding="utf-8"?>
<ssdList version="2.0" xmlns="http://www.isapi.org/ver20/XMLSchema">
  <ssd><!--optional, see XML-ssd for details--></ssd>
</ssdList>
```

See Also

XML_ssd

XML_SSdsmARTTest

XML message about SSD S.M.A.R.T detection parameters

```
<?xml version="1.0" encoding="utf-8"?>
<SSdsmARTTest version="2.0" xmlns="http://www.isapi.org/ver20/XMLSchema">
  <testType><!--optional, xs:string, self-test type: "short", "expanded", "conveyance"--></testType>
</SSdsmARTTest>
```

XML_SSD_SmartTestStatus

JSON message about SSD S.M.A.R.T detection status

```
<?xml version="1.0" encoding="utf-8"?>
<SmartTestStatus version="2.0" xmlns="http://www.isapi.org/ver20/XMLSchema">
  <id><!--required, xs:string, SSD ID--></id>
  <temprature><!--required, xs:integer, temperature, unit: degree--></temprature>
  <powerOnDay><!--required, xs:integer, hour of use, unit: day(s)--></powerOnDay>
  <selfEvaluaiingStatus><!--required, xs:string, self-evaluation status: "ok", "error"--></selfEvaluaiingStatus>
  <allEvaluaiingStatus><!--required, xs:string, overall evaluation: "functional"-in good condition, "badsectors"-bad
block, "fault"-error--></allEvaluaiingStatus>
  <selfTestPercent><!--required, xs:integer, self-test progress--></selfTestPercent>
  <selfTestStatus><!--required, xs:string, self-test status: "ok", "aborted", "interrupted", "failed", "unkown",
"electronic_element_error", "servo_error", "read_failed", "progress"-in progress, "not_tested", "not_recognized"--></
selfTestStatus>
  <testType><!--required, xs:string, self-test type: "short", "expanded", "conveyance"--></testType>
  <TestResultList><!--required, "Size=30"-->
    <TestResult><!--list-->
      <attributeID><!--required, xs:string, attribute ID--></attributeID>
      <attributeName><!--optional, xs:string, attribute name--></attributeName>
      <status><!--required, xs:string, status: "ok"-normal, "illegal"-error--></status>
      <flags><!--required, xs:integer, attribute operation mark--></flags>
```

```
<thresholds><!--required, xs:integer, threshold--></thresholds>
<value><!--required, xs:string, current value--></value>
<worst><!--required, xs:integer, the worst value--></worst>
<rawValue><!--required, xs:integer, raw value--></rawValue>
</TestResult>
</TestResultList>
</SmartTestStatus>
```

XML_SSH

SSH message in XML format

```
<SSH version="2.0" xmlns="http://www.isapi.org/ver20/XMLSchema">
  <enabled><!--req, xs:boolean--></enabled>
</SSH>
```

XML_staticRouteList

staticRouteList message in XML format

```
<staticRouteList version="2.0" xmlns="http://www.isapi.org/ver20/XMLSchema">
  <staticRoute><!--opt-->
    <id>
      <!--req, xs: integer, this node must be configured when deleting-->
    </id>
    <enabled><!--req, xs: boolean--></enabled>
    <dstIpAddr><!--req, xs: string--></dstIpAddr>
    <dstNetmask><!--req, xs: string--></dstNetmask>
    <dstGateway><!--req, xs: string--></dstGateway>
    <ifName><!--req, xs: string, 0: G1, 1: G2--></ifName>
  </staticRoute>
</staticRouteList>
```

XML_storage

XML message about storage working mode

```
<?xml version="1.0" encoding="utf-8"?>
  <storage version="2.0" xmlns="http://www.isapi.org/ver20/XMLSchema">
    <hddList><!--optional, the contents are same to the message XML_hddList (related URI: /ISAPI/ContentMgmt/Storage/hdd)--></hddList>
    <nasList><!--optional, the contents are same to the message XML_nasList (related URI: /ISAPI/ContentMgmt/Storage/nas)--></nasList>
    <workMode><!--optional, xs:string, working mode: "group", "quota", "extract--></workMode>
    <ssdList><!--optional, the contents are same to the message XML_ssdList (related URI: /ISAPI/ContentMgmt/Storage/ssd)--></ssdList>
  </storage>
```

XML_storageExtension

XML message about storage strategy.

```
<storageExtension xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0" >
  <LoopEnable,opt="true,false">
    <!--optional, xs:boolean, whether to enable overwriting-->
  </LoopEnable>
  <enableDormant>
    <!--optional, xs:boolean-->
  </enableDormant>
  <packDuration>
    <!--optional, xs:integer, unit: minute-->
  </packDuration>
  <logStorageMode>
    <!--optional, xs:string, log storage mode: "system"-system default mode, "custom"-custom mode-->
  </logStorageMode>
  <logStorageId>
    <!--dependent, xs:integer, disk ID (saving logs in a disk) or array number (saving logs in disk arrays), which are valid
only when logStorageMode is "custom"; -->
  </logStorageId>
  <logStorageCycle>
    <!--dependent, xs:integer, log storage period, which is valid only when logStorageMode is "custom"-->
  </logStorageCycle>
</storageExtension>
```

XML_StreamingChannel

StreamingChannel message in XML format

```
<StreamingChannel version="2.0" xmlns="http://www.isapi.org/ver20/XMLSchema">
  <id><!--req,xs:string;id--></id>
  <channelName><!--req,xs:string--></channelName>
  <enabled><!--req,xs:boolean--></enabled>
  <Transport><!--req-->
    <maxPacketSize><!--opt, xs: integer--></maxPacketSize>
    <audioPacketLength><!--opt, xs: integer--></audioPacketLength>
    <audioInboundPacketLength><!--opt, xs: integer--></audioInboundPacketLength>
    <audioInboundPortNo><!--opt, xs: integer--></audioInboundPortNo>
    <videoSourcePortNo><!--opt, xs: integer--></videoSourcePortNo>
    <audioSourcePortNo><!--opt, xs: integer--></audioSourcePortNo>
    <ControlProtocolList><!--req, protocol types for streaming-->
      <ControlProtocol><!--req-->
        <streamingTransport>
          <!--req,xs:string,"HTTP,RTSP,SHTTP,SRTP"-->
        </streamingTransport>
      </ControlProtocol>
    </ControlProtocolList>
    <Unicast><!--opt-->
```

```

<enabled><!--req, xs: boolean--></enabled>
<interfaceID><!--opt, xs: string--></interfaceID>
<rtpTransportType>
  <!--opt, xs: string, "RTP/UDP, RTP/TCP"-->
</rtpTransportType>
</Unicast>
<Multicast><!--opt-->
  <enabled><!--req, xs: boolean--></enabled>
  <userTriggerThreshold><!--opt, xs: integer--></userTriggerThreshold>
  <destIPAddress><!--dep, xs: string--></destIPAddress>
  <videoDestPortNo><!--opt, xs: integer--></videoDestPortNo>
  <audioDestPortNo><!--opt, xs: integer--></audioDestPortNo>
  <destIPv6Address><!--dep, xs: string--></destIPv6Address>
  <ttl><!--opt, xs: integer--></ttl>
  <activeMulticastEnabled>
    <!--opt, xs: boolean, "true,false", whether to enable active multicast, which is mutual exclusion with passive
multicast-->
  </activeMulticastEnabled>
  <packagingFormat><!--opt, xs: string, container format--></packagingFormat>
  <FecInfo><!--opt-->
    <fecRatio>
      <!--req, read-only, xs: integer, extra bandwidth occupation ratio of forward error correction (FEC) data, the value
is between 0 and 100, the default value is 0-->
    </fecRatio>
    <fecDestPortNo>
      <!--opt, xs: integer, Port No. of FEC multicast, the default port No. is specified by device-->
    </fecDestPortNo>
  </FecInfo>
</Multicast>
<Security><!--opt-->
  <enabled><!--req, xs: boolean--></enabled>
  <certificateType><!--req, xs: string, opt="digest,digest/baisc" def="digest"--></certificateType>
  <SecurityAlgorithm><!--dependency, this node is valid when certificateType is "digest" or "digest/basic"-->
    <algorithmType><!--optional, xs:string, algorithm type: "MD5", "SHA256", "MD5/SHA256"--></algorithmType>
  </SecurityAlgorithm>
</Security>
<SRTPMulticast><!--opt-->
  <SRTPVideoDestPortNo><!--opt, xs:integer--></SRTPVideoDestPortNo>
  <SRTPAudioDestPortNo><!--opt, xs:integer--></SRTPAudioDestPortNo>
</SRTPMulticast>
</Transport>
<Video><!--opt-->
  <enabled><!--req,xs:boolean--></enabled>
  <videoInputChannelID>
    <!--req,xs:string;id-->
  </videoInputChannelID>
  <videoCodecType>
    <!--req,xs:string,"MPEG4,MJPEG,3GP,H.264,HK.264,MPNG,SVAC,H.265"-->
  </videoCodecType>
  <videoResolutionWidth>
    <!--req,xs:integer-->
  </videoResolutionWidth>

```

```

<videoResolutionHeight>
  <!--req,xs:integer-->
</videoResolutionHeight>
<videoQualityControlType>
  <!--opt,xs:string,"CBR,VBR"-->
</videoQualityControlType>
<constantBitRate>
  <!--dep, xs:integer, constant bit rate, unit: kbps-->
</constantBitRate>
<vbrUpperCap>
  <!--dep, xs:integer, upper limit of variable bit rate, unit: kbps-->
</vbrUpperCap>
<vbrLowerCap>
  <!--dep, xs:integer, lower limit of variable bit rate, unit: kbps-->
</vbrLowerCap>
<maxFrameRate>
  <!--req, xs:integer, maximum frame rate, the value is multiplied by 100 to be returned-->
</maxFrameRate>
<keyFrameInterval><!--opt, xs:integer, milliseconds--></keyFrameInterval>
<rotationDegree><!--opt, xs:integer, degrees, 0..360--></rotationDegree>
<mirrorEnabled><!--opt, xs:boolean--></mirrorEnabled>
<snapshotImageType><!--opt, xs:string, "JPEG,GIF,PNG"--></snapshotImageType>
<Mpeg4Profile><!--dep, xs:string, "SP,ASP"--></Mpeg4Profile>
<H264Profile><!--dep, xs:string, "Baseline,Main,High, Extended"--></H264Profile>
<SVACProfile><!--dep, xs:string, "Baseline,Main,High,Extended"--></SVACProfile>
<GovLength><!--opt, xs:integer--></GovLength>
<SVC>
  <enabled><!--req, xs:boolean--></enabled>
  <SVCMode><!--dep, xs:string, "manual,auto"--></SVCMode>
</SVC>
<smoothing><!--opt, xs:integer--></smoothing>
<SmartCodec><!--dep, -->
  <enabled><!--req, xs:boolean--></enabled>
</SmartCodec>
<vbrAverageCap><!--dep, xs:integer, in kbps--></vbrAverageCap>
<IntelligentInfoDisplayMethod>
  <!--opt, intelligent information displaying method, xs:string, "player,non-player"-->
</IntelligentInfoDisplayMethod>
</Video>
<Audio><!--opt-->
  <enabled><!--req, xs:boolean--></enabled>
  <audioInputChannelID><!--req, xs:string;id--></audioInputChannelID>
  <audioCompressionType>
    <!--req, xs:string, "G.711alaw,G.711ulaw,G.726,G.729,G.729a,G.729b,PCM,MP3,AC3,AAC,ADPCM, MP2L2"-->
  </audioCompressionType>
  <audioInboundCompressionType>
    <!--opt, xs:string, "G.711alaw,G.711ulaw,G.726,G.729,G.729a,G.729b,PCM,MP3,AC3,AAC,ADPCM,MP2L2"-->
  </audioInboundCompressionType>
  <audioBitRate><!--opt, xs:integer, in kbps--></audioBitRate>
  <audioSamplingRate><!--opt, xs:float, in kHz--></audioSamplingRate>
  <audioResolution><!--opt, xs:integer, in bits--></audioResolution>
  <VoiceChanger><!--opt, xs:integer, -12..0..12-->

```

```
<enabled><!--req, xs:boolean--></enabled>
<level><!--req, xs:integer, "-12..12"--></level>
</VoiceChanger>
</Audio>
<enableCABAC><!--opt, xs:boolean--></enableCABAC>
<subStreamRecStatus><!--opt, xs:boolean--></subStreamRecStatus>
<customStreamEnable><!--opt, xs:boolean, whether the stream is custom stream: "true"-yes, this node is not
returned-no--></customStreamEnable>
</StreamingChannel>
```

XML_StreamingChannelList

StreamingChannelList message in XML format

```
<StreamingChannelList version="2.0" xmlns="http://www.isapi.org/ver20/XMLSchema">
  <StreamingChannel/><!--opt, see details in XML_StreamingChannel-->
</StreamingChannelList>
```

See Also

XML_StreamingChannel

XML_StreamingStatus

StreamingStatus message in XML format

```
<StreamingStatus version="2.0" xmlns="http://www.isapi.org/ver20/XMLSchema">
  <totalStreamingSessions><!--req, xs: integer, number of streaming sessions--></totalStreamingSessions>
  <StreamingSessionStatusList/><!--dep, this node is valid only when the number of sessions is larger than 0, refer to
XML_StreamingSessionStatusList for details-->
</StreamingStatus>
```

See Also

XML_StreamingSessionStatusList

XML_StreamingSessionStatusList

StreamingSessionStatusList message in XML format

```
<StreamingSessionStatusList version="2.0" xmlns="http://www.isapi.org/ver20/XMLSchema">
  <StreamingSessionStatus>
    <clientAddress><!--req-->
      <ipAddress><!--dep, xs:string--></ipAddress>
      <ipv6Address><!--dep, xs:string--></ipv6Address>
    </clientAddress>
  </StreamingSessionStatus>
</StreamingSessionStatusList>
```

XML_SubscribeEvent

SubscribeEvent message in XML format

```
<SubscribeEvent version="2.0" xmlns="http://www.isapi.org/ver20/XMLSchema" >
  <heartbeat>
    <!--optional, xs:integer, heartbeat interval, unit: second, the default value is 30s-->
  </heartbeat>
  <eventMode>
    <!--required, xs:string, "all"-upload all alarms/events, "list"-upload specified alarm/event-->
  </eventMode>
  <EventList>
    <Event><!--uploading mode of specified alarm/event, this node exists only when eventMode is "list"-->
      <type>
        <!--required, xs:string, alarm/event types, which are obtained from the capability, refer to Alarm/Event Types for Subscription for its values-->
      </type>
      <minorAlarm>
        <!--opt, xs:string, minor alarm type: "0x400,0x401,0x402,0x403", see details in Access Control Event Type. This node is required when type is "AccessControllerEvent"-->
      </minorAlarm>
      <minorException>
        <!--opt, xs:string, minor exception type: "0x400,0x401,0x402,0x403", see details in Access Control Event Type. This node is required when type is "AccessControllerEvent"-->
      </minorException>
      <minorOperation>
        <!--opt, xs:string, minor operation type: "0x400,0x401,0x402,0x403", see details in Access Control Event Type. This node is required when type is "AccessControllerEvent"-->
      </minorOperation>
      <minorEvent>
        <!--opt, xs:string, minor event type: "0x01,0x02,0x03,0x04", see details in Access Control Event Type. This node is required when type is "AccessControllerEvent"-->
      </minorEvent>
      <pictureURLType>
        <!--opt, xs:string, alarm picture format: "binary"-binary, "localURL"-device local URL, "cloudStorageURL"-cloud storage URL-->
      </pictureURLType>
    </Event>
  </EventList>
  <channels>
    <!--optional, xs:string, event linked channel information, and multiple channels can be linked, each channel is separated by comma, e.g., "1,2,3,4..."-->
  </channels>
  <channels>
    <!--optional, xs:string, specify channels (each channel is separated by comma, e.g., "1,2,3,4...") to be armed, this node does not exist if you want to arm all channels, and if this node exists, the sub node <channels> in the node <Event> is invalid-->
  </channels>
  <identityKey max="64"/>
  <!--opt, xs: string, interaction command of subscription, supports subscribing comparison results of face picture
```


library (importing with this command), the maximum length is 64-->
</SubscribeEvent>

XML_SubscribeEventCap

SubscribeEventCap capability message in XML format

```
<SubscribeEventCap version="2.0" xmlns="http://www.isapi.org/ver20/XMLSchema">
  <heartbeat min="" max="" />
  <!--optional, heartbeat time interval, unit: second-->
  <format opt="xml,json"/><!--req, supported message format-->
  <channelMode opt="all,list" />
  <!--required, channel subscription mode: "all"-subscribe events/alarms of all channels, "list"-subscribe events/
alarms of specific channels-->
  <eventMode opt="all,list" />
  <!--required, event subscription mode: "all"-subscribe all event types (must be supported), "list"-subscribe specific
event types, if "list" is returned, "all" will also be returned-->
  <!--if both the channelMode and eventMode returns "all", it indicates that the device does not support subscribing
event/alarm by event type or channel-->
  <EventList><!--required, dependent, upload mode of specified alarms/events, it is valid only when eventMode is
"list"-->
    <Event><!--required-->
      <type><!--required, xs:string, refer to Supported Alarm/Event Types for details--></type>
      <minorAlarm opt="0x400,0x401,0x402,0x403">
        <!--opt, xs:string, minor alarm type, see details in Access Control Event Type. This node is required when type is
"AccessControllerEvent"-->
      </minorAlarm>
      <minorException opt="0x400,0x401,0x402,0x403">
        <!--opt, xs:string, minor exception type, see details in Access Control Event Type. This node is required when type
is "AccessControllerEvent"-->
      </minorException>
      <minorOperation opt="0x400,0x401,0x402,0x403">
        <!--opt, xs:string, minor operation type, see details in Access Control Event Type. This node is required when type
is "AccessControllerEvent"-->
      </minorOperation>
      <minorEvent opt="0x01,0x02,0x03,0x04">
        <!--opt, xs:string, minor event type, see details in Access Control Event Type. This node is required when type is
"AccessControllerEvent"-->
      </minorEvent>
      <pictureURLType opt="binary,localURL,cloudStorageURL" def="" />
      <!--opt, xs:string, alarm picture format: "binary"-binary, "localURL"-device local URL, "cloudStorageURL"-cloud
storage URL, and the def is followed by the default format-->
    </Event>
  </EventList>
  <pictureURLType opt="binary,localURL,cloudStorageURL" def="" />
  <!--opt, xs:string, alarm picture format: "binary"-binary picture, "localURL"-device local URL, "cloudStorageURL"-
cloud storage URL. This node is the method of uploading all pictures related to the event. If this node is applied,
<b>pictureURLType</b> in <Event> is invalid; otherwise, pictures will be uploaded using the default method returned by the
device capability. For front-end devices, the default method is uploading binary pictures; for back-end devices, the
default method is by device local URL-->
  <ChangedUploadSub><!--message subscription-->
```

```
<interval/><!--opt, xs:integer, lifecycle of arming GUID, the default value is 5 minutes, unit: second. The device will
generate new GUID for the arming connection after it is disconnected for the set lifecycle-->
<StatusSub>
  <all/><!-- opt, xs:boolean, whether to subscribe all events-->
  <channel/><!--opt, xs:boolean, whether to subscribe channel status. This node is not required when <all> is
"true"-->
  <hd/><!--opt, xs:boolean, whether to subscribe disk status. This node is not required when <all> is "true"-->
  <capability/><!--opt, xs:boolean, whether to subscribe capability change status. This node is not required when
<all> is "true"-->
</StatusSub>
</ChangedUploadSub>
<identityKey max="64"/>
<!--opt, xs: string, interaction command of subscription, supports subscribing comparison results of face picture
library (importing with this command), the maximum length is 64-->
</SubscribeEventCap>
```

XML_SubscribeEventResponse

SubscribeEventResponse message in XML format

```
<SubscribeEventResponse>
  <id><!--req, xs:integer, subscription ID--></id>
  <FailedEventList>
    <!--opt, list of subscription failed events. When subscription failed, it should be returned, and the upper layer can
check whether all event/alarm subscriptions are succeeded via the existence of node FailedEventList-->
    <Event>
      <type>
        <!--req, xs:string, refer to Supported Alarm/Event Types for details-->
      </type>
      <minorAlarm>
        <!--opt, xs:string, minor alarm type: "0x400,0x401,0x402,0x403", see details in Access Control Event Type. This
node is required when type is "AccessControllerEvent"-->
      </minorAlarm>
      <minorException>
        <!--opt, xs:string, minor exception type: "0x400,0x401,0x402,0x403", see details in Access Control Event Type.
This node is required when type is "AccessControllerEvent"-->
      </minorException>
      <minorOperation>
        <!--opt, xs:string, minor operation type: "0x400,0x401,0x402,0x403", see details in Access Control Event Type.
This node is required when type is "AccessControllerEvent"-->
      </minorOperation>
      <minorEvent>
        <!--opt, xs:string, minor event type: "0x01,0x02,0x03,0x04", see details in Access Control Event Type. This node is
required when type is "AccessControllerEvent"-->
      </minorEvent>
      <pictureURLType>
        <!--opt,xs:string, opt="binary,localURL,cloudStorageURL", alarm picture transmission mode: "binary"-binary,
"localURL"-device local URL, "cloudStorageURL"-cloud storage URL-->
      </pictureURLType>
      <channels>
        <!--opt, xs:string, "1,2,3,4...", event related channel ID, supports multiple channels, and the channel ID is
```

```
separated by commas-->
</channels>
<subStatusCode>
  <!--req, string, subscription failure error code-->
</subStatusCode>
</Event>
</FailedEventList>
</SubscribeEventResponse>
```

XML_SupplementLight

SupplementLight message in XML format

```
<SupplementLight><!--opt-->
<mode><!--opt, xs: string, adjustment mode of supplement light, opt="schedule,off,on,auto"--></mode>
<Schedule>
  <TimeRange><!--req-->
    <beginTime><!--req, xs: time, ISO8601 time--></beginTime>
    <endTime><!-- req, xs: time, ISO8601 time--></endTime>
  </TimeRange>
</Schedule>
<brightnessLimit>
  <!--dep, xs: integer, brightness of supplement light, which is between 0 and 100; the brightness can be adjust when
<bmode> is set to "on"-->
</brightnessLimit>
<supplementLightMode>
  <!--opt, xs: string, illumination mode: "mixed"-hybrid, "whitelight"-white light, "close"-disabled-->
</supplementLightMode>
<irLightBrightness>
  <!--dep, xs: integer, brightness of IR supplement light, which is between 0 and 100; this node is valid only when
<supplementLightMode> is set to "mixed" and <mixedLightBrightnessRegulatMode> is set to "manual"-->
</irLightBrightness>
<mixedLightBrightnessRegulatMode>
  <!--dep, xs: string, brightness adjustment mode of hybrid supplement light, opt="manual,auto"; this node is valid
only when <supplementLightMode> is set to "mixed"-->
</mixedLightBrightnessRegulatMode>
<highIrLightBrightness>
  <!--dep, xs: integer, brightness of far IR light, which is between 0 and 100; this node cannot be configured together
with node <irLightBrightness>-->
</highIrLightBrightness>
<highWhiteLightBrightness>
  <!--dep, xs: integer, brightness of far white light, which is between 0 and 100; this node cannot be configured
together with node <brightnessLimit>-->
</highWhiteLightBrightness>
<lowIrLightBrightness>
  <!--dep, xs: integer, brightness of near IR light, which is between 0 and 100; this node cannot be configured together
with node <irLightBrightness>-->
</lowIrLightBrightness>
<lowWhiteLightBrightness>
  <!--dep, xs: integer, brightness of near white light, which is between 0 and 100; this node cannot be configured
together with node <brightnessLimit>-->
```

```
</lowWhiteLightBrightness>
<whiteLightBrightness><!--dep, xs: integer, white light brightness, which is between 0 and 100--></
whiteLightBrightness>
  <irLightbrightnessLimit><!--dep, xs:integer, IR light brightness limit, range: [0,100]; this node is valid when the value
of mode "auto"--></irLightbrightnessLimit>
  <whiteLightbrightnessLimit><!--dep, xs:integer, white light brightness limit, range: [0,100]; this node is valid when
the value of mode "auto"--></whiteLightbrightnessLimit>
  <unscheduledBrightness><!--optional, xs:integer, brightness in the unscheduled time, by default it is 0 (the
supplement light is disabled) and the value is between 0 and 100--></unscheduledBrightness>
</SupplementLight>
```

XML_SyncSignalOutputList

SyncSignalOutputList message in XML format

```
<SyncSignalOutputList version="2.0" xmlns="http://www.isapi.org/ver20/XMLSchema">
  <SyncSignalOutput>
    <IOWorkMode>
      <!--req, xs:string, IO output mode: "flashLight"-strobe light control mode, "polarizer"-polarizer control mode,
"continuousLight"-solid light control mode, "flashLightSwitch"-strobe light switching mode (used to switch between
the strobe supplement light mode and IR mode). When this node is set to "polarizer", only <outputStatus>,
<detectBrightnessEnable>, <brightnessThreshold>, <flashEnabled>, <startHour>, <startMinute>, <endHour>, and
<endMinute> are valid, and the <outputStatus> can only be set to "high" or "low". When this node is set to
"continuousLight", only <detectBrightnessEnable>, <brightnessThreshold>, <flashEnabled>, <startHour>,
<startMinute>, <endHour>, and <endMinute> are valid-->
    </IOWorkMode>
    <id>
      <!--req, xs:integer, number of IO outputs, which is between 1 and 8-->
    </id>
    <defaultStatus>
      <!--req, xs:string, IO default status: "high, low"-->
    </defaultStatus>
    <outputStatus>
      <!--req,xs:string, IO effective status: "high, low, pulse"-->
    </outputStatus>
    <aheadTime>
      <!--req, xs:integer, IO pre-output time, unit: microsecond-->
    </aheadTime>
    <timeDelay>
      <!--req, xs:integer, IO effective duration, unit: microsecond-->
    </timeDelay>
    <freqMultiplyulti>
      <!--req, xs:integer, frequency multiplication, which is between 1 and 15-->
    </freqMultiplyulti>
    <dutyRate>
      <!--req, xs:integer, duty ratio, which is between 0 and 40, unit: %-->
    </dutyRate>
    <postFlashEnable>
      <!--req, xs:boolean, checkpoint output: 0-enable, 1-disable-->
    </postFlashEnable>
    <illegalFlashEnable>
```

```
<!--req, xs:boolean, violation output: 0/"true"-enable, 1/"false"-disable-->
</illegalFlashEnable>
<videoFlashEnable>
  <!--req, xs:boolean, video output: 0/"true"-enable, 1/"false"-disable-->
</videoFlashEnable>
<detectBrightnessEnable>
  <!--req, xs:boolean, whether to enable flash light for automatic brightness detection-->
</detectBrightnessEnable>
<brightnessThreshold>
  <!--dep, xs:integer, brightness threshold of the enabled flash light, which is between 0 and 100-->
</brightnessThreshold>
<flashEnabled>
  <!--req, xs:boolean, whether to enable flash light-->
</flashEnabled>
<startHour>
  <!--dep, xs:integer, start time in hour-->
</startHour>
<startMinute>
  <!--dep, xs:integer, start time in minute-->
</startMinute>
<endHour>
  <!--dep, xs:integer, end time in hour-->
</endHour>
<endMinute>
  <!--dep, xs:integer, end time in minute-->
</endMinute>
<plateBrightness>
  <!--req, xs:boolean, whether to enable flash light by license plate brightness: 0-disable, 1-enable. Only one node
among <detectBrightness>, <flashEnabled>, <manualBrightnessEnable> and <plateBrightness> can be enabled-->
</plateBrightness>
<incrBrightEnable>
  <!--opt, xs:boolean, whether to enable brightness enhancement mode (for solid light mode)-->
</incrBrightEnable>
<incrBrightTime>
  <!--req, xs:integer, brightness enhancement duration. This node is valid only when <incrBrightEnable> is "true"-->
</incrBrightTime>
<incrBrightPercent>
  <!--req, xs:integer, percentage of brightness enhancement, which is between 0 and 100. This node is valid only
when <incrBrightEnable> is "true"-->
</incrBrightPercent>
<brightness>
  <!--dep, xs:integer, solid light brightness (for solid light mode), which is between 0 and 100-->
</brightness>
<delayCaptureTime>
  <!--dep, xs:integer, delayed capture time, which is between 1 and 1000, unit: millisecond. This node is valid only
when <incrBrightEnable> is "true"-->
</delayCaptureTime>
<manualBrightnessEnable>
  <!--req, xs:boolean, whether to enable adjusting brightness manually-->
</manualBrightnessEnable>
<manualBrightness>
  <!--dep, xs:integer, brightness adjusted manually, which is between 0 and 100-->
```

```
</manualBrightness>  
</SyncSignalOutput>  
</SyncSignalOutputList>
```

XML_TargetEnhancement

TargetEnhancement message in XML format

```
<TargetEnhancement version="2.0" xmlns="http://www.isapi.org/ver20/XMLSchema">  
  <enabled><!--req,xs:boolean--></enabled>  
</TargetEnhancement>
```

XML_TargetEnhancementCap

TargetEnhancementCap message in XML format

```
<TargetEnhancementCap version="2.0" xmlns="http://www.isapi.org/ver20/XMLSchema">  
  <enabled opt="true,false"><!--req, xs:boolean--></enabled>  
</TargetEnhancementCap>
```

XML_Telnetd

Telnetd message in XML format

```
<Telnetd version="2.0" xmlns="http://www.isapi.org/ver20/XMLSchema">  
  <enabled>  
    <!--req, xs:boolean, whether to enable telnet: "true, false"-->  
  </enabled>  
  <radarMsgEnabled>  
    <!--req, xs:boolean, whether to enable uploading radar debugging information: "true, false"-->  
  </radarMsgEnabled>  
  <illegalLoginLock>  
    <!--req, xs:boolean, "true, false"-->  
  </illegalLoginLock>  
</Telnetd>
```

XML_tempRange

tempRange message in XML format

```
<?xml version="1.0" encoding="utf-8"?>  
<TempRange version="2.0" xmlns="http://www.isapi.org/ver20/XMLSchema">  
  <mode>  
    <!--req, xs:string, temperature range mode: "automatic"-automatic mode, "manual"-manual mode-->  
  </mode>  
  <temperatureUpperLimit>
```

```
<!--dep, xs:float, maximum temperature, corrects to one decimal, the minimum value is "-20", the maximum value is "550", unit: Celsius degree; its value should be larger than temperatureLowerLimit; it is valid when the value of mode is "manual"-->
</temperatureUpperLimit>
<temperatureLowerLimit>
  <!--dep, xs:float, minimum temperature, corrects to one decimal, the minimum value is "-20", the maximum value is "550", unit: Celsius degree; its value should be smaller than temperatureUpperLimit; it is valid when the value of mode is "manual"-->
</temperatureLowerLimit>
</TempRange>
```

XML_TextOverlay

TextOverlay message in XML format

```
<TextOverlay version="2.0" xmlns="http://www.isapi.org/ver20/XMLSchema">
  <id><!--req, xs:string, ID--></id>
  <enabled><!--req, xs:boolean--></enabled>
  <positionX><!--req, xs:float--></positionX>
  <positionY><!--req, xs:float--></positionY>
  <displayText><!--req, xs:string--></displayText>
</TextOverlay>
```

XML_TextOverlayList

TextOverlayList message in XML format

```
<TextOverlayList version="2.0" xmlns="http://www.isapi.org/ver20/XMLSchema">
  <TextOverlay><!--opt, see details in the message of XML_TextOverlay-->
</TextOverlayList>
```

See Also

XML_TextOverlay

XML_ThermalCap

XML message about thermal capability

```
<ThermalCap version="2.0" xmlns="http://www.std-cgi.org/ver20/XMLSchema">
  <isSupportFireDetection><!--optional, xs:boolean, whether supports fire detection--></isSupportFireDetection>
  <isSupportThermometry><!--optional, xs:boolean, whether supports temperature measurement--></
isSupportThermometry>
  <isSupportRealtimeThermometry><!--optional, xs:boolean, whether supports uploading real-time temperature
measurement data--></isSupportRealtimeThermometry>
  <isFireFocusZoomSupport><!--optional, xs:boolean, whether supports visible light lens zooming--></
isFireFocusZoomSupport>
  <isSupportManualRanging>
```

```
<!--optional, xs:boolean, this node will be returned if one or more channels of device supports this function, see
detailed channel capability in the response information-->
</isSupportManualRanging>
<isSupportPower><!--optional, xs:boolean, whether supports power on/off capability--></isSupportPower>
<isSupportRealtimeTempHumi><!--optional, xs:boolean, whether supports real-time detection of temperature and
humidity--></isSupportRealtimeTempHumi>
<ManualThermCap>
  <manualThermRuleNum>
    <!--optional, xs:integer, the max. number of supported rules for manual temperature measurement. If this node is
not returned, it indicates manual temperature measurement is not supported-->
    </manualThermRuleNum>
  </ManualThermCap>
<isSupportManualThermBasic>
  <!--optional, xs:boolean, whether supports basic configuration of manual temperature measurement-->
</isSupportManualThermBasic>
<isSupportFireShieldMask><!--optional, xs:boolean--></isSupportFireShieldMask>
<isSupportSmokeShieldMask><!--optional, xs:boolean--></isSupportSmokeShieldMask>
<isSupportThermometryMode>
  <!--optional, xs:boolean, whether the device supports the configuration of temperature measurement mode-->
</isSupportThermometryMode>
<isSupportThermalPip>
  <!--optional, xs:boolean, whether the device supports the PIP configuration-->
</isSupportThermalPip>
<isSupportThermalIntelRuleDisplay><!--optional, xs:boolean, whether supports VCA rule configuration--></
isSupportThermalIntelRuleDisplay>
<AlgVersionInfo><!--opt, whether supports getting the version information of thermal algorithms library-->
  <thermometryAlgName min = "1" max = "128">
    <!--read-only, xs:string, version information of temperature measurement algorithms library-->
    </thermometryAlgName>
  <shipsAlgName min = "1" max = "128"><!--read-only, xs:string, version name of ship detection algorithms library--
></shipsAlgName>
</AlgVersionInfo>
<isSupportFaceThermometry><!--optional, xs:boolean, whether supports temperature screening configuration--></
isSupportFaceThermometry>
<isSupportThermalBlackBody><!--optional, xs:boolean, whether supports black body configuration--></
isSupportThermalBlackBody>
<isSupportThermalStreamParam><!--optional, xs:boolean, whether supports stream configuration--></
isSupportThermalStreamParam>
<isSupportBodyTemperatureCompensation>
  <!--optional, xs:boolean, whether supports temperature compensation configuration-->
</isSupportBodyTemperatureCompensation>
<isSupportTemperatureCorrection><!--optional, xs:boolean, whether device supports temperature measurement
correction--></isSupportTemperatureCorrection>
<isSupportClickToThermometry><!--optional, xs:boolean, whether device supports clicking to detect temperature--
></isSupportClickToThermometry>
<isSupportThermometryHistorySearch><!--optional, xs:boolean--></isSupportThermometryHistorySearch>
<isSupportBurningPrevention><!--optional, xs:boolean, whether device supports burning prevention--></
isSupportBurningPrevention>
<isSupportTemperatureCollection><!--optional, xs:boolean, whether device supports temperature ANR--></
isSupportTemperatureCollection>
<isSupportJpegPicWithAppendData>
  <!--optional, xs:boolean, whether device supports getting JPEG picture with pixel-to-pixel temperature
```



```
measurement data. If supports, it is returned and values true, if not support, it is not returned-->
</isSupportJpegPicWithAppendData>
<isSupportRealTimethermometryForHTTP>
  <!--optional, xs:boolean, whether device supports real-time temperature measurement. If supports, it is returned
and its value is true, if not support, it is not returned-->
</isSupportRealTimethermometryForHTTP>
<isSupportShipsDetectionWithScene>
  <!--optional, xs:boolean, whether device supports ship detection by scene, this node and isSupportShipsDetection
in XML_SmartCap are mutually exclusive-->
</isSupportShipsDetectionWithScene>
<isSupportthermometryOffLineCapture>
  <!--optional, xs:boolean, whether device supports offline capture. If supports, this node returned and its value is
true; if not, it is not returned-->
</isSupportthermometryOffLineCapture>
<isSupportThermalTemperatureCorrect>
  <!--optional, xs:boolean, whether device supports temperature calibration (related URI: /ISAPI/Thermal/channels/
<ID>/temperatureCorrect?format=json)-->
</isSupportThermalTemperatureCorrect>
<isSupportFaceSnapThermometry>
  <!--optional, xs:boolean, whether device supports uploading captured face picture with temperature information:
true-support, no return-not support-->
</isSupportFaceSnapThermometry>
</ThermalCap>
```

Remarks

When getting thermal product capabilities, **isSupportShipsDetectionWithScene** has a higher priority than **isSupportShipsDetection**. That is, firstly check if the node **isSupportShipsDetectionWithScene** exists and its value is "true", that indicates ship detection according to scene is supported, otherwise, check the if the node **isSupportShipsDetection** exists.

XML_Time

XML message about time parameters

```
<Time version="2.0" xmlns="http://www.isapi.org/ver20/XMLSchema">
  <timeMode><!--required, xs:string, timing mode: "manual, NTP, local, satellite, timecorrect, platform"--></
timeMode>
  <localTime>
    <!--required, xs:datetime, ISO 8601 time format, device time set manually, e.g.: 2018-02-01T19:54:04. This node is
required when <timemode> is "manual" or "local"-->
  </localTime>
  <timeZone>
    <!--required, xs:string, POSIX time zone based on CST for NTP synchronization, e.g.,
CST-8:00:00DST00:30:00,M4.1.0/02:00:00,M10.5.0/02:00:00; this node is valid when <timemode> is "manual", "local"
or "NTP"-->
  </timeZone>
  <satelliteInterval><!--dependent, xs:integer, unit: minute--></satelliteInterval>
  <isSummerTime><!--optional, xs:boolean, whether the time returned by the current device is that in the DST
(daylight saving time) mode: true, false--></isSummerTime>
  <platformType>
```

```
<!--dependent, xs: string, platform type: "EZVIZ"-Hik-Connect; it is valid only when the value of timeMode is  
"platform"-->  
</platformType>  
</Time>
```

XML_TimeTaskList

TimeTaskList message in XML format

```
<TimeTaskList size="">  
  <enabled><!--req, xs: boolean--></enabled>  
  <Parktime min="" max=""><!--req, xs:integer, seconds--></Parktime>  
  <TimeTaskBlock>  
    <dayOfWeek><!--req, xs: integer, day of the week based on ISO8601, "1"-Monday, ...--></dayOfWeek>  
    <TimeTaskRange>  
      <TaskID min="" max=""><!--req, xs: string; ID--></TaskID>  
      <beginTime><!--req, xs: time, ISO8601 time--></beginTime>  
      <endTime><!--req, xs: time, ISO8601 time--></endTime>  
      <Task>  
        <TaskType  
opt="disable,autoscan,framescan,randomscan,panoramascan,patrol,pattern,preset,tiltscan,periodreboot,periodadjust,  
auxoutput">  
          <!--req, xs: strings-->  
        </TaskType>  
        <patrolTaskNum min="" max=""><!--dep, xs: integer, from 0 to 8--></patrolTaskNum>  
        <patternTaskNum min="" max=""><!--dep, xs: integer, from 0 to 8--></patternTaskNum>  
        <presetTaskNum min="" max=""><!--dep, xs: integer, from 0 to 8--></presetTaskNum>  
        <auxoutputTaskNum min="" max=""><!--dep, xs: integer, from 0 to 8--></  
auxoutputTaskNum>  
      </Task>  
    </TimeTaskRange>  
  </TimeTaskBlock>  
  <isSupportTimeTaskCopy><!--req, xs: boolean--></isSupportTimeTaskCopy>  
</TimeTaskList>
```

XML_trackDailyParam

trackDailyParam message in XML format

```
<trackDailyParam version="2.0" xmlns="http://www.isapi.org/ver20/XMLSchema">  
  <year><!--req, xs: integer, year--></year>  
  <monthOfYear><!--req, xs: integer, month of the year--></monthOfYear>  
</trackDailyParam>
```

XML_trackDailyDistribution

trackDailyDistribution message in XML format

```
<trackDailyDistribution version="1.0" xmlns="http://www.isapi.com/ver20/XMLSchema">
  <dayList>
    <day>
      <id><!--req, xs: integer, ID--></id>
      <dayOfMonth><!--req, xs: integer, day of the month, starts from 1st--></dayOfMonth>
      <record><!--req, xs: boolean, true-with video, false-no video--></record>
      <recordType>
        <!--req, xs: string, record type: time-continuous recording; event-record based on event-->
      </recordType>
    </day>
  </dayList>
</trackDailyDistribution>
```

XML_TrackList

TrackList message in XML format

```
<TrackList version="2.0" xmlns="http://www.isapi.org/ver20/XMLSchema">
  <Track/><!--opt, recording schedule list, refer to the message XML_Track for details-->
</TrackList>
```

See Also

[*XML_Track*](#)

XML_Track

Track message in XML format

```
<Track version="2.0" xmlns="http://www.isapi.org/ver20/XMLSchema">
  <id><!--req, xs: integer--></id>
  <Channel><!--req, xs: integer--></Channel>
  <Enable><!--req, xs: boolean--></Enable>
  <Description><!--req, xs: string--></Description>
  <TrackGUID><!--req, xs: string--></TrackGUID>
  <Size><!--opt, xs: integer--></Size>
  <Duration min="" max=""><!--opt, xs: string--></Duration>
  <DefaultRecordingMode><!--req, xs: string--></DefaultRecordingMode>
  <LoopEnable><!--opt, xs: string--></LoopEnable>
  <SrcDescriptor>
    <SrcGUID><!--req, xs: string--></SrcGUID>
    <SrcChannel><!--req, xs: integer--></SrcChannel>
    <StreamHint><!--req, xs: string--></StreamHint>
    <SrcDriver><!--req, xs: string--></SrcDriver>
    <SrcType><!--req, xs: string--></SrcType>
    <SrcUrl><!--req, xs: string--></SrcUrl>
    <SrcUrlMethods><!--req, xs: string--></SrcUrlMethods>
    <SrcLogin><!--req, xs: string--></SrcLogin>
  </SrcDescriptor>
```

```
<TrackSchedule>
  <ScheduleBlockList>
    <ScheduleBlock>
      <ScheduleBlockGUID><!--req, xs: string--></ScheduleBlockGUID>
      <ScheduleBlockType><!--req, xs: string--></ScheduleBlockType>
      <ScheduleAction>
        <id><!--req, xs: integer--></id>
        <ScheduleActionStartTime>
          <DayOfWeek><!--req, xs: string--></DayOfWeek>
          <TimeOfDay><!--req, xs: string--></TimeOfDay>
        </ScheduleActionStartTime>
        <ScheduleActionEndTime>
          <DayOfWeek><!--req, xs: string--></DayOfWeek>
          <TimeOfDay><!--req, xs: string--></TimeOfDay>
        </ScheduleActionEndTime>
        <ScheduleDSTEnable><!--req, xs: boolean--></ScheduleDSTEnable>
        <Description><!--req, xs: string--></Description>
        <Actions>
          <Record><!--req, xs: boolean--></Record>
          <Log><!--req, xs: boolean--></Log>
          <SaveImg><!--req, xs: boolean--></SaveImg>
          <ActionRecordingMode><!--req, xs: string--></ActionRecordingMode>
        </Actions>
      </ScheduleAction>
    </ScheduleBlock>
  </ScheduleBlockList>
</TrackSchedule>
<CustomExtensionList>
  <CustomExtension>
    <CustomExtensionName><!--req, xs: string--></CustomExtensionName>
    <enableSchedule><!--req, xs: boolean--></enableSchedule>
    <SaveAudio><!--req, xs: boolean--></SaveAudio>
    <PreRecordTimeSeconds><!--req, xs: integer--></PreRecordTimeSeconds>
    <PostRecordTimeSeconds><!--req, xs: integer--></PostRecordTimeSeconds>
    <HolidaySchedule>
      <ScheduleBlock>
        <ScheduleBlockGUID><!--req, xs: string--></ScheduleBlockGUID>
        <ScheduleBlockType><!--req, xs: string--></ScheduleBlockType>
      </ScheduleBlock>
    </HolidaySchedule>
  </CustomExtension>
</CustomExtensionList>
<IntelligentRecord><!--opt, xs:boolean, whether to enable VCA recording function: 0-no, 1-yes--></IntelligentRecord>
<delayTime><!--opt, xs:integer, capture delay time, unit: second--></delayTime>
<durationEnabled><!--opt, xs:boolean, whether to enable video expiry time. If this function is not supported, this
node will not be returned. If this function is supported, the video expiry date will be set by the node <Duration>--></
durationEnabled>
</Track>
```

XML_Tracking

Tracking message in XML format

```
<Tracking version="2.0" xmlns="http://www.std-cgi.org/ver20/XMLSchema">
  <enabled>
    <!--req, xs:boolean-->
  </enabled>
  <mode>
    <!--dep, xs:string, "auto,manual"-->
  </mode>
  <trackingTime>
    <!--dep, xs:integer-->
  </trackingTime>
  <normalizedScreenSize>
    <!--req, ro-->
    <normalizedScreenWidth>
      <!--req, ro, xs:integer-->
    </normalizedScreenWidth>
    <normalizedScreenHeight>
      <!--req, ro, xs:integer-->
    </normalizedScreenHeight>
  </normalizedScreenSize>
  <ManualRegionCoordinatesList min="" max="">
    <!--dep-->
    <RegionCoordinates>
      <!--opt-->
      <positionX>
        <!--req, xs:integer, X-coordinate-->
      </positionX>
      <positionY>
        <!--req, xs:integer, Y-coordinate-->
      </positionY>
    </RegionCoordinates>
  </ManualRegionCoordinatesList>
  <Schedule><!--opt, schedule, this node is valid only when enabled is "true". By default the schedule is from 00:00 to 24:00 if no schedule is configured-->
    <TimeRange>
      <beginTime>
        <!--UTC time, e.g.: "2018-03-13T19:42:27+08:00"--->
      </beginTime>
      <endTime>
        <!--UTC time, e.g.: "2018-03-13T19:42:27+08:00"--->
      </endTime>
    </TimeRange>
  </Schedule>
</Tracking>
```

XML_TwoWayAudioChannel

TwoWayAudioChannel message in XML format

```
<TwoWayAudioChannel version="2.0" xmlns="http://www.isapi.org/ver20/XMLSchema">
  <id>
    <!--req, xs:string, two-way audio channel ID-->
  </id>
  <enabled>
    <!--req, xs:boolean, whether to enable two-way audio: "true"-yes, "false"-no-->
  </enabled>
  <audioCompressionType>
    <!--req, xs:string, audio output encoding type: "G.711alaw,G.711ulaw,G.726,G.729,G.729a,G.
729b,PCM,MP3,AC3,AAC,ADPCM,MP2L2"-->
  </audioCompressionType>
  <audioInboundCompressionType>
    <!--opt, xs:string, audio input encoding type: "G.711alaw,G.711ulaw,G.726,G.729,G.729a,G.
729b,PCM,MP3,AC3,AAC,ADPCM"-->
  </audioInboundCompressionType>
  <speakerVolume>
    <!--opt, xs:integer, input volume-->
  </speakerVolume>
  <microphoneVolume>
    <!--opt, xs:integer, output volume-->
  </microphoneVolume>
  <noisereduce>
    <!--opt, xs:boolean, whether to enable noise reduction: "true, false"-->
  </noisereduce>
  <audioBitRate>
    <!--opt, xs:integer, audio frame rate, unit: kbs-->
  </audioBitRate>
  <audioInputType>
    <!--opt, xs:string, audio input type: "MicIn, LineIn"-->
  </audioInputType>
  <associateVideoInputs><!--opt-->
    <enabled>
      <!--req, xs:boolean-->
    </enabled>
    <videoInputChannelList>
      <!--req-->
      <videoInputChannelID>
        <!--opt, xs:string, ID-->
      </videoInputChannelID>
    </videoInputChannelList>
  </associateVideoInputs>
  <lineOutForbidden>
    <!--read-only, xs:boolean, whether the audio output is not supported, if this node is not returned or the value is
"false", it represents that audio output is supported; if the value is "true", it represents that audio output is not
supported-->
  </lineOutForbidden>
  <micInForbidden>
```

```
<!--read-only, xs: boolean, whether the audio input is not supported, if this node is not returned or the value is
"false", it represents that audio input is supported; if the value is "true", it represents that audio input is not
supported-->
</micInForbidden>
</TwoWayAudioChannel>
```

XML_TwoWayAudioChannelCap

TwoWayAudioChannelCap capability message in XML format

```
<TwoWayAudioChannelCap version="2.0" xmlns="http://www.isapi.org/ver20/XMLSchema">
  <id opt="1">
    <!--req, xs:string, two-way audio channel ID-->
  </id>
  <enabled opt="true,false">
    <!--req, xs:boolean, whether to enable two-way audio: "true"-yes, "false"-no-->
  </enabled>
  <audioCompressionType opt="G.711alaw,G.711ulaw,G.726,G.729,G.729a,G.
729b,PCM,MP3,AC3,AAC,ADPCM,MP2L2">
    <!--req, xs:string, audio output encoding type-->
  </audioCompressionType>
  <audioInboundCompressionType opt=" G.711alaw,G.711ulaw,G.726,G.729,G.729a,G.
729b,PCM,MP3,AC3,AAC,ADPCM">
    <!--opt, xs:string, audio input encoding type-->
  </audioInboundCompressionType>
  <speakerVolume opt="0-100">
    <!--opt, xs:integer, input volume, which is between 0 and 100-->
  </speakerVolume>
  <microphoneVolume opt="0-100">
    <!--opt, xs:integer, output volume, which is between 0 and 100-->
  </microphoneVolume>
  <noisereduce opt="true,false">
    <!--opt, xs:boolean, whether to enable noise reduction: "true, false"-->
  </noisereduce>
  <audioBitRate opt="32,64,128">
    <!--opt, xs:integer, audio frame rate, unit: kbs-->
  </audioBitRate>
  <audioInputType opt="MicIn,LineIn">
    <!--opt, xs:string, audio input type-->
  </audioInputType>
  <associateVideoInputs>
    <!--opt-->
    <enabled opt="true,false">
      <!--req, xs:boolean-->
    </enabled>
    <videoInputChannelList>
      <!--req-->
      <videoInputChannelID opt="1">
        <!--opt, xs:string, ID-->
      </videoInputChannelID>
    </videoInputChannelList>
```

```
</associateVideoInputs>
</TwoWayAudioChannelCap>
```

XML_TwoWayAudioChannelList

TwoWayAudioChannelList message in XML format

```
<TwoWayAudioChannelList version="2.0" xmlns="http://www.isapi.org/ver20/XMLSchema">
  <TwoWayAudioChannel/><!--opt, see details in the message of XML_TwoWayAudioChannel-->
</TwoWayAudioChannelList>
```

See Also

XML_TwoWayAudioChannel

XML_TwoWayAudioSession

TwoWayAudioSession message in XML format

```
<TwoWayAudioSession version="2.0" xmlns="http://www.isapi.org/ver20/XMLSchema">
  <sessionId><!--req, xs: string, communication session ID--></sessionId>
</TwoWayAudioSession>
```

XML_upgradeStatus

upgradeStatus message in XML format

```
<upgradeStatus version="2.0" xmlns="http://www.isapi.org/ver20/XMLSchema">
  <upgrading><!--ro, req, xs:boolean--></upgrading>
  <percent><!--ro, req, xs:integer, upgrading progress, the value is between 0 and 100--></percent>
</upgradeStatus>
```

XML_UPnP

UPnP message in XML format

```
<UPnP version="2.0" xmlns="http://www.isapi.org/ver20/XMLSchema">
  <enabled><!--req-->
  <ports><!--opt-->
</UPnP>
```

XML_User

User message in XML format


```
<User version="2.0" xmlns="http://www.isapi.org/ver20/XMLSchema">
  <id><!--req, xs:integer, the value is between 1 and 16, the ID of the administrator account is 1 and the administrator
account cannot be deleted--></id>
  <enabled><!--opt, xs:boolean, whether to enable the user: "true"-yes, "false"-no--></enabled>
  <userName><!--req, xs:string, the sensitive information should be encrypted--></userName>
  <password><!--wo, req, xs:string, the sensitive information should be encrypted--></password>
  <keypadPassword><!--wo, opt, xs:string, keypad password, the sensitive information should be encrypted--></
keypadPassword>
  <loginPassword><!--wo, dep, xs:string, this node depends on security in the URL and it is required when security
exists. The sensitive information should be encrypted, and the password is the administrator password--></
loginPassword>
  <userOperateType><!--opt, xs:integer, user operation type: "1"-network user, "2"-keypad user, "3"-network user and
keypad user. When this node is set to 1 or NULL, password is valid, and password is required when the method is
POST, otherwise it is optional. When this node is set to 2, keypadPassword is valid, and keypadPassword is required
when the method is POST, otherwise it is optional. When this node is set to 3, both password and keypadPassword
are valid, and they are required when the method is POST, otherwise they are optional--></userOperateType>
  <bondIpAddressList><!--opt-->
    <bondIpAddress><!--opt-->
      <id><!--req, xs:integer--></id>
      <ipAddress><!--dep, xs:string--></ipAddress>
      <ipv6Address><!--dep, xs:string--></ipv6Address>
    </bondIpAddress>
  </bondIpAddressList>
  <bondMacAddressList><!--opt-->
    <bondMacAddress><!--opt-->
      <id><!--req, xs:integer--></id>
      <macAddress><!--opt, xs:string--></macAddress>
    </bondMacAddress>
  </bondMacAddressList>
  <userLevel><!--opt, xs:string, "Administrator,Operator,Viewer"--></userLevel>
  <attribute><!--opt-->
    <inherent><!--xs:boolean--></inherent>
  </attribute>
</User>
```

XML_userCheck

userCheck message in XML format

```
<userCheck version="2.0" xmlns="http://www.isapi.org/ver20/XMLSchema">
  <statusValue><!--req, xs:integer, status code: "200"-succeeded, "401"-failed--></statusValue>
  <statusString><!--opt, xs:string, status: "OK", "Unauthorized"--></statusString>
  <isDefaultPassword><!--opt, xs:boolean--></isDefaultPassword>
  <isRiskPassword><!--opt, xs:boolean--></isRiskPassword>
  <isActivated><!--opt, xs:boolean--></isActivated>
  <lockStatus><!--opt, xs:string, locking status: "unlock", "locked"--></lockStatus>
  <unlockTime><!--opt, xs:integer, unlocking remaining time, unit: second--></unlockTime>
  <retryLoginTime><!--opt, xs:integer, remaining login attempts--></retryLoginTime>
</userCheck>
```

XML_UserList

UserList message in XML format

```
<UserList version="2.0" xmlns="http://www.isapi.org/ver20/XMLSchema">
  <User/><!--opt, see details in the message of XML_User-->
</UserList>
```

See Also

XML_User

XML_UserPermission

UserPermission message in XML format

```
<UserPermission version="2.0" xmlns="http://www.isapi.org/ver20/XMLSchema">
  <id><!--req, xs:string--></id>
  <userID><!--req, xs:string, user ID--></userID>
  <userType>
    <!--req, xs:string, user type: "admin"-administrator, which has all permissions and can review and edit user's
    permission, "operator"-operator, which has default permissions, "viewer"-viewer, which has default permissions,
    "installer", "manufacturer"-->
  </userType>
  <localPermission/>
    <!--opt, local permission, see details in the message of XML_localPermission -->
  <remotePermission/>
    <!--opt, remote permission, see details in the message of XML_remotePermission -->
</UserPermission>
```

See Also

XML_localPermission

XML_remotePermission

XML_UserPermissionCap

UserPermissionCap capability message in XML format

```
<UserPermissionCap version="2.0" xmlns="http://www.isapi.org/ver20/XMLSchema">
  <userType><!--req, xs:string, "admin"-administrator, "operator", "viewer", "installer", "manufacturer"--></userType>
  <localPermissionCap><!--opt, see details in the message of XML_localPermissionCap--></localPermissionCap>
  <remotePermissionCap><!--opt, see details in the message of XML_remotePermissionCap--></
remotePermissionCap>
</UserPermissionCap>
```

See Also

XML_localPermissionCap

XML_remotePermissionCap

XML_UserPermissionList

UserPermissionList message in XML format

```
<UserPermissionList version="2.0" xmlns="http://www.isapi.org/ver20/XMLSchema">
  <UserPermission/>
  <!--opt, user permission, see details in the message of XML_UserPermission-->
</UserPermissionList>
```

See Also

XML_UserPermission

XML_VideoPic_CMSearchDescription

XML message about search conditions of video and picture

```
<CMSearchDescription version="2.0" xmlns="http://www.isapi.org/ver20/XMLSchema">
  <searchID>
    <!--required, xs:string, search ID, e.g., {812F04E0-4089-11A3-9A0C-0305E82C2906}-->
  </searchID>
  <trackIDList>
    <trackID>
      <!--required, xs:integer, ID, XX01-search for the main stream video of channel XX, XX03-search for the pictures of
channel XX-->
    </trackID>
  </trackIDList>
  <timeSpanList>
    <timeSpan>
      <startTime><!--required, xs:time, ISO8601 time, start time of search, e.g.: 2017-08-02T00:00:00Z--></startTime>
      <endTime><!--required, xs:time, ISO8601 time, end time of search, e.g.: 2017-08-02T23:59:59Z--></endTime>
    </timeSpan>
  </timeSpanList>
  <searchTypeList>
    <searchKeyword><!--required, xs:string, keyword list for search, the maximum length is 32 and the minimum length
is 1--></searchKeyword>
  </searchTypeList>
  <contentTypeList>
    <contentType><!--required, xs:string, file type to be searched for: "video", "audio", "metadata"-picture, "text",
"mixed", "other"--></contentType>
    <pircamID><!--optional, list, xs:integer, pircam search ID. This node is valid only when the search type is pircam.
Types of pictures captured by pircams in the device may be different, and this node is used to distinguish between
different pircams--></pircamID>
  </contentTypeList>
  <uploadState><!--optional, xs:string, uploading status: all, uploaded, nonUploaded, uploadFailed--></uploadState>
  <lockState><!--optional, xs:string, locking status: lock, unlock, all--></lockState>
  <searchResultPostion><!--required, xs:integer, start position of search--></searchResultPostion>
  <maxResults><!--optional, xs:integer, maximum number of returned results--></maxResults>
```

```

<metadataList>
  <metadataDescriptor>
    <!--required, xs:string, description of file to be searched for, format: "recordType.meta.hikvision.com/
    <eventType>", <eventType> is the event type to which related video should be searched for; for details, see the table
    below in remarks. For example, to search for the video/picture of motion detection, the value of contentType is
    "video"/"metadata" and the value of metadataDescriptor is "recordType.meta.hikvision.com/MOTION"-->
    <!--to search for all videos, the value of contentType is "video" and value of metadataDescriptor is
    "recordType.meta.hikvision.com"; to search for all pictures, the value of contentType is "metadata" and value of
    metadataDescriptor is "recordType.meta.hikvision.com/allPic"-->
    <!--the supported video/picture type of device is returned in the recordSearchType/pictureSearchType of
    XML_RacmCap (related URI: /ISAPI/ContentMgmt/capabilities)-->
  </metadataDescriptor>
  <SearchProperty>
    <plateSearchMask>
      <!--optional, xs:string, ranges from 1 to 31-->
    </plateSearchMask>
    <stateOrProvince>
      <!--optional, xs:integer-->
    </stateOrProvince>
    <country>
      <!--optional, xs:string, country or region ID: 0-the algorithm library does not support the country or region ID,1-
      (CZ-Czech Republic),2-(FRA-France),3-(DE-Germany),4-(E-Spain),5-(IT-Italy),6-(NL-Netherlands),7-(PL-Poland),8-(SVK-
      Slovakia), 9-(BY-Belorussia), 10-(MDA-Moldova),11-(RU-Russia),12-(UA-Ukraine), 0xff-(All)-->
    </country>
    <subType><!--optional, xs:string, sub type: all,motorVehicle,nonMotorVehicle,pedestrian--></subType>
    <capTypeDescriptor><!--optional, xs:string, capability type descriptor: anyType, event, evidence--></
  capTypeDescriptor>
</SearchProperty>
</metadataList>
</CMSearchDescription>

```

Remarks

eventType	Description
AllEvent	All events
safetyHelmet	Hard hat detection
smokeDetection	Smoke detection
leavePosition	Absence detection
peopleNumChange	The number of people changed
linedetection	Line crossing detection
fielddetection	Intrusion detection
regionEntrance	Region entrance detection
regionExiting	Region exiting detection
loitering	Loitering detection

eventType	Description
group	People gathering detection
rapidMove	Fast moving detection
unattendedBaggage	Unattended baggage detection
attendedBaggage	Object removal detection
CMR	Scheduled recording or capture
ALARM	Alarm
EDR	Motion detection or alarm
ALARMANDMOTION	Motion detection and alarm
Command	Command is triggered
pir	PIR
wlsensor	Wireless alarm
callhelp	Calling alarm
facedetection	Face detection
parking	Parking
vehicleDetection	Vehicle detection
manual	Manual
manualSnapShot	Manual capture
playSnapShot	Capture of playback
dredgerDetection	Dredger detection
accessController	Access controller event
securityControlPanel	Security control panel event
violentMotion	Violent motion alarm
advReachHeight	Climbing detection
toiletTarry	In-Toilet overtime
audioAbnormal	Sudden change of sound intensity
standUp	Standing up
getUp	Getting up
vehicleMonitor	Vehicle arming
playCellphone	Playing cellphone

eventType	Description
retention	Overstay detection
failDown	People falling down alarm
sleepOnduty	Sleeping on duty alarm
allPerimeterEvent	All perimeter events
allBehaviorEvent	All behavior analysis events
running	People running
spacingChange	Distance changing detection
faceSnapModeling	Face capture modeling
reverseEntrance	Reverse entering
studentsStoodUp	Student standing up detection
ATMPanel	ATM panel mode
ATMSurround	ATM environment mode
ATMFace	ATM human face mode
ATMSafetyCabin	ATM safety cabin mode

XML_WDR

WDR message in XML format

```
<WDR version="2.0" xmlns="http://www.isapi.org/ver20/XMLSchema">
  <mode><!--req, xs: string, WDR mode: "open,close,auto"--></mode>
  <WDRLevel><!--opt, xs: integer--></WDRLevel>
  <WDRContrastLevel><!--opt, xs: integer--></WDRContrastLevel>
  <WDRLevel1><!--opt, xs: integer--></WDRLevel1>
</WDR>
```

XML_WhiteBalance

WhiteBalance message in XML format

```
<WhiteBalance version="2.0" xmlns="http://www.isapi.org/ver20/XMLSchema">
  <WhiteBalanceStyle/><!--req, xs:string, "auto, manual, indoor, outdoor, autotrace, onece, sodiumlight,
  mercurylight,auto0, atuo1, fluorescent, natural, warm, incandescent"-->
  <WhiteBalanceRed/><!--dep,xs:integer, this node depends on <WhiteBalanceStyle>-->
  <WhiteBalanceBlue/><!--dep,xs:integer, this node depends on <WhiteBalanceStyle>-->
</WhiteBalance>
```

Remarks

Setting **WhiteBalanceRed** and **WhiteBalanceBlue** is available only when **WhiteBalanceStyle** is set to "manual".

XML_WirelessServer

WirelessServer message in XML format

```
<WirelessServer version="2.0" xmlns="http://www.isapi.org/ver20/XMLSchema">
  <wifiApEnabled><!--opt,xs:boolean,"true,false"--></wifiApEnabled>
  <broadcastEnabled><!--opt,xs:boolean,"true,false"--></broadcastEnabled>
  <wlanShareEnabled><!--opt,xs:boolean,"true,false"--></wlanShareEnabled>
  <ssid><!--opt, xs:string--></ssid>
  <WirelessSecurity><!--req-->
    <securityMode>
      <!--opt, xs: string, security mode: "disable,WEP,WPA-personal,WPA2-personal,WPA-RADIUS,WPA-enterprise,WPA2-
enterprise"-->
    </securityMode>
    <WEP><!--dep, depends on <securityMode>-->
      <authenticationType>
        <!--req, xs: string, authentication type: "open,sharedkey,auto"-->
      </authenticationType>
      <defaultTransmitKeyIndex>
        <!--req, xs: integer-->
      </defaultTransmitKeyIndex>
      <wepKeyLength><!--opt, xs: integer, the key length should be 64 and 128--></wepKeyLength>
      <EncryptionKeyList>
        <encryptionKey>
          <!--req, xs: hexBinary, WEP encryption key in hexadecimal format-->
        </encryptionKey>
      </EncryptionKeyList>
    </WEP>
    <WPA><!--dep, depends on <securityMode>-->
      <algorithmType>
        <!--req, xs: string, algorithm type: "TKIP,AES,TKIP/AES"-->
      </algorithmType>
      <sharedKey><!--opt, xs: string, shared key used in WPA--></sharedKey>
      <wpaKeyLength><!--opt, xs: integer, the key length is between 8 and 63--></wpaKeyLength>
      <defaultPassword><!--opt, xs: boolean--></defaultPassword>
    </WPA>
  </WirelessSecurity>
  <DHCPEnabled><!--opt,xs:boolean,"true,false"--></DHCPEnabled>
  <ipVersion><!--opt, xs:string, opt="v4,v6"--></ipVersion>
  <HostIpAddress><!--opt-->
    <ipAddress><!--dep, xs:string--></ipAddress>
    <ipv6Address><!--dep, xs:string--></ipv6Address>
  </HostIpAddress>
  <IPMask><!--opt-->
    <subnetMask><!--dep, xs:string, subnet mask for IPv4 address--></subnetMask>
```

```
<bitMask><!--dep, xs:integer, bitmask IPv6 address--></bitMask>
</IPMask>
<AddressPool><!--opt-->
  <startIPv4Address><!--dep, xs:string--></startIPv4Address>
  <endIPv4Address><!--dep, xs:string--></endIPv4Address>
  <startIPv6Address><!--dep, xs:string--></startIPv6Address>
  <endIPv6Address><!--dep, xs:string--></endIPv6Address>
</AddressPool>
<DNSAddressList><!--opt-->
  <DNSAddress><!--opt>
    <id><!--opt,xs:string,start from 1--></id>
    <ipAddress><!--dep, xs:string--></ipAddress>
    <ipv6Address><!--dep, xs:string--></ipv6Address>
  </DNSAddress>
</DNSAddressList>
<GatewayAddress>
  <ipAddress><!--dep, xs:string--></ipAddress>
  <ipv6Address><!--dep, xs:string--></ipv6Address>
</GatewayAddress>
<wifiApModeType>
  <!--opt, xs: string, current wireless Access Point (AP) mode: "true,false,auto", if this node is returned, it indicates
that the device supports auto mode-->
  </wifiApModeType>
</WirelessServer>
```

Remarks

For Client supports auto AP, the node <<**wifiApModeType**> is valid; for Client does not supports auto AP, the node <**wifiApEnabled**> is valid; the values of these two nodes will effect each other, and when <**wifiApModeType**> equals to "auto", the value of <**wifiApEnabled**> is "true".

XML_WirelessServerStatus

WirelessServerStatus message in XML format

```
<WirelessServerStatus version="1.0" xmlns="http://www.isapi.org/ver20/XMLSchema">
  <wifiApEnabled>
    <!--dep, xs:boolean-->
  </wifiApEnabled>
  <linkDevices>
    <!--dep, xs:integer, maximum number of linked devices-->
  </linkDevices>
  <DeviceInfoList>
    <!--opt-->
    <DeviceInfo>
      <!--opt-->
      <IpAddress>
        <!--req-->
        <IpAddress>
          <!--dep, xs:string-->
        </IpAddress>
      </DeviceInfo>
    </DeviceInfoList>
  </WirelessServerStatus>
```



```
<ipv6Address>
  <!--dep, xs:string-->
</ipv6Address>
</IpAddress>
<hostName>
  <!--opt, xs:string-->
</hostName>
<macAddress>
  <!--opt, xs:string-->
</macAddress>
</DeviceInfo>
</DeviceInfoList>
</WirelessServerStatus>
```

XML_WirelessStatus

WirelessStatus message in XML format

```
<WirelessStatus version="1.0" xmlns="http://www.isapi.org/ver20/XMLSchema">
  <enabled>
    <!--dep, xs:boolean-->
  </enabled>
  <connectionEnabled>
    <!--dep, xs:boolean-->
  </connectionEnabled>
  <IpAddress>
    <!--req-->
    <ipAddress>
      <!--dep, xs:string-->
    </ipAddress>
    <ipv6Address>
      <!--dep, xs:string-->
    </ipv6Address>
  </IpAddress>
  <SubnetMask>
    <!--req-->
    <ipAddress>
      <!--dep, xs:string-->
    </ipAddress>
    <ipv6Address>
      <!--dep, xs:string-->
    </ipv6Address>
  </SubnetMask>
  <RouterAddress>
    <!--req-->
    <ipAddress>
      <!--dep, xs:string-->
    </ipAddress>
    <ipv6Address>
      <!--dep, xs:string-->
    </ipv6Address>
  </RouterAddress>
</WirelessStatus>
```

```
</RouterAddress>
<DNSAddressList>
  <!--opt-->
  <DNSAddress>
    <!--opt-->
    <id>
      <!--opt, xs:string, DNS address ID, which starts from 1-->
    </id>
    <ipAddress>
      <!--dep, xs:string-->
    </ipAddress>
    <ipv6Address>
      <!--dep, xs:string-->
    </ipv6Address>
  </DNSAddress>
</DNSAddressList>
</WirelessStatus>
```

XML_Wireless

Wireless message in XML format

```
<Wireless version="2.0" xmlns="http://www.isapi.org/ver20/XMLSchema">
  <enabled><!--req, xs:boolean--></enabled>
  <wirelessNetworkMode>
    <!--opt, xs:string, "infrastructure,adhoc"-->
  </wirelessNetworkMode>
  <channel><!--opt, xs:string, "1 to 14,auto"--></channel>
  <ssid><!--opt, xs:string--></ssid>
  <wmmEnabled><!--opt, xs:boolean--></wmmEnabled>
  <WirelessSecurity><!--opt-->
    <securityMode>
      <!--opt, xs:string, "disable,WEP,WPA-personal,WPA2-personal,WPA-RADIUS, WPA-enterprise,WPA2-enterprise, WPA/
WPA2"-->
    </securityMode>
    <WEP><!--dep, depends on <securityMode-->
      <authenticationType>
        <!--req, xs:string, "open,sharedkey,auto"-->
      </authenticationType>
      <defaultTransmitKeyIndex><!--req, xs:integer--></defaultTransmitKeyIndex>
      <wepKeyLength><!--opt, xs:integer "64,128"--></wepKeyLength>
      <EncryptionKeyList>
        <encryptionKey>
          <!--req, xs:hexBinary, WEP encryption key in hexadecimal format-->
        </encryptionKey>
      </EncryptionKeyList>
    </WEP>
    <WPA><!--dep, depends on <securityMode-->
      <algorithmType><!--req, xs:string, "TKIP,AES,TKIP/AES"--></algorithmType>
      <sharedKey><!--req, xs:string, pre-shared key used in WPA--></sharedKey>
      <wpaKeyLength><!--req, xs:integer, "8-63"--></wpaKeyLength>
```

```
</WPA>
</WirelessSecurity>
<workScene><!--opt, xs:string, "computerRoom,monitorTerminal"--></workScene>
<protocol><!--req,xs:string, "802.11ac"--></protocol>
<protocolRealTime><!--req,xs:string, real-time mode--></protocolRealTime>
<hideSsid><!--opt, xs:boolean--></hideSsid>
<ChannelConfig><!--opt-->
  <width><!--opt, xs:string, "auto,20,40,80", bandwidth, unit: MHz--></width>
  <autoWidth>
    <!--dep,ro, xs:string, it is valid only when width is "auto"-->
  </autoWidth>
  <channel><!--opt, dep, xs:string, channel frequency, unit: MHz--></channel>
  <autoChannel><!--dep,ro,xs:string, it is valid only when channel is "auto"--></autoChannel>
  <transmitPower> <!--opt, xs:integer, "9,12,15,18,21,24,27", power, unit: dBm--><transmitPower>
  <transmitPowerRealTime> <!--opt, xs:integer, real-time power, unit: dBm--><transmitPowerRealTime>
  <countryID><!--opt,xs:integer, country code--></countryID>
</ChannelConfig>
</Wireless>
```

XML_WPS

WPS message in XML format

```
<WPS version="2.0" xmlns="http://www.isapi.org/ver20/XMLSchema">
  <enable><!--req, xs:boolean--></enable>
</WPS>
```

XML_WpsApPincode

WpsApPincode message in XML format

```
<WpsApPincode version="2.0" xmlns="http://www.isapi.org/ver20/XMLSchema">
  <ssid><!--req, xs:string--></ssid>
  <pinCode><!--req, xs:string--></pinCode>
</WpsApPincode>
```

XML_VCAResource

Message about intelligent resources switch in XML format

```
<VCAResource version="2.0" xmlns="http://www.isapi.org/ver20/XMLSchema">
  <type>
    <!--required, xs:string,
    "basicBehavior,fullBehavior,facesnapBehavior,facesnap,TFS,smartVehicleDetection,smartHVTDDetection,smart,judicial,s
    mart264AndRoadDetection,smart264AndFaceDetection,smart264AndHeatMap,smartIntelligentMonitor,smartTrafficD
    ataCollection,roadDetection,humanRecognition,perimeterCapture,vehicleDetection,HVTDDetection,mixedTargetDetecti
    on,trackingCaptureMode,nonTrackingCaptureMode,close,faceHumanModelingContrast,cityManagement,teacherBeha
```

```
    vior,12MPLiveView,personQueueDetection,verticalPeopleCounting,safetyHelmet,faceCounting,personArming"-->
  </type>
</VCAResource>
```

XML_VideoCap

VideoCap message in XML format

```
<VideoCap version="2.0" xmlns="http://www.isapi.org/ver20/XMLSchema">
  <videoInputPortNums><!--opt, xs: integer--><videoInputPortNums>
  <videoOutputPortNums><!--opt, xs: integer--><videoOutputPortNums>
  <isSupportHeatmap><!--opt, xs: boolean, whether to support heat map function--></isSupportHeatmap>
  <isSupportCounting><!--opt, xs: boolean--></isSupportCounting>
  <countingType><!--dep, xs: string, "human,object"--></countingType>
  <isSupportPreviewSwitch><!--opt, xs: boolean, whether to support live view switch--></isSupportPreviewSwitch>
  <isSupportRecodStatus>
    <!--opt, xs: boolean, whether supports getting recording status-->
  </isSupportRecodStatus>
  <isSupportPrivacyMask>
    <!--opt, xs: boolean, whether supports priavte mask function-->
  </isSupportPrivacyMask>
  <isSupportBinocularPreviewSwitch>
    <!--opt, xs: boolean, whether supports auto-switch during the live view of dual lens camera-->
  </isSupportBinocularPreviewSwitch>
  <isSupportCalibCheck>
    <!--opt, xs: boolean, whether supports calibration verification-->
  </isSupportCalibCheck>
  <isSupportPIP><!--opt, xs: boolean, "true, false"--></isSupportPIP>
  <channelFlexible opt="name,enable,online,linknum">
    <!--capability of getting channel status by condition-->
  </channelFlexible >
  <isSupportFocusVideoMode>
    <!--opt, xs: boolean, video focus mode for installation and debug-->
  </isSupportFocusVideoMode>
  <isSupportExternalChannel>
    <!--opt, xs: boolean, whether supports extending analog channel-->
  </isSupportExternalChannel>
  <isSupportMultiChannelCounting>
    <!--opt, xs: boolean, whether supports people counting of multiple channels-->
  </isSupportMultiChannelCounting>
  <isSupportCountingCollection>
    <!--opt, xs:boolean, whether supports people counting data replenishment-->
  </isSupportCountingCollection>
  <isSupportHeatmapCollection>
    <!--opt, xs:boolean, whether supports heat map data replenishment-->
  </isSupportHeatmapCollection>
  <OSDLanguage opt="GBK,EUC-KR" def="GBK"/>
  <isSupportInitLens><!--req, xs:boolean, whether to support initializing lens--><isSupportInitLens>
  <isSupportOneFocus><!--req, xs:boolean, whether to support one-touch focusing--><isSupportOneFoucs>
  <notSupportFocus><!--req, xs:boolean, the focus capability is not supported--><notSupportFoucs>
  <notSupportIris><!--req, xs:boolean, the iris capability is not supported--><notSupportIris>
```

```
<isSupportCapturePicOverlays><!--opt, xs:boolean, whether to support text overlay on the captured picture--></isSupportCapturePicOverlays>
<isSupportMergePicOverlays><!--opt, xs:boolean, whether to support text overlay on the composite picture--></isSupportMergePicOverlays>
</VideoCap>
```

XML_VideoInput

VideoInput message in XML format

```
<VideoInput version="2.0" xmlns="http://www.isapi.org/ver20/XMLSchema">
  <VideoInputChannelList/><!--op, see details in the message of XML_VideoInputChannelList-->
</VideoInput>
```

See Also

XML_VideoInputChannelList

XML_VideoInputChannel

VideoInputChannel message in XML format

```
<VideoInputChannel version="2.0" xmlns="http://www.isapi.org/ver20/XMLSchema">
  <id>
    <!--req, xs:string-->
  </id>
  <inputPort>
    <!--req, xs:string-->
  </inputPort>
  <videoInputEnabled>
    <!--opt, xs:boolean-->
  </videoInputEnabled>
  <name>
    <!--opt, xs:string-->
  </name>
  <videoFormat>
    <!--opt, xs:string, "PAL, NTSC"-->
  </videoFormat>
  <portType>
    <!--opt, xs:string, "SDI, OPT, VGA, HDMI, YPbPr"-->
  </portType>
  <resDesc>
    <!--opt, xs:string-->
  </resDesc>
</VideoInputChannel>
```

XML_VideoInputChannelList

VideoInputChannelList message in XML format

```
<VideoInputChannelList version="2.0"
xmlns="http://www.isapi.org/ver20/XMLSchema">
  <VideoInputChannel/><!--opt, see details in the message of XML_VideoInputChannel-->
</VideoInputChannelList>
```

See Also

XML_VideoInputChannel

XML_VideoInputChannelsCap

XML message about the capability of configuring video input channels in a batch

```
<VideoInputChannelsCap version="2.0" xmlns="http://www.isapi.org/ver20/XMLSchema">
  <VideoInputChannelList>
    <VideoInputChannel>
      <id><!--required, xs:string, video input channel ID--></id>
      <inputPort><!--required, xs:string, port No. of the video input channel--></inputPort>
      <videoInputEnabled opt="true,false"><!--optional, xs:boolean, whether to enable the video input channel.
Currently, the attribute "opt" will only be returned with "true" (i.e., opt="true"), which indicates that disabling this
channel is not supported--></videoInputEnabled>
      <name><!--optional, xs:string, video input channel name--></name>
      <videoFormat opt="PAL,NTSC"><!--optional, xs:string, video standard: "PAL", "NTSC"--></videoFormat>
      <portType opt="SDI,OPT,VGA,HDMI,YPbPr"><!--optional, xs:string, port type: "SDI", "OPT", "VGA", "HDMI",
"YPbPr"--></portType>
      <resDesc><!--optional, xs:string, resource description--></resDesc>
    </VideoInputChannel>
  </VideoInputChannelList>
</VideoInputChannelsCap>
```

XML_VideoOutput

VideoOutput message in XML format

```
<VideoOutput version="2.0" xmlns="http://www.isapi.org/ver20/XMLSchema">
  <VideoOutputChannelList/><!--opt, refer to the message XML_VideoOutputChannelList for details-->
</VideoOutput>
```

See Also

XML_VideoOutputChannelList

XML_VideoOutputChannel

VideoOutputChannel message in XML format

```
<VideoOutputChannel version="2.0" xmlns="http://www.isapi.org/ver20/XMLSchema">
  <id><!--req, xs: string; ID--></id>
  <type><!--req, xs: string, video output type: "VGA,CVBS,HDMI,Spot,SDI"--></type>
  <menu><!--dep, read-only-->
    <mirrorMenu><!--req, xs: boolean--></mirrorMenu>
  </menu>
  <mode><!--opt,xs:string, video output mode: "close,clip,scale,open,SDI_1080P25,
  ...,HDMI_1080P,HDMI_720P,HDMI_2160P"--></mode>
  <resolution><!--opt, xs: string; video solution: "1920*1080/60HZ,1280*720/50HZ,..."--></resolution>
</VideoOutputChannel>
```

XML_VideoOutputChannelList

VideoOutputChannelList message in XML format

```
<VideoOutputChannelList version="2.0" xmlns="http://www.isapi.org/ver20/XMLSchema">
  <VideoOutputChannel/><!--opt, refer to the message XML_VideoOutputChannel for details-->
</VideoOutputChannelList>
```

See Also

XML_VideoOutputChannel

XML_VideoOverlay

VideoOverlay message in XML format

```
<VideoOverlay version="2.0" xmlns="http://www.isapi.org/ver20/XMLSchema">
  <normalizedScreenSize><!--req, reference value of the region coordinates, which is 704*576 for PAL and 704*480 for
  NTSC-->
    <normalizedScreenWidth><!--ro, req, xs: integer--></normalizedScreenWidth>
    <normalizedScreenHeight><!--ro, req, xs: integer--></normalizedScreenHeight>
  </normalizedScreenSize>
  <attribute><!--opt, OSD properties-->
    <transparent><!--req, xs: boolean--></transparent>
    <flashing><!--req, xs: boolean--></flashing>
  </attribute>
  <TextOverlayList>
    <TextOverlay><!--text overlay parameter-->
      <id><!-- req, xs: string; text ID--></id>
      <enabled><!--req, xs: boolean, whether to enable text overlay--></enabled>
      <positionX><!--req, xs: float, x-coordinate--></positionX>
      <positionY><!-- req, xs: float, y-coordinate--></positionY>
      <displayText><!-- req, xs: string, displayed text contents--></displayText>
```

```

</TextOverlay>
</TextOverlayList>
<DateTimeOverlay>
  <enabled><!--req, xs: boolean, whether to enable date and time overlay--></enabled>
  <positionX><!--req, xs: integer; x-coordinate--></positionX>
  <positionY><!--req, xs: integer; y-coordinate--></positionY>
  <dateStyle>
    <!--opt, xs: string, date display format: YYYY-MM-DD, MM-DD-YYYY, DD-MM-YYYY, CHR-YYYY-MM-DD, CHR-MM-DD-
    YYYY, CHR-DD-MM-YYYY, CHR-YYYY/MM/DD, CHR-MM/DD/YYYY, CHR-DD/MM/YYYY-->
  </dateStyle>
  <timeStyle><!--opt, xs: string, time format: 12hour, 24hour--></timeStyle>
  <displayWeek><!--opt, xs: boolean, whether to display day of the week--></displayWeek>
</DateTimeOverlay>
<channelNameOverlay>
  <enabled><!--req, xs: boolean, whether to enable channel name overlay--></enabled>
  <positionX><!--req, xs: integer; x-coordinate--></positionX>
  <positionY><!--req, xs: integer; y-coordinate--></positionY>
</channelNameOverlay>
<fontSize opt="adaptive,16*16,32*32,48*48,64*64,80*80,96*96,112*112,128*128"><!--opt, xs: string, font size of
video OSD: "adaptive,16*16,32*32,48*48,64*64,80*80,96*96,112*112,128*128", unit: pixel. Currently only 16*16
and 32*32 are supported--></fontSize>
<frontColorMode opt="auto,customize"><!--opt, string, front color mode: "auto,customize"--></frontColorMode>
<frontColor><!--dep, xs: hexBinary, front color--></frontColor>
<alignment><!--opt, xs:string, "customize,alignRight,alignLeft"--></alignment>
<BatteryPowerOverlay/><!--opt-->
<DeviceStatusDisplay>
  <!--opt, display control of the device status on the video of portable speed dome-->
  <batteryCapacityDisplay>
    <!--opt, xs:boolean, whether to display battery capacity-->
  </batteryCapacityDisplay>
  <chargingStatusDisplay>
    <!--opt, xs:boolean, whether to display charging status-->
  </chargingStatusDisplay>
  <bluetoothStatusDisplay>
    <!--opt, xs:boolean, whether to display bluetooth status-->
  </bluetoothStatusDisplay>
  <dialStatusDisplay>
    <!--opt, xs:boolean, whether to display dial-up status-->
  </dialStatusDisplay>
</DeviceStatusDisplay>
<boundary><!--opt, xs:integer--></boundary>
<alignment><!--opt, xs: string, alignment mode: "customize, alignRight, alignLeft"--></alignment>
<publicSecurity><!--req, xs: boolean--></publicSecurity>
</VideoOverlay>

```

XML_VideoSourceActivation

XML Message about Parameters of Activating Network Device

```

<?xml version="1.0" encoding="utf-8"?>
<VideoSourceActivation version="2.0" xmlns="http://www.isapi.org/ver20/XMLSchema">

```



```
<raterat><!--required, xs:string--></raterat>
<passwordType>
  <!--required, xs:string, password types: "default"-NVR password, "selfDefine", "IPCActivatePasswd"-activation
password-->
</passwordType>
<password><!--depend, xs:string--></password>
<ActivationDevice>
  <mode><!--required, xs:string, "channel, descriptor"--></mode>
  <channel><!--depend-->
    <dynVideoInputChannelID><!--required, xs:integer--></dynVideoInputChannelID>
  </channel>
  <descriptor><!--depend-->
    <macAddress><!--required, xs:string--></macAddress>
    <addressingFormatType><!--required, xs:string, "ipaddress, hostname"--></addressingFormatType>
    <raterat><!--depend, xs:string--></raterat>
    <ipAddress><!--depend, xs:string--></ipAddress>
    <ipv6Address><!--depend, xs:string--></ipv6Address>
    <portNo><!--required, xs:integer--></portNo>
    <proxyProtocol><!--required, xs:string, "HIKVISION, SONY, ISAPI, ONVIF, ..."--></proxyProtocol>
  </descriptor>
</ActivationDevice>
</VideoSourceActivation>
```

XML_VideoSourceActivationCapability

XML Message about Capability of Activating Network Device

```
<?xml version="1.0" encoding="utf-8"?>
<VideoSourceActivationCapability version="2.0" xmlns="http://www.isapi.org/ver20/XMLSchema">
  <ActivateByChannel>
    <proxyProtocol opt="HIKVISION"/>
  </ActivateByChannel>
  <ActivateByAddress>
    <proxyProtocol opt="HIKVISION"/>
  </ActivateByAddress>
</VideoSourceActivationCapability>
```

XML_VideoSourceList

XML Message about Searched Network Devices

```
<VideoSourceList version="2.0" xmlns="http://www.isapi.org/ver20/XMLSchema">
  <VideoSourceDescriptor><!--list-->
    <id><!--req, xs: string;id--></id>
    <proxyProtocol><!--req, xs: string, "HIKVISION, SONY, ISAPI, ONVIF"--></proxyProtocol>
    <addressingFormatType><!--req, xs: string, address type: "ipaddress, hostname"--></addressingFormatType>
    <hostName><!--dep, xs: string--></hostName>
    <ipAddress><!--dep, xs: string--></ipAddress>
    <subnetMask><!--opt, xs: string, subnet mask for IPv4 address--></subnetMask>
```

```
<ipv6Address><!--dep, xs: string--></ipv6Address>
<bitMask><!--opt, xs: integer, bitmask IPv6 address--></bitMask>
<serialNumber><!--opt, xs: string--></serialNumber>
<macAddress><!--opt, xs: string--></macAddress>
<firmwareVersion><!--opt, req, xs: string--></firmwareVersion>
<managePortNo><!--opt, xs: integer--></managePortNo>
<userName><!--opt, xs: string--></userName>
<password><!--opt, xs: string--></password>
<srcInputPortNums><!--req, xs: integer--></srcInputPortNums>
<deviceId><!--dep, xs: string--></deviceId>
<deviceModel><!-- opt, xs: string, device model--></deviceModel>
</VideoSourceDescriptor>
</VideoSourceList>
```

XML_WebCertificate

WebCertificate message in XML format

```
<WebCertificate version="2.0" xmlns="http://www.isapi.org/ver20/XMLSchema">
  <CertificateType>
    <!--required, xs:string, authentication type: "basic", "digest", "digest/baisc"-->
  </CertificateType>
  <SecurityAlgorithm><!--dependent, this node is valid when certificateType is "digest" or "digest/basic"-->
    <algorithmType>
      <!--optional, xs:string, algorithm type: "MD5", "SHA256", "MD5/SHA256"-->
    </algorithmType>
  </SecurityAlgorithm>
</WebCertificate>
```

XML_WirelessDialInterface

WirelessDialInterface message in XML format

```
<WirelessDialInterface version="1.0" xmlns="http://www.isapi.org/ver20/XMLSchema">
  <id> <!--req, xs:string, only one network card is supported--> </id>
  <Dial/> <!--opt, refer to the URI: /ISAPI/System/Network/WirelessDial/Interfaces/<ID>/dial -->
  <Schedule> <!-- opt, refer to the URI: /ISAPI/System/Network/WirelessDial/Interfaces/<ID>/schedule -->
  <Dialstatus/> <!-- opt, refer to the URI: /ISAPI/System/Network/WirelessDial/Interfaces/<ID>/dialstatus -->
  <messageConfig/> <!-- opt, refer to the URI: /ISAPI/System/Network/WirelessDial/Interfaces/<ID>/messageConfig -->
  <messageList/> <!-- opt -->
</WirelessDialInterface>
```

XML_WirelessDialInterfaceList

WirelessDialInterfaceList message in XML format

```
<WirelessDialInterfaceList version="1.0" xmlns="http://www.isapi.org/ver20/XMLSchema">
  <WirelessDialInterface/>
</WirelessDialInterfaceList>
```

XML_ZoomFocus

ZoomFocus message in XML format

```
<ZoomFocus version="2.0" xmlns="http://www.isapi.org/ver20/XMLSchema">
  <pqrsZoom/><!--opt, xs: integer, zoom coordinates of Sony zoom camera module-->
  <mnstFocus/><!--opt, xs: integer, focus coordinates of Sony zoom camera module-->
</ZoomFocus>
```

B.2 Log Types for ISAPI

There are four major log types, i.e., alarm log, exception log, operation log, and event log. And each major type contains multiple minor types, see details in the following contents.

Alarm Logs

Log Type	Description
shortCircuit	Short Circuit Alarm
brokenCircuit	Open Circuit Alarm
alarmReset	Alarm Reset
alarmNormal	Return to Normal
passwordError	Incorrect Password (3 Tmes in a Row)
idCardIllegally	Invalid Card ID
keyPADRemove	Keypad Tampered
keyPADRemoveRestore	Keypad Restored
devRemove	Device Tampered
devRemoveRestore	Device Restored
belowAlarmLimit1	Sensor Value is Lower than Alarm Limit Value 1
belowAlarmLimit2	Sensor Value is Lower than Alarm Limit Value 2
belowAlarmLimit3	Sensor Value is Lower than Alarm Limit Value 3
belowAlarmLimit4	Sensor Value is Lower than Alarm Limit Value 4
aboveAlarmLimit1	Sensor Value is Higher than Alarm Limit Value 1

Log Type	Description
aboveAlarmLimit2	Sensor Value is Higher than Alarm Limit Value 2
aboveAlarmLimit3	Sensor Value is Higher than Alarm Limit Value 3
aboveAlarmLimit4	Sensor Value is Higher than Alarm Limit Value 4
UrgencyBtnON	Panic Button Triggered
UrgencyBtnOFF	Panic Button Restored
virtualDefenceBandit	Virtual Zone Burglary Alarm
virtualDefenceFire	Virtual Zone Fire Alarm
virtualDefenceUrgent	Virtual Zone Panic Alarm
motDetStart	Motion Detection Alarm Started
motDetStop	Motion Detection Alarm Stopped
hideAlarmStart	Device Blocked
hideAlarmStop	Device Blocking Alarm Restored
UPSAlarm	UPS Alarm
electricityMeterAlarm	Coulombmeter Alarm
switchPowerAlarm	Switch Power Supply Alarm
GasDetectSys	Gas Detection Alarm
transformerTempAlarm	Transformer Temperature Alarm
tempHumiAlarm	Temperature and Humidity Sensor Alarm
UPSAlarmRestore	UPS Alarm Restored
electricityMeterAlarmRestore	Coulombmeter Alarm Restored
switchPowerAlarmRestore	Switch Power Supply Alarm Restored
GasDetectSysRestore	Gas Detection Alarm Restored
transformerTempAlarmRestore	Transformer Temperature Alarm Restored
tempHumiAlarmRestore	Temperature-Humidity Sensor Alarm Restored
waterLevelSensorAlarm	Temperature-Humidity Sensor Alarm Restored
waterLevelSensorAlarmRestore	Flood Sensor Restored
dustNoiseAlarm	Dust and Noise Sensor Alarm
dustNoiseAlarmRestore	Dust and Noise Sensor Alarm Restored
environmentalLogger	Environmental Data Collector Alarm

Log Type	Description
environmentalLoggerAlarm	Environmental Data Collector Restored
triggerTemper	Detector Tampered
triggerTemperRestore	Detector Restored
emergencyCallHelp	Panic Alarm
emergencyCallHelpRestore	Panic Alarm Restored
consult	Consultation Alarm
consultRestore	Consultation Alarm Restored
deviceMoveAlarm	Device Motion Alarm
deviceMoveAlarmRestore	Device Motion Alarm Restored
earlyWarningAlarm	Early Warning Zone Alarm
earlyWarningAlarmRestore	Early Warning Zone Restored
warningAlarm	Warning Zone Alarm
warningAlarmRestore	Warning Zone Restored
wirelessOutputModTamperEvident	Wireless Output Expander Tampered
wirelessOutputModTamperEvidentReset	Wireless Output Expander tamper Restored
wirelessRepeaterTamperEvident	Wireless Repeater Tampered
wirelessRepeaterTamperEvidentReset	Wireless Repeater tamper Restored
wirelessSirenTamperEvident	Wireless Siren Tampered
wirelessSirenTamperEvidentReset	Wireless Siren Tamper Restored
wirelessKeypadTamperEvident	Wireless Keypad Tampered
wirelessKeypadTamperEvidentReset	Wireless Keypad Tamper Restored
wirelessCardReaderTamperEvident	Wireless Card Reader Tampered
wirelessCardReaderTamperEvidentReset	Wireless Card Reader Tamper Restored
softZoneMedicalAlarm	Virtual Zone Medical Alarm
accessControllerEvent	Access Controller Event
videoIntercomEvent	Video Intercom Event
GJDEvent	GJD Security Control Panel Event
LuminateEvent	LUMINITE Security Control Panel Event
OPTEXEvent	OPTEX Security Control Panel Event

Log Type	Description
cameraDetectorEvent	Detector Event
securityControlPanelEvent	Security Control Panel Event
RS-485AlarmInputModuleEvident	RS-485 Zone Module Tampered
RS-485AlarmInputModuleTamperReset	RS-485 Zone Module Tampering Reset
RS-485WirelessReceiverTamperEvident	RS-485 Wireless Receiver Module Tampered
RS-485WirelessReceiverTamperEvidentReset	RS-485 Wireless Receiver Module Tampering Reset
dredgerDetectionAlarm	Dredger Detection Alarm
crossLineAlarm	Line Crossing Alarm
crossLineAlarmRestore	Line Crossing Alarm Restored
HFPDAlarmStart	High Frequently Appeared Person Alarm Started
HFPDAlarmStop	High Frequently Appeared Person Alarm Stopped
LFPDAlarmStart	Low Frequency Person Alarm Started
LFPDAlarmStop	Low Frequency Person Alarm Stopped
safetyHelmetAlarmStart	Hard Hat Detection Alarm Started
safetyHelmetAlarmStop	Hard Hat Detection Alarm Stopped
dataPrealarm	Traffic Pre-alarm
playCellphoneStart	Playing Cellphone Alarm Started
playCellphoneStop	Playing Cellphone Alarm Stopped
sleepOnDutyStart	Sleeping on Duty Alarm Started
sleepOnDutyStop	Sleeping on Duty Alarm Stopped
vibrationDetectionStart	Vibration Detection Alarm Started
vibrationDetectionStop	Vibration Detection Alarm Stopped
fireEscapeDetectionStart	Fire Engine Access Detection Started
fireEscapeDetectionStop	Fire Engine Access Detection Ended
takingElevatorDetectionStart	Elevator Detection Started
takingElevatorDetectionStop	Elevator Detection Ended
unregisteredStreetVendorStart	Unlicensed Business Vendor Detection Started
unregisteredStreetVendorStop	Unlicensed Business Vendor Detection Ended

Log Type	Description
stallOutsideShopStart	Business Outside Store Detection Started
stallOutsideShopStop	Business Outside Store Detection Ended
stallOccupyingRoadStart	Business on Sidewalk Detection Started
stallOccupyingRoadStop	Business on Sidewalk Detection Ended
illegalHeapStart	Pile Goods or Materials in Chaos Detection Started
illegalHeapStop	Pile Goods or Materials in Chaos Detection Ended
illParkofNonMotorVehicleStart	Non-Motor Vehicles Parking in Chaos Detection Started
illParkofNonMotorVehicleStop	Non-Motor Vehicles Parking in Chaos Detection Ended
illegalOutdoorAdvertisementStart	Unauthorized Outdoor Advertisement Detection Started
illegalOutdoorAdvertisementStop	Unauthorized Outdoor Advertisement Detection Ended
packGarbageStart	Packaged Garbage Detection Started
packGarbageStop	Packaged Garbage Detection Ended
stallUnderUmbrellaStart	Running Business with Patio Umbrella Detection Started
stallUnderUmbrellaStop	Running Business with Patio Umbrella Detection Ended
dustbinOverflowStart	Overflowing Dustbin Detection Started
dustbinOverflowStop	Overflowing Dustbin Detection Ended
exposeGarbageStart	Exposed Garbage Detection Started
exposeGarbageStop	Exposed Garbage Detection Ended
hangClothingAlongStreetStart	Hanging Clothes On Street Detection Started
hangClothingAlongStreetStop	Hanging Clothes On Street Detection Ended
ATMPanelStart	ATM Panel Alarm Started
ATMPanelStop	ATM Panel Alarm Ended
ATMSurroundStart	ATM Surround Alarm Started
ATMSurroundStop	ATM Surround Alarm Ended

Log Type	Description
ATMFaceStart	ATM Face Alarm Started
ATMFaceStop	ATM Face Alarm Ended
ATMSafetyCabinStart	ATM Safety Cabin Alarm Started
ATMSafetyCabinStop	ATM Safety Cabin Alarm Ended
soundIntensityMutation	Sudden Increase of Sound Intensity Detection
soundIntensityMutationStop	Sudden Increase of Sound Intensity Detection Ended
soundIntensitySteepFall	Sudden Decrease of Sound Intensity Detection
soundIntensitySteepFallStop	Sudden Decrease of Sound Intensity Detection Ended
moveAlarm	Motion Alarm
moveAlarmRestored	Motion Alarm Restored
lowTemperatureAlarm	Low Temperature Alarm
lowTemperatureAlarmRestored	Low Temperature Alarm Restored
highTemperatureAlarm	High Temperature Alarm
highTemperatureAlarmRestored	High Temperature Alarm Restored

Exception Logs

Log Type	Description
powerOn	Power on
powerOff	Power off
WDTRreset	WDT Reset
lowBatteryVoltage	Low Battery Voltage
ACLoss	AC Power Disconnected
ACRestore	AC Power Restored
RTCException	RTC Real-time Clock Exception
netFailure	Network Disconnected
netRestore	Network Connected
telLineBroken	Telephone Line Disconnected
telLineRestore	Telephone Line Connected

Log Type	Description
expanderBusLoss	Bus Expander Disconnected
expanderBusRestore	Bus Expander Connected
keypadBusLoss	Keypad Expander Disconnected
keypadBusRestore	Keypad Expander Connected
sensorFailure	Analog Sensor Fault
sensorRestore	Analog Sensor Restored
RS485DisConnect	RS-485 Channel Disconnected
RS485Connect	RS-486 Channel Connected
batteryVoltageRestore	Battery Voltage Restored
wiredNetAbnormal	Wired Network Exception
wiredNetRestore	Wired Network Restored
GPRSAbnormal	GPRS Exception
GPRSRestore	GPRS Restored
3GAbnormal	3G Network Exception
3GRestore	3G Network Restored
SIMCardAbnormal	SIM Card Exception
SIMCardRestore	SIM Card Restored
VILost	Video Loss
illegalAccess	Illegal Login
HDFull	HDD Full
HDError	HDD Error
DCDLost	MODEM Disconnected
IPConflict	IP Address Conflicted
netbroken	Network Disconnected
recError	Recording Error
VIError	Video Input Exception(Only for Analog Channel)
formatHDDError	Remote HDD Formatting Failed
USBError	USB Communication Error
USBRestore	USB Communication Error Restored

Log Type	Description
printError	Printer Error
printRestore	Printer Error Restored
subsystemCommunicationError	Sub-board Communication Error
IPCIpconflict	Network Camera IP Address Conflicted
VIMisMatch	Video Standard Mismatches
MCURestart	MCU Restarted
GprsMouleFault	GPRS Module Fault
telephoneFault	Telephone Module Fault
wifiAbnormal	Wi-Fi Exception
wifiRestore	Wi-Fi Restored
RFAbornal	RF Exception
RFRestore	RF Restored
detectorOnline	Detector Connected
detectorOffline	Detector Disconnected
detectorBatteryNormal	Detector Battery Restored
detectorBatteryLow	Detector Battery Low
dataTrafficOverflow	Cellular Network Data Exceeded
radarSignalFault	Radar Transmitter Fault
radarSignalFaultRestore	Radar Transmitter Restored
wirelessOutputModOffline	Wireless Output Expander Disconnected
wirelessOutputModOnline	Wireless Output Expander Connected
wirelessRepeaterOffline	Wireless Repeater Disconnected
wirelessRepeaterOnline	Wireless Repeater Connected
triggerOffline	Trigger Disconnected
triggerOnline	Trigger Connected
wirelessSirenOffline	Wireless Siren Disconnected
wirelessSirenOnline	Wireless Siren Connected
sirenLowPower	Siren Battery Low
sirenPowerRecovery	Siren Battery Restored

Log Type	Description
ipcDisconnect	Network Camera Disconnected
ipcConnectRecovery	Network Camera Connected
sendMailFailed	Sending Email Failed
eventUploadException	Uploading Event Failed or Uploaded Event Lost
keyfobLowPower	Low Keyfob Battery
keyfobPowerRecovery	Normal Keyfob Battery
detectorOvertime	Detector Heartbeat Timed Out
detectorOvertimeRecovery	Detector Heartbeat Timeout Restored
wSirenOvertime	Wireless Siren Heartbeat Timed Out
wSirenOvertimeRecovery	Wireless Siren Heartbeat Timeout Restored
wOutputOvertime	Wireless Output Module Heartbeat Timed Out
wOutputOvertimeRecovery	Wireless Output Module Heartbeat Timeout Restored
wRepeaterOvertime	Wireless Repeater Heartbeat Timed Out
wRepeaterOvertimeRecovery	Wireless Repeater Heartbeat Timeout Restored
rfJamming	RF Wireless Communication Blocked
rfJammingRecovery	RF Wireless Communication Blocking Restored
batteryMiss	Storage Battery Loss
batteryMissRecovery	Storage Battery Restored
ARCUploadFailed	Uploading to ARC Failed
ARCUploadSucceeded	Uploaded to ARC
ARCUploadRecovery	Uploading to ARC Restored
wirelessKeypadOffline	Wireless Keypad Disconnected
wirelessKeypadOnline	Wireless Keypad Connected
wirelessCardReaderOffline	Wireless Card Reader Disconnected
wirelessCardReaderOnline	Wireless Card Reader Connected
keypadLowPower	Low Keypad Battery
keypadLowPowerRecovery	Low Keypad Battery Recovered
cardReaderLowPower	Low Card Reader Battery

Log Type	Description
cardReaderLowPowerRecovery	Low Card Reader Battery Recovered
wKeypadOvertime	Wireless Keypad Heartbeat Timed Out
wKeypadOvertimeRecovery	Wireless Keypad Heartbeat Timeout Recovered
wCardReaderOvertime	Wireless Card Reader Heartbeat Timed Out
wCardReaderOvertimeRecovery	Wireless Card Reader Heartbeat Timeout Recovered
ATSFailed	ATS Failed
ATSRecovery	ATS Recovered
LANPathFailed	Wired or Wireless Connection Failed
LANPathRecovery	Wired or Wireless Connection Recovered
mobileNetPathFailed	Mobile Network Connection Failed
mobileNetPathRecovery	Mobile Network Connection Recovered
RS-485AlarmInputModuleDisconnected	RS-485 Zone Module Offline
RS-485AlarmInputModuleConnected	RS-485 Zone Module Online
RS-485WirelessReceiverDisconnected	RS-485 Wireless Receiver Module Offline
RS-485WirelessReceiverConnected	RS-485 Wireless Receiver Module Online
keypadDisconnected	Keypad Offline
keypadConnected	Keypad Online
overvoltage	High Supply Voltage
undervoltage	Low Supply Voltage
highHDDTemperature	HDD High Temperature
lowHDDTemperature	HDD Low Temperature
hdImpact	HDD Impact
hdBadBlock	HDD Bad Sector
severeHDFailure	HDD Severe Fault
safetyHelmetException	Hard Hat Detection Exception
ezvizUpgradeException	Hik-Connect Upgrade Exception

Operation Logs

Log Type	Description
guard	Normal Arming
unguard	Normal Disarming
bypass	Bypass
duressAccess	Duress
localReboot	Local Reboot
remoteReboot	Remote Reboot
localUpgrade	Local Upgrade
remoteUpgrade	Remote Upgrade
recoveryDefaultParam	Restore Default Settings
outputAlarm	Remote Alarm Output Control
accessOpen	Access Control: Open
accessClose	Access Control : Closed
sirenOpen	Siren: On
sirenClose	Siren: Off
modZoneConfig	Zone Settings
modAlarmoutConfig	Alarm Output Settings
modAnalogConfig	Sensor Settings
RS485Config	RS-485 Channel Settings
phoneConfig	Dialing Settings
addAdmin	Added Administrator
modAdminParam	Edited Administrator
delAdmin	Deleted Administrator
addNetUser	Added DVR/NVR Operator
modNetUserParam	Edited DVR/NVR Operator
delNetUser	Deleted DVR/NVR Operator
addOperator	Added Camera Operator
modOperatorPw	Edited Camera Operator Password

Log Type	Description
delOperator	Deleted Camera Operator Password
addKeyPadUser	Added Keypad/Card Reader User
delKeyPadUser	Deleted Keyboard/Card Reader User
remoteUserLogin	Remote Login
remoteUserLogout	Remote Logout
remoteGuard	Remote Arming
remoteUnguard	Remote Disarming
modHostConfig	Edited Control Panel Settings
restoreBypass	Bypass Restored
alarmOutOpen	Turned on Output
alarmOutClose	Turned off Output
modSubsystemParam	Edited Subsystem Parameters
groupBypass	Group Bypass
groupBypassRestore	Group Bypass Restored
modGprsParam	Edited GPRS Parameters
modNetReportParam	Edited Network Report Settings
modReportMode	Edited Uploading Mode
modGatewayParam	Edited Access Control Settings
remoteStartRec	Remote: Started Recording
remoteStopRec	Remote: Stopped Recording
transChanStart	Transparent Transmission Started
transChanStop	Transparent Transmission Stopped
startVoiceTalk	Two-way Audio Started
stopVoiceTalk	Two-way Audio Terminated
remotePlayByFile	Remote: Playback or Downloaded by File
remotePlayByTime	Remote: Playback by Time
remotePTZCtrl	Remote: PTZ Control
remoteLockFile	Remote: Locked File
remoteUnlockFile	Remote: Unlocked File

Log Type	Description
remoteFormatHd	Remote: Formatted HDD
remoteDownloadCfgFile	Remote: Exported Configuration Files
remoteUploadCfgFile	Remote: Imported Configuration Files
remoteDownloadRecFile	Remote: Exported File
stayArm	Stay Arming
quickArm	Instant Arming
keyswitchArm	Key Zone Arming
keyswitchDisarm	Key Zone Disarming
clearAlarm	Alarm Cleared
modFaultConfig	Edited System Fault Settings
modAlarmOutConfig	Edited Event Alarm Output Settings
searchExternalModule	Searched for External Module
registerExternalModule	Re-registered External Module
closeKeypadAlarm	Disabled Keypad Beep
mod3GConfig	Edited Mobile Parameters
modPrintConfig	Edited Printer Parameters
SDCardFormat	Formatted SD Card
upgradeSubsystem	Upgraded Sub-board
planArmConfig	Arming/Disarming Schedule Configuration
phoneArm	SMS Arming
phoneStayArm	SMS Stay Arming
phoneQuickArm	SMS Instant Arming
phoneDisarm	SMS Disarming
phoneClearAlarm	SMS Alarm Cleared
whiteConfig	Whitelist Settings
timeTriggerConfig	Enabled/Disabled Trigger Configuration by Schedule
pictureConfig	Capture Settings
tamperConfig	Zone Tamper-Proof Settings
remoteKeypadUpgrade	Remote: Upgraded Keypad

Log Type	Description
singlePartionArmORDisarm	Single-Zone Arming/Disarming
cardConfiguration	Card Settings
cardAramORDisarm	Arming/Disarming by Card
expendNetCenterConfig	Extension Network Center Settings
netCardConfig	NIC Settings
DDNSConfig	DDNS Settings
RS485BusConfig	RS-485 Bus Settings
RS485BusReRegistration	RS-485 Bus Re-registration
remoteOpenElectricLock	Remote: Unlocked
remoteCloseElectricLock	Remote: Locked
localOpenElectricLock	Local: Unlocked
localCloseElectricLock	Local: Locked
openAlarmLamp	Remote: Turned On Alarm Lamp
closeAlarmLamp	Remote: Turned Off Strobe
temporaryPassword	Operation Record of Temporary Password
oneKeyAwayArm	One-Push Away Arming
oneKeyStayArm	One-Push Stay Arming
remoteDelAllVideoAnalysisTask	Empty All Video Analysis Tasks Remotely
singleZoneArm	Single-Zone Arming
singleZoneDisarm	Single-Zone Disarming
HiDDNSConfig	HiDDNS Settings
remoteKeypadUpdata	Remote: Upgraded Keypad
zoneAddDetector	Added Detector
zoneDelDetector	Deleted Detector
qureyDetectorSignal	Checked Detector Signal Strength on Security Control Panel
qureyDetectorBattery	Checked Detector Remaining Battery on Security Control Panel
setDetectorGuard	Detector Arming

Log Type	Description
setDetectorUnguard	Detector Disarming
setWifiParm	Wi-Fi Settings
voiceOpen	Audio On
voiceClose	Mute
functionKeyEnable	Enabled Function Key
functionKeyDisable	Disabled Panel Function Button
readCard	Swiped Patrol Card
localDeviceActive	Activated Device Remotely
localFactoryDefault	Restored Factory Settings Locally
remoteFactoryDefault	Restored Factory Settings Remotely
addWirelessOutputMod	Added Wireless Output Module
delWirelessOutputMod	Deleted Wireless Output Module
addWirelessRepeater	Added Wireless Repeater
delWirelessRepeater	Deleted Wireless Repeater
telListConfig	Mobile Phone Number Settings
searchRFSignal	Checked RF Signal
addWirelessSiren	Added Wireless Siren
delWirelessSiren	Deleted Wireless Siren
flowConfig	Cellular Data Limit Settings
addRemoter	Added Keyfob
delRemoter	Deleted Keyfob
addCard	Added Card
delCard	Deleted Card
remoteAddIpc	Added Network Camera
remoteDelIpc	Deleted Network Camera
remoteSetIpc	Edited Network Camera
localAddressFilterConfig/ remoteAddressFilterConfig	Local/Remote Address Filter Configuration
enterProgramMode	Programming Mode Enabled for Keypad

Log Type	Description
existProgramMode	Programming Mode Disabled for Keypad
localIOTCfgFileInput	Local operation: import IoT configuration file
localIOTCfgFileOutput	Local operation: export IoT configuration file
remoteIOTCfgFileInput	Remote operation: import IoT configuration file
remoteIOTCfgFileOutput	Remote operation: export IoT configuration file
localIOTAdd	Local operation: add IoT channel
remoteIOTAdd	Remote operation: add IoT channel
localIOTDelete	Local operation: delete IoT channel
remoteIOTDelete	Remote operation: delete IoT channel
localIOTSet	Local operation: configure IoT channel
remoteIOTSet	Remote operation: configure IoT channel
armWithFault	Armed with Fault
entryDelay	Entering and Exiting Delay
modArmConfig	Edit Arming Parameters
modCertificateStandard	Edit Authentication Standard
entryPaceTest	Pacing Mode Entered
exitPaceTest	Pacing Mode Exited
addNetOperator	Add Operator
modNetOperator	Edit Operator Information
delNetOperator	Delete Operator
addNetInstaller	Add Installer
modNetInstaller	Edit Installer Information
delNetInstaller	Delete Installer
addManufacturer	Add Manufacturer
modManufacturer	Edit Manufacturer Information
delManufacturer	Delete Manufacturer
upgradeSucceeded	Upgraded
upgradeFailed	Upgrading Failed
zoneDisabled	Zone Shielded

Log Type	Description
localCfgSecurity	Security Parameter Configured Locally
remoteCfgSecurity	Security Parameter Configured Remotely
remoteGetParaSecurity	Security Parameters Obtained Remotely
delRS-485InputModule	RS-485 Zone Module Deleted
delRS-485OutputModule	RS-485 Output Module Deleted
delRS-485WirelessReceiver	RS-485 Wireless Receiver Module Deleted
enrollRS-485InputModule	RS-485 Zone Module Registered
enrollRS-485OutputModule	RS-485 Output Module Registered
delRS-485OutputModule	RS-485 Output Module Deleted
enrollRS-485WirelessReceiver	RS-485 Wireless Receiver Module Registered
enrollKeypad	Keypad Registered
delKeypad	Keypad Deleted
scheduledAngleCalibration	Scheduled Angle Calibration
addZone	Added Zone
modZone	Edited Zone
delZone	Deleted Zone
addAlarmLine	Added Trigger Line
modAlarmLine	Edited Trigger Line
delAlarmLine	Deleted Trigger Line
remoteHFPDconfig/localHFPDconfig	Remote/Local Configuration of Frequently Appeared Person Detection
remoteLFPDconfig	Remote Configuration of Low Frequency Person Detection
modifyUserPassword	Password Changed
logOut	Logged Out
indicatorStatusSettings	Indicator Switch Settings
wiredNetworkSettings	Wired Network Settings
notificationSettings	Message Notification Settings
alarmCenterSettings	Alarm Receiving Center Settings

Log Type	Description
videoRecordStrategySettings	Video Recording Strategy Settings
cameraLinkageSettings	Camera Linkage Settings
armingDisarmingScheduleSettings	Arming and Disarming Schedule Settings
alarmSpeedSettings	Speed Threshold Settings for Triggering Alarms
videoTrackingSwitchSettings	Video Tracking Switching Settings
frequencySettings	Frequency Band Settings
masterSlaveTrackingSettings	Master-Slave Tracking Settings
parkingPointSettings	Parking Point Settings
administratorEdited	Administrator Parameters Edited
securityConfigured	Security Settings
relayParametersEdited	Relay Parameters Edited
radarSensitivitySettings	Radar Sensitivity Settings
detectAngandDistanceSettings	Detector Angle and Range Settings
masterSlaveRadarSettings	Master Radar and Slave Radar Settings
remoteCheckTime	Remote: Manual Time Synchronization
remoteParamSimpleDefault	Remote: Partly Restore to Default Settings
remoteParamFactoryDefault	Remote: Restore All to Default Settings
localAutoSwitchConfig	Configure Auto Power On or Off Locally
remoteAutoSwitchConfig	Configure Auto Power On or Off Remotely
remoteCfgWirelessDialParam	Configure wireless dial-up parameters remotely
localCfgWirelessDialParam	Configure wireless dial-up parameters locally
remoteCfgWirelessSmsParam	Configure wireless message parameters remotely
localCfgWirelessSmsParam	Configure wireless message parameters locally
remoteCfgWirelessSmsSelfHelp	Configure SMS self-service parameters remotely
localCfgWirelessSmsSelfHelp	Configure SMS self-service parameters locally
remoteCfgWirelessNetFlowParam	Configure wireless traffic parameters remotely
localCfgWirelessNetFlowParam	Configure wireless traffic parameters locally
scaleCfg	Scale Settings
radarTrailCfg	Radar Pattern Settings

Log Type	Description
MapImportCfg	Map Importing Settings
radarCalibrationCfg	Radar Calibration Settings
LocalEzvizOperation	Local EZVIZ Operations
RemoteEzvizOperation	Remote EZVIZ Operations
SSHEnabled	SSH Enabled
SSHDisabled	SSH Disabled
installationModeEntered	Installation Mode Enabled
installationModeExited	Installation Mode Disabled
diagnosisModeConfigured	Diagnosis Mode Configured
fileExported	File Exported
audioFileUploaded	Audio File Uploaded
audioFileDeleted	Audio File Deleted
PIRCAMCapture	PIRCAM Captured
SIPIntercomStarted	SIP Intercom Started
SIPIntercomEnded	SIP Intercom Ended
enrollmentModeEntered	Registration Mode Enabled
enrollmentModeExited	Registration Mode Disabled
videoAudioSelfCheckStarted	Self-Test of Audio and Video Started
videoAudioSelfCheckStopped	Self-Test of Audio and Video Stopped
cardReaderunlocked	Card Reader Unlocked
cardReaderlocked	Card Reader Locked
videoAudioSelfCheckEnded	Self-Test of Audio and Video Ended
previewStart	Live View Started
previewStop	Live View Stopped
remoteDelAllVideoAnalysisTask	All Video Analysis Tasks Cleared (One-Touch)
localSSDOperateStart	Local SSD operation (firmware operations) started
localSSDOperateStop	Local SSD operation (firmware operations) ended
remoteSSDOperateStart	Remote SSD operation (firmware operations) started
remoteSSDOperateStop	Remote SSD operation (firmware operations) ended

Log Type	Description
AITargetBPAdd	Reference picture for AI target comparison added
AITargetBPDelete	Reference picture for AI target comparison deleted
AITargetBPSearch	Reference picture for AI target comparison AI searched for
AITargetBPUpdate	Reference picture for AI target comparison updated
AIRuleConfigTrigger	AI rule linkage configured
AudioFileImport	Audio file imported
AudioFileDownload	Audio file downloaded

Event Logs

Log Type	Description
SDKSchool	SDK Synchronization
selfTimeSchool	Time Synchronization by Schedule
insertSubsystem	Plugged in Sub-board
pullOutSubsystem	Pulled out Sub-board
autoArm	Auto Arming
autoDisarm	Auto Disarming
triggerOn	Activated Trigger by Schedule
triggerOff	Deactivated Trigger by Schedule
autoArmFailed	Auto Arming Failed
autoDisarmFailed	Auto Disarming Failed
triggerOnFailed	Activating Trigger Failed
triggerOffFailed	Deactivating Trigger Failed
mandatoryAlarm	Forced Arming
keyPADlocked	Keypad Locked
keyPADunlocked	Keypad Unlocked
insetUSB	Plugged in USB Flash Drive
pulloutUSB	Removed USB Flash Drive
lateRemind	Late to Disarm

Log Type	Description
keypadUnlocked	Unlocked Keypad
timeSynchronization	Time Synchronization
armFailed	Arming Failed
ARCStart	ARC Connected
locked	Locked

Additional Logs

Log Type	Description
doubleVerificationPass	Double Verification Completed
hdFormatStart	Formatting HDD Started
hdFormatStop	Formatting HDD Stopped
wirelessRunningStatus	Wireless Network Running Status

B.3 Supported Alarm/Event Types

This part lists the alarm or event types that can be received or subscribed via Intelligent Security API and the corresponding values to be set for getting the details.

Alarm/Event Type	Value of <type>	Alarm/Event Details
Intrusion Detection	fielddetection	Refer to <i>Intelligent Security API (Behavior Analysis) _Developer Guide</i> for details.
Line Crossing Detection	linedetection	
Region Entrance Detection	regionEntrance	
Region Exiting Detection	regionExiting	
Loitering Detection	loitering	
People Gathering Detection	group	
Fast Moving Detection	rapidMove	
Parking Detection	parking	
Unattended Baggage Detection	unattendedBaggage	

Alarm/Event Type	Value of <type>	Alarm/Event Details
Object Removal Detection	attendedBaggage	
Absence Alarm	leavePosition	
People Number Changed Alarm	peopleNumChange	
Fast Moving Alarm	running	
Violent Motion Alarm	violentMotion	
People Falling Down Alarm	failDown	
Overstay Alarm	retention	
People Counting Alarm	PeopleCounting	Refer to <i>Intelligent Security API (People Counting) _ Developer Guide</i> for details
Temperature Measurement Alarm	TMA	Refer to <i>Intelligent Security API (Thermal) _ Developer Guide</i> for details
Temperature Measurement Pre-Alarm	TMPA	
Temperature Difference Alarm	TDA	
Fire Source Detection	fireDetection	
Smoke Detection	smokeDetection	
Fire and Smoke Detection	smokeAndFireDetection	
Ship Detection	shipsDetection	
Face Detection Alarm	facedetection	Refer to <i>Intelligent Security API (Facial) _ Developer Guide</i> for details.
Face Capture Alarm	faceCapture	
Face Picture Comparison Result	alarmResult	
Stranger Detection Alarm	whiteFaceContrast	
High Frequently Person Detection	HFPD	

Alarm/Event Type	Value of <type>	Alarm/Event Details
Low Frequency Person Detection	LFPD	
Waiting Time Detection Alarm	personQueueTime	Refer to <i>Intelligent Security API (Queue Management) _Developer Guide</i> for details
People Queuing-Up Alarm	personQueueCounting	
Multi-Target-Type Detection Alarm	mixedTargetDetection	Refer to <i>Intelligent Security API (Multi-Target-Type Detection) _Developer Guide</i> for details
Traffic Incident Alarm	AID	Refer to <i>Intelligent Security API (Traffic Enforcement) _Developer Guide</i> for details
Traffic Violation Enforcement Alarm	TFS	
Traffic Data Statistics Alarm	TPS	
ANPR Alarm	ANPR	Refer to <i>Intelligent Security API (ANPR) _Developer Guide</i> for details
Blacklist and Whitelist ANPR Alarm	vehicleMatchResult	
Vehicle Feature Recognition Alarm	vehicleRcogResult	
Radar Measurement Data	radarMeasurement	
Heat Map Alarm	heatmap	Refer to <i>Intelligent Security API (Heat Map) _Developer Guide</i> for details
Face and ID Card Recognition Alarm	cardMatch	Refer to <i>Intelligent Security API (Access Control on Person) _Developer Guide</i> for details
Hot Spare Exception Alarm	hotSpare	The alarm or event details will be provided in the further versions.
Overspeed Alarm	overSpeed	The alarm or event details will be provided in the further versions.
Temperature Too High Alarm	highTempAlarm	The alarm or event details will be provided in the further versions.
Acceleration Exception Alarm	abnormalAcceleration	The alarm or event details will be provided in the further versions.

Alarm/Event Type	Value of <type>	Alarm/Event Details
Collision Alarm	collision	The alarm or event details will be provided in the further versions.
Rollover Detection	rollover	The alarm or event details will be provided in the further versions.
Abnormal Driving Behavior Detection	abnormalDriving	The alarm or event details will be provided in the further versions.
GPS Information Uploaded	GPSUpload	The alarm or event details will be provided in the further versions.
Video Standard Mismatched	PALMismatch	The alarm or event details will be provided in the further versions.
Alarm Input Alarm	IO	
Motion Detection Alarm	VMD	<i>XML_EventNotificationAlert_MotionDetectionMsg</i>
Video Loss Alarm	videoloss	<i>XML_EventNotificationAlert_VideoLossAlarmMsg</i>
Video Tampering Alarm	shelteralarm	<i>XML_EventNotificationAlert_VideoTamperingAlarmMsg</i>
Defocus Alarm	defocus	<i>XML_EventNotificationAlert_DefocusDetectionAlarmMsg</i>
Audio Exception Alarm	audioexception	<i>XML_EventNotificationAlert_AudioExceptionAlarmMsg</i>
Scene Changed Alarm	scenechangedetection	<i>XML_EventNotificationAlert_SceneChangeDetectionAlarmMsg</i>
PIR Alarm	PIR	<i>XML_EventNotificationAlert_PIRAlarmMsg</i>
HDD Full	diskfull	The alarm or event details will be provided in the further versions.
HDD Error	diskerror	<i>XML_EventNotificationAlert_HDDErrorAlarmMsg</i>
Network Disconnected	nicbroken	The alarm or event details will be provided in the further versions.
IP Address Conflicted	ipconflict	The alarm or event details will be provided in the further versions.
Illegal Login Alarm	illaccess	<i>XML_EventNotificationAlert_IllegalLoginAlarmMsg</i>

Alarm/Event Type	Value of <type>	Alarm/Event Details
PoE Power Exception Alarm	poe	The alarm or event details will be provided in the further versions.
Video Exception Alarm	videoException	The alarm or event details will be provided in the further versions.
Recording Exception Alarm	recordException	The alarm or event details will be provided in the further versions.
HDD Unformatted Alarm	diskUnformat	The alarm or event details will be provided in the further versions.
Supply Voltage Exception Alarm	voltageinstable	<i>JSON_EventNotificationAlert_voltageinstable</i>
HDD High Temperature	highHDTemperature	<i>JSON_EventNotificationAlert_HDDHighTemperatureEventMsg</i>
HDD Low Temperature	lowHDTemperature	<i>JSON_EventNotificationAlert_HDDLowTemperatureEventMsg</i>
HDD Impact	hdImpact	<i>JSON_EventNotificationAlert_HDDImpactEventMsg</i>
HDD Bad Sector	hdBadBlock	<i>JSON_EventNotificationAlert_HDDBadSectorEventMsg</i>
HDD Severe Fault	severeHDFailure	<i>JSON_EventNotificationAlert_HDDSevereFaultEventMsg</i>
Certificate Expiry Alarm	certificateRevocation	<i>JSON_EventNotificationAlert_CertificateExpiryAlarmMsg</i>
Network Traffic Pre-alarm	dataPrealarm	<i>JSON_EventNotificationAlert_NetworkTrafficPrealarmMsg</i>
Auto Person Arming	personArmingTrack	Refer to <i>Intelligent Security API (Person Arming)_Developer Guide</i> for details
Manual Person Arming	manualPersonArmingTrack	

B.4 Response Codes of Text Protocol

The response codes returned during the text protocol integration is based on the status codes of HTTP. 7 kinds of status codes are predefined, including 1 (OK), 2 (Device Busy), 3 (Device Error), 4 (Invalid Operation), 5 (Invalid Message Format), 6 (Invalid Message Content), and 7 (Reboot)

Required). Each kind of status code contains multiple sub status codes, and the response codes are in a one-to-one correspondence with the sub status codes.

StatusCode=1

SubStatusCode	Error Code	Description
ok	0x1	Operation completed.
riskPassword	0x10000002	Risky password.
armProcess	0x10000005	Arming process.

StatusCode=2

Sub Status Code	Error Code	Description
noMemory	0x20000001	Insufficient memory.
serviceUnavailable	0x20000002	The service is not available.
upgrading	0x20000003	Upgrading.
deviceBusy	0x20000004	The device is busy or no response.
reConnectIpc	0x20000005	The video server is reconnected.
transferUpgradePackageFailed	0x20000006	Transmitting device upgrade data failed.
startUpgradeFailed	0x20000007	Starting upgrading device failed.
getUpgradeProcessfailed.	0x20000008	Getting upgrade status failed.
certificateExist	0x2000000B	The Authentication certificate already exists.

StatusCode=3

Sub Status Code	Error Code	Description
deviceError	0x30000001	Hardware error.
badFlash	0x30000002	Flash operation error.
28181Uninitialized	0x30000003	The 28181 configuration is not initialized.
socketConnectError	0x30000005	Connecting to socket failed.

Sub Status Code	Error Code	Description
receiveError	0x30000007	Receive response message failed.
deletePictureError	0x3000000A	Deleting picture failed.
pictureSizeExceedLimit	0x3000000C	Too large picture size.
clearCacheError	0x3000000D	Clearing cache failed.
updateDatabasError	0x3000000F	Updating database failed.
searchDatabaseError	0x30000010	Searching in the database failed.
writeDatabaseError	0x30000011	Writing to database failed.
deleteDatabaseError	0x30000012	Deleting database element failed.
searchDatabaseElementError	0x30000013	Getting number of database elements failed.
cloudAutoUpgradeException	0x30000016	Downloading upgrade packet from cloud and upgrading failed.
HBPException	0x30001000	HBP exception.
UDEPException	0x30001001	UDEP exception
elasticSearchException	0x30001002	Elastic exception.
kafkaException	0x30001003	Kafka exception.
HBaseException	0x30001004	Hbase exception.
sparkException	0x30001005	Spark exception.
yarnException	0x30001006	Yarn exception.
cacheException	0x30001007	Cache exception.
trafficException	0x30001008	Monitoring point big data server exception.
faceException	0x30001009	Human face big data server exception.
SSDFileSystemsIsError	0x30001013	SSD file system error (Error occurs when it is non-Ext4 file system)

Sub Status Code	Error Code	Description
insufficientSSDCapacityForFPD	0x30001014	Insufficient SSD space for person frequency detection
wifiException	0x3000100A	Wi-Fi big data server exception
structException	0x3000100D	Video parameters structure server exception.
calibrationTimeout	0x30002051	Calibration timed out.
captureTimeout	0x30006000	Data collection timed out.
lowScore	0x30006001	Low quality of collected data.
uploadingFailed	0x30007004	Uploading failed.

StatusCode=4

Sub Status Code	Error Code	Description
notSupport	0x40000001	Not supported.
lowPrivilege	0x40000002	No permission.
badAuthorization	0x40000003	Authentication failed.
methodNotAllowed	0x40000004	Invalid HTTP method.
notSetHdiskRedund	0x40000005	Setting spare HDD failed.
invalidOperation	0x40000006	Invalid operation.
notActivated	0x40000007	Inactivated.
hasActivated	0x40000008	Activated.
certificateAlreadyExist	0x40000009	The certificate already exists.
operateFailed	0x4000000F	Operation failed.
USBNotExist	0x40000010	USB device is not connected.
upgradePackageMorethan2GB	0x40001000	Up to 2GB upgrade package is allowed to be uploaded.
IDNotExist	0x40001001	The ID does not exist.
synchronizationError	0x40001003	Synchronization failed.
synchronizing	0x40001004	Synchronizing.
importError	0x40001005	Importing failed.
importing	0x40001006	Importing.

Sub Status Code	Error Code	Description
fileAlreadyExists	0x40001007	The file already exists.
invalidID	0x40001008	Invalid ID.
backupnodeNotAllowe Log	0x40001009	Accessing to backup node is not allowed.
exportingError	0x4000100A	Exporting failed.
exporting	0x4000100B	Exporting.
exportEnded	0x4000100C	Exporting stopped.
exported	0x4000100D	Exported.
IPOccupied	0x4000100E	The IP address is already occupied.
IDAlreadyExists	0x4000100F	The ID already exists.
exportItemsExceedLimi t	0x40001010	No more items can be exported.
noFiles	0x40001011	The file does not exist.
beingExportedByAnoth erUser	0x40001012	Being exported by others.
needReAuthentication	0x40001013	Authentication is needed after upgrade.
unitAddNotOnline	0x40001015	The added data analysis server is offline.
unitControl	0x40001016	The data analysis server is already added.
analysis unitFull	0x40001017	No more data analysis server can be added.
unitIDError	0x40001018	The data analysis server ID does not exist.
unitExit	0x40001019	The data analysis server already exists in the list.
unitSearch	0x4000101A	Searching data analysis server in the list failed.
unitNotOnline	0x4000101B	The data analysis server is offline.
unitInfoError	0x4000101C	Getting data analysis server information failed.
unitGetNodeInfoError	0x4000101D	Getting node information failed.
unitGetNetworkInfoErr or	0x4000101E	Getting the network information of data analysis server failed
unitSetNetworkInfoErr or	0x4000101F	Setting the network information of data analysis server failed

Sub Status Code	Error Code	Description
setSmartNodeInfoError	0x40001020	Setting node information failed.
setUnitNetworkInfoError	0x40001021	Setting data analysis server network information failed.
unitRestartCloseError	0x40001022	Rebooting or shutting down data analysis server failed.
virtualIPnotAllowed	0x40001023	Adding virtual IP address is not allowed.
unitInstalled	0x40001024	The data analysis server is already installed.
badSubnetMask	0x40001025	Invalid subnet mask.
uintVersionMismatched	0x40001026	Data analysis server version mismatches.
deviceModelMismatched	0x40001027	Adding failed. Device model mismatches.
unitAddNotSelf	0x40001028	Adding peripherals is not allowed.
noValidUnit	0x40001029	No valid data analysis server.
unitNameDuplicate	0x4000102A	Duplicated data analysis server name.
deleteUnitFirst	0x4000102B	Delete the added data analysis server of the node first.
getLocalInfoFailed	0x4000102C	Getting the server information failed.
getClientAddedNodeFailed	0x4000102D	Getting the added node information of data analysis server failed.
taskExit	0x4000102E	The task already exists.
taskInitError	0x4000102F	Initializing task failed.
taskSubmitError	0x40001030	Submitting task failed.
taskDelError	0x40001031	Deleting task failed.
taskPauseError	0x40001032	Pausing task failed.
taskContinueError	0x40001033	Starting task failed.
taskSeverNoCfg	0x40001035	Full-text search server is not configured.
taskPicSeverNoCfg	0x40001036	The picture server is not configured.
taskStreamError	0x40001037	Streaming information exception.
taskRecSDK	0x40001038	History recording is not supported.

Sub Status Code	Error Code	Description
taskCasaError	0x4000103A	Cascading is not supported.
taskVCARuleError	0x4000103B	Invalid VCA rule.
taskNoRun	0x4000103C	The task is not executed.
unitLinksNoStorageNode	0x4000103D	No node is linked with the data analysis server. Configure the node first.
searchFailed	0x4000103E	Searching video files failed.
searchNull	0x4000103F	No video clip.
userScheOffline	0x40001040	The task scheduler service is offline.
updateTypeUnmatched	0x40001041	The upgrade package type mismatches.
userExist	0x40001043	The user already exists.
userCannotDelAdmin	0x40001044	The administrator cannot be deleted.
userInexistence	0x40001045	The user name does not exist.
userCannotCreatAdmin	0x40001046	The administrator cannot be created.
monitorCamExceed	0x40001048	Up to 3000 cameras can be added.
monitorCunitOverLimit	0x40001049	Adding failed. Up to 5 lower-levels are supported by the control center.
monitorReginOverLimit	0x4000104A	Adding failed. Up to 5 lower-levels are supported by the area.
monitorArming	0x4000104B	The camera is already armed. Disarm the camera and try again.
monitorSyncCfgNotSet	0x4000104C	The system parameters are not configured.
monitorFdSyncing	0x4000104E	Synchronizing. Try again after completing the synchronization.
monitorParseFailed	0x4000104F	Parsing camera information failed.
monitorCreatRootFailed	0x40001050	Creating resource node failed.
deleteArmingInfo	0x40001051	The camera is already . Disarm the camera and try again.
cannotModify	0x40001052	Editing is not allowed. Select again.

Sub Status Code	Error Code	Description
cannotDel	0x40001053	Deletion is not allowed. Select again.
deviceExist	0x40001054	The device already exists.
IPErrorConnectFailed	0x40001056	Connection failed. Check the network port.
cannotAdd	0x40001057	Only the capture cameras can be added.
serverExist	0x40001058	The server already exists.
fullTextParamError	0x40001059	Incorrect full-text search parameters.
storParamError	0x4000105A	Incorrect storage server parameters.
picServerFull	0x4000105B	The storage space of picture storage server is full.
NTPUnconnect	0x4000105C	Connecting to NTP server failed. Check the parameters.
storSerConnectFailed	0x4000105D	Connecting to storage server failed. Check the network port.
storSerLoginFailed	0x4000105E	Logging in to storage server failed. Check the user name and password.
searchSerConnectFailed	0x4000105F	Connecting to full-text search server failed. Check the network port.
searchSerLoginFailed	0x40001060	Logging in to full-text search server failed. Check the user name and password.
kafkaConnectFailed	0x40001061	Connecting to Kafka failed. Check the network port.
mgmtConnectFailed	0x40001062	Connecting to system failed. Check the network port.
mgmtLoginFailed	0x40001063	Logging in to system failed. Check the user name and password.
TDAConnectFailed	0x40001064	Connecting to traffic data access server failed. Checking the server status.
86sdkConnectFailed	0x40001065	Connecting to listening port of iVMS-8600 System failed. Check the parameters.
nameExist	0x40001066	Duplicated server name.
batchProcessFailed	0x40001067	Processing in batch failed.
IDNotExist	0x40001068	The server ID does not exist.

Sub Status Code	Error Code	Description
serviceNumberReachedLimit	0x40001069	No more service can be added.
invalidServiceType.	0x4000106A	Invalid service type.
clusterGetInfo	0x4000106B	Getting cluster group information failed.
clusterDelNode	0x4000106C	Deletion node failed.
clusterAddNode	0x4000106D	Adding node failed.
clusterInstalling	0x4000106E	Creating cluster...Do not operate.
clusterUninstall	0x4000106F	Reseting cluster...Do not operate.
clusterInstall	0x40001070	Creating cluster failed.
clusterIpError	0x40001071	Invalid IP address of task scheduler server.
clusterNotSameSeg	0x40001072	The master node and slave node must be in the same network segment.
clusterVirIpError	0x40001073	Automatically getting virtual IP address failed. Enter manually.
clusterNodeUnadd	0x40001074	The specified master(slave) node is not added.
clusterNodeOffline	0x40001075	The task scheduler server is offline.
nodeNotCurrentIP	0x40001076	The analysis node of the current IP address is required when adding master and slave nodes.
addNodeNetFailed	0x40001077	Adding node failed. The network disconnected.
needTwoMgmtNode	0x40001078	Two management nodes are required when adding master and slave nodes.
ipConflict	0x40001079	The virtual IP address and data analysis server's IP address conflicted.
ipUsed	0x4000107A	The virtual IP address has been occupied.
cloudAnalyseOnline	0x4000107B	The cloud analytic server is online.
virIP&mainIPnotSameNetSegment	0x4000107C	The virtual IP address is not in the same network segment with the IP address of master/slave node.
getNodeDispatchInfoFailed	0x4000107D	Getting node scheduler information failed.
unableModifyManagementNetworkIP	0x4000107E	Editing management network interface failed. The analysis board is in the cluster.

Sub Status Code	Error Code	Description
notSpecifyVirtualIP	0x4000107F	Virtual IP address should be specified for master and slave cluster.
armingFull	0x40001080	No more device can be armed.
armingNoFind	0x40001081	The arming information does not exist.
disArming	0x40001082	Disarming failed.
getArmingError	0x40001084	Getting arming information failed.
refreshArmingError	0x40001085	Refreshing arming information failed.
ArmingPlateSame	0x40001086	The license plate number is repeatedly armed.
ArmingParseXLSError	0x40001087	Parsing arming information file failed.
ArmingTimeError	0x40001088	Invalid arming time period.
ArmingSearchTimeError	0x40001089	Invalid search time period.
armingRelationshipReachesLimit	0x4000108A	No more relation can be created.
duplicateArmingName	0x4000108B	The relation name already exists.
noMoreArmingListAdded	0x4000108C	No more blacklist library can be armed.
noMoreCamerasAdded	0x4000108D	No more camera can be armed.
noMoreArmingListAddedWithCamera	0x4000108E	No more library can be linked to the camera.
noMoreArmingPeriodAdded	0x4000108F	No more time period can be added to the arming schedule.
armingPeriodsOverlapped	0x40001090	The time periods in the arming schedule are overlapped.
noArmingAlarmInfo	0x40001091	The alarm information does not exist.
armingAlarmUnRead	0x40001092	Getting number of unread alarms failed.
getArmingAlarmError	0x40001093	Getting alarm information failed.
searchByPictureTimedOut	0x40001094	Searching picture by picture timeout. Search again.
comparisonTimeRangeError	0x40001095	Comparison time period error.

Sub Status Code	Error Code	Description
selectMonitorNumberUpperLimit	0x40001096	No more monitoring point ID can be filtered.
noMoreComparisonTasksAdded	0x40001097	No more comparison task can be executed at the same time.
GetComparisonResultFailed	0x40001098	Getting comparison result failed.
comparisonTypeError	0x40001099	Comparison type error.
comparisonUnfinished	0x4000109A	The comparison is not completed.
facePictureModelInvalid	0x4000109B	Invalid face model.
duplicateLibraryName.	0x4000109C	The library name already exists.
noRecord	0x4000109D	No record found.
countingRecordsFailed.	0x4000109E	Calculate the number of records failed.
getHumanFaceFrameFailed	0x4000109F	Getting face thumbnail from the picture failed.
modelingFailed.	0x400010A0	Modeling face according to picture URL failed.
1V1FacePictureComparisonFailed	0x400010A1	Comparison 1 VS 1 face picture failed.
libraryArmed	0x400010A2	The blacklist library is armed.
licenseExceedLimit	0x400010A3	Dongle limited.
licenseExpired	0x400010A4	Dongle expired.
licenseDisabled	0x400010A5	Unavailable dongle.
licenseNotExist	0x400010A6	The dongle does not exist.
SessionExpired	0x400010A7	Session expired .
beyondConcurrentLimit	0x400010A8	Out of concurrent limit.
stopSync	0x400010A9	Synchronization stopped.
getProgressFailed	0x400010AA	Getting progress failed.
uploadExtraCaps	0x400010AB	No more files can be uploaded.
timeRangeError	0x400010AC	Time period error.
dataPortNotConnected	0x400010AD	The data port is not connected.

Sub Status Code	Error Code	Description
addClusterNodeFailed	0x400010AE	Adding to the cluster failed. The device is already added to other cluster.
taskNotExist	0x400010AF	The task does not exist.
taskQueryFailed	0x400010B0	Searching task failed.
modifyTimeRuleFailed	0x400010B2	The task already exists. Editing time rule is not allowed.
modifySmartRuleFailed	0x400010B3	The task already exists. Editing VAC rule is not allowed.
queryHistoryVideoFailed	0x400010B4	Searching history video failed.
addDeviceFailed	0x400010B5	Adding device failed.
addVideoFailed	0x400010B6	Adding video files failed.
deleteAllVideoFailed	0x400010B7	Deleting all video files failed.
createVideoIndexFailed	0x400010B8	Indexing video files failed.
videoCheckTypeFailed	0x400010B9	Verifying video files types failed.
configStructuredAddressFailed	0x400010BA	Configuring IP address of structured server failed.
configPictureServerAddressFailed	0x400010BB	Configuring IP address of picture stored server failed.
storageServiceIPNotExist	0x400010BD	The storage server IP address does not exist.
syncBackupDatabaseFailed	0x400010BE	Synchronizing slave database failed. Try again.
syncBackupNTPTimeFailed	0x400010BF	Synchronizing NTP time of slave server failed.
clusterNotSelectLoopbackAddress	0x400010C0	Loopback address is not supported by the master or slave cluster.
addFaceRecordFailed	0x400010C1	Adding face record failed.
deleteFaceRecordFailed	0x400010C2	Deleting face record failed.
modifyFaceRecordFailed	0x400010C3	Editing face record failed.

Sub Status Code	Error Code	Description
queryFaceRecordFailed	0x400010C4	Searching face record failed.
faceDetectFailed	0x400010C5	Detecting face failed.
libraryNotExist	0x400010C6	The library does not exist.
blackListQueryExporting	0x400010C7	Exporting matched blacklists.
blackListQueryExported	0x400010C8	The matched blacklists are exported.
blackListQueryStopExporting	0x400010C9	Exporting matched blacklists is stopped.
blackListAlarmQueryExporting	0x400010CA	Exporting matched blacklist alarms.
blackListAlarmQueryExported	0x400010CB	The matched blacklists alarms are exported.
blackListAlarmQueryStopExporting	0x400010CC	Exporting matched blacklist alarms is stopped.
getBigDataCloudAnalysisFailed	0x400010CD	Getting big data cloud analytic information failed.
setBigDataCloudAnalysisFailed	0x400010CE	Configuring big data cloud analytic failed.
submitMapSearchFailed	0x400010CF	Submitting search by picture task failed.
controlRelationshipNotExist	0x400010D0	The relation does not exist.
getHistoryAlarmInfoFailed	0x400010D1	Getting history alarm information failed.
getFlowReportFailed	0x400010D2	Getting people counting report failed.
addGuardFailed	0x400010D3	Adding arming configuration failed.
deleteGuardFailed	0x400010D4	Deleting arming configuration failed.
modifyGuardFailed	0x400010D5	Editing arming configuration failed.
queryGuardFailed	0x400010D6	Searching arming configurations failed.
uploadUserSuperCaps	0x400010D7	No more user information can be uploaded.

Sub Status Code	Error Code	Description
bigDataServerConnectFailed	0x400010D8	Connecting to big data server failed.
microVideoCloudRequestInfoBuildFailed	0x400010D9	Adding response information of micro video cloud failed.
microVideoCloudResponseInfoBuildFailed	0x400010DA	Parsing response information of micro video cloud failed.
transcodingServerRequestInfoBuildFailed	0x400010DB	Adding response information of transcoding server failed.
transcodingServerResponseInfoParseFailed	0x400010DC	Parsing response information of transcoding server failed.
transcodingServerOffline	0x400010DD	Transcoding server is offline.
microVideoCloudOffline	0x400010DE	Micro video cloud is offline.
UPSServerOffline	0x400010DF	UPS monitor server is offline.
statisticReportRequestInfoBuildFailed	0x400010E0	Adding response information of statistics report failed.
statisticReportResponseInfoParseFailed	0x400010E1	Parsing response information of statistics report failed.
DisplayConfigInfoBuildFailed	0x400010E2	Adding display configuration information failed.
DisplayConfigInfoParseFailed	0x400010E3	Parsing display configuration information failed.
DisplayConfigInfoSaveFailed	0x400010E4	Saving display configuration information failed.
notSupportDisplayConfigType	0x400010E5	The display configuration type is not supported.
passError	0x400010E7	Incorrect password.
upgradePackageLarge	0x400010EB	Too large upgrade package.
sessionUserReachesLimit	0x400010EC	No more user can log in via session.
ISO8601TimeFormatError	0x400010ED	Invalid ISO8601 time format.

Sub Status Code	Error Code	Description
clusterDissolutionFailed	0x400010EE	Deleting cluster failed.
getServiceNodeInfoFailed	0x400010EF	Getting service node information failed.
getUPSInfoFailed	0x400010F0	Getting UPS configuration information failed.
getDataStatisticsReportFailed	0x400010F1	Getting data statistic report failed.
getDisplayConfigInfoFailed	0x400010F2	Getting display configuration failed.
namingAnalysisBoardNotAllowed	0x400010F3	Renaming analysis board is not allowed.
onlyDrawRegionsOfConvexPolygon	0x400010F4	Only drawing convex polygon area is supported.
bigDataServerResponseInfoParseFailed	0x400010F5	Parsing response message of big data service failed.
bigDataServerReturnFailed	0x400010F6	No response is returned by big data service.
microVideoReturnFailed	0x400010F7	No response is returned by micro video cloud service.
transcodingServerReturnFailed	0x400010F8	No response is returned by transcoding service.
UPSServerReturnFailed	0x400010F9	No response is returned by UPS monitoring service.
forwardingServerReturnFailed	0x400010FA	No response is returned by forwarding service.
storageServerReturnFailed	0x400010FB	No response is returned by storage service.
cloudAnalysisServerReturnFailed	0x400010FC	No response is returned by cloud analytic service.
modelEmpty	0x400010FD	No model is obtained.
mainAndBackupNodeCannotModifyManagementNetworkInterfaceIP	0x400010FE	Editing the management interface IP address of master node and backup node is not allowed.

Sub Status Code	Error Code	Description
IDTooLong	0x400010FF	The ID is too long.
pictureCheckFailed	0x40001100	Detecting picture failed.
pictureModelingFailed	0x40001101	Modeling picture failed.
setCloudAnalsisDefaultProvinceFailed	0x40001102	Setting default province of cloud analytic service failed.
InspectionAreasNumberExceedLimit	0x40001103	No more detection regions can be added.
picturePixelsTooLarge	0x40001105	The picture resolution is too high.
picturePixelsTooSmall	0x40001106	The picture resolution is too low.
storageServiceIPEmpty	0x40001107	The storage server IP address is required.
bigDataServerRequestInfoBuildFail	0x40001108	Creating request message of big data service failed.
analysisITimedOut	0x40001109	Analysis time out.
high-performanceModeDisabled.	0x4000110A	Please enable high-performance mode.
configuringUPSMonitoringServerTimedOut	0x4000110B	Configuring the UPS monitoring server time out. Check IP address.
cloudAnalysisRequestInformationBuildFailed	0x4000110C	Creating request message of cloud analytic service failed.
cloudAnalysisResponseInformationParseFailed	0x4000110D	Parsing response message of cloud analytic service failed.
allCloudAnalysisInterfaceFailed	0x4000110E	Calling API for cloud analytic service failed.
cloudAnalysisModelCompareFailed	0x4000110F	Model comparison of cloud analytic service failed.
cloudAnalysisFacePictureQualityRatingFailed	0x40001110	Getting face quality grading of cloud analytic service failed.
cloudAnalysisExtractFeaturePointsFailed	0x40001111	Extracting feature of cloud analytic service failed.
cloudAnalysisExtractPropertyFailed	0x40001112	Extracting property of cloud analytic service failed.

Sub Status Code	Error Code	Description
getAddedNodeInformationFailed	0x40001113	Getting the added nodes information of data analysis server failed.
noMoreAnalysisUnitsAdded	0x40001114	No more data analysis servers can be added.
detectionAreaInvalid	0x40001115	Invalid detection region.
shieldAreaInvalid	0x40001116	Invalid shield region.
noMoreShieldAreasAdded	0x40001117	No more shield region can be drawn.
onlyAreaOfRectangleShapeAllowed	0x40001118	Only drawing rectangle is allowed in detection area.
numberReachedLimit	0x40001119	Number reached the limit.
wait1~3MinutesGetIPAfterSetupDHCP	0x4000111A	Wait 1 to 3 minutes to get IP address after configuring DHCP.
plannedTimeMustbeHalfAnHour	0x4000111B	Schedule must be half an hour.
oneDeviceCannotBuildCluster	0x4000111C	Creating master and backup cluster requires at least two devices.
updatePackageFileNotUploaded	0x4000111E	Upgrade package is not uploaded.
highPerformanceTasksNotSupportDrawingDetectionRegions	0x4000111F	Drawing detection area is not allowed under high-performance mode.
controlCenterIDDoesNotExist	0x40001120	The control center ID does not exist.
regionIDDoesNotExist	0x40001121	The area ID does not exist.
licensePlateFormatError	0x40001122	Invalid license plate format.
managementNodesNotSupportThisOperation	0x40001123	The operation is not supported.
searchByPictureResourceNotConfiged	0x40001124	The conditions for searching picture by picture are not configured.

Sub Status Code	Error Code	Description
videoFileEncapsulationFormatNotSupported	0x40001125	The video container format is not supported.
videoPackageFailure	0x40001126	Converting video container format failed.
videoCodingFormatNotSupported	0x40001127	Video coding format is not supported.
monitorOfDeviceArmingdeleteArmingInfo	0x40001129	The camera is armed. Disarm it and try again.
getVideoSourceTypeFailed	0x4000112A	Getting video source type failed.
smartRulesBuildFailed	0x4000112B	Creating VAC rule failed.
smartRulesParseFailed	0x4000112C	Parsing VAC rule failed.
timeRulesBuildFailed	0x4000112D	Creating time rule failed.
timeRulesParseFailed	0x4000112E	Parsing time rule failed.
monitoInfoInvalid	0x4000112F	Invalid camera information.
addingFailedVersionMismatches	0x40001130	Adding failed. The device version mismatches.
theInformationReturnedAfterCloudAnalysisIsEmpty	0x40001131	No response is returned by the cloud analytic service.
selectingIpAddressOfHostAndSpareNodeFailedCheckTheStatus	0x40001132	Setting IP address for master node and backup node failed. Check the node status.
theSearchIdDoesNotExist	0x40001133	The search ID does not exist.
theSynchronizationIdDoesNotExist	0x40001134	The synchronization ID does not exist.
theUserIdDoesNotExist	0x40001136	The user ID does not exist.
theIndexCodeDoesNotExist	0x40001138	The index code does not exist.
theControlCenterIdDoesNotExist	0x40001139	The control center ID does not exist.
theAreaIdDoesNotExist	0x4000113A	The area ID does not exist.

Sub Status Code	Error Code	Description
theArmingLinkageIdDoesNotExist	0x4000113C	The arming relationship ID does not exist.
theListLibraryIdDoesNotExist	0x4000113D	The list library ID does not exist.
invalidCityCode	0x4000113E	Invalid city code.
synchronizingThePasswordOfSpareServerFailed	0x4000113F	Synchronizing backup system password failed.
editingStreamingTypeIsNotSupported	0x40001140	Editing streaming type is not supported.
switchingScheduledTaskToTemporaryTaskIsNotSupported	0x40001141	Switching scheduled task to temporary task is not supported.
switchingTemporaryTaskToScheduledTaskIsNotSupported	0x40001142	Switching temporary task to scheduled task is not supported.
theTaskIsNotDispatchedOrItIsUpdating	0x40001143	The task is not dispatched or is updating.
thisTaskDoesNotExist	0x40001144	This task does not exist in the cloud analytic service.
duplicatedSchedule	0x40001145	Schedule period cannot be overlapped.
continuousScheduleWithSameAlgorithmTypeShouldBeMerged	0x40001146	The continuous schedule periods with same algorithm type should be merged.
invalidStreamingTimeRange	0x40001147	Invalid streaming time period.
invalidListLibraryType	0x40001148	Invalid list library type.
theNumberOfMatchedResultsShouldBeLargerThan0	0x40001149	The number of search results should be larger than 0.
invalidValueRangeOfSimilarity	0x4000114A	Invalid similarity range.
invalidSortingType	0x4000114B	Invalid sorting type.

Sub Status Code	Error Code	Description
noMoreListLibraryCanBeLinkedToTheDevice	0x4000114C	No more lists can be added to one device.
InvalidRecipientAddressFormat	0x4000114D	Invalid address format of result receiver.
creatingClusterFailedTheDongleIsNotPluggedIn	0x4000114E	Insert the dongle before creating cluster.
theURLIsTooLong	0x4000114F	No schedule configured for the task.
noScheduleIsConfiguredForTheTask	0x40001150	No schedule configured for the task.
theDongleIsExpired	0x40001151	Dongle has expired.
dongleException	0x40001152	Dongle exception.
invalidKey	0x40001153	Invalid authorization service key.
decryptionFailed	0x40001154	Decrypting authorization service failed.
encryptionFailed	0x40001155	Encrypting authorization service failed.
AuthorizeServiceResponseError	0x40001156	Authorization service response exception.
incorrectParameter	0x40001157	Authorization service parameters error.
operationFailed	0x40001158	Operating authorization service error.
noAnalysisResourceOrNoDataInTheListLibrary	0x40001159	No cloud analytic resources or no data in the list library.
calculationException	0x4000115A	Calculation exception.
allocatingList	0x4000115B	Allocating list.
thisOperationIsNotSupportedByTheCloudAnalytics	0x4000115C	This operation is not supported by the cloud analytic service.
theCloudAnalyticsIsInterrupted	0x4000115D	The operation of cloud analytic service is interrupted.
theServiceIsNotReady	0x4000115E	The service is not ready.
searchingForExternalApiFailed	0x4000115F	Searching external interfaces failed.
noOnlineNode	0x40001160	No node is online.

Sub Status Code	Error Code	Description
noNodeAllocated	0x40001161	No allocated node.
noMatchedList	0x40001162	No matched list.
allocatingFailedTooManyFacePictureLists	0x40001163	Allocation failed. Too many lists of big data service.
searchIsNotCompletedSearchAgain	0x40001164	Current searching is not completed. Search again.
allocatingListIsNotCompleted	0x40001165	Allocating list is not completed.
searchingForCloudAnalyticsResultsFailed	0x40001166	Searching cloud analytic service overtime.
noDataOfTheCurrentLibraryFound	0x40001167	No data in the current library. Make sure there is data in the Hbase.
noFacePictureLibraryIsArmed	0x40001168	No face picture library is armed for big data service.
noAvailableDataSlicingVersionInformationArmedFirstAndSliceTheData	0x40001169	Invalid standard version information.
duplicatedOperationDataSlicingIsExecuting	0x4000116A	Slicing failed. Duplicated operation.
slicingDataFailedNoArmedFacePictureLibrary	0x4000116B	Slicing failed. No arming information in the face big data.
GenerateBenchmarkFileFailedSlicingAgain	0x4000116C	Generating sliced file failed. Slice again.
NonprimaryNodesProhibitedFromSlicingData	0x4000116D	Slicing is not allowed by the backup node.
NoReadyNodeToClusterServers	0x4000116E	Creating the cluster failed. No ready node.
NodeManagementServicesOffline	0x4000116F	The node management server is offline.
theCamera(s)OfTheControlCenterAreAlreadyArmed.DisarmThemFirst	0x40001170	Some cameras in control center are already armed. Disarm them and try again.

Sub Status Code	Error Code	Description
theCamera(s)OfTheAreaAreAlreadyArmed.DisarmThemFirst	0x40001171	Some cameras in this area are already armed. Disarm them and try again.
configuringHigh-frequencyPeopleDetectionFailed	0x40001172	Configuring high frequency people detection failed.
searchingForHigh-frequencyPeopleDetectionLogsFailed.	0x40001173	Searching detection event logs of high-frequency people detection failed.
gettingDetailsOfSearchHigh-frequencyPeopleDetectionLogsFailed.	0x40001174	Getting the search result details of high frequency alarms failed.
theArmedCamerasAlreadyExistInTheControlCenter	0x40001175	Some cameras in control center are already armed.
disarmingFailedTheCamerasAreNotArmed	0x40001177	Disarming failed. The camera is not armed.
noDataReturned	0x40001178	No response is returned by the big data service.
preallocFailure	0x40001179	Pre-allocating algorithm resource failed.
overDogLimit	0x4000117A	Configuration failed. No more resources can be pre-allocated.
analysisServicesDoNotSupport	0x4000117B	Not supported.
commandAndDispatchServiceError	0x4000117C	Scheduling service of cloud analytic service error.
engineModuleError	0x4000117D	Engine module of cloud analytic service error.
streamingServiceError	0x4000117E	Streaming component of cloud analytic service error.
faceAnalysisModuleError	0x4000117F	Face analysis module of cloud analytic service error.
vehicleAnalysisModuleError	0x40001180	Vehicle pictures analytic module of cloud analytic service error.

Sub Status Code	Error Code	Description
videoStructuralAnalysisModuleError	0x40001181	Video structuring module of cloud analytic service error.
postprocessingModuleError	0x40001182	Post-processing module of cloud analytic service error.
frequentlyAppearedPersonAlarmsAlreadyConfiguredForListLibrary	0x40001183	High frequency alarm is already armed for blacklist library.
creatingListLibraryFailed	0x40001184	Creating list library failed.
invalidIdentityKeyOfListLibrary	0x40001185	Invalid identity key of list library.
noMoreDevicesCanBeArmed	0x40001186	No more camera can be added.
settingAlgorithmTypeForDeviceFailed	0x40001187	Allocating task resource failed.
gettingHighFrequencyPersonDetectionAlarmInformationFailed	0x40001188	Setting high frequency alarm failed.
invalidSearchCondition	0x40001189	Invalid result.
theTaskIsNotCompleted	0x4000118B	The task is not completed.
resourceOverRemainLimit	0x4000118C	No more resource can be pre-allocated.
frequentlyAppearedPersonAlarmsAlreadyConfiguredForTheCameraDisarmFirstAndTryAgain	0x4000118D	The high frequency alarm of this camera is configured. Delete the arming information and try again.
switchTimeDifflessLimit	0x4000123b	Time difference between power on and off should be less than 10 minutes.
associatedFaceLibNumOverLimit	0x40001279	Maximum number of linked face picture libraries reached.
noMorePeopleNumChangeRulesAdded	0x4000128A	Maximum number of people number changing rules reached.

Sub Status Code	Error Code	Description
noMoreViolentMotionRulesAdded	0x4000128D	Maximum number of violent motion rules reached.
noMoreLeavePositionRulesAdded	0x4000128E	Maximum number of leaving position rules reached.
SMRDiskNotSupportRaid	0x40001291	SMR disk does not support RAID.
OnlySupportHikAndCustomProtocol	0x400012A3	IPv6 camera can only be added via Device Network SDK or custom protocols.
vehicleEnginesNoResource	0x400012A6	Insufficient vehicle engine resources.
noMoreRunningRulesAdded	0x400012A9	Maximum number of running rules reached.
noMoreGroupRulesAdded	0x400012AA	Maximum number of people gathering rules reached.
noMoreFailDownRulesAdded	0x400012AB	Maximum number of people falling down rules reached.
noMorePlayCellphoneRulesAdded	0x400012AC	Maximum number of playing cellphone rules reached.
ruleEventTypeDuplicate	0x400012C8	Event type duplicated.
noMoreRetentionRulesAdded	0x400015AD	Maximum number of people retention rules reached.
noMoreSleepOnDutyRulesAdded	0x400015AE	Maximum number of sleeping on duty rules reached.
polygonNotAllowedCrossing	0x400015C2	Polygons are not allowed to cross.
AITargetBPCaptureFail	0x400019C5	Capturing reference picture for AI target comparison failed.
AITargetBPToDSPFail	0x400019C6	Sending reference picture to DSP for AI target comparison failed.
AITargetBPDuplicateName	0x400019C7	Duplicated name of reference picture for AI target comparison.
audioFileNameWrong	0x400019D0	Incorrect audio file name.

Sub Status Code	Error Code	Description
audioFileImportFail	0x400019D1	Importing audio file failed.
alreadyRunning	0x40002026	The application program is running.
notRunning	0x40002027	The application program is stopped.
packNotFound	0x40002028	The software packet does not exist.
alreadyExist	0x40002029	The application program already exists.
noMemory	0x4000202A	Insufficient memory.
invalidLicense	0x4000202B	Invalid License.
noClientCertificate	0x40002036	The client certificate is not installed.
noCACertificate	0x40002037	The CA certificate is not installed.
authenticationFailed	0x40002038	Authenticating certificate failed. Check the certificate.
clientCertificateExpired	0x40002039	The client certificate is expired.
clientCertificateRevocation	0x4000203A	The client certificate is revoked.
CACertificateExpired	0x4000203B	The CA certificate is expired.
CACertificateRevocation	0x4000203C	The CA certificate is revoked.
connectFail	0x4000203D	Connection failed.
loginNumExceedLimit	0x4000203F	No more user can log in.
HDMIResolutionIllegal	0x40002040	The HDMI video resolution cannot be larger than that of main and sub stream.
hdFormatFail	0x40002049	Formatting HDD failed.
formattingFailed	0x40002056	Formatting HDD failed.
encryptedFormattingFailed	0x40002057	Formatting encrypted HDD failed.
wrongPassword	0x40002058	Verifying password of SD card failed. Incorrect password.
audiolsPlayingPleaseWait	0x40002067	Audio is playing. Please wait.
twoWayAudioInProgressPleaseWait	0x40002068	Two-way audio in progress. Please wait.

Sub Status Code	Error Code	Description
calibrationPointNumFull	0x40002069	The maximum number of calibration points reached.
completeTheLevelCalibrationFirst	0x4000206A	The level calibration is not set.
completeTheRadarCameraCalibrationFirst	0x4000206B	The radar-camera calibration is not set.
pointsOnStraightLine	0x4000209C	Calibrating failed. The calibration points cannot be one the same line.
TValueLessThanOrEqualZero	0x4000209D	Calibration failed. The T value of the calibration points should be larger than 0.
HBDLibNumOverLimit	0x40002092	The number of human body picture libraries reaches the upper limit
theShieldRegionError	0x40002093	Saving failed. The shielded area should be the ground area where the shielded object is located.
theDetectionAreaError	0x40002094	Saving failed. The detection area should only cover the ground area.
invalidLaneLine	0x40002096	Saving failed. Invalid lane line.
enableITSFunctionOfThisChannelFirst	0x400020A2	Enable ITS function of this channel first.
noCloudStorageServer	0x400020C5	No cloud storage server
NotSupportWithVideoTask	0x400020F3	This function is not supported.
noDetectionArea	0x400050df	No detection area
armingFailed	0x40008000	Arming failed.
disarmingFailed	0x40008001	Disarming failed.
clearAlarmFailed	0x40008002	Clearing alarm failed.
bypassFailed	0x40008003	Bypass failed.
bypassRecoverFailed	0x40008004	Bypass recovery failed.
outputsOpenFailed	0x40008005	Opening relay failed.
outputsCloseFailed	0x40008006	Closing relay failed.
registerTimeOut	0x40008007	Registering timed out.

Sub Status Code	Error Code	Description
registerFailed	0x40008008	Registering failed.
addedByOtherHost	0x40008009	The peripheral is already added by other security control panel.
alreadyAdded	0x4000800A	The peripheral is already added.
armedStatus	0x4000800B	The partition is armed.
bypassStatus	0x4000800C	Bypassed.
zoneNotSupport	0x4000800D	This operation is not supported by the zone.
zoneFault	0x4000800E	The zone is in fault status.
pwdConflict	0x4000800F	Password conflicted.
audioTestEntryFailed	0x40008010	Enabling audio test mode failed.
audioTestRecoveryFailed	0x40008011	Disabling audio test mode failed.
addCardMode	0x40008012	Adding card mode.
searchMode	0x40008013	Search mode.
addRemoterMode	0x40008014	Adding keyfob mode.
registerMode	0x40008015	Registration mode.
exDevNotExist	0x40008016	The peripheral does not exist.
theNumberOfExDevLimited	0x40008017	No peripheral can be added.
sirenConfigFailed	0x40008018	Setting siren failed.
chanCannotRepeatedBinded	0x40008019	This channel is already linked by the zone.
inProgramMode	0x4000801B	The keypad is in programming mode.
inPaceTest	0x4000801C	In pacing mode.
arming	0x4000801D	Arming.
masterSlavelsEnable	0x4000802c	The master-slave relationship has taken effect, the slave radar does not support this operation.
forceTrackNotEnabled	0x4000802d	Mandatory tracking is disabled.
isNotSupportZoneConfigByLocalArea	0x4000802e	This area does not support the zone type.

Sub Status Code	Error Code	Description
alarmLineCross	0x4000802f	Trigger lines are overlapped.
zoneDrawingOutOfRange	0x40008030	The drawn zone is out of detection range.
alarmLineDrawingOutOfRange	0x40008031	The drawn alarm trigger line is out of detection range.
hasTargetInWarningArea	0x40008032	The warning zone already contains targets. Whether to enable mandatory arming?
radarModuleConnectFail	0x40008033	Radar module communication failed.
importCfgFilePasswordErr	0x40008034	Incorrect password for importing configuration files.
overAudioFileNumLimit	0x40008038	The number of audio files exceeds the limit.
audioFileNameIsLong	0x40008039	The audio file name is too long.
audioFormatIsWrong	0x4000803a	The audio file format is invalid.
audioFileIsLarge	0x4000803b	The size of the audio file exceeds the limit.
pircamCapTimeOut	0x4000803c	Capturing of pircam timed out.
pircamCapFail	0x4000803d	Capturing of pircam failed.
pircamIsCaping	0x4000803e	The pircam is capturing.
audioFileHasExisted	0x4000803f	The audio file already exists.
subscribeTypeErr	0x4000a016	This metadata type is not supported to be subscribed.
startAppFail	/	Starting running application program failed.
yuvconflict	/	The raw video stream conflicted.
overMaxAppNum	/	No more application program can be uploaded.
noFlash	/	Insufficient flash.
noFlash	/	The platform mismatches.

Status Code=5

Sub Status Code	Error Code	Description
badXmlFormat	0x50000001	Invalid XML format.

StatusCode=6

Sub Status Code	Error Code	Description
badParameters	0x60000001	Invalid parameter.
badHostAddress	0x60000002	Invalid host IP address.
badXmlContent	0x60000003	Invalid XML content.
badIPv4Address	0x60000004	Invalid IPv4 address.
badIPv6Address	0x60000005	Invalid IPv6 address.
conflictIPv4Address	0x60000006	IPv4 address conflicted.
conflictIPv6Address	0x60000007	IPv6 address conflicted.
badDomainName	0x60000008	Invalid domain name.
connectSreverFail	0x60000009	Connecting to server failed.
conflictDomainName	0x6000000A	Domain name conflicted.
badPort	0x6000000B	Port number conflicted.
portError	0x6000000C	Port error.
exportErrorData	0x6000000D	Importing data failed.
badNetMask	0x6000000E	Invalid sub-net mask.
badVersion	0x6000000F	Version mismatches.
badDevType	0x60000010	Device type mismatches.
badLanguage	0x60000011	Language mismatches.
incorrentUserNameOrPasswor d	0x600000012	Incorrect user name or password.
invalidStoragePoolOfCloudServ er	0x600000013	Invalid storage pool. The storage pool is not configured or incorrect ID.
noFreeSpaceOfStoragePool	0x600000014	Storage pool is full.
riskPassword	0x600000015	Risky password.
UnSupportCapture	0x600000016	Capturing in 4096*2160 or 3072*2048 resolution is not supported when H.264+ is enabled.

Sub Status Code	Error Code	Description
userPwdLenUnder8	0x60000023	At least two kinds of characters, including digits, letters, and symbols, should be contained in the password.
userPwdNameSame	0x60000025	Duplicated password.
userPwdNameMirror	0x60000026	The password cannot be the reverse order of user name.
beyondARGSRangeLimit	0x60000027	The parameter value is out of limit.
DetectionLineOutOfDetectionRegion	0x60000085	The rule line is out of region.
DetectionRegionError	0x60000086	Rule region error. Make sure the rule region is convex polygon.
DetectionRegionOutOfCountingRegion	0x60000087	The rule region must be marked as red frame.
PedalAreaError	0x60000088	The pedal area must be in the rule region.
DetectionAreaABError	0x60000089	The detection region A and B must be in the a rule frame.
ABRegionCannotIntersect	0x6000008a	Region A and B cannot be overlapped.
customHBPIDError	0x6000008b	Incorrect ID of custom human body picture library
customHBPIDRepeat	0x6000008c	Duplicated ID of custom human body picture library
dataVersionsInHBDLibMismatches	0x6000008d	Database versions mismatches of human body picture library
invalidHBPID	0x6000008e	Invalid human body picture PID
invalidHBDID	0x6000008f	Invalid ID of human body picture library
humanLibraryError	0x60000090	Error of human body picture library

Sub Status Code	Error Code	Description
humanLibraryNumError	0x60000091	No more human body picture library can be added
humanImagesNumError	0x60000092	No more human body picture can be added
noHumanInThePicture	0x60000093	Modeling failed, no human body in the picture
analysisEnginesNoResourceError	0x60001000	No analysis engine.
analysisEnginesUsageExcced	0x60001001	The engine usage is overloaded.
PicAnalysisNoResourceError	0x60001002	No analysis engine provided for picture secondary recognition.
analysisEnginesLoadingError	0x60001003	Initializing analysis engine.
analysisEnginesAbnormaError	0x60001004	Analysis engine exception.
analysisEnginesFacelibImporting	0x60001005	Importing pictures to face picture library. Failed to edit analysis engine parameters.
analysisEnginesAssociatedChannel	0x60001006	The analysis engine is linked to channel.
smdEncodingNoResource	0x60001007	Insufficient motion detection encoding resources.
smdDecodingNoResource	0x60001008	Insufficient motion detection decoding resources.
diskError	0x60001009	HDD error.
diskFull	0x6000100a	HDD full.
facelibDataProcessing	0x6000100b	Handling face picture library data.
capturePackageFailed	0x6000100c	Capturing packet failed.
capturePackageProcessing	0x6000100d	Capturing packet.
noSupportWithPlaybackAbstract	0x6000100e	This function is not supported. Playback by video synopsis is enabled.

Sub Status Code	Error Code	Description
insufficientNetworkBandwidth	0x6000100f	Insufficient network bandwidth.
tapeLibNeedStopArchive	0x60001010	Stop the filing operation of tape library first.
identityKeyError	0x60001011	Incorrect interaction command.
identityKeyMissing	0x60001012	The interaction command is lost.
noSupportWithPersonDensityDetect	0x60001013	This function is not supported. The people density detection is enabled.
ipcResolutionOverflow	0x60001014	The configured resolution of network camera is invalid.
ipcBitrateOverflow	0x60001015	The configured bit rate of network camera is invalid.
tooGreatTimeDifference	0x60001016	Too large time difference between device and server.
noSupportWithPlayback	0x60001017	This function is not supported. Playback is enabled.
channelNoSupportWithSMD	0x60001018	This function is not supported. Motion detection is enabled.
channelNoSupportWithFD	0x60001019	This function is not supported. Face capture is enabled.
illegalPhoneNumber	0x6000101a	Invalid phone number.
illegalCertificateNumber	0x6000101b	Invalid certificate No.
linkedCameraOutLimit	0x6000101c	Connecting camera timed out.
achieveMaxChannelLimit	0x6000101e	No more channels are allowed.
humanMisInfoFilterEnabledChannelNumError	0x6000101f	No more channels are allowed to enable preventing false alarm.
humanEnginesNoResource	0x60001020	Insufficient human body analysis engine resources.
taskNumberOverflow	0x60001021	No more tasks can be added.

Sub Status Code	Error Code	Description
collisionTimeOverflow	0x60001022	No more comparison duration can be configured.
invalidTaskID	0x60001023	Invalid task ID.
eventNotSupport	0x60001024	Event subscription is not supported.
invalidEZVIZSecretKey	0x60001034	Invalid verification code for Hik-Connect.
needDoubleVerification	0x60001042	Double verification required
noDoubleVerificationUser	0x60001043	No double verification user
timeSpanNumOverLimit	0x60001044	Max. number of time buckets reached
channelNumOverLimit	0x60001045	Max. number of channels reached
noSearchIDResource	0x60001046	Insufficient searchID resources
noSupportDeleteStrangerLib	0x60001051	Deleting stranger library is not supported
noSupportCreateStrangerLib	0x60001052	Creating stranger library is not supported
behaviorAnalysisRuleInfoError	0x60001053	Behavior analysis rule parameters error.
safetyHelmetParamError	0x60001054	Hard hat parameters error.
OneChannelOnlyCanBindOneEngine	0x60001077	No more engines can be bound.
engineTypeMismatch	0x60001079	Engine type mismatched.
badUpgradePackage	0x6000107A	Invalid upgrade package.
AudioFileNameDuplicate	0x60001135	Duplicated audio file name.
CurrentAudioFileAIRuleInUseAlreadyDelete	0x60001136	The AI rule linkage related to current audio file has been deleted.
TransitionUseEmmc	0x60002000	Starting device failed. The EMMC is overused.
AdaptiveStreamNotEnabled	0x60002001	The stream self-adaptive function is not enabled.

Sub Status Code	Error Code	Description
AdaptiveStreamAndVariableBitrateEnabled	0x60002002	Stream self-adaptive and variable bitrate function cannot be enabled at the same time.
noSafetyHelmetRegion	0x60002023	The hard hat detection area is not configured (if users save their settings without configuring the arming area, they should be prompted to configure one).
unclosedSafetyHelmet	0x60002024	The hard hat detection is enabled (If users save their settings after deleting the arming area, they should be prompted to disable hard hat detection first and then delete the arming area).
width/heightRatioOfPictureError	0x6000202C	The width/height ratio of the uploaded picture should be in the range from 1:2 to 2:1.
PTZNotInitialized	0x6000202E	PTZ is not initialized.
PTZSelfChecking	0x6000202F	PTZ is self-checking.
PTZLocked	0x60002030	PTZ is locked.
advancedParametersError	0x60002031	Auto-switch interval in advanced parameters cannot be shorter than parking tolerance for illegal parking detection in speed dome rule settings.
resolutionError	0x60005003	Invalid resolution
deployExceedMax	0x60006018	The arming connections exceed the maximum number.
detectorTypeMismatch	0x60008000	The detector type mismatched.
nameExist	0x60008001	The name already exists.
uploadImageSizeError	0x60008016	The size of the uploaded picture is larger than 5 MB.

Sub Status Code	Error Code	Description
laneAndRegionOverlap	/	The lanes are overlapped.
unitConfigurationNotInEffect	/	Invalid unit parameter.
ruleAndShieldingMaskConflict	/	The line-rule region overlaps with the shielded area.
wholeRuleInShieldingMask	/	There are complete temperature measurement rules in the shielded area.
LogDiskNotSetReadOnlyInGroupMode	0x60001100	The log HDD in the HDD group cannot be set to read-only.
LogDiskNotSetRedundancyInGroupMode	0x60001101	The log HDD in the HDD group cannot be set to redundancy.

StatusCode=7

SubStatusCode	Error Code	Description
rebootRequired	0x70000001	Reboot to take effect.

